

TI-Nspire CAS OS3 and Casio ClassPad version



ESSENTIAL

Mathematical Methods **3 & 4 CAS**

ENHANCED

MICHAEL EVANS

KAY LIPSON

PETER JONES

SUE AVERY

CAS calculator material prepared in collaboration with

Jan Honnens

David Hibbard

Russell Brown



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Introduction

This book provides a complete course for Mathematical Methods Units 3 and 4 CAS. It has been written as a teaching text, with understanding as its chief aim and with ample practice offered through the worked examples and exercises. All the work has been trialled in the classroom, and the approaches offered are based on classroom experience.

The book contains three revision chapters. These provide multiple-choice questions and extended-response questions. Chapter 20 contains 42 extended-response questions which may be used for revision.

Use of a CAS calculator has been included throughout the text and there is also an appendix that provides an introduction to the use of the calculator. The use of matrices to describe transformations, solve systems of linear equations and in the study of Markov sequences is fully integrated. The study of families of functions is also incorporated throughout the text.

Extended-response questions that require a CAS calculator have been incorporated. These questions are indicated by the use of a CAS calculator icon.

The TI-Nspire calculator instructions have been completed by Jan Honnens and the Casio ClassPad instructions have been completed by David Hibbard.

The TI-Nspire instructions are written for operating system 3.0 but can be used with other versions.

The Casio ClassPad instructions are written for operating system 3 or above.





Author profiles

Michael Evans now works at the Australian Mathematical Sciences Institute based at the University of Melbourne. He was Head of Mathematics at Scotch College for many years and has been heavily involved in curriculum development at both Victorian state and national levels. Michael is a highly experienced writer of mathematical texts and is a lead author on a number of the texts in the Essential VCE Mathematics series.

Peter Jones has been an active supporter of school mathematics over many years. Peter is a highly experienced writer of mathematical texts and he is the lead author on two of the Essential Mathematics textbooks. His area of expertise is applied statistics.

Professor Kay Lipson's experience extends through teaching mathematics and statistics at both secondary and tertiary level, as well as extensive periods as a VCE examiner. At the time of publication she was the Academic Dean for Swinburne University Online.

The late Sue Avery was an experienced VCE Mathematics teacher and a key contributor to the Essential VCE Mathematics series. She conducted student seminars on VCE preparation in Maths Methods and Specialist Maths and had also been involved in maths research work for the VCAA.

CHAPTER

6

Trigonometric functions

Objectives

- To measure angles in degrees and radians.
- To define the circular functions, sine, cosine and tangent.
- To explain the symmetry properties of circular functions.
- To find exact values of circular functions.
- To sketch the graphs of circular functions.
- To solve problems with circular functions.

6.1 Review of circular (trigonometric) functions

Measuring angles in degrees and radians

The diagram shows a unit circle, i.e. a circle of radius 1 unit.

The circumference of the unit circle is $2\pi \times 1$
 $= 2\pi$ units




clockwise direction round the




anticlockwise direction round the

Using the TI-Nspire

Use the **Calc**→**Calculate** derivative to find the derivative of $f(x) = |x| - 1$.

Store $\odot$ this derivative as $d_f(x)$ as shown.

Tap a **Graphs** application ($\odot$ →**TI-84**)
 Graphs and let
 $f(x) = \text{abs}(x) - 1$
 $F(x) = d_f(x)$
 Select an appropriate window setting
 ($\odot$ →**Window**/**Zoom**/**Window**
 Settings).

The two graphs are shown. The graph of the derivative has **medium line weight**, available from the **Attributes** menu
 ($\odot$ →**Action**→**Attributes**).

Interactive-Define.
The graphs of the function and its derivative are graphed, each with a different style.
Note: Edit, Cut & Paste can be used to transfer the function and its derivative as required within and across program areas.

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Chapter 2 Graphs of Functions

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Exercise 1B

1 State the domain and range for the relations represented by each of the following graphs:

2 Sketch the graph of each of the following relations and state the domain and range of each:

a $\{(x, y) : y = x^2 + 1\}$	b $\{(x, y) : x^2 + y^2 = 9\}$
c $\{(x, y) : y + 12y = 24, x \geq 0\}$	d $y = \sqrt{x}$
e $\{(x, y) : y^2 = -x, x \in [0, 5]\}$	f $y = x^2 + 2, x \in [0, 4]$
g $\{(x, y) : y = 3x - 2, x \in [-1, 2]\}$	h $y = 4 - x^3$

3 Which of the following relations are functions? State the domain and range for each.

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Essential Mathematical Methods 1B & C

Chapter summary

- A specific continuous random variable is described by the **normal probability density function**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$
 where μ and σ are the mean and standard deviation of the normal random variable X .
- The centre of the normal distribution is determined by the mean, μ , and its width is determined by the standard deviation, σ .
- The **68-95-99.75 rule** states that, for a normal distribution:
 - 1 Approximately 68% of the values lie within one standard deviation of the mean.
 - 2 Approximately 95% of the values lie within two standard deviations of the mean.
 - 3 Approximately 99.75% of the values lie within three standard deviations of the mean.
- If x is a normally distributed random variable with mean μ and standard deviation σ then to **standardise** we first subtract the mean μ from x and then divide the result by the standard deviation σ . The standardised value is denoted z and has a mean $\mu = 0$ and standard deviation $\sigma = 1$.

$$z = \frac{x - \mu}{\sigma}$$
- A **calculator** gives values for the cumulative distribution function of the normal distribution — that is, areas under the standard normal curve for many different values of the standard variable.
- A **calculator** gives the inverse normal values for a random variable — that is, values corresponding to specified areas under the normal curve.

Multiple-choice questions

- 1 The diagram shows the graph of a normal distribution with mean μ and standard deviation σ . Which of the following statements is true?
 - A $\mu = 4$ and $\sigma = 3$
 - B $\mu = 2$ and $\sigma = 4$
 - C $\mu = 4$ and $\sigma = 2$
 - D $\mu = 3$ and $\sigma = 2$
- 2 $P(Z < 1.45)$, where Z is a standard normal random variable
 - A 0.1994 B 0.8066 C 0.0775 D 0.9225
- 3 If Z is a standard normal random variable, $P(Z < c)$
 - A 0.6745 B -0.6745 C 0.3867 D -0.3867

$2i, (3, 4), (2, 3)$
 $0, 1), (1, 5), (2, -4)$
 $2, -2), (2, 4), (4, 6)$
 $3, (2, 4), (3, 4)$

e $\{[x, 4]; x \in \mathbb{R}\}$
f $y = -2x + 4$
i $\{(x, y); x^2 + y^2 = 16\}$

$x) = 3x^2 - 2$.

b State the range of g .

$g(x) = 2x^3 + 2x - 6$.

i and $f(-3)$.

b Evaluate $g(-1)$, $g(2)$ and $g(3)$.

in terms of x .

i $x - 2$ **ii** $g(-2x)$ **iv** $g(x + 2)$ **v** $g(x^2)$

APPENDIX B

Computer Algebra System (TI-Nspire OS3.0)

1 Introduction to the TI-Nspire

This Appendix is written for operating system 2.2.1 for the TI-Nspire Calculator, Touchpad and CX calculators. The CX calculator and the function of some of the keys is shown in the diagram below. Similar diagrams for the Cxpad and Touchpad calculators are available at <http://www.cambridge.edu.sg/nspire/>

The diagram shows a TI-Nspire CX calculator with the following callouts:

- 1 removes menus and submenus (bottom left)
- 2 cursor lock (bottom left)
- 3 multi-touch mouse (bottom left)
- 4 close to menu (left side)
- 5 multi-touch calculator in split screen (left side)
- 6 Touchpad (Touchpad) (center)
- 7 accesses submenus (bottom left)
- 8 displays page number (bottom left)
- 9 displays page menu (bottom left)
- 10 displays home page to select applications, menus, My Documents and start a new document (top right)
- 11 options for each subfunction (top right)
- 12 contextual menus (bottom right)
- 13 menu pointer (bottom right)
- 14 touchpad - displays a character (bottom right)
- 15 freeze function (bottom right)
- 16 complete commands and displays results (bottom right)

This Appendix is written for operating systems 2.0/1 for the TI-Nspire Clickpad, Touchpad and CX calculators. The CX calculator and the function of some of the keys is shown in the diagram below. Similar diagrams for the Clickpad and Touchpad calculators are available at <http://www.cambridge.edu.au/bo/>.

Diagram illustrating touch gestures on a smartphone screen:

- 1. Tap: select items and disengage boxes.
- 2. Tap: undo last move.
- 3. Tap: redo last move.
- 4. Tap: move to next entry box (slide).
- 5. Tap: launch applications in split screen.
- 6. Newfangled Touchpad:
 - a. accesses secondary (system) commands.
 - b. displays page number.
 - c. displays preview page.
 - d. displays next page.

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- print-ready skillsheets to revise the prerequisite knowledge and skills required for the chapter
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Additional resources ...

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Functions and relations

Objectives

- To understand and use the **notation of sets**, including the symbols $\in$, $\subseteq$, $\cap$, $\cup$, $\emptyset$ and $\setminus$.
- To use the notation for **sets of numbers**.
- To understand the concept of **relation**.
- To understand the terms **domain** and **range**.
- To understand the concept of **function**.
- To understand the term **one-to-one**.
- To understand the terms **implied domain**, **restriction of a function**, **hybrid function**, and **odd and even functions**.
- To understand the **modulus function**.
- To understand and use **sums** and **products of functions**.
- To define **composite functions**.
- To understand and find **inverse functions**.
- To apply a knowledge of functions to **solving problems**.

In this chapter, notation that will be used throughout the book will be introduced. The language introduced in this chapter is necessary for expressing important mathematical ideas precisely. If you are working with a CAS calculator it is appropriate to work through the first sections of the appropriate Computer Algebra System Appendix.

1.1 Set notation

Set notation is used widely in mathematics and in this book it is employed where appropriate. This section summarises much of the set notation you will need.

A **set** is a collection of objects. The objects that are in the set are known as the **elements** or members of the set. If x is an element of a set A we write $x \in A$. This can also be read as ‘ x is a member of the set A ’ or ‘ x belongs to A ’ or ‘ x is in A ’.

The notation $x \notin A$ means x is **not** an element of A .

For example: $2 \notin$ set of odd numbers.

A set B is called a **subset** of a set A if and only if $x \in B$ implies $x \in A$.

To indicate that B is a subset of A , we write $B \subseteq A$. This expression can also be read as ‘ B is contained in A ’ or ‘ A contains B ’.

The set of elements common to two sets A and B is called the **intersection** of A and B and is denoted by $A \cap B$. Thus $x \in A \cap B$ if and only if $x \in A$ and $x \in B$.

If the sets A and B have no elements in common, we say A and B are **disjoint**, and write $A \cap B = \emptyset$.

The set $\emptyset$ is called the **empty** set or **null** set.

The **union** of sets A and B , written $A \cup B$, is the set of elements that are either in A or in B . This does not exclude objects that are elements of both A and B .

Example 1

$A = \{1, 2, 3, 7\}$; $B = \{3, 4, 5, 6, 7\}$

Find:

a $A \cap B$

b $A \cup B$

Solution

a $A \cap B = \{3, 7\}$

b $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

Note: In this example, $3 \in A$ and $5 \notin A$ and $\{2, 3\} \subseteq A$.

Finally, the **set difference** of two sets A and B is denoted $A \setminus B$, where:

$$A \setminus B = \{x : x \in A, x \notin B\}$$

e.g., for A and B in Example 1, $A \setminus B = \{1, 2\}$ and $B \setminus A = \{4, 5, 6\}$

There will be a further discussion of set notation in Chapter 14, which will provide the additional notation necessary for the study of probability.

Sets of numbers

The elements of the set $\{1, 2, 3, 4, \dots\}$ are called the **natural numbers**. The set of natural numbers will be denoted by N .

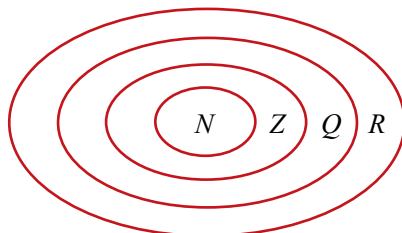
The elements of $\{\dots, -2, -1, 0, 1, 2, \dots\}$ are called **integers**. The set of integers will be denoted by Z .

The numbers of the form $\frac{p}{q}$ with p and q integers, $q \neq 0$, are called **rational numbers**. The rational numbers may be characterised by the property that each rational number may be written as a terminating or recurring decimal. The set of rational numbers will be denoted by Q .

The real numbers that are not rational numbers are called **irrational** (e.g., π and $\sqrt{2}$).

The set of real numbers will be denoted by R .

It is clear that $N \subseteq Z \subseteq Q \subseteq R$ and this may be represented by the diagram:



Note: $\{x: 0 < x < 1\}$ is the set of all real numbers between 0 and 1.

$\{x: x > 0, x \text{ rational}\}$ is the set of all positive rational numbers.

$\{2n: n = 0, 1, 2, \dots\}$ is the set of all even numbers.

Among the most important subsets of R are the **intervals**. The following is an exhaustive list of the various types of intervals and the standard notation for them. We suppose that a and b are real numbers and that $a < b$:

$$\begin{array}{ll} (a, b) = \{x: a < x < b\} & [a, b] = \{x: a \leq x \leq b\} \\ (a, b] = \{x: a < x \leq b\} & [a, b) = \{x: a \leq x < b\} \\ (a, \infty) = \{x: x > a\} & [a, \infty) = \{x: x \geq a\} \\ (-\infty, b) = \{x: x < b\} & (-\infty, b] = \{x: x \leq b\} \end{array}$$

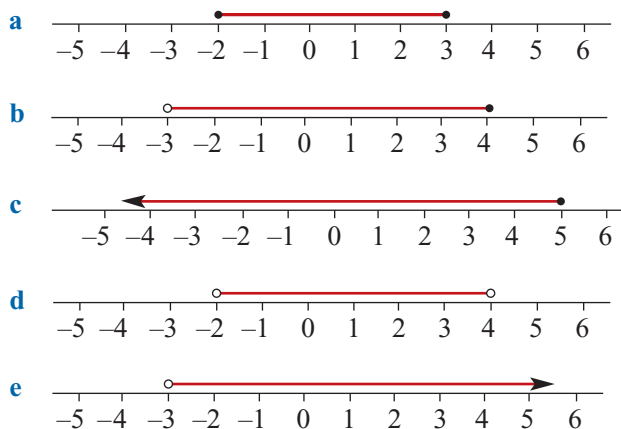
Intervals may be represented by diagrams, as shown in Example 2.

Example 2

Illustrate each of the following intervals of the real numbers on a number line:

a $[-2, 3]$ **b** $(-3, 4]$ **c** $(-\infty, 5]$ **d** $(-2, 4)$ **e** $(-3, \infty)$

Solution



The 'closed' circle indicates that the number is included.

The 'open' circle indicates that the number is not included.

The following are also subsets of the real numbers for which there are special notations:

$$R^+ = \{x: x > 0\}$$

$$R^- = \{x: x < 0\}$$

$R \setminus \{0\}$ is the set of real numbers excluding 0.

$$Z^+ = \{x: x \in Z, x > 0\}$$

The cartesian plane is denoted by R^2 where $R^2 = \{(x, y): x \in R \text{ and } y \in R\}$

Exercise 1A

1 For $X = \{2, 3, 5, 7, 9, 11\}$, $Y = \{7, 9, 15, 19, 23\}$ and $Z = \{2, 7, 9, 15, 19\}$, find:

- a** $X \cap Y$ **b** $X \cap Y \cap Z$ **c** $X \cup Y$ **d** $X \setminus Y$
e $Z \setminus Y$ **f** $X \cap Z$ **g** $[-2, 8] \cap X$ **h** $(-3, 8] \cap Y$
i $(2, \infty) \cap Y$ **j** $(3, \infty) \cup Y$

2 For $X = \{a, b, c, d, e\}$ and $Y = \{a, e, i, o, u\}$, find:

- a** $X \cap Y$ **b** $X \cup Y$ **c** $X \setminus Y$ **d** $Y \setminus X$

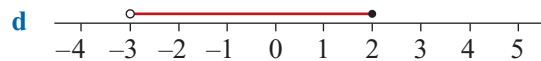
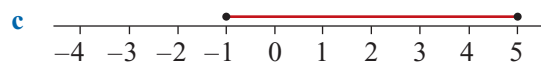
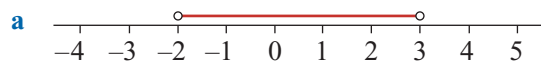
3 For $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $B = \{2, 4, 6, 8, 10\}$ and $C = \{1, 3, 6, 9\}$, find:

- a** $B \cap C$ **b** $B \setminus C$ **c** $A \setminus B$ **d** $(A \setminus B) \cup (A \setminus C)$
e $A \setminus (B \cap C)$ **f** $(A \setminus B) \cap (A \setminus C)$ **g** $A \setminus (B \cup C)$ **h** $A \cap B \cap C$

4 Use the appropriate interval notation, i.e. $[a, b]$, (a, b) etc., to describe each of the following sets:

- a** $\{x: -3 \leq x < 1\}$ **b** $\{x: -4 < x \leq 5\}$ **c** $\{y: -\sqrt{2} < y < 0\}$
d $\left\{x: -\frac{1}{\sqrt{2}} < x < \sqrt{3}\right\}$ **e** $\{x: x < -3\}$ **f** R^+
g R^- **h** $\{x: x \geq -2\}$

5 Describe each of the following subsets of the real number line using the notation $[a, b]$, (a, b) , etc.:



6 Illustrate each of the following intervals on a number line:

- a** $(-3, 2]$ **b** $(-4, 3)$ **c** $(-\infty, 3)$
d $[-4, -1]$ **e** $[-4, \infty)$ **f** $[-2, 5)$

7 For each of the following, use one number line on which to represent the sets:

- a** $[-3, 6], [2, 4], [-3, 6] \cap [2, 4]$ **b** $[-3, 6], R \setminus [-3, 6]$
c $[-2, \infty), (-\infty, 6], [-2, \infty) \cap (-\infty, 6]$ **d** $(-8, -2), R^- \setminus (-8, -2)$

1.2 Identifying and describing relations and functions

An **ordered pair**, denoted (a, b) , is a pair of elements a and b in which a is considered to be the first element and b the second. In this section, only ordered pairs of real numbers are considered.

Two ordered pairs (a, b) and (c, d) are equal if $a = c$ and $b = d$.

A **relation** is a set of ordered pairs. The following are examples of relations:

$$S = \{(1, 1), (1, 2), (3, 4), (5, 6)\}$$

$$T = \{(-3, 5), (4, 12), (5, 12), (7, -6)\}$$

Every relation determines two sets defined as follows:

- The **domain** of a relation S is the set of all first elements of the ordered pairs in S .
- The **range** of a relation S is the set of all second elements of the ordered pairs in S .

In the above examples:

- domain of $S = \{1, 3, 5\}$; range of $S = \{1, 2, 4, 6\}$
- domain of $T = \{-3, 4, 5, 7\}$; range of $T = \{5, 12, -6\}$

A relation may be defined by a rule which pairs the elements in its domain and range. Thus the set

$$\{(x, y): y = x + 1, x \in \{1, 2, 3, 4\}\}$$

is the relation

$$\{(1, 2), (2, 3), (3, 4), (4, 5)\}$$

When the domain of a relation is not explicitly stated, it is understood to consist of all real numbers for which the defining rule has meaning. For example:

$$S = \{(x, y): y = x^2\}$$

is assumed to have domain R and

$$T = \{(x, y): y = \sqrt{x}\}$$

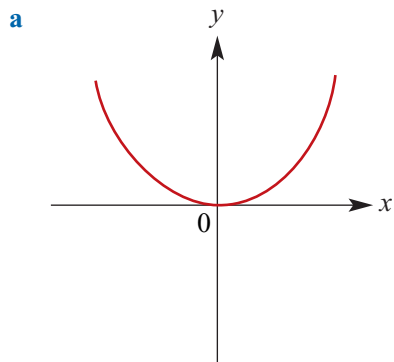
is assumed to have domain $[0, \infty)$.



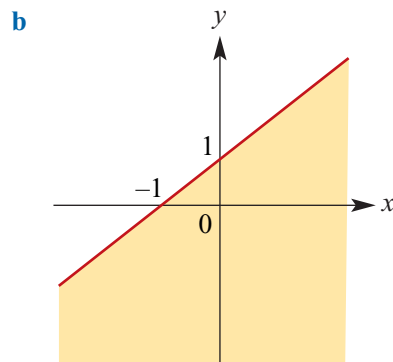
Example 3

Sketch the graph of each of the following relations and state the domain and range of each.

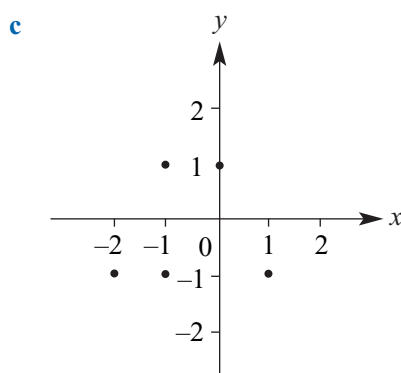
- | | |
|---|--|
| a $\{(x, y): y = x^2\}$ | b $\{(x, y): y \leq x + 1\}$ |
| c $\{(-2, -1), (-1, -1), (-1, 1), (0, 1), (1, -1)\}$ | d $\{(x, y): x^2 + y^2 = 1\}$ |
| e $\{(x, y): 2x + 3y = 6, x \geq 0\}$ | f $\{(x, y): y = 2x - 1, x \in [-1, 2]\}$ |

Solution

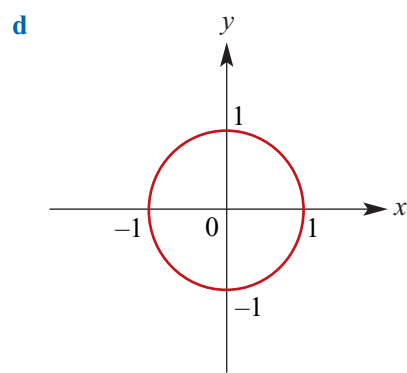
Domain = $\mathbb{R}$; range = $\mathbb{R}^+ \cup \{0\}$



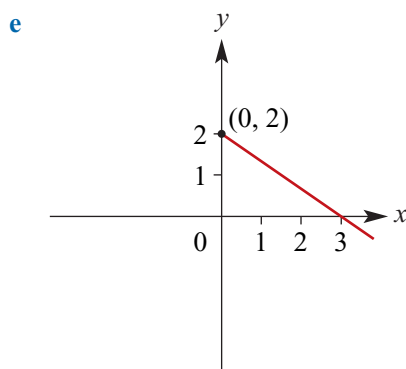
Domain = $\mathbb{R}$; range = $\mathbb{R}$



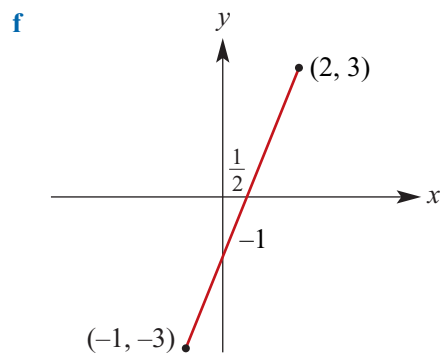
Domain = $\{-2, -1, 0, 1\}$;
range = $\{-1, 1\}$



Domain = $\{x: -1 \leq x \leq 1\}$;
range = $\{y: -1 \leq y \leq 1\}$



Domain = $[0, \infty)$;
range = $(-\infty, 2]$



Domain = $[-1, 2]$;
range = $[-3, 3]$

Sometimes the set notation is not used in the specification of a relation.

For the above example:

a is written as $y = x^2$

b is written as $y \leq x + 1$

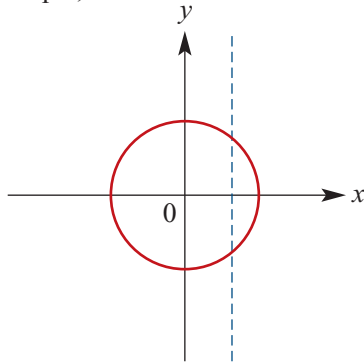
e is written as $2x + 3y = 6, x \geq 0$

A **function** is a relation such that no two ordered pairs of the relation have the same first element. For instance, in Example 3, **a**, **e** and **f** are functions but **b**, **c** and **d** are not.

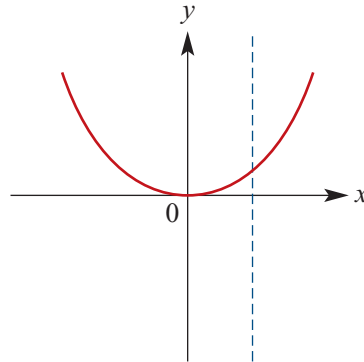
Let S be a relation with domain D . A simple geometric test to determine if S is a function is as follows.

Consider the graph of S . If all vertical lines with equations $x = a$, $a \in D$, cut the graph of S only once, then S is a function.

For example,



$x^2 + y^2 = 1$ is not a function



$y = x^2$ is a function

Functions are usually denoted by lower case letters such as f , g , h .

The definition of a function tells us that for each x in the domain of f there is a unique element, y , in the range such that $(x, y) \in f$. The element y is called the **image** of x under f or the **value** of f at x and is denoted by $f(x)$ (read ‘ f of x ’).

If $(x, y) \in f$, then x is called a **pre-image** of y .

This gives an alternative way of writing functions.

- 1** For the function $\{(x, y): y = x^2\}$, write:

$$f: R \rightarrow R, f(x) = x^2$$

- 2** For the function $\{(x, y): y = 2x - 1, x \in [0, 4]\}$ write:

$$f: [0, 4] \rightarrow R, f(x) = 2x - 1$$

- 3** For the function $\left\{(x, y): y = \frac{1}{x}\right\}$, write:

$$f: R \setminus \{0\} \rightarrow R, f(x) = \frac{1}{x}$$

If the domain is R we often just write the rule, for example in **1** $f(x) = x^2$.

Note that in using the notation $f: X \rightarrow Y$, X is the domain but Y is not necessarily the range. It is a set that contains the range and is called the **codomain**. With this notation for functions the domain of f is written as $\text{dom } f$ and range of f as $\text{ran } f$.

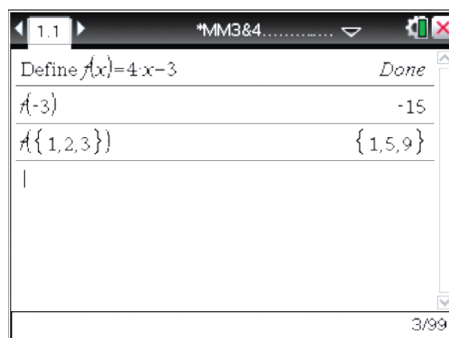
Using the TI-Nspire

Function notation can be used with a CAS calculator.

Use $\text{[menu]} > \text{Actions} > \text{Define}$ to define the function $f(x) = 4x - 3$.

Type $f(-3)$ followed by **enter** to evaluate $f(-3)$.

Type $f(\{1, 2, 3\})$ followed by **enter** to evaluate $f(1)$, $f(2)$ and $f(3)$.



Using the Casio ClassPad

Function notation can be used with a CAS calculator.

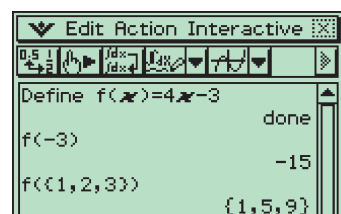
In $\text{Main} > \text{Interactive-Define}$ and enter the function name, variable and expression as shown.

See page 10 for a screen showing the

Define window.

Enter $f(-3)$ in the entry line and tap [EXE] .

In the entry line, type $f(\{1,2,3\})$ to obtain the values of $f(1)$, $f(2)$ and $f(3)$.



Example 4

If $f(x) = 2x^2 + x$, find $f(3)$, $f(-2)$ and $f(x - 1)$.

Solution

$$\begin{aligned}
 f(3) &= 2(3)^2 + 3 = 21 \\
 f(-2) &= 2(-2)^2 - 2 = 6 \\
 f(x - 1) &= 2(x - 1)^2 + x - 1 \\
 &= 2(x^2 - 2x + 1) + (x - 1) \\
 &= 2x^2 - 3x + 1
 \end{aligned}$$

Example 5

If $f(x) = 2x + 1$, find $f(-2)$ and $f\left(\frac{1}{a}\right)$, $a \neq 0$.

Solution

$$f(-2) = 2(-2) + 1 = -3$$

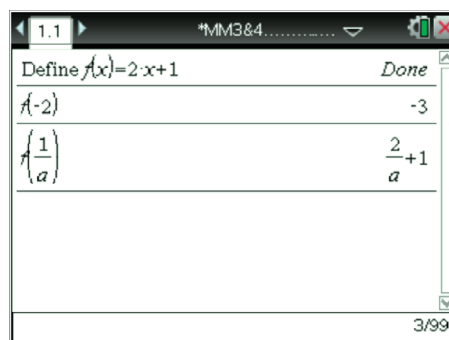
$$f\left(\frac{1}{a}\right) = 2\left(\frac{1}{a}\right) + 1 = \frac{2}{a} + 1$$

Using the TI-Nspire

Use $\left(\text{menu}\right) > \text{Actions} > \text{Define}$ to define the function $f(x) = 2x + 1$.

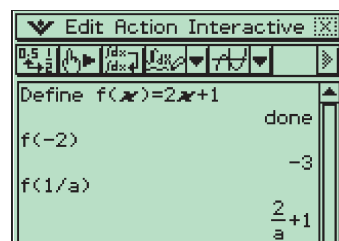
Type $f(-2)$ followed by **enter** to evaluate $f(-2)$.

Type $f\left(\frac{1}{a}\right)$ followed by **enter** to evaluate $f\left(\frac{1}{a}\right)$.

**Using the Casio ClassPad**

In $\left(\text{Main}\right)$ tap **Interactive–Define** and enter the function name f , variable x and expression $2x + 1$.

Now complete $f(-2)$ and $f\left(\frac{1}{a}\right)$.

**Example 6**

Consider the function defined by $f(x) = 2x - 4$ for all $x \in \mathbb{R}$.

- Find the value of $f(2)$, $f(-1)$ and $f(t)$.
- For what values of t is $f(t) = t$?
- For what values of x is $f(x) \geq x$?
- Find the pre-image of 6.

Solution

$$\begin{aligned}
 \text{a} \quad f(2) &= 2(2) - 4 \\
 &= 0 \\
 f(-1) &= 2(-1) - 4 \\
 &= -6 \\
 f(t) &= 2t - 4
 \end{aligned}$$

$$\begin{aligned}
 \text{b} \quad f(t) &= t \\
 2t - 4 &= t \\
 t - 4 &= 0 \\
 \therefore t &= 4
 \end{aligned}$$

$$\begin{aligned}
 \text{c} \quad f(x) &\geq x \\
 2x - 4 &\geq x \\
 x - 4 &\geq 0 \\
 \therefore x &\geq 4
 \end{aligned}$$

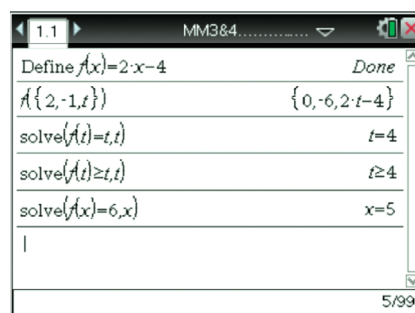
$$\begin{aligned}
 \text{d} \quad f(x) &= 6 \\
 2x - 4 &= 6 \\
 x &= 5 \\
 5 &\text{ is the pre-image of 6.}
 \end{aligned}$$

Using the TI-Nspire

Use $\text{[menu]} > \text{Actions} > \text{Define}$ to define the function and $\text{[menu]} > \text{Algebra} > \text{Solve}$ to solve as shown.

The symbol $\geq$ can be found using $\text{[ctrl]} + \text{[=]}$ and select $\geq$ or use $\text{[ctrl]} + \text{[menu]} > \text{Symbols}$.

On the Clickpad (grey handheld) you can use $\text{[ctrl]} + \text{[>]}$ or $\text{[>]} + \text{[=]}$.

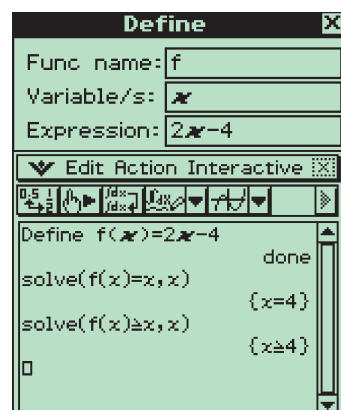


Using the Casio ClassPad

Tap **Interactive–Define** and enter the function name, variable and expression as shown.

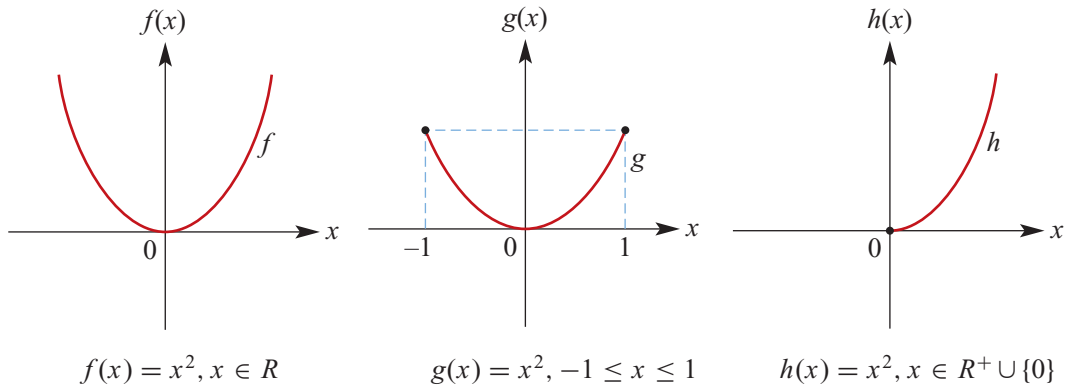
Enter and highlight $f(x) = x$, tap **Interactive–Equation/inequality–solve** and ensure the variable is set as x .

To enter the inequality, press [Keyboard] and look in the $\text{[math]} \text{OPTN}$ to find the $\geq$ symbol.



Restriction of a function

Consider the following functions:



The different letters, f , g and h , used to name the functions, emphasise the fact that there are three different functions even though they each have the same rule. They are different because they are defined for different domains. We call g and h restrictions of f since their domains are subsets of the domain of f .

Example 7

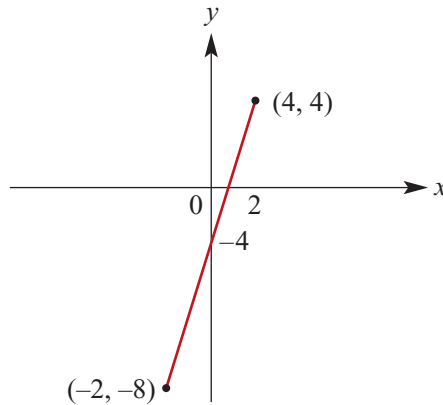
For each of the following, sketch the graph and state the range:

a $f: [-2, 4] \rightarrow \mathbb{R}, f(x) = 2x - 4$

b $g: (-1, 2] \rightarrow \mathbb{R}, g(x) = x^2$

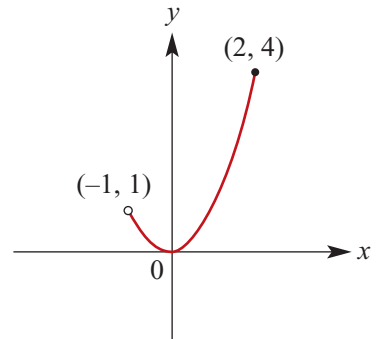
Solution

a



Range = $[-8, 4]$

b

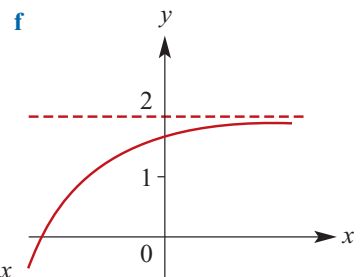
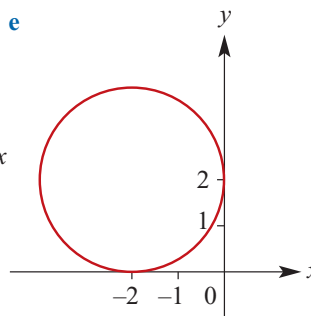
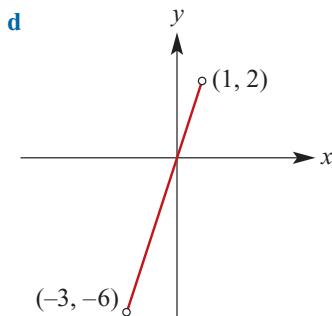
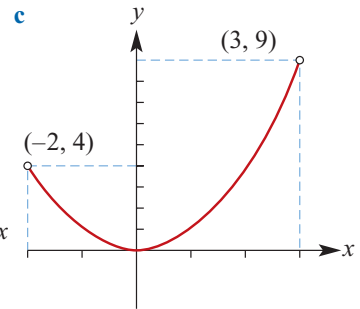
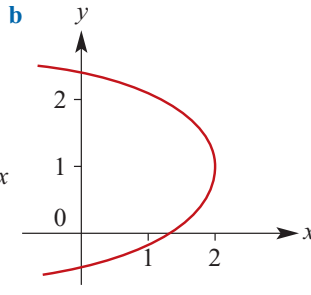
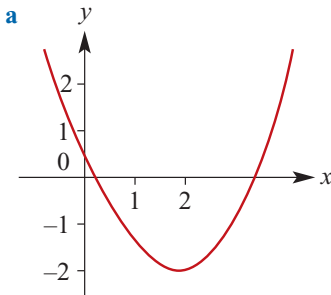


Range = $[0, 4]$

Exercise 1B



- 1 State the domain and range for the relations represented by each of the following graphs:



- 2 Sketch the graph of each of the following relations and state the domain and range of each:

a $\{(x, y): y = x^2 + 1\}$

b $\{(x, y): x^2 + y^2 = 9\}$

c $\{(x, y): 3x + 12y = 24, x \geq 0\}$

d $y = \sqrt{2x}$

e $\{(x, y): y = 5 - x, x \in [0, 5]\}$

f $y = x^2 + 2, x \in [0, 4]$

g $y = 3x - 2, x \in [-1, 2]$

h $y = 4 - x^2$

- 3 Which of the following relations are functions? State the domain and range for each.

a $\{(-1, 1), (-1, 2), (1, 2), (3, 4), (2, 3)\}$

b $\{(-2, 0), (-1, -1), (0, 3), (1, 5), (2, -4)\}$

c $\{(-1, 1), (-1, 2), (-2, -2), (2, 4), (4, 6)\}$

d $\{(-1, 4), (0, 4), (1, 4), (2, 4), (3, 4)\}$

e $\{(x, 4): x \in \mathbb{R}\}$

f $\{(2, y): y \in \mathbb{Z}\}$

g $y = -2x + 4$

h $y \geq 3x + 2$

i $\{x, y): x^2 + y^2 = 16\}$

- 4 Consider the function $g(x) = 3x^2 - 2$.

a Find $g(-2)$, $g(4)$.

b State the range of g .

- 5 Let $f(x) = 2x^2 + 4x$ and $g(x) = 2x^3 + 2x - 6$.

a Evaluate $f(-1)$, $f(2)$ and $f(-3)$.

b Evaluate $g(-1)$, $g(2)$ and $g(3)$.

c Express the following in terms of x :

i $f(-2x)$

ii $f(x - 2)$

iii $g(-2x)$

iv $g(x + 2)$

v $g(x^2)$

- 6 Consider the function $f(x) = 2x - 3$. Find:
- a** the image of 3 **b** the pre-image of 11 **c** $\{x: f(x) = 4x\}$
- 7 Consider the functions $g(x) = 6x + 7$ and $h(x) = 3x - 2$. Find:
- a** $\{x: g(x) = h(x)\}$ **b** $\{x: g(x) > h(x)\}$ **c** $\{x: h(x) = 0\}$
- 8 Rewrite each of the following using the $f: X \rightarrow Y$ notation:
- a** $\{(x, y): y = 2x + 3\}$ **b** $\{(x, y): 3y + 4x = 12\}$
c $\{(x, y): y = 2x - 3, x \geq 0\}$ **d** $y = x^2 - 9, x \in \mathbb{R}$
e $y = 5x - 3, 0 \leq x \leq 2$
- 9 Sketch the graphs of each of the following and state the range of each:
- a** $y = x + 1, x \in [2, \infty)$ **b** $y = -x + 1, x \in [2, \infty)$
c $y = 2x + 1, x \in [-4, \infty)$ **d** $y = 3x + 2, x \in (-\infty, 3)$
e $y = x + 1, x \in (-\infty, 3]$ **f** $y = 3x - 1, x \in [-2, 6]$
g $y = -3x - 1, x \in [-5, -1]$ **h** $y = 5x - 1, x \in (-2, 4)$
- 10 For $f(x) = 2x^2 - 6x + 1$ and $g(x) = 3 - 2x$:
- a** Evaluate $f(2), f(-3), f(-2)$. **b** Evaluate $g(-2), g(1)$ and $g(-3)$.
c Express the following in terms of a :
i $f(a)$ **ii** $f(a + 2)$ **iii** $g(-a)$ **iv** $g(2a)$
v $f(5 - a)$ **vi** $f(2a)$ **vii** $g(a) + f(a)$ **viii** $g(a) - f(a)$
- 11 For $f(x) = 3x^2 + x - 2$, find:
- a** $\{x: f(x) = 0\}$ **b** $\{x: f(x) = x\}$ **c** $\{x: f(x) = -2\}$
d $\{x: f(x) > 0\}$ **e** $\{x: f(x) > x\}$ **f** $\{x: f(x) \leq -2\}$
- 12 For $f(x) = x^2 + x$, find:
- a** $f(-2)$ **b** $f(2)$ **c** $f(-a)$ in terms of a **d** $f(a) + f(-a)$ in terms of a
e $f(a) - f(-a)$ in terms of a **f** $f(a^2)$ in terms of a
- 13 For $g(x) = 3x - 2$, find:
- a** $\{x: g(x) = 4\}$ **b** $\{x: g(x) > 4\}$ **c** $\{x: g(x) = a\}$
d $\{x: g(-x) = 6\}$ **e** $\{x: g(2x) = 4\}$ **f** $\left\{x: \frac{1}{g(x)} = 6\right\}, g(x) \neq 0$
- 14 Find the value of k for each of the following if $f(3) = 3$, where:
- a** $f(x) = kx - 1$ **b** $f(x) = x^2 - k$ **c** $f(x) = x^2 + kx + 1$
d $f(x) = \frac{k}{x}$ **e** $f(x) = kx^2$ **f** $f(x) = 1 - kx^2$
- 15 Find the values of x for which the given functions have the given value:
- a** $f(x) = 5x - 4, f(x) = 2$ **b** $f(x) = \frac{1}{x}, f(x) = 5$
c $f(x) = \frac{1}{x^2}, f(x) = 9$ **d** $f(x) = x + \frac{1}{x}, f(x) = 2$
e $f(x) = (x + 1)(x - 2), f(x) = 0$

1.3 Types of functions and maximal domains

One-to-one and many-to-one functions

A function f is said to be **one-to-one** if for $a, b \in \text{dom } f$, $a \neq b$, then $f(a) \neq f(b)$. In other words, f is called one-to-one if every image under f has a unique pre-image.

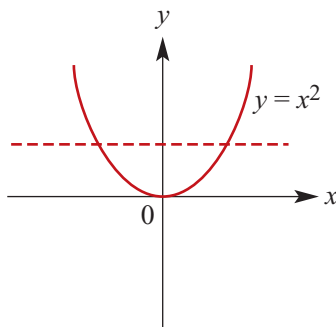
The function $f(x) = 2x + 1$ is a one-to-one function. The function $f(x) = x^2$ is not a one-to-one function as, for example, $f(-3) = 9$ and $f(3) = 9$; i.e., 9 does not have a unique pre-image.

The function $f(x) = 5$ is not a one-to-one function as there are infinitely many pre-images of 5.

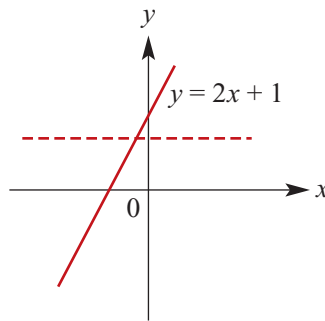
The function $f(x) = x^3$ is a one-to-one function.

A geometric test for a function to be one-to-one is as follows.

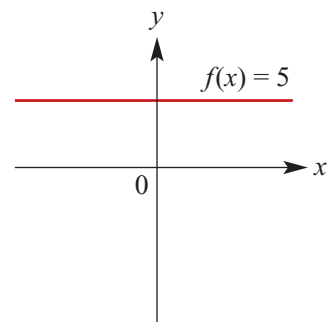
If for any $a \in \text{ran } f$ the horizontal line, $y = a$, crosses the graph of f at only one point, the function is one-to-one.



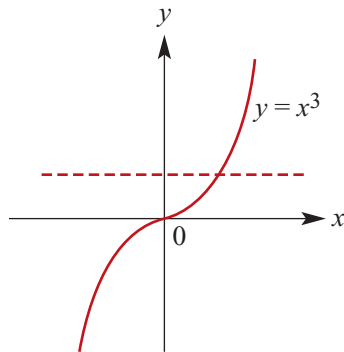
not one-to-one



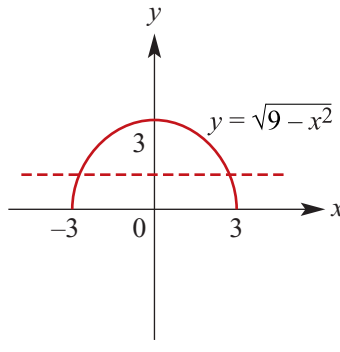
one-to-one



not one-to-one



one-to-one



not one-to-one

A function that is not one-to-one is **many-to-one**.

Implied domains (maximal domains)

If the domain is unspecified, then the domain is the largest subset of R for which the rule is defined. When the domain is not explicitly stated, it is implied by the rule.

Thus for the function, $f(x) = \sqrt{x}$ the implied domain (maximal domain) is $[0, \infty)$. We write:

$$f: [0, \infty) \rightarrow R, f(x) = \sqrt{x}$$

Example 8

Find the implied domain of the functions with the following rules:

a $f(x) = \frac{2}{2x - 3}$

b $g(x) = \sqrt{5 - x}$

c $h(x) = \sqrt{x - 5} + \sqrt{8 - x}$

d $f(x) = \sqrt{x^2 - 7x + 12}$

Solution

a $f(x)$ is not defined when $2x - 3 = 0$, i.e. when $x = \frac{3}{2}$.

Thus the implied domain is $R \setminus \left\{ \frac{3}{2} \right\}$.

b $g(x)$ is defined when $5 - x \geq 0$, i.e. when $x \leq 5$.

Thus the implied domain is $(-\infty, 5]$.

c $h(x)$ is defined when $x - 5 \geq 0$ and $8 - x \geq 0$, i.e. when $x \geq 5$ and $x \leq 8$.

Thus the implied domain is $[5, 8]$.

d $f(x)$ is defined when $x^2 - 7x + 12 \geq 0$.

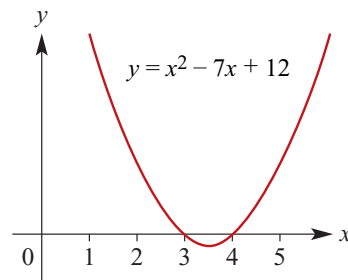
$$x^2 - 7x + 12 \geq 0$$

$$\text{is equivalent to } (x - 3)(x - 4) \geq 0.$$

Therefore, $x \geq 4$ or $x \leq 3$.

Thus the implied domain is

$$(-\infty, 3] \cup [4, \infty).$$



Hybrid functions

Example 9

Sketch the graph of the function f given by:

$$f(x) = \begin{cases} -x - 1 & \text{for } x < 0 \\ 2x - 1 & \text{for } 0 \leq x \leq 1 \\ \frac{1}{2}x + \frac{1}{2} & \text{for } x \geq 1 \end{cases}$$

Solution

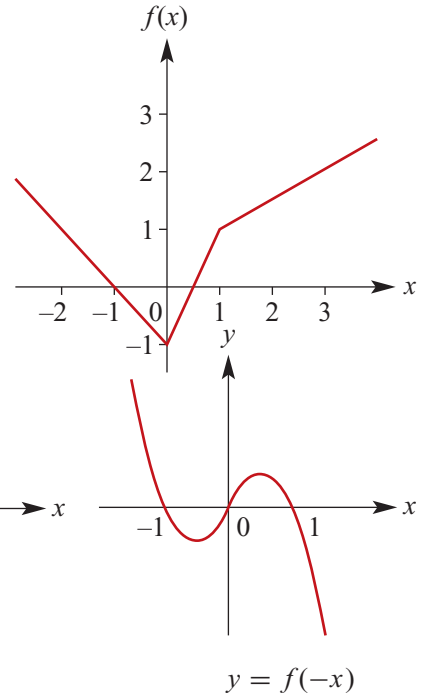
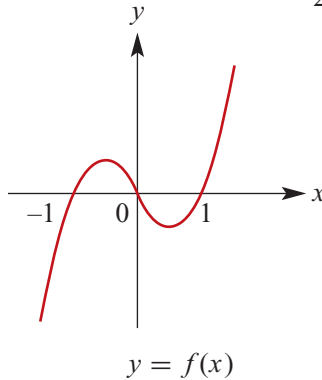
Functions like this, which have different rules for different subsets of the domain, are called **hybrid functions**.

Odd and even functions

An **odd** function has the property that $f(-x) = -f(x)$.

For example, $f(x) = x^3 - x$ is an odd function

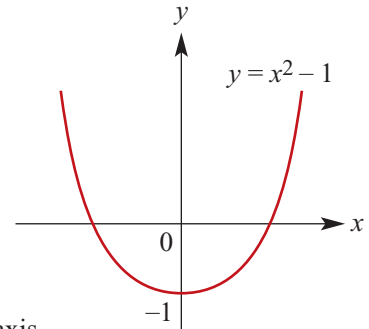
$$\begin{aligned}\text{since } f(-x) &= (-x)^3 - (-x) \\ &= -x^3 + x \\ &= -f(x)\end{aligned}$$



An **even** function has the property that $f(-x) = f(x)$.

For example, $f(x) = x^2 - 1$ is an even function

$$\begin{aligned}\text{since } f(-x) &= (-x)^2 - 1 \\ &= x^2 - 1 \\ &= f(x)\end{aligned}$$



The graphs of even functions are symmetrical about the y -axis.

The properties of odd and even functions often facilitate the sketching of graphs.

Exercise 1C

1 State which of the following functions are one-to-one:

a $\{(2, 3), (3, 4), (5, 4), (4, 6)\}$

b $\{(1, 2), (2, 3), (3, 4), (4, 6)\}$

c $\{(x, y): y = x^2 + 2\}$

d $\{(x, y): y = 2x + 4\}$

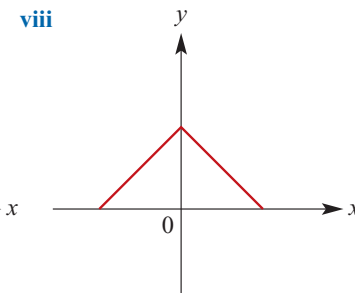
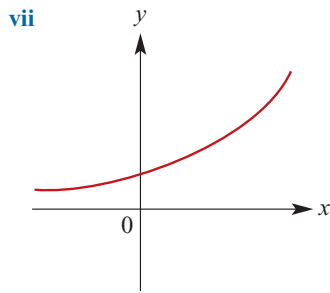
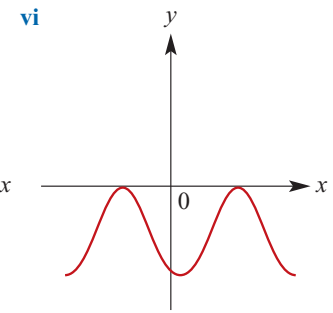
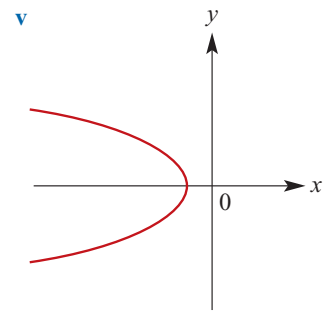
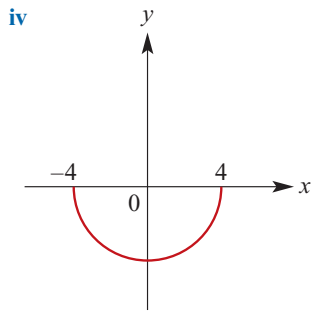
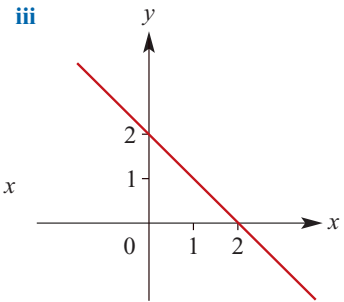
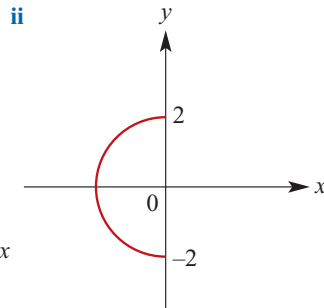
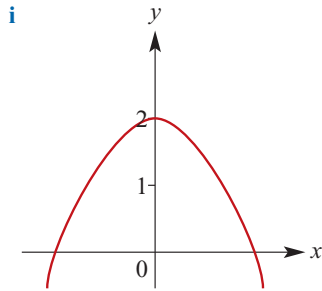
e $f(x) = 2 - x^2$

f $y = x^2, x \geq 1$

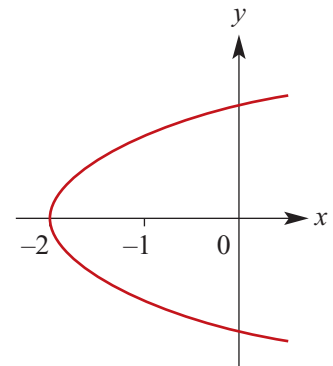
2 The following are graphs of relations.

a State which are the graphs of a function.

b State which are the graphs of a one-to-one function.



3 The graph of the relation $\{(x, y): y^2 = x + 2, x \geq -2\}$ is shown. From this relation, form two functions and specify the range of each.



- 4 a** Draw the graph of $g: R \rightarrow R$, $g(x) = x^2 + 2$.
b By restricting the domain of g , form two one-to-one functions that have the same rule as g .

- 5** State the largest possible domain and range for the functions defined by the rule:

a $y = 4 - x$ **b** $y = \sqrt{x}$ **c** $y = x^2 - 2$ **d** $y = \sqrt{16 - x^2}$
e $y = \frac{1}{x}$ **f** $y = 4 - 3x^2$ **g** $y = \sqrt{x - 3}$

- 6** Each of the following is the rule of a function. In each case write down the implied domain and the range.

a $y = 3x + 2$ **b** $y = x^2 - 2$ **c** $f(x) = \sqrt{9 - x^2}$ **d** $g(x) = \frac{1}{x - 1}$

- 7** Find the implied domain for each of the following rules:

a $f(x) = \frac{1}{x - 3}$ **b** $f(x) = \sqrt{x^2 - 3}$ **c** $g(x) = \sqrt{x^2 + 3}$
d $h(x) = \sqrt{x - 4} + \sqrt{11 - x}$ **e** $f(x) = \frac{x^2 - 1}{x + 1}$
f $h(x) = \sqrt{x^2 - x - 2}$ **g** $f(x) = \frac{1}{(x + 1)(x - 2)}$ **h** $h(x) = \sqrt{\frac{x - 1}{x + 2}}$
i $f(x) = \sqrt{x - 3x^2}$ **j** $h(x) = \sqrt{25 - x^2}$ **k** $f(x) = \sqrt{x - 3} + \sqrt{12 - x}$

- 8** Which of the following functions are odd, even or neither?

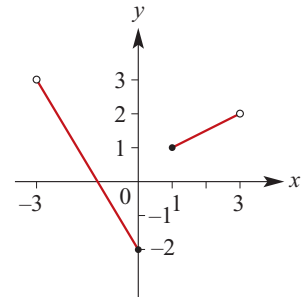
a $f(x) = x^4$ **b** $f(x) = x^5$ **c** $f(x) = x^4 - 3x$
d $f(x) = x^4 - 3x^2$ **e** $f(x) = x^5 - 2x^3$ **f** $f(x) = x^4 - 2x^5$

- 9 a** Sketch the graph of the function:

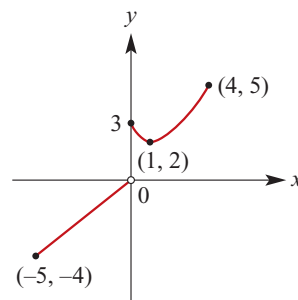
$$f(x) = \begin{cases} -2x - 2, & x < 0 \\ x - 2, & 0 \leq x < 2 \\ 3x - 6, & x \geq 2 \end{cases}$$

- b** What is the range of f ?

- 10** State the domain and range of the function for which the graph is shown.



- 11** State the domain and range of the function for which the graph is shown.



- 12 a** Sketch the graph of the function with rule:

$$f(x) = \begin{cases} 2x + 6 & 0 < x \leq 2 \\ -x + 5 & -4 \leq x \leq 0 \\ -4 & x < -4 \end{cases}$$

- b** State the domain and range of the function.

- 13 a** Sketch the graph of the function with rule:

$$g(x) = \begin{cases} x^2 + 5 & x > 0 \\ 5 - x & -3 \leq x \leq 0 \\ 8 & x < -3 \end{cases}$$

- b** State the range of the function.

- 14** Given that

$$f(x) = \begin{cases} \frac{1}{x}, & x > 3 \\ 2x, & x \leq 3 \end{cases}$$

find:

- a** $f(-4)$ **b** $f(0)$ **c** $f(4)$ **d** $f(a + 3)$ in terms of a
e $f(2a)$ in terms of a **f** $f(a - 3)$ in terms of a

- 15** Given that

$$f(x) = \begin{cases} \sqrt{x-1}, & x \geq 1 \\ 4, & x < 1 \end{cases}$$

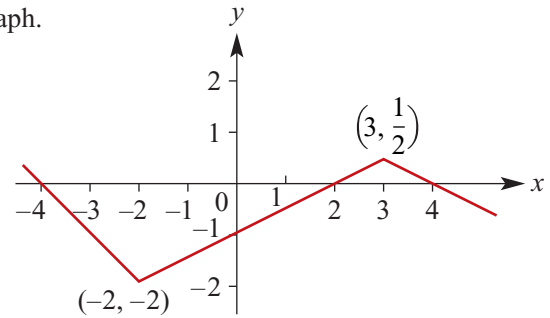
find:

- a** $f(0)$ **b** $f(3)$ **c** $f(8)$
d $f(a + 1)$ in terms of a **e** $f(a - 1)$ in terms of a

- 16** Sketch the graph of the function:

$$g(x) = \begin{cases} -x - 2, & x < -1 \\ \frac{x-1}{2}, & -1 \leq x < 1 \\ 3x - 3, & x \geq 1 \end{cases}$$

17 Specify the function illustrated by the graph.



1.4 The modulus function

The **modulus** or absolute value of a real number x is denoted by $|x|$ and is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

It may also be defined as $|x| = \sqrt{x^2}$.

For example, $|5| = 5$ and $|-5| = 5$.

The function $|x|$ has the following properties:

- $|ab| = |a||b|$
- $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$
- $|a + b| \leq |a| + |b|$. If a and b are both non-negative or both non-positive, then equality holds.
- If $a \geq 0$, $|x| \leq a$ is equivalent to $-a \leq x \leq a$.
- If $a \geq 0$, $|x - k| \leq a$ is equivalent to $k - a \leq x \leq k + a$.

Example 10

Evaluate each of the following:

- | | |
|---------------------------------|-----------------------|
| a i $ -3 \times 2 $ | ii $ -3 \times 2 $ |
| b i $\left \frac{-4}{2}\right $ | ii $\frac{ -4 }{ 2 }$ |
| c i $ -6 + 2 $ | ii $ -6 + 2 $ |

Solution

a i $|-3 \times 2| = |-6| = 6$

ii $|-3| \times |2| = 3 \times 2 = 6$

Note: $|-3 \times 2| = |-3| \times |2|$

b i $\left|\frac{-4}{2}\right| = |-2| = 2$

ii $\frac{|-4|}{|2|} = \frac{4}{2} = 2$

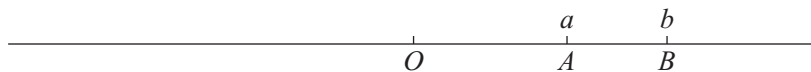
Note: $\left|\frac{-4}{2}\right| = \frac{|-4|}{|2|}$

c i $|-6 + 2| = |-4| = 4$

ii $|-6| + |2| = 6 + 2 = 8$

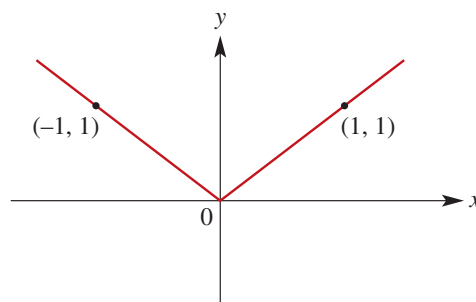
Note: $|-6 + 2| \neq |-6| + |2|$

Consider two points A and B on a number line:



On a number line the distance between points A and B is $|a - b| = |b - a|$. Thus $|x - 2| \leq 3$ can be read as ‘on the number line, the distance of x from 2 is less than or equal to 3’, and $|x| \leq 3$ can be read as ‘on the number line, the distance of x from the origin is less than or equal to 3’. Note that $|x| \leq 3$ is equivalent to $-3 \leq x \leq 3$ or $x \in [-3, 3]$.

The graph of the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is as shown here.



Note that $|x| = |-x|$, i.e. $|x|$ is an even function.

Example 11

Illustrate each of the following sets on a number line and represent the sets using interval notation.

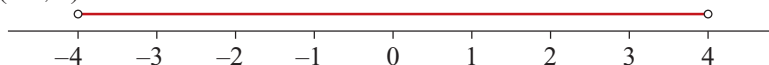
a $\{x: |x| < 4\}$

b $\{x: |x| \geq 4\}$

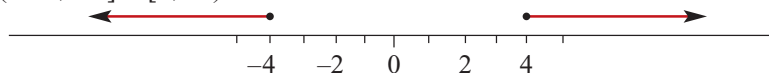
c $\{x: |x - 1| \leq 4\}$

Solution

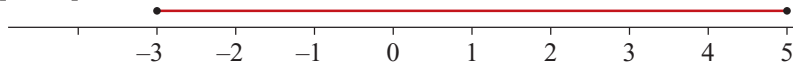
a $(-4, 4)$



b $(-\infty, -4] \cup [4, \infty)$



c $[-3, 5]$



Example 12

Sketch the graphs of each of the following functions and state the range of each of the functions:

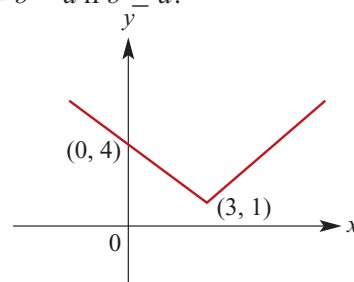
a $f(x) = |x - 3| + 1$

b $f(x) = -|x - 3| + 1$

Solution

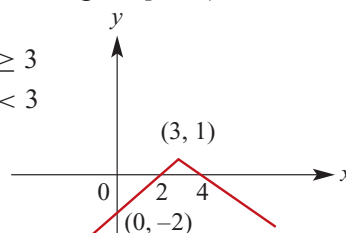
First, note that $|a - b| = a - b$ if $a \geq b$ and $|a - b| = b - a$ if $b \geq a$.

$$\begin{aligned} \text{a } f(x) &= |x - 3| + 1 = \begin{cases} x - 3 + 1 & \text{if } x \geq 3 \\ 3 - x + 1 & \text{if } x < 3 \end{cases} \\ &= \begin{cases} x - 2 & \text{if } x \geq 3 \\ 4 - x & \text{if } x < 3 \end{cases} \end{aligned}$$



Range = $[1, \infty)$

$$\begin{aligned} \text{b } f(x) &= -|x - 3| + 1 = \begin{cases} -(x - 3) + 1 & \text{if } x \geq 3 \\ -(3 - x) + 1 & \text{if } x < 3 \end{cases} \\ &= \begin{cases} -x + 4 & \text{if } x \geq 3 \\ -2 + x & \text{if } x < 3 \end{cases} \end{aligned}$$



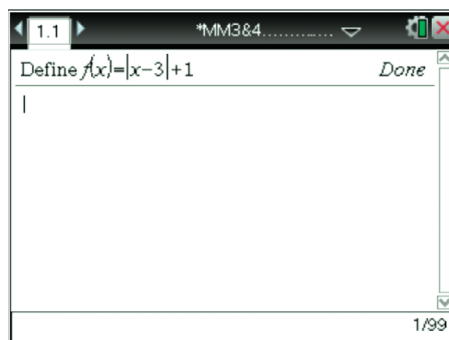
Range = $(-\infty, 1]$

Using the TI-Nspire

Complete as follows:

Define $f(x) = \text{abs}(x - 3) + 1$

The absolute value function can be obtained by typing **abs**, found as a command in the catalog (Ⓢ) (1) (A) or found as a template using (ctrl)+(menu)>**Math Templates**. This can also be found by using the template key (Ⓢ).

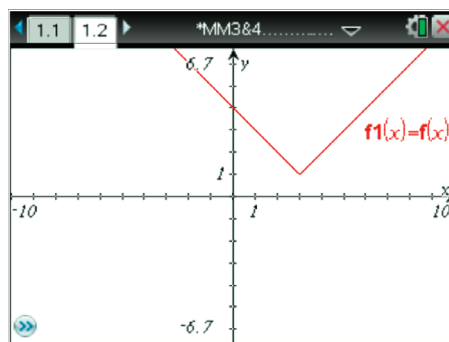


Open a **Graphs** application

(ctrl)+(I)>**Graphs**) and let $f1(x) = f(x)$.

Press **enter** to obtain the graph.

Note that the expression **abs** $(x - 3) + 1$ could have been entered directly for $f1(x)$.



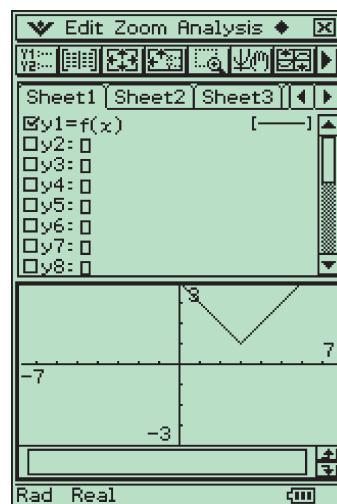
Using the Casio ClassPad

Tap **Interactive-Define** and enter the function name, variable and function as shown.

```
Define f(x)=abs(x-3)+1
done
```

To enter the absolute value, press  and look in the $\boxed{\text{mth}}$ OPTN to find the $|x|$ symbol.

In  enter $f(x)$ into $y1$, tick the box to select and tap  to create the graph.



Note that the expression could be directly entered in the ' $y1 =$ ' line but this gives you greater flexibility to use the function in other ways if required.

Exercise 1D

1 Evaluate each of the following:

- | | | |
|----------------------------|-------------------------------|-------------------------------|
| a $ -5 + 3$ | b $ -5 + -3 $ | c $ -5 - -3 $ |
| d $ -5 - -3 - 4$ | e $ -5 - -3 - -4 $ | f $ -5 + -3 - -4 $ |

2 On a number line, illustrate each of the following sets and represent the sets using interval notation:

- | | | |
|-------------------------------|----------------------------------|----------------------------------|
| a $\{x: x < 3\}$ | b $\{x: x \geq 5\}$ | c $\{x: x - 2 \leq 1\}$ |
| d $\{x: x - 2 < 3\}$ | e $\{x: x + 3 \geq 5\}$ | f $\{x: x + 2 \leq 1\}$ |

3 Sketch the graphs of each of the following functions and state the range of each of the functions:

- | | |
|-------------------------------|--------------------------------|
| a $f(x) = x - 4 + 1$ | b $f(x) = - x + 3 + 2$ |
| c $f(x) = x + 4 - 1$ | d $f(x) = 2 - x - 1 $ |

1.5 Sums and products of functions

The domain of f is denoted by $\text{dom } f$ and the domain of g by $\text{dom } g$. Let f and g be functions such that $\text{dom } f \cap \text{dom } g \neq \emptyset$. The sum, $f + g$, and the product, fg , as functions on $\text{dom } f \cap \text{dom } g$ are defined by:

$$1 \quad (f + g)(x) = f(x) + g(x) \quad \text{and} \quad 2 \quad (fg)(x) = f(x)g(x)$$

The domain of both $f + g$ and fg is the intersection of the domains of f and g , i.e. the values of x for which both f and g are defined.

Graphing sums of functions will be discussed in Section 3.9.

Example 13

If $f(x) = \sqrt{x-2}$ for all $x \geq 2$ and $g(x) = \sqrt{4-x}$ for all $x \leq 4$, find:

a $f + g$ **b** $(f + g)(3)$ **c** fg **d** $(fg)(3)$

Solution

a $\text{dom } f \cap \text{dom } g = [2, 4]$

$$\begin{aligned} (f + g)(x) &= f(x) + g(x) \\ &= \sqrt{x-2} + \sqrt{4-x} \end{aligned}$$

$$\text{dom } (f + g) = [2, 4]$$

b $(f + g)(3) = \sqrt{3-2} + \sqrt{4-3}$
 $= 2$

c $(fg)(x) = f(x)g(x)$
 $= \sqrt{(x-2)(4-x)}$
 $\text{dom } (fg) = [2, 4]$

d $(fg)(3) = \sqrt{(3-2)(4-3)}$
 $= 1$

Exercise 1E

1 For each of the following, find $(f + g)(x)$ and $(fg)(x)$ and state the domain for both $f + g$ and fg :

a $f(x) = 3x, g(x) = x + 2$

b $f(x) = 1 - x^2$ for all $x \in [-2, 2]$ and $g(x) = x^2$ for all $x \in \mathbb{R}^+$

c $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$ for $x \in [1, \infty)$

d $f(x) = x^2, x \geq 0$ and $g(x) = \sqrt{4-x}, 0 \leq x \leq 4$

2 Functions f, g, h , and k are defined by:

i $f(x) = x^2 + 1, x \in \mathbb{R}$

ii $g(x) = x, x \in \mathbb{R}$

iii $h(x) = \frac{1}{x^2}, x \neq 0$

iv $k(x) = \frac{1}{x}, x \neq 0$

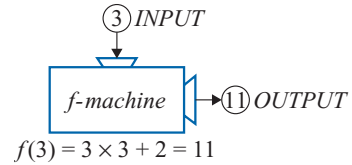
a State which of the above functions are odd and which are even.

b Form the functions of $f + h, fh, g + k, gk, f + g, fg$, stating which are odd and which are even.

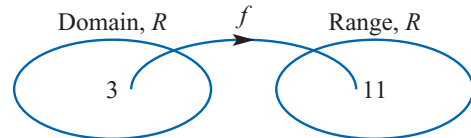
1.6 Composite functions

A function may be considered to be similar to a machine for which the input (domain) is processed to produce an output (range).

For example, the following diagram represents an ‘ f -machine’ where $f(x) = 3x + 2$



An alternative diagram is:



With many processes, more than one machine operation is required to produce an output. Suppose an output is the result of one function being applied after another,

e.g., $f(x) = 3x + 2$
followed by $g(x) = x^2$

This is illustrated diagrammatically on the right.

A new function h is formed.

The rule for h is $h(x) = (3x + 2)^2$

The diagram shows $f(3) = 11$ and then $g(11) = 121$.

This may be written:

$$h(3) = g(f(3)) = g(11) = 121$$

Similarly, $h(-2) = g(f(-2)) = g(-4) = 16$

h is said to be the **composition** of g with f .

This is written $h = g \circ f$ (read ‘composition of f followed by g ’) and the rule for h is defined by $h(x) = g(f(x))$. In the example we have considered:

$$\begin{aligned} h(x) &= g(f(x)) \\ &= g(3x + 2) \\ &= (3x + 2)^2 \end{aligned}$$

The domain of the function $h = g \circ f = \text{domain of } f$.

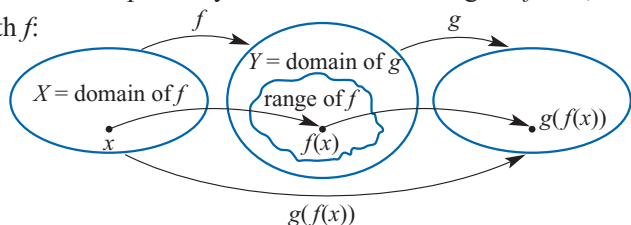
In general for the composition of g with f to be defined, $\text{range of } f \subseteq \text{domain of } g$.

When this composition (or composite function) of g with f is defined it is denoted $g \circ f$.

For functions f and g with domains X and Y respectively and such that the range of $f \subseteq Y$, we define the composite function of g with f :

$g \circ f: X \rightarrow R$, where

$$g \circ f(x) = g(f(x))$$



Example 14

Find both $f \circ g$ and $g \circ f$, stating the domain and range of each where:

$$f: R \rightarrow R, f(x) = 2x - 1 \quad \text{and} \quad g: R \rightarrow R, g(x) = 3x^2$$

Solution

To determine the existence of a composite function, it is useful to form a table of domains and ranges.

	Domain	Range
g	R	$R^+ \cup \{0\}$
f	R	R

$f \circ g$ is defined since $\text{ran } g \subseteq \text{dom } f$, and $g \circ f$ is defined since $\text{ran } f \subseteq \text{dom } g$.

$$\begin{aligned} f \circ g(x) &= f(g(x)) \\ &= f(3x^2) \\ &= 2(3x^2) - 1 \\ &= 6x^2 - 1 \end{aligned}$$

and $\text{dom } f \circ g = \text{dom } g = R$ and $\text{ran } f \circ g = [-1, \infty)$

$$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= g(2x - 1) \\ &= 3(2x - 1)^2 \\ &= 12x^2 - 12x + 3 \end{aligned}$$

$$\begin{aligned} \text{dom } g \circ f &= \text{dom } f \\ &= R \end{aligned}$$

$$\text{ran } g \circ f = [0, \infty)$$

It can be seen from this example that in general $f \circ g \neq g \circ f$.

Using the TI-Nspire

Define $f(x) = 2x - 1$ and $g(x) = 3x^2$.

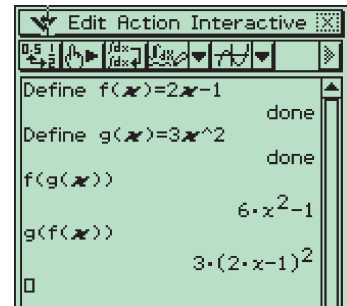
The rules for $f \circ g$ and $g \circ f$ can now be found using $f(g(x))$ and $g(f(x))$.

1.1 *MM3&4	
Define $f(x)=2 \cdot x-1$	Done
Define $g(x)=3 \cdot x^2$	Done
$f(g(x))$	$6 \cdot x^2-1$
$g(f(x))$	$3 \cdot (2 \cdot x-1)^2$
4/99	

Using the Casio ClassPad

Define $f(x) = 2x - 1$ and $g(x) = 3x^2$.

The rules for $f \circ g$ and $g \circ f$ can now be found using $f(g(x))$ and $g(f(x))$.



Example 15

For the functions $g(x) = 2x - 1$, $x \in R$ and $f(x) = \sqrt{x}$, $x \geq 0$:

- State which of $f \circ g$ and $g \circ f$ is defined.
- For the composite function that is defined, state the domain and rule.

Solution

- Range of $f \subseteq$ domain of g
 but range of $g \not\subseteq$ domain of f .
 $\therefore g \circ f$ is defined but $f \circ g$ is not defined.

- $$g \circ f(x) = g(f(x))$$

$$= g(\sqrt{x})$$

$$= 2\sqrt{x} - 1$$

$$\text{dom } g \circ f = \text{dom } f = R^+ \cup \{0\}$$

	Domain	Range
g	R	R
f	$R^+ \cup \{0\}$	$R^+ \cup \{0\}$

Example 16

For the functions $f(x) = x^2 - 1$, $x \in R$, and $g(x) = \sqrt{x}$, $x \geq 0$:

- State why $g \circ f$ is not defined.
- Define a restriction f^* of f such that $g \circ f^*$ is defined and find $g \circ f^*$.

Solution

- a** Range of $f \not\subseteq$ domain of g .
 $\therefore g \circ f$ is not defined.

	Domain	Range
f	R	$[-1, \infty)$
g	$R^+ \cup \{0\}$	$R^+ \cup \{0\}$

- b** For $g \circ f$ to be defined,
 range of $f \subseteq$ domain of g , i.e. range of
 $f \subseteq R^+ \cup \{0\}$. For range of f to be a
 subset of $R^+ \cup \{0\}$, the domain of f
 must be restricted to a subset of:

$$\{x: x \leq -1\} \cup \{x: x \geq 1\}, \text{ or } R \setminus (-1, 1).$$

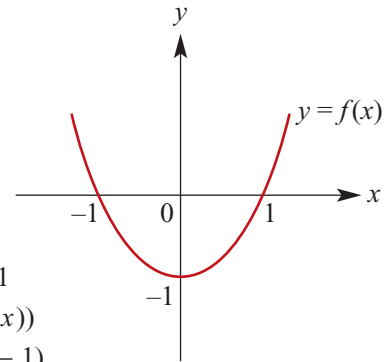
So we define f^* by:

$$\begin{aligned} f^*: R \setminus (-1, 1) &\rightarrow R, f^*(x) = x^2 - 1 \\ g \circ f^*(x) &= g(f^*(x)) \\ &= g(x^2 - 1) \\ &= \sqrt{x^2 - 1} \end{aligned}$$

$$\text{dom } g \circ f^* = \text{dom } f^* = R \setminus (-1, 1)$$

The composite function $g \circ f^*$ is:

$$g \circ f^*: R \setminus (-1, 1) \rightarrow R, g \circ f^*(x) = \sqrt{x^2 - 1}$$



Compositions involving the modulus function

Functions with rules of the form $y = |f(x)|$ and $y = f(|x|)$ are considered in this section.

Functions of the form $y = |f(x)|$

$|f|$ is the composition $g \circ f$ where $g(x) = |x|$. The function f is applied first and then the modulus function. The following observation enables the graph of functions with rule of the form $y = |f(x)|$ to be sketched if the graph of $y = f(x)$ is known:

$$|f(x)| = f(x) \text{ if } f(x) \geq 0 \quad \text{and} \quad |f(x)| = -f(x) \text{ if } f(x) < 0$$

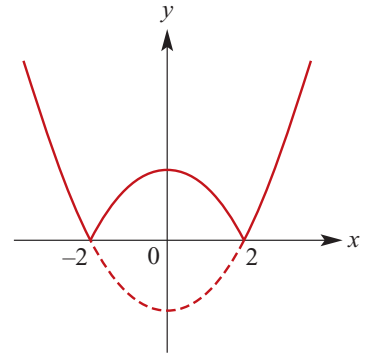
Example 17

Sketch the graphs of each of the following:

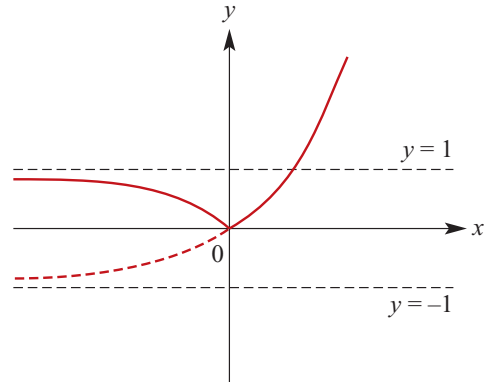
- a** $y = |x^2 - 4|$ **b** $y = |2^x - 1|$

Solution

- a** The graph of $y = x^2 - 4$ is drawn and the negative part reflected in the x -axis.



- b** The graph of $y = 2^x - 1$ is drawn and the negative part reflected in the x -axis.

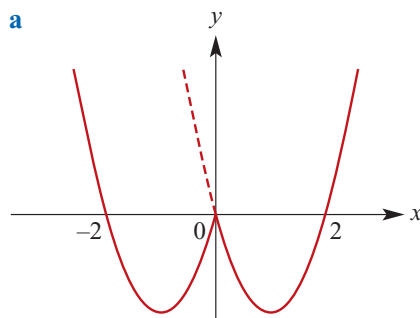
**Functions of the form $y = f(|x|)$**

The graphs of functions with rules of the form $y = f(|x|)$ where $x \in \mathbb{R}$ are sketched by reflecting the graph of $y = f(x)$, for $x \geq 0$, in the y -axis. The function with rule $f(|x|)$ is the result of the composition $f \circ g$ where $g(x) = |x|$.

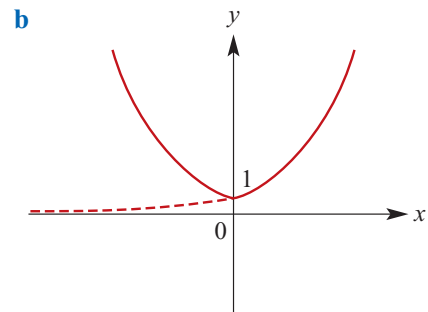
Example 18

Sketch the graphs of each of the following:

- a** $y = |x|^2 - 2|x|$ (This is the rule for the function $f \circ g$ where $f(x) = x^2 - 2x$ and $g(x) = |x|$.)
- b** $y = 2^{|x|}$ (This is the rule for the function $f \circ g$ where $f(x) = 2^x$ and $g(x) = |x|$.)

Solution

The graph of $y = x^2 - 2x$, $x \geq 0$, is reflected in the y -axis.



The graph of $y = 2^x$, $x \geq 0$, is reflected in the y -axis.

Exercise 1F



- 1 For each of the following, find $f(g(x))$ and $g(f(x))$:

a $f(x) = 2x - 1, g(x) = 2x$ c $f(x) = 2x - 1, g(x) = 2x - 3$ e $f(x) = 2x^2 + 1, g(x) = x - 5$	b $f(x) = 4x + 1, g(x) = 2x + 1$ d $f(x) = 2x - 1, g(x) = x^2$ f $f(x) = 2x + 1, g(x) = x $
--	---

- 2 For the functions $f(x) = 2x - 1$ and $h(x) = 3x + 2$, find:

a $f \circ h(x)$	b $h(f(x))$	c $f \circ h(2)$	d $h \circ f(2)$
e $f(h(3))$	f $h(f(-1))$	g $f \circ h(0)$	

- 3 For the functions $f(x) = x^2 + 2x$ and $h(x) = 3x + 1$, find:

a $f \circ h(x)$	b $h \circ f(x)$	c $f \circ h(3)$
d $h \circ f(3)$	e $f \circ h(0)$	f $h \circ f(0)$

- 4 For the functions $h: R \setminus \{0\} \rightarrow R, h(x) = \frac{1}{x^2}$ and $g: R^+ \rightarrow R, g(x) = 3x + 2$, find:

a $h \circ g$ (state rule and domain)	b $g \circ h$ (state rule and domain)
c $h \circ g(1)$	d $g \circ h(1)$

- 5 f and g are the functions given by $f: R \rightarrow R, f(x) = x^2 - 4$ and $g: R^+ \cup \{0\} \rightarrow R, g(x) = \sqrt{x}$.

a State the ranges of f and g .	b Find $f \circ g$, stating its range.
c Explain why $g \circ f$ does not exist.	

- 6 Let f and g be functions given by:

$$f: R \setminus \{0\} \rightarrow R, f(x) = \frac{1}{2} \left(\frac{1}{x} + 1 \right)$$

$$g: R \setminus \left\{ \frac{1}{2} \right\} \rightarrow R, g(x) = \frac{1}{2x - 1}$$

Find:

- | | |
|----------------------|---|
| a $f \circ g$ | b $g \circ f$, and state the range in each case |
|----------------------|---|
- 7 The functions f and g are defined by $f: R \rightarrow R, f(x) = x^2 - 2$ and $g: \{x: x \geq 0\} \rightarrow R$, where $g(x) = \sqrt{x}$.

a Explain why $g \circ f$ does not exist.	b Find $f \circ g$ and sketch its graph.
--	---

 - 8 **a** For $f(x) = 4 - x$ and $g(x) = |x|$, find $f \circ g$ and $g \circ f$ and sketch the graphs of each of these functions.
b For $f(x) = 9 - x^2$ and $g(x) = |x|$, find $f \circ g$ and $g \circ f$ and sketch the graphs of each of these functions.
c For $f: R \setminus \{0\} \rightarrow R, f(x) = \frac{1}{x}$ and $g: R \setminus \{0\} \rightarrow R, g(x) = |x|$, find $f \circ g$ and $g \circ f$ and sketch the graphs of each of these functions.

- 9 $f: \{x: x \leq 3\} \rightarrow R, f(x) = 3 - x$ and $g: R \rightarrow R, g(x) = x^2 - 1$
- Show that $f \circ g$ is not defined.
 - Define a restriction g^* of g such that $f \circ g^*$ is defined and find $f \circ g^*$.
- 10 $f: R^+ \rightarrow R, f(x) = x^{-\frac{1}{2}}$ and $g: R \rightarrow R, g(x) = 3 - x$
- Show that $f \circ g$ is not defined.
 - By suitably restricting the domain of g , obtain a function g_1 such that $f \circ g_1$ is defined.
- 11 Let $f: R \rightarrow R, f(x) = x^2$ and let $g: \{x: x \leq 3\} \rightarrow R, g(x) = \sqrt{3 - x}$. State with reasons whether:
- $f \circ g$ exists
 - $g \circ f$ exists
- 12 Let $f: S \rightarrow R, f(x) = \sqrt{4 - x^2}$ and S be the set of all real values of x for which $f(x)$ is defined. Let $g: R \rightarrow R, g(x) = x^2 + 1$.
- Find S .
 - Find the range of f and the range of g .
 - State whether or not $f \circ g$ and $g \circ f$ are defined and give a reason for each assertion.
- 13 Let a be a positive number, let $f: [2, \infty) \rightarrow R, f(x) = a - x$ and let $g: (-\infty, 1] \rightarrow R, g(x) = x^2 + a$. Find all values of a for which $f \circ g$ and $g \circ f$ both exist.



1.7 Inverse functions

If f is a one-to-one function, then for each number y in the range of f there is exactly one number, x , in the domain of f such that $f(x) = y$.

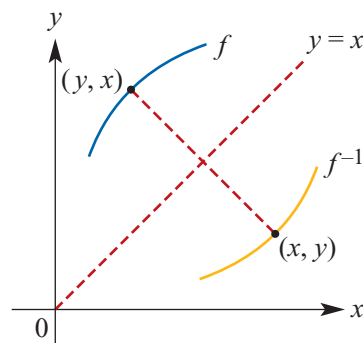
Thus if f is a one-to-one function, a new function f^{-1} , called the inverse of f , may be defined by:

$$f^{-1}(x) = y \text{ if } f(y) = x, \text{ for } x \in \text{ran } f, y \in \text{dom } f$$

It is not difficult to see what the relation between f and f^{-1} means geometrically. The point (x, y) is on the graph of f^{-1} if the point (y, x) is on the graph of f . Therefore, to get the graph of f^{-1} from the graph of f , the graph of f is to be reflected in the line $y = x$.

From this the following is evident:

$$\begin{aligned} \text{dom } f^{-1} &= \text{ran } f \\ \text{ran } f^{-1} &= \text{dom } f \end{aligned}$$



A function has an inverse function if and only if it is one-to-one.

$$\begin{aligned} \text{We note } f \circ f^{-1}(x) &= x, \text{ for all } x \in \text{dom } f^{-1} \\ f^{-1} \circ f(x) &= x, \text{ for all } x \in \text{dom } f \end{aligned}$$

Example 19

Find the inverse function f^{-1} of the function $f(x) = 2x - 3$.

Solution**Method 1**

The graph of f has equation $y = 2x - 3$ and the graph of f^{-1} has equation $x = 2y - 3$, i.e. x and y are interchanged. Solve for y .

$$\begin{aligned} x + 3 &= 2y \\ \text{and} \quad y &= \frac{1}{2}(x + 3) \\ \therefore f^{-1}(x) &= \frac{1}{2}(x + 3) \end{aligned}$$

$$\begin{aligned} \text{and dom } f^{-1} &= \text{ran } f \\ &= R \end{aligned}$$

Method 2

We require f^{-1} such that:

$$\begin{aligned} f(f^{-1}(x)) &= x \\ \therefore 2f^{-1}(x) - 3 &= x \\ \therefore f^{-1}(x) &= \frac{1}{2}(x + 3) \end{aligned}$$

$$\begin{aligned} \text{and dom } f^{-1} &= \text{ran } f \\ &= R \end{aligned}$$

Example 20

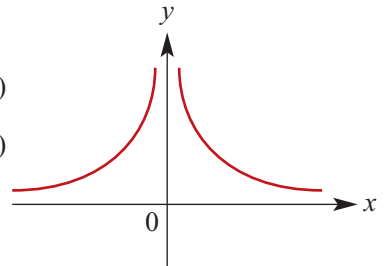
f is the function defined by $f(x) = \frac{1}{x^2}$, $x \in R \setminus \{0\}$. Define a suitable restriction for f , f^* , such that f^{*-1} exists.

Solution

f is not a one-to-one function. Therefore the inverse function f^{-1} is not defined. The following restricted functions of f are one-to-one.

$$\begin{aligned} f_1: (0, \infty) &\rightarrow R, f_1(x) = \frac{1}{x^2} & \text{Range } f_1 &= (0, \infty) \\ f_2: (-\infty, 0) &\rightarrow R, f_2(x) = \frac{1}{x^2} & \text{Range } f_2 &= (0, \infty) \end{aligned}$$

Let f^* be f_1 and determine f_1^{-1} .



Method 1Interchanging x and y :

$$x = \frac{1}{y^2}$$

$$y^2 = \frac{1}{x}$$

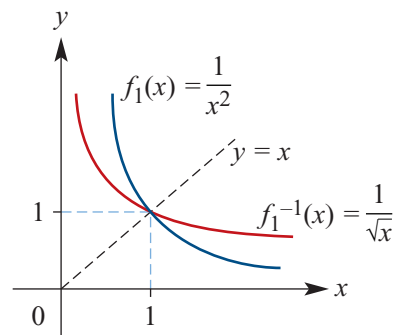
$$\therefore y = \pm \frac{1}{\sqrt{x}}$$

$$\text{But range } f_1^{-1} = \text{domain } f_1$$

$$= (0, \infty)$$

$$\therefore f_1^{-1} = \frac{1}{\sqrt{x}}, \text{ ran } f_1^{-1} = (0, \infty) \text{ and dom } f_1^{-1} = \text{ran } f_1 = (0, \infty)$$

$$\therefore f_1^{-1}: (0, \infty) \rightarrow R, f_1^{-1}(x) = \frac{1}{\sqrt{x}}$$

**Method 2**We require f_1^{-1} such that:

$$f_1(f_1^{-1}(x)) = x$$

$$\therefore \frac{1}{[f_1^{-1}(x)]^2} = x$$

$$\therefore f_1^{-1}(x) = \pm \frac{1}{\sqrt{x}}$$

$$\text{But range } f_1^{-1} = \text{domain } f_1$$

$$= (0, \infty)$$

$$\therefore f_1^{-1} = \frac{1}{\sqrt{x}}, \text{ ran } f_1^{-1} = (0, \infty) \text{ and dom } f_1^{-1} = \text{ran } f_1 = (0, \infty)$$

$$\therefore f_1^{-1}: (0, \infty) \rightarrow R, f_1^{-1}(x) = \frac{1}{\sqrt{x}}$$

Exercise 1G**1** For each of the following, find the rule for the inverse:

a $f: R \rightarrow R, f(x) = x - 4$

b $f: R \rightarrow R, f(x) = 2x$

c $f: R \rightarrow R, f(x) = \frac{3x}{4}$

d $f: R \rightarrow R, f(x) = \frac{3x-2}{4}$

2 Find the inverse of each of the following functions, stating the domain and range for each:

a $f: [-2, 6] \rightarrow R, f(x) = 2x - 4$

b $g(x) = \frac{1}{9-x}, x > 9$

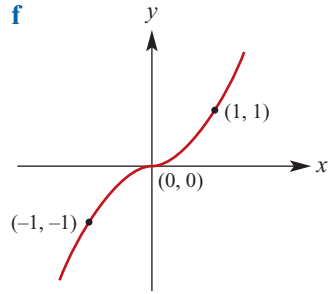
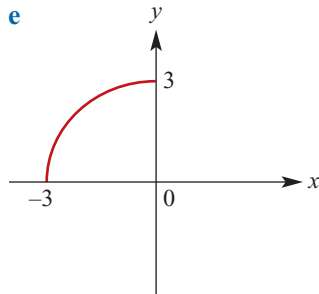
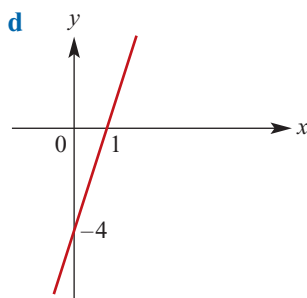
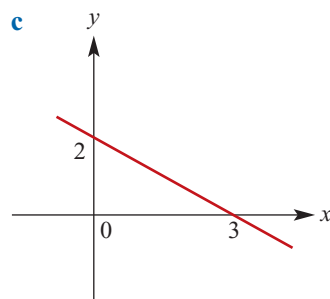
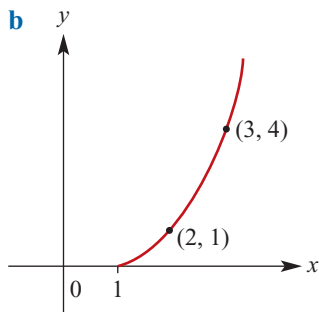
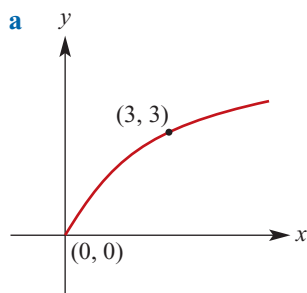
c $h(x) = x^2 + 2, x \geq 0$

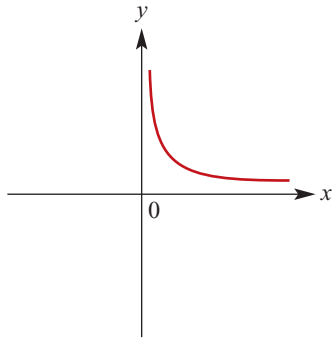
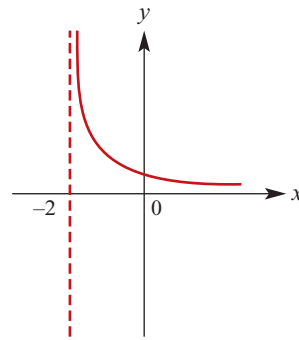
d $f: [-3, 6] \rightarrow R, f(x) = 5x - 2$

e $g: (1, \infty) \rightarrow R, g(x) = x^2 - 1$

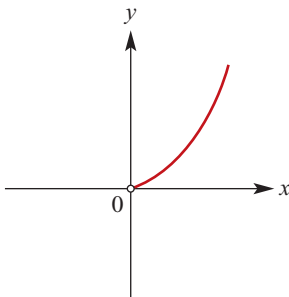
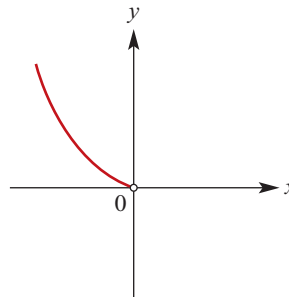
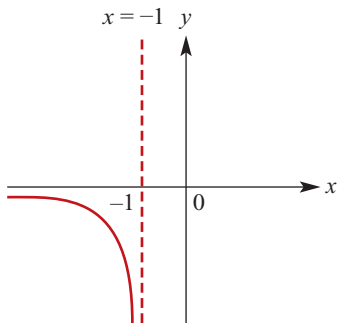
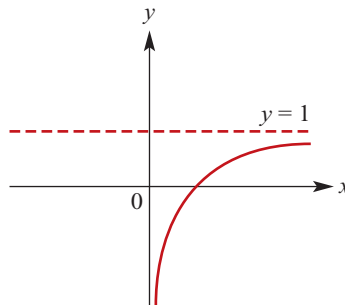
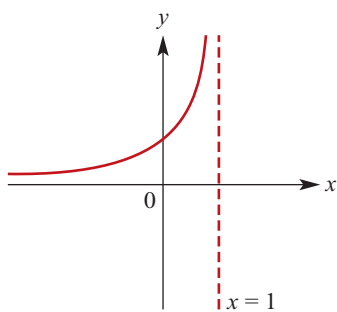
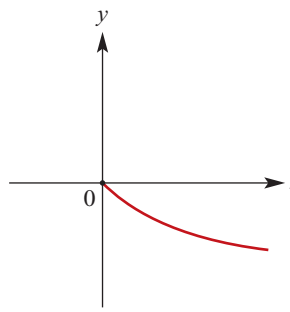
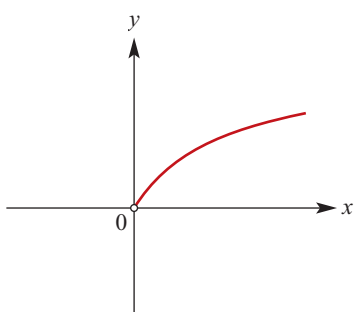
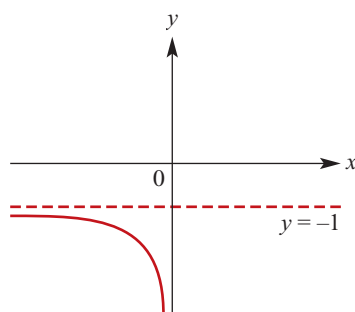
f $h: R^+ \rightarrow R, h(x) = \sqrt{x}$

- 3 Consider the function $g: [-1, \infty) \rightarrow \mathbb{R}$, $g(x) = x^2 + 2x$.
- a** Find g^{-1} , stating the domain and range. **b** Sketch the graph of g^{-1} .
- 4 Let $f: S \rightarrow \mathbb{R}$, where $S = \{x: 0 \leq x \leq 3\}$ and $f(x) = 3 - 2x$. Find $f^{-1}(2)$ and the domain of f^{-1} .
- 5 Find the inverse of each of the following functions, stating the domain and range of each:
- a** $f: [-1, 3] \rightarrow \mathbb{R}$, $f(x) = 2x$ **b** $f: [0, \infty) \rightarrow \mathbb{R}$, $f(x) = 2x^2 - 4$
c $\{(1, 6), (2, 4), (3, 8), (5, 11)\}$ **d** $h: \mathbb{R}^- \rightarrow \mathbb{R}$, $h(x) = \sqrt{-x}$
e $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + 1$ **f** $g: (-1, 3) \rightarrow \mathbb{R}$, $g(x) = (x + 1)^2$
g $g: [1, \infty) \rightarrow \mathbb{R}$, $g(x) = \sqrt{x - 1}$ **h** $h: [0, 2] \rightarrow \mathbb{R}$, $h(x) = \sqrt{4 - x^2}$
- 6 For each of the following functions, sketch the graph of the function and on the same set of axes sketch the graph of the inverse function. For each of the functions state the rule, domain and range of the inverse. It is advisable to draw in the line with equation $y = x$ for each set of axes.
- a** $y = 2x + 4$ **b** $f(x) = \frac{3 - x}{2}$
c $f: [2, \infty) \rightarrow \mathbb{R}$, $f(x) = (x - 2)^2$ **d** $f: [1, \infty) \rightarrow \mathbb{R}$, $f(x) = (x - 1)^2$
e $f: (-\infty, 2] \rightarrow \mathbb{R}$, $f(x) = (x - 2)^2$ **f** $f: \mathbb{R}^+ \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$
g $f: \mathbb{R}^+ \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x^2}$ **h** $h(x) = \frac{1}{2}(x - 4)$
- 7 Copy each of the following graphs and on the same set of axes draw the inverse of each of the corresponding functions:



g**h**

8 Match each of the graphs of **a**, **b**, **c** and **d** with its inverse.

a**b****c****d****A****B****C****D**

- 9 a Let $f: A \rightarrow R$, $f(x) = \sqrt{3-x}$. If A is the set of all real values of x for which $f(x)$ is defined, find A .
- b Let $g: [b, 2] \rightarrow R$ where $g(x) = 1 - x^2$. If b is the smallest real value such that g has an inverse function, find b and $g^{-1}(x)$.

1.8 Applications

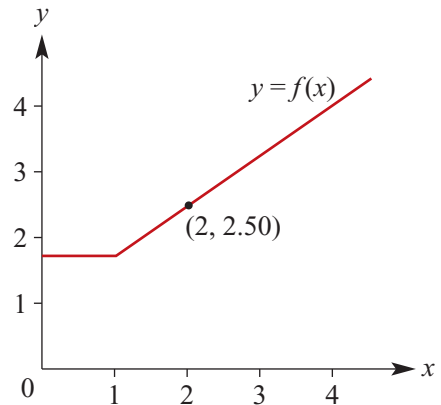
Example 21

The cost of a taxi trip in a particular city is \$1.75 up to and including 1 km. After 1 km the passenger pays an additional 75 cents per kilometre. Find the function f which describes this method of payment and sketch the graph of $y = f(x)$.

Solution

Let x denote the length of the trip.

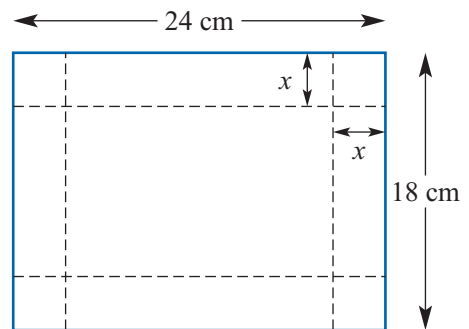
$$\text{Then } f(x) = \begin{cases} 1.75 & \text{for } 0 \leq x \leq 1 \\ 1.75 + 0.75(x - 1) & \text{for } x > 1 \end{cases}$$



Example 22

A rectangular piece of cardboard has dimensions 18 cm by 24 cm. Four squares each x cm by x cm are cut from the corners. An open box is formed by folding up the flaps.

Find a function for V , which gives the volume of the box in terms of x , and state the domain of the function.



Solution

The dimensions of the box will be:

$$24 - 2x, 18 - 2x \text{ and } x$$

The volume of the box is determined by the function:

$$V(x) = (24 - 2x)(18 - 2x)x$$

For the box to be formed:

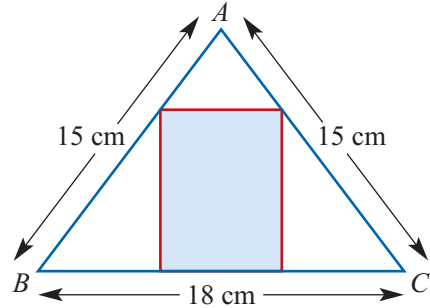
$$24 - 2x > 0 \text{ and } 18 - 2x > 0 \text{ and } x > 0$$

Therefore, $x < 12$ and $x < 9$ and $x > 0$. Therefore the domain is $(0, 9)$.

Example 23

A rectangle is inscribed in an isosceles triangle with the dimensions as shown.

Find an area of the rectangle function and state the domain.



Solution

Let the height of the rectangle be y cm and the width $2x$ cm.

The height (h cm) of the triangle can be determined by Pythagoras' theorem:

$$h = \sqrt{15^2 - 9^2} = 12$$

In the diagram to the right, $AYX \sim ABD$ (i.e., triangle AYX is similar to triangle ABD).

$$\text{Therefore } \frac{x}{9} = \frac{12-y}{12}$$

$$\text{and } \frac{12x}{9} = 12 - y$$

$$\text{which implies } y = 12 - \frac{12x}{9}$$

$$\text{The area of the rectangle, } A = 2xy \\ \text{and } A(x) = 2x \left(12 - \frac{12x}{9} \right)$$

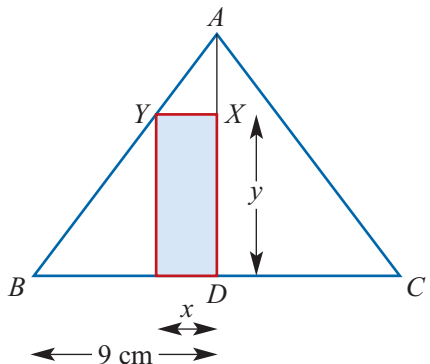
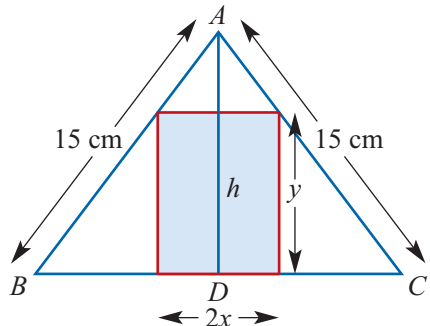
For the rectangle to be formed,

$$x > 0 \text{ and } 12 - \frac{12x}{9} > 0$$

$$\therefore x > 0 \text{ and } x < 9$$

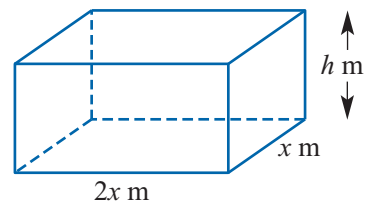
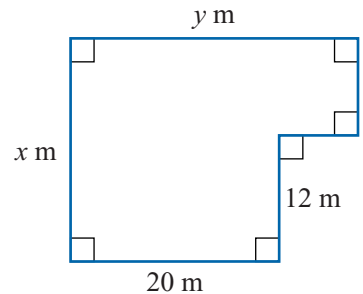
The domain of the function is $(0, 9)$.

$$\begin{aligned} A: (0, 9) \rightarrow R, \quad A(x) &= 2x \left(12 - \frac{12x}{9} \right) \\ &= \frac{24x}{9} (9 - x) \end{aligned}$$



Exercise 1H

- 1 The cost of a taxi trip in a particular city is \$4.00 up to and including 2 km. After 2 km the passenger pays an additional \$2.00 per kilometre. Find the function f which describes this method of payment and sketch the graph of $y = f(x)$, where x is the number of kilometres travelled. (Use a continuous model.)
- 2 A rectangular piece of cardboard has dimensions 20 cm by 36 cm. Four squares each x cm by x cm are cut from the corners. An open box is formed by folding up the flaps. Find a function for V , which gives the volume of the box in terms of x , and state the domain for the function.
- 3 The dimensions of an enclosure are shown. The perimeter of the enclosure is 160 m.
 - a Find a rule for the area, $A \text{ m}^2$, of the enclosure in terms of x .
 - b State a suitable domain of the function $A(x)$.
 - c Sketch the graph of A against x .
 - d Find the maximum possible area of the enclosure and state the corresponding values of x and y .
- 4 A cuboid tank is open at the top and the internal dimensions of its base are x m and $2x$ m. The height is h m. The volume of the tank is V cubic metres and the volume is fixed. Let $S \text{ m}^2$ denote the internal surface area of the tank.
 - a Find S in terms of:
 - i x and h
 - ii V and x
 - b State the maximal domain for the function defined by the rule in a ii.
 - c If $2 < x < 15$ find the maximum value of S if $V = 1000 \text{ m}^3$.
- 5 A man walks at a speed of 2 km/h for 45 minutes and then runs at 4 km/h for 30 minutes. Let S km be the distance the man has run after t minutes. The distance travelled can be described by:



$$S(t) = \begin{cases} at & \text{if } 0 \leq t \leq c \\ bt + d & \text{if } c < t \leq e \end{cases}$$



- a Find the values a, b, c, d, e .
 - b Sketch the graph of $S(t)$ against t .
 - c State the range of the function.
- 6 Suppose Australia Post charged the following rates for airmail letters to Africa: \$1.20 up to 20 g; \$2.00 over 20 g and up to 50 g; \$3.00 over 50 g and up to 150 g.
 - a Write a cost function, C (\$), in terms of the mass, m (g) for letters up to 150 g.
 - b Sketch the graph of the function, stating the domain and range.

Chapter summary

- A **relation** is a set of ordered pairs.
- The **domain** of a relation S is the set of all first elements of the ordered pairs in S . The domain of f is denoted by **dom** f .
- The **range** of a relation S is the set of all the second elements of the ordered pairs in S . The range of f is denoted by **ran** f .
- A **function** is a relation such that no two ordered pairs of the relation have the same first element.
- For each x in the domain of a function f there is a unique element, y , in the range such that $(x, y) \in f$. The element y is called the **image** of x under f or the **value** of f at x and is denoted by $f(x)$.
- A function f is said to be **one-to-one** if for $a, b \in \text{dom } f$, $a \neq b$, then $f(a) \neq f(b)$.
- A function which is not one-to-one is **many-to-one**.
- If the domain of a function is unspecified, then the domain is the largest subset of R for which the rule is defined. This set is called the **implied domain** or **maximal domain** of the function rule.
- A function f is **odd** if $f(-x) = -f(x)$ for all x in the domain of f .
- A function f is **even** if $f(-x) = f(x)$ for all x in the domain of f .
- Let f and g be functions such that $\text{dom } f \cap \text{dom } g \neq \emptyset$. The **sum** $f + g$, and the **product**, fg , as functions on $\text{dom } f \cap \text{dom } g$ are defined by:

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (fg)(x) = f(x) \cdot g(x)$$

- If f is a one-to-one function, a new function, f^{-1} , called the **inverse** of f , may be defined by:

$$f^{-1}(x) = y \text{ if } f(y) = x, \text{ for } x \in \text{ran } f, y \in \text{dom } f$$

- For a function f and its inverse f^{-1} :

$$\begin{aligned} \text{dom } f^{-1} &= \text{ran } f \\ \text{ran } f^{-1} &= \text{dom } f \end{aligned}$$

- The inverse of any relation may be defined. The inverse relation is not a function unless the initial relation is a one-to-one function.
- For a relation $S = \{(a, b)\}$ the inverse relation is $\{(b, a)\}$.
- The modulus or absolute value of a real number x is denoted by $|x|$ and is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

It may also be defined as $|x| = \sqrt{x^2}$

■ The function $|x|$ has the following properties:

- $|ab| = |a||b|$
- $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$
- $|a + b| \leq |a| + |b|$. If a and b are both non-negative or both non-positive, then equality holds.
- If $a \geq 0$, $|x| \leq a$ is equivalent to $-a \leq x \leq a$
- If $a \geq 0$, $|x - k| \leq a$ is equivalent to $k - a \leq x \leq k + a$
- The composition of functions f and g is denoted by $f \circ g$. The rule is given by $f(g(x))$. The domain of $f \circ g$ is the domain of g . The composition $f \circ g$ is defined if the range of $g \subseteq$ the domain of f .

Multiple-choice questions

- For the function with rule $f(x) = \sqrt{6 - 2x}$, which of the following is the maximal domain?
A $(-\infty, 6]$ **B** $[3, \infty)$ **C** $(-\infty, 6]$ **D** $(3, \infty)$ **E** $(-\infty, 3]$
- For $f: [-1, 3) \rightarrow R$, $f(x) = -x^2$, the range is:
A R **B** $(-9, 0]$ **C** $(-\infty, 0]$ **D** $(-9, -1]$ **E** $[-9, 0]$
- For $f(x) = 3x^2 + 2x$, $f(2a) =$
A $20a^2 + 4a$ **B** $6a^2 + 2a$ **C** $6a^2 + 4a$ **D** $36a^2 + 4a$ **E** $12a^2 + 4a$
- For $f(x) = 2x - 3$, $f^{-1}(x) =$
A $2x + 3$ **B** $\frac{1}{2}x + 3$ **C** $\frac{1}{2}x + \frac{3}{2}$ **D** $\frac{1}{2x - 3}$ **E** $\frac{1}{2}x - 3$
- For $f: (a, b] \rightarrow R$, $f(x) = 10 - x$ where $a < b$ the range is:
A $(10 - a, 10 - b)$ **B** $(10 - a, 10 - b]$ **C** $(10 - b, 10 - a)$
D $(10 - b, 10 - a]$ **E** $[10 - b, 10 - a)$

- 6 For the function with rule

$$f(x) = \begin{cases} x^2 + 5 & x \geq 3 \\ -x + 6 & x < 3 \end{cases}$$

the value of $f(a + 3)$, where a is a negative real number, is:

- A** $a^2 + 6a + 14$ **B** $-a + 9$ **C** $-a + 3$
D $a^2 + 14$ **E** $a^2 + 8a + 8$
- Which one of the following sets is a possible domain for the function with rule $f(x) = (x + 3)^2 - 6$ if the inverse function is to exist?
A R **B** $[-6, \infty)$ **C** $(-\infty, 3]$ **D** $[6, \infty)$ **E** $(-\infty, 0]$
 - If $f(x) = 3x^2$ and $g(x) = 2x + 1$, then $f(g(a))$ is equal to:
A $12a^2 + 3$ **B** $12a^2 + 12a + 3$ **C** $6a^2 + 1$
D $6a^2 + 4$ **E** $4a^2 + 4a + 1$

9 For which one of the following functions does an inverse function not exist?

A $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x - 4$

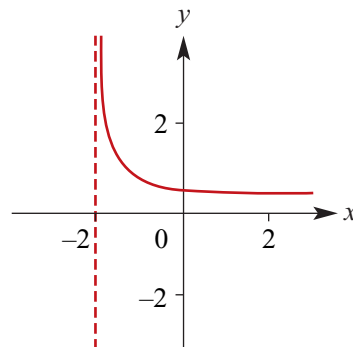
B $g: [-4, 4] \rightarrow \mathbb{R}, g(x) = \sqrt{16 - x^2}$

C $h: [0, \infty) \rightarrow \mathbb{R}, h(x) = -\frac{1}{5}x^2$

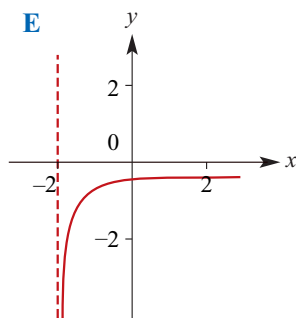
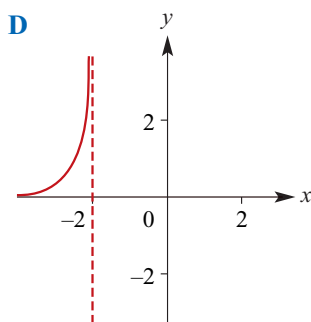
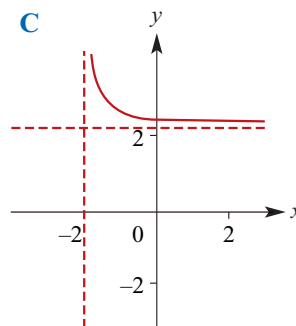
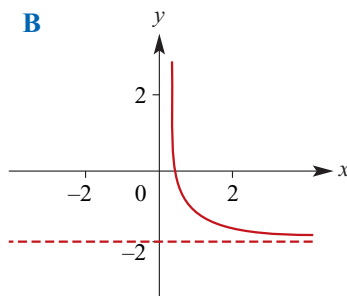
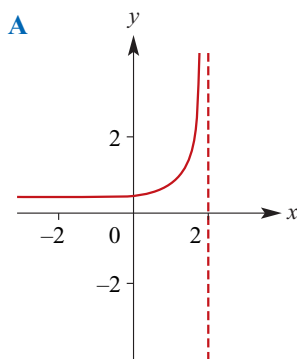
D $p: \mathbb{R}^+ \rightarrow \mathbb{R}, p(x) = \frac{1}{x^2}$

E $q: \mathbb{R} \rightarrow \mathbb{R}, q(x) = 2x^3 - 5$

10 The graph of the function f with rule $y = f(x)$ is shown on the right.



Which one of the following is most likely to be the graph of the inverse function of f ?



Short-answer questions (technology-free)

1 Sketch the graph of each of the following relations and state the implied domain and range:

a $f(x) = x^2 + 1$

b $f(x) = 2x - 6$

c $\{(x, y): x^2 + y^2 = 25\}$

d $\{(x, y): y \geq 2x + 1\}$

e $\{(x, y): y < x - 3\}$

- 2 For the function $g: [0, 5] \rightarrow R$, $g(x) = \frac{x+3}{2}$:
- a** Sketch the graph of $y = g(x)$. **b** State the range of g .
c Find g^{-1} , stating the domain and range of g^{-1} .
d Find $\{x: g(x) = 4\}$. **e** Find $\{x: g^{-1}(x) = 4\}$.
- 3 For $g(x) = 5x + 1$, find:
- a** $\{x: g(x) = 2\}$ **b** $\{x: g^{-1}(x) = 2\}$ **c** $\left\{x: \frac{1}{g(x)} = 2\right\}$
- 4 Sketch the graph of the function f , for which
- $$f(x) = \begin{cases} x+1 & \text{for } x > 2 \\ x^2 - 1 & \text{for } 0 \leq x \leq 2 \\ -x^2 & \text{for } x < 0 \end{cases}$$
- 5 Find the implied domains for each of the following:
- a** $f(x) = \frac{1}{2x-6}$ **b** $g(x) = \frac{1}{\sqrt{x^2-5}}$ **c** $h(x) = \frac{1}{(x-1)(x+2)}$
d $h(x) = \sqrt{25-x^2}$ **e** $f(x) = \sqrt{x-5} + \sqrt{15-x}$ **f** $h(x) = \frac{1}{3x-6}$
- 6 For $f(x) = (x+2)^2$ and $g(x) = x-3$, find $(f+g)(x)$ and $(fg)(x)$.
- 7 For $f: [1, 5] \rightarrow R$, $f(x) = (x-1)^2$, $g: R \rightarrow R$, $g(x) = 2x$, find $f+g$ and fg .
- 8 For $f: [3, \infty) \rightarrow R$, $f(x) = x^2 - 1$, find f^{-1} .
- 9 For $f(x) = 2x+3$ and $g(x) = -x^2$ find:
- a** $(f+g)(x)$ **b** $(fg)(x)$ **c** $\{x: (f+g)(x) = 0\}$
- 10 Let $f: (-\infty, 2] \rightarrow R$, $f(x) = 3x - 4$. On the one set of axes, sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$.

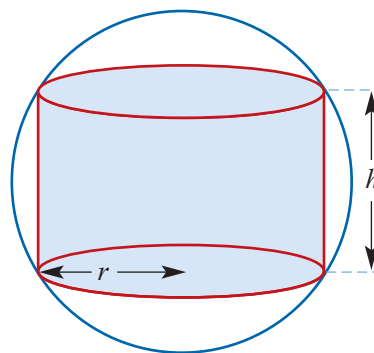
Extended-response questions

- 1 Self-Travel, a car rental firm, has two methods of charging for car rental:
- i** method 1: \$64 per day + 25 cents per kilometre
ii method 2: \$89 per day with unlimited travel.
- a** Write a rule for each method if x kilometres per day are travelled and the cost in dollars is C_1 using method 1 and C_2 using method 2.
b Draw the graph of each, using the same axes.
c Determine, from the graph, the distance that must be travelled per day if method 2 is cheaper than method 1.
- 2 Express the total surface area, S , of a cube as a function of:
- a** the length x of an edge **b** the volume V of the cube
- 3 Express the area of an equilateral triangle as a function of:
- a** the length s of each side **b** the altitude h

- 4 The base of a 3 m ladder leaning against a wall is x metres from the wall.
- Express the distance, d , from the top of the ladder to the ground as a function of x and sketch the graph of the function.
 - State the domain and range of the function.
- 5 A car travels half the distance of a journey at an average speed of 80 km/h and half at an average speed of x km/h.
- Define a function, S , which gives the average speed for the total journey as a function of x .
- 6 A cylinder is inscribed in a sphere with a radius of length 6 cm.

For the cylinder:

- Define a function, V_1 , which gives the volume of the cylinder as a function of the height (h).
(State the rule and domain.)
- Define a function, V_2 , which gives the volume of the cylinder as a function of the radius of the cylinder (r).
(State the rule and domain.)



- 7 Let $f: R \rightarrow R$ and $g: R \rightarrow R$, where $f(x) = x + 1$ and $g(x) = 2 + x^3$.
- State why $g \circ f$ exists and find $g \circ f(x)$.
 - State why $g \circ f$ is a function and find $(g \circ f)^{-1}(10)$.
- 8 A function f is defined as follows:
- $$f(x) = \begin{cases} x^2 - 4, & \text{for } x \in (-\infty, 2) \\ x, & \text{for } x \in [2, \infty) \end{cases}$$
- Sketch the graph of f .
 - Find the value of:
 - $f(-1)$
 - $f(3)$
 - Given $g: S \rightarrow R$ where $g(x) = f(x)$, find the largest set S so that the inverse of g exists and $-1 \in S$.
 - If $h(x) = 2x$, find $f(h(x))$ and $h(f(x))$.
- 9 Find the rule for the area, $A(t)$, enclosed by the graph of the function:



$$f(x) = \begin{cases} 3x, & 0 \leq x \leq 1 \\ 3, & x > 1 \end{cases}$$

the x -axis, the y -axis and the vertical line $x = t$ ($t \geq 0$). State the domain and range of the function.

10 Let $f: R \setminus \left\{ \frac{-d}{c} \right\} \rightarrow R$, $f(x) = \frac{ax+b}{cx+d}$

a Find the inverse function f^{-1} .

b Find the inverse function when:

i $a = 3, b = 2, c = 3, d = 1$

ii $a = 3, b = 2, c = 2, d = -3$

iii $a = 1, b = -1, c = -1, d = -1$

iv $a = -1, b = 1, c = 1, d = 1$

c Determine the possible values of a, b, c and d if $f = f^{-1}$.

11 The radius of the incircle of the right-angled triangle

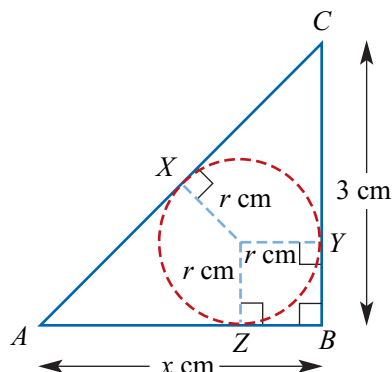
ABC is r cm. Find:

a i YB in terms of r

ii ZB in terms of r

iii AZ in terms of r and x

iv CY



b Use the geometric results $CY = CX$ and $AX = AZ$ to find an expression for r in terms of x .

c i Find r when $x = 4$.

ii Find x when $r = 0.5$.

d Use a CAS calculator to investigate the possible values r can take.

12 Let $f(x) = \frac{px+q}{x+r}$ where $x \in R \setminus \{-r, r\}$.

a If $f(x) = f(-x)$ for all x , show that $f(x) = p$ for $x \in R \setminus \{-r, r\}$.

b If $f(-x) = -f(x)$ for $x \neq 0$, find the rule for $f(x)$ in terms of q .

c If $p = 3, q = 8$ and $r = -3$:

i Find the inverse function of f .

ii Find the values of x for which $f(x) = x$.

13 a Let $f(x) = \frac{x+1}{x-1}$.

i Find $f(2), f(f(2)), f(f(f(2)))$.

ii Find $f(f(x))$.

b Let $f(x) = \frac{x-3}{x+1}$.

Find $f(f(x)), f(f(f(x)))$.



Revising linear functions and matrices

Objectives

- To **revise**:
 - methods for **solving linear equations**
 - methods for solving **simultaneous linear equations**
 - calculating the **gradient** of a straight line
 - interpreting and using the **general equation** of a straight line: $y = mx + c$
 - a method for determining the gradient of a line **perpendicular** to a given line
 - finding the **distance between** two points
 - finding the **midpoint** of a straight line
 - calculating the **angle between** two intersecting straight lines
 - matrix arithmetic
- To apply a knowledge of linear functions to **solving problems**.

It is assumed that the material in this chapter has been covered by students in *Essential Mathematical Methods 1 & 2*. The chapter provides a framework for revision with worked examples and practice exercises.

2.1 Linear equations

Exercise 2A



1 Solve the following linear equations:

a $3x - 4 = 2x + 6$

b $8x - 4 = 3x + 1$

c $3(2 - x) - 4(3 - 2x) = 14$

d $\frac{3x}{4} - 4 = 17$

e $6 - 3y = 5y - 62$

f $\frac{2}{3x - 1} = \frac{3}{7}$

$$\text{g} \quad \frac{2x-1}{3} = \frac{x+1}{4}$$

$$\text{h} \quad \frac{2(x-1)}{3} - \frac{x+4}{2} = \frac{5}{6}$$

$$\text{i} \quad 4y - \frac{3y+4}{2} + \frac{1}{3} = \frac{5(4-y)}{3}$$

$$\text{j} \quad \frac{x+1}{2x-1} = \frac{3}{4}$$

- 2 Solve each of the following pairs of simultaneous linear equations:

$$\text{a} \quad \begin{aligned} x-4 &= y \\ 4y-2x &= 8 \end{aligned}$$

$$\text{b} \quad \begin{aligned} 9x+4y &= 13 \\ 2x+y &= 2 \end{aligned}$$

$$\text{c} \quad \begin{aligned} 7x &= 18+3y \\ 2x+5y &= 11 \end{aligned}$$

$$\text{d} \quad \begin{aligned} 5x+3y &= 13 \\ 7x+2y &= 16 \end{aligned}$$

$$\text{e} \quad \begin{aligned} 19x+17y &= 0 \\ 2x-y &= 53 \end{aligned}$$

$$\text{f} \quad \begin{aligned} \frac{x}{5} + \frac{y}{2} &= 5 \\ x-y &= 4 \end{aligned}$$

- 3 An aircraft, used for fire spotting, flies from its base to locate a fire at an unknown distance, x km away. It travels straight to the fire and back, averaging 240 km/h for the outward trip and 320 km/h for the return trip. If the plane was away for 35 minutes, find the distance, x km.
- 4 A group of hikers is to travel x km by bus at an average speed of 48 km/h to an unknown destination. They then plan to walk back along the same route at an average speed of 4.8 km/h and to arrive back 24 hours after setting out in the bus. If they allow 2 h for lunch and rest, how far must the bus take them?
- 5 The length of a rectangle is 4 cm more than the width. If the length were to be decreased by 5 cm and the width decreased by 2 cm, the perimeter would be 18 cm. Calculate the dimensions of the rectangle.
- 6 In a basketball game a field goal scores two points and a free throw scores one point. John scored 11 points in the game and David 19 points. David scored the same number of free throws as John but twice as many field goals. How many field goals did each score?
- 7 The weekly wage, $\$w$, of a salesman consists of a fixed amount of $\$800$ and then $\$20$ for each unit he sells.
- If he sells n units a week, find a rule for his weekly wage, w , in terms of the number of units sold.
 - Find his wage if he sells 30 units.
 - How many units does he sell if his weekly wage is $\$1620$?
- 8 Water flows into a tank at a rate of 15 litres per minute. At the beginning the tank contained 250 litres.
- Write an expression for the volume, V litres, of water in the tank at time t minutes.
 - How many litres of water are there in the tank after an hour?
 - The tank has a capacity of 5000 litres. How long does it take to fill?
- 9 A tank contains 10 000 litres of water. Water flows out at a rate of 10 litres per minute.
- Write an expression for the volume, V litres, of water in the tank at time t minutes.
 - How many litres of water are there in the tank after an hour?
 - How long does it take for the tank to empty?

- 10** The cost, $\$C$, of hiring diving equipment is $\$100$ plus $\$25$ per hour.
- Write a rule which gives the total charge, $\$C$, of hiring the equipment for t hours (assume that parts of hours are paid for proportionately).
 - Find the cost of hiring the equipment for:
 - 2 hours
 - 2 hours 30 minutes
 - For how many hours can the equipment be hired if the following amounts are available?
 - $\$375$
 - $\$400$

2.2 Linear literal equations and simultaneous linear literal equations

A literal equation in x is an equation whose solution will be expressed in terms of pronumerals rather than numbers.

$2x + 5 = 7$ is an equation whose solution is the number 1.

In the literal equation $ax + b = c$, the solution is $x = \frac{c - b}{a}$.

Literal equations are solved in the same way as solving numerical equations or transposing formulas. Essentially, the literal equation is transposed to make x the subject.

Example 1

Solve the following for x .

a $px - q = r$ **b** $ax + b = cx + d$ **c** $\frac{a}{x} = \frac{b}{2x} + c$

Solution

a $px - q = r$ **b** $ax + b = cx + d$

$$px = r + q \qquad ax - cx = d - b$$

$$x = \frac{r + q}{p} \qquad x(a - c) = \frac{d - b}{a - c}$$

$$x = \frac{d - b}{a - c}$$

c $\frac{a}{x} = \frac{b}{2x} + c$

Multiply both sides of the equation by $2x$.

$$2a = b + 2xc$$

$$2a - b = 2xc$$

$$\frac{2a - b}{2c} = x$$

Simultaneous literal equations

Simultaneous literal equations are solved by the methods of solution of simultaneous equations, i.e. substitution and elimination.

Example 2

Solve each of the following pairs of simultaneous equations for x and y .

a $y = ax + c$
 $y = bx + d$

b $ax - y = c$
 $x + by = d$

Solution

a $ax + c = bx + d$
 $\therefore ax - bx = d - c$
 $x(a - b) = d - c$
 $x = \frac{d - c}{a - b}$

and therefore

$$y = a \left(\frac{d - c}{a - b} \right) + c$$

$$= \frac{ad - bc}{a - b}$$

b $ax - y = c \dots (1)$

$x + by = d \dots (2)$

Multiply (1) by b :

$abx - by = cb \dots (1')$

Add (1') and (2):

$$abx + x = cb + d$$

$$x(ab + 1) = cb + d$$

$$x = \frac{cb + d}{ab + 1}$$

Substitute in (1):

$$a \left(\frac{cb + d}{ab + 1} \right) - y = c$$

$$\therefore y = a \left(\frac{cb + d}{ab + 1} \right) - c$$

$$= \frac{ad - c}{ab + 1}$$

Exercise 2B

1 Solve each of the following for x :

a $ax + n = m$

c $\frac{ax}{b} + c = 0$

e $mx + n = nx - m$

g $\frac{b}{x - a} = \frac{2b}{x + a}$

i $-b(ax + b) = a(bx - a)$

k $\frac{x}{a} - 1 = \frac{x}{b} + 2$

m $\frac{p - qx}{t} + p = \frac{qx - t}{p}$

b $ax + b = bx$

d $px = qx + 5$

f $\frac{1}{x + a} = \frac{b}{x}$

h $\frac{x}{m} + n = \frac{x}{n} + m$

j $p^2(1 - x) - 2pqx = q^2(1 + x)$

l $\frac{x}{a - b} + \frac{2x}{a + b} = \frac{1}{a^2 - b^2}$

n $\frac{1}{x + a} + \frac{1}{x + 2a} = \frac{2}{x + 3a}$

2 For the simultaneous equations $ax + by = p$ and $bx - ay = q$, show that $x = \frac{ap + bq}{a^2 + b^2}$
 and $y = \frac{bp - aq}{a^2 + b^2}$

3 For the simultaneous equations $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$, show that $x = y = \frac{ab}{a+b}$

4 Solve each of the following pairs of simultaneous equations for x and y .

a $ax + y = c$

$x + by = d$

c $ax + by = t$

$ax - by = s$

e $(a+b)x + cy = bc$

$(b+c)y + ax = -ab$

b $ax - by = a^2$

$bx - ay = b^2$

d $ax + by = a^2 + 2ab - b^2$

$bx + ay = a^2 + b^2$

f $3(x-a) - 2(y+a) = 5 - 4a$

$2(x+a) + 3(y-a) = 4a - 1$

5 Write s in terms of a only in the following pairs of equations:

a $s = ah$

$h = 2a + 1$

c $as = a + h$

$h + ah = 1$

e $s = h^2 + ah$

$h = 3a^2$

g $s = 2 + ah + h^2$

$h = a - \frac{1}{a}$

b $s = ah$

$h = a(2 + h)$

d $as = s + h$

$ah = a + h$

f $as = a + 2h$

$h = a - s$

h $3s - ah = a^2$

$as + 2h = 3a$

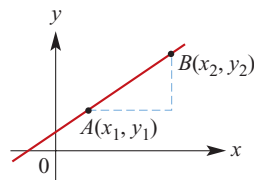
2.3 Linear coordinate geometry

The following is a summary of the material that is assumed to have been covered in *Essential Mathematical Methods 1 & 2*.

Gradient of a straight line joining two points



Gradient, $m = \frac{y_2 - y_1}{x_2 - x_1}$



The general equation of a straight line

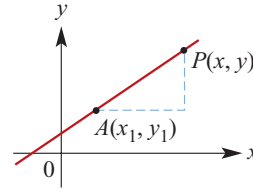
$$y = mx + c$$

where m is the gradient and c is the value of the intercept on the y -axis.

Equation of a straight line passing through a given point (x_1, y_1) and having a gradient, m

Equation of line is:

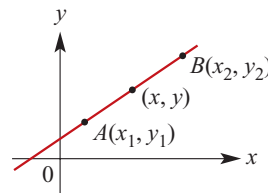
$$y - y_1 = m(x - x_1)$$



Equation of a line passing through two given points (x_1, y_1) and (x_2, y_2)

Equation of line is:

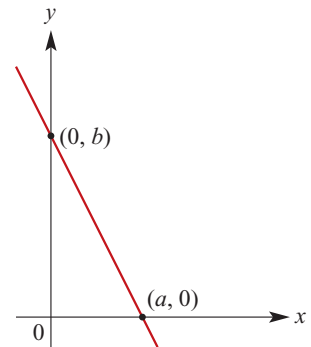
$$y - y_1 = m(x - x_1) \text{ where } m = \frac{y_2 - y_1}{x_2 - x_1}$$



The intercept form of the equation of a straight line

For a line passing through the points $(a, 0)$ and $(0, b)$, the equation is:

$$\frac{x}{a} + \frac{y}{b} = 1$$



Tangent of the angle of slope (θ)

$$\tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

where θ is the angle the line makes with the positive direction of the x -axis.

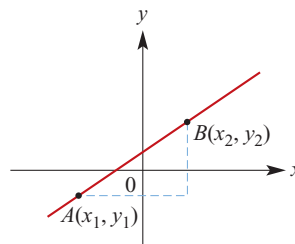
Product of gradients of two perpendicular straight lines

If two straight lines are perpendicular to each other, the product of their gradients is -1 .

$$m_1 m_2 = -1$$

Distance between two points

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

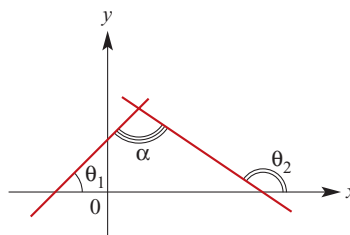


Midpoint

The midpoint of a straight line joining (x_1, y_1) and (x_2, y_2) is the point $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

The angle α between intersecting straight lines

$$\alpha = \theta_2 - \theta_1$$



Example 3

A fruit and vegetable wholesaler sells 30 kg of hydroponic tomatoes for \$148.50 and 55 kg of hydroponic tomatoes for \$247.50. Find a linear model for the cost, C dollars, to buy x kg of hydroponic tomatoes. How much would 20 kg of tomatoes cost?

Solution

Let $(x_1, C_1) = (30, 148.5)$ and $(x_2, C_2) = (55, 247.5)$.

The equation of the straight line is given by:

$$C - C_1 = m(x - x_1) \quad \text{where} \quad m = \frac{C_2 - C_1}{x_2 - x_1}$$

$$\text{Now } m = \frac{247.5 - 148.5}{55 - 30} = 3.96 \quad \text{so} \quad C - 148.5 = 3.96(x - 30)$$

Therefore the straight line is given by the equation $C = 3.96x + 29.7$

Substitute $x = 20$:

$$C = 3.96 \times 20 + 29.7 = 108.9$$

It would cost \$108.90 to buy 20 kg of tomatoes.

Exercise 2C

- 1 Find the coordinates of M , the midpoint of AB , where A and B have the following coordinates:

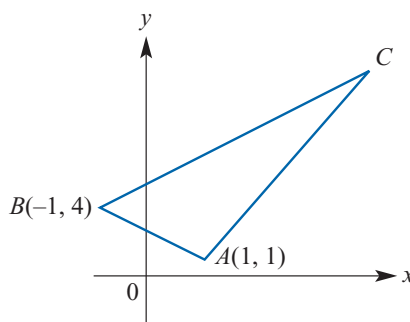
a $A(1, 4), B(5, 11)$

b $A(-6, 4), B(1, -8)$

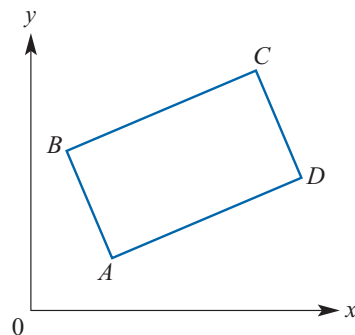
c $A(-1, -6), B(4, 7)$

- 2 Use $y = mx + c$ to sketch the graph of each of the following:
- a** $y = 3x - 3$ **b** $y = -3x + 4$ **c** $3y + 2x = 12$
d $4x + 6y = 12$ **e** $3y - 6x = 18$ **f** $8x - 4y = 16$
- 3 Find the equations of the following straight lines:
- a** gradient $+2$, passing through $(4, 2)$ **b** gradient -3 , passing through $(-3, 4)$
c passing through the points $(1, 3)$ and $(4, 7)$
d passing through the points $(-2, -3)$ and $(2, 5)$
- 4 Use the intercept method to find the equation of the straight lines passing through:
- a** $(-3, 0)$ and $(0, 2)$ **b** $(4, 0)$ and $(0, 6)$
c $(-4, 0)$ and $(0, -3)$ **d** $(0, -2)$ and $(6, 0)$
- 5 Write each of the following in intercept form and hence draw their graphs:
- a** $3x + 6y = 12$ **b** $4y - 3x = 12$
c $4y - 2x = 8$ **d** $\frac{3}{2}x - 3y = 9$
- 6 A printing firm charges \$35 for printing 600 sheets of headed notepaper and \$46 for printing 800 sheets. Find a linear model for the charge, C dollars, for printing n sheets. How much would they charge for printing 1000 sheets?
- 7 An electronic bank teller registered \$775 after it had counted 120 notes and \$975 after it had counted 160 notes.
- a** Find a formula for the sum registered, $(\$C)$, in terms of the number of notes (n) counted.
b Was there a sum already on the register when counting began?
c If so, how much?
- 8 Find the distance between each of the following pairs of points:
- a** $(2, 6), (3, 4)$ **b** $(5, 1), (6, 2)$ **c** $(-1, 3), (4, 5)$
d $(-1, 7), (1, -11)$ **e** $(-2, -6), (2, -8)$ **f** $(0, 4), (3, 0)$
- 9 **a** Find the equation of the straight line which passes through the point $(1, 6)$ and is:
- i** parallel to the line with equation $y = 2x + 3$
ii perpendicular to the line with equation $y = 2x + 3$
b Find the equation of the straight line which passes through the point $(2, 3)$ and is:
- i** parallel to the line with equation $4x + 2y = 10$
ii perpendicular to the line with equation $4x + 2y = 10$
- 10 Find the equation of the line which passes through the point of intersection of the lines $y = x$ and $x + y = 6$ and which is perpendicular to the line with equation $3x + 6y = 12$.
- 11 The length of the line joining $A(2, -1)$ and $B(5, y)$ is 5 units. Find y .

- 12** Find the equation of the line passing through the point $(-1, 3)$ which is:
- a** parallel to the lines with equations below **b** perpendicular to the lines with equations below.
- i** $2x + 5y - 10 = 0$ **ii** $4x + 5y + 3 = 0$
- 13** Find the angle that the lines joining the given points make with the positive direction of the x -axis:
- a** $(-4, 1), (4, 6)$ **b** $(2, 3), (-4, 6)$ **c** $(5, 1), (-1, -8)$ **d** $(-4, 2), (2, -8)$
- 14** Find the acute angle between the lines $y = 2x + 4$ and $y = -3x + 6$
- 15** Given the points $A(a, 3)$, $B(-2, 1)$ and $C(3, 2)$, find the possible value of a if the length of AB is twice the length of BC .
- 16** Three points have coordinates $A(1, 7)$, $B(7, 5)$ and $C(0, -2)$. Find:
- a** the equation of the perpendicular bisector of AB
- b** the point of intersection of this perpendicular bisector and BC .
- 17** The point (h, k) lies on the line $y = x + 1$ and is 5 units from the point $(0, 2)$. Write down two equations connecting h and k and hence find the possible values of h and k .
- 18** P and Q are the points of intersection of the line $\frac{y}{2} + \frac{x}{3} = 1$ with the x and y axes respectively. The gradient of QR is $\frac{1}{2}$, where R is the point with x -coordinate $2a$, $a > 0$.
- a** Find the y -coordinate of R in terms of a .
- b** Find the value of a if the gradient of PR is -2 .
- 19** The figure shows a triangle ABC with $A(1, 1)$, and $B(-1, 4)$.
The gradients of AB , AC and BC are $-3m$, $3m$ and m respectively.
- a** Find the value of m .
- b** Find the coordinates of C .
- c** Show that $AC = 2AB$.



- 20** In the rectangle $ABCD$, A and B are the points $(4, 2)$ and $(2, 8)$ respectively. Given that the equation of AC is $y = x - 2$, find:
- a** the equation of BC
- b** the coordinates of C
- c** the coordinates of D
- d** the area of rectangle $ABCD$.



- 21 $ABCD$ is a parallelogram, lettered anticlockwise, such that A and C are the points $(-1, 5)$ and $(5, 1)$ respectively.
- Find the coordinates of the midpoint of AC .
 - Given that BD is parallel to the line whose equation is $y + 5x = 2$, find the equation of BD .
 - Given that BC is perpendicular to AC , find:
 - the equation of BC
 - the coordinates of B
 - the coordinates of D .

2.4 Applications of linear functions

Example 4

There are two possible methods for paying gas bills:

- method A: a fixed charge of \$25 per quarter + 50c per unit of gas used
- method B: a fixed charge of \$50 per quarter + 25c per unit of gas used.

Determine the number of units which must be used before method B becomes cheaper than method A.

Solution

Let C_1 = charge in \$ using method A

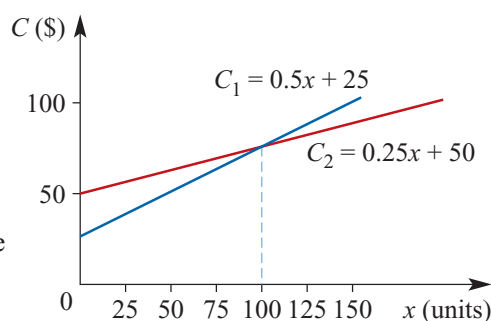
C_2 = charge in \$ using method B

x = number of units of gas used.

Now $C_1 = 25 + 0.5x$

$C_2 = 50 + 0.25x$

It can be seen from the graph that if the number of units exceeds 100, then method B is cheaper.



The solution could also be obtained by solving simultaneous linear equations:

$$\begin{aligned}
 C_1 &= C_2 \\
 25 + 0.5x &= 50 + 0.25x \\
 0.25x &= 25 \\
 x &= 100
 \end{aligned}$$

Exercise 2D

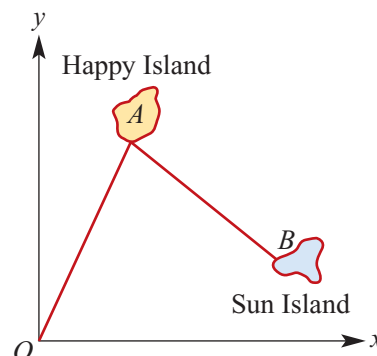


- 1 A car journey of 300 km lasts 4 h. Part of this journey is on a freeway at an average speed of 90 km/h. The rest is on country roads at an average speed of 70 km/h.

Let T be the time (in hours) spent on the freeway.

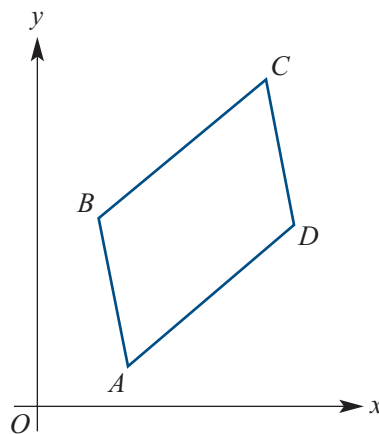
- In terms of T , state the number of hours travelling on country roads.

- b** State:
- i** the distance travelled on the freeway in terms of T
 - ii** the distance travelled on country roads in terms of T .
- c** Find:
- i** T
 - ii** the distance travelled on each type of road.
- 2** A farmer measured the quantity of water in a storage tank 20 days after it was filled and found it contained 3000 litres. After a further 15 days it was measured again and found to have 1200 litres in it. Assume that the amount of water in the tank decreases at a constant rate.
- a** Find the relation between L , the number of litres of water in the tank, and t , the number of days after the tank was filled.
 - b** How much water does the tank hold when it is full?
 - c** Sketch the graph of L against t for a suitable domain.
 - d** State this domain.
 - e** How long does it take for the tank to empty?
 - f** At what rate does the water leave the tank?
- 3** On a small island two rival taxi firms have the following fare structures:
- firm A: fixed charge of \$1 plus 40 cents per kilometre
 - firm B: 60 cents per kilometre, no fixed charge.
- a** Find expressions for C_A , the charge of firm A in terms of n , the number of kilometres travelled, and C_B , the charge of firm B in terms of the number of kilometres travelled.
 - b** Sketch the graph of the charges of each of the firms against the number of kilometres travelled on the one set of axes.
 - c** Find the distance for which both firms charge the same amount.
 - d** On a new set of axes sketch the graph of $D = C_A - C_B$ against n and explain what this graph represents.
- 4** A boat leaves from O to sail to two islands. The boat arrives at a point A on Happy Island with coordinates $(10, 22.5)$ (units are in kilometres).
- a** Find the equation of the line through points O and A .
 - b** Find the distance OA to the nearest metre.
The boat arrives at Sun Island at point B .
The coordinates of point B are $(23, 9)$.
 - c** Find the equation of line AB .
 - d** A third island lies on the perpendicular bisector of line segment AB . Its port is denoted by C . It is known that the x -coordinate of C is 52.
Find the y -coordinate of the point C .



- 5 $ABCD$ is a parallelogram with coordinates $A(2, 2)$, $B(1.5, 4)$ and $C(6, 6)$.

- a Find the gradient of:
 - i line AB
 - ii line AD
- b Find the equation of:
 - i line BC
 - ii line CD
- c Find the equation of the diagonals AC and BD .
- d Find the coordinates of the point of intersection of the diagonals.



2.5 Review of matrix arithmetic

A matrix is a rectangular array of numbers.

- Two matrices **A** and **B** are equal when:
- each has the same number of rows and the same number of columns
 - they have the same number or element at corresponding positions.

$$\begin{bmatrix} 3 & 6 & 5 \\ -6 & 10 & 6 \\ 12 & 1 & 0 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \text{ implies } a = 3, b = 6, c = 5, \text{ etc.}$$

- The size or **dimension** of a matrix is described by specifying the number of rows (m) and the number of columns (n). The dimension is written $m \times n$. The dimensions of the following matrices in order are:

$$3 \times 2, \quad 1 \times 4, \quad 3 \times 3, \quad 1 \times 1$$

$$\begin{bmatrix} -1 & 2 \\ -3 & 4 \\ 5 & 6 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 & 5 & 6 \end{bmatrix} \quad \begin{bmatrix} \sqrt{2} & \pi & 3 \\ 0 & 0 & 1 \\ \sqrt{2} & 0 & \pi \end{bmatrix} \quad [5]$$

- Addition will be defined for two matrices only when they have the same dimension. The sum is found by adding corresponding elements.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

Subtraction is defined in a similar way.

- If $\mathbf{A}$ is an $m \times n$ matrix and k is a real number, $k\mathbf{A}$ is defined to be an $m \times n$ matrix whose elements are k times the corresponding element of $\mathbf{A}$.

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

- If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times r$ matrix, then the product $\mathbf{AB}$ is the $m \times r$ matrix whose entries are determined as follows.

To find the entry in row i and column j of $\mathbf{AB}$, single out row i in matrix $\mathbf{A}$ and column j in matrix $\mathbf{B}$. Multiply the corresponding entries from the row and column and then add up the resulting products.

The product $\mathbf{AB}$ is defined only if the number of columns of $\mathbf{A}$ is the same as the number of rows of $\mathbf{B}$.

$$\text{For } \mathbf{A} = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 4 & 0 \\ 1 & 2 \\ 0 & 3 \end{bmatrix}$$

$\mathbf{A}$ is a 2×3 matrix and $\mathbf{B}$ is a 3×2 matrix. Therefore $\mathbf{AB}$ is a 2×2 matrix.

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 1 & 2 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 4 + 3 \times 1 + 4 \times 0 & 2 \times 0 + 3 \times 2 + 4 \times 3 \\ 5 \times 4 + 6 \times 1 + 7 \times 0 & 5 \times 0 + 6 \times 2 + 7 \times 3 \end{bmatrix} = \begin{bmatrix} 11 & 18 \\ 26 & 33 \end{bmatrix} \end{aligned}$$

- The identity matrix for the family of $n \times n$ matrices is the matrix with ones in the ‘top left’ to ‘bottom right’ diagonal and zeros in all other positions. This is denoted by $\mathbf{I}$.
- If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same dimension and $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ then $\mathbf{A}$ is said to be the inverse of $\mathbf{B}$ and $\mathbf{B}$ is said to be the inverse of $\mathbf{A}$. The inverse of $\mathbf{A}$ is denoted by $\mathbf{A}^{-1}$.

$$\text{■ If } \mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ then } \mathbf{A}^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

$\det(\mathbf{A}) = ad - bc$ is the **determinant** of matrix $\mathbf{A}$.

- A square matrix is said to be **regular** if its inverse exists. Those square matrices which do not have an inverse are called **singular** matrices. The determinant of a singular matrix is 0.

Using the TI-Nspire

Entering matrices using the template

2-by-2 matrices are easiest entered using the 2-by-2 matrix template $\left(\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}\right)$ (ctrl) + $\left(\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}\right)$ on the Clickpad), as shown.

Notice that there is also a template for entering m -by- n matrices.

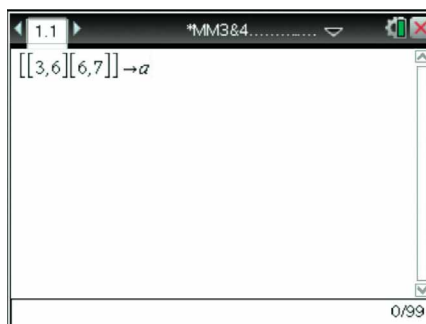
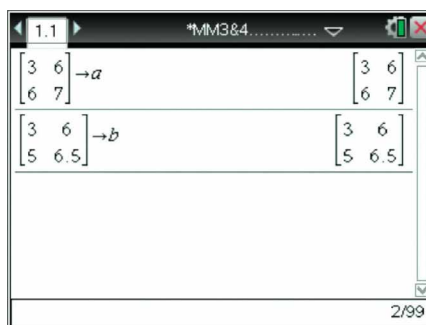
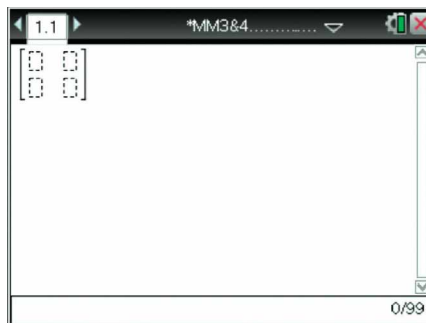
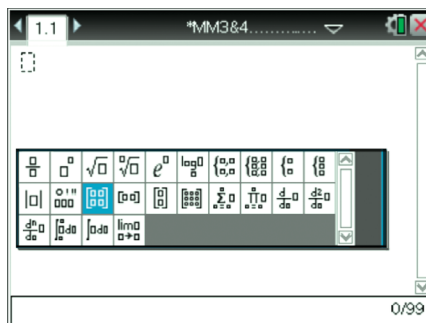
The matrix template can also be obtained using (ctrl) + (menu) > **Math Templates**

To enter the matrix $A = \begin{bmatrix} 3 & 6 \\ 6 & 7 \end{bmatrix}$, use the touch pad to move between the entries of the 2 by 2 matrix template and **store** (ctrl) (var) the matrix as a .

Define the matrix $B = \begin{bmatrix} 3 & 6 \\ 5 & 6.5 \end{bmatrix}$ in a similar way.

Entering matrices directly

To enter matrix A without using the template, enter the matrix row by row as $[[3, 6] [6, 7]]$ and **store** (ctrl) (var) the matrix as a .



Addition, subtraction and multiplication by a scalar

Once A and B are defined as above, $A + B$, $A - B$ and kA can easily be determined.

$a+b$	$\begin{bmatrix} 6 & 12 \\ 11 & 13.5 \end{bmatrix}$
$a-b$	$\begin{bmatrix} 0 & 0 \\ 1 & 0.5 \end{bmatrix}$
$k \cdot a$	$\begin{bmatrix} 3 \cdot k & 6 \cdot k \\ 6 \cdot k & 7 \cdot k \end{bmatrix}$

Matrix multiplication

Multiplication of $A = \begin{bmatrix} 3 & 6 \\ 6 & 7 \end{bmatrix}$, and

$$B = \begin{bmatrix} 3 & 6 \\ 5 & 6.5 \end{bmatrix}.$$

The products AB and BA are shown.

$a \cdot b$	$\begin{bmatrix} 39 & 57 \\ 53 & 81.5 \end{bmatrix}$
$b \cdot a$	$\begin{bmatrix} 45 & 60 \\ 54 & 75.5 \end{bmatrix}$

Inverse and determinants

The operation of matrix inverse is obtained by raising the matrix to the power of -1 .

The **determinant** command

((**menu**)>**Matrix and Vector**>

Determinant) is used as shown.

a^{-1}	$\begin{bmatrix} -7 & 2 \\ 15 & 5 \end{bmatrix}$
$\det(a)$	-15

Using the Casio ClassPad

Matrix editor

To enter matrices, in  press , then in the **2D CALC** screen tap one of the matrix entry buttons    to enter row, column or square matrices.

To add an extra column to a matrix tap .

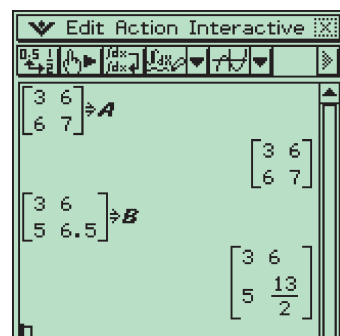
To add an extra row tap .

To expand in both directions, tap .

Enter the matrices A and B as shown using

 to store them as variables in the **2D**

OPTN tab menu.

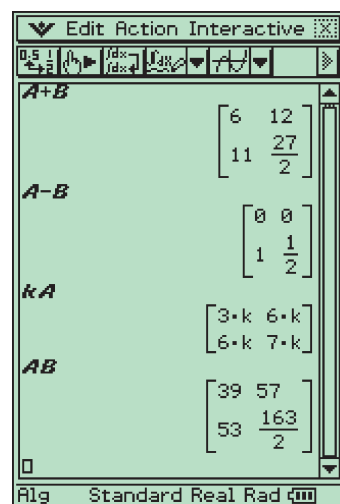


Addition, subtraction, multiplication by a scalar

Once A and B are defined as above, $A + B$, $A - B$ and kB can be determined.

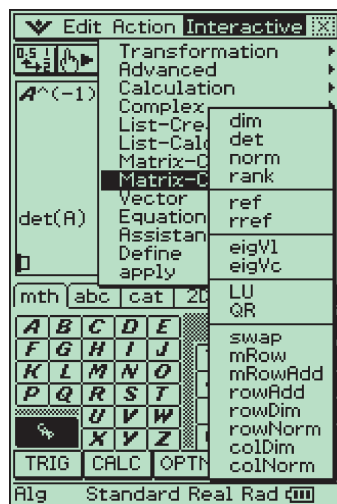
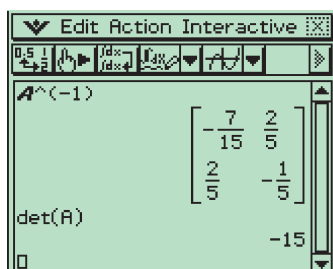
Multiplication

Multiplication AB is shown. Verify for yourself, by entering BA , that multiplication is not commutative.



Inverse and determinant

To find the inverse of a matrix enter A^{-1} . To find the determinant of the matrix, tap **Interactive – Matrix – Calculation – det** and enter the matrix A .



Example 5

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$ find:

a $\det(A)$ **b** A^{-1} **c** X if $AX = \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix}$ **d** Y if $YA = \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix}$

Solution

a $\det(A) = 3 \times 6 - 2 = 16$ **b** $A^{-1} = \frac{1}{16} \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix}$

c $AX = \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix}$

Multiply both sides (from the left) by A^{-1} .

$$A^{-1}AX = A^{-1} \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix}$$

$$\therefore IX = X = \frac{1}{16} \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 16 & 32 \\ 16 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\mathbf{d} \quad \mathbf{YA} = \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix}$$

Multiply both sides (from the right) by $\mathbf{A}^{-1}$.

$$\mathbf{YAA}^{-1} = \frac{1}{16} \begin{bmatrix} 5 & 6 \\ 7 & 2 \end{bmatrix} \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore \mathbf{YI} = \mathbf{Y} = \frac{1}{16} \begin{bmatrix} 24 & 8 \\ 40 & -8 \end{bmatrix}$$

$$\therefore \mathbf{Y} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{5}{2} & -\frac{1}{2} \end{bmatrix}$$

Exercise 2E

- 1 For the matrices $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -2 & -2 \\ 3 & 2 \end{bmatrix}$ find:

a $\mathbf{A} + \mathbf{B}$	b $\mathbf{AB}$	c $\mathbf{BA}$	d $\mathbf{A} - \mathbf{B}$	e $k\mathbf{A}$	f $2\mathbf{A} + 3\mathbf{B}$
g $\mathbf{A} - 2\mathbf{B}$	h $\det(\mathbf{A})$	i $\mathbf{A}^{-1}$	j $\det(\mathbf{B})$	k $\mathbf{B}^{-1}$	l $\det(\mathbf{AB})$
- 2 Find the inverse of the following regular matrices (k is any non-zero real number):

a $\begin{bmatrix} 3 & -1 \\ 4 & -1 \end{bmatrix}$	b $\begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix}$	c $\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
---	--	---
- 3 If $\mathbf{A}, \mathbf{B}$ are the regular matrices $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, find $\mathbf{A}^{-1}$, $\mathbf{B}^{-1}$.
 Also find $\mathbf{AB}$ and hence find, if possible, $(\mathbf{AB})^{-1}$.
 Also find from $\mathbf{A}^{-1}$, $\mathbf{B}^{-1}$, the products $\mathbf{A}^{-1}\mathbf{B}^{-1}$ and $\mathbf{B}^{-1}\mathbf{A}^{-1}$. What do you notice?
- 4 Consider the matrix $\mathbf{A} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$.

a Find $\mathbf{A}^{-1}$.	b If $\mathbf{AX} = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$, find $\mathbf{X}$.	c If $\mathbf{YA} = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$, find $\mathbf{Y}$.
-----------------------------------	--	--
- 5 If $\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 4 & -1 \\ 2 & 2 \end{bmatrix}$ and $\mathbf{C} = \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}$, find:

a $\mathbf{X}$ such that $\mathbf{AX} + \mathbf{B} = \mathbf{C}$	b $\mathbf{Y}$ such that $\mathbf{YA} + \mathbf{B} = \mathbf{C}$
---	---

- 6 If $\mathbf{A}$ is a 2×2 matrix, $a_{12} = a_{21} = 0$, $a_{11} \neq 0$, $a_{22} \neq 0$ then show that $\mathbf{A}$ is regular and find $\mathbf{A}^{-1}$.

7 For $\mathbf{A} = \begin{bmatrix} 3 & 6 & 5 \\ -6 & 10 & 6 \\ 12 & 1 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 6 \\ 2 & 1 & 0 \end{bmatrix}$ find, using a calculator:

a $\mathbf{A} + \mathbf{B}$ **b** $\mathbf{AB}$ **c** $\mathbf{BA}$ **d** $3\mathbf{A}$ **e** $\mathbf{B}^{-1}$
f $\mathbf{A} - \mathbf{B}$ **g** $\mathbf{A} - 2\mathbf{B}$ **h** $\mathbf{A} + 3\mathbf{B}$ **i** $(\mathbf{AB})^{-1}$ **j** $\mathbf{B}^{-1}\mathbf{A}^{-1}$

8 For $\mathbf{A} = \begin{bmatrix} 3 & 6 & 5 & -6 \\ 10 & 6 & 12 & 1 \\ 0 & 4 & 1 & 6 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 6 & 2 & 1 \\ 0 & 7 & 8 & 0 \end{bmatrix}$ find, using a calculator:

a $\mathbf{AB}$ **b** $\mathbf{BA}$ **c** $3\mathbf{A}$ **d** $\mathbf{B}^{-1}$ **e** $\mathbf{A} - \mathbf{B}$ **f** $\mathbf{A} - 2\mathbf{B}$

- 9 Solve each of the following matrix equations for $\mathbf{X}$:

a $\mathbf{AX} = \mathbf{C}$ **b** $\mathbf{AX} + \mathbf{D} = \mathbf{C}$

where $\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 6 \\ 2 & 1 & 0 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$

- 10 Solve each of the following matrix equations for $\mathbf{X}$:

a $\mathbf{AX} = \mathbf{C}$ **b** $\mathbf{AX} + \mathbf{D} = \mathbf{C}$

where $\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 6 & 2 & 1 \\ 0 & 7 & 8 & 0 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 1 \end{bmatrix}$

2.6 Solving systems of linear simultaneous equations in two variables

Inverse matrices can be used to solve certain sets of simultaneous linear equations. Consider the equations:

$$3x - 2y = 5$$

$$5x - 3y = 9$$

This can be written as:

$$\begin{bmatrix} 3 & -2 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

If $\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 5 & -3 \end{bmatrix}$ the determinant of $\mathbf{A}$ is $3(-3) - 5(-2) = 1$ which is not zero and so $\mathbf{A}^{-1}$ exists.

$$\mathbf{A}^{-1} = \begin{bmatrix} -3 & 2 \\ -5 & 3 \end{bmatrix}$$

Multiplying the matrix equation $\begin{bmatrix} 3 & -2 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$ on both sides on the left-hand side by $\mathbf{A}^{-1}$ and using the fact that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ yields the following:

$$\mathbf{A}^{-1} \left(\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} \right) = \mathbf{A}^{-1} \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$\therefore \mathbf{I} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ since } \mathbf{A}^{-1} \begin{bmatrix} 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

This is the solution to the simultaneous equations.

Check by substituting $x = 3$, $y = 2$ in the equations.

When dealing with simultaneous linear equations in two variables which represent straight lines that are parallel, then a singular matrix results.

For example the system

$$\begin{aligned} x + 2y &= 3 \\ -2x - 4y &= 6 \end{aligned}$$

has associated matrix equation

$$\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Note that the determinant of $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} = 1 \times -4 - (-2 \times 2) = 0$

There is no unique solution to the system of equations.

Example 6

Solve the simultaneous equations

$$3x - 2y = 6$$

$$7x + 4y = 7$$

Solution

The matrix equation is $\begin{bmatrix} 3 & -2 \\ 7 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$

Let $\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 7 & 4 \end{bmatrix}$

Then $\mathbf{A}^{-1} = \frac{1}{26} \begin{bmatrix} 4 & 2 \\ -7 & 3 \end{bmatrix}$

and $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{26} \begin{bmatrix} 4 & 2 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} = \frac{1}{26} \begin{bmatrix} 38 \\ -21 \end{bmatrix}$

Using the TI-Nspire

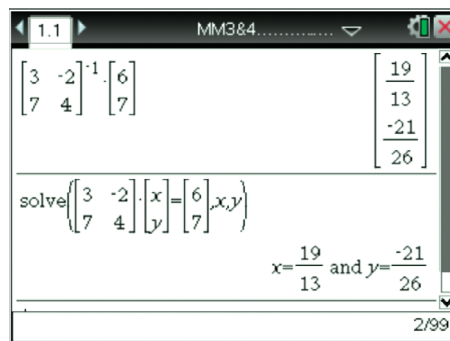
Enter the matrices as shown.

Both the 2-by-2 matrix template and the 2-by-1 matrix template can be found in the Math Templates, $\text{Ⓜ} \text{Ⓢ} (\text{ctrl}) + \text{Ⓜ} \text{Ⓢ} \text{Ⓢ}$ on the Clickpad).

The matrix template can also be obtained using $\text{Ⓜ} \text{Ⓢ} + \text{Ⓜ} \text{Ⓢ} > \text{Math Templates}$

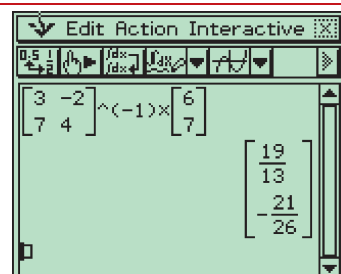
Note: it is also possible to use

solve $\left(\begin{bmatrix} 3 & -2 \\ 7 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}, x, y \right)$
to find the values of x and y .



Using the Casio ClassPad

Enter $\begin{bmatrix} 3 & -2 \\ 7 & 4 \end{bmatrix}^{(-1)} \times \begin{bmatrix} 6 \\ 7 \end{bmatrix}$



Remember that a 2×2 matrix is said to be singular if its determinant is equal to 0. The matrix being singular can correspond to one of two situations:

- 1 There are infinitely many solutions.
- 2 There is no solution.

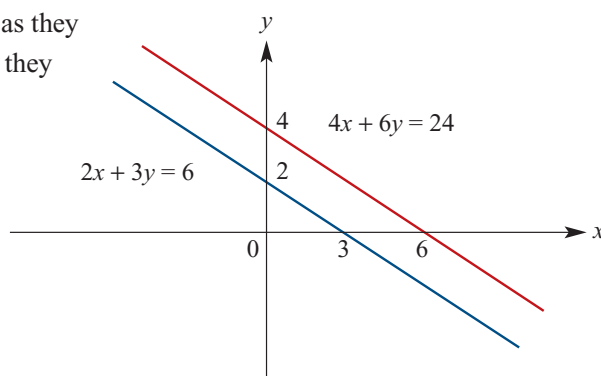
Example 7

Explain why the simultaneous equations $2x + 3y = 6$ and $4x + 6y = 24$ have no solution.

Solution

The equations have no solution as they correspond to parallel lines and they are different lines. There is no point of intersection.

Each of the lines has gradient $-\frac{2}{3}$.



The matrix of the coefficients of x and y is $\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$ and the determinant of this matrix is 0. That is, the matrix is singular.

If, for a system of two linear equations with two variables x and y , the 2×2 matrix of the coefficients of x and y is singular, then the system has either no solutions as discussed above or infinitely many solutions. This second case arises when the two equations represent the same line.

Example 8

The simultaneous equations $2x + 3y = 6$ and $4x + 6y = 12$ have infinitely many solutions. Describe these solutions through the use of a parameter.

Solution

Using the TI-Nspire

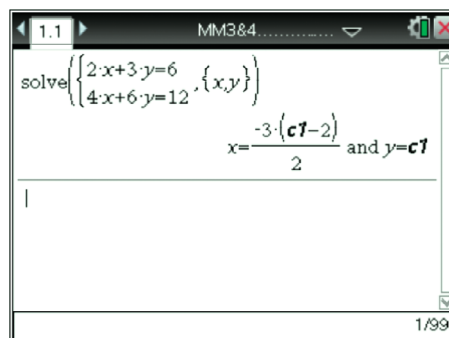
The parameter is a third variable. Note that the two equations represent the same straight line. They both have gradient $-\frac{2}{3}$ and y -intercept 2.

Let λ be this third variable.

In this case let $y = \lambda$. Then $x = \frac{-3(\lambda - 2)}{2}$ and the line can be described by

$$\left\{ \left(\frac{-3(\lambda - 2)}{2}, \lambda \right) : \lambda \in R \right\}.$$

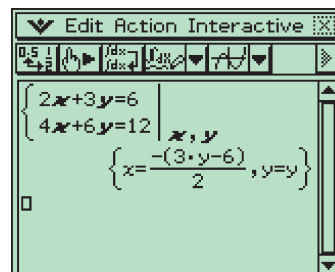
This may seem to make the situation unnecessarily complicated, but it is the solution given by the calculator, as shown opposite. The variable **c1** takes the place of λ .



Using the Casio ClassPad

The ClassPad does not introduce a third variable.

The solution is given in terms of y .



Example 9

Consider the simultaneous linear equations

$$(m - 2)x + y = 2 \text{ and } mx + 2y = k$$

Find the values of m and k such that the system of equations has:

- a** a unique solution **b** no solution **c** infinitely many solutions

Solution

Using a CAS calculator to find the solution

$$x = \frac{4 - k}{m - 4} \quad \text{and} \quad y = \frac{k(m - 2) - 2m}{m - 4}$$

- a** The solution is unique if $m \neq 4$ and k can be any real number.
b If $m = 4$, the equations become

$$2x + y = 2 \quad \text{and} \quad 4x + 2y = k$$

There is no solution if $m = 4$ and $k \neq 4$.

- c** If $m = 4$ and $k = 4$ there are infinitely many solutions as the equations are the same.

This method of expressing a solution generalises to the more complicated situation in three dimensions. This is also discussed in the next section.

Again it is noted that for a system of linear equations in two unknowns, the matrix of the coefficients of x and y being singular corresponds to either no solutions (parallel lines) or infinitely many solutions (the same line).

Exercise 2F

- Solve each of the pairs of simultaneous linear equations using matrix methods:
 - $3x + 2y = 6$ and $x - y = 7$
 - $2x + 6y = 0$ and $y - x = 2$
 - $4x - 2y = 7$ and $5x + 7y = 1$
 - $2x - y = 6$ and $4x - 7y = 5$
- Explain why the simultaneous equations $2x + 3y = 6$ and $4x + 6y = 10$ have no solution.
- The simultaneous equations $x - y = 6$ and $2x - 2y = 12$ have infinitely many solutions. Describe these solutions through the use of a parameter.
- Find the value of m for which the simultaneous equations

$$(m + 3)x + my = 12$$

$$(m - 1)x + (m - 3)y = 7$$

have no solution.

- 5 Find the value of m for which the simultaneous equations

$$3x + my = 5 \text{ and } (m + 2)x + 5y = m \text{ have:}$$

- a** an infinite number of solutions **b** no solutions

- 6 Consider the simultaneous equations

$$mx + 2y = 8$$

$$4x - (2 - m)y = 2m$$

- a** Find the values of m for which there are:

- i** no solutions **ii** infinitely many solutions

- b** Solve the equations in terms of m , for suitable values of m .

- 7 **a** Solve the simultaneous equations $2x - 3y = 4$ and $x + ky = 2$, where k is a constant.

- b** Find the value of k for which there is not a unique solution.

- 8 Find the values of b and c for which the equations $x + 5y = 4$ and $2x + by = c$ have:

- a** a unique solution **b** an infinite set of solutions **c** no solution

2.7 Simultaneous linear equations with more than two variables

Consider the general linear system of three equations in three unknowns:

$$ax + by + cz = d$$

$$ex + fy + gz = h$$

$$kx + my + nz = p$$

It can be written as a matrix equation:

$$\begin{bmatrix} a & b & c \\ e & f & g \\ k & m & n \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d \\ h \\ p \end{bmatrix}$$

$$\text{Let } \mathbf{A} = \begin{bmatrix} a & b & c \\ e & f & g \\ k & m & n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} d \\ h \\ p \end{bmatrix}$$

The equation is $\mathbf{AX} = \mathbf{B}$

We recall that for 3×3 matrices $\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{DI} = \mathbf{D} = \mathbf{ID}$ for all 3×3 matrices $\mathbf{D}$.

If the inverse $\mathbf{A}^{-1}$ exists (this is not always the case) the equation can be solved by multiplying $\mathbf{AX}$, and $\mathbf{B}$, on the left by $\mathbf{A}^{-1}$, and $\mathbf{A}^{-1}(\mathbf{AX}) = \mathbf{A}^{-1}\mathbf{B}$ and $\mathbf{A}^{-1}(\mathbf{AX}) = (\mathbf{A}^{-1}\mathbf{A})\mathbf{X} = \mathbf{IX} = \mathbf{X}$, where $\mathbf{I}$ is the identity matrix for 3×3 matrices.

Hence $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$, which is a formula for the solution of the system. Of course it depends on the inverse $\mathbf{A}^{-1}$ existing, but once $\mathbf{A}^{-1}$ is found then equations of the form $\mathbf{AX} = \mathbf{B}$ can be solved for all possible 3×1 matrices $\mathbf{B}$.

In this course you are not required to find the inverse of a 3×3 matrix by hand but an understanding of matrix arithmetic is necessary.

Example 10

Consider the system of three equations in three unknowns:

$$2x + y + z = -1$$

$$3y + 4z = -7$$

$$6x + z = 8$$

Using matrix methods to solve the system of equations.

Solution

Enter 3×3 matrix $\mathbf{A}$ and 3×1 matrix $\mathbf{B}$ into the calculator.

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 4 \\ 6 & 0 & 1 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} -1 \\ -7 \\ 8 \end{bmatrix}$$

The equations can be written as the matrix equation:

$$\mathbf{AX} = \mathbf{B}$$

Multiply both sides by $\mathbf{A}^{-1}$.

$$\mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{B}$$

$$\mathbf{IX} = \mathbf{A}^{-1}\mathbf{B}$$

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

$$\mathbf{X} = \begin{bmatrix} 1 \\ -5 \\ 2 \end{bmatrix} \text{ Hence } x = 1, y = -5 \text{ and } z = 2$$

It should be noted that, just as for two equations in two unknowns, there is a geometric interpretation for three equations in three unknowns. There is only a unique solution if the equations represent three planes intersecting at a point.

A CAS calculator can be used to solve systems of three equations in the same way that was used for two simultaneous equations. The solution of equations without CAS is considered in Exercise 2G.

Example 11

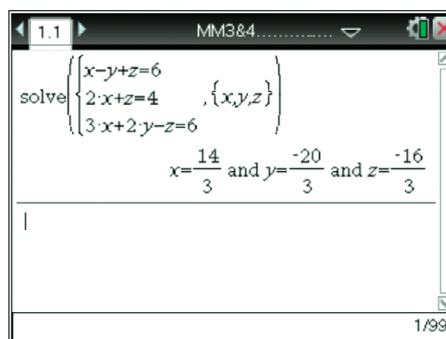
Solve the linear simultaneous equations

$$x - y + z = 6, 2x + z = 4, 3x + 2y - z = 6 \text{ for } x, y \text{ and } z.$$

Solution

Using the TI-Nspire

The simultaneous equations template
(**menu**) > **Algebra** > **Solve System of Equations** > **Solve System of Equations**)
has been used here, but the result could
also have been obtained using
solve($x - y + z = 6$ and $2x + z = 4$
and $3x + 2y - z = 6, \{x, y, z\}$).



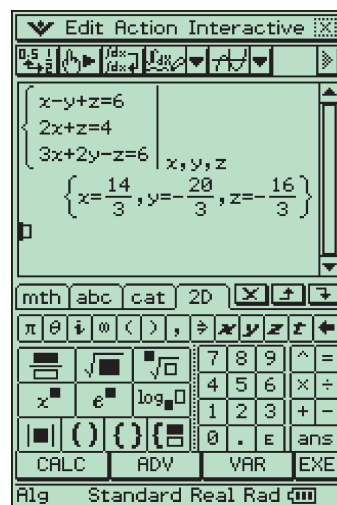
Using the Casio ClassPad

Press **Keyboard** and select the **2D** menu (if necessary tap **2D**).

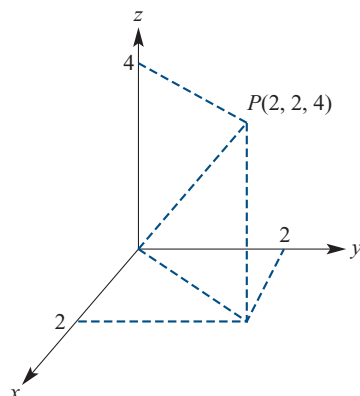
Tap **{** to enter two simultaneous equations and again to add a third.

At the bottom right, enter the variables x, y and z .

The solution is as shown.



As a linear equation in two variables defines a line, a linear equation in three variables defines a plane. The coordinate axes in three dimensions are drawn as shown. The point $P(2, 2, 4)$ is as marked.



An equation of the form $ax + by + cz = e$ defines a plane. For example the equation $x - y + z = 6$ corresponds to the graph shown here.

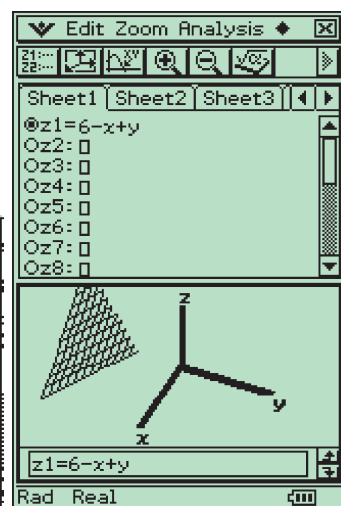
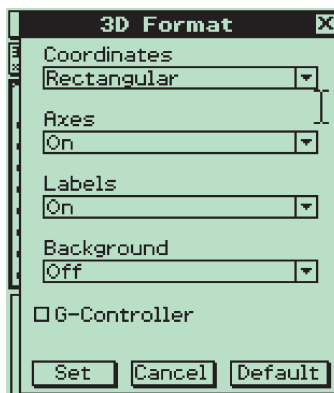
Using the Casio ClassPad



Select  from the main menu and enter $z1 = 6 - x + y$. Tap to select the graph then tap  to produce the graph.

If labels and axes do not appear, tap  then **3D format** and select the settings shown on the right.

The graph may be zoomed and viewed from various angles by dragging the stylus on the screen. A full-screen view is available by selecting the graph frame (bold border) and tapping .



The solution of simultaneous linear equations in three variables can correspond to

- a point
- a line
- a plane

There also may be no solution.

The situations are as shown:

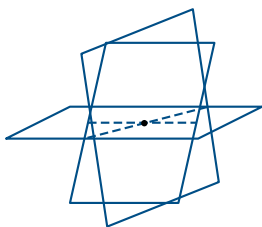


Diagram 1
Intersection at a point

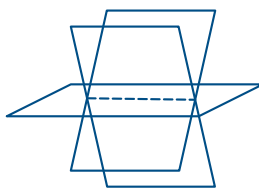


Diagram 2
Intersection, a line



Diagram 3
No intersection

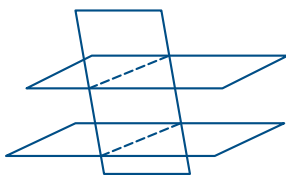


Diagram 4
No common intersection

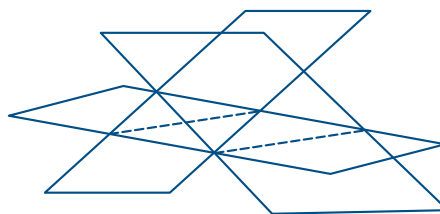


Diagram 5
No common intersection

Example 12

The simultaneous equations $x + 2y + 3z = 13$, $-x - 3y + 2z = 2$ and $-x - 4y + 7z = 17$ have infinitely many solutions. Describe these solutions through the use of a parameter.

Solution

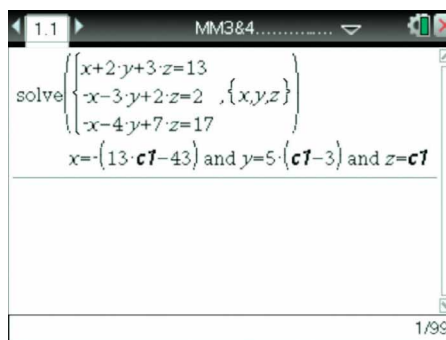
The equations have no unique solution. For example, the point $(-9, 5, 4)$ satisfies all three equations but it is certainly not the only one. We use the CAS calculator to find the solution in terms of a fourth variable, λ . The calculator uses the parameter c for the parameter λ .

Using the TI-Nspire

Let $z = \lambda$, then $y = 5(\lambda - 3)$ and $x = 43 - 13\lambda$

If $\lambda = 4$, $x = -9$, $y = 5$ and $z = 4$.

Note that as z increases by 1 then y increases by 5 and x decreases by 13. All of the points which satisfy the equations lie on a straight line. The situation is similar to that shown in diagram 2 above.

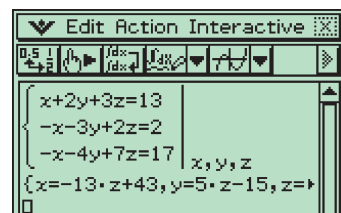


Using the Casio ClassPad

The solution returned is:

$$x = -13z + 43, y = 5(z - 3), z = z$$

The ClassPad does not introduce a parameter.



Exercise 2G

1 Solve each of the following sets of simultaneous equations using matrix methods:

a $2x + 3y - z = 12$	b $x + 2y + 3z = 13$	c $x + y = 5$
$2y + z = 7$	$-x - y + 2z = 2$	$y + z = 7$
$2y - z = 5$	$-x + 3y + 4z = 26$	$z + x = 12$

d $x - y - z = 0, 5x + 20z = 50, 10y - 20z = 30$

2 Consider the simultaneous equations $x + 2y - 3z = 4$ and $x + y + z = 6$

a Subtract the second equation from the first to find y in terms of z .

b Let $z = \lambda$. Solve the equations to give the solution in terms of λ .

3 Consider the simultaneous equations

$$x + 2y + 3z = 13 \dots (1)$$

$$-x - 3y + 2z = 2 \dots (2)$$

$$-x - 4y + 7z = 17 \dots (3)$$

a Add equation (2) to equation (1) and subtract equation (2) from equation (3).

b Comment on the equations obtained in **a**.

c Let $z = \lambda$ and find y in terms of λ .

d Substitute for z and y in terms of λ in equation (1) to find x in terms of λ .

4 Solve each of the following pairs of simultaneous equations, giving your answer in terms of a parameter λ . Use the technique introduced in Question 2.

a $x - y + z = 4$	b $2x - y + z = 6$	c $4x - 2y + z = 6$
$-x + y + z = 6$	$x - z = 3$	$x + y + z = 4$

5 The system of equations $x + y + z + w = 4, x + 3y + 3z = 2, x + y + 2z - w = 6$ has infinitely many solutions. Describe this family of solutions and give the unique solution when $w = 6$.

6 Find all solutions of the following sets of equations:

a $3x - y + z = 4$ and $x + 2y - z = 2$ and $-x + y - z = -2$

b $x - y - z = 0$ and $3y + 3z = -5$

c $2x - y + z = 0$ and $y + 2z = 2$

7 **a** Find the inverse of the matrix $\begin{bmatrix} 2 & a & -1 \\ 3 & 4 & -(a+1) \\ 10 & 8 & a-4 \end{bmatrix}$ in terms of a .

b For what values of a does the inverse not exist?

c Find the value of a for which there are infinitely many solutions to the equations

$$2x + ay - z = 0$$

$$3x + 4y - (a+1)z = 13$$

$$10x + 8y + (a-4)z = 26$$

d For the value of a found in **c**, solve the equations.

8 Find a value of p for which the system of equations

$$3x + 2y - z = 1 \text{ and } x + y + z = 2 \text{ and } px + 2y - z = 1$$

has more than one solution.

(**Hint:** Find the inverse of the matrix of coefficients in terms of p .)

Solve the system of equations for this value of p .

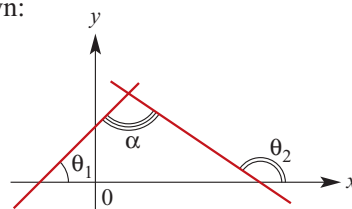


Chapter summary

- Gradient of a straight line joining two points (x_1, y_1) and (x_2, y_2) is $m = \frac{y_2 - y_1}{x_2 - x_1}$.
- Different forms for the equation of a straight line:

$y = mx + c$	where m is the gradient and c is the y -axis intercept
$y - y_1 = m(x - x_1)$	where m is the gradient and (x_1, y_1) is a point on the line
$\frac{x}{a} + \frac{y}{b} = 1$	where $(a, 0)$ and $(0, b)$ are points on the line intersecting the axes.
- Tangent of the angle of slope (θ), $\tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$, where θ is the angle the line makes with the positive direction of the x -axis.
- If two straight lines are perpendicular to each other, the product of their gradients is -1 , i.e. $m_1 m_2 = -1$.
- Distance between two points (x_1, y_1) and $(x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Midpoint of a straight line joining (x_1, y_1) and (x_2, y_2) is the point $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.
- The angle α between intersecting straight lines is as shown:

$$\alpha = \theta_2 - \theta_1$$



- A matrix is a rectangular array of numbers.
- Two matrices **A** and **B** are equal when:
 - each has the same number of rows and the same number of columns
 - they have the same number or element at corresponding positions.
- The size or **dimension** of a matrix is described by specifying the number of rows (m) and the number of columns (n). The dimension is written $m \times n$.
- Addition will be defined for two matrices only when they have the same dimension. The sum is found by adding corresponding elements.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a + e & b + f \\ c + g & d + h \end{bmatrix}$$

Subtraction is defined in a similar way.

- If **A** is an $m \times n$ matrix and k is a real number, $k\mathbf{A}$ is defined to be an $m \times n$ matrix whose elements are k times the corresponding element of **A**.

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

- If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times r$ matrix, then the product $\mathbf{AB}$ is the $m \times r$ matrix whose entries are determined as follows.

To find the entry in row i and column j of $\mathbf{AB}$, single out row i in matrix $\mathbf{A}$ and column j in matrix $\mathbf{B}$. Multiply the corresponding entries from the row and column and then add up the resulting products.

The product $\mathbf{AB}$ is defined only if the number of columns of $\mathbf{A}$ is the same as the number of rows of $\mathbf{B}$.

- The identity matrix $\mathbf{I}$ satisfies the property that $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$
- If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same dimension and $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ then $\mathbf{A}$ is said to be the inverse of $\mathbf{B}$ and $\mathbf{B}$ is said to be the inverse of $\mathbf{A}$. The inverse of $\mathbf{A}$ is denoted $\mathbf{A}^{-1}$.

- If $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $\mathbf{A}^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$

$\det(\mathbf{A}) = ad - bc$ is the **determinant** of matrix $\mathbf{A}$.

- A square matrix is said to be **regular** if its inverse exists. Those square matrices that do not have an inverse are called **singular** matrices.
- Simultaneous equations can be solved using inverse matrices, For example

$$ax + by = c$$

$$dx + ey = f$$

can be written as $\begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ f \end{bmatrix}$ and $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ d & e \end{bmatrix}^{-1} \begin{bmatrix} c \\ f \end{bmatrix}$

Multiple-choice questions

- 1 A straight line has gradient $-\frac{1}{2}$ and passes through $(1, 4)$. The equation of the line is:

A $y = x + 4$

B $y = 2x + 2$

C $y = 2x + 4$

D $y = -\frac{1}{2}x + 4$

E $y = -\frac{1}{2}x + \frac{9}{2}$

- 2 The line $y = -2x + 4$ passes through a point $(a, 3)$. The value of a is:

A $-\frac{1}{2}$

B 2

C $-\frac{7}{2}$

D -2

E $\frac{1}{2}$

- 3 The gradient of a line that is perpendicular to the line shown could be:

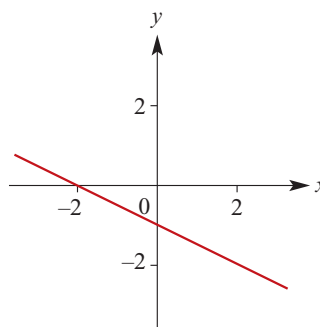
A 1

B $\frac{1}{2}$

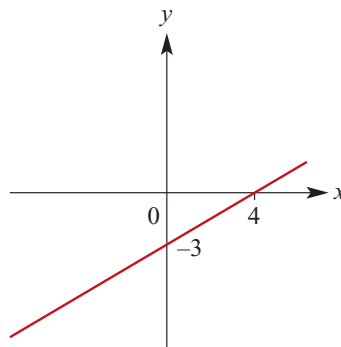
C $-\frac{1}{2}$

D 2

E -2



- 4 The coordinates of the midpoint of AB where A has coordinates $(1, 7)$ and B has coordinates $(-3, 10)$ are:
A $(-2, 3)$ **B** $(-1, 8)$ **C** $(-1, 8.5)$
D $(-1, 3)$ **E** $(-2, 8.5)$
- 5 The solution of the two simultaneous equations $ax - 5by = 11$ and $4ax + 10by = 2$ for x and y , in terms of a and b , is:
A $x = -\frac{10}{a}, y = -\frac{21}{5b}$ **B** $x = \frac{4}{a}, y = -\frac{7}{5b}$
C $x = \frac{13}{5a}, y = -\frac{42}{25b}$ **D** $x = \frac{13}{2a}, y = -\frac{9}{10b}$
E $x = -\frac{3}{a}, y = -\frac{14}{5b}$
- 6 The gradient of the line passing through $(3, -2)$ and $(-1, 10)$ is:
A -3 **B** -2 **C** $-\frac{1}{3}$ **D** 4 **E** 3
- 7 If two lines $-2x + y - 3 = 0$ and $ax - 3y + 4 = 0$ are parallel, then a equals:
A 6 **B** 2 **C** $\frac{1}{3}$ **D** $\frac{2}{3}$ **E** -6
- 8 A straight line passes through $(-1, -2)$ and $(3, 10)$. The equation of the line is:
A $y = 3x - 1$ **B** $y = 3x - 4$ **C** $y = 3x + 1$
D $y = \frac{1}{3}x + 9$ **E** $y = 4x - 2$
- 9 The length of the line segment connecting $(1, 4)$ and $(5, -2)$ is:
A 10 **B** $2\sqrt{13}$ **C** 12 **D** 50 **E** $2\sqrt{5}$
- 10 The function with graph as shown has the rule:
A $f(x) = 3x - 3$
B $f(x) = -\frac{3}{4}x - 3$
C $f(x) = \frac{3}{4}x - 3$
D $f(x) = \frac{4}{3}x - 3$
E $f(x) = 4x - 4$



Short-answer questions (technology-free)

- 1 Solve the following linear equations:
a $3x - 2 = 4x + 6$ **b** $\frac{x+1}{2x-1} = \frac{4}{3}$ **c** $\frac{3x}{5} - 7 = 11$ **d** $\frac{2x+1}{5} = \frac{x-1}{2}$
- 2 Solve each of the following pairs of simultaneous linear equations:
a $y = x + 4$ **b** $\frac{x}{4} - \frac{y}{3} = 2$
 $5y + 2x = 6$ $y - x = 5$

3 Sketch the graph of the relations:

a $3y + 2x = 5$ **b** $x - y = 6$ **c** $\frac{x}{2} + \frac{y}{3} = 1$

4 **a** Find the equation of the straight line which passes through (1, 3) and has gradient -2 .

b Find the equation of the straight line which passes through (1, 4) and (3, 8).

c Find the equation of the straight line which is perpendicular to the line with equation $y = -2x + 6$ and which passes through the point (1, 1).

d Find the equation of the straight line which is parallel to the line with equation $y = 6 - 2x$ and which passes through the point (1, 1).

5 Find the distance between the points with coordinates $(-1, 6)$ and $(2, 4)$.

Extended-response questions

1 A firm manufacturing jackets finds that it is capable of producing 100 jackets per day, but it can only sell all of these if the charge to wholesalers is no more than \$50 per jacket. On the other hand, at the current price of \$75 per jacket, only 50 can be sold per day.

Assume that the graph of price, $\$P$, against number sold per day N is a straight line.

a Sketch the graph of P against N .

b Find the equation of the straight line.

c Use the equation to find:

i the price at which 88 jackets per day could be sold

ii the number of jackets that should be manufactured to sell at \$60 each.

2 A new town was built 10 years ago to house the workers of a woollen mill established in a remote country area. Three years after the town was built, it had a population of 12 000 people. Business in the wool trade steadily grew, and eight years after the town was built the population had swelled to 19 240.

a Assuming the population growth can be modelled by a linear relationship, find a suitable relation for the population, p , in terms of t , the number of years since the town was built.

b Sketch the graph of p against t , and interpret the p -axis intercept.

c Find the current population of the town.

d Calculate the average rate of growth of the town.

3 $ABCD$ is a quadrilateral with angle ABC a right angle. D lies on the perpendicular bisector of AB . The coordinates of A and B are (7, 2) and (2, 5) respectively. The equation of line AD is $y = 4x - 26$.

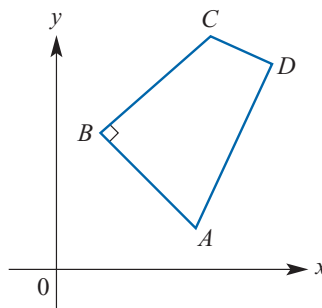
a Find the equation of the perpendicular bisector of line segment AB .

b Find the coordinates of point D .

c Find the gradient of line BC .

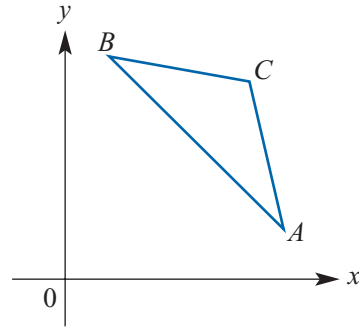
d Find the value of the second coordinate c of the point $C(8, c)$.

e Find the area of quadrilateral $ABCD$.



- 4 Triangle ABC is isosceles with $BC = AC$. The coordinates of the vertices are $A(6, 1)$ and $B(2, 8)$.

- a Find the equation of the perpendicular bisector of AB .
- b If the x -coordinate of C is 3.5, find the y -coordinate of C .
- c Find the length of AB .
- d Find the area of triangle ABC .



- 5 If $A = (-4, 6)$ and $B = (6, -7)$, find:
- a the coordinates of the midpoint of AB
 - b the length of AB
 - c the distance between A and B
 - d the equation of AB
 - e the equation of the perpendicular bisector of AB
 - f the coordinates of P , where $P \in AB$ and $AP : PB = 3 : 1$
 - g the coordinates of P , where $P \in AB$ and $AP : AB = 3 : 1$
- 6 A chemical manufacturer has an order for 500 litres of a 25% acid solution (i.e. 25% by volume is acid). Solutions of 30% and 18% are available in stock.
- a How much acid is required to produce 500 litres of 25% acid solution?
 - b The manufacturer wishes to make up the 500 litres from a mixture of 30% and 18% solutions.

Let x denote the amount of 30% solution required.

Let y denote the amount of 18% solution required.

Use simultaneous equations in x and y to determine the amount of each solution required.

- 7 Bronwyn and Noel have a clothing warehouse in Summerville. They are supplied by three contractors, Brad, Flynn and Elizabeth. The matrix below shows the number of dresses, slacks and shirts a worker, for each of the contractors, can produce in a week. The number produced varies because of the different equipment used by the contractors.

	Brad Flynn Elizabeth		
Dresses	5	6	10
Slacks	3	4	5
Shirts	2	6	5

The warehouse requires 310 dresses, 175 slacks and 175 shirts in a week. How many workers should each contractor employ to meet this requirement exactly?

Families of functions

Objectives

- To consider functions with equation $y = A f(n(x + c)) + b$, where A , n , c and $b \in \mathbb{R}$, for

$$f(x) = x^n, \text{ where } n \text{ is a non-zero rational number}$$

$$\text{and } f(x) = |x|$$

- To use **dilations**, **reflections** and **translations** to sketch graphs of such functions.
- To **determine the rule** for the function of such graphs.
- To use **addition of ordinates** to sketch such graphs.
- To find and graph the **inverse relations** of such functions.
- To use matrices to describe transformations.

3.1 Functions with rule $f(x) = x^n$

In this section functions of the form $f(x) = x^n$, where n is a rational number, are considered. These functions are called **power** functions. We need to use calculus to study all aspects of these functions, but at this stage we can consider some members of this family as an important addition to the functions already introduced.

$f(x) = x^n$ where n is a non-zero rational number

When $n = 1$, $f(x) = x$, i.e. the basic linear function is formed.

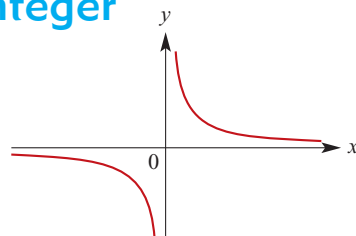
When $n = 2$ and $n = 3$, $f(x) = x^2$ and x^3 respectively. These functions are part of the family of functions of the form $f(x) = x^n$ where n is a positive integer. It is appropriate to delay the introduction of this family until Chapter 4.

$f(x) = x^n$ where n is a negative integer

$$\begin{aligned} \text{When } n = -1, f(x) &= x^{-1} \\ &= \frac{1}{x} \end{aligned}$$

The maximal domain of this function is $\mathbb{R} \setminus \{0\}$.

The graph of the function is as shown.



Asymptotes:

- There is a horizontal asymptote with equation $y = 0$

As $x \rightarrow \infty$, $\frac{1}{x} \rightarrow 0^+$, i.e. from the positive side.

As $x \rightarrow -\infty$, $\frac{1}{x} \rightarrow 0^-$, i.e. from the negative direction.

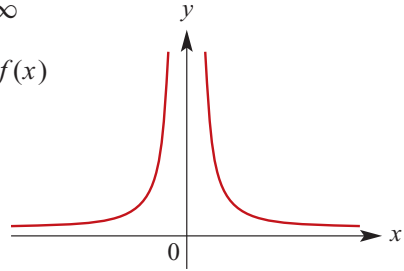
- There is a vertical asymptote with equation $x = 0$

As $x \rightarrow 0^+$, i.e. from the positive direction, $\frac{1}{x} \rightarrow \infty$

As $x \rightarrow 0^-$, i.e. from the negative side, $\frac{1}{x} \rightarrow -\infty$

- $f(x) = \frac{1}{x}$ is an odd function, since $f(-x) = -f(x)$

When $n = -2$, $f(x) = x^{-2} = \frac{1}{x^2}$. The maximal domain of this function is $\mathbb{R} \setminus \{0\}$. The graph of the function is as shown on the right.

*Asymptotes:*

- There is a horizontal asymptote with equation $y = 0$

As $x^2 \rightarrow \infty$, $\frac{1}{x^2} \rightarrow 0^+$, i.e. from the positive side.

- There is a vertical asymptote with equation $x = 0$

As $x \rightarrow 0^+$, i.e. from the positive direction, $\frac{1}{x^2} \rightarrow \infty$

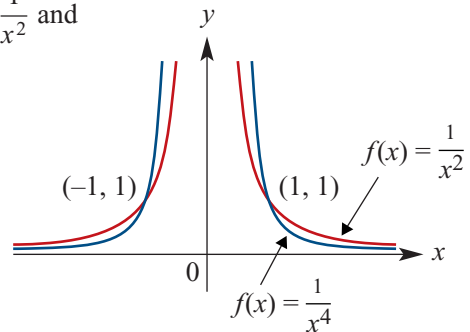
As $x \rightarrow 0^-$, i.e. from the negative side, $\frac{1}{x^2} \rightarrow \infty$

- $f(x) = \frac{1}{x^2}$ is an even function, since $f(-x) = f(x)$

In the diagram on the right, the graphs of $f(x) = \frac{1}{x^2}$ and $f(x) = \frac{1}{x^4}$ are shown on the one set of axes.

- The graphs intersect at the points with coordinates $(1, 1)$ and $(-1, 1)$.

- Note that $\frac{1}{x^2} > \frac{1}{x^4}$ for $x > 1$ and $x < -1$,
and $\frac{1}{x^2} < \frac{1}{x^4}$ for $0 < x < 1$ and $-1 < x < 0$



In general, if n and m are positive even integers with $n < m$:

- The graphs of $f(x) = \frac{1}{x^n}$ and $f(x) = \frac{1}{x^m}$ intersect at the points with coordinates $(1, 1)$ and $(-1, 1)$.

- $\frac{1}{x^n} > \frac{1}{x^m}$ for $x > 1$ and $x < -1$, and $\frac{1}{x^n} < \frac{1}{x^m}$ for $0 < x < 1$ and $-1 < x < 0$

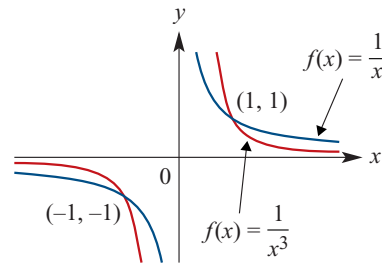
- When n is a positive even integer, $f(x) = \frac{1}{x^n}$ is an even function, since $f(-x) = f(x)$

In the diagram below, the graphs of $f(x) = \frac{1}{x}$ and $f(x) = \frac{1}{x^3}$ are shown on the one set of axes.

■ The graphs intersect at the points with coordinates $(1, 1)$ and $(-1, -1)$.

■ Note that $\frac{1}{x} > \frac{1}{x^3}$ for $x > 1$ and $-1 < x < 0$,

and $\frac{1}{x} < \frac{1}{x^3}$ for $0 < x < 1$ and $x < -1$



In general, if n and m are positive odd integers with $n < m$:

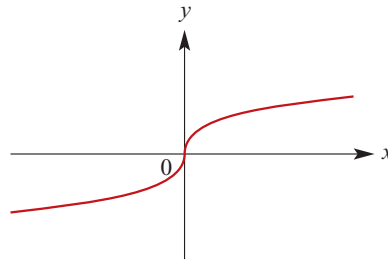
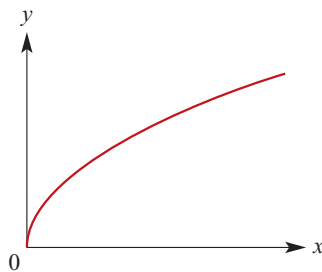
■ The graphs of $f(x) = \frac{1}{x^n}$ and $f(x) = \frac{1}{x^m}$ intersect at the points with coordinates $(1, 1)$ and $(-1, -1)$.

■ $\frac{1}{x^n} > \frac{1}{x^m}$ for $x > 1$ and $-1 < x < 0$, and $\frac{1}{x^n} < \frac{1}{x^m}$ for $0 < x < 1$ and $x < -1$

■ When n is a positive odd integer, $f(x) = \frac{1}{x^n}$ is an odd function, since $f(-x) = -f(x)$

$f(x) = x^{\frac{1}{n}}$ where n is a positive integer

First recall that $x^{\frac{1}{n}} = \sqrt[n]{x}$. In particular $x^{\frac{1}{2}} = \sqrt{x}$. When n is even, the maximal domain is $[0, \infty)$ and when n is odd, the maximal domain is $\mathbb{R}$. The graphs of $f(x) = \sqrt{x} = x^{\frac{1}{2}}$ and $f(x) = \sqrt[3]{x} = x^{\frac{1}{3}}$ are as shown.



■ When n is a positive odd integer, $f(x) = x^{\frac{1}{n}}$ is an odd function, since $f(-x) = -f(x)$.

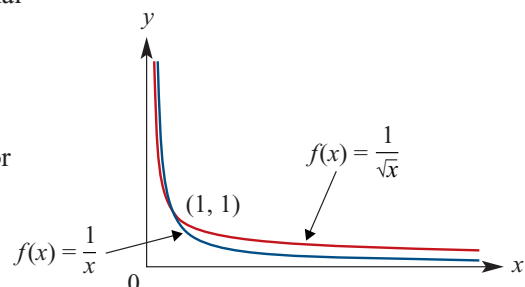
$f(x) = x^{-\frac{1}{n}}$ where n is a positive integer

The expression $x^{-\frac{1}{n}} = \frac{1}{x^{\frac{1}{n}}} = \frac{1}{\sqrt[n]{x}}$. The maximal

domain of the function f with rule $f(x) = \frac{1}{\sqrt[n]{x}}$

is $(0, \infty)$ if n is even, and $\mathbb{R} \setminus \{0\}$ if n is odd.

The graphs of $y = \frac{1}{\sqrt{x}}$ and $y = \frac{1}{x}$ are shown for $x \in (0, \infty)$ on the one set of axes.



- The graphs intersect at the point with coordinates (1, 1).
- Note that $\frac{1}{\sqrt{x}} > \frac{1}{x}$ for $x > 1$, and $\frac{1}{\sqrt{x}} < \frac{1}{x}$ for $0 < x < 1$

Asymptotes:

- For each of the graphs there is a horizontal asymptote with equation $y = 0$ and a vertical asymptote with equation $x = 0$.

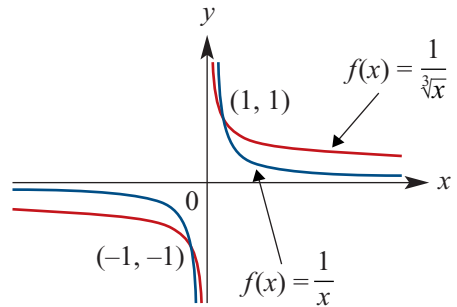
The graphs of $y = \frac{1}{\sqrt[3]{x}}$ and $y = \frac{1}{x}$ are shown for $x \in \mathbb{R} \setminus \{0\}$ on the one set of axes.

- The graphs intersect at the points with coordinates (1, 1) and (-1, -1).

- $\frac{1}{\sqrt[3]{x}} > \frac{1}{x}$ for $x > 1$ and $-1 < x < 0$, and

$$\frac{1}{\sqrt[3]{x}} < \frac{1}{x} \text{ for } 0 < x < 1 \text{ and } x < -1$$

- When n is a positive odd integer, $f(x) = x^{-\frac{1}{n}}$ is an odd function, since $f(-x) = -f(x)$



$f(x) = x^{\frac{p}{q}}$ where p and q are non-zero integers and the highest common factor $(p, q) = 1$

The special cases where $p = \pm 1$ have been considered above.

$x^{\frac{p}{q}}$ is defined as $(x^{\frac{1}{q}})^p$.

It is necessary to have the fraction $\frac{p}{q}$ in simplest form. For example, $(-1)^{\frac{2}{6}} \neq (-1)^{\frac{1}{3}}$ since $(-1)^{\frac{2}{6}} = ((-1)^2)^{\frac{1}{6}} = 1^{\frac{1}{6}} = 1$, but $(-1)^{\frac{1}{3}} = -1$.

An investigation of these graphs with your calculator is worthwhile. Every case will not be listed here.

Example 1

For each of the following, use your calculator to help sketch the graph of $y = f(x)$ for the maximal domain. State this maximal domain and the range corresponding to this domain. Also state the equations of asymptotes if appropriate, and whether the function is odd, even or neither.

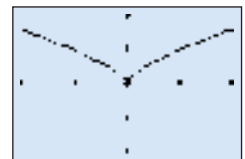
- | | | |
|------------------------------------|------------------------------------|------------------------------------|
| a $f(x) = x^{\frac{2}{3}}$ | b $f(x) = x^{-\frac{2}{3}}$ | c $f(x) = x^{\frac{3}{2}}$ |
| d $f(x) = x^{-\frac{3}{2}}$ | e $f(x) = x^{\frac{3}{5}}$ | f $f(x) = x^{-\frac{3}{5}}$ |

Solution

- a** The maximal domain of $f(x) = x^{\frac{2}{3}}$ is $\mathbb{R}$.

The range is $[0, \infty)$.

$f(x) = x^{\frac{2}{3}}$ is an even function, since
 $f(-x) = f(x)$



- b** The maximal domain of $f(x) = x^{-\frac{2}{3}}$ is $\mathbb{R} \setminus \{0\}$.

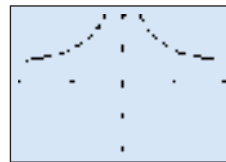
The range is $(0, \infty)$.

The asymptotes have equations $x = 0$ and $y = 0$

Note that $x^{-\frac{2}{3}} = \frac{1}{x^{\frac{2}{3}}}$

As $x^{\frac{2}{3}} \rightarrow 0$, $\frac{1}{x^{\frac{2}{3}}} \rightarrow \infty$ and as $x^{\frac{2}{3}} \rightarrow \infty$, $\frac{1}{x^{\frac{2}{3}}} \rightarrow 0$

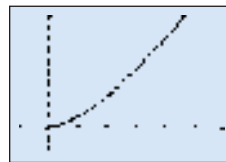
$f(x) = x^{-\frac{2}{3}}$ is an even function, since $f(-x) = f(x)$



- c** The maximal domain of $f(x) = x^{\frac{3}{2}}$ is $[0, \infty)$.

The range is $[0, \infty)$.

$f(x) = x^{\frac{3}{2}}$ is neither odd nor even.



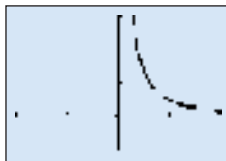
- d** The maximal domain of $f(x) = x^{-\frac{3}{2}}$ is $(0, \infty)$.

The range is $(0, \infty)$. The asymptotes have equations $x = 0$ and $y = 0$. Note that

$x^{-\frac{3}{2}} = \frac{1}{x^{\frac{3}{2}}}$

As $x^{\frac{3}{2}} \rightarrow 0^+$, $\frac{1}{x^{\frac{3}{2}}} \rightarrow \infty$ and as $x^{\frac{3}{2}} \rightarrow \infty$, $\frac{1}{x^{\frac{3}{2}}} \rightarrow 0$

$f(x) = x^{-\frac{3}{2}}$ is neither odd nor even.

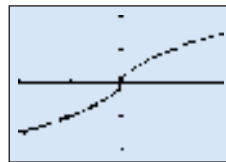


- e** The maximal domain of $f(x) = x^{\frac{3}{5}}$ is $\mathbb{R}$.

The range is $\mathbb{R}$.

$f(x) = x^{\frac{3}{5}}$ is an odd function, since

$f(-x) = -f(x)$



- f** The maximal domain of $f(x) = x^{-\frac{3}{5}}$ is $\mathbb{R} \setminus \{0\}$.

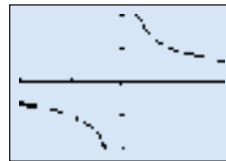
The range is $\mathbb{R} \setminus \{0\}$.

The asymptotes have equations $x = 0$ and $y = 0$

Note that $x^{-\frac{3}{5}} = \frac{1}{x^{\frac{3}{5}}}$.

As $x^{\frac{3}{5}} \rightarrow 0^+$, $\frac{1}{x^{\frac{3}{5}}} \rightarrow \infty$ and as $x^{\frac{3}{5}} \rightarrow \infty$, $\frac{1}{x^{\frac{3}{5}}} \rightarrow 0^+$

$f(x) = x^{-\frac{3}{5}}$ is an odd function, since $f(-x) = -f(x)$



Exercise 3A

- 1** For the function with equation $f(x) = \frac{1}{x^4}$:

- state the maximal domain, the corresponding range and the equations of any asymptotes
- sketch the graph without using your calculator.

2 For each of the following rules of functions:

- i state the maximal domain of the function, the corresponding range and the equations of any asymptotes
- ii sketch the graph using your calculator to assist you.

a $f(x) = \frac{1}{\sqrt{x}}$

b $f(x) = x^{\frac{3}{5}}$

c $f(x) = x^{-\frac{3}{5}}$

d $f(x) = \frac{1}{\sqrt[3]{x}}$

e $f(x) = \sqrt[5]{x}$

3 a Find $\{x: x^{\frac{3}{2}} > x^2\}$

b Find $\{x: x^{-\frac{3}{2}} < x^{-2}\}$

4 For each of the following, state whether the function is odd, even or neither:

a $f(x) = \frac{1}{x}$

b $f(x) = \frac{1}{x^2}$

c $f(x) = \sqrt[3]{x}$

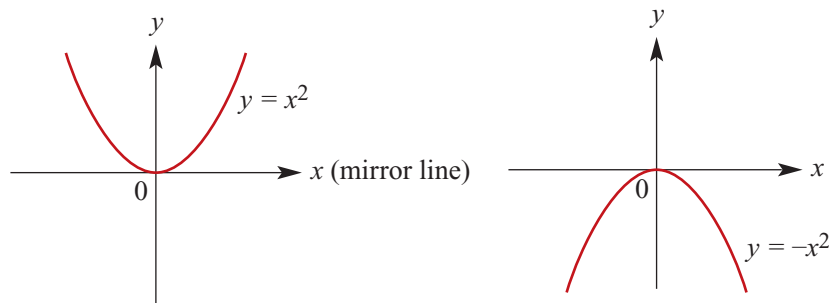
d $f(x) = \frac{1}{\sqrt[3]{x}}$

e $f(x) = x^{\frac{2}{3}}$

f $f(x) = x^{\frac{5}{7}}$

Introducing transformations of functions

Many graphs of functions can be described as transformations of graphs of other functions, or ‘movements’ of graphs about the cartesian plane. For example, the graph of the function $y = -x^2$ can be considered as a reflection, in the x -axis, of the graph of the function $y = x^2$.



Formally, a **transformation** is a one-to-one function (or mapping) from R^2 to R^2 . A good understanding of transformations, combined with knowledge of the ‘simplest’ function and its graph in each family, provides an important tool with which to sketch graphs and identify rules of more complicated functions.

There are three basic types of transformations that are considered in this course: **dilations from the coordinate axes**, **reflections in the coordinate axes** and **translations**. A graph of a function may be transformed to the graph of another function by a dilation from the x - or y -axis, a reflection in either the x - or y -axis, a translation in the positive or negative direction of the x - or y -axis, or a combination of these. The following three sections consider dilations, reflections and translations separately.

3.2

Dilations

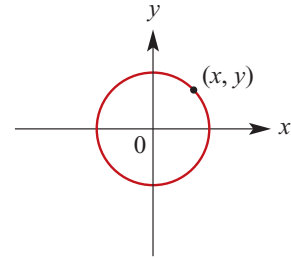
A transformation which, for example, dilates each point in the plane by a factor of 2 from the x -axis can be described as ‘multiplying the y -coordinate of each point in the plane by 2’ and can be written as $(x, y) \rightarrow (x, 2y)$. This is read as ‘the ordered pair (x, y) is mapped onto the



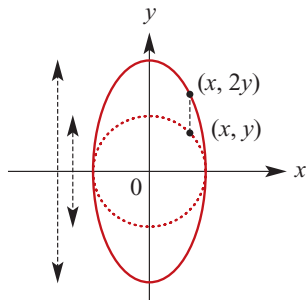
ordered pair $(x, 2y)$ '. The dilation is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(a, 2b)$.

Similarly, a transformation which dilates each point in the plane by a factor of 3 from the y -axis can be described as 'multiplying the x -coordinate of each point in the plane by 3' and can be written as $(x, y) \rightarrow (3x, y)$. This is read as 'the ordered pair (x, y) is mapped onto the ordered pair $(3x, y)$ '. The dilation is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(3a, b)$.

A dilation of a graph of a function can be thought of as the graph 'stretching away from' or 'shrinking towards' an axis. Consider dilating the graph of, say, a circle in various ways, and observe the effect on a general point with coordinates (x, y) on a circle.

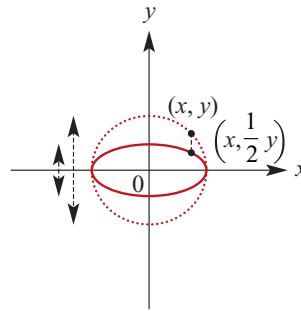


1 A dilation of factor 2 from the x -axis



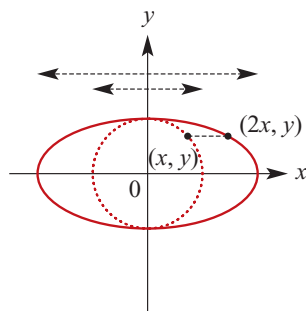
The graph is 'stretched' to twice the height. The point (x, y) is mapped onto $(x, 2y)$,
i.e. $(x, y) \rightarrow (x, 2y)$

2 A dilation of factor $\frac{1}{2}$ from the x -axis



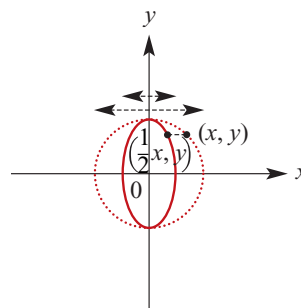
The graph is 'shrunk' to half the height. The point (x, y) is mapped onto $(x, \frac{1}{2}y)$,
i.e. $(x, y) \rightarrow (x, \frac{1}{2}y)$

3 A dilation of factor 2 from the y -axis



The graph is 'stretched' to twice the width. The point (x, y) is mapped onto $(2x, y)$,
i.e. $(x, y) \rightarrow (2x, y)$

4 A dilation of factor $\frac{1}{2}$ from the y -axis



The graph is 'shrunk' to half the width. The point (x, y) is mapped onto $(\frac{1}{2}x, y)$,
i.e. $(x, y) \rightarrow (\frac{1}{2}x, y)$

Example 2

Determine the rule of the image when the graph of $y = \frac{1}{x^2}$ is dilated by a factor of 4:

- a** from the x -axis **b** from the y -axis

Solution

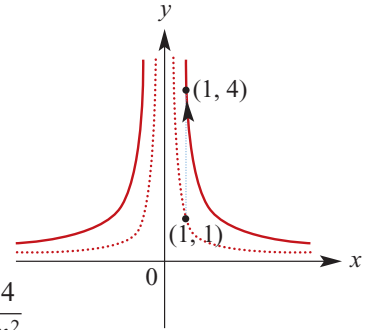
- a** $(x, y) \rightarrow (x, 4y)$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = x$, $y' = 4y$

Rearranging gives $x = x'$, $y = \frac{y'}{4}$

Therefore, $y = \frac{1}{x^2}$ becomes $\frac{y'}{4} = \frac{1}{(x')^2}$

So the rule of the transformed function is $y = \frac{4}{x^2}$



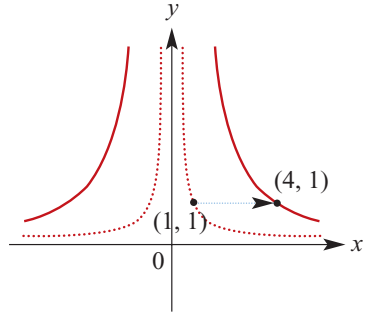
- b** $(x, y) \rightarrow (4x, y)$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = 4x$, $y' = y$

Rearranging gives $x = \frac{x'}{4}$, $y = y'$

Therefore, $y = \frac{1}{x^2}$ becomes $y' = \frac{1}{\left(\frac{x'}{4}\right)^2}$

So the rule of the transformed function is $y = \frac{16}{x^2}$



In general, a **dilation** of factor a , where $a > 0$, **from the x -axis** is a transformation that maps $(x, y) \rightarrow (x, ay)$. To find the equation of the **image** of $y = f(x)$, under the **dilation** $(x, y) \rightarrow (x, ay)$, replace y with $\frac{y}{a}$, i.e. the image of the graph with equation $y = f(x)$ under the dilation $(x, y) \rightarrow (x, ay)$ is the graph with equation $\frac{y}{a} = f(x)$, which is more commonly written as $y = af(x)$

In general, a **dilation** of factor b , where $b > 0$, **from the y -axis** is a transformation that maps $(x, y) \rightarrow (bx, y)$. To find the equation of the **image** of $y = f(x)$, under the **dilation** $(x, y) \rightarrow (bx, y)$, replace x with $\frac{x}{b}$, i.e. the image of the graph with equation $y = f(x)$ under the dilation $(x, y) \rightarrow (bx, y)$ is the graph with equation $y = f\left(\frac{x}{b}\right)$

Example 3

Determine the factor of dilation when the graph of $y = \sqrt{3x}$ is obtained by dilating the graph of $y = \sqrt{x}$:

- a** from the y -axis **b** from the x -axis

Solution

- a** Note that a dilation from the y -axis ‘changes’ the x -values. Write the transformed function as $y' = \sqrt{3x'}$, where (x', y') are the coordinates of the image of (x, y) .

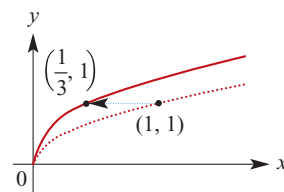
Therefore, $x = 3x'$ (‘changed’ x) and $y = y'$

Rearranging gives $x' = \frac{x}{3}$ and $y' = y$

So the mapping is given by $(x, y) \rightarrow \left(\frac{x}{3}, y\right)$ and

the graph of $y = \sqrt{x}$ is dilated by a factor

of $\frac{1}{3}$ from the y -axis to produce the graph of $y = \sqrt{3x}$



- b** Note that a dilation from the x -axis ‘changes’ the y -values. Write the transformed function as $\frac{y'}{\sqrt{3}} = \sqrt{x'}$, where (x', y') are the coordinates of the image of (x, y) .

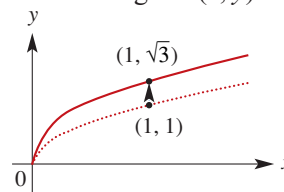
Therefore $x = x'$ and $y = \frac{y'}{\sqrt{3}}$ (‘changed’ y)

Rearranging gives $x' = x$ and $y' = \sqrt{3}y$

So the mapping is given by $(x, y) \rightarrow (x, \sqrt{3}y)$

and the graph of $y = \sqrt{x}$ is dilated by a factor

of $\sqrt{3}$ from the x -axis to produce the graph of $y = \sqrt{3x}$

**Exercise 3B**

- 1** Sketch the graph of each of the following:

a $y = \frac{4}{x}$

b $y = \frac{1}{2x}$

c $y = \sqrt{3x}$

d $y = \frac{2}{x^2}$

- 2** For $y = f(x) = \frac{1}{x^2}$, sketch the graph of each of the following:

a $y = f(2x)$

b $y = 2f(x)$

c $y = f\left(\frac{x}{2}\right)$

d $y = 3f(x)$

e $y = f(5x)$

f $y = f\left(\frac{x}{4}\right)$

- 3** Sketch the graphs of each of the following on the one set of axes:

a $y = \frac{1}{x}$

b $y = \frac{3}{x}$

c $y = \frac{3}{2x}$

- 4** Sketch the graph of the function $f: \mathbb{R}^+ \rightarrow \mathbb{R}$, $f(x) = 3\sqrt{x}$

- 5** State a transformation which maps the graphs of $y = f(x)$ to $y = f_1(x)$ for each of the following:

a i $f(x) = \frac{1}{x^2}$

ii $f_1(x) = \frac{5}{x^2}$

b i $f(x) = \sqrt{x}$

ii $f_1(x) = 4\sqrt{x}$

c i $f(x) = \sqrt{x}$

ii $f_1(x) = \sqrt{5x}$

6 Write down the equation of the rule when the graph of each of the functions below is transformed by:

- i a dilation of factor 4 from the x -axis ii a dilation of factor $\frac{2}{3}$ from the x -axis
 iii a dilation of factor $\frac{1}{2}$ from the y -axis iv a dilation of factor 5 from the y -axis
 a $y = |x|$ b $y = \sqrt[3]{x}$ c $y = \frac{1}{x^3}$ d $y = \frac{1}{x^4}$
 e $y = \frac{1}{\sqrt[3]{x}}$ f $y = x^{\frac{2}{3}}$ g $y = \frac{1}{x^{\frac{3}{4}}}$

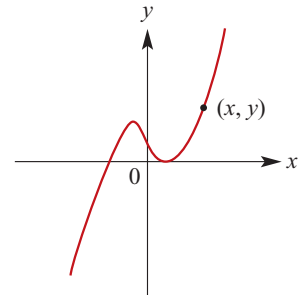


3.3 Reflections

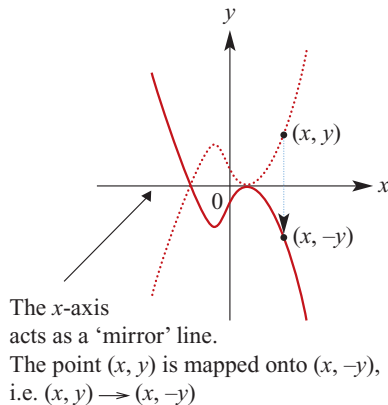
A transformation which, for example, reflects each point in the plane in the x -axis can be described as ‘multiplying the y -coordinate of each point in the plane by -1 ’ and can be written as $(x, y) \rightarrow (x, -y)$. This is read as ‘the ordered pair (x, y) is mapped onto the ordered pair $(x, -y)$ ’. The reflection is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(a, -b)$.

Similarly, a transformation which reflects each point in the plane in the y -axis can be described as ‘multiplying the x -coordinate of each point in the plane by -1 ’ and can be written as $(x, y) \rightarrow (-x, y)$. This is read as ‘the ordered pair (x, y) is mapped onto the ordered pair $(-x, y)$ ’. The reflection is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(-a, b)$.

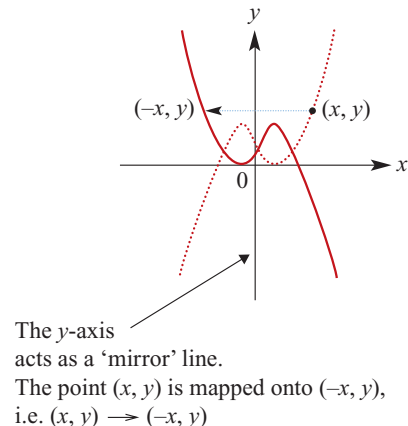
This course of study considers reflections in the x - or y -axis only. (**Note:** The special case where the graph of a function is reflected in the line $y = x$ produces the graph of the inverse relation and is discussed separately in Section 3.10.) Consider reflecting the graph of the function shown here in each axis, and observe the effect on a general point (x, y) on the graph.



1 A reflection in the x -axis:



2 A reflection in the y -axis:



Example 4

Find the rule of the function obtained from the graph of the function with equation $y = \sqrt{x}$ by a reflection:

- a** in the x -axis **b** in the y -axis

Solution

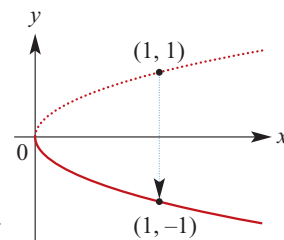
- a** Note that a reflection in the x -axis changes the y -values, so $(x, y) \rightarrow (x, -y)$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = x, y' = -y$

Rearranging gives $x = x', y = -y'$

So $y = \sqrt{x}$ becomes $(-y') = \sqrt{x'}$

The rule of the transformed function is $y = -\sqrt{x}$



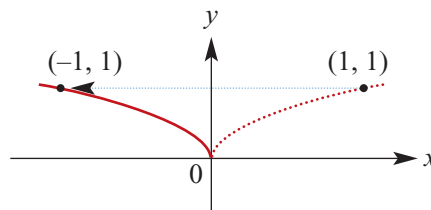
- b** Note that a reflection in the y -axis changes the x -values, so $(x, y) \rightarrow (-x, y)$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = -x, y' = y$

Rearranging gives $x = -x', y = y'$

So $y = \sqrt{x}$ becomes $y' = \sqrt{-x'}$

The rule of the transformed function is $y = \sqrt{-x}$



In general, a **reflection in the x -axis** is the transformation that maps $(x, y) \rightarrow (x, -y)$. To find the equation of the **image** of $y = f(x)$, under the **reflection** $(x, y) \rightarrow (x, -y)$, replace y with $-y$; i.e. the image of the graph with equation $y = f(x)$ under the reflection $(x, y) \rightarrow (x, -y)$ is the graph with equation $-y = f(x)$, which is more commonly written as $y = -f(x)$

In general, a **reflection in the y -axis** is the transformation that maps $(x, y) \rightarrow (-x, y)$. To find the equation of the **image** of $y = f(x)$, under the **reflection** $(x, y) \rightarrow (-x, y)$, replace x with $-x$; i.e. the image of the graph with equation $y = f(x)$ under the reflection $(x, y) \rightarrow (-x, y)$ is the graph with equation $y = f(-x)$

Exercise 3C

- 1** Sketch the graphs, and state the domain, of:

a $y = -\sqrt{x}$ **b** $y = \sqrt{-x}$

- 2** State a transformation which maps the graph of $y = f(x)$ to $y = f_1(x)$, where $f(x) = \sqrt{x}$ and $f_1(x) = \sqrt{-x}$

3 Find the equation of the rule when the graph of each of the functions below is transformed by:

i a reflection in the x -axis

ii a reflection in the y -axis

a $y = |x|$

b $y = \sqrt[3]{x}$

c $y = \frac{1}{x^3}$

d $y = \frac{1}{x^4}$

e $y = \frac{1}{\sqrt[3]{x}}$

f $y = x^{\frac{2}{3}}$

g $y = \frac{1}{x^{\frac{3}{4}}}$

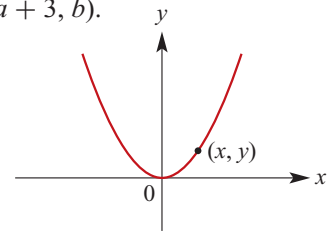
4 Identify whether the functions given in Question 3 are odd, even or neither.

3.4 Translations

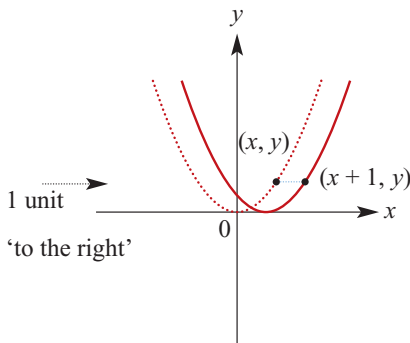
A transformation which, for example, translates each point in the plane 2 units in the positive direction of the y -axis can be described as ‘adding 2 units to the y -coordinate of each point in the plane’ and can be written as $(x, y) \rightarrow (x, y + 2)$. This is read as ‘the ordered pair (x, y) is mapped onto the ordered pair $(x, y + 2)$ ’. The translation is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(a, b + 2)$.

Similarly, a transformation which translates each point in the plane 3 units in the positive direction of the x -axis can be described as ‘adding 3 units to the x -coordinate of each point in the plane’ and can be written as $(x, y) \rightarrow (x + 3, y)$. This is read as ‘the ordered pair (x, y) is mapped onto the ordered pair $(x + 3, y)$ ’. The translation is a one-to-one mapping from R^2 to R^2 and uniquely links any ordered pair (a, b) to the ordered pair $(a + 3, b)$.

A translation moves each point on the graph the same distance in the same direction. Consider translating the graph of the function shown here in various ways, and observe the effect on a general point (x, y) on the graph.

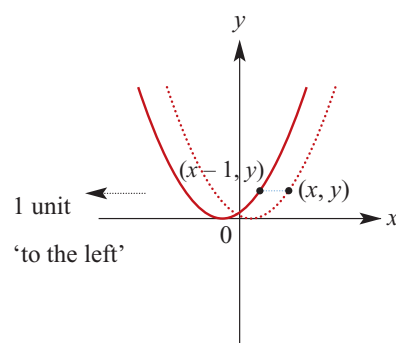


a A translation of 1 unit in the positive direction of the x -axis:



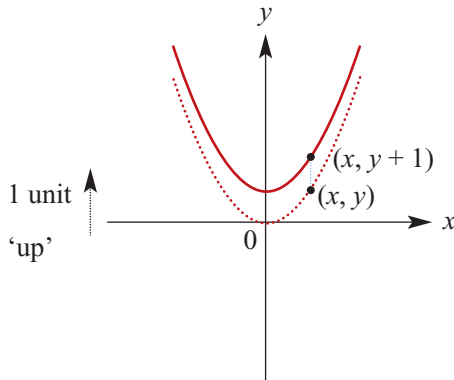
The point (x, y) is mapped onto $(x + 1, y)$,
i.e. $(x, y) \rightarrow (x + 1, y)$

b A translation of 1 unit in the negative direction of the x -axis:



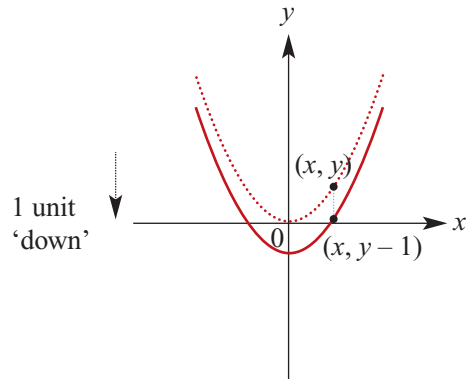
The point (x, y) is mapped onto $(x - 1, y)$,
i.e. $(x, y) \rightarrow (x - 1, y)$

- c A translation of 1 unit in the positive direction of the y -axis:



The point (x, y) is mapped onto $(x, y + 1)$,
i.e. $(x, y) \rightarrow (x, y + 1)$

- d A translation of 1 unit in the negative direction of the y -axis:



The point (x, y) is mapped onto $(x, y - 1)$,
i.e. $(x, y) \rightarrow (x, y - 1)$

In general, the **translation** of h units ($h > 0$) in the **positive direction of the x -axis** and k units ($k > 0$) in the **positive direction of the y -axis** is the transformation that maps $(x, y) \rightarrow (x + h, y + k)$. To find the equation of the **image** of $y = f(x)$, under the **translation** $(x, y) \rightarrow (x + h, y + k)$, replace x with $x - h$ and y with $y - k$, i.e. the image of the graph with equation $y = f(x)$ under the translation $(x, y) \rightarrow (x + h, y + k)$ is the graph with equation $y - k = f(x - h)$, which is more commonly written as $y = f(x - h) + k$

Example 5

Find the equation of the image when the graph of $y = |x|$ is transformed by the following sequence of transformations:

- a translation of 4 units in the positive direction of the x -axis, and
- a translation of 3 units in the negative direction of the y -axis.

Solution

$$(x, y) \rightarrow (x + 4, y - 3)$$

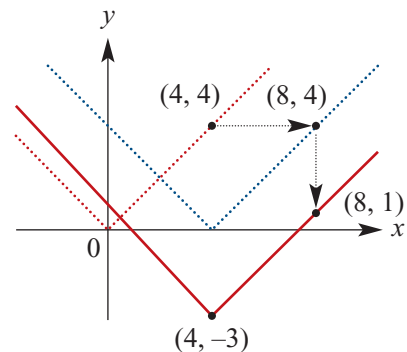
Let (x', y') be the coordinates of the image of (x, y) , so $x' = x + 4$, $y' = y - 3$

Rearranging gives $x = x' - 4$, $y = y' + 3$

So $y = |x|$ becomes $y' + 3 = |x' - 4|$

The rule of the transformed function

$$\text{is } y = |x - 4| - 3$$



Exercise 3D

- 1 Sketch the graphs of each of the following. Label asymptotes and axis intercepts, and state the domain and range.

a $y = \frac{1}{x} + 3$

b $y = \frac{1}{x^2} - 3$

c $y = \frac{1}{(x+2)^2}$

d $y = \sqrt{x-2}$

e $y = \frac{1}{x-1}$

f $y = \frac{1}{x} - 4$

g $y = \frac{1}{x+2}$

h $y = \frac{1}{x-3}$

i $f(x) = \frac{1}{(x-3)^2}$

j $f(x) = \frac{1}{(x+4)^2}$

k $f(x) = \frac{1}{x-1} + 1$

l $f(x) = \frac{1}{x-2} + 2$

m $y = \frac{1}{x^2} - 4$

- 2 For $y = f(x) = \frac{1}{x}$, sketch the graph of each of the following. Label asymptotes and axis intercepts.

a $y = f(x-1)$

b $y = f(x) + 1$

c $y = f(x+3)$

d $y = f(x) - 3$

e $y = f(x+1)$

f $y = f(x) - 1$

- 3 State a transformation which maps the graphs of $y = f(x)$ to $y = f_1(x)$ for each of the following:

a i $f(x) = x^2$

ii $f_1(x) = (x+5)^2$

b i $f(x) = \frac{1}{x}$

ii $f_1(x) = \frac{1}{x} + 2$

c i $f(x) = \frac{1}{x^2}$

ii $f_1(x) = \frac{1}{x^2} + 4$

- 4 Write down the equation of the rule when the graph of each of the functions below is transformed by:

i a translation of 7 units in the positive direction of the x -axis, and 1 unit in the positive direction of the y -axis

ii a translation of 2 units in the negative direction of the x -axis, and 6 units in the negative direction of the y -axis

iii a translation of 2 units in the positive direction of the x -axis, and 3 units in the negative direction of the y -axis

iv a translation of 1 unit in the negative direction of the x -axis, and 4 units in the positive direction of the y -axis

a $y = |x|$

b $y = \sqrt[3]{x}$

c $y = \frac{1}{x^3}$

d $y = \frac{1}{x^4}$

e $y = \frac{1}{\sqrt[3]{x}}$

f $y = x^{\frac{2}{3}}$

g $y = \frac{1}{x^{\frac{3}{4}}}$



Combinations of transformations

In the previous three sections each of the three types of transformations was considered separately. In the remainder of this chapter we look at situations where a graph may have been transformed by any combination of dilations, reflections and translations.

Example 6

Find the rule of the image when the graph of the function with rule $y = \sqrt{x}$ is translated 6 units in the negative direction of the x -axis, reflected in the y -axis and dilated by a factor of 2 from the x -axis.

Solution

The translation of 6 units in the negative direction of the x -axis maps $(x, y) \rightarrow (x - 6, y)$. The reflection in the y -axis maps $(x - 6, y) \rightarrow -(x - 6), y)$. The dilation by a factor of 2 from the x -axis maps $-(x - 6), y) \rightarrow -(x - 6), 2y)$. In summary, $(x, y) \rightarrow -(x - 6), 2y)$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = -(x - 6)$ and $y' = 2y$

Rearranging gives $x = -x' + 6$ and $y = \frac{y'}{2}$

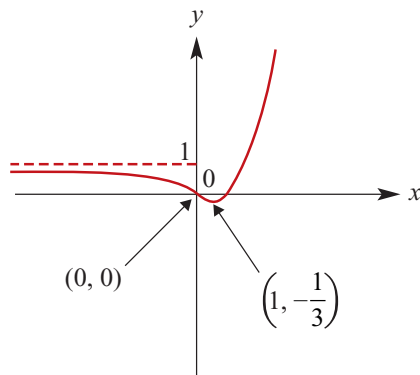
Therefore, $y = \sqrt{x}$ becomes $\frac{y'}{2} = \sqrt{-x' + 6}$

The rule of the transformed function is $y = 2\sqrt{6 - x}$

Example 7

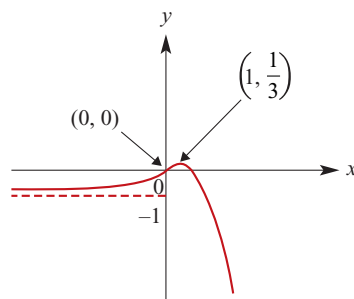
Sketch the graph of the image of the graph shown under the following sequence of transformations:

- a reflection in the x -axis
- a dilation of factor 3 from the x -axis
- a translation of 2 units in the positive direction of the x -axis and 1 unit in the positive direction of the y -axis.

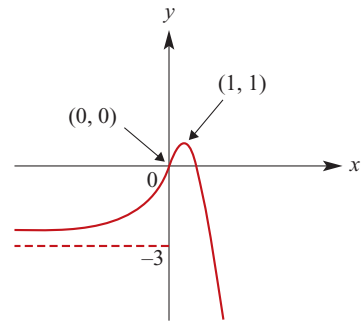


Solution

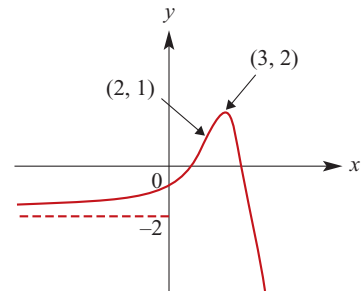
Consider each transformation separately and sketch the graph at each stage. A reflection in the x -axis produces the following graph:



Next consider the dilation of factor 3 from the x -axis:



Finally, apply the translation of 2 units in the positive direction of the x -axis and 1 unit in the positive direction of the y -axis.



Example 8

For the graph of $y = x^2$:

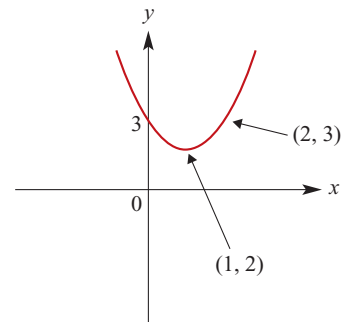
a Sketch the graph of the image under the sequence of transformations:

- a translation of 1 unit in the positive direction of the x -axis and 2 units in the positive direction of the y -axis
- a dilation of factor 2 from the y -axis
- a reflection in the x -axis.

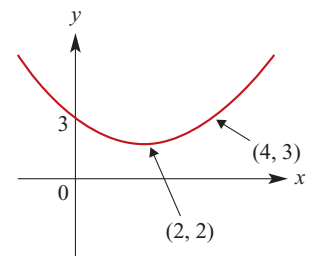
b State the rule of the image.

Solution

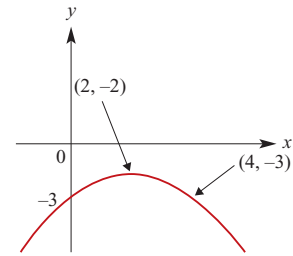
a Consider each transformation separately and sketch the graph at each stage. The translation produces the following graph:



Next consider the dilation of factor 2 from the y -axis:



Finally, apply the reflection in the x -axis:



b The mapping representing the transformations is:

$$(x, y) \rightarrow (x + 1, y + 2) \rightarrow (2(x + 1), y + 2) \rightarrow (2(x + 1), -(y + 2))$$

Let (x', y') be the coordinates of the image of (x, y) , so $x' = 2(x + 1)$ and $y' = -(y + 2)$

Rearranging gives $x = \frac{1}{2}(x' - 2)$ and $y = -y' - 2$

Therefore, $y = x^2$ becomes $-y' - 2 = \left(\frac{1}{2}(x' - 2)\right)^2$

The rule of the transformed function is $y = -\frac{1}{4}(x - 2)^2 - 2$

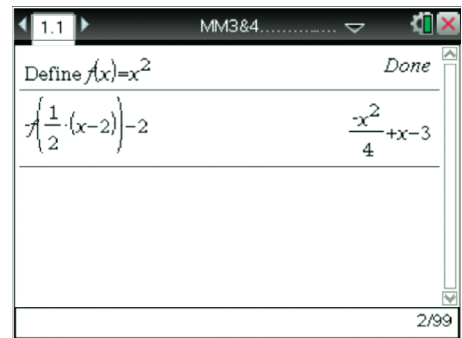
Using the TI-Nspire

Define $f(x) = x^2$

The rule for the new function is

$$-f\left(\frac{1}{2}(x - 2)\right) - 2.$$

The calculator gives the equation of the image of the graph under this sequence of transformations.

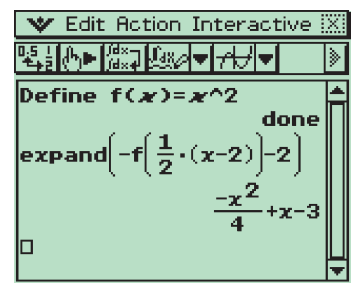


Using the Casio ClassPad

Define $f(x) = x^2$.

The rule for the new function is

$$-f\left(\frac{1}{2}(x - 2)\right) - 2.$$



Exercise 3E



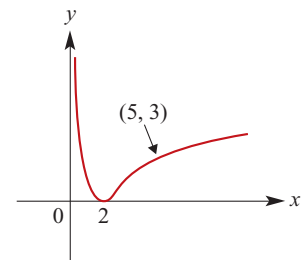
1 Find the rule of the image when the graph of each of the functions listed below undergoes the following sequences of transformations:

- i** a dilation of factor 2 from the x -axis, followed by a reflection in the x -axis, followed by a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis
- ii** a dilation of factor 2 from the x -axis, followed by a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis, followed by a reflection in the x -axis
- iii** a reflection in the x -axis, followed by a dilation of factor 2 from the x -axis, followed by a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis
- iv** a reflection in the x -axis, followed by a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis, followed by a dilation of factor 2 from the x -axis
- v** a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis, followed by a dilation of factor 2 from the x -axis, followed by a reflection in the x -axis
- vi** a translation 3 units in the positive direction of the x -axis and 4 units in the negative direction of the y -axis, followed by a reflection in the x -axis, followed by a dilation of factor 2 from the x -axis

a $y = x $	b $y = \sqrt[3]{x}$	c $y = \frac{1}{x^3}$	d $y = \frac{1}{x^4}$
e $y = \frac{1}{\sqrt[3]{x}}$	f $y = x^{\frac{2}{3}}$	g $y = \frac{1}{x^{\frac{3}{4}}}$	

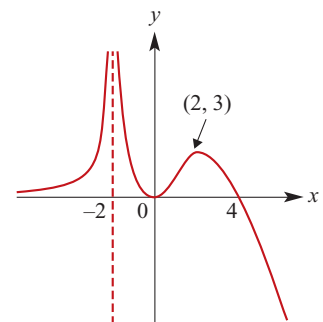
2 Sketch the graph of the image of the graph shown under the following sequence of transformations:

- a reflection in the x -axis
- a dilation of factor 2 from the x -axis
- a translation of 3 units in the positive direction of the x -axis and 4 units in the positive direction of the y -axis



3 Sketch the graph of the image of the graph shown under the following sequence of transformations:

- a reflection in the y -axis
- a translation of 2 units in the negative direction of the x -axis and 3 units in the negative direction of the y -axis
- a dilation of factor 2 from the y -axis



- 4 For the graph of $y = |x|$:
- Sketch the graph of the image under the sequence of transformations:
 - a dilation of factor 2 from the x -axis
 - a translation of 2 units in the negative direction of the x -axis and 1 unit in the negative direction of the y -axis
 - a reflection in the x -axis.
 - State the rule of the image.
- 5 For the graph of $y = x^{\frac{1}{3}}$:
- Sketch the graph of the image under the sequence of transformations:
 - a reflection in the y -axis
 - a translation of 1 unit in the positive direction of the x -axis and 2 units in the negative direction of the y -axis
 - a dilation of factor $\frac{1}{2}$ from the y -axis.
 - State the rule of the image.

3.6 Determining transformations to sketch graphs

By considering a rule for a graph as a combination of transformations of a more ‘simple’ rule, we are able to readily sketch graphs of many apparently ‘complicated’ functions.

Example 9

Identify a sequence of transformations that maps the graph of the function $y = \frac{1}{x}$ onto the graph of the function $y = \frac{4}{x+5} - 3$, and use this to sketch the graph of $y = \frac{4}{x+5} - 3$, stating the equations of asymptotes and the coordinates of axes intercepts.

Solution

Rearrange the rule of the function of the transformed graph into the form $\frac{y' + 3}{4} = \frac{1}{x' + 5}$ (the ‘shape’ of $y = \frac{1}{x}$), where (x', y') are the coordinates of the image of (x, y) .

Therefore $x = x' + 5$ and $y = \frac{y' + 3}{4}$. Rearranging gives $x' = x - 5$ and $y' = 4y - 3$.

So the mapping is given by $(x, y) \rightarrow (x - 5, 4y - 3)$ which identifies the sequence of transformations as:

- a dilation of factor 4 from the x -axis, followed by a translation of 3 units in the negative direction of the y -axis, and

- a translation of 5 units in the negative direction of the x -axis.

Axes intercepts are found in the usual way, as below.

$$\begin{aligned}\text{When } x = 0, \quad y &= \frac{4}{5} - 3 \\ &= -2\frac{1}{5}\end{aligned}$$

$$\begin{aligned}\text{When } y = 0, \quad \frac{4}{x+5} - 3 &= 0 \\ \therefore 4 &= 3x + 15 \\ \therefore 3x &= -11\end{aligned}$$

$$\text{and} \quad x = -\frac{11}{3}$$

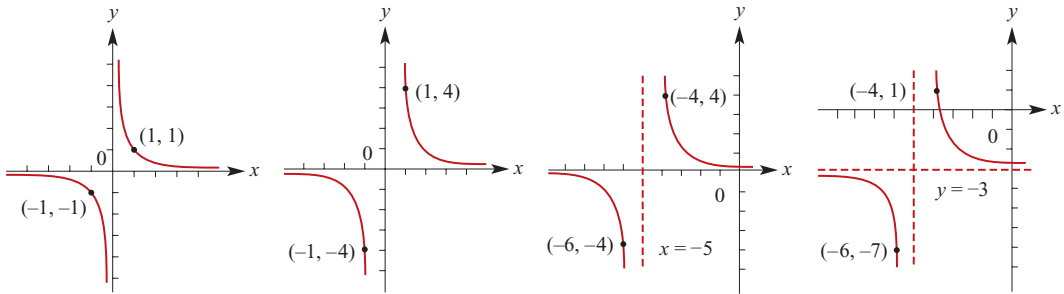
Transformations are shown below:

Original $y = \frac{1}{x}$

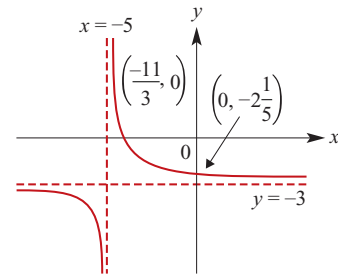
1 Dilation from x -axis:

2 Translation in the negative direction of the x -axis:

3 Translation in the negative direction of the y -axis:



The result, with intercepts marked, is:



Once you have done a few of these types of exercises, you can identify the transformations more quickly by carefully observing the rule of the transformed graph and relating it to the ‘shape’ of the simplest function in its family. Consider the following examples.

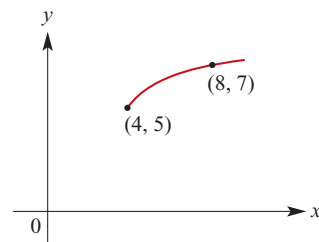
Example 10

Sketch the graph of $y = \sqrt{x-4} + 5$

Solution

The graph is obtained from the graph of $y = \sqrt{x}$ through two translations:

- 4 units in the positive direction of the x -axis
- 5 units in the positive direction of the y -axis.

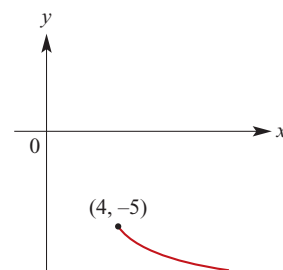
**Example 11**

Sketch the graph of $y = -\sqrt{x-4} - 5$

Solution

The graph is obtained from the graph of $y = \sqrt{x}$ by:

- a translation of 4 units in the positive direction of the x -axis, and
- a reflection in the x -axis, followed by a translation of 5 units in the negative direction of the y -axis.

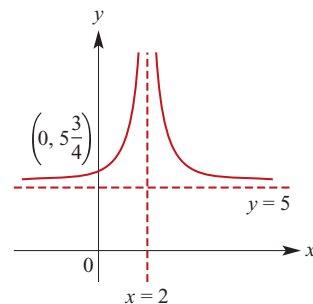
**Example 12**

Sketch of graph of $y = \frac{3}{(x-2)^2} + 5$

Solution

This is obtained from the graph of $y = \frac{1}{x^2}$ by:

- a dilation of factor 3 from the x -axis, followed by a translation of 5 units in the positive direction of the y -axis, and
- a translation of 2 units in the positive direction of the x -axis.



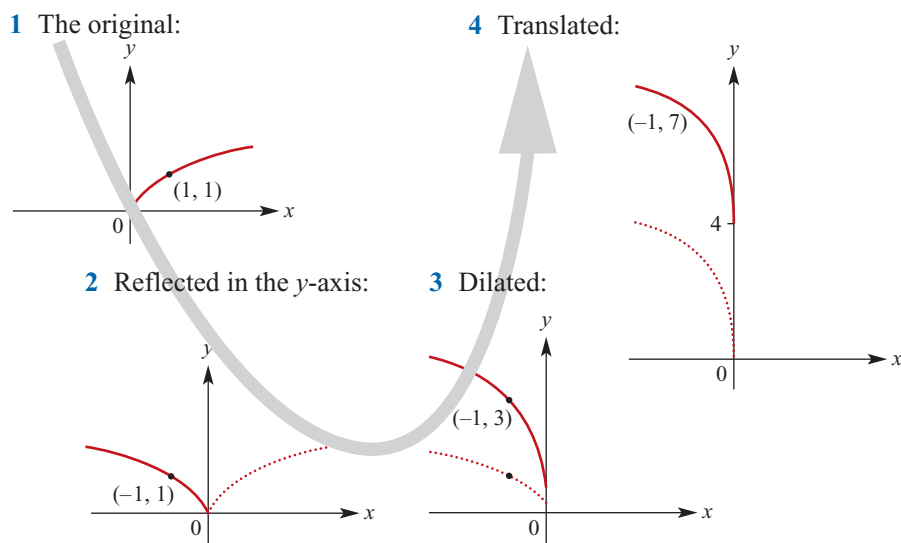
Example 13

Sketch the graph of $y = 3\sqrt{-x} + 4$

Solution

This is obtained from the graph of $y = \sqrt{x}$ by:

- a reflection in the y -axis, and
- a dilation of factor 3 from the x -axis, followed by a translation of 4 units in the positive direction of the y -axis.



In general, the function given by the equation $y = Af(n(x+c)) + b$, where $b, c \in \mathbb{R}^+$ and $A, n \in \mathbb{R}$, represents a transformation of the graph of $y = f(x)$ by:

- a dilation of factor $|A|$ from (and if $A < 0$ a reflection in) the x -axis, followed by a translation of b units in the positive direction of the y -axis, and
- a dilation of factor $\left|\frac{1}{n}\right|$ from (and if $n < 0$ a reflection in) the y -axis, followed by a translation of c units in the negative direction of the x -axis.

Exercise 3F

- 1 In each case below, state the sequence of transformations required to transform the graph of the first equation into the graph of the second equation:

a $y = \frac{1}{x}, y = \frac{2}{x-1} + 3$

b $y = \frac{1}{x^2}, y = \frac{-3}{(x+4)^2} - 7$

c $y = \frac{1}{x^3}, y = \frac{4}{(1-x)^3} - 5$

d $y = \sqrt[3]{x}, y = 2 - \sqrt[3]{x+1}$

e $y = \frac{1}{\sqrt{x}}, y = \frac{2}{\sqrt{-x}} + 3$

f $y = \frac{2}{3-x} + 4, y = \frac{1}{x}$

- 2** Sketch the graph of each of the following without using your calculator. State the equations of asymptotes and axes intercepts. State the range of each function.

$$\begin{array}{lll} \mathbf{a} & f(x) = \frac{3}{x-1} & \mathbf{b} \quad g(x) = \frac{2}{x+1} - 1 \quad \mathbf{c} \quad h(x) = \frac{3}{(x-2)^2} \\ \mathbf{d} & f(x) = \frac{2}{(x-1)^2} - 1 & \mathbf{e} \quad h(x) = \frac{-1}{x-3} \quad \mathbf{f} \quad f(x) = \frac{-1}{x+2} + 3 \\ \mathbf{g} & f(x) = \frac{2}{(x-3)^3} + 4 & \end{array}$$

- 3** Sketch the graph of each of the following without using your calculator. State the range of each.

$$\begin{array}{lll} \mathbf{a} & y = -\sqrt{x-3} & \mathbf{b} \quad y = -\sqrt{x-3} + 2 \quad \mathbf{c} \quad y = \sqrt{2(x+3)} \\ \mathbf{d} & y = \frac{1}{2x-3} & \mathbf{e} \quad y = 5\sqrt{x+2} \quad \mathbf{f} \quad y = -5\sqrt{x+2} - 2 \\ \mathbf{g} & y = \frac{-3}{x-2} & \mathbf{h} \quad y = \frac{-2}{(x+2)^2} - 4 \quad \mathbf{i} \quad y = \frac{3}{2x} - 5 \\ \mathbf{j} & y = \frac{5}{2x} + 5 & \mathbf{k} \quad y = 2|x-3| + 5 \end{array}$$

- 4** Use your calculator to help you sketch the graph of each of the following. State the range of each.

$$\mathbf{a} \quad y = -\sqrt[3]{x+2} + 7 \quad \mathbf{b} \quad y = \frac{4}{\sqrt[3]{x-1}} + 2 \quad \mathbf{c} \quad y = -(x+1)^{\frac{3}{4}} - 6$$

- 5 a** Show that $\frac{3x+2}{x+1} = 3 - \frac{1}{x+1}$
and hence, without using your calculator, sketch the graph of:

$$f: R \setminus \{-1\} \rightarrow R, f(x) = \frac{3x+2}{x+1}$$

- b** Show that $\frac{4x-5}{2x+1} = 2 - \frac{7}{2x+1}$
and hence, without using your calculator, sketch the graph of:

$$f: R \setminus \left\{-\frac{1}{2}\right\} \rightarrow R, f(x) = \frac{4x-5}{2x+1}$$

$$\text{Note: } f(x) = 2 - \frac{7}{2\left(x + \frac{1}{2}\right)}$$

- 6** Sketch the graph of each of the following without using your calculator. State the range of each.

$$\begin{array}{lll} \mathbf{a} & y = \frac{2}{x-3} + 4 & \mathbf{b} \quad y = \frac{4}{3-x} + 4 \quad \mathbf{c} \quad y = \frac{2}{(x-1)^2} + 1 \\ \mathbf{d} & y = 2\sqrt{x-1} + 2 & \mathbf{e} \quad y = -3\sqrt{x-4} + 1 \quad \mathbf{f} \quad y = 5\sqrt{2x+4} + 1 \end{array}$$

3.7 Using matrices for transformations

A summary of some of the transformations and their rules which were introduced earlier in this chapter is presented here. Suppose (x', y') is the image of (x, y) under the mapping in the first column of the table below.

Mapping	Rule
Reflection in the x -axis	$x' = x = x + 0y$ $y' = -y = 0x - y$
Reflection in the y -axis	$x' = -x = -x + 0y$ $y' = y = 0x + y$
Dilation by factor k from the y -axis	$x' = kx = kx + 0y$ $y' = y = 0x + y$
Dilation by factor k from the x -axis	$x' = x = x + 0y$ $y' = ky = 0x + ky$
Reflection in the line $y = x$	$x' = y = 0x + y$ $y' = x = x + 0y$
Translation defined by a vector $\begin{bmatrix} a \\ b \end{bmatrix}$	$x' = x + a$ $y' = y + b$

The first five mappings are special cases of a general kind of mapping defined by

$$\begin{aligned}x' &= ax + by \\ y' &= cx + dy\end{aligned}$$

where a, b, c, d are real numbers.

These equations can be rewritten as

$$\begin{aligned}x' &= a_{11}x + a_{12}y \\ y' &= a_{21}x + a_{22}y\end{aligned}$$

which yields the equivalent matrix equation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

A transformation of the form

$$(x, y) \rightarrow (a_{11}x + a_{12}y, a_{21}x + a_{22}y)$$

is called a linear transformation.

The notation $T: R^2 \rightarrow R^2$ is often used to indicate that a transformation is a mapping from the Cartesian plane into the Cartesian plane. The rule can then be defined through the use of matrices. Some questions formed in this way are given in Chapter 20.

The first five transformations above can be defined by a 2×2 matrix. This is shown in the table below.

Mapping	Matrix
Reflection in the x -axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection in the y -axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Dilation by factor k from the y -axis	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
Dilation of factor k from the x -axis	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
Reflection in the line $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Example 14

Find the image of the point $(2, 3)$ under

a a reflection in the x -axis **b** a dilation of factor k from the y -axis

Solution

$$\text{a } \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}. \text{ Therefore } (2, 3) \rightarrow (2, -3)$$

$$\text{b } \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2k \\ 3 \end{bmatrix}. \text{ Therefore } (2, 3) \rightarrow (2k, 3)$$

Example 15

Consider a linear transformation such that $(1, 0) \rightarrow (3, -1)$ and $(0, 1) \rightarrow (-2, 4)$. Find the image of $(-3, 5)$.

Solution

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \text{ and } \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$\therefore a_{11} = 3, a_{21} = -1 \text{ and } a_{12} = -2, a_{22} = 4$$

i.e. the transformation can be defined by the 2×2 matrix $\begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix}$

Let $(-3, 5) \rightarrow (x', y')$

$$\therefore \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \times -3 + -2 \times 5 \\ -1 \times -3 + 4 \times 5 \end{bmatrix} = \begin{bmatrix} -19 \\ 23 \end{bmatrix}$$

$$\therefore (-3, 5) \rightarrow (-19, 23)$$

The image of $(-3, 5)$ is $(-19, 23)$.

Note that non-linear mappings cannot be represented by a matrix in the way indicated above.

Thus for the translation defined by $(0, 0) \rightarrow (a, b)$

$$x' = x + a$$

$$y' = y + b$$

While this cannot be represented by a square matrix, the defining equations

$$\text{suggest } \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

where the 'sum' has the definition:

$$\text{for each } x, y, a, b \text{ in } R, \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} x + a \\ y + b \end{bmatrix}$$

Composition of mappings

Consider a linear transformation defined by the matrix $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ composed with a

linear transformation defined by the matrix $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$

The composition consists of the transformation of $\mathbf{A}$ being applied first and then the transformation of $\mathbf{B}$.

The matrix of the resulting composition is the product

$$\mathbf{BA} = \begin{bmatrix} b_{11}a_{11} + b_{12}a_{21} & b_{11}a_{12} + b_{12}a_{22} \\ b_{21}a_{11} + b_{22}a_{21} & b_{21}a_{12} + b_{22}a_{22} \end{bmatrix}$$

Example 16

Find the image of the point $(2, -3)$ under a reflection in the x -axis followed by a dilation of factor k from the y -axis.

Solution

Matrix multiplication gives the matrix. Let $\mathbf{A}$ be the transformation reflection in the x -axis and $\mathbf{B}$ the transformation dilation of factor k from the y -axis. Then the required transformation is defined by the product

$$\mathbf{BA} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & -1 \end{bmatrix} \text{ and } \mathbf{BA} \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 2k \\ 3 \end{bmatrix}$$

Example 17

Express the composition of the transformations, dilation of factor k from the y -axis followed by a translation defined by the matrix $\mathbf{C} = \begin{bmatrix} a \\ b \end{bmatrix}$, mapping a point (x, y) to a point (x', y') as a matrix equation. Hence find x and y in terms of x' and y' respectively.

Solution

Let $\mathbf{A}$ be the dilation transformation, $\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}$, and $\mathbf{X}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$

The equation is $\mathbf{AX} + \mathbf{C} = \mathbf{X}'$

Then $\mathbf{AX} = \mathbf{X}' - \mathbf{C}$ and hence $\mathbf{X} = \mathbf{A}^{-1}(\mathbf{X}' - \mathbf{C})$

$$\text{Now } \mathbf{A} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(\mathbf{A}) = k \text{ and therefore } \mathbf{A}^{-1} = \frac{1}{k} \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} \frac{1}{k} & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} \frac{1}{k} & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} x' \\ y' \end{bmatrix} - \begin{bmatrix} a \\ b \end{bmatrix} \right) = \begin{bmatrix} \frac{1}{k} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x' - a \\ y' - b \end{bmatrix} = \begin{bmatrix} \frac{1}{k}(x' - a) \\ y' - b \end{bmatrix}$$

$$\text{Hence } x = \frac{1}{k}(x' - a) \text{ and } y = y' - b$$

Transforming graphs

The notation is now applied to transforming graphs. The notation is consistent with the notation introduced earlier in this chapter.

Example 18

A transformation is defined by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. Find the equation of the image of the graph of the quadratic equation $y = x^2 + 2x + 3$ under this transformation.

Solution

As before the transformation maps $(x, y) \rightarrow (x', y')$.

Using matrix notation,
$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

It can be written as the matrix equation $\mathbf{TX} = \mathbf{X}'$

Now multiply both sides of the equation by $\mathbf{T}^{-1}$.

Therefore $\mathbf{T}^{-1}\mathbf{TX} = \mathbf{T}^{-1}\mathbf{X}'$

and
$$\mathbf{X} = \mathbf{T}^{-1}\mathbf{X}'$$

Therefore
$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ \frac{1}{2}y' \end{bmatrix}$$

$x = x'$ and $y = \frac{y'}{2}$

The curve with equation $y = x^2 + 2x + 3$ is mapped to the curve with equation $\frac{y'}{2} = (x')^2 + 2x' + 3$

This makes quite hard work of an easy problem, but it demonstrates a procedure that can be used for any transformation defined by a 2×2 non-singular matrix.

Example 19

A transformation is described through the equation $\mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{X}'$ where $\mathbf{T} = \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix}$ and

$\mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Find the image of the straight line with equation $y = 2x + 5$ under the transformation.

Solution

First solve the matrix equation for $\mathbf{X}$.

$$\mathbf{T}^{-1}\mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{T}^{-1}\mathbf{X}'$$

$$\mathbf{X} + \mathbf{B} = \mathbf{T}^{-1}\mathbf{X}'$$

$$\text{and } \mathbf{X} = \mathbf{T}^{-1}\mathbf{X}' - \mathbf{B}$$

Therefore
$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{y'}{2} - 1 \\ -\frac{x'}{3} - 2 \end{bmatrix}$$

Therefore $x = \frac{y'}{2} - 1$ and $y = -\frac{x'}{3} - 2$

The straight line with equation $y = 2x + 5$ is transformed to the straight line with equation $-\frac{x'}{3} - 2 = 2\left(\frac{y'}{2} - 1\right) + 5$

Rearranging gives $y' = -\frac{x'}{3} - 5$

Exercise 3G

- 1 Using matrix methods find the image of the point $(1, -2)$ under each of the following transformations:
 - a dilation of factor 3 from the x -axis
 - b dilation of factor 2 from the y -axis
 - c reflection in the x -axis
 - d reflection in the y -axis
 - e reflection in the line $y = x$

- 2 Find the matrix that determines the composition of transformations, in the given order:
 - reflection in the x -axis
 - dilation of factor 2 from the x -axis

- 3 Write down the matrix of each of the following transformations:
 - a reflection in the line $x = 0$
 - b reflection in the line $y = x$
 - c reflection in the line $y = -x$
 - d dilation of factor 2 from the x -axis
 - e dilation of factor 3 from the y -axis

- 4 Express the composition of the transformations, dilation of factor 3 from the x -axis followed by a translation defined by the matrix $\mathbf{C} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, mapping a point (x, y) to a point (x', y') as a matrix equation. Hence find x and y in terms of x' and y' respectively.

- 5 A transformation is defined by the matrix $\begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix}$. Find the equation of the image of the graph of the quadratic equation $y = x^2 + x + 2$ under this transformation.

- 6 A transformation is defined by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$. Find the equation of the image of the graph of the cubic equation $y = x^3 + 2x$ under this transformation.

- 7 A transformation is defined by the matrix $\begin{bmatrix} 0 & 3 \\ -2 & 0 \end{bmatrix}$. Find the equation of the image of the graph of the straight line with equation $y = 2x + 3$ under this transformation.

- 8 A transformation is defined by the matrix $\begin{bmatrix} 0 & 2 \\ -3 & 0 \end{bmatrix}$. Find the equation of the image of the graph of the straight line with equation $y = -2x + 4$ under this transformation.

- 9 A transformation is described through the equation $\mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{X}'$ where $\mathbf{T} = \begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Find the image of the straight line with equation $y = -2x + 6$ under the transformation.

- 10** A transformation is described through the equation $\mathbf{TX} + \mathbf{B} = \mathbf{X}'$ where $\mathbf{T} = \begin{bmatrix} 0 & -3 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$. Find the image of the straight line with equation $y = -2x + 6$ under the transformation.
- 11** A transformation is described through the equation $\mathbf{TX} + \mathbf{B} = \mathbf{X}'$ where $\mathbf{T} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Find the image of the curve with equation $y = -2x^3 + 6x$ under the transformation.
- 12** A transformation is described through the equation $\mathbf{TX} + \mathbf{B} = \mathbf{X}'$ where $\mathbf{T} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. Find the image of the curve with equation $y = -2x^3 + 6x^2 + 2$ under the transformation.

3.8 Determining the rule for a function of a graph

Given sufficient information about a curve, a rule for the function of the graph may be determined. For example, if the coordinates of two points on a hyperbola of the form

$$y = \frac{a}{x} + b$$

are known, the rule for the hyperbola may be found, i.e. the values of a and b may be found.

Sometimes a more specific rule is known. For example, the curve may be a dilation of $y = \sqrt{x}$. It is then known to be of the $y = a\sqrt{x}$ family, and the coordinates of one point (with the exception of the origin) will be enough to determine the value for a .

Example 20

It is known that the points $(1, 5)$ and $(4, 2)$ lie on a curve with the equation $y = \frac{a}{x} + b$. Find the values of a and b .

Solution

$$\text{When } x = 1, y = 5, \text{ therefore } 5 = a + b \quad (1)$$

$$\text{and when } x = 4, y = 2, \text{ therefore } 2 = \frac{a}{4} + b \quad (2)$$

$$\begin{aligned} \text{Subtract (2) from (1):} \quad 3 &= \frac{3a}{4} \\ \therefore a &= 4 \end{aligned}$$

$$\text{Substitute in (1) to find } b: \quad 5 = 4 + b$$

$$\text{Therefore } b = 1$$

$$\text{and } y = \frac{4}{x} + 1$$

Example 21

It is known that the points $(2, 1)$ and $(10, 6)$ lie on a curve with equation $y = a\sqrt{x-1} + b$. Find the values of a and b .

Solution

When $x = 2$, $y = 1$.

$$\text{Therefore } 1 = a\sqrt{1} + b, \text{ i.e. } 1 = a + b \quad (1)$$

When $x = 10$, $y = 6$.

$$\text{Therefore } 6 = a\sqrt{9} + b, \text{ i.e. } 6 = 3a + b \quad (2)$$

$$\text{Subtract (1) from (2): } 5 = 2a$$

$$\therefore a = \frac{5}{2}$$

$$\text{Substitute in (1) to find } b: 1 = \frac{5}{2} + b$$

$$\text{Therefore } b = -\frac{3}{2}$$

and

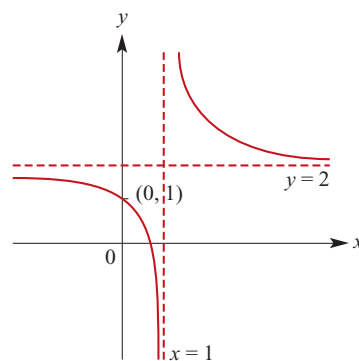
$$y = \frac{5}{2}\sqrt{x-1} - \frac{3}{2}$$

Exercise 3H

- 1 The graph shown has the rule:

$$y = \frac{A}{x+b} + B$$

Find the values of A , b and B .

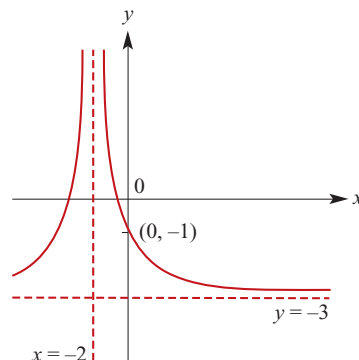


- 2 The points with coordinates $(1, 5)$ and $(16, 11)$ lie on a curve which has a rule of the form $y = A\sqrt{x} + B$. Find A and B .
- 3 The points with coordinates $(1, 1)$ and $(0.5, 7)$ lie on a curve which has a rule of the form $y = \frac{A}{x^2} + B$. Find the values of A and B .

- 4 The graph shown has the rule:

$$y = \frac{A}{(x+b)^2} + B$$

Find the values of A , b and B .



- 5 The points with coordinates $(1, -1)$ and $(2, \frac{3}{4})$ lie on a curve which has a rule of the form $y = \frac{a}{x^3} + b$. Find the values of a and b .
- 6 The points with coordinates $(-1, -8)$ and $(1, -2)$ lie on a curve which has a rule of the form $y = \frac{a}{\sqrt[3]{x}} + b$. Find the values of a and b .

3.9 Addition of ordinates

In Chapter 1 it was established that for functions f and g a new function $f + g$ can be defined by:

$$(f + g)(x) = f(x) + g(x)$$

$$\text{dom}(f + g) = \text{dom}(f) \cap \text{dom}(g)$$

In this section graphing by addition of ordinates is considered.

Example 22

Sketch the graphs of $f(x) = x + 1$ and $g(x) = 3 - 2x$ and hence the graph of $(f + g)(x)$.

Solution

If f and g are functions defined by:

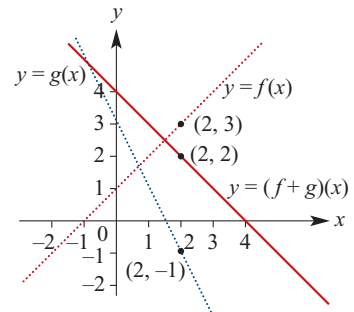
$$f(x) = x + 1$$

and $g(x) = 3 - 2x$

then $(f + g)(x) = f(x) + g(x)$
 $= 4 - x$

We note $(f + g)(2) = f(2) + g(2) = 3 + -1 = 2$,

i.e. the ordinates are added.



Now check that the same principle applies for other points on the graphs. A table of values can be a useful aid to find points that lie on the graph of $y = (f + g)(x)$.

x	$f(x)$	$g(x)$	$(f + g)(x)$
-1	0	5	5
0	1	3	4
$\frac{3}{2}$	$\frac{5}{2}$	0	$\frac{5}{2}$
2	3	-1	2

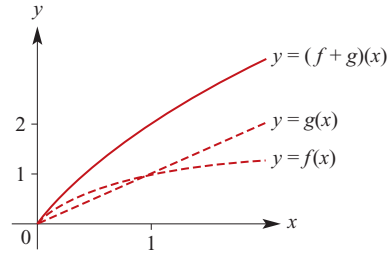
The table shows that the points $(-1, 5)$, $(0, 4)$, $(\frac{3}{2}, \frac{5}{2})$ and $(2, 2)$ lie on the graph of $y = (f + g)(x)$

Example 23

Sketch the graph of $y = (f + g)(x)$ where $f(x) = \sqrt{x}$ and $g(x) = x$

Solution

It can be seen that the function with rule $(f + g)(x) = \sqrt{x} + x$ is defined by the addition of the two functions f and g .

**Exercise 31**

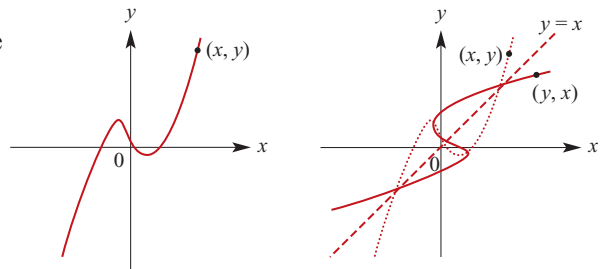
- 1 Sketch the graph of $f: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}$, $f(x) = \sqrt{x} + x$ using addition of ordinates.
- 2 Sketch the graph of $f: [-2, \infty) \rightarrow \mathbb{R}$, $f(x) = \sqrt{x+2} + x$ using addition of ordinates.
- 3 Sketch the graph of $f: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}$, $f(x) = -\sqrt{x} + x$ using addition of ordinates.
- 4 Sketch the graph of $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x} + \frac{1}{x^2}$ using addition of ordinates.
- 5 For each of the following sketch the graph of $f + g$:
 - a $f: [-2, \infty) \rightarrow \mathbb{R}$, $f(x) = \sqrt{2+x}$, $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = -2x$
 - b $f: (-\infty, 2]$, $f(x) = \sqrt{2-x}$, $g: [-2, \infty) \rightarrow \mathbb{R}$, $g(x) = \sqrt{x+2}$

3.10 Graphing inverse functions

A transformation which reflects each point in the plane in the line $y = x$ can be described through ‘interchanging the x - and y -coordinates of each point in the plane’ and can be written as $(x, y) \rightarrow (y, x)$. This is read as ‘the ordered pair (x, y) is mapped onto the ordered pair (y, x) ’. The reflection is a one-to-one mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ and uniquely links any ordered pair (a, b) to the ordered pair (b, a) .

This special case where the graph of a function is reflected in the line $y = x$ produces the graph of the inverse relation.

Consider reflecting the graph of the function shown here in the line $y = x$, and observe the effect on a general point (x, y) on the graph.



The line $y = x$ acts as a ‘mirror’ line. The point (x, y) is mapped onto (y, x) , i.e. $(x, y) \rightarrow (y, x)$

In general, a **reflection in the line $y = x$** is the transformation that maps $(x, y) \rightarrow (y, x)$. To find the equation of the **image** of $y = f(x)$, under the **reflection** $(x, y) \rightarrow (y, x)$, replace x with y and y with x ; i.e. the image of the graph of $\{(x, y): y = f(x)\}$ under the reflection $(x, y) \rightarrow (y, x)$ is the graph of $\{(y, x): y = f(x)\}$ and is called the inverse relation of $y = f(x)$.

Families of functions, with $A, B, b \in R, A \neq 0$

One-to-one functions

$$f(x) = \frac{A}{(x+b)^n} + B, \quad \text{where } n \text{ is a positive odd integer, and}$$

$$g(x) = A(x+b)^{\frac{p}{q}} + B \quad \text{where } p \text{ and } q \text{ are integers, } p \text{ odd}$$

and where the highest common factor of $p, q = 1$

are one-to-one functions, and therefore have inverses that are functions. See Section 1.7. Simple examples of such functions (other than polynomials which will be explored in Chapter 4) are:

$$f(x) = \frac{1}{x}, g(x) = \frac{1}{x^3}, h(x) = \frac{1}{x^5}, \dots$$

$$f(x) = x^{\frac{1}{2}}, g(x) = x^{\frac{1}{3}}, h(x) = x^{\frac{1}{4}}, r(x) = x^{\frac{3}{2}}, s(x) = x^{\frac{3}{4}}, v(x) = x^{\frac{3}{5}}, w(x) = x^{\frac{5}{3}}, \dots$$

$$f(x) = \frac{1}{x^{\frac{1}{2}}}, g(x) = \frac{1}{x^{\frac{1}{3}}}, h(x) = \frac{1}{x^{\frac{1}{4}}}, r(x) = \frac{1}{x^{\frac{3}{2}}}, s(x) = \frac{1}{x^{\frac{3}{4}}}, v(x) = \frac{1}{x^{\frac{3}{5}}}, w(x) = \frac{1}{x^{\frac{5}{3}}}, \dots$$

Example 24

Find the inverse of the function with rule $f(x) = 3\sqrt{x+2} + 4$ and sketch both functions on one set of axes, clearly showing the exact coordinates of intersection of the two graphs.

Solution

Consider $x = 3\sqrt{y+2} + 4$

$$\text{Solve for } y: \frac{x-4}{3} = \sqrt{y+2}$$

$$\text{which implies } y = \left(\frac{x-4}{3}\right)^2 - 2$$

$$\therefore f^{-1}(x) = \left(\frac{x-4}{3}\right)^2 - 2$$

and as the domain of f^{-1} = range of f

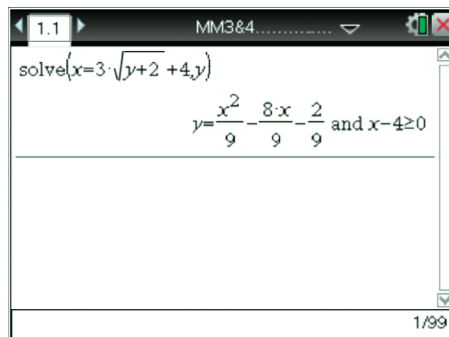
$$f^{-1}: [4, \infty) \rightarrow R, f^{-1}(x) = \left(\frac{x-4}{3}\right)^2 - 2$$

Using the TI-Nspire

To find the rule for the inverse of

$$y = 3\sqrt{x+2} + 4, \text{ enter}$$

solve ($x = 3\sqrt{y+2} + 4, y$).



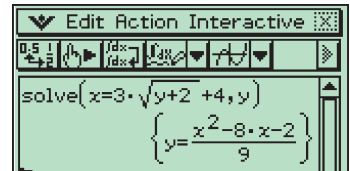
Using the Casio ClassPad

To find the rule for the inverse of

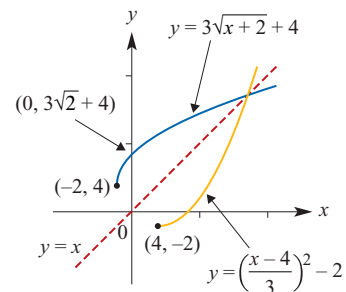
$f(x) = 3\sqrt{x+2} + 4$, enter and highlight

$x = 3\sqrt{y+2} + 4$.

Tap **Interactive–Equation/inequality–solve** and set the variable as y .



Note: The graph of f^{-1} is obtained by reflecting the graph of f in the line $y = x$. The graph of $y = f^{-1}(x)$ is obtained from the graph of $y = f(x)$ by applying the transformation $(x, y) \rightarrow (y, x)$.



The graphs meet where $f(x) = x = f^{-1}(x)$. Points of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are usually found by solving either $f(x) = x$ or $f^{-1}(x) = x$, rather than the more complicated equation $f(x) = f^{-1}(x)$. (Note that points of intersection between the graphs of $y = f(x)$ and $y = f^{-1}(x)$ that do not lie on the line $y = x$ also sometimes exist.)

In this particular example, it is simpler to solve $f^{-1}(x) = x$

That is, $\left(\frac{x-4}{3}\right)^2 - 2 = x, x > 4$

$$\left(\frac{x-4}{3}\right)^2 = x + 2$$

$$x^2 - 17x - 2 = 0$$

$$\therefore x = \frac{17 \pm \sqrt{17^2 + (4 \times 2)}}{2}$$

As $x > 4$, only the positive solution is valid.

The two graphs meet at the point $\left(\frac{17 + \sqrt{297}}{2}, \frac{17 + \sqrt{297}}{2}\right) \approx (17.12, 17.12)$

Example 25

Express $\frac{x+4}{x+1}$ in the form $\frac{a}{x+b} + c$ and hence find the inverse of the function

$f(x) = \frac{x+4}{x+1}$. Sketch both functions on the one set of axes.

Solution

$$\frac{x+4}{x+1} = \frac{3+x+1}{x+1} = \frac{3}{x+1} + \frac{x+1}{x+1} = \frac{3}{x+1} + 1$$

Consider $x = \frac{3}{y+1} + 1$

Solve for y : $x - 1 = \frac{3}{y+1}$

and thus $y + 1 = \frac{3}{x-1}$

and $y = \frac{3}{x-1} - 1$

The range of f is $R \setminus \{1\}$, and thus:

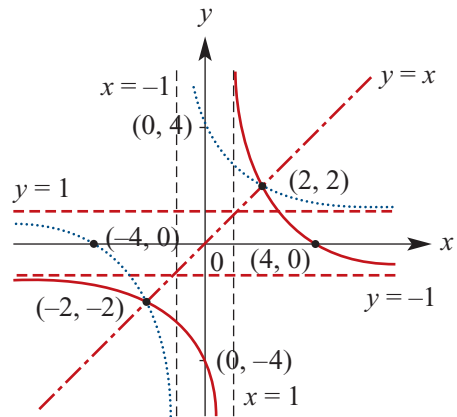
$$f^{-1}: R \setminus \{1\} \rightarrow R, f^{-1}(x) = \frac{3}{x-1} - 1$$

Note: The graph of f^{-1} is obtained by reflecting the graph of f in the line $y = x$. The two graphs meet where

$$\frac{3}{x+1} + 1 = x, x \neq -1,$$

i.e. where $x = \pm 2$

The two graphs meet at the points with coordinates $(2, 2)$ and $(-2, -2)$.



Many-to-one functions

$$f(x) = \frac{A}{(x+b)^n} + B, \quad \text{where } n \text{ is a positive even integer, and}$$

$$g(x) = A(x+b)^{\frac{p}{q}} + B \quad \text{where } p \text{ and } q \text{ are integers, } p \text{ even, } q \text{ odd}$$

and where the highest common factor of $p, q = 1$

are many-to-one functions, and therefore have inverse relations that are not functions.

Simple examples of such functions (other than polynomials which will be explored in Chapter 4) are:

$$f(x) = \frac{1}{x^2}, g(x) = \frac{1}{x^4}, h(x) = \frac{1}{x^6}, \dots$$

$$f(x) = x^{\frac{2}{3}}, g(x) = x^{\frac{2}{5}}, h(x) = x^{\frac{4}{3}}, \dots$$

$$f(x) = x^{-\frac{2}{3}}, g(x) = x^{-\frac{4}{5}}, h(x) = x^{-\frac{6}{5}}, \dots$$

Inverse relations will be discussed in Chapter 7.

Exercise 3J

- 1 Find the inverse function of each of the following, and sketch the graph of the inverse function:

a $f: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}, f(x) = \sqrt{x} + 2$ **b** $f: \mathbb{R} \setminus \{3\} \rightarrow \mathbb{R}, f(x) = \frac{1}{x-3}$

c $f: [2, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x-2} + 4$ **d** $f: \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R}, f(x) = \frac{3}{x-2} + 1$

e $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}, f(x) = \frac{5}{x-1} - 1$ **f** $f: (-\infty, 2] \rightarrow \mathbb{R}, f(x) = \sqrt{2-x} + 1$

- 2 For each of the following functions, find the inverse function and state its domain:

a $g(x) = \frac{3}{x}$

b $g(x) = \sqrt[3]{x+2} - 4$

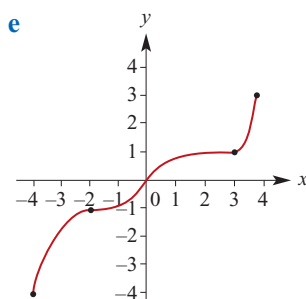
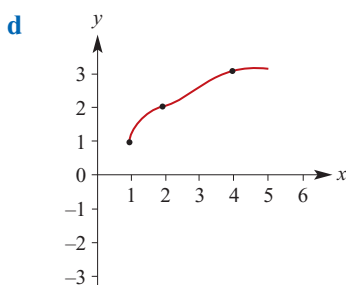
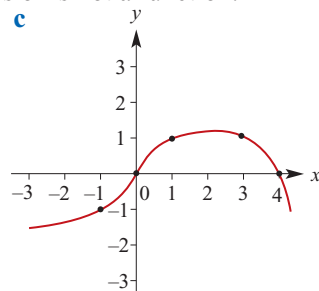
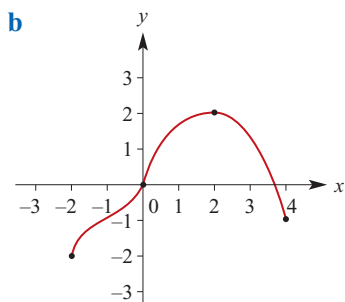
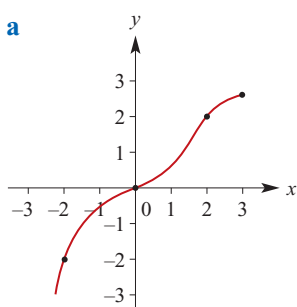
c $h(x) = 2 - \sqrt{x}$

d $f(x) = \frac{3}{x} + 1$

e $h(x) = 5 - \frac{2}{(x-6)^3}$

f $g(x) = \frac{1}{(x-1)^{\frac{3}{4}}} + 2$

- 3 For each of the following, copy the graph onto a grid and sketch the graph of the inverse on the same set of axes. In each case state whether the inverse is or is not a function.



4 Find the rule for the inverse of the following functions:

a $f: R \setminus \{1\} \rightarrow R, f(x) = \frac{x+1}{x-1}$ **b** $f: [2, \infty) \rightarrow R, f(x) = \sqrt{x-2}$

c $f: R \setminus \left\{ \frac{2}{3} \right\} \rightarrow R, f(x) = \frac{2x+3}{3x-2}$

5 Let $f: S \rightarrow R$ be given by $f(x) = \frac{x+3}{2x-1}$ where $S = R \setminus \left\{ \frac{1}{2} \right\}$

a Show that $f \circ f$ is defined.

b Find $f \circ f(x)$ and sketch the graph of $f \circ f$.

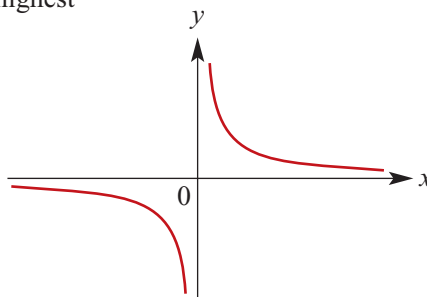
c Write down the inverse of f .



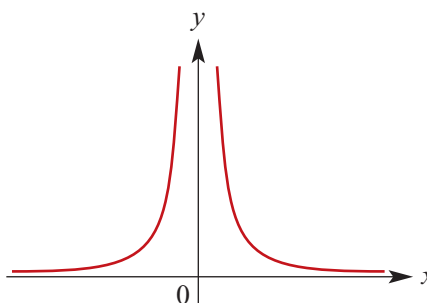
Chapter summary

- Functions of the form $f(x) = \frac{1}{x^n}$, where n is a positive odd integer, and $f(x) = x^{-\frac{p}{q}}$, where p and q are both positive, odd integers with highest common factor 1, have domain $R \setminus \{0\}$ and range $R \setminus \{0\}$. They are one-to-one and odd functions with asymptotes at $x = 0$ and $y = 0$. The points $(-1, -1)$ and $(1, 1)$ lie on the graph. Examples include $f(x) = \frac{1}{x}$, $g(x) = \frac{1}{x^3}$, $h(x) = \frac{1}{x^5}$ and

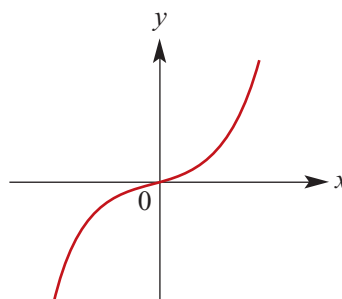
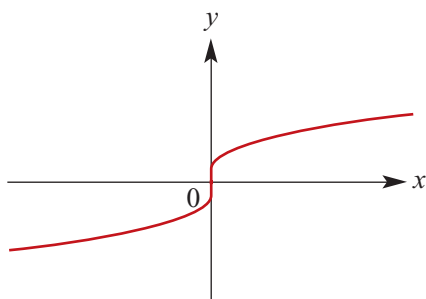
$$r(x) = \frac{1}{x^{\frac{1}{7}}}$$



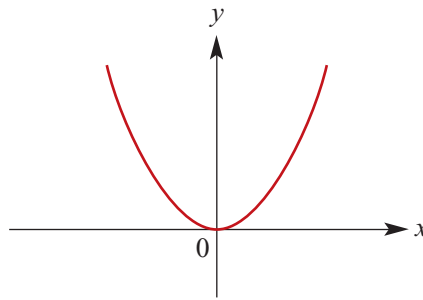
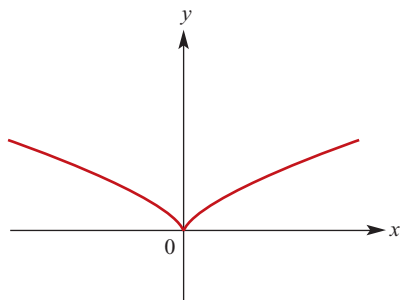
- Functions of the form $f(x) = \frac{1}{x^n}$, where n is a positive even integer, and $f(x) = x^{-\frac{p}{q}}$, where p is a positive even integer and q is a positive odd integer with highest common factor 1, have domain $R \setminus \{0\}$ and range R^+ . They are many-to-one and even functions with asymptotes at $x = 0$ and $y = 0$. The points $(-1, 1)$ and $(1, 1)$ lie on the graph. Examples include $f(x) = \frac{1}{x^2}$, $g(x) = \frac{1}{x^4}$, $h(x) = \frac{1}{x^{\frac{2}{3}}}$ and $r(x) = \frac{1}{x^{\frac{6}{5}}}$



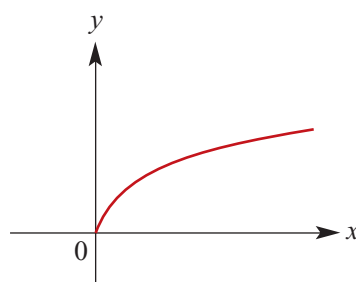
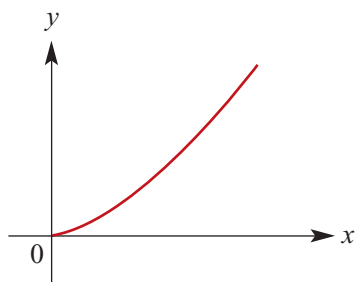
- Functions of the form $f(x) = x^{\frac{p}{q}}$, where p and q are both positive odd integers with highest common factor 1, have domain R and range R . They are one-to-one and odd functions. The points $(-1, -1)$, $(0, 0)$ and $(1, 1)$ lie on the graph. Examples include $f(x) = x^3$, $g(x) = x^{\frac{1}{3}}$, $h(x) = x^{\frac{3}{7}}$ and $r(x) = x^{\frac{5}{3}}$



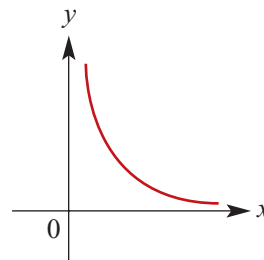
- Functions of the form $f(x) = x^{\frac{p}{q}}$, where p is a positive even integer and q is a positive odd integer with highest common factor 1, have domain R and range $[0, \infty)$. They are many-to-one and even functions. The points $(-1, 1)$ and $(1, 1)$ lie on the graph. Examples include $f(x) = x^2$, $g(x) = x^{\frac{2}{3}}$ and $h(x) = x^{\frac{4}{3}}$



- Functions of the form $f(x) = x^{\frac{p}{q}}$, where p is a positive odd integer and q is a positive even integer with highest common factor 1, have domain $[0, \infty)$ and range $[0, \infty)$ and are one-to-one. The point $(1, 1)$ lies on the graph. Examples include $f(x) = x^{\frac{1}{2}}$, $g(x) = x^{\frac{1}{4}}$, $h(x) = x^{\frac{3}{2}}$ and $r(x) = x^{\frac{3}{4}}$



- Functions of the form $f(x) = x^{-\frac{p}{q}}$, where p is a positive odd integer and q is a positive even integer with highest common factor 1, have domain R^+ and range R^+ and asymptotes at $x = 0$ and $y = 0$. They are one-to-one functions. The point $(1, 1)$ lies on the graph. Examples include $f(x) = \frac{1}{x^{\frac{1}{2}}}$, $g(x) = \frac{1}{x^{\frac{1}{4}}}$, $h(x) = \frac{1}{x^{\frac{3}{2}}}$ and $r(x) = \frac{1}{x^{\frac{5}{4}}}$



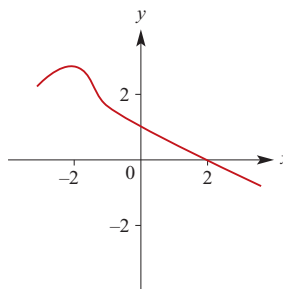
- The **dilation** of factor a ($a > 0$) from the x -axis is the transformation which maps $(x, y) \rightarrow (x, ay)$
- To find the equation of the image of the curve with equation $y = f(x)$ under the **dilation** $(x, y) \rightarrow (x, ay)$, replace y by $\frac{y}{a}$ in $y = f(x)$, i.e. the image of the graph of $y = f(x)$ under the dilation $(x, y) \rightarrow (x, ay)$ is the graph of $\frac{y}{a} = f(x)$ or $y = af(x)$
- The **dilation** of factor b ($b > 0$) from the y -axis is the transformation which maps $(x, y) \rightarrow (bx, y)$
- To find the equation of the image of the curve with equation $y = f(x)$ under the **dilation** $(x, y) \rightarrow (bx, y)$, replace x by $\frac{x}{b}$ in $y = f(x)$, i.e. the image of the graph of $y = f(x)$ under the dilation $(x, y) \rightarrow (bx, y)$ is the graph of $y = f\left(\frac{x}{b}\right)$

- To find the equation of the image of the curve with equation $y = f(x)$ under the **reflection** in the x -axis, $(x, y) \rightarrow (x, -y)$, replace y by $-y$ in $y = f(x)$, i.e. the image of the graph of $y = f(x)$ under the **reflection** $(x, y) \rightarrow (x, -y)$ is the graph of $y = -f(x)$
- To find the equation of the image of the curve with equation $y = f(x)$ under the **reflection** in the y -axis, $(x, y) \rightarrow (-x, y)$, replace x by $-x$ in $y = f(x)$, i.e. the image of the graph of $y = f(x)$ under the **reflection** $(x, y) \rightarrow (-x, y)$ is the graph of $y = f(-x)$
- The **translation** of h units ($h > 0$) in the positive direction of the x -axis and k units ($k > 0$) in the positive direction of the y -axis is the transformation that maps $(x, y) \rightarrow (x + h, y + k)$
- To find the equation of the curve with the equation $y = f(x)$ under the **translation** $(x, y) \rightarrow (x + h, y + k)$, replace x by $x - h$ and y by $y - k$ in $y = f(x)$, i.e. the image of the graph of $y = f(x)$ under the translation $(x, y) \rightarrow (x + h, y + k)$ is the graph of $y - k = f(x - h)$ or $y = f(x - h) + k$
- To find the equation of the **inverse relation** of $y = f(x)$, under the **reflection** $(x, y) \rightarrow (y, x)$, replace x with y and y with x , i.e. the image of the graph with equation $y = f(x)$ under the reflection $(x, y) \rightarrow (y, x)$ is the graph with equation $x = f(y)$
- A function which is one-to-one has an inverse function. A function which is many-to-one has an inverse relation that is not a function.

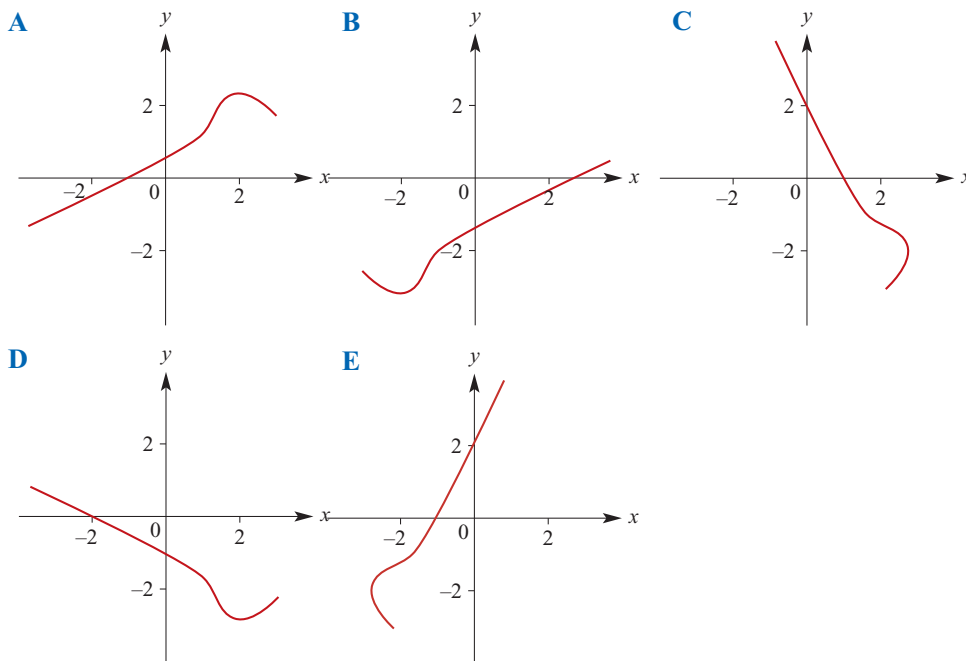
Transformation	Matrix
Reflection in the x -axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection in the y -axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Dilation by factor k from the y -axis	$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
Dilation of factor k from the x -axis	$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
Reflection in the line $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Multiple-choice questions

- 1 The graph of the function with rule $y = f(x)$ is shown below.



Which one of the following is most likely to be the graph of the inverse function?



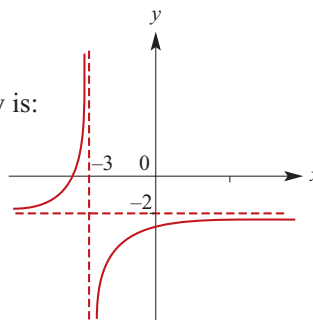
- 2 The graph of the function with rule $y = |x|$ is reflected in the x -axis and then translated 4 units in the negative direction of the x -axis and 3 units in the negative direction of the y -axis. The rule for the new function is:

- A $y = |-x + 4| - 3$ B $y = -|x - 4| + 3$ C $y = -|x - 3| + 4$
 D $y = |-x - 4| + 3$ E $y = -|x + 4| - 3$

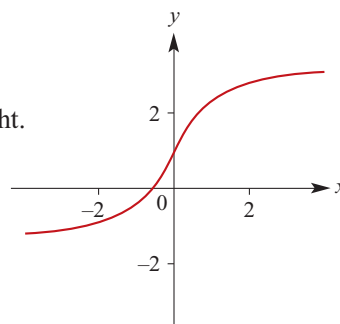
- 3 The graph of the function with rule $y = \frac{a}{x+b} + c$ is shown on the right.

A possible set of values for a , b and c respectively is:

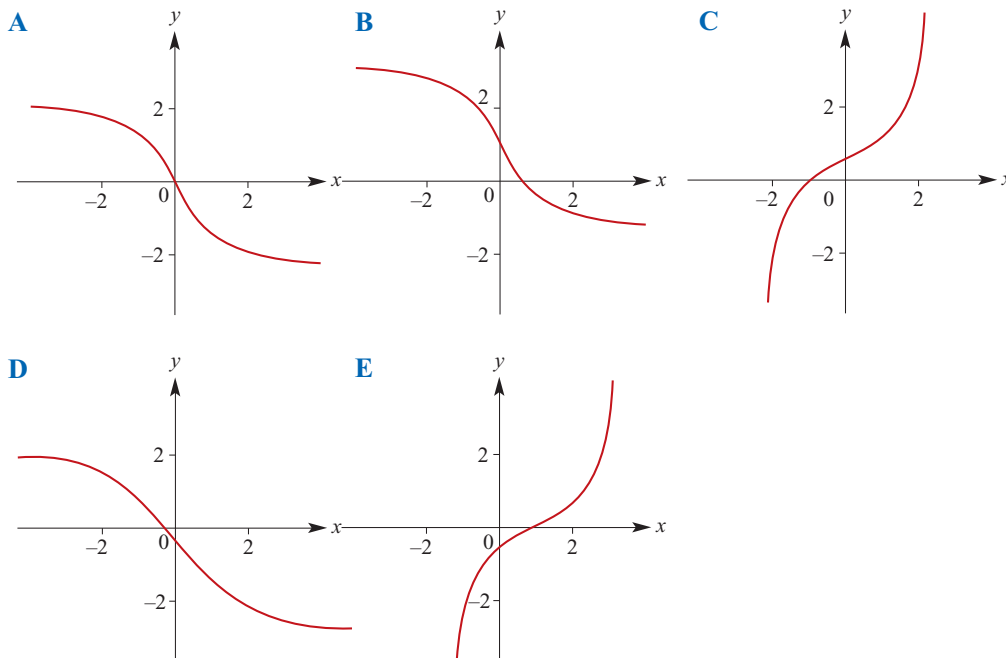
- A $-1, 3, 2$
 B $1, 2, -3$
 C $-1, -3, -2$
 D $-1, 3, -2$
 E $1, 2, -3$



- 4 The graph of the function f is obtained from the graph of the function with equation $y = x^{\frac{1}{3}}$ by a reflection in the y -axis followed by a dilation of 5 units from the x -axis. The rule for f is:
- A** $f(x) = -5x^{\frac{1}{3}}$ **B** $f(x) = \frac{1}{5}(-x)^{\frac{1}{3}}$ **C** $f(x) = (-5x)^{\frac{1}{3}}$
- D** $f(x) = -\frac{1}{5}x^{\frac{1}{3}}$ **E** $f(x) = -5(-x)^{\frac{1}{3}}$
- 5 A function with rule $f(x) = \frac{1}{x^4}$ can be defined on different domains. Which one of the following does not give the correct range for the given domain?
- A** $\text{dom } f = [-1, -0.5], \text{ran } f = [1, 16]$
- B** $\text{dom } f = [-0.5, 0) \cup (0, 0.5], \text{ran } f = [16, \infty)$
- C** $\text{dom } f = (-0.5, 0.5) \setminus \{0\}, \text{ran } f = (16, \infty)$
- D** $\text{dom } f = [-0.5, 1] \setminus \{0\}, \text{ran } f = [1, 16]$
- E** $\text{dom } f = [0.5, 1), \text{ran } f = (1, 16]$
- 6 The function with rule $y = f(x)$ is shown on the right.



Which one of the following could be the graph of the function with rule $y = f(-x)$?



- 7 Let $g(x) = \frac{3}{(x+1)^3} - 2$. The equations of the asymptotes of the inverse function g^{-1} are:
- A** $x = -2, y = 1$ **B** $x = -2, y = -1$ **C** $x = 1, y = -2$
- D** $x = -1, y = -2$ **E** $x = 2, y = -1$

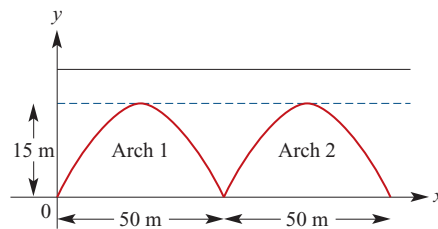
- 8 The equations of the vertical and horizontal asymptotes of the graph with equation $\frac{-2}{(x+3)^4} - 5$ are:
A $x = 3, y = -5$ **B** $x = -5, y = -3$ **C** $x = -3, y = -5$
D $x = -2, y = -5$ **E** $x = -3, y = 5$
- 9 The function $g: R \setminus \{3\} \rightarrow R$, where $g(x) = \frac{1}{x-3} + 2$ has an inverse g^{-1} . The rule and domain of g^{-1} are:
A $g^{-1}(x) = \frac{1}{x-2} + 3, \text{ dom } g^{-1} = R \setminus \{2\}$
B $g^{-1}(x) = \frac{1}{x-2} + 3, \text{ dom } g^{-1} = R \setminus \{3\}$
C $g^{-1}(x) = \frac{1}{x+2} - 3, \text{ dom } g^{-1} = R \setminus \{2\}$
D $g^{-1}(x) = \frac{-1}{x+2} - 3, \text{ dom } g^{-1} = R \setminus \{3\}$
E $g^{-1}(x) = \frac{1}{x-2} + 3, \text{ dom } g^{-1} = R \setminus \{-3\}$
- 10 Which one of the following functions does not have an inverse function?
A $f: [0, \infty) \rightarrow R, f(x) = |x - 2|$ **B** $f: R \rightarrow R, f(x) = x^3$
C $f: [-3, 3] \rightarrow R, f(x) = \sqrt{9 - x}$ **D** $f: R \rightarrow R, f(x) = x^{\frac{1}{3}} + 4$
E $f: R \rightarrow R, f(x) = 3x + 7$

Short-answer questions (technology-free)

- 1 State the maximal domain and range of each of the following:
a $f(x) = \frac{1}{x} + 2$ **b** $f(x) = 3 - 2\sqrt{3x - 2}$ **c** $f(x) = \frac{4}{(x-2)^2} + 3$
d $h(x) = 4 - \frac{3}{x-2}$ **e** $f(x) = \sqrt{x-2} - 5$
- 2 Sketch the graphs of each of the following. Label any asymptotes and axes intercepts. State the range of each:
a $f: R \setminus \{0\} \rightarrow R, f(x) = \frac{1}{x} - 3$ **b** $h: (2, \infty) \rightarrow R, f(x) = \frac{1}{x-2}$
c $f: R \setminus \{1\} \rightarrow R, f(x) = \frac{2}{x-1} - 3$ **d** $h: (2, \infty), f(x) = \frac{-3}{2-x} + 4$
e $f: R \setminus \{1\}, h(x) = 1 - \frac{1}{x-1}$
- 3 Sketch the graphs of each of the following:
a $f(x) = 2\sqrt{x-3} + 1$ **b** $g(x) = \frac{3}{(x-2)^2} - 1$ **c** $h(x) = \frac{-3}{(x-2)^2} - 1$
- 4 The points with coordinates (1, 3) and (3, 7) lie on the curve with equation of the form $y = \frac{a}{x} + b$. Find the values of a and b .
- 5 Find the inverse of the function with the rule $f(x) = \sqrt{x-2} + 4$ and sketch both functions on the one set of axes.
- 6 Find the inverse of the function with the rule $f(x) = \frac{x-2}{x+1}$

Extended-response questions

- 1 Consider the function $f: D \rightarrow R$, for which the rule is $f(x) = \frac{24}{x+2} - 6$, where D is the maximal domain for f .
 - a Find D .
 - b Describe a set of transformations which, when applied to the graph of $y = \frac{1}{x}$, produce the graph of $y = f(x)$. Specify the order in which these transformations are to be applied.
 - c Find the coordinates of the points where the graph of f cuts the axes.
 - d Let $g: (-2, \infty) \rightarrow R$, $g(x) = f(x)$.
 - i Find the rule for g^{-1} , the inverse of g .
 - ii Write down the domain of g^{-1} .
 - iii Find the values of x for which $g(x) = x$ and hence the values of x for which $g(x) = g^{-1}(x)$.
 - iv Sketch the graphs of $y = g(x)$ and $y = g^{-1}(x)$ on the one set of axes.
- 2 Consider the function $f: D \rightarrow R$, for which the rule is $f(x) = 4 - 2\sqrt{2x+6}$, where D is the maximal domain for f .
 - a Find D .
 - b Describe a set of transformations which, when applied to the graph of $y = \sqrt{x}$, produce the graph of $y = f(x)$. Specify the order in which these transformations are to be applied.
 - c Find the coordinates of the points where the graph of f cuts the axes.
 - d Find the rule for f^{-1} , the inverse of f . e Find the domain for f^{-1} .
 - f Find the value(s) of x for which $f(x) = x$ and hence the values of x for which $f(x) = f^{-1}(x)$.
 - g Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the one set of axes.
- 3
 - a
 - i Find the dilation from the x -axis which takes $y = x^2$ to the parabola with its vertex at the origin and which passes through the point $(25, 15)$.
 - ii State the rule which reflects this dilated parabola in the x -axis.
 - iii State the rule which takes the reflected parabola of ii to a parabola with x -intercepts $(0, 0)$ and $(50, 0)$ and vertex $(25, 15)$.
 - iv State the rule which takes the curve of $y = x^2$ to the parabola defined in iii.
 - b The plans for the entrance of a new building involve twin parabolic arches as shown in the diagram.
 - i From the results of a, give the equation for the curve of arch 1.
 - ii Find the translation which maps the curve of arch 1 to the curve of arch 2.
 - iii Find the equation of the curve of arch 2.





- c** The architect wishes to have flexibility in her planning and so wants to develop an algorithm for determining the equations of the curves given arch width m metres and height n metres.
- i** Find the rule for the transformation which takes the graph of $y = x^2$ to the current arch 1 with these dimensions.
 - ii** Find the equation for the curve of arch 1.
 - iii** Find the equation for the curve of arch 2.
- 4** Consider the function $g: D \rightarrow R$, for which the rule is $g(x) = \frac{3}{(3x-4)^2} + 6$, where D is the maximal domain of g .
- a** Find D .
 - b** Find the smallest value of a such that $f: (a, \infty) \rightarrow R$, $f(x) = g(x)$ is a one-to-one function.
 - c** Find the inverse function for f .
 - d** Find the value of x for which $f(x) = f^{-1}(x)$
 - e** On the one set of axes sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$
- 5 a** Sketch the curve with equation $f(x) = \frac{50}{20-x}$, $x \neq 20$
- b** If $g(x) = \frac{50x}{20-x}$:
- i** Show that $g(x) = \frac{1000}{20-x} - 50$
 - ii** Sketch the graph of $y = g(x)$
 - iii** Show that $g(x) = 20f(x) - 50$
- c** Find the rule for the function g^{-1} .
- 6** When the transformation with rule $(x, y) \rightarrow (y, x)$ (a reflection in the line $y = x$) is applied to the graph of the one-to-one function, f , the resulting image has rule $y = f^{-1}(x)$, i.e. the graph of the inverse function is obtained.
- a** For the graph of $y = f(x)$, find the rule for the image of f , in terms of $f^{-1}(x)$, for each of the following sequences of transformations:
- i**
 - a translation of 3 units in the positive direction of x -axis
 - a translation of 5 units in the positive direction of the y -axis
 - a reflection in the line $y = x$
 - ii**
 - a reflection in the line $y = x$
 - a translation of 3 units in the positive direction of x -axis
 - a translation of 5 units in the positive direction of the y -axis
 - iii**
 - a dilation of factor 3 from the x -axis
 - a dilation of factor 5 from the y -axis
 - a reflection in the line $y = x$

- iv ■ a reflection in the line $y = x$
 - a dilation of factor 5 from the y -axis
 - a dilation of factor 3 from the x -axis
- b Find the image of the graph of $y = f(x)$, in terms of $f^{-1}(x)$, under the transformation with rule $(x, y) \rightarrow (ay + b, cx + d)$, where a, b, c and d are positive constants, and describe this transformation in words.

Polynomial functions

Objectives

- To be able to use the technique of equating coefficients.
- To introduce the **functions of the form** $f(x) = a(x + h)^n + k$ and to sketch graphs of this form through the use of **transformations**.
- To **divide polynomials**.
- To use the **factor theorem** to solve cubic equations and quartic equations.
- To use the **remainder theorem**.
- To draw and use **sign diagrams**.
- To **find equations** for given graphs of polynomials.
- To apply **polynomial functions** to problem solving.

4.1 Polynomials

In an earlier chapter, linear functions were discussed. This family of functions is a member of a larger family of polynomial functions.

A function with rule $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, where $a_0, a_1, a_2, a_3, \dots, a_n$ are real constants and n is a positive integer, is called a **polynomial** in x over the real numbers. The numbers $a_0, a_1, a_2, a_3, \dots, a_n$ are called the **coefficients** of the polynomial. Assuming $a_n \neq 0$, the term a_nx^n is called the **leading term**. The integer n of the leading term is the **degree** of the polynomial.

For example:

- $f(x) = a_0 + a_1x$ is a degree one polynomial (a linear function).
- $f(x) = a_0 + a_1x + a_2x^2$ is a degree two polynomial (a quadratic function).
- $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ is a degree three polynomial (a cubic function).
- $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ is a degree four polynomial (a quartic function).

The polynomials above are written with ascending powers of x . They can also be written with descending powers of x , for example:

$$f(x) = 3x + 1, f(x) = x^2 + 2x + 3, f(x) = x^3 + 4x^2 + 3x + 1$$

Polynomials are often written in factorised form, e.g. $(3x + 2)^2, 4(x - 1)^3 + 2$.

Although it is fairly simple to expand such polynomials when the degree is small, say two or three, the binomial theorem discussed in Appendix A facilitates the expansion of polynomials of larger degree.

For example:

$$\begin{aligned}(2x + 3)^4 &= \binom{4}{0} (2x)^4 (3)^0 + \binom{4}{1} (2x)^3 (3)^1 + \binom{4}{2} (2x)^2 (3)^2 \\ &\quad + \binom{4}{3} (2x)^1 (3)^3 + \binom{4}{4} (2x)^0 (3)^4 \\ &= 16x^4 + 96x^3 + 216x^2 + 216x + 81\end{aligned}$$

Equating coefficients

If $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and $Q(x) = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m$ are equal, i.e. if $P(x) = Q(x)$ for all x , then the degree of $P(x) = \text{degree of } Q(x)$ and $a_0 = b_0, a_1 = b_1, a_2 = b_2, \dots$, etc.

Example 1

If $x^2 + 6x + 4 = a(x + 3)^2 + b$ for all $x \in R$, find the values of a and b .

Solution

Expanding the right-hand side of the equation gives:

$$\begin{aligned}a(x + 3)^2 + b &= a(x^2 + 6x + 9) + b \\ &= ax^2 + 6ax + 9a + b\end{aligned}$$

If $x^2 + 6x + 4 = ax^2 + 6ax + 9a + b$ for all $x \in R$, then by equating coefficients:

(coefficient of x^2)	$1 = a$
(coefficient of x)	$6 = 6a$
(coefficient of x^0 , the constant term of the polynomial)	$4 = 9a + b$

Hence $a = 1$ and $b = -5$.

Example 2

- a** If $x^3 + 3x^2 + 3x + 8 = a(x + 1)^3 + b$ for all $x \in R$, find the values of a and b .
b Show that $x^3 + 6x^2 + 6x + 8$ cannot be written in the form $a(x + c)^3 + b$ for real numbers a, b and c .

Solution

- a** The expansion of the right-hand side of the equation gives:

$$\begin{aligned}a(x + 1)^3 + b &= a(x^3 + 3x^2 + 3x + 1) + b \\ &= ax^3 + 3ax^2 + 3ax + a + b\end{aligned}$$

If $x^3 + 3x^2 + 3x + 8 = ax^3 + 3ax^2 + 3ax + a + b$ for all $x \in R$, equating coefficients gives:

$$\begin{array}{ll} \text{(coefficient of } x^3) & 1 = a \\ \text{(coefficient of } x^2) & 3 = 3a \\ \text{(coefficient of } x) & 3 = 3a \\ \text{(coefficient of } x^0, \text{ the constant term of the polynomial)} & 8 = a + b \end{array}$$

Hence $a = 1$ and $b = 7$.

b The expansion of the right-hand side of the equation gives:

$$\begin{aligned} a(x + c)^3 + b &= a(x^3 + 3cx^2 + 3c^2x + c^3) + b \\ &= ax^3 + 3cax^2 + 3c^2ax + c^3a + b \end{aligned}$$

If $x^3 + 6x^2 + 6x + 8 = ax^3 + 3cax^2 + 3c^2ax + c^3a + b$ for all $x \in R$:

$$\begin{array}{llll} \text{(coefficient of } x^3) & 1 = a & (1) \\ \text{(coefficient of } x^2) & 6 = 3ca & (2) \\ \text{(coefficient of } x) & 6 = 3c^2a & (3) \\ \text{(coefficient of } x^0, \text{ the constant term of the polynomial)} & 8 = c^3a + b & (4) \end{array}$$

From (1), $a = 1$

From (2), $c = 2$

For (3), the right-hand side $3c^2a$ equals 12, which is a contradiction.

Division of polynomials

The division of polynomials was introduced in *Essential Mathematical Methods 1 & 2 CAS*.

The general result for polynomial division is:

For non-zero polynomials, $P(x)$ and $D(x)$, if $P(x)$ (the dividend) is divided by $D(x)$ (the divisor), then there are unique polynomials, $Q(x)$ (the quotient) and $R(x)$ (the remainder), such that

$$P(x) = D(x)Q(x) + R(x)$$

Either the degree of $R(x) < D(x)$, or $R(x) = 0$

When $R(x) = 0$, then $D(x)$ is called a divisor of $P(x)$ and $P(x) = D(x)Q(x)$

The following example illustrates the process of dividing.

Example 3

Divide $3x^4 - 9x^2 + 27x - 8$ by $x - 2$.

Solution

$$\begin{array}{r}
 3x^3 + 6x^2 + 3x + 33 \\
 x - 2 \overline{) 3x^4 + 0x^3 - 9x^2 + 27x - 8} \\
 \underline{3x^4 - 6x^3} \\
 6x^3 - 9x^2 \\
 \underline{6x^3 - 12x^2} \\
 3x^2 + 27x \\
 \underline{3x^2 - 6x} \\
 33x - 8 \\
 \underline{33x - 66} \\
 58
 \end{array}$$

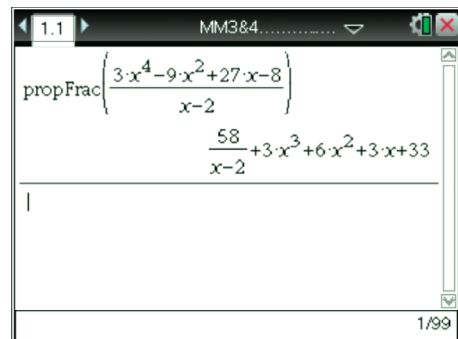
Thus $3x^4 - 9x^2 + 27x - 8 = (x - 2)(3x^3 + 6x^2 + 3x + 33) + 58$ or, equivalently,

$$\frac{3x^4 - 9x^2 + 27x - 8}{x - 2} = 3x^3 + 6x^2 + 3x + 33 + \frac{58}{x - 2}$$

In this example $3x^4 - 9x^2 + 27x - 8$ is the dividend, $x - 2$ is the divisor and the remainder is 58.

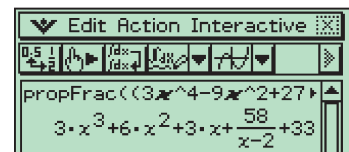
Using the TI-Nspire

Use **propFrac** from (menu) > **Algebra** > **Fraction Tools** > **Proper Fraction** as shown.



Using the Casio ClassPad

Enter and highlight $(3x^4 - 9x^2 + 27x - 8)/(x - 2)$ and tap **Interactive**–**Transformation**–**propFrac**.



The remainder theorem and factor theorem

The following two results are recalled from *Essential Mathematical Methods 1 & 2 CAS*.

The remainder theorem

If a polynomial $P(x)$ is divided by $ax + b$ the remainder is $P\left(\frac{-b}{a}\right)$.

This is easily proved by observing if

$$P(x) = D(x)(ax + b) + R(x) \text{ and } x = \frac{-b}{a}$$

$$P\left(\frac{-b}{a}\right) = D\left(\frac{-b}{a}\right)\left(a \times \frac{-b}{a} + b\right) + R\left(\frac{-b}{a}\right)$$

and thus

$$P\left(\frac{-b}{a}\right) = R\left(\frac{-b}{a}\right)$$

The factor theorem

An immediate consequence of the remainder theorem is the **factor theorem**.

$ax + b$ is a factor of the polynomial $P(x)$ if and only if $P\left(\frac{-b}{a}\right) = 0$.

Example 4

Find the remainder when $P(x) = 3x^3 + 2x^2 + x + 1$ is divided by $2x + 1$.

Solution

By the remainder theorem the remainder is:

$$\begin{aligned} P\left(-\frac{1}{2}\right) &= 3\left(-\frac{1}{2}\right)^3 + 2\left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right) + 1 \\ &= -\frac{3}{8} + \frac{2}{4} - \frac{1}{2} + 1 \\ &= -\frac{3}{8} + 1 = \frac{5}{8} \end{aligned}$$

Example 5

Given that $x + 1$ and $x - 2$ are factors of $6x^4 - x^3 + ax^2 - 6x + b$, find the values of a and b .

Solution

Let $P(x) = 6x^4 - x^3 + ax^2 - 6x + b$.

By the factor theorem, $P(-1) = 0$ and $P(2) = 0$.

Hence,

$$6 + 1 + a + 6 + b = 0 \quad (1)$$

$$\text{and } 96 - 8 + 4a - 12 + b = 0 \quad (2)$$

Rearranging gives:

$$a + b = -13 \quad (1')$$

$$4a + b = -76 \quad (2')$$

Subtract (1') from (2'):

$$3a = -63$$

Therefore $a = -21$ and, from (1'), $b = 8$.

Solving polynomial equations

The factor theorem may be used in the solution of equations.

Example 6

Factorise $P(x) = x^3 - 4x^2 - 11x + 30$ and hence solve the equation $x^3 - 4x^2 - 11x + 30 = 0$

Solution

$$P(1) = 1 - 4 - 11 + 30 \neq 0$$

$$P(-1) = -1 - 4 + 11 + 30 \neq 0$$

$$P(2) = 8 - 16 - 22 + 30 = 0$$

$\therefore x - 2$ is a factor.

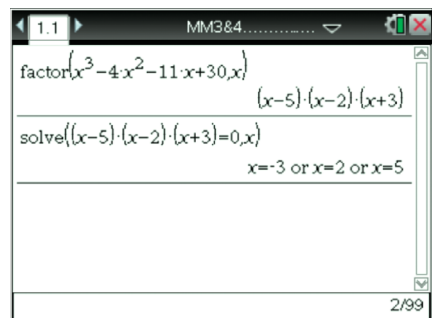
Dividing $x^3 - 4x^2 - 11x + 30$ by $x - 2$ reveals that:

$$\begin{aligned} P(x) &= (x - 2)(x^2 - 2x - 15) \\ &= (x - 2)(x - 5)(x + 3) \end{aligned}$$

$$\begin{aligned} \text{Therefore } x - 2 = 0 \quad \text{or} \quad x - 5 = 0 \quad \text{or} \quad x + 3 = 0 \\ \therefore x = 2 \quad \text{or} \quad x = 5 \quad \text{or} \quad \text{or } x = -3 \end{aligned}$$

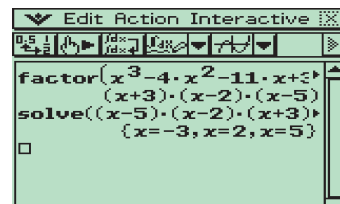
Using the TI-Nspire

Use **factor** ($\text{menu} > \text{Algebra} > \text{Factor}$) and **solve** ($\text{menu} > \text{Algebra} > \text{Solve}$) as shown.



Using the Casio ClassPad

Enter and highlight $x^3 - 4x^2 - 11x + 30$
then tap **Interactive–Transformation–factor**.



Copy and paste the answer to the next entry line and tap **Interactive–Equation/inequality–solve** and ensure the variable set is x .

Exercise 4A



- 1 Find the values of A and B such that $A(x + 3) + B(x + 2) = 4x + 9$ for all real numbers.
- 2 Find the values of A , B and C in each of the following if:
 - a $x^2 - 4x + 10 = A(x + B)^2 + C$ for all $x \in R$
 - b $4x^2 - 12x + 14 = A(x + B)^2 + C$ for all $x \in R$
 - c $x^3 - 9x^2 + 27x - 22 = A(x + B)^3 + C$ for all $x \in R$
- 3 For each of the following, divide the first term by the second:
 - a $2x^3 - 7x^2 + 15x - 3, x - 3$
 - b $5x^5 + 13x^4 - 2x^2 - 6, x + 1$
 - c $x^4 - 9x^3 + 25x^2 - 8x - 2, x^2 - 2$
- 4
 - a Find the remainder when $x^3 + 3x - 2$ is divided by $x + 2$.
 - b Find the value of a for which $(1 - 2a)x^2 + 5ax + (a - 1)(a - 8)$ is divisible by $(x - 2)$ but not by $(x - 1)$.
- 5 Given that $f(x) = 6x^3 + 5x^2 - 17x - 6$:
 - a Find the remainder when $f(x)$ is divided by $x - 2$.
 - b Find the remainder when $f(x)$ is divided by $x + 2$.
 - c Factorise $f(x)$ completely.
- 6
 - a Prove that the expression $x^3 + (k - 1)x^2 + (k - 9)x - 7$ is divisible by $x + 1$ for all values of k .
 - b Find the value of k for which the expression has a remainder of 12 when divided by $x - 2$.
- 7 The function $f(x) = 2x^3 + ax^2 - bx + 3$ has a factor $(x + 3)$. When $f(x)$ is divided by $(x - 2)$, the remainder is 15.
 - a Calculate the values of a and b .
 - b Find the other two factors of $f(x)$.

- 8 The expression $4x^3 + ax^2 - 5x + b$ leaves remainders of -8 and 10 when divided by $(2x - 3)$ and $(x - 3)$ respectively. Calculate the values of a and b .
- 9 Find the remainder when $(x + 1)^4$ is divided by $x - 2$
- 10 Let $P(x) = x^5 - 3x^4 + 2x^3 - 2x^2 + 3x + 1$
- Show that neither $(x - 1)$ nor $(x + 1)$ is a factor of $P(x)$
 - Given that $P(x)$ can be written in the form $(x^2 - 1)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are constants, hence or otherwise, find the remainder when $P(x)$ is divided by $x^2 - 1$
- 11
 - Show that both $(x - \sqrt{3})$ and $(x + \sqrt{3})$ are factors of $x^4 + x^3 - x^2 - 3x - 6$
 - Hence write down one quadratic factor of $x^4 + x^3 - x^2 - 3x - 6$, and find a second quadratic factor.
- 12 Solve each of the following equations for x :
- $(2 - x)(x + 4)(x - 2)(x - 3) = 0$
 - $x^3(2 - x) = 0$
 - $(2x - 1)^3(2 - x) = 0$
 - $(x + 2)^3(x - 2)^2 = 0$
 - $x^4 - 4x^2 = 0$
 - $x^4 - 9x^2 = 0$
 - $12x^4 + 11x^3 - 26x^2 + x + 2 = 0$
 - $x^4 + 2x^3 - 3x^2 - 4x + 4 = 0$
 - $6x^4 - 5x^3 - 20x^2 + 25x - 6 = 0$
- 13 Find the x -axis intercepts and the y -axis intercepts of the graphs of each of the following:
- $y = x^3 - x^2 - 2x$
 - $y = x^3 - 2x^2 - 5x + 6$
 - $y = x^3 - 4x^2 + x + 6$
 - $y = 2x^3 - 5x^2 + x + 2$
 - $y = x^3 + 2x^2 - x - 2$
 - $y = 3x^3 - 4x^2 - 13x - 6$
 - $y = 5x^3 + 12x^2 - 36x - 16$
 - $y = 6x^3 - 5x^2 - 2x + 1$
 - $y = 2x^3 - 3x^2 - 29x - 30$
- 14 The expressions $px^4 - 5x + q$ and $x^4 - 2x^3 - px^2 - qx - 8$ have a common factor $x - 2$. Find the values of p and q .
- 15 Find the remainder when $f(x) = x^4 - x^3 + 5x^2 + 4x - 36$ is divided by $x + 1$.
- 16 Factorise each of the following polynomials using a calculator to help find at least one linear factor:
- $x^3 - 11x^2 - 125x + 1287$
 - $x^3 - 9x^2 - 121x + 1089$
 - $2x^3 - 9x^2 - 242x + 1089$
 - $4x^3 - 367x + 1287$
- 17 Factorise each of the following:
- $x^4 - x^3 - 43x^2 + x + 42$
 - $x^4 + 4x^3 - 27x - 108$

- 18** Factorise each of the following polynomials, using a calculator to help find at least one linear factor:

a $2x^4 - 25x^3 + 57x^2 + 9x + 405$

b $x^4 + 13x^3 + 40x^2 + 81x + 405$

c $x^4 + 3x^3 - 4x^2 + 3x - 135$

d $x^4 + 4x^3 - 35x^2 - 78x + 360$

4.2 Quadratic functions

The following is a summary of material assumed to have been covered in *Essential Mathematical Methods 1 & 2 CAS*.

- The general expression of a quadratic function is $y = ax^2 + bx + c$, $x \in R$.

- To sketch the graph of a quadratic function (called a parabola) use the following:

- If $a > 0$, the function has a minimum value.
- If $a < 0$, the function has a maximum value.
- The value of c gives the y -axis intercept.
- The equation of the axis of symmetry is $x = -\frac{b}{2a}$
- The x -axis intercepts are determined by solving the equation $ax^2 + bx + c = 0$

- A quadratic equation may be solved by:

- factorising

e.g., $2x^2 + 5x - 12 = 0$

$$(2x - 3)(x + 4) = 0$$

$$\therefore x = \frac{3}{2} \text{ or } -4$$

- completing the square

e.g., $x^2 + 2x - 4 = 0$

Add and subtract $\left(\frac{b}{2}\right)^2$ to 'complete the square'.

$$x^2 + 2x + 1 - 1 - 4 = 0$$

$$\therefore (x + 1)^2 - 5 = 0$$

$$\therefore (x + 1)^2 = 5$$

$$\therefore x + 1 = \pm\sqrt{5}$$

$$\therefore x = -1 \pm \sqrt{5}$$

- using the general quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

e.g., $-3x^2 - 12x - 7 = 0$

$$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-7)}}{2(-3)}$$

$$= \frac{6 \pm \sqrt{15}}{-3}$$





- Using the discriminant of the quadratic function $f(x) = ax^2 + bx + c$
- If $b^2 - 4ac > 0$, the graph of the function has two x -axis intercepts.
 - If $b^2 - 4ac = 0$, the graph of the function touches the x -axis.
 - If $b^2 - 4ac < 0$, the graph of the function does not intersect the x -axis.

Example 7

Sketch the graph of $f(x) = -3x^2 - 12x - 7$ by using the quadratic formula to calculate the x -axis intercepts.

Solution

Since $c = -7$ the y -axis intercept is $(0, -7)$.

Find the turning point coordinates.

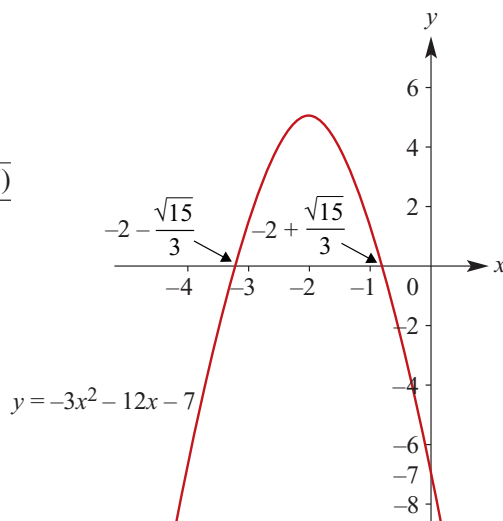
$$\begin{aligned}\text{Axis of symmetry, } x &= -\frac{b}{2a} \\ &= -\frac{(-12)}{2 \times -3} \\ &= -2\end{aligned}$$

$$\begin{aligned}\text{and } f(-2) &= -3(-2)^2 - 12(-2) - 7 \\ &= 5\end{aligned}$$

$\therefore$ turning point coordinates are $(-2, 5)$.

Calculate the x -axis intercepts.

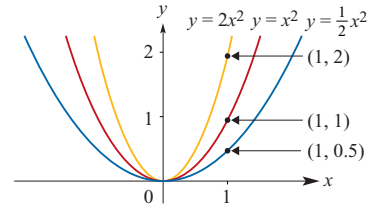
$$\begin{aligned}-3x^2 - 12x - 7 &= 0 \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-7)}}{2(-3)} \\ &= \frac{12 \pm \sqrt{60}}{-6} \\ &= \frac{12 \pm 2\sqrt{15}}{-6} \\ &= \frac{6 \pm \sqrt{15}}{-3} \\ &\approx \frac{6 \pm 3.87}{-3} \text{ (to 2nd decimal place)} \\ &= -3.29 \text{ or } -0.71\end{aligned}$$



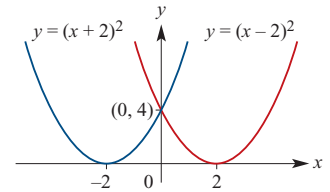
Use of 'completing the square' to sketch quadratics

For $a > 0$, the graph of the function $f(x) = ax^2$ is obtained from the graph of $f(x) = x^2$ by a dilation of factor a from the x -axis.

The graphs on the right are those of $y = x^2$, $y = 2x^2$ and $y = \frac{1}{2}x^2$, i.e. $a = 1, 2$ and $\frac{1}{2}$.



For $h > 0$, the graph of $f(x) = (x + h)^2$ is obtained from the graph of $f(x) = x^2$ by a translation of h units in the negative direction of the x -axis.



For $h < 0$, the graph of $f(x) = (x + h)^2$ is obtained from the graph of $f(x) = x^2$ by a translation of $-h$ units in the positive direction of the x -axis.

The graphs of $y = (x + 2)^2$ and $y = (x - 2)^2$ are shown:

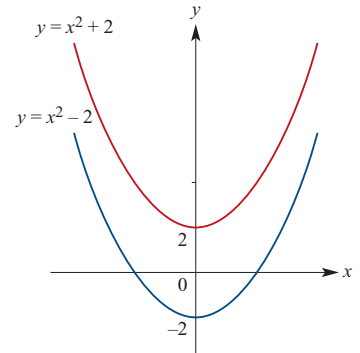
For $k > 0$, the graph of

$$f(x) = x^2 + k$$

is obtained from the graph of

$$f(x) = x^2$$

by a translation of k units in the positive direction of the y -axis.



For $k < 0$, the translation is in the negative direction of the y -axis.

For example, the graph of the function

$$f(x) = (x - 3)^2 + 2$$

is obtained by translating the graph of the function

$$f(x) = x^2$$

by 3 units in the positive direction of the x -axis and 2 units in the positive direction of the y -axis.

Note that the vertex is at $(3, 2)$.

The graph of the function

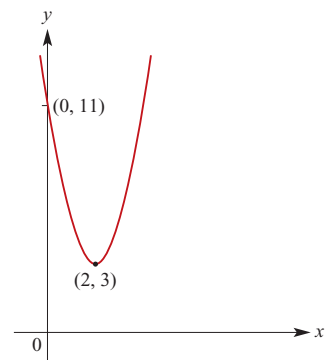
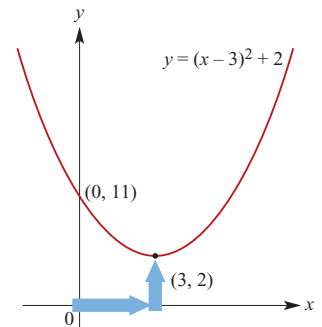
$$f(x) = 2(x - 2)^2 + 3$$

is obtained from the graph of

$$f(x) = x^2$$

by the following transformations:

- dilation of factor 2 from the x -axis
- translation of 2 units in the positive direction of the x -axis
- translation of 3 units in the positive direction of the y -axis.





By completing the square, any quadratic function can be written in the form:

$$y = a(x + h)^2 + k$$

Example 8

Solve $2x^2 - 4x - 5 = 0$ by expressing $2x^2 - 4x - 5$ in the form $y = a(x + h)^2 + k$

Use this to help you sketch the graph of $f(x) = 2x^2 - 4x - 5$

Solution

$$\begin{aligned} f(x) &= 2x^2 - 4x - 5 \\ &= 2\left(x^2 - 2x - \frac{5}{2}\right) \\ &= 2\left(x^2 - 2x + 1 - 1 - \frac{5}{2}\right) \text{ (adding and subtracting } \left(\frac{b}{2}\right)^2 \text{ to 'complete'} \\ &= 2\left[(x^2 - 2x + 1) - \frac{7}{2}\right] \text{ the square')} \\ &= 2\left[(x - 1)^2 - \frac{7}{2}\right] \\ &= 2(x - 1)^2 - 7 \end{aligned}$$

Therefore, the graph of $f(x) = 2x^2 - 4x - 5$ may be obtained from the graph of $y = x^2$ by the following transformations:

- dilation of factor 2 from the x -axis
- translation of 1 unit in the positive direction of the x -axis
- translation of 7 units in the negative direction of the y -axis.

The x -axis intercepts can also be determined by this process.

To solve $0 = 2x^2 - 4x - 5$

$$\therefore 0 = 2(x - 1)^2 - 7$$

$$\therefore (x - 1)^2 = \frac{7}{2}$$

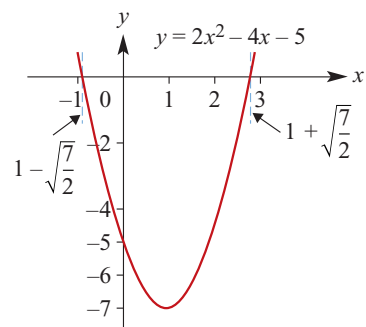
$$\therefore x - 1 = \pm\sqrt{\frac{7}{2}}$$

$$\therefore x = 1 \pm \sqrt{\frac{7}{2}}$$

$$\begin{aligned} \therefore x &= 1 + \sqrt{\frac{7}{2}} & \text{or} & & x &= 1 - \sqrt{\frac{7}{2}} \\ &\approx 2.87 & \text{or} & & &\approx -0.87 \end{aligned}$$

This information can now be used to sketch the graph.

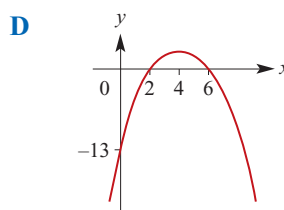
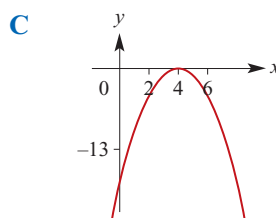
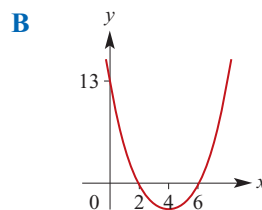
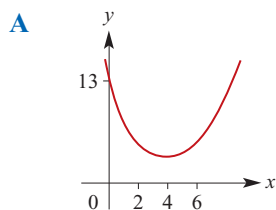
- Since $c = -5$ the y -axis intercept is $(0, -5)$.
- Turning point coordinates are $(1, -7)$.
- x -axis intercepts are $(2.87, 0)$ and $(-0.87, 0)$ to two decimal places.



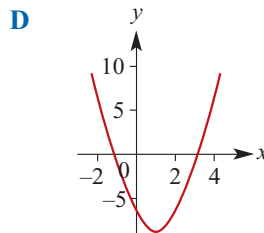
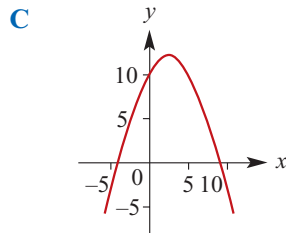
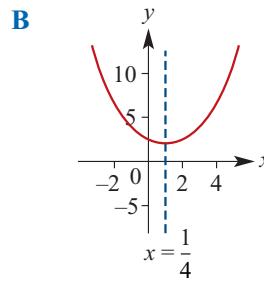
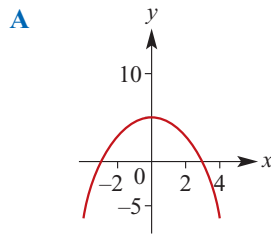
Exercise 4B



- 1 Without sketching the graphs of the following functions, determine whether they cross, touch or do not intersect the x -axis:
- a** $f(x) = x^2 - 5x + 2$ **b** $f(x) = -4x^2 + 2x - 1$ **c** $f(x) = x^2 - 6x + 9$
d $f(x) = 8 - 3x - 2x^2$ **e** $f(x) = 3x^2 + 2x + 5$ **f** $f(x) = -x^2 - x - 1$
- 2 Sketch the graphs of the following functions:
- a** $f(x) = 2(x - 1)^2$ **b** $f(x) = 2(x - 1)^2 - 2$ **c** $f(x) = -2(x - 1)^2$
d $f(x) = 4 - 2(x + 1)^2$ **e** $f(x) = 4 + 2\left(x + \frac{1}{2}\right)^2$ **f** $f(x) = 2(x + 1)^2 - 1$
g $f(x) = 3(x - 2)^2 - 4$ **h** $f(x) = (x + 1)^2 - 1$ **i** $f(x) = 5x^2 - 1$
j $f(x) = 2(x + 1)^2 - 4$
- 3 Express each of the following functions in the form $y = a(x + h)^2 + k$ and hence find the maximum or minimum value and the range in each case:
- a** $f(x) = x^2 + 3x - 2$ **b** $f(x) = x^2 - 6x + 8$ **c** $f(x) = 2x^2 + 8x - 6$
d $f(x) = 4x^2 + 8x - 7$ **e** $f(x) = 2x^2 - 5x$ **f** $f(x) = 7 - 2x - 3x^2$
g $f(x) = -2x^2 + 9x + 11$
- 4 Sketch the graphs of the following functions, clearly labelling the intercepts and turning points:
- a** $y = -x^2 + 2x$ **b** $y = x^2 - 6x + 8$ **c** $y = -2x^2 + 8x - 6$
d $y = -x^2 - 5x - 6$ **e** $f(x) = x^2 + 3x - 2$ **f** $f(x) = 2x^2 + 4x - 7$
g $f(x) = 5x^2 - 10x - 1$ **h** $f(x) = -2x^2 + 4x - 1$
i $y = 2.5x^2 + 3x + 0.3$ **j** $y = -0.6x^2 - 1.3x - 0.1$
- 5 **a** Which of the graphs shown could represent the graph of the equation $y = (x - 4)^2 - 3$?
b Which graph could represent $y = 3 - (x - 4)^2$?



- 6 Which of the curves shown could be defined by each of the following?
- a** $y = \frac{1}{3}(x + 4)(8 - x)$ **b** $y = x^2 - \frac{x}{2} + 1$
c $y = -10 + 2(x - 1)^2$ **d** $y = \frac{1}{2}(9 - x^2)$



- 7 For which values of m does the equation $mx^2 - 2mx + 3 = 0$ have:
- a** two solutions for x ? **b** one solution for x ?
- 8 Show that the equation $(k + 1)x^2 - 2x - k = 0$ has a solution for all values of k .
- 9 For which values of k does the equation $kx^2 - 2kx = 5$ have:
- a** two solutions for x ? **b** one solution for x ?
- 10 Show that the equation $ax^2 - (a + b)x + b = 0$ has a solution for all values of a and b .



4.3 Determining the rule for a parabola

Given sufficient information about a curve, a rule for the function of the graph may be determined.

For example, if the coordinates of three points on a parabola of the form

$$y = ax^2 + bx + c$$

are known, the rule for the parabola may be found, i.e. the values of a , b and c may be found.

Sometimes a more specific rule is known. For example, the curve may be a dilation of $y = x^2$. It is then known to be of the $y = ax^2$ family, and the coordinates of one point (with the exception of the origin) will be enough to determine the value for a .

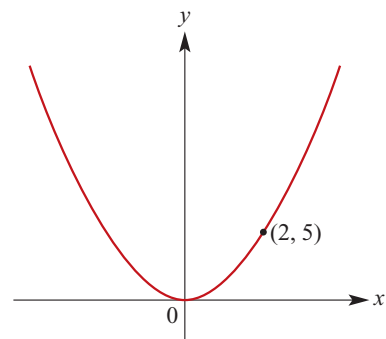
In the following it is assumed that each of the graphs is that of a parabola, and each rule is that of a quadratic function in x .

- 1** This is of the form $y = ax^2$ (since the graph has its vertex at the origin).

When $x = 2$, $y = 5$

$$\therefore 5 = a(2)^2 \Rightarrow a = \frac{5}{4}$$

The rule is $y = \frac{5}{4}x^2$



- 2 This is of the form $y = ax^2 + c$ (since the graph is symmetric about the y -axis).

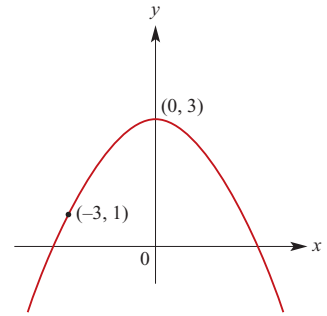
For $(0, 3)$ $3 = a(0) + c$ implies $c = 3$

For $(-3, 1)$ $1 = a(-3)^2 + 3$

$$1 = 9a + 3$$

$$\therefore a = -\frac{2}{9}$$

$\therefore$ the rule is $y = -\frac{2}{9}x^2 + 3$



- 3 This is of the form $y = ax(x - 3)$

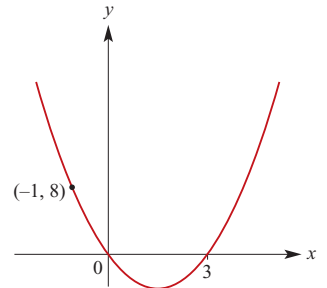
As the point $(-1, 8)$ is on the parabola:

$$8 = -a(-1 - 3)$$

And hence $4a = 8$

Therefore $a = 2$

The rule is $y = 2x(x - 3)$



- 4 This is of the form $y = ax^2 + bx + c$

As the point with coordinates $(-1, 0)$ is on the parabola:

$$0 = a - b + c \quad (1)$$

The y -axis intercept is 2 and therefore $c = 2$

As the point with coordinates $(1, 2)$ is on the parabola:

$$2 = a + b + c \quad (3)$$

Substitute $c = 2$ in (1) and (3) $0 = a - b + 2$

$$\text{From (1)} \quad -2 = a - b \quad (1a)$$

$$\text{From (3)} \quad 0 = a + b \quad (3a)$$

Subtract (3a) from (1a) $-2 = -2b$

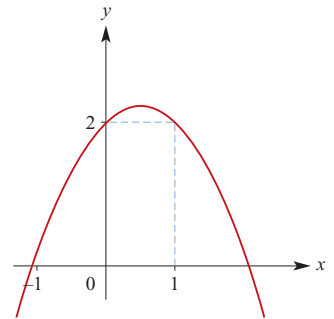
$$\therefore b = 1$$

Substitute $b = 1$ and $c = 2$ in (1).

$$\therefore 0 = a - 1 + 2$$

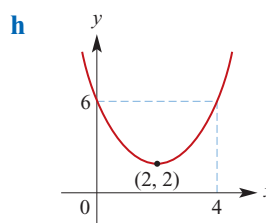
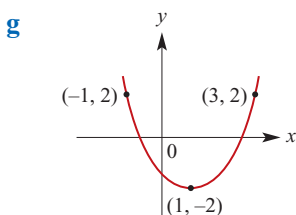
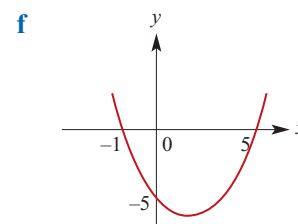
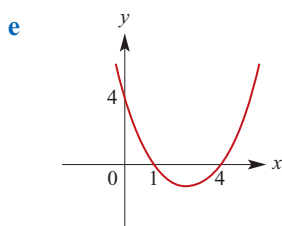
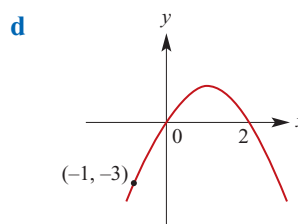
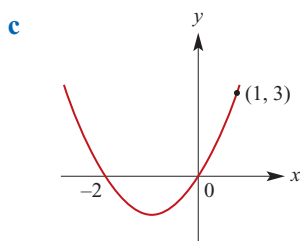
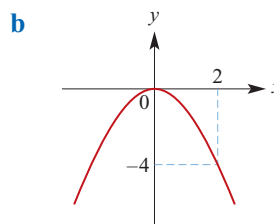
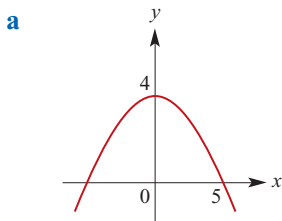
$$0 = a + 1 \Rightarrow a = -1$$

$\therefore$ the quadratic rule is $y = -x^2 + x + 2$

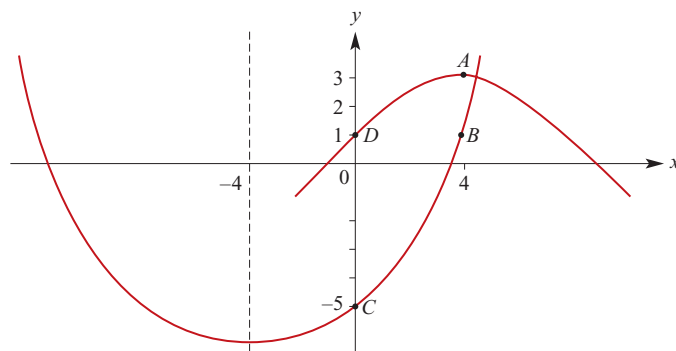


Exercise 4C

1 Determine the equation of each of the following parabolas:



2 Find quadratic expressions representing the two curves shown in the diagram, given that the coefficient of x in each case is 1. A is $(4, 3)$, B is $(4, 1)$, C is $(0, -5)$ and D is $(0, 1)$.



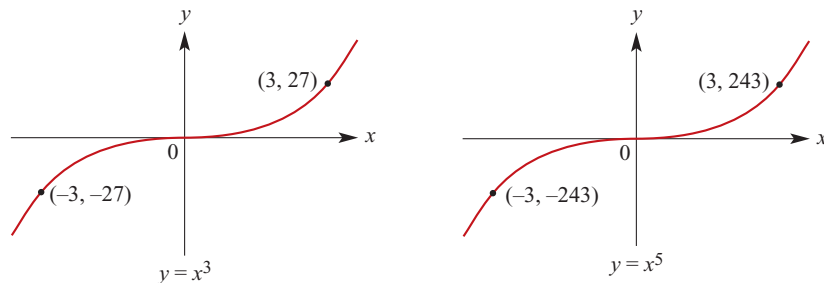
3 The graph of the quadratic function $f(x) = A(x + b)^2 + B$ has a vertex at $(-2, 4)$ and passes through the point $(0, 8)$. Find the values of A , b and B .

4.4 Functions of the form $f: R \rightarrow R$, $f(x) = a(x + h)^n + k$, where n is a natural number

In the previous section it was shown that every quadratic polynomial can be written in the form $a(x + h)^2 + k$. This is not true for polynomials of higher degree. This was shown in Example 2 of this chapter. However, there are many polynomials that can be written in this form. In Chapter 3 the family of power functions was introduced. In this section the sub-family of power functions with rules of the form $f(x) = x^n$ where n is a natural number are considered.

$f(x) = x^n$ where n is an odd positive integer

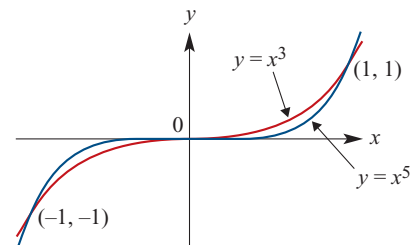
The diagrams below shows the graphs of $y = x^3$ and $y = x^5$



The diagram on the right shows both functions graphed on the one set of axes for a smaller domain.

The following properties can be observed for a function $f(x) = x^n$ where n is an odd integer:

- $f(0) = 0$
- $f(1) = 1$ and $f(-1) = -1$
- $-f(x) = f(-x)$, i.e. f is an odd function.
 $f(-x) = (-x)^n = (-1)^n(x)^n = -f(x)$ as n is odd.
- As $x \rightarrow \infty, f(x) \rightarrow \infty$
- As $x \rightarrow -\infty, f(x) \rightarrow -\infty$



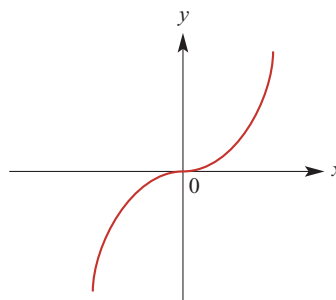
From *Essential Mathematical Methods 1 & 2 CAS* you will recall that the gradient function of $f(x) = x^n$ has rule nx^{n-1} . Hence the gradient is 0 when $x = 0$. As $n - 1$ is even for n odd, $nx^{n-1} > 0$ for all non-zero x . That is, the gradient of the graph of $y = f(x)$ is positive for all non-zero x and zero when $x = 0$. Recall that the stationary point at $x = 0$, for functions of this form, is called a stationary point of inflexion.

Comparing the graphs of $y = x^n$ and $y = x^m$ where n and m are odd and $n > m$

- $x^n < x^m$ when $0 < x < 1$ ($x^n - x^m = x^m(x^{n-m} - 1) < 0$)
- $x^n = x^m$ when $x = 0$
- $x^n > x^m$ when $-1 < x < 0$ ($x^n - x^m = x^m(x^{n-m} - 1) > 0$ as $n - m$ is even, m is odd and x is negative and greater than -1)

- $x^n = x^m$ for $x = 1$ and $x^n = x^m$ for $x = -1$
- $x^n > x^m$ for $x > 1$
- $x^n < x^m$ for $x < -1$

It should be noted that the appearance of graphs is dependent on the scales on the y - and x -axes. Odd power functions are often depicted as shown.



Transformations of graphs of functions with rule $f(x) = x^n$ where n is an odd positive integer

Transformations of these graphs result in graphs with rules of the form $y = a(x + h)^n + k$ where a , h and k are real constants.



Example 9

Sketch the graph of $f(x) = (x - 2)^3 + 1$

Solution

This can be done by noting that a translation $(x, y) \rightarrow (x + 2, y + 1)$ maps the graph of $y = x^3$ to $y = (x - 2)^3 + 1$

Note: The point with coordinates $(2, 1)$ is a point of zero gradient.

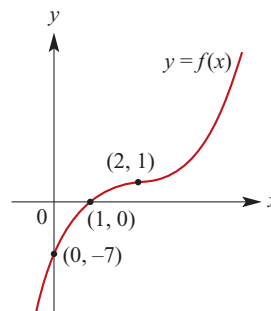
For the axes intercepts, consider this:

- When $x = 0$, $y = (-2)^3 + 1 = -7$
- When $y = 0$, $0 = (x - 2)^3 + 1$

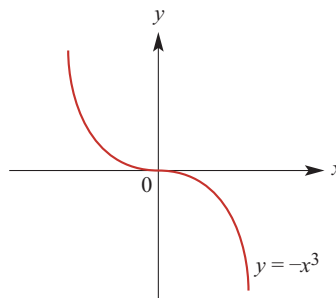
$$\therefore -1 = (x - 2)^3$$

$$\therefore -1 = x - 2$$

and hence $x = 1$



The reflection in the x -axis described by the transformation rule $(x, y) \rightarrow (x, -y)$ and applied to $y = x^3$ results in the graph of $y = -x^3$



Example 10

Sketch the graph of $y = -(x - 1)^3 + 2$

Solution

A reflection in the x -axis followed by a translation with rule $(x, y) \rightarrow (x + 1, y + 2)$ applied to the graph of $y = x^3$ results in the graph $y = -(x - 1)^3 + 2$

Note: $(1, 2)$ is a point of zero gradient.

For the axes intercepts consider this:

■ When $x = 0$, $y = -(-1)^3 + 2 = 3$

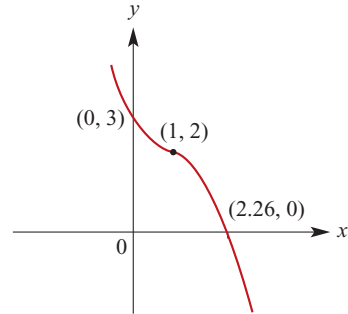
■ When $y = 0$, $0 = -(x - 1)^3 + 2$

$$\therefore (x - 1)^3 = 2$$

$$\therefore x - 1 = 2^{\frac{1}{3}}$$

$$\text{and hence } x = 1 + 2^{\frac{1}{3}}$$

$$\approx 2.26$$

**Example 11**

Sketch the graph of $y = 2(x + 1)^3 + 2$

Solution

A dilation of factor 2 from the x -axis followed by the translation with rule

$$(x, y) \rightarrow (x - 1, y + 2)$$

maps the graph of $y = x^3$ to the graph of

$$y = 2(x + 1)^3 + 2$$

Note: $(-1, 2)$ is a point of zero gradient.

For the axes intercepts consider this:

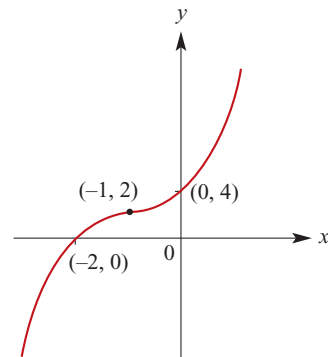
■ When $x = 0$, $y = 2 + 2 = 4$

■ When $y = 0$, $0 = 2(x + 1)^3 + 2$

$$\therefore -1 = (x + 1)^3$$

$$\therefore -1 = x + 1$$

$$\text{and hence } x = -2$$

**Example 12**

The graph of $y = a(x + h)^3 + k$ has a point of zero gradient at $(1, 1)$ and passes through the point $(0, 4)$. Find the values of a , h and k .

Solution

$$h = -1 \text{ and } k = 1$$

Therefore as the graph passes through $(0, 4)$:

$$4 = -a + 1$$

$$\therefore a = -3$$

Example 13

- a** Find the rule for the image of the graph of $y = x^5$ under the following sequence of transformations:
- reflection in the y -axis
 - dilation of factor 2 from the y -axis
 - translation of 2 units in the positive direction of the x -axis and 3 units in the positive direction of the y -axis
- b** Find a sequence of transformations which takes the graph of $y = x^5$ to the graph of $y = 6 - 2(x + 5)^5$

Solution

- a** $(x, y) \rightarrow (-x, y) \rightarrow (-2x, y) \rightarrow (-2x + 2, y + 3)$

Let (x', y') be the image of (x, y) under this transformation.

Hence $x' = -2x + 2$ and $y' = y + 3$

Thus $-2x = x' - 2$, which implies $x = \frac{x' - 2}{-2}$ and $y = y' - 3$

Therefore, the graph of $y = x^5$ maps to the graph of $y' - 3 = \left(\frac{x' - 2}{-2}\right)^5$,

i.e., to the graph of $y = -\frac{1}{32}(x - 2)^5 + 3$

- b** Rearrange $y = 6 - 2(x + 5)^5$ to $\frac{y' - 6}{-2} = (x' + 5)^5$

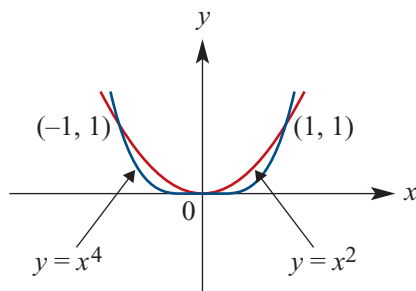
Therefore, $y = \frac{y' - 6}{-2}$ and $x = x' + 5$ and $y' = -2y + 6$ and $x' = x - 5$

The sequence of transformations is:

- reflection in the x -axis
- dilation of factor 2 from the x -axis
- translation of 5 units in the negative direction of the x -axis and 6 units in the positive direction of the y -axis.

$f(x) = x^n$ where n is an even positive integer

The graphs of $y = x^2$ and $y = x^4$ are shown on the one set of axes.



The following properties can be observed for a function $f(x) = x^n$ where n is an even integer:

- $f(0) = 0$
- $f(1) = 1$ and $f(-1) = 1$
- $f(x) = f(-x)$, i.e. f is an even function, $f(-x) = (-x)^n$
 $= (-1)^n(x)^n$
 $= f(x)$ as n is even
- As $x \rightarrow \infty$, $f(x) \rightarrow \infty$
- As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

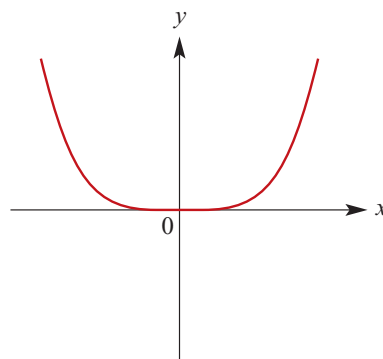
From *Essential Mathematical Methods 1 & 2 CAS* you will recall that the gradient function of $f(x) = x^n$ has rule nx^{n-1} . Hence the gradient is 0 when $x = 0$. As $n - 1$ is odd for n even, $nx^{n-1} > 0$ for $x > 0$ and $nx^{n-1} < 0$ for $x < 0$. That is, the gradient of the graph of $y = f(x)$ is positive for x positive, negative for x negative and zero when $x = 0$.

Comparing the graphs of $y = x^n$ and $y = x^m$ where n and m are even and $n > m$

- $x^n < x^m$ when $0 < x < 1$ $(x^n - x^m = x^m(x^{n-m} - 1) < 0)$
- $x^n = x^m$ when $x = 0$
- $x^n < x^m$ when $-1 < x < 0$ $(x^n - x^m = x^m(x^{n-m} - 1) < 0$ as $n - m$ is even, m is even and x is negative and greater than -1)
- $x^n = x^m$ for $x = 1$ and $x^n = x^m$ for $x = -1$
- $x^n > x^m$ for $x > 1$
- $x^n > x^m$ for $x < -1$

It should be noted that the appearance of graphs is dependent on the scales on the y - and x -axes.

Power functions of even degree are often depicted as shown.



Example 14

The graph of $y = a(x + h)^4 + k$ has a turning point at $(2, 2)$ and passes through the point $(0, 4)$. Find the values of a and h and k .

Solution

$$h = -2 \text{ and } k = 2$$

Therefore as the graph passes through $(0, 4)$:

$$4 = 16a + 2$$

$$\therefore a = \frac{1}{8}$$

Exercise 4D

- Sketch the graphs of each of the following. State the coordinates of the point of zero gradient and the axes intercepts.

a $f(x) = 2x^3$	b $g(x) = -2x^3$	c $h(x) = x^5 + 1$
d $f(x) = x^3 - 4$	e $f(x) = (x + 1)^3 - 8$	f $f(x) = 2(x - 1)^3 - 2$
g $g(x) = -2(x - 1)^3 + 2$	h $h(x) = 3(x - 2)^3 - 4$	i $f(x) = 2(x - 1)^3 + 2$
j $h(x) = -2(x - 1)^3 - 4$	k $f(x) = (x + 1)^5 - 32$	l $f(x) = 2(x - 1)^5 - 2$
- The graph of $y = a(x + h)^3 + k$ has a point of zero gradient at (0, 4) and passes through the point (1, 1). Find the values of a , h and k .
- The graph of $y = a(x + h)^4 + k$ has a turning point at (1, 7) and passes through the point (0, 23). Find the values of a , h and k .
- Find the equation of the image of $y = x^3$ under each of the following transformations:
 - a dilation of factor 3 from the x -axis
 - a translation with rule $(x, y) \rightarrow (x - 1, y + 1)$
 - a reflection in the x -axis followed by a translation with rule

$$(x, y) \rightarrow (x + 2, y - 3)$$
 - a dilation of factor 2 from the x -axis followed by a translation with rule

$$(x, y) \rightarrow (x - 1, y - 2)$$
 - a dilation of factor 3 from the y -axis.
- By applying suitable transformations to $y = x^4$, sketch the graph of each of the following:

a $y = 3(x - 1)^4 - 2$	b $y = -2(x + 2)^4$	c $y = (x - 2)^4 - 6$
d $y = 2(x - 3)^4 - 1$	e $y = 1 - (x + 4)^4$	f $y = -3(x - 2)^4 - 3$

4.5



The general cubic function

Not all cubic functions are of the form $f(x) = a(x + h)^3 + k$. In this section the general cubic function is considered. The form of a general cubic function is:

$$f(x) = ax^3 + bx^2 + cx + d$$

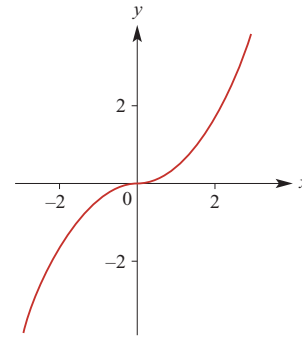


It is impossible to fully investigate cubic functions without the use of calculus. Cubic functions will be revisited in Chapter 9.

The ‘shapes’ of cubic graphs vary. Below is a gallery of cubic graphs demonstrating the variety of ‘shapes’ that are possible.

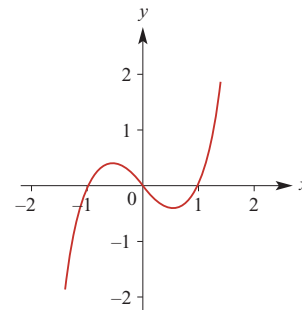
1 $f(x) = x^3 + x$

Note: $(0, 0)$ is not a point of zero gradient.
There is one root, 0.



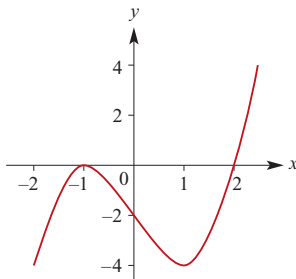
2 $f(x) = x^3 - x$

Note: The turning points do not occur symmetrically between consecutive x -axis intercepts as they do for quadratics. Differential calculus must be used to determine them. There are three roots: 1, 0 and -1 .

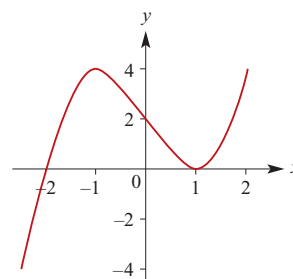


3 $f(x) = x^3 - 3x - 2$

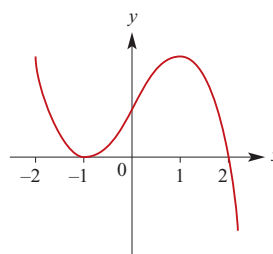
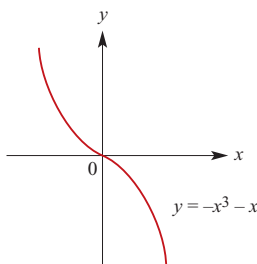
Note: There are two roots: -1 and 2 .



4 $f(x) = x^3 - 3x + 2$

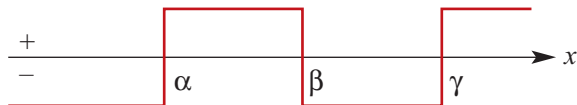


5 The graphs of $f(x) = -x^3 - x$ and $f(x) = -x^3 + 3x + 2$ are shown. They are the reflection in the y -axis of the graphs of 1 and 4 respectively.



Sign diagrams

A sign diagram is a number line diagram that shows when an expression is positive or negative. For a cubic function with rule $f(x) = (x - \alpha)(x - \beta)(x - \gamma)$ when $\gamma > \beta > \alpha$, the sign diagram is as shown:



Example 15

Draw a sign diagram for the expression $x^3 - 4x^2 - 11x + 30$

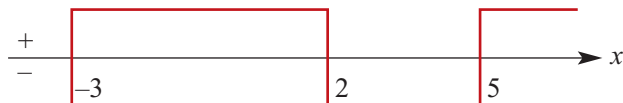
Solution

From Example 6, $f(x) = (x - 2)(x - 5)(x + 3)$

We note: $f(x) > 0$ for $x > 5$
 $f(x) < 0$ for $2 < x < 5$
 $f(x) > 0$ for $-3 < x < 2$
 $f(x) < 0$ for $x < -3$

Also, $f(2) = f(5)$
 $= f(-3)$
 $= 0$

Hence the sign diagram may be drawn.



Example 16

For the cubic function with rule $f(x) = -x^3 + 19x - 30$:

- Sketch the graph of $y = f(x)$ using a calculator to find the values of the coordinates of the turning points, correct to two decimal places.
- Sketch the graph of $y = \frac{1}{2}f(x - 1)$

Solution

$$\begin{aligned} \text{a } f(x) &= -x^3 + 19x - 30 = (3 - x)(x - 2)(x + 5) \\ &= -(x + 5)(x - 2)(x - 3) \end{aligned}$$

We note: $f(x) < 0$ for $x > 3$
 $f(x) > 0$ for $2 < x < 3$
 $f(x) < 0$ for $-5 < x < 2$
 $f(x) > 0$ for $x < -5$

$$\begin{aligned}
 \text{Also, } f(-5) &= f(2) \\
 &= f(3) \\
 &= 0
 \end{aligned}$$

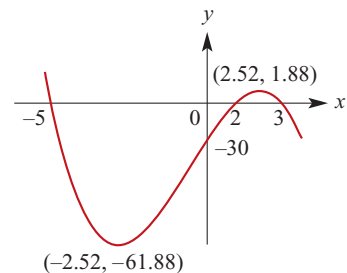
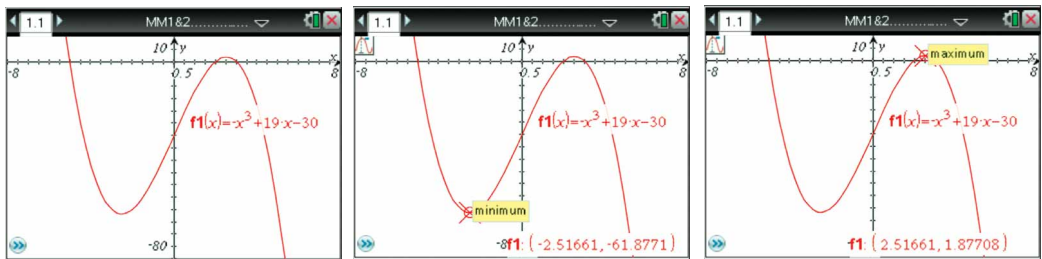
Hence the sign diagram may be drawn.



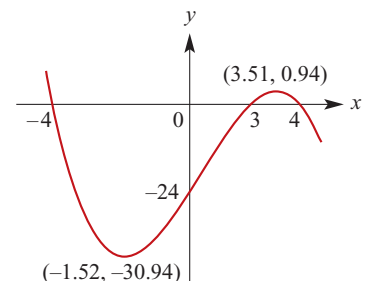
The x -axis intercepts are at $x = -5$, $x = 2$ and $x = 3$ and the y -axis intercept is at $y = -30$.

The window has to be adjusted carefully to see the x -axis intercepts and the turning points.

The local minimum and maximum turning points can be found using a CAS calculator. The method for this was first discussed in *Essential Mathematical Methods 1 & 2 CAS*. It is also discussed in the Appendices for the TI-Nspire and the CASIO ClassPad, at the back of this book. The screens for the TI-Nspire are shown here.



- b** The rule for the transformation is $(x, y) \rightarrow \left(x + 1, \frac{1}{2}y\right)$. By the transformation
- $(-5, 0) \rightarrow (-4, 0)$, $(2, 0) \rightarrow (3, 0)$,
 $(3, 0) \rightarrow (4, 0)$, $(0, -30) \rightarrow (1, -15)$,
 $(2.52, 1.88) \rightarrow (3.52, 0.94)$ and
 $(-2.52, -61.88) \rightarrow (-1.52, -30.94)$



Exercise 4E

- 1 Draw sign diagrams for each of the following expressions:
 - a $(3 - x)(x - 1)(x - 6)$
 - b $(3 + x)(x - 1)(x + 6)$
 - c $(x - 5)(x + 1)(2x - 6)$
 - d $(4 - x)(5 - x)(1 - 2x)$
- 2 a Use a calculator to plot the graph of $y = f(x)$ where $f(x) = x^3 - 2x^2 + 1$
 b On the same screen plot the graphs of:
 - i $y = f(x - 2)$
 - ii $y = f(x + 2)$
 - iii $y = 3f(x)$



4.6 Polynomials of higher degree

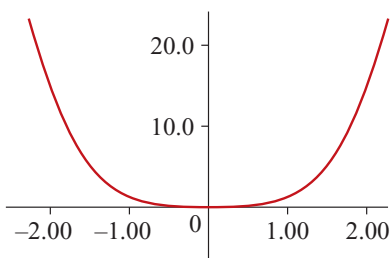
The techniques that have been developed for cubic functions may now be applied to quartic functions and in general to functions of higher degree. It is clear that a polynomial $P(x)$ of degree n has at most n solutions for the equation $P(x) = 0$. It is possible for the graphs of polynomials of even degree to have no x -axis intercepts, for example $P(x) = x^2 + 1$, but graphs of polynomials of odd degree have at least one x -axis intercept.

The general form for a quartic function is:

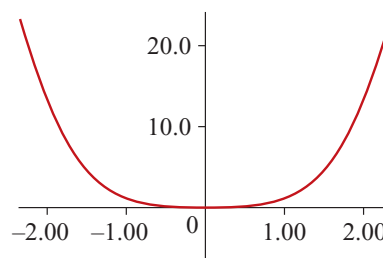
$$f(x) = ax^4 + bx^3 + cx^2 + dx + e$$

A gallery of quartic functions produced with a graphing package is shown below.

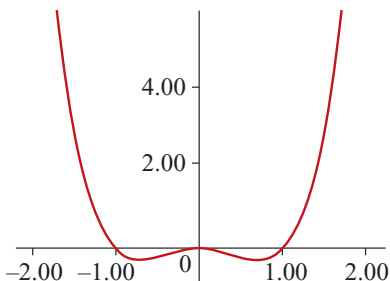
1 $f(x) = x^4$



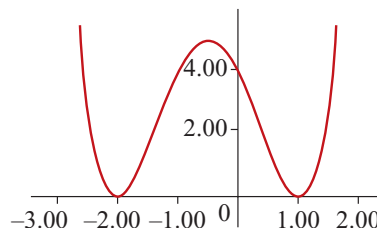
2 $f(x) = x^4 + x^2$



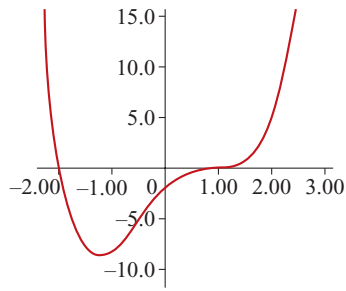
3 $f(x) = x^4 - x^2$



4 $f(x) = (x - 1)^2(x + 2)^2$



5 $f(x) = (x - 1)^3(x + 2)$



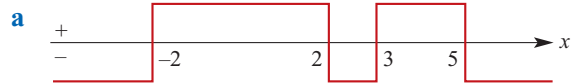
Example 17

Draw a sign diagram for the quartic expressions:

a $(2 - x)(x + 2)(x - 3)(x - 5)$

b $x^4 + x^2 - 2$

Solution



b Let $P(x) = x^4 + x^2 - 2$

$$P(1) = 1 + 1 - 2$$

$$= 0$$

$\therefore x - 1$ is a factor.

$$\begin{array}{r} x^3 + x^2 + 2x + 2 \\ x - 1 \overline{) x^4 + 0x^3 + x^2 + 0x - 2} \end{array}$$

$$\underline{x^4 - x^3}$$

$$x^3 + x^2$$

$$\underline{x^3 - x^2}$$

$$2x^2 + 0x$$

$$\underline{2x^2 - 2x}$$

$$2x - 2$$

$$\underline{2x - 2}$$

$$0$$

$$\therefore P(x) = (x - 1)(x^3 + x^2 + 2x + 2)$$

$$= (x - 1)[x^2(x + 1) + 2(x + 1)]$$

$$= (x - 1)(x + 1)(x^2 + 2)$$



Example 18

Find the coordinates of the points where the graph of $y = p(x)$, $p(x) = x^4 - 2x^2 + 1$, crosses the x - and y -axes, and hence sketch the graph.

Solution

$$p(1) = 1 - 2 + 1 = 0$$

$\therefore x - 1$ is a factor.

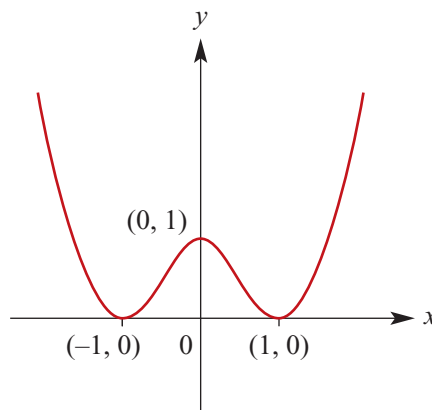
$$\begin{aligned} p(x) &= (x - 1)(x^3 + x^2 - x - 1) \\ &= (x - 1)[x^2(x + 1) - (x + 1)] \\ &= (x - 1)(x + 1)(x^2 - 1) \\ &= (x - 1)^2(x + 1)^2 \end{aligned}$$

Alternatively, note that:

$$\begin{aligned} p(x) &= (x^2 - 1)^2 \\ &= [(x - 1)(x + 1)]^2 \\ &= (x - 1)^2(x + 1)^2 \end{aligned}$$

Therefore, the x -axis intercepts are $(1, 0)$ and $(-1, 0)$.

When $x = 0$, $y = 1$.

**Exercise 4F**

- 1 a Use a calculator to plot the graph of $y = f(x)$ where

$$f(x) = x^4 - 2x^3 + x + 1$$

- b On the same screen plot the graphs of:

i $y = f(x - 2)$

ii $y = f(2x)$

iii $y = f\left(\frac{x}{2}\right)$

- 2 The graph of $y = 9x^2 - x^4$ is as shown. Sketch the graph of each of the following by applying suitable transformations.

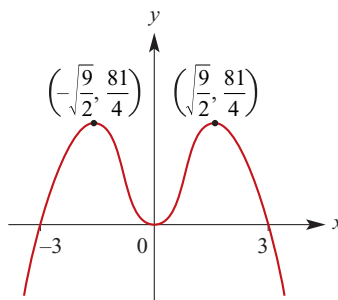
a $y = 9(x - 1)^2 - (x - 1)^4$ b $y = 18x^2 - 2x^4$

c $y = 18(x + 1)^2 - 2(x + 1)^4$

d $y = 9x^2 - x^4 - \frac{81}{4}$

e $y = 9x^2 - x^4 + 1$

(Do not find the x -axis intercepts for e.)



4.7 Determining rules for the graphs of polynomials

It is first worth noting that the graph of a polynomial function of degree n is completely determined by any $n + 1$ points on the curve.

For example, for a cubic function with rule $y = f(x)$, if it is known that $f(a_1) = b_1$, $f(a_2) = b_2$, $f(a_3) = b_3$, $f(a_4) = b_4$, then the rule can be determined.

Finding the rule for a parabola has been discussed in Section 4.3 of this chapter.

Example 19

For the cubic function with rule $f(x) = ax^3 + bx^2 + cx + d$, it is known that the points with coordinates $(-1, -18)$, $(0, -5)$, $(1, -4)$ and $(2, -9)$ lie on the graph of the cubic. Find the values of a , b , c and d .

Solution

The following equations can be formed.

$$-a + b - c + d = -18 \quad (1)$$

$$d = -5 \quad (2)$$

$$a + b + c + d = -4 \quad (3)$$

$$8a + 4b + 2c + d = -9 \quad (4)$$

Adding (1) and (3) gives $2b + 2d = -22$ and as $d = -5$, $b = -6$.

There are now only two unknowns.

$$(3) \text{ becomes } a + c = 7 \quad (3')$$

$$\text{and } (4) \text{ becomes } 8a + 2c = 20 \quad (4')$$

Multiply (3') by 2 and subtract from (4').

$$6a = 6 \text{ which gives } a = 1 \text{ and } c = 6$$



Using the TI-Nspire

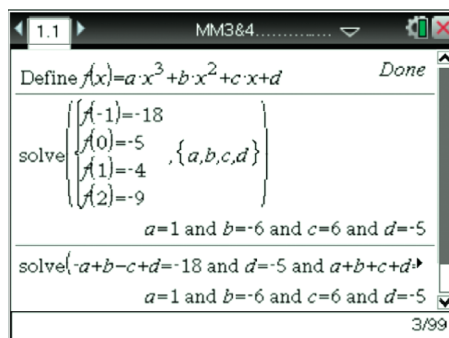
Define $f(x) = ax^3 + bx^2 + cx + d$ and then use the simultaneous equation template (**menu**) **> Algebra > Solve System of Equations > Solve System of Equations**) to solve given $f(-1) = -18$, $f(0) = -5$, $f(1) = -4$ and $f(2) = -9$

Alternatively, enter

solve($-a + b - c + d = -18$ and $d = -5$ and $a + b + c + d = -4$ and $8a + 4b + 2c + d = -9$, $\{a, b, c, d\}$)

The word 'and' can be typed directly or found in the catalog (**menu**) **(1) [A]**).

Both methods are shown on the screen to the right.



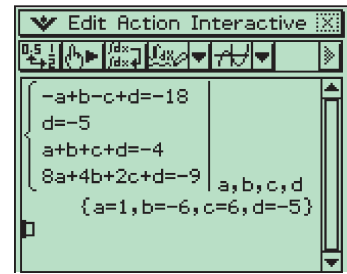
Using the Casio ClassPad

Press **Keyboard** and select **2D** menu (if necessary tap $\rightarrow$). Tap **EQ** three times to produce the template to enter four simultaneous equations. Enter the equations and set the variables in the variable box to a, b, c, d then tap **EXE**.

An alternative method is to define

$$f(x) = ax^3 + bx^2 + cx + d$$

then enter $f(-1) = -18$, $f(0) = 5$, $f(1) = -4$ and $f(2) = -9$ as the simultaneous equations to be solved with variables a, b, c, d as above.



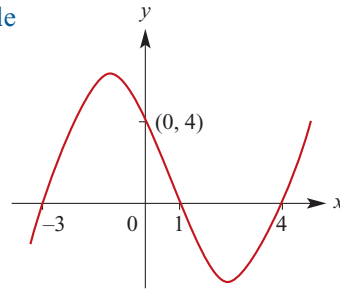
The following examples provide further procedures for finding the rules of cubic functions. It should be noted that a similar facility is available for quartics.

Example 20

The graph shown is that of a cubic function. Find the rule for this cubic function.

Solution

From the graph the function can be seen to be of the form $y = a(x - 4)(x - 1)(x + 3)$.



It remains to find the value of a . The point with coordinates $(0, 4)$ is on the graph. Hence:

$$4 = a(-4)(-1)3$$

$$\therefore a = \frac{1}{3}$$

and
$$y = \frac{1}{3}(x - 4)(x - 1)(x + 3)$$

Example 21

The graph shown is that of a cubic function.
Find the rule for this cubic function.

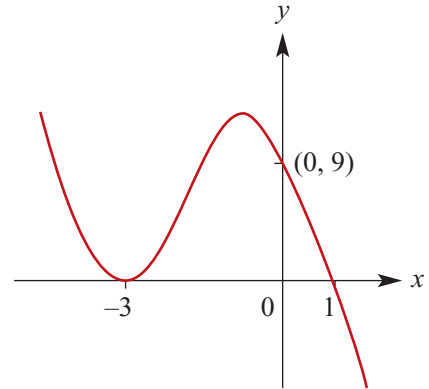
Solution

From the graph it can be seen to be of the form $y = k(x - 1)(x + 3)^2$. It remains to find the value of k . As the value $(0, 9)$ is on the graph:

$$9 = k(-1)(9)$$

$$\therefore k = -1$$

and
$$y = -(x - 1)(x + 3)^2$$



Note: Because the graph reveals that there are two factors, $(x + 3)$ and $(x - 1)$, there must be a third linear factor $(ax + b)$, but this implies that there is an intercept $-\frac{b}{a}$.

Thus $ax + b = k_1(x + 3)$ or $k_2(x - 1)$. A consideration of the signs reveals that $(x + 3)$ is a repeated factor.

Example 22

The graph of a cubic function passes through the point with coordinates $(0, 1)$, $(1, 4)$, $(2, 17)$ and $(-1, 2)$. Find the rule for this cubic function.

Solution

The cubic function will have a rule of the form:

$$y = ax^3 + bx^2 + cx + d$$

The values of a , b , c and d have to be determined.

As the point $(0, 1)$ is on the graph, $d = 1$.

By using the points $(1, 4)$, $(2, 17)$ and $(-1, 2)$, three simultaneous equations are produced:

$$4 = a + b + c + 1$$

$$17 = 8a + 4b + 2c + 1$$

$$2 = -a + b - c + 1$$

These become:

$$3 = a + b + c \quad (1)$$

$$16 = 8a + 4b + 2c \quad (2)$$

$$1 = -a + b - c \quad (3)$$

Add (1) and (3) to find:

$$2b = 4$$

i.e., $b = 2$

Substitute in (1) and (2):

$$1 = a + c \quad (4)$$

$$8 = 8a + 2c \quad (5)$$

Multiply (4) by 2 and subtract from (5).

Hence $6 = 6a$

and $a = 1$

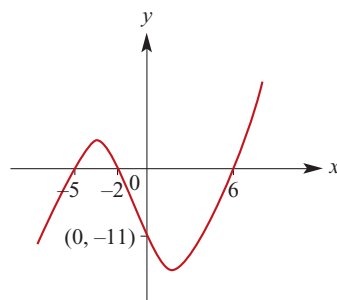
From (4) $c = 0$

Thus $y = x^3 + 2x^2 + 1$

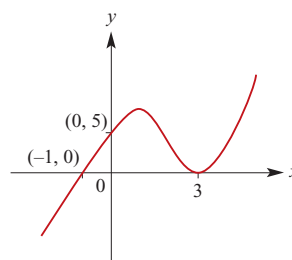
Exercise 4G



- 1 Determine the rule for the cubic function with graph as shown:



- 2 Determine the rule for the cubic function with graph as shown:



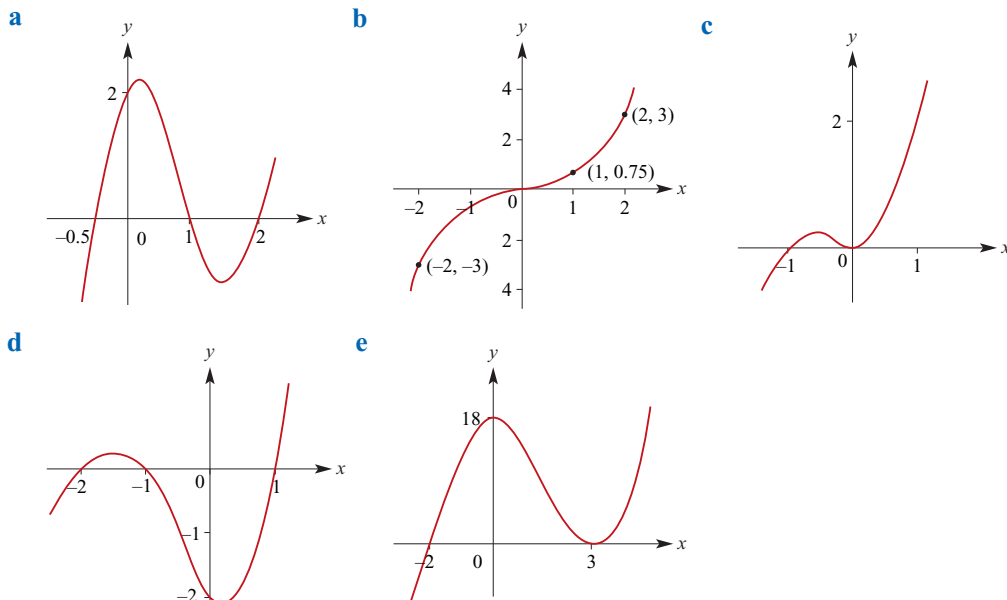
- 3 Find the rule for the cubic function that passes through the following points:

a $(0, 1), (1, 3), (-1, -1)$ and $(2, 11)$

b $(0, 1), (1, 1), (-1, 1)$ and $(2, 7)$

c $(0, -2), (1, 0), (-1, -6)$ and $(2, 12)$

4 Find expressions which define the following cubic curves:



5 Find the equation of the cubic function for which the graph passes through the points with coordinates:

a (0, 135), (1, 156), (2, 115), (3, 0)

b (-2, -203), (0, 13), (1, 25), (2, -11)

6 Find the equation of the quartic function for which the graph passes through the points with coordinates:

a (-1, 43), (0, 40), (2, 70), (6, 1618), (10, 670)

b (-3, 119), (-2, 32), (-1, 9), (0, 8), (1, 11)

c (-3, 6), (-1, 2), (1, 2), (3, 66), (6, 1227)

4.8 Solution of literal equations and systems of equations

Literal equations

Solving literal linear equations and simultaneous equations was undertaken in Section 2.2. In this section other non-linear expressions are considered. They certainly can be solved with a CAS calculator but full setting out is shown here.

Example 23

Solve each of the following literal equations for x :

a $x^2 + kx + k = 0$ **b** $x^3 - 3ax^2 + 2a^2x = 0$ **c** $x(x^2 - a) = 0$

Solution

- a**
- Completing the square gives

$$\begin{aligned}
 x^2 + kx + \frac{k^2}{4} + k - \frac{k^2}{4} &= 0 \\
 \left(x + \frac{k}{2}\right)^2 &= \frac{k^2}{4} - k \\
 x + \frac{k}{2} &= \pm \sqrt{\frac{k^2 - 4k}{4}} \\
 x &= \frac{-k \pm \sqrt{k^2 - 4k}}{2}
 \end{aligned}$$

A real solution exists only for $k^2 - 4k \geq 0$; that is for $k \geq 4$ or $k \leq 0$

- b**
- $x^3 - 3ax^2 + 2a^2x = 0$

$$x(x^2 - 3ax + 2a^2) = 0$$

$$x(x - a)(x - 2a) = 0$$

$$x = 0 \text{ or } x = a \text{ or } x = 2a$$

- c**
- $x(x^2 - a) = 0$
- implies
- $x = 0$
- or
- $x = \sqrt[3]{a}$
- or
- $x = -\sqrt[3]{a}$

In the following, the property that for suitable values of a , b and an odd natural number n , $b^n = a$ is equivalent to $b = a^{\frac{1}{n}}$ and also $b = \sqrt[n]{a}$. Also $b^{\frac{p}{q}} = a$ is equivalent to $b = a^{\frac{q}{p}}$ for suitable values of a and b and integers q and p .

Care must be taken with the application of these. For example $x^2 = 2$ is equivalent to $x = \pm\sqrt{2}$ and $x^{\frac{2}{3}} = 4$ is equivalent to $x = 8$ or $x = -8$.

Example 24

Solve each of the following equations for x :

a $ax^3 - b = c$

b $x^{\frac{3}{5}} = a$

c $x^{\frac{1}{n}} = a$ where n is a natural number and x is a positive real number

d $a(x + b)^3 = c$

e $ax^{\frac{1}{5}} = b$

f $x^5 - c = d$

Solution

a $ax^3 - b = c$

$$ax^3 = b + c$$

$$x^3 = \frac{b + c}{a}$$

$$x = \sqrt[3]{\frac{b + c}{a}}$$

or $x = \left(\frac{b + c}{a}\right)^{\frac{1}{3}}$

b $x^{\frac{3}{5}} = a$ is equivalent to $x = a^{\frac{5}{3}}$

c $x^{\frac{1}{n}} = a$ is equivalent to $x = a^n$

d $a(x + b)^3 = c$

$$(x + b)^3 = \frac{c}{a}$$

$$x + b = \left(\frac{c}{a}\right)^{\frac{1}{3}}$$

$$x = \left(\frac{c}{a}\right)^{\frac{1}{3}} - b$$

$$\begin{aligned} \text{e } ax^{\frac{1}{5}} &= b \\ x^{\frac{1}{5}} &= \frac{b}{a} \\ x &= \left(\frac{b}{a}\right)^5 \end{aligned}$$

$$\begin{aligned} \text{f } x^5 - c &= d \\ x^5 &= c + d \\ x &= (c + d)^{\frac{1}{5}} \end{aligned}$$

Simultaneous equations

In this section, methods for finding the coordinates of the points of intersection of different graphs are discussed.

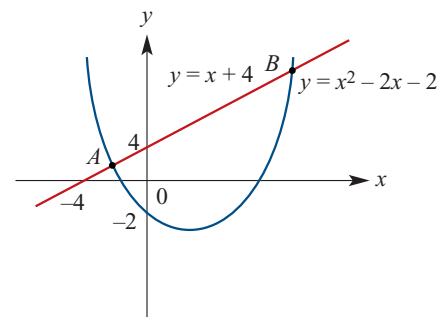
Example 25

Find the coordinates of the points of intersection of the parabola with equation $y = x^2 - 2x - 2$ with the straight line with equation $y = x + 4$

Solution

Consider $x + 4 = x^2 - 2x - 2$

$$\begin{aligned} \text{Then } 0 &= x^2 - 3x - 6 \\ \therefore x &= \frac{3 \pm \sqrt{9 - 4 \times -6 \times 1}}{2} \\ &= \frac{3 \pm \sqrt{33}}{2} \end{aligned}$$



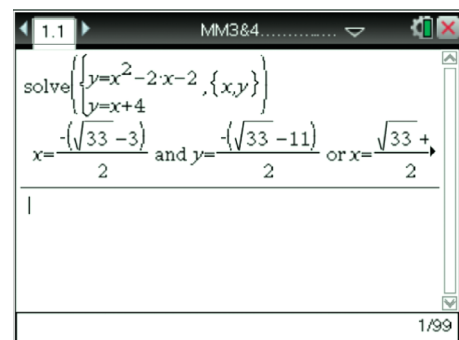
The points of intersection have coordinates

$$A\left(\frac{3 - \sqrt{33}}{2}, \frac{11 - \sqrt{33}}{2}\right) \text{ and } B\left(\frac{3 + \sqrt{33}}{2}, \frac{11 + \sqrt{33}}{2}\right)$$

Using the TI-Nspire

Use the simultaneous equation template
(**menu**) > **Algebra** > **Solve System of Equations** > **Solve System of Equations**
and complete as shown.

Use the **up arrow** ($\blacktriangle$) to move up to the answer and use the **right arrow** ($\blacktriangleright$) to display the remaining part of the answer.

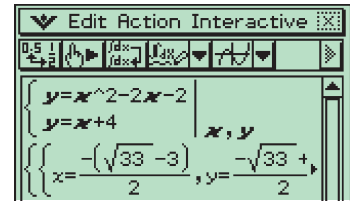


Using the Casio ClassPad

Using the simultaneous equation entry template, enter the equations

$y = x^2 - 2x - 2$ and $y = x + 4$ and set the variables as x, y .

Note the ► at the end of the answer line indicates that you must scroll to the right to see all the solutions.



Example 26

Find the coordinates of the points of intersection of the circle with equation $(x - 4)^2 + y^2 = 16$ and the line with equation $x - y = 0$

Solution

Rearrange $x - y = 0$ to make y the subject.

Substitute $y = x$ into the equation of the circle.

$$\text{i.e.,} \quad (x - 4)^2 + x^2 = 16$$

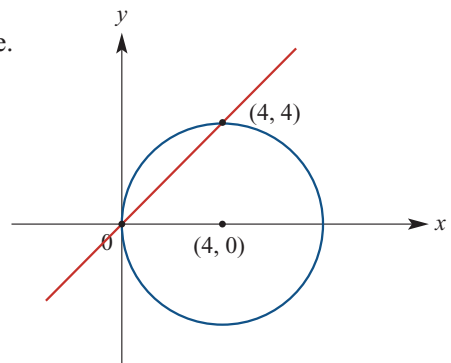
$$\therefore x^2 - 8x + 16 + x^2 = 16$$

$$\text{i.e.,} \quad 2x^2 - 8x = 0$$

$$2x(x - 4) = 0$$

$$x = 0 \text{ or } x = 4$$

The points of intersection are $(0, 0)$ and $(4, 4)$.



Example 27

Find the point of contact of the line with equation $\frac{1}{9}x + y = \frac{2}{3}$ and the curve with equation $xy = 1$

Solution

Rewrite the equations as $y = -\frac{1}{9}x + \frac{2}{3}$ and $y = \frac{1}{x}$

$$\text{Consider} \quad -\frac{1}{9}x + \frac{2}{3} = \frac{1}{x}$$

$$\therefore -x + 6 = \frac{9}{x}$$

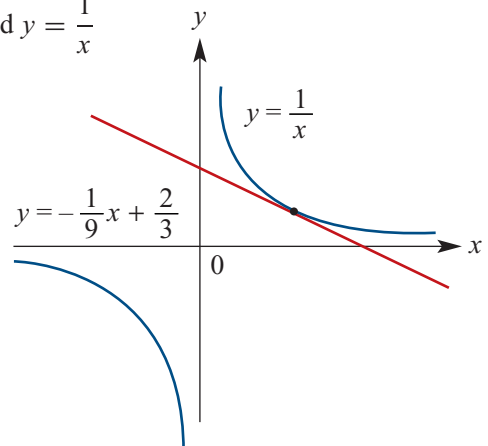
$$\text{and} \quad -x^2 + 6x = 9$$

$$\text{Therefore} \quad x^2 - 6x + 9 = 0$$

$$\text{and} \quad (x - 3)^2 = 0$$

$$\text{i.e. } x = 3$$

The point of intersection is $\left(3, \frac{1}{3}\right)$.

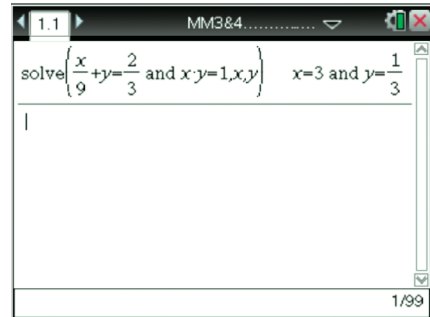


Using the TI-Nspire

Use **(menu) > Algebra > Solve** to solve as shown.

Note that the multiplication sign is required between x and y .

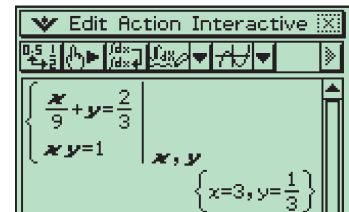
The simultaneous equation solver template **(menu) > Algebra > Solve System of Equations > Solve System of Equations** could also have been used in this example.



Using the Casio ClassPad

In the simultaneous equation entry screen, enter the equations $\frac{x}{9} + y = \frac{2}{3}$ and $xy = 1$ and set the variables as x, y .

Note that for this example, since the fraction entry is on the same screen as the simultaneous equation in the keyboard menu, the fractions can be entered using the fraction entry key.



Exercise 4H

- Solve each of the following literal equations for x :
 - $kx^2 + x + k = 0$
 - $x^3 - 7ax^2 + 12a^2x = 0$
 - $x(x^3 - a) = 0$
 - $x^2 - kx + k = 0$
 - $x^3 - ax = 0$
 - $x^4 - a^4 = 0$
 - $(x - a)^5(x - b) = 0$
 - $(a - x)^4(a - x^3)(x^2 - a) = 0$
- Solve each of the following equations for x :
 - $ax^3 - b = c$
 - $x^{\frac{3}{7}} = a$
 - $x^{\frac{1}{n}} + c = a$ where n is a natural number and x is a positive real number
 - $a(x + b)^3 = c$
 - $ax^{\frac{1}{3}} = b$
 - $x^3 - c = d$
- Find the coordinates of the points of intersection for each of the following:
 - $y = x^2$
 $y = x$
 - $y - 2x^2 = 0$
 $y - x = 0$
 - $y = x^2 - x$
 $y = 2x + 1$

- 4 Find the coordinates of the points of intersection for each of the following:

a $x^2 + y^2 = 178$ $x + y = 16$	b $x^2 + y^2 = 125$ $x + y = 15$	c $x^2 + y^2 = 185$ $x - y = 3$
d $x^2 + y^2 = 97$ $x + y = 13$	e $x^2 + y^2 = 106$ $x - y = 4$	

- 5 Find the coordinates of the points of intersection for each of the following:

a $x + y = 28$ $xy = 187$	b $x + y = 51$ $xy = 518$	c $x - y = 5$ $xy = 126$
-------------------------------------	-------------------------------------	------------------------------------

- 6 Find the coordinates of the points of intersection of the straight line with equation $y = 2x$ and the circle with equation $(x - 5)^2 + y^2 = 25$

- 7 Find the coordinates of the points of intersection of the curves with equation $y = \frac{1}{x-2} + 3$ and $y = x$

- 8 Find the coordinates of the points of intersection of the line with equation $\frac{y}{4} - \frac{x}{5} = 1$ and the circle with equation $x^2 + 4x + y^2 = 12$

- 9 Find the coordinates of the points of intersection of the curve with equation $y = \frac{1}{x+2} - 3$ and the line with equation $y = -x$

- 10 Find the coordinates of the point where the line $4y = 9x + 4$ touches the parabola with equation $y^2 = 9x$

- 11 Find the coordinates of the point where the line with equation $y = 2x + 3\sqrt{5}$ touches the circle $x^2 + y^2 = 9$

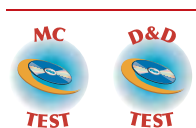
- 12 Find the coordinates of the point where the straight line with equation $y = \frac{1}{4}x + 1$ touches the curve with equation $y = -\frac{1}{x}$

- 13 Find the coordinates of the points of intersection of the curve with equation $y = \frac{2}{x-2}$ and the line $y = x - 1$

- 14 Solve the simultaneous equations:

a $5x - 4y = 7$ and $xy = 6$
b $2x + 3y = 37$ and $xy = 45$
c $5x - 3y = 18$ and $xy = 24$

- 15** What is the condition for $x^2 + ax + b$ to be divisible by $x + c$?
- 16** Solve the simultaneous equations $y = x + 2$ and $y = \frac{160}{x}$
- 17 a** Solve the simultaneous equations $y = mx$ and $y = \frac{1}{x} + 5$ for x in terms of m .
- b** Find the value of m for which the graphs of $y = mx$ and $y = \frac{1}{x} + 5$ touch and give the coordinates of this point.
- c** For which values of m do the graphs not meet?



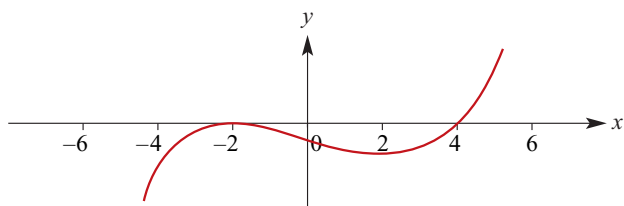
Chapter summary

- A function $P(x) = a_0 + a_1x + \cdots + a_nx^n$, where $a_0, a_1, \dots, a_n$ are constants, is called a **polynomial** in x . The numbers $a_0, a_1, \dots, a_n$ are the **coefficients**. Assuming $a_n \neq 0$, the term a_nx^n is the **leading term**. The integer n is the **degree** of the polynomial.
 - Degree one polynomials are called **linear** functions.
 - Degree two polynomials are called **quadratic** functions.
 - Degree three polynomials are called **cubic** functions.
 - Degree four polynomials are called **quartic** functions.
- **The remainder theorem:** If the polynomial $P(x)$ is divided by $ax + b$, the remainder is $P\left(-\frac{b}{a}\right)$
- **The factor theorem:** $(ax + b)$ is a factor of $P(x)$ if and only if $P\left(-\frac{b}{a}\right) = 0$
- To sketch the graph of a quadratic function $f(x) = ax^2 + bx + c$ (called a parabola), use the following:
 - If $a > 0$, the function has a minimum value.
 - If $a < 0$, the function has a maximum value.
 - The value of c gives the y -axis intercept.
 - The equation of the axis of symmetry is $x = -\frac{b}{2a}$
 - The x -axis intercepts are determined by solving the equation $ax^2 + bx + c = 0$
- A quadratic equation may be solved by:
 - factorising
 - completing the square
 - using the general quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Using the discriminant of the quadratic function $f(x) = ax^2 + bx + c$:
 - If $b^2 - 4ac > 0$, the graph of the function has two x -axis intercepts.
 - If $b^2 - 4ac = 0$, the graph of the function touches the x -axis.
 - If $b^2 - 4ac < 0$, the graph of the function does not intercept the x -axis.
- By completing the square, any quadratic function can be written in the form $y = A(x + b)^2 + B$
 From this it can be seen that the graph of any quadratic function may be obtained by a composition of transformations applied to the graph of $y = x^2$.

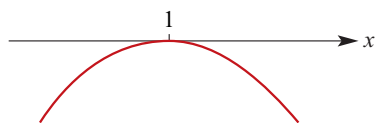
Multiple-choice questions

- 1 The equation $5x^2 - 10x - 2$ in turning point form $a(x + h)^2 + k$ by completing the square is:
- A** $(5x + 1)^2 + 5$ **B** $(5x - 1)^2 - 5$ **C** $5(x - 1)^2 - 5$
D $5(x + 1)^2 - 2$ **E** $5(x - 1)^2 - 7$

- 2 The value(s) of m that will give the equation $mx^2 + 6x - 3 = 0$ two real roots is/are:
A $m = -3$ **B** $m = 3$ **C** $m = 0$ **D** $m > -3$ **E** $m < -3$
- 3 $x^3 + 27$ factorised over R is equal to:
A $(x + 3)^3$ **B** $(x - 3)^3$ **C** $(x + 3)(x^2 - 6x + 9)$
D $(x - 3)(x^2 + 3x + 9)$ **E** $(x + 3)(x^2 - 3x + 9)$
- 4 The equation of the graph shown is:

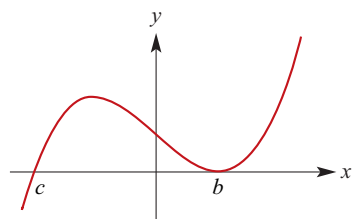


- A** $y = x(x - 2)(x + 4)$ **B** $y = x(x + 2)(x - 4)$ **C** $y = (x + 2)^2(x - 4)$
D $y = (x + 2)(x - 4)^2$ **E** $y = (x + 2)^2(x - 4)^2$
- 5 $x - 1$ is a factor of $x^3 + 3x^2 - 2ax + 1$. The value of a is:
A 2 **B** 5 **C** $\frac{2}{5}$ **D** $-\frac{2}{5}$ **E** $\frac{5}{2}$
- 6 $6x^2 - 8xy - 8y^2$ is equal to:
A $(3x + 2y)(2x - 4y)$ **B** $(3x - 2y)(6x + 4y)$ **C** $(6x - 4y)(x + 2y)$
D $(3x - 2y)(2x + 4y)$ **E** $(6x + y)(x - 8y)$
- 7 A part of the graph of the third-degree polynomial function $f(x)$, near the point $(1, 0)$, is shown below.



Which of the following could be the rule for $f(x)$?

- A** $f(x) = x^2(x - 1)$ **B** $f(x) = (x - 1)^3$ **C** $f(x) = -x(x - 1)^2$
D $f(x) = x(x - 1)^2$ **E** $f(x) = -x(x + 1)^2$
- 8 The coordinates of the turning point of the graph of the function $p(x) = 3((x - 2)^2 + 4)$ are:
A $(-2, 12)$ **B** $(-2, 4)$ **C** $(2, -12)$ **D** $(2, 4)$ **E** $(2, 12)$
- 9 The diagram below shows part of the graph of a polynomial function.



A possible equation for the rule of the function is:

A $y = (x + c)(x - b)^2$

B $y = (x - b)(x - c)^2$

C $y = (x - c)(b - x)^2$

D $y = -(x - c)(b - x)^2$

E $y = (x + b)^2(x - c)$

- 10** The number of roots to the equation $(x^2 + a)(x - b)(x + c) = 0$, where a, b and $c \in \mathbb{R}^+$, is:

A 0

B 1

C 2

D 3

E 4

Short-answer questions (technology-free)

- 1** Sketch the graphs of each of the following quadratic functions. Clearly indicate coordinates of the vertex and axes intercepts.

a $h(x) = 3(x - 1)^2 + 2$

b $h(x) = (x - 1)^2 - 9$

c $f(x) = x^2 - x + 6$

d $f(x) = x^2 - x - 6$

e $f(x) = 2x^2 - x + 5$

f $h(x) = 2x^2 - x - 1$

- 2** The points with coordinates $(1, 1)$ and $(2, 5)$ lie on a parabola with equation of the form $y = ax^2 + b$. Find the values of a and b .

- 3** Solve the equation $3x^2 - 2x - 10 = 0$ by using the quadratic formula.

- 4** Sketch the graphs of each of the following. State the coordinates of the point of zero gradient and the axes intercepts:

a $f(x) = 2(x - 1)^3 - 16$

b $g(x) = -(x + 1)^3 + 8$

c $h(x) = -(x + 2)^3 - 1$

d $f(x) = (x + 3)^3 - 1$

e $f(x) = 1 - (2x - 1)^3$

- 5** Draw a sign diagram for each of the following:

a $y = (x + 2)(2 - x)(x + 1)$

b $y = (x - 3)(x + 1)(x - 1)$

c $y = x^3 + 7x^2 + 14x + 8$

d $y = 3x^3 + 10x^2 + x - 6$

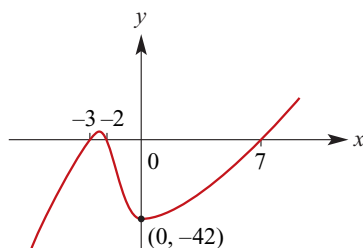
- 6** Without actually dividing, find the remainder when the first polynomial is divided by the second:

a $x^3 + 3x^2 - 4x + 2, x + 1$

b $x^3 - 3x^2 - x + 6, x - 2$

c $2x^3 + 3x^2 - 3x - 2, x + 2$

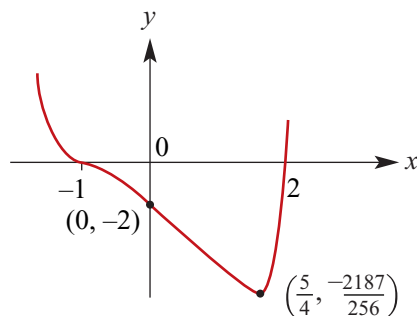
- 7** Determine the rule for the cubic function shown in the graphs:



- 8 The graph of $f(x) = (x + 1)^3(x - 2)$ is shown.

Sketch the graph of:

- a $y = f(x - 1)$
- b $y = f(x + 1)$
- c $y = f(2x)$
- d $y = f(x) + 2$



- 9 Find the rule for the cubic function, the graph of which passes through the points $(1, 1)$, $(2, 4)$, $(3, 9)$ and $(0, 6)$.

Extended-response questions

- 1 The rate of flow of water, R mL/min, into a vessel is described by the quartic expression $R = kt^3(20 - t)$, $0 \leq t \leq 20$, where t minutes is the time elapsed from the beginning of the flow. The graph is shown.

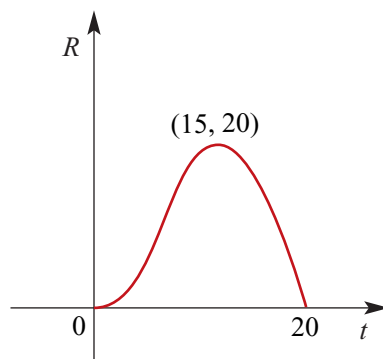
- a Find the value of k .
- b Find the rate of flow when $t = 10$.
- c The flow is adjusted so that a new expression for flow is:

$$R_{\text{new}} = 2kt^3(20 - t), 0 \leq t \leq 20$$

- i Sketch the graph of R_{new} against t for $0 \leq t \leq 20$.
- ii Find the rate of flow when $t = 10$.
- d Water is allowed to run from the vessel and it is found that the rate of flow from the vessel is given by

$$R_{\text{out}} = -k(t - 20)^3(40 - t) \text{ for } 20 \leq t \leq 40$$

- i Sketch the graph of R_{out} against t for $20 \leq t \leq 40$.
 - ii Find the rate of flow when $t = 30$.
- Hint:* Note that the graph of R_{new} against t is given by a dilation of factor 2 from the x -axis. The graph of R_{out} against t is given by a translation with rule $(t, R) \rightarrow (t + 20, R)$, followed by a reflection in the t -axis.



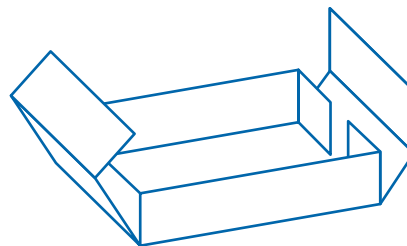
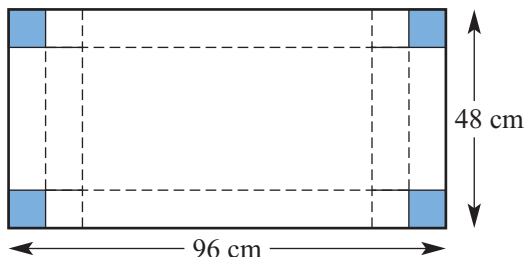
- 2 A large gas container is being deflated. The volume V (in m^3) at time t hours is given by:

$$V = 4(9 - t)^3 \text{ where } 0 \leq t \leq 9$$

- a Find the volume when:
 - i $t = 0$
 - ii $t = 9$
- b Sketch the graph of V against t for $0 \leq t \leq 9$.
- c At what time is the volume 512 m^3 ?



- 3** A reinforced box is made by cutting congruent squares of side length x cm from the four corners of a rectangular piece of cardboard that measures 48 cm by 96 cm. The flaps are folded up.

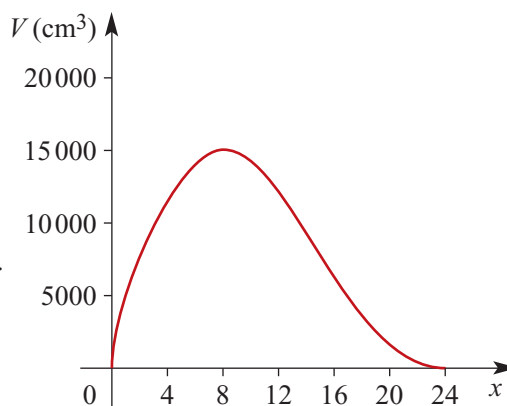


- a** Find an expression for V , the volume of the box formed.

- b** A graph of V against x is as shown.

- i** What is the domain of the function V ?
ii From the graph, find the maximum volume of the box and the value of x for which this occurs (approximate values are required).

- c** Find the volume of the box when $x = 10$.
d It is decided that $0 \leq x \leq 5$. Find the maximum volume possible.
e If $5 \leq x \leq 15$, what is the minimum volume of the box?



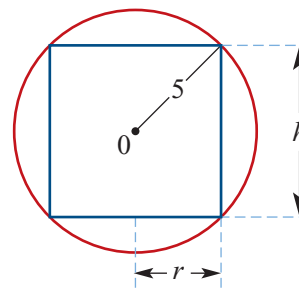
- 4** A hemispherical bowl of radius 6 cm contains water. The volume of water in the hemispherical bowl where the depth of the water is x cm and is given by:

$$V = \frac{1}{3}\pi x^2(18 - x) \text{ cm}^3$$

- a** Find the volume of water when:
i $x = 2$ **ii** $x = 3$ **iii** $x = 4$
b Find the volume when the hemispherical bowl is full.
c Sketch the graph of V against x .
d Find the depth of water when the volume is equal to $\frac{325\pi}{3} \text{ cm}^3$.

- 5** A metal worker is required to cut a circular cylinder from a solid sphere of radius 5 cm. A cross-section of the sphere and the cylinder is shown in the diagram.

- a** Express r in terms of h , where r cm is the radius of the cylinder and h cm is the height of the cylinder. Hence show that the volume, $V \text{ cm}^3$, of the cylinder is given by $V = \frac{1}{4}\pi h(100 - h^2)$.



- b** Sketch the graph of V against h for $0 < h < 10$.

The coordinates of the maximum point are $\approx (5.77, 302.3)$.

- c** Find the volume of the cylinder if $h = 6$.

- d** Find the height and radius of the cylinder if the volume of the cylinder is $48\pi \text{ cm}^3$.

- 6** An open tank is to be made from a sheet of metal 84 cm by 40 cm by cutting congruent squares of side length x cm from each of the corners.

- a** Find the volume $V \text{ cm}^3$ of the box in terms of x .

- b** State the maximal domain for V when it is considered as a function of x .

- c** Plot the graph of V against x using a calculator.

- d** Find the volume of the tank when:

i $x = 2$

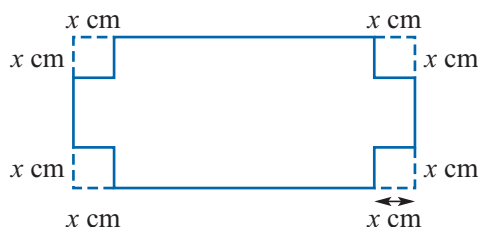
ii $x = 6$

iii $x = 8$

iv $x = 10$

- e** Find the value(s) of x , correct to two decimal places, for which the capacity of the tank is 10 litres.

- f** Find, correct to two decimal places, the maximum capacity of the tank in cubic centimetres.



- 7** The rectangle is defined by vertices B and C on the curve with equation $y = 16 - x^2$ and vertices A and D on the x -axis.

- a i** Find the area, A , of the rectangle in terms of x .

- ii** State the implied domain for the function defined by the rule given in **i**.

- b** Find:

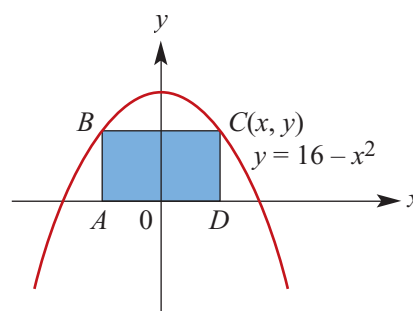
- i** the value of A when $x = 3$

- ii** the value, correct to two decimal places, of x when $A = 25$

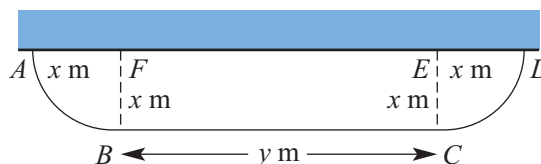
- c** A cuboid has volume V given by the rule $V = xA$

- i** Find V in terms of x .

- ii** Find the value, correct to two decimal places, of x such that $V = 100$.



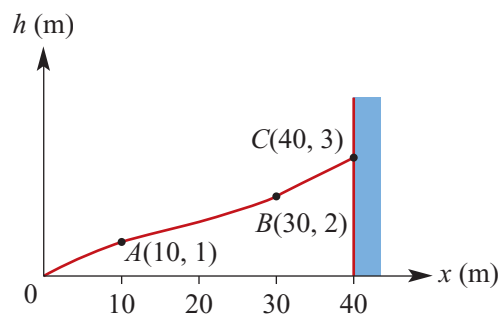
- 8** The plan of a garden adjoining a wall is shown. The rectangle $BCEF$ is of length y m and width x m. The borders of the two end sections are quarter circles of radius x m and centres at E and F . A fence is erected along the curves AB and CD and the straight line CB .



- a Find the area A of the garden in terms of x and y .
- b If the length of the fence is 100 m, find:
 - i y in terms of x
 - ii A in terms of x
 - iii the maximal domain of the function for which the rule has been obtained in ii
- c Find, correct to two decimal places, the value(s) of x if the area of the garden is to be 1000 m^2 .
- d It is decided to build the garden up to a height of $\frac{x}{50}$ metres. If the length of the fence is 100 m, find correct to two decimal places:
 - i the volume $V \text{ m}^3$ of soil needed in terms of x
 - ii the volume $V \text{ m}^3$ of soil needed for a garden of area 1000 m^2
 - iii the value(s) of x for which 500 m^3 of soil is required



- 9 A mound of earth is piled up against a wall. The cross-section is as shown. The coordinates of several points on the surface are given.



- a Find the equation for the cubic function for which the graph passes through the points O , A , B and C .
- b For what value of x is the height of the mound 1.5 metres?
- c The coefficient of x^3 for the function is 'small'. Consider the quadratic formed when the x^3 term is deleted. Compare the graph of the resulting quadratic function with the graph of the cubic function.
- d The mound moves and the curve describing the cross-section now passes through the points $O(0, 0)$, $A(10, 0.3)$, $B(30, 2.7)$ and $D(40, 2.8)$.

Find the equation of the cubic function for which the graph passes through these points.

- e Let $y = f(x)$ be the function obtained in a.
 - i Sketch the graph of the hybrid function:

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq 40 \\ f(80 - x) & 40 < x \leq 80 \end{cases}$$

- ii Comment on the appearance of the graph of $y = g(x)$

Exponential and logarithmic functions

Objectives

- To graph **exponential** and **logarithmic functions**.
- To graph transformations of the graphs of **exponential** and **logarithmic functions**.
- To introduce **Euler's number**.
- To revise the **index** and **logarithm laws**.
- To solve **exponential** and **logarithmic equations**.
- To find rules for the graphs of **exponential** and **logarithmic functions**.
- To find inverses of **exponential** and **logarithmic functions**.
- To apply **exponential functions** to physical occurrences of exponential growth and decay.

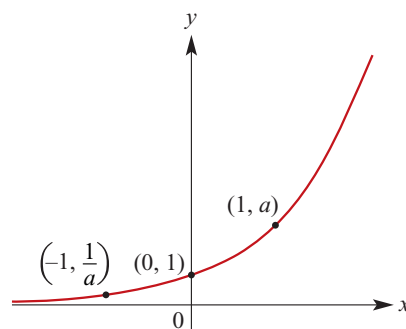
5.1 Exponential functions

The function, $f(x) = a^x$, where $a \in \mathbb{R}^+ \setminus \{1\}$, is called an **exponential function** (or index function).

The graph of $f(x) = a^x$ is shown.

The features of the graph of the exponential function with rule $f(x) = a^x$ are:

- $f(-1) = \frac{1}{a}$
- $f(0) = 1$
- $f(1) = a$
- The x -axis is a horizontal asymptote.
As $x \rightarrow -\infty$, $f(x) \rightarrow 0^+$.
- The maximal domain is $\mathbb{R}$.
- The range of the function is $\mathbb{R}^+$.
- An exponential function is a one-to-one function.



Graphing transformations of the graph of $f(x) = a^x$

Translations

If the transformation, a translation with mapping $(x, y) \rightarrow (x + h, y + k)$, is applied to the graph of $y = a^x$, the image has equation $y = a^{x-h} + k$. The horizontal asymptote has equation $y = k$. The images of the points with coordinates $\left(-1, \frac{1}{a}\right)$, $(0, 1)$ and $(1, a)$ are $\left(-1 + h, \frac{1}{a} + k\right)$, $(h, 1 + k)$ and $(1 + h, a + k)$ respectively. The range of the image is (k, ∞) .

Example 1

Sketch the graph, and state the range, of $y = 2^{x-1} + 2$

Solution

A translation of 1 unit in the positive direction of the x -axis and 2 units in the positive direction of the y -axis is applied to the graph of $y = 2^x$

The equation of the asymptote is $y = 2$

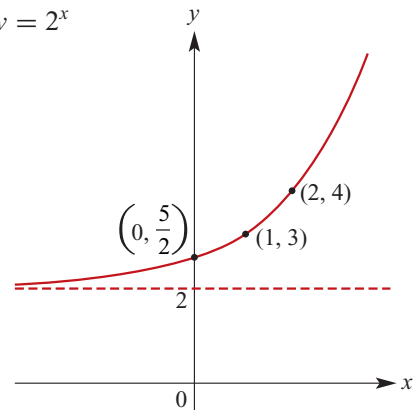
The mapping is $(x, y) \rightarrow (x + 1, y + 2)$

$$\left(-1, \frac{1}{2}\right) \rightarrow \left(0, \frac{5}{2}\right)$$

$$(0, 1) \rightarrow (1, 3)$$

$$(1, 2) \rightarrow (2, 4)$$

The range of the function is $(2, \infty)$.



Reflections

If the transformation, a **reflection in the x -axis** determined by the mapping $(x, y) \rightarrow (x, -y)$, is applied to the graph of $y = a^x$, the image has equation $y = -a^x$. The horizontal asymptote has equation $y = 0$. The images of the points with coordinates $\left(-1, \frac{1}{a}\right)$, $(0, 1)$ and $(1, a)$ are $\left(-1, -\frac{1}{a}\right)$, $(0, -1)$ and $(1, -a)$ respectively. The range of the image is $(-\infty, 0)$.

Example 2

Sketch the graph of $y = -3^x$

Solution

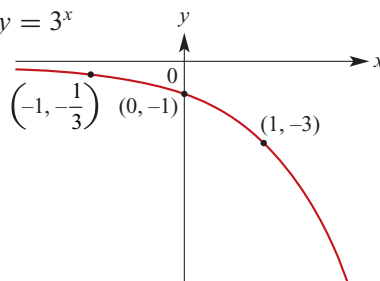
A reflection in the x -axis is applied to the graph of $y = 3^x$

The mapping is $(x, y) \rightarrow (x, -y)$

$$\left(-1, \frac{1}{3}\right) \rightarrow \left(-1, -\frac{1}{3}\right)$$

$$(0, 1) \rightarrow (0, -1)$$

$$(1, 3) \rightarrow (1, -3)$$



If the transformation, a **reflection in the y -axis** determined by the mapping $(x, y) \rightarrow (-x, y)$, is applied to the graph of $y = a^x$, the image has equation $y = a^{-x}$ (or $y = \frac{1}{a^x}$ or $y = \left(\frac{1}{a}\right)^x$). The horizontal asymptote has equation $y = 0$. The images of the points with coordinates $\left(-1, \frac{1}{a}\right)$, $(0, 1)$ and $(1, a)$ are $\left(1, \frac{1}{a}\right)$, $(0, 1)$ and $(-1, a)$ respectively. The range of the image is $(0, \infty)$.

Example 3

Sketch the graph of $y = 6^{-x}$

Solution

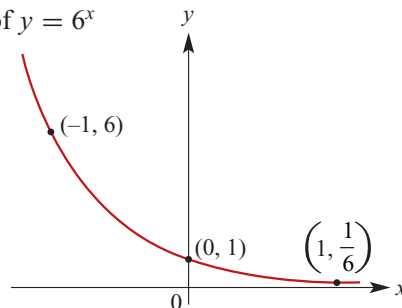
A reflection in the y -axis is applied to the graph of $y = 6^x$

The mapping is $(x, y) \rightarrow (-x, y)$

$$\left(-1, \frac{1}{6}\right) \rightarrow \left(1, \frac{1}{6}\right)$$

$$(0, 1) \rightarrow (0, 1)$$

$$(1, 6) \rightarrow (-1, 6)$$

**Dilations**

If the transformation, a **dilation of factor k ($k > 0$) from the x -axis** determined by the mapping $(x, y) \rightarrow (x, ky)$, is applied to the graph of $y = a^x$, the image has equation $y = ka^x$. The horizontal asymptote has equation $y = 0$. The images of the points with coordinates $\left(-1, \frac{1}{a}\right)$, $(0, 1)$ and $(1, a)$ are $\left(-1, \frac{k}{a}\right)$, $(0, k)$ and $(1, ka)$ respectively. The range of the image is $(0, \infty)$.

Example 4

Sketch the graph of each of the following:

a $y = 3(5)^x$

b $y = (0.2)(8)^x$

Solution

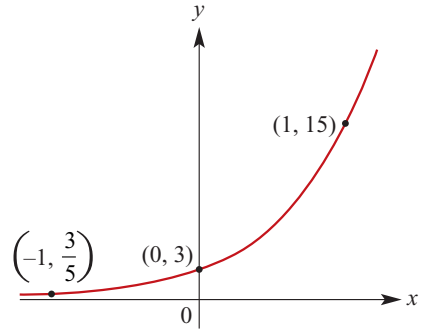
- a** A dilation of factor 3 from the x -axis is applied to the graph of $y = 5^x$

The mapping is $(x, y) \rightarrow (x, 3y)$

$$\left(-1, \frac{1}{5}\right) \rightarrow \left(-1, \frac{3}{5}\right)$$

$$(0, 1) \rightarrow (0, 3)$$

$$(1, 5) \rightarrow (1, 15)$$



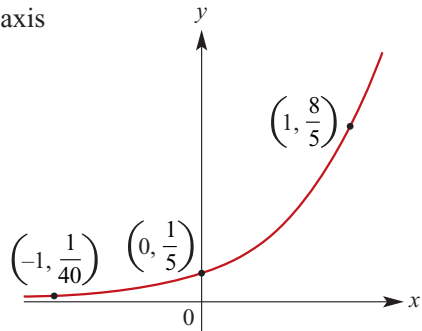
- b** A dilation of factor 0.2 (or $\frac{1}{5}$) from the x -axis is applied to the graph of $y = 8^x$

The mapping is $(x, y) \rightarrow \left(x, \frac{1}{5}y\right)$

$$\left(-1, \frac{1}{8}\right) \rightarrow \left(-1, \frac{1}{40}\right)$$

$$(0, 1) \rightarrow \left(0, \frac{1}{5}\right)$$

$$(1, 8) \rightarrow \left(1, \frac{8}{5}\right)$$



If the transformation, a **dilation of factor k ($k > 0$) from the y -axis** determined by the mapping $(x, y) \rightarrow (x, ky)$, is applied to the graph of $y = a^x$, the image has equation $y = a^{\frac{x}{k}}$. The horizontal asymptote has equation $y = 0$. The images of the points with coordinates $\left(-1, \frac{1}{a}\right)$, $(0, 1)$ and $(1, a)$ are $\left(-k, \frac{1}{a}\right)$, $(0, 1)$ and (k, a) respectively. The range of the image is $(0, \infty)$.

Example 5

Sketch the graph of each of the following:

a $y = 9^{\frac{x}{2}}$

b $y = 2^{3x}$

Solution

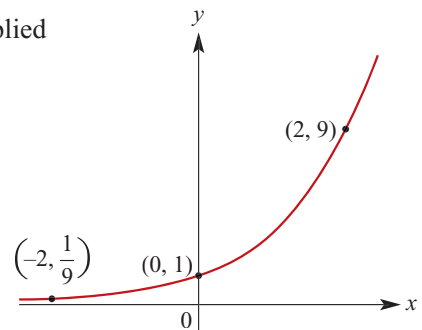
- a** A dilation of factor 2 from the y -axis is applied to the graph of $y = 9^x$

The mapping is $(x, y) \rightarrow (x, 2y)$

$$\left(-1, \frac{1}{9}\right) \rightarrow \left(-1, \frac{2}{9}\right)$$

$$(0, 1) \rightarrow (0, 2)$$

$$(1, 9) \rightarrow (1, 18)$$



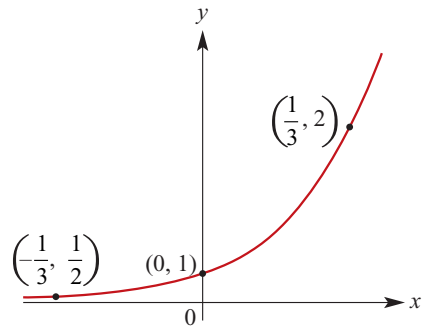
- b** A dilation of factor $\frac{1}{3}$ from the y -axis is applied to the graph of $y = 2^x$

The mapping is $(x, y) \rightarrow \left(\frac{1}{3}x, y\right)$

$$\left(-1, \frac{1}{2}\right) \rightarrow \left(-\frac{1}{3}, \frac{1}{2}\right)$$

$$(0, 1) \rightarrow (0, 1)$$

$$(1, 2) \rightarrow \left(\frac{1}{3}, 2\right)$$



Combinations of transformations

Example 6

Sketch the graph, and state the range, of each of the following:

a $y = 2^{-x} + 3$

b $y = 4^{3x} - 1$

c $y = -10^{x-1} - 2$

Solution

- a** The transformations, a reflection in the y -axis and a translation of 3 units in the positive direction of the y -axis,

are applied to the graph of $y = 2^x$

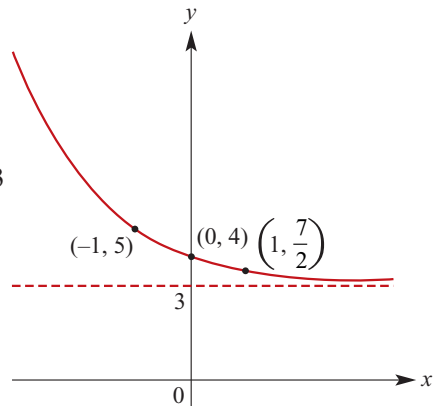
The equation of the asymptote is $y = 3$

The mapping is $(x, y) \rightarrow (-x, y + 3)$

$$\left(-1, \frac{1}{2}\right) \rightarrow \left(1, \frac{7}{2}\right)$$

$$(0, 1) \rightarrow (0, 4)$$

$$(1, 2) \rightarrow (-1, 5)$$



The range of the function is $(3, \infty)$.

- b** The transformations, a dilation of factor $\frac{1}{3}$ from the y -axis followed by a translation of 1 unit in the negative direction of the y -axis, are applied to the graph of $y = 4^x$

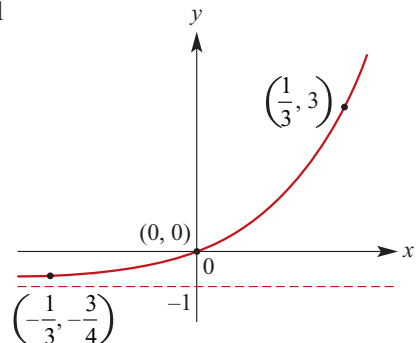
The equation of the asymptote is $y = -1$

The mapping is $(x, y) \rightarrow \left(\frac{1}{3}x, y - 1\right)$

$$\left(-1, \frac{1}{4}\right) \rightarrow \left(-\frac{1}{3}, -\frac{3}{4}\right)$$

$$(0, 1) \rightarrow (0, 0)$$

$$(1, 4) \rightarrow \left(\frac{1}{3}, 3\right)$$



The range of the function is $(-1, \infty)$.

- c** The transformations, a reflection in the x -axis followed by a translation of 1 unit in the positive direction of the x -axis and 2 units in the negative direction of the y -axis, are applied to the graph of $y = 10^x$.

The equation of the asymptote is $y = -2$

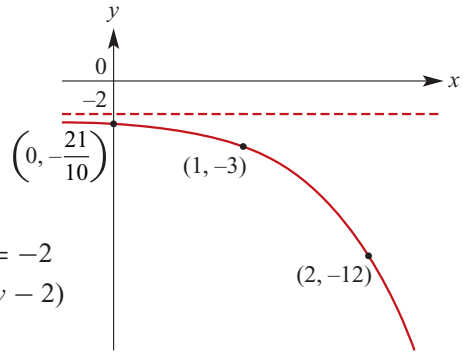
The mapping is $(x, y) \rightarrow (x + 1, -y - 2)$

$$\left(-1, \frac{1}{10}\right) \rightarrow \left(0, -\frac{21}{10}\right)$$

$$(0, 1) \rightarrow (1, -3)$$

$$(1, 10) \rightarrow (2, -12)$$

The range of the function is $(-\infty, -2)$.



Exercise 5A

- For each of the following, use the one set of axes to sketch the graphs (and label asymptotes) of:

a $y = 2^x$ and $y = 3^x$	b $y = 2^{-x}$ and $y = 3^{-x}$
c $y = 5^x$ and $y = -5^x$	d $y = (1.5)^x$ and $y = -(1.5)^x$
- Sketch the graph of each of the following (labelling asymptotes), and state the range of each:

a $y = 2^{x+1} - 2$	b $y = \left(\frac{1}{2}\right)^{x-2}$	c $y = \left(\frac{1}{2}\right)^x - 1$
d $y = \left(\frac{1}{2}\right)^{x-2} + 1$	e $y = 2^{x-2} + 2$	f $y = \left(\frac{1}{2}\right)^{x-2} + 2$
- Sketch the graph of each of the following (labelling asymptotes), and state the range of each:

a $y = 3^x$	b $y = 3^x + 1$	c $y = 1 - 3^x$
d $y = \left(\frac{1}{3}\right)^x$	e $y = 3^{-x} + 2$	f $y = \left(\frac{1}{3}\right)^x - 1$
- For $f(x) = 2^x$, sketch the graph of each of the following, labelling asymptotes where appropriate:

a $y = f(x + 1)$	b $y = f(x) + 1$	c $y = f(-x) + 2$	d $y = -f(x) - 1$
e $y = f(3x)$	f $y = f\left(\frac{x}{2}\right)$	g $y = 2f(x - 1) + 1$	h $y = f(x - 2)$
- Sketch the graph of each of the following (labelling asymptotes), and state the range of each:

a $y = 10^x - 1$	b $y = 10^{\frac{x}{10}} + 1$	c $y = 2 \times 10^x - 20$
d $y = 1 - 10^{-x}$	e $y = 10^{x+1} + 3$	f $y = 2 \times 10^{\frac{x}{10}} + 4$

- 6 A bank offers cash loans at .04% interest per day compounded daily. A loan of \$10 000 is taken and the interest payable at the end of x days is given by $C_1 = 10\,000 [(1.0004)^x - 1]$
- Plot the graph of C_1 against x .
 - Find the interest at the end of:
 - 100 days
 - 300 days
 - After how many days is the interest payable \$1000?
 - A loan company offers \$10 000 with a charge of \$4.25 a day being made. The amount charged after x days is given by $C_2 = 4.25x$
 - Plot the graph of C_2 against x (using the same window as in **a**).
 - Find the smallest value of x for which $C_2 < C_1$.
- 7 If you invest \$100 at 2% per day, compounding daily, the amount of money you would have after x days is given by $y = 100(1.02)^x$ dollars. For how many days would you have to invest to double your money?
- 8 **a**
 - Graph $y = 2^x$, $y = 3^x$ and $y = 5^x$ on the same set of axes.
 - For what values of x is $2^x > 3^x > 5^x$?
 - For what values of x is $2^x < 3^x < 5^x$?
 - For what values of x is $2^x = 3^x = 5^x$?**b** Repeat part **a** for $y = \left(\frac{1}{2}\right)^x$, $y = \left(\frac{1}{3}\right)^x$ and $y = \left(\frac{1}{5}\right)^x$
- c** Use your answers to parts **a** and **b** to sketch the graph of $y = a^x$ for:
- $a > 1$
 - $a = 1$
 - $0 < a < 1$

5.2 The exponential function, $f(x) = e^x$

In the previous section the family of exponential functions $f(x) = a^x$, $a \in \mathbb{R}^+ \setminus \{1\}$, was explored. One member of this family is of such importance in mathematics that it is known as the **exponential function**. This function has the rule $f(x) = e^x$, where e is Euler's number, named after an eighteenth century Swiss mathematician.

Euler's number is defined as:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

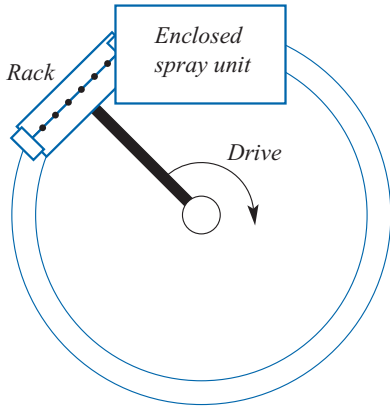
To see what the value of e might be, we could try large values of n and a calculator to evaluate $\left(1 + \frac{1}{n}\right)^n$.

Try $n = 100$	then $\left(1 + \frac{1}{100}\right)^{100} = (1.01)^{100} = 2.7048\dots$
$n = 1000$	$(1.001)^{1000} = 2.7169\dots$
$n = 10\,000$	$(1.0001)^{10\,000} = 2.7181\dots$
$n = 100\,000$	$(1.00001)^{100\,000} = 2.71826\dots$
$n = 1\,000\,000$	$(1.000001)^{1\,000\,000} = 2.71828\dots$

As n is taken larger and larger it can be seen that $\left(1 + \frac{1}{n}\right)^n$ approaches a limiting value (≈ 2.71828). Like π , e is irrational: $e = 2.7182818284590452353\dots$

Investigation into the production of glass marbles

A method of producing high quality glass marbles has been proposed. A rack holding small silica ‘cones’ threaded on a wire will circulate around the track as shown in the diagram. When the rack enters the spray unit it will be subjected to a fine spray of a liquid glass substance. It takes 1 minute to produce a marble.



A marble produced by a single passage around the unit will take 1 minute and the volume will be increased by 100%, i.e. doubled. However, such a large increase in volume, at this slow speed, will tend to produce misshapen marbles. This suggests that the rack should be speeded up. We shall investigate what happens to the volume of the marble as the rack is speeded up and try to answer the question, ‘Is there a maximum volume reached if the rack speeds up indefinitely?’

Let V = volume of the marble at time t .

Also let the original marble volume equal V_0 .

For 1 passage per minute $V = 2 \times V_0$.

Now assume that if the rack is speeded up to do 2 passages/minute then the growth in volume is 50% for each passage; that is:

$$V = \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{2}\right) \quad V = \left(1 + \frac{1}{2}\right)^2 \quad V = 2.25 V_0$$

and similarly:

$$\text{for 4 passages,} \quad V = \left(1 + \frac{1}{4}\right)^4 V_0 = 2.441 \dots V_0$$

$$\text{for 8 passages,} \quad V = \left(1 + \frac{1}{8}\right)^8 V_0 = 2.565 \dots V_0$$

$$\text{for 16 passages,} \quad V = \left(1 + \frac{1}{16}\right)^{16} V_0 = 2.637 \dots V_0$$

$$\text{for 64 passages,} \quad V = \left(1 + \frac{1}{64}\right)^{64} V_0 = 2.697 \dots V_0$$

$$\text{for } n \text{ passages,} \quad V = \left(1 + \frac{1}{n}\right)^n V_0$$

As the rack speeds up, n is taken larger and larger, and it can be seen that $\left(1 + \frac{1}{n}\right)^n$ approaches a limiting value, i.e.

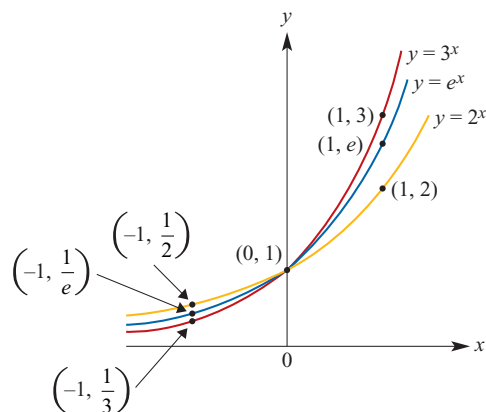
$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n V_0 \\ &= e V_0 \end{aligned}$$

So the maximum volume of the marble if the rack speeds up indefinitely is $e V_0$.

Graphing $y = e^x$

The graph of $y = e^x$ is as shown.

The graphs of $y = 2^x$ and $y = 3^x$ are shown on the same set of axes:



Exercise 5B

- Sketch the graph of each of the following and state the range:

a $f(x) = e^x + 1$	b $f(x) = 1 - e^x$	c $f(x) = 1 - e^{-x}$
d $f(x) = e^{-2x}$	e $f(x) = e^{x-1} - 2$	f $f(x) = 2e^x$
g $h(x) = 2(1 + e^x)$	h $h(x) = 2(1 - e^{-x})$	i $g(x) = 2e^{-x} + 1$
j $h(x) = 2e^{x-1}$	k $f(x) = 3e^{x+1} - 2$	l $h(x) = 2 - 3e^x$
- Solve each of the following equations using a calculator. Give answers correct to three decimal places.

a $e^x = x + 2$	b $e^{-x} = x + 2$	c $x^2 = e^x$	d $x^3 = e^x$
------------------------	---------------------------	----------------------	----------------------
- Using a calculator plot the graph of $y = f(x)$ where $f(x) = e^x$
 - Using the same screen plot the graphs of:

i $y = f(x - 2)$	ii $y = f\left(\frac{x}{3}\right)$	iii $y = f(-x)$
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5.3 Exponential equations

In this section the one-to-one property of exponential functions is exploited to solve exponential equations. This property can be stated as:

$$a^x = a^y \text{ implies } x = y$$

Example 7

Find the value of x for which:

- a** $4^x = 256$ **b** $3^{x-1} = 81$

Solution

a $4^x = 256$
 $4^x = 4^4$
 $\therefore x = 4$

b $3^{x-1} = 81$
 $3^{x-1} = 3^4$
 $\therefore x - 1 = 4$
 $x = 5$

Index laws

The solution of equations may also require an application of one or more of the index laws and these are stated here:

- To multiply two numbers in exponent form with the same base, **add** the exponents:

$$a^m \times a^n = a^{m+n}$$

- To divide two numbers in exponent form with the same base, **subtract** the exponents:

$$a^m \div a^n = a^{m-n}$$

- To raise the power of a to another power, **multiply** the exponents:

$$(a^m)^n = a^{m \times n}$$

$$a^0 = 1$$

Example 8

Find the value of x for which:

a $5^{2x-4} = 25^{-x+2}$

b $9^x = 12 \times 3^x - 27$

Solution

$$\begin{aligned} \text{a} \quad 5^{2x-4} &= 25^{-x+2} \\ &= (5^2)^{-x+2} \\ &= 5^{-2x+4} \\ \therefore 2x-4 &= -2x+4 \\ 4x &= 8 \\ x &= 2 \end{aligned}$$

$$\begin{aligned} \text{b} \quad (3^x)^2 &= 12 \times 3^x - 27 \\ \text{Let } y &= 3^x. \\ \therefore y^2 &= 12y - 27 \\ y^2 - 12y + 27 &= 0 \\ (y-3)(y-9) &= 0 \\ \therefore y-3 &= 0 \text{ or } y-9 = 0 \\ y &= 3 \text{ or } y = 9 \\ \therefore 3^x &= 3 \text{ or } 3^x = 3^2 \\ x &= 1 \text{ or } x = 2 \end{aligned}$$

Exercise 5C



- 1 Simplify the following expressions:

a $3x^2y^3 \times 2x^4y^6$

b $\frac{12x^8}{4x^2}$

c $18x^2y^3 \div 3x^4y$

d $(4x^4y^2)^2 + 2(x^2y)^4$

e $(4x^0)^2$

f $15(x^5y^{-2})^4 \div 3(x^4y)^{-2}$

g $\frac{3(2x^2y^3)^4}{2x^3y^2}$

h $(8x^3y^6)^{\frac{1}{3}}$

i $\frac{x^2 + y^2}{x^{-2} + y^{-2}}$

- 2 Solve for x in each of the following:

a $3^x = 81$

b $81^x = 9$

c $4^x = 256$

d $625^x = 5$

e $32^x = 8$

f $5^x = 125$

g $16^x = 1024$

h $2^{-x} = \frac{1}{64}$

i $5^{-x} = \frac{1}{625}$

- 3 Solve for n in each of the following:

a $5^{2n} \times 25^{2n-1} = 625$

b $4^{2n-2} = 1$

c $4^{2n-1} = \frac{1}{256}$

$$\text{d } \frac{3^{n-2}}{9^{2-n}} = 27$$

$$\text{g } 27^{n-2} = 9^{3n+2}$$

$$\text{j } 2^{n-1} \times 4^{2n+1} = 16$$

$$\text{e } 2^{2n-2} \times 4^{-3n} = 64$$

$$\text{h } 8^{6n+2} = 8^{4n-1}$$

$$\text{k } (27 \times 3^n)^n = 27^n \times 3^{\frac{1}{4}}$$

$$\text{f } 2^{n-4} = 8^{4-n}$$

$$\text{i } 125^{4-n} = 5^{6-2n}$$

4 Solve for x :

$$\text{a } 3^{2x} - 12(3^x) + 27 = 0$$

$$\text{b } 3^{2x} - 2(3^x) - 3 = 0$$

$$\text{c } 5^{2x} - 23(5^x) - 50 = 0$$

$$\text{d } 5^{2x} - 10(5^x) + 25 = 0$$

$$\text{e } 2^{2x} = 6(2^x) - 8$$

$$\text{f } 8(3^x) - 6 = 2(3^{2x})$$

$$\text{g } 2^{2x} - 20(2^x) = -64$$

$$\text{h } 4^{2x} - 5(4^x) = -4$$

$$\text{i } 3(3^{2x}) = 28(3^x) - 9$$

$$\text{j } 7(7^{2x}) = 8(7^x) - 1$$

5.4 Logarithmic functions

The exponential function $f(x) = a^x$, where $a \in \mathbb{R}^+ \setminus \{1\}$, is a one-to-one function. Therefore, there exists an inverse function (see Section 1.7). To find the rule of the inverse function, do the following.

Let $x = a^y$

Therefore $y = \log_a x$

Therefore $f^{-1}(x) = \log_a x$

The following definition was used to find the rule of the inverse:

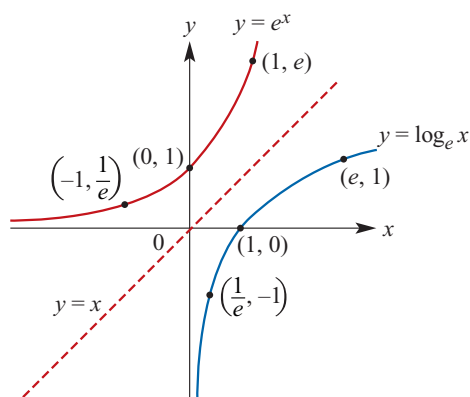
$$\log_a x = y \text{ if } a^y = x$$

For example:

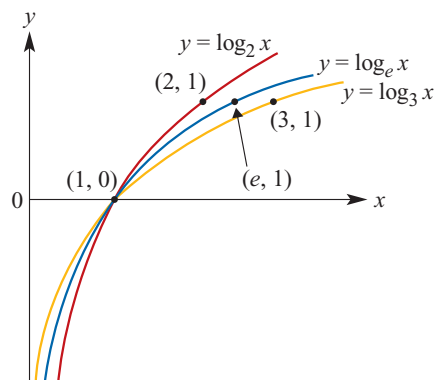
$$\log_2 8 = 3 \text{ is equivalent to the statement } 2^3 = 8$$

$$\log_{10} 0.1 = -1 \text{ is equivalent to the statement } 10^{-1} = 0.1$$

The graphs of $y = e^x$ and its inverse function $y = \log_e x$ are shown on the one set of axes.



The graphs of $y = \log_2 x$, $y = \log_e x$ and $y = \log_3 x$ are shown on the one set of axes.



The features of the graph of the logarithmic function with rule $f(x) = \log_a x$ are:

■ $f(1) = 0$

■ $f(a) = 1$

- The y -axis is a vertical asymptote. As $x \rightarrow 0^+$, $f(x) \rightarrow -\infty$.
- The maximal domain is R^+ .
- The range of the function is R .
- A logarithmic function is a one-to-one function.

Note: The function with rule $f(x) = \log_e x$ is known as the natural logarithm function.

Logarithm laws

We use the index laws to establish rules for computations with logarithms.

- Let $a^x = m$ and $a^y = n$, where m, n and a are positive real numbers.

$$\begin{aligned}\therefore mn &= a^x \times a^y \\ &= a^{x+y}\end{aligned}$$

$\log_a(mn) = x + y$ and since $x = \log_a m$ and $y = \log_a n$ it follows that:

$$\log_a(mn) = \log_a m + \log_a n$$

For example:

$$\begin{aligned}\log_{10} 200 + \log_{10} 5 &= \log_{10} (200 \times 5) \\ &= \log_{10} (1000) \\ &= 3\end{aligned}$$

- $\frac{m}{n} = \frac{a^x}{a^y} = a^{x-y}$

$$\therefore \log_a \left(\frac{m}{n} \right) = x - y$$

and so $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$

For example:

$$\begin{aligned}\log_2 32 - \log_2 8 &= \log_2 \frac{32}{8} \\ &= \log_2 4 \\ &= 2\end{aligned}$$

- If $m = 1$

$$\begin{aligned}\log_a \left(\frac{1}{n} \right) &= \log_a 1 - \log_a n \\ &= -\log_a n\end{aligned}$$

$$\therefore \log_a \left(\frac{1}{n} \right) = -\log_a n$$

- $m^p = (a^x)^p$
 $= a^{xp}$

$$\log_a (m^p) = xp$$

and so $\log_a (m^p) = p \log_a m$

For example:

$$\begin{aligned}3 \log_2 5 &= \log_2 (5^3) \\ &= \log_2 125\end{aligned}$$

Example 9

Without using a calculator, simplify the following:

$$2 \log_{10} 3 + \log_{10} 16 - 2 \log_{10} \frac{6}{5}$$

Solution

$$\begin{aligned} 2 \log_{10} 3 + \log_{10} 16 - 2 \log_{10} \frac{6}{5} &= \log_{10} 3^2 + \log_{10} 16 - \log_{10} \left(\frac{6}{5}\right)^2 \\ &= \log_{10} 9 + \log_{10} 16 - \log_{10} \frac{36}{25} \\ &= \log_{10} \left(9 \times 16 \times \frac{25}{36}\right) \\ &= 2 \end{aligned}$$

Logarithmic equations

Example 10

Solve each of the following equations for x :

a $\log_2 x = 5$

b $\log_2 (2x - 1) = 4$

c $\log_e (3x + 1) = 0$

Solution

a $\log_2 x = 5$
 $\therefore x = 2^5$
 $\therefore x = 32$

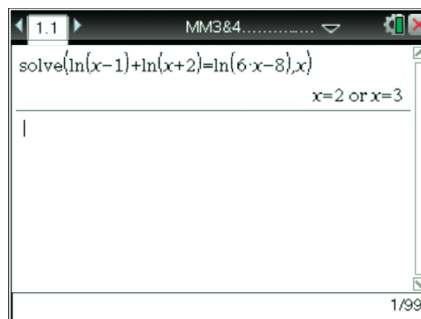
b $\log_2 (2x - 1) = 4$
 $\therefore 2x - 1 = 2^4$
 $\therefore 2x = 17$
 $\therefore x = \frac{17}{2}$

c $\log_e (3x + 1) = 0$
 $\therefore 3x + 1 = e^0$
 $\therefore 3x = 1 - 1$
 $\therefore x = 0$

Using the TI-Nspire

Use **Solve** from the **Algebra** menu as shown.

$\ln(x) = \log_e(x)$, the logarithm with base e , is available on the keypad by pressing $\text{(ctr)} \text{(e}^x\text{)}$. Logarithms with other bases are obtained by pressing the **log** key ($\text{(ctr)} \text{(10}^x\text{)}$) and completing the template.



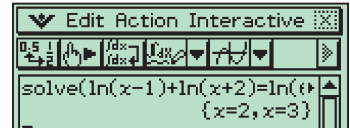
Using the Casio ClassPad

Enter and highlight

$$\ln(x-1) + \ln(x+2) = \ln(6x-8)$$

tap **Interactive–Equation/inequality–**

solve. Ensure the variable is set to x .



Example 11

Solve each of the following equations for x :

a $\log_x 27 = \frac{3}{2}$

b $\log_e(x-1) + \log_e(x+2) = \log_e(6x-8)$

c $\log_2 x - \log_2(7-2x) = \log_2 6$

Solution

a $\log_x 27 = \frac{3}{2}$
 is equivalent to $x^{\frac{3}{2}} = 27$
 By inspection $x = 9$

b $\log_e(x-1) + \log_e(x+2) = \log_e(6x-8)$
 $\therefore \log_e(x-1)(x+2) = \log_e(6x-8)$
 $\therefore x^2 + x - 2 = 6x - 8$
 $\therefore x^2 - 5x + 6 = 0$
 $\therefore (x-3)(x-2) = 0$
 $\therefore x = 3$ or $x = 2$

Note: The solutions must satisfy $x-1 > 0$, $x+2 > 0$ and $6x-8 > 0$, i.e. $x > \frac{4}{3}$.
 Therefore both of these solutions are allowable.

c $\log_2 x - \log_2(7-2x) = \log_2 6$
 $\therefore \log_2 \frac{x}{7-2x} = \log_2 6$
 $\therefore \frac{x}{7-2x} = 6$
 $\therefore x = 42 - 12x$
 $\therefore 13x = 42$
 $\therefore x = \frac{42}{13}$

Graphing transformations of the graph of $f(x) = \log_a x$

Example 12

Sketch the graph of $y = 3 \log_e 2x$

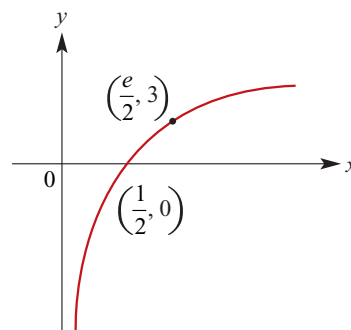
Solution

This is obtained from the graph of $y = \log_e x$ by a dilation of factor 3 from the x -axis and a dilation of factor $\frac{1}{2}$ from the y -axis.

The mapping is $(x, y) \rightarrow \left(\frac{1}{2}x, 3y\right)$

$$(1, 0) \rightarrow \left(\frac{1}{2}, 0\right)$$

$$(e, 1) \rightarrow \left(\frac{1}{2}e, 3\right)$$



Example 13

Sketch the graph, and state the domain, of each of the following:

a $y = \log_2(x - 5) + 1$

b $y = -\log_3(x + 4)$

Solution

- a** The graph of $y = \log_2(x - 5) + 1$ is obtained from the graph of $y = \log_2 x$ by a translation of 5 units in the positive direction of the x -axis and 1 unit in the positive direction of the y -axis.

The equation of the asymptote is $x = 5$

The mapping is $(x, y) \rightarrow (x + 5, y + 1)$

$$(1, 0) \rightarrow (6, 1)$$

$$(2, 1) \rightarrow (7, 2)$$

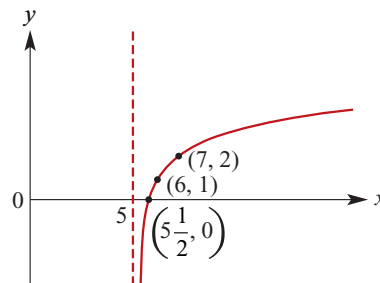
The domain of the function is $(5, \infty)$.

When $y = 0$, $\log_2(x - 5) + 1 = 0$

$$\log_2(x - 5) = -1$$

$$x - 5 = 2^{-1}$$

$$x = 5\frac{1}{2}$$



- b** The graph of $y = -\log_3(x + 4)$ is obtained from the graph of $y = \log_3 x$ by a translation of 4 units in the negative direction of the x -axis and a reflection in the x -axis.

The equation of the asymptote is $x = -4$

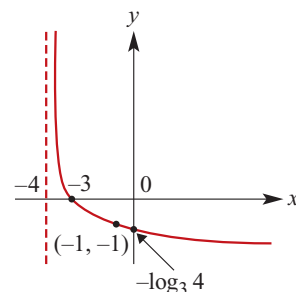
The mapping is $(x, y) \rightarrow (x - 4, -y)$

$$(1, 0) \rightarrow (-3, 0)$$

$$(3, 1) \rightarrow (-1, -1)$$

The domain of the function is $(-4, \infty)$.

$$\begin{aligned}\text{When } x = 0, \quad y &= -\log_3(0 + 4) \\ &= -\log_3 4\end{aligned}$$



Example 14

Sketch the graph of $y = 2 \log_e(x + 5) - 3$.

Solution

The graph of $y = 2 \log_e(x + 5) - 3$ is obtained from the graph of $y = \log_e x$ by a dilation of factor 2 from the x -axis followed by a translation of 5 units in the negative direction of the x -axis and 3 units in the negative direction of the y -axis.

The equation of the asymptote is $x = -5$

The mapping is $(x, y) \rightarrow (x - 5, 2y - 3)$

$$(1, 0) \rightarrow (-4, -3)$$

$$(e, 1) \rightarrow (e - 5, -1)$$

The domain of the function is $(-5, \infty)$.

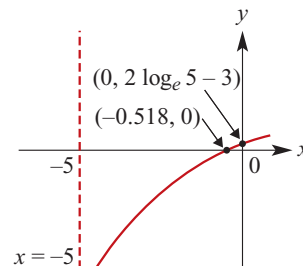
$$\begin{aligned}\text{When } x = 0, \quad y &= 2 \log_e(0 + 5) - 3 \\ &= 2 \log_e 5 - 3\end{aligned}$$

When $y = 0$,

$$2 \log_e(x + 5) - 3 = 0$$

$$\therefore \log_e(x + 5) = \frac{3}{2}$$

$$\begin{aligned}\text{and} \quad x + 5 &= e^{\frac{3}{2}} \\ \therefore x &= e^{\frac{3}{2}} - 5 \\ &\approx -0.518\end{aligned}$$



Exercise 5D



1 Evaluate each of the following:

a $\log_{10} 1000$

b $\log_2 \frac{1}{16}$

c $\log_{10} 0.001$

d $\log_2 64$

e $\log_{10} 1\,000\,000$

f $\log_2 \frac{1}{128}$

2 Express the following as the logarithm of a single term:

a $\log_e 2 + \log_e 3$

b $\log_e 32 - \log_e 8$

c $\log_e 10 + \log_e 100 + \log_e 1000$

d $\log_e \frac{1}{2} + \log_e 14$

e $\log_e \frac{1}{3} + \log_e \frac{1}{4} + \log_e \frac{1}{5}$

f $\log_e uv + \log_e uv^2 + \log_e uv^3$

g $2 \log_e x + 5 \log_e x$

h $\log_e (x + y) + \log_e (x - y) - \log_e (x^2 - y^2)$

3 Solve each of the following equations for x :

a $\log_{10} x = 2$

b $2 \log_2 x = 8$

c $\log_e (x - 5) = 0$

d $\log_2 x = 6$

e $2 \log_e (x + 5) = 6$

f $\log_e (2x) = 0$

g $\log_e (2x + 3) = 0$

h $\log_{10} x = -3$

i $2 \log_2 (x - 4) = 10$

4 Solve each of the following equations for x :

a $\log_{10} x = \log_{10} 3 + \log_{10} 5$

b $\log_e x = \log_e 15 - \log_e 3$

c $\log_e x = \frac{2}{3} \log_e 8$

d $\log_e x + \log_e (2x - 1) = 0$

e $2 \log_e x - \log_e (x - 1) = \log_e (x + 3)$

5 Express each of the following as the logarithm of a single term:

a $\log_{10} 9 + \log_{10} 3$

b $\log_2 24 - \log_2 6$

c $\frac{1}{2} \log_{10} a - \frac{1}{2} \log_{10} b$

d $1 + \log_{10} a - \frac{1}{3} \log_{10} b$

e $\frac{1}{2} \log_{10} 36 - \frac{1}{3} \log_{10} 27 - \frac{2}{3} \log_{10} 64$

6 Without using your calculator, evaluate each of the following:

a $\log_{10} 5 + \log_{10} 2$

b $\log_{10} 5 + 3 \log_{10} 2 - \log_{10} 4$

c $\log_2 \sqrt{2} + \log_2 1 + 2 \log_2 2$

d $2 \log_{10} 5 + 2 \log_{10} 2 + 1$

e $4 \log_{10} 2 - \log_{10} 16$

7 Simplify the following expressions:

a $\log_3 \left(\frac{1}{3^x} \right)$

b $\log_2 x - 2 \log_2 y + \log_2 (xy^2)$

c $\log_e (x^2 - y^2) - \log_e (x - y) - \log_e (x + y)$

8 Solve each of the following equations for x :

a $\log_e (x^2 - 2x + 8) = 2 \log_e x$

b $\log_e (5x) - \log_e (3 - 2x) = 1$

9 Solve each of the following equations for x :

a $\log_e(x) + \log_e(3x + 1) = 1$

b $8e^{-x} - e^x = 2$

10 Solve for x :

$$2 \log_e(x) + \log_e 4 = \log_e(9x - 2)$$

11 Given that $\log_a N = \frac{1}{2}(\log_a 24 - \log_a 0.375 - 6 \log_a 3)$, find the value of N .

12 Sketch the graphs of each of the following. Label the axes intercepts and asymptotes. State the maximal domain and range of each.

a $y = 2 \log_e(x - 3)$

b $y = \log_e(x + 3) - 2$

c $y = 2 \log_e(x + 1) - 1$

d $y = 2 + \log_e(3x - 2)$

e $y = -2 \log_e(x + 2)$

f $y = -2 \log_e(x - 2)$

g $y = 1 - \log_e(x + 1)$

h $y = \log_e(2 - x)$

i $y + 1 = \log_e(4 - 3x)$

13 Sketch the graphs of each of the following. Label the axes intercepts and asymptotes. State the maximal domain of each.

a $y = \log_2 2x$

b $y = \log_{10}(x - 5)$

c $y = -\log_{10} x$

d $y = \log_{10}(-x)$

e $y = \log_{10}(5 - x)$

f $y = 2 \log_2 2x + 2$

g $y = -2 \log_2 3x$

h $y = \log_{10}(-x - 5) + 2$

i $y = 4 \log_2(-3x)$

j $y = 2 \log_2(2 - x) - 6$

14 Solve each of the following equations using a calculator. Give answers correct to three decimal places.

a $-x + 2 = \log_e x$

b $\frac{1}{3} \log_e(2x + 1) = -\frac{1}{2}x + 1$

15 **a** Using a calculator plot the graph of $y = f(x)$ where $f(x) = \log_e x$

b Using the same screen plot the graphs of:

i $y = f(-x)$ **ii** $y = -f(x)$ **iii** $y = f\left(\frac{x}{3}\right)$ **iv** $y = f(3x)$

5.5 Determining rules for graphs of exponential and logarithmic functions

In previous chapters, we considered establishing rules for graphs of some functions. In this chapter, we consider similar questions for exponential and logarithmic functions.

Example 15

The rule for the function of the graph is of the form $y = ae^x + b$. Find the values of a and b .

Solution

When $x = 3$, $y = 22$ and when $x = 0$, $y = 6$

$$\therefore 6 = ae^0 + b \quad (1)$$

and $22 = ae^3 + b \quad (2)$

Subtract (1) from (2):

$$16 = a(e^3 - e^0)$$

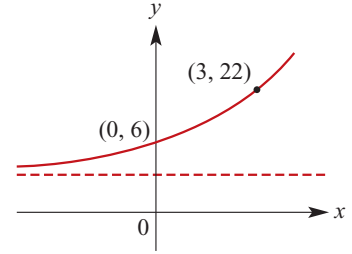
$$\therefore 16 = a(e^3 - 1)$$

$$\begin{aligned} \text{Therefore } a &= \frac{16}{e^3 - 1} \\ &\approx 0.8383 \end{aligned}$$

From equation (1), $b = 6 - a$

$$\begin{aligned} &= 6 - \frac{16}{e^3 - 1} \\ &= \frac{6e^3 - 22}{e^3 - 1} \\ &\approx 5.1617 \end{aligned}$$

Therefore $y \approx 0.8383e^x + 5.1617$

**Example 16**

The rule for the function of the graph shown is of the form $y = a \log_e(x + b)$. Find the values of a and b .

Solution

When $x = 5$, $y = 0$ and when $x = 8$, $y = 1$

$$\therefore 0 = a \log_e(5 + b) \quad (1)$$

and $1 = a \log_e(8 + b) \quad (2)$

From (1) $\log_e(5 + b) = 0$

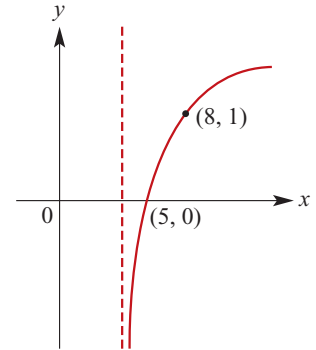
$$\therefore 5 + b = e^0$$

and $b = -4$

From (2) $1 = a \log_e 4$

$$\begin{aligned} \therefore a &= \frac{1}{\log_e 4} \\ &\approx 0.7213 \end{aligned}$$

$$\therefore y \approx 0.7213 \log_e(x - 4)$$

**Example 17**

Given that $y = Ae^{bt}$ and $y = 6$ when $t = 1$ and $y = 8$ when $t = 2$, find the values of b and A .

Solution

When $t = 1$, $y = 6$

Thus $6 = Ae^b$ (1)

When $t = 2$, $y = 8$

Thus $8 = Ae^{2b}$ (2)

Divide (2) by (1):

$$\frac{4}{3} = e^b$$

$$\therefore b = \log_e \frac{4}{3}$$

Substitute in (1):

$$6 = Ae^{\log_e \frac{4}{3}}$$

$$\therefore 6 = \frac{4}{3}A$$

$$\therefore A = \frac{18}{4} = \frac{9}{2}$$

Hence $y = \frac{9}{2}e^{(\log_e \frac{4}{3})t}$ $\left(= \frac{9}{2} \left(\frac{4}{3} \right)^t \right)$

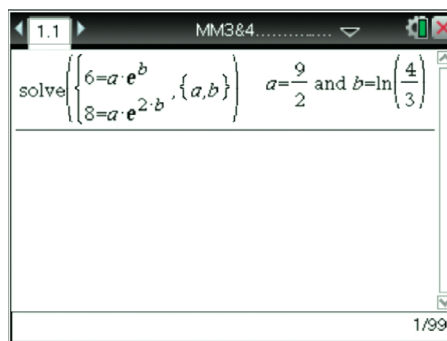
$$y \approx \frac{9}{2}e^{0.288t}$$

Using the TI-Nspire

Use **menu** > **Algebra** > **Solve System of Equations** > **Solve System of Equations** and complete as shown.

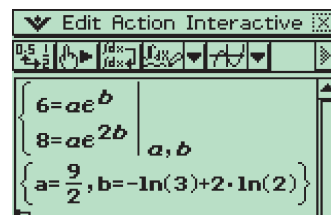
$\ln(x) = \log_e(x)$, the logarithm with base e , is available on the keypad by pressing **(ctrl)(e^x)**. Logarithms to other bases are obtained by pressing the log key **(ctrl)(log)** and completing the template.

Note: Do not use the 'e' from the alpha keys. It will be treated as a variable.



Using the Casio ClassPad

Using the simultaneous equations template complete as shown.

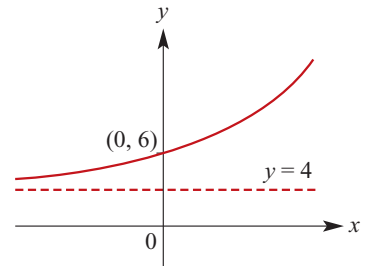


Exercise 5E

- 1 The graph shown has rule:

$$y = ae^x + b$$

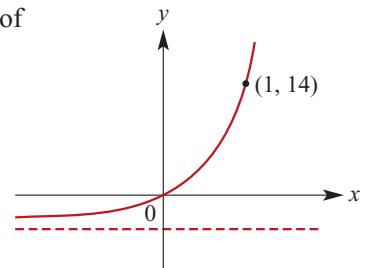
Find the values of a and b .



- 2 The rule for the function for which the graph is shown is of the form:

$$y = ae^x + b$$

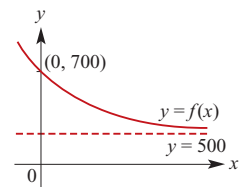
Find the values of a and b .



- 3 The rule for the function f is of the form:

$$f(x) = ae^{-x} + b$$

Find the values of a and b .

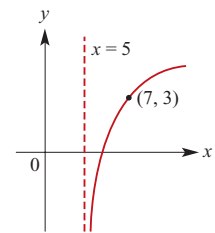


- 4 Find the values of a and b such that the graph of $y = ae^{-bx}$ goes through the points $(3, 50)$ and $(6, 10)$.

- 5 The rule of the graph shown is of the form:

$$y = a \log_2(x - b)$$

Find the values of a and b .



- 6 Find the values of a and b such that the graph of $y = ae^{bx}$ goes through the points $(3, 10)$ and $(6, 50)$.
- 7 Find the values of a and b such that the graph of $y = a \log_2 x + b$ goes through the points $(8, 10)$ and $(32, 14)$.
- 8 Find the values of a and b such that the graph of $y = a \log_2(x - b)$ passes through the points $(5, 2)$ and $(7, 4)$.

- 9 The points (3, 10) and (5, 12) lie on the graph of the function with rule $y = a \log_e (x - b) + c$. The graph has a vertical asymptote with equation $x = 1$. Find the values of a , b and c .
- 10 The graph of the function with rule $f(x) = a \log_e (-x) + b$ passes through the points $(-2, 6)$ and $(-4, 8)$. Find the values of a and b .

5.6 Change of base and solution of exponential equations

It is often useful to change the base of an exponential or logarithmic function, particularly to base 10 or e since these are the only ones available on the calculator.

To change the base of $\log_a x$ from a to b ($a > 0$ and $b > 0$ and $a, b \neq 1$), we use the definition that $y = \log_a x$ implies $a^y = x$

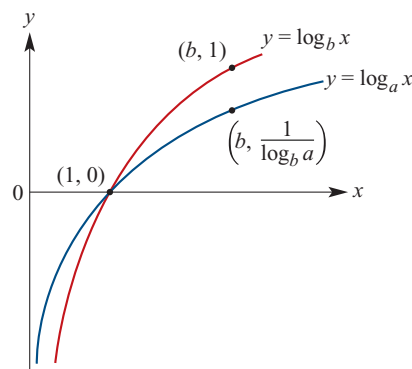
$$\text{Taking } \log_b \text{ of both sides: } \log_b a^y = \log_b x$$

$$\text{Therefore } y \log_b a = \log_b x$$

$$\text{Therefore } y = \frac{1}{\log_b a} \log_b x$$

$$\text{Since } y = \log_a x \quad \log_a x = \frac{1}{\log_b a} \log_b x \quad \text{or} \quad \log_a x = \frac{\log_b x}{\log_b a}$$

This demonstrates that the graph of $y = \log_a x$ can be obtained from the graph of $y = \log_b x$ by a dilation of factor $\frac{1}{\log_b a}$ from the x -axis.



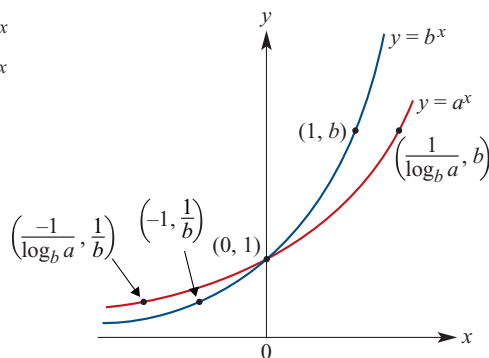
A similar process shows that $y = a^x$ can be written as $x \log_b a = \log_b y$.

$$\text{Rearranging to make } y \text{ the subject: } y = b^{(\log_b a)x}$$

$$\text{Since } y = a^x: \quad a^x = b^{(\log_b a)x}$$

This demonstrates that the graph of $y = a^x$ can be obtained from the graph of $y = b^x$ by a dilation of factor $\frac{1}{\log_b a}$ from the y -axis.

The statement $\log_a x = \frac{\log_b x}{\log_b a}$ can be used to simplify expressions, as in the following examples.



Example 18

Simplify: **a** $\frac{\log_e 27}{\log_e 3}$ **b** $\frac{\log_2 1024}{\log_2 4}$

Solution

$$\begin{array}{ll} \text{a} \quad \frac{\log_e 27}{\log_e 3} = \log_3 27 & \text{b} \quad \frac{\log_2 1024}{\log_2 4} = \log_4 1024 \\ & = 3 \qquad \qquad \qquad = 5 \end{array}$$

Example 19

Evaluate, correct to four significant figures by using change of base:

a $\log_2 10$ **b** $\log_{\frac{1}{2}} 6$

Solution

$$\begin{array}{ll} \text{a} \quad \log_2 10 = \frac{\log_e 10}{\log_e 2} & \text{b} \quad \log_{\frac{1}{2}} 6 = \frac{\log_{10} 6}{\log_{10}(\frac{1}{2})} \\ \approx 3.322 & \approx -2.585 \end{array}$$

Note: These can be evaluated directly with the calculator

Example 20

If $\log_2 6 = k \log_2 3 + 1$, find the value of k .

Solution

$$\begin{aligned} \log_2 6 &= k \log_2(3) + 1 \\ &= \log_2 3^k + \log_2 2 \\ &= \log_2 (2 \times 3^k) \end{aligned}$$

Therefore $6 = 2 \times 3^k$

$$\begin{aligned} 3 &= 3^k \\ k &= 1 \end{aligned}$$

Example 21

Solve for x if $2^x = 11$, expressing the answer to two decimal places.

Solution

Take the $\log_{10}$ of both sides of the equation. This can also be completed by finding $\log_2 11$ directly with a calculator.

$$\begin{aligned} \text{Therefore} \quad \log_{10} 2^x &= \log_{10} 11 \\ \text{i.e.} \quad x \log_{10} 2 &= \log_{10} 11 \\ \text{Therefore} \quad x &= \frac{\log_{10} 11}{\log_{10} 2} \\ &\approx 3.46 \end{aligned}$$

Note: $\log_2 11 = 3.45943 \dots$

Example 22

Solve $3^{2x-1} = 28$, expressing the answer to three decimal places.

Solution

$$\begin{aligned}
 \log_e 3^{2x-1} &= \log_e 28 & \text{or} & \text{Take } \log_3 \text{ of both sides.} \\
 (2x-1)\log_e 3 &= \log_e 28 & 2x-1 &= \log_3 28 \\
 2x-1 &= \frac{\log_e 28}{\log_e 3} & 2x &= \log_3(28) + 1 \\
 2x &= \frac{\log_e 28}{\log_e 3} + 1 & x &= \frac{1}{2}(\log_3(28) + 1) \\
 & & & \approx 2.017 \\
 x &= \frac{1}{2} \left(\frac{\log_e 28}{\log_e 3} + 1 \right) \\
 &\approx 2.017
 \end{aligned}$$

Example 23

Solve $\{x : 0.7^x \geq 0.3\}$, expressing the answer to three decimal places.

Solution

Taking $\log_{10}$ of both sides:

$$\begin{aligned}
 \log_{10} 0.7^x &\geq \log_{10} 0.3 \\
 x \log_{10} 0.7 &\geq \log_{10} 0.3 \\
 x &\leq \frac{\log_{10} 0.3}{\log_{10} 0.7} \quad (\text{Note the sign change.}) \\
 x &\leq \frac{-0.5229}{-0.1549} \\
 x &\leq 3.376
 \end{aligned}$$

Exercise 5F

- 1 Use your calculator to solve each of the following equations, correct to two decimal places:

a $2^x = 6$

b $3^x = 0.7$

c $3^x = 11$

d $4^x = 5$

e $2^{-x} = 5$

f $0.2^x = 3$

g $5^x = 3^{x-1}$

h $8^x = 2005^{x+1}$

i $3^{x-1} = 8$

j $0.3^{x+2} = 0.7$

k $2^{x-1} = 3^{x+1}$

l $1.4^{x+2} = 25(0.9)^x$

m $5^x = 2^{2x-2}$

n $2^{\frac{1}{2}(x+2)} = 3^{x-1}$

o $2^{x+1} \times 3^{x-1} = 100$

- 2 Solve for x using a calculator. Express your answer correct to two decimal places.

a $2^x < 7$

b $3^x > 6$

c $0.2^x > 3$

d $3^{x-2} \leq 8$

e $0.2^x \leq 0.4$

3 Solve each of the following equations for x :

a $2^x = 5$

b $3^x = 7$

c $3^{2x} - 3^{x+2} + 8 = 0$

4 Simplify:

a $\frac{\log_{10} 1024}{\log_{10} 2}$

b $\frac{\log_5 216}{\log_5 6}$

c $\frac{\log_4 81}{\log_4 3} + \log_4 256$

5 Evaluate, correct to four decimal places, by first expressing in terms of $\log_e$:

a $\log_3 26$

b $\log_2 57$

c $\log_4 18$

d $\log_5 99$

e $\log_{\frac{2}{5}} 72$

f $\log_{\frac{1}{3}} 67 + \log_{\frac{1}{3}} \left(\frac{1}{27} \right)$

6 **a** If $a \log_2 7 = 3 - \log_6 14$, find the value of a , correct to three significant figures.

b If $\log_3 18 = \log_{11} k$, find the value of k , correct to one decimal place.

7 Prove that $\log_b a + \log_c b + \log_a c = \frac{1}{\log_a b} + \frac{1}{\log_b c} + \frac{1}{\log_c a}$

8 Prove that if $\log_r p = q$ and $\log_q r = p$, then $\log_q p = pq$

9 If $u = \log_9 x$, find in terms of u :

a x

b $\log_9 (3x)$

c $\log_x 81$

10 Solve the equation $\log_5 x = 16 \log_x 5$

11 Given that $q^p = 25$, find $\log_5 q$ in terms of p .

5.7 Inverses

It has been observed that $f(x) = \log_a x$ and $g(x) = a^x$ are inverse functions. In this section this observation is used to find inverses of related functions, and to transform equations. An important consequence is the following:

$$\begin{aligned} \log_a a^x &= x & \text{for all } x \in R \\ a^{\log_a x} &= x & \text{for } x \in R^+ \end{aligned}$$

Example 24

Find the inverse of the function $f: R \rightarrow R$, $f(x) = e^x + 2$ and state the domain and range of the inverse function.

Solution

Recall that the transformation ‘a reflection in the line $y = x$ ’ is given by the mapping $(x, y) \rightarrow (y, x)$.

Consider $x = e^y + 2$

Then $x - 2 = e^y$

and $y = \log_e (x - 2)$

i.e., the inverse function has rule $f^{-1}(x) = \log_e (x - 2)$

The domain of f^{-1} = the range of $f = (2, \infty)$.

The range of f^{-1} = the domain of $f = R$.

Example 25

Rewrite the equation $y = 2 \log_e(x) + 3$ with x as the subject.

Solution

$$y = 2 \log_e(x) + 3$$

$$\text{Therefore } \frac{y-3}{2} = \log_e x$$

$$\text{and } x = e^{\frac{y-3}{2}}$$

Example 26

Find the inverse of the function $f: (1, \infty) \rightarrow R$, $f(x) = 2 \log_e(x-1) + 3$. State the domain and range of the inverse.

Solution

$$\text{Consider } x = 2 \log_e(y-1) + 3$$

$$\text{Therefore } \frac{x-3}{2} = \log_e(y-1)$$

$$\text{and } y-1 = e^{\frac{x-3}{2}}$$

$$\text{Therefore } y = e^{\frac{x-3}{2}} + 1$$

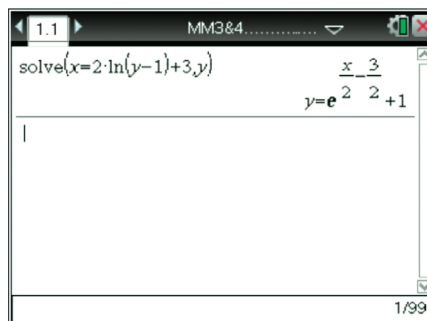
$$\text{Hence } f^{-1}(x) = e^{\frac{x-3}{2}} + 1$$

$$\begin{aligned} \text{The domain of } f^{-1} &= \text{the range of } f \\ &= R \end{aligned}$$

$$\text{The range of } f^{-1} = (1, \infty).$$

Using the TI-Nspire

Use **Solve** from the **Algebra** menu as shown.

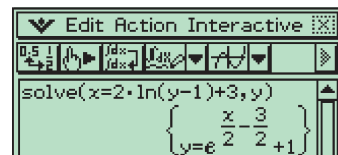


Using the Casio ClassPad

Enter and highlight $x = 2 \ln(y - 1) + 3$ then tap

Interactive–Equation/inequality–solve and

ensure the variable is set to y .



Example 27

Rewrite the equation $P = Ae^{kt}$ with t as the subject.

Solution

$$P = Ae^{kt}$$

Taking logarithms to the base e of both sides:

$$\begin{aligned}\log_e P &= \log_e (Ae^{kt}) \\ \therefore \log_e P &= \log_e A + \log_e e^{kt} \\ \therefore t &= \frac{1}{k}(\log_e P - \log_e A) \\ &= \frac{1}{k} \log_e \left(\frac{P}{A} \right)\end{aligned}$$

Exercise 5G

- On the one set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ where $f: R \rightarrow R$, $f(x) = e^{-x} + 3$
- On the one set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ where: $f: (1, \infty) \rightarrow R$, $f(x) = \log_e(x - 1)$
- Find the inverse of each of the following functions and state the domain and range in each case:
 - $f: R^+ \rightarrow R$, where $f(x) = \log_e 2x$
 - $f: R^+ \rightarrow R$, where $f(x) = 3 \log_e(2x) + 1$
 - $f: R \rightarrow R$, where $f(x) = e^x + 2$
 - $f: R \rightarrow R$, where $f(x) = e^{x+2}$
 - $f: \left(-\frac{1}{2}, \infty\right) \rightarrow R$, where $f(x) = \log_e(2x + 1)$
 - $f: \left(-\frac{2}{3}, \infty\right) \rightarrow R$, where $f(x) = 4 \log_e(3x + 2)$
 - $f: \{x: x > -1\} \rightarrow R$, $f(x) = \log_{10}(x + 1)$
 - $f: R \rightarrow R$, $f(x) = 2e^{x-1}$

- 4** The function f has the rule $f(x) = 1 - e^{-x}$
- a** Sketch the graph of f . **b** Find the domain of f^{-1} and $f^{-1}(x)$.
c Sketch the graph of f^{-1} on the same set of axes as the graph of f .
- 5** Let $f: R \rightarrow R$, $f(x) = 5e^{2x} - 3$
- a** Sketch the graph of f . **b** Find the inverse function f^{-1} .
c Sketch the graph of f^{-1} on the same set of axes as the graph of f .
- 6** Let $f: R^+ \rightarrow R$, $f(x) = 2 \log_e(x) + 1$
- a** Sketch the graph of f .
b Find the inverse function f^{-1} and state the range.
c Sketch the graph of f^{-1} on the same set of axes as the graph of f .
- 7** For each of the formulas, make the pronumeral in brackets the subject:
- a** $y = 2 \log_e(x) + 5$ (x) **b** $P = Ae^{-6x}$ (x) **c** $y = ax^n$ (n)
d $y = 5 \times 10^x$ (x) **e** $y = 5 - 3 \log_e(2x)$ (x) **f** $y = 6x^{2n}$ (n)
g $y = \log_e(2x - 1)$ (x) **h** $y = 5(1 - e^{-x})$ (x)
- 8** For $f: R \rightarrow R$, $f(x) = 2e^x - 4$:
- a** Find the inverse function f^{-1} .
b Find the coordinates of the points of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$
- 9** For $f: R \rightarrow R$, $f(x) = 2 \log_e(x + 3) + 4$:
- a** Find the inverse function f^{-1} .
b Find the coordinates of the points of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$
- 10 a** Using a calculator, for each of the following plot the graphs of $y = f(x)$ and $y = g(x)$, together with the line $y = x$, on the one set of axes.
- i** $f(x) = \log_e x$ and $g(x) = e^x$
ii $f(x) = 2 \log_e(x) + 3$ and $g(x) = e^{\frac{x-3}{2}}$
iii $f(x) = \log_{10} x$ and $g(x) = 10^x$
- b** Use your answers to part **a** to comment on the relationship between $f(x) = a \log_b x + c$ and $g(x) = b^{\frac{x-c}{a}}$

5.8 Exponential growth and decay

Exponential and logarithmic functions are used to model many physical occurrences. It will be shown in Chapter 11 that if a quantity increases or decreases at a rate which is, at any time, proportional to the quantity present, then the quantity present at time t is given by the **law of exponential change**.

Let A be the quantity at time t . Then, $A = A_0 e^{kt}$, where A_0 is a constant.

Growth: $k > 0$ Decay: $k < 0$

The number k is the **rate constant** of the equation.

Physical situations where this is applicable include:

- the growth of a cell
- population growth
- continuously compounded interest
- radioactive decay
- Newton's law of cooling.

Example 28

A bank pays 10% interest compounded annually. You invest \$1000. How does this \$1000 grow as a result of the interest added?

Solution

Set out in tabular form.

End of year		Amount, A
	\$1000	\$1000.00
1	$1000 \times 110\% = 1000 \left(1 + \frac{10}{100}\right) = 1000 \left(1 + \frac{10}{100}\right)$	\$1100.00
2	$(1000 \times 110\%) 110\% = 1000 \left(1 + \frac{10}{100}\right) \left(1 + \frac{10}{100}\right) = 1000 \left(1 + \frac{10}{100}\right)^2$	\$1210.00
3	$(1000 \times 110\%) 110\% \times 110\% = 1000 \left(1 + \frac{10}{100}\right)^3$	\$1331.00
4	$= 1000 \left(1 + \frac{10}{100}\right)^4$	\$1464.10
↓		
10	$= 1000 \left(1 + \frac{10}{100}\right)^{10}$	\$2593.74
↓		
n	$= 1000 \left(1 + \frac{10}{100}\right)^n = 1000 (1.1)^n$	$\$1000(1.1)^n$

$$A = 1000(1.1)^n, n \in N \cup \{0\}$$

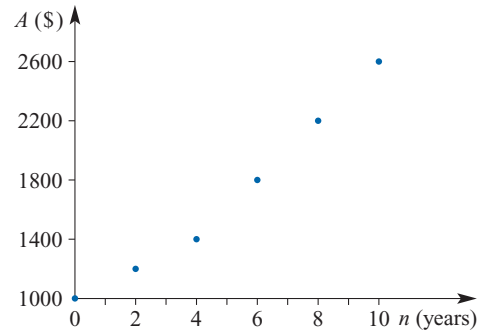
In general, if P = original investment

A = amount the investment grows to after n years

r = compound interest rate $r\%$ per annum

n = number of years invested

$$\text{then } A = P \left(1 + \frac{r}{100}\right)^n$$



Example 29

The population of a town was 8000 at the beginning of 1992 and 15 000 at the end of 1999. Assume that the growth is exponential.

- Find the population at the end of 2001.
- In what year will the population be double that of 1999?

Solution

- Let P be the population at time t years (measured from 1 January 1992).

$$\text{Then } P = 8000e^{kt}$$

At the end of 1999, $t = 8$ and $P = 15\,000$.

$$\therefore 15\,000 = 8000e^{8k}$$

$$\therefore \frac{15}{8} = e^{8k}$$

$$\therefore k = \frac{1}{8} \log_e \frac{15}{8} \\ \approx 0.079$$

The rate of increase is 7.9% per annum.

Note: The approximation 0.079 was not used in the calculations which follow. The value for k was held in the calculator.

When $t = 10$

$$P = 8000e^{10k}$$

$$\approx 17\,552.6049$$

$$\approx 17\,550$$

The population is approximately 17 550.

- When does $P = 30\,000$? Consider the equation:

$$30\,000 = 8000e^{kt}$$

$$\therefore \frac{30\,000}{8000} = e^{kt}$$

$$\therefore \frac{15}{4} = e^{kt}$$

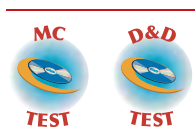
$$\therefore 3.75 = e^{kt}$$

$$\therefore t = \frac{1}{k} \log_e 3.75 \\ \approx 16.82$$

The population reaches 30 000 approximately 16.82 years after the beginning of 1992, i.e. during the year 2008.

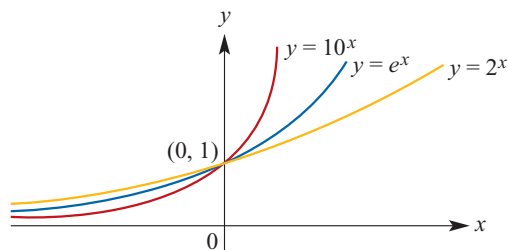
Exercise 5H

- 1 In the initial period of its life a particular species of tree grows in the manner described by the rule $d = d_0 10^{mt}$ where d is the diameter of the tree in centimetres, t years after the beginning of this period. The diameter after 1 year is 52 cm and after 3 years, 80 cm. Calculate the values of the constants d_0 and m .
- 2 The number of bacteria in a certain culture at time, t weeks, is given by the rule $N = N_0 e^{kt}$. If when $t = 2$, $N = 101$ and when $t = 4$, $N = 203$ calculate the values of N_0 and k .
- 3 The number of people, N , who have a particular disease at time t years is given by $N = N_0 e^{kt}$
 - a If the number initially is 20 000 and the number decreases by 20% each year, find:
 - i the value of N_0
 - ii the value of k
 - b How long does it take for 5000 people to be infected?
- 4 Polonium-210 is a radioactive substance. The decay of polonium-210 is described by the formula $M = M_0 e^{-kt}$, where M is the mass in grams of polonium-210 left after t days, and M_0 and k are constants. At time $t = 0$, $M = 10$ g and at $t = 140$, $M = 5$ g.
 - a Find the values of M_0 and k .
 - b What will be the mass of the polonium-210 after 70 days?
 - c After how many days is the mass remaining 2 g?



Chapter summary

- **Sketch graphs** of $y = a^x$, e.g. $a = 2$ or 10 or e , and transformations of these graphs:



- **Index laws**

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

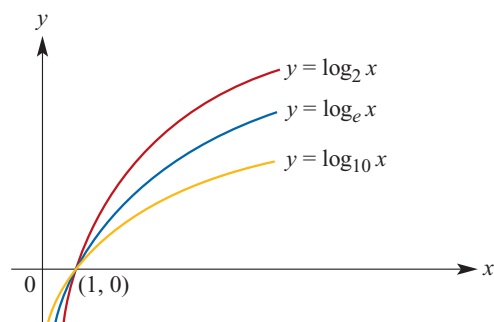
$$(a^m)^n = a^{m \times n}$$

- **Logarithms**

$$\log_a x = y \text{ if } a^y = x$$

The inverse function of $f: R \rightarrow R$, $f(x) = a^x$ is $f^{-1}: R^+ \rightarrow R$, $f^{-1}(x) = \log_a x$

- **Sketch graphs** of $y = \log_a x$, e.g. $a = 2$ or 10 or e , and transformations of these graphs:



- **Logarithm laws**

$$\log_a (mn) = \log_a m + \log_a n$$

$$\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$$

$$\log_a \left(\frac{1}{n} \right) = -\log_a n$$

$$\log_a (m^p) = p \log_a m$$

- **Change of base**

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = b^{(\log_b a)x}$$

- **Inverse properties**

$$\log_a a^x = x$$

$$\text{and } a^{\log_a x} = x$$

■ Law of exponential change

Let A be the quantity at time t . Then:

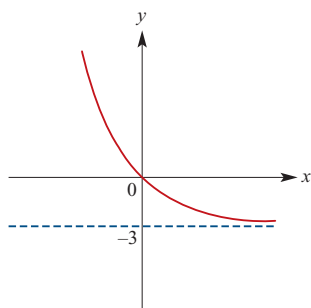
$$A = A_0 e^{kt}, \text{ where } A_0 \text{ is a constant.}$$

Growth: $k > 0$ Decay $k < 0$

The number k is the **rate constant** of the equation.

Multiple-choice questions

- If $4 \log_b |x| = \log_b 16 + 8$, then x is equal to:
A 24 **B** ± 6 **C** $\pm 2b^2$ **D** 6 **E** $2b^2$
- The expression $\log_e (4e^{3x})$ is equal to:
A $\log_e (e^{12x})$ **B** $\log_e 12 + x$ **C** $3x \log_e 4$ **D** $\log_e 4 + 3x$ **E** $12x$
- The expression $3^{\log_3(x-4)}$ is equal to:
A $\frac{x}{4}$ **B** $x - 4$ **C** $3(x - 4)$ **D** $3^x - 3^4$ **E** $\log_3 x - \log_3 4$
- Let $f: A \rightarrow R$, $f(x) = e^{2x}$, $g: B \rightarrow R$, $g(x) = \frac{1}{x+1}$, and $h: C \rightarrow R$,
 $h(x) = e^{2x} + \frac{1}{x+1}$, where A , B and C are the largest domains for which f , g and h respectively are defined. Which one of the following statements is true?
A $A \neq C$ and $\text{ran}(g) = \text{ran}(h)$ **B** $A = B$ and $\text{ran}(f) \neq \text{ran}(h)$
C $A \neq C$ and $\text{ran}(f) = \text{ran}(h)$ **D** $B = C$ and $\text{ran}(g) = \text{ran}(h)$
E $B = C$ and $\text{ran}(g) \neq \text{ran}(h)$
- If $x = 5$ is a solution of the equation $\log_{10}(kx - 3) = 2$, then the exact value of k is:
A $\frac{103}{5}$ **B** $\frac{\log_{10} 2 + 3}{5}$ **C** 2 **D** 5 **E** 21
- $3^{(4 \log_3 x + \log_3 4x)}$ is equal to:
A $8x$ **B** $x^4 + 4x$ **C** $4x^5$ **D** 3^{8x} **E** $\log_3 4x^5$
- The solution of the equation $3x = 10^{-0.3x}$ is closest to:
A 0.83 **B** 0.28 **C** 0 **D** 0.30 **E** 0.91
- The graph of the function with equation $y = ae^{-x} + b$ is shown below.



The values of a and b respectively are:

- A** 3, -3 **B** -3, 3 **C** -3, -3 **D** 0, -3 **E** -3, 0

- 9 Which one of the following statements is not true of the graph of the function $f: R^+ \rightarrow R, f(x) = \log_5 x$?
- A** The domain is R^+ . **B** The range is R . **C** It passes through the point $(5, 0)$.
D It has a vertical asymptote with equation $x = 0$.
E The slope of the tangent at any point on the graph is positive.
- 10 If $3 \log_2 x - 7 \log_2 (x - 1) = 2 + \log_2 y$, then y is equal to:
- A** $\frac{3x}{28(x-1)}$ **B** $\frac{1}{4x^4}$ **C** $3 - 4x$
D $\frac{x^3}{4(x-1)^7}$ **E** $x^3 - (x-1)^7 - 4$

Short-answer questions (technology-free)

- 1 Sketch the graphs of each of the following. Label asymptotes and axes intercepts.
- a** $f(x) = e^x - 2$ **b** $g(x) = 10^{-x} + 1$ **c** $h(x) = \frac{1}{2}(e^x - 1)$
d $f(x) = 2 - e^{-x}$ **e** $f(x) = \log_e(2x + 1)$ **f** $h(x) = \log_e(x - 1) + 1$
g $g(x) = -\log_e(x - 1)$ **h** $f(x) = -\log_e(1 - x)$
- 2 **a** For $f: R \rightarrow R, f(x) = e^{2x} - 1$, find f^{-1} .
b For $f: (2, \infty) \rightarrow R, f(x) = 3 \log_e(x - 2)$, find f^{-1} .
c For $f: (-1, \infty) \rightarrow R, f(x) = \log_{10}(x + 1)$, find f^{-1} .
d For $f: R^+ \rightarrow R, f(x) = 2^x + 1$, find f^{-1} .
- 3 For each of the following, find y in terms of x :
- a** $\log_e y = (\log_e x) + 2$ **b** $\log_{10} y = \log_{10} x + 1$ **c** $\log_2 y = 3 \log_2 x + 4$
d $\log_{10} y = -1 + 5 \log_{10} x$ **e** $\log_e y = 3 - \log_e x$ **f** $\log_e y = 2x - 3$
- 4 Solve each of the following equations for x , expressing your answers in terms of logarithms of base e :
- a** $3^x = 11$ **b** $2^x = 0.8$ **c** $2^x = 3^{x+1}$
- 5 Solve each of the following for x :
- a** $2^{2x} - 2^x - 2 = 0$ **b** $\log_e(3x - 1) = 0$ **c** $\log_{10}(2x) + 1 = 0$
d $10^{2x} - 7 \times 10^x + 12 = 0$
- 6 The graph of the function with equation $y = 3 \log_2(x + 1) + 2$ intersects the axes at the points $(a, 0)$ and $(0, b)$. Find the exact values of a and b .
- 7 The graph of the function with equation $f(x) = 5 \log_{10}(x + 1)$ passes through the point $(k, 6)$. Find the value of k .
- 8 Find the exact value of x for which $4e^{3x} = 287$.
- 9 Find the value of x in terms of a for which $3 \log_a x = 3 + \log_a 8$.
- 10 For the function $f: (4, \infty) \rightarrow R, f(x) = \log_3(x - 4)$, state the domain of the inverse function f^{-1} .
- 11 The graph of the function with equation $f(x) = e^{2x} - 3ke^x + 5$ intersects the axes at $(0, 0)$ and $(a, 0)$ and has a horizontal asymptote at $y = b$. Find the exact values of a, b and k .

Extended-response questions

- 1 A liquid cools from its original temperature of 90°C to a temperature $T^{\circ}\text{C}$ in x minutes. Given that $T = 90(0.98)^x$, find:

a the value of T when $x = 10$ **b** the value of x when $T = 27$

- 2 The population of a village at the beginning of the year 1800 was 240. The population increased so that, after a period of n years, the new population was $240(1.06)^n$. Find:

a the population at the beginning of 1820
b the year in which the population first reached 2500

- 3 The value, $\$V$, of a particular car can be modelled by the equation:

$$V = ke^{-\lambda t}$$

where t years is the age of the car.

The car's original price was $\$22\,497$, and after 1 year it is valued at $\$18\,000$.

a State the value of k and calculate λ , giving your answer to two decimal places.
b Find the value of the car when it is 3 years old.

- 4 The value $\$M$ of a particular house in a certain area during the period 1988 to 1994 can be modelled by the equation $M = Ae^{-pt}$ where t is the time in years after 1 January 1988. The value of the house on 1 January 1988 was $\$65\,000$ and its value on 1 January 1989 was $\$61\,000$.

a State the value of A and calculate the value of p , correct to two significant figures.
b What was the value of the house in 1993? Give your answer to the nearest 100.

- 5 There are two species of insects living in a suburb: the *Asla bibla* and the *Cutus pius*. The number of *Asla bibla* alive at time t days after 1 January 2000 is given by:

$$N_A(t) = 10\,000 + 1000t \quad 0 \leq t \leq 15$$

The number of *Cutus pius* alive at time t days after 1 January 2000 is given by:

$$N_C(t) = 8000 + 3 \times 2^t \quad 0 \leq t \leq 15$$

a With a calculator plot the graphs of $y = N_A(t)$ and $y = N_C(t)$ on the one screen.

b i Find the coordinates of the point of intersection of the two graphs.

ii At what time is $N_C(t) = N_A(t)$?

iii What is the number of each species of insect at this time?

c i Show that $N_A(t) = N_C(t)$ if and only if $t = \frac{1}{\log_{10} 2} \left[3 + \log_{10} \left(\frac{2+t}{3} \right) \right]$

ii Plot the graphs of $y = x$ and $y = \frac{1}{\log_{10} 2} \left[3 + \log_{10} \left(\frac{2+x}{3} \right) \right]$ and find the coordinates of the point of intersection.



d It is found by observation that the model for *Cutus pius* doesn't quite work.

It is known that the model for the population of *Asla bibla* is satisfactory.

The form of the model for *Cutus pius* is $N_C(t) = 8000 + c \times 2^t$.

Find the value of c , correct to two decimal places, if it is known that $N_A(15) = N_C(15)$.

6 The number of a type of bacteria is modelled by the formula $n = A(1 - e^{-Bt})$ where n is the size of the population at time t hours. A and B are positive constants.

a When $t = 2$, $n = 10\,000$ and when $t = 4$, $n = 15\,000$.

i Show that $2e^{-4B} - 3e^{-2B} + 1 = 0$.

ii Use the substitution $a = e^{-2B}$ to show that $2a^2 - 3a + 1 = 0$

iii Solve this equation for a . **iv** Find the exact value of B .

v Find the exact value of A .

b Sketch the graph of n against t .

c After how many hours is the population of bacteria 18 000?

7 The barometric pressure, P cm, of mercury at a height h km above sea level is given by $P = 75(10^{-0.15h})$. Find:

a P when $h = 0$

b P when $h = 10$

c h when $P = 60$

8 A radioactive substance is decaying such that the amount A g at time t years is given by the formula $A = A_0 e^{kt}$. If when $t = 1$, $A = 60.7$ and when $t = 6$, $A = 5$, find the values of the constants A_0 and k .

9 In a chemical reaction the amount (x g) of a substance that has reacted is given by:

$$x = 8(1 - e^{-0.2t})$$

where t is the time from the beginning of the reaction, in minutes.

a Sketch the graph of x against t .

b Find the amount of substance that has reacted after:

i 0 minutes

ii 2 minutes

iii 10 minutes

c Find the time when exactly 7 g of the substance has reacted.

10 Newton's law of cooling for a body placed in a medium of constant temperature states:

$$T - T_s = (T_0 - T_s)e^{-kt}$$

where:

T is the temperature (in $^{\circ}\text{C}$) of the body at time t (in minutes)

T_s is the temperature of the surrounding medium, and

T_0 is the initial temperature of the body.

An egg at 96°C is placed in a sink of water at 15°C to cool. After 5 minutes the egg's temperature is found to be 40°C . (Assume that the temperature of the water does not change.)

a Find the value of k .

b Find the temperature of the egg when $t = 10$.

c How long does it take for the egg to reach a temperature of 30°C ?

- 11** The population of a colony of small, interesting insects is modelled by the following hybrid function:

$$N(t) = \begin{cases} 20 e^{0.2t} & 0 \leq t \leq 50 \\ 20 e^{10} & 50 \leq t \leq 70 \\ 10 e^{10}(e^{70-t} + 1) & t > 70 \end{cases}$$

where t is the number of days.

- a** Sketch the graph of $N(t)$ against t .
- b** Find:
- i** $N(10)$ **ii** $N(40)$ **iii** $N(60)$ **iv** $N(80)$
- c** Find the number of days for the population to reach:
- i** 2968 **ii** 21 932

Circular functions

Objectives

- To measure angles in **degrees and radians**.
- To define the circular functions **sine, cosine and tangent**.
- To explore the **symmetry properties** of circular functions.
- To find **exact values** of circular functions.
- To sketch the **graphs of circular functions**.
- To **solve problems** with circular functions.

6.1 Review of circular (trigonometric) functions

Measuring angles in degrees and radians

The diagram shows a unit circle, i.e. a circle of radius 1 unit.

$$\begin{aligned}\text{The circumference of the unit circle} &= 2\pi \times 1 \\ &= 2\pi \text{ units}\end{aligned}$$

$\therefore$ the distance in an anti-clockwise direction around the circle from:

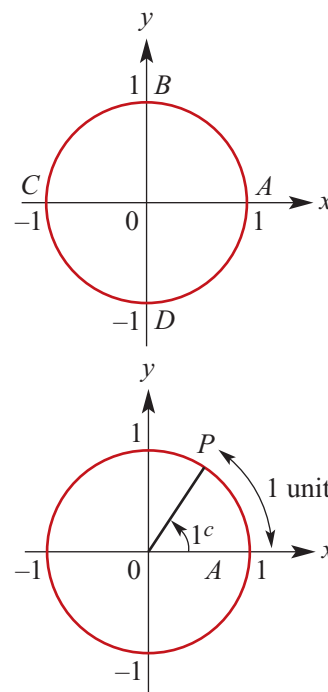
$$A \text{ to } B = \frac{\pi}{2} \text{ units}$$

$$A \text{ to } C = \pi \text{ units}$$

$$A \text{ to } D = \frac{3\pi}{2} \text{ units}$$

Definition of a radian

In moving around the circle a distance of 1 unit from A to P , the angle POA is defined. The measure of this angle is 1 radian.



One radian (written 1^c) is the angle **subtended** at the centre of the unit circle by an arc of length 1 unit.

Note: Angles formed by moving **anti-clockwise** around the circumference of the unit circle are defined as **positive**. Those formed by moving in a **clockwise** direction are said to be **negative**.

Degrees and radians

The angle, in radians, swept out in one revolution of a circle is $2\pi^c$.

$$\therefore 2\pi^c = 360^\circ$$

$$\therefore \pi^c = 180^\circ$$

$$\therefore 1^c = \frac{180^\circ}{\pi} \quad \text{or} \quad 1^\circ = \frac{\pi^c}{180}$$

Example 1

Convert 30° to radians.

Solution

$$\text{Since } 1^\circ = \frac{\pi^c}{180}$$

$$\begin{aligned} \therefore 30^\circ &= \frac{30 \times \pi^c}{180} \\ &= \frac{\pi^c}{6} \end{aligned}$$

Example 2

Convert $\frac{\pi^c}{4}$ to degrees.

Solution

$$\text{Since } 1^c = \frac{180^\circ}{\pi}$$

$$\begin{aligned} \therefore \frac{\pi^c}{4} &= \frac{\pi \times 180^\circ}{4 \times \pi} \\ &= 45^\circ \end{aligned}$$

Note: Often the symbol for radian, c , is omitted. For example, angle 45° is written as $\frac{\pi}{4}$, rather than $\frac{\pi^c}{4}$.

Exercise 6A

1 Express the following angles in radian measure in terms of π :

a 50°

b 136°

c 250°

d 340°

e 420°

f 490°

2 Express, in degrees, the angles with the following radian measures:

a $\frac{\pi}{3}$ b $\frac{5\pi}{6}$ c $\frac{4\pi}{3}$ d $\frac{7\pi}{9}$ e 3.5π f $\frac{7\pi}{5}$

3 Use a calculator to convert each of the following angles from radians to degrees:

a 0.8 b 1.64 c 2.5 d 3.96 e 4.18 f 5.95

4 Use a calculator to express each of the following in radian measure. (Give your answer correct to two decimal places.)

a 37° b 74° c 115° d 122.25° e 340° f 132.5°

Defining circular functions: sine, cosine and tangent

Considering the unit circle.

The position of point P on the circle can be described by relating the cartesian coordinates x and y and the angle, θ . The point P on the circumference corresponding to an angle θ is written $P(\theta)$.

Many different angles will give the same point, P , on the circle so the relation linking an angle to the coordinates is a many-to-one function. There are, in fact, two functions involved and they are called **sine** and **cosine** and are defined as follows:

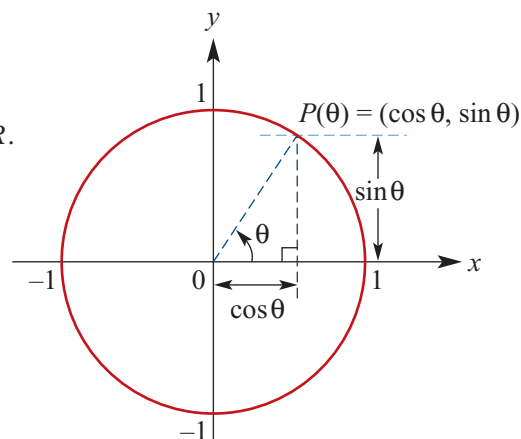
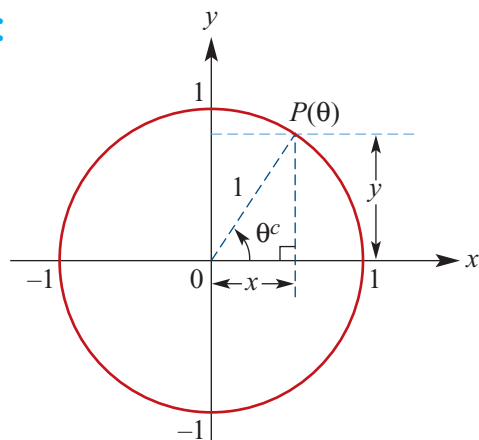
■ The x -coordinate of P , $x = \cos \theta$, $\theta \in R$.

■ The y -coordinate of P , $y = \sin \theta$, $\theta \in R$.

Note: These functions are usually written in an abbreviated form as follows:

■ $x = \cos \theta$

■ $y = \sin \theta$



Example 3

Evaluate $\sin \pi$ and $\cos \pi$.

Solution

In moving through an angle of π , the position is $P(\pi)$, which is $(-1, 0)$.

$$\therefore \cos \pi = -1$$

$$\sin \pi = 0$$

Example 4

Evaluate $\sin -\frac{3\pi}{2}$ and $\cos -\frac{\pi}{2}$.

Solution

$$\sin -\frac{3\pi}{2} = 1$$

$$\cos -\frac{\pi}{2} = 0$$

Example 5

With a calculator evaluate, correct to two decimal places:

a $\sin 1.8$

b $\cos -2.6$

c $\sin 3.8$

Solution

a 0.97

b -0.86

c -0.61

Again consider the unit circle.

If a tangent to the unit circle, at A , is drawn, then the y -coordinate of C , the point of intersection of the extension of OP and the tangent, is called **tangent θ** (abbreviated to $\tan \theta$).

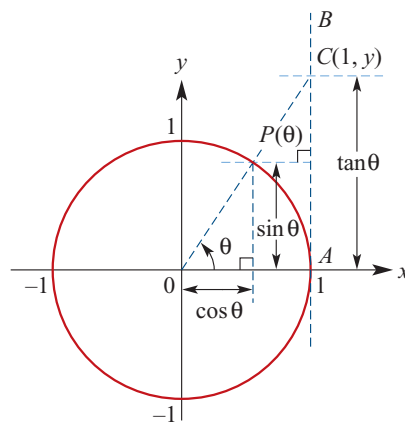
By considering the similar triangles OPD and OCA :

$$\frac{\tan \theta}{1} = \frac{\sin \theta}{\cos \theta}$$

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta}$$

Now when $\cos \theta = 0$, $\tan \theta$ is undefined hence $\tan \theta$ is undefined when:

$$\theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots \therefore \text{domain of } \tan = R \setminus \{\theta : \cos \theta = 0\}$$

**Example 6**

Evaluate, using a calculator:

a $\tan 1.3$

b $\tan 1.9$

c $\tan -2.8$

d $\tan 59^\circ$

e $\tan 138^\circ$

Solution

a $\tan 1.3 = 3.6$

b $\tan 1.9 = -2.93$

c $\tan -2.8 = 0.36$

d $\tan 59^\circ = 1.66$

e $\tan 138^\circ = -0.9$

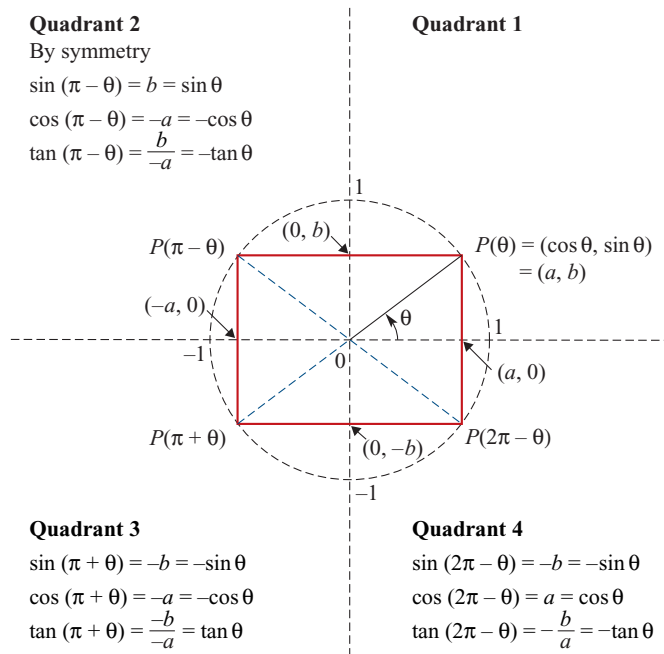
(Your calculator is in **Radian** mode for **a**, **b** and **c** and **Degree** mode for **d** and **e**.)

Exercise 6B

- For each of the following angles, t , determine the values of $\sin t$ and $\cos t$:
 - $t = 0$
 - $t = \frac{3\pi}{2}$
 - $t = -\frac{5\pi}{2}$
 - $t = -\frac{7\pi}{2}$
 - $t = 3\pi$
 - $t = -4\pi$
- Evaluate each of the following:
 - $\tan(-\pi)$
 - $\tan \frac{5\pi}{2}$
 - $\tan\left(-\frac{7\pi}{2}\right)$
 - $\tan 2\pi$
 - $\tan\left(-\frac{\pi}{2}\right)$
- Evaluate each of the following:
 - $\cos(23\pi)$
 - $\cos\left(\frac{49\pi}{2}\right)$
 - $\cos(-35\pi)$
 - $\cos\left(\frac{-45\pi}{2}\right)$
 - $\cos(-24\pi)$
 - $\cos(20\pi)$
- Evaluate each of the following using a calculator. (Give answers correct to two decimal places.)
 - $\sin 1.7$
 - $\sin 2.6$
 - $\sin 4.2$
 - $\cos 0.4$
 - $\cos 2.3$
 - $\cos(-1.8)$
 - $\sin(-1.7)$
 - $\sin(-3.6)$
 - $\tan 1.6$
 - $\tan(-1.2)$
 - $\tan 3.9$
 - $\tan(-2.5)$

Symmetry properties of circular functions

The following relationships can be observed:



Note: These relationships are true for all values of θ .

Signs of circular functions

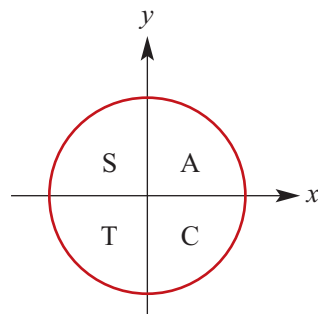
These symmetry properties can be summarised for the signs of $\sin$, $\cos$ and $\tan$ for the four quadrants as follows:

1st quadrant: all are positive (A).

2nd quadrant: $\sin$ is positive (S).

3rd quadrant: $\tan$ is positive (T).

4th quadrant: $\cos$ is positive (C).



Example 7

If $\sin \theta = 0.5$ and $\cos \alpha = 0.6$, find the value of:

a $\sin(\pi - \theta)$

b $\cos(2\pi - \alpha)$

Solution

By symmetry:

a $\sin(\pi - \theta) = \sin \theta$
 $= 0.5$

b $\cos(2\pi - \alpha) = \cos \alpha$
 $= 0.6$

Example 8

For the following find two values of x in the range $0 \leq x \leq 360$:

a $\sin x^\circ = -0.3$

b $\cos x^\circ = -0.7$

Solution

a First solve the equation $\sin x^\circ = 0.3$

Make sure that your calculator is in degree mode.

If $\sin x^\circ = 0.3$

$x = 17.46$

Now the value of $\sin$ is negative for $P(x)$ in the 3rd and 4th quadrants.

From the symmetry relationships:

3rd quadrant $x = 180 + 17.46$
 $= 197.46$

4th quadrant $x = 360 - 17.46$
 $= 342.54$

$\therefore$ if $\sin x^\circ = -0.3$, $x = 197.46, 342.54$

b First solve the equation $\cos x^\circ = 0.7$, i.e. $x = 45.57$

Now the value of $\cos$ is negative for $P(x)$ in the 2nd and 3rd quadrants.

2nd quadrant $x = 180 - 45.57$
 $= 134.43$

3rd quadrant $x = 180 + 45.57$
 $= 225.57$

$\therefore$ if $\cos x^\circ = -0.7$, $x = 134.43, 225.57$

Exercise 6C

- 1 If $\sin \theta = 0.52$, $\cos x = 0.68$ and $\tan \alpha = 0.4$ find the value of:

a $\sin(\pi - \theta)$	b $\cos(\pi + x)$	c $\sin(2\pi + \theta)$	d $\tan(\pi + \alpha)$
e $\sin(\pi + \theta)$	f $\cos(2\pi - x)$	g $\tan(2\pi - \alpha)$	h $\cos(\pi - x)$
- 2 Find all values of x between 0 and 2π for which:

a $\sin x = 0.6$	b $\cos x = 0.8$	c $\sin x = -0.45$	d $\cos x = -0.2$
-------------------------	-------------------------	---------------------------	--------------------------
- 3 Find all values θ between 0 and 360 for which:

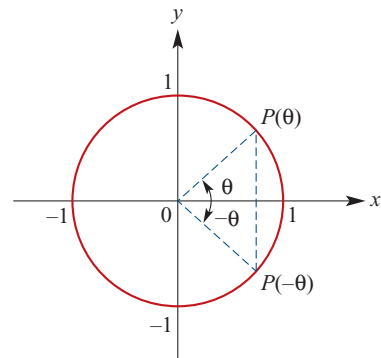
a $\sin \theta^\circ = 0.3$	b $\cos \theta^\circ = 0.4$	c $\sin \theta^\circ = -0.8$	d $\cos \theta^\circ = -0.5$
------------------------------------	------------------------------------	-------------------------------------	-------------------------------------

Further symmetry properties

Negative of angles

By symmetry:

$$\begin{aligned}\cos(-\theta) &= \cos \theta \\ \sin(-\theta) &= -\sin \theta \\ \tan(-\theta) &= \frac{-\sin \theta}{\cos \theta} \\ &= -\tan \theta\end{aligned}$$

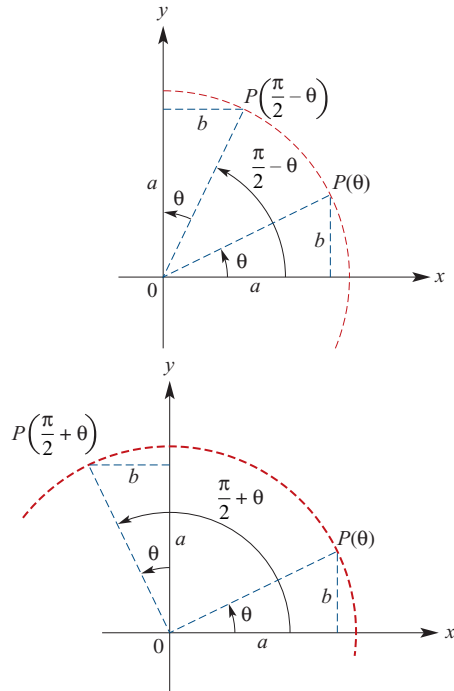


Complementary relationships

$$\begin{aligned}\sin\left(\frac{\pi}{2} - \theta\right) &= a \\ \text{and since } a &= \cos \theta \\ \therefore \sin\left(\frac{\pi}{2} - \theta\right) &= \cos \theta\end{aligned}$$

Similarly:

$$\begin{aligned}\cos\left(\frac{\pi}{2} - \theta\right) &= b \\ \text{and since } b &= \sin \theta \\ \therefore \cos\left(\frac{\pi}{2} - \theta\right) &= \sin \theta \\ \sin\left(\frac{\pi}{2} + \theta\right) &= a \\ &= \cos \theta \\ \cos\left(\frac{\pi}{2} + \theta\right) &= -b \\ &= -\sin \theta\end{aligned}$$



Example 9

If $\sin \theta = 0.3$ and $\cos \psi = 0.8$, find the values of:

a $\sin\left(\frac{\pi}{2} - \psi\right)$	b $\cos\left(\frac{\pi}{2} + \theta\right)$	c $\sin(-\theta)$
--	--	--------------------------

Solution

$$\text{a } \sin\left(\frac{\pi}{2} - \psi\right) = \cos \psi \\ = 0.8$$

$$\text{b } \cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta \\ = -0.3$$

$$\text{c } \sin(-\theta) = -\sin \theta \\ = -0.3$$

Exercise 6D

If $\sin x = 0.3$, $\cos \alpha = 0.6$ and $\tan \theta = 0.7$, find the values of:

- | | | | |
|--|---|--|---|
| 1 $\cos(-\alpha)$ | 2 $\sin\left(\frac{\pi}{2} + \alpha\right)$ | 3 $\tan(-\theta)$ | 4 $\cos\left(\frac{\pi}{2} - x\right)$ |
| 5 $\sin(-x)$ | 6 $\tan\left(\frac{\pi}{2} - \theta\right)$ | 7 $\cos\left(\frac{\pi}{2} + x\right)$ | 8 $\sin\left(\frac{\pi}{2} - \alpha\right)$ |
| 9 $\sin\left(\frac{3\pi}{2} + \alpha\right)$ | 10 $\cos\left(\frac{3\pi}{2} - x\right)$ | | |

Exact values of circular functions

A calculator can be used to find the values of the circular functions for different values of θ . For many values of θ the calculator gives an approximation. We consider some values of θ such that $\sin$, $\cos$ and $\tan$ can be calculated exactly.

Exact values for 0 (0°) and $\frac{\pi}{2}$ (90°)

From the unit circle:

when $\theta = 0$, $\sin \theta = 0$

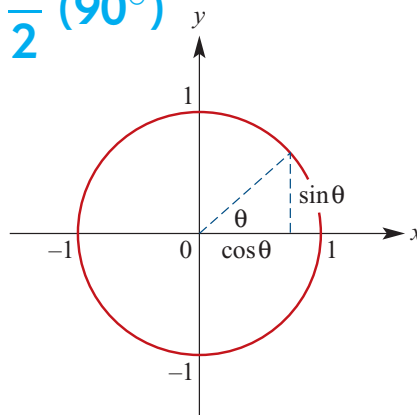
$$\cos \theta = 1$$

$$\tan \theta = 0$$

when $\theta = \frac{\pi}{2}$, $\sin \frac{\pi}{2} = 1$

$$\cos \frac{\pi}{2} = 0$$

$$\tan \frac{\pi}{2} \text{ is undefined.}$$

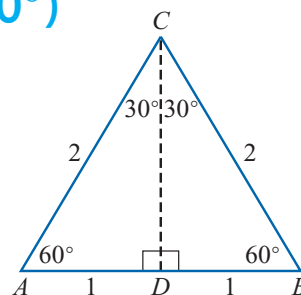
**Exact values for $\frac{\pi}{6}$ (30°) and $\frac{\pi}{3}$ (60°)**

Consider an equilateral triangle ABC of side length 2 units.

In $\triangle ACD$:

by the theorem of Pythagoras:

$$DC = \sqrt{AC^2 - AD^2} \\ = \sqrt{3}$$



$$\begin{aligned}\sin 30^\circ &= \frac{AD}{AC} \\ &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\cos 30^\circ &= \frac{CD}{AC} \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

$$\begin{aligned}\tan 30^\circ &= \frac{AD}{CD} \\ &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}\sin 60^\circ &= \frac{CD}{AC} \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

$$\begin{aligned}\cos 60^\circ &= \frac{AD}{AC} \\ &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\tan 60^\circ &= \frac{CD}{AD} \\ &= \frac{\sqrt{3}}{1} \\ &= \sqrt{3}\end{aligned}$$

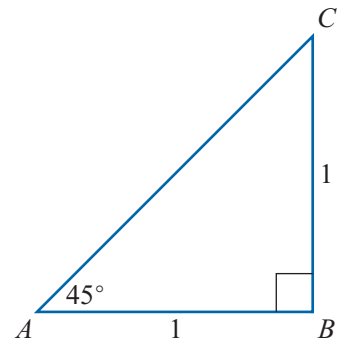
Exact values for $\frac{\pi}{4}$ (45°)

$$\begin{aligned}AC &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2}\end{aligned}$$

$$\begin{aligned}\sin 45^\circ &= \frac{BC}{AC} \\ &= \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}\cos 45^\circ &= \frac{AB}{AC} \\ &= \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}\tan 45^\circ &= \frac{BC}{AB} \\ &= 1\end{aligned}$$



As an aid to memory the exact values for circular functions can be tabulated.

Summary

θ (θ°)	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$ (30°)	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$ (45°)	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$ (60°)	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$ (90°)	1	0	undefined

Example 10

Evaluate:

a $\cos \frac{5\pi}{4}$

b $\sin \frac{11\pi}{6}$

Solution

$$\begin{aligned} \mathbf{a} \quad \cos \frac{5\pi}{4} &= \cos \left(\pi + \frac{\pi}{4} \right) \\ &= -\cos \frac{\pi}{4} \text{ (by symmetry)} \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad \sin \frac{11\pi}{6} &= \sin \left(2\pi - \frac{\pi}{6} \right) \\ &= -\sin \frac{\pi}{6} \text{ (by symmetry)} \\ &= -\frac{1}{2} \end{aligned}$$

Example 11Find all values of θ between 0 and 360 for which:

a $\cos \theta^\circ = \frac{\sqrt{3}}{2}$

b $\sin \theta^\circ = -\frac{1}{2}$

c $\cos \theta^\circ - \frac{1}{\sqrt{2}} = 0$

Solution

- a** $\cos \theta^\circ$ is positive
 $\therefore P(\theta^\circ)$ lies in the 1st or 4th quadrant.

$$\begin{aligned} \cos \theta^\circ &= \frac{\sqrt{3}}{2} \\ \theta &= 30 \text{ or } 360 - 30 \\ &= 30 \text{ or } 330 \end{aligned}$$

- b** $\sin \theta^\circ$ is negative
 $\therefore P(\theta^\circ)$ is in the 3rd or 4th quadrant.

$$\begin{aligned} \sin \theta^\circ &= -\frac{1}{2} \\ \theta &= 180 + 30 \text{ or } 360 - 30 \\ &= 210 \text{ or } 330 \end{aligned}$$

c $\cos \theta^\circ - \frac{1}{\sqrt{2}} = 0$

$$\therefore \cos \theta^\circ = \frac{1}{\sqrt{2}}$$

and since $\cos \theta^\circ$ is positive, $P(\theta^\circ)$ lies in the 1st or 4th quadrant.

$$\begin{aligned} \therefore \cos \theta^\circ &= \frac{1}{\sqrt{2}} \\ \theta &= 45 \text{ or } 360 - 45 \\ &= 45 \text{ or } 315 \end{aligned}$$

Example 12

Find all solutions to the equation $\sin \theta = \frac{1}{2}$ for $\theta \in [0, 4\pi]$.

Solution

We refer to the graph of $y = \sin \theta$ that was studied in *Essential Mathematical Methods 1 & 2 CAS*. The graph of $y = \sin \theta$ is reintroduced in the following sections.

It is clear from the graph that there are four solutions in the interval $[0, 4\pi]$.

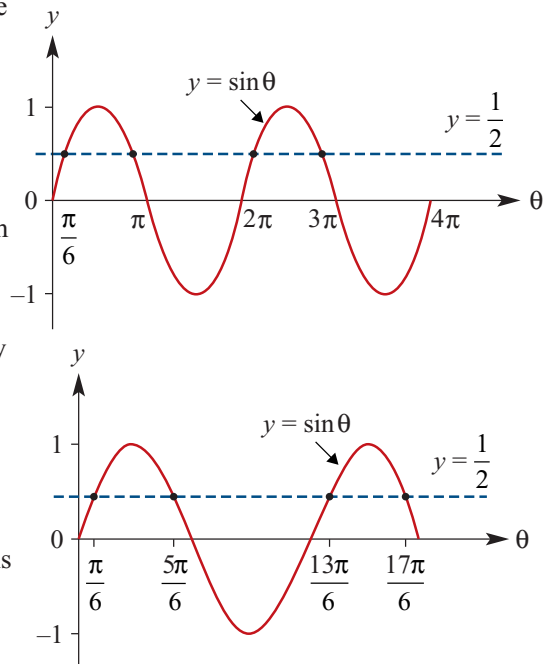
The solution for $x \in [0, \frac{\pi}{2}]$ is $x = \frac{\pi}{6}$.

This solution can be obtained from a knowledge of exact values or using $\sin^{-1}$ on your calculator.

The second solution is obtained by symmetry. The function is positive in the second quadrant and $\sin(\pi - \theta) = \sin \theta$. Therefore $x = \frac{5\pi}{6}$ is the second solution.

It can be seen that further solutions can be achieved by adding 2π , as $\sin \theta = \sin(\theta + 2\pi)$.

Thus $\theta = \frac{13\pi}{6}$ and $\frac{17\pi}{6}$ are also solutions.

**Example 13**

For $\pi < x < \frac{3\pi}{2}$ with $\cos x = -\sin \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).

Solution

$$\cos x = \sin \left(\frac{\pi}{2} - x \right) = -\sin \frac{\pi}{6}$$

$$\text{Hence } \sin \left(\frac{\pi}{2} - x \right) = \sin \left(\frac{7\pi}{6} \right) \text{ as } \sin(\pi + \theta) = \sin(-\theta)$$

$$\text{Solving the equation } \frac{\pi}{2} - x = \frac{7\pi}{6} \text{ gives } x = -\frac{2\pi}{3}$$

This is one solution of the equation $\sin \left(\frac{\pi}{2} - x \right) = \sin \left(\frac{7\pi}{6} \right)$. There are infinitely many.

$$\text{Thus a solution in the interval } \pi < x < \frac{3\pi}{2} \text{ is } x = \frac{4\pi}{3}.$$

Exercise 6E



1 Without using a calculator evaluate the sin, cos and tan of each of the following:

- a** 150° **b** 225° **c** 405° **d** -120° **e** -315° **f** -30°

2 Write down the exact values of:

- a** $\sin \frac{3\pi}{4}$ **b** $\cos \frac{2\pi}{3}$ **c** $\cos \frac{7\pi}{6}$ **d** $\sin \frac{5\pi}{6}$
e $\cos \frac{4\pi}{3}$ **f** $\sin \frac{5\pi}{4}$ **g** $\sin \frac{7\pi}{4}$ **h** $\cos \frac{5\pi}{3}$
i $\cos \frac{11\pi}{3}$ **j** $\sin \frac{200\pi}{3}$ **k** $\cos -\frac{11\pi}{3}$ **l** $\sin \frac{25\pi}{3}$
m $\sin -\frac{13\pi}{4}$ **n** $\cos -\frac{20\pi}{3}$ **o** $\sin \frac{67\pi}{4}$ **p** $\cos \frac{68\pi}{3}$

3 Find, without using a calculator, all the values of θ between 0 and 360 for each of the following:

- a** $\cos \theta^\circ = \frac{1}{2}$ **b** $\sin \theta^\circ = \frac{\sqrt{3}}{2}$ **c** $\sin \theta^\circ = -\frac{1}{2}$
d $2 \cos \theta^\circ + 1 = 0$ **e** $2 \sin \theta^\circ = \sqrt{3}$ **f** $2 \cos \theta^\circ = -\sqrt{3}$

4 Solve each of the following for $x \in [-\pi, \pi]$:

- a** $\sin x = -\frac{1}{2}$ **b** $\cos x = \frac{\sqrt{3}}{2}$ **c** $\cos x = -\frac{\sqrt{3}}{2}$

- 5 **a** For $\frac{\pi}{2} < x < \pi$ with $\sin x = \sin \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).
b For $\frac{\pi}{2} < x < \pi$ with $\cos x = -\cos \frac{\pi}{6}$, find the value of x (do not evaluate $\cos \frac{\pi}{6}$).
c For $\frac{\pi}{2} < x < \pi$ with $\cos x = -\sin \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).
d For $\frac{\pi}{2} < a < \pi$ with $\cos a = -\sin b$ where $0 < b < \frac{\pi}{2}$, find a in terms of b .
e For $\frac{\pi}{2} < a < \pi$ with $\sin a = \cos b$ where $0 < b < \frac{\pi}{2}$, find a in terms of b .

- 6 **a** For $\pi < x < \frac{3\pi}{2}$ with $\sin x = -\sin \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).
b For $\pi < x < \frac{3\pi}{2}$ with $\cos x = -\sin \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).
c For $\pi < x < \frac{3\pi}{2}$ with $\cos x = -\cos \frac{\pi}{6}$, find the value of x (do not evaluate $\sin \frac{\pi}{6}$).
d For $\pi < a < \frac{3\pi}{2}$ with $\cos a = -\sin b$ where $0 < b < \frac{\pi}{2}$, find a in terms of b .
e For $\pi < a < \frac{3\pi}{2}$ with $\sin a = -\sin b$ where $0 < b < \frac{\pi}{2}$, find a in terms of b .
f For $\pi < a < \frac{3\pi}{2}$ with $\tan a = \tan b$ where $0 < b < \frac{\pi}{2}$, find a in terms of b .

7 Find, without using a calculator, all the values of x between 0 and 2π for each of the following:

- a** $\sqrt{2} \sin x - 1 = 0$ **b** $\sqrt{2} \cos x + 1 = 0$ **c** $2 \cos x + \sqrt{3} = 0$
d $2 \sin x + 1 = 0$ **e** $1 - \sqrt{2} \cos x = 0$ **f** $4 \cos x + 2 = 0$

- 8 Find, without using a calculator, all the values of x between 0 and 2π for each of the following:

a $\sqrt{2} \sin 2x + 1 = 0$

b $2 \cos 3x + \sqrt{3} = 0$

c $4 \sin 2x - 2 = 0$

d $\sqrt{2} \cos 3x - 1 = 0$

e $10 \sin 3x - 5 = 0$

f $2 \sin 2x = \sqrt{2}$

g $4 \cos 3x = -2\sqrt{3}$

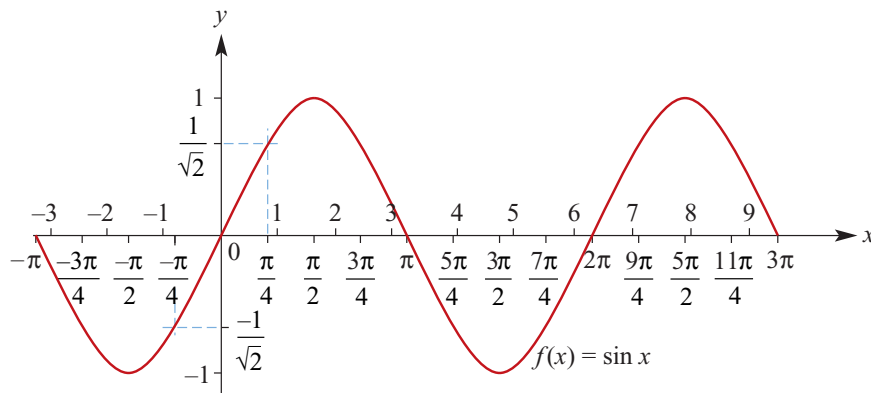
h $-2 \sin 3x = \sqrt{2}$

i $4 \cos 2x = -2$

6.2 Graphs of sine and cosine

Graphs of sine functions

The graph of $f(x) = \sin x$ is given below. It has been plotted for $-\pi \leq x \leq 3\pi$.

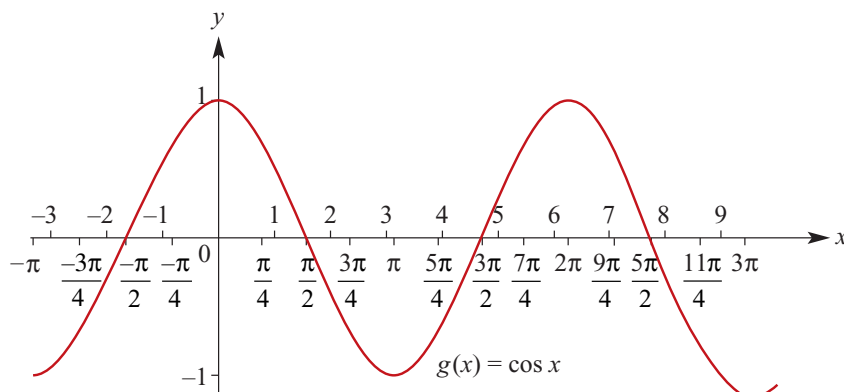


Observations from the graph

- The graph repeats itself after an interval of 2π units, i.e. $f(x + 2k\pi) = f(x)$ for all $x \in \mathbb{R}, k \in \mathbb{Z}$. A function which repeats itself regularly is called a **periodic** function and the interval between the repetitions is called the **period** of the function. Thus $\sin x$ has period of 2π units.
- The maximum and minimum values of $\sin x$ are 1 and -1 respectively. The distance between the mean position and the maximum position is called the **amplitude**. The graph of $f(x) = \sin x$ has an amplitude of 1.

Graphs of cosine functions

The graph of $g(x) = \cos x$ for $-\pi \leq x \leq 3\pi$ is as shown.



6.3 Transformations applied to graphs of $y = \sin x$ and $y = \cos x$

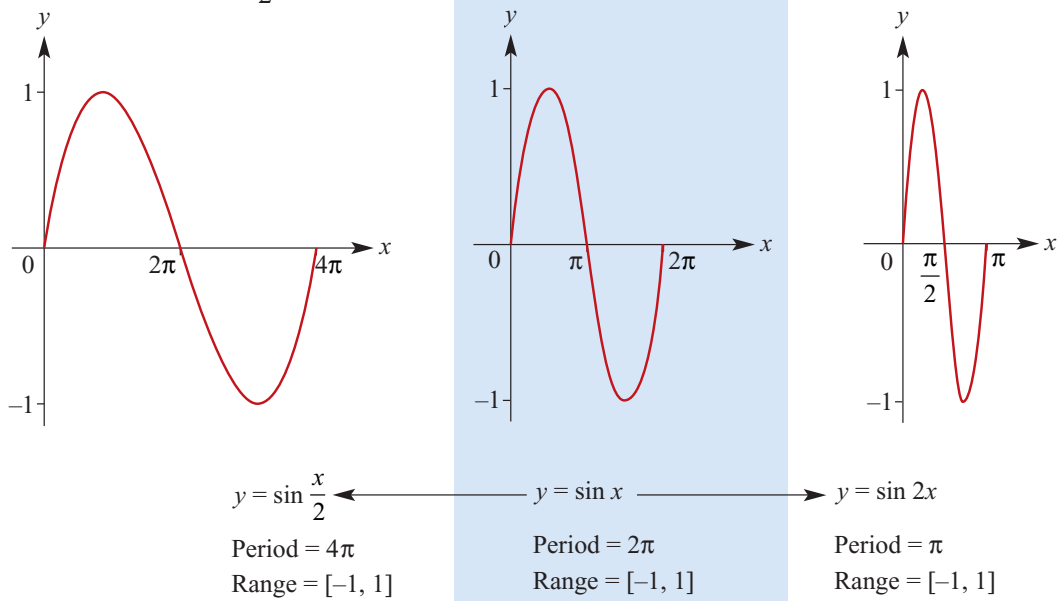
Dilations



A dilation of factor 2 from the y -axis has the rule $(x, y) \rightarrow (2x, y)$.

Hence $(0, 0) \rightarrow (0, 0)$, $(\frac{\pi}{2}, 1) \rightarrow (\pi, 1)$ and $(\pi, 0) \rightarrow (2\pi, 0)$. When this transformation is applied to $y = \sin x$ it will be ‘stretched out’ parallel to the x -axis. Let $(x, y) \rightarrow (x', y')$ under this transformation. Then $x' = 2x$ and $y' = y$, and thus $x = \frac{x'}{2}$ and $y = y'$. Hence $y = \sin x$ is mapped to $y = \sin \frac{x}{2}$.

A dilation of factor $\frac{1}{2}$ from the y -axis will map $y = \sin x$ to $y = \sin 2x$.

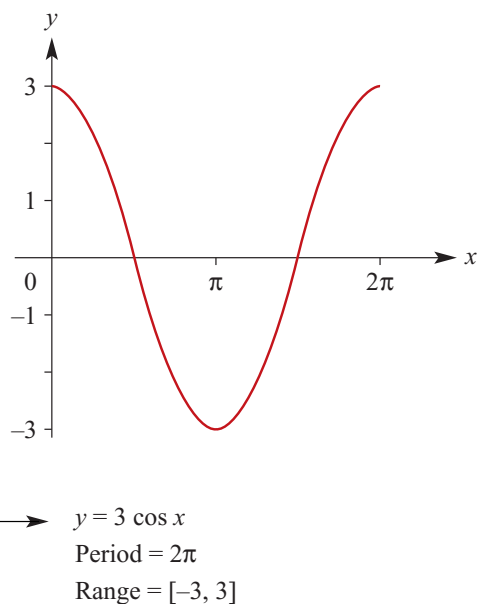
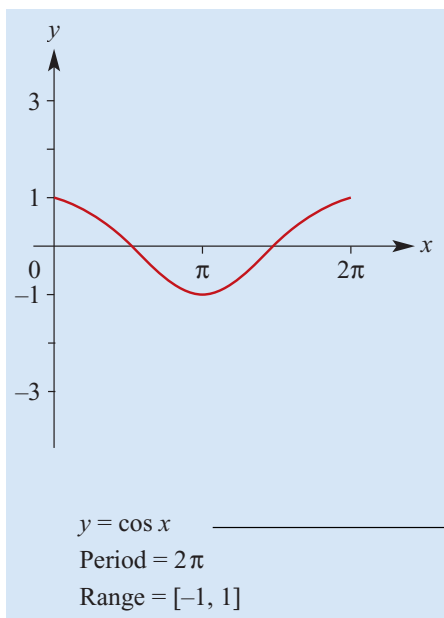


In general:

- | | |
|---|---|
| ■ $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin(nx)$ | ■ $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \cos(nx)$ |
| ■ Period = $\frac{2\pi}{ n }$ | ■ Period = $\frac{2\pi}{ n }$ |
| ■ Amplitude = 1 | ■ Amplitude = 1 |
| ■ Range = $[-1, 1]$ | ■ Range = $[-1, 1]$ |

A dilation of factor 3 from the x -axis has rule $(x, y) \rightarrow (x, 3y)$. Hence $(0, 1) \rightarrow (0, 3)$, $(\frac{\pi}{2}, 0) \rightarrow (\frac{\pi}{2}, 0)$ and $(\pi, -1) \rightarrow (\pi, -3)$. When this transformation is applied to $y = \cos x$, it will be ‘stretched out’ parallel to the y -axis.

$y = \cos x$ is mapped to $y = 3 \cos x$.



In general:

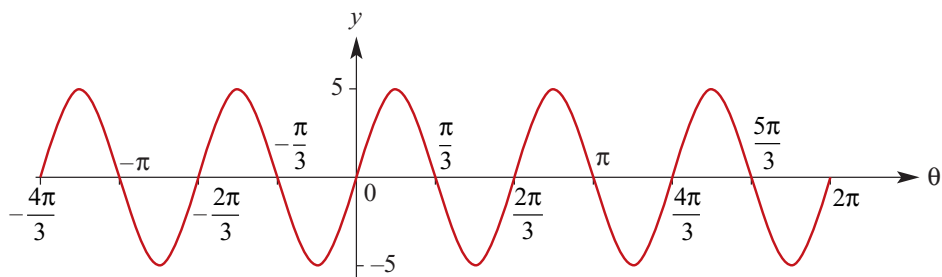
- | | |
|---|---|
| ■ $f: R \rightarrow R, f(x) = a \sin(nx)$ | ■ $f: R \rightarrow R, f(x) = a \cos(nx)$ |
| ■ Period = $\frac{2\pi}{ n }$ | ■ Period = $\frac{2\pi}{ n }$ |
| ■ Amplitude = $ a $ | ■ Amplitude = $ a $ |
| ■ Range = $[- a , a]$ | ■ Range = $[- a , a]$ |

Example 14

Sketch the graph of $y = 5 \sin 3\theta$ for $-\frac{4\pi}{3} \leq \theta \leq 2\pi$.

Solution

The amplitude = 5
 The period = $\frac{2\pi}{3}$



The θ -axis intercepts can also be found by solving the equation.

$$5 \sin 3\theta = 0$$

$$\therefore \sin 3\theta = 0$$

$$\therefore 3\theta = -4\pi, -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi$$

$$\theta = -\frac{4\pi}{3}, -\pi, -\frac{2\pi}{3}, -\frac{\pi}{3}, 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi$$

Exercise 6F

- 1 Write down **i** the period and **ii** the amplitude of each of the following:

a $3 \sin \theta$

b $2 \sin 3\theta$

c $\frac{1}{2} \cos 2\theta$

d $2 \sin \frac{1}{3}\theta$

e $3 \cos 4\theta$

f $\frac{1}{2} \sin \theta$

g $3 \cos \frac{1}{2}\theta$

- 2 Sketch the graph of each of the following, showing one complete cycle. State the amplitude and period.

a $y = 2 \sin 3\theta$

b $y = 2 \cos 2\theta$

c $y = 3 \sin \frac{1}{3}\theta$

d $y = \frac{1}{3} \cos 2\theta$

e $y = 3 \sin 4\theta$

- 3 Sketch the graph of $f: R \rightarrow R$, $f(x) = 5 \cos 3x$ for $-\pi \leq x \leq \pi$.

- 4 Sketch the graph of $f: R \rightarrow R$, $f(x) = \frac{1}{2} \sin 2x$ for $-\pi \leq x \leq 2\pi$.

- 5 Sketch the graph of $f: R \rightarrow R$, $f(x) = 2 \cos \frac{3x}{2}$ for $0 \leq x \leq 2\pi$.

- 6 Find the equation of the image of the graph of $y = \sin x$ under a dilation of factor 2 from the x -axis followed by a dilation of factor 3 from the y -axis.

- 7 Find the equation of the image of the graph of $y = \cos x$ under a dilation of factor $\frac{1}{2}$ from the x -axis followed by a dilation of factor 3 from the y -axis.

- 8 Find the equation of the image of the graph of $y = \sin x$ under a dilation of factor $\frac{1}{2}$ from the x -axis followed by a dilation of factor 2 from the y -axis.

Reflections in the axes

The function with rule $f(x) = \sin x$ is an odd function, i.e. $f(-x) = -f(x)$. A reflection in the y -axis gives the ‘same’ result as a reflection in the x -axis when applied to the graph of $y = \sin x$.

The function with rule $f(x) = \cos x$ is an even function, i.e. $f(-x) = f(x)$. The graph of $f(x) = \cos x$ is mapped onto itself when reflected in the y -axis.

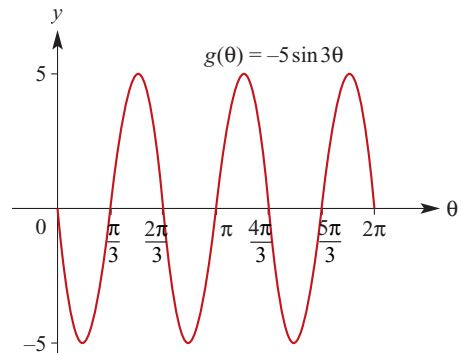
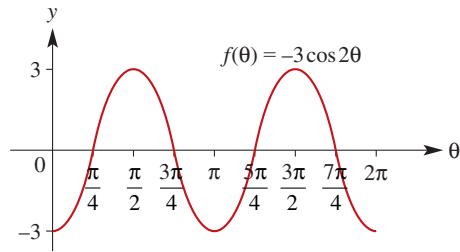
Example 15

Sketch the graph of:

- a $f(\theta) = -3 \cos 2\theta$ for $0 \leq \theta \leq 2\pi$
 b $g(\theta) = 5 \sin(-3\theta)$ for $0 \leq \theta \leq 2\pi$

Solution

- a Period = π
 Amplitude = 3
 b Period = $\frac{2\pi}{3}$
 Amplitude = $|-5|$
 $= 5$

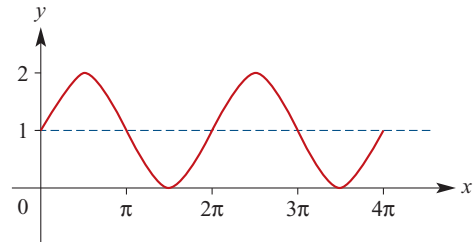


Translations

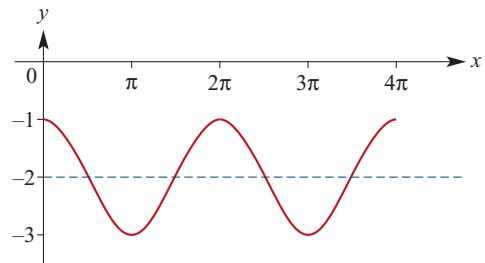
Translations in the direction of the y-axis

The graph of $y = \sin x + 1$ is obtained from the graph of $y = \sin x$ by a translation of 1 unit in the positive direction of the y-axis.

$$\begin{aligned}\text{Period} &= 2\pi \\ \text{Range} &= [0, 2] \\ \text{Amplitude} &= 1\end{aligned}$$

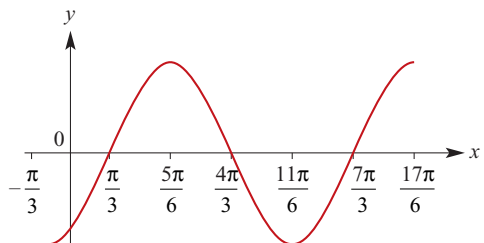


The graph of $y = \cos 2x - 2$ is obtained from the graph of $y = \cos 2x$ by a translation of 2 units in the negative direction of the y-axis.

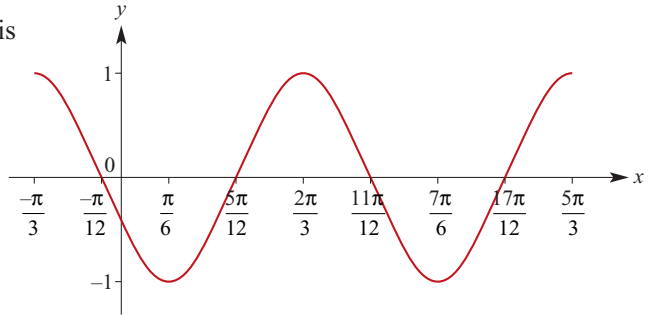


Translations in the direction of the x-axis

The graph of $y = \sin\left(x - \frac{\pi}{3}\right)$ is obtained from the graph of $y = \sin x$ by a translation of $\frac{\pi}{3}$ units in the positive direction of the x-axis.



The graph of $y = \cos 2\left(x + \frac{\pi}{3}\right)$ is obtained from the graph of $y = \cos 2x$ by a translation of $\frac{\pi}{3}$ units in the negative direction of the x -axis.



Example 16

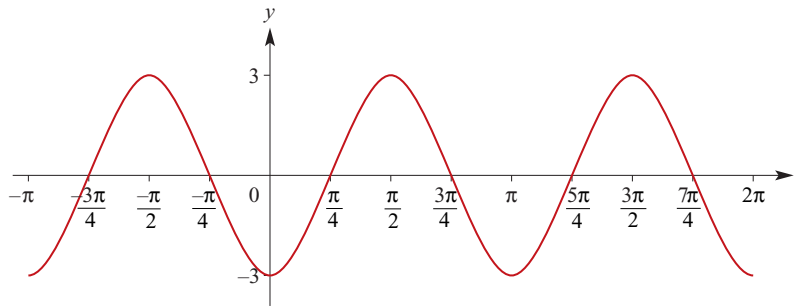
On separate axes sketch the graphs of:

a $y = 3 \sin 2\left(t - \frac{\pi}{4}\right)$ for $-\pi \leq t \leq 2\pi$ **b** $y = 3 \cos 3\left(t + \frac{\pi}{3}\right)$ for $-\pi \leq t \leq \pi$

Solution

a **Note:** the transformations applied to $y = \sin t$ are:

- a dilation of factor 3 from the x -axis
- a dilation of factor $\frac{1}{2}$ from the y -axis
- a translation of $\frac{\pi}{4}$ in the positive direction of the x -axis.



$$\text{Period} = \pi$$

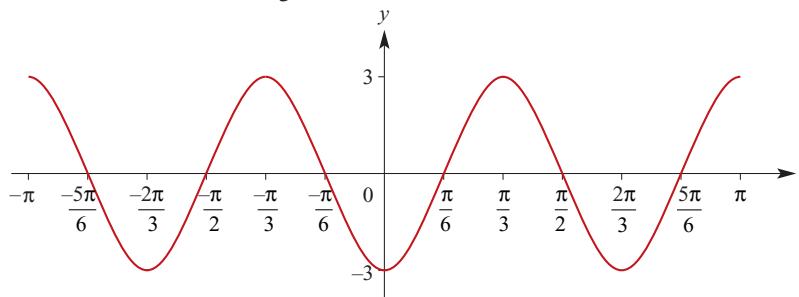
$$\text{Amplitude} = 3$$

$$\text{Range} = [-3, 3]$$

Note: This is the graph of $y = -3 \cos 2t$.

b **Note:** the transformations applied to $y = \cos t$ are:

- a dilation of factor 3 from the x -axis
- a dilation of factor $\frac{1}{3}$ from the y -axis
- a translation of $\frac{\pi}{3}$ in the negative direction of the x -axis.



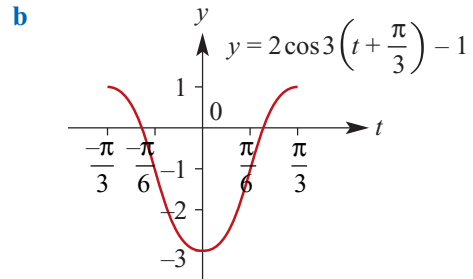
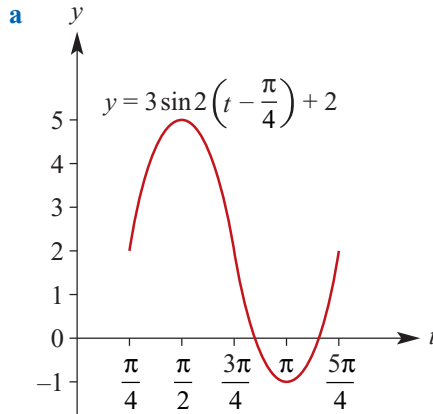
Sketching graphs of $y = a \sin n(t \pm \varepsilon) \pm b$ and $y = a \cos n(t \pm \varepsilon) \pm b$

Example 17

On separate axes sketch the graphs of:

a $y = 3 \sin 2 \left(t - \frac{\pi}{4} \right) + 2, \frac{\pi}{4} \leq t \leq \frac{5\pi}{4}$ **b** $y = 2 \cos 3 \left(t + \frac{\pi}{3} \right) - 1, -\frac{\pi}{3} \leq t \leq \frac{\pi}{3}$

Solution



Observations

- The graph of $y = 3 \sin 2 \left(t - \frac{\pi}{4} \right) + 2$ is the same shape as the graph of $y = 3 \sin 2 \left(t - \frac{\pi}{4} \right)$, but it is translated 2 units in the positive direction of the y -axis.
- Similarly, the graph of $y = 2 \cos 3 \left(t + \frac{\pi}{3} \right) - 1$ is the same shape as the graph of $y = 2 \cos 3 \left(t + \frac{\pi}{3} \right)$, but it is translated 1 unit in the negative direction of the y -axis.

Exercise 6G

- 1 Sketch the graph of each of the following, showing one complete cycle. State the period, amplitude and range in each case.

a $y = 2 \sin \left(\theta - \frac{\pi}{3} \right)$	b $y = \sin 2(\theta - \pi)$
c $y = 3 \sin 2 \left(\theta + \frac{\pi}{4} \right)$	d $y = \sqrt{3} \sin 3 \left(\theta - \frac{\pi}{2} \right)$
e $y = 2 \sin 3x + 1$	f $y = 3 \cos 2 \left(x + \frac{\pi}{2} \right) - 1$
g $y = \sqrt{2} \sin 2 \left(\theta - \frac{\pi}{6} \right) + 2$	h $y = 3 - 4 \sin 2x$ i $y = 2 - 3 \cos 2 \left(\theta - \frac{\pi}{2} \right)$
- 2 Find the equation of the image of the graph of $y = \cos x$ under:
 - a** a dilation of factor $\frac{1}{2}$ from the x -axis, followed by a dilation of factor 3 from the y -axis, followed by a translation of $\frac{\pi}{4}$ units in the positive direction of the x -axis.

- b** a dilation of factor 2 from the x -axis, followed by a translation of $\frac{\pi}{4}$ units in the positive direction of the x -axis.
- c** a dilation of factor $\frac{1}{3}$ from the x -axis, followed by a reflection in the x -axis, followed by a translation of $\frac{\pi}{3}$ units in the positive direction of the x -axis.

Finding axis intercepts

Example 18

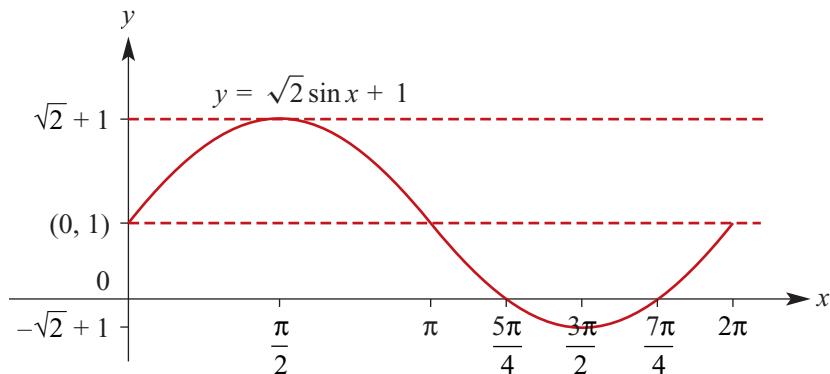
Sketch the graphs of each of the following for $x \in [0, 2\pi]$. Clearly indicate axis intercepts.

a $y = \sqrt{2} \sin x + 1$ **b** $y = 2 \cos 2x - 1$ **c** $y = 2 \sin 2\left(x - \frac{\pi}{3}\right) - \sqrt{3}$

Solution

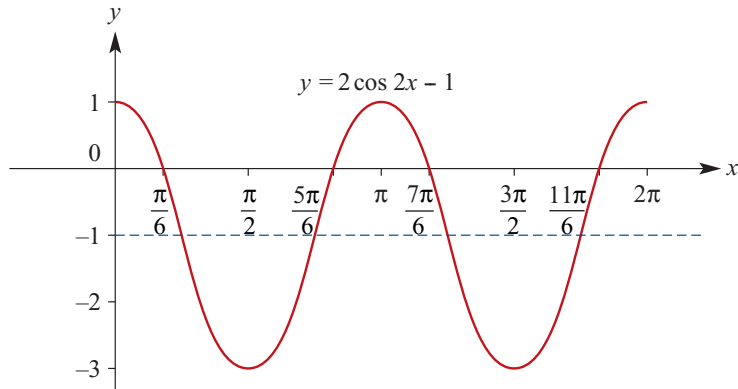
- a** To determine the axis intercepts the equation $\sqrt{2} \sin x + 1 = 0$ must be solved.

$$\begin{aligned}\sqrt{2} \sin x + 1 &= 0 \\ \therefore \sin x &= -\frac{1}{\sqrt{2}} \\ \therefore x &= \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4} \\ \therefore x &= \frac{5\pi}{4}, \frac{7\pi}{4} \\ \therefore \text{intercepts are } &\left(\frac{5\pi}{4}, 0\right), \left(\frac{7\pi}{4}, 0\right)\end{aligned}$$



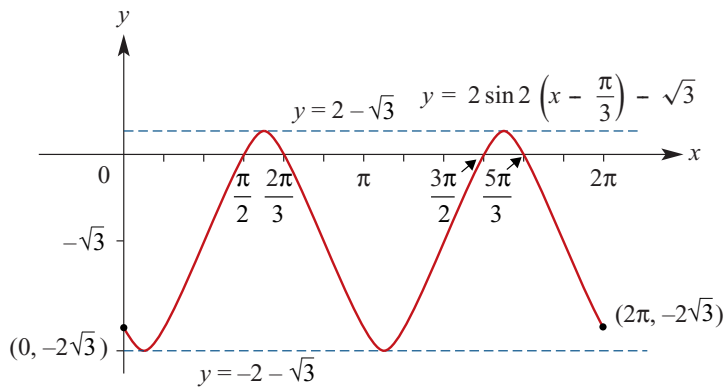
$$\begin{aligned}
 \text{b } 2 \cos 2x - 1 &= 0 \\
 \therefore \cos 2x &= \frac{1}{2} \\
 \therefore 2x &= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \\
 \therefore x &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}
 \end{aligned}$$

$$\therefore \text{intercepts are } \left(\frac{\pi}{6}, 0\right), \left(\frac{5\pi}{6}, 0\right), \left(\frac{7\pi}{6}, 0\right), \left(\frac{11\pi}{6}, 0\right)$$



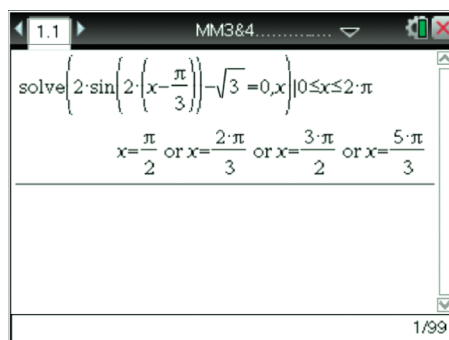
$$\begin{aligned}
 \text{c } \sin 2\left(x - \frac{\pi}{3}\right) &= \frac{\sqrt{3}}{2} \\
 \therefore 2\left(x - \frac{\pi}{3}\right) &= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3} \\
 \therefore x - \frac{\pi}{3} &= \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3} \\
 &= \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}
 \end{aligned}$$

$$\therefore \text{intercepts are } \left(\frac{\pi}{2}, 0\right), \left(\frac{2\pi}{3}, 0\right), \left(\frac{3\pi}{2}, 0\right), \left(\frac{5\pi}{3}, 0\right)$$



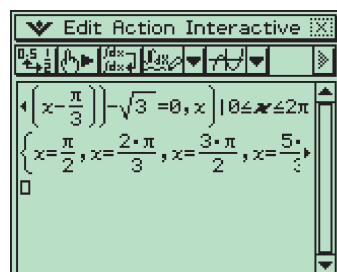
Using the TI-Nspire

Use **Solve** from the **Algebra** menu as shown.



Using the Casio ClassPad

To find the x -axis intercepts, enter $2\sin\left(2\left(x - \frac{\pi}{3}\right)\right) - \sqrt{3} = 0 \mid 0 \leq x \leq 2\pi$. Highlight the *equation part only*, then tap **Interactive–Equation/inequality–solve** to find the solutions.



Exercise 6H



- 1 Sketch the graphs of each of the following for $x \in [0, 2\pi]$. List the x -axis intercepts of each graph for this interval.

a $y = 2 \cos x + 1$

b $y = 2 \cos 2x - \sqrt{3}$

c $y = \sqrt{2} \cos x - 1$

d $y = 2 \cos x - 2$

e $y = \sqrt{2} \cos\left(x - \frac{\pi}{4}\right) + 1$

- 2 Sketch the graphs of each of the following for $x \in [-2\pi, 2\pi]$.

a $y = 2 \sin 3x - 2$

b $y = 2 \sin\left(x - \frac{\pi}{4}\right) + 1$

c $y = 2 \sin 2x - 3$

d $y = 1 - 2 \sin x$

e $y = 2 \cos 3\left(x - \frac{\pi}{4}\right)$

f $y = 2 \cos\left(3x - \frac{\pi}{4}\right)$

g $y = 1 - \cos 2x$

h $y = -1 - \sin x$

6.4 Addition of ordinates

Example 19

Using the same scale and axes sketch the graphs of $y_1 = 2 \sin x$ and $y_2 = 3 \cos 2x$ for $0 \leq x \leq 2\pi$. Use addition of ordinates to sketch the graph of $y = 2 \sin x + 3 \cos 2x$

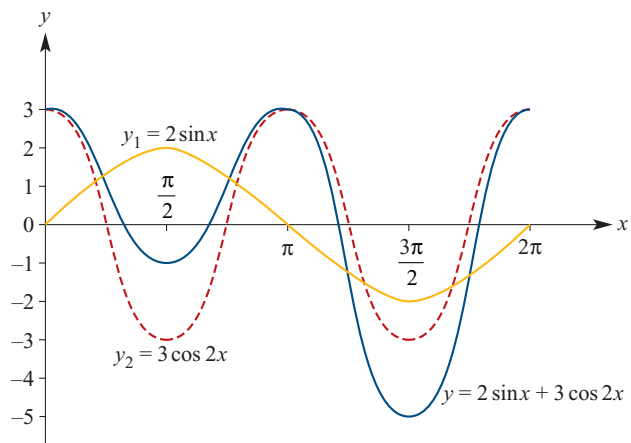
Solution

The graphs of $y_1 = 2 \sin x$ and $y_2 = 3 \cos 2x$ are shown. To obtain points on the graph of $y = 2 \sin x + 3 \cos 2x$, the process of addition of ordinates is used.

Let $y = y_1 + y_2$ when $y_1 = 2 \sin x$ and $y_2 = 3 \cos 2x$

$$\begin{aligned} \text{For example, at } x = 0, \quad y &= 0 + 3 = 3 \\ x = \frac{\pi}{4}, \quad y &= \frac{2}{\sqrt{2}} + 0 = \frac{2}{\sqrt{2}} \\ x = \frac{\pi}{2}, \quad y &= 2 - 3 = -1 \\ x = \pi, \quad y &= 0 + 3 = 3 \\ x = \frac{3\pi}{2}, \quad y &= -2 - 3 = -5 \end{aligned}$$

and so on.



Exercise 6I

Use addition of ordinates to sketch the graphs of each of the following for $\theta \in [-\pi, \pi]$:

- 1 $y = \sin \theta + 2 \cos \theta$
- 2 $y = 2 \cos 2\theta + 3 \sin 2\theta$
- 3 $y = \frac{1}{2} \cos 2\theta - \sin \theta$
- 4 $y = 3 \cos \theta + \sin 2\theta$
- 5 $y = 2 \sin \theta - 4 \cos \theta$

6.5 Determining the rule for graphs of circular functions

In previous chapters, the procedures for finding the rule for a graph known to be a polynomial, exponential or logarithmic function were introduced. In this section, we consider the procedures for finding the rules for graphs of functions known to be of the form $f(t) = A \sin(nt + b)$.

Example 20

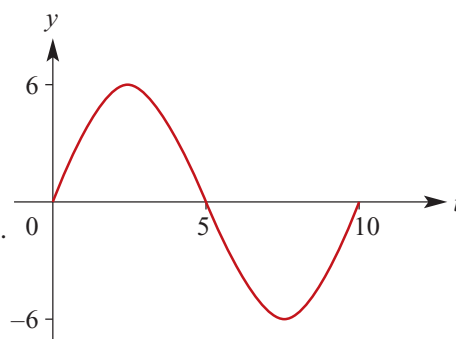
A function has rule $f(t) = A \sin(nt)$. The amplitude is 6; the period is 10. Find A and n and sketch the graph of $y = f(t)$ for $0 \leq t \leq 10$.

Solution

$$\begin{aligned}\text{The period} &= \frac{2\pi}{n} \\ &= 10 \\ \therefore n &= \frac{\pi}{5}\end{aligned}$$

The amplitude is 6 and therefore $A = 6$.

The function has rule $f(t) = 6 \sin \frac{\pi t}{5}$.



Example 21

The graph shown is that of a function with rule:

$$y = A \sin(nt) + b$$

Find A and n and b .

Solution

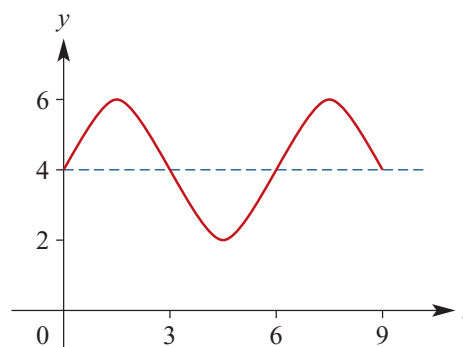
The amplitude is 2, and therefore $A = 2$.

The period = 6

Therefore $\frac{2\pi}{n} = 6$ and $n = \frac{\pi}{3}$.

The 'centrelines' is $y = 4$, and therefore $b = 4$.

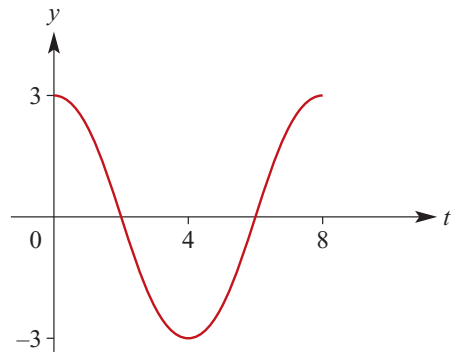
The function is $y = 2 \sin \left(\frac{\pi t}{3} \right) + 4$



Exercise 6J

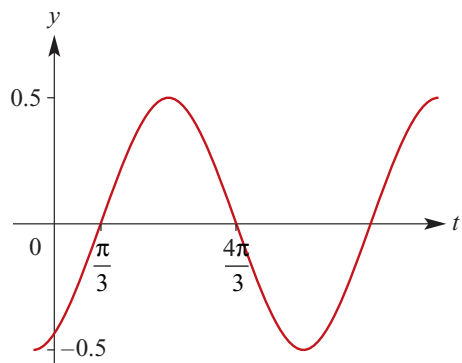
- 1 The graph shown has rule of the form $y = A \cos(nt)$.

Find the values of A and n .



- 2 The graph shown has rule of the form $y = A \sin(t + \epsilon)$.

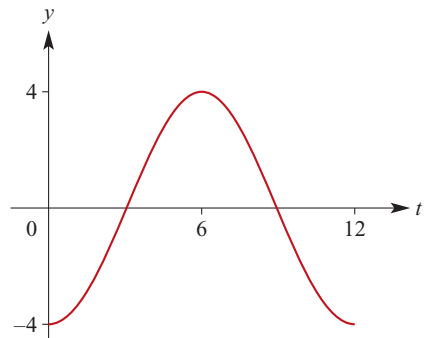
Find the values of A and ϵ .



- 3 A function with rule $y = A \sin(nt) + b$ has range $[2, 8]$ and period $\frac{2\pi}{3}$.
Find the values of A , n and b .

- 4 The graph shown has rule of the form $y = A \cos(nt)$.

Find the values of A and n .



- 5 A function with rule $y = A \sin(nt + \epsilon)$ has the following properties:

- range = $[-4, 4]$
- period = 8
- when $t = 2$, $y = 0$.

Find values for A , n and ϵ .

- 6 A function with rule $y = A \sin(nt + \epsilon)$ has the following properties:

- range = $[-2, 2]$
- period = 6

■ when $t = 1, y = 1$.

Find values for A, n and ϵ .

7 A function with rule $y = A \sin(nt + \epsilon) + d$ has the following properties:

■ range = $[-2, 6]$

■ period = 8

■ when $t = 2, y = 2$.

Find values for A, n and ϵ .

8 A function with rule $y = A \sin(nt + \epsilon) + d$ has the following properties:

■ range = $[0, 4]$

■ period = 6

■ when $t = 1, y = 3$.

Find values for A, n, d and ϵ .

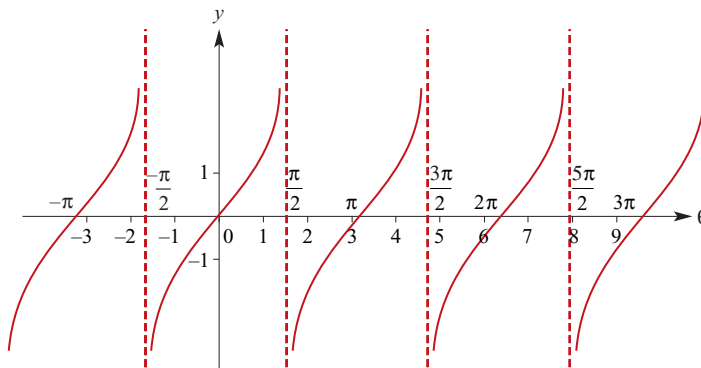
6.6 The function $\tan \theta$



As previously discussed, $\tan \theta = \frac{\sin \theta}{\cos \theta}$ where $\cos \theta \neq 0$. It can be seen from this that the vertical asymptotes of the graph of $y = \tan \theta$ occur where $\cos \theta = 0$, i.e. for values of $\theta = \dots -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots$. These values can be described by $\theta = (2k + 1)\frac{\pi}{2}$ where $k \in \mathbb{Z}$.

A table of values for $y = \tan \theta$ is given below:

θ	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π	$\frac{9\pi}{4}$	$\frac{5\pi}{2}$	$\frac{11\pi}{4}$	3π
y	0	1	undefined	-1	0	1	undefined	-1	0	1	undefined	-1	0	1	undefined	-1	0



Note: $\theta = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$ and $\frac{5\pi}{2}$ are asymptotes.

Observations from the graph

■ The graph repeats itself every π units, i.e. the period of $\tan$ is π .

■ Range of $\tan$ is $\mathbb{R}$.

■ The vertical asymptotes have the equation $\theta = (2k + 1)\frac{\pi}{2}$ where $k \in \mathbb{Z}$.

Example 22

Sketch the graph of each of the following for $x \in [-\pi, \pi]$.

a $y = 3 \tan 2x$

b $y = -2 \tan 3x$

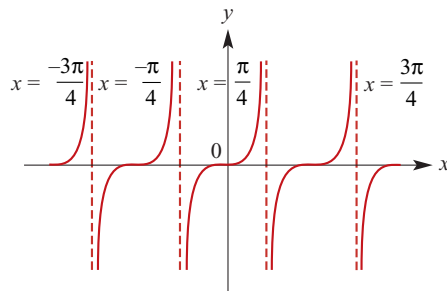
Solution

a $y = 3 \tan 2x$

Period $= \frac{\pi}{2}$

Asymptotes $x = \frac{(2k+1)\pi}{4}, k \in \mathbb{Z}$

Axes intercepts $x = \frac{k\pi}{2}, k \in \mathbb{Z}$

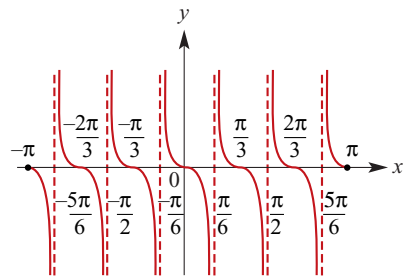


b $y = -2 \tan 3x$

Period $= \frac{\pi}{3}$

Asymptotes $x = \frac{(2k+1)\pi}{6}, k \in \mathbb{Z}$

Axes intercepts $x = \frac{k\pi}{3}, k \in \mathbb{Z}$

**Example 23**

Sketch the graph of $y = 3 \tan \left(2x - \frac{\pi}{3} \right)$ for $\frac{\pi}{6} \leq x \leq \frac{13\pi}{6}$.

Solution

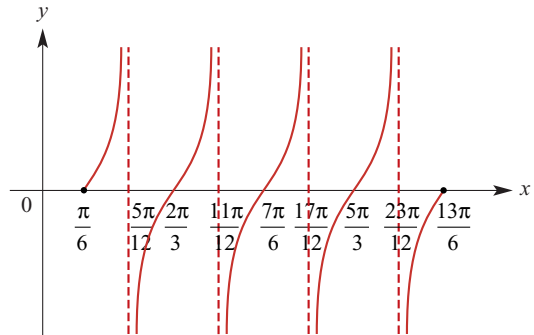
Consider $y = 3 \tan \left[2 \left(x - \frac{\pi}{6} \right) \right]$

The transformations are:

- a dilation of factor 3 from the x -axis
- a dilation of factor $\frac{1}{2}$ from the y -axis
- a translation of $\frac{\pi}{6}$ units in the positive direction of the x -axis.

The period of the function is $\frac{\pi}{2}$.

The range of the function is $\mathbb{R}$.



In general:

- $f: \mathbb{R} \setminus \left\{ x: x = \frac{(2k+1)\pi}{n}, k \in \mathbb{Z} \right\} \rightarrow \mathbb{R}, f(x) = a \tan(nx)$
- Period $= \frac{\pi}{|n|}$
- Range $= \mathbb{R}$

For a translation of b units in the positive direction of the x -axis of $f(x) = a \tan(nx)$ the asymptotes will have equations:

$$x = \frac{(2k+1)\pi}{n} + b$$

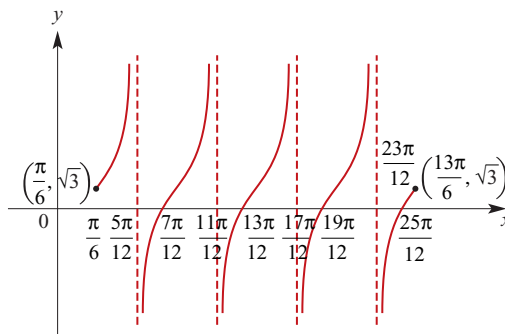
Example 24

Sketch the graph of $y = 3 \tan\left(2x - \frac{\pi}{3}\right) + \sqrt{3}$ for $\frac{\pi}{6} \leq x \leq \frac{13\pi}{6}$.

Solution

The transformations are:

- a dilation of factor 3 from the x -axis
- a dilation of factor $\frac{1}{2}$ from the y -axis
- a translation of $\frac{\pi}{6}$ units in the positive direction of the x -axis
- a translation of $\sqrt{3}$ units in the positive direction of the y -axis.



The graph is the graph obtained in Example 23 translated $\sqrt{3}$ units in the positive direction of the y -axis.

The axes intercepts are determined by solving the equation:

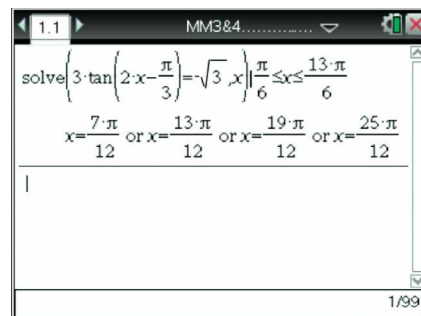
$$\begin{aligned} 3 \tan\left(2x - \frac{\pi}{3}\right) + \sqrt{3} &= 0 \text{ for } \frac{\pi}{6} \leq x \leq \frac{13\pi}{6} \\ \therefore \tan\left(2x - \frac{\pi}{3}\right) &= -\frac{\sqrt{3}}{3} = \frac{-1}{\sqrt{3}} \\ \therefore 2x - \frac{\pi}{3} &= \frac{5\pi}{6} \text{ or } \frac{11\pi}{6} \text{ or } \frac{17\pi}{6} \text{ or } \frac{23\pi}{6} \\ 2x &= \frac{7\pi}{6} \text{ or } \frac{13\pi}{6} \text{ or } \frac{19\pi}{6} \text{ or } \frac{25\pi}{6} \\ x &= \frac{7\pi}{12} \text{ or } \frac{13\pi}{12} \text{ or } \frac{19\pi}{12} \text{ or } \frac{25\pi}{12} \end{aligned}$$

Using the TI-Nspire

To find the x -axis intercepts,

Enter

$$\text{solve}\left(3 \tan\left(2x - \frac{\pi}{3}\right) = -\sqrt{3}, x\right) \mid \frac{\pi}{6} \leq x \leq \frac{13\pi}{6}$$

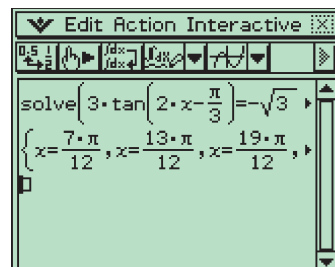


Using the Casio ClassPad

To find the x -axis intercepts, enter

$$3 \tan \left(2x - \frac{\pi}{3} \right) = -\sqrt{3} \mid \frac{\pi}{6} \leq x \leq \frac{13\pi}{6}$$

then highlight *the equation part only*, then tap **Interactive–Equation/inequality–solve** to find the solutions.



Example 25

Solve the equation $\tan \left[\frac{1}{2} \left(x - \frac{\pi}{4} \right) \right] = -1$ for $x \in [-2\pi, 2\pi]$.

Solution

$$\tan \left[\frac{1}{2} \left(x - \frac{\pi}{4} \right) \right] = -1$$

$$\text{implies } \frac{1}{2} \left(x - \frac{\pi}{4} \right) = \frac{3\pi}{4} \text{ or } \frac{-\pi}{4}$$

$$x - \frac{\pi}{4} = \frac{3\pi}{2} \text{ or } \frac{-\pi}{2}$$

$$\text{Therefore } x = \frac{7\pi}{4} \text{ or } \frac{-\pi}{4}$$

Solution of equations of the form $\sin nx = k \cos nx$

If $\sin nx = k \cos nx$, then $\tan nx = k$.

Example 26

Solve the equation $\sin 2x = \cos 2x$ for $x \in [0, 2\pi]$.

Solution

$$\sin 2x = \cos 2x$$

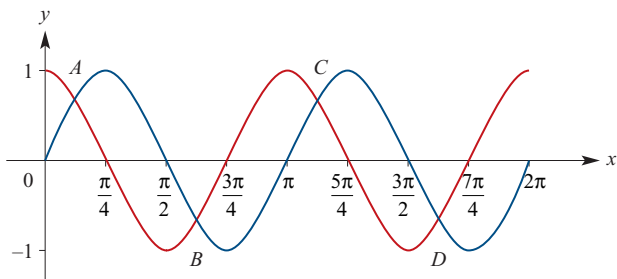
implies

$$\tan 2x = 1$$

$$\therefore 2x = \frac{\pi}{4} \text{ or } \frac{5\pi}{4} \text{ or } \frac{9\pi}{4} \text{ or } \frac{13\pi}{4} \quad (\text{remember tan is positive for the first and third quadrants})$$

$$\therefore x = \frac{\pi}{8} \text{ or } \frac{5\pi}{8} \text{ or } \frac{9\pi}{8} \text{ or } \frac{13\pi}{8}$$

This can be shown graphically.



The points of intersection, A , B , C and D , occur when:

$$x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8} \text{ and } \frac{13\pi}{8} \text{ respectively}$$

Example 27

On the same set of axes, sketch the graphs of $y = \sin x$ and $y = \cos x$ for $x \in [0, 2\pi]$ and find the coordinates of the points of intersection.

Solution

$$\sin x = \cos x$$

implies

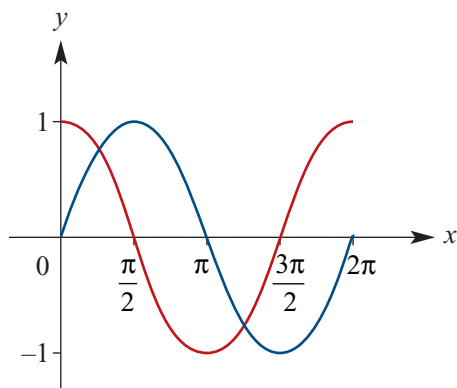
$$\tan x = 1$$

$$\therefore x = \frac{\pi}{4} \text{ or } \frac{5\pi}{4}$$

The function $\tan$ is positive in the first and third quadrant.

The coordinates of the points of intersection are:

$$\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right) \text{ and } \left(\frac{5\pi}{4}, -\frac{1}{\sqrt{2}}\right)$$



We recall the exact values of $\tan$:

$$\tan 0 = 0, \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}, \tan \frac{\pi}{4} = 1, \tan \frac{\pi}{3} = \sqrt{3}$$

and the symmetry properties:

- $\tan(\pi - \theta) = -\tan \theta$
- $\tan(\pi + \theta) = \tan \theta$
- $\tan(2\pi - \theta) = -\tan \theta$
- $\tan(-\theta) = -\tan \theta$

Example 28

Solve the equation $\sin 3x = \frac{1}{\sqrt{3}} \cos 3x$ for $x \in [-\pi, \pi]$.

Solution

$$\frac{\sin 3x}{\cos 3x} = \frac{1}{\sqrt{3}}$$

$$\text{implies } \tan 3x = \frac{1}{\sqrt{3}}$$

$$\therefore 3x = \frac{\pi}{6} \text{ or } \frac{7\pi}{6} \text{ or } \frac{13\pi}{6} \text{ or } -\frac{11\pi}{6} \text{ or } -\frac{5\pi}{6} \text{ or } -\frac{17\pi}{6}$$

$$\therefore x = \frac{\pi}{18} \text{ or } \frac{7\pi}{18} \text{ or } \frac{13\pi}{18} \text{ or } -\frac{11\pi}{18} \text{ or } -\frac{5\pi}{18} \text{ or } -\frac{17\pi}{18}$$

Exercise 6K

1 State the period for each of the following:

a $\tan 3\theta$ **b** $\tan \frac{\theta}{2}$ **c** $\tan \frac{3\theta}{2}$ **d** $\tan(\pi\theta)$ **e** $\tan\left(\frac{\pi\theta}{2}\right)$

2 Sketch the graph of each of the following for $x \in (0, 2\pi)$:

a $y = \tan 2x$ **b** $y = 2 \tan 3x$ **c** $y = 2 \tan\left(x + \frac{\pi}{4}\right)$
d $y = 3 \tan x + 1$ **e** $y = 2 \tan\left(x + \frac{\pi}{2}\right) + 1$ **f** $y = 3 \tan 2\left(x - \frac{\pi}{4}\right) - 2$

3 Sketch the graph of $y = -2 \tan(\pi\theta)$ for $-2 \leq \theta \leq 2$.

4 Sketch the graph of $y = \tan(-2\theta)$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

5 Solve each of the following equations for $x \in [0, 2\pi]$:

a $\sqrt{3} \sin x = \cos x$ **b** $\sin 4x = \cos 4x$ **c** $\sqrt{3} \sin 2x = \cos 2x$
d $-\sqrt{3} \sin 2x = \cos 2x$ **e** $\sin 3x = -\cos 3x$ **f** $\sin x = 0.5 \cos x$
g $\sin x = 2 \cos x$ **h** $\sin 2x = -\cos 2x$ **i** $\cos 3x = \sqrt{3} \sin 3x$
j $\sin 3x = \sqrt{3} \cos 3x$

6 **a** On the same set of axes sketch the graphs of $y = \cos 2x$ and $y = -\sin 2x$ for $x \in [-\pi, \pi]$.

b Find the coordinates of the points of intersection.

c On the same set of axes sketch the graph of $y = \cos 2x - \sin 2x$.

7 **a** On the same set of axes sketch the graphs of $y = \cos x$ and $y = \sqrt{3} \sin x$ for $x \in [0, 2\pi]$.

b Find the coordinates of the points of intersection.

c On the same set of axes sketch the graph of $y = \cos x + \sqrt{3} \sin x$.

8 Solve each of the following equations for $0 \leq x \leq 2\pi$:

a $\tan\left(2x - \frac{\pi}{4}\right) = \sqrt{3}$

b $3 \tan 2x = -\sqrt{3}$

c $\tan\left(3x - \frac{\pi}{6}\right) = -1$

9 A function with rule $y = A \tan nt$ has the following properties:

■ asymptotes have equations $t = (2k + 1)\frac{\pi}{6}$ where $k \in \mathbb{Z}$

■ when $t = \frac{\pi}{12}$, $y = 5$.

Find values for A and n .

10 A function with rule $y = A \tan(nt)$ has the following properties:

■ period is 2

■ when $t = \frac{1}{2}$, $y = 6$.

Find values for A and n .

6.7 General solution of circular function equations

Solution of circular function equations has been discussed for functions over a restricted domain. In this section, we consider the general solutions of such equations over the maximal domain for each function.

In the following the convention is that:

■ $\cos^{-1}$ has range $[0, \pi]$

■ $\sin^{-1}$ has range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

■ $\tan^{-1}$ has range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

For example $\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$ and $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

Also

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \text{ and } \sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

If a circular function equation has one or more solutions in one ‘cycle’, then it will have corresponding solutions in each ‘cycle’ of its domain; i.e., there will be an infinite number of solutions.

For example, if $\cos x = a$, then the solution in the interval $[0, \pi]$ is given by:

$$x = \cos^{-1}(a)$$

By the symmetry properties of the cosine function, other solutions are given by:

$$-\cos^{-1}(a), \pm 2\pi + \cos^{-1}(a), \pm 2\pi - \cos^{-1}(a), \pm 4\pi + \cos^{-1}(a), \pm 4\pi - \cos^{-1}(a),$$

and so on.

In general, if $\cos(x) = a$,

$$x = 2n\pi \pm \cos^{-1}(a), \text{ where } n \in Z \text{ and } a \in [-1, 1]$$

Similarly, if $\tan(x) = a$,

$$x = n\pi + \tan^{-1}(a), \text{ where } n \in Z \text{ and } a \in R$$

If $\sin(x) = a$,

$$x = 2n\pi + \sin^{-1}(a) \text{ or } x = (2n+1)\pi - \sin^{-1}(a), \text{ where } n \in Z \text{ and } a \in [-1, 1]$$

Note: An alternative and more concise way to express the general solution of $\sin(x) = a$ is:

$$x = n\pi + (-1)^n \sin^{-1}(a), \text{ where } n \in Z \text{ and } a \in [-1, 1]$$

Example 29

Find the general solution to each of the following equations:

a $\cos(x) = 0.5$

b $\sqrt{3} \tan(3x) = 1$

c $2 \sin(x) = \sqrt{2}$

Solution

a $x = 2n\pi \pm \cos^{-1}(0.5)$

$$= 2n\pi \pm \frac{\pi}{3}$$

$$= \frac{(6n \pm 1)\pi}{3}, n \in Z$$

b $\tan(3x) = \frac{1}{\sqrt{3}}$

$$3x = n\pi + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= n\pi + \frac{\pi}{6}$$

$$= \frac{(6n+1)\pi}{6}$$

$$x = \frac{(6n+1)\pi}{18}, n \in Z$$

c $\sin(x) = \frac{1}{\sqrt{2}}$

$$x = 2n\pi + \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 2n\pi + \frac{\pi}{4}$$

$$= \frac{(8n+1)\pi}{4}, n \in Z$$

or $x = (2n+1)\pi - \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$$= (2n+1)\pi - \frac{\pi}{4}$$

$$= \frac{(8n+3)\pi}{4}, n \in Z$$

Using the TI-Nspire

Make sure the calculator is in Radian mode.

- a** Use **Solve** from the **Algebra** menu and complete as shown.

Note the use of $\frac{1}{2}$ rather than 0.5 to ensure that the answer is exact.

TI-Nspire calculator screen showing the solve command for $\cos(x) = \frac{1}{2}$. The screen displays the equation $\text{solve}\left(\cos(x) = \frac{1}{2}, x\right)$ and the solutions $x = \frac{(6 \cdot n + 1) \cdot \pi}{3}$ or $x = \frac{(6 \cdot n - 1) \cdot \pi}{3}$. The status bar at the bottom indicates 1/99.

- b** Complete as shown.

TI-Nspire calculator screen showing the solve command for $\sqrt{3} \cdot \tan(3x) = 1$. The screen displays the equation $\text{solve}(\sqrt{3} \cdot \tan(3x) = 1, x)$ and the solution $x = \frac{(6 \cdot n + 1) \cdot \pi}{18}$. The status bar at the bottom indicates 1/99.

- c** Complete as shown.

TI-Nspire calculator screen showing the solve command for $\sqrt{2} \cdot \sin(x) = 1$. The screen displays the equation $\text{solve}(\sqrt{2} \cdot \sin(x) = 1, x)$ and the solutions $x = 2 \cdot n \cdot \pi + \frac{3 \cdot \pi}{4}$ or $x = 2 \cdot n \cdot \pi + \frac{\pi}{4}$. The status bar at the bottom indicates 1/99.

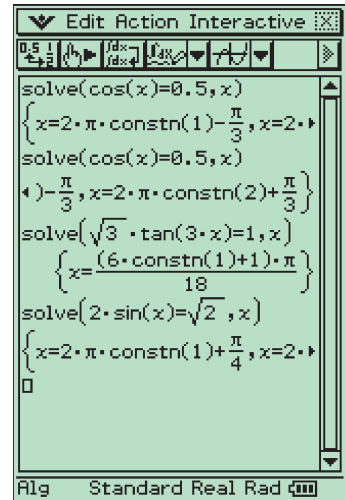
Using the Casio ClassPad

Make sure the calculator is in Radian mode.

- a** Enter and highlight $\cos(x) = 0.5$ then tap **Interactive–Equation/inequality–solve** and ensure the variable is set to x .

Note that in the screen shown, this equation has been solved twice and the solution scrolled to the right to show the other solution on the second occasion.

- b** Enter and highlight $\sqrt{3} \tan(3x) = 1$ and follow the instructions in part **a**.



- c** Enter and highlight $2 \sin(x) = \sqrt{2}$ and follow the instructions in part **a**.

The calculator uses the notation $\text{constn}(1)$ and $\text{constn}(2)$ to represent constants when general solutions are found. The answers for **a** should be read

$$x = 2n\pi - \pi/3 \text{ and } x = 2n\pi + \pi/3.$$

Example 30

Find the first three positive solutions to each of the following equations:

- a** $\cos(x) = 0.5$ **b** $\sqrt{3} \tan(3x) = 1$ **c** $2 \sin(x) = \sqrt{2}$

Solution

- a** The general solution (from Example 29) is given by $x = \frac{(6n \pm 1)\pi}{3}, n \in \mathbb{Z}$

$$\text{When } n = 0, x = \pm \frac{\pi}{3}, \text{ and when } n = 1, x = \frac{5\pi}{3} \text{ or } x = \frac{7\pi}{3}$$

$$\text{The first three positive solutions of } \cos(x) = 0.5 \text{ are } x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

- b** The general solution (from Example 29) is given by $x = \frac{(6n + 1)\pi}{18}, n \in \mathbb{Z}$

$$\text{When } n = 0, x = \frac{\pi}{18}, \text{ and when } n = 1, x = \frac{7\pi}{18}, \text{ and when } n = 2, x = \frac{13\pi}{18}$$

$$\text{The first three positive solutions of } \sqrt{3} \tan(3x) = 1 \text{ are } x = \frac{\pi}{18}, \frac{7\pi}{18}, \frac{13\pi}{18}$$

- c** The general solution (from Example 29) is given by $x = \frac{(8n + 1)\pi}{4}$ or

$$x = \frac{(8n + 3)\pi}{4}, n \in \mathbb{Z}$$

$$\text{When } n = 0, x = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}, \text{ and when } n = 1, x = \frac{9\pi}{4} \text{ or } \frac{11\pi}{4}$$

$$\text{The first three positive solutions of } 2 \sin(x) = \sqrt{2} \text{ are } x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}$$

Example 31

Find the general solution for each of the following

a $\sin\left(x - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

b $\tan\left(2x - \frac{\pi}{3}\right) = 1$

Solution

a $x - \frac{\pi}{3} = n\pi + (-1)^n \sin^{-1}\left(\frac{\sqrt{3}}{2}\right), \text{ where } n \in \mathbb{Z}$

$$x = n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{3}$$

$$x = n\pi + \frac{2\pi}{3} \text{ if } n \text{ is even and } x = n\pi \text{ if } n \text{ is odd.}$$

b $\tan\left(2x - \frac{\pi}{3}\right) = 1$

$$2x - \frac{\pi}{3} = n\pi + \frac{\pi}{4}$$

$$2x = n\pi + \frac{\pi}{4} + \frac{\pi}{3}$$

$$x = \frac{1}{2} \left(n\pi + \frac{7\pi}{12} \right)$$

$$= \left(\frac{12n + 7}{24} \right) \pi, \text{ where } n \in \mathbb{Z}$$

Exercise 6L

1 Evaluate each of the following for:

i $n = 1$

ii $n = 2$

iii $n = -2$

a $2n\pi \pm \cos^{-1}(1)$

b $2n\pi \pm \cos^{-1}\left(-\frac{1}{2}\right)$

2 Find the general solution to each of the following equations:

a $\cos(x) = 0.5$

b $2 \sin(3x) = \sqrt{3}$

c $\sqrt{3} \tan(x) = 3$

3 Find the first two positive solutions to each of the following equations:

a $\sin(x) = 0.5$

b $2 \cos(2x) = \sqrt{3}$

c $\sqrt{3} \tan(2x) = -3$

4 It was found that a trigonometric equation had general solution

$$x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right) \text{ where } n \in \mathbb{Z}. \text{ Find the solutions for the equation in the interval } [-2\pi, 2\pi].$$

5 It was found that a trigonometric equation had general solution $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{2}\right)$ where $n \in \mathbb{Z}$. Find the solutions for the equation in the interval $[-\pi, 2\pi]$.

6 Find the general solution of each of the following:

a $\cos 2 \left(x + \frac{\pi}{3} \right) = \frac{1}{2}$ **b** $2 \tan 2 \left(x + \frac{\pi}{4} \right) = 2\sqrt{3}$ **c** $2 \sin \left(x + \frac{\pi}{3} \right) = -1$

7 Find the general solution to $2 \cos \left(2x + \frac{\pi}{4} \right) = \sqrt{2}$, and hence find all the solutions for x in the interval $(-2\pi, 2\pi)$.

8 Find the general solution to $\sqrt{3} \tan \left(\frac{\pi}{6} - 3x \right) - 1 = 0$, and hence find all the solutions for x in the interval $[-\pi, 0]$.

9 Find the general solution to $2 \sin (4\pi x) + \sqrt{3} = 0$, and hence find all the solutions for x in the interval $[-1, 1]$.

6.8 Identities

We develop several useful identities in this section.

The Pythagorean identity

Consider a point, $P(\theta)$, on the unit circle.

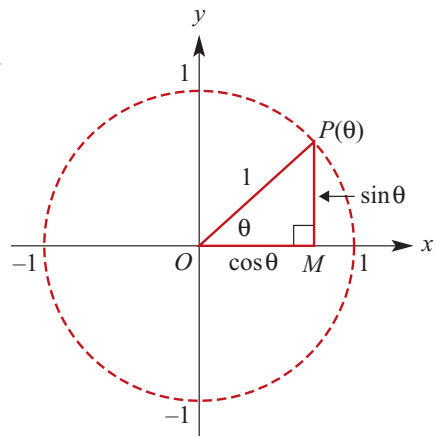
By the theorem of Pythagoras:

$$OP^2 = OM^2 + MP^2$$

$$\therefore 1 = (\cos \theta)^2 + (\sin \theta)^2$$

Now $(\cos \theta)^2$ and $(\sin \theta)^2$ may be written as $\cos^2 \theta$ and $\sin^2 \theta$.

$$\therefore 1 = \cos^2 \theta + \sin^2 \theta$$



Since this is true for all values of θ , it is called an identity. In particular this is called the Pythagorean identity:

$$\cos^2 \theta + \sin^2 \theta = 1$$

Addition formulas

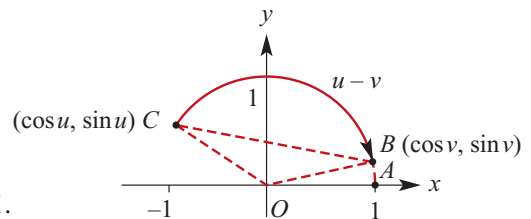
Consider a unit circle.

Let arc length $AB = v$ units

arc length $AC = u$ units

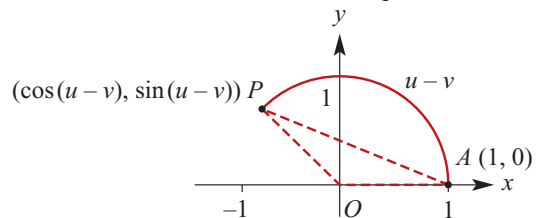
$\therefore$ arc length $CB = u - v$ units

Rotate $\triangle OCB$ so that B is coincident with A .



The point P has coordinates

$(\cos(u - v), \sin(u - v))$.



Since the triangles CBO and PAO are congruent, $CB = PA$.

Apply the coordinate distance formula:

$$\begin{aligned} CB^2 &= (\cos u - \cos v)^2 + (\sin u - \sin v)^2 \\ &= 2 - 2(\cos u \cos v + \sin u \sin v) \\ PA^2 &= [\cos(u - v) - 1]^2 + (\sin(u - v) - 0)^2 \\ &= 2 - 2(\cos(u - v)) \end{aligned}$$

Equating these:

$$\begin{aligned} 2 - 2(\cos u \cos v + \sin u \sin v) &= 2 - 2[\cos(u - v)] \\ \therefore \cos(u - v) &= \cos u \cos v + \sin u \sin v \end{aligned}$$

Replacing v with $-v$, $\cos(u - (-v)) = \cos u \cos(-v) + \sin u \sin(-v)$

From symmetry properties, $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$

$$\therefore \cos(u + v) = \cos u \cos v - \sin u \sin v$$

Example 32

Evaluate $\cos 75^\circ$.

Solution

$$\begin{aligned} \cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \end{aligned}$$

Replace u with $\frac{\pi}{2} - u$ in $\cos(u - v)$:

$$\therefore \cos\left[\left(\frac{\pi}{2} - u\right) - v\right] = \cos\left(\frac{\pi}{2} - u\right) \cos v + \sin\left(\frac{\pi}{2} - u\right) \sin v$$

Applying symmetry properties, $\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$

$$\text{and } \cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$$

$$\therefore \cos\left[\frac{\pi}{2} - (u + v)\right] = \sin u \cos v + \cos u \sin v$$

$$\therefore \sin(u + v) = \sin u \cos v + \cos u \sin v$$

Replacing v with $-v$, $\sin(u - v) = \sin u \cos(-v) + \cos u \sin(-v)$

$$\therefore \sin(u - v) = \sin u \cos v - \cos u \sin v$$

Double angle formulas

$$\cos(u + v) = \cos u \cos v - \sin u \sin v$$

Replacing v with u :

$$\cos(u + u) = \cos u \cos u - \sin u \sin u$$

$$\begin{aligned}\cos 2u &= \cos^2 u - \sin^2 u \\ &= 2 \cos^2 u - 1 \\ &= 1 - 2 \sin^2 u\end{aligned}$$

$$\text{since } \cos^2 u = 1 - \sin^2 u$$

$$\text{and } \sin^2 u = 1 - \cos^2 u$$

Similarly replacing v with u in $\sin(u + v) = \sin u \cos v + \cos u \sin v$:

$$\therefore \sin 2u = \sin u \cos u + \cos u \sin u$$

$$\sin 2u = 2 \sin u \cos u$$

$$\text{Replacing } v \text{ with } u \text{ in } \tan(u + v) = \frac{\tan u + \tan v}{1 - \tan u \tan v}:$$

$$\therefore \tan(u + u) = \frac{\tan u + \tan u}{1 - \tan u \tan u}$$

$$\tan 2u = \frac{2 \tan u}{1 - \tan^2 u}$$

Example 33

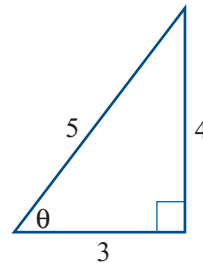
If $\tan \theta = \frac{4}{3}$ and $0 < \theta < \frac{\pi}{2}$, evaluate $\sin 2\theta$.

Solution

$$\sin \theta = \frac{4}{5} \text{ and } \cos \theta = \frac{3}{5}$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$\begin{aligned}&= 2 \times \frac{4}{5} \times \frac{3}{5} \\ &= \frac{24}{25}\end{aligned}$$



Exercise 6M

The following questions require the Pythagorean identity.

- 1 Given that $\cos x = \frac{3}{5}$ and $\frac{3\pi}{2} < x < 2\pi$, find $\sin x$ and $\tan x$.

2 Given that $\sin x = \frac{5}{13}$ and $\frac{\pi}{2} < x < \pi$, find $\cos x$ and $\tan x$.

3 Given that $\cos x = \frac{1}{5}$ and $\frac{3\pi}{2} < x < 2\pi$, find $\sin x$ and $\tan x$.

6.9 Applications of circular functions

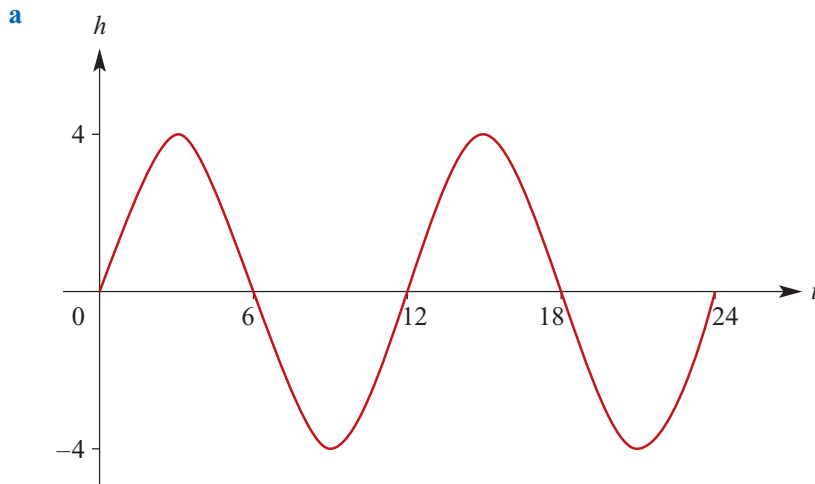
Example 34

It is suggested that the height $h(t)$ metres of the tide above mean sea level on 1 January at Warnung is given approximately by the rule:

$$h(t) = 4 \sin\left(\frac{\pi}{6}t\right), \text{ where } t \text{ is the number of hours after midnight}$$

- Draw the graph of $y = h(t)$ for $0 \leq t \leq 24$.
- When was high tide?
- What was the height of the high tide?
- What was the height of the tide at 8 a.m.?
- A boat can only cross the harbour bar when the tide is at least 1 m above mean sea level. When could the boat cross the harbour bar on 1 January?

Solution



Note: period $= 2\pi \div \frac{\pi}{6} = 12$

b High tide occurs when $h(t) = 4$.

$$4 \sin\left(\frac{\pi}{6}t\right) = 4$$

$$\text{implies } \sin\left(\frac{\pi}{6}t\right) = 1$$

$$\therefore \frac{\pi}{6}t = \frac{\pi}{2}, \frac{5\pi}{2}$$

$$\therefore t = 3, 15$$

i.e. high tide occurs at 0300 (3 a.m.) and 1500 (3 p.m.).

c The high tide has height 4 metres above the mean height.

$$\begin{aligned}\mathbf{d} \quad h(8) &= 4 \sin\left(\frac{8\pi}{6}\right) \\ &= 4 \sin\left(\frac{4\pi}{3}\right) \\ &= 4 \times \frac{-\sqrt{3}}{2} \\ &= -2\sqrt{3}\end{aligned}$$

The water is $2\sqrt{3}$ metres below the mean height at 8 a.m.

e First, consider $4 \sin \frac{\pi}{6}t = 1$

$$\text{Thus } \sin \frac{\pi}{6}t = \frac{1}{4}$$

$$\therefore \frac{\pi}{6}t = 0.2527, 2.889, 6.5359, 9.172$$

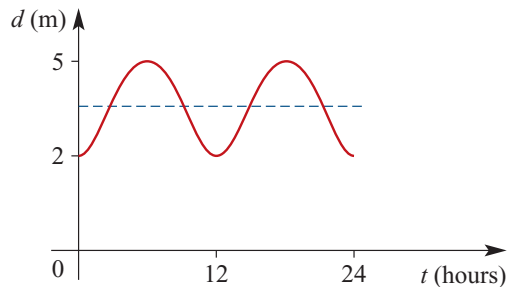
$$\therefore t = 0.4826, 5.5174, 12.4824, 17.5173$$

i.e. the water is at height 1 m at 0029, 0531, 1229 and 1731.

Thus the boat can pass across the harbour bar between 0029 and 0531 and between 1229 and 1731.

Exercise 6N

- 1** The graph shows the distance $d(t)$ of the tip of the hour hand of a large clock from the ceiling at time t hours.



a $d(t)$ is the rule for a sinusoidal function. Find:

- i** the amplitude
- ii** the period
- iii** the rule for $d(t)$
- iv** the length of the hour hand

b At what times is the distance less than 3.5 metres from the ceiling?

- 2** The water level on a beach wall is a sinusoidal function (i.e. has a rule of the form $y = a \sin(nt + \epsilon) + b$).

In this case, the function $d(t) = 6 + 4 \cos\left(\frac{\pi}{6}t - \frac{\pi}{3}\right)$ where t is the number of hours after midnight and d is the depth of the water, in metres.

- a** Sketch the graph of $d(t)$ for $0 \leq t \leq 24$.
- b** What is the earliest time of day at which the water is at its highest?
- c** When is the water 2 m up the wall?

- 3** In a tidal river, the time between high tides is 12 hours. The average depth of water in the port is 5 m; at high tide the depth is 8 m. Assume that the depth of water is given by

$$h(t) = A \sin(nt + \epsilon) + b$$

where t is the number of hours after 12:00 noon.

At 12:00 noon there is a high tide.

- a Find the values of A , n , b and ϵ .
 - b At what times is the depth of the water 6 m?
 - c Sketch the graph of $y = h(t)$ for $0 \leq t \leq 24$.
- 4 A particle moves on a straight line, Ox , and its distance x metres from O at time t (s) is given by $x = 3 + 2 \sin 3t$.
- a Find its greatest distance from O .
 - b Find its least distance from O .
 - c Find the times at which it is 5 m from O for $0 \leq t \leq 5$.
 - d Find the times at which it is 3 m from O for $0 \leq t \leq 3$.
 - e Describe the motion of the particle.
- 5 The temperature $A^\circ\text{C}$ inside a house at t hours after 4 a.m. is given by $A = 21 - 3 \cos\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$, and the temperature $B^\circ\text{C}$ outside the house at the same time is given by $B = 22 - 5 \cos\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$.
- a Find the temperature inside the house at 8 a.m.
 - b Write down an expression for $D = A - B$, the difference between the inside and outside temperatures.
 - c Sketch the graph of D for $0 \leq t \leq 24$.
 - d Determine when the inside temperature is less than the outside temperature.
- 6 Passengers on a ferris wheel ride access their seats from a platform 5 m above the ground. As each seat is filled the ferris wheel moves around so that the next seat can be filled. Once all seats are filled the ride begins and lasts for 6 minutes. The height h m of Isobel's seat above the ground t seconds after the ride has begun is given by the equation $h = 15 \sin(10t - 45)^\circ + 16.5$
- a Use a calculator to sketch the graph of the equation for the first 2 minutes of the ride.
 - b How far above the ground is Isobel's seat at the commencement of the ride?
 - c After how many seconds does Isobel's seat pass the access platform?
 - d How many times will her seat pass the access platform in the first 2 minutes?
 - e How many times will her seat pass the access platform during the entire ride?
- Due to a malfunction the ride stops abruptly 1 minute and 40 seconds into the ride.
- f How far above the ground is Isobel stranded?
 - g If Isobel's brother Hamish had a seat 1.5 m above the ground at the commencement of the ride, how far above the ground was Hamish stranded?



Chapter summary

■ Symmetry properties of circular functions:

$$\sin(\pi - \theta) = \sin \theta$$

$$\cos(\pi - \theta) = -\cos \theta$$

$$\tan(\pi - \theta) = -\tan \theta$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\sin(2\pi - \theta) = -\sin \theta$$

$$\cos(2\pi - \theta) = \cos \theta$$

$$\tan(2\pi - \theta) = -\tan \theta$$

$$\sin(\pi + \theta) = -\sin \theta$$

$$\cos(\pi + \theta) = -\cos \theta$$

$$\tan(\pi + \theta) = \tan \theta$$

$$\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta$$

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

■ Exact values of trigonometric functions:

$$\sin 0 = 0$$

$$\sin 0^\circ = 0$$

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\cos 0 = 1$$

$$\cos 0^\circ = 1$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 0 = 0$$

$$\tan 0^\circ = 0$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\sin \frac{\pi}{2} = 1$$

$$\sin 90^\circ = 1$$

$$\tan \frac{\pi}{3} = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3}$$

$$\cos \frac{\pi}{2} = 0$$

$$\cos 90^\circ = 0$$

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

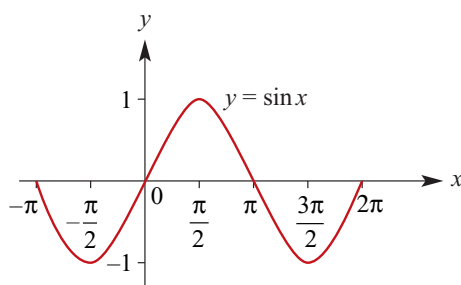
$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan \frac{\pi}{4} = 1$$

$$\tan 45^\circ = 1$$

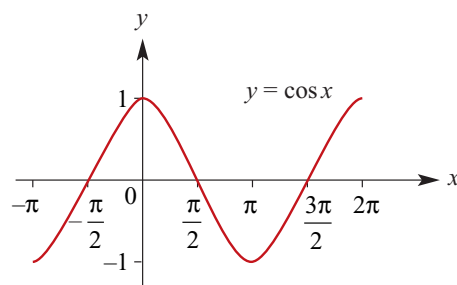
■ Graphs of $y = \sin x$ and $y = \cos x$ and transformations of these graphs:



period = 2π

amplitude = 1

range = $[-1, 1]$



period = 2π

amplitude = 1

range = $[-1, 1]$

- For $y = a \cos(nx)$ and $y = a \sin(nx)$:

$$\text{period} = \frac{2\pi}{|n|}$$

$$\text{amplitude} = |a|$$

$$\text{range} = [-|a|, |a|]$$

- Pythagorean identity:

$$\cos^2 \theta + \sin^2 \theta = 1$$

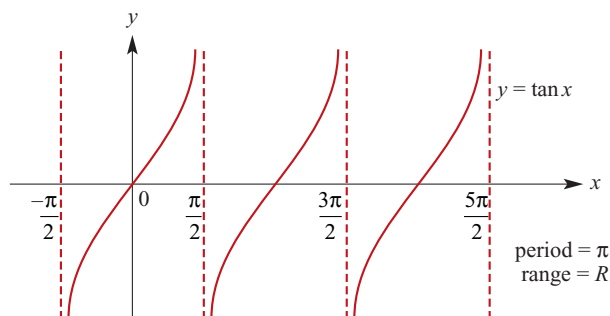
- The graph of $y = \tan x$:

- For $y = a \tan(nx)$:

■ $\text{period} = \frac{\pi}{|n|}$

■ $\text{range} = R$

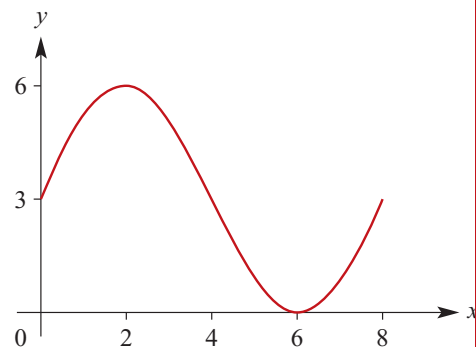
■ asymptotes have equations $x = \frac{(2k+1)\pi}{n}, k \in Z$



Multiple-choice questions

- The period of the graph of $y = 3 \sin\left(\frac{1}{2}x - \pi\right) + 4$ is:
A π **B** 3 **C** 4π **D** $\pi + 4$ **E** 2π
- The equation of the image of the graph of $y = \sin x$ under a transformation of a dilation of factor $\frac{1}{2}$ from the y -axis followed by a translation of $\frac{\pi}{4}$ in the positive direction of the x -axis is:
A $y = \sin\left(\frac{1}{2}x + \frac{\pi}{4}\right)$ **B** $y = \sin\left(\frac{1}{2}x - \frac{\pi}{4}\right)$ **C** $y = 2 \sin\left(x - \frac{\pi}{4}\right)$
D $y = \sin\left(2x - \frac{\pi}{4}\right)$ **E** $y = \sin\left[2\left(x - \frac{\pi}{4}\right)\right]$
- The function $f: R \rightarrow R$, where $f(x) = a \sin(bx) + c$, where a, b and c are positive constants, has period:
A a **B** b **C** $\frac{2\pi}{a}$ **D** $\frac{2\pi}{b}$ **E** $\frac{b}{2\pi}$
- The equation $3 \sin x - 1 = b$, where b is a positive real number, has one solution in the interval $(0, 2\pi)$. The value of b is:
A 2 **B** 0.2 **C** 3 **D** 5 **E** 6
- The range of the graph of $y = f(x)$, where $f(x) = 5 \cos\left(2x - \frac{\pi}{3}\right) - 7$ is:
A $[-12, -2]$ **B** $[-7, 7]$ **C** $(-2, 5)$ **D** $[-2, 5]$ **E** $[-2, 12]$
- Let $f(x) = p \cos(5x) + q$ where $p > 0$. Then $f(x) \leq 0$ for all values of x if:
A $q \geq 0$ **B** $-p \leq q \leq p$ **C** $p \leq -q$ **D** $p \geq q$ **E** $-q \leq p$

- 7 The vertical distance of a point on a wheel from the ground as it rotates is given by $D(t) = 3 - 3 \sin 6\pi t$, where t is the time in seconds. The time in seconds for a full rotation of the wheel is:
- A** $\frac{1}{6\pi}$ **B** $\frac{1}{3}$ **C** 6π **D** $\frac{1}{3\pi}$ **E** 3
- 8 A sequence of transformations which takes the graph of $y = \cos x$ to the graph of $y = -2 \cos \left(\frac{x}{3}\right)$ is:
- A** a dilation of factor $\frac{1}{3}$ from the x -axis, followed by dilation of factor $\frac{1}{2}$ from the y -axis, followed by a reflection in the x -axis
- B** a dilation of factor $\frac{1}{2}$ from the x -axis, followed by dilation of factor 3 from the y -axis, followed by a reflection in the y -axis
- C** a dilation of factor 2 from the x -axis, followed by dilation of factor 3 from the y -axis, followed by a reflection in the x -axis
- D** a dilation of factor 3 from the x -axis, followed by dilation of factor 2 from the y -axis, followed by a reflection in the x -axis
- E** a dilation of factor 2 from the x -axis, followed by dilation of factor $\frac{1}{3}$ from the y -axis, followed by a reflection in the x -axis
- 9 The equation of the image of $y = \cos x$ under a transformation of a dilation of factor 2 from the x -axis, followed by a translation of $\frac{\pi}{4}$ in the positive direction of the x -axis is:
- A** $y = \cos \left(\frac{1}{2}x + \frac{\pi}{4}\right)$ **B** $y = \cos \left(\frac{1}{2}x - \frac{\pi}{4}\right)$ **C** $y = 2 \cos \left(x + \frac{\pi}{4}\right)$
- D** $y = 2 \sin \left(x - \frac{\pi}{4}\right)$ **E** $y = 2 \cos \left(x - \frac{\pi}{4}\right)$
- 10 Which of the following is likely to be the rule for the graph of the circular function shown?
- A** $y = 3 + 3 \cos \frac{\pi x}{4}$
- B** $y = 3 + 3 \sin \frac{\pi x}{4}$
- C** $y = 3 + 3 \sin 4\pi x$
- D** $y = 3 + 3 \cos \frac{x}{4}$
- E** $y = 3 + 3 \sin \frac{x}{4}$



Short-answer questions (technology-free)

- Solve each of the following equations for $x \in [-\pi, 2\pi]$:

a $\sin x = \frac{1}{2}$	b $2 \cos x = -1$	c $2 \cos x = \sqrt{3}$
d $\sqrt{2} \sin x + 1 = 0$	e $4 \sin x + 2 = 0$	f $\sin 2x + 1 = 0$
g $\cos 2x = \frac{-1}{\sqrt{2}}$	h $2 \sin 3x - 1 = 0$	
- Sketch the graphs of each of the following, showing one cycle. Clearly label axes intercepts.

a $f(x) = \sin 3x$	b $f(x) = 2 \sin 2x - 1$	c $g(x) = 2 \sin 2x + 1$
d $f(x) = 2 \sin\left(x - \frac{\pi}{4}\right)$	e $f(x) = 2 \sin \frac{\pi x}{3}$	f $h(x) = 2 \cos \frac{\pi x}{4}$
- Solve each of the following equations for $x \in [0, 360]$:

a $\sin x^\circ = 0.5$	b $\cos(2x)^\circ = 0$	c $2 \sin x^\circ = -\sqrt{3}$
d $\sin(2x + 60)^\circ = \frac{-\sqrt{3}}{2}$	e $2 \sin(\frac{1}{2}x)^\circ = \sqrt{3}$	
- Sketch the graphs of each of the following, showing one cycle. Clearly label axes intercepts.

a $y = 2 \sin\left(x + \frac{\pi}{3}\right) + 2$	b $y = -2 \sin\left(x + \frac{\pi}{3}\right) + 1$
c $y = 2 \sin\left(x - \frac{\pi}{4}\right) + \sqrt{3}$	d $y = -3 \sin x$
e $y = \sin\left(x - \frac{\pi}{6}\right) + 3$	f $y = 2 \sin\left(x - \frac{\pi}{2}\right) + 1$
- Sketch, on the same set of axes, the curves $y = \cos x$ and $y = \sin 2x$ for the interval $0 \leq x \leq 2\pi$, labelling each curve carefully. State the number of solutions in this interval of the equations:

a $\sin 2x = 0.6$	b $\sin 2x = \cos x$	c $\sin 2x - \cos x = 1$
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- Sketch on separate axes for $0^\circ \leq x^\circ \leq 360^\circ$:

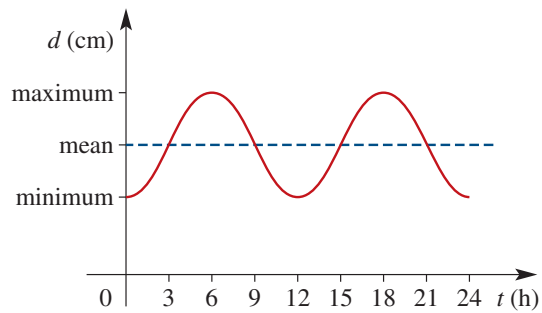
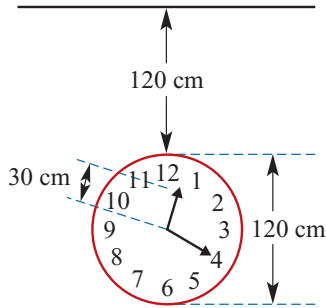
a $y = 3 \cos x^\circ$	b $y = \cos 2x^\circ$	c $y = \cos(x - 30)^\circ$
-------------------------------	------------------------------	-----------------------------------
- Solve each of the following for $x \in [-\pi, \pi]$:

a $\tan x = \sqrt{3}$	b $\tan x = -1$	c $\tan 2x = -1$	d $\tan(2x) + \sqrt{3} = 0$
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Extended-response questions

- In a tidal river, the time between high tide and low tide is 6 hours. The average depth of water at a point in the river is 4 metres; at high tide the depth is 5 metres.
 - Sketch the graph of the depth of water at the point for the time interval 0 to 24 hours if the relationship between time and depth is sinusoidal and there is high tide at noon.
 - If a boat requires a depth of 4 metres of water in order to sail, how many hours before noon can it enter the point in the river and by what time must it leave if it is not to be stranded?
 - If a boat requires a depth of 3.5 metres of water in order to sail, at what time before noon can it enter the point in the river and by what time must it leave if it is not to be stranded?

2



A clock hangs 120 cm below a ceiling. The clock has a diameter of 120 cm, and the hour hand is 30 cm long. The graph shows the distance from the ceiling to the tip of the hour hand over a 24-hour period.

- a** What are the values for the maximum, minimum and mean distance?
b An equation that determines this curve is of the form:

$$y = A \sin(nt + \epsilon) + b$$

Find the values of A , n , ϵ and b .

- c** Find the distance from the ceiling to the tip of the hour hand at:
i 2:00 a.m. **ii** 11:00 p.m.
d Find the times in the morning at which the tip of the hour hand is 200 cm below the ceiling.

- 3** A weight is suspended from a spring as shown.

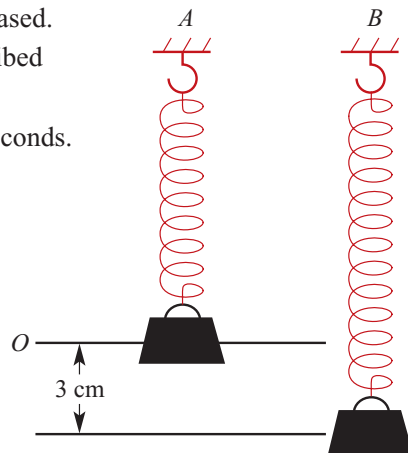
The weight is pulled down 3 cm from O and released.

The vertical displacement from O at time t is described

by a function of the form $y = a \cos nt$.

Let y cm be the vertical displacement at time t seconds.

t	0	0.5	1
y	-3	3	-3



It was also noted that the centre of the weight went no further than 3 cm from the centre O .

- a** Find the values of a and n .
b Sketch the graph of y against t .
c Find when the centre of the weight is first:
i 1.5 cm above O **ii** 1.5 cm below O
d When does the weight first reach a point 1 cm below O ?

- 4 The manager of a reservoir and its catchment area has noted that the inflow of water into the reservoir is very predictable and in fact models the inflow with a curve with rule of the form $y = a \sin(nt + \epsilon) + b$.

The following observations were made:

- The average inflow is $100\,000 \text{ m}^3/\text{day}$.
 - The minimum daily flow is $80\,000 \text{ m}^3/\text{day}$.
 - The maximum daily flow is $120\,000 \text{ m}^3/\text{day}$, and this occurs on 1 May ($t = 121$) each year.
- a Find the values of a , b and n and the smallest possible positive value for ϵ .
 - b Sketch the graph of y against t .
 - c Find the times of year when the inflow per day is:
 - i $90\,000 \text{ m}^3/\text{day}$
 - ii $110\,000 \text{ m}^3/\text{day}$
 - d Find the inflow rate on 1 June.

- 5 The number of hours of daylight at a point on the Antarctic Circle is given approximately by $d = 12 + 12 \cos \frac{1}{6} \pi \left(t + \frac{1}{3} \right)$ where t is the number of months that have elapsed since 1 January.

- a Find d i on 21 June ($t \approx 5.7$), and ii on 21 March ($t \approx 2.7$)
- b When will there be 5 hours of daylight?

- 6 The depth, $D(t)$ m, of water at the entrance to a harbour at t hours after midnight on a particular day is given by $D(t) = 10 + 3 \sin \left(\frac{\pi t}{6} \right)$, $0 \leq t \leq 24$.

- a Sketch the graph of $D(t)$ for $0 \leq t \leq 24$.
- b Find the value of t for which $D(t) \geq 8.5$.
- c Boats that need a depth of w m are permitted to enter the harbour only if the depth of the water at the entrance is at least w m for a continuous period of 1 hour. Find, correct to one decimal place, the largest value of w that satisfies this condition.

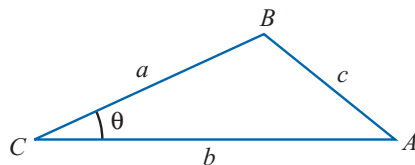
- 7 The depth of water at the entrance to a harbour t hours after high tide is D m, where $D = p + q \cos(rt)^\circ$ for suitable constants p , q , r . At high tide the depth is 7 m; at low tide, 6 hours later, the depth is 3 m.

- a Show that $r = 30$ and find the values of p and q .
- b Sketch the graph of D against t for $0 \leq t \leq 12$.
- c Find how soon after low tide a ship that requires a depth of at least 4 m of water will be able to enter the harbour.

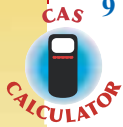
- 8 The area of a triangle is given by $A = \frac{1}{2}ab \sin \theta$ and the perimeter is given by

$$P = a + b + \sqrt{a^2 + b^2 - 2ab \cos \theta}$$

- a For $a = b = 10$ and $\theta = \frac{\pi}{3}$, find:
 - i the area of the triangle
 - ii the perimeter of the triangle

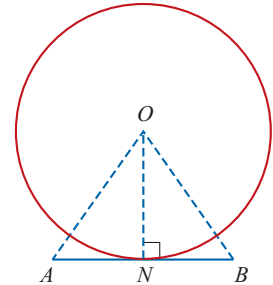


- b** For $a = b = 10$, find the value(s) of θ for which $A = P$. (Give value(s) correct to two decimal places.)
- c** Show graphically that for $a = b = 6$, $P > A$ for all θ .
- d** Assume $\theta = \frac{\pi}{2}$. If $a = 6$, find the value of b such that $A = P$.
- e** For $a = 10$, $b = 6$, find the value(s) of θ for which $A = P$.
- f** If $a = b$ and $\theta = \frac{\pi}{3}$, find the value of a such that $A = P$.



- 9** AB is one side of a regular n -sided polygon that circumscribes a circle, i.e. each edge of the polygon is tangent to the circle. The circle has radius of 1.

- a** Show that the area of triangle OAB is $\tan\left(\frac{\pi}{n}\right)$.
- b** Show that the area, A , of the polygon is given by $A = n \tan\left(\frac{\pi}{n}\right)$.
- c** Use a calculator to help sketch the graph of $A(x) = x \tan\left(\frac{\pi}{x}\right)$ for $x \geq 3$. Label the horizontal asymptote.
- d** What is the difference in area of the polygon and the circle when:
- i** $n = 3?$ **ii** $n = 4?$ **iii** $n = 12?$ **iv** $n = 50?$
- e** State the area of an n -sided polygon that circumscribes a circle of radius r cm.
- f i** Find a formula for the n -sided regular polygon that can be inscribed in a circle of radius 1.
- ii** Sketch the graph of this function for $x \geq 3$.



Functions revisited

Objectives

- To define **sums** and **products of functions** and graph these new functions.
- To define **composite functions** and graph these new functions.
- To understand and find **inverse functions** and **relations**.
- To be able to recognise the general form of possible models of data.
- To apply a knowledge of functions to **solving problems**.

7.1 Operations on functions

In the previous six chapters, families of functions have been introduced and compositions and inverses have been considered. In this section, we revisit these concepts with all the functions of Mathematical Methods (CAS) at our disposal.

Example 1

For $g(x) = |x|$ and $f(x) = \sin x$:

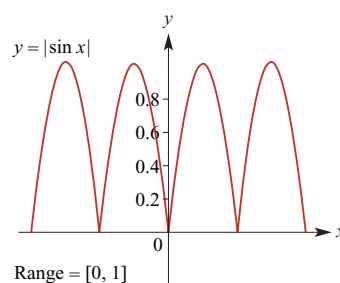
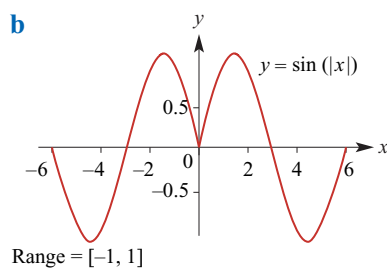
- a Find the rules for $f \circ g(x)$ and $g \circ f(x)$
- b Sketch the graphs of $y = f \circ g(x)$ and $y = g \circ f(x)$ for $x \in [-2\pi, 2\pi]$ and state the range of each of these composite functions.

Solution

a $f \circ g(x) = f(g(x)) = f(|x|) = \sin(|x|)$

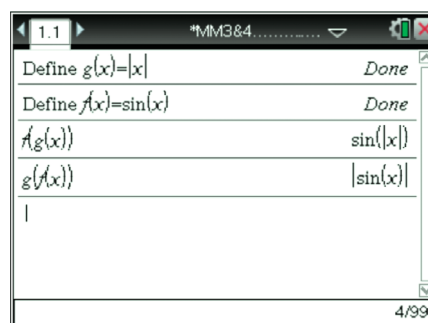
$g \circ f(x) = g(f(x)) = g(\sin x) = |\sin x|$

b

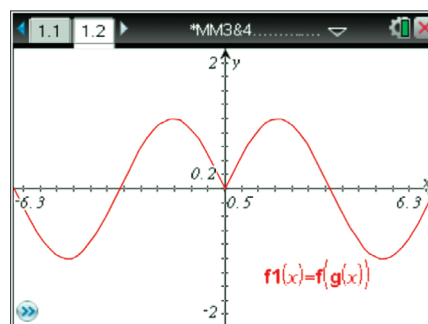


Using the TI-Nspire

Define $g(x) = \text{abs}(x)$ and $f(x) = \sin(x)$ by using **define** ((menu) > **Actions** > **Define**) Enter $f(g(x))$ and $g(f(x))$ as shown.



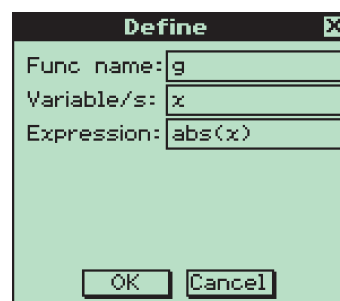
Open a **Graphs** application ((ctrl) + ① > **Graphs**) and let $f1(x) = f(g(x))$ and $f2(x) = g(f(x))$. Select an appropriate **window setting** ((menu) > **Window/Zoom** > **Window Settings**). The graph of $f(g(x))$ is shown.



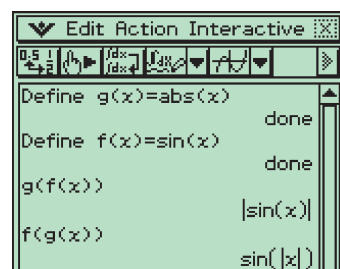
Using the Casio ClassPad

Define $g(x) = \text{abs}(x)$ and $f(x) = \sin x$ using **Interactive-Define**.

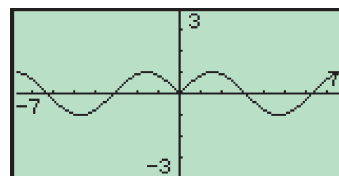
Note that although there is a function $|■|$ in the **2D** menu for entering absolute value, this cannot be used within the Define box and you must type $\text{abs}(x)$.



Enter $f(g(x))$ and $g(f(x))$ to find $f \circ g(x)$ and $g \circ f(x)$.



Enter $f(g(x))$ and $g(f(x))$ into line $y1 =$ and $y2 =$ in the  menu. Tick the box for the graph you want and tap  to produce the graph. The screen shows $f(g(x))$.



Example 2

For $g(x) = |x|$ and $f(x) = \log_e x$:

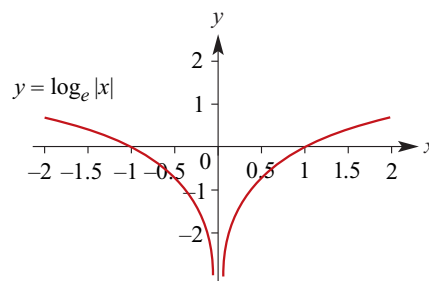
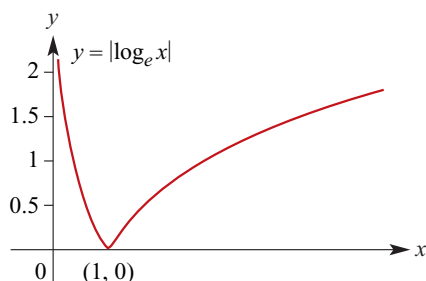
- State the maximal domain for g for $f \circ g$ to exist and the maximal domain for f for $g \circ f$ to exist.
- Find the rules for $f \circ g(x)$ and $g \circ f(x)$
- Sketch the graphs of $y = f \circ g(x)$ and $y = g \circ f(x)$

Solution

- For $f \circ g$ to exist the range of g must be a subset of the domain of f . The maximal domain of f is R^+ and hence $f \circ g$ is defined for $x \in R \setminus \{0\}$, i.e. the maximal domain of $f \circ g$ is $R \setminus \{0\}$. For $g \circ f$ to exist the range of f must be a subset of the domain of g . The maximal domain of g is R and hence $g \circ f$ is defined for $x \in R^+$, i.e. the maximal domain of $g \circ f$ is R^+ .
- $$f \circ g(x) = f(g(x)) = f(|x|) = \log_e |x|$$

$$g \circ f(x) = g(f(x)) = g(\log_e x) = |\log_e x|$$

c



It is sometimes useful to be able to express a given function as a composition of two functions. This will be seen for differentiation in Chapter 9.

Example 3

Express each of the following as the composition of two functions:

- a** $h(x) = e^{x^2}$ **b** $h(x) = \sin |x|$ **c** $h(x) = (x^2 - 2)^n$ where n is a natural number

Solution

a $h(x) = e^{x^2}$

Choose $f(x) = e^x$ and $g(x) = x^2$

Then $h(x) = f \circ g(x)$

b $h(x) = \sin |x|$

Choose $f(x) = \sin x$ and $g(x) = |x|$

Then $h(x) = f \circ g(x)$

c $h(x) = (x^2 - 2)^n$ where n is a natural number

Choose $f(x) = x^n$ and $g(x) = x^2 - 2$

Then $h(x) = f \circ g(x)$

This is not the only possible choice of functions, but the ‘natural’ choice is made.

Example 4

Let $f(x) = e^{2x}$ and $g(x) = \frac{1}{\sqrt{x}}$ for $x \in R^+$.

Find f^{-1} , g^{-1} , $f \circ g$, $g \circ f$, $(f \circ g)^{-1}$, $(g \circ f)^{-1}$, $f^{-1} \circ g^{-1}$ and $g^{-1} \circ f^{-1}$

Solution

■ $f^{-1}(x) = \frac{1}{2} \log_e x, x \in R^+$

■ $g^{-1}(x) = \frac{1}{x^2}, x \in R^+$

■ $f \circ g(x) = f\left(\frac{1}{\sqrt{x}}\right) = e^{2x \cdot \frac{1}{2}} = e^{\frac{2}{\sqrt{x}}}, x \in R^+$

■ $g \circ f(x) = g(e^{2x}) = \frac{1}{e^x} = e^{-x}, x \in R$

■ For $(f \circ g)^{-1}$, let $x = e^{\frac{2}{\sqrt{y}}}$

Then $\log_e x = \frac{2}{\sqrt{y}}$ and $y = \left(\frac{2}{\log_e(x)}\right)^2$

$(f \circ g)^{-1}(x) = \left(\frac{2}{\log_e(x)}\right)^2, x \in (1, \infty)$

■ $g^{-1} \circ f^{-1}(x) = g^{-1}\left(\frac{1}{2} \log_e x\right)$
 $= \left(\frac{2}{\log_e(x)}\right)^2, x \in (1, \infty)$

■ For $(g \circ f)^{-1}$, let $x = \frac{1}{e^y}$ and $y = -\log_e x$

$(g \circ f)^{-1}(x) = -\log_e x, x \in R^+$

■ $f^{-1} \circ g^{-1}(x) = f^{-1}\left(\frac{1}{x^2}\right)$
 $= \frac{1}{2} \log_e \frac{1}{x^2}$
 $= -\log_e x, x \in R^+$

Note that $(g \circ f)^{-1}(x) = f^{-1} \circ g^{-1}(x)$ and $(f \circ g)^{-1}(x) = g^{-1} \circ f^{-1}(x)$

Exercise 7A

- 1** For each of the following, where $g(x) = |x|$:

- i** Find the rules $f \circ g(x)$ and $g \circ f(x)$

- ii** Find the range of $y = f \circ g(x)$ and $y = g \circ f(x)$ (and state the maximal domain for each of the composite functions to exist).
- a** $f(x) = 3 \sin 2x$ **b** $f(x) = -2 \cos 2x$ **c** $f(x) = e^x$
d $f(x) = e^{2x} - 1$ **e** $f(x) = -2e^x - 1$ **f** $f(x) = \log_e 2x$
g $f(x) = \log_e (x - 1)$ **h** $f(x) = -\log_e x$
- 2** Express each of the following as the composition of two functions:
- a** $h(x) = e^{x^3}$ **b** $h(x) = \cos |2x|$
c $h(x) = (x^2 - 2x)^n$ where n is a natural number **d** $h(x) = \cos (x^2)$
e $h(x) = \cos^2 x$ **f** $h(x) = (x^2 - 1)^4$ **g** $h(x) = \log_e x^2$
h $h(x) = |\cos 2x|$ **i** $h(x) = (x^2 - 2x)^3 - 2(x^2 - 2x)$
- 3** The functions f and g are defined by $f: R \rightarrow R$, $f(x) = e^{4x}$ and $g: R^+ \rightarrow R$, $g(x) = 2\sqrt{x}$. Find each of the following:
- a** $g \circ f(x)$ **b** $(g \circ f)^{-1}(x)$ **c** $f \circ g^{-1}(x)$
- 4** The functions f and g are defined by $f: R \rightarrow R$, $f(x) = e^{-2x}$ and $g: R \rightarrow R$, $g(x) = x^3 + 1$.
- a** Find the inverse function of each of these functions.
b Find the rules $f \circ g(x)$ and $g \circ f(x)$ and state the range of each of these composite functions.
- 5** The function f is defined by $f: (-1, \infty) \rightarrow R$, $f(x) = \frac{1}{x+1}$.
- a** Find f^{-1} . **b** Solve the equation $f(x) = f^{-1}(x)$ for x .
- 6** The functions f and g are defined by $f: (-1, \infty) \rightarrow R$, $f(x) = \log_e (x + 1)$ and $g: (-1, \infty) \rightarrow R$, $g(x) = x^2 + 2x$.
- a** Define f^{-1} and g^{-1} , giving their rules and domains.
b Find the rule for $f \circ g(x)$.
- 7** The functions f and g are defined by $f: (0, \infty) \rightarrow R$, $f(x) = \log_e x$ and $g: (0, \infty) \rightarrow R$, $g(x) = \frac{1}{x}$. Find $f \circ g(x)$ and simplify $f(x) + f \circ g(x)$.
- 8** The functions g and h are defined by $g: R \rightarrow R$, $g(x) = 5x^2 + 3$ and $h: [3, \infty) \rightarrow R$, $h(x) = \sqrt{\frac{x-3}{5}}$. Find $h(g(x))$.
- 9** For $f(x) = (x - 4)(x - 6)$ and $g(x) = x^2 - 4$:
- a** Find $f(g(x))$ and $g(f(x))$ **b** Solve the equation $g(f(x)) - f(g(x)) = 158$ for x .
- 10** For $f(x) = 4 - x^2$, solve the equation $f(f(x)) = 0$ for x .
- 11** For $f(x) = e^x - e^{-x}$, show that:
- a** $f(-x) = -f(x)$ **b** $[f(x)]^3 = f(3x) - 3f(x)$
- 12** Let $g: R \rightarrow R$, $\log_e g(x) = ax + b$. Given that $g(0) = 1$ and $g(1) = e^6$, find a and b and hence $g(x)$.

7.2 Inverse relations

In Chapter 1, we showed that for one-to-one functions an inverse function can be found. In fact, the inverse of any relation may be defined. The inverse relation is not a function, however, unless the initial relation is a one-to-one function.

Recall that a relation S is a set of ordered pairs. If the set S is described as $\{(a, b)\}$, i.e. $S = \{(a, b)\}$, then the inverse relation T is described as $\{(b, a) : (a, b) \in S\}$.

For the relation

$$\{(1, 2), (1, 3), (4, 2), (4, 3), (1, 5)\}$$

the inverse relation is

$$\{(2, 1), (3, 1), (2, 4), (3, 4), (5, 1)\}$$

In general the inverse relation can be found by reversing the first and second coordinates.

For example, the function $f: R \rightarrow R$, $f(x) = x^2$ can be written as $\{(x, y) : y = x^2\}$.

The inverse relation is $\{(x, y) : x = y^2\}$.

The function $f: R \rightarrow R$, $f(x) = x^2 + 2$ can be written as $\{(x, y) : y = x^2 + 2\}$, and the inverse relation is $\{(x, y) : x = y^2 + 2\}$.

The relation $\{(x, y) : x^2 + y^2 = 4\}$ has inverse $\{(x, y) : y^2 + x^2 = 4\}$, which is the same set.

The relation $\{(x, y) : y = \pm\sqrt{x}\}$ has inverse $\{(x, y) : x = \pm\sqrt{y}\}$, which is equal to $\{(x, y) : y = x^2\}$.

The functions of the families

$$f(x) = A(x + b)^2 + B \quad A, B, b \in R, A \neq 0$$

$$g(x) = \frac{A}{(x + b)^2} + B \quad A, B, b \in R, A \neq 0$$

$$h(x) = A|x + b| + B \quad A, B, b \in R, A \neq 0$$

are many-to-one functions and therefore the corresponding inverse relations are not functions.

Example 5

Find the inverse relation of the function $f(x) = 4(x + 1)^2$ and sketch the graph of both relations on the one set of axes.

Solution

Consider $x = 4(y + 1)^2$

Solving for y : $\frac{x}{4} = (y + 1)^2$

$$\therefore y + 1 = \pm \sqrt{\frac{x}{4}}$$

$$\text{and } y = -1 \pm \sqrt{\frac{x}{4}}$$

The inverse relation is:

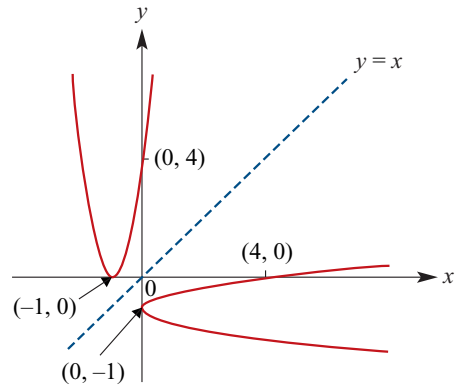
$$\left\{ (x, y) : y = -1 \pm \frac{1}{2}\sqrt{x} \right\}$$

The graphs of the original function and its inverse relation do not meet.

If the function f is restricted to form a new function

$$f^* : [-1, \infty) \rightarrow \mathbb{R}, f^*(x) = 4(x + 1)^2$$

f^* is a one-to-one function and therefore the inverse is also a function.

**Example 6**

Find the inverse of the function $h(x) = e^{|x|}$

Solution

The function is not one-to-one. The range of $h(x) = e^{|x|}$ is $[1, \infty)$. The domain of the inverse is $[1, \infty)$.

Let $y = e^{|x|}$. Consider $x = e^{|y|}$

Therefore $\log_e x = |y|$ and $y = \pm \log_e x$ for $x \geq 1$

Example 7

Find the inverse of the function with the rule $h(x) = x^{\frac{2}{3}} + 16$

Solution

$h(x) = x^{\frac{2}{3}} + 16$. Consider $x = y^{\frac{2}{3}} + 16$.

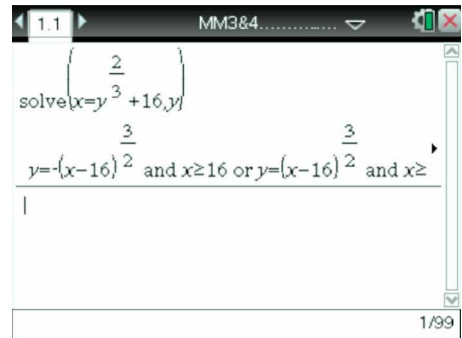
Then $y^{\frac{2}{3}} = x - 16$ and $y^2 = (x - 16)^3$.

Therefore $y = \pm \sqrt{(x - 16)^3}$

Using the TI-Nspire

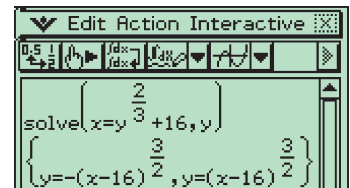
Complete as shown.

Use the **up arrow** ($\blacktriangle$) to move up to the answer and use the **right arrow** ($\blacktriangleright$) to display the remaining part of the answer.



Using the Casio ClassPad

Enter and highlight $x = y^{2/3} + 16$, then tap **Interactive–Equation/inequality–solve** and ensure the variable is set to y .



Example 8

Find the inverse of the function $g: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}, g(x) = \frac{1}{(x-1)^2}$

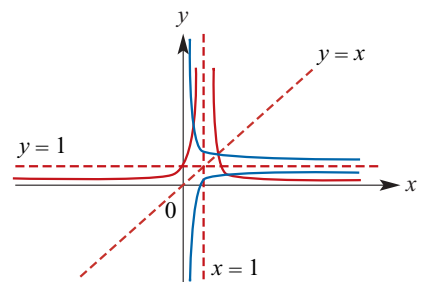
Solution

The function is not one-to-one. The range of g is $\mathbb{R}^+$. Therefore the domain of the inverse relation is $\mathbb{R}^+$.

$$\text{Consider } x = \frac{1}{(y-1)^2}$$

$$\text{Therefore } (y-1)^2 = \frac{1}{x} \text{ and } y-1 = \pm \frac{1}{\sqrt{x}}$$

$$\text{Therefore } y = 1 \pm \frac{1}{\sqrt{x}}$$



Exercise 7B



- 1 Let $f: [a, \infty) \rightarrow R$, $f(x) = (x - 1)^2 + 2$
 - a Find the smallest value of a such that f is one-to-one.
 - b With this value of a , state the range of f .
 - c Sketch the graph of f .
 - d Find f^{-1} and state the domain and range of f^{-1} .
 - e Sketch the graph of f and f^{-1} on the one set of axes.
- 2 Find the inverse relation of each of the following functions:

a $f(x) = 2x^2$	b $f(x) = 2x^2 - 1$	c $f(x) = (x + 1)^2$
d $g(x) = (x + 1)^2 - 1$	e $h(x) = 2(x - 1)^2 + 1$	f $f(x) = 1 - (x - 1)^2$
- 3 Find the inverse relation of each of the following functions, and state its domain:

a $g: R \setminus \{2\} \rightarrow R$, $g(x) = \frac{1}{(x - 2)^2}$	b $g: R \setminus \{1\} \rightarrow R$, $g(x) = \frac{1}{(x - 1)^2} + 1$	c $g: R \setminus \{-1\} \rightarrow R$, $g(x) = \frac{2}{(x + 1)^2}$
d $f(x) = x - 1 $	e $f(x) = x - 2 + 3$	f $f(x) = x + 1 - 2$
- 4 a For $f(x) = 2x^{\frac{3}{5}} + 6$ find $f^{-1}(x)$ and state the domain of f^{-1} .
 b For $f(x) = 2x^{\frac{2}{5}} + 6$ find the rule and domain for the inverse relation.

7.3 Sums and products of functions and addition of ordinates

In Chapter 1, we established that for functions f and g new functions $(f + g)$ and (fg) can be defined by:

- $(f + g)(x) = f(x) + g(x)$ $(fg)(x) = f(x)g(x)$
- $\text{dom}(f + g) = \text{dom}(f) \cap \text{dom}(g)$ $\text{dom}(fg) = \text{dom}(f) \cap \text{dom}(g)$

In Chapter 3, we considered graphing by addition of ordinates. The new functions that have been defined in Chapters 4 to 6 may now be included.

Example 9

For $f(x) = \cos x$ and $g(x) = e^{-x}$:

- a Find the rules for $(f + g)(x)$ and $(fg)(x)$
- b Evaluate $(f + g)(0)$ and $(fg)(0)$

Solution

- a $(f + g)(x) = \cos x + e^{-x}$ and $(fg)(x) = e^{-x} \cos x$
- b $(f + g)(0) = 1 + 1 = 2$ and $(fg)(0) = 1 \times 1 = 1$

Example 10

Sketch the graph of

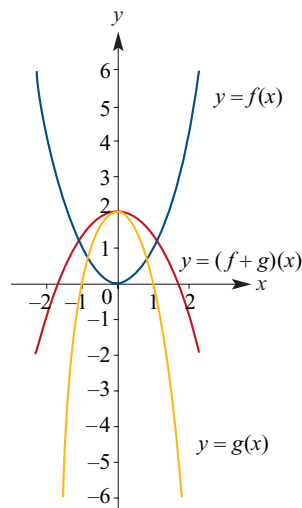
$$y = (f + g)(x)$$

where $f(x) = x^2$

and $g(x) = 2 - 2x^2$

Solution

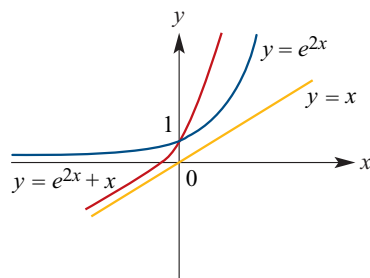
It can be seen that the function with rule $(f + g)(x) = 2 - x^2$ is defined by the addition of the two functions f and g .

**Example 11**

For $f(x) = x$ and $g(x) = e^{2x}$ sketch the graph of $y = (f + g)(x)$

Solution

Note that $(f + g)(0) = 0 + e^0 = 1$ and $(f + g)(x) = 0$ implies $e^{2x} + x = 0$. This cannot be solved analytically but a calculator can be used to find that a numerical solution is -0.43 , correct to two decimal places. Also note that as $x \rightarrow -\infty$, $(f + g)(x) \rightarrow x$ from 'above'.

**Exercise 7C**

1 For $f(x) = e^{-2x}$ and $g(x) = -2x$:

a Find the rules for:

i $(f + g)$

ii (fg)

b Evaluate:

i $(f + g)\left(\frac{-1}{2}\right)$

ii $(fg)\left(\frac{-1}{2}\right)$

2 For $f(x) = \sin\left(\frac{\pi x}{2}\right)$ and $g(x) = -2x$:

a Find the rules for:

i $(f + g)$

ii (fg)

b Evaluate:

i $(f + g)(1)$

ii $(fg)(1)$

3 For $f(x) = \cos\left(\frac{\pi x}{2}\right)$ and $g(x) = e^x$:

a Find the rules for:

i $(f + g)$

ii (fg)

b Evaluate:

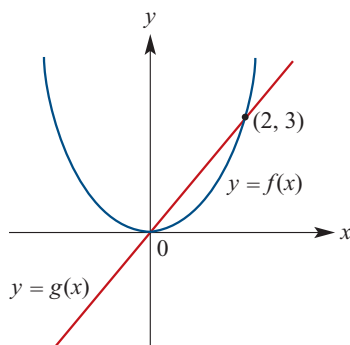
i $(f + g)(0)$

ii $(fg)(0)$

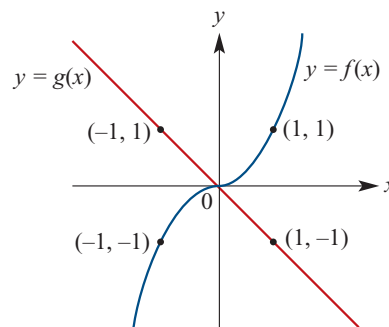
4 Sketch the graph of $f(x) = x^2$ and $g(x) = 3x + 2$ on the one set of axes and hence, using addition of ordinates, sketch the graph of $y = x^2 + 3x + 2$.

5 Copy and add the graph of $y = (f + g)(x)$ using addition of ordinates.

a



b



6 For each of the following, sketch the graph of $f + g$

a $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2, g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = 3$

b $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 + 2x, g: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}, g(x) = \sqrt{x}$

c $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -x^2, g: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}, g(x) = \sqrt{x}$

7 Sketch the graph of $f(x) = e^{-2x}$ and $g(x) = -2x$ on the one set of axes and hence, using addition of ordinates, sketch the graph of $y = e^{-2x} - 2x$

8 Sketch the graph of $f(x) = 2e^{2x}$ and $g(x) = x + 2$ on the one set of axes and hence, using addition of ordinates, sketch the graph of $y = 2e^{2x} + x + 2$

9 For each of the following, find the rule and sketch the graph of:

i $(f + g)(x)$

ii $(fg)(x)$

a $f(x) = |x|$ and $g(x) = x$

b $f(x) = |x|$ and $g(x) = -x$

c $f(x) = -|x|$ and $g(x) = x$

7.4 Identities with function notation

Many of the properties which have been investigated for the functions introduced in the first six chapters may be expressed using function notation.

For example the rules for logarithms

$$\log_e(x) + \log_e(y) = \log_e(xy)$$

$$\log_e\left(\frac{x}{y}\right) = \log_e(x) - \log_e(y)$$

can be written in the following way if $f = \log_e$:

$$\begin{aligned} f(x) + f(y) &= f(xy) \\ f(x) - f(y) &= f\left(\frac{x}{y}\right) \end{aligned}$$

Example 12

- a** For the function with rule $f(x) = 2x$, verify that $f(x + y) = f(x) + f(y)$ for all x and y .
b For the function with rule $f(x) = x + 2$ verify that $f(x + y) \neq f(x) + f(y)$ for all x and y .

Solution

a $f(x + y) = 2(x + y) = 2x + 2y = f(x) + f(y)$

b $f(x + y) = (x + y) + 2 = x + y + 2 = x + 1 + y + 1 - 1 = f(x) + f(y) - 1$

If $f(x) + f(y) - 1 = f(x) + f(y)$ then $-1 = 0$, which is a contradiction.

Example 13

If $f(x) = \frac{1}{x}$ show that $f(x) + f(y) = (x + y)f(xy)$ for x and y non-zero real numbers.

Solution

$$f(x) + f(y) = \frac{1}{x} + \frac{1}{y} = \frac{y + x}{xy} = (x + y) \times \frac{1}{xy} = (x + y)f(xy)$$

Example 14

For the function $f(x) = e^x$, give the functional identities for $f(x + y)$ and $f(x - y)$.

Solution

$$f(x + y) = e^{x+y} = e^x e^y = f(x)f(y)$$

$$f(x - y) = e^{x-y} = \frac{e^x}{e^y} = \frac{f(x)}{f(y)}$$

Example 15

For the function $f(x) = \cos x$, give a counterexample to show that $f(x + y) \neq f(x) + f(y)$ for all x and y .

Solution

For $x = 0$ and $y = \pi$:

$$f(0 + \pi) = f(\pi) = -1$$

$$f(0) + f(\pi) = 1 + -1 = 0$$

Exercise 7D

- 1 For $f(x) = 2x$, find an equivalent expression for $f(x - y)$ in terms of $f(x)$ and $f(y)$.
- 2 For $f(x) = 2x + 3$, show that $f(x + y)$ can be written in the form $f(x) + f(y) + a$ and give the value of a .
- 3 A function g satisfies the property that $[g(x)]^2 = g(x)$. Find the possible values of $g(x)$.
- 4 For the function with rule $f(x) = |x|$, give a counterexample to show that $f(x + y) \neq f(x) + f(y)$ for all x and y .
- 5 For the function with rule $f(x) = \sin x$, give a counterexample to show that $f(x + y) \neq f(x) + f(y)$ for all x and y .
- 6 For the function with rule $f(x) = \frac{1}{x^2}$, show that $f(x) + f(y) = (x^2 + y^2)f(xy)$
- 7 **a** For the function with rule $h(x) = x^2$, give a counterexample to show that $h(x + y) \neq h(x) + h(y)$ for all x and y .
b Show that $h(x + y) = h(x) + h(y)$ implies $x = 0$ or $y = 0$
- 8 Show that for $g(x) = 2^{3x}$, $g(x + y) = g(x)g(y)$
- 9 Show that the functions with rule of the form $f(x) = x^n$ where n is a natural number satisfy the identities $f(xy) = f(x)f(y)$ and $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$
- 10 For the function with rule $f(x) = ax$ where $a \in \mathbb{R} \setminus \{0, 1\}$, give a counterexample to show that $f(xy) \neq f(x)f(y)$ for all x and y .

7.5 Families of functions and solving literal equations

This section demonstrates a different use of parameters. They can be used to discuss families of relations.

Here are some familiar families of relations:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |mx|, \text{ where } m \in \mathbb{R}.$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = ax^3 \text{ where } a \in \mathbb{R} \setminus \{0\}.$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |mx + 2| \text{ where } m \in \mathbb{R}.$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = ke^{mx} \text{ where } m \in \mathbb{R} \setminus \{0\} \text{ and } k \in \mathbb{R} \setminus \{0\}.$$

Their use makes it possible to describe general properties.

What can be said in general about each of these? The family of functions of the form $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |mx + 2|$ where $m \in \mathbb{R}$ is explored in the following example.

Example 16

For $f: R \rightarrow R, f(x) = |mx + 2|$ where $m \in R^+$:

- a Find the x -axis intercept.
- b For which values of m is the x -axis intercept less than -2 ?
- c Find the inverse relation of f .
- d Find the equation of the line perpendicular to the graph at the point with coordinates $(0, 2)$.

Solution

a $|mx + 2| = 0$ implies $mx = -2$ and $x = -\frac{2}{m}$. The x -axis intercept is $-\frac{2}{m}$.

b If $-\frac{2}{m} < -2$, then $\frac{2}{m} > 2$

Multiply both sides of the inequality by m ($m > 0$).

$$2 > 2m$$

Therefore $m < 1$

Therefore the x -axis intercept is less than -2 for $0 < m < 1$

c Consider $x = |my + 2|$ and solve for y .

If $my + 2 \geq 0$, i.e. $y \geq \frac{-2}{m}$

then $x = my + 2$

$$my = x - 2 \text{ and } y = \frac{x - 2}{m}$$

$$\text{Therefore } y = \frac{x}{m} - \frac{2}{m}$$

If $my + 2 < 0$, i.e. $y < \frac{-2}{m}$

then $x = -my - 2$

$$y = -\frac{x}{m} - \frac{2}{m}$$

d The perpendicular line has gradient $-\frac{1}{m}$.

$$\text{The equation is determined as } y - 2 = -\frac{1}{m}x$$

$$\text{and the gradient-intercept form is } y = -\frac{1}{m}x + 2$$

The following example demonstrates how literal equations can be formed and solved with the exponential and logarithmic functions discussed in the previous two chapters.

Example 17

Solve each of the following for x . All constants are positive reals.

a $ae^{bx} - c = 0$ **b** $\log_e(x - a) = b$ **c** $\log_e(cx - a) = 1$

Solution

a $e^{bx} = \frac{c}{a}$

$$bx = \log_e\left(\frac{c}{a}\right)$$

$$x = \frac{1}{b} \log_e\left(\frac{c}{a}\right)$$

b $\log_e(x - a) = b$

$$x - a = e^b$$

$$x = e^b + a$$

c $\log_e(cx - a) = 1$

$$cx - a = e$$

$$cx = a + e$$

$$x = \frac{a + e}{c}$$

Example 18

The graph of a quadratic function passes through the points (1, 6) and (2, 4). Find the coefficients of the quadratic rule in terms of c , the y -axis intercept of the graph.

Solution

Let $f(x) = ax^2 + bx + c$ be a function in this family. Then $f(1) = 6$ and $f(2) = 4$

The following equations are obtained:

$$a + b + c = 6 \text{ and } 4a + 2b + c = 4$$

Solving these gives:

$$a = \frac{c - 8}{2} \quad b = \frac{20 - 3c}{2}$$

The equation of the quadratic in terms of c is:

$$y = \frac{c - 8}{2}x^2 + \frac{20 - 3c}{2}x + c$$

Example 19

A transformation $T: R^2 \rightarrow R^2$ is defined by the matrix equation $\mathbf{AX} + \mathbf{C} = \mathbf{X}'$, where

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 0 & k \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } \mathbf{X}' = \begin{bmatrix} x' \\ y' \end{bmatrix} \text{ where } k \text{ is a non-zero real number.}$$

a Solve the equation for $\mathbf{X}$.

b Find the image of the curve with equation $y = \frac{1}{x}$ under this transformation.

c Find the value of k if the image passes through the origin.

Solution

a $\mathbf{AX} + \mathbf{C} = \mathbf{X}'$

Therefore $\mathbf{X} = \mathbf{A}^{-1}(\mathbf{X}' - \mathbf{C})$

$$= \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{k} \end{bmatrix} \begin{bmatrix} x' - 5 \\ y' - 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{5}(x' - 5) \\ \frac{1}{k}(y' - 2) \end{bmatrix}$$

Therefore $x = \frac{x' - 5}{5}$ and $y = \frac{y' - 2}{k}$

b The image has equation $\frac{y' - 2}{k} = \frac{5}{x' - 5}$

and $y = \frac{5k}{x - 5} + 2$

c If the graph passes through the origin, $0 = -k + 2$ and $k = 2$

Exercise 7E

- 1 Consider $f: R \rightarrow R$, $f(x) = mx - 4$ where $m \in R \setminus \{0\}$.
 - a** Find the x -axis intercept.
 - b** For which values of m is the x -axis intercept less than or equal to 1?
 - c** Find the inverse function of f .
 - d** Find the coordinates of the point of intersection of the graph of $y = f(x)$ with the graph of $y = x$
 - e** Find the equation of the line perpendicular to the line at the point with coordinates $(0, -4)$.
- 2 Consider $f: R \rightarrow R$, $f(x) = -2x + c$ where $c \in R$.
 - a** Find the x -axis intercept.
 - b** For which values of c is the x -axis intercept less than or equal to 1?
 - c** Find the inverse function of f .
 - d** Find the coordinates of the point of intersection of the graph of $y = f(x)$ with the graph of $y = x$
 - e** Find the equation of the line perpendicular to the line at the point with coordinates $(0, c)$.
- 3 Consider the family of quadratics with rule of the form $y = x^2 - bx$ where b is a non-zero real number.
 - a** Find the x -axis intercepts.

- b** Find the coordinates of the vertex of the parabola.
 - c i** Find the coordinates of the points of intersection of the graph of $y = x^2 - bx$ with the line $y = -x$ in terms of b .
 - ii** For what value(s) of b is there one intersection point?
 - iii** For what value(s) of b are there two intersection points?
- 4 a** The graph of $f(x) = x^2$ is translated to the graph of $y = f(x + h)$. Find the possible values of h if $f(1 + h) = 8$
- b** The graph of $f(x) = x^2$ is transformed to the graph of $y = f(ax)$. Find the possible values of a if the graph of $y = f(ax)$ passes through the point with coordinates $(1, 8)$.
- c** The quadratic with equation $y = ax^2 + bx$ has vertex with coordinates $(1, 8)$. Find the values of a and b .
- 5** Consider the family of functions with rule of the form $f(x) = \sqrt{2a - x}$, where a is a positive real number.
 - a** State the maximal domain of f .
 - b** Find the coordinates of the point of intersection of the graph of $y = f(x)$ with the graph of $y = x$
 - c** For what value of a does the line with equation $y = x$ intersect the graph of $y = f(x)$ at the point with coordinates $(1, 1)$?
 - d** For what value of a does the line with equation $y = x$ intersect the graph of $y = f(x)$ at the point with coordinates $(2, 2)$?
 - e** For what value of a does the line with equation $y = x$ intersect the graph of $y = f(x)$ at the point with coordinates (c, c) where c is a positive real number?
- 6** Consider the function with rule $f(x) = |x^2 - ax|$.
 - a** State the coordinates of the x -axis intercepts.
 - b** State the coordinates of the y -axis intercept.
 - c** Find the maximum value of the function in the interval $[0, a]$.
 - d** Find the possible values of a for which the point $(-1, 4)$ lies on the graph of $y = f(x)$
- 7** Solve each of the following for x . All constants are positive reals.
 - a** $-ae^{bx} + c = 0$ **b** $c \log_e (x + a) = b$
 - c** $\log_e (cx - a) = 0$ **d** $e^{ax+b} = c$
- 8** Consider the functions with rule of the form $f(x) = c \log_e (x - a)$ where a is a positive constant.
 - a** State the equation of the vertical asymptote.
 - b** State the coordinates of the x -axis intercept.
 - c** State the coordinates of the point where the graph crosses the line $y = 1$
 - d** If the graph of the function crosses the line $y = 1$ when $x = 2$, find the value of c in terms of a .

- 9 Consider the family of functions with rule $f(x) = e^{x-1} - b$ where $b > 0$.
- State the equation of the horizontal asymptote.
 - State the coordinates of the x -axis intercept.
 - Give the values of b for which the x -axis intercept is:
 - at the origin
 - a negative number
- 10 The graph of a quadratic function passes through the points $(-1, 6)$ and $(1, 4)$. Find the coefficients of the quadratic rule in terms of c , the y -axis intercept of the graph.
- 11 The graph of a cubic function passes through the points $(-1, 6)$, $(1, -2)$ and $(2, 4)$. Find the coefficients of the cubic rule in terms of d , the y -axis intercept of the graph.
- 12 A quadratic function has rule $y = \left(\frac{c-8}{2}\right)x^2 + \left(\frac{20-3c}{2}\right)x + c$. Find the values of c for which:
- the graph of $y = f(x)$ touches the x -axis
 - the graph of $y = f(x)$ has two distinct x -axis intercepts
- 13 The graph of a cubic function passes through the points $(-2, 8)$, $(1, 1)$ and $(3, 4)$. Find the coefficients of the quadratic rule in terms of d , the y -axis intercept of the graph.
- 14 A transformation is defined by the matrix equation $\mathbf{AX} + \mathbf{C} = \mathbf{X}'$, where $\mathbf{A} = \begin{bmatrix} -4 & 0 \\ 0 & k \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}$ and $\mathbf{X}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ where k is a non-zero real number.
- Solve the equation for $\mathbf{X}$.
 - Find the image of the curve with equation $y = \frac{1}{x}$ under this transformation.
 - Find the value of k if the image passes through the origin.
- 15 A transformation is defined by the matrix equation $\mathbf{AX} + \mathbf{C} = \mathbf{X}'$, where $\mathbf{A} = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} a \\ -2 \end{bmatrix}$, $\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}$, and $\mathbf{X}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ where a is a non-zero real number.
- Solve the equation for $\mathbf{X}$.
 - Find the image of the curve with equation $y = 2^x$ under this transformation.
 - Find the value of a if the image passes through the origin.



Chapter summary

- Let f and g be functions such that $\text{dom } f \cap \text{dom } g \neq \emptyset$. The **sum**, $f + g$, and the **product**, fg , as functions on $\text{dom } f \cap \text{dom } g$ are defined by:

- $(f + g)(x) = f(x) + g(x)$
- $fg(x) = f(x) \cdot g(x)$

- In general for the composition of g with f to be defined, the range of $f \subseteq \text{domain of } g$. When this composition (or composite function) of g with f is defined it is denoted $g \circ f$

For functions f and g with domains X and Y respectively and such that the range of $f \subseteq Y$, the composite function of g with f is defined by:

$$g \circ f: X \rightarrow R, \text{ where } g \circ f(x) = g(f(x))$$

- If f is a one-to-one function, a new function, f^{-1} , called the **inverse** of f , may be defined by:

$$f^{-1}(x) = y \text{ if } f(y) = x, \text{ for } x \in \text{ran } f, y \in \text{dom } f$$

For a function f and its inverse f^{-1} :

$$\text{dom } f^{-1} = \text{ran } f$$

$$\text{ran } f^{-1} = \text{dom } f$$

The inverse of any relation may be defined. The inverse relation is not a function unless the initial relation is a one-to-one function.

For a relation $S = \{(a, b)\}$ the inverse relation is $\{(b, a)\}$.

Multiple-choice questions

- 1 The graph of the function with rule $h(x) = \frac{x^4 + 2}{x^2}$ can be drawn by adding the ordinates of the graphs of two functions f and g . The rules for f and g could be:
 - A $f(x) = x^4, g(x) = \frac{2}{x^2}$
 - B $f(x) = x^2, g(x) = \frac{2}{x^2}$
 - C $f(x) = x^4 + 2, g(x) = x^2$
 - D $f(x) = x^4 + 2, g(x) = \frac{2}{x^2}$
 - E $f(x) = x^2, g(x) = 2$
- 2 Which one of the following functions is not a one-to-one function?
 - A $f: R^+ \rightarrow R, f(x) = \frac{1}{x^2}$
 - B $f: R \rightarrow R, f(x) = x^3$
 - C $f: R \rightarrow R, f(x) = 10^x$
 - D $f: R^+ \rightarrow R, f(x) = \log_{10} x$
 - E $f: R \rightarrow R, f(x) = \cos x$
- 3 The function f has domain R and rule $f(x) = \frac{4}{x^2} + 9$. The rule of the inverse relation is:
 - A $y = 2 \pm \frac{2}{\sqrt{x-3}}$
 - B $y = 3 \pm \frac{2}{\sqrt{x-9}}$
 - C $y = 3 \pm \frac{2}{\sqrt{x}}$
 - D $y = \pm \frac{2}{\sqrt{9-x}}$
 - E $y = \pm \frac{2}{\sqrt{x-9}}$

- 4 The function f has domain R and rule $f(x) = |x| + 2$. The domain and rule of the inverse are:
- A** $[-\infty, 4]$ and $x = |y| + 2$ **B** $[0, \infty)$ and $x = \pm y + 2$
C R and $x = \pm y + 2$ **D** $[2, \infty)$ and $x = |y| + 2$
E $R \setminus \{2\}$ and $x - y = 2$
- 5 The range of the function $f: R \rightarrow R$, $f(x) = 3|\cos x| - 1$ is:
- A** $[1, 4]$ **B** $[-1, 2]$ **C** $[-4, 2]$ **D** $[-2, 1]$ **E** $[0, 3]$
- 6 For the function with equation $f(x) = e^x$, which one of the following is **not** correct for all positive real x and y ?
- A** $f(x + y) = f(x)f(y)$ **B** $f^{-1}(xy) = f^{-1}(x) + f^{-1}(y)$
C $f^{-1}(x^y) = yf^{-1}(x)$ **D** $f^{-1}(1) = 0$ **E** $f^{-1}(x) = \frac{1}{f(x)}$
- 7 If $f(x) = \cos x$ and $g(x) = 3x^2$ then $g\left(f\left(\frac{\pi}{3}\right)\right)$ is equal to:
- A** $\cos \frac{\pi^2}{9}$ **B** $\frac{1}{\sqrt{2}}$ **C** 1 **D** $\frac{3}{4}$ **E** $\frac{4}{3}$
- 8 Which of the following is not an even function?
- A** $f: R \rightarrow R$, $f(x) = |2x|$ **B** $f: R \rightarrow R$, $f(x) = \cos^2 x$
C $f: R \rightarrow R$, $f(x) = \cos x$ **D** $f: R \rightarrow R$, $f(x) = 4x^2 - 3$
E $f: R \rightarrow R$, $f(x) = (x - 2)^2$
- 9 It is known that the graph of the function with rule $y = 2ax + \cos 2x$ has an x -axis intercept when $x = \pi$. The value of a is:
- A** 2 **B** $\frac{1}{2\pi}$ **C** 2π **D** -2π **E** $-\frac{1}{2\pi}$
- 10 If, for $x > 5$, $g(x) = \log_e(x - 5)$ and $2[g(x)] = g(f(x))$ then $f(x)$ is equal to:
- A** $5x - 8$ **B** $x^2 - 10x + 30$ **C** $5x^2$ **D** $(2x - 10)^2$ **E** $2x - 2$

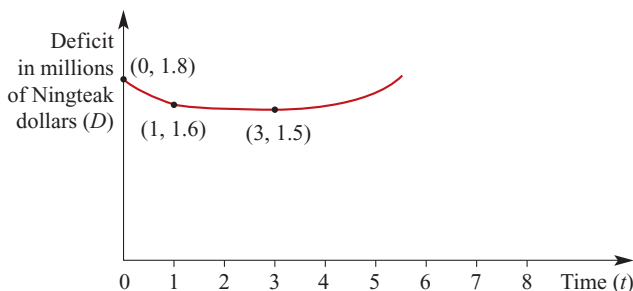
Short-answer questions (technology-free)

- 1 For each of the following, where $g(x) = |x|$:
- Find the rules $f \circ g(x)$ and $g \circ f(x)$
 - Find the range of $y = f \circ g(x)$ and $y = g \circ f(x)$ (and state the maximal domain for each of the composite functions to exist).
- a** $f(x) = 3 \cos 2x$ **b** $f(x) = \log_e 3x$
c $f(x) = \log_e(2 - x)$ **d** $f(x) = -\log_e 2x$
- 2 Express each of the following as the composition of two functions:
- a** $h(x) = \cos |2x|$ **b** $h(x) = (x^2 - x)^n$ where n is a natural number
c $h(x) = \log_e(\sin x)$ **d** $h(x) = -2 |\sin 2x|$
e $h(x) = (x^2 - 3x)^4 - 2(x^2 - 3x)^2$

- 3 For $f(x) = 2 \cos\left(\frac{\pi x}{2}\right)$ and $g(x) = e^{-x}$:
- Find the rules for:
 - $(f + g)$
 - (fg)
 - Evaluate:
 - $(f + g)(0)$
 - $(fg)(0)$
- 4 Let $f: [a, \infty) \rightarrow \mathbb{R}$, $f(x) = -(3x - 2)^2 + 3$
- Find the smallest value of a such that f is one-to-one.
 - With this value of a , state the range of f .
 - Sketch the graph of f .
 - Find f^{-1} and state the domain and range of f^{-1} .
 - Sketch the graph of f and f^{-1} on the one set of axes.
- 5 Find the inverse relation of each of the following functions:
- $f(x) = (3x - 2)^2 + 4$
 - $f(x) = -2x^2 + 3$

Extended-response questions

- 1 The deficit of a government department in Ningteak, a small monarchy east of Africa, is shown below. The deficit is continually assessed over a period of 8 years. The graph shown below is that of the deficit over these 8 years:



The graph is read as follows: 'The deficit at the beginning of the 8-year period is \$1.8 million. At the end of the third year the deficit is \$1.5 million, and this is the smallest deficit for the period, $0 \leq t \leq 8$ '.

- Find the rule for D in terms of t , assuming that it is of the form $D = at^2 + bt + c$
 - Use this model to predict the deficit at the end of 8 years.
- 2 The rainfall, R , recorded during a very rainy day in North Queensland, was as follows:
- | | |
|---------------|-----------------|
| At 4:00 a.m. | 7.5 mm per hour |
| At 8:00 a.m. | 9 mm per hour |
| At 10:00 a.m. | 8 mm per hour |
- Assume a quadratic rule of the form $R = at^2 + bt + c$ is applicable for $0 \leq t \leq 12$ where $t = 0$ is 4:00 a.m. Use the quadratic model to predict the rate of rainfall at 12:00 noon. At what time was the rate of rainfall greatest?

- 3 Consider $f: R^+ \rightarrow R$, $f(x) = e^{-x}$ and $g: (-\infty, 1) \rightarrow R$, $g(x) = \frac{1}{x-1}$
- State the range of f and g .
 - Find f^{-1} and g^{-1} .
 - Find $g \circ f$
 - Sketch the graph of $y = g \circ f(x)$
 - Find $(g \circ f)^{-1}$
 - Sketch the graph of $y = (g \circ f)^{-1}(x)$
- 4 a For $f: [5, \infty) \rightarrow R$, $f(x) = \sqrt{x-3}$:
- Sketch the graph of $y = f(x)$ for $x \in [5, \infty)$.
 - State the range.
 - Find f^{-1} .
- b For $h: [4, \infty) \rightarrow R$, $h(x) = \sqrt{x-p}$ with inverse function h^{-1} that has domain $[1, \infty)$:
- Find p .
 - Find the rule for h^{-1} .
 - Sketch the graphs of $y = h(x)$ and $y = h^{-1}(x)$ on the one set of axes.
- 5 Let $f: (0, \pi) \rightarrow R$ with $f(x) = \sin x$ and $g: [1, \infty) \rightarrow R$ and $g(x) = \frac{1}{x}$
- Find the range of f .
 - Find the range of g .
 - Give a sufficient reason why $f \circ g$ is defined and find $f \circ g(x)$
 - State, with sufficient reason, whether $g \circ f$ is defined.
 - Find g^{-1} , giving its domain and range.
 - Give a sufficient reason why $g^{-1} \circ f$ is defined and find $g^{-1} \circ f(x)$. Also state the domain and range of this function.
- 6 A population of insects is determined by a rule of the form

$$n = \frac{c}{1 + ae^{-bt}} \quad t \geq 0$$

where n is the number of insects alive at time t days.

- Consider the population for $c = 5790$, $a = 4$ and $b = 0.03$.
 - Find the equation of the horizontal asymptotes by considering values of n as t becomes large.
 - Find n when $t = 0$.
 - Sketch the graph of the function.
 - Find the exact value of t for which $n = 4000$.



- Use your calculator to find values of a , b and c such that the population growth yields the following table:

t	1	10	100
n	1500	2000	5000

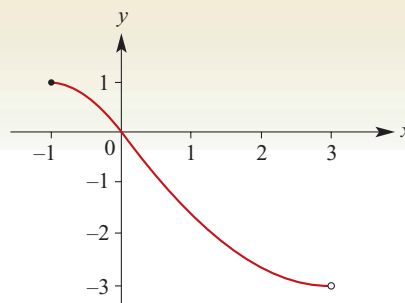
- Sketch the graph for this population.

Revision of Chapters 1–7

8.1 Multiple-choice questions

- 1 The domain of the function whose graph is shown is:

A $(-3, 1]$
B $(-1, 3]$
C $[1, 3]$
D $[-1, 3)$
E $(-1, 3)$



- 2 Which of the following sets of ordered pairs does not represent a function where y is the value of the function?

A $\{(x, y): x = 2y^2, x \geq 0\}$ **B** $\{(x, y): y = \frac{1}{x}, x \in \mathbb{R} \setminus \{0\}\}$
C $\{(x, y): y = 2x^3 + 3, x \in \mathbb{R}\}$ **D** $\{(x, y): y = 3x^2 + 7, x \in \mathbb{R}\}$
E $\{(x, y): y = e^x - 1, x \in \mathbb{R}\}$

- 3 The implied (largest possible) domain for the function with the rule $y = \frac{1}{\sqrt{2-x}}$ is:

A $\mathbb{R} \setminus \{2\}$ **B** $(-\infty, 2)$ **C** $(2, \infty)$ **D** $(-\infty, 2]$ **E** $\mathbb{R}^+$

- 4 If $f(x) = \frac{x}{x-1}$ then $f\left(-\frac{1}{a}\right)$, in simplified form, is equal to:

A $\frac{1}{-1-a}$ **B** -1 **C** 0 **D** $\frac{a^2}{1-a}$ **E** $\frac{1}{a+1}$

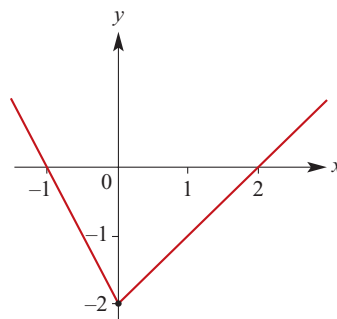
- 5 The graph shown has the equation:

A $y = \begin{cases} x-2, & x > 0 \\ -2x-2, & x \leq 0 \end{cases}$ **B** $y = \begin{cases} 2x-2, & x \geq 0 \\ -2x-2, & x < 0 \end{cases}$

C $y = \begin{cases} x - 2, & x > 0 \\ -2x - 1, & x \leq 0 \end{cases}$

D $y = \begin{cases} x + 2, & x > 0 \\ -2x - 2, & x \leq 0 \end{cases}$

E $y = \begin{cases} x - 2, & x > 0 \\ -x - 2, & x \leq 0 \end{cases}$



- 6** If $f: [0, 2\pi] \rightarrow \mathbb{R}$ where $f(x) = \sin 2x$ and $g: [0, 2\pi] \rightarrow \mathbb{R}$ where $g(x) = 2 \sin x$, then the value of $(f + g)\left(\frac{3\pi}{2}\right)$ is:

A 2 **B** 0 **C** -1 **D** 1 **E** -2

- 7** If $f(x) = 3x + 2$ and $g(x) = 2x^2$, then $f(g(3))$ equals:

A 36 **B** 20 **C** 56 **D** 144 **E** 29

- 8** If $f(x) = 3x^2$, $0 \leq x \leq 6$ and $g(x) = \sqrt{2 - x}$, $x \leq 2$, the domain of $f + g$ is:

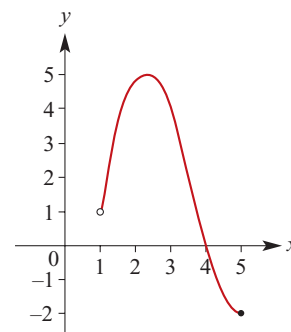
A $[0, 2]$ **B** $[0, 6]$ **C** $(-\infty, 2]$ **D** $\mathbb{R}^+ \cup \{0\}$ **E** $[2, 6]$

- 9** If $g(x) = 2x^2 + 1$ and $f(x) = 3x + 2$, then the rule of the product function $fg(x)$ equals:

A $2x^2 + 3x + 3$ **B** $6x^3 + 4x^2 + 3x + 2$
C $6x^3 + 3$ **D** $6x^3 + 2x^2 + 3$
E $6x^3 + 2$

- 10** The domain of the function whose graph is shown is:

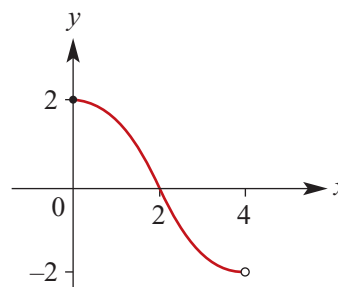
A $[1, 5]$
B $(1, 5]$
C $(-2, 5]$
D $(1, 5)$
E $(-2, 5)$



- 11** The implied domain for the function with equation $y = \sqrt{4 - x^2}$ is:

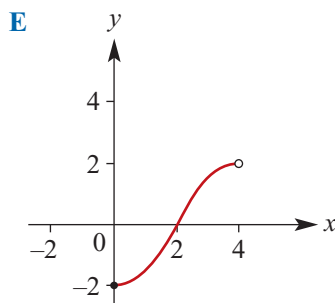
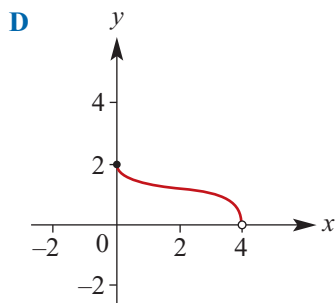
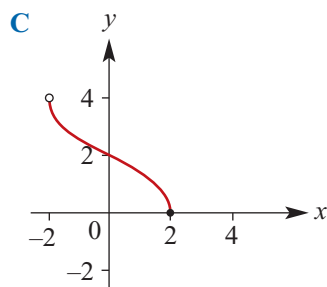
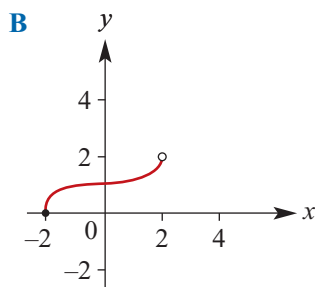
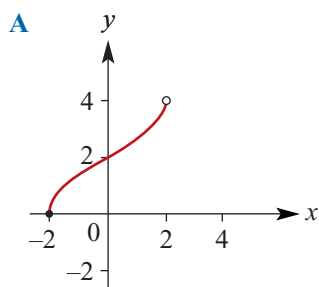
A $[2, \infty)$ **B** $\{x: -2 < x < 2\}$ **C** $[-2, 2]$
D $(-\infty, 2]$ **E** $\mathbb{R}^+$

- 12** The graph of the function with rule $y = f(x)$ is shown.



(cont'd.)

Which one of the following graphs is the graph of the inverse of f ?



13 The graph shown has the rule:

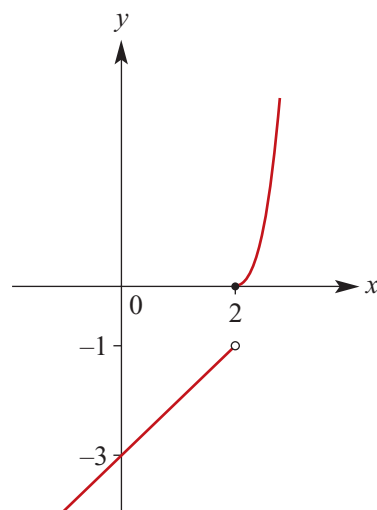
A $y = \begin{cases} (x-2)^2, & x \geq 2 \\ x-3, & x < 2 \end{cases}$

B $y = \begin{cases} x-2, & x \geq 2 \\ x-3, & x < 2 \end{cases}$

C $y = \begin{cases} (2-x)^2, & x \geq 2 \\ 2x-3, & x < 2 \end{cases}$

D $y = \begin{cases} (x-2)^2, & x < 2 \\ 2x-3, & x \geq 2 \end{cases}$

E $y = \begin{cases} (x-2)^2, & x \geq 2 \\ 2x-3, & x < 2 \end{cases}$



14 The inverse, f^{-1} , of the function $f: [2, 3] \rightarrow \mathbb{R}$, $f(x) = 2x - 4$ is:

A $f^{-1}: [0, 2] \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{x}{2} + 4$ **B** $f^{-1}: [3, 2] \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{x+4}{2}$

C $f^{-1}: [2, 3] \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{1}{2x-4}$ **D** $f^{-1}: [0, 2] \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{1}{2x-4}$

E $f^{-1}: [0, 2] \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{x+4}{2}$

15 f is the function defined by $f(x) = \frac{1}{x^2+2}$, $x \in \mathbb{R}$. A suitable restriction for f , f^* such that f^{*-1} exists, would be:

A $f^*: [-1, 1] \rightarrow \mathbb{R}$, $f^*(x) = \frac{1}{x^2+2}$ **B** $f^*: \mathbb{R} \rightarrow \mathbb{R}$, $f^*(x) = \frac{1}{x^2+2}$

C $f^*: [-2, 2] \rightarrow R, f^*(x) = \frac{1}{x^2 + 2}$ **D** $f^*: [0, \infty) \rightarrow R, f^*(x) = \frac{1}{x^2 + 2}$

E $f^*: [-1, \infty) \rightarrow R, f^*(x) = \frac{1}{x^2 + 2}$

- 16** Let $h: [a, 2] \rightarrow R$ where $h(x) = 2x - x^2$. If a is the smallest real value such that h has an inverse function, h^{-1} , then a equals:

A -1 **B** 0 **C** 1 **D** -2 **E** $\frac{1}{2}$

- 17** If $f(x) = 3x - 2, x \in R$, then $f^{-1}(x)$ equals:

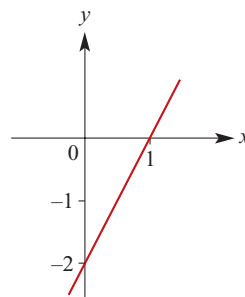
A $\frac{1}{3x - 2}$ **B** $3x + 2$ **C** $\frac{1}{3}(x - 2)$ **D** $3x + 6$ **E** $\frac{1}{3}(x + 2)$

- 18** The solution of the equation $2x = \frac{3x}{2} - 4$ is:

A 4 **B** -2 **C** -8 **D** 1 **E** 2

- 19** The graph shows:

A $y + 2 = x$
B $y = 2x - 2$
C $y + 2x + 2 = 0$
D $y = -2x + 2$
E $y - 2 = x$



- 20** If $\frac{2(x - 1)}{3} - \frac{x + 4}{2} = \frac{5}{6}$, then x equals:

A 5 **B** $\frac{7}{5}$ **C** $\frac{21}{5}$ **D** 21 **E** 3

- 21** The equation of the line that passes through the points $(-2, 3)$ and $(4, 0)$ is:

A $2y = x + 4$ **B** $y = -\frac{1}{2}x - 2$ **C** $2y + x = 4$
D $y = \frac{1}{2}x - 2$ **E** $2y - x = 4$

- 22** If the angle between the lines $2y = 8x + 10$ and $3x - 6y = 22$ is θ , then $\tan \theta$ is best approximated by:

A 1.17 **B** 1.40 **C** 2 **D** 0.86 **E** 1

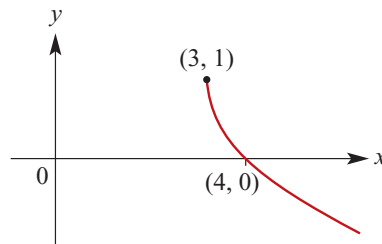
- 23** The line with equation $y = \frac{4}{5}x - 4$ meets the x -axis at A and the y -axis at B . If O is the origin, the area of the triangle OAB is:

A $3\frac{1}{5}$ square units **B** $9\frac{2}{5}$ square units **C** 10 square units
D 15 square units **E** 20 square units

- 24** If the equations $2x - 3y = 12$ and $3x - 2y = 13$ are simultaneously true, then $x + y$ equals:
A -5 **B** -1 **C** 0 **D** 1 **E** 5
- 25** If the graphs of the relations $7x - 6y = 20$ and $3x + 4y = 2$ are drawn on the same pair of axes, the x -coordinate of the point of intersection is:
A -2 **B** -1 **C** 1 **D** 2 **E** 3
- 26** A possible equation for the graph shown is:
A $y - 3 = \frac{1}{x - 1}$
B $y + 3 = \frac{1}{x + 1}$
C $y - 3 = \frac{1}{x + 1}$
D $y - 4 = \frac{1}{x + 1}$
E $y = \frac{1}{x - 1} - 3$
-
- 27** The function given by $f(x) = \frac{1}{x+3} - 2$ has the range given by:
A $R \setminus \{-2\}$ **B** R **C** $R \setminus \{3\}$ **D** $R \setminus \{2\}$ **E** $R \setminus \{-3\}$
- 28** A parabola has its vertex at $(2, 3)$. A possible equation for this parabola is:
A $y = (x + 2)^2 + 3$ **B** $y = (x - 2)^2 - 3$ **C** $y = (x + 2)^2 - 3$
D $y = (x - 2)^2 + 3$ **E** $y = 3 - (x + 2)^2$
- 29** Which one of the following is an even function of x ?
A $f(x) = 3x + 1$ **B** $f(x) = x^3 - x$ **C** $f(x) = (1 - x)^2$
D $f(x) = -x^2$ **E** $f(x) = x^3 + x^2$
- 30** The graph of $y = 3\sqrt{x+2}$ can be obtained from the graph of $y = \sqrt{x}$ by:
A a translation $(x, y) \rightarrow (x - 2, y)$ followed by a dilation of factor 3 from the x -axis
B a translation $(x, y) \rightarrow (x + 2, y)$ followed by a dilation of factor $\frac{1}{3}$ from the x -axis
C a translation $(x, y) \rightarrow (x + 3, y)$ followed by a dilation of factor 3 from the y -axis
D a translation $(x, y) \rightarrow (x - 2, y)$ followed by a dilation of factor 3 from the y -axis
E a translation $(x, y) \rightarrow (x + 2, y)$ followed by a dilation of factor 3 from the y -axis
- 31** A function with rule $f(x) = 3\sqrt{x-2} + 1$ has maximal domain:
A $(-\infty, 2)$ **B** $[1, \infty)$ **C** $(2, \infty)$ **D** $[-2, \infty)$ **E** $[2, \infty)$

- 32** A possible equation of the graph shown is:

- A** $y = 2\sqrt{x-3} + 1$
B $y = -2\sqrt{x-3} + 1$
C $y = \sqrt{x-3} + 1$
D $y = -\sqrt{x-3} + 1$
E $y = -2\sqrt{x-3} + 2$



- 33** The range of the function $f: R \setminus \{2\} \rightarrow R, f(x) = \frac{3}{(x-2)^2} + 4$ is:

- A** $(3, 4]$ **B** $(-\infty, 4)$ **C** $[3, 4)$ **D** $[4, \infty)$ **E** $(4, \infty)$

- 34** If $3x^2 + kx + 1 = 0$ when $x = 1$, then k equals:

- A** -4 **B** -1 **C** 1 **D** 4 **E** 0

- 35** The quadratic equation whose roots are 5 and -7 is:

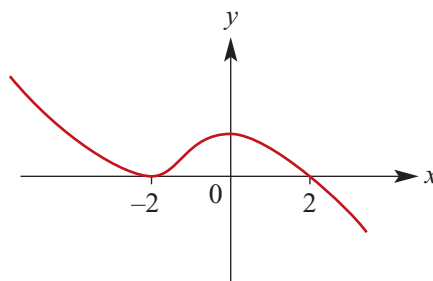
- A** $x^2 + 2x - 35 = 0$ **B** $x^2 - 2x - 35 = 0$ **C** $x^2 + 12x - 35 = 0$
D $x^2 - 12x - 35 = 0$ **E** $-x^2 + 12x + 35 = 0$

- 36** If $x^3 - 5x^2 + x + k$ is divisible by $x + 1$, then k equals:

- A** -7 **B** -5 **C** -2 **D** 5 **E** 7

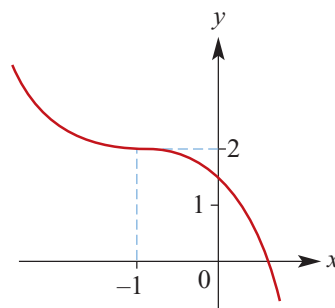
- 37** Which one of the following could be the equation of the graph shown?

- A** $y = x(x-2)(x+2)$
B $y = -x(x+2)(x-2)$
C $y = -(x+2)^2(x-2)$
D $y = (x-2)^2(x+2)$
E $y = x(x-2)^2$



- 38** The graph shown is:

- A** $y + 2 = -2(x+1)^3$
B $y - 2 = 2(x-1)^3$
C $y = x^3 + 2$
D $y = -\frac{1}{2}(x+1)^3 + 2$
E $y = 2(x-1)^3 - 2$



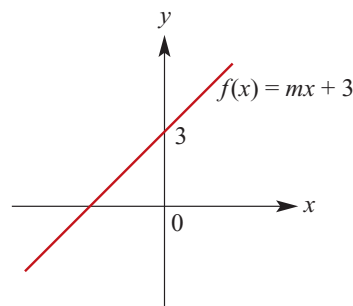
- 39** $P(x) = x^3 + 2x^2 - 5x - 6$ has the factors:

- A** $(x-1)(x-2)(x+3)$ **B** $(x+1)(x+2)(x+3)$
C $(x+1)(x-2)(x+3)$ **D** $(x+1)(x-2)(x-3)$
E $(x-1)(x-2)(x-3)$

- 40** If $P(x) = 2x^3 - 2x^2 + 3x + 1$, when $P(x)$ is divided by $(x-2)$ the remainder is:

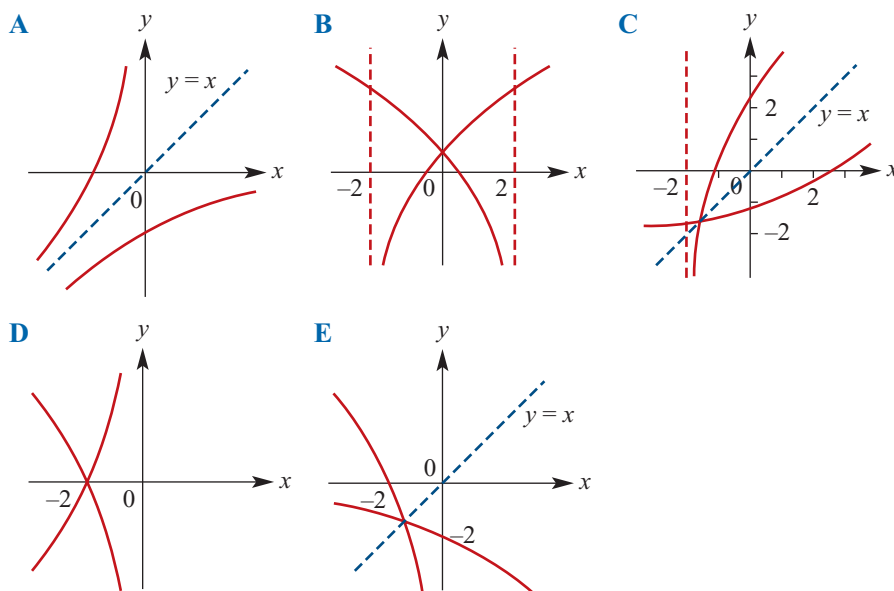
- A** 31 **B** 15 **C** 1 **D** -2 **E** -29

- 41 The graph shown is that of the function $f(x) = mx + 3$, where m is a constant. The inverse f^{-1} is defined as $f^{-1}: R \rightarrow R$, $f^{-1}(x) = ax + b$, where a and b are constants. Which one of the following statements is true?

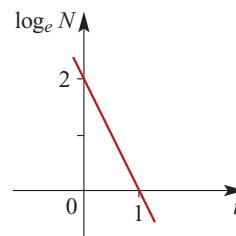


- A** $a = \frac{-3}{m}, b = \frac{1}{m}$ **B** $a < 0$ and $b < 0$ **C** $a = -m, b = 3$ **D** $a > 0$ and $b > 0$ **E** $a = \frac{1}{m}, b = \frac{-3}{m}$
- 42 If $x^3 + 2x^2 + ax - 4$ has a remainder 1 when divided by $x + 1$, then a equals:
A -8 **B** -4 **C** -2 **D** 0 **E** 2
- 43 Which of these equations is represented by the graph shown?
- A** $y = (x + 2)^2(x - 2)$ **B** $y = 16 - x^4$ **C** $y = (x^2 - 4)^2$ **D** $y = (x + 2)^2(2 - x)$ **E** $y = x^4 - 16$
- 44 The function $f: R \rightarrow R$ where $f(x) = e^{-x} + 1$ has an inverse function f^{-1} . The domain of f^{-1} is:
A $(0, \infty)$ **B** R **C** $[1, \infty)$ **D** $(1, \infty)$ **E** $[0, \infty)$
- 45 The function $f: R^+ \rightarrow R$ where $f(x) = 2 \log_e x + 1$ has an inverse function f^{-1} . The rule for f^{-1} is given by:
A $f^{-1}(x) = 2e^{x-1}$ **B** $f^{-1}(x) = e^{\frac{1}{2}(x-1)}$ **C** $f^{-1}(x) = e^{\frac{x}{2}-1}$
D $f^{-1}(x) = 2e^{x+1}$ **E** $f^{-1}(x) = \frac{1}{2}e^{x-1}$
- 46 Let $f: R \rightarrow R$ where $f(x) = e^{-x}$ and $g: (-1, \infty) \rightarrow R$ where $g(x) = \log_e(x + 2)$. The function with the rule $y = f(g(x))$ has the range:
A $(1, \infty)$ **B** $(0, 1)$ **C** $(0, 1]$ **D** $[1, \infty)$ **E** $[0, 1]$
- 47 The function $g: R \rightarrow R$ where $g(x) = e^x - 1$ has an inverse whose rule is given by:
A $f^{-1}(x) = \frac{1}{e^x - 1}$ **B** $f^{-1}(x) = -\log_e(x + 1)$ **C** $f^{-1}(x) = \log_e(x - 1)$
D $f^{-1}(x) = \log_e(1 - x)$ **E** $f^{-1}(x) = \log_e(x + 1)$
- 48 The function $f: [4, \infty) \rightarrow R$ where $f(x) = \log_e(x - 3)$ has an inverse. The domain of this inverse is:
A $[0, \infty)$ **B** $(0, \infty)$ **C** $[4, \infty)$ **D** $(3, \infty)$ **E** R

- 49** The function $f: R \rightarrow R$ where $f(x) = e^{x-1}$ has an inverse whose rule is given by $f^{-1}(x) =$
A $e^{-(x-1)}$ **B** $-\log_e x$ **C** $1 + \log_e x$ **D** $\log_e (x + 1)$ **E** $\log_e (x - 1)$
- 50** The function $f: R^+ \rightarrow R$ where $f(x) = \log_e \frac{x}{2}$ has an inverse function f^{-1} . The rule for f^{-1} is given by $f^{-1}(x) =$
A $e^{\frac{1}{2}x}$ **B** $\log_e \left(\frac{2}{x}\right)$ **C** $\frac{1}{2}e^{\frac{x}{2}}$ **D** $2e^x$ **E** $\frac{1}{\log_e \frac{x}{2}}$
- 51** For what values of x is the function f with the rule $f(x) = -2 + \log_e (3x - 2)$ defined?
A $(-2, \infty)$ **B** $\left(\frac{2}{3}, \infty\right)$ **C** $[-2, \infty)$ **D** $\left[\frac{2}{3}, \infty\right)$ **E** $(2, \infty)$
- 52** The graphs of the function $f: (-2, \infty) \rightarrow R$ where $f(x) = 2 + \log_e (x + 2)$ and its inverse f^{-1} are best shown by which one of the following?



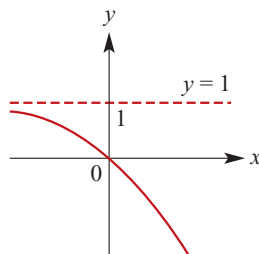
- 53** For $\log_2 8x + \log_2 2x = 6$, $x =$
A 1.5 **B** ± 1.5 **C** 2 **D** ± 2 **E** 6.4
- 54** The equation $\log_{10} x = y(\log_{10} 3) + 1$ is equivalent to the equation:
A $x = 10(3^y)$ **B** $x = 30^y$ **C** $x = 3^y + 10$
D $x = y^3 + 10$ **E** $x = 10y^3$
- 55** The graph indicates that the relationship between N and t is:



- A** $N = 2 - e^{-2t}$
B $N = e^{2-2t}$
C $N = e^{2t} + 2$
D $N = \frac{e^{-2t}}{100}$
E $N = -2e^{2t}$

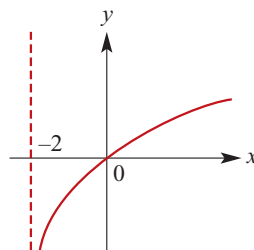
56 A possible equation for the graph is:

- A $y = 1 - e^x$
 B $y = 1 - e^{-x}$
 C $y = 1 + e^x$
 D $y = 1 + e^{-x}$
 E $y = e^{-x} - 1$



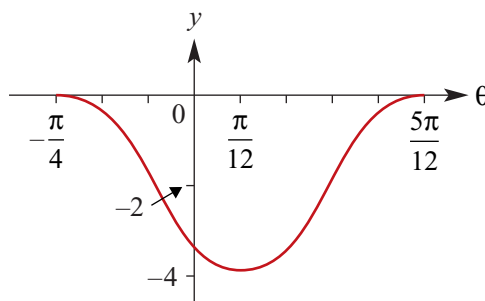
57 A possible equation for the graph is:

- A $y = \log_e (x - 2)$
 B $y = \log_e \frac{1}{2}(x + 2)$
 C $y = \log_e 2(x + 1)$
 D $y = 2 \log_e (x + 1)$
 E $y = \frac{1}{2} \log_e (x + 2)$



58 A possible equation for the graph shown is:

- A $y = 2 \cos 3 \left(\theta + \frac{\pi}{4} \right) - 4$
 B $y = 2 \cos 2 \left(\theta + \frac{\pi}{4} \right) - 2$
 C $y = 2 \sin 3 \left(\theta + \frac{\pi}{4} \right) - 2$
 D $y = 2 \cos 3 \left(\theta + \frac{\pi}{4} \right) - 2$
 E $y = 2 \cos 3 \left(\theta - \frac{\pi}{4} \right) - 2$



59 The function $f: R \rightarrow R$ where $f(x) = 2 - 3 \cos 2 \left(\theta + \frac{\pi}{2} \right)$ has range:

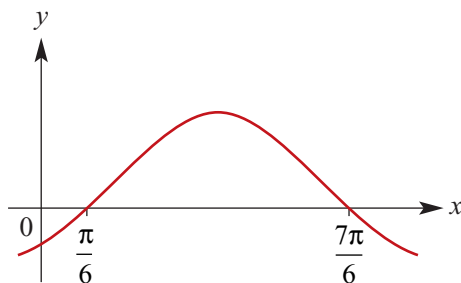
- A $[-3, 5]$ B $[2, 5]$ C R D $[-1, 5]$ E $[-3, 2]$

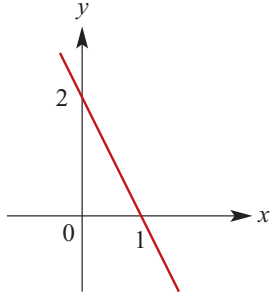
60 Two values between 0 and 2π for which $2 \sin \theta + \sqrt{3} = 0$ are:

- A $\frac{\pi}{3}, \frac{2\pi}{3}$ B $60^\circ, 240^\circ$ C $\frac{2\pi}{3}, \frac{5\pi}{3}$ D $\frac{4\pi}{3}, \frac{5\pi}{3}$ E $\frac{7\pi}{6}, \frac{11\pi}{6}$

61 A possible equation for the graph shown is:

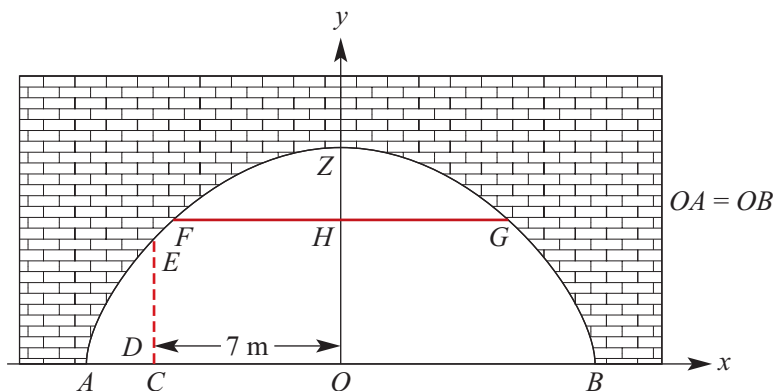
- A $y = \sin \left(x - \frac{\pi}{6} \right)$
 B $y = \sin \left(x + \frac{\pi}{6} \right)$
 C $y = -\sin \left(x - \frac{\pi}{6} \right)$
 D $y = \cos \left(x - \frac{\pi}{6} \right)$
 E $y = \cos \left(x + \frac{\pi}{6} \right)$



- 62** The function $f: R \rightarrow R$, $f(x) = 3 \sin 2x$ has:
- A** amplitude 3 and period π **B** amplitude 2 and period $\frac{\pi}{2}$
C amplitude 1 and period $\frac{\pi}{2}$ **D** amplitude $\frac{3}{2}$ and period 2π
E amplitude $1\frac{1}{2}$ and period 2π
- 63** The function $f: R \rightarrow R$, where $f(x) = 3 \sin 2x$ has range:
- A** $[0, 3]$ **B** $[-2, 2]$ **C** $[2, 3]$ **D** $[-3, 3]$ **E** $[-1, 5]$
- 64** Consider the polynomial $p(x) = (x - 2a)^2(x + a)(x^2 + a)$ where $a > 0$. The equation $p(x) = 0$ has exactly:
- A** 1 distinct real solution **B** 2 distinct real solutions
C 3 distinct real solutions **D** 4 distinct real solutions
E 5 distinct real solutions
- 65** The gradient of a straight line perpendicular to the line shown is:
- A** 2
B -2
C $-\frac{1}{2}$
D $\frac{1}{2}$
E 3
- 
- 66** The graph of a function f whose rule is $y = f(x)$ has exactly one asymptote for which the equation is $y = 6$. The inverse function f^{-1} exists. The inverse function will have:
- A** a horizontal asymptote with equation $y = 6$
B a vertical asymptote with equation $x = 6$
C a vertical asymptote with equation $x = \frac{-1}{6}$
D a horizontal asymptote with equation $y = -6$ **E** no asymptote
- 67** The function $f: R \rightarrow R$, $f(x) = a \sin(bx) + c$ where a, b and c are positive constants has period:
- A** a **B** b **C** c **D** $\frac{2\pi}{a}$ **E** $\frac{2\pi}{b}$
- 68** The functions $f: [18, 34] \rightarrow R$, $f(x) = 2x - 4$ and $g: R^+ \rightarrow R$, $g(x) = \log_2 x$ are used to define the composite function $g \circ f$. The range of $g \circ f$ is:
- A** $[2, \infty)$ **B** $\left[\frac{3}{2}, \infty\right)$ **C** $[5, 6]$ **D** R^+ **E** R
- 69** The rule for the inverse relation of the function with rule $y = x^2 - 4x + 5$ and domain R is:
- A** $y = 2 \pm \sqrt{x+1}$ **B** $y^2 = 2x + 5$ **C** $y = 2 \pm \sqrt{x-1}$
D $y = \sqrt{4x-5}$ **E** $y = 4x - 5$
- 70** The range of the function with rule $y = -3|\sin 2x| + 3$ is:
- A** $[0, 3]$ **B** $[0, 6]$ **C** $[-3, 3]$ **D** $[0, 6]$ **E** $[-3, 6]$

8.2 Extended-response questions

- 1 An arch is constructed as shown.

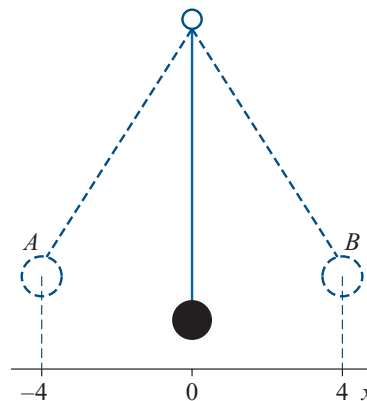


The height of the arch is 9 metres ($OZ = 9$ m). The width of the arch is 20 metres ($AB = 20$ m). The equation of the curve is of the form $y = ax^2 + b$, taking axes as shown.

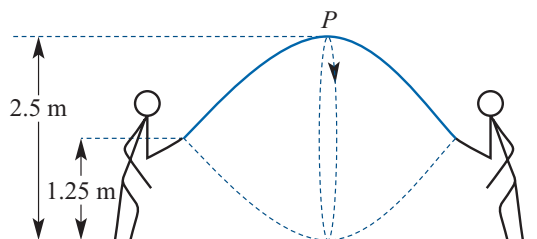
- a Find values of a and b .
 - b A man of height 1.8 m stands at C ($OC = 7$ m). How far above his head is the point E on the arch? (That is, find the distance DE .)
 - c A horizontal bar FG is placed across the arch as shown. The height, OH , of the bar above the ground is 6.3 m. Find the length of the bar.
- 2 a The expression $2x^3 + ax^2 - 72x - 18$ leaves a remainder of 17 when divided by $x + 5$. Determine the value of a .
- b Solve the equation: $2x^3 = x^2 + 5x + 2$
- c i Given that the expression $x^2 - 5x + 7$ leaves the same remainder whether divided by $x - b$ or $x - c$, where $b \neq c$, show that $b + c = 5$
- ii Given further that $4bc = 21$ and $b > c$, find the values of b and c .
- 3 As a pendulum swings, its horizontal position x , measured from the central position, varies from -4 cm (at A) to 4 cm (at B). x is given by the rule:

$$x = -4 \sin \pi t$$

- a Sketch the graph of x against t for $t \in [0, 2]$.
- b Find the horizontal position of the pendulum for:
 - i $t = 0$
 - ii $t = \frac{1}{2}$
 - iii $t = 1$
- c Find the first time that the pendulum has horizontal position $x = 2$.
- d Find the period of the pendulum, i.e. the time it takes to go from A to B and back to A .



- 4 Two people are rotating a skipping rope. The rope is held 1.25 m above the ground. It reaches a height of 2.5 m above the ground, and just touches the ground.



The vertical position, y m, of the point P on the rope at time t seconds is given by the rule:

$$y = -1.25 \cos(2\pi t) + 1.25$$

- a** Find y when:
- i** $t = 0$ **ii** $t = \frac{1}{2}$ **iii** $t = 1$
- b** How long does it take for one revolution of the rope?
- c** Sketch the graph of y against t .
- d** Find the first time that the point P on the rope is 2.00 metres above the ground.
- 5 The population of a country is found to be growing *continuously* at an annual rate of 2.96% after 1 January 1950. The population t years after 1 January 1950 is given by the formula:

$$p(t) = 150 \times 10^6 e^{kt}$$

- a** Find the value of k . **b** Find the population on 1 January 1950.
- c** Find the expected population on 1 January 2000.
- d** After how many years would the population be 300×10^6 ?
- 6 A football is kicked so that it leaves the player's foot with a velocity of V m/s. The horizontal distance travelled by the football after being kicked is given by the formula:

$$x = \frac{V^2 \sin 2\alpha}{10}$$

where α is the angle of projection.

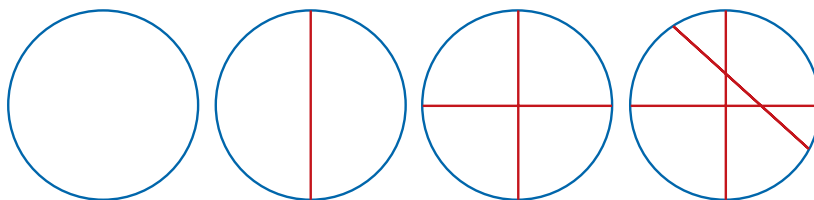
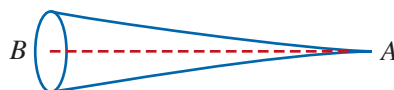
- a** Find the distance the ball is kicked if $V = 25$ m/s and $\alpha = 45^\circ$.
- b** For $V = 20$, sketch the graph of x against α for $0 \leq \alpha \leq 90^\circ$.
- c** If the ball goes 30 m and the initial velocity is 20 m/s, find the angle of projection.
- 7 A large urn was filled with water. It was turned on, and the water was heated until its temperature reached 95°C . This occurred at exactly 2:00 p.m., at which time the urn was turned off and the water began to cool. The temperature of the room where the urn was located remained constant at 15°C . Commencing at 2:00 p.m. and finishing at midnight, Jenny measured the temperature of the water every hour on the hour for the next 10 hours and recorded the results.

At 4:00 p.m. Jenny recorded the temperature of the water to be 55°C . She found that the temperature (T degrees Celsius) of the water could be described by the equation:

$$T = Ae^{-kt} + 15, \text{ where } 0 \leq t \leq 10$$

where t is the number of hours after 2:00 p.m.

- a** Find the values of A and k .
- b** Find the temperature at midnight.
- c** At what time did Jenny first record a temperature less than 24°C ?
- d** Sketch the graph of T against t .
- 8** On an overnight interstate train an electrical fault meant that the illumination in two carriages, A and B , was affected. Before the fault occurred the illumination in carriage A was I units and $0.66I$ units in carriage B . Every time the train stopped the illumination in carriage A reduced by 17% and by 11% in carriage B .
- a** Write down exponential expressions for the expected illumination in each carriage after the train had stopped for the n th time.
- b** At some time after the fault occurred the illumination in both carriages was approximately the same. At how many stations did the train stop before this occurred?
- 9** The diagram shows a conical glass fibre. The circular cross-sectional area at end B is 0.02 mm^2 . The cross-sectional area diminishes by a factor of $(0.92)^{\frac{1}{10}}$ per metre length of the fibre. The total length is 5 m.
- a** Write down a rule for the cross-sectional area of the fibre at a distance x m from B .
- b** What is the cross-sectional area of the fibre at a point one-third of its length from B ?
- c** The fibre is constructed in such a way that the strength increases in the direction B to A . At a distance x m from B the strength is given by the rule $S = (0.92)^{10-3x}$. If the load the fibre will take at each point before breaking is given by load = strength $\times$ cross-sectional area, write down an expression, in terms of x , for the load the fibre will stand at a distance x m from B .
- d** A piece of glass fibre that will have to carry loads of up to $0.02 \times (0.92)^{2.5}$ units is needed. How much of the 5 m fibre could be used with confidence for this purpose?
- 10** A pizza is divided by a number of straight cuts as shown. The table shows the largest number of pieces $f(n)$ into which it is divided by n cuts.



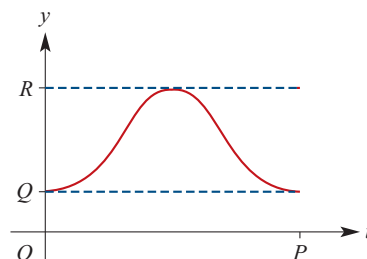
n	0	1	2	3	4	5
$f(n)$	1	2	4	7		

- a** Find a quadratic model for this data.
- b** Use your model to find the greatest number of pizza pieces produced by:
- i** 4 straight cuts **ii** 5 straight cuts
- c** Check your answers to **b** by drawing diagrams.

- 11 a The graph is of one complete cycle of:

$$y = h - k \cos\left(\frac{\pi t}{6}\right)$$

- i How many units long is OP ?
 ii Express OQ , OR in terms of h and k .



- b The number of hours of daylight on the 21st of each month in a city in the northern hemisphere is given by the table:

x	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
y	7.5	8.2	9.9	12.0	14.2	15.8	16.5	15.9	14.3	12.0	9.8	8.1	7.5

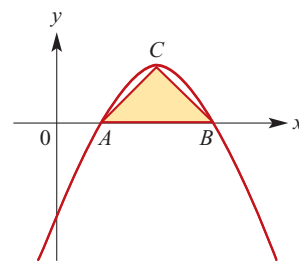
Using suitable scales, plot these points and draw a curve through them. Call December month 0, January month 1, etc., and treat all months as of equal length.

- c Find the values of h and k so that your graph is approximately that of:

$$y = h - k \cos\left(\frac{\pi t}{6}\right)$$

- 12 a In the figure $y = 1 - a(x - 3)^2$ intersects the x -axis at A and B . Point C is the vertex of the curve and a is a positive constant.

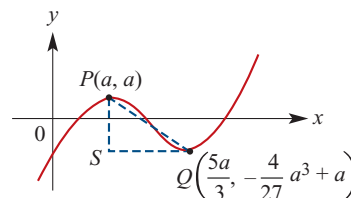
- i Find the coordinates of A and B in terms of a .
 ii Find the area of triangle ABC in terms of a .



- b The graph shown has rule:

$$y = (x - a)^2(x - 2a) + a \text{ where } a > 0$$

- i Use a calculator to sketch the graph for $a = 1, 2, 3$
 ii Find the values of a for which $\frac{-4}{27}a^3 + a = 0$
 iii Find the values of a for which $\frac{-4}{27}a^3 + a < 0$
 iv Find the value of a for which $\frac{-4}{27}a^3 + a = -1$
 v Find the value of a for which $\frac{-4}{27}a^3 + a = 1$
 vi Plot the graphs $y = (x - a)^2(x - 2a) + a$ for the values of a obtained in iv and v.
 c Triangle PSQ is a right-angled triangle.
 i Give the coordinates of S .
 ii Find the length of PS and SQ in terms of a .
 iii Give the area of triangle PSQ in terms of a .
 iv Find the value of a for which the area of the triangle is 4.
 v Find the value of a for which the area of the triangle is 1500.



Differentiation of polynomials, power functions and rational functions

Objectives

- To understand the concept of **limit**.
- To understand the **definition of differentiation**.
- To understand and use the notation for the **derivative** of a polynomial function.
- To find the **gradient** of a curve of a polynomial function by calculating its derivative.
- To differentiate functions having **negative integer powers**.
- To understand and use the **chain rule**.
- To differentiate **rational powers**.
- To understand and use the **product rule**.
- To understand and use the **quotient rule**.
- To deduce the **graph of the gradient function** from the graph of a function.

In this chapter, we review some of the important ideas and results that have been introduced in earlier studies in calculus. The chain rule, product rule and quotient rule are introduced in this chapter.

9.1 The gradient of a curve at a point

First, we develop a technique for calculating the rate of change in polynomial functions. To illustrate this, we consider this introductory example.

On Planet X , an object falls a distance of y metres in t seconds where $y = 0.8t^2$

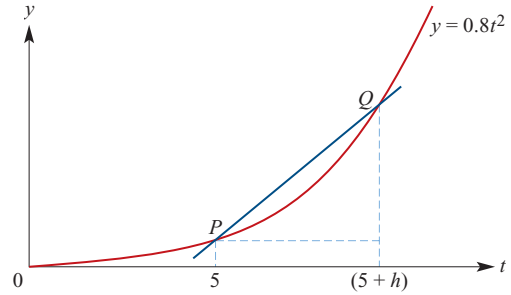
(**Note:** On Earth the commonly used model is $y = 4.9t^2$)

Can a general expression for the speed of such an object after t seconds be found?

The gradient of chord PQ approximates the speed of the object. The ‘shorter’ this chord is made, the better the approximation.

Let P be the point on the curve where $t = 5$.

Let Q be the point on the curve corresponding to h seconds after $t = 5$, i.e. Q is the point on the curve where $t = 5 + h$



$$\begin{aligned}\text{The gradient of chord } PQ &= \frac{0.8[(5+h)^2 - 5^2]}{(5+h) - 5} \\ &= \frac{0.8[(5+h)^2 - 5^2]}{h} \\ &= 0.8(10+h)\end{aligned}$$

The table gives the gradient for different values of h . Use your calculator to check these.

h	Gradient of PQ
0.7	8.56
0.6	8.48
0.5	8.40
0.4	8.32
0.3	8.24
0.2	8.16
0.1	8.08

If the values of h are taken to be of smaller and smaller magnitude, it is found that the gradient of chord PQ gets closer and closer to 8. The gradient at the point where $t = 5$ is 8.

Thus the speed of the object at the moment $t = 5$ is 8 m/s.

The speed of the object at the moment $t = 5$ is the limiting value of the gradient of PQ as Q approaches P .

Next, a formula for the speed of the object at any time t is needed. Let P be the point $(t, 0.8t^2)$ on the curve and Q be the point $(t+h, 0.8(t+h)^2)$.

$$\begin{aligned}\text{The gradient of chord } PQ &= \frac{0.8[(t+h)^2 - t^2]}{(t+h) - t} \\ &= 0.8(2t+h)\end{aligned}$$

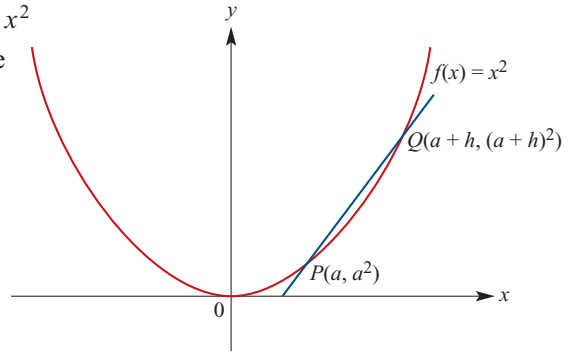
From this an expression for the speed can be found. Consider the limit as h approaches 0; that is, the value of $0.8(2t+h)$ as h becomes arbitrarily small.

The speed at time t is $1.6t$ metres per second. (The gradient of the curve at the point corresponding to time t is $1.6t$.) Now that a result that gives the speed of an object at any time t has been found, the gradient of similar functions can be investigated.

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$

The gradient of the chord PQ in the figure

$$\begin{aligned} &= \frac{(a+h)^2 - a^2}{a+h-a} \\ &= \frac{a^2 + 2ah + h^2 - a^2}{a+h-a} \\ &= 2a + h \end{aligned}$$



and the gradient at P can be seen to be $2a$. The limit as h approaches 0 of

$(2a + h)$ is $2a$. It can be seen that there is nothing special about a . So if x is a real number a similar formula holds.

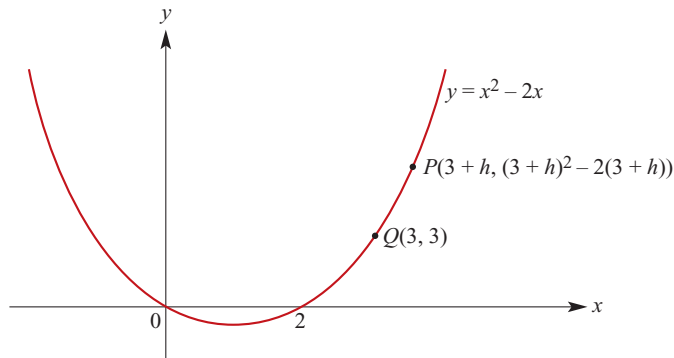
It is said that $2x$ is the **derivative** of x^2 **with respect to** x or, more briefly, the **derivative** of x^2 is $2x$.

The straight line that passes through P and that has gradient $2a$ is said to be the **tangent** to the curve at P .

From the discussion at the beginning of the chapter it was found that the derivative of $0.8t^2$ is $1.6t$.

Example 1

Find the gradient of $y = x^2 - 2x$ at the point Q with coordinates $(3, 3)$.



Solution

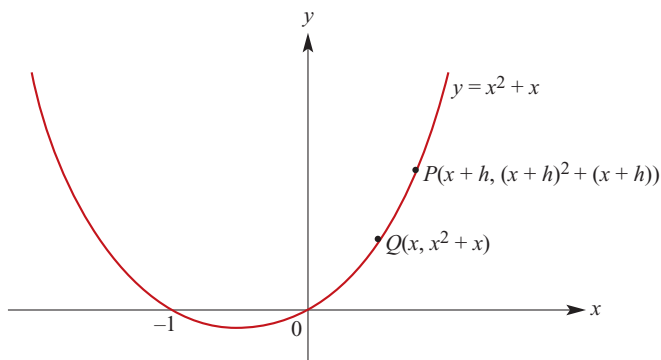
Consider chord PQ .

$$\begin{aligned} \text{Gradient of chord } PQ &= \frac{(3+h)^2 - 2(3+h) - 3}{3+h-3} \\ &= \frac{9 + 6h + h^2 - 6 - 2h - 3}{3+h-3} \\ &= \frac{4h + h^2}{h} \\ &= 4 + h \end{aligned}$$

From this it can be deduced that the gradient at the point $(3, 3)$ is 4.

Example 2

Find the derivative of $x^2 + x$ with respect to x .

**Solution**

$$\begin{aligned}
 \text{The gradient of chord } PQ &= \frac{(x+h)^2 + (x+h) - (x^2 + x)}{x+h-x} \\
 &= \frac{x^2 + 2xh + h^2 + x + h - (x^2 + x)}{h} \\
 &= \frac{2xh + h^2 + h}{h} \\
 &= 2x + h + 1
 \end{aligned}$$

From this it can be seen that the derivative of $x^2 + x$ is $2x + 1$

The notation for limit as h approaches 0 of $2x + h + 1$ is $\lim_{h \rightarrow 0} 2x + h + 1$

Example 3

Find $\lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{(x+h) - x}$

Solution

$$\begin{aligned}
 \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{(x+h) - x} &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3h^2x + h^3 - x^3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3x^2h + 3h^2x + h^3}{h} \\
 &= \lim_{h \rightarrow 0} 3x^2 + 3hx + h^2 \\
 &= 3x^2
 \end{aligned}$$

Note: It has been found that the derivative of x^3 is $3x^2$.

Example 4

Find:

a $\lim_{h \rightarrow 0} 22x^2 + 20xh + h$

b $\lim_{h \rightarrow 0} \frac{3x^2h + 2h^2}{h}$

c $\lim_{h \rightarrow 0} 3x$

d $\lim_{h \rightarrow 0} 4$

Solution

a $\lim_{h \rightarrow 0} 22x^2 + 20xh + h = 22x^2$

b $\lim_{h \rightarrow 0} \frac{3x^2h + 2h^2}{h} = \lim_{h \rightarrow 0} 3x^2 + 2h = 3x^2$

c $\lim_{h \rightarrow 0} 3x = 3x$

d $\lim_{h \rightarrow 0} 4 = 4$

Exercise 9A

1 For a curve with equation $y = x^2 + 5x$:

- a Find the gradient of the chord PQ where P is the point $(2, 14)$ and Q is the point $(2 + h, (2 + h)^2 + 5(2 + h))$.
- b Find the gradient of PQ where $h = 0.1$.
- c From the result of a find the gradient of the curve at the point $(2, 14)$.

2 Find:

- | | |
|---|--|
| a $\lim_{h \rightarrow 0} \frac{4x^2h^2 + xh + h}{h}$ | b $\lim_{h \rightarrow 0} \frac{2x^3h - 2xh^2 + h}{h}$ |
| c $\lim_{h \rightarrow 0} 40 - 50h$ | d $\lim_{h \rightarrow 0} 5h$ |
| e $\lim_{h \rightarrow 0} 5$ | f $\lim_{h \rightarrow 0} \frac{30h^2x^2 + 20h^2x + h}{h}$ |
| g $\lim_{h \rightarrow 0} \frac{3h^2x^3 + 2hx + h}{h}$ | h $\lim_{h \rightarrow 0} 3x$ |
| i $\lim_{h \rightarrow 0} \frac{3x^3h - 5x^2h^2 + hx}{h}$ | j $\lim_{h \rightarrow 0} 6x - 7h$ |

3 For the curve with equation $y = x^3 - x$:

- a Find the gradient of the chord PQ where P is the point $(1, 0)$ and Q is the point $((1 + h), (1 + h)^3 - (1 + h))$.
- b From the result of a find the gradient of the curve at the point $(1, 0)$.

4 If $f(x) = x^2 - 2$, simplify $\frac{f(x + h) - f(x)}{h}$. Hence find the derivative of $x^2 - 2$.

5 Let P and Q be points on the curve $y = x^2 + 2x + 5$ at which $x = 2$ and $x = 2 + h$ respectively. Express the gradient of the line PQ in terms of h and hence find the gradient of the curve $y = x^2 + 2x + 5$ at $x = 2$.

9.2

The derived function



In this section, we develop a notation for the derivative of a function.

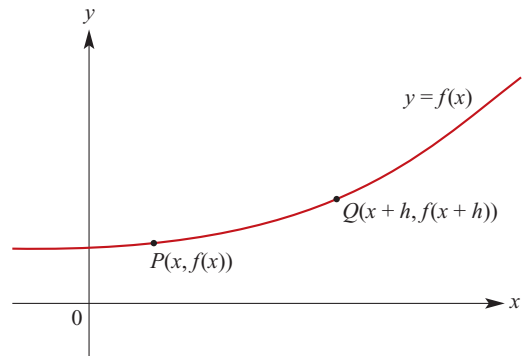
Consider the graph $y = f(x)$ of the function $f: R \rightarrow R$.

The gradient of the chord PQ

$$\begin{aligned}
 &= \frac{f(x + h) - f(x)}{x + h - x} \\
 &= \frac{f(x + h) - f(x)}{h}
 \end{aligned}$$

Therefore, the gradient of the graph at P is given by:

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$



This limit gives a rule for a new function called the **derived function** (or **derivative function**) and is denoted by f' where:

$$f': R \rightarrow R \text{ and } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The function f is said to be **differentiable** at a point $(a, f(a))$ if $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.

Example 5

Find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ for each of the following:

a $f(x) = 3x^2 + 2x + 2$ **b** $f(x) = x^3 + 2$

Solution

$$\begin{aligned} \text{a } \frac{f(x+h) - f(x)}{h} &= \frac{3(x+h)^2 + 2(x+h) + 2 - (3x^2 + 2x + 2)}{h} \\ &= \frac{3(x^2 + 2xh + h^2) + 2(x+h) + 2 - (3x^2 + 2x + 2)}{h} \\ &= \frac{3x^2 + 6xh + 3h^2 + 2x + 2h + 2 - 3x^2 - 2x - 2}{h} \\ &= \frac{6xh + 3h^2 + 2h}{h} = 6x + 3h + 2 \end{aligned}$$

$$\text{Therefore, } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 6x + 3h + 2 = 6x + 2$$

$$\begin{aligned} \text{b } \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^3 + 2 - (x^3 + 2)}{h} \\ &= \frac{x^3 + 3x^2h + 3xh^2 + h^3 + 2 - x^3 - 2}{h} \\ &= \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \end{aligned}$$

$$\text{Therefore, } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 = 3x^2$$

Using the TI-Nspire

Find the derivative of $f(x) = 3x^2 + 2x + 2$ by first principles.

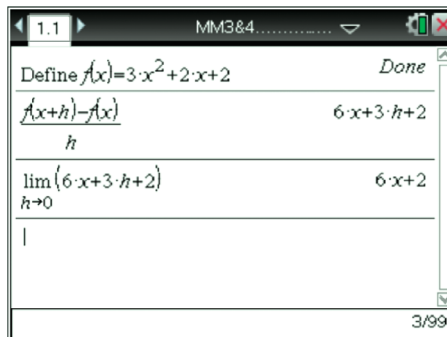
Firstly, **define** $f(x) = 3x^2 + 2x + 2$

Then calculate the gradient of the chord,

$$\frac{f(x+h) - f(x)}{h}$$

Finally, use $\left(\text{menu}\right) > \text{Calculus} > \text{Limit}$ and complete as shown to find the limit.

The limit template can also be obtained through the 2D templates, $\left(\text{2D template}\right)$ (or $\left(\text{ctrl}\right) + \left(\text{2D template}\right)$ on the Clickpad)

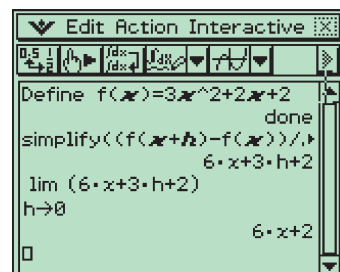


Using the Casio ClassPad

Find the derivative of

$f(x) = 3x^2 + 2x + 2$ by first principles.

Define $f(x)$ using **Interactive – Define** and then enter and highlight $(f(x+h) - f(x))/h$ and tap **Interactive – Transformation – simplify** to find the expression. Copy and paste the answer to a new line, highlight it, and then tap **Interactive – Calculation – lim**.



Finding $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ for a given function is often referred to as finding the derivative of $f(x)$ with respect to x by **first principles**.

In this chapter only polynomial and rational functions are considered. For polynomial functions the derived function always exists and is defined for every number in the domain of f .

In Section 9.10, we discuss the existence of the derivative.

The following can be deduced from the work of the previous section:

- For $f(x) = x^2$, $f'(x) = 2x$
- For $f(x) = x^3$, $f'(x) = 3x^2$
- For $f(x) = x^4$, $f'(x) = 4x^3$
- For $f(x) = 1$, $f'(x) = 0$

This gives the following general result:

For $f(x) = x^n$, $f'(x) = nx^{n-1}$, $n = 1, 2, 3, \dots$

For $f(x) = 1$, $f'(x) = 0$

From the previous section it can be seen that for k , a constant:

- If $f(x) = kx^n$, then $f'(x) = knx^{n-1}$

If $g(x) = kf(x)$, where k is a constant, then $g'(x) = kf'(x)$

That is, **the derivative of a number multiple is the multiple of the derivative**.

For example, for $g(x) = 5x^2$, the derived function $g'(x) = 5(2x) = 10x$

Another important rule for differentiation is:

If $f(x) = g(x) + h(x)$, then $f'(x) = g'(x) + h'(x)$

That is, **the derivative of the sum is the sum of the derivatives**.

For example, for $f(x) = x^2 + 2x$ the derived function $f'(x) = 2x + 2$

Example 6

Find the derivative of $x^5 - 2x^3 + 2$

Solution

If $f(x) = x^5 - 2x^3 + 2$ then:

$$\begin{aligned} f'(x) &= 5x^4 - 2(3x^2) + 2(0) \\ &= 5x^4 - 6x^2 \end{aligned}$$

Example 7

Find the derivative of $f(x) = 3x^3 - 6x^2 + 1$ and $f'(1)$

Solution

$$\begin{aligned} f'(x) &= 3(3x^2) - 6(2x) + 1(0) \\ &= 9x^2 - 12x \\ f'(1) &= 9 - 12 \\ &= -3 \end{aligned}$$

Example 8

Find the gradient of the curve determined by the rule $f(x) = 3x^3 - 6x^2 + 1$ at the point $(1, -2)$.

Solution

Now $f'(x) = 9x^2 - 12x$ and $f'(1) = 9 - 12 = -3$

The gradient of the curve is -3 at the point $(1, -2)$.

An alternative notation for the derivative is the following:

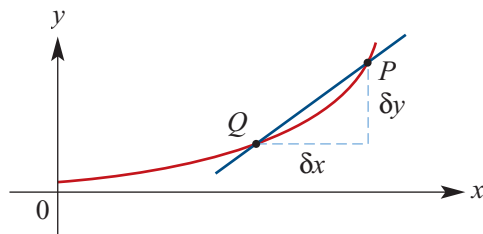
■ If $y = x^3$, then the derivative can be denoted by $\frac{dy}{dx}$, so that $\frac{dy}{dx} = 3x^2$

In general, if y is a function of x , the derivative of y with respect to x is denoted $\frac{dy}{dx}$ and, with the use of a different symbol z , where z is a function of t , the derivative of z with respect to t is $\frac{dz}{dt}$.

In this notation d is not a factor and cannot be cancelled. This came about because in the eighteenth century the standard diagram for finding the limiting gradient was labelled as in the figure shown. (' δ ' is the lower case Greek letter for 'd', and is pronounced *delta*.)

' δx ' means a difference in x .

' δy ' means a difference in y .



For example:

- If $x = t^3 + t$, then $\frac{dx}{dt} = 3t^2 + 1$
- If $y = t^2$, then $\frac{dy}{dt} = 2t$
- If $y = \frac{1}{3}x^3 + x^2$, then $\frac{dy}{dx} = x^2 + 2x$

Example 9

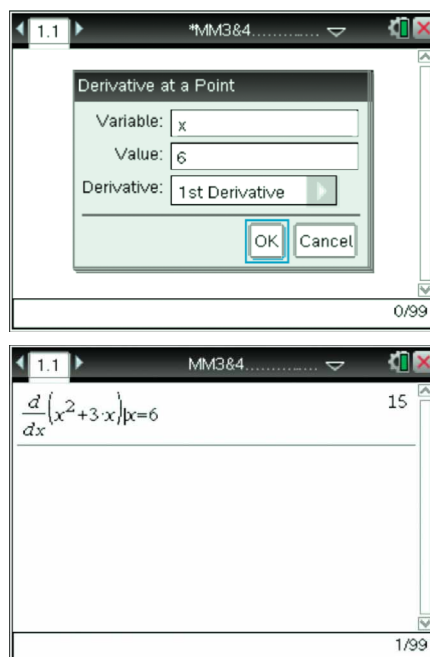
Use a CAS calculator to find the gradient of $f(x) = x^2 + 3x$ at the point on the graph where $x = 6$.

Solution

Using the TI-Nspire

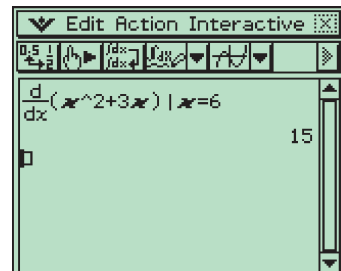
Find the derivative and evaluate for $x = 6$.

Use **menu** > **Calculus** > **Derivative at a Point** and complete as shown.



Using the Casio ClassPad

Press the keyboard button and select the **2D** **CALC** menu screen. Tap $\frac{d}{dx}$ and **2D**. Enter the derivative with respect to x **2D** as shown. The $|$ symbol is found in **math** **OPTN** menu screen.



Example 10

The planned path for a flying saucer leaving a planet is defined by the equation:

$$y = \frac{1}{4}x^4 + \frac{2}{3}x^3 \text{ for } x > 0$$

The units are kilometres. (The x -axis is horizontal, and the y -axis vertical.)

- a** What will be the direction of motion when the x -value is:
i 2? **ii** 3?
- b** Find a point on the path where the flying saucer's path is inclined at 45° to the positive x -axis.
- c** Are there any other points on the path that satisfy the situation described in **b**?

Solution

a $\frac{dy}{dx} = x^3 + 2x^2$

i When $x = 2$, $\frac{dy}{dx} = 8 + 8$
 $= 16$
 $(\tan^{-1} 16)^\circ = 86.42^\circ$ (to the x -axis)

ii When $x = 3$, $\frac{dy}{dx} = 27 + 18$
 $= 45$
 $(\tan^{-1} 45)^\circ = 88.73^\circ$ (to the x -axis)

- b, c** When the flying saucer is flying at 45° to the direction of the x -axis, the gradient of the curve of its path is given by $\tan 45^\circ$.

Thus to find the point at which this happens we consider the equation:

$$\begin{aligned}\frac{dy}{dx} &= \tan 45^\circ \\ \therefore x^3 + 2x^2 &= 1 \\ \therefore x^3 + 2x^2 - 1 &= 0 \\ \therefore (x+1)(x^2+x-1) &= 0 \\ \therefore x &= -1 \text{ or } x = \frac{-1 \pm \sqrt{5}}{2}\end{aligned}$$

The only acceptable solution is $x = \frac{-1 + \sqrt{5}}{2}$ ($x = 0.62$), as the other two possibilities give negative values for x and we are only considering positive values for x .

Example 11

- a** For $y = (x+3)^2$, find $\frac{dy}{dx}$. **b** For $z = (2t-1)^2(t+2)$, find $\frac{dz}{dt}$.
- c** For $y = \frac{x^2+3x}{x}$, find $\frac{dy}{dx}$.

Solution

- a** It is first necessary to write $y = (x + 3)^2$ in expanded form.

$$\therefore y = x^2 + 6x + 9$$

$$\text{and } \frac{dy}{dx} = 2x + 6$$

- b** First expanding:

$$\begin{aligned} z &= (4t^2 - 4t + 1)(t + 2) \\ &= 4t^3 - 4t^2 + t + 8t^2 - 8t + 2 \\ &= 4t^3 + 4t^2 - 7t + 2 \\ \text{and } \frac{dz}{dt} &= 12t^2 + 8t - 7 \end{aligned}$$

- c** First dividing by x :

$$y = x + 3$$

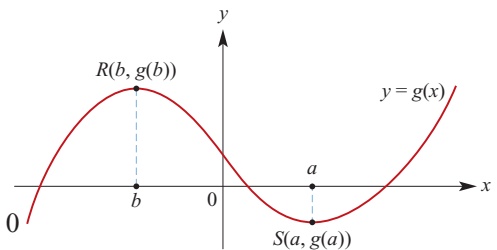
$$\therefore \frac{dy}{dx} = 1$$

Gradients can, of course, be negative or zero. They are not always positive.

At a point $(a, g(a))$ of the graph $y = g(x)$ the gradient is $g'(a)$.

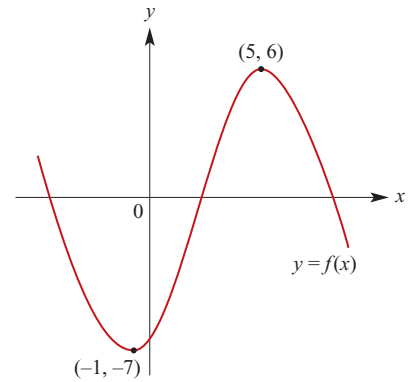
Some features of the graph are:

- For $x < b$ the gradient is positive, i.e. $g'(x) > 0$
- For $x = b$ the gradient is zero, i.e. $g'(b) = 0$
- For $b < x < a$ the gradient is negative, i.e. $g'(x) < 0$
- For $x = a$ the gradient is zero, i.e. $g'(a) = 0$
- For $x > a$ the gradient is positive, i.e. $g'(x) > 0$

**Example 12**

For the graph of $f: \mathbb{R} \rightarrow \mathbb{R}$ find:

- a** $\{x: f'(x) > 0\}$
b $\{x: f'(x) < 0\}$
c $\{x: f'(x) = 0\}$

**Solution**

- a** $\{x: f'(x) > 0\} = \{x: -1 < x < 5\} = (-1, 5)$
b $\{x: f'(x) < 0\} = \{x: x < -1\} \cup \{x: x > 5\} = (-\infty, -1) \cup (5, \infty)$
c $\{x: f'(x) = 0\} = \{-1, 5\}$

Exercise 9B

- 1** In each of the following, find $f'(x)$ by finding $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$:

- a** $f(x) = 5x^2$ **b** $f(x) = 3x + 2$ **c** $f(x) = 5$
d $f(x) = 3x^2 + 4x + 3$ **e** $f(x) = 5x^3 - 5$ **f** $f(x) = 5x^2 - 6x$

2 For each of the following, find $f'(x)$:

a $f(x) = x^5$

b $f(x) = 4x^7$

c $f(x) = 6x$

d $f(x) = 5x^2 - 4x + 3$

e $f(x) = 4x^3 + 6x^2 + 2x - 4$

f $f(x) = 5x^4 + 3x^3$

g $f(x) = -2x^2 + 4x + 6$

h $f(x) = 6x^3 - 2x^2 + 4x - 6$

3 For each of the following, find $\frac{dy}{dx}$:

a $y = -2x$

b $y = 7$

c $y = 5x^3 - 3x^2 + 2x + 1$

d $y = \frac{2}{5}(x^3 - 4x + 6)$

e $y = (2x + 1)(x - 3)$

f $y = 3x(2x - 4)$

g $y = \frac{10x^7 + 2x^2}{x^2}, x \neq 0$

h $y = \frac{9x^4 + 3x^2}{x}, x \neq 0$

4 **a** Differentiate $y = (2x - 1)^2$ with respect to x .

b For $y = \frac{x^3 + 2x^2}{x}, x \neq 0$, find $\frac{dy}{dx}$

c Given that $y = 2x^3 - 6x^2 + 18x$, find $\frac{dy}{dx}$

Hence show that $\frac{dy}{dx} > 0$ for all x .

d Given that $y = \frac{x^3}{3} - x^2 + x$, find $\frac{dy}{dx}$. Hence show that $\frac{dy}{dx} \geq 0$ for all x .

5 At the points on the following curves corresponding to given values of x , find the y -coordinate and the gradient:

a $y = x^2 + 2x + 1, x = 3$

b $y = x^2 - x - 1, x = 0$

c $y = 2x^2 - 4x, x = -1$

d $y = (2x + 1)(3x - 1)(x + 2), x = 4$

e $y = (2x + 5)(3 - 5x)(x + 1), x = 1$

f $y = (2x - 5)^2, x = 2\frac{1}{2}$

6 For the function, $f(x) = 3(x - 1)^2$, find the value(s) of x for which:

a $f(x) = 0$

b $f'(x) = 0$

c $f'(x) > 0$

d $f'(x) < 0$

e $f'(x) = 10$

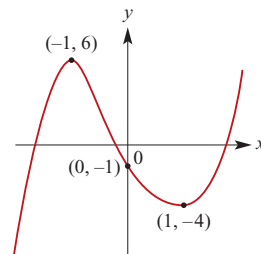
f $f(x) = 27$

7 For the graph of $y = h(x)$ illustrated, find:

a $\{x: h'(x) > 0\}$

b $\{x: h'(x) < 0\}$

c $\{x: h'(x) = 0\}$

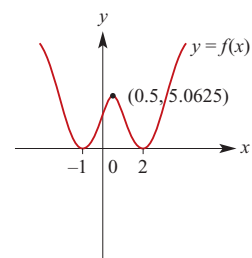


8 For the graph of $y = f(x)$ shown, find:

a $\{x: f'(x) > 0\}$

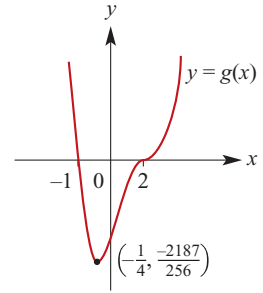
b $\{x: f'(x) < 0\}$

c $\{x: f'(x) = 0\}$



9 For the graph of $y = g(x)$ shown, find:

- a $\{x: g'(x) > 0\}$
- b $\{x: g'(x) < 0\}$
- c $\{x: g'(x) = 0\}$



10 Find the coordinates of the parabola $y = x^2 - 2x - 8$ at which:

- a the gradient is zero
- b the tangent is parallel to $y = 2x + 6$
- c the tangent is parallel to $3x + 2y = 8$

11 Find the coordinates of the points on the curves given by the following equations at which the gradient has the given values:

- a $y = 2x^2 - 4x + 1$; gradient = -6
- b $y = 4x^3$; gradient = 48
- c $y = x(5 - x)$; gradient = 1
- d $y = x^3 - 3x^2$; gradient = 0

9.3 Differentiating x^n where n is a negative integer

In this section, we add new functions to the family of functions for which we can find the derived functions. In particular, we consider functions that involve linear combinations of powers of x where a power may be a negative integer, for example:

- $f: R \setminus \{0\} \rightarrow R, f(x) = x^{-1}$
- $f: R \setminus \{0\} \rightarrow R, f(x) = 2x + x^{-1}$
- $f: R \setminus \{0\} \rightarrow R, f(x) = x + 3 + x^{-2}$

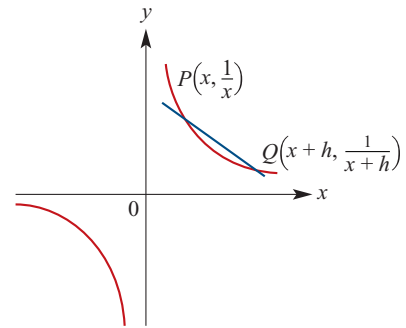
Note that we are reintroducing the function notation. This notation emphasises the need for consideration of domain.

Example 13

Let $f: R \setminus \{0\} \rightarrow R, f(x) = \frac{1}{x}$. Find $f'(x)$ using the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Solution

$$\begin{aligned}
 \text{Gradient of chord } PQ &= \frac{f(x+h) - f(x)}{x+h-x} \\
 &= \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\
 &= \frac{x - (x+h)}{(x+h)x} \times \frac{1}{h} \\
 &= \frac{-h}{(x+h)x} \times \frac{1}{h} \\
 &= \frac{-1}{(x+h)x}
 \end{aligned}$$

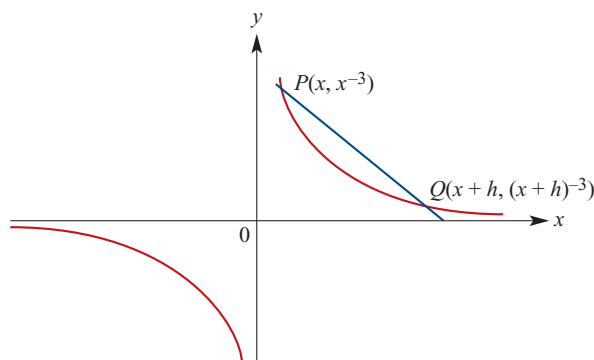


The gradient of the curve at $P = \lim_{h \rightarrow 0} \frac{-1}{(x+h)x} = \frac{-1}{x^2} = -x^{-2}$ and $f'(x) = -x^{-2}$

Example 14

Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = x^{-3}$. Find $f'(x)$ using the definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Solution

$$\begin{aligned} \text{Gradient of chord } PQ &= \frac{(x+h)^{-3} - x^{-3}}{h} \\ &= \frac{x^3 - (x+h)^3}{(x+h)^3 x^3} \times \frac{1}{h} \\ &= \frac{x^3 - (x^3 + 3x^2h + 3xh^2 + h^3)}{(x+h)^3 x^3} \times \frac{1}{h} \\ &= \frac{-3x^2h - 3xh^2 - h^3}{(x+h)^3 x^3} \times \frac{1}{h} \\ &= \frac{-3x^2 - 3xh - h^2}{(x+h)^3 x^3} \end{aligned}$$

$$\begin{aligned} \text{The gradient of the curve at } P &= \lim_{h \rightarrow 0} \frac{-3x^2 - 3xh - h^2}{(x+h)^3 x^3} \\ &= \frac{-3x^2}{x^6} \\ &= \frac{-3}{x^4} \\ &= -3x^{-4} \text{ and } f'(x) = -3x^{-4} \end{aligned}$$

The generalisation of the result found in Section 9.2 can now be stated:

For $f(x) = x^n$, $f'(x) = nx^{n-1}$, n a non-zero integer.

For $f(x) = 1$, $f'(x) = 0$

Note that for $n \leq -1$, the domain of f can be taken to be $\mathbb{R} \setminus \{0\}$ and for $n \geq 1$ we take the domain of f to be $\mathbb{R}$.

Example 15

Find the derivative of $x^4 - 2x^{-3} + x^{-1} + 2$, $x \neq 0$

Solution

$$\begin{aligned}\text{If } f(x) &= x^4 - 2x^{-3} + x^{-1} + 2, x \neq 0 \\ f'(x) &= 4x^3 - 2(-3x^{-4}) + (-x^{-2}) + 2(0) \\ &= 4x^3 + 6x^{-4} - x^{-2}, x \neq 0\end{aligned}$$

Example 16

Find the derivative f' of $f: R \setminus \{0\} \rightarrow R$, $f(x) = 3x^2 - 6x^{-2} + 1$

Solution

$$\begin{aligned}f': R \setminus \{0\} \rightarrow R, f'(x) &= 3(2x) - 6(-2x^{-3}) + 1(0) \\ &= 6x + 12x^{-3}\end{aligned}$$

Example 17

Find the gradient of the curve determined by the function $f: R \setminus \{0\}$, $f(x) = x^2 + \frac{1}{x}$ at the point $(1, 2)$.

Solution

$$\begin{aligned}f': R \setminus \{0\} \rightarrow R, f'(x) &= 2x + (-x^{-2}) \\ &= 2x - x^{-2} \\ f'(1) &= 2 - 1 \\ &= 1\end{aligned}$$

The gradient of the curve is 1 at the point $(1, 2)$.

Exercise 9C

1 Differentiate each of the following with respect to x :

$$\begin{array}{lll}\text{a } 3x^{-2} + 5x^{-1} + 6 & \text{b } \frac{5}{x^3} + 6x^2 & \text{c } \frac{-5}{x^3} + \frac{4}{x^2} + 1 \\ \text{d } 6x^{-3} + 3x^{-2} & \text{e } \frac{4x^2 + 2x}{x^2}\end{array}$$

2 Find the derivative of each of the following:

$$\begin{array}{lll}\text{a } \frac{2z^2 - 4z}{z^2}, z \neq 0 & \text{b } \frac{6 + z}{z^3}, z \neq 0 & \text{c } 16 - z^{-3}, z \neq 0 \\ \text{d } \frac{4z + z^3 - z^4}{z^2}, z \neq 0 & \text{e } \frac{6z^2 - 2z}{z^4}, z \neq 0 & \text{f } \frac{6}{x} - 3x^2, x \neq 0\end{array}$$

3 Find the x -coordinates of the points on the curve $y = \frac{x^2 - 1}{x}$ at which the gradient of the curve is 5.

4 Given that the curve $y = ax^2 + \frac{b}{x}$ has a gradient of -5 at the point $(2, -2)$, find the value of a and b .

- 5 Find the gradient of the curve $y = \frac{2x-4}{x^2}$ at the point where the curve crosses the x -axis.
- 6 The gradient of the curve $y = \frac{a}{x} + bx^2$ at the point $(3, 6)$ is 7. Calculate the values of a and b .
- 7 For the curve with equation $y = \frac{5}{3}x + kx^2 - \frac{8}{9}x^3$, calculate the possible values of k such that the tangents at the points with x -coordinates 1 and $-\frac{1}{2}$ respectively are perpendicular.
- 8 Find the gradient of each of the following curves at the stated point:
- a $y = x^{-2} + x^3, x \neq 0$ at the point $\left(2, 8\frac{1}{4}\right)$
 - b $y = x^{-2} - \frac{1}{x}, x \neq 0$ at the point $\left(4, \frac{1}{2}\right)$
 - c $y = x^{-2} - \frac{1}{x}, x \neq 0$, at the point $(1, 0)$
 - d $y = x(x^{-1} + x^2 - x^{-3}), x \neq 0$ at the point $(1, 1)$
- 9 a Sketch the graph of $f: R \setminus \{0\} \rightarrow R, f(x) = \frac{2}{x^2}$
- b Let P be the point $(1, 2)$ and Q the point $(1+h, f(1+h))$. Find the gradient of chord PQ .
- c Hence find the gradient of the curve $f(x) = \frac{2}{x^2}$ at $(1, 2)$.

9.4 The chain rule

An expression such as $q(x) = (x^3 + 1)^2$ may be differentiated by expanding and then differentiating each term separately. This method is a great deal more tiresome for an expression such as $q(x) = (x^3 + 1)^{30}$

We express $q(x) = (x^3 + 1)^2$ as the composition of two simpler functions defined by:

$$g(x) = x^3 + 1 (= u) \text{ and } f(u) = u^2 (= y)$$

which are ‘chained’ together as:

$$x \xrightarrow{g} u \xrightarrow{f} y$$

That is, $q(x) = (x^3 + 1)^2$ is expressed as the composition of two functions f and g .

Thus $q = f \circ g$ and $q(x) = f(g(x))$

The chain rule gives a method of differentiating such functions. The chain rule states:

If $q(x) = f(g(x))$, and g is differentiable at x , and f is differentiable at $g(x)$, then the derivative of q exists and

$$q'(x) = f'(g(x))g'(x)$$

Or in the notation of Leibniz, where as above $u = g(x) = x^3 + 1$ and $y = u^2$,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Proof of the chain rule

For the derivative of $f \circ g$ where $x = a$, consider the gradient of the secant:

$$\frac{f \circ g(a+h) - f \circ g(a)}{h} = \frac{f(g(a+h)) - f(g(a))}{h} \\ \times \frac{g(a+h) - g(a)}{g(a+h) - g(a)}, \text{ provided } g(a+h) - g(a) \neq 0$$

Now write $b = g(a)$ and $b+k = g(a+h)$ so that $k = g(a+h) - g(a)$

The expression becomes:

$$\frac{f(b+k) - f(b)}{k} \times \frac{g(a+h) - g(a)}{h}$$

Since g is differentiable at a , g is also continuous at a and so:

$$\lim_{h \rightarrow 0} k = \lim_{h \rightarrow 0} [g(a+h) - g(a)] = 0$$

Thus as h tends to 0 so does k .

And thus $q'(x) = f'(g(x))g'(x)$.

Note that this proof does not hold for a function g such that $g(a+h) - g(a) = 0$ for arbitrarily chosen small h .

Example 18

Differentiate $y = (4x^3 - 5x)^{-2}$

Solution

The differentiation is undertaken using both notations.

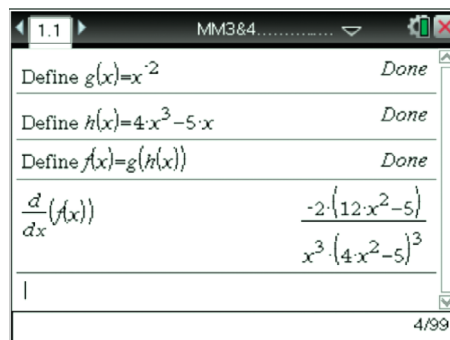
Let $u = 4x^3 - 5x$ Then $y = u^{-2}$ We have $\frac{dy}{du} = -2u^{-3}$ and $\frac{du}{dx} = 12x^2 - 5$ $\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ $= -2u^{-3} \cdot (12x^2 - 5)$ $= \frac{-2(12x^2 - 5)}{(4x^3 - 5x)^3}$	Let $f(x) = g(h(x))$ where $h(x) = 4x^3 - 5x$ and $g(x) = x^{-2}$ Then $f'(x) = g'(h(x))h'(x)$, $g'(x) = -2x^{-3}$ and $h'(x) = 12x^2 - 5$ Then $g'(h(x)) = -2(h(x))^{-3} = -2(4x^3 - 5x)^{-3}$ Therefore $f'(x) = -2(4x^3 - 5x)^{-3} \times (12x^2 - 5)$ $= \frac{-2(12x^2 - 5)}{(4x^3 - 5x)^3}$
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Using the TI-Nspire

Define $g(x)$ and $h(x)$

Define $f(x) = g(h(x))$

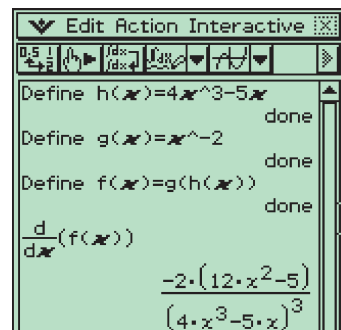
Use $\text{menu} > \text{Calculus} > \text{Derivative}$ and complete as shown.



Using the Casio ClassPad

Define $g(x)$ and $h(x)$, then define

$f(x) = g(h(x))$ and take the derivative of $f(x)$.



Example 19

Find the gradient of the curve with equation $y = \frac{16}{3x^2 + 1}$ at the point (1, 4).

Solution

Let $u = 3x^2 + 1$. Then $y = 16u^{-1}$

$$\text{So } \frac{du}{dx} = 6x \quad \text{and} \quad \frac{dy}{dx} = -16u^{-2}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= -16u^{-2} \cdot 6x \\ &= \frac{-96x}{(3x^2 + 1)^2} \end{aligned}$$

$$\therefore \text{at } x = 1 \text{ the gradient is } \frac{-96}{16} = -6$$

Example 20

Find the derivative of $y = (3x + 4)^{20}$

Solution

Let $u = 3x + 4$. Then $y = u^{20}$

$$\text{So } \frac{du}{dx} = 3 \quad \text{and} \quad \frac{dy}{du} = 20u^{19}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= 20u^{19} \cdot 3 \\ &= 60(3x + 4)^{19} \end{aligned}$$

Example 21

Use the chain rule to prove $\frac{dy}{dx} = nx^{n-1}$ for $y = x^n$ where n is a negative integer.
(Assume the result for n a positive integer and $x > 0$.)

Solution

Let n be a negative integer and $y = x^n$. Then $y = \frac{1}{x^{-n}}$ and $-n$ is a positive integer.

Let $u = x^{-n}$. Then $y = \frac{1}{u} = u^{-1}$

Thus $\frac{dy}{du} = -u^{-2}$ and $\frac{du}{dx} = -nx^{-n-1}$ ($-n$ is a positive integer)

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= -u^{-2} \cdot (-nx^{-n-1}) \\ &= nx^{-n-1}(x^{-n})^{-2} \\ &= nx^{-n-1}(x^{2n}) \\ &= nx^{n-1} \end{aligned}$$

Example 22

Given that $f(x) = (x^2 + 1)^3$, find $f'(x)$.

Solution

Now $f = k \circ g$ where $k(x) = x^3$ and $g(x) = x^2 + 1$

It follows that $k'(x) = 3x^2$ and $g'(x) = 2x$

Since by the chain rule $f'(x) = k'(g(x))g'(x)$ we have in this case

$$f'(x) = 3(g(x))^2 \cdot 2x$$

$$\text{which yields } f'(x) = 6x(x^2 + 1)^2$$

Exercise 9D

1 Differentiate each of the following with respect to x :

a $(x^2 + 1)^4$ **b** $(2x^2 - 3)^5$ **c** $(6x + 1)^4$ **d** $(ax + b)^n$
e $(ax^2 + b)^n$ **f** $(1 - x^2)^{-3}$ **g** $\left(x^2 - \frac{1}{x^2}\right)^{-3}$ **h** $(1 - x)^{-1}$

2 Differentiate each of the following with respect to x :

a $(x^2 + 2x + 1)^3$ **b** $(x^3 + 2x^2 + x)^4$ **c** $\left(6x^3 + \frac{2}{x}\right)^4$ **d** $(x^2 + 2x + 1)^{-2}$

3 Differentiate each of the following with respect to x , giving the answer in terms of $f(x)$ and $f'(x)$:

a $[f(x)]^n$, where n is a positive integer **b** $\frac{1}{f(x)}$ where $f(x) \neq 0$

9.5 Differentiating rational powers $\left(x^{\frac{p}{q}}\right)$

Using the chain rule in the form $\frac{dy}{du} = \frac{dy}{dx} \cdot \frac{dx}{du}$

with $y = u$ we have $1 = \frac{dy}{dx} \cdot \frac{dx}{dy}$

and thus

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}, \text{ for } \frac{dx}{dy} \neq 0$$

Let $y = x^{\frac{1}{n}}$ where $n \in \mathbb{Z} \setminus \{0\}$ and $x > 0$

Then $y^n = x$ and $\frac{dx}{dy} = ny^{n-1}$

From the above results we have

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{ny^{n-1}} \\ &= \frac{1}{n \left(x^{\frac{1}{n}}\right)^{n-1}} \\ &= \frac{1}{n} x^{\frac{1}{n}-1} \end{aligned}$$

For $y = x^{\frac{1}{n}}$, $\frac{dy}{dx} = \frac{1}{n} x^{\frac{1}{n}-1}$; $n \in \mathbb{Z} \setminus \{0\}$ and $x > 0$

If n is odd then the result is also true for $x < 0$, but for both odd and even n the result does not hold when $x = 0$.

This result may now be extended to any rational power.

Let $y = x^{\frac{p}{q}}$, where $p, q \in \mathbb{Z} \setminus \{0\}$.

We write $y = \left(x^{\frac{1}{q}}\right)^p$. Let $u = x^{\frac{1}{q}}$. Then $y = u^p$

$$\begin{aligned} \text{The chain rule yields } \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= pu^{p-1} \cdot \frac{1}{q}x^{\frac{1}{q}-1} \\ &= p\left(x^{\frac{1}{q}}\right)^{p-1} \cdot \frac{1}{q}x^{\frac{1}{q}-1} \\ &= \frac{p}{q}x^{\frac{p}{q}-\frac{1}{q}x^{\frac{1}{q}-1}} \\ &= \frac{p}{q}x^{\frac{p}{q}-1} \end{aligned}$$

If q is odd then the result is also true for $x < 0$, but for both odd and even q the result does not hold when $x = 0$.

Thus we have the result for any non-zero rational power and, in fact, it is true for any non-zero real power:

$$\text{for } f(x) = x^a, f'(x) = ax^{a-1}, \text{ for } x > 0 \text{ and } a \in \mathbb{R}$$

Example 23

Find the derivative of each of the following with respect to x :

a $2x^{-\frac{1}{5}} + 3x^{\frac{2}{7}}$ **b** $\sqrt[3]{x^2 + 2x}$

Solution

$$\begin{aligned} \text{a } \frac{d\left(2x^{-\frac{1}{5}} + 3x^{\frac{2}{7}}\right)}{dx} &= 2 \times \frac{-1}{5}x^{-\frac{6}{5}} + 3 \times \frac{2}{7}x^{-\frac{5}{7}} \\ &= -\frac{2}{5}x^{-\frac{6}{5}} + \frac{6}{7}x^{-\frac{5}{7}} \\ \text{b } \frac{d(\sqrt[3]{x^2 + 2x})}{dx} &= \frac{d\left((x^2 + 2x)^{\frac{1}{3}}\right)}{dx} \\ &= (x^2 + 2) \times \frac{1}{3}(x^2 + 2x)^{-\frac{2}{3}} \quad (\text{chain rule}) \\ &= \frac{2x + 2}{3\sqrt[3]{(x^2 + 2x)^2}} \end{aligned}$$

Exercise 9E

- 1 Find the derivative of each of the following with respect to x :

a $x^{\frac{1}{5}}$	b $x^{\frac{5}{2}}$	c $x^{\frac{5}{2}} - x^{\frac{3}{2}}, x > 0$
d $3x^{\frac{1}{2}} - 4x^{\frac{5}{3}}$	e $x^{-\frac{6}{7}}$	f $x^{-\frac{1}{4}} + 4x^{\frac{1}{2}}$
- 2 Find the gradient of each of the following at the stated value for x :

a $f(x) = x^{\frac{1}{3}}$ where $x = 27$	b $f(x) = x^{\frac{1}{3}}$ where $x = -8$
c $f(x) = x^{\frac{2}{3}}$ where $x = 27$	d $f(x) = x^{\frac{5}{4}}$ when $x = 16$
- 3 Find the derivative of each of the following with respect to x :

a $\sqrt{2x+1}$	b $\sqrt{4-3x}$	c $\sqrt{x^2+2}$
d $\sqrt[3]{4-3x}$	e $\frac{x^2+2}{\sqrt{x}}$	f $3\sqrt{x}(x^2+2x)$
- 4 **a** Show that $\frac{d}{dx}(\sqrt{x^2 \pm a^2}) = \frac{x}{\sqrt{x^2 \pm a^2}}$
b Show that $\frac{d}{dx}(\sqrt{a^2 - x^2}) = \frac{-x}{\sqrt{a^2 - x^2}}$
- 5 If $y = (x + \sqrt{x^2 + 1})^2$, show $\frac{dy}{dx} = \frac{2y}{\sqrt{x^2 + 1}}$
- 6 Find the derivative with respect to x of each of the following:

a $\sqrt{x^2 + 2}$	b $\sqrt[3]{x^3 - 5x}$	c $\sqrt[5]{x^2 + 2x}$
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9.6 Product rule

In the next two sections, we introduce two more rules for differentiation. The first of these is the **product rule**.

Let $F(x) = f(x) \cdot g(x)$

If $f'(x)$ and $g'(x)$ exist, then $F'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$

For example, if $F(x) = (x^2 + 3x)(4x + 5)$, F can be considered as the product of two functions f and g where $f(x) = x^2 + 3x$ and $g(x) = 4x + 5$

The product rule gives:

$$\begin{aligned}
 F'(x) &= f(x) \cdot g'(x) + g(x) \cdot f'(x) \\
 &= (x^2 + 3x) \cdot 4 + (4x + 5) \cdot (2x + 3) \\
 &= 4x^2 + 12x + 8x^2 + 22x + 15 \\
 &= 12x^2 + 34x + 15
 \end{aligned}$$

This could also have been achieved by first multiplying $x^2 + 3x$ by $4x + 5$ and then differentiating.

Proof of the product rule

By the definition of the derivative of F , we have:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h} \end{aligned}$$

Adding and subtracting $f(x+h) \cdot g(x)$:

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x) + [f(x+h) \cdot g(x) - f(x+h) \cdot g(x)]}{h}$$

Regrouping:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{[f(x+h) \cdot g(x+h) - f(x+h) \cdot g(x) + f(x+h) \cdot g(x) - f(x) \cdot g(x)]}{h} \\ F'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)[g(x+h) - g(x)] + g(x)[f(x+h) - f(x)]}{h} \\ &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] \end{aligned}$$

and, since f and g are differentiable:

$$F'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

The product rule is restated:

If $F(x) = f(x) \cdot g(x)$, then

$$F'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

The product rule may also be stated in Leibniz notation and a geometric interpretation is given.

If $y = uv$ where u and v are functions of x

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

δv	$u \delta v$	$\delta u \delta v$
v	uv	$v \delta u$
	u	δu

In the figure above, $y = uv$, δy is the shaded area, as explained below.

$$\begin{aligned} \therefore \delta y &= (u + \delta u)(v + \delta v) - uv \\ &= uv + v \delta u + u \delta v + \delta u \delta v - uv \\ &= v \delta u + u \delta v + \delta u \delta v \\ \therefore \frac{\delta y}{\delta x} &= v \frac{\delta u}{\delta x} + u \frac{\delta v}{\delta x} + \frac{\delta u}{\delta x} \frac{\delta v}{\delta x} \delta x \end{aligned}$$

When $\delta x \rightarrow 0$,

$$\frac{\delta v}{\delta x} = \frac{dv}{dx}, \quad \frac{\delta u}{\delta x} = \frac{du}{dx} \quad \text{and} \quad \frac{\delta y}{\delta x} = \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

Example 24

Differentiate each of the following with respect to x :

a $(2x^2 + 1)(5x^3 + 16)$ **b** $x^3(3x - 5)^4$

Solution

a Let $y = (2x^2 + 1)(5x^3 + 16)$

Let $u = 2x^2 + 1$ and $v = 5x^3 + 16$

Then $\frac{du}{dx} = 4x$ and $\frac{dv}{dx} = 15x^2$

The product rule gives:

$$\begin{aligned} \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= (2x^2 + 1) \cdot 15x^2 + (5x^3 + 16) \cdot 4x \\ &= 30x^4 + 15x^2 + 20x^4 + 64x \\ &= 50x^4 + 15x^2 + 64x \end{aligned}$$

b Let $y = x^3(3x - 5)^4$

Let $u = x^3$ and $v = (3x - 5)^4$

Then $\frac{du}{dx} = 3x^2$ and $\frac{dv}{dx} = 12(3x - 5)^3$ (using the chain rule).

The product rule gives:

$$\begin{aligned} \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= 12x^3(3x - 5)^3 + (3x - 5)^4 \cdot 3x^2 \\ &= (3x - 5)^3 [12x^3 + 3x^2(3x - 5)] \\ &= (3x - 5)^3 [12x^3 + 9x^3 - 15x^2] \\ &= (3x - 5)^3 (21x^3 - 15x^2) \\ &= 3x^2(7x - 5)(3x - 5)^3 \end{aligned}$$

Example 25

For $F: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $F(x) = x^{-3}(10x^2 - 5)^3$, find $F'(x)$.

Solution

Let $f(x) = x^{-3}$ and $g(x) = (10x^2 - 5)^3$

Then $f'(x) = -3x^{-4}$ and $g'(x) = 60x(10x^2 - 5)^2$ (chain rule)

$$\begin{aligned}\therefore F'(x) &= x^{-3} \times 60x(10x^2 - 5)^2 + (10x^2 - 5)^3 \cdot -3x^{-4} \\ &= (10x^2 - 5)^2 [60x^{-2} + (10x^2 - 5) \cdot -3x^{-4}] \\ &= (10x^2 - 5)^2 \left[\frac{60x^2 - 30x^2 + 15}{x^4} \right] \\ &= \frac{(10x^2 - 5)^2(30x^2 + 15)}{x^4}\end{aligned}$$

Exercise 9F

Find the derivative of each of the following with respect to x , using the product rule:

- | | | |
|-----------------------------|------------------------------------|----------------------------|
| 1 $(2x^2 + 6)(2x^3 + 1)$ | 2 $3x^{\frac{1}{2}}(2x + 1)$ | 3 $3x(2x - 1)^3$ |
| 4 $4x^2(2x^2 + 1)^2$ | 5 $(3x + 1)^{\frac{3}{2}}(2x + 4)$ | 6 $(x^2 + 1)\sqrt{2x - 4}$ |
| 7 $x^3(3x^2 + 2x + 1)^{-1}$ | 8 $x^4\sqrt{2x^2 - 1}$ | 9 $x^2\sqrt[3]{x^2 + 2x}$ |

9.7 Quotient rule

Let $F(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$. If $f'(x)$ and $g'(x)$ exist then:

$$F'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

For example, if $F(x) = \frac{x^3 + 2x}{x^5 + 2}$, we see that F can be considered as a quotient of two functions f and g where $f(x) = x^3 + 2x$ and $g(x) = x^5 + 2$

The quotient rule gives:

$$\begin{aligned}F'(x) &= \frac{(x^5 + 2)(3x^2 + 2) - (x^3 + 2x)5x^4}{(x^5 + 2)^2} \\ &= \frac{3x^7 + 6x^2 + 2x^5 + 4 - 5x^7 - 10x^5}{(x^5 + 2)^2} \\ &= \frac{-2x^7 - 8x^5 + 6x^2 + 4}{(x^5 + 2)^2}\end{aligned}$$

Proof of the quotient rule

By the definition of derivative of F , we have:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \right] \times \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x+h)}{g(x+h)g(x)} \times \frac{1}{h} \end{aligned}$$

Adding and subtracting $f(x) \cdot g(x)$ in the numerator:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x+h) + f(x) \cdot g(x) - f(x) \cdot g(x)}{g(x+h) \cdot g(x)} \times \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{g(x)[f(x+h) - f(x)]}{h} - \frac{f(x)[g(x+h) - g(x)]}{h} \right) \times \frac{1}{g(x+h)g(x)} \\ &= \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2} \end{aligned}$$

The quotient rule may be also stated in Leibniz notation:

If $y = \frac{u}{v}$, $v \neq 0$ where u and v are functions of x

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Example 26

Find the derivative of $\frac{x-2}{x^2+4x+1}$ with respect to x .

Solution

$$\text{Let } y = \frac{x-2}{x^2+4x+1}$$

The quotient rule gives:

$$\begin{aligned} \frac{dy}{dx} &= \frac{x^2+4x+1 - (2x+4)(x-2)}{(x^2+4x+1)^2} \\ &= \frac{x^2+4x+1 - (2x^2-8)}{(x^2+4x+1)^2} \\ &= \frac{-x^2+4x+9}{(x^2+4x+1)^2} \end{aligned}$$

Exercise 9G



1 Find the derivative of each of the following with respect to x :

a $\frac{x}{x+4}$

b $\frac{x^2 - 1}{x^2 + 1}$

c $\frac{x^{\frac{1}{2}}}{1+x}$

d $\frac{(x+2)^3}{x^2+1}$

e $\frac{x-1}{x^2+2}$

f $\frac{x^2+1}{x^2-1}$

g $\frac{3x^2+2x+1}{x^2+x+1}$

h $\frac{2x+1}{2x^3+2x}$

2 Find the y -coordinate and the gradient at points on each of the following curves corresponding to the given value of x :

a $y = (2x+1)^4x^2$; $x = 1$

b $y = x^2\sqrt{x+1}$; $x = 0$

c $y = x^2(2x+1)^{\frac{1}{2}}$; $x = 0$

d $y = \frac{x}{x^2+1}$; $x = 1$

e $y = \frac{2x+1}{x^2+1}$; $x = 1$

3 For each of the following, find $f'(x)$:

a $f(x) = (x+1)\sqrt{x^2+1}$

b $f(x) = (x^2+1)\sqrt{x^3+1}$, $x > -1$

c $f(x) = \frac{2x+1}{x+3}$

9.8 The graph of the gradient function

The graph of the derivative function is also considered in Section 9.10 where differentiability is considered.

Consider the quadratic function with rule $y = f(x)$.

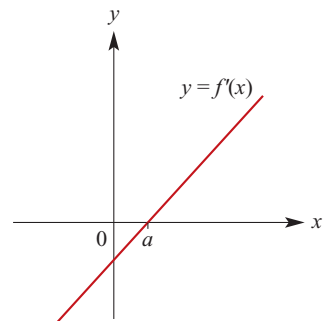
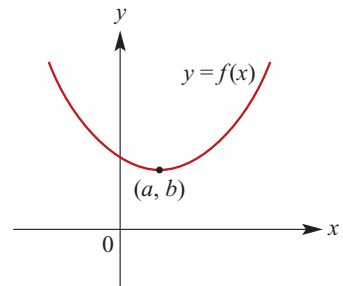
The vertex is at the point with coordinates (a, b) .

For $x < a$, $f'(x) < 0$

For $x = a$, $f'(x) = 0$

For $x > a$, $f'(x) > 0$

The graph of the derivative function with rule $y = f'(x)$ is therefore as shown to the right. The derivative function f' is linear as f is quadratic.



Consider the cubic function with rule $y = g(x)$.

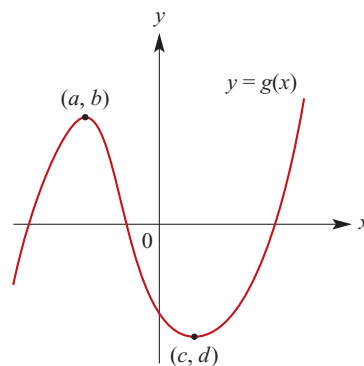
For $x < a$, $f'(x) > 0$

For $x = a$, $f'(x) = 0$

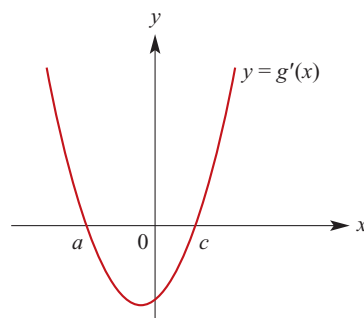
For $a < x < c$, $f'(x) < 0$

For $x = c$, $f'(x) = 0$

For $x > c$, $f'(x) > 0$

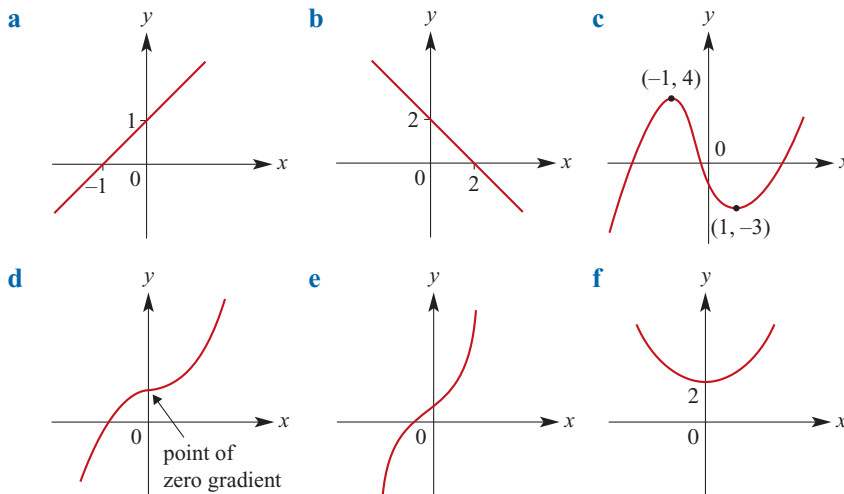


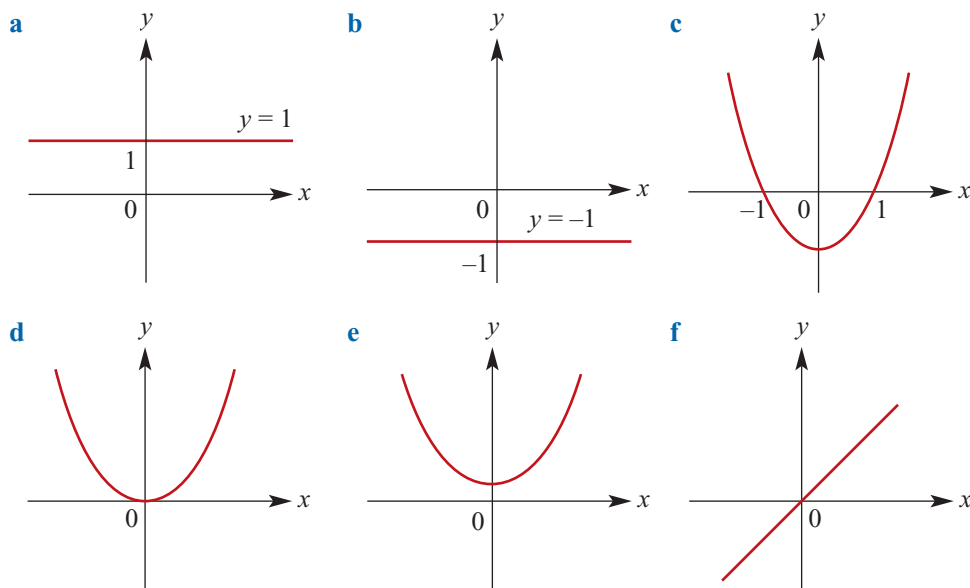
The graph of the derivative function with rule $y = g'(x)$ is therefore as shown to the right. The derivative is known to be quadratic as g is cubic.



Example 27

Sketch the graph of the derivative function for each of the functions of the graphs shown:



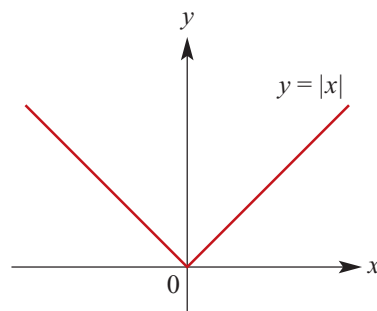
Solution**Note:** Not all features of the graphs are known.

In Section 9.10, differentiability of a function is considered formally. Consider the absolute value function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Consider the gradient of a chord connecting the points $(0, 0)$ and $(h, f(h))$ on the graph of $f(x) = |x|$

$$\begin{aligned} \frac{f(0+h) - f(0)}{h} &= \begin{cases} \frac{h}{h} & h > 0 \\ \frac{-h}{h} & h < 0 \end{cases} \\ &= \begin{cases} 1 & h > 0 \\ -1 & h < 0 \end{cases} \end{aligned}$$



As the value for the gradient is not unique, we say $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ does not exist, i.e., f is not differentiable at $x = 0$.

Note: The gradient to the left of 0 is -1 and to the right of 0 the gradient is 1 . The idea of left and right limits is further explored in Section 9.9.

Example 28

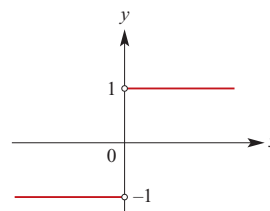
Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$

Sketch the graph of the derivative for a suitable domain.

Solution

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

$f'(x)$ is not defined at $x = 0$.

**Using the TI-Nspire**

The derivative of $|x|$ yields the function **sign**(x).

This gives the result of -1 if $x < 0$ and 1 if $x > 0$.

(Somewhat confusingly it returns ± 1 if $x = 0$)

Note that the function **sign**() can be obtained using **(menu) > Number > Number Tools > Sign**, but it may also be typed directly using the letters on the keyboard.

Open a **Graphs** application

(ctrl) + [I] > Graphs) and let

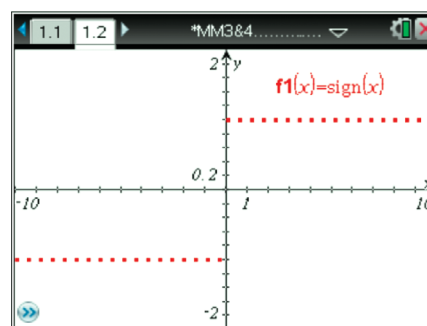
$f1(x) = \text{sign}(x)$.

Select an appropriate **window setting**

(menu) > Window/Zoom > Window Settings).

The graph of **sign**(x) is given with the attributes set to a dotted line.

Expression	Result
$\frac{d}{dx}(x)$	sign (x)
sign (-8)	-1
sign (2)	1
sign (0)	±1



The VCAA has chosen the definition below for the sign or signum function:

$$\text{sgn}(x) = \begin{cases} 1 & \text{when } x > 0 \\ 0 & \text{when } x = 0 \\ -1 & \text{when } x < 0 \end{cases}$$

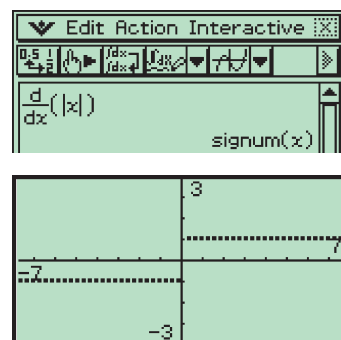
This is different from both the definition used in the TI-Nspire given above and that used in the Casio ClassPad. Students should take care with their use of this function and adhere to the VCAA definition for MM 3&4 (CAS) assessments. In the answers to this book, the corresponding hybrid function is always used.

Using the Casio ClassPad

The derivative of $|x|$ yields the function **signum**(x).

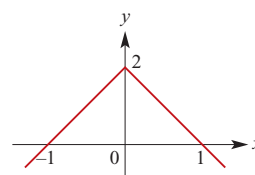
This gives a result of -1 if $x < 0$ and 1 if $x > 0$. (Somewhat confusingly, it returns ± 1 if $x = 0$.)

The graph of **signum**(x) is given with the graph mode set to [.....].



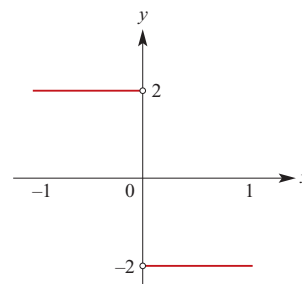
Example 29

Draw a sketch graph of f' where the graph of f is as illustrated. Indicate where f' is not defined.



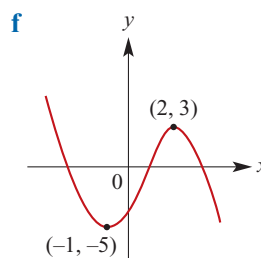
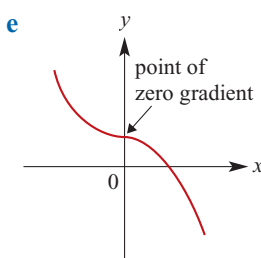
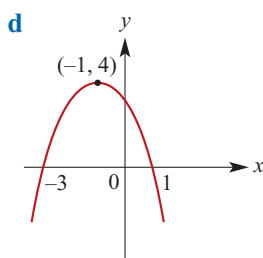
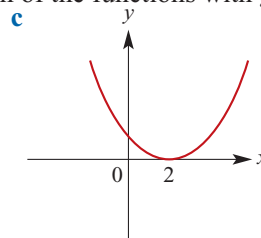
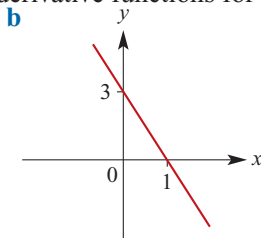
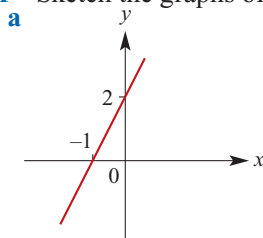
Solution

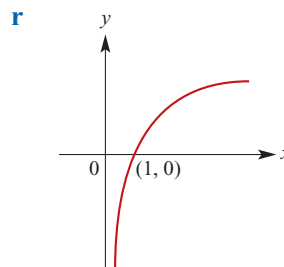
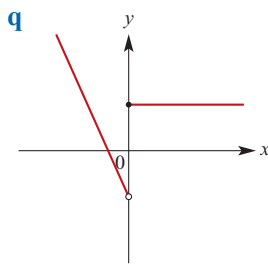
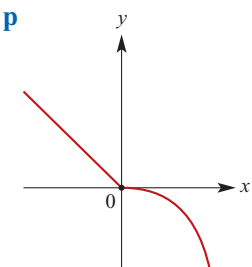
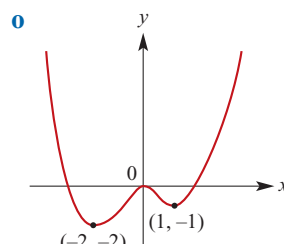
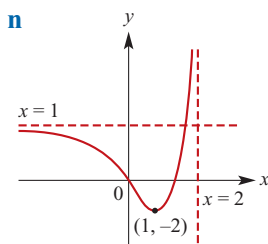
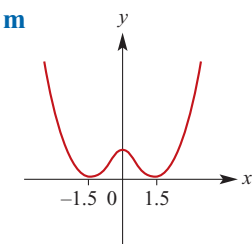
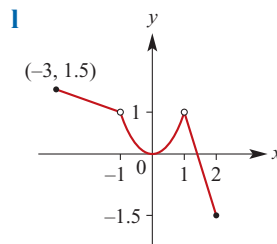
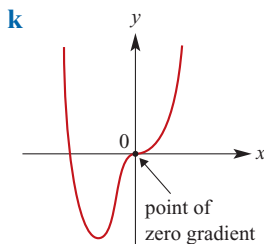
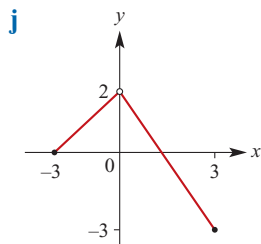
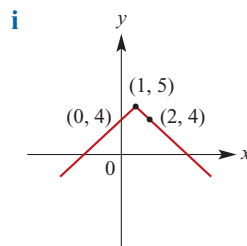
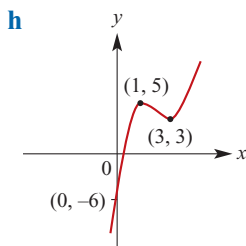
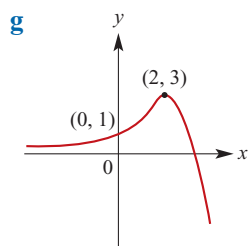
The derivative does not exist at $x = 0$, i.e. the function is not differentiable at $x = 0$.



Exercise 9H

1 Sketch the graphs of the derivative functions for each of the functions with graphs shown:





- 2 a** Use a calculator to plot the graph of $y = f(x)$ where $f(x) = (x^2 - 2x)^2$
- b** Using the same screen plot the graph of $y = f'(x)$. (Do not attempt to determine the rule for $f'(x)$ first.)
- c** Use a calculator to determine $f'(x)$ for:
- i** $x = 0$ **ii** $x = 2$ **iii** $x = 1$ **iv** $x = 4$
- d** For $0 \leq x \leq 1$ find the value of x for which:
- i** $f(x)$ is a maximum **ii** $f'(x)$ is a maximum
- 3** On the same screen plot the graphs of $y = f(x)$ and $y = f'(x)$ where $f(x) = \frac{x^3}{3} - x^2 + x + 1$. Comment.
- 4** On the same screen plot the graphs of $y = g(x)$ and $y = g'(x)$ where $g(x) = x^3 + 2x + 1$. Comment.

- 5 **a** On the same screen plot the graphs of $y = h(x)$ and $y = h'(x)$ where $h(x) = x^4 + 2x + 1$
- b** Find the value(s) of x for which:
- i** $h(x) = 3$ **ii** $h'(x) = 3$
- 6 Plot the graph of $y = f(x)$ and $y = f'(x)$ on the one set of axes where $f(x) = 3x^3 + 4x$
- a** Find the minimum gradient of the graph of $y = f(x)$
- b** Find the values of x for which $f'(x) = 6$
- 7 Consider the function $g(x) = \frac{f(2+x) - f(2)}{x}$ where $-0.5 < x < 0.5$ and $x \neq 0$
- a** Plot the graph of $g(x)$ when:
- i** $f(x) = x^3$ **ii** $f(x) = 4x^2$ **iii** $f(x) = x^4$
- b** Consider the table of values with the increment below. Start at $x = -0.5$
- i** 0.1 **ii** 0.05

9.9 Review of limits and continuity*

Limits

The limit of a function with rule $f(x)$ is said to be the value that $f(x)$ approaches as x approaches a given value. $\lim_{x \rightarrow a} f(x) = p$ means that as x approaches a , $f(x)$ approaches p . An important idea is that it is possible to get as close as desired to p as x approaches a .

Note that $f(x)$ may or may not be defined at $x = a$.

With many functions $f(a)$ is defined, so to evaluate the limit we simply substitute the value a into the rule for the function.

Example 30

If $f(x) = -4x^2$, find $\lim_{x \rightarrow 2} (-4x^2)$

Solution

Since $f(x) = -4x^2$ is defined at $x = 2$:

$$\begin{aligned} \lim_{x \rightarrow 2} (-4x^2) &= -4(2)^2 \\ &= -16 \end{aligned}$$

If the function is not defined at the value where the limit is to be found, the procedure is different.

* This topic is included in *Essential Mathematical Methods 1 & 2 CAS*.

Example 31

For $f(x) = \frac{x^2 - 3x + 2}{x - 2}$, $x \neq 2$, find $\lim_{x \rightarrow 2} f(x)$.

Solution

Observe that $f(x)$ is defined for $x \in \mathbb{R} \setminus \{2\}$.

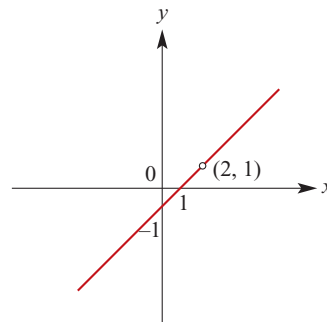
It is apparent that as x takes values closer and closer to 2, regardless of whether x approaches 2 from the left or from the right, the values of $f(x)$ become closer and closer to 1.

That is, $\lim_{x \rightarrow 2} f(x) = 1$

This can be seen by observing:

$$\begin{aligned} f(x) &= \frac{(x-1)(x-2)}{x-2} \\ &= x-1, \quad x \neq 2 \end{aligned}$$

The graph of $f: \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R}$, $f(x) = x - 1$ is shown.



The following are important results that are useful for the evaluation of limits:

- $\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$
i.e., the limit of the sum is the sum of the limits.
- $\lim_{x \rightarrow c} (kf(x)) = k \lim_{x \rightarrow c} f(x)$, k being a given number (non-zero)
- $\lim_{x \rightarrow c} (f(x)g(x)) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x)$
i.e., the limit of the product is the product of the limits.
- $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided $\lim_{x \rightarrow c} g(x) \neq 0$
i.e., the limit of the quotient is the quotient of the limits.

Example 32

Find:

a $\lim_{x \rightarrow 0} x^2 + 2$

b $\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x - 3}$

c $\lim_{x \rightarrow 2} \frac{(x^2 - x - 2)}{x - 2}$

d $\lim_{x \rightarrow 3} (2x + 1)(3x - 2)$

e $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 10}{x^2 - 25}$

Solution

$$\text{a } \lim_{x \rightarrow 0} (x^2 + 2) = \lim_{x \rightarrow 0} x^2 + \lim_{x \rightarrow 0} 2 = 0 + 2 = 2$$

$$\text{b } \lim_{x \rightarrow 3} \frac{x^2 - 3x}{x - 3} = \lim_{x \rightarrow 3} \frac{x(x - 3)}{x - 3} = \lim_{x \rightarrow 3} x = 3$$

$$\text{c } \lim_{x \rightarrow 2} \frac{(x^2 - x - 2)}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

$$\text{d } \lim_{x \rightarrow 3} (2x + 1)(3x - 2) = \lim_{x \rightarrow 3} (2x + 1) \lim_{x \rightarrow 3} (3x - 2) = 7 \times 7 = 49$$

$$\text{e } \lim_{x \rightarrow 3} \frac{x^2 - 7x + 10}{x^2 - 25} = \lim_{x \rightarrow 3} \frac{(x - 2)(x - 5)}{(x + 5)(x - 5)} = \lim_{x \rightarrow 3} \frac{(x - 2)}{(x + 5)} = \frac{1}{8}$$

Limit notation

The notation of limits is used to describe the behaviour of graphs.

Consider $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x^2}$. Observe that as $x \rightarrow 0$, both from the left and the right, $f(x)$ increases without bound. The limit notation for this is $\lim_{x \rightarrow 0} f(x) = \infty$

For $g: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $g(x) = \frac{1}{x}$, the behaviour of $g(x)$ as x approaches 0 from the left is different from the behaviour as x approaches 0 from the right.

With limit notation this is written as:

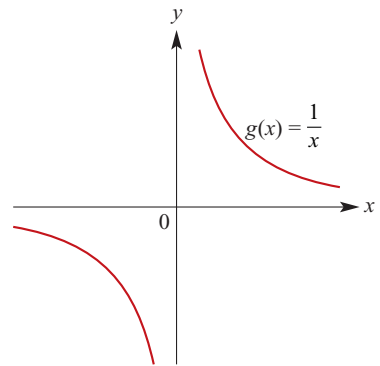
$$\lim_{x \rightarrow 0^-} g(x) = -\infty \text{ and } \lim_{x \rightarrow 0^+} g(x) = \infty$$

Now examine this function as the magnitude of x becomes very large. It can be seen that as x increases without bound through positive values, the corresponding values of $g(x)$ approach zero. Likewise as x decreases without bound through negative values, the corresponding values of $f(x)$ also approach zero.

Symbolically this is written as:

$$\lim_{x \rightarrow \infty} g(x) = 0^+ \text{ and } \lim_{x \rightarrow -\infty} g(x) = 0^-$$

Many functions approach a limiting value, or limit, as x approaches $\pm\infty$.

**Left and right limits**

An idea that is useful in the following discussion is the existence of limits from the left and right.

If the value of $f(x)$ approaches the number p as x approaches a from the right-hand side, then it is written as $\lim_{x \rightarrow a^+} f(x) = p$ and, if the value of $f(x)$ approaches the number p as x approaches a from the left-hand side, it is written as $\lim_{x \rightarrow a^-} f(x) = p$

The limit as x approaches a exists only if the limit from the left and right both exist and are equal. Then:

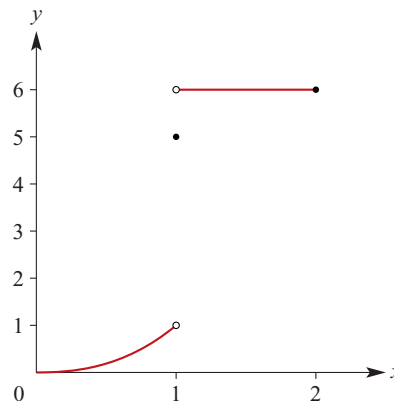
$$\lim_{x \rightarrow a} f(x) = p$$

The following is an example of the limit not existing for a particular value.

$$\text{Let } f(x) = \begin{cases} x^3 & \text{for } 0 \leq x < 1 \\ 5 & \text{for } x = 1 \\ 6 & \text{for } 1 < x \leq 2 \end{cases}$$

It is clear from the graph of f that $\lim_{x \rightarrow 1} f(x)$ does not exist. However, if x is allowed to approach 1 from the left, then $f(x)$ approaches 1. On the other hand, if x is allowed to approach 1 from the right, then $f(x)$ approaches 6.

Also $f(1) = 5$.



Continuity at a point — informal definition

A function with rule $f(x)$ is said to be continuous when $x = a$ if the graph of $y = f(x)$ can be drawn through the point with coordinates $(a, f(a))$ without a break. Otherwise there is said to be a discontinuity at $x = a$.

A more formal definition of continuity is:

A function f is continuous at a point a if $f(a)$, $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ all exist and are equal.

Or, equivalently:

A function f is continuous at the point $x = a$ if the following three conditions are met:

- 1 $f(x)$ is defined at $x = a$
- 2 $\lim_{x \rightarrow a} f(x)$ exists
- 3 $\lim_{x \rightarrow a} f(x) = f(a)$

A function is said to be discontinuous at a point if it is not continuous at that point. We say that a function is continuous everywhere if it is continuous for all real numbers.

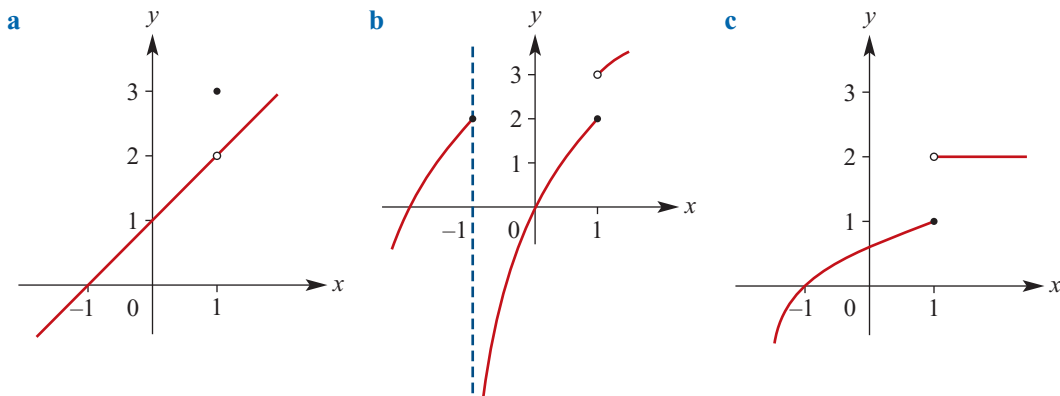
The polynomial functions are all continuous for $\mathbb{R}$. Most of the functions considered in this course are continuous for their domains.

The function with rule $f(x) = \frac{1}{x}$ does have a discontinuity, where $x = 0$, as $f(0)$ is not defined. It is continuous everywhere in its domain.

Hybrid functions, as introduced in Chapter 5, provide examples of functions that have points of discontinuity where the function is defined.

Example 33

State the values for x for which each of the functions whose graphs are shown below have a discontinuity:

**Solution**

- a** Discontinuity at $x = 1$ as $f(1) = 3$ but $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 2$
- b** Discontinuity at $x = -1$, as $f(-1) = 2$ and $\lim_{x \rightarrow -1^-} f(x) = 2$ but $\lim_{x \rightarrow -1^+} f(x) = -\infty$ and a discontinuity at $x = 1$, as $f(1) = 2$ and $\lim_{x \rightarrow 1^-} f(x) = 2$ but $\lim_{x \rightarrow 1^+} f(x) = 3$
- c** Discontinuity at $x = 1$, as $f(1) = 1$ and $\lim_{x \rightarrow 1^-} f(x) = 1$ but $\lim_{x \rightarrow 1^+} f(x) = 2$

Example 34

For each of the functions following, state the values of x for which there is a discontinuity, and use the definition of continuity in terms of $f(a)$, $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ to explain why each is a discontinuity:

- a** $f(x) = \begin{cases} 2x & \text{if } x \geq 0 \\ -2x + 1 & \text{if } x < 0 \end{cases}$
- b** $f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -2x + 1 & \text{if } x < 0 \end{cases}$
- c** $f(x) = \begin{cases} x & \text{if } x \leq -1 \\ x^2 & \text{if } -1 < x < 0 \\ -2x + 1 & \text{if } x \geq 0 \end{cases}$
- d** $f(x) = \begin{cases} x^2 + 1 & \text{if } x \geq 0 \\ -2x + 1 & \text{if } x < 0 \end{cases}$
- e** $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -2x & \text{if } x < 0 \end{cases}$

Solution

- a** $f(0) = 0$ but $\lim_{x \rightarrow 0^-} f(x) = 1$. Therefore a discontinuity at $x = 0$.
- b** $f(0) = 0$ but $\lim_{x \rightarrow 0^-} f(x) = 1$. Therefore a discontinuity at $x = 0$.
- c** $f(-1) = -1$ but $\lim_{x \rightarrow -1^+} f(x) = 1$. Therefore a discontinuity at $x = -1$.
 $f(0) = 1$ but $\lim_{x \rightarrow 0^-} f(x) = 0$. Therefore a discontinuity at $x = 0$.
- d** No discontinuity **e** No discontinuity

Exercise 9I

1 Find the following limits:

a $\lim_{x \rightarrow 2} 17$

b $\lim_{x \rightarrow 6} (x - 3)$

c $\lim_{x \rightarrow \frac{1}{2}} (2x - 5)$

d $\lim_{t \rightarrow -3} \frac{(t+2)}{(t-5)}$

e $\lim_{t \rightarrow 2} \frac{t^2 + 2t + 1}{t + 1}$

f $\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{x}$

g $\lim_{t \rightarrow 1} \frac{t^2 - 1}{t - 1}$

h $\lim_{x \rightarrow 9} \sqrt{x+3}$

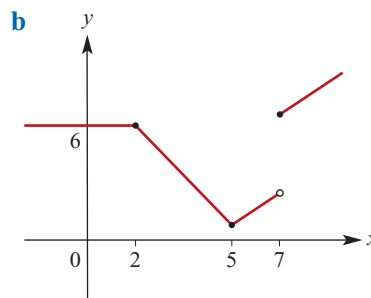
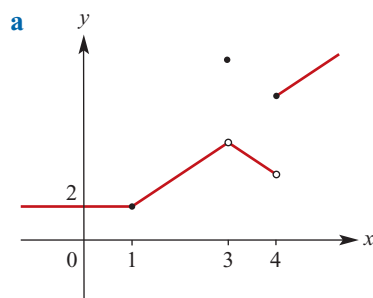
i $\lim_{x \rightarrow 0} \frac{x^2 - 2x}{x}$

j $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

k $\lim_{x \rightarrow 2} \frac{3x^2 - x - 10}{x^2 + 5x - 14}$

l $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x^2 - 6x + 5}$

2 For each of the following graphs, give the values of x for which a discontinuity occurs. Give reasons.



3 For each of the functions following, state the values of x for which there is a discontinuity and use the definition of continuity in terms of $f(a)$, $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ to explain why each stated value of x corresponds to a discontinuity:

a $f(x) = \begin{cases} 3x & \text{if } x \geq 0 \\ -2x + 2 & \text{if } x < 0 \end{cases}$

b $f(x) = \begin{cases} x^2 + 2 & \text{if } x \geq 1 \\ -2x + 1 & \text{if } x < 1 \end{cases}$

c $f(x) = \begin{cases} -x & \text{if } x \leq -1 \\ x^2 & \text{if } -1 < x < 0 \\ -3x + 1 & \text{if } x \geq 0 \end{cases}$

4 The rule of a particular function is given below. For what values of x is the graph of this function continuous?

$$y = \begin{cases} 2, & x < 1 \\ (x-4)^2 - 9, & 1 \leq x < 7 \\ x - 7, & x \geq 7 \end{cases}$$

9.10 Differentiability

If a function is differentiable at x , then it is also continuous at x . The converse, however, is not true. A function that is continuous at x is not necessarily differentiable at x . Consider, for example, the function $f(x) = |x|$ where it was found, in Section 9.8, that:

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

The function $f(x) = |x|$ is not differentiable at the point $x = 0$, although it is continuous at that point.

Example 35

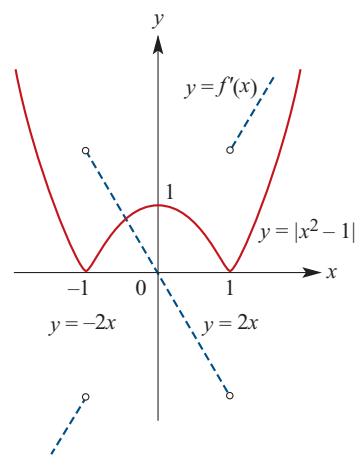
Find the derivative of $f(x) = |x^2 - 1|$ and sketch the graphs of $y = f(x)$ and $y = f'(x)$

Solution

$f(x) = g(h(x))$ where $g(x) = |x|$ and $h(x) = x^2 - 1$

The chain rule gives:

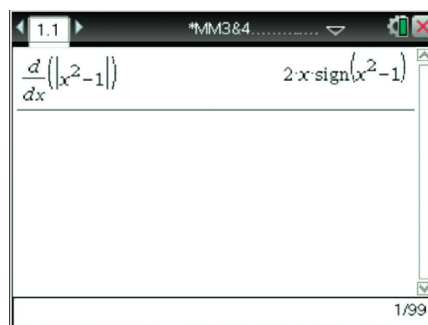
$$\begin{aligned} f'(x) &= g'(h(x)) h'(x) \\ &= 2x \times \begin{cases} 1 & \text{if } h(x) > 0 \\ -1 & \text{if } h(x) < 0 \end{cases} \\ &= \begin{cases} 2x & \text{if } h(x) > 0 \\ -2x & \text{if } h(x) < 0 \end{cases} \\ &= \begin{cases} 2x & \text{if } x^2 - 1 > 0 \\ -2x & \text{if } x^2 - 1 < 0 \end{cases} \\ &= \begin{cases} 2x & \text{if } x > 1 \text{ or } x < -1 \\ -2x & \text{if } -1 < x < 1 \end{cases} \end{aligned}$$



It is clear that the function is not differentiable at $x = 1$ or $x = -1$.

Using the TI-Nspire

Use the **Derivative** template from the **Calculus** menu to find the derivative of $|x^2 - 1|$.

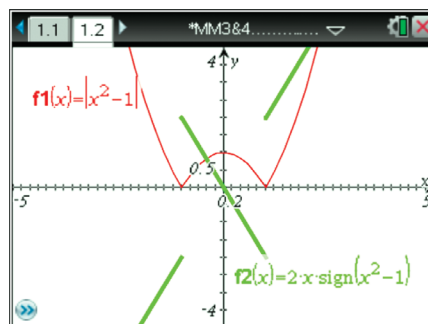


Open a **Graphs** application ($\text{ctrl} + \text{I}$) **Graphs** and let $f1(x) = \text{abs}(x^2 - 1)$ and $f2(x) = 2x \text{ sign}(x^2 - 1)$.

Select an appropriate **window setting** (menu) **> Window/Zoom > Window Settings**).

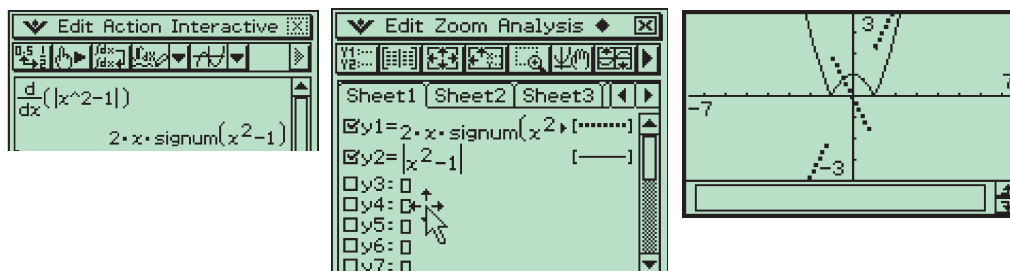
The two graphs are shown. The graph of the derivative has **medium line weight**, available from the **Attributes** menu

(menu) **> Actions > Attributes**).



Using the Casio ClassPad

The derivative of $|x^2 - 1|$ is found with the result graphed in dot mode together with the graph of the function $f(x)$ as a solid line.



Example 36

Find the derivative of $f(x) = |x|^2 - |x|$ and sketch the graph of $y = f(x)$ and $y = f'(x)$

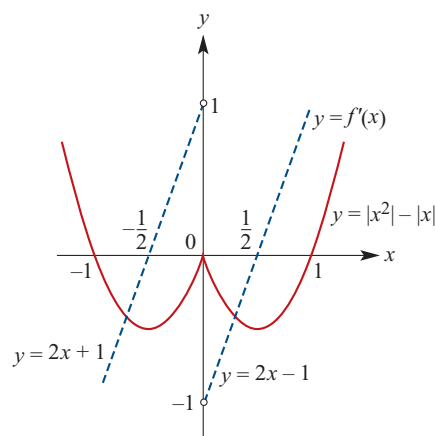
Solution

$f(x) = g(h(x))$ where $g(x) = x^2 - x$ and

$h(x) = |x|$

The chain rule gives:

$$\begin{aligned}
 f'(x) &= g'(h(x))h'(x) \\
 &= (2h(x) - 1)h'(x) \\
 &= (2|x| - 1) \times \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases} \\
 &= \begin{cases} (2|x| - 1) & \text{if } x > 0 \\ -(2|x| - 1) & \text{if } x < 0 \end{cases} \\
 &= \begin{cases} (2|x| - 1) & \text{if } x > 0 \\ -2|x| + 1 & \text{if } x < 0 \end{cases} \\
 &= \begin{cases} 2x - 1 & \text{if } x > 0 \\ 2x + 1 & \text{if } x < 0 \end{cases}
 \end{aligned}$$



This may also be obtained by first writing:

$$\begin{aligned}
 f(x) &= |x|^2 - |x| \\
 &= \begin{cases} x^2 - x & \text{if } x > 0 \\ x^2 + x & \text{if } x < 0 \end{cases}
 \end{aligned}$$

Using the TI-Nspire

Use **(menu) > Calculus > Derivative** to find the derivative of $|x|^2 - |x|$.

Store **(ctrl) (var)** this derivative as $df(x)$ as shown.

Open a **Graphs** application **(ctrl) + [I] >**

Graphs) and let

$$f1(x) = \text{abs}(x)^2 - \text{abs}(x) \text{ and}$$

$$f2(x) = df(x).$$

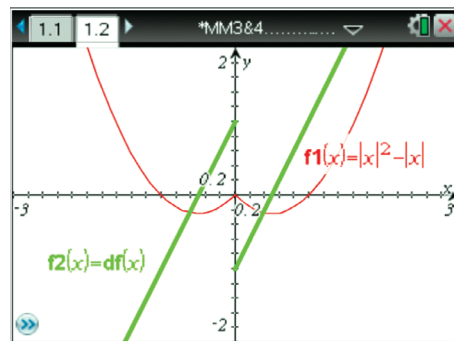
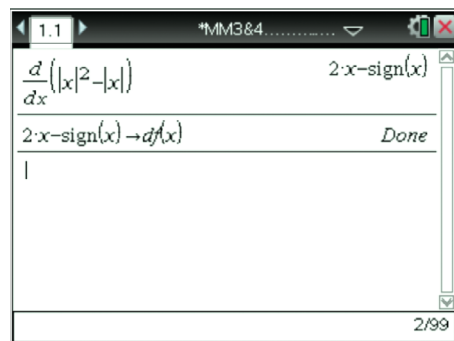
Select an appropriate **window setting**

(menu) > Window/Zoom > Window

Settings).

The two graphs are shown. The graph of the derivative has **medium line weight**, available from the **Attributes** menu

(menu) > Actions > Attributes).



Using the Casio ClassPad

The derivative of $|x|^2 - |x|$ is found using



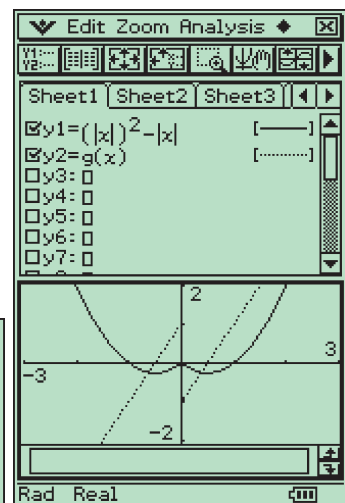
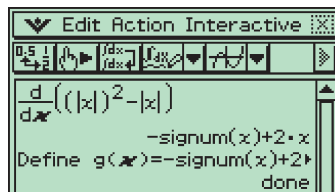
Main. The derivative is defined as $g(x)$ using

Interactive-Define.

The graphs of the function and its derivative are graphed, each with a different style.

Note: Edit, Cut & Paste

can be used to transfer the function and its derivative as required within and across program areas.



There are hybrid functions that are differentiable for R . The smoothness of the 'joins' determines if this is the case.

Example 37

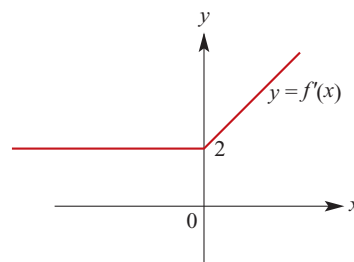
For the function with rule $f(x) = \begin{cases} x^2 + 2x + 1 & \text{if } x \geq 0 \\ 2x + 1 & \text{if } x < 0 \end{cases}$, find $f'(x)$ and sketch the graph of $y = f'(x)$

Solution

$$f'(x) = \begin{cases} 2x + 2 & \text{if } x \geq 0 \\ 2 & \text{if } x < 0 \end{cases}$$

In particular $f'(0)$ is defined and is equal to 2.

Also, $f(0) = 1$. The two sections of the graph join smoothly at $(0, 1)$.

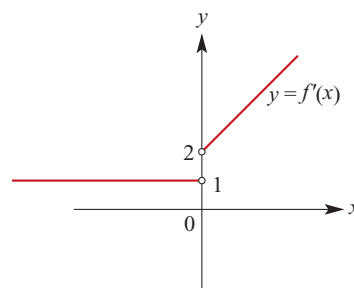
**Example 38**

For the function with rule $f(x) = \begin{cases} x^2 + 2x + 1 & \text{if } x \geq 0 \\ x + 1 & \text{if } x < 0 \end{cases}$, state the set of values for which the derivative is defined, find $f'(x)$ for this set of values and sketch the graph of $y = f'(x)$

Solution

$$f'(x) = \begin{cases} 2x + 2 & \text{if } x > 0 \\ 1 & \text{if } x < 0 \end{cases}$$

$f'(0)$ is not defined as the limits from the left and right are not equal. The function is differentiable for $R \setminus \{0\}$.



To test if a hybrid function is differentiable at a join at $x = a$:

- 1 test whether the function is continuous at $x = a$
- 2 test whether $\lim_{x \rightarrow a^-} f'(x)$ and $\lim_{x \rightarrow a^+} f'(x)$ exist and are equal.

Example 39

For the function with rule $f(x) = x^{\frac{1}{3}}$ state when the derivative is defined and sketch the graph of the derivative function.

Solution

By the rule $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}$. The derivative is not defined at $x = 0$.

This is clear through the definition of limit as:

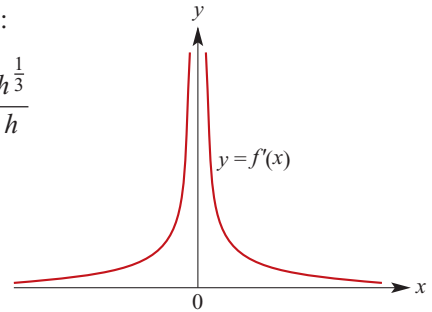
$$f'(0) = \lim_{h \rightarrow 0} \frac{(0+h)^{\frac{1}{3}} - 0^{\frac{1}{3}}}{(0+h) - 0} = \lim_{h \rightarrow 0} \frac{h^{\frac{1}{3}}}{h}$$

and thus the derivative is not defined.

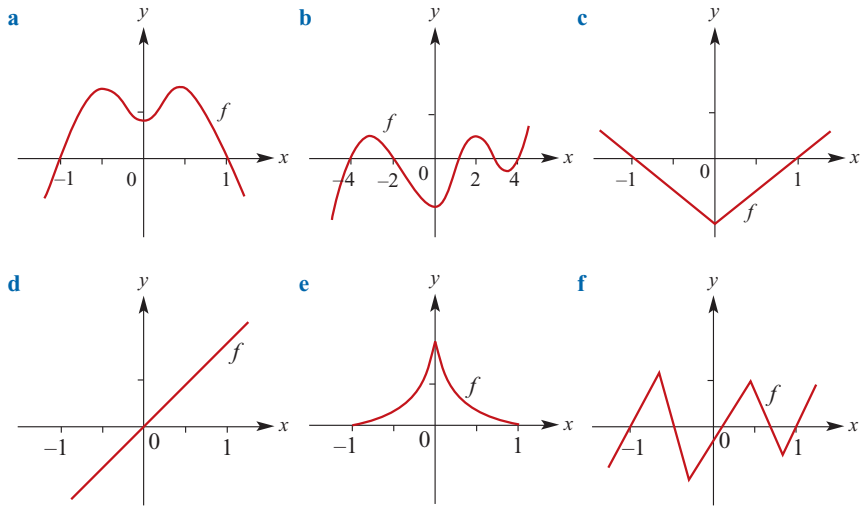
It can be seen that $\lim_{h \rightarrow 0} \frac{h^{\frac{1}{3}}}{h} = \lim_{h \rightarrow 0} \frac{1}{h^{\frac{2}{3}}} = \infty$

The graph of the derivative is as shown.

The function $f(x) = x^{\frac{1}{3}}$ is continuous at $x = 0$ but not differentiable.

**Exercise 9J**

- 1 In each of the figures below, a function graph f is given. Sketch the graph of f' . Your sketch of f' cannot be exact, but $f'(x)$ should be equal to zero at values of x for which the gradient of f is zero; $f'(x)$ should be less than zero where the original graph slopes downward; and so on.



- 2 For the function with rule $f(x) = \begin{cases} -x^2 + 3x + 1 & \text{if } x \geq 0 \\ 3x + 1 & \text{if } x < 0 \end{cases}$, find $f'(x)$ and sketch the graph of $y = f'(x)$
- 3 For the function with rule $f(x) = \begin{cases} x^2 + 2x + 1 & \text{if } x \geq 1 \\ -2x + 3 & \text{if } x < 1 \end{cases}$, state the set of values for which the derivative is defined, find $f'(x)$ for this set of values and sketch the graph of $y = f'(x)$
- 4 For the function with rule $f(x) = \begin{cases} -x^2 - 2x + 1 & \text{if } x \geq -1 \\ -2x + 3 & \text{if } x < -1 \end{cases}$, state the set of values for which the derivative is defined, find $f'(x)$ for this set of values and sketch the graph of $y = f'(x)$

- 5 For each of the following, give the set of values for which the derivative is defined, give the derivative and sketch the graph of the derivative function:

a $f(x) = (x - 1)^{\frac{1}{3}}$ **b** $f(x) = x^{\frac{1}{5}}$ **c** $f(x) = x^{\frac{2}{3}}$ **d** $f(x) = (x + 2)^{\frac{2}{5}}$

- 6 **a** For $f(x) = |x^2 - 4x|$ find $f'(x)$ for $x \in \mathbb{R} \setminus \{0, 4\}$.

- b** For $f(x) = |x|^2 - 4|x|$ find $f'(x)$ for $x \in \mathbb{R} \setminus \{0\}$.

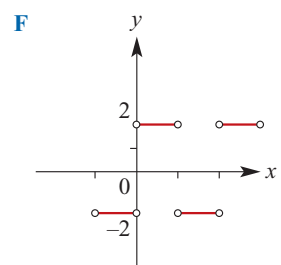
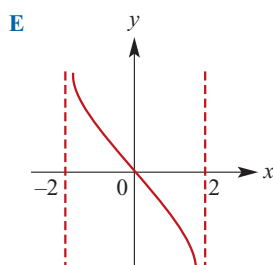
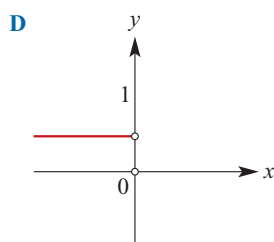
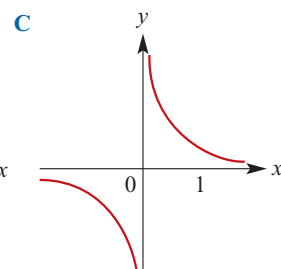
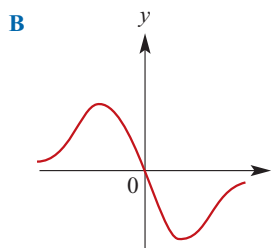
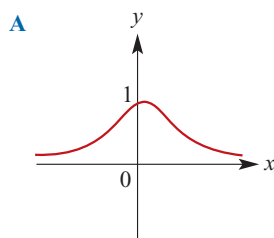
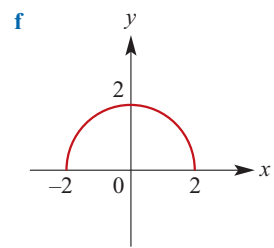
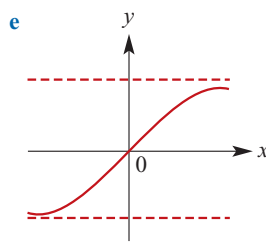
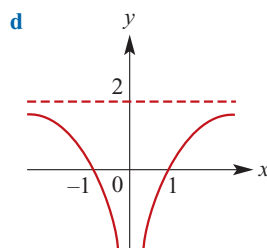
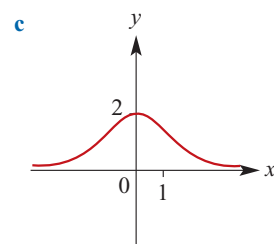
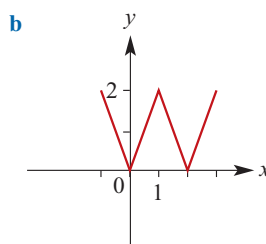
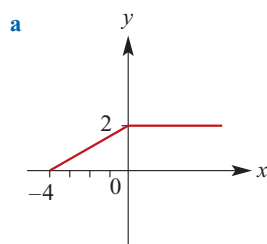
9.11 Miscellaneous exercises

Exercise 9K

- 1 For $y = x^2 + 1$:

- a** Find the average rate of change of y with respect to x over the interval $[3, 5]$.
b Find the instantaneous rate of change of y with respect to x at the point $x = -4$.

- 2 Match the graphs of the functions shown in **a–f** with the graphs of their derivatives **A–F**:



- 3** Let $y = \frac{x^2 - 1}{x^4 - 1}$
- a** Find $\frac{dy}{dx}$. **b** Find $\left\{x: \frac{dy}{dx} = 0\right\}$
- 4** If $f(3) = -2$ and $f'(3) = 5$, find $g'(3)$ if:
- a** $g(x) = 3x^2 - 5f(x)$ **b** $g(x) = \frac{3x + 1}{f(x)}$
- 5** If $f(4) = 6$ and $f'(4) = 2$, find $g'(4)$ if:
- a** $g(x) = \sqrt{x}f(x)$ **b** $g(x) = \frac{f(x)}{x}$
- 6** Given that $f'(x) = \sqrt{3x + 4}$ and $g(x) = x^2 - 1$, find $F'(x)$ if $F(x) = f(g(x))$
- 7** If $f(x) = 2x^2 - 3x + 5$, find:
- a** $f'(x)$ **b** $f'(0)$ **c** $\{x: f'(x) = 1\}$
- 8** If $f(x) = \frac{1}{3x - 1}$, find $f'(2)$.
- 9** If $y = 1 - x^2$ prove that $x \frac{dy}{dx} + 2 = 2y$ for all values of x .
- 10** If $A = 4\pi r^2$, calculate $\frac{dA}{dr}$ when $r = 3$.
- 11** At what point on the graph of $y = 1.8x^2$ is the gradient 1?
- 12** If $y = 3x^2 - 4x + 7$, find the value of x such that $\frac{dy}{dx} = 0$
- 13** If $y = x^3$, prove that $x \frac{dy}{dx} = 3y$ for all values of x .
- 14** If $y = \frac{x^2 + 2}{x^2 - 2}$, find $\frac{dy}{dx}$. **15** If $z = 3y^2 + 4$ and $y = 2x - 1$, find $\frac{dz}{dx}$
- 16** If $y = (5 - 7x)^9$, calculate $\frac{dy}{dx}$. **17** If $y = 3x^{\frac{1}{3}}$, find $\frac{dy}{dx}$ when $x = 27$.
- 18** If $y = \sqrt{5 + x^2}$, find $\frac{dy}{dx}$ when $x = 2$.
- 19** Find $\frac{dy}{dx}$ when $x = 1$, given that $y = (x^2 + 3)(2 - 4x - 5x^2)$
- 20** If $y = \frac{x}{1 + x^2}$, find $\frac{dy}{dx}$ when $x = 1$. **21** If $y = \frac{2 + x}{x^2 + x + 1}$, find $\frac{dy}{dx}$ when $x = 0$.
- 22** Let $f(x) = \frac{1}{2x + 1}$
- a** Use the definition of derivative to find $f'(x)$.
- b** Find the gradient of the curve of f at the point $(0, 1)$.

- 23 Let $f(x) = x^3 + 3x^2 - 1$. Find:
a $\{x: f'(x) = 0\}$ b $\{x: f'(x) > 0\}$ c $\{x: f'(x) < 0\}$
- 24 Let $y = \frac{x}{1-x}$
a Find $\frac{dy}{dx}$. b Write $\frac{dy}{dx}$ in terms of y .
- 25 If $y = (x^2 + 1)^{-\frac{3}{2}}$, find $\frac{dy}{dx}$. 26 If $P = 3s^2 - 3s$, find $\frac{dP}{ds}$ when $s = 1$.
- 27 If $y = x^4$, prove that $x \frac{dy}{dx} = 4y$

The questions below are based on the following two definitions

A function f is said to be **strictly decreasing** on a given set if for all a and b in the set $b > a$ implies $f(b) < f(a)$.

A function f is said to be **strictly increasing** on a given set if for all a and b in the set $b > a$ implies $f(b) > f(a)$.

- 28 a Show that $f: R \rightarrow R$, $f(x) = x^3$ is a strictly increasing function for R by showing $f'(x) > 0$, for all non-zero x and showing that if $b > 0$, then $f(b) > f(0)$ and if $0 > b$ then $f(0) > f(b)$.
b Show that $f: R \rightarrow R$, $f(x) = -x^3$ is a strictly decreasing function for R .
- 29 a Show that $f: [0, \infty) \rightarrow R$, $f(x) = x^2$ is a strictly increasing function.
b Show that $f: (-\infty, 0] \rightarrow R$, $f(x) = x^2$ is a strictly decreasing function.
- 30 For the function $f: R \rightarrow R$, $f(x) = x^2 - x - 12$ show that the largest interval for which f is strictly increasing is $\left[\frac{1}{2}, \infty\right)$.



Chapter summary

- For points $P(x, f(x))$ and $Q(x+h, f(x+h))$ on the graph of $y = f(x)$ the gradient of the chord PQ is:

$$\frac{f(x+h) - f(x)}{h}$$

- The gradient of the graph of $y = f(x)$ at P is defined as:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- For $f: R \rightarrow R$, the derived function is denoted by f' , where $f': R \rightarrow R$ and

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- For $f(x) = x^n$, $f'(x) = nx^{n-1}$, $n = 1, 2, 3, \dots$ and for $f(x) = 1$, $f'(x) = 0$

- If $g(x) = kf(x)$, where k is a constant, then $g'(x) = kf'(x)$

- If $f(x) = g(x) + h(x)$, then $f'(x) = g'(x) + h'(x)$

- If y is a function of x , then the derivative of y with respect to x is denoted by $\frac{dy}{dx}$

- For $f(x) = x^n$, $f'(x) = nx^{n-1}$, n is a non-zero integer.

$$\text{For } f(x) = 1, f'(x) = 0$$

For $n \leq -1$, the domain of f is taken to be $R \setminus \{0\}$, and for $n \geq 1$, the domain of f is taken to be R .

- **The chain rule**

$$\frac{dy}{dx} = \frac{du}{dx} \cdot \frac{dy}{du}$$

and in function notation:

$$(f \circ g)'(x) = f'(g(x))g'(x), \text{ where } f \circ g(x) = f(g(x))$$

- For $f(x) = x^a$, $f'(x) = ax^{a-1}$, for $x > 0$ and $a \in R$.

- **The product rule**

If $F(x) = f(x) \cdot g(x)$, then:

$$F'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

- In Leibniz notation:

If $y = uv$, where, u and v are functions of x

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

- **The quotient rule**

If $F(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$, then:

$$F'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

In Leibniz notation:

$$\text{If } y = \frac{u}{v} \text{ where } u \text{ and } v \text{ are functions of } x, v \neq 0$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

■ The following are important results that are useful for the evaluation of limits:

- $\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$
i.e., the limit of the sum is the sum of the limits.
- $\lim_{x \rightarrow c} (kf(x)) = k \lim_{x \rightarrow c} f(x)$, k being a given number (non-zero)
- $\lim_{x \rightarrow c} (f(x)g(x)) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x)$
i.e., the limit of the product is the product of the limits.
- $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided $\lim_{x \rightarrow c} g(x) \neq 0$
i.e., the limit of the quotient is the quotient of the limits.

■ A function f is continuous at the point $x = a$ if the following three conditions are met:

- 1 $f(x)$ is defined at $x = a$ 2 $\lim_{x \rightarrow a} f(x)$ exists 3 $\lim_{x \rightarrow a} f(x) = f(a)$

■ A function is said to be discontinuous at a point if it is not continuous at that point. We say that a function is continuous everywhere if it is continuous for all real numbers.

■ A function f is said to be differentiable at x if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists.

Multiple-choice questions

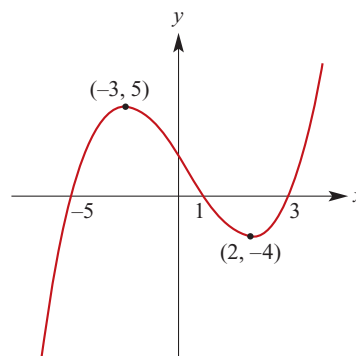
- 1 If $f(x) = \frac{4x^4 - 12x^2}{3x}$, then $f'(x)$ is equal to:
 A $\frac{16x^3 - 24x}{3}$ B $4x^2 - 4$ C $\frac{16x^3 - 24x}{3x}$ D $4x^2 - 8x$ E $\frac{8x^3 - 16x}{3x}$
- 2 For $f: R \setminus \{7\} \rightarrow R$ where $f(x) = 5 + \frac{5}{(7-x)^2}$, $f'(x) > 0$ for:
 A $R \setminus \{7\}$ B R C $x < 7$ D $x > 7$ E $x > 5$
- 3 Let $y = f(g(x))$ where $g(x) = 2x^4$. Then $\frac{dy}{dx}$ is equal to:
 A $8x^3 f'(2x^4)$ B $8x^2 f(4x^3)$ C $8x^4 f(x) f'(x^3)$
 D $2f(x) f'(x^3)$ E $8x^3$
- 4 Which of the following is not true for the curve of $y = f(x)$ where $f(x) = x^{\frac{1}{3}}$?
 A The gradient is defined for all real numbers.
 B The curve passes through the origin.
 C The curve passes through the points with coordinates $(1, 1)$ and $(-1, -1)$.
 D For $x > 0$ the gradient is positive. E For $x > 0$ the gradient is decreasing.

- 5 The graph of the function with rule $y = \frac{k}{2(x^3 + 1)}$ has gradient 1 when $x = 1$. The value of k is:

A 1 B $-\frac{8}{3}$ C $-\frac{1}{2}$ D -4 E $-\frac{1}{4}$

- 6 For the graph shown, the gradient is positive for:

A $-3 < x < 2$
 B $-3 \leq x \leq 2$
 C $x < -3$ or $x > 2$
 D $x \leq -3$ or $x \geq 2$
 E $-3 \leq x \leq 3$



- 7 For the function $f(x) = 4x(2 - 3x)$, $f'(x) < 0$ for:

A $x < \frac{1}{3}$ B $0 < x < \frac{2}{3}$ C $x = \frac{1}{3}$
 D $x > \frac{1}{3}$ E $x = 0, \frac{2}{3}$

- 8 For $y = \sqrt{3 - 2f(x)}$, $\frac{dy}{dx}$ is equal to:

A $\frac{2f'(x)}{\sqrt{3 - 2f(x)}}$ B $\frac{-1}{2\sqrt{3 - 2f(x)}}$ C $\frac{1}{2}\sqrt{3 - 2f'(x)}$
 D $\frac{3}{2} \frac{1}{[3 - 2f'(x)]}$ E $\frac{-f'(x)}{\sqrt{3 - 2f(x)}}$

- 9 The point on the curve defined by the equation $y = (x + 3)(x - 2)$ where the gradient is -7 has coordinates:

A $(-4, 6)$ B $(-4, 0)$ C $(-3, 0)$ D $(-3, -5)$ E $(-2, 0)$

- 10 The function $y = ax^2 - bx$ has a zero gradient only for $x = 2$. The x -axis intercepts of the graph of this function are:

A $\frac{1}{2}, \frac{-1}{2}$ B $0, 4$ C $0, -4$ D $0, \frac{1}{2}$ E $0, \frac{-1}{2}$

Short-answer questions (technology-free)

- 1 Differentiate each of the following with respect to x :

a $x + \sqrt{1 - x^2}$	b $\frac{4x + 1}{x^2 + 3}$	c $\sqrt{1 + 3x}$
d $\frac{2 + \sqrt{x}}{x}$	e $(x - 9)\sqrt{x - 3}$	f $x\sqrt{1 + x^2}$
g $\frac{x^2 - 1}{x^2 + 1}$	h $\frac{x}{x^2 + 1}$	i $(2 + 5x^2)^{\frac{1}{3}}$
j $\frac{2x + 1}{x^2 + 2}$	k $(3x^2 + 2)^{\frac{2}{3}}$	

- 2 Find the gradient of the curve of each of the following functions at the point corresponding to the x -value given:

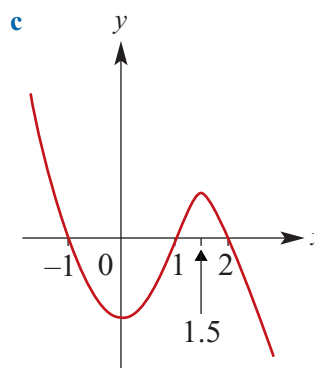
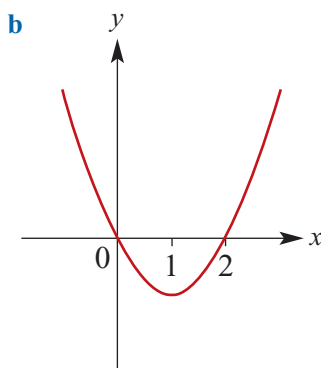
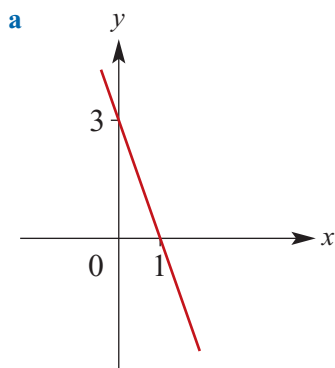
a $y = 3x^2 - 4$; $x = -1$

b $y = \frac{x-1}{x^2+1}$; $x = 0$

c $y = (x-2)^5$; $x = 1$

d $y = (2x+2)^{\frac{1}{3}}$; $x = 3$

- 3 Sketch the graphs of the derivative functions for each of the following functions from the graphs shown:



- 4 Find the derivative of $\left(4x + \frac{9}{x}\right)^2$ and values of x at which the derivative is zero.

5 **a** For $y = \frac{2x-3}{x^2+4}$, show that $\frac{dy}{dx} = \frac{8+6x-2x^2}{(x^2+4)^2}$

- b** Find the values of x for which y and $\frac{dy}{dx}$ are both positive.

- 6 Find the derivative of each of the following, given that f is a differentiable function for all real numbers.

a $xf(x)$ **b** $|f(x)|$, given that $f(x) \geq 0$ only for $x \in [0, 4]$ **c** $\frac{x^2}{[f(x)]^2}$

Extended-response questions

- 1 **a** For the functions f and g that are defined and differentiable for all real numbers, it is known that:

■ $f(1) = 6$, $g(1) = -1$, $g(6) = 7$ and $f(-1) = 8$

■ $f'(1) = 6$, $g'(1) = -2$, $f'(-1) = 2$ and $g'(6) = -1$

Find:

i $(f \circ g)'(1)$

ii $(g \circ f)'(1)$

iii $(fg)'(1)$

iv $(gf)'(1)$

v $\left(\frac{f}{g}\right)'(1)$

vi $\left(\frac{g}{f}\right)'(1)$

- b** It is known that f is a cubic function with rule $f(x) = ax^3 + bx^2 + cx + d$. Find the values of a , b , c and d .

- 2 Let f be a function, which is differentiable for R , with graph defined by the rule $y = f(x)$. It is known that:
- $f'(x) = 0$ for $x = 1$ and $x = 5$
 - $f'(x) > 0$ for $x > 5$ and $x < 1$
 - $f'(x) < 0$ for $1 < x < 5$
 - $f(1) = 6$ and $f(5) = 1$
- a For $y = f(x + 2)$ find the values of x for which:
- i $\frac{dy}{dx} = 0$
 - ii $\frac{dy}{dx} > 0$
- b For $y = f(x - 2)$:
- i Find the values of x for which $\frac{dy}{dx} = 0$
 - ii Find the coordinates of the points on the graph for which $\frac{dy}{dx} = 0$
- c For $y = f(2x)$:
- i Find the values of x for which $\frac{dy}{dx} = 0$
 - ii Find the coordinates of the points on the graph for which $\frac{dy}{dx} = 0$
- d For $y = f\left(\frac{x}{2}\right)$:
- i Find the values of x for which $\frac{dy}{dx} = 0$
 - ii Find the coordinates of the points on the graph for which $\frac{dy}{dx} = 0$
- e For $y = 3f\left(\frac{x}{2}\right)$:
- i Find the values of x for which $\frac{dy}{dx} = 0$
 - ii Find the coordinates of the points on the graph for which $\frac{dy}{dx} = 0$
- 3 Let $f(x) = (x - \alpha)^n(x - \beta)^m$ where m and n are positive integers with $m > n$ and $\beta > \alpha$
- a Solve the equation $f(x) = 0$ for x
 - b Find $f'(x)$.
 - c Solve the equation $f'(x) = 0$ for x
 - d i If m and n are odd, find the set of values for which $f'(x) > 0$
 - ii If m is odd and n is even, find the set of values for which $f'(x) > 0$
- 4 Consider the function with rule $f(x) = \frac{x^n}{1 + x^n}$ where n is a positive even integer.
- a Show that $f(x) = 1 - \frac{1}{x^n + 1}$
 - b Find $f'(x)$.
 - c Show that $0 \leq f(x) < 1$ for all x .
 - d State the set of values for which $f'(x) = 0$
 - e State the set of values for which $f'(x) > 0$
 - f Show that f is an even function.

Applications of differentiation

Objectives

- To be able to find the equation of the **tangent** and the **normal** at a given point on a curve.
- To be able to calculate the **angles between straight lines and curves**.
- To be able to find the **stationary points** on the curves of certain polynomial functions and to state the nature of such points.
- To use differentiation techniques to **sketch graphs of rational functions**.
- To solve **maxima and minima problems**.
- To use the derivative of a function in **rates of change** problems.
- To use the chain rule in the solution of related rates problems.

10.1 Tangents and normals

The derivative of a function is a new function that gives the measure of the gradient at each point of the curve. Having the gradient, we can find the equation of the tangent for a given point on the curve. Suppose (x_1, y_1) is a point on the curve $y = f(x)$. Then if f is differentiable for $x = x_1$, the equation of the tangent at (x_1, y_1) is given by $y - y_1 = f'(x_1)(x - x_1)$.

Example 1



Find the equation of the tangent of the curve $y = x^3 + \frac{1}{2}x^2$ at the point $x = 1$.

Solution

When $x = 1$, $y = \frac{3}{2}$ so $\left(1, \frac{3}{2}\right)$ is a point on the tangent.

Further, $\frac{dy}{dx} = 3x^2 + x$

Thus the gradient of the tangent to the curve at $x = 1$ is 4 and the equation of the tangent is $y - \frac{3}{2} = 4(x - 1)$, which becomes $y = 4x - \frac{5}{2}$.

The **normal** to a curve at a point on the curve is the line that passes through the point and is perpendicular to the tangent at that point.

From earlier work you should know that lines with gradients m_1 and m_2 are perpendicular if, and only if, $m_1 m_2 = -1$.

Thus if a tangent has a gradient m , the normal has gradient $-\frac{1}{m}$.

Example 2

Find the equation of the normal to the curve with equation $y = x^3 - 2x^2$ at the point $(1, -1)$.

Solution

The point $(1, -1)$ is on the normal.

$$\text{Further } \frac{dy}{dx} = 3x^2 - 4x$$

Thus the gradient of the normal at $x = 1$ is $\frac{-1}{-1} = +1$.

Hence the equation of the normal is $y + 1 = 1(x - 1)$

i.e., the equation of the normal is $y = x - 2$

Example 3

Use a CAS calculator to find the equation of the tangent to the curve with equation $y = x^{\frac{3}{2}} - 4x^{\frac{1}{2}}$ at the point on the graph where $x = 4$.

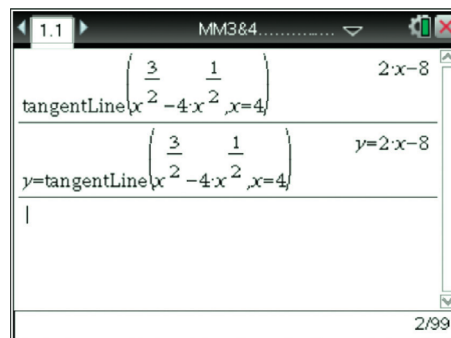
Solution

Using the TI-Nspire

Use $\text{[menu]} > \text{Calculus} > \text{Tangent Line}$ and complete as shown.

The tangent can also be found in a

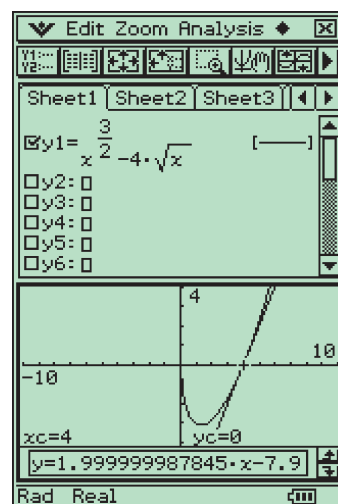
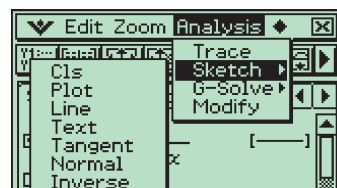
Graphs application.



Using the Casio ClassPad

This can be done in the  menu. Enter the function, select the tick box and tap  to produce the graph. Ensure the graph window is selected (bold border) and tap **Analysis–Sketch–Tangent**. Press the number 4 and OK to sketch the tangent at $x = 4$. The equation of the tangent is shown in the formula line.

Note that the results in  are approximate calculations, not algebraic calculations. Different screen resolutions will give slightly different answers. Scrolling in the formula line should convince you that the answer should be read as $y = 2x - 8$.



Example 4

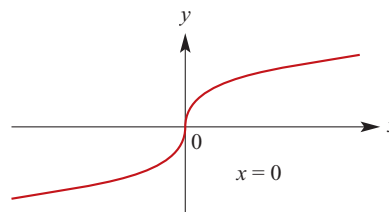
Find the equation of the tangent of:

- a** $f(x) = x^{\frac{1}{3}}$ where $x = 0$ **b** $f(x) = x^{\frac{2}{3}}$ where $x = 0$

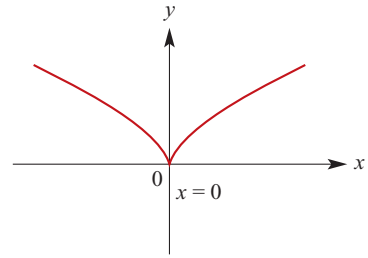
Solution

- a** The derivative of f is not defined at $x = 0$.

For $x \in \mathbb{R} \setminus \{0\}$, $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}$. It is clear that f is continuous at $x = 0$ and $|f'(x)| \rightarrow \infty$ as $x \rightarrow 0$. It can be said that the graph of $y = f(x)$ has a **vertical tangent** at $x = 0$.



- b** The derivative of f is not defined at $x = 0$.
 For $x \in \mathbb{R} \setminus \{0\}$, $f'(x) = \frac{2}{3}x^{-\frac{1}{3}}$. It is clear
 that f is continuous at $x = 0$ and $f'(x) \rightarrow \infty$
 as $x \rightarrow 0^+$ and $f'(x) \rightarrow -\infty$ as $x \rightarrow 0^-$.
 There is a **cusp** at $x = 0$ and the graph of
 $y = f(x)$ has a **vertical tangent** at $x = 0$.



Exercise 10A

- Find the equation of the tangent and of the normal of the curve $y = x^2 - 1$ at the point $(2, 3)$.
- Find the equation of the normal to the curve $y = x^2 + 3x - 1$ at the point where the curve cuts the y -axis.
- Find the equations of the normals to the curve $y = x^2 - 5x + 6$ at the points where it cuts the x -axis.
- Find the equation of the tangent and of the normal of the curve $y = (2x + 1)^9$ at the point $(0, 1)$.
- Find the coordinates of the point on $y = x^2 - 5$ at which the curve has gradient 3. Hence find the value of c for which the line $y = 3x + c$ is tangent to $y = x^2 - 5$.
- Find the equation to **i** the tangent and **ii** the normal at the point corresponding to the given x -value on each of the following curves:

a $y = x^2 - 2$; $x = 1$	b $y = x^2 - 3x - 1$; $x = 0$
c $y = \frac{1}{x}$; $x = -1$	d $y = (x - 2)(x^2 + 1)$; $x = -1$
e $y = \sqrt{3x + 1}$; $x = 0$	f $y = \sqrt{x}$; $x = 1$
g $y = x^{\frac{2}{3}} + 1$; $x = 1$	h $y = x^3 - 8x$; $x = 2$
i $y = x^3 - 3x^2 + 2$; $x = 2$	j $y = 2x^3 + x^2 - 4x + 1$; $x = 1$
- Find the equation to the tangent at the point corresponding to the given x -value on each of the following curves:

a $y = \frac{x^2 - 1}{x^2 + 1}$; $x = 0$	b $y = \sqrt{3x^2 + 1}$; $x = 1$
c $y = \frac{1}{2x - 1}$; $x = 0$	d $y = \frac{1}{(2x - 1)^2}$; $x = 1$
- Find the equation to the tangent at the point where $y = 0$ for each of the following curves:

a $y = (x - 4)^{\frac{1}{3}}$	b $y = (x + 5)^{\frac{2}{3}}$	c $y = (2x + 1)^{\frac{1}{3}}$	d $y = (x + 5)^{\frac{4}{5}}$
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10.2 Angles between curves*

Angle between two straight lines

Two straight lines l_1 and l_2 make angles θ_1 and θ_2 with the positive direction of the x -axis. We write

$$\text{gradient of } l_1 = m_1 = \tan \theta_1$$

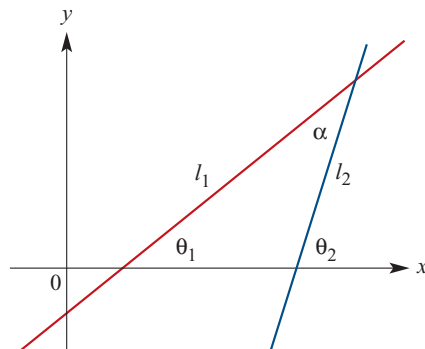
$$\text{gradient of } l_2 = m_2 = \tan \theta_2$$

The angle, α , between the lines is given by:

$$\alpha = \theta_2 - \theta_1 \text{ (see the diagram)}$$

We note that $\tan \alpha = \tan (\theta_2 - \theta_1)$

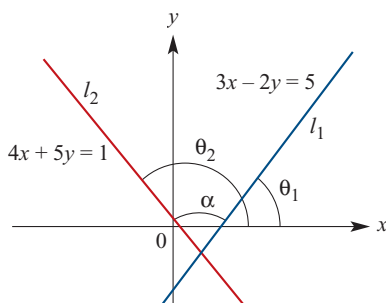
$$= \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_1 \tan \theta_2}$$



Example 5

Two straight lines l_1 and l_2 have equations $3x - 2y = 5$ and $4x + 5y = 1$. Find the angle between l_1 and l_2 .

Solution



$$\begin{aligned} \tan \theta_2 &= \frac{-4}{5} \quad \text{and} \quad \tan \theta_1 = \frac{3}{2} \\ \therefore \theta_2 &= 180 - 38.66 \quad \theta_1 = 56.31 \\ &= 141.34 \end{aligned}$$

$$\begin{aligned} \therefore \text{the angle between the two lines is} \\ 85.03^\circ = 85^\circ 2' \end{aligned}$$

Alternative solution

$$\begin{aligned} \text{Gradient of } l_1 &= m_1 \\ &= \frac{3}{2} \end{aligned}$$

$$\begin{aligned} \text{Gradient of } l_2 &= m_2 \\ &= \frac{-4}{5} \end{aligned}$$

* This topic is not listed in the study design, but can be included for the understanding of other topics that are included.

Thus if β is the angle between l_1 and l_2 :

$$\begin{aligned}\tan \beta &= \frac{\frac{3}{2} - \left(\frac{-4}{5}\right)}{1 + \frac{3}{2} \times \frac{-4}{5}} \\ &= \frac{-23}{2}\end{aligned}$$

In this calculation we have found β , the obtuse angle between the lines l_1 and l_2 . The acute angle α between the lines satisfies $\tan \alpha = \frac{23}{2}$ and as before:

$$\begin{aligned}\alpha &= 85.03^\circ \\ &= 85^\circ 2'\end{aligned}$$

The process described above to find the angle between two straight lines may be used to find the angle between two curves as we define the angle between curves to be the angle between the tangents to the curves at the point of intersection.

Example 6

Find the angle between the curve $y = x^2$ and the line $y = x + 12$ at the point of their intersection, $(4, 16)$.

Solution

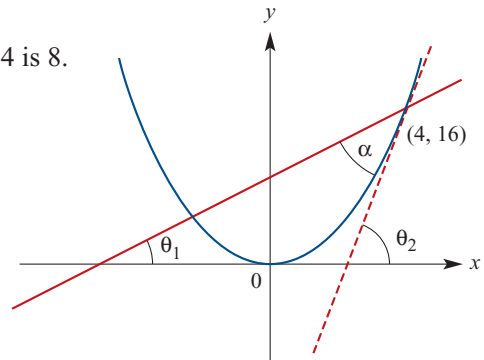
The derivative of x^2 is $2x$.

Therefore the gradient of $y = x^2$ at $x = 4$ is 8.

$$\begin{aligned}\tan \theta_1 &= 1 \quad \text{and} \quad \tan \theta_2 = 8 \\ \theta_2 &= 82.87\end{aligned}$$

$$\begin{aligned}\therefore \theta_1 &= 45 \\ \therefore \alpha &= \theta_2 - \theta_1 \\ &= 82.87 - 45 \\ &= 37.87\end{aligned}$$

i.e. the acute angle between the curve and the line is $37^\circ 52'$.



Example 7

Find the angle between the curve $f(x) = 4 - x^2$ and the curve $g(x) = x^2 - 4$ at the point $(2, 0)$.

Solution

$f'(x) = -2x$ and $g'(x) = 2x$ and therefore

$$f'(2) = -4 \text{ and } g'(2) = 4$$

From the diagram:

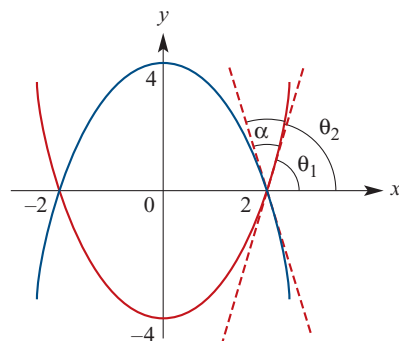
$$\alpha = \theta_2 - \theta_1$$

$$\therefore \alpha = 104.04 - 75.96 \text{ as } \tan \theta_2 = -4$$

$$\text{and } \tan \theta_1 = 4$$

$$= 28.07$$

$\therefore$ the angle between the two parabolas is 28.07° .

**Alternative solution**

$$\tan \alpha = \tan (\theta_2 - \theta_1)$$

$$= \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_2 \tan \theta_1}$$

$$= \frac{-4 - 4}{1 + (-4) \times 4}$$

$$= \frac{-8}{-15} = \frac{8}{15}$$

$$\therefore \alpha = 28.07^\circ$$

Exercise 10B

- Find the magnitude of the acute angle between the lines $2y + 3x = 4$ and $x + y = 5$
- The graphs with equations $y = x^2 - 2x$ and $y = x$ intersect at the point $(3, 3)$. Find the acute angle lying between them at this point.
- The curves with equations $y = 9 - x^2$ and $y = x^2 - 9$ intersect at the point $(3, 0)$. Find the obtuse angle lying between the curves at this point.
- Find the acute angles lying between the following pairs of lines. (Express your answers in the form $\tan^{-1}(\quad)$.)
 - $y = x - 6, y = 3x + 1$
 - $2x + y + 3 = 0, x + y = 0$
 - $3x + 4y - 2 = 0, 2x - 3y - 4 = 0$
- Find the coordinates of the points of intersection of the curves with equations $y = x^3$ and $y = x^2$.
 - Find the angle of intersection at each of these points.
- Find the acute angle of intersection of the curves with equations $y = x^2$ and $y = \frac{1}{x}$
- Find the acute angles of intersection of the given pairs of curves:
 - $y = x^3, y = x$
 - $y = 2x, y = x + x^3$
 - $y = x^2, y = 1$
 - $x = 2, y = x^2$

10.3



Linear approximation

In the graph l is the tangent to the curve at the point $(x, f(x))$.

$f'(x)$ is the gradient of the tangent.

From the diagram it can be seen that when h is small $f(x+h) \approx f(x) + hf'(x)$

This can also be seen by considering the definition of derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\therefore f'(x) \approx \frac{f(x+h) - f(x)}{h} \text{ where } h \text{ is small}$$

$$\therefore f(x+h) \approx hf'(x) + f(x) \text{ where } h \text{ is small}$$

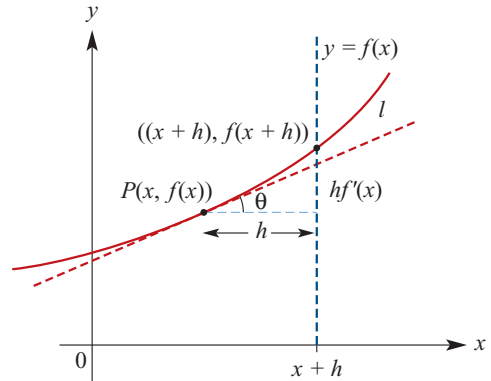
The approximate increase in $f(x)$ is given by $f(x+h) - f(x) = hf'(x)$

With the alternative notation:

$$\delta y \approx \frac{dy}{dx} \delta x$$



The linear approximation of a function may be considered as using the tangent at a point to approximate the curve in an immediate neighbourhood of the point at which the tangent is determined.



Example 8

Given that $f(x) = x^4 - x^3$, find in terms of p the approximate increase in $f(x)$ as x increases from 2 to $2+p$, where p is small.

Solution

$$\begin{aligned} f(2+p) &\approx pf'(2) + f(2) \text{ where } f'(x) = 4x^3 - 3x^2 \\ \therefore f(2+p) &\approx p(4 \times 2^3 - 3 \times 2^2) + (2^4 - 2^3) \\ &= p(20) + 8 \\ &= 20p + 8 \end{aligned}$$

$$\begin{aligned} \text{The approximate increase} &= f(2+p) - f(2) \\ &= 20p + 8 - 8 \end{aligned}$$

$\therefore$ the increase in $f(x)$ is $20p$.

Using Liebniz notation:

$$\begin{aligned} \frac{\delta y}{\delta x} &\approx \frac{dy}{dx} \\ \therefore \delta y &\approx \frac{dy}{dx} \delta x \end{aligned}$$

When $x = 2$

$$\delta y \approx 20p$$

The following table of values indicates the accuracy of the approximation:

p	x	$f(x)$	$f(2) + 20p$
-0.01	1.99	7.8018	7.8
0	2	8	8
0.01	2.01	8.2018	8.2
0.02	2.02	8.4073	8.4
0.03	2.03	8.6164	8.6
↓			
0.1	2.1	10.187	10

In practical problems you are often required to find the percentage change in a quantity resulting from a given change in another quantity. For a function f the **percentage change** between a and $a + h$ is defined to be

$$100 \left(\frac{f(a+h) - f(a)}{f(a)} \right)$$

provided $f(a) \neq 0$.

Using the result that $f(a+h) \approx hf'(a) + f(a)$:

$$\text{the percentage change is approximately equal to } \frac{100hf'(a)}{f(a)}$$

Example 9

The time for a pendulum of length l cm to complete one swing is given by the function with rule $f(l) = k\sqrt{l}$, where k is a constant. If an error is made in the measurement of the length so that the measured length is $2\frac{1}{2}\%$ greater than the actual length, find the approximate percentage error if the function $f(l)$ is used to calculate the time of a swing.

Solution

Let l_1 be the actual length of the pendulum.

$$\text{By the above, the percentage error} = \frac{100hf'(l_1)}{f(l_1)}$$

$$\text{where } f(l_1) = kl_1^{\frac{1}{2}}, f'(l_1) = \frac{1}{2}kl_1^{-\frac{1}{2}} \text{ and } h = \frac{l_1}{40}$$

$$\begin{aligned}
 \therefore \text{the percentage error} &= \left(100 \times \frac{l_1}{40} \times \frac{k}{2l_1^{\frac{1}{2}}} \right) \div kl_1^{\frac{1}{2}} \\
 &= \left(\frac{5l_1^{\frac{1}{2}}k}{4} \right) \times \frac{1}{kl_1^{\frac{1}{2}}} \\
 &= \frac{5}{4}
 \end{aligned}$$

Example 10

Given that $y = 10 - \frac{5}{x}$ and that the value of y increases from 5 by a small amount $\frac{p}{10}$, find in terms of p :

a the approximate change in x

b the corresponding percentage change in x

Solution

a For $y = 10 - \frac{5}{x}$, $\frac{dy}{dx} = \frac{5}{x^2}$
and when $y = 5$, $x = 1$.

$$\begin{aligned}
 \therefore \frac{\delta y}{\delta x} &\approx \frac{5}{1} \\
 &= 5, \text{ close to } x = 1
 \end{aligned}$$

$$\text{i.e. } \frac{1}{5}\delta y \approx \delta x$$

$$\text{Here } \delta y = \frac{p}{10} \text{ and so } \delta x \approx \frac{p}{50}$$

b Percentage change $\approx \frac{\delta x}{x} \times 100\%$
 $= \left(\frac{p}{50} \times 100 \right) \%$
 $= 2p\%$

Example 11

Differentiate $\frac{1}{\sqrt{x}}$ with respect to x and use the result to find an approximate value for $\frac{1}{\sqrt{100.5}}$.

Solution

$$y = x^{-\frac{1}{2}} \text{ and } \frac{dy}{dx} = -\frac{1}{2}x^{-\frac{3}{2}}$$

$$\begin{aligned}
 \text{When } x = 100, y &= \frac{1}{10} \text{ and } \frac{dy}{dx} = -\frac{1}{2}(100)^{-\frac{3}{2}} \\
 &= \frac{-1}{2000}
 \end{aligned}$$

$$\text{Using } \frac{dy}{dx} \approx \frac{\delta y}{\delta x}$$

$$\delta y \approx \frac{dy}{dx} \delta x$$

$$= \frac{-1}{2000} \times 0.5 \text{ when } x = 100 \text{ and } \delta x = 0.5$$

$$\therefore \delta y = -\frac{1}{4000}$$

(cont'd.)

$$\begin{aligned}\text{and } \frac{1}{\sqrt{100.5}} &\approx \frac{1}{10} - \frac{1}{4000} \\ &= 0.09975\end{aligned}$$

Exercise 10C

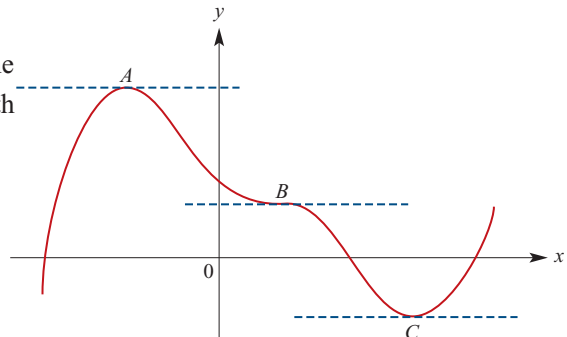
- The equation of a curve is $y = 4x^3 - 8x^2 + 10$
 - Find $\frac{dy}{dx}$.
 - Find the value of $\frac{dy}{dx}$ when $x = 2$.
 - Find the approximate change in y as x increases from 2 to 2.02.
- For $y = x^{-\frac{1}{2}}$:
 - Find $\frac{dy}{dx}$.
 - Determine the value of $\frac{dy}{dx}$ for $x = 100$.
 - Find the change in y as x changes from 100 to 103.
 - Find an approximate value of $\frac{1}{\sqrt{103}}$.
 - Find the change in y as x changes from a to $a + h$ where h is small.
 - Find an expression for $\frac{1}{\sqrt{a+h}}$ where h is small.
- Given that $y = x^4 - 5x^3$, find in terms of q the approximate increase in y as x increases from 2 to $2 + q$ where q is small.
- Given that $y = 6 - 4x + 5x^2$ and that the value of x increases from 5 by a small amount $\frac{p}{20}$, determine in terms of p :
 - the approximate change in y
 - the corresponding percentage change in y
- A curve has equation $y = \frac{5x^2 + 8}{x}$. Find an expression for $\frac{dy}{dx}$ and hence find, in terms of p , where p is small, the approximate increase in y as x increases from 2 to $2 + p$.
- For each of the following, write down an expression for the approximate change, δy , in y when x changes from a to $a + p$ where p is small:

a $y = 3x + 7$	b $y = \sqrt{x}$	c $y = \frac{2}{3x + 1}$
d $y = (2x + 1)^4$	e $y = (6x^2 - 1)^3$	f $y = \frac{x^2 + 2x + 6}{x^2 + 1}$
g $y = \sqrt[3]{x^2 + 10}$	h $y = \frac{6x + 1}{x + 1}$	
- The area of a circular disc increases from $100\pi \text{ cm}^2$ to $101\pi \text{ cm}^2$. Find the corresponding increase in the radius.
- Given that $y = x^{-\frac{1}{3}}$ use calculus to determine an approximate value for $\frac{1}{\sqrt[3]{0.9}}$.

- 9 Given that $\frac{dy}{dx} = 6x - 4$, calculate the approximate change in y when x increases from -2 to -1.97 .
- 10 The radius r of a circle is 5 cm. Find the increase in the area, $A \text{ cm}^2$, of the circle when the radius expands by p cm where p is small.
- 11 The time for a swing, T seconds, of a pendulum of length l m is given by the rule $T = 2\pi\sqrt{\frac{l}{g}}$, where g is a constant.
- a Find $\frac{dT}{dl}$.
- b Find the approximate increase in T when l is increased from 1.6 to 1.7. Give the answer in terms of g .
- 12 A 2% error is made in measuring the radius of a sphere. Find the percentage error in the surface area. (The surface area of a sphere is given by $A = 4\pi r^2$.)
- 13 One side of a rectangle is three times the other. If the perimeter increases by 2%, what is the percentage increase in the area?
- 14 a For $f(x) = \frac{1}{1-x}$, find:
- i $f'(x)$ ii $f'(0)$
- b Show that $f(h) \approx 1 + h$ when h is small. [Use $f(0+h) \approx f'(0)h + f(0)$]
- 15 a For $f(x) = \sqrt{1+x}$, find:
- i $f'(x)$ ii $f'(0)$
- b Show that $f(h) \approx 1 + \frac{h}{2}$ when h is small.
- 16 Show that $f(x) = \frac{1}{2-x}$ can be approximated by $f(x) = \frac{1}{2} + \frac{x}{4}$ for x close to 0.
- 17 The radius of a sphere is measured as 4 cm with a possible error of 0.05. What is the approximate error for:
- a the surface area? (Use $S = 4\pi r^2$) b the volume? (Use $V = \frac{4}{3}\pi r^3$)

10.4 Stationary points

In the previous chapter we have seen that the gradient at a point $(a, g(a))$ of the curve with rule $y = g(x)$ is given by $g'(a)$. A point $(a, g(a))$ on a curve $y = g(x)$ is said to be a stationary point if $g'(a) = 0$. (Equivalently: for $y = g(x)$, $\frac{dy}{dx} = 0$ when $x = a$ implies $((a, g(a))$ is a stationary point.)



For example in the graph shown, there are stationary points at A , B and C . At such points the tangents are parallel to the x -axis (illustrated as dotted lines).

The reason for the name **stationary points** becomes clear if we look at an application to motion of a particle.

Example 12

The displacement, x metres, of a particle moving in a straight line is given by $x = 9t - \frac{1}{3}t^3$, where $0 \leq t \leq 3\sqrt{3}$ and t seconds is the time taken. Find its maximum displacement.

Solution

$\frac{dx}{dt} = 9 - t^2$, and maximum displacement

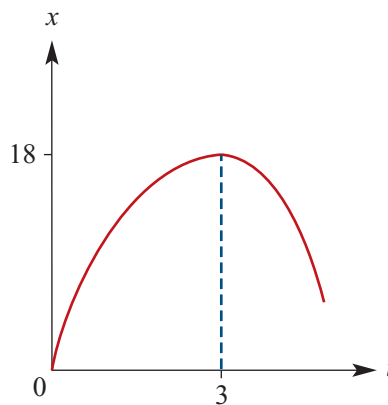
occurs when $\frac{dx}{dt} = 0$

So $t = 3$ or $t = -3$ (but $t = -3$ lies outside the domain).

At $t = 3$, $x = 18$.

Thus the stationary point is $(3, 18)$ and the maximum displacement is 18 metres.

Note that the stationary point occurs when the rate of change of displacement with respect to time (velocity) is zero. The particle stopped moving forward at $t = 3$.



Example 13

Find the stationary points of the following functions:

a $y = 9 + 12x - 2x^2$

b $p = 2t^3 - 5t^2 - 4t + 13$ for $t > 0$

c $y = 4 + 3x - x^3$

d $p = |12x^2 - 6|$

Solution

a $y = 9 + 12x - 2x^2$

$$\frac{dy}{dx} = 12 - 4x$$

A stationary point occurs when $\frac{dy}{dx} = 0$, i.e. when $12 - 4x = 0$

$$\therefore x = 3$$

$$\begin{aligned} \text{When } x = 3, y &= 9 + 12 \times 3 - 2 \times 3^2 \\ &= 27 \end{aligned}$$

Thus the stationary point is at $(3, 27)$.

b $P = 2t^3 - 5t^2 - 4t + 13$

$$\frac{dp}{dt} = 6t^2 - 10t - 4, t > 0$$

$$\frac{dp}{dt} = 0 \text{ implies } 2(3t^2 - 5t - 2) = 0$$

$$\therefore (3t + 1)(t - 2) = 0$$

$$\therefore t = -\frac{1}{3} \text{ or } t = 2$$

but $t > 0$ and therefore the only acceptable solution is $t = 2$.

$$\text{When } t = 2, p = 16 - 20 - 8 + 13 = 1.$$

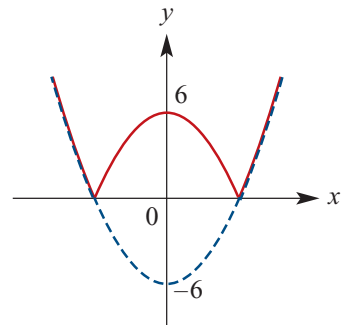
So the corresponding stationary point is $(2, 1)$.

d First consider the graph of $y = 12x^2 - 6$.

It has a stationary point where $\frac{dy}{dx} = 0$,

which is where $x = 0$. The coordinates of the stationary point are $(0, -6)$.

The graph of $y = |12x^2 - 6|$ is as shown. It is clear that the stationary point has coordinates $(0, 6)$.



c $y = 4 + 3x - x^3$

$$\frac{dy}{dx} = 3 - 3x^2$$

$$\frac{dy}{dx} = 0 \text{ implies } 3(1 - x^2) = 0$$

$$\therefore x = \pm 1$$

$\therefore$ stationary points occur at $(1, 6)$ and $(-1, 2)$.

Example 14

The curve with equation $y = x^3 + ax^2 + bx + c$ passes through $(0, 5)$ with stationary point $(2, 7)$. Find a, b, c .

Solution

When $x = 0, y = 5$

Thus $5 = c$

$$\frac{dy}{dx} = 3x^2 + 2ax + b \text{ and at } x = 2, \frac{dy}{dx} = 0$$

$$\text{Therefore } 0 = 12 + 4a + b \quad (1)$$

The point $(2, 7)$ is on the curve and therefore:

$$7 = 2^3 + 2^2a + 2b + 5$$

$$2 = 8 + 4a + 2b$$

$$\therefore 4a + 2b + 6 = 0 \quad (2)$$

Subtract (1) from (2):

$$-b + 6 = 0$$

$$\therefore b = 6$$

Substitute in (1):

$$0 = 12 + 4a + 6$$

$$-18 = 4a$$

$$-\frac{9}{2} = a$$

$$\therefore a = -\frac{9}{2}, b = 6, c = 5$$

Using the TI-Nspire

Use **(menu) > Actions > Define** to define

$$f(x) = x^3 + ax^2 + bx + c$$

Find the **derivative** (**(menu) > Calculus >**

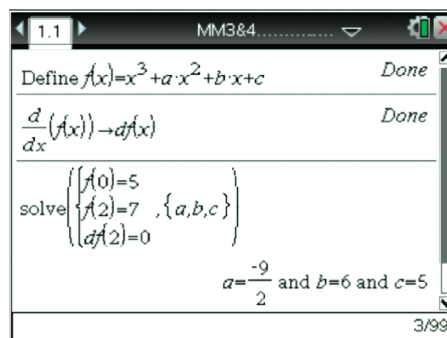
Derivative) of $f(x)$ and **store** (**(ctrl) (var)**) this derivative as $df(x)$ as shown.

Then use the simultaneous equation solver

(**(menu) > Algebra > Solve System of**

Equations > Solve System of Equations)

to find a , b and c if $f(0) = 5$, $f(2) = 7$ and $df(2) = 0$.



Using the Casio ClassPad

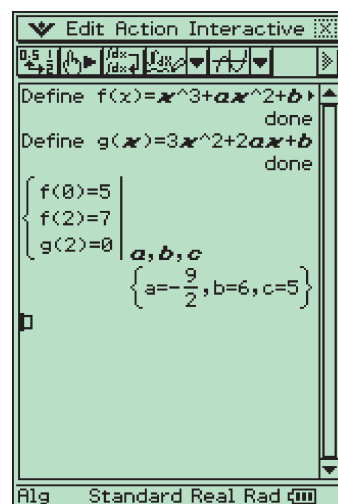
Using **Interactive-Define**, define

$$f(x) = x^3 + ax^2 + bx + c \text{ and}$$

$$g(x) = 3x^2 + 2ax + b.$$

Then turn on the soft keyboard and tap **{** from the **[2D]** menu twice to produce a simultaneous equation template with three lines.

Enter the equations and variables as shown.



Exercise 10D

1 Find the stationary points for each of the following:

a $f(x) = x^3 - 12x$

b $g(x) = 2x^2 - 4x$

c $h(x) = 5x^4 - 4x^5$

d $f(t) = 8t + 5t^2 - t^3$ for $t > 0$

e $g(z) = 8z^2 - 3z^4$

f $f(x) = 5 - 2x + 3x^2$

g $h(x) = x^3 - 4x^2 - 3x + 20, x > 0$

h $f(x) = 3x^4 - 16x^3 + 24x^2 - 10$

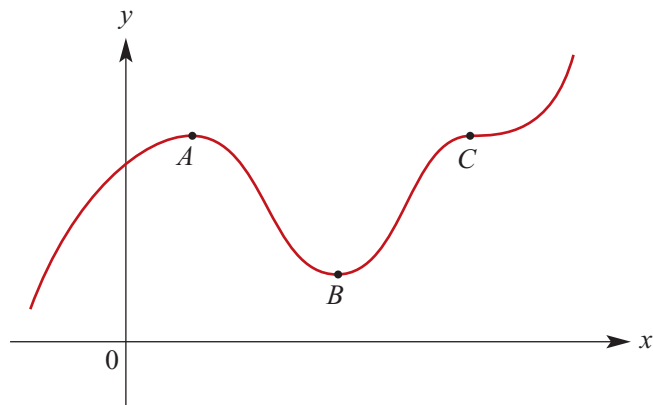
2 **a** The curve with equation $f(x) = x^2 - ax + 9$ has a stationary point when $x = 3$. Find a .

b The curve with equation $h(x) = x^3 - bx^2 - 9x + 7$ has a stationary point when $x = -1$. Find b .

- 3 The tangent to the curve of $y = ax^2 + bx + c$ at the point where $x = 2$ is parallel to the line $y = 4x$. There is a stationary point at $(1, -3)$. Find the value of a , b and c .
- 4 The graph of $y = ax^3 + bx^2 + cx + d$ touches the line $2y + 6x = 15$ at the point $A\left(0, 7\frac{1}{2}\right)$ and has a stationary point at $B(3, -6)$. Find the values of a , b , c , and d .
- 5 The curve with equation $y = ax + \frac{b}{2x - 1}$ has a stationary point at $(2, 7)$. Find:
a the value of a and b **b** the coordinates of the other stationary point
- 6 Find the x -coordinates, in terms of n , of the stationary points of the curve with equation $y = (2x - 1)^n(x + 2)$, where n is a natural number.
- 7 Find the x -coordinates of the stationary points of the curve with equation $y = (x^2 - 1)^n$ where n is an integer greater than 1.
- 8 Find the coordinates of the stationary points of the curve with equation $y = \frac{x}{x^2 + 1}$
- 9 Find the coordinates of the stationary points of the curve with equation:
a $y = |x^2 - 4x|$ **b** $y = |x^3 - 4x|$ **c** $y = |x^3 - 4x^2|$ **d** $y = |x^4 - 4x^2|$

10.5 Types of stationary points

The graph below has three stationary points A , B , C .



- A** The point A is called a **local maximum** point. Notice that immediately to the left of A , the gradient is positive and immediately to right the gradient is negative. A diagram to represent this is:
- B** The point B is called a **local minimum** point. Notice that immediately to the left of B , the gradient is negative and immediately to the right the gradient is positive. A diagram to represent this is:

	+	0	-
gradient	$\nearrow$	—	$\searrow$
	-	0	+
gradient	$\searrow$	—	$\nearrow$

C The point C is called a stationary point of inflexion.

A diagram for this is:

	+	0	+
gradient		—	
	—	0	—
gradient		—	

Clearly it is also possible to have stationary points of inflexion for which our diagram would be:

Stationary points of type A and B are referred to as **turning points**.

Example 15

For the function $f: R \rightarrow R$, $f(x) = 3x^3 - 4x + 1$:

- Find the stationary points and state their nature.
- Sketch the graph.

Solution

- a** $y = f(x)$ has stationary points where $f'(x) = 0$ and $f'(x) = 9x^2 - 4 = 0$ which implies $x = \pm \frac{2}{3}$.

There are stationary points at $\left(-\frac{2}{3}, f\left(-\frac{2}{3}\right)\right)$, and $\left(\frac{2}{3}, f\left(\frac{2}{3}\right)\right)$;

that is, at $\left(-\frac{2}{3}, 2\frac{7}{9}\right)$ and $\left(\frac{2}{3}, -\frac{7}{9}\right)$.

$f'(x)$ is of constant sign on each of:

$$\left\{x: x < -\frac{2}{3}\right\}, \left\{x: -\frac{2}{3} < x < \frac{2}{3}\right\} \text{ and } \left\{x: x > \frac{2}{3}\right\}$$

To calculate the sign of $f'(x)$ on each of these sets, simply choose a representative number in each.

$$\begin{aligned} \text{Thus} \quad f'(-1) &= 9 - 4 = 5 > 0 \\ f'(0) &= 0 - 4 = -4 < 0 \\ f'(1) &= 9 - 4 = 5 > 0 \end{aligned}$$

We can thus put together the following table:

x		$-\frac{2}{3}$		$\frac{2}{3}$	
$f'(x)$	+	0	—	0	+
shape of f		—		—	

$\therefore$ there is a local maximum at $\left(-\frac{2}{3}, 2\frac{7}{9}\right)$ and a local minimum at $\left(\frac{2}{3}, -\frac{7}{9}\right)$.

- b** To sketch the graph of this function we need to find the axis intercepts and investigate the behaviour of the graph for:

$$x > \frac{2}{3} \text{ and } x < -\frac{2}{3}$$

Firstly, $f(0) = 1$, the y -intercept is 1.

Consider $f(x) = 0$, which implies $3x^3 - 4x + 1 = 0$

By inspection (factor theorem) $(x - 1)$ is a factor and by division

$$3x^3 - 4x + 1 = (x - 1)(3x^2 + 3x - 1)$$

Now $(x - 1)(3x^2 + 3x - 1) = 0$ implies that $x = 1$ or $3x^2 + 3x - 1 = 0$

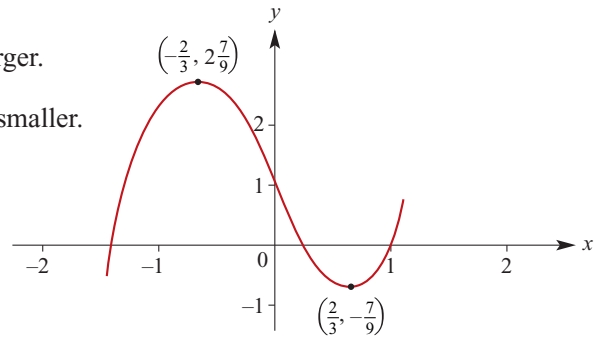
$$\begin{aligned} 3x^2 + 3x - 1 &= 3 \left[\left(x + \frac{1}{2} \right)^2 - \frac{1}{4} - \frac{1}{3} \right] \\ &= 3 \left[\left(x + \frac{1}{2} \right)^2 - \frac{21}{36} \right] \\ &= 3 \left(x + \frac{1}{2} - \frac{1}{6}\sqrt{21} \right) \cdot \left(x + \frac{1}{2} + \frac{1}{6}\sqrt{21} \right) \end{aligned}$$

Thus the x -intercepts are at:

$$x = -\left(\frac{1}{2} + \frac{1}{6}\sqrt{21} \right), \quad x = \frac{1}{6}\sqrt{21} - \frac{1}{2}, \quad x = 1$$

For $x > \frac{2}{3}$, $f(x)$ becomes larger.

For $x < -\frac{2}{3}$, $f(x)$ becomes smaller.



Example 16

Plot the graph of $y = x^3 - 19x + 20$ and determine:

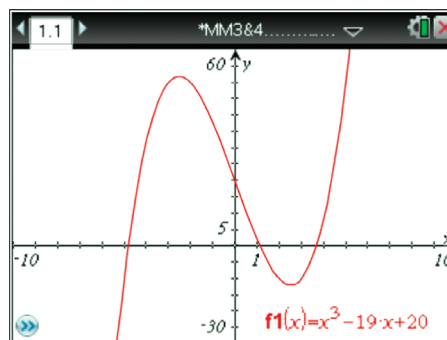
- a** the value of y when $x = -4$ **b** the values of x when $y = 0$
c the value of $\frac{dy}{dx}$ when $x = -1$
d the coordinates of the local maximum.

Once you have plotted the graph of a function, the CAS calculator can be used to determine a number of key values for that graph that include, where appropriate:

- its value at any point ■ the value of its derivative at any point
 ■ its zeros ■ its local maxima ■ its local minima.

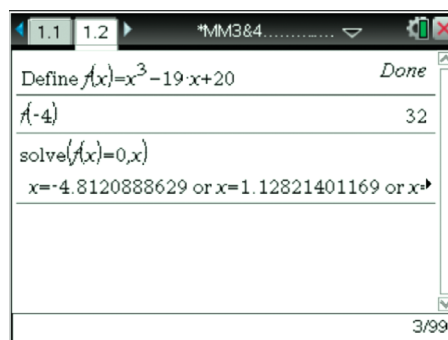
Solution**Using the TI-Nspire**

Graph the equation of
 $y = x^3 - 19x + 20$ in an appropriate
window
 ((menu) > **Window/Zoom** > **Window**
Settings).



Define $f(x) = x^3 - 19x + 20$

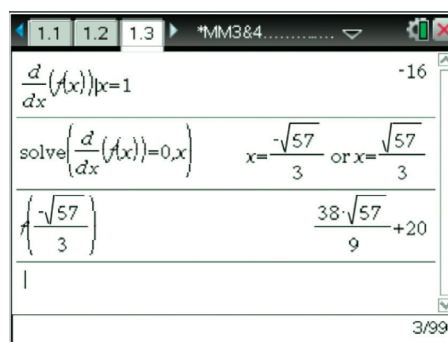
- a** $f(-4) = 32$
b **solve** ($f(x) = 0, x$)



c Find the **derivative** of $f(x)$ at $x = 1$ as shown.

d solve $\left(\frac{d}{dx}(f(x)) = 0, x\right)$ and substitute to find the y -coordinate.

Note: As the function was also defined in the **Graphs** application as $f1(x)$, the variable $f1(x)$ could have been used in place of $f(x)$ in the above calculations.

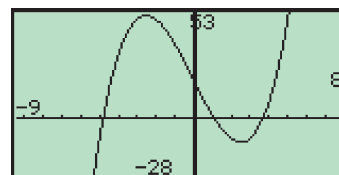


Using the Casio Classpad

Define $y = x^3 - 19x + 20$.

Plot the graph.

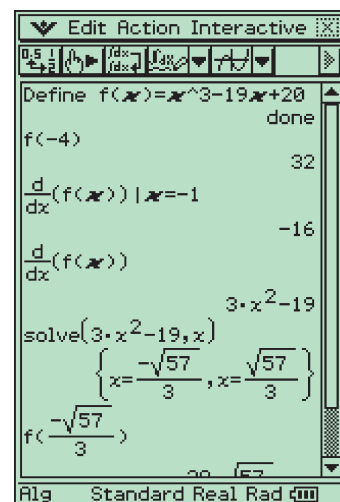
a $f(-4) = 32$



b Enter and highlight $f(x) = 0$ then tap **Interactive–Equation/inequality–solve** and set the variable to x .

c In **2D** **CALC**, tap $\frac{d}{dx}$ and enter x as the variable (solid square) and $f(x)$ as the function (empty square). Enter the value $x = -1$ as shown.

d In the screen the derivative has first been found then copied to a new entry line and solved. The y value can be found by substitution of the solutions into $f(x)$. (Copy and paste for accuracy.)



Example 17

Sketch the graph of $y = |x^3 - 36x|$

Solution

First consider the graph of $y = |x^3 - 36x|$

Intercepts

The x -axis intercepts occur where $x^3 - 36x = 0$

$x(x^2 - 36) = 0$ which implies $x = 0$ or $x = 6$ or $x = -6$

Turning points

$\frac{dy}{dx} = 0$ implies $3x^2 - 36 = 0$

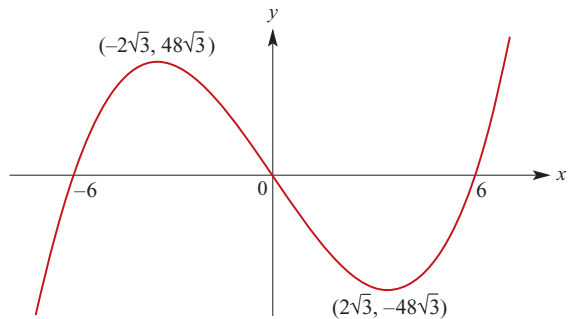
Therefore there are turning points where $x = \pm\sqrt{12} = \pm 2\sqrt{3}$

$$\begin{aligned}\text{When } x = 2\sqrt{3}, y &= (2\sqrt{3})^3 - 36(2\sqrt{3}) \\ &= 24\sqrt{3} - 72\sqrt{3} \\ &= -48\sqrt{3}\end{aligned}$$

$$\begin{aligned}\text{When } x = -2\sqrt{3}, y &= (-2\sqrt{3})^3 - 36(-2\sqrt{3}) \\ &= -24\sqrt{3} + 72\sqrt{3} \\ &= 48\sqrt{3}\end{aligned}$$

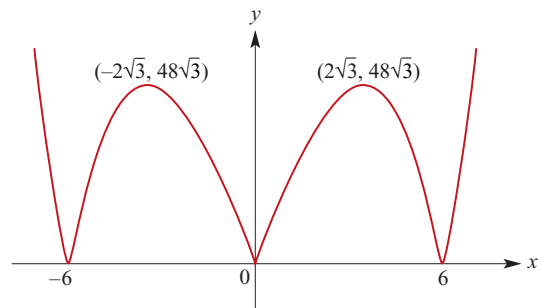
The turning points have coordinates $(2\sqrt{3}, -48\sqrt{3})$ and $(-2\sqrt{3}, 48\sqrt{3})$.

It is clear that $(2\sqrt{3}, -48\sqrt{3})$ is a local minimum and $(-2\sqrt{3}, 48\sqrt{3})$ is a local maximum.



The graph of $y = |x^3 - 36x|$ can now be drawn.

It is clear that $(2\sqrt{3}, 48\sqrt{3})$ and $(-2\sqrt{3}, 48\sqrt{3})$ are local maxima.



Exercise 10E

- 1 Write down the values of x for which the following derived functions are zero and prepare in each case a gradient table, as in Example 15, showing whether the corresponding points on the graph are local maxima, local minima or stationary points of inflexion:

a $f'(x) = 4x^2$

b $f'(x) = (x - 2)(x + 5)$

c $f'(x) = (x + 1)(2x - 1)$

d $f'(x) = -x^2 + x + 12$

e $f'(x) = x^2 - x - 12$

f $f'(x) = 5x^4 - 27x^3$

g $f'(x) = (x - 1)(x - 3)$

h $f'(x) = -(x - 1)(x - 3)$

- 2 Find the stationary points on each of the following curves and state their nature:

a $y = x(x^2 - 12)$

b $y = x^2(3 - x)$

c $y = x^3 - 5x^2 + 3x + 2$

d $y = 3 - x^3$

e $y = 3x^4 + 16x^3 + 24x^2 + 3$

f $y = x(x^2 - 1)$



- 3 Sketch the graphs of each of the following, finding:

i axes intercepts

ii stationary points

a $y = 4x^3 - 3x^4$

b $y = x^3 - 6x^2$

c $y = 3x^2 - x^3$

d $y = x^3 + 6x^2 + 9x + 4$

e $y = (x^2 - 1)^5$

f $y = (x^2 - 1)^4$

- 4 **a** Find the stationary points of the graph $y = 2x^3 + 3x^2 - 12x + 7$, stating the nature of each.
b Show that the graph passes through $(1, 0)$.
c Find the other axes intercepts. **d** Sketch the graph.
- 5 **a** Show that the polynomial $P(x) = x^3 + ax^2 + b$ has a stationary point at $x = 0$ for all a and for all b .
b Given that $P(x)$ has a second stationary point at $(-2, 6)$, find the values of a and b and the nature of both stationary points.
- 6 Sketch the graph of $f(x) = (2x - 1)^5(2x - 4)^4$. Clearly state the coordinates of:
a axes intercepts **b** stationary points
 State the nature of each of the stationary points.
- 7 **a** Sketch the graph of $f(x) = (4x^2 - 1)^6$ and $g(x) = (4x^2 - 1)^5$ on the one set of axes.
b Find:
i $\{x: (4x^2 - 1)^6 > (4x^2 - 1)^5\}$ **ii** $\{x: f'(x) > g'(x)\}$

- 8** Sketch the graph of each of the following. State the axes intercepts and coordinates of stationary points.

a $y = x^3 + x^2 - 8x - 12$

b $y = 4x^3 - 18x^2 + 48x - 290$

- 9** Find the coordinates of the stationary points of each of the following and determine their nature:

a $f(x) = 3x^4 + 4x^3$

b $f(x) = x^4 + 2x^3 - 1$

c $f(x) = 3x^3 - 3x^2 + 12x + 9$

- 10** Consider the function f defined by:

$$f(x) = \frac{1}{8}(x-1)^3(8-3x) + 1$$

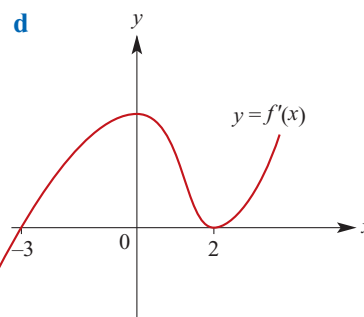
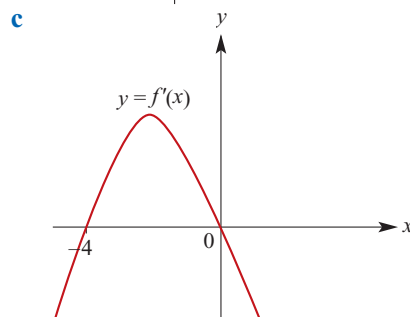
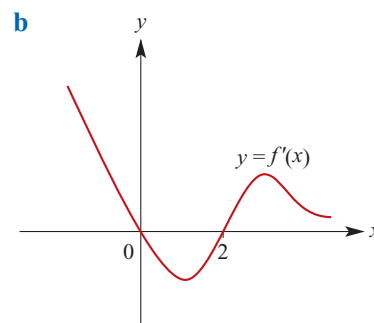
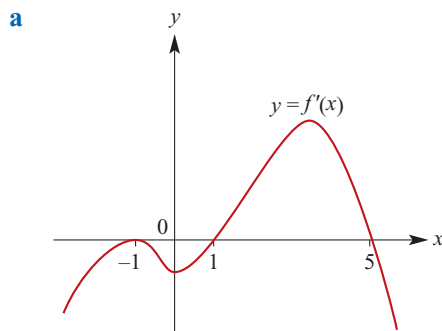
- a** Show that $f(0) = 0$ and $f(3) = 0$.

- b** Show that $f'(x) = \frac{3}{8}(x-1)^2(9-4x)$ and specify the value of x for which $f'(x) \geq 0$.

- c** Sketch the graph of $y = f(x)$

- 11** Sketch the graph of $y = 3x^4 - 44x^3 + 144x^2$, finding the coordinates of all turning points.

- 12** The graphs below show the graph of f' of a function f . Find the values of x for which the graph of $y = f(x)$ has a stationary point and state the nature of the stationary point.



- 13** Find the coordinates of the stationary points, and state the nature of each, for the curve with equation:

a $y = x^4 - 16x^2$

b $y = |x^4 - 16x^2|$

c $y = |x^{2m} - 16x^{2m-2}|$ where m is a positive integer greater than or equal to 2

14 Let $f: R \rightarrow R$, $f(x) = x^3 - 3x^2 - 9x + 11$. Sketch the graph of:

- a $y = f(x)$ b $y = 2f(x)$ c $y = f(x + 2)$
 d $y = f(x - 2)$ e $y = -f(x)$

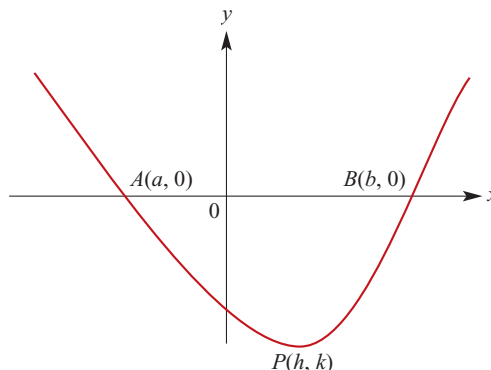
15 Let $f: R \rightarrow R$, $f(x) = 2 + 3x - x^3$. Sketch the graph of:

- a $y = f(x)$ b $y = -2f(x)$ c $y = 2f(x - 1)$
 d $y = f(x) - 3$ e $y = 3f(x + 1)$

16 The graph shown opposite has equation $y = f(x)$. Suppose a dilation of factor p from the x -axis followed by a translation of l units in the positive direction of the x -axis is applied to the graph.

For the graph of the image, state:

- a the axes intercepts
 b the coordinates of the turning point.



17 It is found that the graph of quartic function passes through the points with coordinates $(1, 21)$, $(2, 96)$, $(5, 645)$, $(6, 816)$, $(7, 861)$.

- a Find the rule of the quartic and plot the graph. Determine the turning points and axes intercepts.
 b Plot the derivative graph on the same screen.
 c Find the value of the function when $x = 10$.
 d For what value(s) of x is the value of the function 500?

10.6 Absolute maxima and minima

In the last section local maxima and minima were discussed. These are often not the actual maximum and minimum values of the function. The actual maximum value for a function defined on an interval is called the **absolute maximum**. The corresponding point on the graph of the function is not necessarily a stationary point. The actual minimum value for a function defined on an interval is called the **absolute minimum**. The corresponding point on the graph of the function is not necessarily a stationary point.

M is the absolute maximum value of a continuous function f for an interval $[a, b]$ if $f(x) \leq M$ for all $x \in [a, b]$.

N is the absolute minimum value of a continuous function f for an interval $[a, b]$ if $f(x) \geq N$ for all $x \in [a, b]$.

Example 18

Let $f: [-2, 4] \rightarrow R$, $f(x) = x^2 + 2$. Find the absolute maximum and the absolute minimum value of the function.

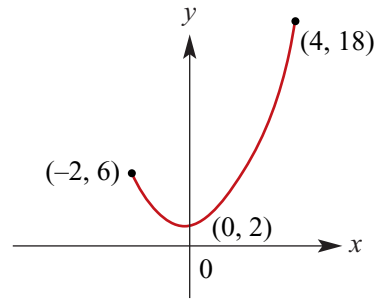
Solution

The minimum value occurs when $x = 0$ and is 2.

The maximum value occurs when $x = 4$ and is 18.

The minimum value occurs at a stationary point of the graph but the endpoint $(4, 18)$ is not a stationary point.

The absolute maximum value is 18 and the absolute minimum value is 2.

**Example 19**

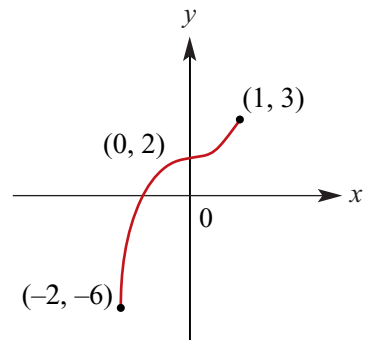
Let $f: [-2, 1] \rightarrow \mathbb{R}$, $f(x) = x^3 + 2$. Find the maximum and minimum value of the function.

Solution

The minimum value occurs when $x = -2$ and is -6 .

The maximum value occurs when $x = 1$ and is 3.

The absolute minimum value and the absolute maximum value do not occur at stationary points.

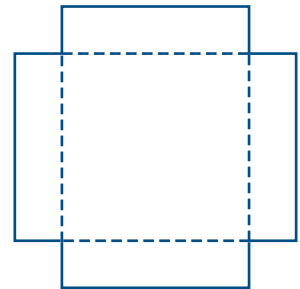
**Example 20**

From a square piece of metal of side length 2 m, four squares are removed as shown in the figure opposite.

The metal is then folded about the dotted lines to give an open box with sides of height x m.

- Show that the volume, $V \text{ m}^3$, is given by

$$V = 4x^3 - 8x^2 + 4x$$
- Sketch the graph of V against x for a suitable domain.
- If the height of the box must be less than 0.3 m, i.e. $x \leq 0.3$, what will be the maximum volume of the box?

**Solution**

- Length of box = $(2 - 2x)$ metres, height = x metres
 $\therefore$ volume = $(2 - 2x)^2 x$
 $= (4 - 8x + 4x^2)x$
 $= 4x^3 - 8x^2 + 4x$ cubic metres

b Let $V = 4x^3 - 8x^2 + 4x$

Local maximum point will occur when $\frac{dV}{dx} = 0$

$$\frac{dV}{dx} = 12x^2 - 16x + 4$$

and $\frac{dV}{dx} = 0$ implies

$$12x^2 - 16x + 4 = 0$$

$$\therefore 3x^2 - 4x + 1 = 0$$

$$\therefore (3x - 1)(x - 1) = 0$$

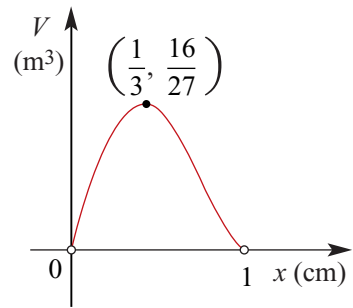
$$\therefore x = \frac{1}{3} \text{ or } x = 1$$

Note: When $x = 1$, the length of the box $= 2 - 2 \times 1$ which is zero.

$\therefore$ the only value to be considered is $x = \frac{1}{3}$

$\therefore$ a local maximum occurs when $x = \frac{1}{3}$

$$\begin{aligned} \text{and volume} &= \left(2 - 2 \times \frac{1}{3}\right)^2 \times \frac{1}{3} \\ &= \frac{16}{9} \times \frac{1}{3} \\ &= \frac{16}{27} \text{ m}^3 \end{aligned}$$



c The local maximum of $V(x)$ defined on $[0, 1]$ was at $\left(\frac{1}{3}, \frac{16}{27}\right)$.

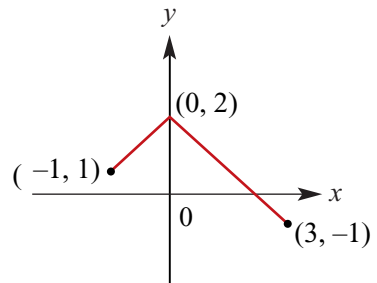
But for the new problem $V'(x) > 0$ for all $x \in [0, 0.3]$ and $\frac{1}{3}$ is not in this interval. Therefore the maximum volume occurs when $x = 0.3$ and is 0.588.

Example 21

For the function $f: [-1, 3] \rightarrow \mathbb{R}$, $f(x) = 2 - |x|$, sketch the graph and state the absolute maximum and minimum value of the function.

Solution

There are no stationary points, but the function has an absolute maximum value of 2 when $x = 0$. It has a minimum value of -1 when $x = 3$.



Exercise 10F

- 1 Let $f: [-3, 2] \rightarrow R$, $f(x) = x^3 + 2x + 3$. Find the absolute maximum and absolute minimum value of the function for its domain.
- 2 Let $f: [-1.5, 2.5] \rightarrow R$, $f(x) = 2x^3 - 6x^2$. Find the absolute maximum and absolute minimum value of the function.
- 3 Let $f: [-2, 6] \rightarrow R$, $f(x) = 2x^4 - 8x^2$. Find the absolute maximum and absolute minimum value of the function.
- 4 Let $f: [-3, 3] \rightarrow R$, $f(x) = 2 - 8x^2$. Find the absolute maximum and absolute minimum value of the function.
- 5 A rectangular block is such that the sides of its base are of length x cm and $3x$ cm. The sum of the lengths of all its edges is 20 cm.
 - a Show that the volume V cm³ is given by $V = 15x^2 - 12x^3$
 - b Find $\frac{dV}{dx}$.
 - c Find the coordinates of the local maximum of the graph of V against x for $x \in [0, 1.25]$.
 - d If $x \in [0, 0.8]$ find the absolute maximum value of V and the value of x for which this occurs.
 - e If $x \in [0, 1]$ find the absolute maximum value of V and the value of x for which this occurs.
- 6 For the variables x , y and z it is known that $x + y = 30$ and $z = xy$
 - a If $x \in [2, 5]$ find the possible values of y .
 - b Find the absolute maximum and minimum values of z .
- 7 A piece of string 10 metres long is cut into two pieces to form two squares.
 - a If one piece of string has length x metres show that the combined area of the two squares is given by $A = \frac{1}{8}(x^2 - 10x + 50)$
 - b Find $\frac{dA}{dx}$.
 - c Find the value of x that makes A a minimum.
 - d If two squares are formed but $x \in [0, 11]$, find the maximum possible area of the two squares.
- 8 Find the absolute maximum and minimum values of the function
 $g: [2.1, 8] \rightarrow R$, $g(x) = x + \frac{1}{x-2}$
- 9 Consider the function $f: [2, 3] \rightarrow R$, $f(x) = \frac{1}{x-1} + \frac{1}{4-x}$.
 - a Find $f'(x)$.
 - b Find the coordinates of the stationary point of the graph of $y = f(x)$.
 - c Find the absolute maximum and absolute minimum of the function.

- 10 Consider the function $f: [0, 3] \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x+1} + \frac{1}{4-x}$.
- Find $f'(x)$.
 - Find the coordinates of the stationary point of the graph of $y = f(x)$
 - Find the absolute maximum and absolute minimum of the function.
- 11 For the function $f: [-1, 8] \rightarrow \mathbb{R}$, $f(x) = 2 - x^{\frac{2}{3}}$, sketch the graph and state the absolute maximum and minimum value of the function.

10.7 Maxima and minima problems

Many practical problem require that some quantity (for example, cost of manufacture or fuel consumption) be **minimised**, that is, be made as small as possible. Other problems require that some quantity (for example, profit on sales or attendance at a concert) be **maximised**, that is, be made as large as possible. We can use differential calculus to solve many of these problems.

Example 22

A farmer has sufficient fencing to make a rectangular pen of perimeter 200 metres. What dimensions will give an enclosure of maximum area?

Solution

Let the required length of fencing be x metres. Then the width is $100 - x$ metres and the area is A square metres where $A = x(100 - x) = 100x - x^2$

The maximum value of A occurs when $\frac{dA}{dx} = 0$

$$\frac{dA}{dx} = 100 - 2x$$

$$\text{and } \frac{dA}{dx} = 0 \text{ implies } x = 50$$

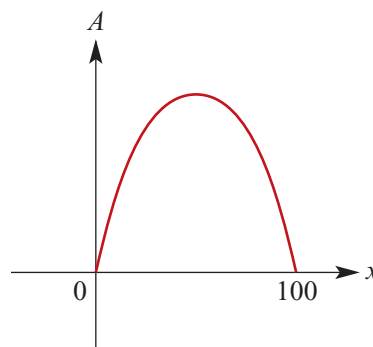
To see that $x = 50$ gives a maximum we may use the gradient chart

x		50	
$\frac{dA}{dx}$	+	0	-
slope	/	—	\

$\therefore$ maximum when $x = 50$ and $A = 50(100 - 50) = 2500$

$\therefore$ maximum area = 2500 square metres

$\therefore$ the corresponding dimensions are 50 m by 50 m.



Example 23

Two variables x and y are such that $x^4y = 8$. A third variable z is defined by $z = x + y$. Find the values of x and y that give z a stationary value and show that this value of z is a minimum.

Solution

Obtain y in terms of x from the equation $x^4y = 8$

$$y = 8x^{-4}$$

Substitute in the equation $z = x + y$

$$z = x + 8x^{-4} \quad (1)$$

Therefore z is in terms of one variable, x .

Differentiate with respect to x :

$$\frac{dz}{dx} = 1 - 32x^{-5}$$

Stationary point occurs where $\frac{dz}{dx} = 0$

$$1 - 32x^{-5} = 0$$

$$\therefore 32x^{-5} = 1$$

$$\therefore x^5 = 32$$

$$\therefore x = 2 \text{ and the corresponding value of } y \text{ is } 8 \times 2^{-4} = \frac{1}{2}$$

Now substitute in equation (1) to find z :

$$\begin{aligned} z &= 2 + \frac{8}{16} \\ &= 2\frac{1}{2} \end{aligned}$$

Determine nature of stationary point by gradient chart:

		2	
$\frac{dz}{dx}$	-		+
	$\searrow$	$-$	$\nearrow$

This gives a minimum at $\left(2, 2\frac{1}{2}\right)$.

Example 24

A cylindrical tin canister closed at both ends has a surface area of 100 cm^2 . Find, correct to two decimal places, the greatest volume it can have. If the radius of the canister can be at most 2 cm, find the greatest volume it can have.

Solution

Let the radius of the circular end of the tin be r cm and the height of the tin h cm. Also let the volume of the tin be $V \text{ m}^3$.

Obtain equations for the surface area and the volume.

$$\text{Surface area: } 2\pi r^2 + 2\pi r h = 100 \quad (1)$$

$$\text{Volume: } V = \pi r^2 h \quad (2)$$

The process we follow now is very similar to Example 23.

Obtain h in terms of r from equation (1):

$$h = \frac{1}{2\pi r}(100 - 2\pi r^2)$$

Substitute in equation (2):

$$\begin{aligned} V &= \pi r^2 \times \frac{1}{2\pi r}(100 - 2\pi r^2) \\ \therefore V &= 50r - \pi r^3 \end{aligned} \quad (3)$$

The stationary point of the graph of $V = 50r - \pi r^3$ occurs when $\frac{dV}{dr} = 0$
 $\frac{dV}{dr} = 0$ implies $50 - 3\pi r^2 = 0$

$$\begin{aligned} \therefore r &= \pm \sqrt{\frac{50}{3\pi}} \\ &\approx 2.3 \end{aligned}$$

But $r = -\sqrt{\frac{50}{3\pi}}$ does not fit the practical situation.

Substitute in equation (3) to find V :

$$V \approx 76.78$$

The maximum volume is 76.78 cm^3 correct to two decimal places.

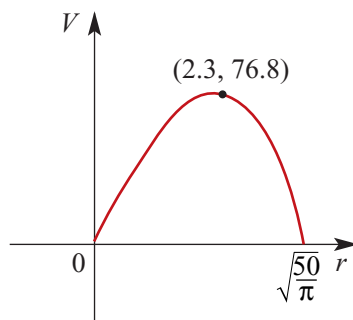
Determine the nature of the stationary point.

	2.3	
+	0	-
/	—	\

giving a maximum at (2.3, 76.8).

It can be observed that the volume is given by a function f with rule $f(r) = 50r - \pi r^3$ and

domain $\left(0, \sqrt{\frac{50}{\pi}}\right)$, giving the graph.



If the greatest radius the canister can have is 2 cm, then the function f has domain $[0, 2]$. It has been seen that $f'(r) > 0$ for all $r \in [0, 2]$. The maximum value occurs when $r = 2$. The maximum volume in this case is $f(2) = 100 - 8\pi \approx 74.87 \text{ cm}^3$.

In some situations the variables may not be continuous. For instance one or the other of them may only take integer values. In such cases it is not strictly valid to use techniques of differentiation to solve the problem. However, in some problems we may model the non-continuous case with a continuous function so that the techniques of differential calculus may be used. Worked examples 25 and 26 illustrate this.

Example 25

A TV cable company has 1000 subscribers who are paying \$5 per month. It can get 100 more subscribers for each \$0.10 decrease in the monthly fee. What rate will yield maximum revenue and what will this revenue be?

Solution

Let x denote the rate.

$$\text{Then the number of subscribers} = 1000 + 100 \left(\frac{5 - x}{0.1} \right)$$

(Note that we are treating a discrete situation with a continuous function.)

$$\begin{aligned} \text{And the revenue} &= x(1000 + 1000(5 - x)) \\ &= 1000(6x - x^2) \end{aligned}$$

Let R denote the revenue.

$$\text{Then } \frac{dR}{dx} = 1000(6 - 2x)$$

$$\frac{dR}{dx} = 0 \text{ implies } 6 - 2x = 0$$

Hence $x = 3$

The gradient chart is:

x		3	
$f'(x)$	+	0	-
the shape of f	/	—	\

Thus for maximum revenue the rate should be \$3 and this gives a total revenue of \$9000.

Example 26

A manufacturer annually produces and sells 10 000 shirts. Sales are uniformly distributed throughout the year. The production cost of each shirt is \$23 and other costs (storage, insurance, interest) depend on the total number of shirts in a production run. (A production run is the number, x , of shirts which are under production at a given time.)

The other costs (annually) are $\$x^{\frac{3}{2}}$. The set-up costs for a production run are \$40. Find the size of a production run which will minimise the other costs for a year.

Solution

The number of production runs per year = $\frac{10\,000}{x}$

The set-up costs for these production runs = $40 \left(\frac{10\,000}{x} \right)$

The total set-up and carrying costs, $C = x^{\frac{3}{2}} + \frac{400\,000}{x}$, $x > 0$

$$\therefore C = x^{\frac{3}{2}} + 400\,000x^{-1}$$

$$\frac{dC}{dx} = \frac{3}{2}x^{\frac{1}{2}} - \frac{400\,000}{x^2}$$

$$\frac{dC}{dx} = 0 \text{ implies}$$

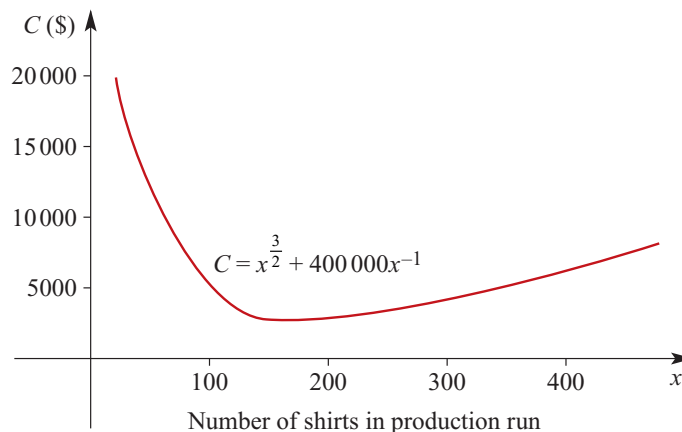
$$\frac{3}{2}x^{\frac{1}{2}} = \frac{400\,000}{x^2}$$

$$\therefore x^{\frac{5}{2}} = \frac{400\,000 \times 2}{3}$$

$$\therefore x \approx 148.04$$

x		148.04	
$\frac{dC}{dx}$	-	0	+
shape	$\searrow$	—	$\nearrow$

Each production run should be 148 items.



Exercise 10G

- 1 Find two positive numbers whose sum is 4 and such that the sum of the cube of the first and the square of the second is as small as possible.
- 2 Find the point on the parabola $y = x^2$ that is closest to the point (3, 0).
- 3 The number of salmon swimming upstream in a river to spawn is approximated by $s(x) = -x^3 + 3x^2 + 360x + 5000$ with x representing the temperature of the water in degrees ($^{\circ}\text{C}$). (This function is valid only if $6 \leq x \leq 20$.) Find the water temperature that produces the maximum number of salmon swimming upstream.
- 4 The number of mosquitoes, $M(x)$ in millions, in a certain area depends on the September rainfall, x , measured in mm, and is approximately given by:

$$M(x) = \frac{1}{30}(50 - 32x + 14x^2 - x^3)$$

Find the rainfall that will produce the maximum and the minimum number of mosquitoes.

- 5 Find the maximum area of a rectangular piece of ground that can be enclosed by 100 m of fencing.
- 6 For $x + y = 100$ prove that the product $P = xy$ is a maximum when $x = y$ and find the maximum value of P .
- 7 A farmer has 4 km of fencing wire and wishes to fence a rectangular piece of land through which flows a straight river, which is to be utilised as one side of the enclosure. How can this be done to enclose as much land as possible?
- 8 Two positive quantities p and q vary in such a way that $p^3q = 9$. Another quantity z is defined by $z = 16p + 3q$. Find values of p and q that make z a minimum.
- 9 A cuboid has a total surface area of 150 cm^2 with a square base of side $x \text{ cm}$.
 - a Show that the height, $h \text{ cm}$, of the cuboid is given by $h = \frac{75 - x^2}{2x}$
 - b Express the volume of the cuboid in terms of x .
 - c Hence determine its maximum volume as x varies.

10.8 Rates of change

The derivative of a function has been used to find the gradient of the corresponding curve. It is clear that the process of differentiation may be used to tackle problems involving rates of change of many kinds.

The derivative of y with respect to x , $\frac{dy}{dx}$, gives the rate of change of y with respect to x . Of course if $\frac{dy}{dx} > 0$ the change is an increase in the value of y corresponding to an increase in x , and if $\frac{dy}{dx} < 0$ the change is a decrease in the value of y corresponding to an increase in x .



Example 27

A balloon which develops a microscopic leak will decrease in volume. Its volume $V \text{ cm}^3$ at time t seconds is $V = 600 - 10t - \frac{1}{100}t^2$, $t > 0$.

- a** Find the rate of change of volume after:
i 10 seconds **ii** 20 seconds
b For how long could the model be valid?

Solution

a $\frac{dV}{dt} = -10 - \frac{t}{50}$

i When $t = 10$,

$$\begin{aligned}\frac{dV}{dt} &= -10 - \frac{1}{5} \\ &= -10\frac{1}{5}\end{aligned}$$

i.e. the volume is decreasing at a rate of $10\frac{1}{5} \text{ cm}^3$ per second.

ii When $t = 20$,

$$\begin{aligned}\frac{dV}{dt} &= -10 - \frac{2}{5} \\ &= -10\frac{2}{5}\end{aligned}$$

i.e. the volume is decreasing at a rate of $10\frac{2}{5} \text{ cm}^3$ per second.

- b** The model will not be meaningful when $V < 0$. Consider $V = 0$.

$$\begin{aligned}600 - 10t - \frac{1}{100}t^2 &= 0 \\ \therefore t &= \frac{10 \pm \sqrt{100 + \frac{1}{100} \times 600 \times 4}}{-0.02} \\ \therefore t &= -1056.78 \text{ or } t = 56.78 \text{ (to two decimal places)} \\ \therefore \text{the model may be suitable for } 0 < t < 56.78\end{aligned}$$

Displacement, velocity and acceleration were introduced for a body moving in a straight line in earlier work.* Displacement was specified with respect to a reference point O on that line. For velocity ($v \text{ m/s}$):

$$v = \frac{ds}{dt}$$

and acceleration ($a \text{ m/s}^2$)

$$a = \frac{dv}{dt}$$

Example 28

A point moves along a straight line so that its distance $x \text{ cm}$ from a point O at time t seconds is given by the formula $x = t^3 - 6t^2 + 9t$. Find:

- a** at what times and in what positions the point will have zero velocity
b its acceleration at those instants **c** its velocity when its acceleration is zero

* Essential Mathematical Methods 1 & 2 CAS

Solution

$$\begin{aligned}
 \text{Velocity} &= v \\
 &= \frac{dx}{dt} \\
 &= 3t^2 - 12t + 9
 \end{aligned}$$

a When $v = 0$,

$$\begin{aligned}
 3(t^2 - 4t + 3) &= 0 \\
 (t - 1)(t - 3) &= 0 \\
 t &= 1 \text{ or } t = 3
 \end{aligned}$$

i.e. the velocity is zero when
 $t = 1$ and $t = 3$ and where
 $x = 4$ and $x = 0$

c The acceleration is zero when $6t - 12 = 0$, i.e. when $t = 2$

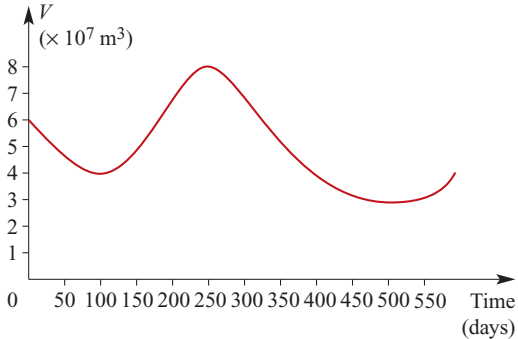
When $t = 2$, the velocity $v = 3 \times 4 - 24 + 9$
 $= -3 \text{ m/s}$

b Acceleration $= \frac{dv}{dt}$
 $= 6t - 12$
 $\therefore$ acceleration $= -6 \text{ m/s}^2$ when $t = 1$ and
 acceleration $= 6 \text{ m/s}^2$ when $t = 3$

Exercise 10H

- Express each of following in symbols:
 - the rate of change of volume (V) with respect to time (t)
 - the rate of change of surface area (S) of a sphere with respect to radius (r)
 - the rate of change of volume (V) of a cube with respect to edge length (x)
 - the rate of change of area (A) with respect to time (t)
 - the rate of change of volume (V) of water in a glass, with respect of depth of water (h)
- If your interest (I) in *Essential Mathematical Methods* can be expressed as $I = \frac{4}{(t+1)^2}$, where t is the time in days measured from the first day of Term 1, how fast is your interest waning when $t = 10$?
- A reservoir is being emptied, and the quantity of water, $V \text{ m}^3$, remaining in the reservoir t days after it starts to empty is given by $V(t) = 10^3(90 - t)^3$.
 - At what rate is the reservoir being emptied at time t ?
 - How long does it take to empty the reservoir?
 - What is the volume of water in the reservoir when $t = 0$?
 - After what time is the reservoir being emptied at $3 \times 10^5 \text{ m}^3/\text{day}$?
 - Sketch the graph of $V(t)$ against t .
 - Sketch the graph of $V'(t)$ against t .
- A coffee percolator allows 1000 mL of water to flow into a filter in 20 minutes. The volume which has flowed into the filter t minutes from the start is given by:

$$V(t) = \frac{1}{160} \left(5t^4 - \frac{t^5}{5} \right), 0 \leq t \leq 20$$

- a** At what rate is the water flowing into the filter at time t minutes?
- b** Sketch the graph of $\frac{dV}{dt}$ against t for $0 \leq t \leq 20$.
- c** When is the rate of flow greatest?
- 5** The graph shown below is that of the volume, $V \text{ m}^3$, of water in a reservoir at time t days.
- a** At what times is the rate of flow from the reservoir $0 \text{ m}^3/\text{day}$?
- b** Find an estimate for the flow at $t = 200$.
- c** Find the average rate of flow for the interval $[100, 250]$.
- d** State the times for which there is net flow into the reservoir.
- 
- 6** A point moves along a straight line so that its distance, $x \text{ cm}$, from a point O at time t seconds is given by $x = 2t^3 - 9t^2 + 12t$. Find:
- a** the velocity v as a function of t
- b** at what times and in what positions the point will have zero velocity
- c** its acceleration at those instants
- d** its velocity when its acceleration is zero
- 7** A particle moves in a straight line such that its position $x \text{ cm}$ from a point O at time t seconds is given by the equation $x = 8 + 2t - t^2$. Find:
- a** its initial position
- b** its initial velocity
- c** when and where the velocity is zero
- d** its acceleration at time t
- 8** A particle is moving in a straight line such that its position $x \text{ cm}$ from a point O at time t seconds is given by $x = \sqrt{2t^2 + 2}$. Find:
- a** the velocity as a function of t
- b** the acceleration as a function of t
- c** the velocity and acceleration when $t = 1$

10.9 Related rates of change

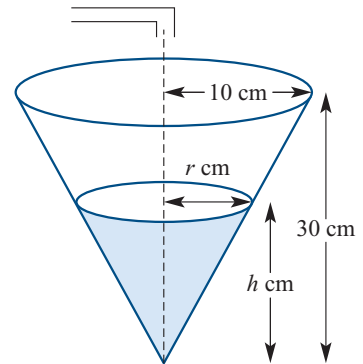
Consider the situation of a container, which is a right circular cone, being filled from a tap.

At time t seconds:

- there are $V \text{ cm}^3$ of water in the cone
- the height of the water in the cone is $h \text{ cm}$
- the radius of the circular water surface is $r \text{ cm}$.

V , h and r change as the water flows in:

- $\frac{dV}{dt}$ is the rate of change of volume with respect to time.
- $\frac{dh}{dt}$ is the rate of change of height with respect to time.
- $\frac{dr}{dt}$ is the rate of change of radius with respect to time.



It is clear that these rates are related to each other. The chain rule is used to establish these relationships.

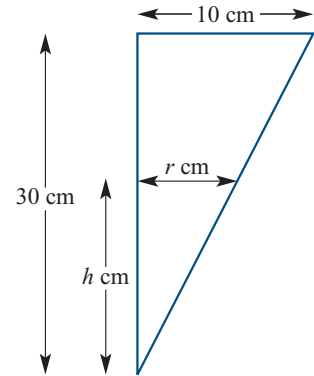
For example, if the height of the cone is 30 cm and the radius of the cone is 10 cm, similar triangles yield

$$\frac{r}{h} = \frac{10}{30}$$

and $h = 3r$

Then the chain rule is used.

$$\begin{aligned}\frac{dh}{dt} &= \frac{dh}{dr} \frac{dr}{dt} \\ &= 3 \cdot \frac{dr}{dt}\end{aligned}$$



The volume of a cone is given in general by $V = \frac{1}{3}\pi r^2 h$ and in this case $V = \pi r^3$ since $h = 3r$. Therefore by using the chain rule again:

$$\begin{aligned}\frac{dV}{dt} &= \frac{dV}{dr} \frac{dr}{dt} \\ &= 3\pi r^2 \cdot \frac{dr}{dt}\end{aligned}$$

The relationships between the rates have been established.

Example 29

The radius of a circular disc is increasing at a constant rate of 0.005 cm/s. Find the rate at which the area of the disc is increasing when the radius is 20 cm.

Solution

Let $A \text{ cm}^2$ be the area of the disc and $r \text{ cm}$ the radius of the disc.

$$A = \pi r^2 \text{ and } \frac{dA}{dr} = 2\pi r$$

$$\begin{aligned}\text{Then } \frac{dA}{dt} &= \frac{dA}{dr} \times \frac{dr}{dt} \\ &= 2\pi r \times 0.005 \\ &= \frac{\pi r}{100}\end{aligned}$$

$$\begin{aligned}\text{When } r = 20, \frac{dA}{dt} &= \frac{20\pi}{100} \\ &= \frac{\pi}{5} \text{ cm}^2/\text{s}\end{aligned}$$

The area of the disc is increasing at a rate of $\frac{\pi}{5} \text{ cm}^2/\text{s}$.

Example 30

Variables x and y are related by the equation $y = \frac{2x - 4}{x}$. Given that x and y are functions of t and that $\frac{dx}{dt} = 4$, find:

- a** $\frac{dy}{dt}$ in terms of x **b** $\frac{dy}{dt}$ when $x = 4$

Solution

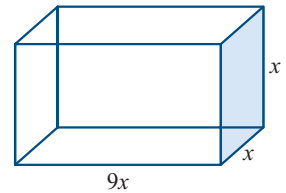
a For $y = \frac{2x - 4}{x}$, $y = 2 - \frac{4}{x}$ and $\frac{dy}{dx} = \frac{4}{x^2}$ **b** When $x = 4$, $\frac{dy}{dt} = \frac{16}{4^2} = 1$

$$\begin{aligned}\text{Therefore } \frac{dy}{dt} &= \frac{dy}{dx} \times \frac{dx}{dt} \\ &= \frac{4}{x^2} \times 4 \\ &= \frac{16}{x^2}\end{aligned}$$

Example 31

The diagram shows a block of ice. Dimensions are in centimetres.

- a** Find an expression for the surface area, $S \text{ cm}^2$, in terms of x .
b Find $\frac{dS}{dx}$.
c The ice is melting so that S is decreasing at a rate of $4 \text{ cm}^2/\text{s}$. Find the rate at which x is changing when $x = 5$.

**Solution**

a $S = 9x^2 + 9x^2 + 9x^2 + 9x^2 + x^2 + x^2$ **b** $\frac{dS}{dx} = 76x$
 $= 38x^2$

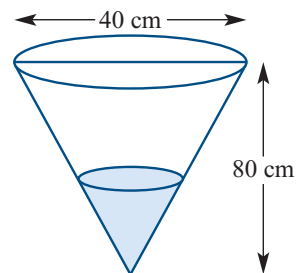
c $\frac{dS}{dt} = \frac{dS}{dx} \times \frac{dx}{dt}$
 $-4 = 76x \times \frac{dx}{dt}$
 $\frac{-4}{76x} = \frac{dx}{dt}$

$$\begin{aligned}\text{When } x = 5, \frac{dx}{dt} &= \frac{-4}{76 \times 5} \\ &= \frac{-1}{95}\end{aligned}$$

x is decreasing at a rate of $\frac{1}{95} \text{ cm/s}$.

Exercise 10I

- 1 A circular patch of oil spreads out from a point on a lake so that the area of the patch grows at a rate of $3 \text{ cm}^2/\text{s}$. At what rate is the radius increasing when the radius of the oil patch is 5 cm?
- 2 The edges of a square are increasing in length at a rate of 0.2 cm/s . Find the rate at which the area of the square is increasing when the length of each side of the square is 20 cm.
- 3 A circular slick is increasing at a rate of $5 \text{ cm}^2/\text{s}$. Find the rate of increase of the radius of the slick when the area is $400\pi \text{ cm}^2$.
- 4 When the depth of liquid in a container is $x \text{ cm}$ the volume is $x(x^2 + 36) \text{ cm}^3$. Liquid is added to the container at a rate of $3 \text{ cm}^3/\text{s}$. Find the rate of change of the depth of liquid at the instant when $x = 11$.
- 5 The surface area of a cube is changing at the rate of $8 \text{ cm}^2/\text{s}$. How fast is the volume changing when the surface area is 60 cm^2 ?
- 6 A vessel has such a shape that when the depth of the water in it is $x \text{ cm}$, the volume, $V \text{ cm}^3$, is given by the rule $V = 108x + x^2$. Water is poured in at a constant rate of $30 \text{ cm}^3/\text{s}$. At what rate is the level of water rising when the depth is 8 cm?
- 7 A melting snowball which is always spherical in shape is decreasing in volume at a constant rate of $8 \text{ cm}^3/\text{min}$. Find the rate at which the radius is changing when the snowball has a radius of 4 cm. (The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.)
- 8 Variables x and y are related by the equation $y = (x^2 - 5)^{10}$. Given that x and y are functions of t and that $\frac{dx}{dt} = 1$, find:
 - a $\frac{dy}{dt}$ in terms of x
 - b $\frac{dy}{dt}$ when $x = 4$
- 9 A species of tree has a perfectly cylindrical trunk. It grows so that the radius of the trunk increases uniformly at a rate of 0.01 metre/year . The height of the trunk remains constant. One particular tree has radius 0.2 metres and height 5.0 metres. Find the rate at which the volume of its trunk is increasing.
- 10 The diagram shows a container in the form of an inverted cone of height 80 cm and diameter 40 cm. Water is poured into the container at the rate of $20 \text{ cm}^3/\text{s}$.
 - a Find the volume, $V \text{ cm}^3$, of water in the container when the depth is $x \text{ cm}$.
 - b Find the rate at which the depth is increasing when $x = 5$.



10.10 Families of functions

Example 32

Consider the function $f(x) = (x - a)^2(x - b)$ where a and b are positive constants with $b > a$.

- Find the derivative of $f(x)$ with respect to x .
- Find the coordinates of the stationary points of the graph of $y = f(x)$
- Show that the stationary point at $(a, 0)$ is always a local maximum.
- Find the values of a and b if the stationary points occur where $x = 3$ and $x = 4$

Solution

- Use a CAS calculator to find that $f'(x) = (x - a)(3x - a - 2b)$
- The coordinates of the stationary points are $(a, 0)$ and $\left(\frac{a + 2b}{3}, \frac{4(a - b)^3}{27}\right)$
- If $x < a$ then $f'(x) > 0$ and if $x > a$ and $x < \frac{a + 2b}{3}$ then $f'(x) < 0$, therefore local maximum.
- $a = 3$ as $a < b$ and $\frac{a + 2b}{3} = 4$ implies $b = \frac{9}{2}$

Example 33

The graph of the function $y = x^3 - 3x^2$ is translated by a units in the positive direction of the x -axis and b units in the positive direction of the y -axis (a and b are positive constants).

- Find the coordinates of the turning points of the graph of $y = x^3 - 3x^2$
- Find the coordinates of the turning points of its image.

Solution

- The turning points have coordinates $(0, 0)$ and $(2, -4)$.
- The turning points of the image are (a, b) and $(2 + a, -4 + b)$.

Example 34

A cubic function has rule $y = ax^3 + bx^2 + cx + d$. It passes through the points $(1, 6)$ and $(10, 8)$ and has turning points where $x = -1$ and $x = 2$.

- Find using matrix methods the values of a, b, c and d .
- Find the equation of the image of the curve under the transformation defined by the matrix equation:

$$\mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{X}' \text{ where } \mathbf{T} = \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Solution

a A CAS calculator has been used for matrix calculations.

The equations obtained are:

$$6 = a + b + c + d$$

$$8 = 1000a + 100b + 10c + d$$

$$\text{And as } \frac{dy}{dx} = 3ax^2 + 2bx + c$$

$$0 = 3a - 2b + c$$

and

$$0 = 12a + 4b + c$$

These can be written as a matrix equation.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1000 & 100 & 10 & 1 \\ 3 & -2 & 1 & 0 \\ 12 & 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \text{Therefore } \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1000 & 100 & 10 & 1 \\ 3 & -2 & 1 & 0 \\ 12 & 4 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 8 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{4}{1593} \\ \frac{-2}{531} \\ \frac{-8}{531} \\ \frac{9584}{1593} \end{bmatrix} \end{aligned}$$

$$\text{Therefore } a = \frac{4}{1593} \quad b = \frac{-2}{531} \quad c = \frac{-8}{531} \quad d = \frac{9584}{1593}$$

b First solve the matrix equation for $\mathbf{X}$.

$$\mathbf{T}^{-1} \mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{T}^{-1} \mathbf{X}'$$

$$\mathbf{X} + \mathbf{B} = \mathbf{T}^{-1} \mathbf{X}'$$

$$\text{and } \mathbf{X} = \mathbf{T}^{-1} \mathbf{X}' - \mathbf{B}$$

$$\text{Therefore } \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{y'}{2} \\ -\frac{x'}{3} \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Therefore } x = \frac{y'}{2} - 1 \text{ and } y = -\frac{x'}{3} - 2$$

The image curve has equation

$$-\frac{x'}{3} = a \left(\frac{y'}{2} - 1 \right)^3 + b \left(\frac{y'}{2} - 1 \right)^2 + c \left(\frac{y'}{2} - 1 \right) + d + 2$$

where a , b , c and d have the values given above.

Example 35

A cubic function f has rule $f(x) = ax^3 + bx^2 + cx$. The graph has a stationary point at $(1, 6)$.

- a Find a and b in terms of c .
- b Find the value of c for which the graph has a stationary point at the point where $x = 2$.

Solution

- a Two equations are obtained.

$$6 = a + b + c \text{ as } f(1) = 6$$

$$f'(x) = 3ax^2 + 2bx + c \text{ and therefore}$$

$$3a + 2b + c = 0$$

$$\text{The solution is } a = c - 12 \text{ and } b = 18 - 2c$$

$$\text{The equation is } f(x) = (c - 12)x^3 + (18 - 2c)x^2 + cx$$

- b $f'(x) = 3(c - 12)x^2 + 2(18 - 2c)x + c$

$$f'(2) = 0$$

$$\text{Therefore } 12(c - 12) + 4(18 - 2c) + c = 0$$

$$5c = 72$$

$$c = \frac{72}{5}$$

Exercise 10J

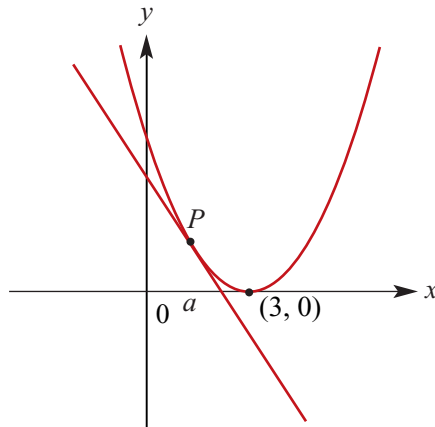
- 1 Consider the function $f(x) = (x - 1)^2(x - b)$ where $b > 1$.
 - a Find the derivative of $f(x)$ with respect to x .
 - b Find the coordinates of the stationary points of the graph of $y = f(x)$
 - c Show that the stationary point at $(1, 0)$ is always a local maximum.
 - d Find the value of b if the stationary points occur where $x = 1$ and $x = 4$.
- 2 Consider the function $f: [0, \infty) \rightarrow R$, defined by

$$f(x) = x^2 - ax^3$$

where a is a real number, $a > 0$.

- a Determine intervals on which f is a decreasing function and the intervals on which f is an increasing function.
- b Find the equation of the tangent to the graph of f at the point $(\frac{1}{a}, 0)$.
- c Find the equation of the normal to the graph of f at the point $(\frac{1}{a}, 0)$.
- d What is the range of f ?

- 3 A line with equation $y = mx + c$ is a tangent to the curve $y = (x - 3)^2$ at a point P where $x = a$, such that $0 < a < 3$.



- a i Find the gradient of the curve where $x = a$, for $0 < a < 3$.
ii Hence express m in terms of a .
 - b State the coordinates of the point P , expressing your answer in terms of a .
 - c Find the equation of the tangent where $x = a$.
 - d Find the x -axis intercept of the tangent.
- 4 a The graph of $f(x) = x^4$ is translated to the graph of $y = f(x + h)$. Find the possible values of h if $f(1 + h) = 16$.
b The graph of $f(x) = x^3$ is transformed to the graph of $y = f(ax)$. Find the possible value of a if the graph of $y = f(ax)$ passes through the point with coordinates $(1, 8)$.
c The quartic function with equation $y = ax^4 - bx^3$ has a turning point with coordinates $(1, 16)$. Find the values of a and b .
- 5 The graph of the function $y = x^4 - 4x^2$ is translated by a units in the positive direction of the x -axis and b units in the positive direction of the y -axis (a and b are positive constants).
a Find the coordinates of the turning points of the graph of $y = x^4 - 4x^2$
b Find the coordinates of the turning points of its image.
- 6 Consider the cubic function with rule $f(x) = (x - a)^2(x - 1)$ where $a > 1$.
a Find the coordinates of the turning points of the graph of $y = f(x)$
b State the nature of each of the turning points.
c Find the equation of the tangent to the curve at the point where:
i $x = 1$ ii $x = a$ iii $x = \frac{a + 1}{2}$
- 7 Consider the quartic function with rule $f(x) = (x - 1)^2(x - b)^2$ with $b > 1$.
a Find the derivative of f .
b Find the coordinates of the turning points of f .
c Find the value of b such that the graph of $y = f(x)$ has a turning point at $(2, 1)$.

- 8** A cubic function has rule $y = ax^3 + bx^2 + cx + d$. It passes through the points (1, 6) and (10, 8) and has turning points where $x = -1$ and $x = 1$.
- a** Find using matrix methods the values of a , b , c and d .
- b** Find the equation of the image of the curve under the transformation defined by the matrix equation:

$$\mathbf{T}(\mathbf{X} + \mathbf{B}) = \mathbf{X}' \text{ where } \mathbf{T} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

- 9** A cubic function f has rule $f(x) = ax^3 + bx^2 + cx$. The graph has a stationary point at (1, 10).
- a** Find a and b in terms of c .
- b** Find the value of c for which the graph has a stationary point at the point where $x = 3$.
- 10** A quartic function f has rule $f(x) = ax^4 + bx^3 + cx^2 + dx$. The graph has a stationary point at (1, 1) and passes through the point (-1, 4).
- a** Find a , b and c in terms of d .
- b** Find the value of d for which the graph has a stationary point at the point where $x = 4$.



Chapter summary

- For point (x, y) on the curve with equation $y = f(x)$, the equation of the tangent to the curve at (x_1, y_1) is

$$y - y_1 = f'(x_1)(x - x_1)$$

and the equation of the normal to the curve at (x_1, y_1) is

$$y - y_1 = -\frac{1}{f'(x_1)}(x - x_1)$$

- Two curves with equations $y = f(x)$ and $y = g(x)$ intersect at (x_1, y_1) and have gradients m_1 and m_2 respectively.

Let θ be the angle between the curves at (x_1, y_1) .

$$\text{Then } \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

- The relationship $f(x + h) \approx f(x) + hf'(x)$ for a small value of h is used to estimate a value of $f(x + h)$ close to a known value $f(x)$.
- The curve with equation $y = f(x)$ has stationary points where $f'(x) = 0$.

The point $(a, f(a))$ is a local maximum if

$$f'(a) = 0$$

and immediately to the left the gradient is positive and immediately to the right the gradient is negative.

The point $(b, f(b))$ is a local minimum if

$$f'(b) = 0$$

and immediately to the left the gradient is negative and immediately to the right the gradient is positive.

The point $(c, f(c))$ is a stationary point of inflexion if

$$f'(c) = 0$$

and immediately to the left and the right the gradient is positive or immediately to the left and the right the gradient is negative.

- M is the absolute maximum value of a function f in an interval $[a, b]$ if $f(x) \leq M$ for all $x \in [a, b]$.
- N is the absolute minimum value of a function f in an interval $[a, b]$ if $f(x) \geq N$ for all $x \in [a, b]$.

Multiple-choice questions

- 1 The line with equation $y = 4x + c$ is a tangent to the curve with equation $y = x^2 - x - 5$. The value of c is:

A $-\frac{45}{4}$

B $-1 + 2\sqrt{2}$

C 2

D $\frac{5}{2}$

E $-\frac{2}{5}$

- 2 The equation of the tangent to the curve of the function with equation $y = x^4$ at the point where $x = 1$ is:
A $y = -4x - 3$ **B** $y = \frac{1}{4}x - 3$ **C** $y = -4x$ **D** $y = \frac{1}{4}x + \frac{5}{4}$ **E** $y = 4x - 3$
- 3 For the function with rule $f(x)$, $f'(a) = f'(b) = 0$, $f'(x) > 0$ for $x \in [a, b]$, $f'(x) < 0$ for $x < a$ and $f'(x) > 0$ for $x > b$. The nature of the stationary points of the graph of $y = f(x)$ at the points with coordinates $(a, f(a))$ and $(b, f(b))$ is:
A local maximum at $(a, f(a))$ and local minimum at $(b, f(b))$
B local minimum at $(a, f(a))$ and local maximum at $(b, f(b))$
C stationary point of inflexion at $(a, f(a))$ and local minimum at $(b, f(b))$
D stationary point of inflexion at $(a, f(a))$ and local maximum at $(b, f(b))$
E local minimum at $(a, f(a))$ and stationary point of inflexion at $(b, f(b))$
- 4 Water is draining from a cone-shaped funnel at a rate of $500 \text{ cm}^3/\text{min}$. The cone has a base radius of 20 cm and a height of 100 cm. Let h cm be the depth of water in the funnel at time t minutes. The rate of decrease of h in cm/min is given by:
A $\frac{3750}{\pi h^2}$ **B** $\frac{12\,500}{\pi h^2}$ **C** $30\pi h^2$ **D** 10 **E** $\frac{24}{\pi}$
- 5 The radius of a sphere is increasing at a rate of 5 cm/min. When the radius is 10 cm, the rate of increase, in cm^3/min , of the volume of the sphere is:
A 2000π **B** $\frac{2000\pi}{3}$ **C** $\frac{4000\pi}{3}$ **D** $\frac{40\pi}{3}$ **E** $\frac{500\pi}{3}$
- 6 The graph of the function with rule $y = f(x)$ has a local maximum at the point with coordinates $(a, f(a))$. The graph also has a local minimum at the origin but no other stationary points. The graph of the function with rule $y = -2f\left(\frac{x}{2}\right) + k$ where k is a positive real number has:
A a local maximum at the point with coordinates $(2a, -2f(a) + k)$
B a local minimum at the point with coordinates $\left(\frac{a}{2}, 2f(a) + k\right)$
C a local maximum at the point with coordinates $\left(\frac{a}{2}, -2f(a) + k\right)$
D a local maximum at the point with coordinates $(2a, -2f(a) - k)$
E a local minimum at the point with coordinates $(2a, -2f(a) + k)$
- 7 For $f(x) = x^3 - x^2 - 1$, the values of x for which the graph of $y = f(x)$ has stationary points are:
A $\frac{2}{3}$ only **B** 0 and $\frac{2}{3}$ **C** 0 and $-\frac{2}{3}$ **D** $-\frac{1}{3}$ and 1 **E** $\frac{1}{3}$ and -1
- 8 Let f be differentiable for all values of x in $[0, 6]$. The graph with equation $y = f(x)$ has a local minimum point at $(2, 4)$. The equation of the tangent at the point with coordinates $(2, 4)$ is:
A $y = 2x$ **B** $x = 2$ **C** $y = 4$ **D** $2x - 4y = 0$ **E** $4x - 2y = 0$
- 9 The volume $V \text{ cm}^3$ of a solid is given by the formula $V = -10x(2x^2 - 6)$ where x cm is a particular measurement. The value of x for which the volume is a maximum is:
A 0 **B** 1 **C** $\sqrt{2}$ **D** $\sqrt{3}$ **E** 2

- 10 The equation of the normal to the curve with equation $y = x^2$ at the point where $x = a$ is:

A $y = \frac{-1}{2a}x + 2 + a^2$ **B** $y = \frac{-1}{2a}x + \frac{1}{2} + a^2$ **C** $y = 2ax - a^2$
D $y = 2ax + 3a^2$ **E** $y = \frac{1}{2a}x + 2 + a^2$

Short-answer questions (technology-free)

- a** Find the equation of the tangent to the curve $y = x^3 - 8x^2 + 15x$ at the point with coordinates $(4, -4)$.

b Find the coordinates of the point where the tangent meets the curve again.
- Find the equation of the tangent to the curve $y = 3x^2$ at the point where $x = a$. If this tangent meets the y -axis at P , find in terms of a the y -coordinate of P .
- Find the equation of the tangent to the curve with equation

$$y = x^3 - 7x^2 + 14x - 8$$

at the point where $x = 1$.

Find the x -coordinate of a second point at which the tangent is parallel to the tangent at $x = 1$.

- Use the formula $A = \pi r^2$ for the area of a circle to find:
 - the average rate at which the area of a circle changes with respect to the radius as the radius increases from $r = 2$ to $r = 3$
 - the instantaneous rate at which the area changes with r when $r = 3$
- Find the stationary points of the graphs for each of the following and state their nature:

a $f(x) = 4x^3 - 3x^4$ **b** $g(x) = x^3 - 3x - 2$ **c** $h(x) = x^3 - 9x + 1$
- Sketch the graph of $y = x^3 - 6x^2 + 9x$
- The derivative of the function $y = f(x)$ is:

$$\frac{dy}{dx} = (x - 1)^2(x - 2)$$

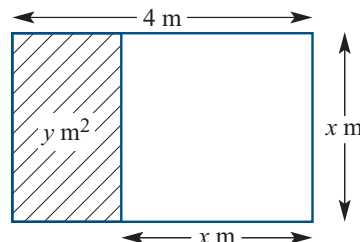
Find the x -coordinate and state the nature of each stationary point.

- Find the equation of the tangent to the curve $y = x^3 - 3x^2 - 9x + 11$ at $x = 2$.
- Let $f: R \rightarrow R$, where $f(x) = 3 + 6x^2 - 2x^3$
Determine the values of x for which the graph of $y = f(x)$ has a positive gradient.
- For what value(s) of x do the graphs of $y = x^3$ and $y = x^3 + x^2 + x - 2$ have the same gradient?
- For the function with rule $f(x) = (x - 1)^{\frac{4}{5}}$:
 - State the values for which the function is differentiable, and find the rule for f' .
 - Find the equations of the tangent at the point $(2, 1)$ and $(0, 1)$.
 - Find the coordinates of the point of intersection of the two tangents.

- 12** A spherical bubble, initially of radius length 1 cm, expands steadily, its radius increases by 1 cm/s and it bursts after 5 seconds.
- Find the rate of increase of volume with respect to the change in radius when the radius is 4 cm.
 - Find the rate of increase of volume with respect to time when the radius is 4 cm.

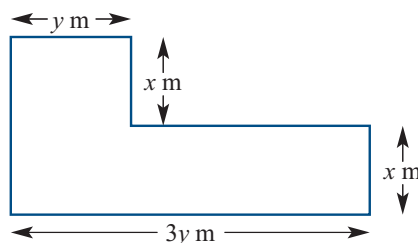
Extended-response questions

- 1** The diagram shows a rectangle with sides 4 m and x m and a square with side x m. The area of the shaded region is $y \text{ m}^2$.



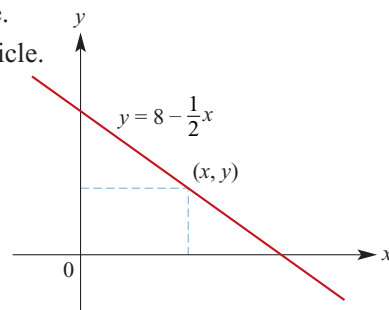
- Find an expression for y in terms of x .
- Find the set of possible values for x .
- Find the maximum value of y and the corresponding value of x .
- Explain briefly why this value of y is a maximum.
- Sketch the graph of y against x .
- State the set of possible values for y .

- 2** A flower bed is to be L-shaped, as shown in the figure, and its perimeter is 48 m.



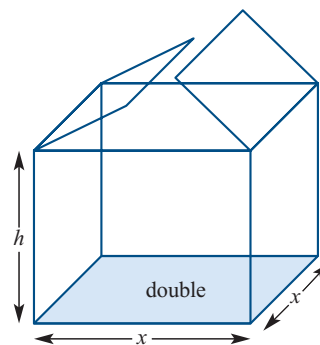
- Write down an expression for the area, $A \text{ m}^2$, in terms of y and x .
 - Find y in terms of x .
 - Write down an expression for A in terms of x .
 - Find the values of x and y that give the maximum area.
 - Find the maximum area.
- 3** It costs $(12 + 0.008x)$ dollars per kilometre to operate a truck at x kilometres per hour. In addition it costs \$14.40 per hour to pay the driver.
- What is the total cost per kilometre if the truck is driven at:
 - 40 km/h?
 - 64 km/h?
 - Write an expression for C , the total cost per kilometre, in terms of x .
 - Sketch the graph of C against x for $0 < x < 120$.
 - At what speed should the truck be driven to minimise the total cost per kilometre?
- 4** A box is to be made from a 10 cm by 16 cm sheet of metal by cutting equal squares out of the corners and bending up the flaps to form the box. Let the lengths of the sides of the squares be x cm and $V \text{ cm}^3$ the volume of the box formed.
- Show that $V = 4(x^3 - 13x^2 + 40x)$
 - State the set of x -values for which the expression for V in terms of x is valid.
 - Find the values of x such that $\frac{dV}{dx} = 0$.
 - Find the dimensions of the box if the volume is to be a maximum.

- e Find the maximum volume of the box.
 f Sketch the graph of V against x for the domain established in b.
- 5 A manufacturer finds that the daily profit, $\$P$, from selling n articles is given by $P = 100n - 0.4n^2 - 160$
- a i Find the value of n which maximises the daily profit.
 ii Find the maximum daily profit.
- b Sketch the graph of P against n . (Use a continuous graph.)
- c State the allowable values for n for a profit to be made.
- d Find the value of n which maximises the profit per article.
- 6 A rectangle has one vertex at the origin, another on the positive x -axis, another on the positive y -axis and a fourth on the line $y = 8 - \frac{x}{2}$
- What is the greatest area the rectangle can have?



- 7 At a factory the time, T seconds, spent in producing a certain size metal component is related to its weight, w kg, by $T = k + 2w^2$, where k is a constant.
- a If a 5 kg component takes 75 seconds to produce, find k .
- b Sketch the graph of T against w .
- c Write down an expression for the average time A (in seconds per kilogram).
- d i Find the weight that yields the minimum average machinery time.
 ii State the minimum average machining time.

- 8 A manufacturer produces cardboard boxes that have a square base. The top of each box consists of a double flap that opens as shown. The bottom of the box has a double layer of cardboard for strength. Each box must have a volume of 12 cubic metres.



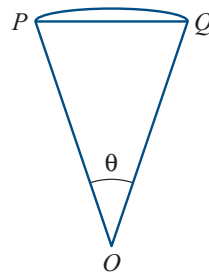
- a Show that C , the area of cardboard required, $= 3x^2 + 4xh$
- b Express C as a function of x only.
- c Sketch the graph of C against x for $x > 0$.
- d i What dimensions of the box will minimise the amount of cardboard used?
 ii What is the minimum area of cardboard used?
- 9 An open tank is to be constructed with a square base and vertical sides to contain 500 m^3 of water. What must be the dimensions of the area of sheet metal used in its construction if this area is to be a minimum?
- 10 A piece of wire of length 1 m is bent into the shape of a sector of a circle of radius a cm and sector angle θ . Let the area of the sector be $A \text{ cm}^2$.
- a Find A in terms of a and θ . b Find A in terms of θ .
- c Find the value of θ for which A is a maximum.
- d Find the maximum area of the sector.

- 11** A piece of wire of fixed length, L cm, is bent to form the boundary $OPQO$ of a sector of a circle. The circle has centre O and radius r cm. The angle of the sector is θ radians.

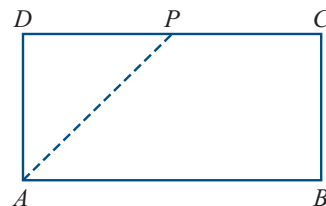
a Show that the area, A cm², of the sector is given by:

$$A = \frac{1}{2}rL - r^2$$

- b** **i** Find a relationship between r and L for which $\frac{dA}{dr} = 0$
ii Find the corresponding value of θ .
iii Determine the nature of the stationary points found in **i**.
c Show that, for the value of θ found in **b ii**, the area of the triangle OPQ is approximately 45.5% of the area of sector OPQ .



- 12** A Queensland resort has a large swimming pool as illustrated with $AB = 75$ m and $AD = 30$ m. A boy can swim at 1 m/s and run at $1\frac{2}{3}$ m/s. He starts at A , swims to a point P on DC , and runs from P to C . He takes 2 seconds to pull himself out of the pool.



a Let $DP = x$ m and the total time be T s.

$$\text{Show that } T = \sqrt{x^2 + 900} + \frac{3}{5}(75 - x) + 2$$

b Find $\frac{dT}{dx}$

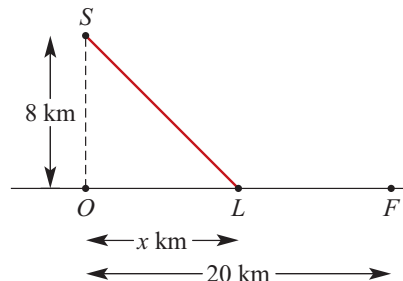
- c** **i** Find the value of x for which the time taken is a minimum.
ii Find the minimum time.

d Find the time taken if the boy runs from A to D and then D to C .

- 13** The point S is 8 km offshore from the point O which is located on the straight shore of a lake, as shown in the diagram. The point F is on the shore, 20 km from O . Contestants race from the start, S , to the finish, F , by rowing in a straight line to some point, L , on the shore and then running along the shore to F . A certain contestant rows at 5 km per hour and runs at 15 km per hour.

a Show that, if the distance OL is x km, the time taken by this contestant to complete the course is (in hours):

$$T(x) = \frac{\sqrt{64 + x^2}}{5} + \frac{20 - x}{15}$$

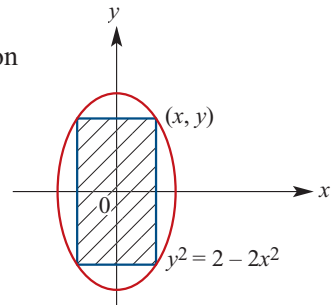


- b** Show that the time taken by this contestant to complete the course has its minimum value when $x = 2\sqrt{2}$. Find this time.

- 14** At noon the captain of a ship sees two fishing boats approaching. One of them is 10 km due east and travelling west at 8 km/h. The other is 6 km due north, travelling south at 6 km/h. At what time will the fishing boats be closest together and how far apart will they be?

- 15** A rectangular beam is to be cut from a non-circular tree trunk whose cross-sectional outline can be represented by the equation $y^2 = 2 - 2x^2$

- Show that the area of the cross-section of the beam is given by $A = 4x\sqrt{2 - 2x^2}$ where x is the half-width of the beam.
- State the possible values for x .
- Find the value of x for which the cross-sectional area of the beam is a maximum and find the corresponding value of y .
- Find the maximum cross-sectional area of the beam.



- 16** An isosceles trapezoid is inscribed in the parabola $y = 4 - x^2$ as illustrated.

- a** Show that the area of the trapezoid is:

$$\frac{1}{2}(4 - x^2)(2x + 4)$$

- b** Show that the trapezoid has its greatest area when

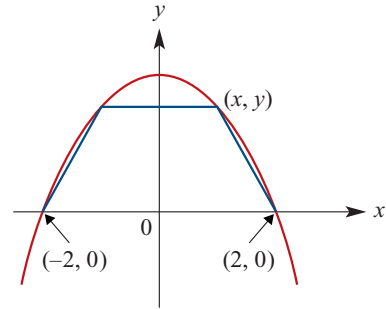
$$x = \frac{2}{3}.$$

- c** Repeat with the parabola $y = a^2 - x^2$

- i** Show that the area, A , of the trapezoid $= (a^2 - x^2)(a + x)$

- ii** Use the product rule to find $\frac{dA}{dx}$.

- iii** Show that a maximum occurs when $x = \frac{a}{3}$.



- 17** It is believed that, for some time after planting in ideal conditions, the area covered by a particular species of ground-cover plant has a rate of increase of y cm²/week, given by $y = -t^3 + bt^2 + ct$ where t is the number of weeks after planting.

- a** Find b and c given the following table of observations:

t	1	2
y	10	24

- Assuming that the model is accurate for the first 8 weeks after planting, when, during this period, is:
 - the area covered by the plant a maximum?
 - the rate of increase in area a maximum?
- According to the model, if the plant covered 100 cm² when planted, what area will it cover after 4 weeks?

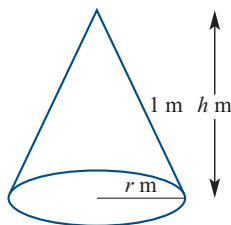
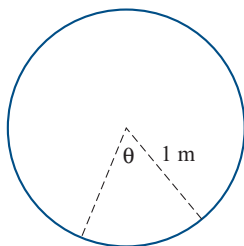
- d** Discuss the implications for the future growth of the plant if the model remains accurate for longer than the first 4 weeks.
- 18** Let $f(x) = x^3 - 3x^2 + 6x - 10$
- a** Find the coordinates of the point on the graph of f for which $f'(x) = 3$.
- b** Express $f'(x)$ in the form $a(x + p)^2 + q$
- c** Hence show that the gradient of f is greater than 3 for all points on the curve of f other than that point found in **a**.
- 19** A curve with equation of the form $y = ax^3 + bx^2 + cx + d$ has zero gradient at the point $\left(\frac{1}{3}, \frac{4}{27}\right)$ and also touches, but does not cross, the x -axis at the point $(1, 0)$.
- a** Find a, b, c and d .
- b** Find the values of x for which the curve has a negative gradient.
- c** Sketch the curve.
- 20** The volume of water, $V \text{ m}^3$, in a reservoir when the depth indicator shows y metres is given by the formula:

$$V = \frac{\pi}{3}[(y + 630)^3 - (630)^3]$$

- a** Find the volume of water in the reservoir when $y = 40$.
- b** Find the rate of change of volume with respect to height, y .
- c** Sketch the graph of V against y for $0 \leq y \leq 60$.
- d** If $y = 60 \text{ m}$ is the maximum depth of the reservoir, find the capacity (m^3) of the reservoir.
- e** If $\frac{dV}{dt} = 20\,000 - 0.005\pi(y + 630)^2$, where t is the time in days from 1 January, sketch the graph of $\frac{dV}{dt}$ against y for $0 \leq y \leq 60$.
- 21** Water is being poured into a flask. The volume, $V \text{ mL}$, of water in the flask at time, t seconds, is given by:

$$V(t) = \frac{3}{4} \left(10t^2 - \frac{t^3}{3} \right), 0 \leq t \leq 20$$

- a** Find the volume of water in the flask when:
- i** $t = 0$ **ii** $t = 20$
- b** Find the rate of flow of water, $\frac{dV}{dt}$, into the flask.
- c** Sketch the graph of $V(t)$ against t for $0 \leq t \leq 20$.
- d** Sketch the graph of $V'(t)$ against t for $0 \leq t \leq 20$.
- e** At what time is the flow greatest and what is the flow at this time?
- 22** A cone is made by cutting out a sector with central angle θ from a circular piece of cardboard of radius 1 m and joining the two cut edges to form a cone of slant height 1 m as shown in the following diagrams.



The volume of a cone is given by the formula $V = \frac{1}{3}\pi r^2 h$

- a** **i** Find r in terms of θ . **ii** Find h in terms of θ .

iii Show that $V = \frac{1}{3}\pi \left(\frac{2\pi - \theta}{2\pi}\right)^2 \sqrt{1 - \left(\frac{2\pi - \theta}{2\pi}\right)^2}$

- b** Find the value of V when $\theta = \frac{\pi}{4}$
c Find the value of θ for which the volume of the cone is 0.3 m^3 .
d **i** Use a calculator to determine the value of θ that maximises the volume of the cone.
ii Find the maximum volume.
e Determine the maximum volume using calculus.

- 23 a** For the function with rule $f(x) = x^3 + ax^2 + bx$ plot the graph of each of the following using a calculator. (Give axes intercepts, coordinates of stationary points and the nature of stationary points.)

- i** $a = 1, b = 1$ **ii** $a = -1, b = -1$ **iii** $a = 1, b = -1$
iv $a = -1, b = 1$

- b** **i** Find $f'(x)$.
ii Solve the equation $f'(x) = 0$ for x , giving your answer in terms of a and b .
c **i** Show that the graph of $y = f(x)$ has exactly one stationary point if $a^2 - 3b = 0$
ii If $b = 3$, find the corresponding value(s) of a which satisfy the condition $a^2 - 3b = 0$. Find the coordinates of the stationary points and state the nature of each.
iii Plot the graph(s) of $y = f(x)$ for these values of a and b using a graphics calculator.
iv Plot the graph of the corresponding derivative functions on the same set of axes.
d State the relationship between a and b if no stationary points exist for the graph of $v = f(x)$

- 24** Consider the function with rule $f(x) = 6x^4 - x^3 + ax^2 - 6x + 8$

- a** **i** If $x + 1$ is a factor of $f(x)$ find the value of a .
ii Using a calculator plot the graph of $y = f(x)$ for this value of a .
b Let $g(x) = 6x^4 - x^3 + 21x^2 - 6x + 8$
i Plot the graph of $y = g(x)$
ii Find the minimum value of $g(x)$ and the value of x for which this occurs.
iii Find $g'(x)$.
iv Using a calculator solve the equation $g'(x) = 0$ for x .



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CALCULATOR

- v Find $g'(0)$ and $g'(10)$.
- vi Find the derivative of $g'(x)$.
- vii Show that the graph of $y = g'(x)$ has no stationary points and thus deduce that $g'(x) = 0$ has only one solution.

25 For the quartic function f , with rule $f(x) = (x - a)^2(x - b)^2$, $a > 0$ and $b > 0$:

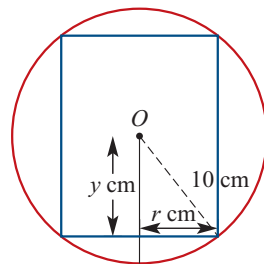
- a Show that $f'(x) = 2(x - a)(x - b)[2x - (b + a)]$.
- b i Solve the equation $f'(x) = 0$ for x . ii Solve the equation $f(x) = 0$ for x .
- c Hence find the coordinates of the stationary points of the graph of $y = f(x)$.
- d Plot the graph of $y = f(x)$ on a calculator for several values of a and b .
- e i If $a = b$, $f(x) = (x - a)^4$. Sketch the graph of $y = f(x)$.
- ii If $a = -b$, find the coordinates of the stationary points.
- iii Plot the graph of $y = f(x)$ for several values of a , given that $a = -b$.

26 For the quartic function f with rule $f(x) = (x - a)^3(x - b)$, $a > 0$ and $b > 0$:

- a Show that $f'(x) = (x - a)^2[4x - (3b + a)]$.
- b i Solve the equation $f'(x) = 0$ ii Solve the equation $f(x) = 0$.
- c Find the coordinates of the stationary points of the graph of $y = f(x)$ and state the nature of the stationary points.
- d Using a calculator plot the graph of $y = f(x)$ for several values of a and b .
- e If $a = -b$ state the coordinates of the stationary points in terms of a .
- f i State the relationship between b and a if there is a local minimum for $x = 0$.
- ii Illustrate this for $b = 1$, $a = -3$ on a calculator.
- g Show that if there is a turning point for $x = \frac{a + b}{2}$, then $b = a$ and $f(x) = (x - a)^4$.

27 A cylinder is to be cut from a sphere. The cross-section through the centre of the sphere is as shown. The radius of the sphere is 10 cm. Let r cm be the radius of the cylinder.

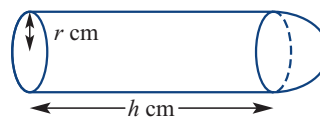
- a i Find y in terms of r and hence the height, h cm, of the cylinder.
- ii The volume of a cylinder is given by $V = \pi r^2 h$. Find V in terms of r .
- b i Plot the graph of V against r using a calculator.
- ii Find the maximum volume of the cylinder and the corresponding values of r and h . (Use a calculator.)
- iii Find the two possible values of r if the volume is 2000.
- c i Find $\frac{dV}{dr}$.
- ii Hence find the exact value of the maximum volume and the volume of r for which this occurs.
- d i Plot the graph of the derivative function $\frac{dV}{dr}$ against r , using a calculator.
- ii From the calculator, find the values of r for which $\frac{dV}{dr}$ is positive.
- iii From the calculator, find the values of r for which $\frac{dV}{dr}$ is increasing.



CAS

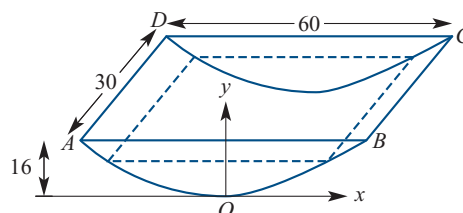
CALCULATOR

- 28** A wooden peg consists of a cylinder of length h cm and a hemispherical cap of radius r cm so that the volume, V cm³, of the peg is given by $V = \pi r^2 h + \frac{2}{3} \pi r^3$

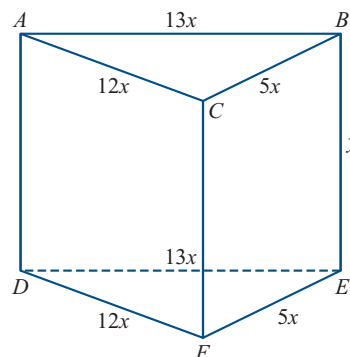


- a** If the surface area of the peg is 100π cm²:
- i** Find h in terms of r .
 - ii** Find V as a function of r .
 - iii** Find the possible values of r (i.e. find the domain of the function defined in **ii**).
 - iv** Find $\frac{dV}{dr}$.
 - v** Sketch the graph of V against r .
- b** If $h = 6$, $V = 6\pi r^2 + \frac{2}{3}\pi r^3$. For $r = 4$:
- i** Show that a small increase of p cm in the radius results in an increase of $80\pi p$ in the volume.
 - ii** Show that a small increase of $q\%$ in the radius will cause an approximate increase of $2.3\% q$ in the volume.

- 29** The diagram shows a container open at the top, whose dimensions in cm are as indicated. The cross-section profile AOB is a parabola whose vertex is at a lowest point O . $ABCD$ is a horizontal rectangle.



- a** Water is poured into the container at a rate of 100 cm³/s. Find the rate at which the level of liquid is rising when the depth of liquid in the container is 9 cm, given that the volume V cm³ of liquid when the depth of water is y cm is given by $V = 300y^{\frac{3}{2}}$.
- b** Find the equation of the parabola.
- c** The water is being poured in at a rate of 100 cm³/s.
- i** Find the rate of change of x with respect to time when $y = 9$.
 - ii** The surface of the water is a rectangle (the rectangle with dashed lines in the diagram). Find the rate of change of the surface area when $y = 9$.
- 30** A triangular prism has dimensions as shown in the diagram. All lengths are in centimetres. The volume of the prism is 3000 cm³.
- a**
- i** Find y in terms of x .
 - ii** Find S cm², the surface area of the prism, in terms of x .
- b**
- i** Find $\frac{dS}{dx}$.
 - ii** Find the minimum surface area, correct to three decimal places.
- c** Given that x is increasing at 0.5 cm/s find the rate at which the surface area is increasing when $x = 10$.



Differentiation of transcendental functions

Objectives

- To differentiate **exponential functions**.
- To differentiate **natural logarithmic functions**.
- To find the derivatives of **circular functions**.
- To apply the differentiation of transcendental functions to **solving problems**.

11.1 Differentiation of e^x

In this section we investigate the derivative of functions of the form $f(x) = a^x$. It is found that the number e , Euler's number, has the special property that $f'(x) = f(x)$ where $f(x) = e^x$.

Let $f: R \rightarrow R$, $f(x) = 2^x$

To find the derivative of $f(x)$ we recall that:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2^{x+h} - 2^x}{h} \\ &= 2^x \lim_{h \rightarrow 0} \frac{2^h - 1}{h} \\ &= 2^x f'(0) \end{aligned}$$

A calculator can be used to see $f'(0) = \lim_{h \rightarrow 0} \frac{2^{0+h} - 2^0}{h} \approx 0.693$. This is done by entering $Y_1 = (2^X - 1)/X$ and considering values of the function for $X \rightarrow 0$.

Thus $f'(x) \approx 0.693 \times 2^x$

Let $g: R \rightarrow R$, $g(x) = 3^x$. Then as above for f it may be shown that $g'(x) = 3^x g'(0)$

It can be shown that $g'(0) \approx 1.0986$ and hence $g'(x) = 3^x g'(0) \approx 1.0986 \times 3^x$. The question arises of the existence of a number, b , between 2 and 3 such that if $h(x) = b^x$, $h'(x) = b^x$, i.e. $h'(0) = 1$

By using a calculator the limit as $h \rightarrow 0$ of $\frac{b^h - 1}{h}$ for various values of b between 2 and 3 can be investigated.

This investigation is continued through the spreadsheet shown below. Start by taking values for b between 2.71 and 2.72 (column A) and finding $f'(0)$ for these values (column B). From these results it may be seen that the required value of b lies between 2.718 and 2.719; in columns D and E the investigation is continued in a similar way with values between 2.718 and 2.719. From this the required value of b is seen to lie between 2.7182 and 2.7183.

	A	B	C	D	E
1	0.00000001	is the value of h			
2		$f'(0)$			$f'(0)$
3	2.710	0.996948635		2.7180	0.99989632129649
4	2.711	0.997317584		2.7181	0.99993311408753
5	2.712	0.997686378		2.7182	0.99996990687856
6	2.713	0.998055039		2.7183	1.00000669966950
7	2.714	0.998423566		2.7184	1.00004347025610
8	2.715	0.998791960		2.7185	1.00008026304720
9	2.716	0.999160221		2.7186	1.00011705583820
10	2.717	0.999528327		2.7187	1.00019061921580
11	2.718	0.999896321		2.7188	1.00022738980240
12	2.719	1.000264160		2.7189	1.00022738980240
13	2.720	1.000631888		2.7190	1.00026416038900

The value of b is in fact e , Euler's number, which was introduced in previous work. Our results can be recorded:

$$\text{for } f(x) = e^x, f'(x) = e^x$$

Consider $y = e^{kx}$ where $k \in R$. The chain rule is used to find the derivative.

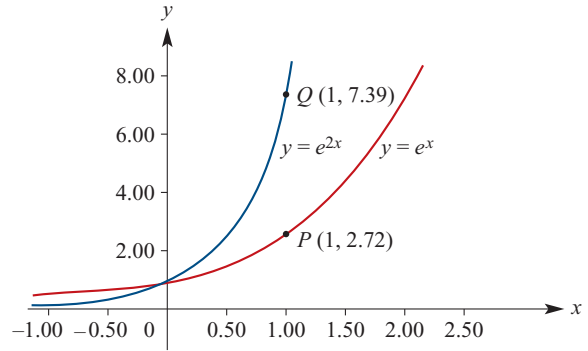
Let $u = kx$. Then $y = e^u$

$$\begin{aligned} \text{The chain rule yields } \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= e^u \cdot k \\ &= ke^{kx} \end{aligned}$$

$$\text{For } f(x) = e^{kx}, k \in R, f'(x) = ke^{kx}$$

The graph illustrates the effect of multiplying x by 2.

The gradient of $y = e^x$ at $P(1, e)$ is e and the gradient of $y = e^{2x}$ at $Q(1, e^2)$ is $2e^2$.



Example 1

Find the derivative of each of the following with respect to x :

a e^{3x}

b e^{-2x}

c e^{2x+1}

d e^{x^2}

e $\frac{1}{e^{2x}} + e^{3x}$

Solution

a Let $y = e^{3x} \frac{dy}{dx} = 3e^{3x}$

c Let $y = e^{2x+1}$
Then $y = e^{2x} \cdot e$
 $= e \cdot e^{2x}$ (index laws)
 $\therefore \frac{dy}{dx} = 2e \cdot e^{2x}$
 $= 2e^{2x+1}$

e Let $y = e^{-2x} + e^{3x}$
 $\left(\text{Note: } \frac{1}{e^{2x}} = e^{-2x} \right)$
Then $\frac{dy}{dx} = -2e^{-2x} + 3e^{3x}$

b Let $y = e^{-2x} \frac{dy}{dx} = -2e^{-2x}$

d Let $y = e^{x^2}$
Let $u = x^2$
Then $y = e^u$ and the chain rule yields:
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot 2x$
 $= 2xe^{x^2}$

Example 2

Differentiate each of the following with respect to x :

a $e^x(2x^2 + 1)$

b $\frac{e^x}{e^{2x} + 1}$

c $e^x \sqrt{x-1}$

Solution

a We use the product rule:

Let $y = e^x(2x^2 + 1)$
Then $\frac{dy}{dx} = e^x(2x^2 + 1) + 4xe^x$
 $= e^x(2x^2 + 4x + 1)$

b We use the quotient rule:

Let $y = \frac{e^x}{e^{2x} + 1}$
Then $\frac{dy}{dx} = \frac{(e^{2x} + 1)e^x - 2e^{2x} \cdot e^x}{(e^{2x} + 1)^2}$
 $= \frac{e^{3x} + e^x - 2e^{3x}}{(e^{2x} + 1)^2}$
 $= \frac{e^x - e^{3x}}{(e^{2x} + 1)^2}$

- c The product rule and chain rule are used:

$$\begin{aligned}
 \text{Let } y &= e^x \sqrt{x-1} \\
 \text{Then } \frac{dy}{dx} &= e^x \sqrt{x-1} + \frac{1}{2} e^x (x-1)^{-\frac{1}{2}} \\
 &= e^x \sqrt{x-1} + \frac{e^x}{2(x-1)^{\frac{1}{2}}} \\
 &= \frac{2e^x(x-1) + e^x}{2(x-1)^{\frac{1}{2}}} \\
 &= \frac{2xe^x - e^x}{2\sqrt{x-1}}
 \end{aligned}$$

In general, the chain rule gives:

$$\text{For } h(x) = e^{f(x)}, h'(x) = f'(x)e^{f(x)}$$

Example 3

Find the gradient of the curve $y = e^{2x} + 4$ at the point:

- a (0, 5) b (1, $e^2 + 4$)

Solution

a $\frac{dy}{dx} = 2e^{2x}$

When $x = 0$, $\frac{dy}{dx} = 2$

$\therefore$ the gradient of $y = e^{2x} + 4$ is 2 at (0, 5)

b When $x = 1$, $\frac{dy}{dx} = 2e^2$

$\therefore$ the gradient of $y = e^{2x} + 4$ is $2e^2$ (14.78 to two decimal places)

Example 4

Find the derivative, with respect to x , of:

- a $e^{f(x)}$ b $f(e^x)$

and evaluate the derivative of each when $x = 2$ if $f(2) = 0$ and $f'(2) = 4$

Solution

a Let $y = e^{f(x)}$ and $u = f(x)$

Therefore $y = e^u$

Then $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ (chain rule)

$$= e^u f'(x)$$

$$= e^{f(x)} f'(x)$$

When $x = 2$, $\frac{dy}{dx} = e^0 \times 4 = 4$

b Let $y = f(e^x)$ and $u = e^x$

Therefore $y = f(u)$

Then $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ (chain rule)

$$= f'(u) \cdot e^x$$

$$= f'(e^x) \cdot e^x$$

$$= f'(e^2) \cdot e^2$$

Exercise 11A

1 Find the derivative of each of the following with respect to x :

a e^{5x}

b $7e^{-3x}$

c $3e^{-4x} + e^x - x^2$

d $\frac{e^{2x} - e^x + 1}{e^x}$

e e^{x^2+3x+1}

f e^{3x^2-x}

g $e^{2x} + e^4 + e^{-2x}$

2 Find $f'(x)$ for each of the following:

a $f(x) = e^x(x^2 + 1)$

b $f(x) = e^{2x}(x^3 + 3x + 1)$

c $f(x) = e^{4x+1}(x+1)^2$

d $f(x) = e^{-4x}\sqrt{(x+1)}, x \geq -1$

3 Find $f'(x)$ for each of the following:

a $f(x) = \frac{e^x}{e^{3x} + 3}$

b $f(x) = \frac{e^x + 1}{e^x - 1}$

c $f(x) = \frac{e^{2x} + 2}{e^{2x} - 2}$

4 Differentiate each of the following with respect to x :

a x^4e^{-2x}

b e^{2x+3}

c $(e^{2x} + x)^{\frac{3}{2}}$

d $\frac{1}{x}e^x$

e $e^{\frac{1}{2}x^2}$

f $(x^2 + 2x + 2)e^{-x}$

5 Find each of the following:

a $\frac{d(e^x f(x))}{dx}$

b the derivative of $\frac{e^x}{f(x)}$ with respect to x

c $\frac{d(e^{f(x)})}{dx}$

d $\frac{d(e^x[f(x)]^2)}{dx}$

11.2 Differentiation of the natural logarithm function

For the function with rule $f(x) = e^{kx}$, $f'(x) = ke^{kx}$

This will be used to find the derivative of the function $g: R^+ \rightarrow R$, $g(x) = \log_e x$

Let $y = \log_e kx$

Then

$$e^y = kx$$

and with x the subject

$$x = \frac{1}{k}e^y$$

From our observation above

$$\frac{dx}{dy} = \frac{1}{k}e^y$$

But $e^y = kx$. Thus

$$\frac{dx}{dy} = \frac{kx}{k} = x$$

$$\therefore \frac{dy}{dx} = \frac{1}{x}$$

For $f: R^+ \rightarrow R$, $f(x) = \log_e(kx)$ $f': R^+ \rightarrow R$, $f'(x) = \frac{1}{x}$

Example 5Find the derivative of each of the following with respect to x :

a $\log_e(5x) \quad x > 0$

b $\log_e(5x + 3) \quad x > -\frac{3}{5}$

Solution

a Let $y = \log_e(5x)$, $x > 0$

Thus $\frac{dy}{dx} = \frac{1}{x}$

or let $u = 5x$. Then $y = \log_e u$

The chain rule gives:

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{u} \times 5 \\ &= \frac{5}{u} \\ &= \frac{1}{x}\end{aligned}$$

b Let $y = \log_e(5x + 3)$, $x > -\frac{3}{5}$
Let $u = 5x + 3$. Then $y = \log_e u$

The chain rule gives:

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{u} \times 5 \\ &= \frac{5}{u} \\ &= \frac{5}{5x + 3}\end{aligned}$$

In general if $y = \log_e(ax + b)$ with $x > -\frac{b}{a}$ then $\frac{dy}{dx} = \frac{a}{ax + b}$ **Example 6**

Find: **a** $\frac{d(\log_e |x|)}{dx}, x \neq 0$

b $\frac{d(\log_e |ax + b|)}{dx}, x \neq -\frac{b}{a}$

Solution

a Let $y = \log_e |x|$

If $x > 0$,

$y = \log_e x$ and $\frac{dy}{dx} = \frac{1}{x}$

If $x < 0$,

then $y = \log_e(-x)$ and using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -1 \times \frac{1}{-x} \\ \text{Hence } \frac{d(\log_e |x|)}{dx} &= \frac{1}{x}\end{aligned}$$

b Let $y = \log_e |ax + b|$

Let $u = ax + b$

Then $y = \log_e |u|$

and the chain rule gives:

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{u}(a) \\ &= \frac{1}{ax + b}(a) \\ &= \frac{a}{ax + b}\end{aligned}$$

In general if $y = \log_e |ax + b|$ with $x \neq -\frac{b}{a}$ then $\frac{dy}{dx} = \frac{a}{ax + b}$

Example 7Differentiate each of the following with respect to x :

a $\log_e(x^2 + 2)$

b $x^2 \log_e x, x > 0$

c $\frac{\log_e x}{x^2}, x > 0$

Solution**a** We use the chain rule.

Let $y = \log_e(x^2 + 2)$ and $u = x^2 + 2$

Then $y = \log_e u$, $\frac{dy}{du} = \frac{1}{u}$ and $\frac{du}{dx} = 2x$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \frac{1}{u} \cdot 2x \\ &= \frac{2x}{x^2 + 2}\end{aligned}$$

b We use the product rule.

Let $y = x^2 \log_e x$

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= 2x \log_e x + x^2 \times \frac{1}{x} \\ &= 2x \log_e x + x\end{aligned}$$

c We use the quotient rule.

Let $y = \frac{\log_e x}{x^2}$

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= \frac{x^2 \times \frac{1}{x} - 2x \log_e x}{x^4} \\ &= \frac{x - 2x \log_e x}{x^4} \\ &= \frac{1 - 2 \log_e x}{x^3}\end{aligned}$$

In general, the chain rule gives:

for $h(x) = \log_e(f(x))$, $h'(x) = \frac{f'(x)}{f(x)}$

Exercise 11B**1** Find the derivative of each of the following with respect to x :

a $y = 2 \log_e x$

b $y = 2 \log_e 2x$

c $y = x^2 + 3 \log_e 2x$

d $y = 3 \log_e x + \frac{1}{x}$

e $y = 3 \log_e(4x) + x$

f $y = \log_e(x + 1)$

g $y = \log_e(|2x + 3|)$

h $y = \log_e(|3 - 2x|)$

i $y = 3 \log_e(|2x - 3|)$

j $y = -3 \log_e\left(\left|\frac{x}{5} - 3\right|\right)$

k $y = 4x - 3 \log_e\left(\left|\frac{x}{2} - 3\right|\right)$

2 For each of the following find $f'(x)$:

a $f(x) = \log_e (x^2 + 1)$

b $f(x) = x \log_e x, x > 0$

c $f(x) = \frac{\log_e x}{x}, x > 0$

d $f(x) = 2x^2 \log_e x, x > 0$

e $f(x) = e^x \log_e x, x > 0$

f $f(x) = x \log_e (-x), x < 0$

g $f(x) = \log_e (e^x)$

h $f(x) = \frac{\log_e x}{x^2 + 1}$

3 Find the y -coordinate and the gradient at the points corresponding to a given value of x on the following curves:

a $y = \log_e x, x > 0; x = e$

b $y = x \log_e x, x > 0; x = e$

c $y = \log_e (x^2 + 1); x = e$

d $y = \log_e (-x), x < 0; x = -e$

e $y = x + \log_e x, x > 0; x = 1$

f $y = \log_e |x - 2|; x = 1$

g $y = \log_e |2x - 1|; x = 0$

h $y = \log_e |x^2 - 1|; x = 0$

4 Find $f'(1)$ if $f(x) = \log_e \sqrt{x^2 + 1}$

5 Differentiate $\log_e (1 + x + x^2)$.

6 If $f(x) = \log_e (x^2 + 1)$, find $f'(3)$.

7 It is known that $f(0) = 2$ and $f'(0) = 4$. Find $\frac{d(\log_e (f(x)))}{dx}$ when $x = 0$.

8 It is known that $f(1) = 2$ and $f'(1) = 4$. Find the derivative of $f(x) \log_e (x)$ when $x = 1$.

11.3 Applications of differentiation of exponential and logarithmic functions

Example 8

Sketch the graph of $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^{x^3}$

Solution

As $x \rightarrow -\infty, f(x) \rightarrow 0$

Axis intercepts

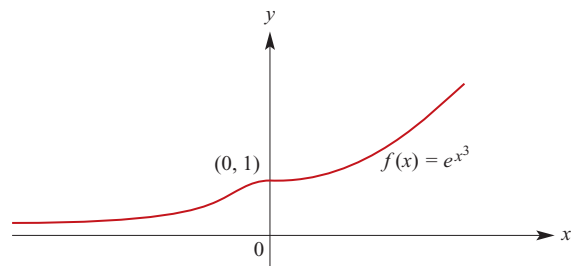
When $x = 0, f(x) = 1$

Turning points

$$f'(x) = 3x^2 e^{x^3}$$

and $f'(x) = 0$ implies $x = 0$

The gradient of f is always greater than or equal to 0 which means that $(0, 1)$ is a stationary point of inflexion.



Example 9

The growth of a population is modelled by the following alternatives:

- a** 24% a year **b** 12% every 6 months **c** 2% a month

If initially the population is 1000, find the population at the end of a year for each of the three alternatives.

Solution

- a** Population at the end of 1 year

$$\begin{aligned} &= 1.24 \times 1000 \\ &= 1240 \end{aligned}$$

- b** Population at the end of 1 year

$$\begin{aligned} &= (1.12)^2 \times 1000 \\ &= 1.2544 \times 1000 \\ &= 1254.4 \quad (= 1254 \text{ as integer values are required}) \end{aligned}$$

- c** Population at the end of one year

$$\begin{aligned} &= (1.02)^{12} \times 1000 \\ &= 1.268\,242 \times 1000 \\ &= 1268.24 \quad (= 1268 \text{ as integer values are required}) \end{aligned}$$

Previously it was seen that the limiting process, indicated in Example 9, leads to the population growth model for a year. If the population is considered to be increasing continuously at a rate of 24% a year then the population growth for a year can be written as the limit as n approaches infinity of $\lim_{n \rightarrow \infty} 1000 \left(1 + \frac{24}{100n}\right)^n$. In Chapter 5 it was stated that

$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$. The similarity is clear and a little algebraic manipulation gives

$$\begin{aligned} &\lim_{n \rightarrow \infty} 1000 \left(1 + \frac{24}{100n}\right)^n \\ &= \lim_{n \rightarrow \infty} 1000 \left(1 + \frac{1}{\frac{100n}{24}}\right)^{\left(\frac{100n}{24}\right)0.24} \\ &= \lim_{n \rightarrow \infty} 1000 \left(1 + \frac{1}{\frac{100n}{24}}\right)^{\frac{100n}{24}} \\ &= 1000e^{0.24} \\ &= 1.271\,25 \times 1000 \\ &= 1271.25 \quad (1271 \text{ as integer values are required}) \end{aligned}$$

After x years the population would be given by

$$\begin{aligned} P(x) &= 1000e^{0.24x} \\ \text{and } P'(x) &= 0.24 \times 1000e^{0.24x} \\ &= 0.24 P(x) \end{aligned}$$

i.e. the population is growing continuously at a rate of 24% of its population at any particular time.

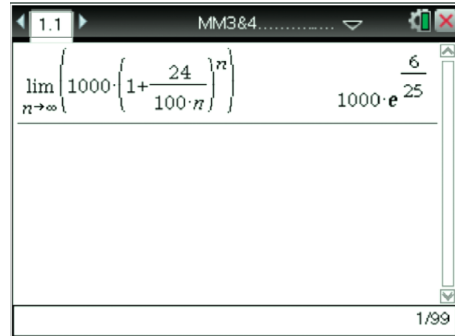
Using the TI-Nspire

Use $\text{menu} > \text{Calculus} > \text{Limit}$ to find the limit

$$\lim_{n \rightarrow \infty} 1000 \left(1 + \frac{24}{100n} \right)^n$$

∞ , the symbol for infinity, can be found using

π or $\text{ctrl} + \text{[]}$ (or $\text{ctrl} + i$ on the Clickpad)



Using the Casio ClassPad

A CAS calculator can be used to evaluate this limit.

Find the limit $\lim_{n \rightarrow \infty} 1000 \left(1 + \frac{24}{100n} \right)^n$.

Enter and highlight the expression

$$1000 \left(1 + \frac{24}{100n} \right)^n$$

Interactive–Calculation–lim and set the variable as n and the point as ∞ .



Example 10

Given that $f(x) = x - e^{2x}$ find in terms of p the approximate increase in $f(x)$ as x increases from 0 to $0 + p$, where p is small.

Solution

$$f'(x) = 1 - 2e^{2x} \text{ and } f'(0) = 1 - 2 = -1$$

$$\therefore f(0 + p) \approx pf'(0) + f(0)$$

$$= -p - 1$$

$$\text{Therefore } f(0 + p) - f(0) \approx -p - 1 - (-1)$$

$$= -p$$

Example 11

Given that $f(x) = e^{\frac{x}{10}}$ find in terms of h the approximate value of $f(10 + h)$, given that h is small.

Solution

$$\begin{aligned} f'(x) &= \frac{1}{10} e^{\frac{x}{10}} \text{ and } f'(10) = \frac{1}{10} e^{\frac{10}{10}} = \frac{e}{10} \\ \therefore f(10 + h) &\approx h \times \frac{e}{10} + e \\ &= e \left(\frac{h}{10} + 1 \right) \end{aligned}$$

Example 12

For $f: (0, \infty) \rightarrow \mathbb{R}$, $f(x) = x \log_e(x)$

- a** Find $f'(x)$. **b** Solve the equation $f(x) = 0$
c Solve the equation $f'(x) = 0$ **d** Sketch the graph of $y = f(x)$

Solution

a $f'(x) = x \times \frac{1}{x} + \log_e(x)$ (product rule)
 $= 1 + \log_e(x)$

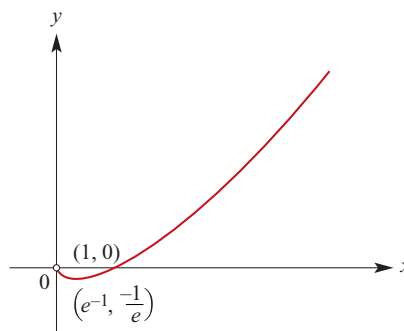
b $f(x) = x \log_e(x)$ and $f(x) = 0$ implies
 $x = 0$ or $\log_e(x) = 0$

Thus as $x \in (0, \infty)$, $x = 1$

c $f'(x) = 0$ implies $1 + \log_e(x) = 0$

Therefore $\log_e(x) = -1$ and $x = e^{-1}$

d When $x = e^{-1}$, $y = e^{-1} \log_e e^{-1} = \frac{-1}{e}$

**Example 13**

A particle moves along a curve with equation $y = \frac{1}{2}(e^{2x} + e^{-2x})$ where $x > 0$. The particle moves so that at time t seconds, its velocity in the positive y -axis direction is 2 units/second,

i.e. $\frac{dy}{dt} = 2$

Find $\frac{dx}{dt}$ when $x = 1$.

Solution

For $y = \frac{1}{2}(e^{2x} + e^{-2x})$, $\frac{dy}{dx} = e^{2x} - e^{-2x}$

Using the chain rule $\frac{dx}{dt} = \frac{dx}{dy} \frac{dy}{dt}$

$$= \frac{1}{e^{2x} - e^{-2x}} \times 2$$

$$= \frac{1}{e^{2x} - \frac{1}{e^{2x}}} \times 2$$

$$= \frac{1}{\frac{e^{4x} - 1}{e^{2x}}} \times 2$$

$$= \frac{2e^{2x}}{e^{4x} - 1}$$

The velocity in the positive direction of the x -axis when $x = 1$ is $\frac{2e^2}{e^4 - 1}$ units per second.

Exercise 11C

- 1 Sketch the graph of $f(x) = e^{-\frac{x^2}{2}}$
- 2 Let $f(x) = x^2 e^x$. Find $\{x: f'(x) < 0\}$.
- 3 Find the values of x for which $100e^{-x^2+2x-5}$ increases as x increases and hence find the maximum value of $100e^{-x^2+2x-5}$.
- 4 Let $f(x) = e^x - 1 - x$
 - a Find the minimum value of $f(x)$.
 - b Hence show $e^x \geq 1 + x$ for all real x .
- 5 Find an equation of the tangent to the graph for each function at the given value of x :

a $f(x) = e^x + e^{-x}; x = 0$	b $f(x) = \frac{e^x - e^{-x}}{2}; x = 0$
c $f(x) = x^2 e^{2x}; x = 1$	d $f(x) = e^{\sqrt{x}}; x = 1$
e $f(x) = x e^{x^2}; x = 1$	f $f(x) = x^2 e^{-x}; x = 2$
- 6 For $f(x) = x + e^{-x}$:
 - a Find the position and nature of any stationary points.
 - b Find, if they exist, the equations of any asymptotes.
 - c Sketch the graph of $y = f(x)$
- 7 A vehicle is travelling in a straight line from point O . Its displacement after t seconds is $0.4e^t$ metres. Find the velocity of the vehicle when $t = 0, t = 1, t = 2$.

- 8** A manufacturing company has a daily output y on day t of a production run given by $y = 600(1 - e^{-0.5t})$.
- Sketch the graph of y against t . (Assume a continuous model.)
 - Find the instantaneous rate of change of output y with respect to t on the 10th day.
- 9** Assume that the number of bacteria present in a culture at time t is given by $N(t)$ where $N(t) = 24te^{-0.2t}$. At what time will the population be at a maximum? Find the maximum population.
- 10** Find $\frac{dy}{dx}$ for:
- $y = e^{-2x}$
 - $y = Ae^{kx}$
- In each case, express your answer in terms of y .
- 11** The mass m kg of radioactive lead remaining in a sample t hours after observations began is given by $m = 2e^{-0.2t}$.
- Find the mass left after 12 hours.
 - Find how long it takes to fall to half of its value at $t = 0$.
 - Find how long it takes for the mass to fall to **i** one-quarter and **ii** one-eighth of its value at $t = 0$.
 - Express the rate of decay as a function of m .
- 12** Given that $y = e^{2x}$, find in terms of q the approximate increase in y as x increases from 0 to q .
- 13** For $f: R \rightarrow R$, $f(x) = e^{ax}$
- Find $f'(x)$.
 - Find an approximation for $f(h)$, where h is small, in terms of h and a .
 - Find an approximation for $f(b+h)$, where h is small, in terms of b , h and a .
- 14** For each of the following write down an expression for the approximate change, δy , in y when x changes from a to $a+p$ where p is small:
- $y = 2e^{\frac{x}{2}}$
 - $y = 3 - 2e^x$
 - $y = xe^x$
 - $y = \frac{x}{e^x}$
- 15** Given that $y = e^{2t} + 1$ and $x = e^t + 1$, find:
- $\frac{dy}{dt}$ and $\frac{dx}{dt}$
 - $\frac{dy}{dx}$ when $t = 0$
- 16** $y = e^x(px^2 + qx + r)$ is such that the tangents at $x = 1$ and $x = 3$ are parallel to the x -axis. The point with coordinates $(0, 9)$ is on the curve. Find the values of p , q and r .
- 17** The volume, $V \text{ cm}^3$, of water in a dish when the depth is $h \text{ cm}$ is given by the rule $V = \frac{\pi}{2}(e^{2h} - 1)$. The depth of the dish is 2.5 cm. If water is being poured in at $5 \text{ cm}^3/\text{s}$, find:
- the rate at which the depth of the water is increasing when the depth is 2 cm
 - the rate at which the depth of the water is increasing when the depth is 2.5 cm

- 18** A particle moves along a curve with equation $y = \frac{1}{2}(e^{2x} + e^{-2x})$ where $x > 0$. The particle moves so that at time t seconds, its velocity in the positive y -axis direction is 2 units/second, i.e. $\frac{dy}{dt} = 2$
Find $\frac{dx}{dt}$ when $x = 1$.
- 19** **a** Let $y = e^{4x^2-8x}$. Find $\frac{dy}{dx}$
b Find the coordinates of the stationary point on the curve of $y = e^{4x^2-8x}$ and state its nature.
c Sketch the graph of $y = e^{4x^2-8x}$
d Find the equation of the normal to the curve of $y = e^{4x^2-8x}$ at the point where $x = 2$
- 20** **a** Find the equation of the tangent and normal of the graph of $f(x) = \log_e x$ at the point $(1, 0)$.
b Find the equation of the tangent and normal of the graph of $f(x) = \log_e |x|$ at the point $(-1, 0)$.
- 21** **a** Find the equation of the tangent of the graph of $f(x) = \log_e 2x$ at the point $\left(\frac{1}{2}, 0\right)$.
b Find the equation of the tangent of the graph of $f(x) = \log_e kx$ at the point $\left(\frac{1}{k}, 0\right)$, ($k \in \mathbb{R}^+$).
c Find the equation of the tangent of the graph of $f(x) = \log_e |kx|$ at the point $\left(\frac{1}{k}, 0\right)$, ($k \in \mathbb{R} \setminus \{0\}$).
- 22** On the same set of axes sketch the graphs of $y = \log_e x$ and $y = \log_e 5x$ and use them to explain why $\frac{d}{dx}(\log_e x) = \frac{d}{dx}(\log_e 5x)$
- 23** **a** For $y = \log_e x$, find the increase in y as x increases from 1 to $1 + p$ where p is small.
b For $y = \log_e x$, find the increase in y as x increases from a to $a + p$ where p is small and $a > 1$.
c For $f(x) = \log_e (x + 1)$ show that $f(h) \approx h$ for h 'close' to 0.
- 24** For $y = \log_e (1 + x^2)$ find the increase in y as x increases from 0 to p where p is a small positive number.
- 25** For $y = \log_e \sqrt{1 + x + x^2}$, find the increase in y as x increases from 0 to p where p is a small positive number.
- 26** Find an approximation for $\log_e (1.01)$.
- 27** If $y = \log_e (t)$ and $x = \log_e (t^2 + 1)$ for $t > 0$, find $\frac{dy}{dt}$ and $\frac{dx}{dt}$ and hence find $\frac{dy}{dx}$ when $t = 1$.

- 28** The volume $V \text{ cm}^3$ of water in a dish when the depth is $h \text{ cm}$ is given by the rule $V = \pi(h+1)[(\log_e(h+1))^2 + h]$. The depth of the dish is 15 cm. If water is being poured in at $5 \text{ cm}^3/\text{s}$, find:
- a** the rate at which the depth of the water is increasing when the depth is 2 cm
 - b** the rate at which the depth of the water is increasing when the depth is 10 cm
- 29** For the function $f: (0, \infty) \rightarrow \mathbb{R}$, $f(x) = x^2 \log_e(x)$:
- a** Find $f'(x)$.
 - b** Solve the equation $f(x) = 0$
 - c** Solve the equation $f'(x) = 0$
 - d** Sketch the graph of $y = f(x)$

11.4 Derivatives of circular functions

The following results, which were established in Chapter 6, will be utilised:

- $\cos^2 \theta + \sin^2 \theta = 1$
- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

The derivative of $\sin k\theta$

We first consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(\theta) = \sin \theta$

Let $P(\theta, \sin \theta)$ and $Q(\theta + h, \sin(\theta + h))$ be points on the graph $f(\theta) = \sin \theta$

$$\begin{aligned} \text{The gradient of the chord } PQ &= \frac{\sin(\theta + h) - \sin \theta}{h} \\ &= \frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h} \\ &= \frac{\sin \theta(\cos h - 1) + \cos \theta \sin h}{h} \\ &= \frac{\sin \theta(\cos h - 1)}{h} + \frac{\cos \theta \sin h}{h} \end{aligned}$$

We consider what happens when $h \rightarrow 0$.

Use your calculator to check the following tables:

h	$\cos h$
0.1	0.995 004
0.05	0.998 750
0.01	0.999 950
0.001	0.999 999

h	$\sin h$	$\frac{\sin h}{h}$
0.1	0.099 834	0.998 334
0.05	0.049 979	0.999 583
0.01	0.009 999	0.999 983
0.001	0.000 999	0.999 999

These results suggest that:

$$\begin{aligned}\lim_{h \rightarrow 0} (\cos h - 1) &= 0 \text{ and } \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \\ \text{Therefore } f'(\theta) &= \lim_{h \rightarrow 0} \left(\frac{\sin \theta (\cos h - 1)}{h} + \frac{\cos \theta \sin h}{h} \right) \\ &= \sin \theta \times 0 + \cos \theta \times 1 \\ &= \cos \theta\end{aligned}$$

$$\begin{aligned}\text{For } f: R \rightarrow R, f(\theta) &= \sin \theta \\ f': R \rightarrow R, f'(\theta) &= \cos \theta\end{aligned}$$

For $f: R \rightarrow R, f(\theta) = \sin k\theta$

We use the chain rule to determine the rule $f'(\theta)$.

Let $y = \sin k\theta$ and $u = k\theta$

$$\begin{aligned}\text{Then } y = \sin u \text{ and } \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\ &= \cos u \cdot k \\ &= k \cos k\theta\end{aligned}$$

$$\begin{aligned}\text{i.e. for } f: R \rightarrow R, f(\theta) &= \sin(k\theta) \\ f': R \rightarrow R, f'(\theta) &= k \cos(k\theta)\end{aligned}$$

The derivative of $\cos k\theta$

We turn our attention to finding the derivative of $\cos k\theta$.

We first note the following:

$$\cos \theta = \sin \left(\frac{\pi}{2} - \theta \right) \quad \text{and} \quad \sin \theta = \cos \left(\frac{\pi}{2} - \theta \right)$$

These results will be used in the following way:

$$\begin{aligned}\text{let } y &= \cos \theta \\ &= \sin \left(\frac{\pi}{2} - \theta \right)\end{aligned}$$

We let $u = \frac{\pi}{2} - \theta$ and therefore $y = \sin u$

$$\begin{aligned}\text{The chain rule gives } \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\ &= \cos u \cdot -1 \\ &= -\cos \left(\frac{\pi}{2} - \theta \right) \\ &= -\sin \theta\end{aligned}$$

We have the following results:

$$\begin{aligned} f: R &\rightarrow R, f(\theta) = \cos \theta \\ f': R &\rightarrow R, f'(\theta) = -\sin \theta \\ \text{and} \\ f: R &\rightarrow R, f(\theta) = \cos(k\theta) \\ f': R &\rightarrow R, f'(\theta) = -k \sin(k\theta) \end{aligned}$$

Notation: $\sin^n \theta = (\sin \theta)^n$ and $\cos^n \theta = (\cos \theta)^n$. For convenience a new function, secant, is introduced. The rule of secant is $\sec \theta = \frac{1}{\cos \theta}$

The derivative of $\tan k\theta$

Let $y = \tan \theta$. We write $y = \frac{\sin \theta}{\cos \theta}$ and apply the quotient rule to find the derivative.

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{\cos \theta \cos \theta - \sin \theta (-\sin \theta)}{(\cos \theta)^2} \\ &= \frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta} \\ &= \frac{1}{\cos^2 \theta} \quad (\text{by the Pythagorean identity}) \\ &= \sec^2 \theta \end{aligned}$$

From this we state the result:

$$\begin{aligned} \text{For } f(\theta) &= \tan k\theta \\ f'(\theta) &= k \sec^2 k\theta \end{aligned}$$

The maximum domain for this function is $R \setminus \left\{ (2n+1)\frac{\pi}{2} : n \in Z \right\}$.

Example 14

Find the derivative of each of the following with respect to θ :

- | | | |
|--------------------------------|---------------------------|--------------------------------|
| a $\sin 2\theta$ | b $\sin^2 2\theta$ | c $\sin^2(2\theta + 1)$ |
| d $\cos^3(4\theta + 1)$ | e $\tan 3\theta$ | f $\tan(3\theta^2 + 1)$ |

Solution

a $2 \cos 2\theta$

b Let $y = \sin^2 2\theta$ and $u = \sin 2\theta$

Then $y = u^2$ and applying the chain rule:

$$\begin{aligned}
 \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\
 &= 2u \cdot 2 \cos 2\theta \\
 &= 4u \cos 2\theta \\
 &= 4 \sin 2\theta \cos 2\theta \\
 &= 2 \sin 4\theta, \text{ as } \sin 4\theta = 2 \sin 2\theta \cos 2\theta
 \end{aligned}$$

c Let $y = \sin^2(2\theta + 1)$ and $u = \sin(2\theta + 1)$

Then $y = u^2$ and applying the chain rule:

$$\begin{aligned}
 \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\
 &= 2u \cdot 2 \cos(2\theta + 1) \\
 &= 4 \sin(2\theta + 1) \cos(2\theta + 1) \\
 &= 2 \sin 2(2\theta + 1), \text{ as } \sin 2(2\theta + 1) = 2 \sin(2\theta + 1) \cos(2\theta + 1) \\
 &= 2 \sin(4\theta + 2)
 \end{aligned}$$

d Let $y = \cos^3(4\theta + 1)$ and $u = \cos(4\theta + 1)$

Then $y = u^3$ and applying the chain rule:

$$\begin{aligned}
 \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\
 &= 3u^2 \cdot -4 \sin(4\theta + 1) \\
 &= -12 \cos^2(4\theta + 1) \sin(4\theta + 1) \\
 &= -6 \sin(8\theta + 2) \cos(4\theta + 1)
 \end{aligned}$$

e $3 \sec^2 3\theta$

f Let $y = \tan(3\theta^2 + 1)$ and $u = 3\theta^2 + 1$

Then $y = \tan u$ and applying the chain rule:

$$\begin{aligned}
 \frac{dy}{d\theta} &= \frac{dy}{du} \cdot \frac{du}{d\theta} \\
 &= \sec^2 u \cdot 6\theta \\
 &= 6\theta \sec^2(3\theta^2 + 1)
 \end{aligned}$$

Example 15Find the y -coordinate and the gradient at the points on the following curves corresponding to the given values of θ :

a $y = \sin \theta$, $\theta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$

b $y = \cos \theta$, $\theta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$

c $y = \tan \theta$, $\theta = 0$ and $\theta = \frac{\pi}{4}$

Solution

$$\begin{aligned} \text{a} \quad \text{For } \theta = \frac{\pi}{4}, y &= \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} \\ \frac{dy}{d\theta} &= \cos \theta \end{aligned}$$

$\therefore$ the gradient at $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ is $\frac{1}{\sqrt{2}}$

For $\theta = \frac{\pi}{2}, y = \sin \frac{\pi}{2} = 1$ and the gradient at $\left(\frac{\pi}{2}, 1\right)$ is 0

$$\begin{aligned} \text{b} \quad \text{For } \theta = \frac{\pi}{4}, y &= \cos \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} \\ \frac{dy}{d\theta} &= -\sin \theta \end{aligned}$$

$\therefore$ the gradient at $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ is $-\frac{1}{\sqrt{2}}$

For $\theta = \frac{\pi}{2}, y = \cos \frac{\pi}{2} = 0$ and the gradient at $\left(\frac{\pi}{2}, 0\right)$ is -1

$$\text{c} \quad \text{For } \theta = 0, y = \tan 0 = 0$$

$$\frac{dy}{d\theta} = \sec^2 \theta$$

$\therefore$ the gradient at $(0, 0)$ is 1

For $\theta = \frac{\pi}{4}, y = \tan \frac{\pi}{4} = 1$ and the gradient at $\left(\frac{\pi}{4}, 1\right)$ is 2

Example 16

Find the derivative of each of the following with respect to x :

$$\text{a} \quad 2x^2 \sin 2x \quad \text{b} \quad \frac{\sin x}{x+1}, x \neq -1 \quad \text{c} \quad e^{2x} \sin(2x+1) \quad \text{d} \quad \cos 4x \sin 2x$$

Solution

$$\text{a} \quad \text{Let } y = 2x^2 \sin 2x$$

Applying the product rule:

$$\frac{dy}{dx} = 4x \sin 2x + 4x^2 \cos 2x$$

$$\text{c} \quad \text{Let } y = e^{2x} \sin(2x+1)$$

Applying the product rule:

$$\begin{aligned} \frac{dy}{dx} &= 2e^{2x} \sin(2x+1) + 2e^{2x} \cos(2x+1) \\ &= 2e^{2x} [\sin(2x+1) + \cos(2x+1)] \end{aligned}$$

$$\text{d} \quad \text{Let } y = \cos 4x \sin 2x$$

Applying the product rule:

$$\frac{dy}{dx} = -4 \sin 4x \sin 2x + 2 \cos 2x \cos 4x$$

$$\text{b} \quad \text{Let } y = \frac{\sin x}{x+1}, x \neq -1$$

Applying the quotient rule:

$$\frac{dy}{dx} = \frac{(x+1) \cos x - \sin x}{(x+1)^2}$$

Exercise 11D

- 1 Find the derivative with respect to x of each of the following:

a $\sin 5x$

b $\cos 5x$

c $\tan 5x$

d $\sin^2 x$

e $\tan(3x + 1)$

f $\cos(x^2 + 1)$

g $\sin^2\left(x - \frac{\pi}{4}\right)$

h $2 \cos x^\circ$

i $3 \sin x^\circ$

j $\tan(3x)^\circ$
- 2 Find the y -coordinate and the gradient at the points corresponding to the given value of x on the following curves:

a $y = \sin 2x, x = \frac{\pi}{8}$

b $y = \sin 3x, x = \frac{\pi}{6}$

c $y = 1 + \sin 3x, x = \frac{\pi}{6}$

d $y = \cos^2 2x, x = \frac{\pi}{4}$

e $y = \sin^2 2x, x = \frac{\pi}{4}$

f $y = \tan 2x, x = \frac{\pi}{8}$
- 3 For each of the following find $f'(x)$:

a $f(x) = 5 \cos x - 2 \sin 3x$

b $f(x) = \cos x + \sin x$

c $f(x) = \sin x + \tan x$

d $f(x) = \tan^2 x$
- 4 Differentiate each of the following with respect to x :

a $x^3 \cos x$

b $\frac{\cos x}{1+x}$

c $e^{-x} \sin x$

d $3x + 2 \cos x$

e $\sin 3x \cos 4x$

f $\tan 2x \sin 2x$

g $12x \sin x$

h $x^2 e^{\sin x}$

i $x^2 \cos^2 x$

j $e^x \tan x$
- 5 For each of the following find $f'(\pi)$:

a $f(x) = \frac{2x}{\cos x}$

b $f(x) = \frac{3x^2 + 1}{\cos x}$

c $f(x) = \frac{e^x}{\cos x}$

d $f(x) = e^x \sin x$

e $f(x) = \frac{\sin x}{x}$

f $f(x) = \cos^2 2x$
- 6 a If $y = -\log_e(\cos x)$, find $\frac{dy}{dx}$ b If $y = -\log_e |\cos x|$, find $\frac{dy}{dx}$
- 7 Find the derivative of each of the following functions:

a $\frac{\sin x + \cos x}{\sin x - \cos x}$

b $e^{\tan x}$

c $x^{\frac{7}{2}} \sin 3x$

11.5 Applications of derivatives of circular functions

Example 17

Find the equation of the tangent of the curve with equation $y = \sin x$ at the point where $x = \frac{\pi}{3}$.

Solution

$$\frac{dy}{dx} = \cos x. \text{ When } x = \frac{\pi}{3}, y = \frac{\sqrt{3}}{2} \text{ and } \frac{dy}{dx} = \frac{1}{2}$$

Therefore the equation of tangent is:

$$y - \frac{\sqrt{3}}{2} = \frac{1}{2} \left(x - \frac{\pi}{3} \right), \text{ which can be written in the form}$$

$$y = \frac{x}{2} - \frac{\pi}{6} + \frac{\sqrt{3}}{2}$$

Example 18

Find the equation of the tangent and normal of the graph of $y = -\cos x$ at the point $\left(\frac{\pi}{2}, 0\right)$.

Solution

We first find the gradient of the curve at this point.

$$\frac{dy}{dx} = \sin x \text{ and when } x = \frac{\pi}{2}, \frac{dy}{dx} = 1$$

$$\therefore \text{the equation of the tangent is } y - 0 = 1 \left(x - \frac{\pi}{2}\right)$$

$$\text{i.e. } y = x - \frac{\pi}{2}$$

The gradient of the normal is -1 and therefore the equation of the normal is:

$$y - 0 = -1 \left(x - \frac{\pi}{2}\right)$$

$$\text{i.e. } y = -x + \frac{\pi}{2}$$

Example 19

Let $f(\theta) = \sin(2\theta)$. If θ is increased by a small amount h find:

- a** an approximation for $f(\theta + h)$ where $\theta = \frac{\pi}{6}$
b an expression for the percentage change for $\theta = \frac{\pi}{6}$

Solution

$$\mathbf{a} \quad f'(\theta) = 2 \cos(2\theta) \text{ and } f'\left(\frac{\pi}{6}\right) = 2 \cos\left(\frac{\pi}{3}\right) = 1$$

$$\begin{aligned} \text{Therefore } f\left(\frac{\pi}{6} + h\right) &\approx h + f\left(\frac{\pi}{6}\right) \\ &= h + \frac{\sqrt{3}}{2} \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad \text{Percentage change} &= \frac{100hf'\left(\frac{\pi}{6}\right)}{f\left(\frac{\pi}{6}\right)} \\ &= \frac{100 \times h \times 1}{\frac{\sqrt{3}}{2}} \\ &= \frac{200h}{\sqrt{3}} \\ &= \frac{200h\sqrt{3}}{3} \end{aligned}$$

$$\text{The percentage change is } \frac{200h\sqrt{3}}{3}.$$

Example 20

The curves with equation $y = \cos x$ and $y = -\sin 2x$ intersect at the point where $x = \frac{7\pi}{6}$. Find the acute angle between the curves at this point.

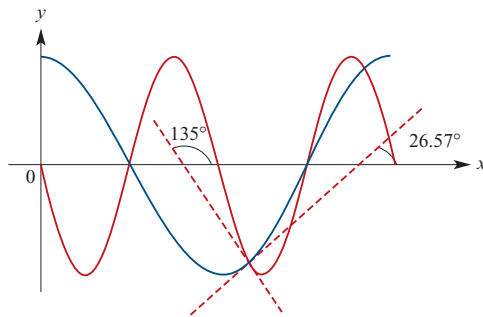
Solution

For $y = \cos x$, $\frac{dy}{dx} = -\sin x$ where $x = \frac{7\pi}{6}$.

For $y = -\sin 2x$, $\frac{dy}{dx} = -2\cos 2x$ where $x = \frac{7\pi}{6}$.

For the curve of $y = -\sin 2x$ at the point where $x = \frac{7\pi}{6}$, the angle of inclination to the positive direction of the x -axis is 135° .

For the curve of $y = \cos x$ at the point where $x = \frac{7\pi}{6}$, the angle of inclination to the positive direction of the x -axis is $\left[\tan^{-1}\left(-\frac{1}{2}\right)\right]^\circ \approx 26.57^\circ$. Therefore the angle between the two curves $\approx 135^\circ - 26.57^\circ \approx 108.43^\circ$. The acute angle is 71.57° .

**Example 21**

Find the local maximum and minimum values of $f(x) = 2 \sin x + 1 - 2 \sin^2 x$ where $0 \leq x \leq 2\pi$.

Solution

Find $f'(x)$ and solve $f'(x) = 0$.

$$f(x) = 2 \sin x + 1 - 2 \sin^2 x$$

$$\begin{aligned}\therefore f'(x) &= 2 \cos x - 4 \sin x \cos x \\ &= 2 \cos x (1 - 2 \sin x)\end{aligned}$$

$$f'(x) = 0 \text{ when } \cos x = 0 \text{ or } 1 - 2 \sin x = 0$$

$$\text{i.e. when } \cos x = 0 \text{ or } \sin x = \frac{1}{2}$$

$$\cos x = 0 \text{ implies } x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\sin x = \frac{1}{2} \text{ implies } x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$f\left(\frac{\pi}{2}\right) = 1, f\left(\frac{3\pi}{2}\right) = -3, f\left(\frac{\pi}{6}\right) = \frac{3}{2}, f\left(\frac{5\pi}{6}\right) = \frac{3}{2}$$

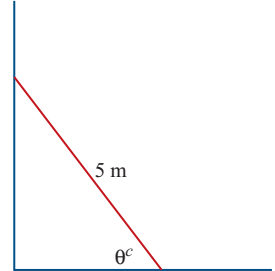
		$\frac{\pi}{6}$		$\frac{\pi}{2}$		$\frac{5\pi}{6}$		$\frac{3\pi}{2}$	
$f''(x)$	+	0	-	0	+	0	-	0	+
shape	/	—	\	—	/	—	\	—	/

$\therefore$ max. at $\left(\frac{\pi}{6}, \frac{3}{2}\right)$ and $\left(\frac{5\pi}{6}, \frac{3}{2}\right)$
 and min. at $\left(\frac{\pi}{2}, 1\right)$ and $\left(\frac{3\pi}{2}, -3\right)$

Example 22

The diagram shows a 5-metre pole leaning against a vertical wall. The lower end of the pole moves away from the wall at a constant speed of a m/s.

- Find for the instant that the foot of the pole is 3 m from the wall
- the speed at which the top of the pole is descending
 - the rate at which the radian measure of the angle defined by the pole and the horizontal between the bottom of the pole and the wall is changing.



Solution

- a** Let h metres be the distance, at time t seconds, from the top of the pole to the horizontal.

Let x metres be the distance, at time t seconds, from the bottom of the pole to the wall.

$$\begin{aligned}
 h &= \sqrt{25 - x^2} \\
 \text{Therefore } \frac{dh}{dx} &= -2x \times \frac{1}{2\sqrt{25 - x^2}} \\
 &= \frac{-x}{\sqrt{25 - x^2}} \\
 \text{and } \frac{dh}{dt} &= \frac{dh}{dx} \times \frac{dx}{dt} \\
 &= \frac{-x}{\sqrt{25 - x^2}} \times a
 \end{aligned}$$

$$\text{Furthermore } x = 3, \frac{dh}{dt} = \frac{-3a}{4}$$

The pole is sliding down the wall at a rate of $\frac{3a}{4}$ m/s.

- b** First it is noted $\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$

$$x = 5 \cos \theta. \text{ Therefore } \frac{dx}{d\theta} = -5 \sin \theta \text{ and } \frac{d\theta}{dx} = \frac{-1}{5 \sin \theta}$$

$$\text{Therefore } \frac{d\theta}{dt} = \frac{-1}{5 \sin \theta} \times a$$

$$\text{When } x = 3, \sin \theta = \frac{4}{5} \text{ and } \frac{d\theta}{dt} = \frac{-a}{4} \text{ radians/second}$$

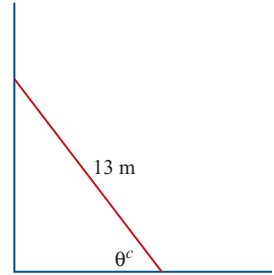
Exercise 11E



- 1 Find the equation of the tangent to each of the following curves at the point corresponding to the given x -value:
 - a $y = \sin 2x, x = 0$
 - b $y = \cos x, x = \frac{\pi}{2}$
 - c $y = \tan x, x = \frac{\pi}{4}$
 - d $y = \tan 2x, x = 0$
 - e $y = \sin x + \sin 2x, x = 0$
 - f $y = x - \tan x, x = \frac{\pi}{4}$
- 2 If $y = 3x + 2 \cos x$, find $\frac{dy}{dx}$ and hence show that y increases as x increases.
- 3 The tangent to the curve with equation $y = \tan 2x$ at the point where $x = \frac{\pi}{8}$ meets the y -axis at the point A . Find the distance OA where O is the origin.
- 4 Find the acute angle of intersection of the curves $y = \cos x$ and $y = e^{-x}$ at the point $(0, 1)$.
- 5 Find the acute angle of intersection of the curves $y = \sin 2x$ and $y = \cos x$ at the point $(\frac{\pi}{2}, 0)$.
- 6 The volume, $V(t)$, of water in a reservoir at time t is given by $V(t) = 3 + 2 \sin \frac{t}{4}$
 - a Find the volume in the reservoir at time $t = 10$.
 - b Find the rate of change of the volume of water in the reservoir at time $t = 10$.
- 7 Let $y = \tan \theta$.
 - a i If θ is increased by a small amount p find an approximation for the increase in y in terms of p and θ .
 ii If $\theta = \frac{\pi}{4}$, find the increase in y in terms of p .
 - b Find an approximate value of $\tan(46^\circ)$ using the results of a.
- 8 a Given $f: R \rightarrow R$, where $f(x) = \cos x$, find $f(\frac{\pi}{4})$ and $f'(\frac{\pi}{4})$
 b Use the results of a to find an approximate value of:
 - i $\cos(\frac{\pi}{4} + h)$ where h is small
 - ii $\cos 0.8$
- 9 For each of the following write down an expression for the approximate change, δy , in y when x changes from a to $a + p$ where p is small:
 - a $y = \cos(2x)$
 - b $y = \sin(\frac{x}{2})$
 - c $y = \tan(2x)$
 - d $y = 1 - \tan(\frac{x}{2})$
 - e $y = \cos(\frac{\pi}{4} - x)$
 - f $y = \sin(\frac{-x}{2})$
- 10 Find the local maximum and minimum values of $f(x)$ for each of the following and state the corresponding x -value of each. (Consider $0 \leq x \leq 2\pi$ only.)
 - a $f(x) = 2 \cos x - (2 \cos^2 x - 1)$
 - b $f(x) = 2 \cos x + 2 \sin x \cos x$
 - c $f(x) = 2 \sin x - (2 \cos^2 x - 1)$
 - d $f(x) = 2 \sin x + 2 \sin x \cos x$

- 11** The diagram shows a 13-metre pole leaning against a vertical wall. The lower end of the pole moves away from the wall at a constant speed of 1 m/s. For the instant that the foot of the pole is 12 m from the wall, find:

- a** the speed at which the top of the pole is descending
- b** the rate at which the radian measure of the angle defined by the pole and the horizontal between the bottom of the pole and the wall is changing



11.6 Miscellaneous exercises

In the following exercise a CAS calculator is to be used for all questions.

Exercise 11F

- Find the derivative of each of the following for the given x -value. Give the answer correct to two decimal places.
 - a** $y = e^{-(\frac{x}{10})^2} \sin x, x = \frac{\pi}{4}$
 - b** $y = \log_e(\sin x), x = \frac{\pi}{4}$
 - c** $y = \sin x + \cos 3x, x = 0.7$
 - d** $y = \log_e(x) + \sin x, x = 1$
- Find the coordinates of the local minima and maxima for $x \in [0, 2\pi]$ for each of the following. (Values are to be given correct to two decimal places.)
 - a** $y = \log_e(x) + \sin(x)$
 - b** $y = \log_e(\sin x)$
 - c** $y = \log_e(2x) + \cos(2x)$
 - d** $y = x^2 + \cos(x)$
 - e** $y = \tan(2x) + \frac{1}{\cos x}$
 - f** $y = \cos^2(2x) \sin x$
- For $y = \tan(2x) + \frac{1}{\cos x}$ find $\frac{dy}{dx}$ and plot the graph of $\frac{dy}{dx}$ against x for $x \in [4, 5.5]$. Use this to confirm the results of **2e**.
- For each of the following, plot the graph of the derivative function for the stated domain:
 - a** $f(x) = \cos^3(2x) \sin(x), [0, 2\pi]$
 - b** $f(x) = e^{\frac{x}{10}} \log_e(x), (0, 10]$
 - c** $f(x) = \sin^3(x) \cos^5(x), [0, 2\pi]$
- a** With a CAS calculator plot the graph of:
 - i** $y = e^x - 2$
 - ii** $y = x$
 - b** Find the coordinates of the points of intersection of $y = e^x - 2$ and $y = x$ from the calculator, correct to two decimal places.
 - c** Find the inverse function of $y = e^x - 2$ and plot this inverse on the same screen as **a**.
 - d** Find $\{x: e^x - 2 > x\}$
 - e**
 - i** The graph of $y = e^x + a$ passes through the origin. Find the value of a .
 - ii** Find the equation of the tangent to $y = e^x - 1$ at the origin.
 - iii** Find the point of intersection of $y = e^x - 1$ and $y = x$
 - iv** Find the inverse of $y = e^x - 1$
 - v** Using a calculator plot the graph of $y = e^x - 1$, the inverse of this function, and $y = x$

6 a With a CAS calculator plot the graph of:

i $y = 2e^x - 2$

ii $y = \log_e \left(\frac{x+2}{2} \right)$

iii $y = x$

b Find the gradient of $y = 2e^x - 2$ and $y = \log_e \left(\frac{x+2}{2} \right)$ at the origin.

c Find the equations of the tangents of $y = 2e^x - 2$ and $y = \log_e \left(\frac{x+2}{2} \right)$ at the origin.

d Consider the function with rule $f(x) = ae^x - a$, $a > 0$

i Find f^{-1}

ii Find the equations of the tangents to the graphs of $y = \log_e \left(\frac{x+a}{a} \right)$ and $y = ae^x - a$ at the origin.

iii Plot the graphs of $y = \log_e \left(\frac{x+a}{a} \right)$ and $y = ae^x - a$ for $a = 3$ and $a = \frac{1}{3}$.

iv Comment on the points of intersection of the two graphs.

11.7 Applications of transcendental functions

Example 23

The number of bacteria, N , in a culture increases at a rate proportional to the number present according to the law $N = N_0 e^{kt}$, where t is the number of hours of growth and k and N_0 are constants.

a Prove that $\frac{dN}{dt}$ is proportional to N .

b If it takes 48 hours for the colony to double in number, find k and hence the rate at which the colony is increasing when $N = 10^4$.

Solution

$$\begin{aligned} \text{a } \frac{dN}{dt} &= kN_0 e^{kt} \\ &= kN \end{aligned}$$

i.e. $\frac{dN}{dt}$ is proportional to N .

b When $t = 0$, $N = N_0$

When $N = 2N_0$, $t = 48$

$$2N_0 = N_0 e^{48k}$$

$$2 = e^{48k}$$

$$\therefore k = \frac{1}{48} \log_e 2$$

$$\approx 0.0144$$

When $N = 10^4$,

$$\frac{dN}{dt} = kN$$

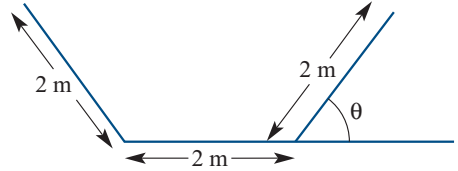
$$= \frac{1}{48} \log_e 2 \times 10^4$$

$$\approx 144.41$$

The colony is increasing at a rate of ≈ 144 per hour.

Example 24

The cross-section of a drain is to be an isosceles trapezoid, three of whose sides are 2 metres long, as shown. Find θ so that the cross-sectional area will be as great as possible and find this maximum.



Solution

$$\begin{aligned}\text{Area of the trapezium} &= \frac{1}{2} \times 2 \sin \theta (2 + 2 + 4 \times \cos \theta) \\ &= \sin \theta (4 + 4 \cos \theta)\end{aligned}$$

Let $A \text{ m}^2$ be the area of the trapezium.

$$\begin{aligned}\text{Then } A &= \sin \theta (4 + 4 \cos \theta) \\ \text{and } \frac{dA}{d\theta} &= \cos \theta (4 + 4 \cos \theta) - 4 \sin^2 \theta \\ &= 4 \cos \theta + 4 \cos^2 \theta - 4(1 - \cos^2 \theta) \\ &= 4 \cos \theta + 8 \cos^2 \theta - 4\end{aligned}$$

The maximum will occur when $\frac{dA}{d\theta} = 0$.
Consider:

$$\begin{aligned}8 \cos^2 \theta + 4 \cos \theta - 4 &= 0 \\ \therefore 2 \cos^2 \theta + \cos \theta - 1 &= 0 \\ \therefore (2 \cos \theta - 1)(\cos \theta + 1) &= 0 \\ \therefore \cos \theta &= \frac{1}{2} \text{ or } \cos \theta = -1\end{aligned}$$

The practical restriction on θ is that $0 < \theta \leq \frac{\pi}{2}$

Therefore the only possible solution is that $\theta = \frac{\pi}{3}$ and a gradient table confirms that $\frac{\pi}{3}$ gives a maximum.

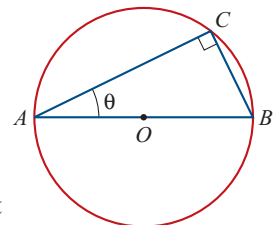
$$\text{When } \theta = \frac{\pi}{3}, A = \frac{\sqrt{3}}{2} (4 + 2) = 3\sqrt{3}$$

i.e. the maximum cross-sectional area is $3\sqrt{3} \text{ m}^2$.

		$\frac{\pi}{3}$	
$A'(\theta)$	+	0	-
shape	/	—	\

Example 25

The figure shows a circular lake, centre O , of radius 2 km. A man swims across the lake from A to C at 3 km/h and then walks around the edge of the lake from C to B at 4 km/h.



a If $\angle BAC = \theta$ radians and the total time taken is T hours, show that

$$T = \frac{1}{3} (4 \cos \theta + 3\theta)$$

- b Find the value of θ for which $\frac{dT}{d\theta} = 0$ and determine whether this gives a maximum or minimum value of T ($0^\circ < \theta^\circ < 90^\circ$).

Solution

a The time taken = $\frac{\text{distance travelled}}{\text{speed}}$

Therefore for the swim the time taken = $\frac{4 \cos \theta}{3}$ hours and the walk takes $\frac{4\theta}{4}$ hours.

$\therefore$ the total time taken = $\frac{1}{3}(4 \cos \theta + 3\theta)$

b $\frac{dT}{d\theta} = \frac{1}{3}(-4 \sin \theta + 3)$

The stationary point occurs where $\frac{dT}{d\theta} = 0$ and $\frac{dT}{d\theta} = \frac{1}{3}(-4 \sin \theta + 3)$

implies $\sin \theta = \frac{3}{4}$

$\therefore \theta = 48^\circ 35'$

If $\theta < 48^\circ 35'$, $T'(\theta) > 0$

if $\theta > 48^\circ 35'$, $T'(\theta) < 0$

$\therefore$ maximum when $\theta = 48^\circ 35'$

When $\sin \theta = \frac{3}{4}$, $T = 1.73$ hours.

If the man swims straight across it takes $1\frac{1}{3}$ hours.

If he goes all the way around the edge it takes approximately 1.57 hours.

Example 26

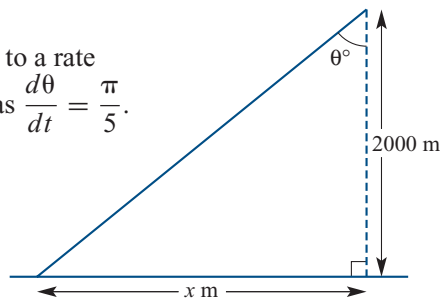
A beacon that makes one revolution every 10 seconds is located on a ship 2 km from a straight shoreline. How fast is the beam moving along the shoreline when it makes an angle of 45° with the shoreline?

Solution

One revolution every 10 seconds is equivalent to a rate of $\frac{\pi}{5}$ radians per second. This can be written as $\frac{d\theta}{dt} = \frac{\pi}{5}$.

Also $x = 2000 \tan \theta$ and $\frac{dx}{d\theta} = 2000 \sec^2 \theta$

By the chain rule $\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$
 $= 2000 \sec^2 \theta \times \frac{\pi}{5}$
 $= 400\pi \sec^2 \theta$



When the angle with the shoreline is 45° , $\theta = \frac{\pi}{4}$

$$\text{and } \frac{dx}{dt} = 400\pi \sec^2 \frac{\pi}{4} \\ = 800\pi$$

The beam is moving along the shoreline at a speed of 800π metres/second.

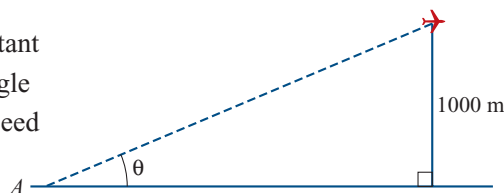
Exercise 11G

- 1 A car tyre is inflated to a pressure of 30 units. Eight hours later it is found to have deflated to 10 units. The pressure P at time t hours is given by:

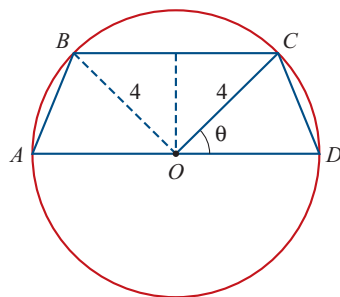
$$P = P_0 e^{-\lambda t}$$

- a Find the values of P_0 and λ . b At what time would the pressure be 8 units?
c Find the rate of loss of pressure at:
i time $t = 0$ ii time $t = 8$

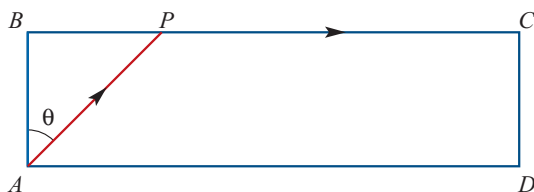
- 2 An aeroplane is flying horizontally at a constant height of 1000 m. At a certain instant the angle of elevation is 30° and decreasing and the speed of the aeroplane is 480 km/h.



- a How fast is θ decreasing at this instant?
(Answer in degrees/s.)
b How fast is the distance between the aeroplane and the observation point changing at this instant?
- 3 $ABCD$ is a trapezium with $AB = CD$, with vertices on the circle and with centre O . AD is a diameter of the circle. The radius of the circle is 4 units.



- a Find BC in terms of θ .
b Find the area of the trapezium in terms of θ and hence find the maximum area.
- 4 The figure shows a rectangular field in which $AB = 300$ m and $BC = 1100$ m.



$$AB = 300 \text{ m} \\ BC = 1100 \text{ m}$$

- a An athlete runs across the field from A to P at 4 m/s. Find the time taken to run from A to P in terms of θ .

b The athlete, on reaching P , immediately runs to C at 5 m/s. Find the time taken to run from P to C in terms of θ .

c Show, using the results from **a** and **b**, that the total time taken, T seconds, is given by

$$T = 220 + \frac{75 - 60 \sin \theta}{\cos \theta}$$

d Find $\frac{dT}{d\theta}$.

e Find the value of θ for which $\frac{dT}{d\theta} = 0$ and show that this is the value of θ for which T is a minimum.

f Find the minimum value of T and the distance of point P from B that will minimise her running time.



Chapter summary

- For $f(x) = e^{kx}$, $f'(x) = ke^{kx}$
- For $f(x) = \log_e kx$, with $kx > 0$, $f'(x) = \frac{1}{x}$
- For $f(x) = \log_e |kx|$, $f'(x) = \frac{1}{x}$
- For $f(x) = \sin kx$, $f'(x) = k \cos kx$
- For $f(x) = \cos kx$, $f'(x) = -k \sin kx$
- For $f(x) = \tan(kx)$, $f'(x) = k \sec^2(kx)$

$$= k \frac{1}{\cos^2(kx)}$$

$$= \frac{k}{\cos^2(kx)}$$

Multiple-choice questions

- The derivative of $e^{-2ax} \cos(ax)$ with respect to x is:
A $-ae^{-2ax} \cos(ax) - 2ae^{-2ax} \sin(ax)$ **B** $ae^{-2ax} \cos(ax) - 2ae^{-2ax} \sin(ax)$
C $-2ae^{-2ax} \cos(ax) - ae^{-2ax} \sin(ax)$ **D** $2ae^{-2ax} \cos(ax) + 2ae^{-2ax} \sin(ax)$
E $-ae^{-2ax} \cos(ax) - 2ae^{-2ax} \sin(ax)$
- For $f(x) = \frac{\cos x}{x-a}$, where a is a constant, find $f'(x)$.
A $\frac{\sin x}{x-a} + \frac{\cos x}{(x-a)^2}$ **B** $-\frac{\sin x}{x-a} - \frac{\cos x}{(x-a)^2}$ **C** $\frac{\sin x}{x-a} - \frac{\cos x}{(x-a)^2}$
D $\frac{x \sin x}{x-a} - \frac{x \cos x}{(x-a)^2}$ **E** $\frac{\sin x}{x} - \frac{\cos x}{x}$
- For $f: R \rightarrow R$, $f(x) = e^x - ex$, the coordinates of the turning point of the graph of $y = f(x)$ are:
A $\left(1, \frac{1}{e}\right)$ **B** $(1, e)$ **C** $(0, 1)$ **D** $(1, 0)$ **E** $(e, 1)$
- The equation of the tangent of $y = e^{ax}$ at the point $\left(\frac{1}{a}, e\right)$ is:
A $y = e^{ax-1} + 1$ **B** $y = ae^{ax}x$ **C** $y = 1 - ae^{ax}$
D $y = \frac{e^2x}{a}$ **E** $y = aex$
- If $z = \log_e(x)$ then δz is approximately equal to:
A $\log_e(x + \delta x)$ **B** $\log_e(\delta x)$ **C** $\frac{1}{\delta x}$ **D** $\frac{\delta x}{x}$ **E** $\frac{1}{x}$
- If $z = \sin x$ and $\sin 1 = a$ then using the linear approximation the value of $\sin(1.1)$ is equal to:
A $0.1\sqrt{1-a^2}$ **B** $0.1 \cos 1$ **C** $0.1\sqrt{1-a^2} + a$ **D** $a + 0.1$ **E** $0.1a$
- Under certain conditions, the number of bacteria, N , in a sample increases with time, t hours, according to the rule $N = 4000e^{0.2t}$. The rate, to the nearest whole number of bacteria per hour, that the bacteria are growing 3 hours from the start is:
A 1458 **B** 7288 **C** 16 068 **D** 80 342 **E** 109 731

- 8 The gradient of the tangent to the curve $y = x^2 \cos 5x$ at the point where $x = \pi$ is:
A $5\pi^2$ **B** $-5\pi^2$ **C** 5π **D** -5π **E** -2π
- 9 The equation of the tangent to the curve with equation $y = e^{-x} - 1$ at the point where the curve crosses the y -axis is:
A $y = x$ **B** $y = -x$ **C** $y = \frac{1}{2}x$ **D** $y = -\frac{1}{2}x$ **E** $y = -2x$
- 10 For $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{ax} - \frac{ax}{e}$, the coordinates of the turning point of the graph of $y = f(x)$ are:
A $\left(-\frac{1}{a}, 0\right)$ **B** $\left(\frac{1}{a}, \frac{1}{e}\right)$ **C** $\left(-\frac{1}{a}, \frac{2}{e}\right)$ **D** $\left(-1, \frac{1}{e}\right)$ **E** $(1, 0)$

Short-answer questions (technology-free)

- 1 Differentiate each of the following with respect to x :
a $\log_e(x^2 + 2)$ **b** $\sin(3x + 2)$ **c** $\cos\left(\frac{x}{2}\right)$ **d** e^{x^2-2x}
e $\log_e(3 - x)$ **f** $\sin(2\pi x)$ **g** $\sin^2(3x + 1)$ **h** $\sqrt{\log_e x}$, $x > 1$
i $\frac{2 \log_e 2x}{x}$ **j** $x^2 \sin(2\pi x)$
- 2 Differentiate each of the following with respect to x :
a $e^x \sin 2x$ **b** $2x^2 \log_e x$ **c** $\frac{\log_e x}{x^3}$ **d** $\sin 2x \cos 3x$
e $\frac{\sin 2x}{\cos 2x}$ **f** $\cos^3(3x + 2)$ **g** $x^2 \sin^2(3x)$
- 3 Find the gradient of each of the following curves for the stated value of x :
a $y = e^{2x} + 1$, $x = 1$ **b** $y = e^{x^2+1}$, $x = 0$
c $y = 5e^{3x} + x^2$, $x = 1$ **d** $y = 5 - e^{-x}$, $x = 0$
- 4 Differentiate each of the following with respect to x :
a e^{ax} **b** e^{ax+b} **c** e^{a-bx} **d** $be^{ax} - ae^{bx}$ **e** $\frac{e^{ax}}{e^{bx}}$
- 5 A vehicle is travelling in a straight line from a point O . Its displacement after t seconds is $0.25e^t$ metres. Find the velocity of the vehicle at $t = 0$, $t = 1$, $t = 2$ and $t = 4$ (seconds).
- 6 The temperature, $\theta^\circ\text{C}$, of material inside a nuclear power station at time t seconds after a reaction begins is given by $\theta = \frac{1}{4}e^{100t}$
a Find the rate of temperature increase at time t .
b Find the rate of increase of temperature when $t = \frac{1}{20}$.
- 7 Find the equation of the tangent of $y = e^x$ at $(1, e)$.
- 8 The diameter of a tree (D cm) t years after 1 January 1990 is given by $D = 50e^{kt}$
a Prove that $\frac{dD}{dt} = cD$ for some constant c .
b If $k = 0.2$, find the rate of increase of D where $D = 100$.
- 9 If $x = 2e^{-2t} + 3e^{-t}$, show that $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = 0$
- 10 Find the minimum value of $e^{3x} + e^{-3x}$.

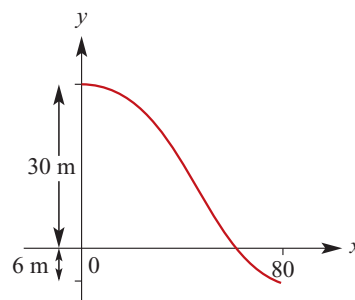
- 11 $y = e^{ax}$ is a solution of the equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$. Find two values of a .
- 12 a Find the equation of the tangent of $y = \log_e x$ at the point $(e, 1)$.
 b Find the equation of the tangent to $y = 2 \sin\left(\frac{x}{2}\right)$ at the point $\left(\frac{\pi}{2}, \sqrt{2}\right)$.
 c Find the equation to the tangent to $y = \cos x$ at the point $\left(\frac{3\pi}{2}, 0\right)$.
 d Find the equation of the tangent of $y = \log_e |x|$ at the point $(-e, 1)$.

Extended-response questions

- 1 A section of a rollercoaster can be described by the rule:

$$y = 18 \cos\left(\frac{\pi x}{80}\right) + 12, 0 \leq x \leq 80$$

- a Find the gradient function, $\frac{dy}{dx}$.
 b Sketch the graph of $\frac{dy}{dx}$ against x .
 c State the coordinates of the point on the track for which the magnitude of the gradient is maximum.



- 2 The kangaroo population in a certain confined region is given by $f(x) = \frac{100\,000}{1 + 100e^{-0.3x}}$ where x is the time in years.
- a Find $f'(x)$
 b Find the rate of growth of the kangaroo population when:
 i $x = 0$ and ii $x = 4$
- 3 Consider the function $f: \{x: x < a\} \rightarrow R, f(x) = 8 \log_e (6 - 0.2x)$ where a is the largest value for which f is defined.
- a What is the value of a ?
 b Find the exact values for the coordinates of the points where the graph of $y = f(x)$ crosses each axis.
 c Find the gradient of the tangent to the graph of $y = f(x)$ at the point where $x = 20$.
 d Find the rule of the inverse function f^{-1} .
 e State the domain of the inverse function f^{-1} .
 f Sketch the graph of $y = f(x)$
- 4 a Using a calculator plot the graphs of $f(x) = \sin x$ and $g(x) = e^{\sin x}$ on the one screen.
 b Find $g'(x)$ and hence find the coordinates of the stationary points of $y = g(x)$ for $x \in [0, 2\pi]$.
 c Give the range of g .
 d State the period of g .



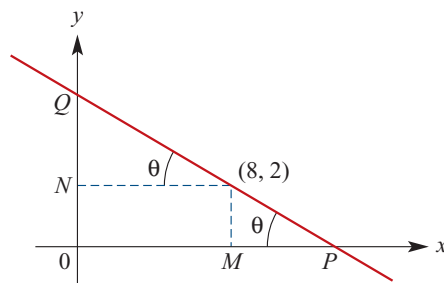


- $$h(x) = af(x - b) + c \quad \text{and} \quad k(x) = ag(x - b) + c$$

d Use the chain rule and properties of transformations to prove that if the tangent of the curve with equation $y = f(x)$ at the point (x_1, y_1) is $y = mx + c$, then the equation of the tangent of the curve with equation $y = af(bx)$ at the point $\left(\frac{x_1}{b}, y_1a\right)$ is $y = a(mb x + c)$

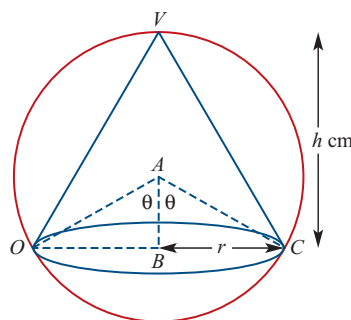
- $$x = \frac{60}{5e^{\lambda t} - 3}, \text{ where } \lambda = \frac{1}{2} \log_e \frac{6}{5}$$

- 7 A straight line is drawn through the point $(8, 2)$ to intersect the positive y -axis at Q and the positive x -axis at P . (In this problem we will determine the minimum value of $OP + OQ$.)



- a** Show that the derivative of $\frac{1}{\tan \theta}$ is $-\operatorname{cosec}^2 \theta$.
- b** Find MP in terms of θ .
- c** Find NQ in terms of θ .
- d** Hence find $OP + OQ$ in terms of θ . Denote $OP + OQ$ by x .
- e** Find $\frac{dx}{d\theta}$.
- f** Find the minimum value for x and the value of θ for which this occurs.
- 8** Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x - e^{-x}$
- a** Find $f'(x)$.
- b** Find $\{x: f(x) = 0\}$
- c** Show that $f'(x) > 0$ for all x .
- d** Sketch the graph of f .
- 

- 9 a Find all values of x for which $(\log_e x)^2 = 2 \log_e x$
 b Find the gradient of each of the curves $y = 2 \log_e x$ and $y = (\log_e x)^2$ at the point $(1, 0)$.
 c Use these results to sketch on one set of axes the graphs of $y = 2 \log_e x$ and $y = (\log_e x)^2$
 d Find $\{x: 2 \log_e x > (\log_e x)^2\}$
- 10 For what value of x is $\frac{\log_e x}{x}$ a maximum? That is, when is the ratio of the logarithm of a number to the number a maximum?
- 11 A cone is inscribed inside a sphere as illustrated. The radius of the sphere is a cm, and the magnitude of $\angle OAB = \text{magnitude of } \angle CAB = \theta$. The height of the cone is denoted by h . The radius of the cone is denoted by r .



- a Find h , the height of the cone in terms of a and θ .
 b Find r , the radius of the cone in terms of a and θ .
 The volume, $V \text{ cm}^3$, of the cone is given by $V = \frac{1}{3} \pi r^2 h$.
 c Use the results from **a** and **b** to show that:
- $$V = \frac{1}{3} \pi a^3 \sin^2 \theta (1 + \cos \theta)$$
- d Find $\frac{dV}{d\theta}$ (a is a constant) and hence find the value of θ for which the volume is a maximum.
 e Find the maximum volume of the cone in terms of a .
- 12 A psychologist hypothesised that the ability of a mouse to memorise during the first 6 months of its life can be modelled by the function f , given by $f: (0, 6] \rightarrow R$, $f(x) = x \log_e x + 1$, i.e. the ability to memorise at age x years is $f(x)$.
- a Find $f'(x)$
 b Find the value of x for which $f'(x) = 0$ and hence find when the mouse's ability to memorise is a minimum.
 c Sketch the graph of f .
 d When is the mouse's ability to memorise a maximum in this period?
- 13 Some bacteria are introduced into a supply of fresh milk. After t hours there are y grams of bacteria present where $y = \frac{Ae^{bt}}{1 + Ae^{bt}}$ (1) and A and b are positive constants.
- a Show that $0 < y < 1$ for all values of t . b Find $\frac{dy}{dt}$ in terms of t .
 c From (1) show that $Ae^{bt} = \frac{y}{1-y}$
 d i Show that $\frac{dy}{dt} = by(1-y)$
 ii Hence, or otherwise, show that the maximum value of $\frac{dy}{dt}$ occurs when $y = 0.5$.
 e If $A = 0.01$ and $b = 0.7$ find when, to the nearest hour, the bacteria will be increasing at the fastest rate.

- 14** A searchlight is located at ground level vertically below the path of an approaching aircraft, which is flying at a constant speed of 400 m/s, at a height of 10 000 metres. If the light is continuously directed at the aircraft, find the rate, in degrees per second, at which the searchlight is turning at the instant when the aircraft is at a horizontal distance of 5000 m from the searchlight. Give your answer correct to three decimal places.
- 15** A rocket, R , rises vertically from level ground at a point A . It is observed from another point B on the ground where B is 10 km from A . When the angle of elevation ABR has the value $\frac{\pi}{4}$ radians, this angle is increasing at the rate of 0.005 radians per second. Find in, km/s, the velocity of the rocket at that instant.
- 16** **a** Find the equation of the tangent to the curve $y = e^x$ at the point $(1, e)$.
b Find the equation of the tangent to the curve $y = e^{2x}$ at the point $\left(\frac{1}{2}, e\right)$.
c Find the equation of the tangent to the curve $y = e^{kx}$ at the point $\left(\frac{1}{k}, e\right)$.
d Show that $y = xke$ is the only tangent to the curve $y = e^{kx}$ which passes through the origin.
e Hence determine for what values of k the equation $e^{kx} = x$ has:
i a unique real root **ii** no real roots
- 17** Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$, $f(x) = \frac{e^x}{x}$
a Find $f'(x)$ **b** Find $\{x: f'(x) = 0\}$
c Find the coordinates of the one stationary point and state its nature.
d i Find $\frac{f'(x)}{f(x)}$ **ii** Find $\lim_{x \rightarrow \infty} \frac{f'(x)}{f(x)}$ and comment.
e Sketch the graph of f .
f The number of birds (n) in an island colony decreased and increased with time (t) years according to the approximate formula

$$n = \frac{ae^{kt}}{t}$$

over some interval of years, where t is measured from 1900 and a and k are constant. If during this period the population was the same in 1965 as it was in 1930, when was it least?

- 18** A culture contains 1000 bacteria and 5 hours later the number has increased to 10 000. The number, N , of bacteria present at any time, t hours, is given by $N = Ae^{kt}$
a Find the values of A and k .
b Find the rate of growth at any time t .
c Show that the rate of growth is proportional to the number of bacteria present at any time.
d Find this rate of growth when:
i $t = 4$ **ii** $t = 50$

- 19 The populations of two ant colonies, A and B , are increasing according to the rules:

■ A : population $= 2 \times 10^4 e^{0.03t}$

■ B : population $= 10^4 e^{0.05t}$

After how many years will their populations:

- a** be equal? **b** be increasing at the same rate?

- 20 The depth, $D(t)$ metres, of water at the entrance to a harbour at t hours after midnight on a particular day is given by:

$$D(t) = 10 + 3 \sin\left(\frac{\pi t}{6}\right), 0 \leq t \leq 24$$

- a** Sketch the graph of $D(t)$ for $0 \leq t \leq 24$.
b Find the values of t for which $D(t) \geq 8.5$.
c Find the rate at which the depth is changing when:

i $t = 3$ **ii** $t = 6$ **iii** $t = 12$

- d i** At what times is the depth increasing most rapidly?

- ii** At what times is the depth decreasing most rapidly?

- 21 A particle on the end of a spring, which is hanging vertically, is oscillating so that its length h metres above the floor after t seconds is given by:

$$y = 0.5 + 0.2 \sin(3\pi t), t \geq 0$$

- a** Find the greatest height above the floor and the time at which this height is first reached.
b Find the period of oscillation. **c** Find the speed of the particle when $t = \frac{1}{3}, \frac{2}{3}, \frac{1}{6}$.

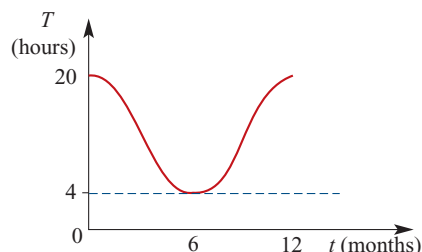
- 22 The length of night on Seal Island varies between 20 hours in midwinter and 4 hours in midsummer. The relationship between T , the number of hours of night, and t , the number of months past the longest night in 1991, is given by

$$T(t) = p + q \cos \pi r t$$

where p , q and r are constants.

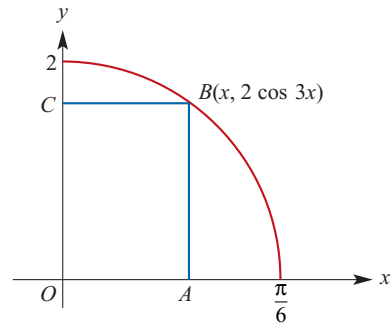
Assume that the year consists of 12 months of equal length.

The graph of T against t is illustrated.



- a** Find the value of:
i r and **ii** p and q
b Find $T'(3)$ and $T'(9)$ and find the rate of change of hours of night with respect to the number of months.
c Find the average rate of change of hours of night from $t = 0$ to $t = 6$.
d After how many months is the rate of change of hours of night a maximum?

- 23** A section of the graph of $y = 2 \cos 3x$ is shown in the diagram.



- a** Show that the area, A , of the rectangle $OABC$ in terms of x is $2x \cos 3x$
- b** **i** Find $\frac{dA}{dx}$.
- ii** Find $\frac{dA}{dx}$ when $x = 0$ and $x = \frac{\pi}{6}$.
- c** **i** On a calculator plot the graph of $A = 2x \cos 3x$ for $x \in [0, \frac{\pi}{6}]$.
- ii** Find the two values of x for which the area of the rectangle is 0.2 square units.
- iii** Find the maximum area of the rectangle and the value of x for which this occurs.
- d** **i** Show that $\frac{dA}{dx} = 0$ is equivalent to $\tan 3x = \frac{1}{3x}$
- ii** Using a calculator plot the graphs of $y = \tan 3x$ and $y = \frac{1}{3x}$ for $x \in (0, \frac{\pi}{6})$ and find the coordinates of the point of intersection.

- 24 a** A population of insects grows according to the model

$$N(t) = 1000 - t + 2e^{\frac{t}{20}} \text{ for } t \geq 0$$

where t is the number of days after 1 January 2000.

- i** Find the rate of growth of the population as a function of t .
- ii** Find the minimum population size and value of t for which this occurs.
- iii** Find $N(0)$. **iv** Find $N(100)$.
- v** Sketch the graph of N against t for $0 \leq t \leq 100$.
- b** It is found that the change in population of another species is given by:

$$N_2(t) = 1000 - t^{\frac{1}{2}} + 2e^{\frac{t^{\frac{1}{2}}}{20}}$$

- i** Find $N_2(0)$. **ii** Find $N_2(100)$.
- iii** Plot the graph of $y = N_2(t)$ for $t \in [0, 5000]$ on a calculator.
- iv** Solve the equation $N_2'(t) = 0$ and hence give the minimum population of this species of insects.

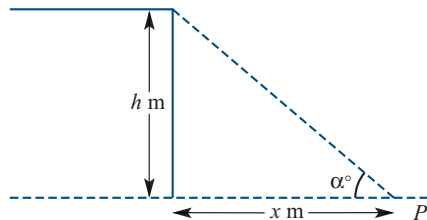
- c** A third model is:

$$N_3(t) = 1000 - t^{\frac{3}{2}} + 2e^{\frac{t}{20}}$$

Use a calculator to:

- i** plot a graph for $0 \leq t \leq 200$
- ii** find the minimum population and the time at which this occurs
- d** **i** For N_3 , find $N_3'(t)$.
- ii** Show $N_3'(t) = 0$ is equivalent to $t = 20 \log_e (15\sqrt{t})$

- 25** The height h m of a cliff is determined by measuring the angle of elevation of the top of the cliff from a point P level with the base of the cliff and at a distance x m from it.

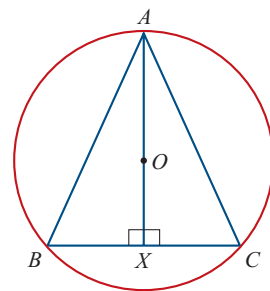


- a** Show that if an error is made in the measurement of the angle by $(\delta\alpha)^\circ$, then the calculation of the height of the cliff yields an error of approximately $\frac{\pi x}{180} \sec^2(\alpha^\circ) \delta\alpha$
- b** Let $x = 50$. Using a calculator plot the graph of:

$$y = \frac{\pi}{18} \times 5 \times \sec^2(\alpha^\circ) \text{ for } 0 < \alpha < 90$$

- c** If the greatest error in the measurement of the angle is $(0.02)^\circ$, what is the greatest possible error for the calculation of the height of the cliff in terms of α ?

- 26** Triangle ABC is isosceles with $AB = AC$. O is the centre of the circumcircle of the triangle. The radius of the circle is R cm. Let $AX = h$ cm and angle $BAX = \alpha^\circ$ where X is the midpoint of BC .



- a**
- i** Find BX in terms of h and α .
 - ii** Apply Pythagoras' theorem to triangle OBX to find R in terms of h and α .
 - iii** Use the result that $1 + \tan^2 \alpha = \sec^2 \alpha$ to show:

$$R = \frac{h}{2} \sec^2 \alpha$$

- b** If the measure of h is exact but that of the semi-vertex angle α° is to have an error of $(\delta\alpha)^\circ$.
- i** Find the error in R , i.e. find δR .
 - ii** Find δR if $h = 0$ and the semi-vertex angle of 30° is measured to be 29.1° .
- 27 a** Consider the curve with equation $y = (2x^2 - 5x)e^{ax}$. If the curve passes through the point with coordinates $(3, 10)$, find the value of a .
- b**
- i** For the curve with equation $y = (2x^2 - 5x)e^{ax}$, find the x -axis intercepts.
 - ii** Use calculus to find the x -values for which there is a turning point, in terms of a .



Integration

Objectives

- To use numerical methods **to determine area**.
- To **integrate functions with rule of the form $f(x) = (ax + b)^n$, $n \in \mathbb{Q}$** .
- To integrate **polynomial functions**.
- To integrate **trigonometric functions**.
- To integrate **exponential functions**.
- To determine **areas under curves** by integration.
- To use integration to **solve problems**.

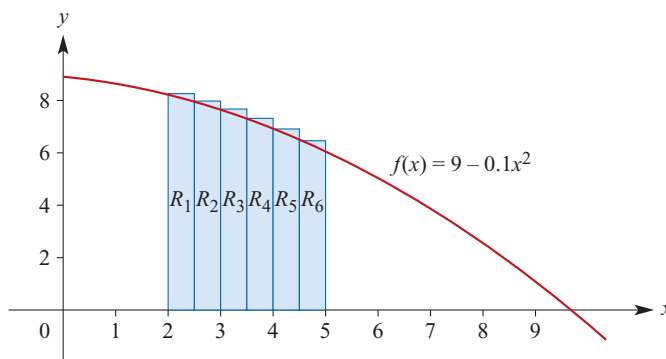
12.1 Approximations leading to the definite integral

In this section we consider three methods for determining the area under a graph.



The left endpoint estimate

First, an approximation is found for the area of the shaded region by dividing the region into rectangles as illustrated. The width of each rectangle is 0.5.



The area of rectangle $R_1 = 0.5 f(2.0) = 4.30$ square units.

The area of rectangle $R_2 = 0.5 f(2.5) = 4.1875$ square units.

The area of rectangle $R_3 = 0.5 f(3.0) = 4.05$ square units.

The area of rectangle $R_4 = 0.5f(3.5) = 3.8875$ square units.

The area of rectangle $R_5 = 0.5f(4.0) = 3.70$ square units.

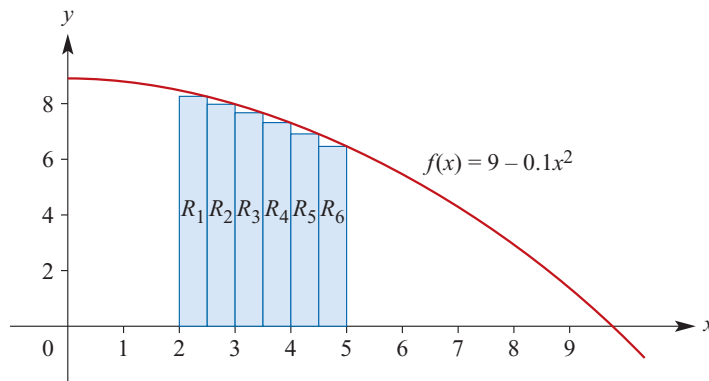
The area of rectangle $R_6 = 0.5f(4.5) = 3.4875$ square units.

The sum of the areas of the rectangles = 23.6125 square units.

The left endpoint estimate will be:

- **larger** than the actual area for a graph which is **decreasing** over the interval.
- **smaller** than the actual area for a graph which is **increasing** over the interval.

The right endpoint estimate



Again the rectangles have width 0.5 and we find:

- The area of rectangle $R_1 = 0.5f(2.5) = 4.1875$ square units.
- The area of rectangle $R_2 = 0.5f(3.0) = 4.05$ square units.
- The area of rectangle $R_3 = 0.5f(3.5) = 3.8875$ square units.

↓

- The area of rectangle $R_6 = 0.5f(5.0) = 3.25$ square units.
- The sum of the areas of the rectangles = 22.5625 square units.

This is called the right endpoint estimate for the area.

For f decreasing over $[a, b]$, left endpoint estimate $\geq$ true area $\geq$ right endpoint estimate.

For f increasing over $[a, b]$, left endpoint estimate $\leq$ true area $\leq$ right endpoint estimate.

From this it can be seen that a further estimate for the area may be achieved by the average of the two estimates:

$$\text{average} = \frac{\text{left endpoint estimate} + \text{right endpoint estimate}}{2}$$

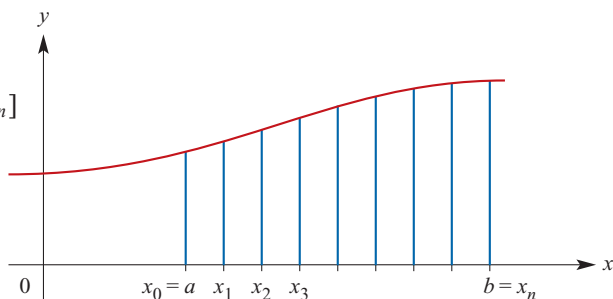
For the example discussed above, average = 23.0875 square units.

It is clear that if narrower strips are chosen then the estimate is closer to the true value. This is time-consuming to do by hand but a computer program or spreadsheet makes the process quite manageable.

If the interval $[a, b]$ on the x -axis is divided into n equal sub-intervals

$$[a, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, x_n]$$

the two methods may be summarised as follows.



The left endpoint estimate:

$$L_n = \frac{b-a}{n} [f(x_0) + f(x_1) + \dots + f(x_{n-1})]$$

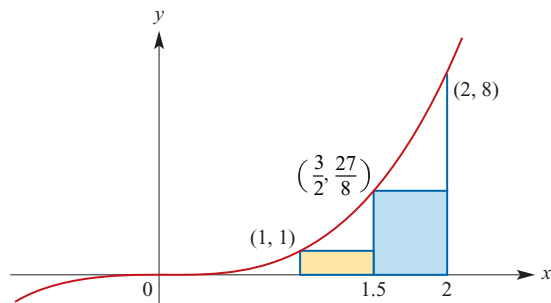
The right endpoint estimate:

$$R_n = \frac{b-a}{n} [f(x_1) + f(x_2) + \dots + f(x_n)]$$

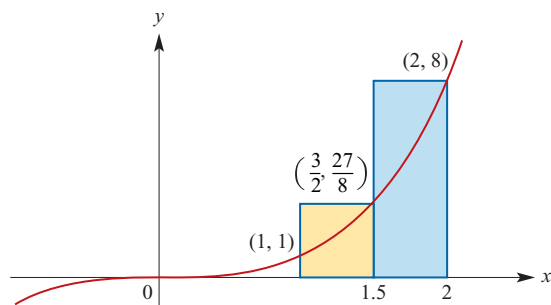
The methods are not limited to situations where the graph is either increasing or decreasing for all the interval, but may be used to determine the area under any continuous curve. These methods may be applied to any continuous function on an interval $[a, b]$ to determine an approximate value of the area.

Example 1

- a** Using the two rectangles shown find an approximation for the shaded area. (This is the left endpoint method.)



- b** Using the two rectangles shown find an approximation for the shaded area. (This is the right endpoint method.)



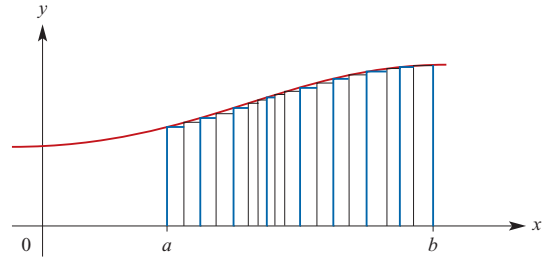
Solution

a Area = $\frac{1}{2} \times 1 + \frac{1}{2} \times \frac{27}{8} = \frac{1}{2} + \frac{27}{16} = \frac{35}{16} = 2\frac{3}{16}$ square units

b Area = $\frac{1}{2} \times \frac{27}{8} + \frac{1}{2} \times 8 = \frac{27}{16} + 4 = 5\frac{11}{16}$ square units

It can be seen from the above that for a continuous function f such that $f(x)$ is positive for all x in the interval $[a, b]$, and if the interval $[a, b]$ is partitioned into arbitrarily small intervals, then the **area** under the curve between $x = a$ and $x = b$ can be defined by this limit process.

In the diagram on the right rectangles formed from partitions are shown. They can be of varying width, but the limiting process must take the width of all the rectangles to approach zero.



The definite integral

Suppose that f is a function that is continuous for a closed interval $[a, b]$ and $f(x)$ is positive for x a member of this interval. Then the area under the curve of $y = f(x)$ from $x = a$ to $x = b$ is called the **definite integral** of $f(x)$ from $x = a$ to $x = b$, and is given the symbol $\int_a^b f(x) dx$. The function f is called the **integrand** and a and b are the lower and upper limits of the integral.

This limiting process can be expressed as

$$\int_a^b f(x) dx = \lim_{\delta x \rightarrow 0} \sum_{i=1}^n f(x_i^*) \delta x_i$$

where the interval $[a, b]$ is partitioned into n subintervals, with the i th subinterval of length δx_i and containing x_i^* , and $\delta x = \max\{\delta x_i : i = 1, 2, \dots, n\}$

When a function is linear or a hybrid composed of linear constituents, then the area may be found using geometric techniques.

Example 2

Evaluate each of the following definite integrals by using an area formula:

a $\int_1^3 (x - 1) dx$

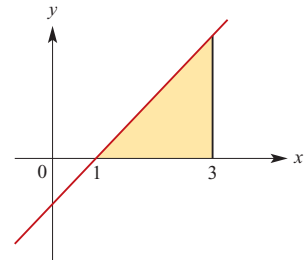
b $\int_{-1}^3 |x - 1| dx$

c $\int_1^2 (x + 1) dx$

Solution

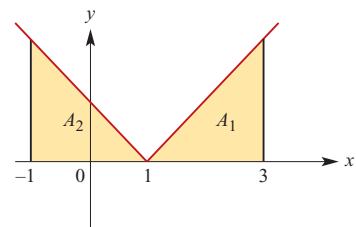
- a** The area of the triangle $= \frac{1}{2} \times 2 \times 2$
 $= 2$ square units

Therefore $\int_1^3 (x - 1) dx = 2$



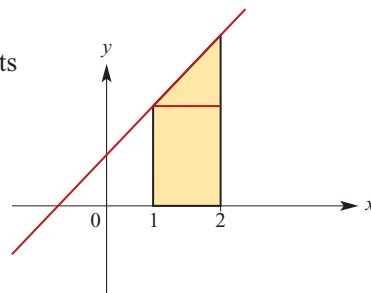
- b** The area $= A_1 + A_2 = 2 + \frac{1}{2} \times 2 \times 2 = 4$ square units

Therefore $\int_{-1}^3 |x - 1| dx = 4$



- c** The required region is a trapezium.

Hence area = $\frac{1}{2} \times 1 \times (2 + 3) = \frac{5}{2}$ square units

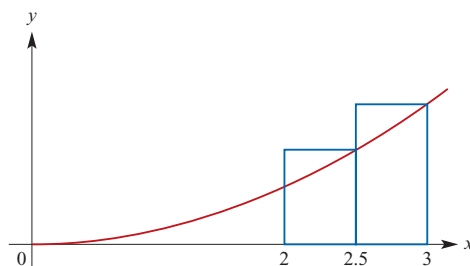


A calculus method for determining areas will be introduced in Section 12.5.

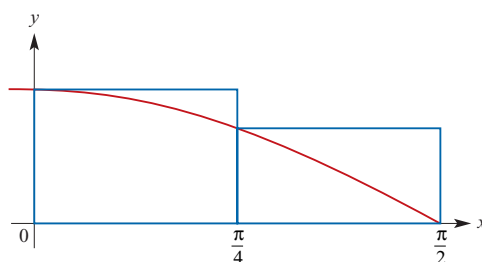
Exercise 12A

- 1** Use two rectangles to approximate the area contained between the curve and the x -axis. Use the method indicated and give your answer correct to two decimal places.

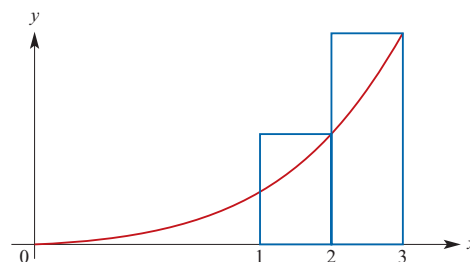
- a** $y = \frac{1}{2}x^2$ between $x = 2$ and $x = 3$
using the right endpoint method



- b** $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{2}$
using the left endpoint method

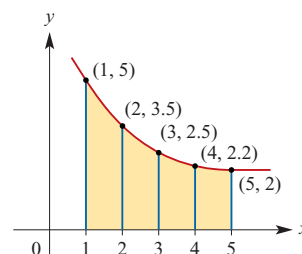


- c** $y = \frac{1}{2}x^3$ between $x = 1$ and $x = 3$
using the right endpoint method



- 2** Calculate:

- a** the left endpoint estimate
b the right endpoint estimate
to find the area of the region shaded in the figure.
(Use sub-intervals as shown.)

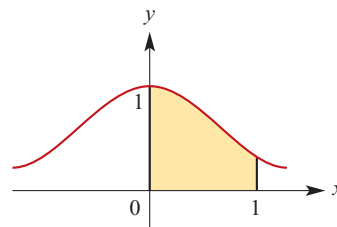


- 3 Calculate an approximation to an area under the graph of $y = x(4 - x)$ between $x = 0$ and $x = 4$ using:

- a** 4 strips of width 1.0 (Do by hand.)
b 20 strips of width 0.2 (You can use a calculator program if you wish).

- 4 The graph is that of $y = \frac{1}{1 + x^2}$. It is known that the area of the shaded region is $\frac{\pi}{4}$.

- a** Apply the right endpoint rule with strips of width 0.25 to find the area under the curve.
b Find an approximate value for π . How could you improve the approximation?

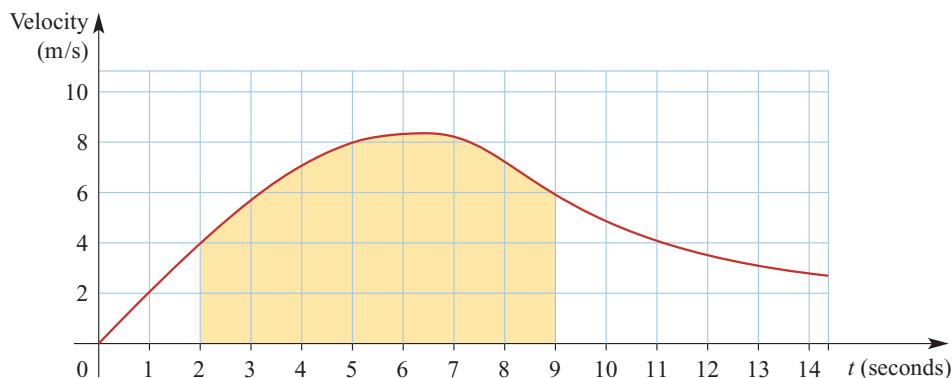


- 5 A table of values is given for the rule $y = f(x)$.

x	0	1	2	3	4	5	6	7	8	9	10
y	3	3.5	3.7	3.8	3.9	3.9	4.0	4.0	3.7	3.3	2.9

Find the area enclosed by the graph of $y = f(x)$, the lines $x = 0$ and $x = 10$, and the x -axis by:

- a** the left endpoint estimate **b** the right endpoint estimate
- 6 The graph shows the velocity (in m/s) of a body at time t seconds.



- a** Use the left endpoint rule to find the area of the shaded region.
b What does this area represent?
- 7 Calculate, by using the right endpoint estimate, an approximation to the area under the graph of $y = 2^x$ between $x = 0$ and $x = 3$, using strips of width 0.5. Write your answer correct to two decimal places.
- 8 Using area formula evaluate each of the following definite integrals:

a $\int_2^5 (x - 2)dx$ **b** $\int_{-1}^5 |x - 2|dx$ **c** $\int_1^2 (2x + 1)dx$

12.2 Antidifferentiation

The derivative of x^2 with respect to x is $2x$. Conversely, given that an unknown expression has derivative $2x$ it is clear that the unknown expression could be x^2 . The process of finding a function from its derivative is called **antidifferentiation**.

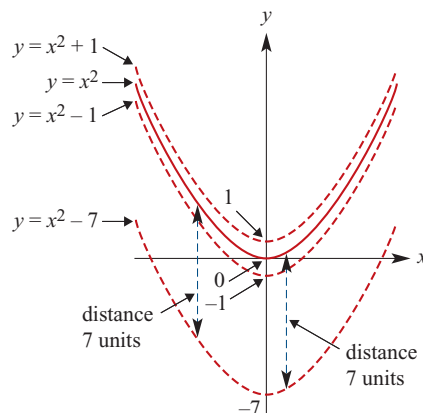
Consider the polynomial functions $f(x) = x^2 + 1$ and $g(x) = x^2 - 7$

Then $f'(x) = 2x$ and $g'(x) = 2x$

i.e. the two different functions have the same derived function.

Both $x^2 + 1$ and $x^2 - 7$ are said to be antiderivatives of $2x$. If two functions have the same derivative on an interval, then they differ by a constant. If two functions have the same derived function, then the graph of one function is obtained by a translation parallel to the y -axis of the other. Several antiderivatives of $2x$ are illustrated on the right.

Each of the graphs is a translation of $y = x^2$ parallel to the y -axis.



The general antiderivative of $2x$ is $x^2 + c$ where c is an arbitrary real number. The notation of Leibniz is used to state this with symbols:

$$\int 2x \, dx = x^2 + c$$

The reason why the symbol is the same as that used for the definite integral of Section 12.1 will become evident in Section 12.5. This is read as ‘the general antiderivative of $2x$ with respect to x is equal to x^2 squared plus c ’ or ‘the *indefinite* integral of $2x$ with respect to x is $x^2 + c$ ’.

To be more precise, the indefinite integral is the set of all antiderivatives and to emphasise this we write:

$$\int 2x \, dx = \{f(x): f'(x) = 2x\} = \{x^2 + c: c \in \mathbb{R}\}$$

The set notation is not commonly used, but it should be clearly understood that there is not a unique antiderivative for a given function. Set notation is not used in the following, but it is advisable to keep it in mind when considering further results.

In general:

$$\text{If } F'(x) = f(x), \\ \int f(x) \, dx = F(x) + c, \text{ where } c \text{ is an arbitrary real number.}$$

The antiderivative of $x, r \neq -1$

From previous work it is known that:

$$\begin{aligned} f(x) = x^3 &\text{ implies } f'(x) = 3x^2 \\ f(x) = x^8 &\text{ implies } f'(x) = 8x^7 \\ f(x) = x^{\frac{3}{2}} &\text{ implies } f'(x) = \frac{3}{2}x^{\frac{1}{2}} \\ f(x) = x^{-4} &\text{ implies } f'(x) = -4x^{-5} \end{aligned}$$

Reversing this process gives:

$$\begin{aligned} \int 3x^2 dx &= x^3 + c, \text{ where } c \text{ is an arbitrary constant} \\ \int 8x^7 dx &= x^8 + c, \text{ where } c \text{ is an arbitrary constant} \\ \int \frac{3}{2}x^{\frac{1}{2}} dx &= x^{\frac{3}{2}} + c, \text{ where } c \text{ is an arbitrary constant} \\ \int -4x^{-5} dx &= x^{-4} + c, \text{ where } c \text{ is an arbitrary constant} \end{aligned}$$

and from this:

$$\begin{aligned} \int x^2 dx &= \frac{1}{3}x^3 + c \\ \int x^7 dx &= \frac{1}{8}x^8 + c \\ \int x^{\frac{1}{2}} dx &= \frac{2}{3}x^{\frac{3}{2}} + c \\ \int x^{-5} dx &= -\frac{1}{4}x^{-4} + c \end{aligned}$$

Generalising, it is seen that:

$$\int x^r dx = \frac{x^{r+1}}{r+1} + c, r \in \mathbb{Q} \setminus \{-1\}$$

Note: This definition can only be applied for suitable values of x for a given value of r . For example if $r = \frac{1}{2}$, $x \in \mathbb{R}^+$ is a suitable restriction.

For $r = -2$, $x \in \mathbb{R} \setminus \{0\}$ and if $r = 3$, $x \in \mathbb{R}$.

The following results are recorded, following immediately from the corresponding results for differentiation in Chapter 9:

$$\begin{aligned} \int f(x) + g(x) dx &= \int f(x) dx + \int g(x) dx \\ \text{and } \int kf(x) dx &= k \int f(x) dx \text{ where } k \text{ is a real number} \end{aligned}$$

Example 3

Find the general antiderivative (indefinite integral) of each of the following:

a $3x^5$ **b** $3x^2 + 4x^{-2} + 3$

Solution

$$\begin{aligned}\text{a } \int 3x^5 dx &= 3 \int x^5 dx \\ &= 3 \times \frac{x^6}{6} + c \\ &= \frac{x^6}{2} + c\end{aligned}$$

$$\begin{aligned}\text{b } \int 3x^2 + 4x^{-2} + 3dx &= 3 \int x^2 dx + 4 \int x^{-2} dx + 3 \int 1 dx \\ &= 3 \left(\frac{x^3}{3} \right) + \frac{4x^{-1}}{-1} + \frac{3x}{1} + c \\ &= x^3 - \frac{4}{x} + 3x + c\end{aligned}$$

Example 4

Find y in terms of x if:

a $\frac{dy}{dx} = \frac{1}{x^2}$

b $\frac{dy}{dx} = 3\sqrt{x}$

c $\frac{dy}{dx} = x^{\frac{3}{4}} + x^{-\frac{3}{4}}$

Solution

$$\begin{aligned}\text{a } \int \frac{1}{x^2} dx &= \int x^{-2} dx \\ &= \frac{x^{-1}}{-1} + c \\ &= -\frac{1}{x} + c \\ \therefore y &= -\frac{1}{x} + c\end{aligned}$$

$$\begin{aligned}\text{b } \int 3\sqrt{x} dx &= 3 \int x^{\frac{1}{2}} dx \\ &= 3 \times \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c \\ &= 2x^{\frac{3}{2}} + c \\ \therefore y &= 2x^{\frac{3}{2}} + c\end{aligned}$$

$$\begin{aligned}\text{c } \int x^{\frac{3}{4}} + x^{-\frac{3}{4}} dx \\ \therefore y &= \frac{4}{7}x^{\frac{7}{4}} + 4x^{\frac{1}{4}} + c\end{aligned}$$

In some situations extra information that makes it possible to determine a unique antiderivative is given.

Example 5

It is known that $f'(x) = x^3 + 4x^2$ and $f(0) = 0$. Find $f(x)$.

Solution

$$\begin{aligned}\int x^3 + 4x^2 dx &= \frac{x^4}{4} + \frac{4x^3}{3} + c \\ \therefore f(x) &= \frac{x^4}{4} + \frac{4x^3}{3} + c\end{aligned}$$

and as $f(0) = 0$, $c = 0$

$$\therefore f(x) = \frac{x^4}{4} + \frac{4x^3}{3}$$

Antidifferentiation can be used to find the equation of a curve from its derivative. That is, given information about the gradient of a curve and a point on the curve the equation of the curve can be found.

Example 6

For the curve $y = f(x)$, $f'(x) = 2x$. If the curve passes through the point $(-1, 4)$, find the equation of the curve.

Solution

$$\begin{aligned} f'(x) = 2x \text{ implies } f(x) &= \int 2x dx \\ \int 2x dx &= \frac{2x^2}{2} + c = x^2 + c \\ \therefore f(x) &= x^2 + c \\ \text{but } f(-1) &= 4 \text{ so } 4 = (-1)^2 + c \\ \therefore c &= 3 \text{ and } f(x) = x^2 + 3 \end{aligned}$$

Exercise 12B

1 Find:

a $\int \frac{1}{2}x^3 dx$ **b** $\int 5x^3 - 2x dx$ **c** $\int \frac{4}{5}x^3 - 3x^2 dx$ **d** $\int (2-z)(3z+1) dz$

2 Find:

a $\int 3x^{-2} dx$ **b** $\int 2x^{-4} + 6x dx$ **c** $\int \sqrt{x}(2+x) dx$
d $\int 3x^{\frac{1}{3}} - 5x^{\frac{5}{4}} dx$ **e** $\int \frac{3z^4 + 2z}{z^3} dz$ **f** $\int 3x^{\frac{3}{4}} - 7x^{\frac{1}{2}} dx$

3 Find y in terms of x for each of the following:

a $\frac{dy}{dx} = 2x - 3$ and $y = 1$ when $x = 1$ **b** $\frac{dy}{dx} = x^3$ and $y = 6$ when $x = 0$
c $\frac{dy}{dx} = x^{\frac{1}{2}} + x$ and $y = 6$ when $x = 4$

4 A curve with equation $y = f(x)$ passes through the point $(2, 0)$ and $f'(x) = 3x^2 - \frac{1}{x^2}$. Find $f(x)$.

5 Find s in terms of t if $\frac{ds}{dt} = 3t - \frac{8}{t^2}$ and $s = 1\frac{1}{2}$ when $t = 1$.

6 The curve $f(x)$ for which $f'(x) = 16x + k$, where k is a constant, has a stationary point at $(2, 1)$. Find:

a the value of k **b** the value of $f(x)$ when $x = 7$

12.3 Antidifferentiation of $(ax + b)^r$

Case 1: $r \geq 1$

For $f: R \rightarrow R$, $f(x) = (ax + b)^r$, $r \in Q$, $r \geq 1$

$$f'(x) = ar(ax + b)^{r-1} \quad (\text{using the chain rule})$$

Case 2: $r < 1$

For $f: S \rightarrow R$, $f(x) = (ax + b)^r$, $r \in Q \setminus \{0\}$, $r < 1$

$$f'(x) = ar(ax + b)^{r-1}$$

for a suitable domain S .

Thus for a suitable domain the following holds:

$$\int (ax + b)^r dx = \frac{1}{a(r+1)}(ax + b)^{r+1} + c, \quad r \neq -1$$

The result does not hold for $r = -1$.

Example 7

Find the general antiderivative of:

a $(3x + 1)^5$

b $(2x - 1)^{-2}$

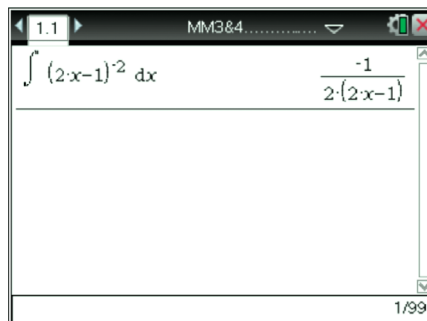
Solution

$$\begin{aligned} \text{a} \quad \int (3x + 1)^5 dx &= \frac{1}{3(5+1)}(3x + 1)^6 + c \\ &= \frac{1}{18}(3x + 1)^6 + c \end{aligned}$$

$$\begin{aligned} \text{b} \quad \int (2x - 1)^{-2} dx &= \frac{1}{2(-2+1)}(2x - 1)^{-1} + c \\ &= -\frac{1}{2}(2x - 1)^{-1} + c \end{aligned}$$

Using the TI-Nspire

Use $\text{menu} > \text{Calculus} > \text{Integral}$ to find the integral of $(2x - 1)^{-2}$.



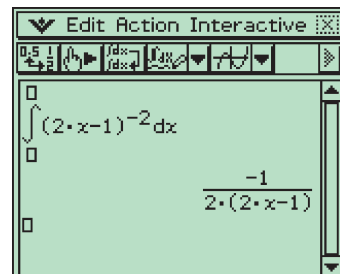
Using the Casio ClassPad

Find the integral of $(2x - 1)^{-2}$.

Enter and highlight the expression, then tap

Interactive–Calculation– $\int$.

Note that the two boxes on the integral allow for definite integrals to be evaluated. This is covered later in the chapter.



Case 3: $r = -1$

But what happens when $r = -1$? In other words, what is $\int \frac{1}{(ax + b)} dx$?

Remember that $\frac{d}{dx}(\log_e x) = \frac{1}{x}$

Thus, $\int \frac{1}{x} dx = \log_e x + c$, provided of course that $x > 0$.

If $x < 0$, then $\frac{d}{dx} \log_e(-x) = -1 \times \frac{1}{-x} = \frac{1}{x}$, so that $\int \frac{1}{x} dx = \log_e(-x)$ for $x < 0$.

Since $|x| = x$ when $x > 0$ and $|x| = -x$ when $x < 0$

$$\int \frac{1}{x} dx = \log_e |x| + c$$

or, more generally,

$$\int \frac{1}{(ax + b)} dx = \frac{1}{a} \log_e |ax + b| + c$$

Example 8

- Find the general antiderivative of $\frac{2}{3x - 2}$
- Given $\frac{dy}{dx} = \frac{3}{x}$ and $y = 10$ when $x = 1$, find an expression for y in terms of x .
- Given $\frac{dy}{dx} = \frac{3}{x}$ and $y = 10$ when $x = -1$, find an expression for y in terms of x .

Solution

$$\text{a} \quad \int \frac{2}{3x-2} dx = \frac{1}{3} \times 2 \log_e |3x-2| + c = \frac{2}{3} \log_e |3x-2| + c$$

$$\text{b} \quad y = \int \frac{3}{x} dx = 3 \log_e |x| + c$$

When $x = 1$, $y = 10$, and therefore

$$10 = 3 \log_e 1 + c = 0 + c \text{ or } c = 10$$

Thus $y = 3 \log_e |x| + 10$

$$\text{c} \quad y = \int \frac{3}{x} dx = 3 \log_e |x| + c$$

When $x = -1$, $y = 10$ and therefore

$$10 = 3 \log_e |-1| + c$$

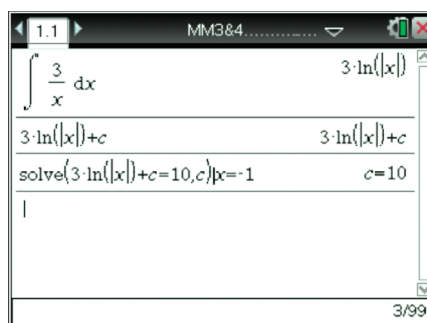
Hence $c = 10$ and therefore $y = \log_e |x| + 10$

Using the TI-Nspire

Use **(menu) > Calculus > Integral** to find an antiderivative of $\frac{3}{x}$.

Add c to find the *general* antiderivative of $\frac{3}{x}$.

Use **solve()** to determine the value of c as shown.



Using the Casio ClassPad

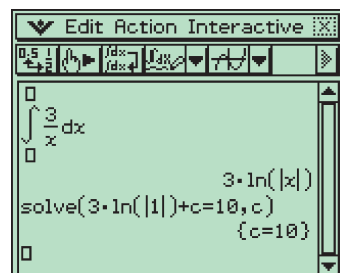
Enter and highlight the expression $\frac{3}{x}$ then tap

Interactive–Calculation– $\int$.

Note that the ClassPad does not add c to the indefinite integral.

Copy and paste the answer to the next line, replace the x with 1 and add the c to complete the equation $3 \ln |1| + c = 0$. Then tap

Interactive–Equation/inequality–solve and ensure the variable is set to c .



Exercise 12C**1** Find:

a $\int (2x - 1)^2 dx$

b $\int (2 - t)^3 dt$

c $\int (5x - 2)^3 dx$

d $\int (3x + 6)^{\frac{1}{2}} dx$

e $\int (3x + 6)^{-\frac{1}{2}} dx$

f $\int (2x - 4)^{\frac{7}{2}} dx$

g $\int (3x + 11)^{\frac{4}{3}} dx$

h $\int \sqrt{(2 - 3x)} dx$

i $\int (5 - 2x)^4 dx$

j $\int (3 - 4x)^{-5} dx$

2 Find an antiderivative of each of the following:

a $\frac{1}{2x}$

b $\frac{1}{3x + 2}$

c $\frac{3}{1 - 4x}$

d $\frac{x}{x + 1}$

e $\frac{x + 1}{x}$

f $\frac{1}{(x + 1)^2}$

g $\frac{(x + 1)^2}{x}$

h $\frac{2x + 1}{x + 1}$

i $\frac{3}{(x - 1)^3}$

3 Find:

a $\int \frac{5}{x} dx$

b $\int \frac{3}{x - 4} dx$

c $\int \frac{10}{2x + 1} dx$

d $\int \frac{6}{5 - 2x} dx$

e $\int 6(1 - 2x)^{-2} dx$

f $\int (4 - 3x)^{-1} dx$

4 Find y in terms of x for each of the following:

a $\frac{dy}{dx} = \frac{1}{2x}$ and $y = 2$ when $x = e^2$

b $\frac{dy}{dx} = \frac{2}{5 - 2x}$ and $y = 10$ when $x = 2$

5 A curve with equation $y = f(x)$ passes through the point $(5 + e, 10)$ and $f'(x) = \frac{10}{x - 5}$. Find the equation of the curve.**6** Given that $\frac{dy}{dx} = \frac{3}{x - 2}$ and $y = 10$ when $x = 0$, find an expression for y in terms of x .**7** Given that $\frac{dy}{dx} = \frac{5}{2 - 4x}$ and $y = 10$ when $x = -2$, find an expression for y in terms of x .**12.4 The antiderivative of e^{kx}**

In Chapter 11 it was found that for

$$f: R \rightarrow R, f(x) = e^{kx}$$

$$f': R \rightarrow R, f'(x) = ke^{kx}$$

Thus $\int e^{kx} dx = \frac{1}{k}e^{kx} + c, k \neq 0$

Example 9

Find the general antiderivative of each of the following:

a e^{4x}

b $e^{5x} + 6x$

c $e^{3x} + 2$

d $e^{-x} + e^x$

Solution

a $\int e^{4x} dx = \frac{1}{4}e^{4x} + c$

b $\int e^{5x} + 6x dx = \frac{1}{5}e^{5x} + 3x^2 + c$

c $\int e^{3x} + 2 dx = \frac{1}{3}e^{3x} + 2x + c$

d $\int e^{-x} + e^x dx = -e^{-x} + e^x + c$

Example 10

If the gradient at a point (x, y) on a curve is given by $5e^{2x}$ and the curve passes through $(0, 7.5)$, find the equation of the curve.

Solution

Let the curve have the equation $y = f(x)$

Then $f'(x) = 5e^{2x}$ and

$$\int 5e^{2x} dx = \frac{5}{2}e^{2x} + c$$

$$\therefore f(x) = \frac{5}{2}e^{2x} + c$$

but $f(0) = 7.5$

$$\begin{aligned}\therefore 7.5 &= \frac{5}{2}e^0 + c \\ &= 2.5 + c\end{aligned}$$

$$\therefore c = 5$$

so that $f(x) = \frac{5}{2}e^{2x} + 5$

Exercise 12D

1 Find:

a $\int e^{2x} - e^{\frac{x}{2}} dx$

b $\int \frac{e^{2x} + 1}{e^x} dx$

c $\int 2e^{3x} - e^{-x} dx$

d $\int 5e^{\frac{x}{3}} - 3e^{\frac{x}{5}} dx$

2 Find y in terms of x for each of the following:

a $\frac{dy}{dx} = e^{2x} - x$ and $y = 5$ when $x = 0$ b $\frac{dy}{dx} = \frac{3 - e^{2x}}{e^x}$ and $y = 4$ when $x = 0$

3 Given that $\frac{dy}{dx} = ae^{-x} + 1$ and that when $x = 0$, $\frac{dy}{dx} = 3$ and $y = 5$, find the value of y when $x = 2$.

4 The curve for which $\frac{dy}{dx} = e^{kx}$, where k is a constant, is such that the tangent at $(1, e^2)$ passes through the origin. Find the gradient of this tangent and hence determine:

- a the value of k b the equation of the curve

12.5 The fundamental theorem of calculus and the definite integral

The integrals that you have learned to evaluate by calculus techniques to date are known as **indefinite integrals** because they are only defined to within an arbitrary constant, for example $\int 3x^2 dx = x^3 + c$. In general terms, we write $\int f(x) dx = G(x) + c$; that is, the integral of $f(x)$ is $G(x)$ plus a constant, where $G(x)$ is the antiderivative of $f(x)$.

Consideration of the **definite integral** will now be resumed and the connection established. In Section 12.1 the following definition was given.

Suppose that f is a function that is continuous for a closed interval $[a, b]$ and $f(x)$ is positive for x , a member of this interval. Then the area under the curve of $y = f(x)$ from $x = a$ to $x = b$ is called the **definite integral** of $f(x)$ from $x = a$ to $x = b$, and is given the symbol $\int_a^b f(x) dx$. The function f is called the integrand and a and b are the upper and lower bounds of the integral.

The fundamental theorem of integral calculus provides a connection between the area definition of the definite integral and the antiderivatives discussed previously. An outline of the proof is given in the final section of this chapter. First, attention must be given to the case when $f(x) < 0$.

The net signed area

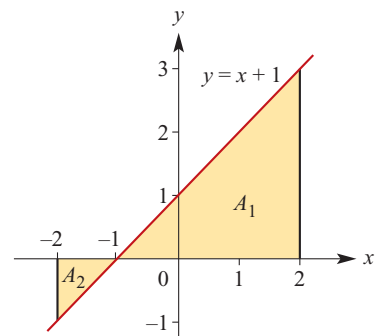
For $f(x) = x + 1$

$$\begin{aligned} A_1 &= \frac{1}{2} \times 3 \times 3 \\ &= 4\frac{1}{2} \quad (\text{area of a triangle}) \end{aligned}$$

$$\begin{aligned} A_2 &= \frac{1}{2} \times 1 \times 1 \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{The total area} &= A_1 + A_2 \\ &= 5 \end{aligned}$$

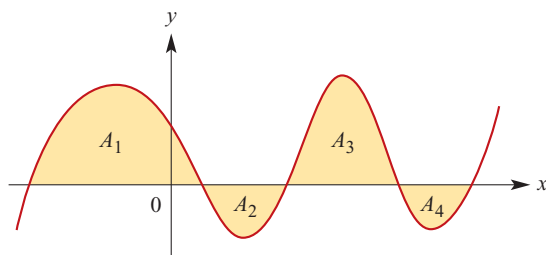
$$\begin{aligned} \text{The net signed area} &= A_1 - A_2 \\ &= 4 \end{aligned}$$



Regions below the x -axis have **negative signed areas**. Regions above the x -axis have **positive signed areas**.

The total area of the shaded region is $A_1 + A_2 + A_3 + A_4$

The net signed area of the shaded region is $A_1 - A_2 + A_3 - A_4$



$\int_a^b f(x)dx$ for any continuous function f , for the interval $[a, b]$, is the net signed area of the regions contained between the curve and the x -axis for $x \in [a, b]$.

It can be determined by a limiting process as discussed briefly in the first section of this chapter.

The fundamental theorem of integral calculus is as follows.

If f is a continuous function on the interval $[a, b]$, then:

$$\int_a^b f(x)dx = G(b) - G(a) \text{ where } G \text{ is any antiderivative of } f.$$

$\int_a^b f(x)dx$ is called the **definite integral** from $x = a$ to $x = b$. The number a is called the **lower limit of integration** and b is called the **upper limit of integration**. The function f is called the **integrand**. To facilitate setting out we sometimes write $G(b) - G(a) = [G(x)]_a^b$.

Example 11

Evaluate the definite integral: $\int_1^2 x dx$

Solution

$$\int x dx = \frac{1}{2}x^2 + c$$

so

$$\begin{aligned} \int_1^2 x dx &= \frac{1}{2} \times 2^2 + c - \left(\frac{1}{2} \times 1^2 + c \right) \\ &= 2 - \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Note that the arbitrary constant cancels out. Because of this we ignore it when evaluating definite integrals. We also use the more compact notation

$G(b) - G(a) = [G(x)]_a^b$ to help with setting out, as shown in the following example:

$$\begin{aligned} \int_1^2 x dx &= \left[\frac{x^2}{2} \right]_1^2 \\ &= \frac{2^2}{2} - \frac{1^2}{2} \\ &= \frac{3}{2} \end{aligned}$$

Example 12

Evaluate the definite integral: $\int_0^1 2e^{-2x} dx$

Solution

$$\begin{aligned}\int_0^1 2e^{-2x} dx &= \left[\frac{2}{-2} e^{-2x} \right]_0^1 \\ &= -1 (e^{-2 \times 1} - e^{-2 \times 0}) \\ &= -1 (e^{-2} - 1) \\ &= 1 - e^{-2}\end{aligned}$$

Properties of the definite integral

- 1 $\int_a^a f(x)dx = 0$
- 2 $\int_a^b f(x)dx = \int_c^b f(x)dx + \int_a^c f(x)dx, a < c < b$
- 3 $\int_a^b kf(x)dx = k \int_a^b f(x)dx$
- 4 $\int_a^b f(x) \pm g(x)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
- 5 $\int_a^b f(x)dx = - \int_b^a f(x)dx$ (Note that this is a definition.)

Example 13

Evaluate each of the following definite integrals:

a $\int_2^3 x^2 dx$ **b** $\int_3^2 x^2 dx$ **c** $\int_0^4 e^{2x} + 1 dx$ **d** $\int_0^1 x^{\frac{1}{2}} + x^{\frac{3}{2}} dx$

e $\int_1^4 2x^{\frac{1}{2}} + e^{\frac{x}{2}} dx$ **f** $\int_{-1}^4 \frac{1}{5-x} dx$ **g** $\int_4^5 \frac{1}{5-2x} dx$

Solution

a

$$\int_2^3 x^2 dx$$

$$= \left[\frac{x^3}{3} \right]_2^3$$

$$= \frac{27}{3} - \frac{8}{3}$$

$$= 9 - 2\frac{2}{3}$$

$$= \frac{19}{3}$$

b

$$\int_3^2 x^2 dx$$

$$= \left[\frac{x^3}{3} \right]_3^2$$

$$= \frac{8}{3} - \frac{27}{3}$$

$$= -\frac{19}{3}$$

c

$$\int_0^4 e^{2x} + 1 dx$$

$$= \left[\frac{1}{2} e^{2x} + x \right]_0^4$$

$$= \frac{1}{2} e^8 + 4 - \left(\frac{1}{2} e^0 + 0 \right)$$

$$= \frac{1}{2} (e^8 + 7)$$

$$\mathbf{d} \quad \int_0^1 x^{\frac{1}{2}} + x^{\frac{3}{2}} dx$$

$$\begin{aligned} &= \left[\frac{2}{3} x^{\frac{3}{2}} + \frac{2}{5} x^{\frac{5}{2}} \right]_0^1 \\ &= \frac{2}{3} + \frac{2}{5} \\ &= \frac{16}{15} \end{aligned}$$

$$\mathbf{e} \quad \int_1^4 2x^{\frac{1}{2}} + e^{\frac{x}{2}} dx$$

$$\begin{aligned} &= \left[\frac{4}{3} x^{\frac{3}{2}} + 2e^{\frac{x}{2}} \right]_1^4 \\ &= \frac{4}{3} \times 8 + 2e^2 - \left(\frac{4}{3} + 2e^{\frac{1}{2}} \right) \\ &= \frac{28}{3} + 2e^2 - 2e^{\frac{1}{2}} \\ &= 2 \left(\frac{14}{3} + e^2 - e^{\frac{1}{2}} \right) \end{aligned}$$

$$\mathbf{f} \quad \int_{-1}^4 \frac{1}{5-x} dx$$

$$\begin{aligned} &= [-\log_e |5-x|]_{-1}^4 \\ &= -(\log_e 1 - \log_e 6) \\ &= \log_e 6 \end{aligned}$$

$$\mathbf{g} \quad \int_4^5 \frac{1}{5-2x} dx$$

$$\begin{aligned} &= -\frac{1}{2} [\log_e |5-2x|]_4^5 \\ &= -\frac{1}{2} (\log_e |-5| - \log_e |-3|) \\ &= -\frac{1}{2} \log_e \frac{5}{3} \\ &= \frac{1}{2} \log_e \frac{3}{5} \end{aligned}$$

Exercise 12E

1 Evaluate each of the following:

$$\mathbf{a} \quad \int_1^2 x^2 dx$$

$$\mathbf{b} \quad \int_{-1}^3 x^3 dx$$

$$\mathbf{c} \quad \int_0^1 x^3 - x dx$$

$$\mathbf{d} \quad \int_{-1}^2 (x+1)^2 dx$$

$$\mathbf{e} \quad \int_1^2 \frac{1}{x^2} dx$$

$$\mathbf{f} \quad \int_1^4 x^{\frac{1}{2}} + 2x^2 dx$$

$$\mathbf{g} \quad \int_0^2 x^3 + 2x^2 + x + 2 dx$$

$$\mathbf{h} \quad \int_1^4 2x^{\frac{3}{2}} + 5x^3 dx$$

2 Evaluate each of the following:

$$\mathbf{a} \quad \int_0^1 (2x+1)^3 dx$$

$$\mathbf{b} \quad \int_0^2 (4x+1)^{-\frac{1}{2}} dx$$

$$\mathbf{c} \quad \int_1^2 (1-2x)^2 dx$$

$$\mathbf{d} \quad \int_0^1 (3-2x)^{-2} dx$$

$$\mathbf{e} \quad \int_0^2 (3+2x)^{-3} dx$$

$$\mathbf{f} \quad \int_{-1}^1 (4x+1)^3 dx$$

$$\mathbf{g} \quad \int_0^1 \sqrt{2-x} dx$$

$$\mathbf{h} \quad \int_3^4 \frac{1}{\sqrt{2x-4}} dx$$

$$\mathbf{i} \quad \int_0^1 \frac{1}{(3+2x)^2} dx$$

3 Evaluate each of the following:

$$\mathbf{a} \quad \int_0^1 e^{2x} dx$$

$$\mathbf{b} \quad \int_0^1 e^{-2x} + 1 dx$$

$$\mathbf{c} \quad \int_0^1 2e^{\frac{x}{3}} + 2 dx$$

$$\mathbf{d} \quad \int_{-2}^2 \frac{e^x + e^{-x}}{2} dx$$

4 Given that $\int_0^4 h(x) dx = 5$, evaluate:

a $\int_0^4 2h(x)dx$

b $\int_0^4 (h(x) + 3)dx$

c $\int_4^0 h(x)dx$

d $\int_0^4 (h(x) + 1)dx$

e $\int_0^4 (h(x) - x)dx$

5 Evaluate each of the following:

a $\int_0^4 \frac{1}{x-6}dx$

b $\int_{-3}^0 \frac{1}{3-2x}dx$

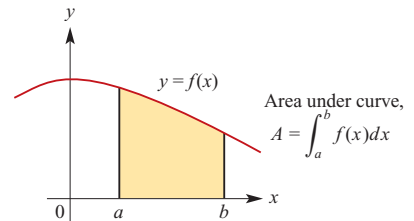
c $\int_5^6 \frac{3}{2x+7}dx$

12.6 Area under a curve

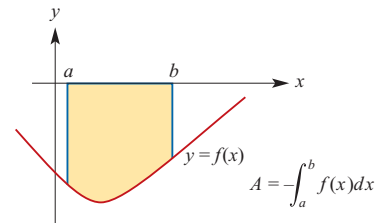
Recall that the definite integral $\int_a^b f(x)dx$ gives the net signed area ‘under’ the curve.

Finding the area of a region

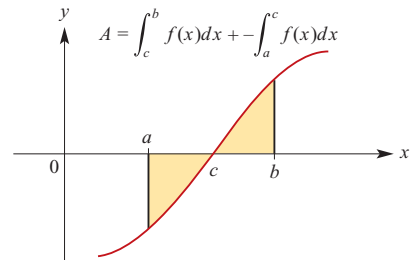
■ If $f(x) \geq 0$ for all $x \in [a, b]$, the area A of the region contained between the curve, the x -axis and the lines $x = a$ and $x = b$ is given by $A = \int_a^b f(x)dx = G(b) - G(a)$ where G is an antiderivative of f .



■ If $f(x) \leq 0$ for all $x \in [a, b]$, the area A of the region contained between the curve, the x -axis and the lines $x = a$ and $x = b$ is given by $A = -\int_a^b f(x)dx$



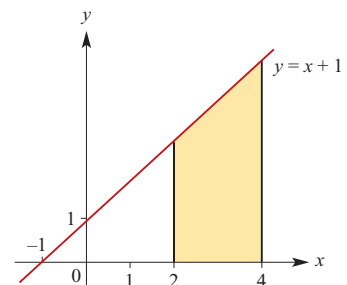
■ If $c \in (a, b)$, $f(c) = 0$ and $f(x) \geq 0$ for $x \in (c, b]$ and $f(x) \leq 0$ for $x \in [a, c)$, then the area A of the shaded region is given by $A = \int_c^b f(x)dx + -\int_a^c f(x)dx$



Note: In determining the areas ‘under’ curves, the sign of $f(x)$ in the given interval is the critical factor.

Example 14

Find the area of the region between the x -axis, the line $y = x + 1$ and the lines $x = 2$ and $x = 4$.
Check the answer by working out the area of the trapezium.



Solution

$$\text{Area} = \int_2^4 x + 1 dx$$

An antiderivative of $x + 1$ is $\frac{x^2}{2} + x$.

$$\begin{aligned}\text{We write } \int_2^4 x + 1 dx &= \left[\frac{x^2}{2} + x \right]_2^4 \\ &= \left(\frac{4^2}{2} + 4 \right) - \left(\frac{2^2}{2} + 2 \right) \\ &= 12 - 4 \\ &= 8\end{aligned}$$

The shaded region has area 8 square units.

$$\begin{aligned}\text{Check: Area of trapezium} &= \text{average height} \times \text{base} \\ &= \frac{3 + 5}{2} \times 2 \\ &= 8 \text{ square units}\end{aligned}$$

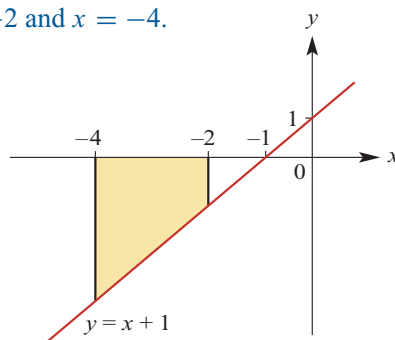
Example 15

Find the area under the line $y = x + 1$ and between $x = -2$ and $x = -4$.

Note: The negative is introduced as the area is the negative of the integral from -4 to -2 .

Solution

$$\begin{aligned}\text{Area} &= - \int_{-4}^{-2} x + 1 dx \\ &= - \left[\frac{x^2}{2} + x \right]_{-4}^{-2} \\ &= -(0 - 4)\end{aligned}$$



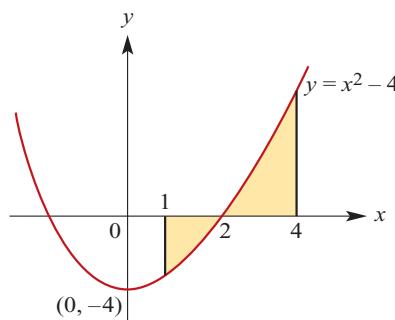
The area of the shaded region is 4 square units.

Example 16

Find the exact area of the shaded region.

Solution

$$\begin{aligned}\text{Area} &= \int_2^4 x^2 - 4 dx + - \int_1^2 x^2 - 4 dx \\ &= \left[\frac{x^3}{3} - 4x \right]_2^4 - \left[\frac{x^3}{3} - 4x \right]_1^2 \\ &= \frac{64}{3} - 16 - \left(\frac{8}{3} - 8 \right) - \left[\left(\frac{8}{3} - 8 \right) - \left(\frac{1}{3} - 4 \right) \right]\end{aligned}$$



$$\begin{aligned}
&= \frac{56}{3} - 8 - \left[\frac{7}{3} - 4 \right] \\
&= \frac{56}{3} - \frac{7}{3} - 4 \\
&= \frac{49}{3} - \frac{12}{3} \\
&= \frac{37}{3}
\end{aligned}$$

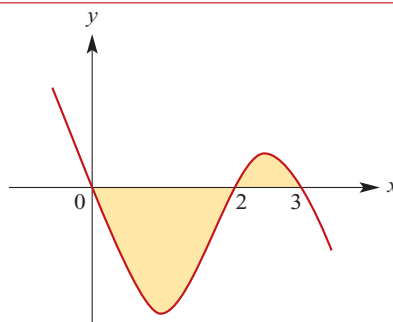
The area is $\frac{37}{3}$ square units.

Example 17

Find the exact area of the regions enclosed by the graph of $y = x(2 - x)(x - 3)$ and the x -axis.

Solution

$$\begin{aligned}
y &= x(-x^2 + 5x - 6) \\
&= -x^3 + 5x^2 - 6x
\end{aligned}$$



Note: There is no need to find the coordinates of the stationary points.

$$\begin{aligned}
\text{Area} &= \int_2^3 (-x^3 + 5x^2 - 6x) dx + - \int_0^2 (-x^3 + 5x^2 - 6x) dx \\
&= \left[-\frac{x^4}{4} + \frac{5x^3}{3} - \frac{6x^2}{2} \right]_2^3 + - \left[-\frac{x^4}{4} + \frac{5x^3}{3} - \frac{6x^2}{2} \right]_0^2 \\
&= \left(-\frac{81}{4} + 45 - 27 \right) - \left(-4 + \frac{40}{3} - 12 \right) - \left(-4 + \frac{40}{3} - 12 \right) \\
&= -\frac{81}{4} + 18 + 32 - \frac{80}{3} \\
&= 50 - \frac{243 + 320}{12} \\
&= 3\frac{1}{12} \\
\text{Area is } 3\frac{1}{12} \text{ square units.}
\end{aligned}$$

Exercise 12F

- 1 Find the exact area of the region(s) bounded by the x -axis and the graph of each of the following:
 - a $y = 3x^2 + 2$ between $x = 0$ and $x = 1$ (sketch the graph first)
 - b $y = x^3 - 8$ between $x = 2$ and $x = 4$ (sketch the graph first)
 - c $y = 4 - x$ between:
 - i $x = 0$ and $x = 4$ (sketch the graph first)
 - ii $x = 0$ and $x = 6$ (sketch the graph first)

- 2** Find the exact area bounded by the x -axis and the graph of each of the following:

a $y = x^2 - 2x$

b $y = (4 - x)(3 - x)$

c $y = (x + 2)(7 - x)$

d $y = x^2 - 5x + 6$

e $y = 3 - x^2$

f $y = x^3 - 6x^2$

- 3** For each of the following sketch a graph to illustrate the region for which the definite integral gives the area:

a $\int_1^4 2x + 1 \, dx$

b $\int_0^3 3 - x \, dx$

c $\int_0^4 x^2 \, dx$

d $\int_{-1}^1 4 - 2x^2 \, dx$

e $\int_2^4 \sqrt{x} \, dx$

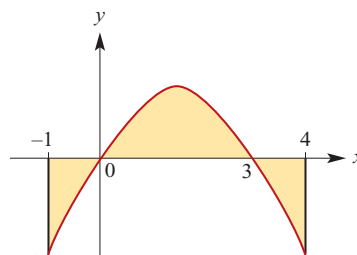
f $\int_0^1 (1 - x)(1 + x)^2 \, dx$

- 4** Find the exact area of the region bounded by the curve $y = 3x + 2x^{-2}$, the lines $x = 2$ and $x = 5$ and the x -axis.

- 5** Sketch the graph of $f(x) = 1 + x^3$ and find the exact area of the region bounded by the curve and the axes.

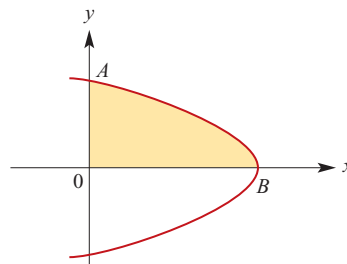
- 6 a** Evaluate $\int_{-1}^4 x(3 - x) \, dx$.

- b** Find the exact area of the shaded region in the figure.



- 7 a** In the figure, the graph of $y^2 = 9(1 - x)$ is shown. Find the coordinates of A and B .

- b** Find the exact area of the shaded region by evaluating $\int_0^b \left(1 - \frac{y^2}{9}\right) \, dy$ for a suitable choice of b .



- 8 a** Show that $a^x = e^{x(\log_e a)}$ for $a > 0$.
b Hence find the derivative and an antiderivative of a^x .
c Hence or otherwise, show that the area under the curve $y = a^x$ ($a \geq 0$) between the lines $x = 0$ and $x = b$ is $\frac{1}{\log_e a}(a^b - 1)$
- 9** Sketch the graph of $f(x) = 4e^{2x} + 3$ and find the exact area of the region enclosed between the curve, the axes and the line $x = 1$.
- 10** Sketch the graph of $y = x(2 - x)(x - 1)$ and find the exact area of the region enclosed between the curve and the x -axis.
- 11** Sketch the graph of $y = \frac{1}{2 - 3x}$ and find the exact area of the region enclosed by the curve, the x -axis and the lines with equations $x = -3$ and $x = -2$.

- 12 Sketch the graph of $y = 2 + \frac{1}{x+4}$ and find the exact area of the region enclosed by the curve, the axes and the line $x = -2$.

12.7 Integration of circular functions

Recall the following results from Chapter 11:

- If $f(x) = \sin(kx + a)$, then $f'(x) = k \cos(kx + a)$
- If $g(x) = \cos(kx + a)$, then $g'(x) = -k \sin(kx + a)$

Thus the following results hold:

$$\int \sin(kx + a) dx = -\frac{1}{k} \cos(kx + a) + c$$

$$\int \cos(kx + a) dx = \frac{1}{k} \sin(kx + a) + c$$

Example 18

Find antiderivatives of each of the following:

a $\sin\left(3x + \frac{\pi}{4}\right)$ **b** $\frac{1}{4} \sin 4x$

Solution

a $-\frac{1}{3} \cos\left(3x + \frac{\pi}{4}\right) + c$ **b** $-\frac{1}{16} \cos 4x + c$

Example 19

Find the exact value of each of the following definite integrals:

a $\int_0^{\frac{\pi}{4}} \sin 2x dx$ **b** $\int_0^{\frac{\pi}{2}} 2 \cos x + 1 dx$

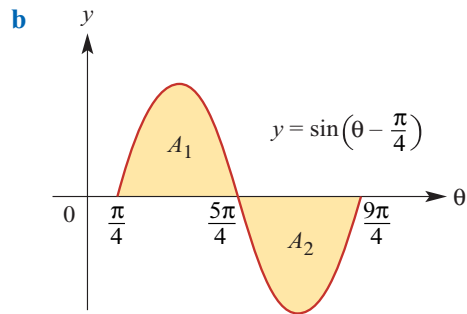
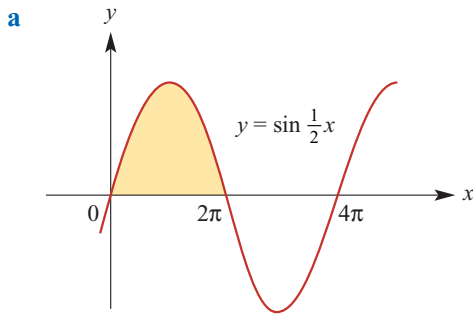
Solution

a
$$\begin{aligned} \int_0^{\frac{\pi}{4}} \sin 2x dx &= \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{4}} \\ &= -\frac{1}{2} \cos \frac{\pi}{2} - \left(-\frac{1}{2} \cos 0 \right) \\ &= 0 + \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

b
$$\begin{aligned} \int_0^{\frac{\pi}{2}} 2 \cos x + 1 dx &= [2 \sin x + x]_0^{\frac{\pi}{2}} \\ &= 2 \sin \frac{\pi}{2} + \frac{\pi}{2} - (2 \sin 0 + 0) \\ &= 2 + \frac{\pi}{2} \end{aligned}$$

Example 20

Find the exact area of the shaded region for each of the graphs:

**Solution**

a Area = $\int_0^{2\pi} \sin \frac{1}{2}x \, dx$

$$= \left[-2 \cos \frac{1}{2}x \right]_0^{2\pi}$$

$$= -2 \cos \pi - (-2 \cos 0)$$

$$= 4$$

$\therefore$ area of shaded region = 4 square units

b Regions A_1 and A_2 must be considered separately.

Area of $A_1 = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin\left(\theta - \frac{\pi}{4}\right) d\theta$

$$= \left[-\cos\left(\theta - \frac{\pi}{4}\right) \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= -(\cos \pi - \cos 0)$$

$$= 2$$

Area of $A_2 = -\int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} \sin\left(\theta - \frac{\pi}{4}\right) d\theta$

$$= \left[\cos\left(\theta - \frac{\pi}{4}\right) \right]_{\frac{5\pi}{4}}^{\frac{9\pi}{4}}$$

$$= \cos 2\pi - \cos \pi$$

$$= 2$$

$\therefore$ area of shaded region = 4 square units

Exercise 12G

1 Find an antiderivative of each of the following:

a $\cos 3x$

b $\sin \frac{1}{2}x$

c $3 \cos 3x$

d $2 \sin \frac{1}{2}x$

e $\sin\left(2x - \frac{\pi}{3}\right)$

f $\cos 3x + \sin 2x$

g $\cos 4x - \sin 4x$

$$\mathbf{h} \quad -\frac{1}{2} \sin 2x + \cos 3x \quad \mathbf{i} \quad -\frac{1}{2} \cos \left(2x + \frac{\pi}{3} \right) \quad \mathbf{j} \quad \sin \pi x$$

2 Find the exact value of each of the following definite integrals:

$$\mathbf{a} \quad \int_0^{\frac{\pi}{4}} \sin x dx$$

$$\mathbf{b} \quad \int_0^{\frac{\pi}{4}} \cos 2x dx$$

$$\mathbf{c} \quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} \cos \theta d\theta$$

$$\mathbf{d} \quad \int_0^{\frac{\pi}{2}} \sin \theta + \cos \theta d\theta$$

$$\mathbf{e} \quad \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta$$

$$\mathbf{f} \quad \int_0^{\frac{\pi}{3}} (\cos 3\theta + \sin 3\theta) d\theta$$

$$\mathbf{g} \quad \int_0^{\frac{\pi}{3}} \left[\cos 3\theta + \sin \left(\theta - \frac{\pi}{3} \right) \right] d\theta$$

$$\mathbf{h} \quad \int_0^{\pi} \left(\sin \frac{x}{4} + \cos \frac{x}{4} \right) dx$$

$$\mathbf{i} \quad \int_0^{\frac{\pi}{4}} \sin \left(2x - \frac{\pi}{3} \right) dx$$

$$\mathbf{j} \quad \int_0^{\pi} \left(\cos 2x - \sin \frac{x}{2} \right) dx$$

3 Calculate the exact area of the region bounded by the curve $y = \sin \frac{1}{2}x$, the x -axis and the line $x = \frac{\pi}{2}$.

4 Draw a graph to illustrate the area given by the definite integral and evaluate the integral for each of the following:

$$\mathbf{a} \quad \int_0^{\frac{\pi}{4}} \cos x dx$$

$$\mathbf{b} \quad \int_0^{\frac{\pi}{3}} \sin 2x dx$$

$$\mathbf{c} \quad \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos 2x dx$$

$$\mathbf{d} \quad \int_0^{\frac{\pi}{2}} \cos \theta + \sin \theta d\theta$$

$$\mathbf{e} \quad \int_0^{\frac{\pi}{2}} \sin 2\theta + 1 d\theta$$

$$\mathbf{f} \quad \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta$$

5 Find the exact value of each of the following definite integrals:

$$\mathbf{a} \quad \int_0^{\frac{\pi}{2}} \sin \left(2x + \frac{\pi}{4} \right) dx$$

$$\mathbf{b} \quad \int_0^{\frac{\pi}{3}} \cos \left(3x + \frac{\pi}{6} \right) dx$$

$$\mathbf{c} \quad \int_0^{\frac{\pi}{3}} \cos \left(3x + \frac{\pi}{3} \right) dx$$

$$\mathbf{d} \quad \int_0^{\frac{\pi}{4}} \cos(3\pi - x) dx$$

6 Sketch the curve $y = 2 + \sin 3x$ for the interval $0 \leq x \leq \frac{2\pi}{3}$ and calculate the exact area enclosed by the curve, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{3}$.

12.8 Miscellaneous exercises

Example 21

If $f(x) = \log_e(x^2 + 1)$:

$\mathbf{a}$ Show that $f'(x) = \frac{2x}{x^2 + 1}$

$\mathbf{b}$ Hence evaluate $\int_0^2 \frac{x}{x^2 + 1} dx$

Solution

a Let $y = \log_e(x^2 + 1)$
 and $v = x^2 + 1$
 Then $\frac{dv}{dx} = 2x$
 and $y = \log_e v$
 $\therefore \frac{dy}{dv} = \frac{1}{v}$
 $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx}$
 $= \frac{1}{v} \cdot 2x$
 i.e. $f'(x) = \frac{2x}{x^2 + 1}$

b $\int_0^2 \frac{x}{x^2 + 1} dx = \frac{1}{2} \int_0^2 \left(\frac{2x}{x^2 + 1} \right) dx$
 $= \frac{1}{2} [\log_e(x^2 + 1)]_0^2$
 $= \frac{1}{2} [\log_e 5 - \log_e 1]$
 $= \frac{1}{2} \log_e 5$

Example 22

If $f(x) = \frac{\cos x}{\sin x}$:

a Show that $f'(x) = \frac{-1}{\sin^2 x}$

b Hence evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sin^2 x} dx$

Solution

a Using the quotient rule:

$$\begin{aligned} f'(x) &= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} \\ &= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} \\ &= \frac{-1}{\sin^2 x} \end{aligned}$$

b $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sin^2 x} dx = -\left[\frac{\cos x}{\sin x} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$
 $= -\frac{\cos \frac{\pi}{2}}{\sin \frac{\pi}{2}} + \frac{\cos \frac{\pi}{4}}{\sin \frac{\pi}{4}}$
 $= 1$

Example 23

a If $f(x) = x \log_e(kx)$, find $f'(x)$ and hence find $\int \log_e(kx) dx$ where k is a positive real constant.

b If $f(x) = x^2 \log_e(kx)$, find $f'(x)$ and hence find $\int x \log_e(kx) dx$ where k is a positive real constant.

Solution

a $f'(x) = \log_e(kx) + x \times \frac{1}{x} = \log_e(kx) + 1$

Antidifferentiating both sides of the equation with respect to x gives:

$$\begin{aligned} \int f'(x) dx &= \int \log_e(kx) dx + \int 1 dx \\ x \log_e(kx) + c_1 &= \int \log_e(kx) dx + x + c_2 \end{aligned}$$

Therefore $\int \log_e(kx) dx = x \log_e(kx) - x + c_1 - c_2$

And $\int \log_e(kx) dx = x \log_e(kx) - x + c$

$$\begin{aligned} \text{b } f'(x) &= 2x \log_e(kx) + x^2 \times \frac{1}{x} \\ &= 2x \log_e(kx) + x \end{aligned}$$

Antidifferentiating both sides of the equation with respect to x gives:

$$\begin{aligned} \int f'(x) dx &= \int 2x \log_e(kx) dx + \int x dx \\ x^2 \log_e(kx) + c_1 &= \int 2x \log_e(kx) dx + \frac{x^2}{2} + c_2 \\ \text{Therefore } \int x \log_e(kx) dx &= \frac{1}{2} x^2 \log_e(kx) - \frac{x^2}{4} + c \end{aligned}$$

It is not possible to find rules for antiderivatives of all functions, for example e^{-x^2} . It is possible to find approximations for definite integrals that involve functions for which a rule for the antiderivative cannot be determined.

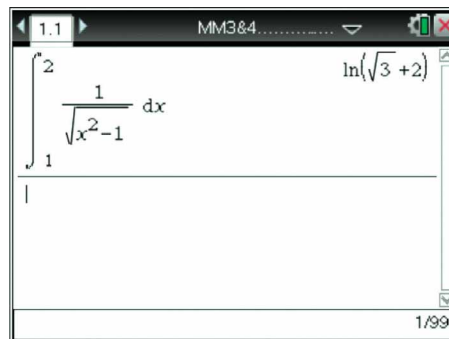
In other cases a CAS calculator can be used to find an antiderivative of a function for which the rule does exist, but students are not required to be able to find this rule by hand. The following three examples illustrate these cases. A TI-Nspire is used but the CASIO ClassPad can also be used to achieve these results.

Example 24

Find $\int_1^2 \frac{1}{\sqrt{x^2-1}} dx$.

Solution

Use the **Integral** template from the **Calculus** menu and complete as shown.

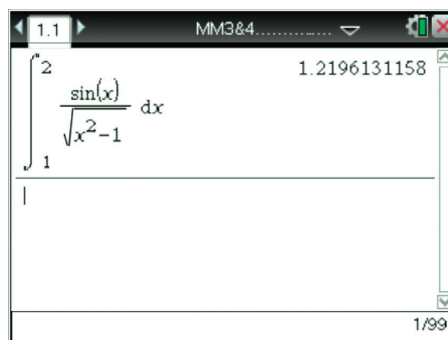


Example 25

Find $\int_1^2 \frac{\sin(x)}{\sqrt{x^2-1}} dx$, correct to two decimal places.

Solution

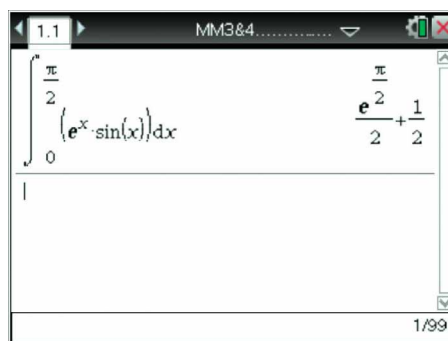
Use the **Integral** template from the **Calculus** menu (**menu**) > **Calculus** > **Integral** and complete as shown. Note that it is impossible to find an antiderivative of $\frac{\sin(x)}{\sqrt{x^2-1}}$, so the calculator returns an approximate (decimal) answer.

**Example 26**

Find $\int_0^{\frac{\pi}{2}} e^x \sin(x) dx$.

Solution

Use the **Integral** template from the **Calculus** menu and complete as shown.

**Exercise 12H**

1 Find the exact value of each of the following:

a $\int_1^4 \sqrt{x} dx$

c $\int_0^8 \sqrt[3]{x} dx$

e $\int_1^2 e^{2x} + \frac{4}{x} dx$

g $\int_0^{\pi} \sin \frac{x}{4} + \cos \frac{x}{4} dx$

i $\int_1^4 \left(2 + \frac{1}{x}\right)^2 dx$

b $\int_{-1}^1 (1+x)^2 dx$

d $\int_0^{\frac{\pi}{3}} \cos 2x - \sin \frac{1}{2}x dx$

f $\int_0^{\frac{\pi}{2}} (\sin 2x + \cos 3x) dx$

h $\int_0^{\frac{\pi}{2}} 5x + \sin 2x dx$

j $\int_0^1 x^2(1-x) dx$



- 2 Find the exact area of the region between the graph of $f(x) = \sin x$, the x -axis and the line $x = \frac{\pi}{3}$.

- 3 Find the value of the following definite integrals, correct to two decimal places:

a $\int_0^{20} 10 \cos\left(\frac{\pi x}{40}\right) e^{\frac{x}{80}} dx$

b $\int_2^5 \frac{e^x}{(x-1)^2} dx$

c $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\cos x}{(x-1)^2} dx$

d $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{100 \cos x}{x^2} dx$

e $\int_0^{\pi} e^{\left(\frac{x}{10}\right)^2} \sin x dx$

f $\int_0^{\frac{\pi}{4}} \cos^3(x) e^{-x} dx$

4 a Show that $\frac{2x+3}{x-1} = 2 + \frac{5}{x-1}$

b Hence evaluate $\int_2^4 \frac{2x+3}{x-1} dx$

5 a Show that $\frac{5x-4}{x-2} = 5 + \frac{6}{x-2}$

b Hence evaluate $\int_3^4 \frac{5x-4}{x-2} dx$

6 a If $y = \left(1 - \frac{1}{2}x\right)^8$, find $\frac{dy}{dx}$. Hence or otherwise, find $\int \left(1 - \frac{1}{2}x\right)^7 dx$

b If $y = \log_e |\cos x|$, find $\frac{dy}{dx}$. Hence evaluate $\int_0^{\frac{\pi}{3}} \tan x dx$

7 Find a function f such that $f'(x) = \sin\left(\frac{1}{2}x\right)$ and $f\left(\frac{4\pi}{3}\right) = 2$

- 8 For each of the following, find $f(x)$:

a $f'(x) = \cos 2x$ and $f(\pi) = 1$

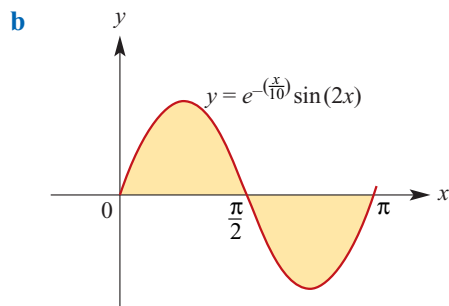
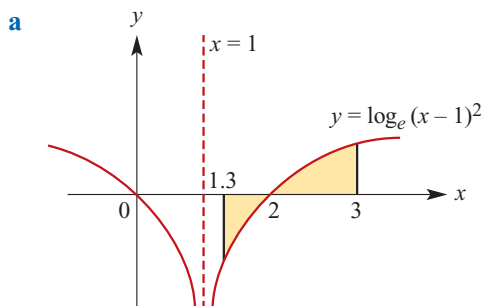
b $f'(x) = \frac{3}{x}$ and $f(1) = 6$

c $f'(x) = e^{\frac{x}{2}}$ and $f(0) = 1$

9 Find $\frac{d}{dx} (x \sin 3x)$, and hence evaluate $\int_0^{\frac{\pi}{6}} x \cos 3x dx$

- 10 The curve with equation $y = a + b \sin \frac{\pi x}{2}$ passes through the points $(0, 1)$ and $(3, 3)$. Find a and b . Find the area of the region enclosed between this curve, the x -axis and the lines $x = 0$ and $x = 1$.

- 11 Find the area of the shaded region for each of the following shaded areas, correct to three decimal places:



- 12** Evaluate $\int_0^\pi e^{-\left(\frac{x}{10}\right)} \sin(2x) dx$, correct to four decimal places.
- 13** The gradient of a curve with equation $y = f(x)$ is given by $f'(x) = x + \sin 2x$ and $f(0) = 1$. Find $f(x)$.
- 14** Let $f(x) = g'(x)$ and $h(x) = k'(x)$, where $g(x) = (x^2 + 1)^3$ and $k(x) = \sin x^2$. Find:
- a** $\int f(x) dx$ **b** $\int h(x) dx$ **c** $\int f(x) + h(x) dx$
- d** $\int -f(x) dx$ **e** $\int (f(x) - 4) dx$ **f** $\int 3h(x) dx$
- 15 a** Differentiate $\frac{\sin x}{\cos x}$ and hence find an antiderivative of $\frac{1}{\cos^2 x}$
- b** Differentiate $\log_e(3x^2 + 7)$ and hence evaluate $\int_0^2 \frac{x}{(3x^2 + 7)} dx$
- c** Find the derivatives of $x + \sqrt{1 + x^2}$ and $\log_e(x + \sqrt{1 + x^2})$
By simplifying your last result if necessary, evaluate $\int_0^1 \frac{1}{\sqrt{1 + x^2}} dx$
- 16** Sketch the graph of $y = \frac{2}{x-1} + 4$ and evaluate $\int_2^3 \frac{2}{x-1} + 4 dx$
Indicate on your graph the region for which you have determined the area.
- 17** Sketch the graph of $y = \sqrt{2x-4} + 1$ and evaluate $\int_2^3 \sqrt{2x-4} + 1 dx$
Indicate on your graph the region for which you have determined the area.
- 18** Evaluate each of the following:
- a** $\int_3^4 \sqrt{x-2} dx$ **b** $\int_0^2 \sqrt{2-x} dx$ **c** $\int_0^1 \frac{1}{3x+1} dx$
- d** $\int_1^2 \frac{1}{2x-1} + 3 dx$ **e** $\int_{2.5}^3 \sqrt{2x-5} - 6 dx$ **f** $\int_3^4 \frac{1}{\sqrt{x-2}} dx$

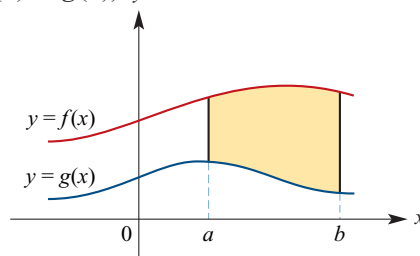
12.9 Area of a region between two curves

Let f and g be continuous on $[a, b]$ such that:

$$f(u) \geq g(u) \text{ for } u \in [a, b], (f(a) \geq g(a), f(b) \geq g(b))$$

Then the area of the region bounded by the curves and the lines $x = a$ and $x = b$ can be calculated by evaluating:

$$\int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b f(x) - g(x) dx$$



Example 27

Find the area of the region bounded by the parabola $y = x^2$ and the line $y = 2x$.

Solution

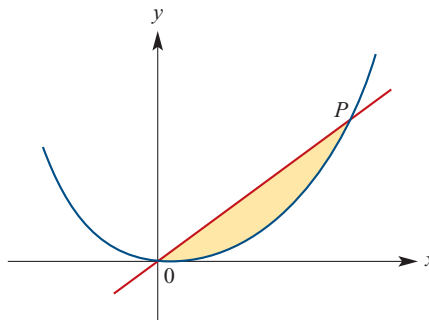
We first find the coordinates of the point P .

For $2x = x^2$, $x(x - 2) = 0$, which implies $x = 0$ or $x = 2$.

Therefore the coordinates of P are $(2, 4)$.

$$\begin{aligned}\text{The required area} &= \int_0^2 2x - x^2 dx \\ &= \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= 4 - \frac{8}{3} \\ &= \frac{4}{3}\end{aligned}$$

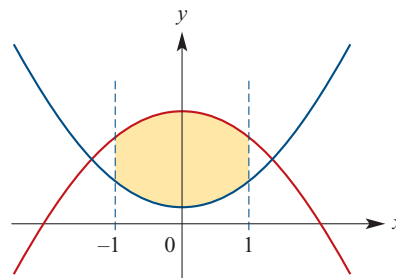
The area is $\frac{4}{3}$ square units.

**Example 28**

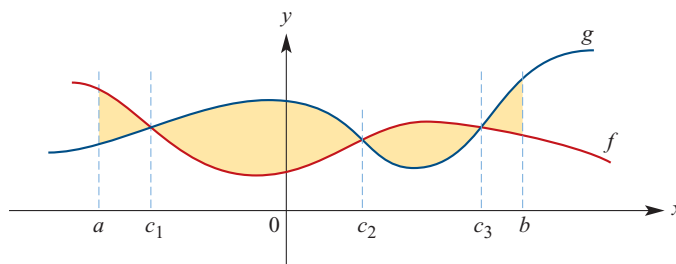
Calculate the area of the region enclosed between the curves with equations $y = x^2 + 1$ and $y = 4 - x^2$, and the lines $x = 1$ and $x = -1$.

Solution

$$\begin{aligned}\text{Required area} &= \int_{-1}^1 4 - x^2 - (x^2 + 1) dx \\ &= \int_{-1}^1 (3 - 2x^2) dx \\ &= \left[3x - \frac{2x^3}{3} \right]_{-1}^1 \\ &= \frac{14}{3}\end{aligned}$$



In Examples 27 and 28, the graph of one function is always ‘above’ the graph of the other for the intervals considered. What happens when the graphs cross?



To find the area of the shaded region we must consider the intervals $[a, c_1]$, $[c_1, c_2]$, $[c_2, c_3]$ and $[c_3, b]$ separately. Thus, the shaded area is given by:

$$\int_a^{c_1} f(x) - g(x) dx + \int_{c_1}^{c_2} g(x) - f(x) dx + \int_{c_2}^{c_3} f(x) - g(x) dx + \int_{c_3}^b g(x) - f(x) dx$$

Example 29

Find the area of the region enclosed by the graphs of $f(x) = x^3$ and $g(x) = x$

Solution

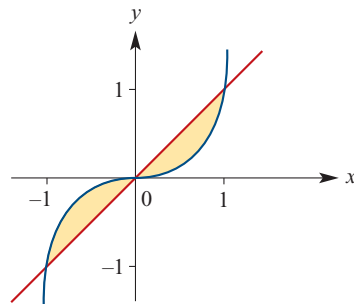
The graphs intersect where $f(x) = g(x)$

$$\begin{aligned} x^3 &= x \\ \therefore x^3 - x &= 0 \\ \therefore x(x^2 - 1) &= 0 \\ \therefore x &= 0 \text{ or } x = \pm 1 \end{aligned}$$

We see that $f(x) \geq g(x)$ for $-1 \leq x \leq 0$
and $f(x) \leq g(x)$ for $0 \leq x \leq 1$

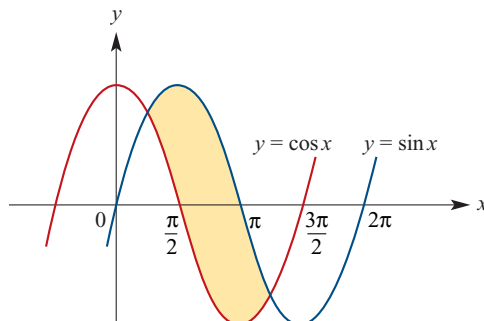
Thus the area is given by:

$$\begin{aligned} \int_{-1}^0 f(x) - g(x) dx + \int_0^1 g(x) - f(x) dx &= \int_{-1}^0 x^3 - x dx + \int_0^1 x - x^3 dx \\ &= \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 \\ &= -\left(-\frac{1}{4}\right) + \frac{1}{4} \\ &= \frac{1}{2} \end{aligned}$$



Example 30

Find the area of the shaded region of the graph.



Solution

To find the coordinates of the point of intersection:

$$\sin x = \cos x, \tan x = 1, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\begin{aligned} \text{Area} &= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx \\ &= [-\cos x - \sin x]_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \\ &= -\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \\ &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \\ &= \frac{4}{\sqrt{2}} \\ &= 2\sqrt{2} \end{aligned}$$

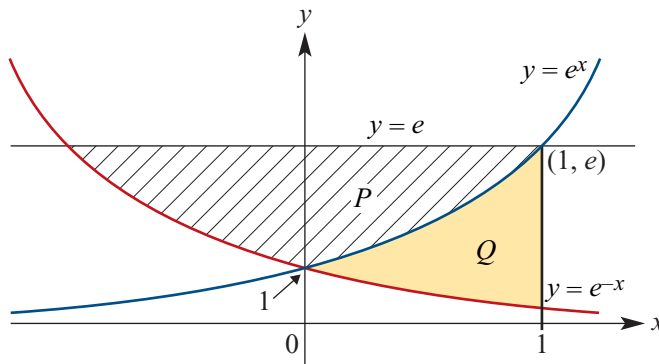
$\therefore$ area of shaded region $= 2\sqrt{2}$ square units

Exercise 121

- 1 Find the exact area of the region bounded by the graphs of $f(x) = 5 - x^2$ and $g(x) = (x - 1)^2$
- 2 Find the exact area of the region bounded by the graphs of $y = 12 - x - x^2$ and $y = x + 4$
- 3 Find the exact area of the region bounded by the graphs with equations:

a $y = x + 3$ and $y = 12 + x - x^2$	b $y = 3x + 5$ and $y = x^2 + 1$
c $y = 3 - x^2$ and $y = 2x^2$	d $y = x^2$ and $y = 3x$
e $y^2 = x$ and $x - y = 2$	

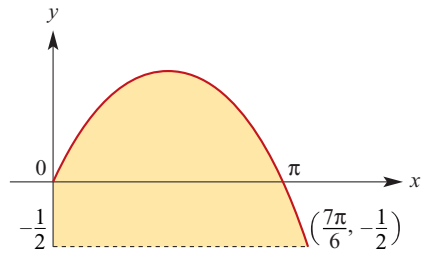
4



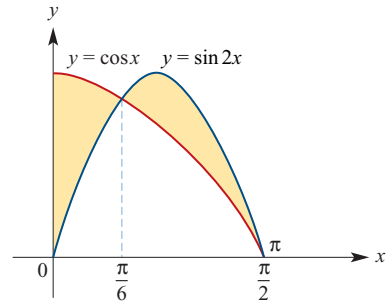
Find the areas of:

- a** the region P
b the region Q

- 5 The figure shows part of the curve $y = \sin x$. Calculate the area of the shaded region, correct to three decimal places.



- 6 Using the same axes sketch the curves $y = \sin x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$. Calculate the smaller of the two areas enclosed by the curves.
- 7 Find the area of the shaded region.



- 8 Find the coordinates of P , the point of intersection of the curves $y = e^x$ and $y = 2 + 3e^{-x}$. If these curves cut the y -axis at points A and B respectively, calculate the area bounded by AB and the arcs AP and BP . Give your answer correct to three decimal places.

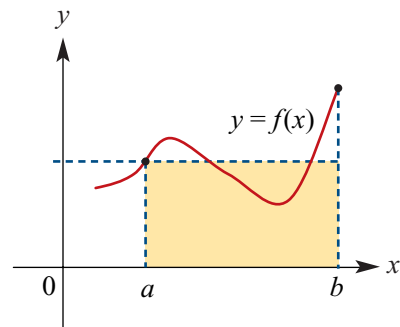
12.10 Applications of integration

Average value of a function

The average value of a function f with rule $y = f(x)$ for an interval $[a, b]$ is defined as:

$$\frac{1}{b-a} \int_a^b f(x) dx$$

In terms of the graph of $y = f(x)$, the average is the height of a rectangle having the same area as the area under the graph for the interval $[a, b]$.



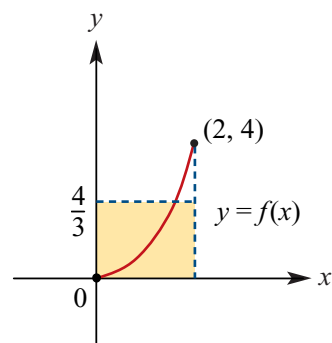
Example 31

Find the average value of $f(x) = x^2$ for the interval $[0, 2]$. Illustrate with a horizontal line determined by this value.

Solution

$$\begin{aligned}\text{Average} &= \frac{1}{2-0} \int_0^2 f(x) dx \\ &= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 \\ &= \frac{1}{2} \times \frac{8}{3} \\ &= \frac{4}{3}\end{aligned}$$

Note: The area of the rectangle $= \int_0^2 f(x) dx$

**Kinematics**

We may be given a rule for acceleration at time t and by the use of antidifferentiation with respect to t and some additional information we can deduce rules for both velocity and position.

Example 32

A body starts from O and moves in a straight line. After t seconds ($t \geq 0$) its velocity (v cm/s) is given by $v = 2t - 4$. Find:

- its position x in terms of t
- its position after 3 seconds
- its average velocity in the first 3 seconds
- the distance travelled in the first 3 seconds
- its average speed in the first 3 seconds

Solution

- a** Antidifferentiate with respect to t to find the expression for position x m at time t seconds:

$$x = t^2 - 4t + c$$

When $t = 0$, $x = 0$ and therefore $c = 0$:

$$\therefore x = t^2 - 4t$$

- b** When $t = 3$, $x = -3$. The body is 3 units to the left of O .

c Average velocity $= \frac{-3 - 0}{3} = -1$ m/s

- d** $v = 0$ when $2t - 4 = 0$, i.e. when $t = 2$

When $t = 2$, $x = -4$

Therefore the body goes from $x = 0$ to $x = -4$ in the first 2 seconds and then back to -3 in the next second. It has travelled 5 m in the first 3 seconds.

- e** Average speed is $\frac{5}{3}$ m/s.

Example 33

A particle starts from rest 3 metres from a fixed point and moves in a straight line with an acceleration of $a = 6t + 8$. Find its position and velocity at time t seconds.

Solution

$$a = \frac{dv}{dt} = 6t + 8$$

By antidifferentiating $v = 3t^2 + 8t + c$

At $t = 0$, $v = 0$ and so $c = 0$

$$\therefore v = 3t^2 + 8t$$

By antidifferentiating again $x = t^3 + 4t^2 + d$

At $t = 0$, $x = 3$ and so $d = 3$

$$\therefore x = t^3 + 4t^2 + 3$$

Example 34

A stone is projected vertically upward from the top of a building 20 m high with an initial velocity of 15 m/s.

Find:

- the time taken for the stone to reach its maximum height
- the maximum height reached by the stone
- the time taken for the stone to reach the ground
- the velocity of the stone as it hits the ground

In this case we only consider the stone's motion in a vertical direction so we can consider it as rectilinear motion. Also we will assume that the acceleration due to gravity is approximately -10 m/s^2 (note that downward is considered the negative direction).

Solution

Given that $a = -10$

$$\therefore v = -10t + c$$

At $t = 0$, $v = 15$

$$\therefore v = -10t + 15$$

$$\therefore x = -5t^2 + 15t + d$$

At $t = 0$, $x = 20$

$$\therefore x = -5t^2 + 15t + 20$$

- a** The stone will reach its maximum height when $v = 0$

$$\therefore -10t + 15 = 0$$

which implies $t = 1.5$

b At $t = 1.5$,
$$x = -5(1.5)^2 + 15(1.5) + 20$$
$$= 31.25$$

The maximum height reached by the stone is 31.25 metres.

- c** The stone reaches the ground when $x = 0$

$$-5t^2 + 15t + 20 = 0$$

$$-5(t^2 - 3t - 4) = 0$$

$$-5(t - 4)(t + 1) = 0$$

$$\therefore t = 4 \text{ (solution of } t = -1 \text{ is rejected since } t \geq 0)$$

i.e. the stone takes 4 seconds to reach the ground.

d At $t = 4$,
$$v = -10(4) + 15$$
$$= -25$$

i.e. velocity on impact is -25 m/s.

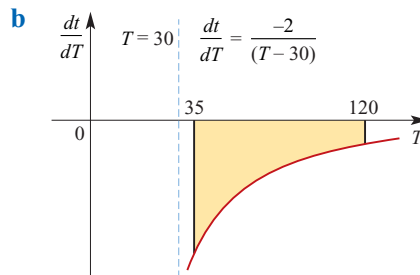
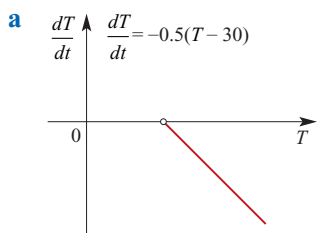
Other applications

Example 35

The rate of change of temperature with respect to time of a liquid which has been boiled and then allowed to cool is given by $\frac{dT}{dt} = -0.5(T - 30)$, where T is the temperature ($^{\circ}\text{C}$) at time t (min).

- a** Sketch the graph of $\frac{dT}{dt}$ against T for $T > 30$.
- b** Sketch the graph of $\frac{dt}{dT}$ against T for $T > 30$.
- c** **i** Find the area of the region enclosed between the graph of **b**, the x -axis and the lines $T = 35$ and $T = 120$. Give your answer correct to two decimal places.
- ii** What does this area represent?

Solution



c **i** Area $= - \int_{35}^{120} \frac{-2}{(T - 30)} dT$
$$= 5.78$$

- ii** The area represents the time taken for the temperature to change from 35°C to 120°C .

Example 36

The function $f: R^+ \cup \{0\} \rightarrow R$, $f(x) = \log_e(x+1)$

a Find f^{-1} and sketch the graphs of f and f^{-1} on the one set of axes.

b Find the exact value of the area $\int_0^{\log_e 2} f^{-1}(x) dx$

c Find the exact value of $\int_0^1 f(x) dx$

Solution

a Let $x = \log_e(y+1)$

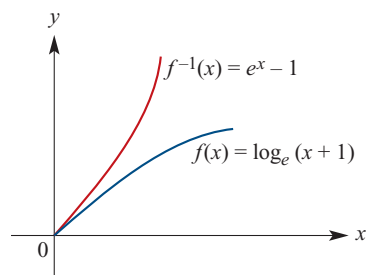
$$\therefore e^x = y + 1$$

$$\therefore y = e^x - 1$$

$\therefore$ the inverse function is:

$$f^{-1}: R^+ \cup \{0\} \rightarrow R, f^{-1}(x) = e^x - 1$$

$$\begin{aligned} \mathbf{b} \quad \int_0^{\log_e 2} f^{-1}(x) dx &= \int_0^{\log_e 2} e^x - 1 dx \\ &= [e^x - x]_0^{\log_e 2} \\ &= e^{\log_e 2} - \log_e 2 - (e^0) \\ &= 2 - \log_e 2 - 1 \\ &= 1 - \log_e 2 \end{aligned}$$



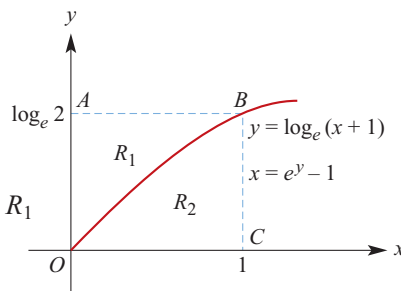
c Area of rectangle $OABC = \log_e 2$

$$\begin{aligned} \text{Area of region } R_1 &= \int_0^{\log_e 2} e^y - 1 dy \\ &= [e^y - y]_0^{\log_e 2} \\ &= 1 - \log_e 2 \quad (\text{see b}) \end{aligned}$$

Area of region R_2 = area of rectangle

$$\begin{aligned} &OABC - \text{area of region } R_1 \\ &= \log_e 2 - (1 - \log_e 2) \\ &= 2 \log_e 2 - 1 \end{aligned}$$

$$\therefore \int_0^1 f(x) dx = 2 \log_e 2 - 1$$

**Exercise 12J**

1 Find the average value of each of the following functions for the stated interval:

a $f(x) = x(2-x)$, $x \in [0, 2]$

b $f(x) = \sin(x)$, $x \in [0, \pi]$

c $f(x) = \sin(x)$, $x \in \left[0, \frac{\pi}{2}\right]$

d $f(x) = \sin(nx)$, $x \in \left[0, \frac{2\pi}{n}\right]$

e $f(x) = e^x + e^{-x}$, $x \in [-2, 2]$

- 2 A body is cooling and its temperature, T , in degrees Celsius after t minutes is given by $T = 50e^{-\frac{t}{2}}$. What is its average temperature over the first 10 minutes of cooling?
- 3 Find the average speed over the given interval for each of the following speed functions. For each of them sketch a graph and mark in the average as a horizontal line. Time is in seconds and speed in metres per second.
 - a $v = 20t$ $[0, 5]$
 - b $v = 24 \sin\left(\frac{1}{4}\pi t\right)$ $[0, 4]$
 - c $v = 5(1 - e^{-t})$ $[0, 5]$
- 4 A body falls from rest. Its velocity in metres per second at time t seconds is given by $v = 9.8t$. Find the average velocity of the body over the first 3 seconds of its motion.
- 5 Find the mean value of $x(a - x)$ from $x = 0$ to $x = a$
- 6 A quantity of gas expands under pressure p N/m² according to the law $pv^{0.9} = 300$ where v m³ is the volume of gas under pressure p N/m².
 - a What is the average pressure as the volume changes from $\frac{1}{2}$ m² to 1 m²?
 - b If the change in volume in terms of t is given by $v = 3t + 1$, what is the average pressure as the time changes from 0 to 1?
- 7 A body starts from O and moves in a straight line. After t seconds ($t \geq 0$) its velocity (v cm/s) is given by $v = 2t - 3$. Find:
 - a its position x in terms of t
 - b its position after 3 seconds
 - c its average velocity in the first 3 seconds
 - d the distance travelled in the first 3 seconds
 - e its average speed in the first 3 seconds
- 8 The velocity (v m/s) at time t seconds ($t \geq 0$) of a particle is given by $v = 2t^2 - 8t + 6$. It is initially 4 m to the right of a point O . Find:
 - a its displacement and acceleration at any time
 - b its displacement when the velocity is zero
 - c its acceleration when the velocity is zero
- 9 A body moves in a straight line with an acceleration of 8 m/s². If after 1 second it passes through O and after 3 seconds is 30 metres from O , find its initial displacement relative to O .
- 10 A body moves in a straight line so that its acceleration (a m/s²) after time t seconds ($t \geq 0$) is given by $a = 2t - 3$. If the initial position of the body is 2 m to the right of a point O and its velocity is 3 m/s, find the particle's position and velocity after 10 seconds.
- 11 A body is projected vertically upwards with a velocity of 25 m/s. (Its acceleration due to gravity is -10 m/s².) Find:
 - a the particle's velocity at any time
 - b its height above the point of projection at any time

- c the time it takes to reach its maximum height
- d the maximum height reached
- e the time taken to return to the point of projection

12 The function $f: R^+ \rightarrow R$, $f(x) = \log_e(2x)$

a Find f^{-1} and sketch the graphs of f and f^{-1} on the one set of axes.

b Find the exact value of the area $\int_0^{\log_e 4} f^{-1}(x) dx$.

c Find the exact value of $\int_{\frac{1}{2}}^2 f(x) dx$.

13 Heat escapes from a storage tank so that the rate of heat loss C in kilojoules per day is given by $\frac{dH}{dt} = 1 + \frac{3}{4} \sin\left(\frac{\pi t}{60}\right)$ for $0 \leq t \leq 200$ where $H(t)$ is the total accumulated heat loss at time t days after midday on 1 April.

a Sketch the graph of $\frac{dH}{dt}$ against t for $0 \leq t \leq 200$.

b Find the values of t for which the rate of heat loss, i.e. $\frac{dH}{dt}$, is greater than 1.375.

c Find the values of t for which the rate of heat loss reaches its maximum.

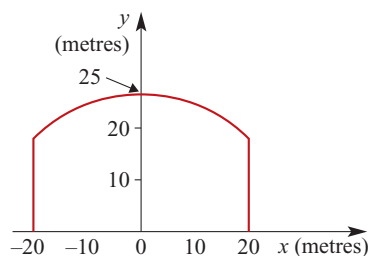
d Find the heat lost between

i $t = 0$ and $t = 120$

ii $t = 0$ and $t = 200$



14 The roof of an exhibition hall has the shape of the function $f: [-20, 20] \rightarrow R$ where $f(x) = 25 - 0.02x^2$. The hall is 80 metres long. A cross-section of the hall is shown in the figure. An air conditioning company wishes to find the volume of the hall so that a suitable system may be installed. Find this volume.



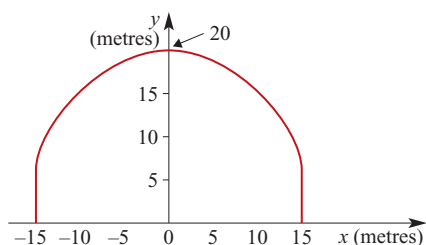
15 An aircraft hangar has the cross-section illustrated.

The roof has the shape of the function

$f: [-15, 15] \rightarrow R$ where $f(x) = 20 - 0.06x^2$

a Find the area of the cross-section.

b Find the volume of the hangar if it is 100 metres long.

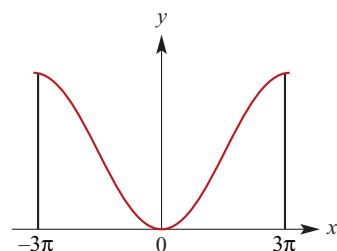


16 A long trough whose cross-section is parabolic is $1\frac{1}{2}$ metres wide at the top and 2 metres deep. Find the depth of water when it is half-full.

17 A sculpture has cross-section as shown.

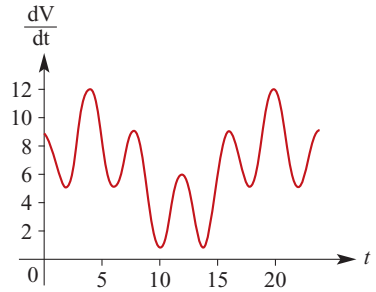
The equation of the curve is $y = 3 - 3 \cos\left(\frac{x}{3}\right)$ for $x \in [-3\pi, 3\pi]$. All measurements are in metres.

a Find the maximum value of the function and hence the height of the sculpture.



- b** The sculpture has a flat metal finish on one face, which in the diagram is represented by the region between the curve and the x -axis. Find the area of this region.
- c** There is a strut that meets the right side of the curve at right angles and passes through the point $(9, 0)$.
 - i** Find the equation of the normal to the curve where $x = a$.
 - ii** Find, correct to three decimal places, the value of a if the normal passes through $(9, 0)$.

- 18** The graph shows the number of litres/minute of water flowing through a pipe against number of minutes since the machine started. The pipe is attached to the machine which requires the water for cooling.



The curve has equation:

$$\frac{dV}{dt} = 3 \left[\cos\left(\frac{\pi t}{2}\right) + \sin\left(\frac{\pi t}{8}\right) + 2 \right]$$

- a** What is the rate of flow of water when:
 - i** $t = 0$? **ii** $t = 2$? **iii** $t = 4$?
- b** Find, correct to three decimal places, the maximum and minimum flow through the pipe.
- c** Find the volume of water which flows through the pipe in the first 8 minutes.

12.11 The fundamental theorem of calculus

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function such that $f(x) \geq 0$ for all $x \in [a, b]$.

We define geometrically the function $F(x)$ by saying that it is the measure of the area under the curve between a and x . We thus have $F(a) = 0$.

It will be shown that $F'(x) = f(x)$.

Consider the quotient $\frac{F(x+h) - F(x)}{h}$, $h > 0$.

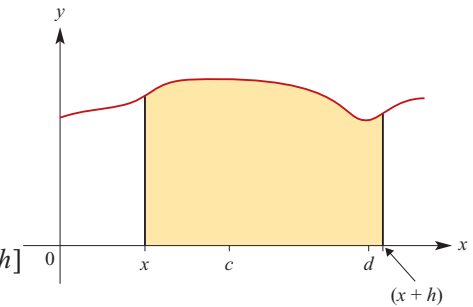
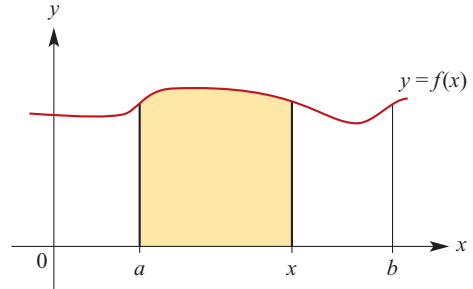
By our definition of $F(x)$, $F(x+h) - F(x)$ is the area between x and $x+h$.

Let c be the point in $[x, x+h]$ such that $f(c) \geq f(z)$ for all $z \in [x, x+h]$ and let d be the point in the same interval such that $f(d) \leq f(z)$ for all $z \in [x, x+h]$.

Thus $f(d) \leq f(z) \leq f(c)$ for all $z \in [x, x+h]$

Therefore $hf(d) \leq F(x+h) - F(x) \leq hf(c)$

i.e. the shaded region has an area less than the area of the rectangle with base h and height $f(c)$ and an area greater than the area of the rectangle with base h and height $f(d)$.



$$\text{Dividing by } h \quad f(d) \leq \frac{F(x+h) - F(x)}{h} \leq f(c)$$

As $h \rightarrow 0$ both $f(c)$ and $f(d)$ approach $f(x)$. Thus we have shown $F'(x) = f(x)$.

We know that if $G(x)$ is an antiderivative of $f(x)$ then $F(x) = G(x) + k$, where k is a constant.

Let $x = a$. We then have:

$$0 = F(a) = G(a) + k, \text{ i.e. } k = -G(a)$$

Thus $F(x) = G(x) - G(a)$ and letting $x = b$ yields

$$F(b) = G(b) - G(a) \quad (1)$$

The area under the curve $y = f(x)$ between a and b is $G(b) - G(a)$ where $G(x)$ is an antiderivative of $f(x)$.

A similar argument could be used if $f(x) < 0$ for all $x \in [a, b]$, but in this case we must take F to be the negative of the area under the curve.

We thus have by (1):

$$\int_a^b f(x)dx = G(b) - G(a) \text{ where } G \text{ is an antiderivative of } f$$

This result holds generally and is known as the **fundamental theorem of integral calculus**.

Finally the limit of a sum is considered in a special case. It is provided here as an indication of how this process can be undertaken in general.

We first introduce a notation to help us express sums. We do this through examples.

$$\begin{aligned} \sum_{i=1}^5 x_i &= x_1 + x_2 + x_3 + x_4 + x_5 \\ \sum_{i=1}^3 y_i^2 &= y_1^2 + y_2^2 + y_3^2 \\ \sum_{i=1}^4 x_i f(x_i) &= x_1 f(x_1) + x_2 f(x_2) + x_3 f(x_3) + x_4 f(x_4) \\ \sum_{i=1}^n x_i &= x_1 + x_2 + x_3 + \cdots + x_n \end{aligned}$$

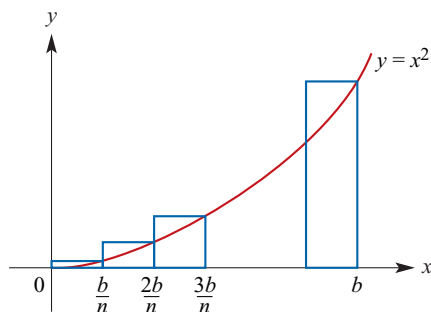
(Σ is the upper case of the Greek letter 'sigma' which is used in mathematics to denote *sum*.)

Consider the graph of $y = x^2$.

Divide the interval $[0, b]$ into n sub-intervals.

This gives a sequence of closed intervals.

$$\left[0, \frac{b}{n}\right], \left[\frac{b}{n}, \frac{2b}{n}\right], \left[\frac{2b}{n}, \frac{3b}{n}\right], \dots, \left[\frac{(n-1)b}{n}, b\right]$$



We consider the sum of the areas of the rectangles for which the height of each rectangle is defined by the right endpoint of the interval.

$$\begin{aligned}
 \text{Sum} &= \frac{b-a}{n} \left[f\left(\frac{b-a}{n}\right) + f\left(\frac{2b-a}{n}\right) + f\left(\frac{3b-a}{n}\right) + \cdots + f(b) \right] \\
 &= \frac{b-a}{n} \left[\frac{b^2}{n^2} + \frac{4b^2}{n^2} + \frac{9b^2}{n^2} + \cdots + b^2 \right] \\
 &= \frac{b^3}{n^3} [1 + 4 + 9 + \cdots + n^2] \\
 &= \frac{b^3}{n^3} \sum_{i=1}^n i^2
 \end{aligned}$$

There is a rule for working out the sum of the first n square numbers. This is:

$$\sum_{i=1}^n i^2 = \frac{n}{6}(n+1)(2n+1)$$

$$\begin{aligned}
 \therefore \text{right endpoint estimate} &= \frac{b^3}{n^3} \times \frac{n}{6}(n+1)(2n+1) \\
 &= \frac{b^3}{6n^2}(2n^2 + 3n + 1) \\
 &= \frac{b^3}{6} \left(2 + \frac{3}{n} + \frac{1}{n^2} \right)
 \end{aligned}$$

As n becomes very large, the terms $\frac{3}{n}$ and $\frac{1}{n^2}$ become very small.

$$\text{We write } \lim_{n \rightarrow \infty} \frac{b^3}{6} \left(2 + \frac{3}{n} + \frac{1}{n^2} \right) = \frac{b^3}{3}.$$

We read this as ‘the limit of the sum as n approaches infinity is $\frac{b^3}{3}$ ’.

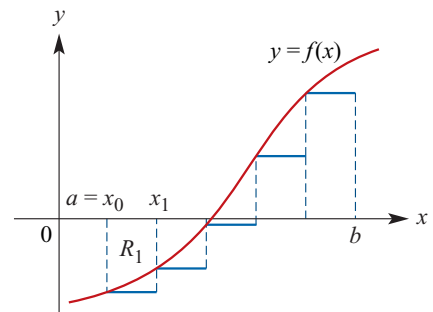
It can also be shown that working with the left endpoint rectangles for n rectangles and considering the limit as n approaches infinity gives $\frac{b^3}{3}$. (This result was first achieved by Archimedes.)

This may also be applied in general to a continuous function f , in an interval $[a, b]$. For convenience we will consider an increasing function.

We note that to calculate the area of rectangle R_1 we calculate $(x_1 - x_0)f(x_0)$ and as $f(x_0) < 0$ the result is a negative. We have found the signed area of R_1 .

If an interval $[a, b]$ is divided into n sub-intervals and the rectangles to be used to find the area have their ‘height’ determined by the left endpoint of that interval then the area is given by:

$$\frac{b-a}{n} \sum_{i=0}^{n-1} f(x_i)$$



If the limit as $n \rightarrow \infty$ exists, we can make the following definition:

$$\lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{i=0}^{n-1} f(x_i) = \int_a^b f(x) dx$$

The procedure indicated could also have started with the right endpoint estimate being used, as the left and right endpoint estimates will converge to the one limit as n approaches infinity. Definite integrals may be defined as the limit of suitable sums and the fundamental theorem of calculus holds true under this definition.



Chapter summary

- To find the general antiderivative:

If $F'(x) = f(x)$

then $\int f(x)dx = F(x) + c$, where c is an arbitrary real number.

- $\int x^r dx = \frac{x^{r+1}}{r+1} + c, r \in \mathbb{Q} \setminus \{-1\}$

- $\int f(x) + g(x)dx = \int f(x)dx + \int g(x)dx$

- $\int kf(x)dx = k \int f(x)dx$ where k is a real number

- $\int e^{kx} dx = \frac{1}{k} e^{kx} + c, k \neq 0$

$$\int \sin kx dx = -\frac{1}{k} \cos kx + c, k \neq 0$$

$$\int \cos kx dx = \frac{1}{k} \sin kx + c, k \neq 0$$

$$\int \frac{1}{kx+b} dx = \frac{1}{k} \log_e |kx+b| + c, k \neq 0$$

- The fundamental theorem of integral calculus states that:

$\int_a^b f(x)dx = G(b) - G(a)$ where G is any antiderivative of f and $\int_a^b f(x)dx$ is called the definite integral from a to b .

The number a is called the lower limit of integration and b is called the upper limit of integration. The function f is called the integrand.

- If $f(x) \geq 0$ for all $x \in [a, b]$, the area of the region contained between the curve, the x -axis and the lines $x = a$ and $x = b$ is given by $\int_a^b f(x)dx$

If $f(x) \leq 0$ for all $x \in [c, b]$ the area of the region contained between the curve, the x -axis and the

lines $x = c$ and $x = b$ is given by $-\int_c^b f(x)dx$

If $c \in [a, b]$, $f(c) = 0$ and $f(x) \geq 0$ for $x \in (c, b]$ and $f(x) \leq 0$ for $x \in [a, c)$, then the area of the shaded region is given by:

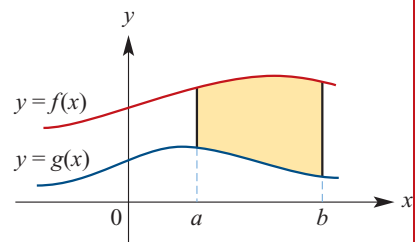
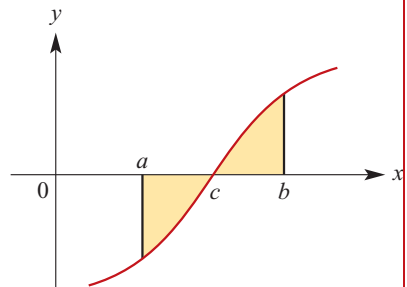
$$\int_c^b f(x)dx + \left\{ -\int_a^c f(x)dx \right\}$$

- To find the area of the shaded region bounded by the curves and the lines $x = a$ and $x = b$ use

$$\int_a^b f(x)dx - \int_a^b g(x)dx = \int_a^b f(x) - g(x)dx$$

where f and g are continuous on $[a, b]$ such that

$$f(u) > g(u) \text{ for } u \in [a, b], (f(a) \geq g(a), f(b) \geq g(b))$$



- If the interval $[a, b]$ on the x -axis is divided into n equal sub-intervals

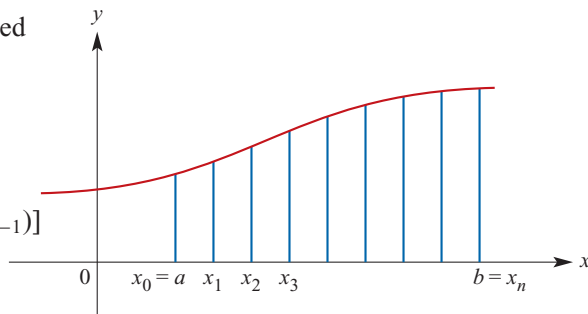
$$[a, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, x_n]$$

the left endpoint estimate is:

$$L_n = \frac{b-a}{n} [f(x_0) + f(x_1) + \dots + f(x_{n-1})]$$

The right endpoint estimate is:

$$R_n = \frac{b-a}{n} [f(x_1) + f(x_2) + \dots + f(x_n)]$$



- The average value of an integrable function f for an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$.

Multiple-choice questions

1 $\int_0^2 3f(x) + 2dx =$

A $3 \int_0^2 f(x)dx + 3x$ B $3 \int_0^2 f(x)dx + x$ C $3 \int_0^2 f(x)dx + 4$

D $3f'(x) + 4$ E $\int_0^2 f(x)dx + 4$

- 2 For a and b positive real constants the antiderivative of $\sqrt{(ax-b)^3}$, $x > \frac{b}{a}$ is:

A $\frac{3}{2}(ax-b)^{\frac{1}{2}} + c$ B $\frac{1}{5\sqrt{(ax-b)}} + c$ C $\frac{2}{5a}(ax-b)^{\frac{5}{2}} + c$

D $\frac{1}{5a}\sqrt{(ax-b)^5} + c$ E $\frac{2}{5\sqrt{(ax-b)^5}} + c$

- 3 An expression using integral notation for the area of the shaded region shown is:

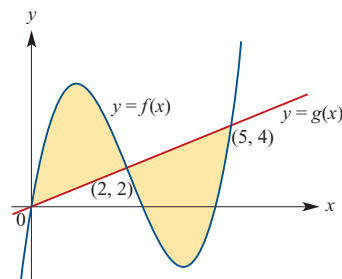
A $\int_0^5 f(x) - g(x)dx$

B $\int_0^2 f(x) - g(x)dx + \int_2^5 g(x) - f(x)dx$

C $\int_0^4 f(x) - g(x)dx$

D $\int_0^2 f(x) - g(x)dx + \int_2^4 f(x) - g(x)dx$

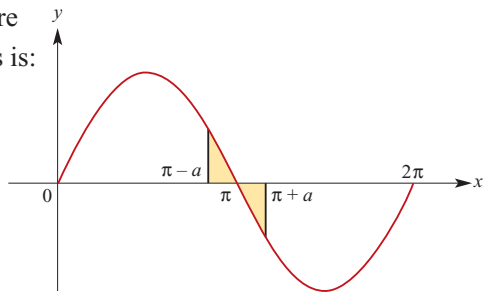
E $\int_0^2 f(x) - g(x)dx + \int_2^4 g(x) - f(x)dx$



- 4 $\int_a^b c dx$ in terms of a , b and c , where a , b and c are distinct real number constants, is equal to:

A ca B $cb - ca$ C $ca - b$ D $cb - a$ E $c(a+b)$

- 5 An expression for y if $\frac{dy}{dx} = \frac{ax}{2} + 1$ and $y = 1$ when $x = 0$ is:
A $y = \frac{ax^2}{4} + x + 1$ **B** $y = a$ **C** $y = ax^2 + x - 1$
D $y = ax^2 + x + a$ **E** $y = ax^2 + ax + a$
- 6 The function f such that $f'(x) = -6 \sin 3x$ and $f\left(\frac{2\pi}{3}\right) = 3$ is:
A $-18 \cos 3x + 21$ **B** $-2 \cos 2x + 5$ **C** $-2 \sin 3x + 1$
D $2 \cos 3x + 1$ **E** $2 \sin 4x + 3$
- 7 The area of the region enclosed by the curve $y = e^{5x} - 2 \sin 4x$, the x -axis and the lines $x = -1$ and $x = 1$, correct to two decimal places, is:
A 0.17 **B** 29.55 **C** 29.68 **D** 29.85 **E** 30.02
- 8 Given $\frac{dy}{dx} = ae^{-x} + 2$ and that when $x = 0$, $\frac{dy}{dx} = 5$ and $y = 1$, then when $x = 2$, $y =$
A $-\frac{3}{e^2} + 2$ **B** $-\frac{3}{e^2} + 4$ **C** $-\frac{3}{e^2} + 8$ **D** $3e^2 + 4$ **E** $3e^2 + 8$
- 9 The rate of the flow of water from a tap follows the rule $R(t) = 5e^{-0.1t}$ where $R(t)$ litres per minute is the rate of flow after t minutes. The number of litres, to the nearest litre, which flowed out in the first 3 minutes is:
A 0 **B** 5 **C** 13 **D** 50 **E** 153
- 10 The graph represents the function $y = \sin x$ where $0 \leq x \leq 2\pi$. The total area of the shaded regions is:
A $\frac{1}{2} \cos a$ **B** $2 \cos a$
C $\frac{1}{2}(1 - \cos a)$ **D** $2(1 - \cos a)$
E $2 \sin^2 a$



Short-answer questions (technology-free)

- 1 Evaluate each of the following definite integrals:

a $\int_2^3 x^3 dx$

b $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \theta d\theta$

c $\int_a^{4a} (a^{\frac{1}{2}} - x^{\frac{1}{2}}) dx$, where a is a positive constant.

d $\int_1^4 \frac{3}{\sqrt{x}} - 5\sqrt{x} - x^{-\frac{3}{2}} dx$

e $\int_0^{\frac{\pi}{4}} \cos 2\theta d\theta$

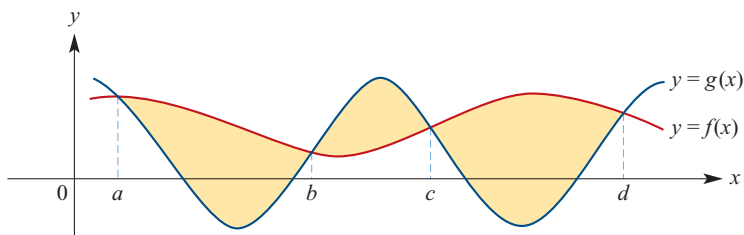
f $\int_1^e \frac{1}{x} dx$

g $\int_0^{\frac{\pi}{2}} \sin 2\left(\theta + \frac{\pi}{4}\right) d\theta$

h $\int_0^{\pi} \sin 4\theta d\theta$

- 2 Find $\int_{-1}^2 x + 2f(x) dx$ if $\int_{-1}^2 f(x) dx = 5$

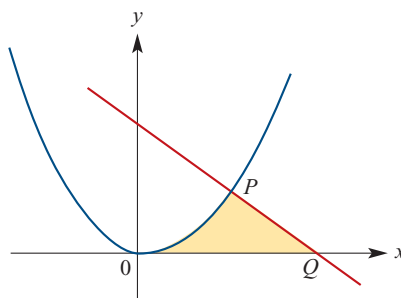
- 3 Find $\int_1^5 f(x)dx$ if $\int_0^1 f(x)dx = -2$ and $\int_0^5 f(x)dx = 1$
- 4 Find $\int_3^{-2} f(x)dx$ if $\int_{-2}^1 f(x)dx = 2$ and $\int_1^3 f(x)dx = -6$
- 5 Evaluate $\int_0^2 (x+1)^7 dx$.
- 6 Evaluate $\int_0^1 (3x+1)^3 dx$.
- 7 Find $\int_0^3 f(3x)dx$ if $\int_0^9 f(x)dx = 5$
- 8 Find $\int_0^1 f(3x+1)dx$ if $\int_1^4 f(x)dx = 5$
- 9 Set up a sum of definite integrals that represents the total shaded area between the curves $y = f(x)$ and $y = g(x)$



- 10 The figure shows the curve $y = x^2$ and the straight line $2x + y = 15$

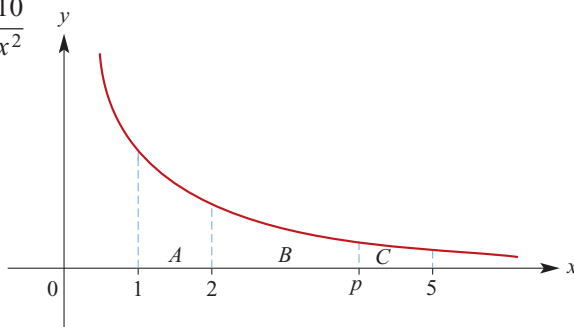
Find:

- a the coordinates of P and Q
- b the area of the shaded region

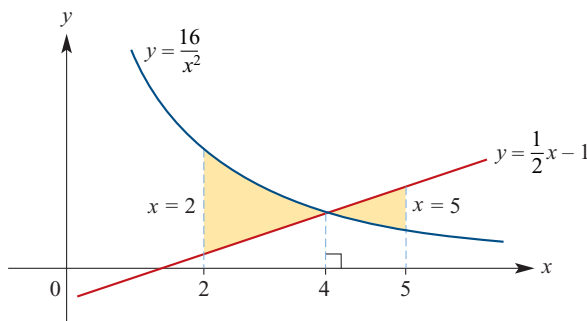


- 11 The figure shows part of the curve $y = \frac{10}{x^2}$
- Find:

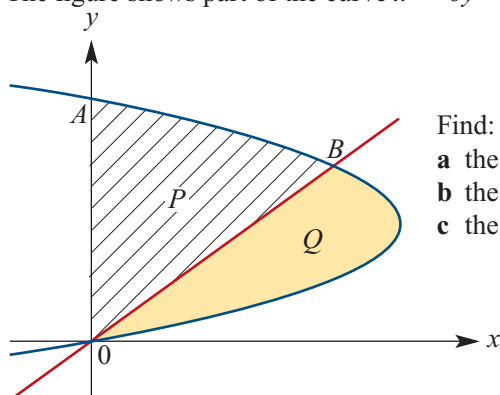
- a the area of region A
- b the value of p for which the regions B and C are of equal area



- 12 Find the area of the shaded region:



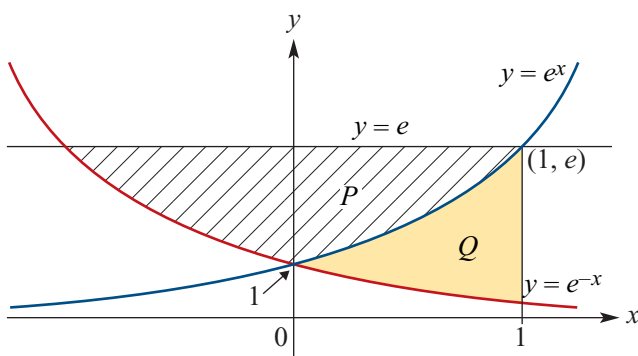
- 13 The figure shows part of the curve $x = 6y - y^2$ and part of the line $y = x$



Find:

- a** the coordinates of A and B
b the area of region P 
c the area of region Q 

14



Find the areas of:

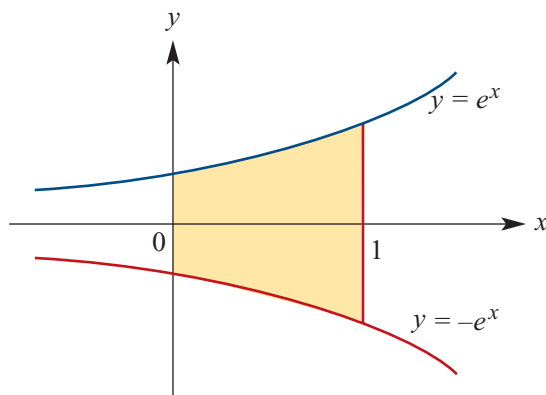
- a** the region P 
b the region Q 

- 15 **a** Sketch the graph of $y = e^x + 1$ and clearly indicate, by shading the region, the area given by $\int_0^2 e^x + 1 dx$.
b Evaluate $\int_0^2 e^x + 1 dx$.
- 16 **a** Sketch the graph of $y = e^{-x}$ and $y = e^x$ on the one set of axes and clearly indicate, by shading the region, the area given by:

$$\int_0^2 e^{-x} dx + \int_{-2}^0 e^x dx$$

- b** Evaluate $\int_0^2 e^{-x} dx + \int_{-2}^0 e^x dx$.

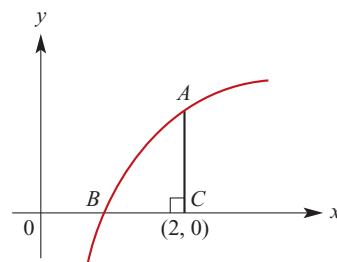
- 17 **a** Evaluate $\int_0^1 e^x dx$.
b By symmetry find the area of the region shaded in the figure.



- 18** Sketch the graph of $f(x) = 2e^{2x} + 3$ and find the area of the region enclosed between the curve, the axes and the line $x = 1$
- 19** Sketch the graph of $y = x(x - 2)(x + 1)$ and find the area of the region contained between the graph and the x -axis. (Do not attempt to find the coordinates of the turning points.)
- 20** Evaluate each of the following definite integrals:
- | | |
|---|---|
| <p>a $\int_0^2 e^{-x} + x dx$</p> | <p>b $\int_{-2}^{-1} x + \frac{1}{x-1} dx$</p> |
| <p>c $\int_0^{\frac{\pi}{2}} \sin x + x dx$</p> | <p>d $\int_{-4}^{-5} e^x + \frac{1}{2-2x} dx$</p> |

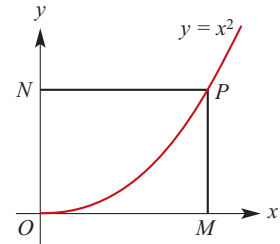
Extended-response questions

- 1** The rate of flow of water from a reservoir is given by $\frac{dV}{dt} = 1000 - 30t^2 + 2t^3$, $0 \leq t \leq 15$ where V is measured in millions of litres and t is the number of hours after the sluice gates are opened.
- Find the rate of flow (in million litres/hour) when $t = 0$ and $t = 2$.
 - Find:
 - the times when the rate of flow is a maximum
 - the maximum flow
 - Sketch the graph of $\frac{dV}{dt}$ against t for $0 \leq t \leq 15$.
 - Find the area beneath the graph between $t = 0$ and $t = 10$.
 - What does this area represent?
- 2** The population of penguins on an island off the coast of Tasmania is increasing steadily. The rate of growth is given by the function $R: [0, \infty) \rightarrow R$, $R(t) = 10 \log_e(t + 1)$. The rate is measured in number of penguins per year. The date 1 January 1875 coincides with $t = 0$.
- Find the rate of growth of penguins when $t = 5$, $t = 10$, $t = 100$.
 - Sketch the graph of R for $[0, \infty)$.
 - Find the inverse function R^{-1} .
 - Find the area under the graph of $y = R(t)$ between $t = 0$ and $t = 100$. (Use the inverse function to help find this area.)
 - What does this area represent?
- 3** The diagram shows part of the curve with equation
- $$y = x - \frac{1}{x^2}$$
- C is the point $(2, 0)$.
- Find:
- the equation of the tangent to the curve at point A
 - the coordinates of the point T where this tangent meets the x -axis

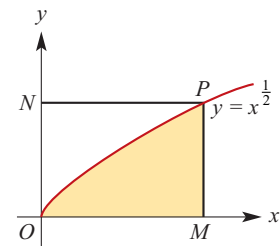


- c the coordinates of the point B where the curve meets the x -axis
- d the area of the region enclosed by the curve and the lines AT and BT
- e the ratio of the area found in part d to the area of the triangle ATC

- 4 a In the figure, P is a point on the curve $y = x^2$.
Prove that the curve divides the rectangle $OMPN$ into two regions whose areas are in the ratio 2 : 1.



- b In the figure, P is a point on the curve $y = x^{\frac{1}{2}}$. Prove that the area of the shaded region is two-thirds of the area of the rectangle $OMPN$.



- c For the curve $y = x^n$, P is a point on the curve with PM and PN the perpendiculars from P to the x -axis and the y -axis respectively. Prove that the area of the region enclosed between PM , the x -axis and the curve is equal to $\frac{1}{n+1}$ of the area of the rectangle $OMPN$.

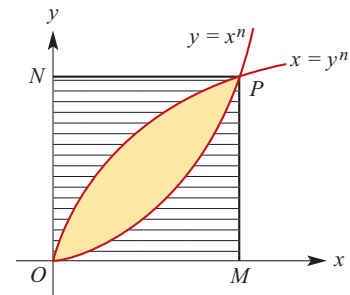
- 5 a Find the area enclosed between the parabolas $y^2 = x$ and $x^2 = y$.
b Show that the curves with equations $y = x^n$ and $y^n = x$ intersect at $(1, 1)$ (where $n = 1, 2, 3, \dots$).

- c Show that the area of the region contained between the curves $y = x^n$ and $y^n = x$ is $\frac{n-1}{n+1}$

- d Find the area of the region indicated by horizontal shading in the diagram.

- e Use your result from c to find the area of the region between the curves for $n = 10, 100, 1000$.

- f Describe the result for n very large.



- 6 It is thought that the average temperature θ of a piece of charcoal in a barbecue will increase at a rate $\frac{d\theta}{dt}$ given by $\frac{d\theta}{dt} = e^{2.6t}$ where θ is in degrees and t is in minutes.
- a If the charcoal starts at a temperature of 30°C , find the expected average temperature of the charcoal after 3 minutes.
 - b Sketch the graph of θ against t .
 - c At what time does the temperature of the charcoal reach 500°C ?
 - d Find the average rate of increase of temperature from $t = 1$ to $t = 2$.

- 7 It is believed that the velocity of a certain atomic particle t microseconds after a collision will be given by the expression

$$\frac{dx}{dt} = ve^{-t}, \quad v = 5 \times 10^4 \text{ m/s}$$

where x stands for distance travelled in metres.

- a** What is the initial velocity of the particle?
b What happens to the velocity at $t \rightarrow \infty$ (i.e. t becomes very large)?
c How far will the particle travel between $t = 0$ and $t = 20$?
d Find an expression for x in terms of t . **e** Sketch the graph of x against t .
 8 **a** Differentiate $e^{-3x} \sin 2x$ and $e^{-3x} \cos 2x$ with respect to x .
b Hence show that

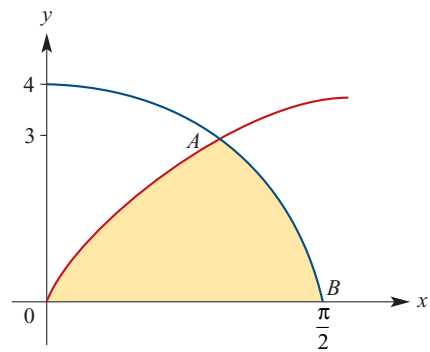
$$e^{-3x} \sin 2x + c_1 = -3 \int e^{-3x} \sin 2x dx + 2 \int e^{-3x} \cos 2x dx$$

$$\text{and } e^{-3x} \cos 2x + c_2 = -3 \int e^{-3x} \cos 2x dx - 2 \int e^{-3x} \sin 2x dx$$

- c** Use the two equations from **b** to determine $\int e^{-3x} \sin 2x dx$.

- 9 The curves $y = 3 \sin x$, $y = 4 \cos x$ ($0 \leq x \leq \frac{\pi}{2}$) intersect at a point A .

- a** If $x = a$ at the point of intersection of the two curves:
i Find $\tan a$.
ii Hence find $\sin a$ and $\cos a$.
b Hence find the area of the shaded region in the diagram.

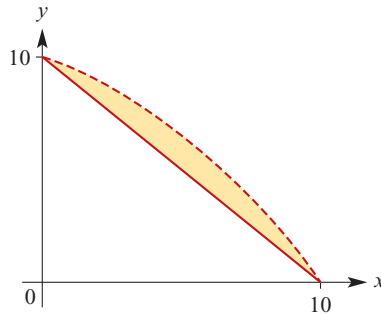
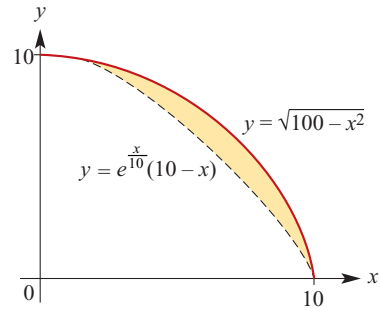


- 10 **a** If $y = x \log_e x$ find $\frac{dy}{dx}$. Hence find the value of $\int_1^e \log x dx$.
b If $y = x(\log_e x)^n$, where n is a positive integer, find $\frac{dy}{dx}$.
c Let $I_n = \int_1^e (\log_e x)^n dx$. For $n > 1$, show that $I_n + nI_{n-1} = e$.
d Hence find the value of $\int_1^e (\log_e x)^3 dx$.
 11 The curves $y^2 = ax$ and $x^2 = by$, where a and b are both positive, intersect at the origin and at the point (r, s) . Find r and s in terms of a and b . Prove that the two curves divide the rectangle whose corners are the points $(0, 0)$, $(0, s)$, (r, s) , $(r, 0)$ into three regions of equal area.

- 12** A teacher attempts to draw a quarter circle on the white board. The circle is of radius 10. However, the first attempt results in a curve with rule $y = e^{\frac{x}{10}}(10 - x)$.

The circle has equation $y = \sqrt{100 - x^2}$

- Find $\frac{dy}{dx}$ for both functions.
- Find the gradient of each of the functions when $x = 0$.
- Find the gradient of $y = e^{\frac{x}{10}}(10 - x)$ when $x = 10$.
- Find the area of the shaded region correct to two decimal places using a calculator.
- Find the percentage error for the calculation of the area of the quarter circle.
- The teacher draws in a chord from $(0, 10)$ to $(10, 0)$. Find the area of the shaded region using a calculator.
 - Use the result that the derivative of $e^{\frac{x}{10}}(10 - x)$ is $-e^{\frac{x}{10}} + \frac{1}{10}e^{\frac{x}{10}}(10 - x)$ to find $\int_0^{10} e^{\frac{x}{10}}(10 - x) dx$ by analytic techniques.
 - Find the exact area of the original shaded region and compare it to the answer of **d**.



- 13** A water-cooling device has a system of water circulation for the first 30 minutes of its operation. The circulation follows the following sequence:

- For the first three minutes water is flowing in.
- For the second three minutes water is flowing out.
- For the third three minutes water is flowing in.

This pattern is continued for the first 30 minutes. The rate of flow of water is given by the function

$$R(t) = 10e^{-\frac{t}{10}} \sin\left(\frac{\pi t}{3}\right)$$

where $R(t)$ litres/minute is the rate of flow at time t minutes. Initially there are 4 litres of water in the device.

- Find:
 - $R(0)$
 - $R(3)$
- Find $R'(t)$
- Solve the equation $R'(t) = 0$ for $t \in [0, 12]$.
 - Find the coordinates of the stationary points of $y = R(t)$ for $t \in [0, 12]$.
- Solve the equation $R(t) = 0$ for $t \in [0, 12]$.
- Sketch the graph of $y = R(t)$ for $t \in [0, 12]$.

- f i** How many litres of water flowed into the device for $t \in [0, 3]$?
- ii** How many litres of water flowed out of the device for $t \in [3, 6]$?
- iii** How many litres of water are in the device when $t = 6$? (Remember there are initially 4 litres of water.)
- g** How many litres of water are there in the container when $t = 30$?
- 14 a** Sketch the graph of $f(x) = 2 \sin x - 1$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
- b** Evaluate $\int_0^{\frac{\pi}{6}} f(x) dx$ and indicate the area given by this integral on the graph of **a**.
- c** Find the inverse function f^{-1} .
- d** Evaluate $\int_0^1 f^{-1}(x) dx$ and indicate the area given by this integral on the sketch graph of **a**.
- 15 a** Use the identities $\cos 2x = 2 \cos^2 x - 1$ and $\cos 2x = 1 - 2 \sin^2 x$ to show that:
- $$\frac{1 - \cos 2x}{1 + \cos 2x} = \sec^2 x - 1$$
- b** Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{1 + \cos 2x} dx$

Revision of Chapters 9–12

13.1 Multiple-choice questions

- 1 If $y = \frac{x^4 + x}{x^2}$, then $\frac{dy}{dx}$ equals:
A $\frac{4x^3 + 2x}{2x}$ **B** $x^2 + 1$ **C** $2x$ **D** $\frac{2x^3 + 1}{x^2}$ **E** $2x - \frac{1}{x^2}$
- 2 If $f(x) = x^5 + x^3 + x$, the value of $f'(1)$ is:
A 0 **B** 1 **C** 2 **D** 9 **E** -3
- 3 If $y = (4 - 9x^4)^{\frac{1}{2}}$, then $\frac{dy}{dx}$ equals:
A $-\frac{9}{2}(4 - 9x^4)^{-\frac{1}{2}}$ **B** $\frac{1}{2}(4 - 9x^4)^{-\frac{1}{2}}$ **C** $2(4 - 9x^4)^{-\frac{1}{2}}$
D $-3x(4 - 9x^4)^{-\frac{1}{2}}$ **E** $-18x^3(4 - 9x^4)^{-\frac{1}{2}}$
- 4 The gradient of the curve with equation $y = \sin 2x + 1$ at $(0, 1)$ is:
A 1 **B** -1 **C** 0 **D** 2 **E** -2
- 5 $\frac{d}{dx}(e^{x^2+1})$ is:
A $2x$ **B** $2xe^{x^2+1}$ **C** $2xe^{2x}$ **D** $(x^2 + 1)e^{x^2}$ **E** $(x^2 + 1)e^{x^2}$
- 6 The derivative of $\frac{1}{1+x}$ is:
A $\frac{1}{(1+x)^2}$ **B** $\frac{1}{1-x}$ **C** $\frac{-1}{(1+x)^2}$ **D** 1 **E** $\frac{1}{2}$
- 7 P and Q lie on the curve $y = x^3$. The x -coordinates of P and Q are 2 and $(2 + h)$ respectively. The gradient of chord PQ is:
A $\frac{h^3 - 8}{h - 2}$ **B** $12 + 6h$ **C** 12 **D** $\frac{(2 + h)^3 - 8}{h}$ **E** $12 + 6h + h^2$

- 8 If $f(x)$ equals $\frac{3}{x}$, then $\frac{[f(x+h) - f(x)]}{h}$ is equal to:

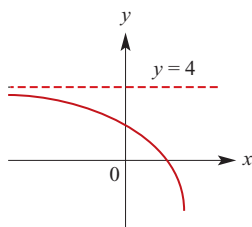
A $-\frac{3}{x(x+h)}$ B $\frac{3}{x^2}$ C $\frac{-3}{x^2}$ D $\frac{-3}{h(x+h)}$ E $f'(x)$

- 9 The gradient of $y = ce^{2x}$ is equal to 11 when $x = 0$. The value of c is:

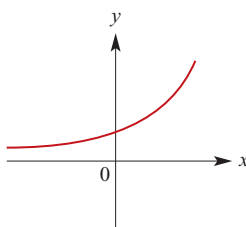
A 0 B 1 C 5 D 5.5 E $5e^{-2}$

- 10 $f: R \rightarrow R$, $f(x) = 4 - e^{-2x}$. The graph of $f'(x)$ is best represented by:

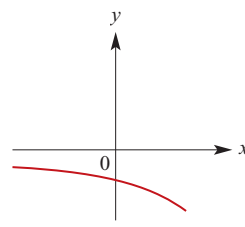
A



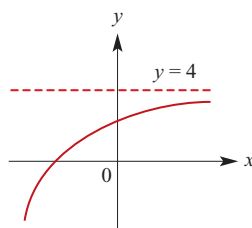
B



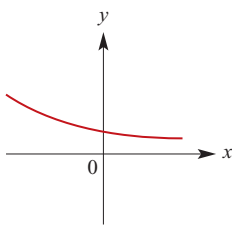
C



D

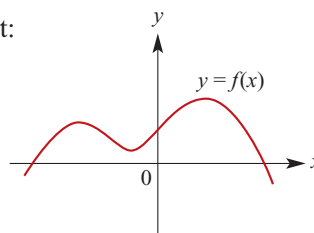


E



- 11 For the graph of $y = f(x)$ shown, $f'(x) = 0$ at:

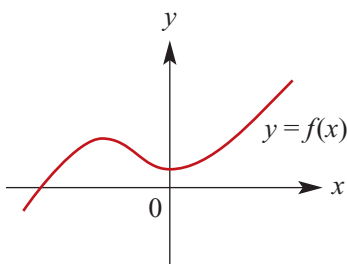
A 3 points B 2 points
C 5 points D 0 points
E none of these



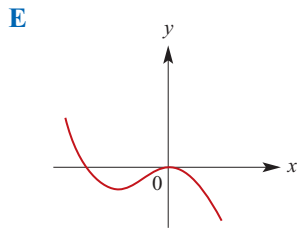
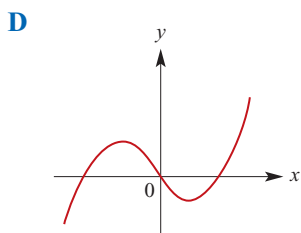
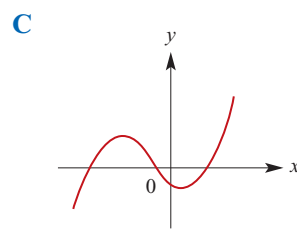
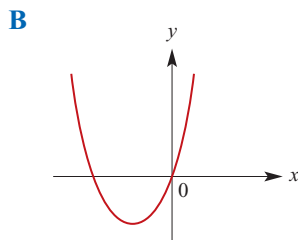
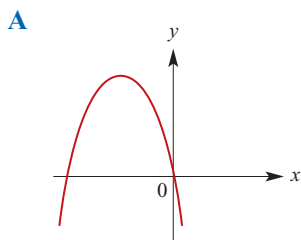
- 12 The graph of $y = bx^2 - cx$ crosses the x -axis at the point $(4, 0)$. The gradient at the point is 1. The value of c is:

A 8 B 1 C 4 D -8 E 2

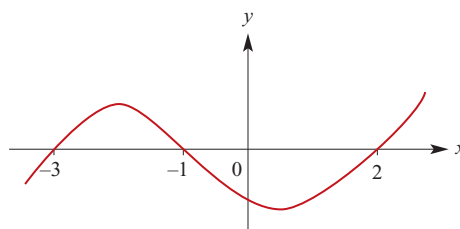
13



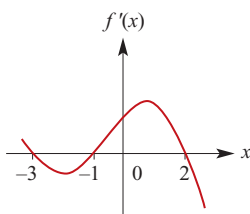
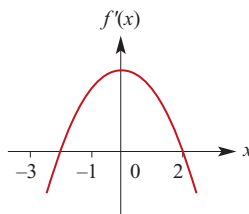
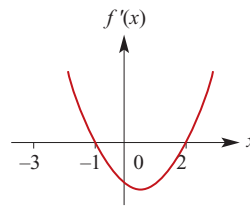
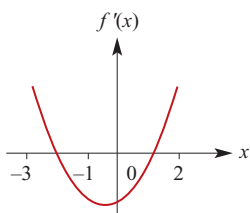
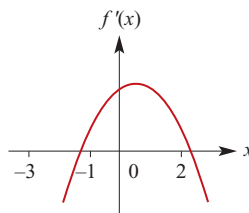
The graph that best represents the graph of $y = f'(x)$ is:



- 14** If $y = (3x^4 - 2)^4$, then $\frac{dy}{dx}$ equals:
A $x^4(3x^4 - 2)^3$ **B** $4(3x^4 - 2)^3$ **C** $12x^{12}$
D $(12x^2 - 2)^4$ **E** $48x^3(3x^4 - 2)^3$
- 15** $f'(x) = g'(x)$ and $f(x) = 3x^2 + 2$. If $g(2) = 29$, $g(x) =$
A $3x^3 + 5$ **B** $3x^2 - 3$ **C** $\frac{x^3}{3} + 2x$ **D** $3x^2 + 17$ **E** $6x + 17$
- 16** For $f(x) = e^{kx} + e^{-kx}$, $f'(x) > 0$ for:
A $x \in \mathbb{R}$ **B** $x \geq 0$ **C** $x < 0$ **D** $x \leq 0$ **E** $x > 0$
- 17** If $y = 3x^2 + 2x - \frac{4}{x^2}$, then $\frac{dy}{dx}$ is equal to:
A $6x + 2 - \frac{8}{x^3}$ **B** $6x + 2 + \frac{8}{x^3}$ **C** $6x + 2 + \frac{8}{x}$
D $x^3 + x^2 + \frac{4}{x}$ **E** $x^3 + x^2 - \frac{4}{x}$
- 18** Rainwater is being collected in a water tank. The volume, $V \text{ m}^3$, of water in the tank after time, t minutes, is given by $V = 2t^2 + 3t + 1$. The average rate of change of volume of water between times $t = 2$ and $t = 4$, in m^3/min is:
A 11 **B** 13 **C** 15 **D** 17 **E** 19
- 19** $P(x, f(x))$ and $Q((x+h), f(x+h))$ are two points on the curve with equation $f(x) = x^2 - 2x + 1$. The gradient of the line joining P and Q is given by:
A $2x - 2$ **B** $2xh - 4x - 2h + 2$ **C** $2xh - 2h - h^2$
D $2xh - 2h + h^2$ **E** $2x - 2 + h$
- 20** The graph of $y = f(x)$ is shown.



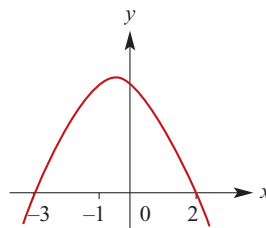
A possible graph of the gradient function f' with rule given by $f'(x)$ is:

A**B****C****D****E**

- 21** The graph of the derived function f' given by $y = f'(x)$ is shown.

The function is increasing as:

- A** $\{x: x \geq 0\}$
B $\{x: -3 \leq x \leq 2\}$
C $\{x: x \geq 2\}$
D $\{x: x \leq -3\} \cup \{x: x \geq 2\}$
E $\{x: x \leq 0\}$



- 22** Which one of the following gives the gradient of the tangent to a curve with the equation $y = f(x)$ at the point $x = 2$?

- A** $\frac{f(x+h) - f(x)}{h}$ **B** $f(2+h) - f(2)$ **C** $\frac{f(2+h) - f(2)}{h}$
D $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ **E** $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$

- 23** The derivative of $\frac{e^{2x} + e^{-2x}}{e^x}$ is:

- A** $e^x + e^{-3x}$ **B** $e^x - 3e^{-3x}$ **C** $xe^x - 3xe^{-3x}$
D $\frac{2e^{2x} - e^{-2x}}{e^x}$ **E** $\frac{e^{3x} - 3e^{-x}}{e^{x^2}}$

- 24** The maximum value of $-x^2 + 4x + 3$ is:

- A** 2 **B** 3 **C** $2 + 2\sqrt{7}$ **D** 7 **E** 15

- 25** The equation of a tangent to a curve with equation $y = 1 + e^{2x}$ at the point $(0, 2)$ is:

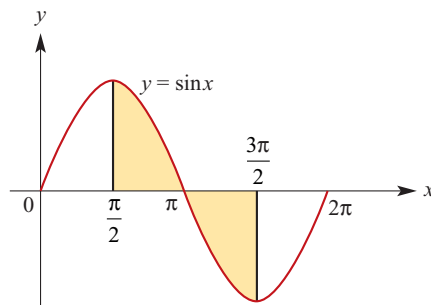
- A** $y = 2e^{2x}$ **B** $y = 2x + 2$ **C** $y = \frac{-1}{2}x + 2$
D $y = 2$ **E** $y = 2e^{2x} + 2$

- 26** The graph of the curve with equation $y = x^2 - x^3$ has stationary points where x is equal to:

- A** 0 and $\frac{2}{3}$ **B** 0 and 1 **C** -1 and 0 **D** 0 and $\frac{3}{2}$ **E** 2 and -3

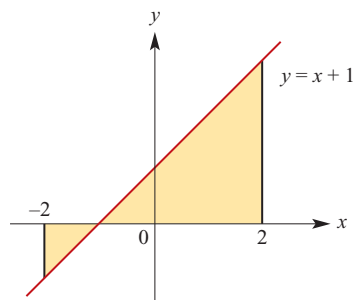
- 27 The derivative of $\frac{4x^2 + 6}{x}$ is:
A $8x + 6$ **B** $8x - 6$ **C** $\frac{4x^2 + 6}{x^2}$ **D** $\frac{4x^2 - 6}{x^2}$ **E** $\frac{8x^3 - 6}{x^2}$
- 28 If $f(x) = 4x^3 - 3x + 7 - \frac{2}{x}$, then $f'(1)$ is equal to:
A -2 **B** 18 **C** 7 **D** 11 **E** 14
- 29 Given that $f'(x) = x^2 + \frac{1}{x}$ and $f(1) = \frac{1}{3}$, then $f(x)$ is equal to:
A $\frac{x^3}{3} + \log_e x$ **B** $\frac{x^3}{3} \log_e x + \frac{2}{3}$ **C** $\frac{x^3}{3} - \log_e x - \frac{1}{3}$
D $\frac{-x^3}{3} + \log_e x + \frac{2}{3}$ **E** $\frac{x^3}{3} - \log_e x + \frac{1}{3}$
- 30 If $y = F(x)$ and $\frac{dy}{dx} = f(x)$, then $\int_2^3 f(x) dx$ is equal to:
A $f(3) - f(2)$ **B** $F'(3) - F'(2)$ **C** $F(3) - F(2)$
D $f(x) + c$ **E** $F(3) - f(2)$
- 31 The straight line $y = 2x + 1$ cuts the curve $y = e^x + 1$ at $(0, 1)$. At this point the acute angle between the line and the curve is θ where $\tan \theta$ is equal to:
A $\frac{1}{3}$ **B** 3 **C** 0 **D** -1 **E** 1
- 32 The area of the shaded region of the graph is given by:

- A** $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin x \, dx$
B $\int_{\pi}^{\frac{3\pi}{2}} \sin x \, dx + \int_{\frac{\pi}{2}}^{\pi} \sin x \, dx$
C $\int_{\frac{3\pi}{2}}^{\pi} \sin x \, dx + \int_{\frac{\pi}{2}}^{\pi} \sin x \, dx$
D $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin x \, dx + \int_{\pi}^{\frac{\pi}{2}} \sin x \, dx$
E $\pi \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin^2 x \, dx$



- 33 The area of the shaded region of the graph is given by:

- A** $\int_0^2 x + 1 \, dx - \int_2^0 x + 1 \, dx$
B $\int_{-2}^2 x + 1 \, dx$
C $\int_0^2 x + 1 \, dx + \int_{-2}^0 x + 1 \, dx$
D $\int_{-1}^2 x + 1 \, dx - \int_{-2}^{-1} x + 1 \, dx$
E $\int_{-1}^2 x + 1 \, dx + \int_{-2}^{-1} x + 1 \, dx$



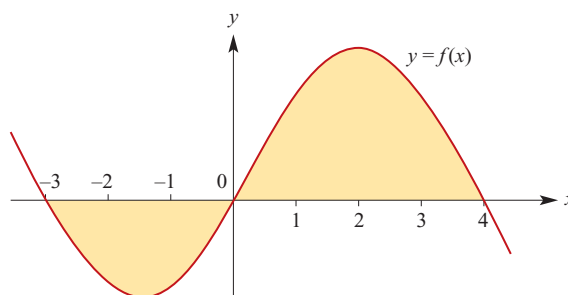
- 34 If $\frac{dy}{dx} = \frac{1}{x^2}$ and $y = 2$ when $x = 1$, then:
A $y = \frac{-1}{x}$ **B** $y = \frac{-1}{x} + 3$ **C** $y = \frac{-2}{x^3}$ **D** $y = \frac{2}{x^3}$ **E** $y = \frac{1}{x} + 1$

- 35 If $\int_0^{36} \frac{dx}{2x+9} = \log_e k$, then k is:

A 3 **B** $\frac{9}{2}$ **C** $6\sqrt{2}$ **D** 9 **E** 81

- 36 The area of the shaded region of the graph below is given by:

- A** $\int_{-3}^f f(x) dx$
B $\int_{-3}^0 f(x) dx + \int_0^4 f(x) dx$
C $\int_{-3}^1 f(x) dx + \int_1^4 f(x) dx$
D $\int_4^0 f(x) dx + \int_{-3}^0 f(x) dx$
E none of these



- 37 $\int x^2 - \frac{1}{x^2} + \sin x dx$ is:

- A** $\frac{x^3}{3} + \frac{1}{x} + \cos x + c$ **B** $\frac{x^3}{3} - \frac{2}{x} - \cos x + c$
C $2x - \frac{3}{x^2} + \cos x + c$ **D** $\frac{x^3}{3} + \frac{1}{x} - \cos x + c$
E $2x + \frac{2}{x^3} + \cos x + c$

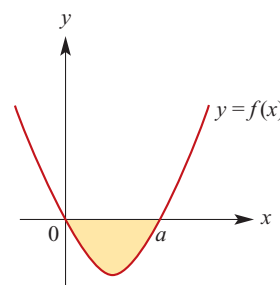
- 38 The area bounded by the curve $y = \frac{1}{3-x}$, the x -axis, the y -axis and the line $x = 2$ is:
A $\log_e 3$ **B** $\log_e \frac{1}{3}$ **C** $-\log_e (3-x)$ **D** $\log_e 2$ **E** $\log_e \frac{1}{2}$

- 39 If $\int_a^b \sin 2x dx = 0$, then:

- A** $b = \frac{3\pi}{4}, a = \frac{\pi}{4}$ **B** $b = \frac{\pi}{2}, a = 0$ **C** $b = \pi, a = \frac{\pi}{2}$
D $b = \frac{\pi}{6}, a = \frac{\pi}{3}$ **E** $b = \pi, a = \frac{\pi}{4}$

- 40 The area of the shaded region of the graph is given by:

- A** $\int_0^a -f(x) dx$ **B** $\int_0^a f(x) dx$
C $\int_0^a (x - f(x)) dx$ **D** $\int_0^a f(x) - x dx$
E none of these



- 41 An antiderivative of $x^2 - \frac{1}{x}$ is:

- A** $2x - \frac{2}{x^2}$ **B** $\frac{x^3}{3} - \log_e x$ **C** $x^3 + \frac{1}{x^2}$
D $x^3 - \log_e x$ **E** none of these

42 The function $f(x) = x^3 + 3x^2 - 9x + 7$ is increasing only when:

- A $x > 0$ B $-3 < x < 1$ C $x < -1$ or $x > 3$
 D $x < -3$ or $x > 1$ E $-3 < x < 1$

43 For $f(x) = \frac{\sin x}{x}$, $f'(x) =$

- A $\cos x$ B $\frac{x \cos x - \sin x}{x^2}$ C $\frac{x \cos x - \sin x}{x}$
 D $\frac{\sin x - x \cos x}{x^2}$ E $\frac{x \cos x - \sin x}{\sin^2 x}$

44 If $y = \log_e(\cos 2x)$ for $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ is equal to:

- A $\frac{2}{\cos 2x}$ B $-\frac{2 \sin 2x}{\cos 2x}$ C $\frac{1}{x} \cos 2x - 2 \sin 2x \log_e x$
 D $\frac{\sin 2x}{2 \cos 2x}$ E $\frac{1}{\cos 2x}$

45 If $f'(x) = \sin 2x$ and $f(0) = 3$ then:

- A $f(x) = -\frac{1}{2} \cos 2x + 3$ B $f(x) = \frac{1}{2} \cos 2x + 3$
 C $f(x) = -\frac{1}{2} \cos 2x + 3\frac{1}{2}$ D $f(x) = -\frac{1}{2} \cos 2x + 2\frac{1}{2}$
 E $f(x) = \frac{1}{2} \cos 2x + 2\frac{1}{2}$

46 The equation of the tangent to the curve with equation $y = 4e^{3x} - x$ at the point $(0, 4)$ is:

- A $y = 12x + 4$ B $y = -4x + 4$ C $y = 4$
 D $y = 11x + 4$ E $y = 4x + 4$

47 The function $f(x) = x^3 - x^2 - x + 2$ has a local minimum at the point:

- A $(-1, 0)$ B $(1, 1)$ C $(2, 0)$ D $(-1, 1)$ E $(1, 0)$

48 $\frac{d}{dx} \left(\frac{x-1}{\sqrt{x}} \right)$ equals:

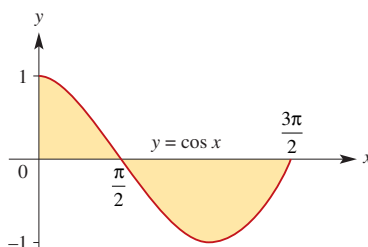
- A $2\sqrt{x}$ B $\frac{x+1}{x\sqrt{x}}$ C $\frac{3x-1}{2\sqrt{x}}$ D $\frac{x+1}{2x\sqrt{x}}$ E $\frac{3x-1}{2x\sqrt{x}}$

49 $\frac{d}{dx}(e^{\cos x}) =$

- A $e^{\cos x}$ B $e^{\cos x} \cdot \sin x$ C $-e^{\cos x} \cdot \sin x$
 D $e^{\sin x}$ E $e^{\sin x} \cdot \cos x$

50 The total area, in square units, of the shaded regions is:

- A 3
 B -1
 C 1
 D 2
 E -2



- 51 Given that $x + y = 1$, the maximum value of $P = x^2 + xy - y^2$ occurs for x equal to:
A 2 **B** -1 **C** $\frac{3}{2}$ **D** 1 **E** $\frac{2}{3}$
- 52 The gradient of the normal to the curve of $y = e^{-\cos x}$ at the point where $x = \frac{\pi}{3}$ is:
A $\frac{\sqrt{3}}{2e^{\frac{1}{2}}}$ **B** $\frac{-2e^{\frac{1}{2}}}{\sqrt{3}}$ **C** $\frac{1}{2e^{\frac{\sqrt{3}}{2}}}$ **D** $\frac{2e^{\frac{1}{2}}}{\sqrt{3}}$ **E** $-e^{-\frac{1}{2}}$
- 53 $\int_0^{\frac{\pi}{2}} (\cos x + \sin x) dx$ equals:
A -2 **B** -1 **C** 1 **D** $\frac{\pi}{2}$ **E** 2
- 54 Let f be differentiable for all values of x in $[0, 2]$. The graph with equation $y = f(x)$ has a maximum point at $(1, 3)$. The equation of the tangent at $(1, 3)$ is:
A $x + 3y = 0$ **B** $x = 1$ **C** $y = 3$ **D** $x - 3y = 0$ **E** $3x + y = 0$
- 55 The equation of the tangent to the curve with equation $y = 4 - x^2$ at the point $(1, 3)$ is:
A $y = 3x$ **B** $y = -x + 4$ **C** $y = -2x + 5$
D $y = x + 2$ **E** $y = 2x + 1$
- 56 The maximum value of $P = -x^2 + 6x + 4$ is:
A 3 **B** $-6 + 2\sqrt{5}$ **C** 4 **D** 13 **E** 24
- 57 The straight line $y = x + 3$ cuts the parabola $y = x^2 - 2x + 3$ at the point $(3, 0)$. The angle between the line and the parabola at this point is θ , where $\tan \theta$ is:
A 3 **B** $\frac{3}{4}$ **C** $\frac{3}{5}$ **D** -2 **E** 1
- 58 The graph of the function whose rule is $f(x) = x^3 - x^2 - 1$ has stationary points when x equals:
A $\frac{2}{3}$ only **B** 0 and $\frac{2}{3}$ **C** 0 and $-\frac{2}{3}$ **D** $-\frac{1}{3}$ and 1 **E** $\frac{1}{3}$ and -1
- 59 If $\frac{f(2+h) - f(2)}{h} = h^2 + 6h + 12$, then $f'(2)$ equals:
A 4 **B** 6 **C** 10 **D** 12 **E** 28
- 60 If $f(x) = \log_e 3x$, then $f'(1)$ is:
A $\frac{1}{3}$ **B** $\log_e 3$ **C** 1 **D** $3 \log_e 3$ **E** 3
- 61 If $f(x) = \log_e (3x)$, then $f'(1)$ is equal to:
A $\frac{1}{3}$ **B** 1 **C** 3 **D** $-\log_e 3$ **E** $3 \log_e 3$
- 62 If $y = x^2 e^x$, then the minimum value of y is:
A -2 **B** 0 **C** $4e^{-2}$ **D** $-4e^{-2}$ **E** e
- 63 If $z = \log_e (y)$, then δz is approximately equal to:
A $\log_e (y + \delta y)$ **B** $\log_e (\delta y)$ **C** $\frac{1}{\delta y}$ **D** $\frac{\delta y}{y}$ **E** $\frac{1}{y}$

64 If $f(x) = a \sin(3x)$ where a is constant and $f'(\pi) = 2$, then a is equal to:

- A -3 B $-\frac{3}{2}$ C $\frac{3}{2}$ D $\frac{2}{3}$ E $-\frac{2}{3}$

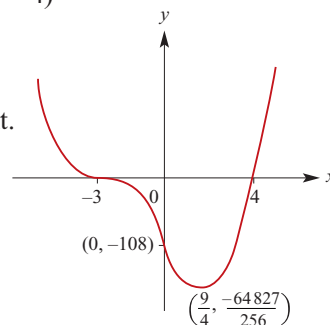
65 An antiderivative of $\frac{1}{(2x-5)^{\frac{5}{2}}}$ is equal to:

- A $\frac{-3}{(2x-5)^{\frac{5}{2}}}$ B $\frac{-1}{3(2x-5)^{\frac{3}{2}}}$ C $\frac{5}{(2x-5)^{\frac{5}{2}}}$
 D $\frac{7}{2(2x-5)^{\frac{7}{2}}}$ E $\frac{1}{3(2x-5)^{\frac{3}{2}}}$

66 The graph shown is of a function with rule $y = (x+3)^3(x-4)$

Which of the following is *not* true?

- A $\frac{dy}{dx} = 0$ when $x = \frac{9}{4}$ and $x = -3$ and at no other point.
 B There is only one turning point on the graph.
 C The x -axis is a tangent to the graph where $x = -3$.
 D There is only one stationary point on the graph.
 E $y \geq \frac{-64\,827}{256}$ for all values of x .



13.2 Extended-response questions

- The amount of salt (s grams) in 100 litres of salt solution at time t minutes is given by $s = 50 + 30e^{-\frac{1}{5}t}$
 - Find the amount of salt in the mixture after 10 minutes.
 - Sketch the graph of s against t for $t > 0$.
 - Find the rate of change of salt solution at time t (in terms of t).
 - Find the rate of change of salt solution at time t (in terms of s).
 - Find the concentration (grams/litre) of salt at time $t = 0$.
 - Find the value of t for which the salt solution first reaches a concentration of 0.51 grams/litre.
- A medium is kept at a constant temperature T of 20°C . A body is placed in this medium. The temperature at time t min of this body is given by:

$$T = 40e^{-0.36t} + 20, t \geq 0$$

- Find the initial temperature of the body.
 - Sketch the graph of T against t , $t \geq 0$.
 - Find the rate of change of temperature with respect to time (in terms of t).
 - Find the rate of change of temperature with respect to time (in terms of T).
- A certain food is susceptible to contamination from bacterial spores of two types, F and G . In order to kill the spores the food is subjected to a temperature of 120°C . The number

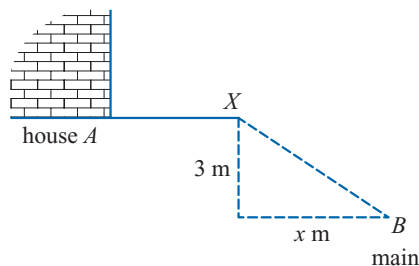
of live spores after t minutes can be approximated by $f(t) = 1000e^{-0.5t}$ for F -type spores and $g(t) = 1200e^{-0.7t}$ for G -type spores.

- a** Find the time required to kill 50% of the F -type spores.
 - b** Find the total number of live spores of both types when $t = 0$, and find the percentage of these that are still alive when $t = 5$.
 - c** Find the rate at which the total number of live spores is decreasing when $t = 5$.
 - d** Find the value for t for which the number of live F -type spores and the number of live G -type spores are equal.
 - e** On the same set of axes sketch the graphs of $f(t)$ and $g(t)$ for $t \geq 0$.
- 4** A body falls from rest in a medium and its velocity, V metres/second, after t seconds is given by $V = 100(1 - e^{-0.2t})$
 - a** Sketch the graph of V against t , $t > 0$.
 - b** Express the acceleration at any instant:
 - i** in terms of t
 - ii** in terms of V
 - c** Find the value of t for which the velocity of the body is 80 m/s.
 - 5** A body, which is at a higher temperature than its surroundings, cools according to Newton's Law of Cooling, $T = T_0e^{-kt}$, where T_0 is the original excess of temperature and T is the excess of temperature after time t minutes.
 - a** Prove the $\frac{dT}{dt}$ is proportional to T .
 - b** If the original temperature of the body is 100°C , the temperature of its surroundings is 30°C and the body cools to 70°C in 20 minutes, find the value of k correct to three decimal places.
 - c** At what rate is the temperature decreasing after 30 minutes?
 - 6** Suppose that the spread of a cold virus through a population is such that the proportion $p(t)$ of the population which has had the virus up to time t days after its introduction into the population is given by:

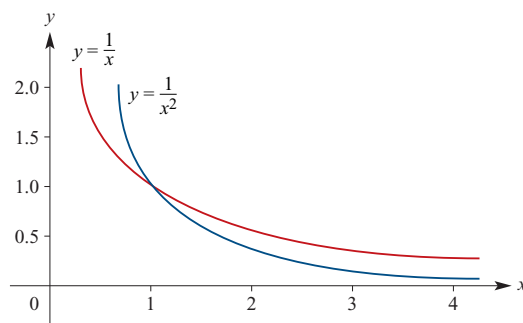
$$p(t) = 0.2 - 0.2e^{\frac{-t}{20}} + 0.1e^{\frac{-t}{10}} \text{ for } t \geq 0$$
 - a**
 - i** Find, correct to four decimal places, the proportion of the population which has had the virus up to 10 days after its introduction.
 - ii** Find the proportion of the population that eventually catches the virus.
 - b** The number of new cases on day t is proportional to $p'(t)$. Find how long after the introduction of the virus the number of new cases per day is at a maximum.
 - 7** A manufacturer determines that the total cost, \$ C per annum, of producing a product is given by $C = 0.05x^2 + 5x + 500$ where x is the number of units produced per annum. At what level of output will the average cost per unit be a minimum? (Use a continuous function to model this discrete situation.)
 - 8** A real estate firm owns the Shantytown Apartments which consist of 70 garden-type apartments. The owner can find a tenant for all the apartments at \$500 each per month. However, for every \$20 a month increase there will be two vacancies with no possibility of filling them. What price per apartment will maximise monthly revenue? (Use a continuous function to model this discrete situation.)

- 9 A curve with equation $y = (ax + b)^{-2}$ has a y -axis intercept $\left(0, \frac{1}{4}\right)$ and at this point the gradient is $-\frac{3}{4}$. Find the value(s) of a and the value(s) of b and sketch the graph.
- 10 The amount of liquid, $V \text{ m}^3$, in a large pool at time t days given by:
- $$V = \frac{5 \times 10^4}{(t+1)^2}, t \geq 0$$
- Find the initial volume of the pool.
 - Find the rate of change of volume with respect to time when $t = 1$.
 - Find the average rate of change for the interval $t = 1$ to $t = 4$.
 - When is the amount of water in the pool less than 1 cubic metre?
 - Sketch the graph of V against t for $t \geq 0$.
- 11 Each week a factory produced N thousand bottle tops and the cost of production is reckoned to be $\$1000C$ where $C = (N^3 + 16)^{\frac{1}{4}}$
- Sketch the graph of C against N . (Use a continuous function to model this discrete situation.)
 - Calculate $\frac{dC}{dN}$.
 - What does $\frac{dC}{dN}$ represent?
- 12 The cost of running a ship at a constant speed of $V \text{ km/h}$ is $160 + \frac{1}{100}V^3$ dollars per hour.
- Find the cost of a journey of 1000 km at a speed of 10 km/h.
 - Find the cost, C dollars, of a journey of 1000 km at a speed of $V \text{ km/h}$.
 - Sketch the graph of C against V .
 - Find the most economical speed for the journey, and the minimum cost.
 - If the ship were to have a maximum speed of 16 km/h, find what the minimum cost would be.
- 13 **a** A camper is situated on an island shore at a point A , 12 km from the nearest point B on the straight shore of the mainland. He wishes to reach a town C , 30 km along the shore from B in the least possible time. If he can row his boat at 5 km/h and walk at 8 km/h, how far along the shore from B towards C should he land?
- b** Repeat **a** if C is only 24 km from B .

- 14 To connect a house to a gas supply a pipe must be installed connecting a point A on the house to a point B on the main. B is 3 m below ground level and at a horizontal distance of 4 m from the building. If it costs $\$25$ per metre to lay pipe underground and $\$10$ on the surface, find the length of pipe which should be on the surface to minimise costs.

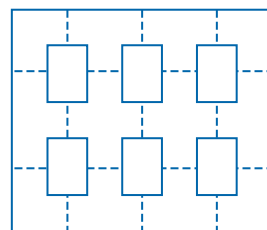


- 15** Let $g: R^+ \rightarrow R, g(x) = \frac{1}{x}$
and $h: R^+ \rightarrow R, h(x) = \frac{1}{x^2}$



- a** Find $\{x: g(x) > h(x)\}$.
 - b** Find $\{x: g'(x) > h'(x)\}$, i.e. the set of x for which the gradient of g is greater than the gradient of h .
 - c** On one set of axes sketch the graphs of $f: R^+ \rightarrow R, f(x) = \frac{1}{x^3}$ and $h: R^+ \rightarrow R, h(x) = \frac{1}{x^2}$. Find $\{x: h(x) > f(x)\}$ and $\{x: h'(x) > f'(x)\}$.
 - d** For $f_1: R^+ \rightarrow R, f_1(x) = \frac{1}{x^n}$ and $f_2: R^+ \rightarrow R, f_2(x) = \frac{1}{x^{n+1}}$, find $\{x: f_1(x) > f_2(x)\}$ and $\{x: f_1'(x) > f_2'(x)\}$.
- 16** **a** Find the points $P\left(x, \frac{1}{x}\right)$ on the curve $y = \frac{1}{x}$ for which the distance OP is a minimum (O is the point $(0, 0)$).
- b** Find the points $P\left(x, \frac{1}{x^2}\right)$ on the curve $y = \frac{1}{x^2}$ for which the distance OP is a minimum.
- c** Find the points $P\left(x, \frac{1}{x^n}\right)$ on the curve $y = \frac{1}{x^n}$, n a positive integer, for which the distance OP is minimum.

- 17** The figure represents an intended basic design for a workshop wall which is to have six equal windows spaced so that all the dotted lines are to be 2 metres long. The total area of window space is to be 36 square metres.



- a** Express the total area, A square metres, of brickwork as a function of the window height, x metres.
 - b** Sketch the graph of A against x .
 - c** Find the dimensions of each window which will give a minimum amount of brickwork.
 - d** If building regulations require that either dimension, height or width, of a window shall not be less than 1 metre, find the maximum amount of brickwork that could be used.
- 18** A company produces items at a cost price of \$2.00 per item. Market research indicates that the likely number of items sold per month will be $\frac{800}{p^2}$ where p dollars is the selling price of each item. Find the value of p for which the company would expect to maximise its total monthly profit, and the corresponding number of items sold.

- 19 a** Sketch the graph of the equation $y = x^2 - a^2$. Label the points A, B at which it cuts the x -axis. Write down the coordinates of A and B .
- b** Find the area of the region between the x -axis and the graph.
- c** Draw a rectangle $ABCD$ on your sketch, lying *below* the x -axis, whose area is equal to the area found in part **b** of this question. What is the length of the side BC ?
- d** If the vertex of the parabola is at point V , calculate the ratio:

$$\frac{\text{length of } BC}{\text{length of } OV}$$

- 20 a** Calculate $\int_{-3}^1 (1 - t^2) dt$ and illustrate the region of the cartesian plane for which this integral gives the signed area.
- b** Show that $\int_a^1 (1 - t^2) dt = 0$ implies $a^3 - 3a + 2 = 0$
- c** Find the values of a for which $\int_a^1 (1 - t^2) dt = 0$

- 21** The rate of flow of water into a tank is given by

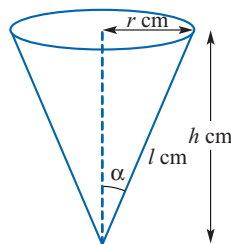
$$\frac{dV}{dt} = 10e^{-(t+1)}(5 - t), \quad 0 \leq t \leq 5$$

where V litres is the amount of water in the tank at time t minutes. Initially the tank is empty.



- a i** Find the initial rate of flow of water into the tank.
- ii** Find the value of t for which $\frac{dV}{dt} = 0$
- iii** Find the time, to the nearest second, when the rate is 1 litre/minute.
- iv** Find the first time, to the nearest second, when $\frac{dV}{dt} < 0.1$
- b** Find the amount of water in the tank when $t = 5$.
- c** Find the time at which there are 10 litres of water in the tank (to the nearest second).

- 22** A cone has a radius r cm, height h cm, vertical angle α and slant edge l cm. The area of the curved surface area, S cm², is given by the formula $S = \pi rl$ (Assume α is measured in radians.)



- a** Find S in terms of h and α .
- b i** If $h = 10$, find $\frac{dS}{d\alpha}$ in terms of α .
- ii** If an error of $(\delta\alpha)$ is made in the measurement of α , find the approximation for the error in S .
- iii** Find the percentage error in S .
- iv** Show that the percentage error is a minimum when $\sin \alpha = \frac{1}{\sqrt{3}}$
- v** Use a calculator to find the value of α for which the curved surface area is 200 cm². Give the answer in degrees.



Discrete random variables and their probability distributions

Objectives

- To review the basic concept of **probability**.
- To define **discrete random variables**.
- To define the **probability distribution** for a discrete random variable.
- To calculate and interpret **expectation (mean), median and mode** for a discrete random variable.
- To calculate and interpret **variance and standard deviation** for a discrete random variable.
- To illustrate the property that for many random variables **approximately 95% of the distribution is within two standard deviations of the mean**.

14.1 Review of probability



If a six-sided die is rolled the possible outcomes are the numbers 1, 2, 3, 4, 5, 6. These are generally listed as the elements of a set, and the set of all elements under consideration is ϵ , the **sample space**. Thus, in this case:

$$\epsilon = \{1, 2, 3, 4, 5, 6\}$$



An **event** is a subset of the sample space, usually denoted by a capital letter. If the event A is defined as ‘an even number when a fair six-sided die is rolled’, we write:

$$A = \{2, 4, 6\}$$



If A and B are two events the **union** of A and B , denoted $A \cup B$, is equivalent to either event A or event B or both occurring. Thus, if event A is ‘an even number when a fair six-sided die is rolled’ and event B is a ‘number greater than 2 when a fair six-sided die is rolled’, then

$$A = \{2, 4, 6\}, B = \{3, 4, 5, 6\}, \text{ and}$$

$$A \cup B = \{2, 3, 4, 5, 6\}$$

The **intersection** of A and B , denoted by $A \cap B$, is equivalent to both event A and event B occurring. Thus, using the events A and B already described:

$$A \cap B = \{4, 6\}$$

Probability is a numerical measure of the chance of a particular event occurring and may be determined experimentally or by symmetry.

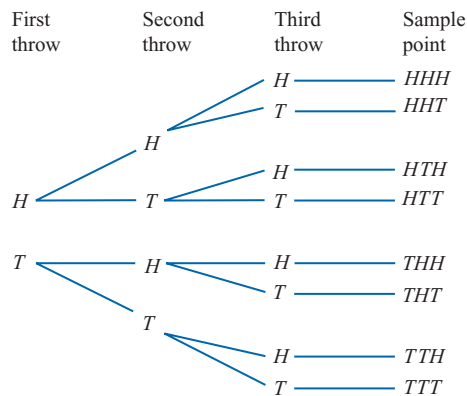
In some experiments it is helpful to list the elements of the sample space systematically by means of a tree diagram.

Example 1

Find the sample space when three coins are tossed and the outcomes noted.

Solution

To list the elements of the sample space, construct a tree diagram:



Each path along the branches of the tree identifies a sample point, giving the sample space as:

$$\varepsilon = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Example 2

Enrolment records show that the local school has 800 students: 450 female and 350 male. What is the probability that a randomly selected student will be female?

Solution

Since there are 450 female students in the total of 800, then:

$$\begin{aligned} \Pr(\text{Female}) &= \frac{450}{800} \\ &= \frac{9}{16} \end{aligned}$$

Whatever method is used to determine the probability of an event, the following rules will hold:

- $0 \leq \Pr(A) \leq 1$ for all events $A \subseteq \epsilon$
- $\Pr(\epsilon) = 1$
- $\Pr(\emptyset) = 0$ where $\emptyset$ represents the null set
- $\Pr(A') = 1 - \Pr(A)$, where A' is the complement of A
- $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$ (the addition rule)

When two events have no outcomes in common, i. e. they cannot occur together, they are called **mutually exclusive** events. In this case $\Pr(A \cap B) = 0$, and the addition rule becomes:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) \quad (\text{the addition rule when } A \text{ and } B \text{ are mutually exclusive})$$

Example 3

If one card is chosen at random from a well-shuffled deck of 52 cards, what is the probability that the card is:

- a** an ace? **b** not a heart? **c** either an ace or a heart?

Solution

- a** Let A be the event ‘the card drawn is an ace’. A standard deck of cards contains four aces so

$$\begin{aligned} \Pr(A) &= \frac{4}{52} \\ &= \frac{1}{13} \end{aligned}$$

- b** Let H be the event ‘the card drawn is a heart’. Since there are 13 cards in each suit

$$\begin{aligned} \Pr(H) &= \frac{13}{52} \\ &= \frac{1}{4} \end{aligned}$$

and thus

$$\begin{aligned} \Pr(H') &= 1 - \Pr(H) \\ &= 1 - \frac{1}{4} \\ &= \frac{3}{4} \end{aligned}$$

- c** Using the addition rule:

$$\begin{aligned} \Pr(A \cup H) &= \Pr(A) + \Pr(H) - \Pr(A \cap H) \quad (\text{the addition rule}) \\ \text{Now } \Pr(A \cap H) &= \frac{1}{52} \quad (\text{this event corresponds to drawing the Ace of Hearts}) \\ \text{so } \Pr(A \cup H) &= \frac{1}{13} + \frac{1}{4} - \frac{1}{52} \\ &= \frac{4}{13} \end{aligned}$$

Probabilities associated with compound events are sometimes able to be calculated more easily from a probability table, sometimes called a **Karnaugh map**.

Example 4

Simone visits the dentist every 6 months for a checkup. The probability that she will need her teeth cleaned is 0.35, the probability that she will need a filling is 0.1 and the probability that she will need both is 0.05.

- What is the probability that she will not have her teeth cleaned on a visit, but will have a filling?
- What is the probability that she will not need either of these treatments?

Solution

The information in the question may be entered into a table as shown (we will use C to represent ‘cleaning’ and F to represent ‘filling’).

	F	F'	
C	0.05 $(C \cap F)$	$(C \cap F')$	0.35 (C)
C'	$(C' \cap F)$	$(C' \cap F')$	(C')
	0.1 (F)	(F')	1 (ϵ)

All the empty cells in the table may now be filled in by subtraction. The completed table is:

	F	F'	
C	0.05 $(C \cap F)$	0.3 $(C \cap F')$	0.35 (C)
C'	0.05 $(C' \cap F)$	0.6 $(C' \cap F')$	0.65 (C')
	0.1 (F)	0.9 (F')	1 (ϵ)

Thus, the probability that she will not have her teeth cleaned on a visit but will have a filling is $\Pr(C' \cap F) = 0.05$, while the probability that she will not need either of these treatments is $\Pr(C' \cap F') = 0.6$.

Conditional probability

The probability of an event A occurring when it is known that some event B has occurred is called conditional probability and is written $\Pr(A|B)$. This is usually read as ‘the probability of A given B ’, and can be thought of as a means of adjusting probability in the light of new information.

Sometimes, the probability of an event occurring is not affected by the previous outcome of a similar random experiment. For example, the probability of a coin showing a head when it is tossed is independent of whether the previous coin toss resulted in a head or a tail. Thus,

$\Pr(\text{head}|\text{head on the previous toss})$

$$= \Pr(\text{head}|\text{tail on the previous toss})$$

$$= \Pr(\text{head})$$

For other situations, however, a previous result may alter the probability. For example, the probability of rain today given that it rained yesterday will generally be different from the probability that it will rain today given that it didn't rain yesterday.

Conditional probability is defined as follows.

$\Pr(A|B)$ describes the conditional probability of event A occurring given that event B has already occurred, and

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

or $\Pr(A \cap B) = \Pr(A|B) \Pr(B)$ (the multiplication rule)

The probabilities associated with multi-stage experiments can be calculated by constructing an appropriate tree diagram and by multiplying along the relevant branches (from the multiplication rule).

Example 5

Consider the information in Example 4. Given that Simone had her teeth cleaned on a particular visit to the dentist, what is the probability that she also needed a filling?

Solution

The probability that Simone needed a filling, given that she has her teeth cleaned, is:

$$\begin{aligned}\Pr(F|C) &= \frac{\Pr(F \cap C)}{\Pr(C)} \\ &= \frac{0.05}{0.35} \\ &= \frac{1}{7}\end{aligned}$$

Example 6

As part of an evaluation of the school canteen all students from Years 10, 11 and 12 at a Senior Secondary College were asked to rate the canteen as poor, good or excellent. The following results were obtained:

	Year			
Rating	10	11	12	Total
Poor	30	20	10	60
Good	80	65	35	180
Excellent	60	65	35	160
Total	170	150	80	400

What is the probability that a student chosen at random from the Senior Secondary College:

- a** is in Year 12?
- b** is in Year 12 and rates the canteen as excellent?
- c** is in Year 12, given that they have rated the canteen as excellent?
- d** rated the canteen as excellent, given that they are in Year 12?

Solution

Let T represent the event 'the student is in Year 12' and E the event 'the rating is excellent'.

- a** From the table we can see that there are 80 students in Year 12 and 400 students altogether. Thus:

$$\begin{aligned}\Pr(T) &= \frac{80}{400} \\ &= \frac{1}{5}\end{aligned}$$

- b** From the table, there are 35 students in Year 12 who also rate the canteen as excellent. Thus:

$$\begin{aligned}\Pr(T \cap E) &= \frac{35}{400} \\ &= \frac{7}{80}\end{aligned}$$

- c** Once again, this probability can be determined directly from the table. A total of 160 students rated the canteen as excellent, and of these 35 were in Year 12.

Thus

$$\begin{aligned}\Pr(T|E) &= \frac{35}{160} \\ &= \frac{7}{32}\end{aligned}$$

- d** Using the table, there are 80 students in Year 12, 35 of whom rated the canteen as excellent. Thus:

$$\begin{aligned}\Pr(E|T) &= \frac{35}{80} \\ &= \frac{7}{16}\end{aligned}$$

Note that the answers to parts **c** and **d** could also have been determined using the rule for conditional probability, but here it is easier to determine the probability directly from the table.

Independent events

Two events are independent if

$$\Pr(A|B) = \Pr(A)$$

so whether or not B has occurred has no effect on the probability of A occurring.

This means that:

$$\Pr(A \cap B) = \Pr(A) \Pr(B) \text{ if } A \text{ and } B \text{ are independent.}$$

The converse is also true. That is:

$$\text{if } \Pr(A \cap B) = \Pr(A)\Pr(B) \text{ then } A \text{ and } B \text{ are independent.}$$

Example 7

The probability that Monica remembers to do her homework is 0.7, while the probability that Patrick remembers to do his homework is 0.4. If these events are independent, then what is the probability that:

- a** both will do their homework?
- b** Monica will do her homework but Patrick forgets?

Solution

Let M be the event ‘Monica does her homework’ and P the event ‘Patrick does his homework’. Since these events are independent:

$$\begin{array}{ll} \text{a} \quad \Pr(M \cap P) = \Pr(M) \Pr(P) & \text{b} \quad \Pr(M \cap P') = \Pr(M) \Pr(P') \\ & = 0.7 \times 0.4 \\ & = 0.28 \end{array} \qquad \begin{array}{ll} & = 0.7 \times 0.6 \\ & = 0.42 \end{array}$$

The probabilities associated with multi-stage experiments can be calculated by constructing an appropriate tree-diagram, and then multiplying along the relevant branches (from the multiplication rule).

Example 8

Adrienne, Regan and Michael are doing the dishes. Since Adrienne is the oldest, she washes the dishes 40% of the time. Regan and Michael each wash 30% of the time. When Adrienne washes the probability of at least one dish being broken is 0.01, when Regan washes the probability is 0.02, and when Michael washes the probability is 0.03. Their parents don’t know who is washing the dishes one particular night.

- What is the probability that at least one dish will be broken?
- Given that at least one dish is broken, what is the probability that the person washing was Michael?

Solution

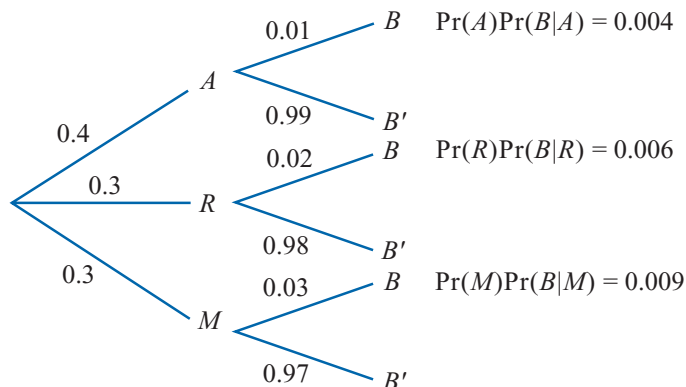
Let A be the event ‘Adrienne washes the dishes’, R the event ‘Regan washes the dishes’ and M be the event ‘Michael washes the dishes’. From the question we can thus write:

$$\Pr(A) = 0.4, \Pr(R) = 0.3, \Pr(M) = 0.3$$

Let B be the event that at least one dish is broken. Then

$$\Pr(B|A) = 0.01, \Pr(B|R) = 0.02, \Pr(B|M) = 0.03$$

This information can be summarised in a tree diagram as shown:



- a** Now, the probability of at least one dish being broken is given by:

$$\begin{aligned}
 \Pr(B) &= \Pr(B \cap A) + \Pr(B \cap R) + \Pr(B \cap M) \\
 &= \Pr(A) \Pr(B|A) + \Pr(R) \Pr(B|R) + \Pr(M) \Pr(B|M) \\
 &= 0.004 + 0.006 + 0.009 \\
 &= 0.019
 \end{aligned}$$

- b** The required probability is $\Pr(M|B)$, and

$$\begin{aligned}
 \Pr(M|B) &= \frac{\Pr(M \cap B)}{\Pr(B)} \\
 &= \frac{0.009}{0.019} \\
 &= \frac{9}{19}
 \end{aligned}$$

Exercise 14A

- A blank six-sided die is marked with a 1 on two sides, a 2 on one side, and a 3 on the remaining three sides. Find the probability that when the die is rolled:
 - a 3 shows
 - a 2 or a 3 shows
- An insurance company analysed the records of 500 drivers to determine the relationship between age and accidents in the last year. The results were:

Age	Accidents in the last year				
	0	1	2	3	Over 3
Under 20	19	35	25	17	10
20–29	30	45	33	39	17
30–39	40	33	15	6	2
40–49	18	15	10	3	1
Over 49	21	25	17	13	11



What is the probability that a driver chosen from this group at random:

- is under 20 and has had three accidents in the last year?
 - is from 40–49 years and has had no accidents in the last year?
 - is from 20–29 years old?
 - has had more than three accidents?
- At a particular university the probability that an Arts student studies a language is 0.3, literature is 0.6, and both is 0.25. What is the probability that:
 - a student studies either a language or literature?
 - a student studies neither a language nor literature?

- 4 A computer manufacturer notes that 5% of the computers are returned owing to faulty disk drives, 2% are returned owing to faulty keyboards, and 0.3% are returned because both disk drives and keyboards are faulty. Find the probability that the next computer returned will have:
- a either a faulty disk drive or faulty keyboard
 - b a faulty disk drive and a working keyboard
- 5 A new drug has been released and produces some minor side effects. 8% of users suffer only a loss of sleep and 12% of users suffer only bouts of nausea. 75% of users will have no side effects at all. What percentage of users will suffer from both loss of sleep and nausea?
- 6 In a particular town the probability that an adult owns a car is 0.7, while the probability that an adult owns a car and is employed is 0.6. If a randomly selected adult is found to own a car, what is the probability that he or she is also employed?
- 7 A group of 1000 people, eligible to vote, were asked their age and preference in an upcoming election, with the following results:

	18–25 years	26–40 years	Over 40 years	Total
Candidate <i>A</i>	200	100	85	385
Candidate <i>B</i>	250	230	50	530
No preference	50	20	15	85
Total	500	350	150	1000

What is the probability that a person chosen from this group at random:

- a is 18–25 years old?
 - b prefers candidate *A*?
 - c is 18–25 years old, given that they prefer candidate *A*?
 - d prefers candidates *A*, given that they are in the 18–25 years age group?
- 8 Nathan knows that his probability of kicking more than four goals on a wet day is 0.3, while on a dry day it is 0.6. The probability that it will be wet on the day of the next game is 0.7. Calculate the probability that Nathan will kick more than four goals in the next game.
- 9 Find the probability that, in three tosses of a fair coin, there are three heads, given that there is at least one head.
- 10 The chance that a harvest is poorer than average is 0.5, but if it is known that a certain disease *D* is present, this probability increases to 0.8. The disease *D* is present in 30% of the harvests. Find the probability that, when the harvest is observed to be poorer than average, the disease *D* is present.
- 11 The test used to determine if a person suffers from a particular disease is not perfect. The probability of a person with the disease returning a positive result is 0.95, while the probability of a person without the disease returning a positive result is 0.02. The probability that a randomly selected person has the disease is 0.03. What is the probability that a randomly selected person will return a positive result?

- 12** Cycling to school each morning, a girl goes through three sets of traffic lights. The probability of stopping at the first set is 0.6. If she stops at any one set, the probability that she has to stop at the next is 0.9. If she doesn't have to stop at any one set, the probability that she doesn't have to stop at the next is 0.7. Use a tree diagram to find the probability that:
- a** she has to stop at all three lights **b** she stops only at the second set of lights
c she stops at exactly one set of lights
- 13** There are four red socks and two blue socks in a drawer. Two socks are removed at random. What is the probability of obtaining:
- a** two red socks? **b** two blue socks? **c** one of each colour?
- 14** A car salesperson was interested in the relationship between the size of the car a customer purchased and their marital status. From the sales records, the following table was constructed:

Size of car	Marital status		
	Married	Single	Total
Large	60	20	80
Medium	100	60	160
Small	90	70	160
Total	250	150	400

What is the probability that a person chosen at random from this group:

- a** drives a small car? **b** is single and drives a small car?
c is single, given that they drive a small car?
d drives a small car, given that they are single?
- 15** Jenny has two boxes of chocolates. Box *A* contains three white chocolates and four dark chocolates. Box *B* contains two white chocolates and five dark chocolates. Jenny first chooses a box at random and then selects a chocolate at random from it. Find the probability that:
- a** Jenny selects a white chocolate
b given that Jenny selects a white chocolate, it was chosen from Box *A*.
- 16** For a particular petrol station 30% of customers buy 'super' and 60% buy unleaded and 10% diesel. When a customer buys super there is a 25% chance they will fill the tank. Customers buying unleaded have a 20% chance they will fill the tank. Of those buying diesel, 70% fill their tank.
- a** What is the probability that when a car leaves the petrol station it will not have a full tank?
b Given that a car leaving the petrol station has a full tank, what is the probability that the tank contains unleaded petrol?
- 17** A bag contains three red, four white and five black balls. If three balls are taken without replacement, what is the probability that they are all the same colour?

14.2 Discrete random variables

Suppose three balls are drawn at random from a jar containing four white and six black balls, with replacement (i. e. each selected ball is replaced before the next draw). The sample space for this random experiment is as follows:

$$\varepsilon = \{WWW, WWB, WBW, BWW, WBB, BWB, BBW, BBB\}$$

Suppose the variable of interest is the number of white balls in the sample. This corresponds to a simpler sample space whose outcomes are numbers. If X represents the number of white balls in the sample then the possible values of X are 0, 1, 2, and 3. X is called a **random variable**. A random variable is a function that assigns a number to each outcome in the sample space ε . A random variable X is said to be **discrete** if it can assume only a countable number of values. The number of white balls in the sample of size three is a discrete random variable which may take one of the values 0, 1, 2, 3. Other examples of discrete random variables are the number of children in a family, and the size of shoes that people may wear. (Note that discrete random variables do not have to take only whole number values.)

Consider again the sample space for the random experiment described above. Each outcome in the sample space corresponds to a value of X :

Experiment outcome	Value of X
WWW	$X = 3$
WWB	$X = 2$
BWW	$X = 2$
BWB	$X = 1$
WBW	$X = 2$
WBB	$X = 1$
BBW	$X = 1$
BBB	$X = 0$

Associated with each event is a probability. Since the individual draws of the ball from the jar are independent events, we can determine the probabilities by multiplying and adding appropriate terms.

Example 9

What is the probability that, if a ball is drawn three times from a jar containing four white and six black balls, with replacement, a white ball will be drawn exactly once (i. e. the situations where $X = 1$ in the table)?

Solution

$X = 1$ corresponds to the outcomes WBB , BWB and BBW . Since there are 10 balls in total, $\Pr(W) = \frac{4}{10} = 0.4$ and $\Pr(B) = \frac{6}{10} = 0.6$.

$$\begin{aligned}\text{Thus } \Pr(X = 1) &= \Pr(WBB) + \Pr(BWB) + \Pr(BBW) \\ &= (0.4 \times 0.6 \times 0.6) + (0.6 \times 0.4 \times 0.6) + (0.6 \times 0.6 \times 0.4) \\ &= 0.432\end{aligned}$$

Exercise 14B

- 1 Which of the following random variables are discrete?
 - a The number of people in your family
 - b Waist measurement c Shirt size
 - d The number of tosses of a die required to throw a six
- 2 Which of the following random variables are discrete?
 - a Your age b Your height to the nearest centimetre
 - c The time you will wait to be served at the bank
 - d The number of people in the queue at the bank
- 3 A fair coin is tossed three times and the number of heads noted.
 - a List the sample space.
 - b List the possible values of the random variable X , the number of heads, together with the corresponding sample outcomes.
 - c Find $\Pr(X \geq 2)$.
- 4 Two dice are rolled and the numbers noted.
 - a List the sample space.
 - b A random variable Y is defined as the total of the numbers showing on the two dice. List the possible values of Y , together with the corresponding sample outcomes.
 - c Find $\Pr(Y < 5)$. d Find $\Pr(Y = 3 | Y < 5)$.
- 5 A die is weighted as follows:

$$\Pr(2) = \Pr(3) = \Pr(4) = \Pr(5) = 0.2 \qquad \Pr(1) = \Pr(6) = 0.1$$

The die is rolled twice, and the smaller of the numbers showing is noted. Let Y represent this value.

- a List the sample space.
 - b List the possible values of Y .
 - c Find $\Pr(Y = 1)$.
- 6 Suppose three balls are selected at random, with replacement, from a jar containing four white and six black balls. If X is the number of white balls in the sample. Find:
 - a $\Pr(X = 2)$ b $\Pr(X = 3)$ c $\Pr(X \geq 2)$ d $\Pr(X = 3 | X \geq 2)$
- 7 A fair die is rolled twice and the sum of the numbers showing on the two faces recorded, and the following events defined:

A = 'a four on the first roll'

B = 'a four on the second roll'

C = 'the sum of the two numbers is at least eight'

D = 'the sum of the two numbers is at least 10'

- a** List the sample space obtained. **b** Find $\Pr(A)$, $\Pr(B)$, $\Pr(C)$, and $\Pr(D)$.
c Find $\Pr(A|B)$, $\Pr(A|C)$ $\Pr(A|D)$.
d Which of the pairs of events A and B , A and C , or A and D are independent?

14.3 Discrete probability distributions

A probability distribution consists of all values that a random variable can take, together with the probability of each of these values. For example, if a fair die is rolled then the **probability distribution** is:

x	1	2	3	4	5	6
$\Pr(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

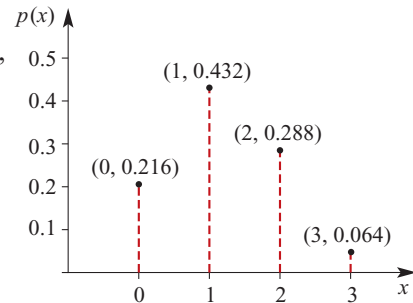
The **probability distribution** of a discrete random variable X is described by a function $p(x) = \Pr(X = x)$. This function is called a **discrete probability function**.

Consider the black and white balls given in Example 9. The probability distribution is:

x	0	1	2	3
$p(x)$	0.216	0.432	0.288	0.064

This may be shown graphically:

Note that the probabilities entered in the table add to 1, which must occur if all values of the random variable have been listed. $\sum p(x)$ is a shorthand way of writing the sum of all values of $p(x)$, and $\sum_{x=a}^{x=b} p(x)$ is the sum of the values of $p(x)$ for x between a and b inclusive. This notation is discussed in Appendix A.



In general:

For *any* discrete probability function, the following must be true:

- 1** The value of $p(x)$ is greater than or equal to zero and less than or equal to one, for every value of X . That is:

$$0 \leq p(x) \leq 1 \text{ for all } x$$

- 2** The sum of all values of $p(x)$ must be exactly one. That is:

$$\sum_x p(x) = 1$$

- 3** To determine the probability that X takes a value in the interval from a to b (including the values a and b) the values of $p(x)$ from $x = a$ to $x = b$ are summed. That is:

$$\Pr(a \leq X \leq b) = \sum_{x=a}^{x=b} p(x)$$

Example 10

Let X be the number of tails showing when a fair coin is tossed three times.

- Find the probability distribution of X and show that all the probabilities sum to 1.
- Find the probability that one or more heads shows.
- Find the probability that more than one head shows.

Solution

- The sample space resulting from tossing a fair coin three times is:

$$\varepsilon = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Now

$$\begin{aligned} p(0) &= \Pr(X = 0) = \Pr(\{TTT\}) &&= \frac{1}{8} \\ p(1) &= \Pr(X = 1) = \Pr(\{HTT, THT, TTH\}) &&= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8} \\ p(2) &= \Pr(X = 2) = \Pr(\{HHT, HTH, THH\}) &&= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8} \\ p(3) &= \Pr(X = 3) = \Pr(\{HHH\}) &&= \frac{1}{8} \end{aligned}$$

Thus the probability distribution of X is:

x	0	1	2	3
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

- The probability that one or more heads shows:

$$\Pr(X \geq 1) = p(1) + p(2) + p(3) = \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$$

- The probability that more than one head shows:

$$\Pr(X > 1) = \Pr(X \geq 2) = p(2) + p(3) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

Exercise 14C

- Which of the following is not a probability distribution?

a

x	1	3	5	7
$p(x)$	0.1	0.3	0.5	0.7

b

x	-1	0	1	2
$p(x)$	0.25	0.25	0.25	0.25

c

x	0.25	0.5	0.75	1.0
$p(x)$	-0.5	-0.25	0.25	0.5

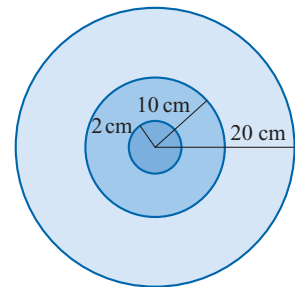
d

x	10	20	30	40
$p(x)$	10%	20%	30%	40%

- 2 A spinner is numbered from 1 to 5, and each of the five numbers is equally likely to come up. Find:
 - a the probability distribution of X , the number showing on the spinner
 - b $\Pr(X \geq 3)$, the probability that the number showing on the spinner is three or more
 - c $\Pr(X \leq 3 | X \geq 3)$
- 3 A coin is known to be biased, so that the probability of obtaining a head on any toss is 0.4. Find the probability distribution of X , the number of heads observed when the coin is tossed twice.
- 4 Three balls are selected from a jar containing four black and six red balls. Find the probability distribution of the number of black balls in the sample:
 - a if the ball chosen is replaced after each selection
 - b if the ball chosen is not replaced after each selection
- 5 Two dice are rolled and the outcomes noted.
 - a List the sample space for this experiment.
 - b Find the probability distribution of X , the sum of the numbers showing on the two dice.
 - c Draw a graph of the probability distribution of X .
 - d Find $\Pr(X \geq 9)$, the probability that the sum of the two numbers showing is nine or more.
 - e Find $\Pr(X \leq 10 | X \geq 9)$.
- 6 Two dice are rolled and the outcomes noted.
 - a List the sample space for this experiment.
 - b Find the probability distribution of Y , the remainder when the larger number showing is divided by the smaller number. (Note that when the two numbers are the same $Y = 0$.)
 - c Draw a graph of the probability distribution of Y .
- 7 Suppose two socks are drawn without replacement from a drawer containing four red and six black socks. Let X represent the number of red socks obtained.
 - a Find the probability distribution for X .
 - b From the probability distribution determine the probability a pair of socks is obtained.

- 8 A dart board consists of three circular sections, with radii of 2 cm, 10 cm and 20 cm respectively as shown in the diagram.

When a dart lands in the centre circle the score is 100 points, in the middle circular section the score is 20 points and in the outer circular section the score is 10 points. Assume that all darts thrown hit the board, each dart is equally likely to land at any point on the dart board, and none lands on the lines.



- a Suppose X is the number of points scored on one throw. Find the probability distribution for X .
- b Let Y represent the total score when two darts are thrown. Find the probability distribution for Y .

- 9 Erin and Nick are going to play a tennis match. Suppose that they each have equal chance of winning any set (0.5) and that they plan to play until one player has won three sets. Let X be the number of sets played until the match is complete.
- Find $\Pr(X = 3)$.
 - List the sample outcomes that correspond to $X = 4$, and use this to find $\Pr(X = 4)$.
 - Hence or otherwise, find $\Pr(X = 5)$.

14.4 Measures of centre and variability for a discrete random variable

From your studies of statistics you may already be familiar with the measures of centre (mean, median and mode) and measures of spread (range, interquartile range, variance and standard deviation). When these are calculated from a set of data they are termed ‘sample statistics’. It is also possible to use the probability distribution to determine the theoretically ‘true’ values of the mean, median and mode. When they are calculated from the probability distribution they are called ‘population parameters’. Determining of the values of these parameters is the topic for this section.

Measure of centre

Expected value (mean value)

When the true mean value of a random variable is determined from the probability distribution it is generally called the **expected value** of the random variable. Expected value has a wide variety of uses and applications. The concept of expected value first arose in gambling problems, where gamblers wished to know how much they could expect to win or lose in the long run, in order to decide whether or not a particular game was a good investment.

Example 11

A person may buy a lucky ticket for \$1.00. They have a 20% chance of winning \$2.00, and a 5% chance of winning \$11.00, otherwise they lose. Is this a good game to play?

Solution

Let W be the amount the person will win. As it costs \$1.00 to play the person can lose \$1.00 ($W = -1$), win \$1.00 ($W = 1$) or win \$10.00 ($W = 10$). Thus, the amount the person may win, W , has a probability distribution given by:

W	-1	1	10
$\Pr(W = w)$	0.75	0.20	0.05

Suppose you played the game 1000 times. You would expect to lose \$1 on about 750 occasions, to win \$1 about 200 times and to win \$10 about 50 times. Thus, you would win about

$$\$ \frac{-1 \times 750 + 1 \times 200 + 10 \times 50}{1000} = -\$0.05 \text{ per game}$$

Thus your ‘expectation’ is to lose 5 cents per game and we write this as:

$$E(W) = -0.05$$

This gives an indication of the worth of the game; in the long run, you would expect to lose about 5 cents a game. This then is what we call the **expected value** of X (or the **mean** of X). It is not the amount we expect to win on any one game (you cannot lose 5 cents in one game!). It is the amount that we expect to win on average per game in the long run.

Example 11 demonstrates how the expected value of a random variable X is determined.

The **expected value** of a random variable X is determined by summing the product of each value of X and the probability that X takes that value.

That is:

$$\begin{aligned} E(X) &= \sum_x x \cdot \Pr(X = x) \\ &= \sum_x x \cdot p(x) \end{aligned}$$

$E(X)$ may be considered as the long-run average value of X . It is generally denoted by the greek letter μ (mu), and is also called the **mean** of X .

Example 12

A coin is biased in favour of heads, so that the probability of obtaining a head on any single toss is 0.6. The coin is tossed three times and the results noted. If X is the number of heads obtained on the three tosses, find the expected value of X , $E(X)$.

Solution

The following probability distribution can be calculated by listing the sample space and determining the value of X and the associated probability for each outcome.

x	0	1	2	3
$\Pr(X = x)$	0.064	0.288	0.432	0.216

$$\begin{aligned} \mu &= E(X) \\ &= \sum_x x \cdot p(x) \\ &= (0 \times 0.064) + (1 \times 0.288) + (2 \times 0.432) + (3 \times 0.216) \\ &= 0.288 + 0.864 + 0.648 \\ &= 1.8 \end{aligned}$$

This means that if the three coins were tossed many times, an average of 1.8 heads per toss would be observed.

Sometimes we wish to find the expected value of a function of X . This is determined by calculating the value of the function for each value of X , and then summing the products of these values and the associated probabilities. Thus, the expected (or mean) value of the function $g(x)$ is defined as:

$$E[g(X)] = \sum_x g(x) \cdot p(x)$$

Example 13

For the random variable X defined in Example 12, find:

a $E(3X + 1)$ **b** $E(X^2)$

Solution

$$\begin{aligned} \text{a } E(3X + 1) &= \sum_x (3x + 1) \cdot p(x) \\ &= (1 \times 0.064) + (4 \times 0.288) + (7 \times 0.432) + (10 \times 0.216) \\ &= 0.064 + 1.152 + 3.024 + 2.16 \\ &= 6.4 \end{aligned}$$

$$\begin{aligned} \text{b } E(X^2) &= \sum_x x^2 \cdot p(x) \\ &= (0^2 \times 0.064) + (1^2 \times 0.288) + (2^2 \times 0.432) + (3^2 \times 0.216) \\ &= 0.288 + 1.728 + 1.944 \\ &= 3.96 \end{aligned}$$

Let us compare the values found in Example 13 with the value of $E(X)$ found in Example 12.

In part **a** we found that $E(3X + 1) = 6.4$, and $3E(X) + 1 = 3 \times 1.8 + 1 = 6.4$. That is:

$$E(3X + 1) = 3E(X) + 1$$

In part **b** we found that $E(X^2) = 3.96$, and $(E(X))^2 = 1.8^2 = 3.24$. That is:

$$E(X^2) \neq (E(X))^2$$

These two examples illustrate an important point concerning the function of expected values.

In general, the expected value of a function of X is not equal to that function of the expected value of X . That is:

$$E[g(X)] \neq g[E(X)]$$

An exception to this rule is when the function is a linear function of X . Then:

$$E(aX + b) = aE(X) + b \quad (a, b \text{ constant})$$

This is illustrated in Example 14.

Example 14

For a \$5 monthly fee a TV repair company guarantees customers a complete service. The company estimates the probability that a customer will require one service call in a month is 0.05, the probability of two calls as 0.01 and the probability of three calls or more as 0.00. Each call costs the repair company \$40. What is the TV repair company's expected monthly gain from such a contract?

Solution

We may summarise the given information in the following table.

Calls	0	1	2	≥ 3
Gain (G)	5	-35	-75	?
$\Pr(G = g)$	0.94	0.05	0.01	0.00

$$\begin{aligned}
 E(G) &= \sum_{g=0}^2 g \cdot \Pr(G = g) \\
 &= 5 \times 0.94 - 35 \times 0.05 - 75 \times 0.01 \\
 &= 2.20
 \end{aligned}$$

Thus, the company can expect to gain \$2.20 per month on each contract sold. An alternative method of solution uses the property of expectation previously mentioned.

Let X = the number of calls received
 and gain, $G = 5 - 40X$
 Thus $E(G) = 5 - 40 \times E(X)$
 $E(X) = 1 \times 0.05 + 2 \times 0.01$
 $= 0.07$
 Thus $E(G) = 5 - 40 \times 0.07$
 $= 2.20$ as previously determined

Another very useful property of expectation is that the sum of the expected values of two random variables is equal to the expected value of their sum. That is, if X and Y are any two random variables, then:

$$E(X + Y) = E(X) + E(Y)$$

Median

The median value of a random variable is the middle value of the distribution, the value such that 50% of the distribution is greater than it, and 50% is less than it.

To find the median of the distribution, add the probabilities from the smallest value of X upwards until the sum reaches or exceeds 0.5.

- If the sum exceeds 0.5, then the value of X that is reached is the median value of X , as shown in Example 15.
- If the sum equals 0.5 exactly, then the median is taken as the value halfway between this value and the next, as shown in Example 16.

Example 15

Consider the following probability distribution. Find the median value of X .

x	0	1	2	3
$\Pr(X = x)$	0.064	0.288	0.432	0.216

Solution

$$\Pr(X \leq 0) = 0.064 \leq 0.5$$

$$\begin{aligned}\Pr(X \leq 1) &= 0.064 + 0.288 \\ &= 0.352 \leq 0.5\end{aligned}$$

$$\begin{aligned}\Pr(X \leq 2) &= 0.064 + 0.288 + 0.432 \\ &= 0.784 \geq 0.5\end{aligned}$$

Thus the median value of X for this probability distribution is 2.

Example 16

Consider the following probability distribution. Find the median value of X .

x	1	3	6	8	11
$\Pr(X = x)$	0.1	0.2	0.2	0.2	0.3

Solution

From the table it can be seen that:

$$\Pr(X = 1) + \Pr(X = 3) + \Pr(X = 6) = 0.5$$

Similarly:

$$\Pr(X = 8) + \Pr(X = 11) = 0.5$$

Since $\Pr(X \leq 6) = 0.5$ and $\Pr(X \geq 8) = 0.5$ the median value of the distribution is taken as midway between 6 and 8. That is, for this distribution the median is 7.

Mode

The modal value (or mode) is the most likely or most probable value of the random variable—that is, the value associated with the largest value of the probability.

The mode, M , of X is the value of the random variable such that $\Pr(X = M) \geq \Pr(X = x)$ for all other values of X

Example 17

Consider the following probability distribution. Find the mode of X .

x	0	1	2	3
$\Pr(X = x)$	0.064	0.288	0.432	0.216

Solution

From the table we can see that the highest probability is 0.432, which means that the mode is 2.

Measures of variability: variance and standard deviation

As well as knowing the long-run average value of the random variable (the mean), it is also useful to have a measure of how close to this mean are the values that the random variable may take — that is, a measure of how spread out the probability distribution is. The most useful measures of variability for a discrete random variable are the variance and standard deviation.

The **variance** of the random variable X is a measure of spread of the probability distribution about its mean or expected value μ . It is defined as:

$$\text{Var}(X) = E[(X - \mu)^2]$$

and may be considered as the long-run average value of the square of the distance from X to μ . The variance is usually denoted σ^2 , where σ is the lower-case Greek letter *sigma*.

$$\begin{aligned}\text{From the definition } \text{Var}(X) &= E[(X - \mu)^2] \\ &= \sum_x (x - \mu)^2 \cdot \Pr(X = x)\end{aligned}$$

Since the variance is determined by squaring the distance from X to μ , it is no longer in the units of measurement of the original random variable. A measure of spread in the appropriate unit is found by taking the square root of the variance.

The **standard deviation** of X is defined as:

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

The standard deviation is usually denoted σ .

Example 18

Suppose that a discrete random variable X has the probability distribution shown.

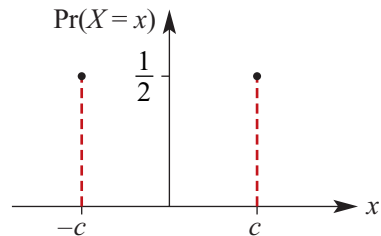
Find the standard deviation of X .

x	c	$-c$
$\Pr(X = x)$	0.5	0.5

Solution

$$\begin{aligned}\mu &= E(X) = \sum_x x \cdot \Pr(X = x) \\ &= (c \times 0.5) + (-c \times 0.5) \\ &= 0\end{aligned}$$

$$\begin{aligned}\sigma^2 &= \text{Var}(X) \\ &= E[(X - \mu)^2] \\ &= E(X^2) \text{ since } \mu = 0 \\ &= \sum_x x^2 \cdot \Pr(X = x) \\ &= c^2 \times 0.5 + (-c)^2 \times 0.5 \\ &= c^2\end{aligned}$$



which is the average of (the distance X is away from μ)²

Therefore, $\sigma = \text{sd}(X) = c$

Using the definition is not always the easiest way to calculate the variance.

An alternative (computational) formula for variance is:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

This is obtained as follows.

$$\text{We already know that } E(aX + b) = aE(X) + b \quad (1)$$

$$\text{and } E(X + Y) = E(X) + E(Y) \quad (2)$$

Hence,

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) + E(-2\mu X) + E(\mu^2) && \text{using (2)} \\ &= E(X^2) - 2\mu E(X) + \mu^2 && \text{using (1)} \\ &= E(X^2) - 2\mu^2 + \mu^2 && \text{since } \mu = E(X) \\ &= E(X^2) - \mu^2 \end{aligned}$$

A useful property of variance is:

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad (a, b \text{ constant})$$

Example 19

If X is a random variable such that $\text{Var}(X) = 9$, find:

a $\text{Var}(3X + 2)$ **b** $\text{Var}(-X)$

Solution

$$\begin{aligned} \text{a } \text{Var}(3X + 2) &= 3^2 \text{Var}(X) \\ &= 9 \times 9 \\ &= 81 \end{aligned}$$

$$\begin{aligned} \text{b } \text{Var}(-X) &= \text{Var}(-1 \times X) \\ &= (-1)^2 \text{Var}(X) \\ &= \text{Var}(X) \\ &= 9 \end{aligned}$$

Interpretation of standard deviation

It would make the standard deviation more meaningful if we could give it an interpretation that relates to the probability distribution. It can be shown that, for many random variables, about 95% of the distribution lies within two standard deviations either side of the mean.

In general, for many random variables X :

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

Example 20

The number of chocolate bars sold by a manufacturer in any month has the following

distribution:

x	100	150	200	250	300	400
$p(x)$	0.05	0.15	0.35	0.25	0.15	0.05

What is the probability that X takes a value in the interval $\mu - 2\sigma$ to $\mu + 2\sigma$?

Solution

First of all the values of μ and σ must be determined.

$$\begin{aligned}\mu &= E(X) = \sum_x x \cdot p(x) \\ &= 5 + 22.5 + 70 + 62.5 + 45 + 20 \\ &= 225\end{aligned}$$

Before determining the standard deviation σ , we need to find the variance σ^2 .

$$\begin{aligned}\text{Var}(X) &= E(X^2) - [E(X)]^2 \\ \text{and } E(X^2) &= \sum_x x^2 \cdot p(x) \\ &= 500 + 3375 + 14\,000 + 15\,625 + 13\,500 + 8000 \\ &= 55\,000\end{aligned}$$

$$\begin{aligned}\text{Thus } \text{Var}(X) &= 55\,000 - (225)^2 \\ &= 4375\end{aligned}$$

$$\begin{aligned}\text{and hence } \text{sd}(X) &= \sqrt{4375} \\ &= 66.14 \text{ (rounded to two decimal places)}\end{aligned}$$

$$\begin{aligned}\text{Thus } \Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) &= \Pr(92.72 \leq X \leq 357.28) \\ &= \Pr(100 \leq X \leq 300) \text{ since } X \text{ only takes the} \\ &\quad \text{values in the table} \\ &= 0.95 \text{ from the probability distribution of } X\end{aligned}$$

Exercise 14D

- 1 For each of the following probability distributions, find the mode, median and mean (expected value):

a

x	1	3	5	7
$p(x)$	0.1	0.3	0.3	0.3

b

x	-1	0	1	2
$p(x)$	0.25	0.25	0.25	0.25

c

x	0	1	2	3	4	5	6	7
$p(x)$	0.09	0.22	0.26	0.21	0.13	0.06	0.02	0.01

d

x	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$p(x)$	0.08	0.13	0.09	0.19	0.20	0.03	0.10	0.18

e

x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

f

x	-3	-2	-1	0	1	2	3
$p(x)$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

2 A random variable X has the probability distribution:

x	1	2	3	4	5
$\Pr(X = x)$	c	0.3	0.1	0.2	0.05

Find:

- | | | | |
|----------|---|----------|--------------------------|
| a | the constant c | b | the mode of X |
| c | the median value of X | d | $E(X)$, the mean of X |
| e | $\text{Var}(X)$ the variance of X , and hence the standard deviation of X | | |
| f | $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$ | | |

3 A random variable X has the probability distribution:

x	1	2	3	4	5
$\Pr(X = x)$	k	$2k$	$3k$	$4k$	$5k$

Find:

- a** the constant k **b** the mode of X
c the median value of X **d** $E(X)$, the expectation of X .
e $\text{Var}(X)$, the variance of X **f** $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

4 A business consultant evaluates a proposed venture as follows. A company stands to make a profit of \$10 000 with a probability 0.15, to make a profit of \$5000 with a probability 0.45, to break even with a probability of 0.25, and to lose \$5000 with a probability of 0.15. Find the expected profit.

5 A spinner is numbered from 0 to 5, and each of the six numbers has an equal chance of coming up. A player who bets \$1 on any numbers wins \$5.00 if that number comes up, otherwise the \$1 is lost. What is the player's expected profit on the game?

6 Suppose that the probability of having a female child is not as high as having a male child, and that the probability distribution for the number of male children in a three-child family has been determined from past records as:

No. of males, x	0	1	2	3
$p(x)$	0.12	0.36	0.38	0.14

What is the mean number of males in a three-child family?

7 Two dice are rolled and the outcomes noted. If X is the sum of the numbers showing on the two dice, find:

- a** $E(X)$, the expectation of X **b** $\text{Var}(X)$, the variance of X
c $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

8 The random variable X represents the number of chocolate chips in a certain brand of cookie, and is known to have the probability distribution:

x	2	3	4	5	6
$p(x)$	0.01	0.25	0.40	0.30	0.04

Calculate:

- a** $E(X)$ **b** $E(X^3)$ **c** $E(5X - 4)$ **d** $E\left(\frac{1}{X}\right)$

- 9 If $\text{Var}(X) = 16$, find:

a $\text{Var}(2X)$ **b** $\text{Var}(X + 2)$ **c** $\text{Var}(1 - X)$ **d** $\text{sd}(3X)$

- 10 The number of heads obtained when a fair coin is tossed six times has the probability distribution:

x	0	1	2	3	4	5	6
$p(x)$	0.0156	0.0937	0.2344	0.3126	0.2344	0.0937	0.0156

Find:

a $E(X)$, the mean of X **b** $\text{Var}(X)$, the variance of X **c** $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

- 11 A discrete random variable X takes values 0, 1, 2, 4, 8 with probabilities as shown in the table:

x	0	1	2	4	8
$\Pr(X = x)$	p	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$

a Find p . **b** Find $E(X)$. **c** Find $\text{Var}(X)$.

- 12 A biased die is such that the probability of any face landing uppermost is proportional to the number on that face. Thus, if X denotes the score obtained in one throw of this die, $\Pr(X = r) = kr$, $r = 1, 2, 3, 4, 5, 6$ where k is a constant.

a Find the value of k . **b** Find $E(X)$. **c** Find $\text{Var}(X)$.

- 13 An unbiased die is in the form of a regular tetrahedron and has its faces numbered 1, 2, 3, 4. When the die is thrown onto a horizontal table, the number on the face in contact with the table is noted. Two such dice are thrown and the score X is found by multiplying these numbers together.

a Give the probability distribution of X .

b Determine the values of:

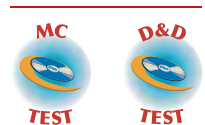
i $\Pr(X > 8)$ **ii** $E(X)$ **iii** $\text{Var}(X)$

- 14 A coin and a six-sided die are thrown simultaneously. The random variable X is defined as follows: if the coin shows a head, then X is the score on the die; if the coin shows a tail, then X is twice the score on the die.

a Find the expected value, μ , of X . **b** Find $\Pr(X < \mu)$. **c** Find $\text{Var}(X)$.

- 15 A player throws a die whose faces are numbered 1 to 6 inclusive. If the player obtains a 6 she throws a die a second time, and in this case her score is the sum of 6 and the second number; otherwise her score is the number first obtained. The player has no more than two throws.

Let X be the random variable denoting the player's score. Write down the probability distribution of X , and determine the mean of X .



- 3 To determine the probability that X takes a value in the interval from a to b (including the values a and b) the values of $p(x)$ from $x = a$ to $x = b$ are summed. That is:

$$\Pr(a \leq X \leq b) = \sum_{x=a}^{x=b} p(x)$$

- The **mode** (M) of a random variable X is the value of X such that:

$$\Pr(X = M) \geq \Pr(X = x) \text{ for all other values of } X$$

- The **median** value of a random variable is the middle value of the distribution, the value such that 50% of the distribution is greater than it, and 50% is less than it.

To find the median of the distribution, add the probabilities from the smallest value of X upwards until the sum reaches or exceeds 0.5.

- If the sum exceeds 0.5, then the highest value accumulated is the median value of X .
 - If the sum equals 0.5 exactly, then the median is taken as the value halfway between this value and the next.
- The **expected value** of a random variable X is determined by summing the product of each value of X and the probability that X takes that value. That is:

$$\begin{aligned} E(X) &= \sum_x x \cdot \Pr(X = x) \\ &= \sum_x x \cdot p(x) \end{aligned}$$

$E(X)$ may be considered as the long-run average value of X . It is generally denoted μ (mu), and is also called the **mean** of X .

- In general, the expected value of a function of X is not equal to that function of the expected value of X . That is:

$$E[g(X)] \neq g[E(X)]$$

An exception to this rule is when the function is a linear function of X . Then:

$$E(aX + b) = aE(X) + b \quad (a, b \text{ constant})$$

- The **variance** of the random variable X is a measure of spread of the probability distribution about its mean or expected value μ . It is defined as:

$$\text{Var}(X) = E[(X - \mu)^2]$$

and may be considered as the long-run average value of the square of the distance from X to μ . The variance is usually denoted σ^2 , where σ is the lower case of the Greek letter sigma.

- The **standard deviation** of X is defined as:

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

The standard deviation is usually denoted σ .

- An alternative (computational) formula for variance is:

$$\text{Var}(X) = E(X)^2 - [E(X)]^2$$

- A useful property of variance is:

$$\text{Var}(aX + b) = a^2 \cdot \text{Var}(X), \quad (a, b \text{ constant})$$

and hence

$$\text{sd}(aX + b) = a \cdot \text{sd}(X), \quad (a, b \text{ constant})$$

- In general, for many random variables X :

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

Multiple-choice questions

- 1 Consider the following table:

x	-2	-1	0	1	2
$\Pr(X = x)$	$2k$	$3k$	0.1	$3k$	$2k$

For the table to represent a probability function, the value of k is:

- A** 0.09 **B** 0.9 **C** 0.01 **D** 0.20 **E** 1

- 2 Suppose that the random variable X has the probability distribution given in the following table:

x	-3	-2	-1	0	1	2	3
$\Pr(X = x)$	0.07	0.15	0.22	0.22	0.17	0.12	0.05

$\Pr(-3 \leq X < 0)$ is equal to:

- A** 0.59 **B** 0.37 **C** 0.22 **D** 0.44 **E** 0.66

Questions 3, 4 and 5 refer to the following probability distribution:

x	1	2	3	4	5
$\Pr(X = x)$	0.46	0.24	0.14	0.09	0.07

- 3 For this probability distribution the mode of X is:

- A** 2 **B** 1 **C** 1.59 **D** 2.07 **E** 5.87

- 4 For this probability distribution the median of X is:

- A** 2 **B** 1 **C** 1.59 **D** 2.07 **E** 5.87

- 5 For this probability distribution the expected value $E(X)$ is:

- A** 2 **B** 1 **C** 1.59 **D** 2.07 **E** 5.87

- 6 For a random variable X , $E(X) = 1.20$ and $E(X^2) = 1.69$. The standard deviation of X is equal to:

- A** 1.3 **B** $\sqrt{3.13}$ **C** 0.25 **D** 0.7 **E** 0.5

- 7 Suppose that a random variable X is such that $E(X) = 100$ and $\text{Var}(X) = 100$. Suppose further that Y is a random variable such that $Y = 3X + 10$. Then:

- A** $E(X) = 300$ and $\text{Var}(X) = 900$ **B** $E(X) = 310$ and $\text{Var}(X) = 300$
C $E(X) = 310$ and $\text{Var}(X) = 900$ **D** $E(X) = 300$ and $\text{Var}(X) = 30$
E $E(X) = 310$ and $\text{Var}(X) = 100\sqrt{3}$

Questions 8 and 9 refer to the following probability distribution of the random variable X , where $0 < p < \frac{1}{3}$:

x	-1	0	1
$\text{Pr}(X = x)$	p	$2p$	$1 - 3p$

- 8 The mean of X is:

- A** 1 **B** 0 **C** $1 - 4p$ **D** $4p$ **E** $1 + 4p$

- 9 If the mode of X is 0, then:

- A** $p = \frac{1}{3}$ **B** $\frac{1}{5} < p < \frac{1}{3}$ **C** $p < \frac{1}{5}$ **D** $p > \frac{1}{5}$ **E** $p = \frac{1}{2}$

- 10 The random variable X has the following probability distribution:

x	-2	0	2
$\text{Pr}(X = x)$	a	b	0.2

If the mean of X is 0.2, then:

- A** $a = 0.2, b = 0.6$ **B** $a = 0.1, b = 0.7$ **C** $a = 0.4, b = 0.4$
D $a = 0.1, b = 0.7$ **E** $a = 0.5, b = 0.3$

Short-answer questions (technology-free)

- If $\text{Pr}(A) = 0.5$, $\text{Pr}(B) = 0.2$ and $\text{Pr}(A \cup B) = 0.7$, are the events A and B mutually exclusive? Explain.
- Show, using a diagram or otherwise, that $\text{Pr}(A \cup B) = 1 - \text{Pr}(A' \cap B')$. How would you describe this relationship in words?
- A box contains five black and four white balls. Find the probability that two balls drawn at random are of different colours if:
 - the first ball drawn is replaced before the second is drawn
 - the balls are drawn without replacement
- A gambler has two coins, A and B ; the probabilities of their turning up heads are 0.8 and 0.4 respectively. One coin is selected at random and tossed twice, and a head and a tail are observed. Find the probability that the coin selected was A .
- A random variable X has the following probability distribution:

x	-1	0	1	2	3	4
$\text{Pr}(X = x)$	k	$2k$	$3k$	$2k$	k	k

Find:

- a** the constant k **b** the mode of X **c** the median value of X
d the expected value of X , $E(X)$ **e** the variance of X , $\text{Var}(X)$

- 6** If X has a probability function given by:

$$p(x) = \frac{1}{4}, x = 2, 4, 16, 64$$

find:

- a** $E(X)$ **b** $E\left(\frac{1}{X}\right)$ **c** $\text{Var}(X)$ **d** $\text{sd}(X)$

- 7** A manufacturer sells cylinders for \$ x each; the cost of the manufacture of each cylinder is \$2.00. If a cylinder is defective it is returned and the purchase money refunded. A returned cylinder is regarded as a total loss to the manufacturer. The probability that a cylinder is returned is $\frac{1}{5}$.

- a** Let P = profit per cylinder. Find the probability function of P .
b Find the mean of P in terms of x .
c How much should the manufacturer sell the cylinders for in order to make a profit in the long term?

- 8** A group of 1000 drivers was classified according to age and the number of accidents they had been involved in during the previous year.

The results are as follows.

	Age < 30	Age ≥ 30
At most one accident	130	170
More than one accident	470	230

- a** Calculate the probability that, if a driver is chosen at random from this group, the driver is aged less than 30 and has had more than one accident.
b Calculate the probability that a randomly chosen driver is aged less than 30 given that he or she has had more than one accident.
- 9** This year, 70% of the population have been immunised against a certain disease. Records indicate that an immunised person has a 5% chance of contracting the disease whereas for a non-immunised person this figure is 60%. Calculate the overall percentage of the population who are expected to contract the disease.
- 10** Given $\Pr(A) = \frac{1}{2}$, $\Pr(B) = \frac{1}{4}$ and $\Pr(A|B) = \frac{1}{6}$, find:
a $\Pr(A \cap B)$ **b** $\Pr(A \cup B)$ **c** $\Pr(A'|B)$ **d** $\Pr(A|B')$

Extended-response questions

- 1** Given the following probability function:

x	1	2	3	4	5	6	7
$\Pr(X = x)$	c	$2c$	$2c$	$3c$	c^2	$2c^2$	$7c^2 + c$

- a** Find c . **b** Evaluate $\Pr(X \geq 5)$. **c** If $\Pr(X \leq k) > 0.5$, find the minimum value of k .

- 2** Janet and Alan are going to play a tennis match. The probability of Janet winning the first set is 0.3. After that, Janet's probability of winning a set is 0.6 if she has won the previous set, but only 0.4 if she has lost it. The match will continue until either Janet or Alan has won two sets.
- Construct a tree diagram to show the possible course of the match.
 - Find the probability that:
 - Janet will win
 - Alan will win
 - Let X be the number of sets played until the match is complete.
 - Find the probability distribution of X .
 - Find the expected number of sets that the match will take, $E(X)$.
 - Given that the match lasted three sets, find the probability that Alan won.
- 3** A manufacturing company has three assembly lines, A , B and C . It has been found that 95% of the products produced on assembly line A will be free from faults, 98% from assembly line B will be free from faults and 99% from assembly line C will be free from faults. Assembly line A produces 50% of the day's output, assembly line B produces 30% of the day's output, and the rest is produced by assembly line C . If an item is chosen at random from the company's stock, find the probability that it:
- was produced by machine A
 - is defective given that it came from machine A
 - is defective
 - was produced by machine A , given that it was found to be defective
- 4** A recent study found that P , the number of passengers per car entering a city on the freeway on a workday morning, is:

p	0	1	2	3	4	5
$\Pr(P = p)$	0.39	0.27	0.16	0.12	0.04	0.02

- Compute $E(P)$, the mean number of passengers per car.
 - Compute $\text{Var}(P)$ and hence the standard deviation of P .
 - Find $\Pr(\mu - 2\sigma \leq P \leq \mu + 2\sigma)$.
 - The fees for cars at a toll booth on the freeway are as follows:
 - Cars carrying no passengers pay \$1 toll.
 - Cars carrying one passenger pay \$0.40 toll.
 - Cars carrying two passengers pay no toll.
 Let T be the toll paid by a randomly selected car on the freeway.
 - Construct the probability distribution for T .
 - Find $E(T)$, the mean toll paid per car.
 - Find $\Pr(\mu - 2\sigma \leq T \leq \mu + 2\sigma)$.
- 5** Five identical cards are placed face down on the table. Three of the cards are marked \$5 and the remaining two are marked \$10. A player picks two cards at random (without replacement) and is paid an amount equal to the sum of the values on the two cards. How much should the player pay to play if this is to be a fair game? (A fair game is considered to be one where $E(X) = 0$.)

- 6 The random variable Y , the number of cars sold per week by a car salesperson, has the following probability distribution:

y	0	1	2	3	4	5	6	7	8
$\Pr(Y = y)$	0.135	0.271	0.271	0.180	0.090	0.036	0.012	0.003	0.002

- a Compute $E(y)$, the mean number of sales per week.
 - b Compute $\text{Var}(Y)$, and hence the standard deviation of Y .
 - c The car salesperson is given a bonus as follows: if less than three cars per week are sold, no bonus is given; if three or four cars are sold, \$100 bonus is given; if more than four cars, a \$200 bonus is given. Let B be the bonus paid to a salesperson.
 - i Construct the probability distribution for B .
 - ii Find $E(B)$, the mean bonus paid.
- 7 A given investment scheme is such that there is a 10% chance of receiving a profit of 40% of the amount invested, a 15% chance of a 30% profit, a 25% chance of a 20% profit, a 20% chance of a 10% profit, a 15% chance of breaking even, a 10% chance of a 10% loss and a 5% chance of 20% loss.
- a Find the mean and standard deviation of the percentage return on the amount invested.
 - b Find the probability that the percentage return on the amount invested is within two standard deviations of the mean.
 - c An investor investing in the scheme pays a brokerage fee of 2% on the amount invested and a tax of 40% on the return (= profit – brokerage) of the investment (assume that a loss results in a tax refund for this investment).
Express the percentage gain in terms of the percentage return on the amount invested and hence find the mean and standard deviation of the percentage gain.
- 8 A concert featuring a popular singer is scheduled to be held in a large open air theatre. The promoter is concerned that rain will cause people to stay away. A weather forecaster predicts that the probability of rain on any day at that particular time of the year is 0.33. If it does not rain the promoter will make a profit of \$250 000 on the concert. If it does rain the profit will be reduced to \$20 000. An insurance company agrees to insure the concert for \$250 000 against rain for a premium of \$60 000. Should the promoter buy the insurance?
- 9 A game is devised as follows: on two rolls of a single die you will lose \$10 if the sum showing is 7, and win \$11 if the sum showing is either 11 or 12. How much should you win or lose if any other sum comes up in order for the game to be fair?
- 10 A new machine is to be developed by a manufacturing company. Prototypes are to be made until one satisfies the specifications of the company. Only then will it go into production. If, however, after three prototypes are made none are satisfactory, then the project is to be abandoned.
It is estimated that the probability a prototype will fail to produce a satisfactory model is 0.35, independent of any other already tested.

- a** Find the probability that:
- i** the first prototype is successful
 - ii** the first is not successful but the second is
 - iii** the first two are not successful but the third is **iv** the project is abandoned
- b** It is estimated that the cost of developing and testing the first prototype is \$7 million and each subsequent prototype developed costs half of the one before. Find the expected cost of the project.
- c** If a machine is developed then it is estimated that the income will be \$20 million. (If the project is abandoned there is no income.) Find the expected profit.
- 11** Alfred and Bertie play a game, each starting with \$100. Two dice are thrown. If the total score is 5 or more then Alfred pays \$ x , where $0 < x \leq 8$, to Bertie. If the total score is 4 or less then Bertie pays \$ $(x + 8)$ to Alfred.
- a** Find the expected value of Alfred's cash after the first game.
 - b** Find the value of x for the game to be fair.
 - c** Given that $x = 3$, find the variance of Alfred's cash after the first game.
- 12** A die is loaded so that the chance of throwing a 1 is $\frac{x}{4}$, the chance of a 2 is $\frac{1}{4}$ and the chance of a 6 is $\frac{1}{4}(1 - x)$. The chance of a 3, 4, or 5 is $\frac{1}{6}$. The die is thrown twice.
- a** Prove that the chance of throwing a total of 7 is $\frac{9x - 9x^2 + 10}{72}$
 - b** Find the value of x which will make this chance a maximum and find this maximum probability.
- 13** A game of chance consists of rolling a disc of diameter 2 cm on a horizontal square board. The board is divided into 25 small squares of side 4 cm. A player wins a prize if, when a disc settles, it lies entirely within any one small square. There is a ridge round the outside edge of the board so that the disc always bounces back, cannot fall off and lies entirely within the boundary of the large square.
- Prizes are awarded as follows:
- | | |
|--|-----|
| ■ centre (the middle square) | 50c |
| ■ inner (the eight squares surrounding the centre) | 25c |
| ■ corner (the four corner squares) | 12c |
| ■ outer (any other smaller square) | 5c |
- When no skill is involved, the centre of the disc may be assumed to be randomly distributed over the accessible region.
- a** Calculate the probability in any one throw of winning:
 - i** 50c **ii** 25c **iii** 12c **iv** 5c **v** no prize
 - b** The proprietor wishes to make a profit in the long run, but is anxious to charge as little as possible to attract customers. He charges C cents, where C is an integer. Find the lowest value of C that will yield a profit.

The binomial distribution

Objectives

- To define **Bernoulli** sequences.
- To review the basic concepts of **the binomial probability distribution**.
- To investigate the **graph** of the binomial probability distribution, and the effect on the graph of variation in the values of the parameters.
- To calculate and interpret the **mean**, **variance** and **standard deviation** for the binomial probability distribution.
- To apply this knowledge of the binomial probability distribution in the solution of probability-related problems.

15.1 Bernoulli sequences and the binomial probability distribution

An experiment often consists of repeated trials, each of which may be considered as having only two possible outcomes. For example, when a coin is tossed the two possible outcomes are ‘head’ and ‘tail’. When a die is rolled, the two possible outcomes are determined by the random variable of interest for the experiment. If the event of interest is a ‘six’, then the two outcomes are ‘six’ and ‘not a six’. A **Bernoulli sequence** is the name used to describe a sequence of trials that possess the following properties:

- Each trial results in one of two outcomes, which are usually designated either a success, S , or a failure, F .
- The probability of success on a single trial, p , is constant for all trials, and thus the probability of failure on a single trial is $(1 - p)$.
- The trials are independent (so that the outcome on any trial is not affected by the outcome of any previous trial).

Consider, for example, a player shooting for a goal in netball. The player can score the goal or miss the goal. If the probability of the player scoring a goal is the same for each attempt at goal, then the player’s shots at goal form a Bernoulli sequence.

Example 1

Suppose that a netball player has a probability of $\frac{1}{2}$ of scoring a goal each time. What is the probability that she will score one goal from her first two attempts?

Solution

If the player scores one goal from her first two attempts, then the sequence of events could be either a goal first (G) followed by a miss (M) or a miss first (M) followed by a goal (G). Thus:

$$\begin{aligned}\Pr(\text{one goal from first two attempts}) &= \Pr(GM) + \Pr(MG) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{2}\end{aligned}$$

The binomial probability distribution

The number of successes in a Bernoulli sequence of n trials is called a **binomial random variable** and is said to have a **binomial probability distribution**.

Example 2

In a certain country the probability of a female child being born is 0.52. What is the probability that a family with four children will have two daughters?

**Solution**

One way for a family with four children to have two daughters is for the two females to be born first, followed by two males:

$FFMM$

where F represents ‘female’ and M represents ‘male’.

The probability of this particular outcome is:

$$0.52 \times 0.52 \times 0.48 \times 0.48 = (0.52)^2(0.48)^2$$

Of course, this is only one of the possible ways in which the family could be arranged. How many different arrangements of $FFMM$ are there? There are $4!$ arrangements of 4 distinct objects in a row. In this case there are *repetitions*, as there are two F ’s and two M ’s. To account for this divide by $2!$ for the F ’s, and $2!$ for the M ’s, giving

$$\frac{4!}{2!2!}$$

which is the expansion of 4C_2 , usually denoted in the binomial distribution as $\binom{4}{2}$.*

Thus, the probability of obtaining exactly two females is given by:

$$\binom{4}{2}(0.52)^2(0.48)^2$$

This logic can be repeated to find the probability of the random variable X taking any value from 0 to n where there are n trials.

In general, the probability of achieving x successes in n independent trials of a binomial experiment is

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

where
$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

The constants that determine the specific form of a probability distribution are called **parameters** of the distribution. If a random variable X has a probability function of this form, then X has a binomial distribution with parameters n and p .

Example 3

Show that the sum of the binomial probabilities is equal to 1.

Solution

The sum of the binomial probabilities is $\sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x}$.

The binomial theorem is discussed in Appendix A. This states:

$$\begin{aligned} (x + a)^n &= \sum_{i=0}^n \binom{n}{i} x^{n-i} a^i \\ &= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} a + \binom{n}{2} x^{n-2} a^2 + \dots + \binom{n}{n} a^n \end{aligned}$$

Now, using the binomial theorem, the sum of the binomial probabilities is given by:

$$\begin{aligned} \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} &= [p + (1 - p)]^n \\ &= (1)^n \\ &= 1 \end{aligned}$$

* See Appendix A for a revision of the work on permutations and combinations introduced in *Essential Mathematical Methods 1 & 2 CAS*.

Example 4

Find the probability of obtaining exactly three heads when a fair coin is tossed seven times, correct to four decimal places.

Solution

Obtaining a head is considered a success here, and the probability of success on each of the seven independent trials is 0.5.

Let X be the number of heads obtained. In this case the parameters are $p = 0.5$ and $n = 7$.

$$\begin{aligned}\Pr(X = 3) &= \binom{7}{3} (0.5)^3 (1 - 0.5)^{7-3} \quad x = 0, 1, \dots, 7 \\ &= 35 \times (0.5)^7 \\ &= 0.2734\end{aligned}$$

Example 5

The probability that a person currently in prison has ever been imprisoned before is 0.72. Find the probability that of five prisoners chosen at random at least three have been imprisoned before, correct to four decimal places.

Solution

If X is the number of prisoners who have been imprisoned before, then

$$\Pr(X = x) = \binom{5}{x} (0.72)^x (0.28)^{5-x} \quad x = 0, 1, \dots, 5$$

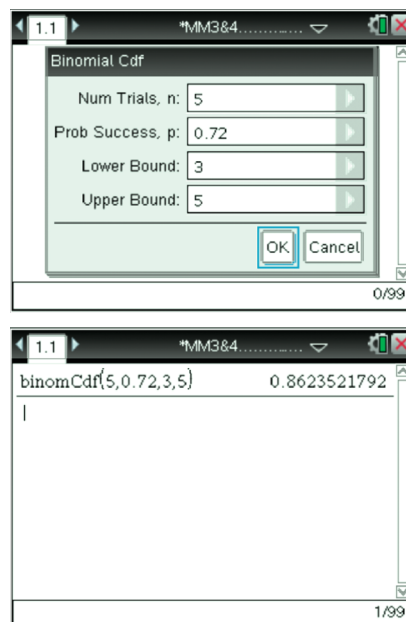
and $\Pr(X \geq 3) = \Pr(X = 3) + \Pr(X = 4) + \Pr(X = 5)$

$$\begin{aligned}&= \binom{5}{3} (0.72)^3 (0.28)^2 + \binom{5}{4} (0.72)^4 (0.28)^1 + \binom{5}{5} (0.72)^5 (0.28)^0 \\ &= 0.8624\end{aligned}$$

Using the TI-Nspire

Use **(menu)** > **Probability** > **Distributions** > **Binomial Cdf** and complete as shown.
Use the **tab** key (**(tab)**) to move between cells.

The result is as shown.



Using the Casio ClassPad

In  tap in the entry line and select

Calc–Distribution. →

In the field below Type, tap the ▼ and scroll to select **Binomial PD**, then tap **Next >>**.

Complete the entry screen as shown then tap **Next >>**.
The calculator returns the answer as shown:

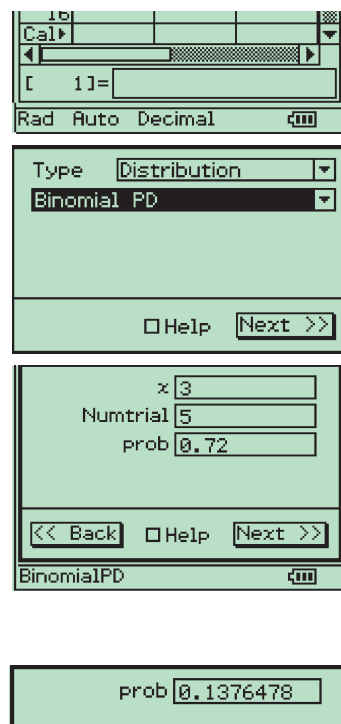
$$\Pr(X = 3) = 0.293.$$

To calculate $\Pr(X \geq 3)$ we use the fact that
 $\Pr(X \geq 3) = 1 - \Pr(X \leq 2)$.

Tap **<< Back**, change the distribution to **Binomial CD** and complete the entries as shown above. The calculator answer is shown.

The required answer is

$$\Pr(X \geq 3) = 1 - 0.1376478 = 0.8623522.$$



Exercise 15A

- 1 Which of the following describe a Bernoulli sequence?
 - a Tossing a fair coin many times
 - b Drawing balls from an urn containing five red and three black balls, replacing the chosen ball each time
 - c Selecting people at random from the population and noting their age
 - d Selecting people at random from the population and noting their sex, male or female
- 2 Evaluate:
 - a $\binom{6}{4}(0.3)^4(0.7)^2$
 - b $\binom{5}{3}(0.2)^3(0.8)^2$
 - c $\binom{5}{2}(0.9)^2(0.1)^3$
- 3 A fair die is rolled six times and the number of twos noted. Find the probability of obtaining:
 - a exactly 3 twos
 - b more than 3 twos
 - c at least 3 twos
- 4 A fair die is rolled five times and the number of sixes that occur in the five rolls is noted. Find the probability of:
 - a the first roll being at six and the rest not
 - b exactly one of the five rolls resulting in a six
- 5 A fair die is rolled 50 times. Find the probability of observing:
 - a exactly 10 sixes
 - b no more than 10 sixes
 - c at least 10 sixes
- 6 Find the probability of getting at least nine successes in 100 trials for which the probability of success is $p = 0.1$.
- 7 A fair coin is tossed 50 times. If X is the number of heads observed, find:
 - a $\Pr(X = 25)$
 - b $\Pr(X \leq 25)$
 - c $\Pr(X \leq 10)$
 - d $\Pr(X \geq 40)$
- 8 A supermarket has four checkouts in operation. A customer is in a hurry and leaves without making a purchase if all the checkouts are busy. At that time of day the probability of each checkout being free is 0.25. Assume that whether or not a checkout is busy is independent of any other checkout. What is the probability that the customer will make a purchase?
- 9 A survey of the population in a particular city found that 40% of people regularly participate in sport. What is the probability of fewer than half of a random sample of six people regularly participating in sport?
- 10 It is known that the probability of a particular drug causing side effects in a person is 0.2. What is the probability that at least two of a random sample of 10 people will experience side effects?
- 11 It is known that the proportion of voters in an electorate who favour a certain candidate is 50%. What is the probability that more than 55% of a sample of 100 voters will favour her?

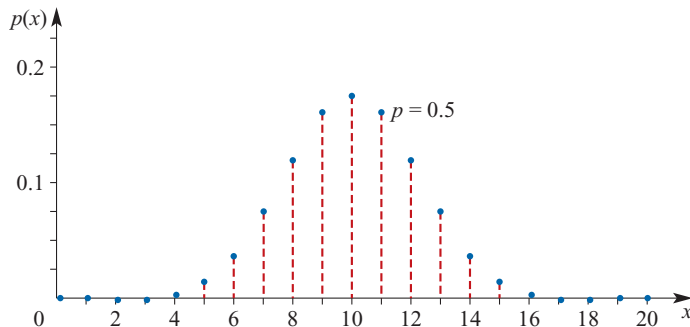
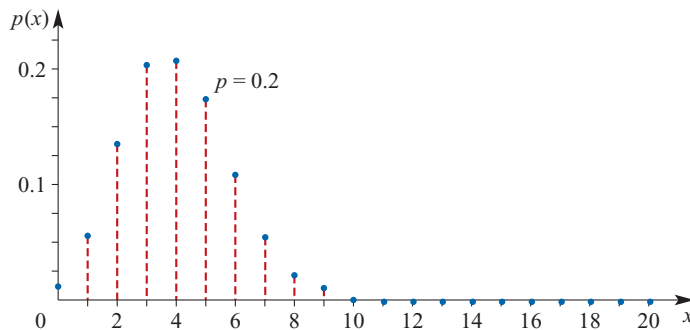
- 12** The manager of a shop knows from experience that 60% of her customers will use a credit card to pay for their purchases. Find the probability that:
- a** the next three customers will use a credit card, and the three after that will not
 - b** three of the next six customers will use a credit card
 - c** at least three of the next six customers will use a credit card
 - d** exactly three of the next six customers will use a credit card, given that at least three of the next six customers use a credit card
- 13** An examination consists of six multiple-choice questions. Each question has four possible answers. At least three correct answers are required to pass the examination. Suppose a student guesses the answer to each question.
- a** What is the probability the student guesses every question correctly?
 - b** What is the probability the student will pass the examination?
- 14** Records show that, on average, six of the 30 days in November will be rainy. Assuming a binomial distribution, with each day in November as an independent trial, find the probability that next November will have, at most, 10 rainy days.
- 15** Of the customers who deal with a car rental company, 45% prefer an automatic car. If there are 35 automatic cars available, what is the probability that the company will not be able to meet the demand for automatic cars in a random group of 80 customers?
- 16** A multiple-choice test has eight questions, each with five possible answers, only one of which is correct. Find the probability that a student who guesses the answer to every question will have:
- a** no correct answers
 - b** six or more correct answers
 - c** every question correct, given they have six or more correct answers
- 17** The probability that a full forward in Australian Rules football will kick a goal from outside the 50-metre line is 0.15. If the full forward has 10 kicks at goal from outside the 50-metre line, find the probability that he will:
- a** kick a goal every time
 - b** kick at least one goal
 - c** kick more than one goal, given that he kicked at least one goal
- 18** A sales representative for a tyre manufacturer claims that the company's steel-belted radial tyres last at least 55 000 kilometres. A tyre dealer decides to check this claim by testing eight of the tyres. If 75% or more of the eight tyres he tests last at least 55 000 kilometres, he will purchase tyres from the sales representative. If, in fact, 90% of the steel-belted radials produced by the manufacturer last at least 55 000 kilometres, what is the probability that the tyre dealer will purchase tyres from the sales representative?
- 19** An examination consists of 25 multiple-choice questions. Each question has four possible answers. At least 13 correct answers are required to pass the examination. Suppose the student guesses the answer to each question.

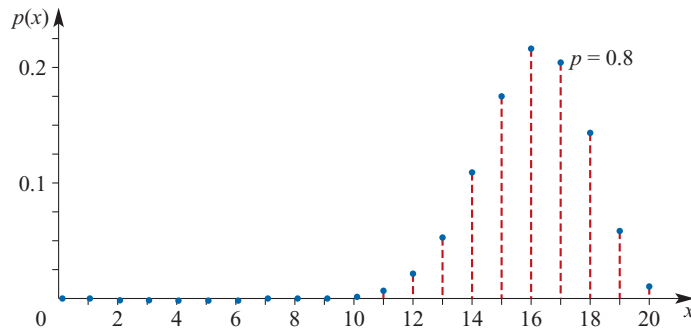
- a** What is the probability that the student guesses exactly 13 question correctly?
b What is the probability the student will pass the examination?
- 20** A multiple-choice test has 20 questions, each with five possible answers, only one of which is correct. Find the probability that a student who guesses the answer to every question will have:
- a** no correct answers **b** 10 or more correct answers
c at least 12 questions correct, given they have 10 or more correct answers

15.2 The graph of the binomial probability distribution

As discussed in Chapter 14, a probability distribution function may be represented as a formula, a table or a graph. Naturally, each of these will vary as the values of the parameters n and p vary. In this section the behaviour of the graph of the binomial probability distribution will be investigated.

Consider for example the graph of the binomial probability distribution for 20 trials ($n = 20$) for differing values of the probability of success p .





By comparing the three graphs it can be seen that:

- When $p = 0.2$ the graph is positively skewed, indicating that mostly from 1 to 8 successes are observed in the 20 trials.
- When $p = 0.5$ the graph is symmetrical (which is as expected when the probability of success is the same as the probability of failure), indicating that mostly from 6 to 14 successes are observed in the 20 trials.
- When $p = 0.8$ the graph is negatively skewed, indicating that mostly from 12 to 19 successes are observed in the 20 trials.

A method for plotting a binomial distribution with a CAS calculator can be found in Appendix B.

Exercise 15B

- 1 From the graphs on page 518 and above, find the ‘most likely’ outcome from a binomial experiment when:
 - a $n = 20$ and $p = 0.2$
 - b $n = 20$ and $p = 0.5$
 - c $n = 20$ and $p = 0.8$
- 2 The following table gives the binomial distribution for $n = 10$ and various values of p given to two decimal places:

x	0	1	2	3	4	5	6	7	8	9	10
$p = 0.3$	0.03	0.12	0.23	0.27	0.20	0.10	0.04	0.01	0.00	0.00	0.00
$p = 0.6$	0.00	0.00	0.01	0.04	0.11	0.20	0.25	0.21	0.12	0.04	0.01
$p = 0.9$	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.06	0.19	0.39	0.35

Sketch the graph of the binomial distribution for each of the values of p on the same axes, and write a description of the plots obtained.

- 3 The following table gives the binomial distribution for $n = 6$ and $p = 0.5$ and for $n = 10$ and $p = 0.5$. (Note that because of rounding error the probabilities do not sum exactly to 1.)

x	0	1	2	3	4	5	6	7	8	9	10
$n = 6$ $p = 0.5$	0.02	0.09	0.23	0.31	0.23	0.09	0.02	0.00	0.00	0.00	0.00
$n = 10$ $p = 0.5$	0.0	0.01	0.04	0.12	0.21	0.25	0.21	0.12	0.04	0.01	0.00

Sketch the graph of the binomial distribution for each set of values of n and p on the same axes, and write a description of the plots obtained.

- 4 Plot the probability distribution function

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 8$ and $p = 0.25$.

- 5 Plot the probability distribution function

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 12$ and $p = 0.35$.

- 6 a Plot the probability distribution function

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 10$ and $p = 0.2$.

- b On the same axes plot

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 10$ and $p = 0.8$, using a different plotting symbol.

- c Compare the two distributions.

- 7 a Plot the probability distribution function

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 20$ and $p = 0.2$.

- b On the same axes plot

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 20$ and $p = 0.5$, using a different plotting symbol.

- c On the same axes plot

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

for $n = 20$ and $p = 0.8$, using a different plotting symbol.

- d Comment on the effect of the value of p on the shape of the distribution.

15.3 Expectation and variance

How many heads would you expect to obtain, on average, if a fair coin was tossed 10 times? While the exact number of heads in the 10 tosses would vary, and could theoretically take values from 0 to 10, it seems reasonable that the long-run average number of heads would be 5. It turns out that this is correct. That is, for a binomial random variable when $n = 10$ and $p = 0.5$:

$$E(X) = \sum_x x \cdot p(x) = 5$$

In general, the expected value for a binomial random variable is equal to the number of trials multiplied by the probability of success. The variance may also be calculated from the parameters n and p .

In general, if X is the number of successes in n trials, each with probability of success p , then the expected value and variance of X are:

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1 - p) \end{aligned}$$

While it is not necessary in this course to be familiar with the derivations of these formulae, they are included for completeness. First of all, consider the expected value of X . By definition:

$$\begin{aligned} E(X) &= \sum_x x \cdot p(x) \\ &= \sum_{x=0}^n x \cdot \binom{n}{x} p^x (1 - p)^{n-x} \quad (\text{substituting the distribution formula}) \\ &= \sum_{x=0}^n x \cdot \left(\frac{n!}{x!(n-x)!} \right) p^x (1 - p)^{n-x} \quad \left(\text{expanding } \binom{n}{x} \right) \\ &= \sum_{x=1}^n x \cdot \left(\frac{n!}{x!(n-x)!} \right) p^x (1 - p)^{n-x} \quad (\text{since the first term will equal zero when } x = 0) \\ &= \sum_{x=1}^n x \cdot \left(\frac{n!}{x(x-1)!(n-x)!} \right) p^x (1 - p)^{n-x} \quad (\text{since } x! = x(x-1)!) \\ &= \sum_{x=1}^n \left(\frac{n!}{(x-1)!(n-x)!} \right) p^x (1 - p)^{n-x} \quad (\text{cancelling the } x\text{'s}) \end{aligned}$$

This expression is very similar to the probability function for a binomial random variable, and the sum of a probability function $p(x)$ is equal to 1. Taking factors of n and p from the expression and letting $z = x - 1$ gives:

$$\begin{aligned} E(X) &= np \sum_{x=1}^{n-1} \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x} \\ &= np \sum_{z=1}^{n-1} \binom{n-1}{z} p^z (1-p)^{n-1-z} \end{aligned}$$

Note that $p(z) = \sum_{z=1}^{n-1} \binom{n-1}{z} p^z (1-p)^{n-1-z}$ is the sum of all values of the probability function for a binomial random variable z , which is the number of successes in $n - 1$ trials each with probability of success p , and is equal to 1. Thus, for a binomial random variable:

$$E(X) = np$$

The variance of the binomial random variable may be found using:

$$\text{Var}(X) = \sigma^2 = E(X^2) - \mu^2, \text{ where } \mu = np$$

Thus to find σ^2 , first determine $E(X^2)$.

$$\begin{aligned} E(X^2) &= \sum_{x=0}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n x^2 \left(\frac{n!}{x!(n-x)!} \right) p^x (1-p)^{n-x} \end{aligned}$$

Note that this time x^2 does not appear as a factor of $x!$. The strategy used here is to first determine $E[X(X-1)]$.

$$\begin{aligned} E[X(X-1)] &= \sum_{x=0}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n x(x-1) \left(\frac{n!}{x!(n-x)!} \right) p^x (1-p)^{n-x} \\ &= \sum_{x=2}^n x(x-1) \left(\frac{n!}{x!(n-x)!} \right) p^x (1-p)^{n-x} \end{aligned}$$

since the first and second terms of this sum equal zero (when $x = 0$ and $x = 1$). Again, this expression is very similar to the binomial distribution formula, but is not quite equal to the sum of a probability function $p(x)$. Taking out factors of $n(n-1)p^2$ and letting $z = x - 2$ gives

$$\begin{aligned} E[X(X-1)] &= n(n-1)p^2 \sum_{x=2}^n \left(\frac{(n-2)!}{(x-2)!(n-x)!} \right) p^{x-2} (1-p)^{n-x} \\ &= n(n-1)p^2 \sum_{z=0}^{n-2} \binom{n-2}{z} p^z (1-p)^{n-2-z} \end{aligned}$$

Again, $p(z) = \sum_{z=0}^{n-2} \binom{n-2}{z} p^z (1-p)^{n-2-z}$ is the sum of all values of the probability function for binomial random variable z , which is the number of successes in $n-2$ trials each with probability of success p and is thus equal to 1, and

$$\begin{aligned} E[X(X-1)] &= n(n-1)p^2 \text{ and } E[X(X-1)] = E(X^2) - E(X) \\ n(n-1)p^2 &= E(X^2) - np \\ E(X^2) &= n(n-1)p^2 + np \end{aligned}$$

which is an expression for $E(X^2)$ in terms of n and p as required. Thus:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - \mu^2 \\ &= n(n-1)p^2 + np - (np)^2 \\ &= np(1-p) \end{aligned}$$

Example 6

An examination consists of 30 multiple-choice questions, each question having three possible answers. If a student guesses the answer to every question, how many will she expect to get right?

Solution

Since the number of correct answers is a binomial random variable, with parameters $p = \frac{1}{3}$ and $n = 30$, the student who guesses has an expected result of $\mu = np = 10$ correct answers (not enough to pass if the pass mark is 50%!).

Example 7

The probability of contracting influenza this winter is known to be 0.2. Of the 100 employees at a certain business how many would the owner expect to get influenza? Find the standard deviation of the number who will get influenza and calculate $\mu \pm 2\sigma$. Interpret the interval $[\mu - 2\sigma, \mu + 2\sigma]$ for this example.

Solution

The number of people who get influenza is a binomial random variable, with parameters $p = 0.2$ and $n = 100$. The owner will expect $\mu = np = 20$ of her employees to contract influenza, with a variance

$$\begin{aligned} \sigma^2 &= np(1-p) \\ &= 100 \times 0.2 \times 0.8 \\ &= 16 \end{aligned}$$

and hence a standard deviation $\sigma = \sqrt{16}$

$$= 4$$

$$\begin{aligned}\text{Then } \mu \pm 2\sigma &= 20 \pm (2 \times 4) \\ &= 20 \pm 8 \text{ or from 12 to 28}\end{aligned}$$

Thus the owner of the business knows there is a probability of about 0.95 that from 12 to 28 of her employees will contract influenza this winter.

Exercise 15C

- Find the mean and variance of each of the binomial random variables with parameters:
 - $n = 25$ and $p = 0.2$
 - $n = 10$ and $p = 0.6$
 - $n = 500$ and $p = \frac{1}{3}$
 - $n = 40$ and $p = 20\%$
- When a fair die is rolled six times, find:
 - the expected value for the number of sixes obtained
 - the probability that more than the expected number of sixes is obtained
- The survival rate for a certain disease is 75%. Of the next 50 people who contract the disease, how many would you expect would survive?
- A binomial random variable has a mean 12 and variance 9. Find the parameters n and p , and hence $\Pr(X = 7)$.
- A binomial random variable has a mean 30 and variance 21. Find the parameters n and p , and hence $\Pr(X = 20)$.
- A fair coin is tossed 20 times. Find the mean and standard deviation of the number of heads obtained and calculate $\mu \pm 2\sigma$. Interpret the interval $[\mu - 2\sigma, \mu + 2\sigma]$ for this example.
- Records show that 60% of the students in a certain state attend government schools. If a group of 200 students are to be selected at random, find the mean and standard deviation of the number of students in the group who attend government schools, and calculate $\mu \pm 2\sigma$. Interpret the interval $[\mu - 2\sigma, \mu + 2\sigma]$ for this sample.

15.4 Using the CAS calculator to find the sample size

Example 8

The probability of winning a prize in a game of chance is 0.45. What is the least number of games that must be played to ensure that the probability of winning at least twice is more than 0.95?

Solution

Since the probability of winning each game is the same each time the game is played, this is an example of a binomial distribution, with the probability of success $p = 0.45$. The required answer is value of n such that:

$$\Pr(X \geq 2) > 0.95$$

$$\text{Equivalently, } 1 - \Pr(X < 2) > 0.95$$

$$\therefore \Pr(X < 2) < 0.05$$

$$\begin{aligned}\Pr(X < 2) &= \Pr(X = 0) + \Pr(X = 1) \\ &= \binom{n}{0} 0.45^0 (0.55)^n + \binom{n}{1} 0.45^1 (0.55)^{n-1} \\ &= (0.55)^n + n \times 0.45^1 (0.55)^{n-1} \text{ since } \binom{n}{0} = 1 \text{ and } \binom{n}{1} = n\end{aligned}$$

The answer is the value of n such that:

$$(0.55)^n + n \times 0.45^1 (0.55)^{n-1} < 0.05$$

This is not an equation that can be solved algebraically, but a CAS calculator can be used to solve this equation.

The calculator solves the equation $(0.55)^n + n \times 0.45(0.55)^{n-1} = 0.05$
Numerically $n = 8.4754 \dots$ Thus the game must be played at least nine times to ensure that the probability of winning at least twice is more than 0.95.

Exercise 15D

- 1 The probability of a target shooter hitting the bullseye on any one shot is 0.2.
 - a If the shooter takes five shots at the target, find the probability of:
 - i missing the bullseye every time
 - ii hitting the bullseye at least once
 - b What is the smallest number of shots the shooter should make to ensure a probability of more than 0.95 of hitting the bullseye at least once?
- 2 The probability of winning a prize with a lucky ticket on a wheel of fortune is 0.1.
 - a If a person buys 10 lucky tickets, find the probability of:
 - i winning twice
 - ii winning at least one prize
 - b What is the smallest number of tickets that should be bought to ensure a probability of more than 0.7 of winning at least once?
- 3 Rex is shooting at a target. His probability of hitting the target is 0.6. What is the minimum number of shots needed for the probability of Rex hitting the target exactly five times to be more than 25%?

- 4 Janet is selecting chocolates at random out of a box. She knows that 20% of the chocolates have hard centres. What is the minimum number of chocolates she needs to select to ensure that the probability of choosing exactly three hard centres is more than 10%?
- 5 The probability of winning a prize in a game of chance is 0.35. What is the fewest number of games that must be played to ensure that the probability of winning at least twice is more than 0.9?
- 6 Geoff has determined that his probability of hitting '4' off any ball when playing cricket is 0.07. What is the fewest number of balls he must face to ensure that the probability of hitting more than one '4' is more than 0.8?
- 7 Monique is practising goaling for netball. She knows from past experience that her chances of making any one shot is about 70%. Her coach has asked her to keep practising until she scores 50 goals. How many shots would she need to attempt to ensure that the probability of making at least 50 shots is more than 0.99?

Hint:

Problems of this type can be solved using a spreadsheet. For the TI calculator:

Enter 1 to 100 in column A using fill down. Use the cumulative binomial distribution function obtained from the **Distribution** submenu in column B and fill down until a probability of obtaining 49 or less goals is less than 0.01 is obtained.



Chapter summary

- A **Bernoulli sequence** describes the repetitions of an experiment where:
 - Each trial results in one of two outcomes, which are usually designated either a success, S , or a failure, F .
 - The probability of success on a single trial, p , is constant for all trials (and thus the probability of failure on a single trial is $(1 - p)$).
 - The trials are independent (so that the outcome on any trial is not affected by the outcome of any previous trial).
- If X is the number of successes in n Bernoulli trials, then X is called a **binomial random variable** and is said to have a **binomial probability distribution** with parameters n and p . In general, the probability of observing x successes in n independent trials of a binomial experiment is:

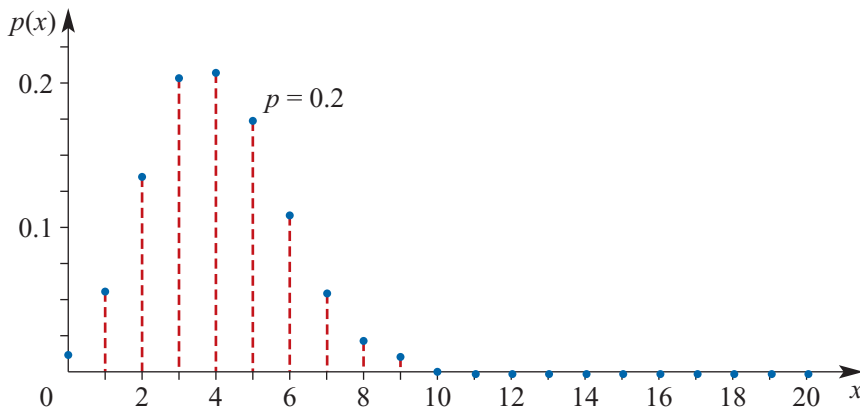
$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

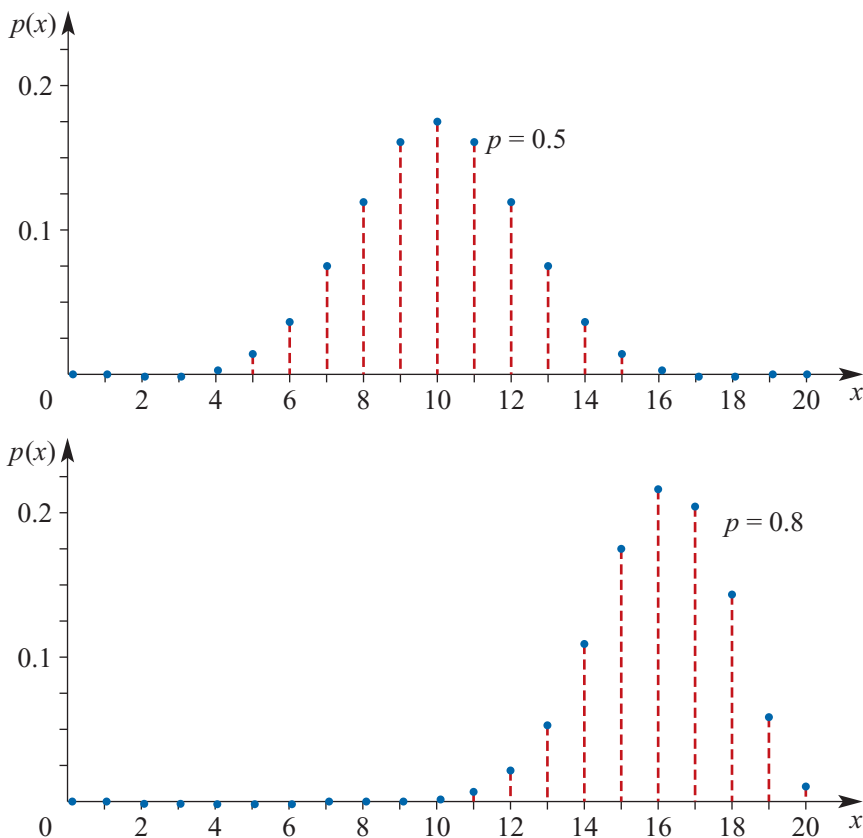
where
$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

- If X has a binomial probability distribution with parameters n and p :

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1 - p) \end{aligned}$$

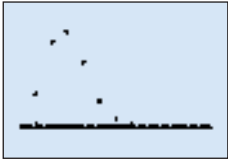
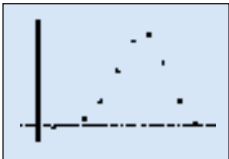
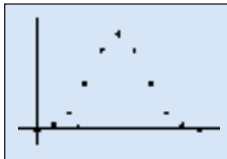
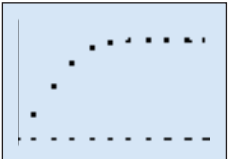
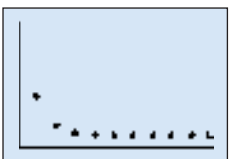
- The shape of the graph of the binomial probability function depends on the values of n and p .





Multiple-choice questions

- A coin is biased so that the probability of a head is 0.6. The probability that exactly three heads will be observed when the coin is tossed five times is:
A 0.6×3 **B** $(0.6)^3$ **C** $(0.6)^3(0.4)^2$
D $10 \times (0.6)^3(0.4)^2$ **E** ${}^5C_3(0.6)^5$
- The probability that the 8:25 train arrives on time is 0.35. What is the probability that the train is on time at least once during a working week (Monday to Friday)?
A $1 - (0.65)^5$ **B** $(0.35)^5$ **C** $1 - (0.35)^5$
D $5 \times (0.35)^1(0.65)^4$ **E** $(0.65)^5$
- A fair die is rolled four times. The probability that a number greater than 4 is observed on two occasions is:
A $\frac{1}{4}$ **B** $\frac{16}{81}$ **C** $\frac{1}{9}$ **D** $\frac{1}{81}$ **E** $\frac{8}{27}$
- The probability a person in a certain town has a tertiary education is 0.4. What is the probability that if 80 people are chosen at random from this town, less than 30 will have a tertiary education?
A 0.7139 **B** 0.2861 **C** 0.0827 **D** 0.3687 **E** 0.3750

- 5 If X is a binomial random variable with parameters $n = 18$ and $p = \frac{1}{3}$, then the mean and variance of X are closest to:
- A $\mu = 6, \sigma^2 = 4$ B $\mu = 9, \sigma^2 = 4$ C $\mu = 6, \sigma^2 = 2$
 D $\mu = 6, \sigma^2 = 16$ E $\mu = 18, \sigma^2 = 6$
- 6 Which one of the following graphs best represents the shape of a binomial probability distribution of the random variable X with 10 independent trials and probability of success 0.7?
- A  B  C 
 D  E 
- 7 Suppose that X is a binomial random variable with mean $\mu = 10$ and standard deviation $\sigma = 2$. Then the probability of success, p , in any trial is:
- A 0.4 B 0.5 C 0.6 D 0.7 E 0.8
- 8 Suppose that X is the number of heads observed when a coin known to be biased towards heads is tossed 10 times. If $\text{Var}(X) = 1.875$, then the probability of a head on any one toss is:
- A 0.25 B 0.55 C 0.75 D 0.65 E 0.80

Questions 9 and 10 refer to the following information:

The probability of Thomas beating William in a set of tennis is 0.24, and Thomas and William decide to play a set of tennis every day for n days.

- 9 What is the fewest number of days on which they should play to ensure that the probability of Thomas winning at least one set is more than 0.95?
- A 7 B 8 C 9 D 10 E 11
- 10 What is the fewest number of days on which they should play to ensure that the probability of Thomas winning at least two sets is more than 0.95?
- A 12 B 18 C 17 D 21 E 14

Short-answer questions (technology-free)

- 1 If X is a binomial random variable with parameters $n = 4$ and $p = \frac{1}{3}$, find:
- a $\Pr(X = 0)$ b $\Pr(X = 1)$ c $\Pr(X \leq 1)$ d $\Pr(X \geq 1)$
- 2 A salesperson knows that 60% of the people who enter a particular shop will make a purchase. What is the probability that of the next three people who enter the shop exactly two will make a purchase?

- 3 If 10% of patients fail to improve on a certain medication, find the probability that of five patients selected at random one or more will fail to show improvement.
- 4 A machine has a probability of 0.1 of manufacturing a defective part.
 - a What is the expected number of defective parts in a random sample of 20 parts manufactured by the machine?
 - b What is the standard deviation of the number of defective parts?
- 5 An experiment consists of four independent trials. The probability of success in a trial is p . Each trial ends in either a success or a failure. Find the probability of each of the following in terms of p :
 - a no successes b one success c at least one success
 - d four successes e at least two successes
- 6 A coin is tossed 10 times. The probability of three heads is $n \times (\frac{1}{2})^{10}$. State the value of n .
- 7 An experiment consists of five independent trials. Each trial results in a success or failure. The probability of success in a trial is p . Find in terms of p the probability of exactly one success given at least one success.
- 8 A dice is tossed five times. What is the probability of obtaining an even number on the uppermost face on three of the tosses?
- 9 In a particular city the probability of rain on any day in June is $\frac{1}{5}$. What is the probability of it raining on three of five days?

Extended-response questions

- 1 In a test to detect learning disabilities, a child is given 10 questions, each of which has possible answers labelled A , B and C . Children with a disability of type 1 almost always answer A or B on every question, while children with a disability of type 2 almost always answer C on every question. Children without either disability have an equal chance of answering A , B or C for each question.
 - a What is the probability that the answers given by a child without either disability will be all A 's and B 's, thereby indicating a type 1 disability?
 - b A child is further tested for type 2 disability if he or she answers C five or more times. What is the probability that a child without either disability will test positive for type 2 disability?
- 2 An inspector takes a random sample of 10 items from a very large batch. If none of the items is defective he accepts the batch, otherwise he rejects the batch. What is the probability that a batch is accepted if the fraction of defective items is 0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1? Plot these probabilities against the corresponding fraction defective. Is the inspection method a good one or not?

- 3 It has been found in the past that 4% of the compact disks produced in a certain factory are defective. A sample of 10 is drawn randomly from each hour's production and the number of defectives is noted.
- In what percentage of these hourly samples would there be at least two defectives?
 - Find the mean and standard deviation of the number of defectives noted, and calculate $\mu \pm 2\sigma$.
 - A particular sample is found to contain three defectives. Would this cause you to have doubts about the production process?
- 4 A pizza company claims that they deliver 90% of orders within 30 minutes. In a particular 2-hour period the supervisor notes that 67 pizzas are ordered and 12 are delivered late. If the delivery company is correct, and 90% of pizza are delivered on time, what is the probability that at least 12 pizzas are delivered late?
- 5 **a** A sample of six objects is to be drawn from a large population in which 20% of the objects are defective. Find the probability that the sample contains:
- three defectives
 - less than three defectives
- b** Another large population contains a proportion p of defective items.
- Write down an expression of p for P , the probability that a sample of six contains exactly two defectives.
 - By differentiating to find $\frac{dP}{dp}$, show that P is greatest when $p = \frac{1}{3}$.
- 6 Groups of six people are chosen at random and the number, x , of people in each group who normally wear glasses is recorded. The results obtained from 200 groups of six are shown in the table:

No. in group wearing glasses (x)	0	1	2	3	4	5	6
No. of occurrences (p)	17	53	65	45	18	2	0

- Calculate, from the above data, the mean value of x .
 - Assuming that the situation can be modelled by a binomial distribution having the same mean as the one calculated above, state the appropriate values for the binomial parameters n and p .
 - Calculate the theoretical frequencies corresponding to those in the table.
- 7 A sampling inspection scheme is devised as follows. A sample of size 10 is drawn at random from a large batch of articles and all 10 articles are tested. If the sample contains less than two faulty articles the batch is accepted; if the sample contains three or more faulty articles, the batch is rejected; but if the sample contains exactly two faulty articles, a second sample of size 10 is taken and tested. If this second sample contains no faulty articles, the batch is accepted; but if it contains any faulty articles, the batch is rejected. Previous experience has shown that 5% of the articles in a batch are faulty.

- a** Find the probability that the batch is accepted after the first sample is taken.
 - b** Find the probability that the batch is rejected.
 - c** Find the expected number of articles to be tested.
- 8** It may be assumed that dates of birth in a large population are distributed throughout the year so that a probability of a randomly chosen person's date of birth being in any particular month may be taken as $\frac{1}{12}$.
 - a** Find the probability that of six people chosen at random exactly two will have birthday in January.
 - b** Find the probability that of eight people chosen at least one will have a birthday in January.
 - c** N people are chosen at random. Find the least value of N so that the probability that at least one will have a birthday in January exceeds 0.9.
- 9** Suppose that, in flight, aeroplane engines fail with probability q , independently of each other, and a plane will complete the flight successfully if at least half of its engines are still working. For what values of q is a two-engine plane to be preferred to a four-engine one?

Markov chains

Objectives

- To use matrices to summarise and solve conditional probability problems.
- To introduce **Markov chains**.
- To use the properties of **Markov chains** to solve problems involving prediction.
- To introduce the concept of the **steady state** of a Markov chain.
- To compare the **length of run** of Bernoulli sequences and Markov chains.

16.1 Using matrices to represent conditional probability

Suppose a netball player has a probability of $\frac{1}{2}$ of scoring a goal on her first attempt. However, it has been found that this player is more likely to score a goal on a subsequent attempt if she scored a goal on the previous attempt, and more likely to miss a goal on a subsequent attempt when she missed the goal on the previous attempt. Let $\Pr(G_i)$ represent the probability that a goal is scored on the i th attempt, and $\Pr(M_i)$ represent the probability that a goal is missed on the i th attempt; then we can summarise the situation as follows:

$$\begin{array}{lll} \Pr(G_1) = \frac{1}{2} & \Pr(G_{i+1}|G_i) = \frac{3}{5} & \Pr(G_{i+1}|M_i) = \frac{1}{3} \\ \Pr(M_1) = \frac{1}{2} & \Pr(M_{i+1}|G_i) = \frac{2}{5} & \Pr(M_{i+1}|M_i) = \frac{2}{3} \end{array}$$

These probabilities can be used to determine the likelihood of certain sequences of events.

Note that $\Pr(G_i) + \Pr(M_i) = 1$.

Example 1

For the situation described above, find the probability that the player scores one goal from her first two attempts.

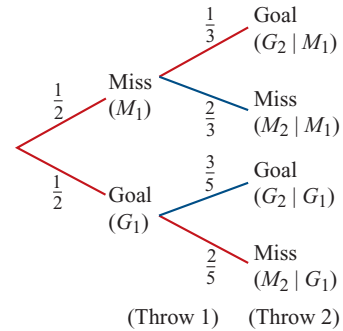
Solution

If the player scores one goal from her first two attempts, then the sequence of events could be either a goal first (G_1) followed by a miss (M_2) or a miss first (M_1) followed by a goal (G_2).

$$\begin{aligned}
 \text{Thus } \Pr(\text{one goal from first two attempts}) &= \Pr(G_1 \cap M_2) + \Pr(M_1 \cap G_2) \\
 &= \Pr(G_1)\Pr(M_2|G_1) + \Pr(M_1)\Pr(G_2|M_1) \\
 &= \frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{1}{3} \\
 &= \frac{2}{10} + \frac{1}{6} \\
 &= \frac{11}{30}
 \end{aligned}$$

A tree diagram is often helpful in completing such problems.

$$\begin{aligned}
 \Pr(\text{one goal from two attempts}) &= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{2}{5} \\
 &= \frac{1}{6} + \frac{2}{10} \\
 &= \frac{11}{30}
 \end{aligned}$$



The red path indicates the ‘pathway’ for this event.

Example 2

An elderly lady likes both Jasmine and Green teas. She drinks no other type and drinks tea every day. She switches from day to day. If she drinks Jasmine one day she drinks Green tea the next day with a probability of $\frac{3}{4}$. If she drinks Green tea one day, then she will drink Jasmine tea the next day with a probability of $\frac{2}{5}$. If she drinks Green tea on a Monday, what is the probability of her drinking Green tea on a Wednesday?

Solution

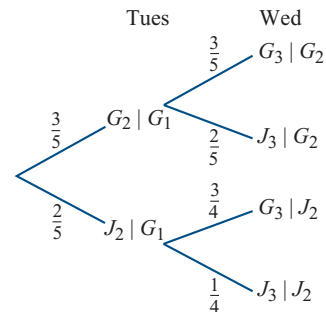
Let J_i be the event ‘Jasmine on the i th day.’

Let G_i be the event ‘Green on the i th day.’

$$\Pr(J_{i+1}|G_i) = \frac{2}{5} \quad \Pr(G_{i+1}|G_i) = \frac{3}{5}$$

$$\Pr(G_{i+1}|J_i) = \frac{3}{4} \quad \Pr(J_{i+1}|J_i) = \frac{1}{4}$$

$$\begin{aligned}
 \Pr(\text{Green tea on Wednesday}) &= \frac{3}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{3}{4} \\
 &= \frac{33}{50}
 \end{aligned}$$



Consider again the information in Example 1. The conditional probabilities that define this problem can be arranged in a matrix as follows:

$$\begin{bmatrix} \Pr(G_{i+1}|G_i) & \Pr(G_{i+1}|M_i) \\ \Pr(M_{i+1}|G_i) & \Pr(M_{i+1}|M_i) \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{3} \end{bmatrix}$$

The numbers in each column add up to 1, because they indicate the only possible outcomes that can occur. This matrix of conditional probabilities is often called a **transition matrix**.

To find $\Pr(G_2)$ and $\Pr(M_2)$ the law of total probability is used. Writing this law in terms of the problem here gives:

$$\Pr(G_2) = \Pr(G_2|G_1)\Pr(G_1) + \Pr(G_2|M_1)\Pr(M_1)$$

and

$$\Pr(M_2) = \Pr(M_2|G_1)\Pr(G_1) + \Pr(M_2|M_1)\Pr(M_1)$$

as before.

These equations can be written as the product of two matrices, as follows:

$$\begin{bmatrix} \Pr(G_2) \\ \Pr(M_2) \end{bmatrix} = \begin{bmatrix} \Pr(G_2|G_1) & \Pr(G_2|M_1) \\ \Pr(M_2|G_1) & \Pr(M_2|M_1) \end{bmatrix} \begin{bmatrix} \Pr(G_1) \\ \Pr(M_1) \end{bmatrix}$$

Since the probability of scoring a goal on the first attempt is $\frac{1}{2}$, then:

$$\begin{bmatrix} \Pr(G_2) \\ \Pr(M_2) \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{7}{15} \\ \frac{8}{15} \end{bmatrix}$$

Example 3

Suppose that the probability of snow on any day is conditional on whether or not it snowed on the preceding day. The probability that it will snow on a particular day given that it snowed on the day before is 0.65, and the probability that it will snow on a particular day given that it did not snow on the day before is 0.3. If the probability that it will snow on Friday is 0.6, what is the probability that it will snow on Saturday?

Solution

Define the events A = snow on Friday

B = snow on Saturday

then: $\Pr(B|A) = 0.65$ and so $\Pr(B'|A) = 0.35$

and $\Pr(B|A') = 0.3$ and so $\Pr(B'|A') = 0.7$

In addition: $\Pr(A) = 0.6$ $\Pr(A') = 0.4$

$$\text{Now } \begin{bmatrix} \Pr(B) \\ \Pr(B') \end{bmatrix} = \begin{bmatrix} \Pr(B|A) & \Pr(B|A') \\ \Pr(B'|A) & \Pr(B'|A') \end{bmatrix} \begin{bmatrix} \Pr(A) \\ \Pr(A') \end{bmatrix}$$

$$\text{Substituting gives } \begin{bmatrix} \Pr(B) \\ \Pr(B') \end{bmatrix} = \begin{bmatrix} 0.65 & 0.3 \\ 0.35 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.51 \\ 0.49 \end{bmatrix}$$

So the probability that it will snow on Saturday is 0.51, and the probability it will not is 0.49.

What then is the probability that it will snow on Sunday? If C is the event that it snows on Sunday, then using the previous matrix representation:

$$\begin{bmatrix} \Pr(C) \\ \Pr(C') \end{bmatrix} = \begin{bmatrix} \Pr(C|B) & \Pr(C|B') \\ \Pr(C'|B) & \Pr(C'|B') \end{bmatrix} \begin{bmatrix} \Pr(B) \\ \Pr(B') \end{bmatrix}$$

The matrix $\begin{bmatrix} \Pr(C|B) & \Pr(C|B') \\ \Pr(C'|B) & \Pr(C'|B') \end{bmatrix}$ is numerically the same as the matrix $\begin{bmatrix} \Pr(B|A) & \Pr(B|A') \\ \Pr(B'|A) & \Pr(B'|A') \end{bmatrix}$ in that it describes in general the probabilities of each outcome given the outcome of the preceding event, which is constant.

From Example 3:

$$\begin{aligned} \begin{bmatrix} \Pr(C) \\ \Pr(C') \end{bmatrix} &= \begin{bmatrix} 0.65 & 0.30 \\ 0.35 & 0.70 \end{bmatrix} \begin{bmatrix} \Pr(B) \\ \Pr(B') \end{bmatrix} = \begin{bmatrix} 0.65 & 0.30 \\ 0.35 & 0.70 \end{bmatrix} \begin{bmatrix} 0.65 & 0.30 \\ 0.35 & 0.70 \end{bmatrix} \begin{bmatrix} \Pr(A) \\ \Pr(A') \end{bmatrix} \\ &= \begin{bmatrix} 0.65 & 0.30 \\ 0.35 & 0.70 \end{bmatrix}^2 \begin{bmatrix} \Pr(A) \\ \Pr(A') \end{bmatrix} = \begin{bmatrix} 0.4785 \\ 0.5215 \end{bmatrix} \end{aligned}$$

This process can be continued so that the probability of it snowing further ahead can be determined by continuing to multiply by the matrix of probabilities.

Example 4

Consider the goal shooter of the netball team from Example 1, where

$$\begin{aligned} \Pr(G_1) &= \frac{1}{2} & \Pr(G_{i+1}|G_i) &= \frac{3}{5} & \Pr(G_{i+1}|M_i) &= \frac{1}{3} \\ \Pr(M_1) &= \frac{1}{2} & \Pr(M_{i+1}|G_i) &= \frac{2}{5} & \Pr(M_{i+1}|M_i) &= \frac{2}{3} \end{aligned}$$

What is the probability she will goal at the fifth attempt? Give your answer correct to four decimal places.

Solution

$$\begin{bmatrix} \Pr(G_5) \\ \Pr(M_5) \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{3} \end{bmatrix}^4 \begin{bmatrix} \Pr(G_1) \\ \Pr(M_1) \end{bmatrix} = \begin{bmatrix} 0.4573 & 0.4522 \\ 0.5427 & 0.5478 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.4548 \\ 0.5452 \end{bmatrix}$$

Thus the probability that the goal shooter scores on her fifth attempt is 0.4548.

Exercise 16A

- 1 Suppose that it is known that the probability of the Liberal party winning an election if they are already in power is 0.6, and the probability of the Labor party winning the election if they are already in power is 0.65.

If the Liberal party is already in power:

- a What is the probability of the Liberal party winning the next two elections?
- b What is the probability of the Labor party winning the next two elections?

- 2 In a certain country the probability of a child being female is 0.65 if the preceding child is female, and 0.44 if the preceding child is male. If F_i is the event that the i th child is female, and M_i is the event that the i th child is male then this can be written in the form:

$$\begin{bmatrix} \Pr(F_{i+1}) \\ \Pr(M_{i+1}) \end{bmatrix} = \mathbf{T} \times \begin{bmatrix} \Pr(F_i) \\ \Pr(M_i) \end{bmatrix}$$

where $\mathbf{T}$ is a 2×2 matrix.

- a Write down the matrix $\mathbf{T}$.
 - b If the probability of a first child being female is 0.52, find the probability that the second child of a family with two children will be female.
- 3 Suppose that the probability of a tennis player winning a game is 0.7 if she has won the preceding game and 0.4 if she has lost the preceding game. If W_i is the event that she wins the i th game, and L_i is the event she loses the i th game:
- a Write down a matrix equation relating $\Pr(W_{i+1})$ and $\Pr(L_{i+1})$ to $\Pr(W_i)$ and $\Pr(L_i)$.
 - b If the probability of her winning the first game is 0.39, find the probability that she wins the second game.
- 4 Suppose that the probability that Jane is late to school is 0.35 if she was late the day before, and her probability of being on time for school is 0.9 if she was on time the day before. If L_i is the event that she is late on the i th day, and T_i is the event she is on time on the i th day:
- a Write down a matrix equation relating $\Pr(L_{i+1})$ and $\Pr(T_{i+1})$ to $\Pr(L_i)$ and $\Pr(T_i)$.
 - b If the probability of her being late on Monday is 0.4, find the probability that she is on time on Tuesday.
- 5 Suppose that the outcome of a rugby game between Australia and England is such that:

$$\begin{array}{lll} \Pr(A_1) = 0.66 & \Pr(A_{i+1}|A_i) = 0.57 & \Pr(A_{i+1}|E_i) = 0.47 \\ \Pr(E_1) = 0.34 & \Pr(E_{i+1}|A_i) = 0.43 & \Pr(E_{i+1}|E_i) = 0.53 \end{array}$$

where A represents the event that Australia wins, and E represents the event that England wins.

If three games are played in a series:

- a What is the probability, correct to three decimal places, that Australia wins all three games?
 - b What is the probability, correct to three decimal places, that England wins all three games?
 - c Write down a matrix equation which relates $\Pr(A_{i+1})$ and $\Pr(E_{i+1})$ to $\Pr(A_i)$ and $\Pr(E_i)$, and use it to determine:
 - i the probability that Australia wins the second game
 - ii the probability that Australia wins the third game
- 6 Suppose that the probability of rain is dependent on whether or not it has rained on a previous day. If it has rained the day before, the probability of rain is 0.43. If it has not rained on the previous day, then the probability of rain is 0.16. Suppose further that the

probability of rain on Monday one week is 0.2. That is:

$$\begin{aligned}\Pr(R_{i+1}|R_i) &= 0.43 & \Pr(R_{i+1}|F_i) &= 0.16 \\ \Pr(R_1) &= 0.2 & \Pr(F_1) &= 0.8\end{aligned}$$

where R_i represents the event that it rains on day i , and F_i represents the event that day i is fine.

- a** What is the probability, correct to three decimal places, that it rains from Monday to Friday inclusive?
- b** What is the probability, correct to three decimal places, that it is fine from Monday to Friday inclusive?
- c** Write down a matrix equation that relates $\Pr(R_{i+1})$ and $\Pr(F_{i+1})$ to $\Pr(R_1)$ and $\Pr(F_1)$, and use it to determine correct to three decimal places:
 - i** the probability that Saturday will be fine
 - ii** the probability that it rains on Sunday

16.2 Markov chains*

Consider a system that is observed at discrete moments of time and, at any of these points in time, can be in any one of a finite number of states. For example, a machine at a point in time can be designated as either *working* or *broken*. The state of the system at time n is denoted as X_n . If the system also has the property that the state in which it will be at a future point in time is dependent only on the state at present, then it is called a **Markov chain**, named after Russian mathematician Andrei Markov (1856–1922).

In a two-state Markov chain it is assumed that:

- Each step (or trial) results in one of two possible outcomes.
- The probability of each possible outcome is *conditional* only on the result of the step immediately preceding it.
- The conditional probabilities for each possible outcome are the same on each occasion (that is, the same matrix is used for each transition).

The simplest form of a Markov chain is one that has only two states. Consider again the machine that at the start of the day can be in either of two states, working or broken. Suppose further that if the machine is broken the probability that it will be fixed by the end of the day is a , and that if the machine is working the probability that it will break is b . Let the state 0 correspond to the machine being broken, and the state 1 correspond to the machine working. If X_n is the state of the machine at time n , then:

$$\begin{aligned}\Pr(\text{machine is working at time } n+1 \text{ if it was broken at time } n) &= \Pr(X_{n+1} = 1|X_n = 0) = a \\ \Pr(\text{machine is broken at time } n+1 \text{ if it was working at time } n) &= \Pr(X_{n+1} = 0|X_n = 1) = b\end{aligned}$$

Since there are only two possible states it follows that:

$$\begin{aligned}\Pr(\text{machine is broken at time } n+1 \text{ if it was broken at time } n) &= \Pr(X_{n+1} = 0|X_n = 0) \\ &= 1 - a \\ \Pr(\text{machine is working at time } n+1 \text{ if it was working at time } n) &= \Pr(X_{n+1} = 1|X_n = 1) \\ &= 1 - b\end{aligned}$$

* The Mathematical Methods Units 3 and 4 CAS study design requires only the study of two-state Markov chains.

These probabilities can be written in matrix form.

$$\mathbf{T} = \begin{bmatrix} \Pr(X_{n+1} = 0|X_n = 0) & \Pr(X_{n+1} = 0|X_n = 1) \\ \Pr(X_{n+1} = 1|X_n = 0) & \Pr(X_{n+1} = 1|X_n = 1) \end{bmatrix} = \begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

The matrix $\mathbf{T}$ is called a **transition matrix** because it describes the transition from one step to the next in a sequence where the probability of an outcome at each step is conditional on the state at the previous step.

In general, a Markov chain is defined by:

- a **transition matrix**, $\mathbf{T}$, which for a situation that has m outcomes or **states** is a square matrix of dimension $m \times m$; the elements of the transition matrix are conditional probabilities
- an **initial state** vector, $\mathbf{S}_0$, which has dimensions $m \times 1$, and which gives information about the Markov chain at the beginning of the sequence, or step 0; the elements of the initial state matrix may be numbers, percentages or the results of an individual trial.

Information about the Markov chain at step n of a sequence is given by the state vector $\mathbf{S}_n$. This can be determined from the transition matrix, $\mathbf{T}$, and the initial state matrix, $\mathbf{S}_0$.

Since $\mathbf{S}_1 = \mathbf{T} \times \mathbf{S}_0$

and $\mathbf{S}_2 = \mathbf{T} \times \mathbf{S}_1 = \mathbf{T} \times (\mathbf{T} \times \mathbf{S}_0) = (\mathbf{T} \times \mathbf{T}) \times \mathbf{S}_0 = \mathbf{T}^2 \times \mathbf{S}_0$

and $\mathbf{S}_3 = \mathbf{T} \times \mathbf{S}_2 = \mathbf{T} \times (\mathbf{T} \times \mathbf{S}_1) = \mathbf{T} \times (\mathbf{T} \times (\mathbf{T} \times \mathbf{S}_0)) = (\mathbf{T} \times \mathbf{T} \times \mathbf{T}) \times \mathbf{S}_0 = \mathbf{T}^3 \times \mathbf{S}_0$

and so on, it follows that:

$$\mathbf{S}_n = \mathbf{T} \times \mathbf{S}_{n-1} = \mathbf{T}^n \times \mathbf{S}_0$$

This gives a general result for Markov chains.

In general, for a Markov chain where:

- $\mathbf{S}_0$ is an $m \times 1$ column matrix that describes the states at step 0
- $\mathbf{T}$ is a corresponding $m \times m$ transition matrix
- $\mathbf{S}_n$ is an $m \times 1$ column matrix giving information about the states at step n of the Markov chain

then $\mathbf{S}_n = \mathbf{T} \times \mathbf{S}_{n-1} = \mathbf{T}^n \times \mathbf{S}_0$

The information in the state matrix $\mathbf{S}_n$ may be numbers, percentages or probabilities.

For a two-state Markov chain:

■ If $\mathbf{S}_0 = \begin{bmatrix} \text{number in state 0 at step 0} \\ \text{number in state 1 at step 0} \end{bmatrix}$ then $\mathbf{S}_n = \begin{bmatrix} \text{estimate of the number in state 0 at step } n \\ \text{estimate of the number in state 1 at step } n \end{bmatrix}$

■ If $\mathbf{S}_0 = \begin{bmatrix} \text{percentage in state 0 at step 0} \\ \text{percentage in state 1 at step 0} \end{bmatrix}$ then $\mathbf{S}_n = \begin{bmatrix} \text{estimate of the percentage in state 0 at step } n \\ \text{estimate of the percentage in state 1 at step } n \end{bmatrix}$

■ If $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, which means that a 'success' has been observed at step 0, or $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, which means that a

'failure' has been observed at step 0, then

$$\mathbf{S}_n = \begin{bmatrix} \text{probability of observing a success at step } n \\ \text{probability of observing a failure at step } n \end{bmatrix}$$

An example where the initial state information is in numbers is given in Example 5.

Example 5

Suppose that a car rental firm has two branches, one in Melbourne and the other in Geelong. Cars are usually rented for a week and returned to the same place. However, the probability that a car rented in Melbourne will be returned to Geelong is known to be 0.1, and the probability that a car rented in Geelong will be returned to Melbourne is 0.2. Initially the company places 85 cars in Melbourne and 40 in Geelong. Determine:

- the transition matrix, $\mathbf{T}$, which can be used to represent this information
- the estimated number of cars in each location at the end of week 4

Solution

- Define the events A_i = car is in Melbourne at step i
 B_i = car is in Geelong at step i

$$\text{Then } \mathbf{T} = \begin{bmatrix} \Pr(A_{i+1}|A_i) & \Pr(A_{i+1}|B_i) \\ \Pr(B_{i+1}|A_i) & \Pr(B_{i+1}|B_i) \end{bmatrix} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

- The estimated number of cars in each location at step 4 is given by $\mathbf{S}_4$ where

$$\mathbf{S}_4 = \mathbf{T}^4 \times \mathbf{S}_0$$

and $\mathbf{S}_0$ describes the initial allocation of the cars in Melbourne and Geelong,

$$\text{so } \mathbf{S}_0 = \begin{bmatrix} 85 \\ 40 \end{bmatrix}$$

$$\begin{aligned} \text{Thus } \mathbf{S}_4 = \mathbf{T}^4 \times \mathbf{S}_0 &= \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}^4 \begin{bmatrix} 85 \\ 40 \end{bmatrix} = \begin{bmatrix} 0.7467 & 0.5066 \\ 0.2533 & 0.4934 \end{bmatrix} \begin{bmatrix} 85 \\ 40 \end{bmatrix} \\ &= \begin{bmatrix} 83.7 \\ 41.3 \end{bmatrix} = \begin{bmatrix} 84 \\ 41 \end{bmatrix} \text{ to the nearest whole numbers} \end{aligned}$$

Thus by the end of week 4 it is estimated that there will be 84 cars in Melbourne and 41 in Geelong.

An example where the initial state information is in percentages is given in Example 6.

Example 6

Suppose that there are only two choices of garage in a country town, and records show that 63% of the time customers will continue to purchase their petrol from garage A if they purchased their petrol from garage A in the previous month, while 78% of the time consumers will continue to purchase their petrol from garage B if they purchased their petrol from garage B in the previous month. Suppose that, in month 0, 33% of customers chose garage A and 67% chose garage B . Find:

- the transition matrix that can be used to represent this information
- the percentage of customers choosing each garage in month 3

Solution

- a** Define the events A_i = customer chooses garage A in month i
 B_i = customer chooses garage B in month i

$$\text{Then } \mathbf{T} = \begin{bmatrix} \Pr(A_{i+1}|A_i) & \Pr(A_{i+1}|B_i) \\ \Pr(B_{i+1}|A_i) & \Pr(B_{i+1}|B_i) \end{bmatrix} = \begin{bmatrix} 0.63 & 0.22 \\ 0.37 & 0.78 \end{bmatrix}$$

- b** The estimated percentage of customers at each garage in month 3 is given by $\mathbf{S}_3$ where

$$\mathbf{S}_3 = \mathbf{T}^3 \times \mathbf{S}_0$$

and $\mathbf{S}_0$ describes the initial percentage of cars choosing garage A and garage B , so

$$\mathbf{S}_0 = \begin{bmatrix} 33 \\ 67 \end{bmatrix}$$

Thus

$$\mathbf{S}_3 = \mathbf{T}^3 \times \mathbf{S}_0 = \begin{bmatrix} 0.63 & 0.22 \\ 0.37 & 0.78 \end{bmatrix}^3 \begin{bmatrix} 33 \\ 67 \end{bmatrix} = \begin{bmatrix} 0.4161 & 0.3472 \\ 0.5839 & 0.6528 \end{bmatrix} \begin{bmatrix} 33 \\ 67 \end{bmatrix} = \begin{bmatrix} 37.0 \\ 63.0 \end{bmatrix}$$

Thus in month 3 it is estimated that 37% of the customers will be using garage A , and 63% garage B .

Sometimes the initial step of the Markov chain, $\mathbf{S}_0$ will not be a percentage as in Example 6, but rather the results of a single trial, as illustrated in the following example.

Example 7

Suppose that the probability that a bus is late, given that it is late the previous day, is 0.15, while the probability that it is on time if it is on time the previous day is 0.90.

- a** Find the transition matrix that can be used to represent this information.
b What is the probability that the bus will be late on day 3:
i if it is on time on day 0?
ii if it is late on day 0?

Give all answers correct to four decimal places.

Solution

- a** This information can be represented by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} 0.15 & 0.10 \\ 0.85 & 0.90 \end{bmatrix}$$

- b** To find the probabilities associated with day 3, calculate $\mathbf{S}_3 = \mathbf{T}^3 \times \mathbf{S}_0$

i If the bus was on time on day 0, then $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\mathbf{S}_3 = \begin{bmatrix} 0.15 & 0.10 \\ 0.85 & 0.90 \end{bmatrix}^3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1054 & 0.1053 \\ 0.8946 & 0.8948 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1053 \\ 0.8948 \end{bmatrix}$$

Thus the probability of the bus being late on day 3 is:

$$\Pr(\text{late}) = 0.1053$$

and the probability of the bus being on time on day 3 is:

$$\Pr(\text{on time}) = 0.8948$$

ii If the bus was late on day 0, then $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\mathbf{S}_3 = \begin{bmatrix} 0.15 & 0.10 \\ 0.85 & 0.90 \end{bmatrix}^3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1054 & 0.1053 \\ 0.8946 & 0.8948 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1054 \\ 0.8946 \end{bmatrix}$$

Thus the probability of the bus being late on day 3 is:

$$\Pr(\text{late}) = 0.1054$$

and the probability of the bus being on time on day 3 is:

$$\Pr(\text{on time}) = 0.8946$$

A three-state Markov chain works in the same way as a two-state Markov chain, except that here the transition matrix will be 3×3 , and the initial state matrix will be 3×1 , as illustrated in the following example.

Note: Three-state Markov chains are not included in the study design for Mathematical Methods Units 3 and 4 CAS.

Example 8

Suppose that a soccer team can win, draw or lose each match. Let state 0 correspond to the team winning, state 1 correspond to the team drawing, and state 2 correspond to the team losing. The probabilities associated with each state are given by the following transition matrix:

$$\begin{array}{c} \begin{matrix} & W & D & L \end{matrix} \\ \begin{matrix} W \\ D \\ L \end{matrix} \begin{bmatrix} 0.7 & 0.5 & 0.4 \\ 0.2 & 0.4 & 0.3 \\ 0.1 & 0.1 & 0.3 \end{bmatrix} \end{array}$$

$W = \text{win}, D = \text{draw}, L = \text{loss}$

- What is the probability that the team wins game 3 of the season if they won the last game of the previous season?
- What is the probability that the team wins game 4 of the season if they drew the last game of the previous season?

Solution

a If team won game 0, then $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\mathbf{S}_3 = \begin{bmatrix} 0.7 & 0.5 & 0.4 \\ 0.2 & 0.4 & 0.3 \\ 0.1 & 0.1 & 0.3 \end{bmatrix}^3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.614 & 0.606 & 0.594 \\ 0.262 & 0.270 & 0.274 \\ 0.124 & 0.124 & 0.132 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.614 \\ 0.262 \\ 0.124 \end{bmatrix}$$

Thus the probability that the team wins game 3 if they win game 0 is 0.614.

b If team drew game 0, then $S_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$S_4 = \begin{bmatrix} 0.7 & 0.5 & 0.4 \\ 0.2 & 0.4 & 0.3 \\ 0.1 & 0.1 & 0.3 \end{bmatrix}^4 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.6104 & 0.6088 & 0.6056 \\ 0.2648 & 0.2664 & 0.2680 \\ 0.1248 & 0.1248 & 0.1264 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.6088 \\ 0.2664 \\ 0.1248 \end{bmatrix}$$

Thus the probability that the team wins game 4 if they drew game 0 is 0.6088.



Exercise 16B

- 1 Consider the transition matrix $T = \begin{bmatrix} 0.8 & 0.5 \\ 0.2 & 0.5 \end{bmatrix}$ and an initial state matrix $S_0 = \begin{bmatrix} 30 \\ 70 \end{bmatrix}$.
 - a Use the relationship $S_n = TS_{n-1}$ to determine:
 - i S_1
 - ii S_2
 - iii S_3
 - b Determine the value of T^5 .
 - c Use the relationship $S_n = T^n S_0$ to determine:
 - i S_2
 - ii S_3
 - iii S_7
- 2 Consider the transition matrix $T = \begin{bmatrix} 0.86 & 0.2 \\ 0.14 & 0.8 \end{bmatrix}$ and an initial state matrix $S_0 = \begin{bmatrix} 180 \\ 200 \end{bmatrix}$.
 - a Use the relationship $S_n = TS_{n-1}$ to determine:
 - i S_1
 - ii S_2
 - iii S_3
 - b Determine the value of T^6 .
 - c Use the relationship $S_n = T^n S_0$ to determine:
 - i S_2
 - ii S_3
 - iii S_5
- 3 A Markov chain has the transition matrix $T = \begin{bmatrix} 0.41 & 0.23 \\ 0.59 & 0.77 \end{bmatrix}$.
 - a If the initial state matrix $S_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ find:
 - i S_1
 - ii S_2
 - iii S_5
 - b If the initial state matrix $S_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ find:
 - i S_1
 - ii S_2
 - iii S_5
- 4 A Markov chain has the transition matrix $T = \begin{bmatrix} \frac{3}{8} & \frac{2}{3} \\ \frac{5}{8} & \frac{1}{3} \end{bmatrix}$.
 - a If the initial state matrix $S_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ find:
 - i S_1
 - ii S_2
 - iii S_7 , correct to four decimal places

- b** If the initial state matrix $S_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ find:
- i** S_1 **ii** S_2 **iii** S_7 , correct to four decimal places
- *5** A Markov chain has the transition matrix $T = \begin{bmatrix} 0.28 & 0.34 & 0.80 \\ 0.04 & 0.48 & 0.19 \\ 0.68 & 0.18 & 0.01 \end{bmatrix}$.
- a** If the initial state matrix $S_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ find:
- i** S_1 **ii** S_2 **iii** S_9
- b** If the initial state matrix $S_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ find:
- i** S_1 **ii** S_2 **iii** S_9
- c** If the initial state matrix $S_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ find:
- i** S_1 **ii** S_2 **iii** S_9
- 6** There are two popular beaches in a seaside resort, Bayside and Surfside. Most people tend to go back to the same beach each day, but there is a probability of 0.15 that a person who goes to Bayside one day will go to Surfside the next, and a probability of 0.2 that a person who goes to Surfside on one day goes to Bayside on the next day. Suppose that on one Monday there are 125 people at Bayside and 240 people at Surfside. Determine:
- a** the transition matrix, T , that which can be used to represent this information
- b** the estimated number of people at each beach on Thursday (give your answers to the nearest whole number)
- 7** Suppose that the initial population of the inner city in a town is 35 000 and that of the suburban area is 65 000. If every year 40% of the inner city population moves to the suburbs, while 30% of the suburb population moves to the inner part of the city, determine:
- a** the transition matrix, T , that can be used to represent this information
- b** The estimated number of people living in the suburbs after 4 years, assuming that the total population remains constant. Give your answer to the nearest 100 people.
- 8** A factory has a large number of machines that can be in one of two states, operating or broken. The probability that an operating machine breaks by the end of the day is 0.08, and the probability that a broken machine is repaired by the end of the day is 0.88. Find:
- a** the transition matrix that can be used to represent this information
- b** the percentage of machines that are operating at the end of day 5, if initially 18% of machines are broken. Give your answer correct to one decimal place.

* The Mathematical Methods Units 3 and 4 CAS study design requires only the study of two-state Markov chains.

- 9** Suppose that there are two secondary schools in a country town. Experience has shown that 9% of students will move from school *A* to school *B* at the end of each year, while 13% of students will move from school *B* to school *A*. Suppose that initially 40% of students attend school *A* and 60% attend school *B*. Find:
- a** the transition matrix that can be used to represent this information
 - b** the percentage of students attending each school at the end of 5 years, correct to one decimal place
- 10** Every year on Anzac Day Collingwood play Essendon in AFL football. Suppose that the probability that Collingwood will win is 0.5 if they have won the year before, and the probability that Essendon will win is 0.72 if they have won the year before.
- a** Find the transition matrix that can be used to represent this information.
 - b** What is the probability that Collingwood will win in Year 5:
 - i** if they won in year 0?
 - ii** if they lost in year 0?
- 11** A tennis player can serve to the forehand or to the backhand of her opponent. If she serves to the forehand, there is an 80% chance that the next serve will be to the backhand, while if she serves to the backhand there is a 70% chance that the next serve will be to the forehand.
- a** Find the transition matrix that can be used to represent this information.
 - b** What is the probability that the tennis player will serve the fourth serve to the backhand:
 - i** if the first serve was to the forehand?
 - ii** if the first serve was to the backhand?
- *12** Suppose that the trees in a forest are assumed to fall into three age groups: young trees (0–15 years), middle-aged trees (16–30 years), and old trees (more than 30 years). A time period is 15 years, and it is assumed that in each time period:
- 10% of young trees, 20% of middle-aged trees and 40% of old trees die
 - surviving trees enter into the next age group; old trees remain old
 - dead trees are replaced by young trees.

This information is summarised in the following transition matrix:

$$\begin{array}{c} Y \quad M \quad O \\ \begin{array}{l} Y \\ M \\ O \end{array} \left[\begin{array}{ccc} 0.1 & 0.2 & 0.4 \\ 0.9 & 0 & 0 \\ 0 & 0.8 & 0.6 \end{array} \right] \end{array}$$

Suppose that 20 000 young trees are planted in the forest.

- a** What is the age distribution of the trees after 15 years (one time period)?
- b** What is the age distribution of the trees after 45 years (three time periods)?

* The Mathematical Methods Units 3 and 4 CAS study design requires only the study of two-state Markov chains.

***13** According to researchers, people can be classified as ‘happy’, ‘neither happy nor sad’ or ‘sad’. However, the way people feel changes from day to day. Suppose that in a particular study:

- 80% of people who are happy today will be happy tomorrow.
 - 15% of people who are happy today will be neither happy nor sad tomorrow
 - 5% of people who are happy today will be sad tomorrow
 - 40% of people who are neither happy nor sad today will be happy tomorrow
 - 30% of people who are neither happy nor sad today will be the same tomorrow
 - 30% of people who are neither happy nor sad today will be sad tomorrow
 - 35% of people who are sad today will be happy tomorrow
 - 30% of people who are sad today will be neither happy nor sad tomorrow
 - 35% of people who are sad today will be sad tomorrow.
- a** Find the transition matrix that can be used to represent this information.
- b** At the beginning of the study 60% of people are identified as ‘happy’, 25% are ‘neither happy nor sad’ and 15% are ‘sad’. Write down a column matrix $\mathbf{S}_0$ that describes this situation.
- c** What percentages of people are expected to be happy, neither happy nor sad, or sad 1 day later?
- d** What percentages of people are expected to be happy, neither happy nor sad, or sad 5 days later?

***14** The temperature on any day can be described as ‘cool’, ‘mild’, ‘warm’ or ‘hot’. The likelihood of a day being in any one of these categories is dependent on the temperature the day before. The probabilities associated with each state are given by the following transition matrix:

$$\begin{array}{c} C \quad M \quad W \quad H \\ \begin{array}{c} C \\ M \\ W \\ H \end{array} \begin{bmatrix} 0.4 & 0.2 & 0.2 & 0.3 \\ 0.3 & 0.3 & 0.2 & 0.1 \\ 0.2 & 0.3 & 0.3 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.5 \end{bmatrix} \end{array}$$

- a** What is the probability that day 3 is hot if it is mild on day 0?
- b** What is the probability that day 3 is hot if it is hot on day 0?

16.3 Steady state of a Markov chain

Consider again the machine in Section 16.2, which at the start of the day can be in either of two states, working or broken. Suppose also that state 0 corresponds to the machine being broken, state 1 corresponds to the machine working, X_n is the state of the machine at time n , and

$$\begin{aligned} \Pr(X_{n+1} = 1 | X_n = 0) &= a & \Pr(X_{n+1} = 0 | X_n = 0) &= 1 - a \\ \Pr(X_{n+1} = 0 | X_n = 1) &= b & \Pr(X_{n+1} = 1 | X_n = 1) &= 1 - b \end{aligned}$$

* The Mathematical Methods Units 3 and 4 CAS study design requires only the study of two-state Markov chains.

Suppose that for a particular machine $a = 0.65$ and $b = 0.09$, so that the transition matrix

$$\mathbf{T} = \begin{bmatrix} 0.35 & 0.09 \\ 0.65 & 0.91 \end{bmatrix}.$$

How does the probability that the machine is working change as the Markov chain progresses and, in particular, what effect does the initial state of the machine have on the probability that the machine is in a particular state on day n ?

If the machine is broken at the beginning, then $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and

$$\mathbf{S}_1 = \begin{bmatrix} 0.35 & 0.09 \\ 0.65 & 0.91 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.35 \\ 0.65 \end{bmatrix}, \text{ so that the probability that the machine is working by the end of day 1 is } 0.65.$$

If the machine is working at the beginning, then $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and

$$\mathbf{S}_1 = \begin{bmatrix} 0.35 & 0.09 \\ 0.65 & 0.91 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.09 \\ 0.91 \end{bmatrix}, \text{ so that the probability that the machine is working by the end of day 1 is } 0.91.$$

What are the probabilities associated with the states of the machine at the end of day 2?

$$\mathbf{T}^2 = \begin{bmatrix} 0.35 & 0.09 \\ 0.65 & 0.91 \end{bmatrix}^2 = \begin{bmatrix} 0.1810 & 0.1134 \\ 0.8190 & 0.8866 \end{bmatrix}$$

If the machine is broken at the beginning, then $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{S}_2 = \mathbf{T}^2 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.181 \\ 0.819 \end{bmatrix}$, so that the probability that the machine is working by the end of day 2 is 0.819.

If the machine is working at the beginning, then $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{S}_2 = \mathbf{T}^2 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1134 \\ 0.8866 \end{bmatrix}$, so that the probability that the machine is working by the end of day 2 is 0.8866.

Continuing in this way, the following table of probabilities can be established (probabilities are given to four decimal places):

Day (n)	$\Pr(X_n = 1 X_0 = 0)$	$\Pr(X_n = 1 X_0 = 1)$
1	0.65	0.91
2	0.819	0.8866
3	0.8629	0.8805
4	0.8744	0.8789
5	0.8773	0.8785
6	0.8781	0.8784
7	0.8783	0.8784
8	0.8784	0.8784
9	0.8784	0.8784

From the table an interesting phenomenon can be observed: by the time the Markov chain reaches day 8, the probability that the machine will be working is the same whether or not the machine was working in the initial state, at least to four decimal places. Furthermore, the probability that the machine is working remains constant, to four decimal places, from step 9 onwards. The limiting value of each probability is called a **steady state** probability of the Markov chain. All two-state Markov chains, with a transition matrix with all non-zero entries, will eventually converge to a steady state.

Using a calculator the long run values for these probabilities can be determined by evaluating $\lim_{n \rightarrow \infty} \begin{bmatrix} 0.35 & 0.09 \\ 0.65 & 0.91 \end{bmatrix}^n$ which gives the steady state probability matrix $\mathbf{T}_s$, and

$$\mathbf{T}_s = \begin{bmatrix} 0.1216 & 0.1216 \\ 0.8784 & 0.8784 \end{bmatrix}, \text{ correct to four decimal places.}$$

Where this limit exists for a two-state Markov chain, it is possible to find general expressions for the steady state probabilities $\Pr(X_n = 0)$ and $\Pr(X_n = 1)$. By definition, the steady state probabilities are such that:

$$\begin{aligned} \Pr(X_{n+1} = 0) &= \Pr(X_n = 0) \\ \text{and} \quad \Pr(X_{n+1} = 1) &= \Pr(X_n = 1) \end{aligned}$$

Suppose that the machine is at the steady state at time n , and consider then the probabilities at time $n + 1$:

$$\begin{bmatrix} \Pr(X_{n+1} = 0) \\ \Pr(X_{n+1} = 1) \end{bmatrix} = \begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix} \begin{bmatrix} \Pr(X_n = 0) \\ \Pr(X_n = 1) \end{bmatrix} = \begin{bmatrix} \Pr(X_n = 0) \\ \Pr(X_n = 1) \end{bmatrix}$$

Expanding gives:

$$(1-a)\Pr(X_n = 0) + b\Pr(X_n = 1) = \Pr(X_n = 0) \quad (1)$$

$$a\Pr(X_n = 0) + (1-b)\Pr(X_n = 1) = \Pr(X_n = 1) \quad (2)$$

From (1)

$$\begin{aligned} (1-a)\Pr(X_n = 0) + b(1 - \Pr(X_n = 0)) &= \Pr(X_n = 0) \\ \therefore \Pr(X_n = 0) - a\Pr(X_n = 0) + b - b\Pr(X_n = 0) &= \Pr(X_n = 0) \\ \therefore \Pr(X_n = 0) - a\Pr(X_n = 0) - b\Pr(X_n = 0) - \Pr(X_n = 0) &= -b \\ \therefore -a\Pr(X_n = 0) - b\Pr(X_n = 0) &= -b \\ \therefore \Pr(X_n = 0)(-a - b) &= -b \\ \therefore \Pr(X_n = 0) &= \frac{b}{a+b} \end{aligned}$$

and hence

$$\Pr(X_n = 1) = \frac{a}{a+b}$$

This can be written in general as follows.

If a two-state Markov chain is described by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

then the steady state probabilities are:

$$\Pr(X_n = 0) = \frac{b}{a+b}$$

$$\Pr(X_n = 1) = \frac{a}{a+b}$$

Example 9

Find the steady state probabilities for the machine previously described, where:

$$\Pr(X_{n+1} = 1 | X_n = 0) = a = 0.65$$

$$\Pr(X_{n+1} = 0 | X_n = 1) = b = 0.09$$

Solution

Substituting in the formulas for the steady state probabilities:

$$\Pr(X_n = 0) = \frac{b}{a+b} = \frac{9}{74}$$

$$\Pr(X_n = 1) = \frac{a}{a+b} = \frac{65}{74}$$

These values are consistent with those shown in the table.

To find the percentages associated with the steady state of a system, the steady state probabilities just need to be expressed as percentages. To find the numbers associated with the steady state of a system, the population is apportioned according to the steady state probabilities. Each of these are illustrated in the following example.

Example 10

For the system described in Example 9:

- a Find, to the nearest per cent, the percentage of machines that would be expected to be working in the long term, if 80% are working and 20% are broken at the beginning of the first day.
- b Find, to the nearest whole number, the number of machines that would be expected to be working in the long term, if 145 are working and 32 are broken at the beginning of the first day.

Solution

- a** From Example 9 the steady state probability that the machine is working is:

$$\Pr(X_n = 1) = \frac{65}{74} \approx 0.8784$$

Thus, the percentage of machines working in the long term is 88%. This percentage is not dependent on the initial state of the system.

- b** Again the steady state probability that the machine is working is:

$$\Pr(X_n = 1) = \frac{65}{74}$$

The total number of machines is $145 + 32 = 177$, so the estimated number of machines working in the long term is $177 \times \frac{65}{74} = 155$ to the nearest whole number.

Once again, this number is not dependent on the initial state of the system.

Convergence to a steady state of the probabilities is also not guaranteed for Markov chains that are three-state or more. Consider for example the following system where:

- a person who catches the train to work one day has a probability of 0.6 of catching the train again the next day, 0.4 of catching the bus the next day, and 0 of driving to work the next day
- a person who catches the bus to work one day has a probability of 0.3 of catching the train the next day, 0.7 of catching the bus again the next day, and 0 of driving to work the next day
- a person who drives to work one day has a probability of 0 of catching the train the next day, 0 of catching the bus again the next day, and 1 of driving to work the next day.

This gives the following transition matrix:

$$\mathbf{T} = \begin{bmatrix} 0.6 & 0.3 & 0 \\ 0.4 & 0.7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculating the probabilities associated with this transition matrix shows that they continue to depend on the initial state of the system, as can be seen in the following table where probabilities are calculated correct to four decimal places.

Step (n)	$\Pr(X_n = 0 X_0 = 0)$	$\Pr(X_n = 0 X_0 = 1)$	$\Pr(X_n = 0 X_0 = 2)$
1	0.6000	0.3000	0
2	0.4800	0.3900	0
3	0.4440	0.4170	0
4	0.4332	0.4251	0
5	0.4300	0.4275	0
6	0.4290	0.4283	0
7	0.4287	0.4285	0
8	0.4286	0.4285	0
9	0.4286	0.4286	0

It can be seen that in this Markov chain, if the system is in state 0 or 1 initially, then the probability of eventually achieving state 0 does not depend on which of these states the system is in, but if the system is in state 2 initially it can never achieve state 0. That is, the Markov chain does not achieve a steady state. To achieve a steady state it is necessary that every state eventually be accessible from any other state.

Exercise 16C

- 1 Suppose that the probability of a darts player winning a game is 0.5 if he won the preceding game and 0.4 if he lost the preceding game. Suppose $X_n = 1$ if the player wins the n th game and $X_n = 0$ if the player loses the n th game.
 - a Write down the transition matrix that describes this situation.
 - b Construct a table giving the values for $\Pr(X_n = 1|X_0 = 0)$ and $\Pr(X_n = 1|X_0 = 1)$ to four decimal places for $n = 1, 2, 3, 4, 5$ and 6.
 - c From the table, find the steady state probability that $\Pr(X_n = 1)$, and hence determine the steady state probability for $\Pr(X_n = 0)$.
 - d Use the formula for steady state probabilities to verify your answer in part c.
- 2 Suppose that the probability that Millie is on time for work is 0.85 if she was on time the day before, and her probability of being on time for work is 0.95 if she was late the day before. Suppose $X_n = 0$ if Millie is on time on the n th day, and $X_n = 1$ if Millie is late on the n th day.
 - a Write down the transition matrix which describes this situation.
 - b Construct a table giving the values for $\Pr(X_n = 0|X_0 = 0)$ and $\Pr(X_n = 0|X_0 = 1)$ to four decimal places for $n = 1, 2, 3, 4, 5$ and 6.
 - c From the table, find the steady state probability that $\Pr(X_n = 0)$, and hence determine the steady state probability for $\Pr(X_n = 1)$.
 - d Use the formula for steady state probabilities to verify your answer in part c.
- 3 Suppose that there are two dentists in a country town, Dr Hatchet and Dr Youngblood. Each year, 10% of the patients of Dr Hatchet move to Dr Youngblood, while 18% of patients move from Dr Youngblood to Dr Hatchet. Suppose that initially 50% of patients go to Dr Hatchet and 50% go to Dr Youngblood. Find the percentage of patients, correct to one decimal place, who will eventually be attending each dentist if this pattern continues indefinitely.
- 4 There are only two choices of supermarket in a country town. It has been found that 74% of customers will continue to purchase their groceries from store A if they purchased their groceries from store A in the previous week, while 86% of consumers will continue to purchase their groceries from store B if they purchased their groceries from store B in the previous week. Suppose that in the previous week (week 0) store A recorded 563 customers and store B recorded 226 customers. Assuming that the total number of customers remains the same, find:

- a** the transition matrix that can be used to represent this information
- b** the number of customers at each store in week 3, to the nearest whole number
- c** the number of customers shopping in each store in the long term, to the nearest whole number
- 5** For the transition matrix $\mathbf{T} = \begin{bmatrix} 0.3 & 0.4 \\ 0.7 & 0.6 \end{bmatrix}$ and an initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 20 \\ 30 \end{bmatrix}$:
- a** Calculate $\mathbf{S}_n = \mathbf{T}^n \mathbf{S}_0$ for $n = 10, 15, 20$ and 50 to find the steady state solution, correct to four decimal places.
- b** Use the steady state formulas to find the result.
- 6** For the transition matrix $\mathbf{T} = \begin{bmatrix} 0.3 & 0.5 \\ 0.7 & 0.5 \end{bmatrix}$ and an initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:
- a** Calculate $\mathbf{S}_n = \mathbf{T}^n \mathbf{S}_0$ for $n = 10, 15, 20$ and 50 .
- b** Use the steady state formulas to find the steady state solution.
- *7** A Markov chain has transition matrix $\mathbf{T} = \begin{bmatrix} 0.15 & 0.45 & 0.2 \\ 0.35 & 0.25 & 0.45 \\ 0.5 & 0.3 & 0.35 \end{bmatrix}$
- a** If the initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ calculate $\mathbf{S}_n = \mathbf{T}^n \mathbf{S}_0$ for $n = 10, 15, 20$ and 25 to find an approximate steady state solution. Give your answer to four decimal places.
- b** If the initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ calculate $\mathbf{S}_n = \mathbf{T}^n \mathbf{S}_0$ for $n = 10, 15, 20$ and 25 to find an approximate steady state solution. Give your answer to four decimal places.
- c** If the initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ calculate $\mathbf{S}_n = \mathbf{T}^n \mathbf{S}_0$ for $n = 10, 15, 20$ and 25 to find an approximate steady state solution. Give your answer to four decimal places.
- d** Compare your answers to parts **a**, **b** and **c**.
- 8** Suppose that cars can be rented at either of two locations, Melbourne or Ballarat. Assume that all cars are rented at the beginning of the week for 1 week. Records show that:
- 70% of cars hired in Melbourne are returned to Melbourne and 30% are returned to Ballarat
 - 75% of cars hired in Ballarat are returned to Ballarat and 25% are returned to Melbourne.
- a** Find the transition matrix that can be used to represent this information.
- b** At the beginning of the week there are 150 cars in Melbourne and 80 in Ballarat. Find how many cars are estimated to be at each location in the long term.

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- 9 According to researchers, people can be classified as happy (H) or sad (S). However, the way people feel changes from day to day. Suppose this situation can be described by the following transition matrix:

$$\begin{array}{c} H \quad S \\ H \begin{bmatrix} 0.6 & 0.1 \\ 0.4 & 0.9 \end{bmatrix} \\ S \end{array}$$

What percentages of people are expected to be happy or sad in the long term?

- *10 Suppose that the trees in a forest are assumed to fall into three age groups: young trees (0–15 years), middle-aged trees (16–30 years) and old trees (more than 30 years). A time period is 15 years, and it is assumed that in each time period:
- 10% of young trees, 20% of middle-aged trees and 40% of old trees die
 - surviving trees enter into the next age group; old trees remain old
 - dead trees are replaced by young trees.

This information is summarised in the following transition matrix:

$$\begin{array}{c} Y \quad M \quad O \\ Y \begin{bmatrix} 0.1 & 0.2 & 0.4 \\ 0.9 & 0 & 0 \\ 0 & 0.8 & 0.6 \end{bmatrix} \\ M \\ O \end{array}$$

Suppose that 20 000 young trees are planted in the forest. What is the eventual age distribution of the trees? Give your answers to the nearest whole number.

- *11 The temperature on any day can be described as ‘cool’, ‘mild’, ‘warm’ or ‘hot’. The likelihood of a day being in any one of these categories is dependent on the temperature the day before. The probabilities associated with each state are given by the following transition matrix:

$$\begin{array}{c} C \quad M \quad W \quad H \\ C \begin{bmatrix} 0.4 & 0.2 & 0.2 & 0.3 \\ 0.3 & 0.3 & 0.2 & 0.1 \\ 0.2 & 0.3 & 0.3 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.5 \end{bmatrix} \\ M \\ W \\ H \end{array}$$

What percentages of days are expected to be cool, mild, warm or hot in the long term, correct to one decimal place?

16.4 Comparing run length for Bernoulli sequences and Markov chains

Consider a Bernoulli sequence, where independent trials can result in either a success or a failure, and a two-state Markov chain, where the outcomes are also designated as either a

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success or a failure. The number of times that the same outcome is observed in sequence is called the **length of run** for the sequence.

In a Bernoulli sequence the question of interest is often the number of failures before the first success is observed. For example, how many tails might be observed when a fair coin is tossed before a head is observed? Possible sequences might be:

H	which is a run of length 0
TH	which is a run of length 1
TTH	which is a run of length 2
TTTH	which is a run of length 3
TTTTH	which is a run of length 4

and so on.

This situation gives rise to a distribution called the **geometric** distribution, which gives the probabilities associated with the number of failures observed before the first success. If p is the probability of success then the geometric distribution is described by the distribution function:

$$\Pr(X = x) = (1 - p)^x p \quad x = 0, 1, 2, \dots$$

Example 11

Find the probability that a fair die will need to be rolled five times before a ‘six’ is observed on the sixth roll.

Solution

Here, $p = \frac{1}{6}$ and

$$\Pr(X = 5) = \left(1 - \frac{1}{6}\right)^5 \frac{1}{6} = 0.0670$$

Example 12

Suppose a biased coin, where $\Pr(H) = 0.9$, is tossed. Find the probability that:

- a three heads in a row are observed before a tail
- b six heads in a row are observed before a tail
- c three tails in a row are observed before a head
- d six tails in a row are observed before a head

Solution

- a The probability of a success $p = 0.1$ and $\Pr(X = 3) = (1 - 0.1)^3 0.1 = 0.0729$
- b The probability of a success $p = 0.1$ and $\Pr(X = 6) = (1 - 0.1)^6 0.1 = 0.053144$
- c The probability of a success $p = 0.9$ and $\Pr(X = 3) = (1 - 0.9)^3 0.9 = 0.0009$
- d The probability of a success $p = 0.9$ and $\Pr(X = 6) = (1 - 0.9)^6 0.9 = 0.0000009$

It can be seen from Example 12 that, in a Bernoulli sequence, the more likely an outcome, the greater the likelihood of a long sequence of that outcome. Although it would be possible to observe runs of heads of length six here, it would be extremely unlikely to observe runs of tails of length six. Even observing three tails in a row is quite an unlikely outcome in that example.

In a Bernoulli sequence where there are two outcomes A and B , then $\Pr(A) = 1 - \Pr(B)$. Thus, it follows that long runs of outcome A would be associated with short runs of the outcome B , and vice versa.

This, however, is not necessarily the case with the length of run for a Markov chain. Since the probabilities at each step are conditional probabilities, it follows that for a two-state chain long runs of both of the outcomes might be observed. Suppose, for example, the probability that it is wet one day given that it was wet the day before is 0.75, whereas the probability that it is dry one day given that it was dry the day before is 0.7, giving the following transition matrix:

$$\mathbf{T} = \begin{bmatrix} \Pr(W_{i+1}|W_i) & \Pr(W_{i+1}|D_i) \\ \Pr(D_{i+1}|W_i) & \Pr(D_{i+1}|D_i) \end{bmatrix} = \begin{bmatrix} 0.75 & 0.3 \\ 0.25 & 0.7 \end{bmatrix}$$

Suppose that the initial state of the system is wet. Then the probability, correct to four decimal places, of x wet days in a row before a dry day is:

x	0	1	2	3	4
$\Pr(X = x)$	0.2500	0.75×0.25 $= 0.1875$	$0.75^2 \times 0.25$ $= 0.1406$	$0.75^3 \times 0.25$ $= 0.1055$	$0.75^4 \times 0.25$ $= 0.0791$

Suppose that the initial state of the system is dry. Then the probability, correct to four decimal places, of y dry days in a row before a wet day is:

y	0	1	2	3	4
$\Pr(Y = y)$	0.3000	0.7×0.3 $= 0.2100$	$0.7^2 \times 0.3$ $= 0.1470$	$0.7^3 \times 0.3$ $= 0.1029$	$0.7^4 \times 0.3$ $= 0.0720$

Since in Markov sequences the probabilities are conditional, it follows that the system may demonstrate either long or short runs of both of the outcomes.

Example 13

For the Markov chain defined by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} \Pr(W_{i+1}|W_i) & \Pr(W_{i+1}|D_i) \\ \Pr(D_{i+1}|W_i) & \Pr(D_{i+1}|D_i) \end{bmatrix} = \begin{bmatrix} 0.75 & 0.3 \\ 0.25 & 0.7 \end{bmatrix}$$

where W_i is the event day i is wet, and D_i is the event day i is dry, what is the probability that, given that the first day is dry:

- there will be a run of 4 wet days in a row before the next dry day?
- there will be a run of 4 more dry days in a row before the next wet day?

Solution

- a** If the initial state is a dry day then the probability of 4 wet days followed by a dry day is given by:

$$0.3 \times 0.75^3 \times 0.25 = 0.0316$$

- b** If the initial state is a dry day then the probability of 4 more dry days followed by a wet day is given by:

$$0.7^4 \times 0.3 = 0.07203$$

Exercise 16D

Give all answers correct to 4 decimal places for this exercise.

- 1** It is known in a certain town that on any evening 35% of people will not be at home. What is the probability that a telemarketer will need to make five phone calls before finding someone at home on the sixth call?
- 2** It is known that the proportion of voters in an electorate who prefer candidate A is 60%. What is the probability that the first four people asked prefer candidate B , and the fifth prefers candidate A ?
- 3** The manager of a shop knows from experience that 60% of her customers will use a credit card to pay for their purchases. Find the probability that:
 - a** the next three customers will use a credit card, and the fourth will pay cash
 - b** the next three customers pay cash, and the fourth will use a credit card
- 4** Suppose a biased coin, where $\Pr(H) = 0.7$ is tossed. Find the probability that:
 - a** three heads in a row are observed before a tail
 - b** six heads in a row are observed before a tail
 - c** three tails in a row are observed before a head
 - d** six tails in a row are observed before a head
- 5** For the Markov chain defined by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} \Pr(W_{i+1}|W_i) & \Pr(W_{i+1}|D_i) \\ \Pr(D_{i+1}|W_i) & \Pr(D_{i+1}|D_i) \end{bmatrix} = \begin{bmatrix} 0.65 & 0.28 \\ 0.35 & 0.72 \end{bmatrix}$$

where W_i is the event day i is wet, and D_i is the event day i is dry, find the probability that, given that the first day is wet:

- a** there will be a run of four more wet days in a row before the next dry day
- b** there will be a run of four dry days in a row before the next wet day

- 6 Consider the goal shooter of a netball team, whose probabilities of goaling (G) or missing (M) are conditional on the outcome of her preceding attempt as follows:

$$\begin{aligned}\Pr(G_{i+1}|G_i) &= \frac{4}{5} & \Pr(G_{i+1}|M_i) &= \frac{1}{3} \\ \Pr(M_{i+1}|G_i) &= \frac{1}{5} & \Pr(M_{i+1}|M_i) &= \frac{2}{3}\end{aligned}$$

Find the probability that:

- a the goal shooter scores a total of six goals in a row before she misses if she scores at her first attempt
 - b the goal shooter misses a total of six attempts in a row before she scores if she misses at her first attempt.
- 7 For the Markov chain defined by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} \Pr(S_{i+1} = 0|S_i = 0) & \Pr(S_{i+1} = 0|S_i = 1) \\ \Pr(S_{i+1} = 1|S_i = 0) & \Pr(S_{i+1} = 1|S_i = 1) \end{bmatrix} = \begin{bmatrix} 0.53 & 0.34 \\ 0.47 & 0.66 \end{bmatrix}$$

where $S_i = 0$ if it does not snow on day i , and where $S_i = 1$ if it does snow on day i . Find the probability that, given that $S_0 = 0$:

- a the next 3 days will snow and on the fourth day it will not snow
 - b the next 3 days will not snow and on the fourth day it will snow
- 8 Suppose that the probability that an operating machine breaks down by the end of the day is 0.10, and the probability that a broken machine is repaired by the end of the day is 0.55. Find:
- a the probability that a machine that breaks down on day 0 will not be working until day 5
 - b the probability that a machine that is working day 0 will break down for the first time on day 5



Chapter summary

- Matrices can be used to represent conditional probabilities and, by multiplication of matrices, problems associated with the law of total probability can be evaluated.
- A two-state **Markov chain** describes a situation of repetitions of an experiment where:
 - each step (or trial) results in one of two possible outcomes
 - the probability of each possible outcome is *conditional* only on the result of the step immediately preceding it
 - the conditional probabilities for each possible outcome are the same on each occasion (that is, the same matrix is used for each transition).
- The state of the system at the beginning of the Markov chain is called the **initial state**, usually denoted $\mathbf{S}_0$ and, for an m -state Markov chain, is a column matrix of dimension $m \times 1$.
- A **transition matrix** $\mathbf{T}$ is a matrix giving the probability for each of the possible outcomes at each step of a Markov chain conditional on each of the possible outcomes at the previous step. For an m -state Markov chain $\mathbf{T}$ is a square matrix of dimension $m \times m$. If X_n is the outcome at time n , then for a two-state Markov chain:

$$\mathbf{T} = \begin{bmatrix} \Pr(X_{n+1} = 0 | X_n = 0) & \Pr(X_{n+1} = 0 | X_n = 1) \\ \Pr(X_{n+1} = 1 | X_n = 0) & \Pr(X_{n+1} = 1 | X_n = 1) \end{bmatrix}$$

- The **transition matrix** $\mathbf{T}$ enables each step of a Markov chain to be predicted from the previous step, according to the rules:

$$\mathbf{S}_{n+1} = \mathbf{T} \times \mathbf{S}_n$$

$$\text{and} \quad \mathbf{S}_n = \mathbf{T}^n \times \mathbf{S}_0$$

- Eventually all two-state Markov chains, with a transition matrix with all non-zero entries, will converge to a state where the probabilities are no longer dependent on the initial state. These probabilities are called the **steady state** probabilities of the Markov chain. If a two-state Markov chain is described by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

then the steady state probabilities are:

$$\Pr(X_n = 0) = \frac{b}{a+b}$$

$$\Pr(X_n = 1) = \frac{a}{a+b}$$

- Convergence to a steady state of the probabilities is not guaranteed for Markov chains that are three-state or more.
- In a Bernoulli sequence, long runs of one outcome are associated with short runs of the alternative outcome.
- In a Markov sequence where the probabilities are conditional, the system may demonstrate either long or short runs of both of the outcomes.

Multiple-choice questions

- 1 Consider the matrices

$$\mathbf{U} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 0.43 \\ 0.45 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 2 & 0 \\ 1 & 1 & 2 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 0.1 & 0.2 \\ 0.2 & 0.4 \end{bmatrix}$$

The matrix that could be a transition matrix for a Markov chain is

A $\mathbf{U}$ **B** $\mathbf{V}$ **C** $\mathbf{W}$ **D** $\mathbf{X}$ **E** $\mathbf{Y}$

- 2 Suppose that a two-state Markov chain is defined by a transition matrix
- $\mathbf{T}$
- and an initial state matrix
- $\mathbf{S}_0$
- . Which of the following is a true statement?

A The dimension of $\mathbf{T}$ is 2×2 and the dimension of $\mathbf{S}_0$ is 2×1 .
B The dimension of $\mathbf{T}$ is 2×1 and the dimension of $\mathbf{S}_0$ is 1×2 .
C The dimension of $\mathbf{T}$ is 2×2 and the dimension of $\mathbf{S}_0$ is 1×2 .
D The dimension of $\mathbf{T}$ is 2×2 and the dimension of $\mathbf{S}_0$ is 1×1 .
E The dimension of $\mathbf{T}$ is 2×2 and the dimension of $\mathbf{S}_0$ is 2×2 .

The following information is needed for Questions 3 to 6:

A Markov chain is defined by a transition matrix $\mathbf{T} = \begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.5 \end{bmatrix}$ and an initial state matrix

$$\mathbf{S}_0 = \begin{bmatrix} 100 \\ 200 \end{bmatrix}.$$

- 3 For this Markov chain,
- $\mathbf{S}_1 =$

A $\begin{bmatrix} 60 \\ 200 \end{bmatrix}$ **B** $\begin{bmatrix} 140 \\ 160 \end{bmatrix}$ **C** $\begin{bmatrix} 160 \\ 140 \end{bmatrix}$ **D** $\begin{bmatrix} 166 \\ 144 \end{bmatrix}$ **E** $\begin{bmatrix} 200 \\ 100 \end{bmatrix}$

- 4 For this Markov chain,
- $\mathbf{T}^2$
- is closest to:

A $\begin{bmatrix} 0.36 & 0.25 \\ 0.16 & 0.25 \end{bmatrix}$ **B** $\begin{bmatrix} 0.56 & 0.55 \\ 0.44 & 0.45 \end{bmatrix}$ **C** $\begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.5 \end{bmatrix}$
D $\begin{bmatrix} 1.2 & 1.0 \\ 0.8 & 1.0 \end{bmatrix}$ **E** not defined

- 5 For this Markov chain,
- $\mathbf{S}_2$
- is closest to:

A $\begin{bmatrix} 160 \\ 140 \end{bmatrix}$ **B** $\begin{bmatrix} 166 \\ 134 \end{bmatrix}$ **C** $\begin{bmatrix} 166.7 \\ 133.3 \end{bmatrix}$ **D** $\begin{bmatrix} 640 \\ 560 \end{bmatrix}$ **E** $\begin{bmatrix} 400 \\ 800 \end{bmatrix}$

- 6 For this Markov chain, the steady state solution is closest to:

A $\begin{bmatrix} 166.5 \\ 133.5 \end{bmatrix}$ **B** $\begin{bmatrix} 166.6 \\ 133.4 \end{bmatrix}$ **C** $\begin{bmatrix} 166.7 \\ 133.3 \end{bmatrix}$ **D** $\begin{bmatrix} 166.8 \\ 133.2 \end{bmatrix}$ **E** $\begin{bmatrix} 166.9 \\ 133.1 \end{bmatrix}$

The following information is needed for Questions 7 and 8:

A factory has a large number of machines that can be in one of two states, operating or broken. The probability that an operating machine breaks down by the end of the day is 0.05 and the probability that a broken machine is repaired by the end of the day is 0.80.

- 7 A transition matrix $\mathbf{T}$ that can be used to represent this information is:

A $\begin{bmatrix} 0.95 & 0.20 \\ 0.05 & 0.80 \end{bmatrix}$ **B** $\begin{bmatrix} 0.05 & 0.20 \\ 0.95 & 0.80 \end{bmatrix}$ **C** $\begin{bmatrix} 0.05 \\ 0.80 \end{bmatrix}$

D $\begin{bmatrix} 0.9125 & 0.3500 \\ 0.0875 & 0.6500 \end{bmatrix}$ **E** $\begin{bmatrix} 0.95 & 0.80 \\ 0.05 & 0.20 \end{bmatrix}$

- 8 If a machine is operating at the end of day 1, then the probability that it is broken by the end of day 6 can be found by evaluating:

A $\mathbf{T}^6 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ **B** $\mathbf{T}^5 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ **C** $\mathbf{T}^6 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ **D** $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \times \mathbf{T}^6$ **E** $\mathbf{T}^6 \times \begin{bmatrix} 0 & 1 \end{bmatrix}$

The following information is needed for Questions 9 and 10:

Suppose that the probability of a squash player winning a point is 0.7 if she has won the preceding point and 0.4 if she has lost the preceding point. Suppose $X_n = 1$ if the player wins the n th point and $X_n = 0$ if the player loses the n th point.

- 9 The steady state probability, $\Pr(X_n = 1)$ is equal to:

A 0.55 **B** 0.57 **C** $\frac{3}{7}$ **D** $\frac{4}{7}$ **E** 0.6

- 10 The probability that the player wins the three points in a row after losing the first point, and then loses the next point: is

A 0.1960 **B** 0.0412 **C** 0.0192 **D** 0.0588 **E** 0.1029

Short-answer questions (technology-free)

- 1 For the transition matrix $\mathbf{T} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$ and an initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, use the relationship $\mathbf{S}_n = \mathbf{T}\mathbf{S}_{n-1}$ to determine:

a $\mathbf{S}_1$ **b** $\mathbf{S}_2$

- 2 For the transition matrix $\mathbf{T} = \begin{bmatrix} 0.7 & 0.5 \\ 0.3 & 0.5 \end{bmatrix}$ and an initial state matrix $\mathbf{S}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ use the relationship $\mathbf{S}_n = \mathbf{T}\mathbf{S}_{n-1}$ to determine:

a $\mathbf{S}_1$ **b** $\mathbf{S}_2$

- 3 Suppose that the probability of a child being male is $\frac{3}{5}$ if the preceding child is male, and $\frac{3}{7}$ if the preceding child is female. If M_i is the event that the i th child is male, and F_i is the event that the i th child is female then this can be written in the form:

$$\begin{bmatrix} \Pr(F_{i+1}) \\ \Pr(M_{i+1}) \end{bmatrix} = \mathbf{T} \times \begin{bmatrix} \Pr(F_i) \\ \Pr(M_i) \end{bmatrix}$$

where $\mathbf{T}$ is a 2×2 matrix. Write down the matrix $\mathbf{T}$.

- 4 Suppose that the probability of snow on Mount Buller is dependent on whether or not it has snowed on the previous day. If it has snowed the day before the probability of snow is 0.8. If it has not snowed on the previous day then the probability of snow is 0.1. If it has snowed on Thursday, what is the probability that it will not snow on the following Saturday?
- 5 Suppose that a football team can win or lose each match. Let state 0 correspond to the team winning and state 1 correspond to the team losing. The probabilities associated with each state are given by the following transition matrix:

$$\begin{bmatrix} 0.64 & 0.44 \\ 0.36 & 0.56 \end{bmatrix}$$

What is the probability that the team lose game 2 of the season if they win game 1?

- 6 Suppose that there are two sports clubs in a town. Each year, 22% of the members of club A move to club B, and while 32% of the members of club B move to club A. Suppose that initially 40% of members belong to club A and 60% to club B. Find the percentage of members who will eventually belong to each club in the long term.
- 7 For the Markov chain defined by the transition matrix:

$$\mathbf{T} = \begin{bmatrix} \Pr(W_{i+1}|W_i) & \Pr(W_{i+1}|D_i) \\ \Pr(D_{i+1}|W_i) & \Pr(D_{i+1}|D_i) \end{bmatrix} = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$$

where W_i is the event day i is wet, and D_i is the event day i is dry, what is the probability that, given that the first day is wet, there will be a run of 3 more wet days in a row before the next dry day?

Extended-response questions

- 1 Suppose that the outcome of a test series between Australia and India follows a Markov sequence. If $X_i = 0$ if India wins match i , and $X_i = 1$ if Australia wins match i then:

$$\begin{aligned} \Pr(X_0 = 0) &= \frac{4}{9} & \Pr(X_{i+1} = 0|X_i = 0) &= \frac{1}{2} & \Pr(X_{i+1} = 0|X_i = 1) &= \frac{1}{3} \\ \Pr(X_0 = 1) &= \frac{5}{9} & \Pr(X_{i+1} = 1|X_i = 0) &= \frac{1}{2} & \Pr(X_{i+1} = 1|X_i = 1) &= \frac{2}{3} \end{aligned}$$

- a What is the probability that Australia wins the first three matches?
- b What is the probability that India wins the first three matches?
- c Write down the transition matrix that describes this situation, and use it to determine the probability that India wins match 3:
- if India wins the first match
 - if Australia wins the first match
- d Find the steady state probability of Australia winning a test match, $\Pr(X_n = 1)$.

- 2** Suppose that there are two choices of transport to the city: bus and train. It has been found that 84% of people will continue to travel by bus if they travelled by bus the previous week, while 77% of people will continue to travel by train if they travelled by train the previous week.
- a** Find the transition matrix that can be used to represent this information.
 - b** Suppose that in week 0, 218 people travelled by bus and 455 people travelled by train. Find the estimated number of people travelling by bus and train in week 3, if the total number of people travelling remains the same.
 - c** Find the long run estimate of the number of customers using each of the two methods of travel if the total number of people travelling remains the same.
 - d** Given that they travelled by bus last week, find the probability that a person will travel by train for the next 3 weeks in a row, and then by bus the following week.
- 3** Suppose that library books can be borrowed for 1 month and returned at either of two branch libraries: Camberwell or Hawthorn. Assume that all books are borrowed at the beginning of the week and returned at the end of the week. Records show that:
- 56% of books borrowed in Camberwell are returned to Camberwell and 44% are returned to Hawthorn
 - 74% of books borrowed in Hawthorn are returned to Hawthorn and 26% are returned to Camberwell.
- a** Find the transition matrix that can be used to represent this information.
 - b** At the beginning of January 30% of the books are in Camberwell and 70% are in Hawthorn. Find:
 - i** the percentage of books estimated to be at each location at the end of week 3, correct to two decimal places
 - ii** the percentage of books estimated to be at each location in the long term, correct to two decimal places

Continuous random variables and their probability distributions

Objectives

- To define **continuous random variables**.
- To specify the distributions for continuous random variables using **probability density functions**.
- To calculate and interpret the **expectation (mean), median, mode, variance** and **standard deviation** for a continuous random variable.
- To relate the **graph** of a probability density function to the values of the **parameters** defining that function.
- To relate probabilities for intervals to the graph of a probability density function.
- To use calculus to calculate probabilities for intervals for a probability density function.
- To use technology to calculate probabilities for intervals for a probability density function.
- To apply knowledge of probability density functions for continuous random variables to solve probability-related problems.

17.1 Continuous random variables

A continuous random variable is one that can take any value in an interval of the real number line. For example, if X is the random variable which takes its values as ‘distance in metres’ that a parachutist lands from a particular marker, then X is a continuous random variable, and here the values which X may take are the non-negative real numbers.

A continuous random variable has no limit as to the accuracy with which it can be measured. For example, let W be the random variable with values ‘a person’s weight in kilograms’, and W_i be the random variable with values ‘a person’s weight in kilograms measured to the i th decimal place’.

Then:	$W_0 = 83$	implies	$82.5 \leq W < 83.5$
	$W_1 = 83.3$	implies	$83.25 \leq W < 83.35$
	$W_2 = 83.28$	implies	$83.275 \leq W < 83.285$
	$W_3 = 83.281$	implies	$83.2805 \leq W < 83.2815$

and so on. Thus, the random variable W cannot take an exact value, since it is always rounded to the limits imposed by the method of measurement used. Hence, the probability of W being exactly equal to a particular value is zero, and this is true for all continuous random variables.

That is,

$$\Pr(W = w) = 0 \quad \text{for all } w$$

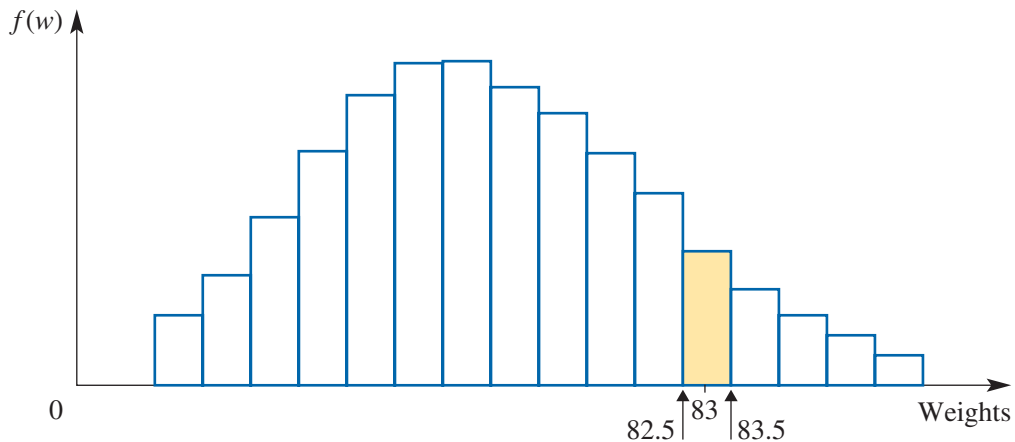
In practice, considering W taking a particular value is equivalent to W taking a value in an appropriate interval.

Thus, from above:

$$\Pr(W_0 = 83) = \Pr(82.5 \leq W < 83.5)$$

To determine the value of this probability you could begin by measuring the weight of a large number of randomly chosen people, and determine the proportion of the people in the group who have weights in this interval.

Suppose after doing this a histogram of weights was obtained as shown.



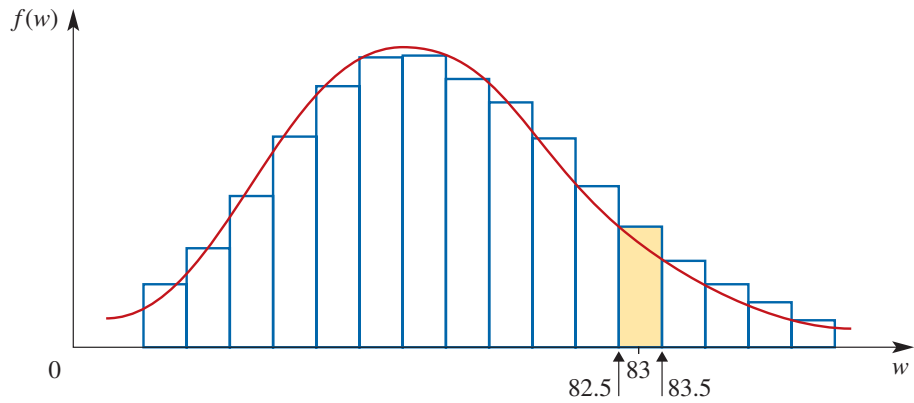
From this histogram:

$$\begin{aligned} \Pr(W_0 = 83) &= \Pr(82.5 \leq W < 83.5) \\ &= \frac{\text{shaded area from 82.5 to 83.5}}{\text{total area}} \end{aligned}$$

If the histogram is scaled so that the total area under the blocks is one, then:

$$\begin{aligned} P(W_0 = 83) &= \Pr(82.5 \leq W < 83.5) \\ &= \text{area under block from 82.5 to 83.5} \end{aligned}$$

Suppose that the sample size gets larger and the class interval width gets smaller. If theoretically this process is continued so that the intervals are arbitrarily small, the histogram can be modelled by a smooth curve, as shown in the following diagram.



The curve obtained here is of great importance for a continuous random variable.

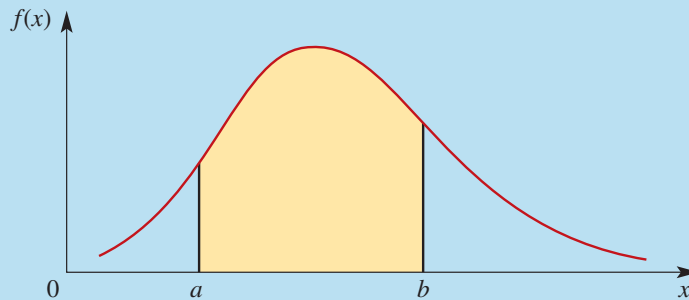
The function f whose graph models the histogram as the number of intervals is increased is called the **probability density function**. The probability density function f is used to describe the probability distribution of a continuous random variable, X .

Now, the probability of interest is no longer represented by the area under the histogram, but the area under the curve. That is:

$$\begin{aligned}\Pr(W_0 = 83) &= \Pr(82.5 \leq W < 83.5) \\ &= \text{area under the graph of the function with rule } f(w) \text{ from } 82.5 \text{ to } 83.5 \\ &= \int_{82.5}^{83.5} f(w)dw\end{aligned}$$

In general, for the continuous random variable X with density function f :

$$\Pr(a < X < b) = \int_a^b f(x)dx \text{ is given by the area of the shaded region.}$$



In order to be a probability density function a function must satisfy certain conditions.

If the range of the continuous random variable X is $[a, b]$ then the domain of its probability density function f is $[a, b]$. The probability density function will satisfy two properties:

- 1** $f(x) \geq 0$ for all $x \in [a, b]$ and **2** $\int_a^b f(x)dx = 1$

Note that $\Pr(X < c) = \Pr(X \leq c)$

$$= \int_a^c f(x)dx$$

The probability density function does not give probabilities and $f(x)$ may take values greater than one. The probabilities are given by areas determined by the graph of f .

In many cases the range of X is an unbounded interval, for example $[1, \infty)$ or indeed R . Therefore, some new notation is necessary.

If the range of X is $[1, \infty)$, then it would be required that the area of the region under the graph of the probability density function be one. This can be expressed as $\int_1^{\infty} f(x)dx = 1$.

This is computed as $\lim_{k \rightarrow \infty} \int_1^k f(x)dx$. If the probability density function f has domain R , then

$$\int_{-\infty}^{\infty} f(x)dx = 1. \text{ This is computed as } \lim_{k \rightarrow \infty} \int_{-k}^k f(x)dx.$$

Any probability density function f with domain $[a, b]$ (or any other interval) may be extended to a function with domain R by defining $f^*(x) = f(x)$ for all $x \in [a, b]$ and $f^*(x) = 0$ for all $x \notin [a, b]$.

This leads to the following:

A probability density function (or its natural extension) satisfies the following two properties:

- 1** $f(x) \geq 0$ for all x **2** $\int_{-\infty}^{\infty} f(x)dx = 1$

Note that, since the probability of X taking any exact value is zero, then:

$$\Pr(a < X < b) = \Pr(a \leq X < b) = \Pr(a < X \leq b) = \Pr(a \leq X \leq b)$$

That is, there is no difference between the numerical values of all of these expressions.

Example 1

Suppose that the random variable X has the density function with the rule:

$$f(x) = \begin{cases} cx & 0 \leq x \leq 2 \\ 0 & \text{if } x > 2 \text{ or } x < 0 \end{cases}$$

- a** Find the value of c that makes f a probability density function.
b Find $\Pr(X > 1.5)$.

Solution

- a** Since f is a probability density function, then:

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= 1 \\ \text{Now } \int_{-\infty}^{\infty} f(x) dx &= \int_0^2 cx dx \text{ since } f(x) = 0 \text{ elsewhere} \\ &= \left[\frac{cx^2}{2} \right]_0^2 \\ &= 2c \end{aligned}$$

Therefore

$$\begin{aligned} 2c &= 1 \\ c &= 0.5 \end{aligned}$$

$$\begin{aligned} \text{b } \Pr(X > 1.5) &= \int_{1.5}^2 0.5x dx \\ &= 0.5 \left[\frac{x^2}{2} \right]_{1.5}^2 \\ &= 0.5 \left(\frac{4}{2} - \frac{2.25}{2} \right) \\ &= 0.4375 \end{aligned}$$

Example 2

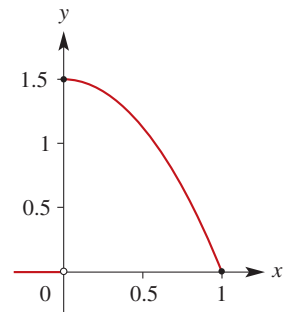
Consider the function f with the rule:

$$f(x) = \begin{cases} 1.5(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{if } x > 1 \text{ or } x < 0 \end{cases}$$

- a** Sketch the graph of f . **b** Show that f is a probability density function.
c Find $\Pr(X > 0.5)$.

Solution

- a** From the form of the function the graph of the probability density function is a parabola with intercepts at $(0, 1.5)$ and $(1, 0)$, as shown.



- b** From the graph it may be seen that $f(x) \geq 0$ for all x as is required.
 The function must also satisfy the other conditions previously noted, namely:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\begin{aligned}
 \text{Now } \int_{-\infty}^{\infty} f(x)dx &= \int_0^1 1.5(1-x^2)dx \text{ since } f(x) = 0 \text{ elsewhere} \\
 &= 1.5 \left[x - \frac{x^3}{3} \right]_0^1 \\
 &= 1.5 \left(1 - \frac{1}{3} \right) \\
 &= 1
 \end{aligned}$$

This is as required. Thus $f(x)$ is a probability density function.

$$\begin{aligned}
 \text{c } \Pr(X > 0.5) &= \int_{0.5}^1 1.5(1-x^2)dx \\
 &= 1.5 \left[x - \frac{x^3}{3} \right]_{0.5}^1 \\
 &= 1.5 \left[\left(1 - \frac{1}{3} \right) - \left(0.5 - \frac{0.125}{3} \right) \right] \\
 &= 0.3125
 \end{aligned}$$

Some intervals for which definite integrals need to be evaluated are of the form $(-\infty, a]$ or $[a, \infty)$ or $(-\infty, \infty)$.

To integrate over the interval $(-\infty, a]$ find $\lim_{k \rightarrow -\infty} \int_k^a f(x)dx$

To integrate over the interval $[a, \infty)$ find $\lim_{k \rightarrow \infty} \int_a^k f(x)dx$

To integrate over the interval $(-\infty, \infty)$ find $\lim_{k \rightarrow \infty} \int_{-k}^k f(x)dx$, where these limits are defined, as shown in Example 3.

Example 3

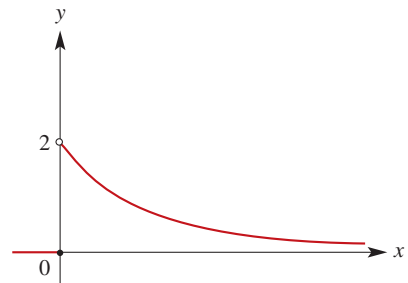
Consider the exponential probability density function f with the rule:

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

- a** Sketch the graph of f . **b** Show that f is a probability density function.
c Find $\Pr(X > 1)$.

Solution

- a** From Chapter 5 you are familiar with the form of the exponential function. In this case, when $x = 0$, $y = 2$, and the values of y decrease as x increases.



- b** It is known that $f(x) \geq 0$ for all x as required.

The second condition is that $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\text{Now } \int_{-\infty}^{\infty} 2e^{-2x} dx = \int_0^{\infty} 2e^{-2x} dx, \text{ since } f(x) = 0 \text{ for } x \leq 0$$

$$\begin{aligned} \text{To evaluate this integral, consider } & \lim_{k \rightarrow \infty} \int_0^k 2e^{-2x} dx \\ &= \lim_{k \rightarrow \infty} \left[\frac{2e^{-2x}}{-2} \right]_0^k \\ &= \lim_{k \rightarrow \infty} [-e^{-2x}]_0^k \\ &= \lim_{k \rightarrow \infty} (-e^{-2k}) - (-e^{-0}) \\ &= 0 + e^{-0} \\ &= 1 \end{aligned}$$

Thus f satisfies the requirements of a probability density function.

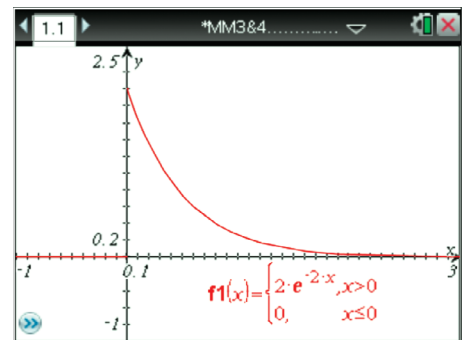
$$\begin{aligned} \text{c } \Pr(X > 1) &= \lim_{k \rightarrow \infty} \int_1^k 2e^{-2x} dx \\ &= \lim_{k \rightarrow \infty} \left[\frac{2e^{-2x}}{-2} \right]_1^k \\ &= \lim_{k \rightarrow \infty} [-e^{-2x}]_1^k \\ &= \lim_{k \rightarrow \infty} (-e^{-2k}) - (-e^{-2}) \\ &= 0 + e^{-2} \\ &= \frac{1}{e^2} \\ &= 0.1353, \text{ correct to four decimal places} \end{aligned}$$

Using the TI-Nspire

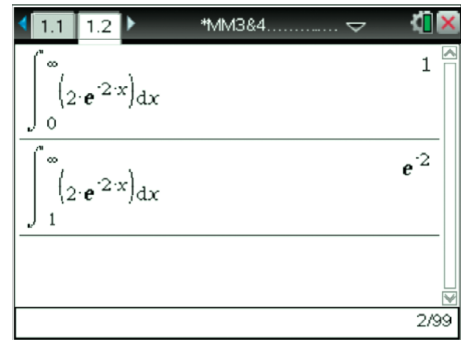
This is an application of integration.

- a** The graph is as shown.

The hybrid function template, $\left\{ \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right\}$, has been used in this example. Access the template using $\textcircled{w}$, $\textcircled{ctr}$ + $\textcircled{up}$ on the Clickpad)



b and **c** The required integrations are performed to achieve the results.
 ∞ , the symbol for infinity, can be found using π or $\text{ctrl} + \infty$ (or $\text{ctrl} + i$ on the Clickpad)

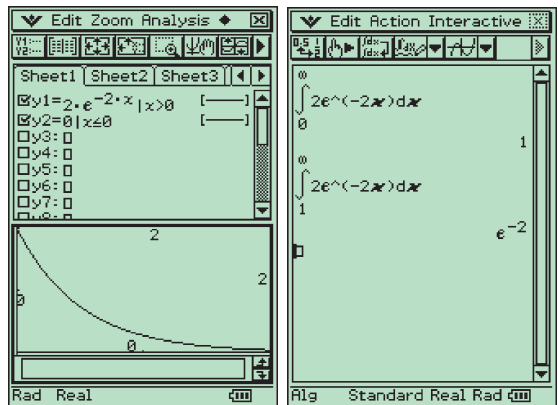


Using the Casio ClassPad

This is an application of integration.

a The graph is as shown.

b and **c** The integrations are performed to achieve the results.



Exercise 17A

- 1 Show that the function f with rule

$$f(x) = \begin{cases} \frac{24}{x^3} & 3 \leq x \leq 6 \\ 0 & x < 3 \text{ or } x > 6 \end{cases}$$

is a probability density function.

- 2 Let X be a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} x^2 + kx + 1 & 0 \leq x \leq 2 \\ 0 & x < 0 \text{ or } x > 2 \end{cases}$$

Determine the constant k such that f is a valid probability density function.



- 3 Consider the random variable X having the probability density function with the rule:

$$f(x) = \begin{cases} 12x^2(1-x) & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

- a** Sketch the graph of $f(x)$. **b** Find $\Pr(X < 0.5)$.
c Shade the region which represents this probability on your sketch graph.
- 4 Consider the random variable Y with probability density function with the rule:

$$f(y) = \begin{cases} ke^{-y} & y \geq 0 \\ 0 & y < 0 \end{cases}$$

Find:

- a** the constant k **b** $\Pr(Y \leq 2)$
- 5 The quarantine period for a certain disease is between 5 and 11 days after contact. The probability of showing the first symptoms at various times during the quarantine period is described by the probability density function:

$$f(t) = \frac{1}{36}(t-5)(11-t)$$

- a** Sketch the graph of the function.
b Find the probability that the symptoms will appear within 7 days of contact.
- 6 A probability model for the mass x kg of a 2-year-old child is given by:

$$f(x) = k \sin \left[\frac{1}{10} \pi (x - 7) \right], 7 \leq x \leq 17$$

- a** Show that $k = \frac{1}{20} \pi$.
b Hence find the percentage of 2-year-olds whose mass is:
i greater than 16 kg **ii** between 12 and 13 kg
- 7 A probability density function for the lifetime t hours of Electra light globes is given by the rule:

$$f(t) = ke^{(-\frac{t}{200})}, t > 0$$

Find:

- a** the value of the constant k
b the probability that a bulb will last more than 1000 hours

- 8 A probability density function has a probability density function given by

$$f(x) = \begin{cases} k(1+x) & -1 \leq x \leq 0 \\ k(1-x) & 0 < x \leq 1 \\ 0 & x < -1 \text{ or } x > 1 \end{cases}$$

where $k > 0$.

- a** Sketch the probability density function. **b** Evaluate k .
c Find the probability that X lies between -0.5 and 0.5 .

- 9 Let X be a continuous random variable with probability density function given by:

$$f(x) = \begin{cases} 3x^2 & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

- a** Sketch the graph of $f(x)$.
b Find $\Pr(0.25 < X < 0.75)$ and sketch this on your graph.

- 10 A random variable X has a probability density function f with the rule:

$$f(x) = \begin{cases} \frac{1}{100}(10+x) & -10 < x \leq 0 \\ \frac{1}{100}(10-x) & 0 < x \leq 10 \\ 0 & x \leq -10 \text{ or } x > 10 \end{cases}$$

- a** Sketch the graph of f . **b** Find $\Pr(-1 \leq X < 1)$.

- 11 The life in hours, X , of a type of light globe has a probability density function with the rule:

$$f(x) = \begin{cases} \frac{k}{x^2} & x > 1000 \\ 0 & x \leq 1000 \end{cases}$$

- a** Evaluate k . **b** Calculate the probability the globe will last at least 2000 hours.

- 12 The weekly demand for petrol, X (in thousands of litres), at a particular service station is a random variable with probability density function:

$$f(x) = \begin{cases} 2\left(1 - \frac{1}{x^2}\right) & 1 \leq x \leq 2 \\ 0 & x < 1 \text{ or } x > 2 \end{cases}$$

- a** Determine the probability that more than 1.5 thousand litres are demanded in 1 week.
b Determine the probability that the demand for petrol in 1 week is less than 1.8 thousand litres, given that it is more than 1.5 thousand litres.

- 13** The length of time, X (in minutes), between the arrival of customers at an autobank is a random variable with probability density function:

$$f(x) = \begin{cases} \frac{1}{5}e^{-\frac{x}{5}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- a** Find the probability that more than 8 minutes elapses between successive customers.
b Find the probability that more than 12 minutes elapses between successive customers, given that more than 8 minutes has passed.
- 14** A random variable X has density function given by:

$$f(x) = \begin{cases} 0.2 & -1 < x \leq 0 \\ 0.2 + 1.2x & 0 < x \leq 1 \\ 0 & x \leq -1 \text{ or } x > 1 \end{cases}$$

- a** Find $\Pr(X \leq 0.5)$. **b** Hence find $\Pr(X > 0.5 | X > 0.1)$.
- 15** The continuous random variable X has probability density function f given by:

$$f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- a** Sketch the graph of f .
b Find:
i $\Pr(X < 0.5)$ **ii** $\Pr(X \geq 1)$ **iii** $\Pr(X \geq 1 | X > 0.5)$
- 16** The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} |ax| & -2 \leq x \leq 2 \\ 0 & x < -2 \text{ or } x > 2 \end{cases}$$

where a is a constant.

- a** Sketch the graph of f . **b** Hence or otherwise, find the value of a .

17.2 Cumulative distribution functions*

Another function of importance in describing a continuous random variable is the cumulative distribution function F , or CDF. For a continuous random variable X , with probability density function f , the cumulative distribution function defined on an interval $[a, b]$ is given by

$$\begin{aligned} f(x) &= \Pr(X \leq x) \\ &= \int_a^x f(t)dt \end{aligned}$$

where f is the probability density function and t is the variable of integration.

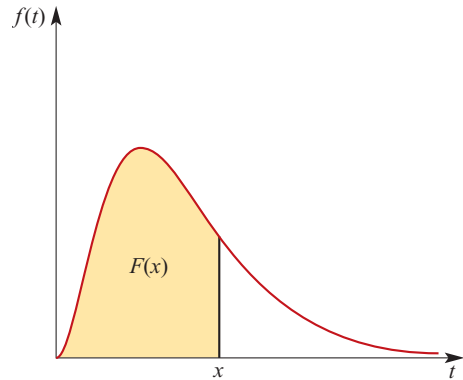
* Cumulative distribution functions are not mentioned explicitly in the Mathematical Methods Units 3 and 4 CAS study design but are useful in our study of continuous probability distributions.

For the probability density function defined on R :

$$\begin{aligned} f(x) &= \Pr(X \leq x) \\ &= \int_{-\infty}^x f(t) dt \end{aligned}$$

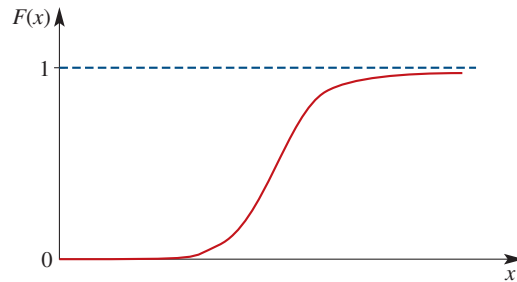
The graphical representation of the relationship between the probability density function and the cumulative distribution function is shown in the diagram on the right.

It is clear that this result is an application of the fundamental theorem of calculus and the derivative of $F(x)$ is $f(x)$, i.e. the derivative of the cumulative distribution function is the density function.



That is, the function F describes the area under the probability density function between the lower bound of the domain of f and x (in the diagram, the lower bound is 0). So, to find $\Pr(X \leq n)$ for a particular random variable X , you would substitute the value of n into the expression for the cumulative distribution function and evaluate.

Thus, the cumulative distribution function for a particular value x gives the probability that the random variable X takes a value less than or equal to that value. As X is a continuous random variable, the cumulative distribution function is also continuous.



There are three important properties of a cumulative distribution function with a probability density function with domain $[a, b]$.

If F is a cumulative distribution function, with probability density function with domain $[a, b]$:

- 1 The probability that the random variable X takes a value less than a is zero. That is:

$$F(a) = 0$$

- 2 The probability that the random variable X takes a value less than or equal to b is 1. That is:

$$F(b) = 1$$

- 3 If x_1 and x_2 are values of x such that $x_1 \leq x_2$, then:

$$\Pr(X \leq x_1) \leq \Pr(X \leq x_2)$$

$$\text{That is } F(x_1) \leq F(x_2)$$

The function F is a **non-decreasing** function.

For a probability density function defined on R , $F(-\infty) = 0$ and $F(\infty) = 1$

Example 4

The time (in seconds) it takes a student to complete a puzzle is a random variable with a density function with the rule:

$$f(x) = \begin{cases} \frac{5}{x^2} & x \geq 5 \\ 0 & x < 5 \end{cases}$$

Find $F(x)$, the cumulative distribution function of X .

Solution

$$\begin{aligned} F(x) &= \int_5^x f(t) dt \\ &= \int_5^x \frac{5}{t^2} \\ &= \left[\frac{-5}{t} \right]_5^x \\ &= \frac{-5}{x} + 1 \end{aligned}$$

That is: $F(x) = 1 - \frac{5}{x}$

The importance of the cumulative distribution function is that probabilities for various intervals can be computed directly from $F(x)$.

Example 5

Use the cumulative distribution function found in Example 4 to find:

- a** the probability that a student takes less than 12 seconds to complete the puzzle
- b** the probability that a student takes between 8 and 10 seconds to complete the puzzle, given that he takes less than 12 seconds

Solution

- a** $\Pr(X \leq 12)$ can be obtained by substituting 12 in the expression for $F(x)$.

$$\begin{aligned} F(12) &= 1 - \frac{5}{12} \\ &= \frac{7}{12} \end{aligned}$$

$$\begin{aligned}
 \text{b } \Pr(8 \leq X \leq 10 | X \leq 12) &= \frac{\Pr(8 \leq X \leq 10 \cap X \leq 12)}{\Pr(X \leq 12)} = \frac{\Pr(8 \leq X \leq 10)}{\Pr(X \leq 12)} \\
 &= \frac{F(10) - F(8)}{F(12)} \\
 &= \frac{\frac{1}{2} - \frac{3}{8}}{\frac{7}{12}} \\
 &= \frac{3}{14}
 \end{aligned}$$

Exercise 17B

- 1 The probability density function for a random variable X is:

$$f(x) = \begin{cases} \frac{1}{5} & 0 < x < 5 \\ 0 & x \leq 0 \text{ or } x \geq 5 \end{cases}$$

- a Find $F(x)$, the cumulative distribution function of X . b Hence find $\Pr(X \leq 3)$.

- 2 A random variable X has a cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x^2} & x \geq 0 \end{cases}$$

- a Sketch the graph of $F(x)$. b Find $\Pr(X \geq 2)$.
c Find $\Pr(X \geq 2 | X < 3)$.

- 3 The continuous random variable X has cumulative distribution function F given by:

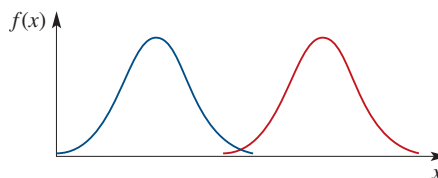
$$F(x) = \begin{cases} 0 & x < 0 \\ kx^2 & 0 \leq x \leq 6 \\ 1 & x > 6 \end{cases}$$

- a Determine the value of the constant k . b Calculate $\Pr\left(\frac{1}{2} \leq X \leq 1\right)$.

17.3 Mean, median and mode for a continuous random variable

The centre is an important summary feature of a probability distribution.

The diagram shows two probability distributions which are identical except for their centres.



More than one measure of centre may be determined for a continuous random variable, and each gives useful information about the random variable under consideration. The most generally useful measure of centre is the mean.

Mean

In the same way as the mean or expected value for a discrete random variable is defined, one can define the **mean** or **expected value** for a continuous random variable.

The mean or expected value of a continuous random variable X is given by:

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

provided the integral exists. As in the case of a discrete random variable the mean is always denoted by the Greek letter μ (mu).

If $f(x) = 0$ for all $x \notin [a, b]$, then $E(X) = \int_a^b xf(x) dx$

This definition is consistent with the definition of the expected value for a discrete random variable. As in the case of a discrete random variable, the mean or expected value of a continuous random variable is not necessarily the value the random variable takes in any particular experiment, but the long-run average value of the variable. As an example, consider the daily demand for petrol at a service station. The mean of this variable tells us the average daily demand for petrol over a very long period of time.

Example 6

Find the expected value of the random variable X which has probability density function with rule:

$$f(x) = \begin{cases} 0.5x & 0 \leq x \leq 2 \\ 0 & x < 0 \text{ or } x > 2 \end{cases}$$

Solution

$$\begin{aligned} \text{By definition} \quad E(X) &= \int_{-\infty}^{\infty} xf(x) dx \\ &= \int_0^2 x \times 0.5x dx \quad \text{since } f(x) = 0 \text{ elsewhere} \\ &= 0.5 \int_0^2 x^2 dx \\ &= 0.5 \left[\frac{x^3}{3} \right]_0^2 \\ &= \frac{4}{3} \end{aligned}$$

The mean of a function of X is defined as follows. (In this case the function of X is denoted as $g(X)$ and is the composition of the random variable X and the function g .)

The expected value of $g(X)$ is

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

provided the integral exists.

Generally, as in the case of a discrete random variable, the expected value of a function of X is not equal to that function of the expected value of X . That is:

$$E\{g(x)\} \neq g\{E(X)\}$$

Example 7

If X is the random variable with probability density function f :

$$f(x) = \begin{cases} 0.5x & 0 \leq x \leq 2 \\ 0 & x < 0 \text{ or } x > 2 \end{cases}$$

Find:

- a** the expected value of X^2 **b** the expected value of e^x (correct to four decimal places)

Solution

$$\begin{aligned} \mathbf{a} \quad E[X^2] &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= \int_0^2 x^2 \times 0.5x dx \quad \text{since } f(x) = 0 \text{ elsewhere} \\ &= 0.5 \int_0^2 x^3 dx \\ &= 0.5 \left[\frac{x^4}{4} \right]_0^2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad E[e^x] &= \int_{-\infty}^{\infty} e^x f(x) dx \\ &= \int_0^2 e^x f(x) dx \quad \text{since } f(x) = 0 \text{ elsewhere} \\ &= \int_0^2 e^x \times 0.5x dx \\ &= 4.195 \text{ (correct to three decimal places)} \end{aligned}$$

A case where the equality does hold is where g is a linear function of X . Then

$$E(aX + b) = aE(X) + b \quad (a, b \text{ constant})$$

Percentiles and the median

Another value of interest is the value of X which bounds a particular area under a probability density function. For example, a teacher may wish determine the mark (p) below which lie

75% of all students' marks. This is called the 75th percentile of the population, and is found by solving:

$$\int_{-\infty}^p f(x)dx = 0.75$$

This can be stated more generally.

The value p of X which is the solution of an equation of the form

$$\int_{-\infty}^p f(x)dx = q$$

is called a *percentile* of the distribution. Here q is the required percentile. (For the example above $q = 75\% = 0.75$.)

Example 8

The duration of telephone calls to the order department of a large company is a random variable X minutes with probability density function:

$$f(x) = \begin{cases} \frac{1}{3}e^{-\frac{x}{3}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Find the value of a such that 90% of phone calls last less than a minutes.

Solution

To find the value of a , solve the equation:

$$\int_0^a \frac{1}{3}e^{-\frac{x}{3}} dx = 0.90$$

$$\text{That is, } \left[-e^{-\frac{x}{3}}\right]_0^a = 0.90$$

$$\therefore 1 - e^{-\frac{a}{3}} = 0.90$$

$$\therefore -\frac{a}{3} = \log_e 0.10$$

$$\begin{aligned} \therefore a &= 3 \log_e (10) \\ &= 6.908 \text{ (correct to three decimal places)} \end{aligned}$$

So 90% of the calls to this company last less than 6.908 minutes.

As for discrete probability distributions, a percentile of special interest is the median, or 50th percentile. The median is the middle value of the distribution. That is, the probability of X taking a value below the median is 0.5, and the probability of X taking a value above the median is 0.5. Thus, if m is the median value of the distribution, then:

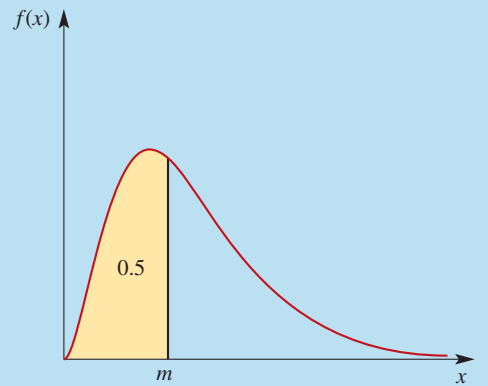
$$\begin{aligned} \Pr(X \leq m) &= \Pr(X > m) \\ &= 0.5 \end{aligned}$$

The median may be determined readily from the cumulative distribution function, as it is the 50th percentile of the distribution. Graphically, the median is the value of the random variable which divides the area under the probability density function in half.

The median is another measure of centre of a continuous probability distribution.

The median, m , of a probability distribution function is the value of X such that:

$$\int_{-\infty}^m f(x)dx = 0.5$$



Example 9

Suppose the probability density function of weekly sales of topsoil, X (in tonnes), is given by the rule:

$$f(x) = \begin{cases} 2(1-x) & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Find the median value of X , and interpret.

Solution

The solution here is the value of m such that:

$$\begin{aligned} \int_0^m 2(1-x) &= 0.5 \\ \therefore 2 \left[x - \frac{x^2}{2} \right]_0^m &= 0.5 \\ \therefore 2m - m^2 &= 0.5 \\ \therefore m^2 - 2m + 0.5 &= 0 \\ m &= 0.293 \text{ or } 1.707 \end{aligned}$$

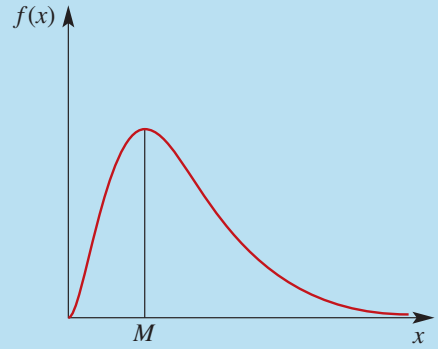
But since $0 \leq x \leq 1$ the median $m = 0.293$ tonnes. That is, in the long run, 50% of weekly sales will be less than 0.293 tonnes, and 50% will be more.

Mode

Another value of interest is the mode, M , of the distribution. Although not actually a measure of centre the mode does tell us the most likely or most probable value of X . Thus, the mode occurs at the maximum value of $f(x)$, and may be located through considering the graph of the probability density function.

If f is the probability density function of X , then the mode, M , is the value of X such that:

$$f(M) \geq f(x) \text{ for all other } x$$



Example 10

Find the mode of the continuous random variable X with probability density function:

$$f(x) = \begin{cases} 12x^2(1-x) & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

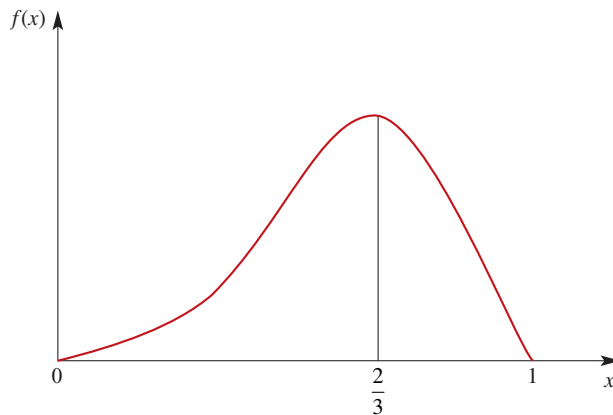
Solution

To find the maximum value of $f(x)$ in the interval $[0,1]$, begin by expanding the expression and then differentiating the function as follows:

$$f(x) = 12x^2 - 12x^3$$

$$\begin{aligned} \text{Thus } f'(x) &= 24x - 36x^2 \\ &= 0 \\ 12x(2 - 3x) &= 0 \\ x &= 0 \text{ or } x = \frac{2}{3} \end{aligned}$$

Consideration of the function shows that $x = 0$ corresponds to a minimum value of $f(x)$, and $x = \frac{2}{3}$ corresponds to a maximum value of $f(x)$. Thus the mode $M = \frac{2}{3}$



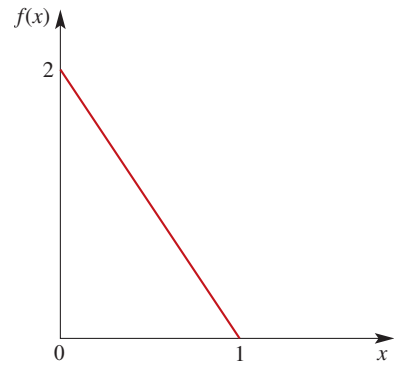
Example 11

If the probability density function of weekly sales of topsoil, X (in tonnes), is as given in Example 9, find the most likely weekly sales.

Solution

The most likely weekly sales correspond to the mode of $f(x)$. Since the probability density function is a linear function of x , and more importantly strictly decreasing, calculus will not help, so the solution will need to be determined graphically.

Consideration of the graph of f shows that the maximum value of $f(x)$ occurs when $x = 0$.

**Exercise 17C**

- 1 Find the mean, $E(X)$, of the continuous random variables with the following probability density functions:

a $f(x) = 2x, 0 < x < 1$
b $f(x) = \frac{1}{2\sqrt{x}}, 0 < x < 1$

c $f(x) = 6x(1 - x), 0 < x < 1$
d $f(x) = \frac{1}{x^2}, x \geq 1$

- 2 Use your calculator to firstly check that each of the following is a probability density function, and then to find the mean, $E(X)$, of the continuous random variables with the following probability density functions:

a $f(x) = \sin x, 0 < x < \frac{\pi}{2}$
b $f(x) = \log_e x, 1 < x < e$

c $f(x) = \frac{1}{\sin^2 x}, \frac{\pi}{4} < x < \frac{\pi}{2}$
d $f(x) = -4x \log_e x, 0 < x < 1$

- 3 A continuous random variable X has the probability density function given by:

$$f(x) = \begin{cases} 2x^3 - x + 1 & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

- a** Find the mean value of X , μ .
b Find the probability that X takes a value less than or equal to the mean.
- 4 Consider the probability density function given by:

$$f(x) = \frac{1}{2\pi}(1 + \cos x), -\pi \leq x \leq \pi$$

Find the expected value of X .

- 5 A random variable Y has a probability density function:

$$f(y) = \begin{cases} Ay & 0 \leq y \leq B \\ 0 & x < 0 \text{ or } x > B \end{cases}$$

Find A and B if the mean of Y is 2.

- 6 A random variable X has the probability density function given by:

$$f(x) = \begin{cases} 12x^2(1-x) & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Find:

a $E\left(\frac{1}{X}\right)$ **b** $E(e^x)$

- 7 The random variable X has a probability density function given by:

$$f(x) = \begin{cases} k & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Find:

a the value of k **b** the median, m , of X

- 8 The time X seconds between arrivals of particles at a radiation counter has been found to have a probability density function f with the rule:

$$f(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x \geq 0 \end{cases}$$

Find:

a $\Pr(X \leq 1)$ **b** $\Pr(1 \leq X \leq 2)$ **c** the median, m , of X

- 9 A continuous random variable has a probability density function given by:

$$f(x) = \begin{cases} 5(1-x)^4 & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Find the median, m , of X (correct to four decimal places).

- 10 Suppose that the time (in minutes) between telephone calls received at a pizza restaurant has probability density function:

$$f(x) = \begin{cases} \frac{1}{4}e^{-\frac{x}{4}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Find the median time between calls.

- 11** The cumulative distribution function* of a continuous random variable X , $F(x)$, is given by:

$$F(x) = \begin{cases} 0 & x < 0 \\ 4x^3 - 3x^4 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Find:

- a** the probability density function of X **b** the mode, M , of X

- 12** A random variable X has a probability density function:

$$f(x) = \begin{cases} \frac{1}{2} \sin x & 0 < x < \pi \\ 0 & x \leq 0 \text{ or } x \geq \pi \end{cases}$$

Find the mode, M , of X .

- 13** A continuous random variable X has the probability density function given by:

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & x < 0 \text{ or } x \geq 2 \end{cases}$$

- a** Find μ , the expected value of X . **b** Find m , the median value of X .
c Find M , the mode of X .

- 14** Let the probability density function of X be:

$$f(x) = \begin{cases} 30x^4(1-x) & 0 < x < 1 \\ 0 & x < 0 \text{ or } x \geq 1 \end{cases}$$

- a** Find the expected value of X , μ .
b Find the median value of X , m , and hence show that the mean is less than the median.

- 15** A probability model for the mass x kg of a 2-year-old child is given by:

$$f(x) = \frac{1}{20} \pi \sin \left[\frac{1}{10} \pi (x - 7) \right], 7 \leq x \leq 17$$

- a** Find the mode of X , M . **b** Find the median value of X , m .

- 16** A random variable X has density function given by:

$$f(x) = \begin{cases} 0.2 & -1 < x \leq 0 \\ 0.2 + 1.2x & 0 < x \leq 1 \\ 0 & x < -1 \text{ or } x > 1 \end{cases}$$

- a** Find μ , the expected value of X . **b** Find m , the median value of X .
c Find M , the mode of X .

* Cumulative distribution functions are not mentioned explicitly in the Mathematical Methods Units 3 and 4 CAS study design but are useful in our study of continuous probability distributions.

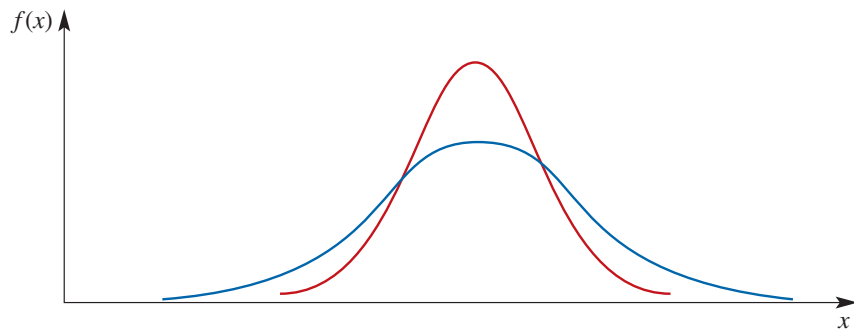
- 17** The exponential probability distribution describes the distribution of the time between random events, such as phone calls. The general form of the exponential distribution with parameter λ is:

$$f(x) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- a** Differentiate kxe^{-kx} and hence find an antiderivative of kxe^{-kx} .
- b** Show that the mean of an exponential random variable is λ .
- c** On the same axes, sketch the graph of an exponential random variable for:
 - i** $\lambda = \frac{1}{2}$ **ii** $\lambda = 1$ **iii** $\lambda = 2$
- d** Describe the effect of the variation of the parameter λ on the graph of the distribution.

17.4 Measures of spread

Another important summary feature of a distribution is variation or spread. The following diagram shows two distributions that are identical except for their spreads.



As in the case of centre, there is more than one measure of spread. The most commonly used is the variance, together with its companion measure, the standard deviation. Others that you may be familiar with are the range and the interquartile range.

Variance and standard deviation

The *variance* of the random variable X is a measure of the spread of a probability distribution about its mean or expected value μ . It is defined as follows.

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \end{aligned}$$

As for discrete random variables the variance is usually denoted σ^2 , where σ is the lower case of the Greek letter sigma.

Variance may be considered as the long-run average value of the square of the distance from the values of X to μ . Since the variance is determined by squaring the distance from the values of X to μ it is no longer in the units of measurement of the original random variable. A measure of spread in the appropriate unit is found by taking the square root of the variance.

The **standard deviation** of X is defined as:

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

The standard deviation is usually denoted σ .

As in the case of discrete random variables, an alternative (computational) formula for variance is generally used.

To calculate variance, use:

$$\text{Var}(X) = E(X^2) - \mu^2$$

The computational form of the expression for variance is derived as follows:

$$\begin{aligned}\text{Var}(X) &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\ &= \int_{-\infty}^{\infty} (x^2 - 2\mu x + \mu^2) f(x) dx \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx - \int_{-\infty}^{\infty} 2\mu x f(x) dx + \int_{-\infty}^{\infty} \mu^2 f(x) dx \\ &= E(X^2) - 2\mu \int_{-\infty}^{\infty} x f(x) dx + \mu^2 \int_{-\infty}^{\infty} f(x) dx \\ \text{Since } \int_{-\infty}^{\infty} x f(x) dx &= \mu \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) dx = 1 \\ \text{Var}(X) &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2\end{aligned}$$

Example 12

Find the variance and standard deviation of the random variable X which has the probability density function f with rule:

$$f(x) = \begin{cases} 0.5x & 0 \leq x \leq 2 \\ 0 & x < 0 \text{ or } x > 2 \end{cases}$$

Solution

Use the computational formula $\text{Var}(X) = E(X^2) - \mu^2$

First, evaluate $E(X^2)$:

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= \int_0^2 x^2 \times 0.5x dx \\ &= 0.5 \int_0^2 x^3 dx \\ &= 0.5 \left[\frac{x^4}{4} \right]_0^2 \\ &= 0.5 \times 4 \\ &= 2 \end{aligned}$$

Now $E(X) = \frac{4}{3}$ (from Example 6)

so $\text{Var}(X) = 2 - \left(\frac{4}{3}\right)^2$

$$= \frac{2}{9}$$

and $\text{sd}(X) = \sqrt{\frac{2}{9}}$

$$= \frac{\sqrt{2}}{3} \text{ or, correct to three decimal places, } 0.471$$

It helps to make the standard deviation more meaningful to give it an interpretation which relates to the probability distribution. As already stated for discrete random variables, it is also the case for many continuous random variables that about 95% of the distribution lies within two standard deviations either side of the mean.

In general, for many continuous random variables X :

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

Example 13

The life of a certain brand of battery, X , is a continuous random variable with mean 50 hours and variance 16 hours. Find an (approximate) interval for the time period for which 95% of the batteries would be expected to last.

Solution

$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$. In this case $\mu = 50$ and $\sigma = \sqrt{16} = 4$, so one could expect 95% of the batteries to last between 42 hours and 58 hours.

Range

The range of a random variable is the difference between the smallest and the largest value the variable may take, and is clearly determined from the domain of the probability density function.

Example 14

Determine the range of the random variable X which has the probability density function:

$$f(x) = \begin{cases} \frac{1}{9}(4x - x^2) & 0 \leq x \leq 3 \\ 0 & x < 0 \text{ or } x > 3 \end{cases}$$

Solution

From the probability density function X may take values from 0 to 3. Thus:

$$\begin{aligned} \text{range of } X &= 3 - 0 \\ &= 3 \end{aligned}$$

Interquartile range

The interquartile range is the range of the middle 50% of the distribution. It is the difference between the 75th percentile, also known as Q3, and the 25th percentile, also known as Q1.

Example 15

Determine the interquartile range of the random variable X which has the probability density function:

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Solution

To find the 25th percentile a , solve:

$$\int_0^a 2x dx = 0.25 \Rightarrow [x^2]_0^a = 0.25 \Rightarrow a^2 = 0.25 \Rightarrow a = \sqrt{0.25} = 0.5$$

To find the 75th percentile b , solve:

$$\int_0^b 2x dx = 0.75 \Rightarrow [x^2]_0^b = 0.75 \Rightarrow b^2 = 0.75 \Rightarrow b = \sqrt{0.75} = 0.866, \\ \text{correct to three decimal places}$$

Thus the interquartile range is $0.866 - 0.5 = 0.366$, correct to three decimal places.

Note that the negative solutions to each of these equations were not appropriate, as $0 \leq x \leq 1$.

Exercise 17D



- 1 A continuous random variable X has a probability density function given by:

$$f(x) = \begin{cases} 3x^2 & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

Find:

- a** a such that $\Pr(X \leq a) = 0.25$ **b** b such that $\Pr(X \leq b) = 0.75$
c the interquartile range of X
- 2 Given that the random variable X has probability density function:

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & x \leq 0 \text{ or } x \geq 1 \end{cases}$$

- a** Find the interquartile range of X .
b Find the variance of X , and hence the standard deviation of X .
- 3 If the random variable X has probability density function given by:
- $$f(x) = \begin{cases} 0.5e^x & x \leq 0 \\ 0.5e^{-x} & x > 0 \end{cases}$$
- a** Sketch the graph of $y = f(x)$
b Find the interquartile range of X , giving your answer correct to three decimal places.
- 4 The continuous random variable X has probability density function given by:

$$f(x) = \begin{cases} \frac{k}{x} & 1 \leq x \leq 9 \\ 0 & x < 1 \text{ or } x > 9 \end{cases}$$

- a** Find the value of k .
b Find the mean and variance of X , giving your answer correct to three decimal places.
- 5 The continuous random variable X has density function f given by:

$$f(x) = \begin{cases} 0 & x < 0 \\ 2 - 2x & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

- a** Find the interquartile range of X . **b** Find the mean and variance of X .
- 6 A random variable X has probability density function f with the rule:

$$f(x) = \begin{cases} 0 & x < 0 \\ 2xe^{-x^2} & x \geq 0 \end{cases}$$

Find the interquartile range of X .

- 7 A random variable X has a probability density function:

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 2 \\ 0 & x \geq 2 \end{cases}$$

- a** Find the interquartile range of X . **b** Find the mean and variance of X .

- 8 The queuing time, X minutes, of a traveller at the ticket office of a large railway station has probability density function, f , defined by:

$$f(x) = \begin{cases} kx(100 - x^2) & 0 \leq x \leq 10 \\ 0 & x > 10 \text{ or } x < 0 \end{cases}$$

Find:

- a** the value of k **b** the mean of the distribution
c the standard deviation of the distribution, correct to two decimal places.

- 9 A probability density function is given by:

$$f(x) = \begin{cases} k(a^2 - x^2) & -a \leq x \leq a \\ 0 & x > a \text{ or } x < -a \end{cases}$$

- a** Find k in terms of a .
b Find the value of a which gives a standard deviation of 2.

- 10 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} |k(3 - x)| & 0 \leq x \leq 6 \\ 0 & x > 6 \text{ or } x < 0 \end{cases}$$

where k is a constant.

- a** Sketch the graph of f . **b** Hence or otherwise, find the value of k .
c Verify that the mean of X is 3. **d** Find $\text{Var}(X)$.

17.5 Properties of mean and variance

It has already been stated that the expected value of a function of X is not equal to that function of the expected value. That is, in general:

$$E\{g(x)\} \neq g\{E(x)\}$$

An exception to this rule is the case where the mean of a linear function of X is required. In this case the mean of the linear function is equal to the linear function of the mean, thus:

For any random variable X

$$E(aX + b) = aE(X) + b$$

The validity of this statement can be readily demonstrated.

$$\begin{aligned}
 E(aX + b) &= \int_{-\infty}^{\infty} (ax + b)f(x)dx \\
 &= \int_{-\infty}^{\infty} axf(x)dx + \int_{-\infty}^{\infty} bf(x)dx \\
 &= a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx \\
 &= aE(X) + b, \text{ since } \int_{-\infty}^{\infty} f(x)dx = 1
 \end{aligned}$$

Consider the variance of a linear function of X , $\text{Var}(aX + b)$.

$$\begin{aligned}
 \text{Var}(aX + b) &= E(aX + b)^2 - [E(aX + b)]^2 \\
 \text{Now} \quad [E(aX + b)]^2 &= [aE(X) + b]^2 = [a\mu + b]^2 = a^2\mu^2 + 2ab\mu + b^2 \\
 \text{and} \quad E(aX + b)^2 &= E(a^2X^2 + 2abX + b^2) \\
 &= a^2E(X^2) + 2ab\mu + b^2 \\
 \text{Thus} \quad \text{Var}(aX + b) &= a^2E(X^2) + 2ab\mu + b^2 - a^2\mu^2 - 2ab\mu - b^2 \\
 &= a^2E(X^2) - a^2\mu^2 \\
 &= a^2\text{Var}(X)
 \end{aligned}$$

That is:

For any random variable X :

$$\text{Var}(ax + b) = a^2 \text{Var}(X)$$

Although initially the absence of b in the result may seem surprising, on reflection it makes sense that adding a constant to X merely translates the probability density function of X , but has no effect on the spread of the resultant distribution. If the random variable X has a probability density function with the rule $f(x)$, then the probability distribution with random variable $X + b$ has a corresponding probability density function with the rule $f(x - b)$.

Similarly, multiplying by a is similar to a dilation by a factor of a , and this is consistent with the answer obtained. However, there has to be an adjustment to determine the probability density function of the random variable aX as the transformation must be area-preserving. This is chosen as the function with the rule $\frac{1}{a}f\left(\frac{x}{a}\right)$.

Thus if $f(x)$ is the probability density function for the random variable X , then the probability density function of the random variable $aX + b$ is $\frac{1}{a}f\left(\frac{x - b}{a}\right)$. The transformation is described by $(x, y) \rightarrow \left(ax + b, \frac{y}{a}\right)$. That is, if a and b are positive, a dilation of factor a from the y -axis and $\frac{1}{a}$ from the x -axis and a translation of b units in the positive direction of the x -axis.

Example 16

Suppose X is a continuous random variable with mean $\mu = 10$ and variance $\sigma^2 = 2$.

- a Find $E(2X + 1)$.
- b Find $\text{Var}(1 - 3X)$.
- c If X has a probability density function with rule $f(x)$, describe the rule for a probability density function g , in terms of f for $2X + 1$.

Solution

- a $E(2X + 1) = 2E(X) + 1 = 2 \times 10 + 1 = 21$
- b $\text{Var}(1 - 3X) = (-3)^2 \text{Var}(X) = 9 \times 2 = 18$
- c The rule is $g(x) = \frac{1}{a} f\left(\frac{x-b}{a}\right)$ where $a = 2$ and $b = 1$. Therefore

$$g(x) = \frac{1}{2} f\left(\frac{x-1}{2}\right)$$

Sums of independent random variables*

In previous work the idea of independent events has been introduced. The term ‘independent’ is also applied to random variables. While a formal definition of independent random variables is beyond the scope of this course, basically two random variables are independent if their joint probability density function is a product of their individual probability density functions. This is also true for their cumulative distribution functions. Knowing random variables are independent allows us to make certain conclusions concerning their mean and variance.

Suppose X and Y are independent random variables; then:

$$\begin{aligned} E(X + Y) &= E(X) + E(Y) \\ \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) \end{aligned}$$

Note that in fact $E(X + Y) = E(X) + E(Y)$ whether or not X and Y are independent, but the same cannot be said for variance.

Example 17

A manufacturing process involves two stages. The time taken to complete stage 1, X hours, is a continuous random variable with mean $\mu = 4$ and standard deviation $\sigma = 1.5$. The time taken to complete stage 2, Y hours, is a continuous random variable with mean $\mu = 7$ and standard deviation $\sigma = 1$. Find the mean and standard deviation of the total time taken to complete the manufacturing process, if the times taken at each stage are independent.

* This section covers material not included in the Mathematical Methods Units 3 and 4 CAS study design.

Solution

The total processing time is given by $X + Y$. To find the mean total processing time:

$$\begin{aligned} E(X + Y) &= E(X) + E(Y) \\ &= 4 + 7 \\ &= 11 \end{aligned}$$

Since X and Y are independent:

$$\begin{aligned} \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) \\ &= (1.5)^2 + (1)^2 \\ &= 2.25 + 1 \\ &= 3.25 \end{aligned}$$

Hence the standard deviation of the total processing time is:

$$\begin{aligned} \text{sd}(X + Y) &= \sqrt{3.25} \\ &= 1.803 \end{aligned}$$

Exercise 17E

- The amount of flour used each day in a bakery is a continuous random variable X with mean of 4 tonnes. The cost of the flour is $C = 300X + 100$. Find $E(C)$.
- For certain glass ornaments the proportion of impurities per ornament, X , is a random variable with a density function given by:

$$f(x) = \begin{cases} \frac{3x^2}{2} + x & 0 \leq x \leq 1 \\ 0 & x < 0 \text{ or } x > 1 \end{cases}$$

The value of each ornament (in dollars) is $V = 100 - 1.5X$

- Find $E(X)$ and $\text{Var}(X)$.
 - Hence find the mean and standard deviation of V .
- Let X be a random variable with probability density function:

$$f(x) = \begin{cases} \frac{3x^2}{2} & -1 \leq x \leq 1 \\ 0 & x < -1 \text{ or } x > 1 \end{cases}$$

Find:

- $E(3X)$ and $\text{Var}(3X)$
 - $E(3 - X)$ and $\text{Var}(3 - X)$
 - the rule of a probability density function for $3X$
- The time spent waiting in line at a fast-food outlet is a continuous random variable X with mean $\mu = 2.5$ minutes and standard deviation $\sigma = 1.5$ minutes. The time taken for the fast food to be prepared and served is a continuous random variable Y with mean $\mu = 5$ minutes and standard deviation $\sigma = 2.5$ minutes.

- a** Find the mean and variance of the total time the customer waits for the order.
 - b** Find the time interval within which about 95% of customers would wait.
- 5** The length of time Joan waits for her train to work each morning is a continuous random variable with mean $\mu = 7$ minutes and standard deviation $\sigma = 2$ minutes. If the time waited each day is independent of the time waited any other day, find the mean and the standard deviation of the total time waited each week (assume 5 working days per week).



Chapter summary

- A **continuous random variable** is one that can take any value in an interval of the real number line.
- A continuous random variable can be described by a **probability density function** f . There are many different probability density functions with different shapes and properties. However, they all have the following fundamental properties:

- For any value of x , the value of $f(x)$ is positive. That is:

$$f(x) \geq 0 \text{ for all } x$$

- The total area enclosed by the graph of $f(x)$ and the x -axis is equal to 1. That is:

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

- The probability of X taking a value in the interval (a, b) is found by determining the area under the probability density curve between a and b . That is:

$$\Pr(a < X < b) = \int_a^b f(x)dx$$

- The mean or expected value of a continuous random variable X which has probability density function $f(x)$ given by

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

provided the integral exists.

- The mean is always denoted by the Greek letter μ (mu).
- If $g(X)$ is a function of X , then the mean or expected value of $g(X)$ is

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

provided the integral exists.

- In general:

$$E[g(X)] \neq g[E(X)]$$

- The median for a continuous random variable X is m such that:

$$\int_{-\infty}^m f(x)dx = 0.5$$

- If $f(x)$ is the probability density function of X , then the mode of M is the value of X such that:

$$f(M) \geq f(x) \text{ for all other } x$$

- For a continuous random variable X with probability density function $f(x)$ the variance is given by

$$\begin{aligned}\text{Var}(X) &= \text{E}[(X - \mu)^2] \\ &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx\end{aligned}$$

provided the integral exists.

- The variance is usually denoted σ^2 , where σ is the lower case of the Greek letter sigma.
- The standard deviation of X is defined as:

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

- To calculate variance, use:

$$\text{Var}(X) = \text{E}(X^2) - \mu^2$$

- In general, for most continuous random variables X :

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

- The interquartile range of X is

$$\text{IQR} = b - a$$

where a and b are such that

$$\int_{-\infty}^a f(x) dx = 0.25 \text{ and } \int_{-\infty}^b f(x) dx = 0.75$$

and where $f(x)$ is the probability density function of X .

- For any random variable X :

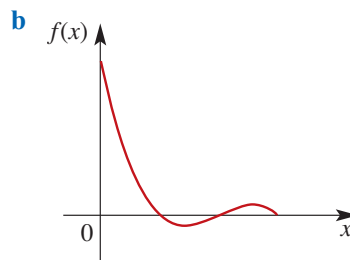
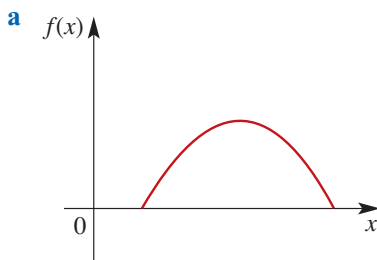
$$\text{E}(aX + b) = a\text{E}(X) + b$$

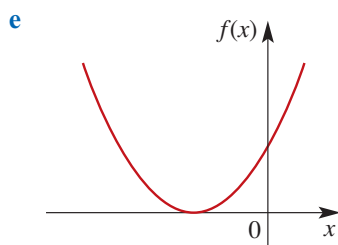
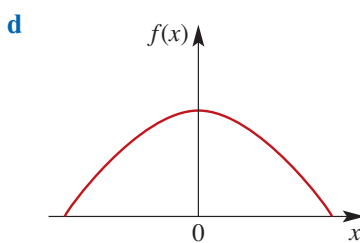
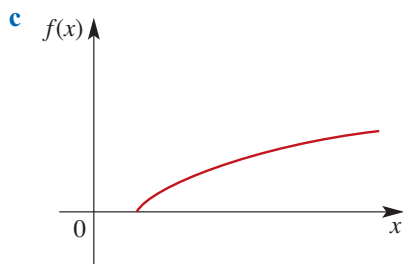
- For any random variable X :

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Multiple-choice questions

- 1 Which of the following graphs could **not** represent a probability density function f ?





- 2** If the function $f(x) = 4x$ represents a probability density function, then which of the following could be the domain of f ?

A $0 < x < 0.25$ **B** $0 < x < 0.5$ **C** $0 < x < 1$
D $0 < x < \frac{1}{\sqrt{2}}$ **E** $\frac{1}{\sqrt{2}} < x < \frac{2}{\sqrt{2}}$

- 3** If a random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{2} \sin x & 0 < x < k \\ 0 & x \geq k \text{ or } x \leq 0 \end{cases}$$

then k is equal to:

A 1 **B** $\frac{\pi}{2}$ **C** 2 **D** π **E** 2π

The following information relates to questions 4, 5 and 6.

A random variable X has probability density function:

$$f(x) = \begin{cases} \frac{3}{4}(x^2 - 1) & 1 < x < 2 \\ 0 & x \leq 1 \text{ or } x \geq 2 \end{cases}$$

- 4** $\Pr(X \leq 1.3)$ is closest to:
A 0.0743 **B** 0.4258 **C** 0.3 **D** 2.25 **E** 0.9258
- 5** The mean of X , $E(X)$, is equal to:
A 1 **B** $\frac{3}{2}$ **C** $\frac{9}{4}$ **D** $\frac{27}{32}$ **E** $\frac{27}{16}$
- 6** The variance of X is:
A $\frac{27}{16}$ **B** $\frac{67}{1280}$ **C** $\frac{81}{16}$ **D** $\frac{81}{256}$ **E** $\frac{729}{256}$

- 7 If a random variable X has a probability density function

$$f(x) = \begin{cases} \frac{x^3}{4} & 0 \leq x \leq 2 \\ 0 & x > 2 \text{ or } x < 0 \end{cases}$$

then the median of X is closest to:

- A** 1.5 **B** 1.4142 **C** 1.6818 **D** 1.2600 **E** 1

- 8 If a random variable X has a probability density function

$$f(x) = \begin{cases} \frac{3}{2}(x-1)(x-2)^2 & 1 \leq x \leq 3 \\ 0 & x < 1 \text{ or } x > 3 \end{cases}$$

then the mode of X is:

- A** 1 **B** 1.333 **C** 2 **D** 2.6 **E** 3

- 9 If the consultation time (in minutes) at a surgery is represented by a random variable X , which has probability density function

$$f(x) = \begin{cases} \frac{x}{40000}(400 - x^2) & 0 \leq x \leq 20 \\ 0 & x < 0 \text{ or } x > 20 \end{cases}$$

then the expected consultation time (in minutes) for three patients is:

- A** $10\frac{2}{3}$ **B** 30 **C** 32 **D** 42 **E** $43\frac{2}{3}$

- 10 Suppose that the top 10% of students will be given an 'A' in mathematics. If the distribution of scores on the examination is a random variable X with probability density function

$$f(x) = \begin{cases} \frac{\pi}{100} \sin\left(\frac{\pi x}{50}\right) & 0 \leq x \leq 50 \\ 0 & x < 0 \text{ or } x > 50 \end{cases}$$

then the minimum score required to be awarded an 'A' is closest to:

- A** 40 **B** 41 **C** 42 **D** 43 **E** 44

Short-answer questions (technology-free)

- 1 The probability density function of X is given by:

$$f(x) = \begin{cases} kx & 1 \leq x \leq \sqrt{2} \\ 0 & \text{otherwise} \end{cases}$$

- a** Find k . **b** Find $\Pr(1 < X < 1.1)$. **c** Find $\Pr(1 < X < 1.2)$

- 2 If the probability density function of X is given by:

$$f(x) = \begin{cases} a + bx^2 & 0 \leq x \leq 1 \\ 0 & x > 1 \text{ or } x < 0 \end{cases}$$

and $E(X) = \frac{2}{3}$, find a and b .

- 3 The probability density function of X is given by:

$$f(x) = \begin{cases} \frac{\sin x}{2} & 0 \leq x \leq \pi \\ 0 & x > \pi \text{ or } x < 0 \end{cases}$$

Find the median and mode of X .

- 4 The cumulative distribution function of X is given by:

$$F(x) = \begin{cases} 1 & x \geq 5 \\ \frac{1}{4}(x-1) & 1 \leq x < 5 \\ 0 & x < 1 \end{cases}$$

- a** Find $\Pr(1 < X < 3)$. **b** Find $\Pr(X > 2 | 1 < X < 3)$. **c** Find $\Pr(X > 4 | X > 2)$.

- 5 Consider the random variable X having the probability density function given by:

$$f(x) = \begin{cases} 12x^2(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- a** Sketch the graph of $y = f(x)$
b Find $\Pr(X < 0.5)$ and illustrate this probability on your sketch graph.

- 6 The probability density function of a random variable X is:

$$f(x) = \begin{cases} kx^2(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- a** Determine k . **b** Find the mode of X .
c Find the probability that X is less than $\frac{2}{3}$.
d Find the probability that X is less than $\frac{1}{3}$, given that X is less than $\frac{2}{3}$.

- 7 Y is a continuous random variable with probability density function:

$$f(y) = \begin{cases} e^{-y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Find:

- a** $\Pr(Y < 2)$ **b** $\Pr(Y < 3 | Y > 1)$

- 8 The cumulative distribution function of X is given by:

$$F(x) = \begin{cases} \log_e x & 1 \leq x \leq e \\ 0 & x > e \text{ or } x < 1 \end{cases}$$

- a** Find the median value of X . **b** Find the interquartile range of X .
c Find the probability density function of X , $f(x)$.

- 9 The amount of fluid, X , in a can of soft drink is a continuous random variable with mean 330 mL and standard deviation 5 mL. Find an (approximate) interval for the amount of soft drink contained in 95% of the cans.

- 10** The weight, X , of cereal in a packet is a continuous random variable with mean 250 g and variance 4 g. Find an (approximate) interval for the weight of cereal contained in 95% of the packets.

Extended-response questions

- 1** The distribution of X , the life of a certain electronic component, in hours, is described by the following probability density function:

$$f(x) = \begin{cases} \frac{a}{100} \left(1 - \frac{x}{100}\right) & 100 < x < 1000 \\ 0 & x \leq 100 \text{ and } x \geq 1000 \end{cases}$$

- a** What is the value of a ? **b** Find the expected value of the life of the components.
c Find the cumulative distribution function, $F(x)$, for this probability density function.
- 2** The continuous random variable X has cumulative distribution function given by:

$$F(x) = \begin{cases} 0 & x < 0 \\ 2x - x^2 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

- a** Find $\Pr(X > 0.5)$.
b Find a such that $\Pr(X < a) = 0.8$.
c Find $E(X)$ and $E(\sqrt{X})$.
- 3** The probability density function of X is given by:

$$f(x) = \begin{cases} \frac{\pi}{20} \cos\left(\frac{\pi}{10}(x-6)\right) & 1 \leq x \leq 11 \\ 0 & x < 1 \text{ or } x > 11 \end{cases}$$

- a** Find the median and interquartile range of X .
b Find the mean and the variance of X .
- 4** A hardware shop sells a certain size nail either in a small packet at \$1.00 per packet, or loose at \$4.00 per kilogram. On any shopping day the number, X , of packets sold is a binomial random variable with number of trials equal to 8 and probability of success of 0.6, and the weight, Y kilograms, of nails sold loose is a continuous random variable with probability density function f given by:

$$f(y) = \begin{cases} \frac{2(y-1)}{25} & 1 \leq y \leq 6 \\ 0 & y < 1 \text{ or } y > 6 \end{cases}$$

- a** Find the probability that the weight of nails sold loose on any shopping day will be between 4 kg and 5 kg.
b Calculate the expected money received on any shopping day from the sale of this size nail in the store.

- 5 The continuous random variable X has the probability density function f , where:

$$f(x) = \begin{cases} \frac{(x-2)}{2} & 2 \leq x \leq 4 \\ 0 & x < 2 \text{ or } x > 4 \end{cases}$$

By first expanding $(X - c)^2$, or otherwise, find two values of c such that:

$$E[(X - c)^2] = \frac{2}{3}$$

- 6 A wholesaler sells material offcuts in 5 m lengths. The amount, in metres, of unusable material are the values of a random variable, X , with probability density function:

$$f(x) = \begin{cases} k(x-1)(3-x) & 1 \leq x \leq 3 \\ 0 & x < 1 \text{ or } x > 3 \end{cases}$$

- a Show that $k = 0.75$.
 - b Find the mean and variance of X .
 - c Find the probability that X is greater than 2.5 m.
- 7 The continuous random variable X has probability density function f , where:

$$f(x) = \begin{cases} \frac{k}{12(x+1)^3} & 0 \leq x \leq 4 \\ 0 & x < 0 \text{ or } x > 4 \end{cases}$$

- a Find k .
- b Evaluate $E(X + 1)$. Hence, find the mean of X .
- c Use your calculator to verify your answer to part b.
- d Find the value of $c > 0$ for which $\Pr(X \leq c) = c$.

- 8 The yield of a variety of corn has probability density function:

$$f(x) = \begin{cases} kx & 0 \leq x < 2 \\ k(4-x) & 2 \leq x < 4 \\ 0 & x < 0 \text{ or } x > 4 \end{cases}$$

- a Find k .
- b Find the expected value, μ , and variance of the yield of corn.
- c Find the probability $\Pr(|X - \mu| < 1)$.
- d Find the value of a such that $\Pr(X > a) = 0.6$, giving your answer correct to one decimal place.



The normal distribution

Objectives

- To introduce the **standard normal distribution**.
- To introduce the family of normal distributions as transformations of the standard normal distribution.
- To consider the effect of variation in the values of the parameters defining the normal distribution on the graph of the probability density function.
- To recognise the **mean, median, mode, variance** and **standard deviation** of a normal distribution.
- To use technology to determine **probabilities for intervals** in the solution of problems where the normal distribution is appropriate.

18.1 The normal distribution



The most useful continuous distribution, and one that occurs frequently, is the normal distribution. The probability density functions of normal random variables are symmetric, single peaked, bell-shaped density curves. Data sets occurring in nature will often have such a bell-shaped distribution, as measurements on many random variables are closely approximated by a normal probability distribution.



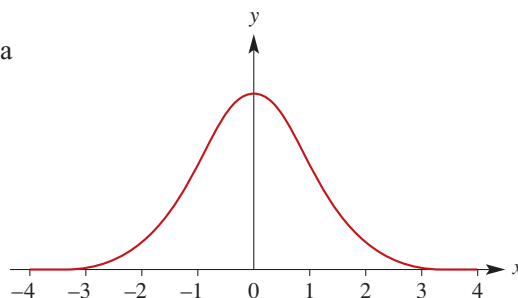
The simplest form of the normal distribution is a random variable with probability density function f with rule:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$



This random variable has a mean $\mu = 0$ and a standard deviation $\sigma = 1$. Because it is the simplest form of the normal distribution it is given a special name, the **standard normal distribution**. The graph of the standard normal distribution is as shown.

The probability density function is symmetric around zero as $f(-x) = f(x)$, i.e. it is an even function.



The line $y = 0$ is an asymptote; that is, as $x \rightarrow \pm\infty$, $y \rightarrow 0$. Almost all of the area under the probability density function lies between ± 3 . It can be seen from the graph that the mean, median and mode of this distribution are the same, and are equal to 0. While the probability density function for the standard normal distribution cannot be integrated exactly, the value of the mean can be verified using the calculator.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\ &= \int_0^{\infty} \frac{x}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx + \int_{-\infty}^0 \frac{x}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \end{aligned}$$

That is, the mean $E(X)$ of the standard normal distribution is 0.

What can be said about the standard deviation of this distribution? Again the calculator must be used.

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

That is, $E(X^2) = 1$

Thus, $\text{Var}(X) = E(X^2) - [E(X)]^2 = 1 - 0 = 1$ and $\text{sd}(X) = \sqrt{\text{Var}(X)} = 1$

A random variable with a standard normal distribution has probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

The standard normal distribution has a mean $\mu = 0$ and a standard deviation $\sigma = 1$

Henceforth, we will denote the random variable of the standard normal distribution by Z .

The normal distribution does not apply just to the special circumstances where the mean is zero and the standard deviation is 1. The curve of the probability density function for a normal distribution with mean μ and standard deviation σ may be obtained from the curve of the standard normal distribution by a transformation with rule:

$$(x, y) \rightarrow \left(\sigma x + \mu, \frac{y}{\sigma} \right)$$

i.e. a dilation of factor σ from the y -axis and a dilation of factor $\frac{1}{\sigma}$ from the x -axis followed by a translation of μ units in the positive direction of the x -axis, for $\mu > 0$. This was discussed for general continuous probability distributions in the previous chapter.

The transformation which maps the curve of the density function of a normal distribution with mean μ and standard deviation σ to the curve of the standard normal distribution is:

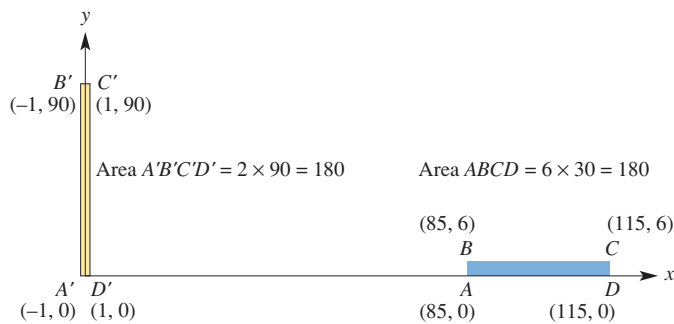
$$(x, y) \rightarrow \left(\frac{x - \mu}{\sigma}, \sigma y \right)$$

That is a translation of μ units in the negative direction of the x -axis, a dilation of factor $\frac{1}{\sigma}$ from the y -axis and a dilation of factor σ from the x -axis.

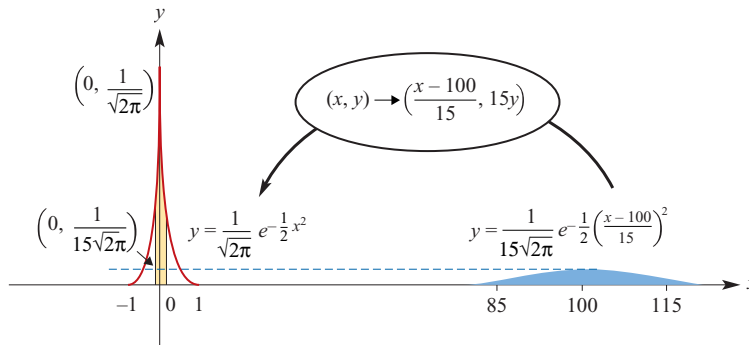
For the example with $\mu = 100$ and $\sigma = 15$:

$$(x, y) \rightarrow \left(\frac{x - 100}{15}, 15y \right)$$

These transformations are area-preserving. For example, the rectangle $ABCD$ is mapped to $A'B'C'D'$. The area of both rectangles is 180 square units.



This property enables the probabilities of any normal distribution to be determined by the probabilities of the standard normal distribution.



The shaded regions are of equal area.

This leads to the general rule for the family of normal probability distributions.

The rule for the general normal probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \mu \in \mathbb{R}, \sigma > 0$$

If X is a normal probability distribution with mean μ and standard deviation σ

$\Pr(X \leq a) = \Pr\left(Z \leq \frac{a - \mu}{\sigma}\right)$, where Z is the random variable of the standard normal.

The general form of the normal density function includes two parameters, μ and σ , which are the mean (μ) and the standard deviation (σ) of that particular distribution.

When a random variable has a distribution described by a normal probability density function the random variable is said to have a **normal distribution**.

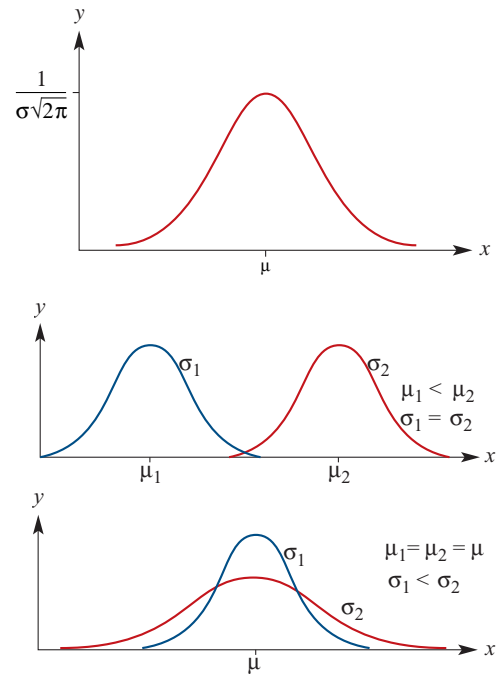
The graph below shows a normal density function.

As with all continuous probability density functions, the normal density function has the fundamental properties that:

- probability corresponds to an area under the curve
- the total area under the curve is 1.

However, it has some additional special properties.

- The normal density function is symmetric and bell-shaped with its **centre** determined by the **mean** of the distribution it represents and its **width** determined by the **standard deviation**. It may be seen from the graph on the right that the graph of $y = f(x)$ is symmetric about the line $x = \mu$, and has a maximum value of $\frac{1}{\sigma\sqrt{2\pi}}$ which occurs when $x = \mu$. Thus the **location** of the curve on the graph is determined by the value of μ and the **steepness of the curve** by the value of σ .



- Irrespective of the values of the mean and standard deviation of a particular normal function the percentage of values that lie within a given number of standard deviations from the mean is always the same.

Example 1

On the same set of axes sketch (a calculator can be used to help) the graphs of the standard normal probability density function and graph of the probability density function of:

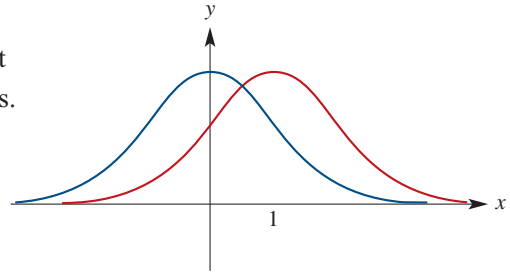
- a the normal distribution with mean 1 and standard deviation 1
- b the normal distribution with mean 1 and standard deviation 2.

Solution

- a** The graph has been translated 1 unit in the positive direction of the x -axis.

The rules of the probability density functions are $y = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and

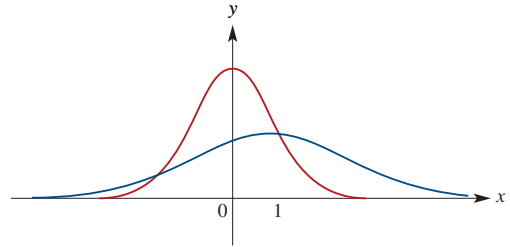
$$y = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2}$$



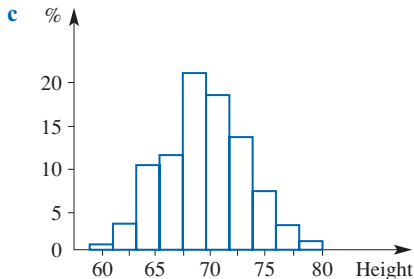
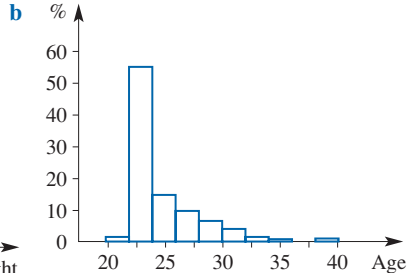
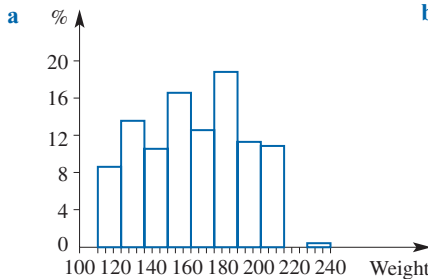
- b** The graph has been dilated from the y -axis by a factor of 2 and from the x -axis by factor $\frac{1}{2}$, and translated 1 unit in the positive direction of the x -axis.

The rules of the probability density functions are $y = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and

$$y = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1}{2}\right)^2}$$

**Exercise 18A**

- 1** Which of the following data distributions appear to be approximately normally distributed?



- 2** The random variables X_1 and X_2 are both normally distributed with means μ_1 and μ_2 , and standard deviations σ_1 and σ_2 , respectively. If $\mu_1 < \mu_2$, and $\sigma_1 < \sigma_2$, sketch both distributions on the same diagram.

3 Consider the normal probability density function:

$$f(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2} \quad x \in R$$

a Use your calculator to find:

$$\int_{-\infty}^{\infty} f(x) dx$$

b i Express $E(X)$ as an integral.

ii Use your calculator to evaluate the integral found in **i**.

c i Write down an expression for $E(X^2)$. **ii** What is the value of $E(X^2)$?

iii What is the value of σ ?

4 Consider the normal probability density function:

$$f(x) = \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x+4}{5}\right)^2} \quad x \in R$$

a Use your calculator to find:

$$\int_{-\infty}^{\infty} f(x) dx$$

b i Express $E(X)$ as an integral.

ii Use your calculator to evaluate the integral found in **i**.

c i Write down an expression for $E(X^2)$. **ii** What is the value of $E(X^2)$?

iii What is the value of σ ?

5 The probability density function of a normal random variable X is:

$$f(x) = \frac{1}{10\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-3}{10}\right)^2}$$

a Write down the mean and the standard deviation of X .

b Sketch the graph of $f(x)$.

6 The probability density function of a normal random variable X is:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+3)^2}$$

a Write down the mean and the standard deviation of X .

b Sketch the graph of $f(x)$.

7 The probability density function of a normal random variable X is:

$$f(x) = \frac{1}{\sqrt{18\pi}} e^{-\frac{1}{2}\left(\frac{x}{3}\right)^2}$$

a Write down the mean and the standard deviation of X .

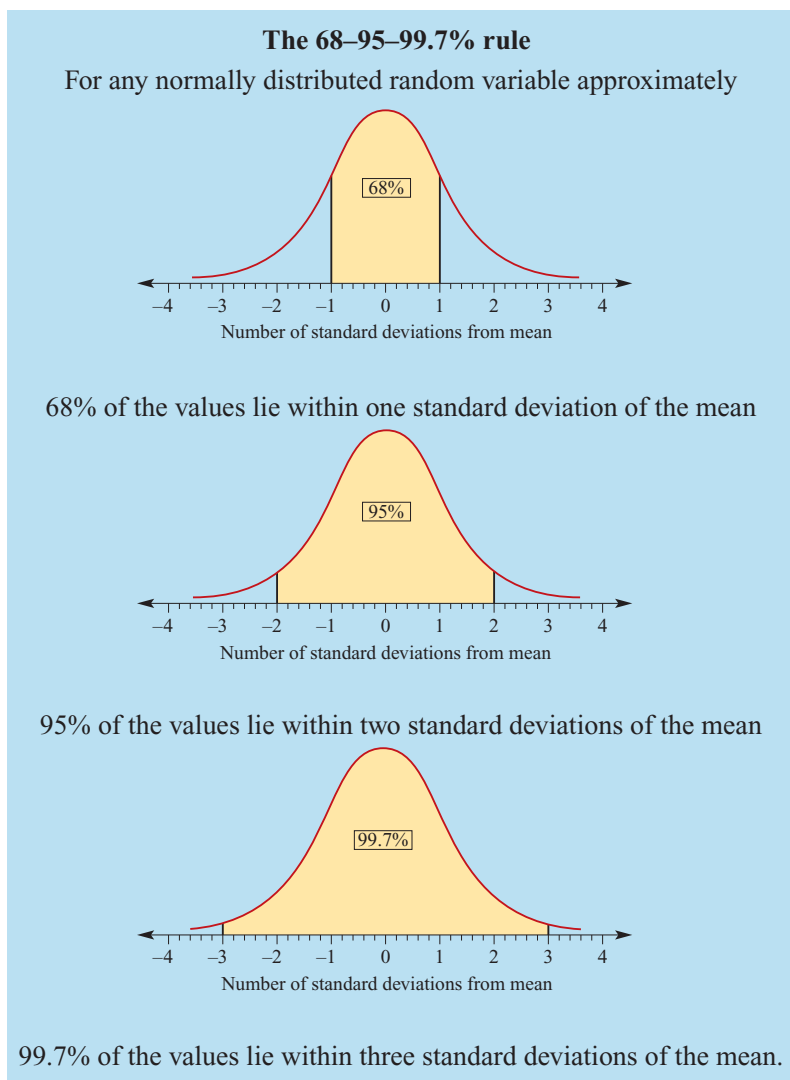
b Sketch the graph of $f(x)$.

- 8 Describe the sequence of transformations which takes the graph of the probability density function of the standard normal distribution to the graph of the probability density function of the normal distribution with:
- a** $\mu = 3$ and $\sigma = 2$ **b** $\mu = 3$ and $\sigma = \frac{1}{2}$ **c** $\mu = -3$ and $\sigma = 2$
- 9 Describe the sequence of transformations which takes the graphs of each of the probability density functions of the normal distribution with the given mean and standard to the graph of the probability density function of the standard normal distribution.
- a** $\mu = 3$ and $\sigma = 2$ **b** $\mu = 3$ and $\sigma = \frac{1}{2}$ **c** $\mu = -3$ and $\sigma = 2$

18.2 Standardisation and the 68–95–99.7% rule



For any normal distribution it can be shown that approximately 68% of the values will lie within one standard deviation of the mean, approximately 95% of the values lie within two standard deviations of the mean and almost all (99.7%) within three standard deviations. This gives rise to what has become known as the **68–95–99.7% rule**.



If we know that a random variable is approximately normally distributed, and we know its mean and standard deviation, then we can use the 68–95–99.7% rule to quickly make some important statements about the way in which the data values are distributed.

Example 2

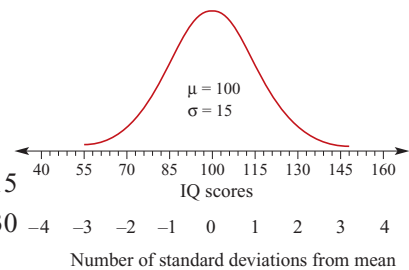
Experience has shown that the scores obtained on a commonly used IQ test can be assumed to be normally distributed with a mean $\mu = 100$ with a standard deviation $\sigma = 15$.

Approximately what percentage of the distribution lies within one, two or three standard deviations of the mean?

Solution

Knowing that the distribution of scores is normally distributed with $\mu = 100$ and $\sigma = 15$, the 68–95–99.7% rule means that approximately:

- 68% of the scores will lie between 85 and 115
- 95% of the scores will lie between 70 and 130
- 99.7% will lie between 55 and 145.



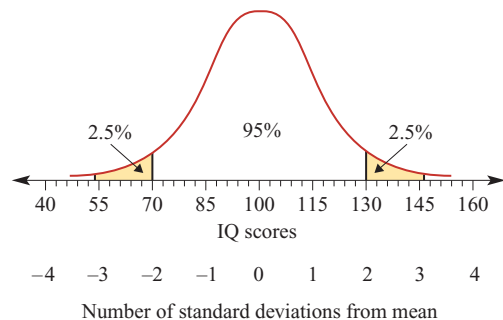
Statements can also be made about the percentage of scores that lie in the tails of the distribution by using the symmetry of the distribution and noting that the total area under curve is 100%.

Example 3

From Example 2, 95% of the scores in the IQ distribution lie between 70 and 130 (that is, within two standard deviations of the mean). What percentage of the distribution is **more** than two standard deviations above or below the mean (in this instance less than 70 or greater than 130)?

Solution

If we focus our attention on the tails of the distribution, we see that 5% of the IQ scores lie outside this region. Using the symmetry of the distribution we can say that 2.5% of the scores are below 70 and 2.5% are above 130. That is, if you obtained a score greater than 130 on this test you would have been in the top 2.5% of the group.



Clearly the standard deviation is a natural measuring stick for normally distributed data. For example, a person who obtained a score of 112 on an IQ test with a mean of $\mu = 100$ and a standard deviation $\sigma = 15$ is less than one standard deviation from the mean. Their score is typical of the group as a whole as it lies well within the middle 68% of scores. In contrast, a

person who scores 133 has done exceptionally well; their score is more than two standard deviations from the mean and this puts them in the top 2.5%. Because of the additional insight provided, it is usual to convert normally distributed data to a new set of units which shows the number of standard deviations each data value lies from the mean of the distribution. These new values are called **standardised values** or **z-values**. To standardise a data value x we first subtract the mean μ of the normal random variable from each score and then divide the result by the standard deviation σ .

$$\text{standardised value} = \frac{\text{data value} - \text{mean of the normal curves}}{\text{standard deviation of the normal curve}}$$

or symbolically

$$z = \frac{x - \mu}{\sigma}$$

Standardised values can be positive or negative:

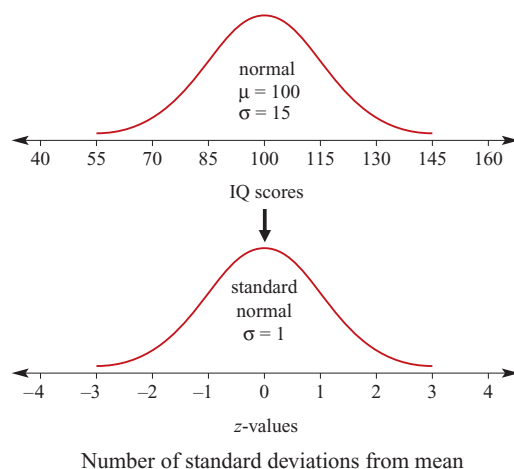
- a **positive** z-value indicates that the data value it represents lies **above** the mean
- a **negative** z-value indicates that the data value lies **below** the mean.

For example, an IQ score of 90 lies **below** the mean and has a standardised value of:

$$\begin{aligned} z &= \frac{x - \mu}{\sigma} \\ &= \frac{90 - 100}{15} \\ &= \frac{-10}{15} \\ &\approx -0.67 \end{aligned}$$

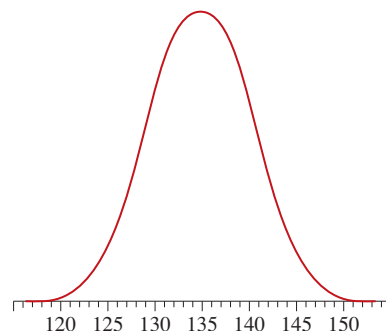
Clearly, not all data values in a normal distribution lie exactly one, two or three standard deviations from the mean. In these circumstances specially prepared tables or a calculator are used to give the required probability. There are as many different normal curves as there are values of μ and σ , but, if the measurement scale is changed to 'standard deviations from the mean' or z-values, all normal curves reduce to the same normal curve with a mean $\mu = 0$ and a standard deviation $\sigma = 1$. This special normal curve is known as the **standard normal**.

The figures on the right show how standardising IQ scores transforms a normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$ into the standard normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$.

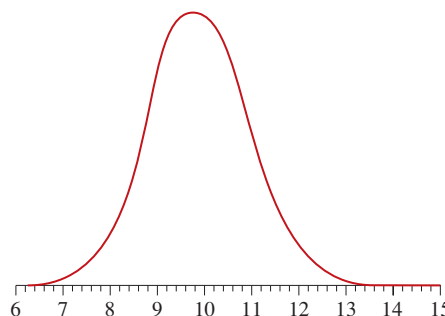


Exercise 18B

- 1 What are the values of the mean and standard deviation of the normal distribution shown?



- 2 What are the values of the mean and standard deviation of the normal distribution shown?



- 3 Experience has shown that the scores obtained on a commonly used IQ test can be assumed to be normally distributed with a mean $\mu = 100$ and a standard deviation $\sigma = 15$. What percentage of scores lies above 115?
- 4 The heights of young women are normally distributed with a mean $\mu = 160$ cm and a standard deviation $\sigma = 8$ cm.
What percentage of the women would you expect to have heights:
- a** between 152 and 168 cm? **b** greater than 168 cm? **c** less than 136 cm?
- 5 Fill in the blanks in the following paragraph:
The age at marriage of males in the US in the 1980s was approximately normally distributed with a mean $\mu = 27.3$ years and a standard deviation $\sigma = 3.1$ years. From this data we can conclude that, in the 1980s about 95% of males married between the ages of . . . and . . .
- 6 Fill in the blanks in the following statement of the 68–95–99.7% rule:
For any normal distribution, about
- 68% of the values lie within . . . standard deviation of the mean.
 - . . . % of the values lie within two standard deviations of the mean.
 - . . . % of the values lie within . . . standard deviations of the mean.
- 7 If you are told that in Australian adults, nostril width is approximately normally distributed with a mean $\mu = 2.3$ cm and a standard deviation $\sigma = 0.3$ cm, find the percentage of people with nostril widths less than 1.7 cm.

- 8 The distribution of IQ scores for the inmates of a certain prison is approximately normal with a mean $\mu = 85$ and a standard deviation $\sigma = 15$.
- What percentage of the prison population have an IQ of 100 or higher?
 - If someone with an IQ of 70 or less can be classified as having special needs, what percentage of the prison population could be classified as having special needs?
- 9 The distribution of heights of navy officers was found to be normal with a mean $\mu = 175$ cm and a standard deviation $\sigma = 5$ cm. Determine:
- the percentage of navy officers with heights between 170 cm and 180 cm
 - the percentage of officers with heights greater than 180 cm
 - the approximate percentage of navy officers with heights greater than 185 cm
- 10 The distribution of blood pressures (systolic) among women of similar ages is normal with a mean of 120 (mm of mercury) and a standard deviation of 10 (mm of mercury). Determine the percentage of women with a systolic blood pressure:
- between 100 and 140
 - greater than 130
 - greater than 120
 - between 90 and 150
- 11 The heights of women are normally distributed with a mean $\mu = 160$ cm and a standard deviation $\sigma = 8$ cm. What is the standardised value for the height of a woman who is:
- 160 cm tall?
 - 150 cm tall?
 - 172 cm tall?
- 12 The length of pregnancy for a human is approximately normally distributed with a mean $\mu = 270$ days and a standard deviation $\sigma = 10$ days. How many standard deviations away from the mean is a pregnancy of length:
- 256 days?
 - 281 days?
 - 305 days?
- 13 Michael scores 85 on the mathematics section of a scholastic aptitude test, the results of which are known to be normally distributed with a mean of 78 and a standard deviation of 5. Cheryl sits for a mathematics ability test and scores 27. These test scores are normally distributed with a mean of 18 and a standard deviation of 6. Assuming that both tests measure the same kind of ability, who has the better score?
- 14 The results obtained by a student in two subjects, Biology and History, showing mark (x), subject mean (μ) and standard deviation (σ) are shown in the table below:

Subject	Mark (x)	Mean (μ)	Standard deviation (σ)	Standardised mark (z)
Biology	77	68.5	4.9	
History	79	75.3	4.1	

Complete the table by calculating the student's standardised mark for each subject and use this information to determine in which subject the student did best *relative* to her peers.

- 15 Three students scored as follows on three different tests in French, English and Mathematics:

Subject	Test	Mark (x)	Mean (μ)	Standard deviation (σ)	Standardised mark (z)
Mary	French	19	15	4	
	English	42	35	8	
	Mathematics	20	20	5	
Steve	French	21	23	4	
	English	39	42	3	
	Mathematics	23	18	4	
Sue	French	15	15	5	
	English	42	35	10	
	Mathematics	19	20	5	

- a Determine the standardised mark for each student on each test.
- b Who is the best student in:
 - i French?
 - ii English?
 - iii Mathematics?
- c Who is the best student overall? Give reasons for your answer.

18.3 Determining normal probabilities

The CAS calculator can be used to determine areas under the standard normal curve for many different z -values, allowing us to find probabilities for values other than one, two or three standard deviations from the mean.

For the **TI-Nspire**, probabilities associated with the normal distribution are found in a **Calculator** application by accessing the **Distributions** menu in the **Probability** menu.

There are two commands which refer to the normal distribution:

- 1 The **Normal Pdf** command gives the value of the probability density function $f(x)$ for a particular value of X .
- 2 The **Normal Cdf** command gives the value of the cumulative density function $F(x)$ — that is, the area under the probability density function, for an interval.

To determine probabilities (areas under the probability density function), select **Normal Cdf**. This pastes the command onto the screen. To complete the command, enter the upper and lower limits for the probability to be calculated. If the distribution is the standard normal, then no more is required.

For the **Casio ClassPad** choose **Statistics** from the **Menu**. Then choose **Distribution** from the **Calc** menu.

Example 4

Suppose that Z is a standard normal random variable (that is, it has a mean $\mu = 0$ and a standard deviation $\sigma = 1$). Find:

- a** $\Pr(-1 < Z < 2.5)$ **b** $\Pr(Z > 1)$

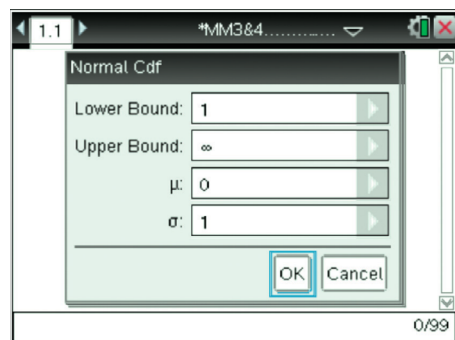
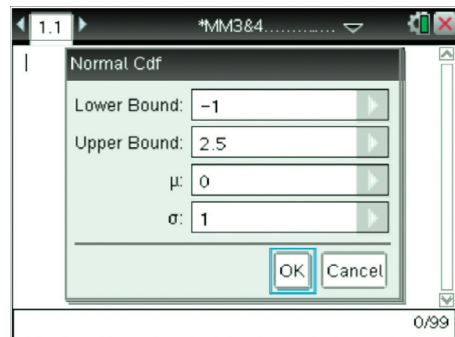
Solution**Using the TI-Nspire**

- a** Use **(menu)** > **Probability** > **Distributions** > **Normal Cdf** and complete as shown. Use the **tab** key (**(tab)**) to move between cells.

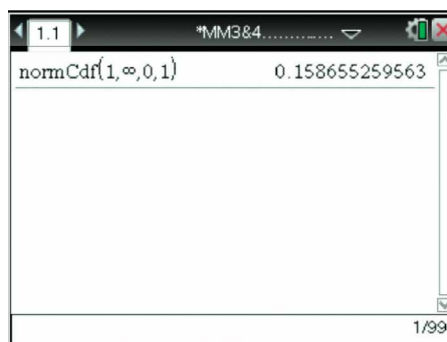
The answer is

$$\Pr(-1 < Z < 2.5) = 0.8351$$

- b** Use **(menu)** > **Probability** > **Distributions** > **Normal Cdf** and complete as shown. ∞ , the symbol for infinity, can be found using **(pi)** or **(ctrl)+(inf)** (or **(ctrl)+(i)** on the Clickpad)



The answer is $\Pr(Z > 1) = 0.1587$



Using the Casio ClassPad

In ▼ tap in the entry line and select

Calc – Distribution.

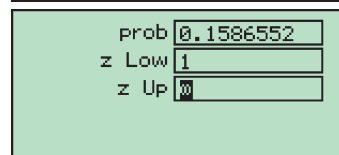
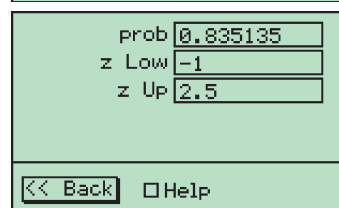
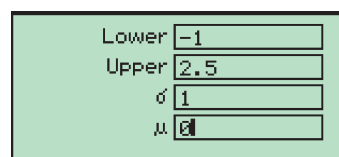
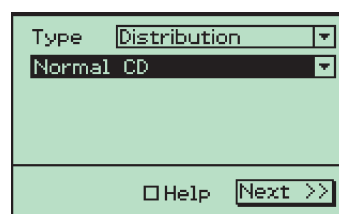
- a** In the field below Type, tap the ▼ and scroll to select **Normal CD**, then tap ▼.

Enter the values shown: Lower = -1 ,
Upper = 2.5 , $\sigma = 1$, $\mu = 0$.

The answer is

$$\Pr(-1 \leq Z \leq 2) = 0.8351.$$

- b** For $\Pr(Z > 1)$ enter Lower = 1 , Upper = ∞ . The answer is
 $\Pr(Z > 1) = 0.1587$.



To find probabilities from other than the standard normal distribution, there are two options. The standardisation formula discussed in the last section could be used to transform any variable to standard normal. The calculator will determine the probability associated with any normal distribution directly, by entering the appropriate values for μ and σ .

Some problems are solved by finding the value of the random variable which corresponds to a specified area under the normal curve. For the TI-Nspire, this can be done through the command **Inverse Normal** selected from the **Distributions** menu in the **Probability** menu.

The calculator can be used to determine **percentiles** of any normal distribution.

Example 5

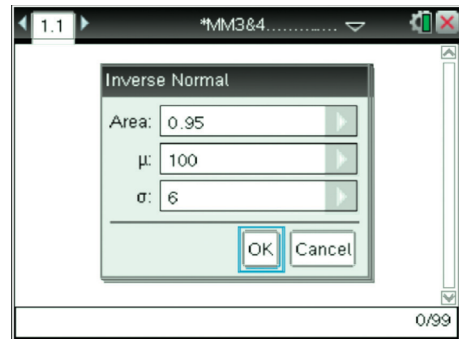
Suppose X is normally distributed with a mean $\mu = 100$ and a standard deviation $\sigma = 6$.

Find:

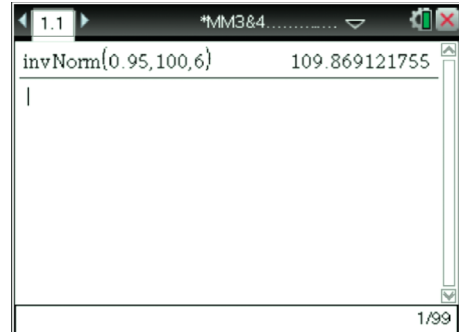
- a k such that $\Pr(X \leq k) = 0.95$
- b c_1 and c_2 , such that $\Pr(c_1 < X < c_2) = 0.95$

Solution**Using the TI-Nspire**

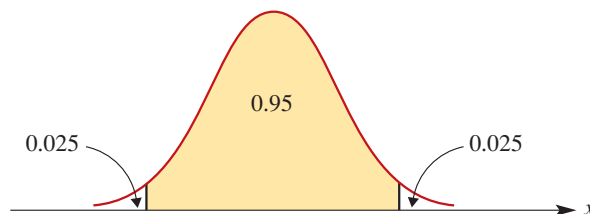
- a Use $\text{menu} > \text{Probability} > \text{Distributions} > \text{Inverse Normal}$ and complete as shown.



The value of k is 109.869



- b Examination of the normal curve shows that there are (infinitely) many intervals which enclose an area of 0.95. By convention, the interval which leaves equal areas in each tail is chosen.



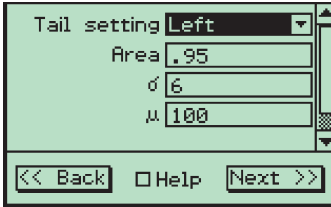
To find c_1 enter the value of 0.025 in the argument of the calculator. Enter 0.975 to find the value of c_2 . It is found that $c_1 = 88.240$ and $c_2 = 111.760$.

Using the ClassPad

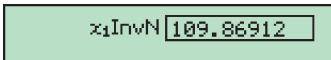
- a** In  tap **Calc – Distribution** and scroll to **Inverse Normal CD**. Enter the values given for μ , σ and the area.

Tail setting is set to Left to indicate we seek the value for which 95% of the area lies to the left.

- b** For $\Pr(Z > 1)$, enter Lower = 1, Upper = ∞ .
The answer is $\Pr(Z > 1) = 0.1587$.



Tail setting: Left
Area: .95
 σ : 6
 μ : 100
Navigation: << Back, Help, Next >>



$x_1\text{InvN}$: 109.86912

Exercise 18C

- Suppose Z is a standard normal random variable (that is, it has a mean $\mu = 0$ and a standard deviation $\sigma = 1$). Find the following probabilities, drawing an appropriate diagram in each case:

a $\Pr(Z < 2)$	b $\Pr(Z < 2.5)$	c $\Pr(Z \leq 2.5)$	d $\Pr(Z < 2.53)$
e $\Pr(Z \geq 2)$	f $\Pr(Z > 1.5)$	g $\Pr(Z \geq 0.34)$	h $\Pr(Z > 1.01)$
- Suppose Z is a standard normal random variable. Find the following probabilities, drawing an appropriate diagram in each case:

a $\Pr(Z > -2)$	b $\Pr(Z > -0.5)$	c $\Pr(Z > -2.5)$	d $\Pr(Z \geq -1.283)$
e $\Pr(Z < -2)$	f $\Pr(Z < -2.33)$	g $\Pr(Z \leq -1.8)$	h $\Pr(Z \leq -0.95)$
- Suppose Z is a standard normal random variable. Find the following probabilities, drawing an appropriate diagram in each case:

a $\Pr(-1 < Z < 1)$	b $\Pr(-2 < Z < 2)$	c $\Pr(-3 < Z < 3)$
----------------------------	----------------------------	----------------------------

How do these results compare with the ‘68%–95%–99.7% rule’ discussed in Section 18.2?
- Suppose Z is a standard normal random variable. Find the following probabilities, drawing an appropriate diagram in each case:

a $\Pr(2 < Z < 3)$	b $\Pr(-1.5 < Z < 2.5)$
c $\Pr(-2 < Z < -1.5)$	d $\Pr(-1.4 < Z < -0.8)$
- Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \leq c) = 0.9$
- Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \leq c) = 0.75$

- 7 Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \leq c) = 0.975$
- 8 Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \geq c) = 0.95$
- 9 Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \geq c) = 0.8$
- 10 Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \leq c) = 0.10$
- 11 Suppose Z is a standard normal random variable. Draw an appropriate diagram and then find the value c such that $\Pr(Z \leq c) = 0.025$
- 12 Let X be a normal random variable with mean $\mu = 100$ and a standard deviation $\sigma = 6$. Find:
a $\Pr(X < 110)$ **b** $\Pr(X < 105)$ **c** $\Pr(X > 110)$ **d** $\Pr(105 < X < 110)$
- 13 Let X be a normal random variable with mean $\mu = 40$ and a standard deviation $\sigma = 5$. Find:
a $\Pr(X < 48)$ **b** $\Pr(X < 36)$ **c** $\Pr(X > 32)$ **d** $\Pr(32 < X < 36)$
- 14 Let X be a normal random variable with mean $\mu = 6$ and a standard deviation $\sigma = 2$. Find:
a c such that $\Pr(X < c) = 0.95$ **b** k such that $\Pr(X < k) = 0.90$
- 15 Let X be a normal random variable with mean $\mu = 10$ and a standard deviation $\sigma = 3$. Find:
a c such that $\Pr(X < c) = 0.50$ **b** k such that $\Pr(X < k) = 0.975$
- 16 Give that X is a normally distributed random variable with mean equal to 22 and standard deviation of 7, find:
a $\Pr(X < 26)$ **b** $\Pr(25 < X < 27)$
c $\Pr(X < 26 | 25 < X < 27)$ **d** c such that $\Pr(X < c) = 0.95$
e k such that $\Pr(X > k) = 0.9$ **f** c_1 and c_2 such that $\Pr(c_1 < X < c_2) = 0.95$
- 17 Let X be a normal random variable with mean $\mu = 10$ and a standard deviation $\sigma = 0.5$. Find:
a $\Pr(X < 11)$ **b** $\Pr(X < 11 | X < 13)$
c c such that $\Pr(X < c) = 0.95$ **d** k such that $\Pr(X < k) = 0.2$
e c_1 and c_2 such that $\Pr(c_1 < X < c_2) = 0.95$

18.4 Solving problems using the normal distribution

The normal distribution can be used to solve many practical problems, as shown in the following examples.

Example 6

The time taken to complete a logical reasoning task is normally distributed with a mean of 55 seconds and a standard deviation of 8 seconds.

- Find the probability, correct to four decimal places, that a randomly chosen person will take less than 50 seconds to complete the task.
- Find the probability, correct to four decimal places, that a randomly chosen person will take less than 50 seconds to complete the task, if it is known that that person took less than 60 seconds to complete the task.

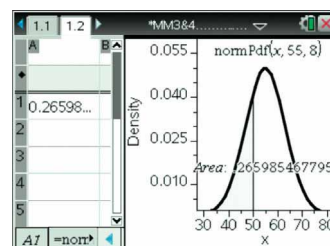
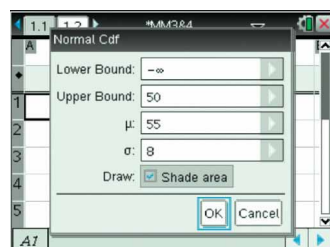
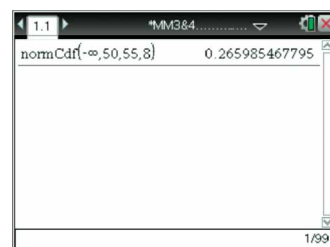
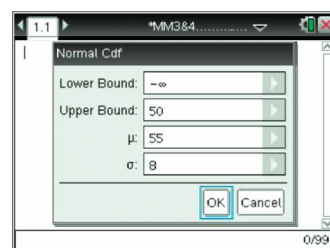
Solution

Using the TI-Nspire

- Use **menu** > **Probability** > **Distributions** > **Normal Cdf** and complete as shown.

The answer is that
 $\Pr(X < 50) = 0.2660$

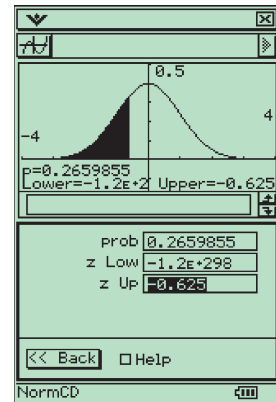
Note: You can also do this in a Lists & Spreadsheet application and plot the graph as shown.



$$\begin{aligned}
 \text{b } \Pr(X < 50 | X < 60) &= \Pr\left(\frac{X < 50 \cap X < 60}{X < 60}\right) \text{ by definition} \\
 &= \Pr\left(\frac{X < 50}{X < 60}\right) \\
 \therefore \Pr(X < 50 | X < 60) &= \frac{0.2660}{0.7340} \\
 &= 0.3624
 \end{aligned}$$

Using the Casio ClassPad

- a** In  tap **Calc – Distribution** and scroll to **Normal CD** in the Type area. Enter the values as given (∞ is found by pressing **Keyboard**). Tap **Next >>** to obtain the answer and the upper and lower bounds converted to z-scores. The answer is $\Pr(X < 50) = 0.2660$. Tap **AB** to obtain a graph of the probability density function.



When the mean and standard deviation of a normal distribution are unknown it is sometimes necessary to transform to the standard normal. This is demonstrated in the following example.

Example 7

Limits of acceptability imposed on the lengths of a certain batch of metal rods are 1.925 cm and 2.075 cm. It is observed that, on average, 5% are rejected as undersized and 5% are rejected as oversized.

Assuming that the lengths are normally distributed find the mean and standard deviation of the distribution.

Solution

It is given that $\Pr(X > 2.075) = 0.05$ and $\Pr(X < 1.925) = 0.05$

Symmetry tells us that the mean is equal to $(2.075 + 1.925) \div 2 = 2$

Transforming to the standard normal gives:

$$\Pr\left(Z > \frac{2.075 - \mu}{\sigma}\right) = 0.05 \text{ and } \Pr\left(Z < \frac{1.925 - \mu}{\sigma}\right) = 0.05$$

$$\text{Also } \Pr\left(Z < \frac{2.075 - \mu}{\sigma}\right) = 0.95$$

Using the inverse normal facility of the calculator gives:

$$\frac{2.075 - \mu}{\sigma} = 1.6448 \dots \text{ and } \frac{1.925 - \mu}{\sigma} = -1.6448 \dots$$

These solve to confirm $\mu = 2$ and substituting in the first of the equations gives:

$$\frac{2.075 - 2}{\sigma} = 1.6448 \dots$$

Solving for σ gives:

$$\sigma = 0.045596 \dots$$

and to four decimal places $\sigma = 0.0456$

Exercise 18D



- 1 Suppose that IQ scores are normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$
 - a What is the probability that a person chosen at random has an IQ:
 - i greater than 110?
 - ii less than 75?
 - iii greater than 130 given that they have an IQ greater than 110?
 - b To join an elite club one has to have an IQ in the top 5% of the population. What IQ score would be necessary to join this club?
- 2 The heights of women are normally distributed with a mean $\mu = 160$ cm and a standard deviation $\sigma = 8$ cm.
 - a What is the probability that a woman chosen at random would be:
 - i taller than 155 cm?
 - ii shorter than 170 cm?
 - iii taller than 170 cm given that she is between 168 and 174 cm tall?
 - b What height would put a woman among the tallest 10% of the population?
 - c What height would put a woman among the shortest 20% of the population?
- 3 The results of a mathematics exam are normally distributed with mean $\mu = 50$ and standard deviation $\sigma = 7$.
 - a What is the probability that a student chosen at random has an exam score:
 - i greater than 60?
 - ii less than 75?
 - iii greater than 60 given that they passed? (Assume a pass mark of 50%.)
 - b The top 15% of the class are to be awarded a distinction. What score would be required to gain a distinction in this exam?
- 4 The lengths of a species of fish are normally distributed with a mean length of 40 cm and a standard deviation of 4 cm. Find the percentage of these fish having lengths:
 - a greater than 45 cm
 - b between 35.5 cm and 45.5 cm

- 5 The weight of cats is normally distributed. It is known that 10% of cats weigh more than 1.8 kg, and 15% of cats weigh less than 1.35 kg. Find the mean and the standard deviation of this distribution.
- 6 The marks of a large number of students in a statistics examination are normally distributed with a mean of 48 marks and a standard deviation of 15 marks.
- a If the pass mark is 53, find the percentage of students who passed the examination.
 - b If 8% of students gained an A on the examination by scoring a mark of at least c , find the value of c .
- 7 The height of a certain population of adult males is normally distributed with mean 176 cm and standard deviation 7 cm.
- a Find the probability that the height of a randomly selected individual will exceed 190 cm.
 - b If two individuals are selected at random, find the probability that both of their heights will exceed 190 cm.
 - c Suppose 10 individuals are selected at random. Find the probability that at least two will have heights that exceed 190 cm.
- 8 The volume of soft drink in a 1 litre bottle is normally distributed. The soft drink company needs to calibrate its filling machine. They don't want to put too much soft drink into each bottle, as it adds to their expense. However, they know they will be fined if more than 2% of bottles are more than 2 millilitres under volume. The standard deviation of the volume dispensed by the filling machine is 2.5 millilitres. What should they choose as the target volume (i.e. the mean of the distribution)? Give your answer to the nearest millilitre.
- 9 The weights of pumpkins sold to a greengrocer are normally distributed with a mean of 1.2 kg and a standard deviation of 0.4 kg. The pumpkins are sold in three sizes:
- | | |
|--------|-----------------------|
| small | under 0.8 kg |
| medium | from 0.8 kg to 1.8 kg |
| large | over 1.8 kg |
- a Find the proportions of pumpkins in each of the three sizes.
 - b The prices of the pumpkins are \$2.80 for a small, \$3.50 for a medium and \$5.00 for a large. Find the expected cost for 100 pumpkins chosen at random from the greengrocer's supply.
- 10 Potatoes are delivered to a chip factory in semitrailer loads. A sample of 1 kg of the potatoes is chosen from each load and tested for starch content. From past experience it is known that the starch content is normally distributed with a standard deviation 2.1.
- a For a semi-trailer with potatoes with mean starch content of 22.0:
 - i What is the probability that the reading is 19.5 or less?
 - ii What reading will be exceeded with probability of 0.98?

- b** If the starch content is greater than 22.0 the potatoes cannot be used for chips, so the semitrailer load is rejected. What is the probability that a load with mean starch content of 18.0 will be rejected?
- 11** The amount of a certain chemical in a type A cell is normally distributed with mean of 10 and a standard deviation of 1, while the amount in a type B cell is normally distributed with a mean of 14 and a standard deviation of 2. To determine whether a cell is a type A or a type B, the amount of chemical in the cell is measured and the cell is classified as a type A if the amount is less than a specified value c , and as being of type B otherwise.
- a** If $c = 12$ calculate the probability that a type A cell will be misclassified, and the probability that a type B cell will be misclassified.
- b** Find the value of c for which the two probabilities of misclassification are equal.



Chapter summary

- A special continuous random variable is described by the **normal** probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where μ and σ are the mean and standard deviation of the normal random variable X .

- The centre of the normal curve is determined by the mean, μ , and its width is determined by the standard deviation, σ .
- The **68–95–99.7% rule** states that, for any normal distribution:
 - 1 Approximately 68% of the values lie within one standard deviation of the mean.
 - 2 Approximately 95% of the values lie within two standard deviations of the mean.
 - 3 Approximately 99.7% of the values lie within three standard deviations of the mean.

- If X is a normally distributed random variable with mean μ and standard deviation σ then to **standardise** we first subtract the mean μ from X and then divide the result by the standard deviation σ . The standardised value is denoted z and has a mean $\mu = 0$ and standard deviation $\sigma = 1$.

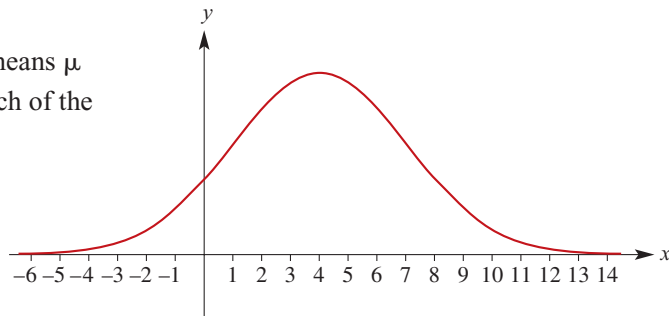
$$z = \frac{x - \mu}{\sigma}$$

- A **calculator** gives the values for the cumulative distribution function of the normal distribution—that is, areas under the standard normal curve for many different values of the random variable.
- A **calculator** gives the inverse normal values for a random variable—that is, values corresponding to specified areas under the normal curve.

Multiple-choice questions

- 1 The diagram shows the graph of a normal distributions with means μ and standard deviations σ . Which of the following statements is true?

- A $\mu = 4$ and $\sigma = 3$
- B $\mu = 3$ and $\sigma = 4$
- C $\mu = 4$ and $\sigma = 2$
- D $\mu = 3$ and $\sigma = 2$
- E $\mu = 4$ and $\sigma = 4$



- 2 $\Pr(Z > 1.45)$, where Z is a standard normal random variable, is:
- A 0.1394 B 0.8606 C 0.0735 D 0.9625 E 0.0925
- 3 If Z is a standard normal random variable, and $\Pr(Z < c) = 0.25$, then the value of c is closest to:
- A 0.6745 B -0.6745 C 0.3867 D 0.5987 E -0.5987

- 4 The random variable X has a normal distribution with mean 12 and variance of 9. If Z is a standard normal random variable then the probability that X is more than 15 is equal to:
- A** $\Pr(Z < 1)$ **B** $\Pr(Z > 1)$ **C** $\Pr\left(Z > \frac{1}{3}\right)$
- D** $1 - \Pr\left(Z > \frac{1}{3}\right)$ **E** $1 - \Pr(Z > 1)$
- 5 If the actual length of an AFL game is normally distributed with a mean of 102 minutes and a standard deviation of 3 minutes, then the percentage of games that last more than 110 minutes is approximately:
- A** 96.2% **B** 81.3% **C** 2.7% **D** 18.7% **E** 0.38%
- 6 If the number of goals Collingwood scores a week is a normally distributed random variable with a mean of 16 and a standard deviation of 2, then in what percentage of their matches (approximately) do they score from 10 to 22 goals?
- A** 5% **B** 16% **C** 68% **D** 95% **E** 99.7%
- 7 The amount of water Steve uses to water the garden is normally distributed with a mean of 100 litres and a standard deviation of 14 litres. On 20% of occasions it takes him more than the k litres to water the garden. What is the value of k ?
- A** 88.2 **B** 110.7 **C** 120.0 **D** 111.8 **E** 114.0
- 8 If X is a normally distributed random variable with mean $\mu = 6$ and standard deviation $\sigma = 3$, then the transformation which maps the curve of the density function of X , $f(x)$ to the curve of the standard normal distribution is:
- A** $(x, y) \rightarrow \left(\frac{x-3}{6}, 6y\right)$ **B** $(x, y) \rightarrow \left(\frac{x-6}{3}, \frac{y}{3}\right)$ **C** $(x, y) \rightarrow \left(\frac{x-6}{3}, 3y\right)$
- D** $(x, y) \rightarrow (3(x+6), 3y)$ **E** $(x, y) \rightarrow \left(3(x+6), \frac{y}{3}\right)$
- 9 The marks achieved by Angie in Politics, Mathematics and Indonesian, together with the mean and the standard deviation for each subject, are given in the following table:

Subject	Mark	Mean (μ)	Standard deviation (σ)
Mathematics	72	72	5
Indonesian	57	59	2
Politics	68	64	4

Which of the following statements is correct?

- A** Angie's best subject was Politics, followed by Mathematics and then Indonesian.
- B** Angie's best subject was Mathematics, followed by Politics and then Indonesian.
- C** Angie's best subject was Politics, followed by Indonesian and then Mathematics.
- D** Angie's best subject was Mathematics, followed by Indonesian and then Politics.
- E** Angie's best subject was Indonesian, followed by Mathematics and then Indonesian.
- 10 Suppose that X is normally distributed with a mean of 11.3 and a standard deviation of 2.9. The values of c_1 and c_2 , such that $\Pr(c_1 < X < c_2) = 0.90$ are closest to:
- A** 5.5, 17.1 **B** 6.08, 16.52 **C** 15.02, 7.58 **D** 6.53, 16.07 **E** 5.62, 16.98

Short-answer questions (technology-free)

- 1 For the standard normal random variable Z , $\Pr(Z \leq a) = p$.
Find in terms of p :
a $\Pr(Z > a)$ **b** $\Pr(Z < -a)$ **c** $\Pr(-a \leq Z \leq a)$
- 2 X is the normal random variable with mean 4 and standard deviation 1. Z is the standard normal random variable.
a If $\Pr(X < 3) = \Pr(Z < a)$, $a = ?$
b If $\Pr(X > 5) = \Pr(Z > b)$, $b = ?$
c $\Pr(X > 4) = ?$
- 3 A normal distribution has mean 8 and standard deviation 3. Give the rule that maps the curve of the density function to the density function of the standard normal.
- 4 X is a normal random variable with mean μ and standard deviation σ . If $\mu < a < b$, $\Pr(X < b) = p$ and $\Pr(X < a) = q$, find:
a $\Pr(X < a | X < b)$ **b** $\Pr(X < 2\mu - a)$ **c** $\Pr(X > b | X > a)$
- 5 X is a normal random variable with mean 4 and standard deviation 2. Write each of the following probabilities in terms of Z .
a $\Pr(X < 5)$ **b** $\Pr(X < 3)$ **c** $\Pr(X > 5)$
d $\Pr(3 < X < 5)$ **e** $\Pr(3 < X < 6)$

In questions 6 to 8 you will use the following:

$$\Pr(Z < 1) = 0.84, \Pr(Z < 2) = 0.98, \Pr(Z < 0.5) = 0.69$$

- 6 A machine produces metal rods with mean diameter 2.5 mm and standard deviation 0.05 mm. Let X be the random variable of the normal distribution. Find:
a $\Pr(X < 2.55)$ **b** $\Pr(X < 2.5)$ **c** $\Pr(X < 2.45)$ **d** $\Pr(2.45 < X < 2.55)$
- 7 Nuts are packed in tins such that the mean weight of the tins is 500 g and the standard deviation is 5 g. The weights are normally distributed with random variable W . Find:
a $\Pr(W > 505)$ **b** $\Pr(500 < W < 505)$ **c** $\Pr(W > 505 | W > 500)$ **d** $\Pr(W > 510)$
- 8 A random variable X has a normal distribution with mean 6 and standard deviation 1. Find:
a $\Pr(X < 6.5)$ **b** $\Pr(6 < X < 6.5)$ **c** $\Pr(6.5 < X < 7)$ **d** $\Pr(5 < X < 7)$
- 9 Suppose that three tests were given in your Mathematics course. The class means and standard deviations, together with your scores, are listed in the table:

Test	μ	σ	Your score
A	50	11	62
B	47	17	64
C	63	8	73

On which test did you do best and on which did you do worst?

Extended-response questions

- 1 A test devised to measure mathematical aptitude gives scores that are normally distributed with a mean of 50 and a standard deviation of 10. If we wish to categorise the results so that the highest 10% of scores are designated as high aptitude, the next 20% as moderate aptitude, the middle 40% as average, the next 20% as little aptitude and the lowest 10% as no aptitude, what ranges of scores will be covered by each of these five categories?

- 2 If X is normally distributed with $\mu = 10$ and $\sigma = 2$, find the value of k such that

$$\Pr(\mu - k \leq X \leq \mu + k) = 0.95$$

- 3 Records kept by a manufacturer of car tyres suggest that the distribution of the mileage from their tyres is normal, with a mean of 60 000 kilometres and a standard deviation of 5000 kilometres.
- What proportion of the company's tyres last:
 - less than 55 000 kilometres?
 - more than 50 000 kilometres but less than 74 000 kilometres?
 - more than 72 000 kilometres, given that they have already lasted more than 60 000 kilometres?
 - The company's advertising manager wishes to claim that '90% of our tyres last longer than c kilometres'. What should c be?
 - What is the probability that a customer buys five tyres at the same time and finds that they all last longer than 72 000 kilometres?
- 4 The owner of a new van complained to the dealer that he was using, on average, 18 litres of petrol to drive 100 km. The dealer pointed out that the 15 litres/100 km referred to in an advertisement was 'just a guide and that actual consumption will vary'. Suppose that the distribution of fuel consumption for this make of van is normal with a mean of 15 litres/100 km and standard deviation 0.75 litres/100 km.
- How probable is a van that uses at least 18 litres/100 km?
 - What does your answer to **a** suggest about the manufacturer's claim?
 - Find c_1 and c_2 such that the van's fuel consumption is more than c_1 but less than c_2 with a probability of 0.95.
- 5 Suppose that L , the useful life (in hours) of a fluorescent tube used for indoor gardening, is normally distributed with a mean of 600 and a standard deviation of 4. The fluorescent tubes are sold in boxes of 10. Find the probability that at least three of the tubes in a randomly selected box last longer than 605 hours.

- 6 The amount of anaesthetic required to cause surgical anaesthesia in patients is normally distributed, with a mean of 50 mg and a standard deviation of 10 mg. The lethal dose is also normally distributed with a mean of 110 mg and a standard deviation of 20 mg. If a dosage that brings 90% of patients to surgical anaesthesia were used, what percentage of patients would be killed by this dose?
- 7 The hardness of a metal may be determined by impressing a hardened point into the surface of the metal and then measuring the depth of penetration of the point. Suppose that the hardness of a particular alloy is normally distributed with mean 70 and standard deviation 3.
- a If a specimen is acceptable only if its hardness is between 65 and 75, what is the probability that a randomly chosen specimen has an acceptable hardness?
 - b If the acceptable range of hardness was $(70 - c, 70 + c)$, for what value of c would 95% of all specimens have acceptable hardness?
 - c If the acceptable range is the same as in **a**, and the hardness of each of 10 randomly selected specimens is independently determined, what is the expected number of acceptable specimens among the 10?
 - d What is the probability that at most eight out of 10 independently selected specimens have a hardness less than 73.84?
 - e The profit on an acceptable specimen is \$20.00, while unacceptable specimens result in a loss of \$5.00. If P dollars is the profit on a randomly selected specimen, find the mean and variance of P .
- 8 The weekly error (in seconds) of a brand of watch is known to be normally distributed. Only those watches with an error of less than 5 seconds are acceptable.
- a Find the mean and standard deviation of the distribution of error if 3% of watches are rejected for losing time and 3% are rejected for gaining time.
 - b Determine the probability that fewer than two watches are rejected in a batch of 10 such watches.
- 9 In a given manufacturing process, components are rejected if they have a particular dimension greater than 60.4 mm or less than 59.7 mm. It is found that 3% are rejected as being too large and 5% are rejected for being too small. Assume that the dimension is normally distributed.
- a Find the mean and standard deviation of the distribution of the dimension, correct to one decimal place.
 - b Use the result of **a** to find the percentage of rejects if the limits for acceptance are changed to 60.3 mm and 59.6 mm.

- 10** A brand of detergent is sold in bottles of two sizes — standard and large. For each size, the content (in litres) of a randomly chosen bottle is normally distributed with mean and standard deviation as given in the table:

	Mean	Standard deviation
Standard bottle	0.760	0.008
Large bottle	1.010	0.009

- a** Find the probability that a randomly chosen standard bottle contains less than 0.75 litres.
- b** Find the probability that a box of 10 randomly chosen standard bottles contains at least three bottles whose contents are each less than 0.75 litres.
- c** Using the result $E(aX - bY) = aE(X) - bE(Y)$ and $\text{Var}(aX - bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$ find the probability that there is more detergent in four randomly chosen standard bottles than in three randomly chosen large bottles. (Assume $aX - bY$ is normally distributed.)

Revision of Chapters 14–18

19.1 Multiple-choice questions

- 1 A box contains 12 red balls and 4 green balls. A ball is selected at random from the box and not replaced, and then a second ball is drawn. The probability that the two balls are both green is equal to:

A $\frac{1}{4}$ B $\frac{1}{16}$ C $\frac{3}{64}$ D $\frac{1}{8}$ E $\frac{1}{20}$

- 2 In a six-item true/false test the probability that a student who guesses will obtain six correct answers is:

A 0.9844 B 0.0278 C 0.5 D 0.0156 E 0.17

Questions 3, 4, 5 and 6 refer to the following probability distribution:

x	4	6	7	9
$\Pr(X = x)$	0.3	0.2	0.1	0.4

- 3 For this probability distribution the mean, $E(X)$, is equal to:

A 6.7 B 0.275 C 6.5 D 2.75 E 2.59

- 4 For this probability distribution the variance, $\text{Var}(X)$, is equal to:

A 19.45 B 4.41 C 6.7 D 2.1 E 0.61

- 5 Let $Y = 2X - 1$, where X has the probability distribution given in the table. The probability distribution of Y is:

A

y	4	6	7	9
$\Pr(Y = y)$	0.3	0.2	0.1	0.4

B

y	8	12	14	18
$\Pr(Y = y)$	0.3	0.2	0.1	0.4

C

y	4	6	7	9
$\Pr(Y = y)$	0.6	0.4	0.2	0.8

D

y	6	10	12	16
$\Pr(Y = y)$	0.3	0.2	0.1	0.4

E

y	7	11	13	17
$\Pr(Y = y)$	0.3	0.2	0.1	0.4

- 6** Let $Z = 4 - X$, where X has the probability distribution given in the table. The variance of Z is:

A 11.59 **B** 2.1 **C** 0.41 **D** 4.41 **E** -0.41

- 7** X has the probability distribution given by:

x	1	2	3	4
$\Pr(X = x)$	$4c^2$	$5c^2$	$4c^2$	$3c^2$

The value of c is:

A 0.5 **B** 0.0263 **C** 0.1622 **D** 0.25 **E** 0.0625

- 8** Suppose a spinner numbered 1, 2, 3, 4, 5, 6 is spun until a '3' appears. The number of spins is noted. The sample space for this random experiment is:

A {1, 2, 3, 4, 5, 6} **B** {0, 1, 2, 3, 4, 5, 6} **C** {1, 2, 3, 4, ...}
D {3} **E** {1, 2, 3}

- 9** Suppose that in Melbourne the probability of the temperature exceeding 30°C on a particular day is 0.6 if the temperature exceeded 30°C on the previous day, and 0.25 if it did not. If the temperature exceeds 30°C on Monday, then the probability that it exceeds 30°C on Wednesday is:

A 0.36 **B** 0.10 **C** 0.60 **D** 0.30 **E** 0.46

- 10** For a random variable X , $E(X) = 11$ and $E(X^2) = 202$. The standard deviation of X is equal to:

A 191 **B** 13.82 **C** 9 **D** 3.72 **E** 81

- 11** A set of test scores have a probability distribution with mean $\mu = 50$ and the standard deviation $\sigma = 10$. Which of the following intervals contains about 95% of the test scores?

A (40, 60) **B** (30, 70) **C** (20, 80) **D** (46.84, 53.16) **E** (43.68, 56.32)

- 12** If three fair coins are tossed what is the probability that there are at least two heads?

A $\frac{1}{3}$ **B** $\frac{6}{7}$ **C** $\frac{1}{4}$ **D** $\frac{1}{2}$ **E** $\frac{1}{8}$

- 13** Let X be a binomial random variable with parameters $n = 400$ and $p = 0.1$. The mean of X , $E(X)$, is equal to:

A 36 **B** 6 **C** 40 **D** 6.32 **E** 360

- 14 Which of the following does not define a binomial variable?
- A A die is tossed 10 times and the number of sixes observed.
 - B A die is rolled until a six is obtained and the number of rolls counted.
 - C A die is rolled five times and the number of even numbers showing observed.
 - D A sample of 20 people is chosen from a population and the number of females counted.
 - E The number of questions answered correctly by a student who is guessing every question is noted.
- 15 Let X be a binomial random variable with parameters $n = 900$ and $p = 0.2$. The standard deviation of X is equal to:
- A 18 B 144 C 180 D 13.42 E 12
- 16 Let X be a binomial random variable with a variance of 9.4248. If $n = 42$ then the probability of success p is equal to:
- A 0.45 B 0.22 C 0.34 D 0.68 E 0.34 or 0.66
- 17 $\binom{7}{5} p^5 (1-p)^2$ is the probability of:
- A exactly two failures B exactly two successes C at least two failures
D exactly five failures E more failures than successes
- 18 The proportion of female students in a particular university is 0.2. A sample of 10 students is chosen at random from the entire student population. What is the probability that the sample contains exactly four female students?
- A 0.0881 B 0.5000 C 0.0328 D 0.0016 E 0.9672
- 19 Jane decides to call five friends to invite each of them to a party. The probability of a friend not being home when Jane calls is 0.4. What is the probability that Jane finds at least one of her friends at home?
- A 0.0778 B 0.9222 C 0.0102 D 0.9898 E 0.0768
- 20 Consider the matrices
- $$\mathbf{U} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 0.4 & 0.5 \\ 0.6 & 0.5 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 0.7 & 0.3 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 0.1 & 0 & 0.2 \\ 0.3 & 0.2 & 0 \\ 0.1 & 0.1 & 0.2 \end{bmatrix}$$
- The matrix that could be a transition matrix for a Markov chain is:
- A $\mathbf{U}$ B $\mathbf{V}$ C $\mathbf{W}$ D $\mathbf{X}$ E $\mathbf{Y}$
- 21 Suppose that a four-state Markov chain is defined by a transition matrix $\mathbf{T}$ and an initial state matrix $\mathbf{S}_0$. Which of the following is a true statement?
- A The dimension of $\mathbf{T}$ is 4×4 and the dimension of $\mathbf{S}_0$ is 4×1 .
 - B The dimension of $\mathbf{T}$ is 2×2 and the dimension of $\mathbf{S}_0$ is 2×1 .
 - C The dimension of $\mathbf{T}$ is 4×1 and the dimension of $\mathbf{S}_0$ is 4×4 .

D The dimension of $\mathbf{T}$ is 4×4 and the dimension of $\mathbf{S}_0$ is 1×1 .

E The dimension of $\mathbf{T}$ is 4×4 and the dimension of $\mathbf{S}_0$ is 4×4 .

- 22** A Markov chain is defined by a transition matrix $\mathbf{T} = \begin{bmatrix} 0.59 & 0.44 \\ 0.41 & 0.56 \end{bmatrix}$ and an initial state

matrix $\mathbf{S}_0 = \begin{bmatrix} 100 \\ 500 \end{bmatrix}$. For this Markov chain, $\mathbf{S}_2 =$

- A** $\begin{bmatrix} 200 \\ 1000 \end{bmatrix}$ **B** $\begin{bmatrix} 279 \\ 321 \end{bmatrix}$ **C** $\begin{bmatrix} 558 \\ 642 \end{bmatrix}$ **D** $\begin{bmatrix} 305.85 \\ 294.15 \end{bmatrix}$ **E** $\begin{bmatrix} 309.8775 \\ 290.1225 \end{bmatrix}$

The following information is needed for Questions 23 and 24:

Suppose that the probability of a badminton player winning a point is 0.77 if he has won the preceding point and 0.44 if he has lost the preceding point, and that $X_n = 1$ if the player wins the n th point and $X_n = 0$ if the player loses the n th point.

- 23** The steady state probability, $\Pr(X_n = 1)$ is equal to:

- A** 0.66 **B** 0.34 **C** $\frac{44}{67}$ **D** $\frac{23}{67}$ **E** 0.77

- 24** The probability that the player wins four points in a row after losing the first point, and then loses the next point, is closest to:

- A** 0.0462 **B** 0.0600 **C** 0.2009 **D** 0.1364 **E** 0.1050

- 25** If a random variable X has probability density function

$$f(x) = \begin{cases} kx^3 + \frac{3}{4}x & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

then k is equal to:

- A** $-\frac{3}{16}$ **B** $\frac{6}{25}$ **C** $-\frac{9}{16}$ **D** $-\frac{1}{8}$ **E** $-\frac{3}{8}$

- 26** If a random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{9}(4x - x^2) & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

then $\Pr(X \leq 2)$ is closest to:

- A** 0.6667 **B** 0.4074 **C** 0.5926 **D** 0.4444 **E** 0.5556

- 27** If a random variable X has probability density function

$$f(x) = \begin{cases} \frac{8}{3}(1-x) & 0 < x < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

then the median of X is equal to:

- A** 0.222 **B** 0.667 **C** 0.250 **D** 1.791 **E** 0.209

- 28 If a random variable X has probability density function

$$f(x) = \begin{cases} \frac{24}{x^3} & 3 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

then the mode of X is closest to:

- A 4.5 B 3 C 6 D 4 E 3.33

- 29 If a random variable X has a probability density function

$$f(x) = \begin{cases} 2\left(1 - \frac{1}{x^2}\right) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

then the mean of X is closest to:

- A 1 B 1.614 C 2 D 1.5 E 0.609

- 30 The probability of obtaining a z -value which falls between $z = -1.0$ and $z = 0$ for a standard normal distribution is approximately:

- A 0.05 B 0.20 C 0.34 D 0.68 E 0.16

- 31 For a normal probability distribution which of the following is true?

- A The mean is always positive.
 B No value can be more than four standard deviations away from the mean.
 C The area under the normal curve is approximately equal to 1.
 D The standard deviation is always positive.
 E The standard deviation is less than the mean.

- 32 If X is a normally distributed random variable with mean $\mu = 2$ and standard deviation $\sigma = 0.5$, then the probability that X is greater than 2.6 is closest to:

- A 0.8849 B 0.9918 C 0.1151 D 0.0082 E 0.0302

- 33 If X is a normally distributed random variable with mean $\mu = 2$ and standard deviation $\sigma = 2$, then the probability that X is less than -2 is given by:

- A 0.1587 B 0.8413 C 0.9772 D 0.1228 E 0.0228

- 34 If X is a normally distributed random variable with mean $\mu = 3$ and variance $\sigma^2 = 0.4$, then the probability that X is greater than -2.73 is given by:

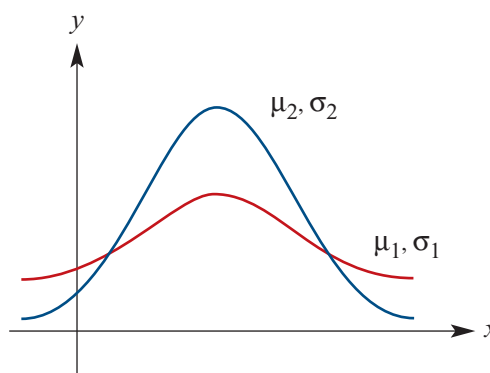
- A 1 B 0 C 0.9115 D 0.0885 E 0.5537

- 35 If X is a normally distributed random variable with mean $\mu = 2$ and variance $\sigma^2 = 4$, then $\Pr(1 < X < 2.5)$ is given by:

- A 0.5987 B 0.2902 C 0.6915 D 0.4013 E 0.3085

- 36** A machine fills cups with cordial when money is inserted in an automatic dispensing machine. If the amount of cordial in the cup is a normally distributed random variable with a mean of 50 mL and a standard deviation of 2 mL, then 90% of the cups contain more than:
- A** 44.87 mL **B** 53.29 mL **C** 46.71 mL **D** 52.56 mL **E** 47.44 mL
- 37** Lengths of blocks of cheese are found to be normally distributed with a mean of 10 cm and a variance of 0.5 cm. Then 95% of the blocks of cheese are shorter than:
- A** 11.39 cm **B** 8.84 cm **C** 11.16 cm **D** 8.61 cm **E** 9.18 cm
- 38** If X is a normally distributed random variable with mean $\mu = 1$ and variance $\sigma^2 = 2.25$ and $\Pr(\mu - k < X < \mu + k) = 0.7$, then $k =$
- A** 1.555 **B** 1.037 **C** 0.787 **D** 0.524 **E** 2.332
- 39** The weight of a packet of biscuits is known to be normally distributed with a mean of 1 kg. If a packet is more than 0.05 kg underweight, it is unacceptable. If it is found that 3% of packets are unacceptable, then the standard deviation of the weight is:
- A** 1.881 **B** 0.027 **C** 10.488 **D** 0.030 **E** 37.616

- 40** Consider the diagram, which shows the probability density functions of two normally distributed random variables, one with a mean μ_1 and a standard deviation σ_1 , and the other with a mean μ_2 and a standard deviation σ_2 .



Which of the following statements is true?

- A** $\mu_1 = \mu_2, \sigma_1 < \sigma_2$ **B** $\mu_1 = \mu_2, \sigma_1 > \sigma_2$ **C** $\mu_1 > \mu_2, \sigma_1 = \sigma_2$ **D** $\mu_1 < \mu_2, \sigma_1 = \sigma_2$ **E** $\mu_1 = \mu_2, \sigma_1 = \sigma_2$
- 41** If the heights of a certain population of men are normally distributed with a mean of 173 cm and a variance of 25 cm, then about 68% of men in the population have heights in the interval (in cm):
- A** (148, 198) **B** (168, 178) **C** (163, 183) **D** (158, 188) **E** (123, 223)

19.2 Extended-response questions

- 1** At each of a series of trials, the probability of the occurrence of a certain event is $\frac{1}{2}$, except that it cannot occur in two consecutive trials.
- a** Show that the probability of it occurring just twice **i** in three trials is $\frac{1}{4}$, and **ii** in four trials is $\frac{1}{2}$.
- b** What is the probability of it occurring just twice in five trials?

- 2 Katia and Mikki play a game in which a fair six-sided die is thrown five times. Katia will receive \$1 from Mikki if there is an odd number of sixes, and Mikki will receive \$ x from Katia if there is an even number of sixes. Find x for the game to be fair. (Count zero as even.)
- 3 Lleyton Hewitt and Roger Federer are playing the final of a tennis tournament. The winner of the match will be the first player to win two sets. The result of each set follows a Markov sequence, as described below:

$$\begin{array}{lll} \Pr(L_1) = 0.45 & \Pr(L_{i+1}|L_i) = 0.6 & \Pr(L_{i+1}|R_i) = 0.45 \\ \Pr(R_1) = 0.55 & \Pr(R_{i+1}|L_i) = 0.4 & \Pr(R_{i+1}|R_i) = 0.55 \end{array}$$

where L represents the event that Lleyton Hewitt wins, and R represents the event that Roger Federer wins.

- What is the probability that Lleyton Hewitt wins the first two sets?
 - What is the probability that Roger Federer wins the first two sets?
 - What is the probability that Lleyton Hewitt will win the match:
 - given that he wins the first set?
 - given that Roger Federer wins the first set?
 - What is the probability that Lleyton Hewitt will win the match?
- 4 A newspaper seller buys papers for 50 cents and sells them for 75 cents, and cannot return unsold papers. Daily demand has the following distribution, and each day's demand is independent of the previous day's demand:

Number of customers	24	25	26	27	28	29	30
Probability	0.05	0.10	0.10	0.25	0.25	0.15	0.10

If the newspaper seller stocks too many papers a loss is incurred. If too few papers are stocked potential profit is lost because of the excess demand. Let s represent the number of newspapers stocked, and X the daily demand.

- If P is the newspaper seller's profit for a particular stock level s , find an expression for P in terms of s and X .
 - Find the expected value of the profit, $E(P)$, when $s = 26$.
 - Hence find an expression for the expected profit when s is unknown.
 - By evaluating the expression for expected profit for different values of s determine how many papers the newspaper seller should stock.
- 5 Show that if X is a random variable with mean μ and variance σ^2 , and $Z = \frac{X - \mu}{\sigma}$, then $E(Z) = 0$ and $\text{Var}(Z) = 1$.
- 6 Anne and Jane play a game against each other, which starts with Anne aiming to throw a bean bag into a circle marked on the ground.
- The probability that the bean bag lands entirely inside the circle is $\frac{1}{2}$, and the probability that it lands on the rim of the circle is $\frac{1}{3}$.

- i Show that the probability that the bag lands entirely outside the circle is $\frac{1}{6}$.
 - ii What is the probability that two successive throws land outside the circle?
 - iii What is the probability that for two successive throws, the first lands on the rim of the circle and the second inside the circle?
 - b Jane then shoots at a target on which she can score 10, 5 or 0. With any one shot the probability that she scores 10 is $\frac{2}{5}$, the probability that she scores 5 is $\frac{1}{10}$, and the probability that she scores 0 is $\frac{1}{2}$. With exactly two shots, what are the probabilities that she scores:
 - i 20? ii 10?
 - c When the bean bag thrown by Anne lands outside the circle, Jane is allowed two shots at her target. If however, the bean bag lands on the rim of the circle Jane has one shot, and if it lands inside the circle Jane is not allowed any shots. Find the probability that Jane scores 10 as a result of any one throw from Anne.
- 7 Suppose that the outcome of a rugby test series between Australia and England follows a Markov sequence. If $X_i = 0$ if England wins game i , and $X_i = 1$ if Australia wins game i then:
- $$\begin{aligned} \Pr(X_0 = 0) &= \frac{5}{8} & \Pr(X_{i+1} = 0 | X_i = 0) &= \frac{3}{4} & \Pr(X_{i+1} = 0 | X_i = 1) &= \frac{1}{2} \\ \Pr(X_0 = 1) &= \frac{3}{8} & \Pr(X_{i+1} = 1 | X_i = 0) &= \frac{1}{4} & \Pr(X_{i+1} = 1 | X_i = 1) &= \frac{1}{2} \end{aligned}$$
- (Assume that the game is played until one team wins and that there are no draws.)
- a Write down the transition matrix that describes this situation, and use it to determine the probability that:
 - i England wins game 3 if they win the first game
 - ii England wins game 5 if Australia wins the first game
 - b Find the steady state probability of Australia winning a game, $\Pr(X_n = 1)$.
- 8 Suppose that there are two dentists in a country town. Records show that, from year to year:
- Of Dr Laslett's patients, 85% of patients will stay with Dr Laslett and 15% will move to Dr Kildare.
 - Of Dr Kildare's patients, 94% of patients will stay with Dr Kildare and 6% will move to Dr Laslett.
- a Find the transition matrix that can be used to represent this information.
 - b At the beginning of the year 65% of the patients go to Dr Laslett and 35% to Dr Kildare. Find:
 - i the percentage of patients estimated to be at each dentist at the end of year 2
 - ii the percentage of patients estimated to be at each dentist after 10 years

(cont'd.)

- c** Assume that 12 000 patients are treated by dentists in the town each year. Neither of the practices can afford to operate if the number of patients seen falls below 4000 annually. Will either of the dentists go out of business and, if so, who and during which year?
- 9** The lifetime of a certain brand of light globe is normally distributed with mean $\mu = 400$ hours and standard deviation $\sigma = 50$ hours.
- a** Find the probability that a randomly chosen light globe will last more than 375 hours.
- b** The light globes are sold in boxes of 10. Find the probability that at least nine of the globes in a randomly selected box will last more than 375 hours.
- 10** A large taxi company determined that the distance travelled annually per taxi is normally distributed with a mean of 80 000 km and a standard deviation of 20 000 km.
- a** What is the probability that a randomly selected taxi will travel between 56 000 and 60 000 kilometres in a year?
- b** What percentage of taxis can be expected to travel either below 48 000 or above 96 000 kilometres in a year?
- c** How many of the 250 taxis in the fleet are expected to travel between 48 000 and 96 000 kilometres in a year?
- d** At least how many kilometres would be travelled by 85% of the taxis?
- 11** The amount of cereal in boxes, packed by a particular machine, is normally distributed with mean μ grams and standard deviation $\sigma = 5$ grams.
- a** Find:
- i** the proportion of boxes that will be underweight (that is, weigh less than 500 grams) when $\mu = 505$ grams
- ii** the value of μ required to ensure that only 1% of boxes are underweight
- b** As a check on the setting of the machine a random sample of five boxes is chosen and the setting changed if more than one of them is underweight. Find the probability that the setting of the machine is changed when $\mu = 505$.
- 12** A random variable X is normally distributed with a mean of $\mu = 2$ and a variance $\sigma^2 = 13$. A random variable Y is also normally distributed with a mean of $\mu = -17$ and a variance $\sigma^2 = \frac{1}{8}$. X and Y are independent.
- a** Find $E(8X + Y)$.
- b** If X and Y are independent random variables, then:
- $$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y), \text{ } a \text{ and } b \text{ are constants}$$
- Find $\text{sd}(8X + Y)$.
- c** Given that a linear combination of two normally distributed random variables is also normally distributed, find $\Pr(8X + Y > -3)$.

- 13** A factory has two machines that produce widgets. The time taken to produce a widget using machine 1, X , is normally distributed with a mean of 10 seconds and a standard deviation of 2 seconds. The time taken to produce a widget using machine 2, Y , has the following probability density function:

$$f(y) = \begin{cases} k(y - 8) & 8 < y < 12 \\ 0 & \text{otherwise} \end{cases}$$

- a**
- i** Find the value of k .
 - ii** Show that machine 1 has a greater probability of producing a widget in less than 11 seconds than machine 2.
 - iii** Find which machine, on average, is quicker in producing widgets.
- b** Suppose that 60% of the widgets manufactured at the factory are produced by machine 1, and 40% by machine 2. If a widget selected at random is known to have been produced in less than 10 seconds, what is the probability that it was produced by machine 1?
- 14** The queuing time, X minutes, at the box office at a movie theatre has probability density function:

$$f(x) = \begin{cases} kx(100 - x^2) & 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

- a** Find:
- i** the value of k
 - ii** the mean of X
 - iii** the probability that a theatregoer will have to queue for more than 3 minutes
 - iv** the probability that a theatregoer will have to queue for more than 3 minutes given that she queues for less than 7 minutes
- b** If 10 moviegoers go independently to the theatre, find the probability that at least five of them will be required to queue for more than 3 minutes.
- 15** Electronic sensors of a certain type fail when they become too hot. The temperature at which a randomly chosen sensor fails is $T^\circ\text{C}$, where T is modelled as a normal random variable with mean μ and standard deviation σ .
- a** In a laboratory test, 98% of a random sample of sensors continued working at a temperature of 80° but only 4% continued working at 104° .
- i** Show the given information on a sketch of the distribution of T .
 - ii** Determine estimates of the values of μ and σ .
- b** More extensive tests confirm that T is normally distributed, but with $\mu = 94.5$ and $\sigma = 5.7$. Use these values in the rest of the question.
- i** Determine what proportion of sensors will operate in boiling water (i.e. at 100°C).
 - ii** The manufacturers wish to quote a safe operating temperature at which 99% of the sensors will work. What temperature should they quote?
- 16** Jam is packed in tins of nominal net weight 1 kg. The actual weight of jam delivered to a tin by the filling machine is normally distributed about the mean weight set on the machine, with a standard deviation of 12 g.

- a** If the machine is set to 1 kg find the probability that a tin chosen at random contains less than 985 g.
- b** It is a legal requirement that no more than 1% of tins contains less than the nominal weight. Find the minimum setting of the filling machine which will meet this requirement.

- 17** The random variable X has the probability function

$$\Pr(X = x) = \begin{cases} \frac{c}{x} & x = 1, 2, \dots, 6 \\ 0 & \text{otherwise} \end{cases}$$

where c is a constant.

Find the value of:

- a** c **b** $E(X)$ **c** $\text{Var}(X)$

- 18** A flight into an airport is declared to be ‘on time’ if it touches down within 3 minutes either side of the advertised arrival time; otherwise it is declared early or late. On any one occasion, the probability that a flight is on time is 0.5 and the probability that it is late is 0.3. The time of arrival on each day of a particular flight is independent of the time of arrival on any other day.

- a** Calculate the probabilities that:
 - i** on any one day the flight arrives early **ii** on any one day it does not arrive late
 - iii** it arrives on time on three consecutive days
 - iv** it arrives late on Monday one week, but is on time for all of the remaining 4 weekdays
- b** In a given week of 5 days find the probability that it is:
 - i** late exactly once **ii** early exactly twice
- c** The airline is reported to the authority if the flight is late on more than two occasions in a 5-day week. Find the probability that this happens.

- 19** In a factory machines A , B , and C are all producing springs of the same length. Of the total production of springs in the factory, machine A produces 35% and machines B and C produce 25% and 40% respectively. Of their production machines A , B and C produce 3%, 6% and 5% defective springs respectively.

- a** Find the probability that:
 - i** a randomly selected spring is produced by machine A and is defective
 - ii** a randomly selected spring is defective
- b** Given that a randomly selected spring is defective find the probability that it was produced by machine C .
- c** Given that a randomly selected spring is not defective find the probability that it was produced by either machine A or machine B .

- 20** An electronic game comes with five batteries. The game only needs four batteries to work, but because the batteries are sometimes faulty the manufacturer includes five of them with the game. Suppose that X is the number of good batteries included with the game. The probability distribution of X is given in the following table:

x	0	1	2	3	4	5
$\Pr(X = x)$	0.01	0.02	0.03	0.04	0.45	0.45

- a** Use the information in the table to:
- i** find μ , the expected value of X
 - ii** find σ , the standard deviation of X correct to one decimal place
 - iii** find, exactly, the proportion of the distribution that lies within two standard deviations of the mean
 - iv** find the probability that a randomly selected game works, i.e. $\Pr(X \geq 4)$
- b** The electronic games are packed in boxes of 20. Whether or not an electronic game in the box will work is independent of any other game in the box working. Let Y be the number of working games in the box.
- i** Name the distribution of Y .
 - ii** Find the expected number of working games in the box.
 - iii** Find the standard deviation of the number of working games in the box.
 - iv** Find the probability that a randomly chosen box will contain at least 19 working games.

Revision of Chapters 1–19

20.1 Multiple-choice questions

- 1 The simultaneous linear equations

$$mx - 2y = 0$$

$$6x - (m + 4)y = 0$$

where k is a real constant have a unique solution provided

- A** $m \in \{-6, 2\}$ **B** $m \in R \setminus \{-6, 2\}$ **C** $m \in \{-2, 6\}$
D $m \in R \setminus \{-2, 6\}$ **E** $m \in R \setminus \{0\}$

- 2 The general solution to the equation $\sin(2x) = 1$ (n is an integer) is

- A** $x = n\pi + \frac{\pi}{4}$ **B** $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{2}$
C $x = 2n\pi + \frac{\pi}{4}$ or $x = 2n\pi - \frac{\pi}{4}$ **D** $x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{4}$
E $x = n\pi + \frac{\pi}{4}$ or $x = 2n\pi + \frac{\pi}{4}$

- 3 Let $f: R \rightarrow R, f(x) = x^3$. Which one of the following is **not** true?

- A** $f(xy) = f(x)f(y)$ **B** $f(x) + f(-x) = 0$ **C** $f(2x) = 8f(x)$
D $f(x - y) = f(x) - f(y)$ **E** $f(y - x) + f(x - y) = 0$

- 4 Which of the following is **not** true about the function $f: R \rightarrow R, f(x) = |5x + 4| - 3$?

- A** The graph of f is continuous everywhere
B The graph of f' is continuous everywhere
C $f(x) \geq -3$ for all values of x
D $f'(x) = 5$ for all $x > 0$
E $f'(x) = -5$ for all $x < -2$

- 5 Let $k = \int_{-6}^{-2} \frac{2}{x} dx$, then e^k is equal to
A $\log_e 3$ **B** 1 **C** $\frac{1}{9}$ **D** 9 **E** $\frac{1}{3}$
- 6 The average value of the function with rule $f(x) = \log_e (x + 2)$ over the interval $[0, 3]$ is
A $\frac{-1}{5}$ **B** $\log_e 6$ **C** $\frac{\log_e 5}{4}$ **D** $\frac{5 \log_e 5 - 4}{4}$ **E** $\frac{5 \log_e 5 - 3 \log_e 3 - 4}{4}$
- 7 The average value of the function $y = \sin (2x)$ over the interval $\left[0, \frac{\pi}{2}\right]$ is
A $\frac{2}{\pi}$ **B** $\frac{\pi}{2}$ **C** 0.5 **D** 0 **E** π
- 8 The linear transformation $T: R^2 \rightarrow R^2$ is defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \end{bmatrix}$
 The equation of the image of the curve $y = x^2$ under T is
A $2y = (x - 5)^2 + 2$ **B** $3y = (x - 5)^2 + 3$ **C** $9y = (x + 5)^2 - 9$
D $9y = (x - 5)^2 + 9$ **E** $y = \left(\frac{x}{5} - 5\right)^2 + 2$
- 9 If $f(x) = e^{3x}$ for all real x , and $[f(x)]^3 = f(y)$, then y is equal to
A e^{9x} **B** $9x$ **C** $3x$ **D** $9x^3$ **E** $(3x)^3$
- 10 The continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} \sin(2x) & 0 \leq x \leq \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

The value of a such that $\Pr(X > a) = 0.25$ is closest to

- A** 0.25 **B** 0.75 **C** 1.04 **D** 1.05 **E** 1.09
- 11 The radius of a sphere is increasing at a rate of 4 cm/min. When the radius is 2 cm, the rate of increase, in cm^3/min , of the volume of the sphere is
A 32π **B** 48π **C** 64π **D** 96π **E** 120π
- 12 The function f is a probability density function with $f(x) = \begin{cases} 1 + 2e^{\frac{x}{k}} & 0 \leq x \leq 2k \\ 0 & \text{otherwise} \end{cases}$
 Hence k is equal to
A $\frac{1}{2}e^{-2}$ **B** $1 + e^2$ **C** e^{-2} **D** $1 - e^{-2}$ **E** 1
- 13 The range of the function $f: \left[0, \frac{\pi}{2}\right) \rightarrow R, f(x) = |\cos(2x) - 2|$ is
A $[1, 3)$ **B** $[-3, -1]$ **C** $(1, 3]$ **D** $(-1, 3]$ **E** $(0, 1]$
- 14 The random variable X has a normal distribution with mean 8 and standard deviation 0.25. If Z has the standard normal distribution, then the probability that X is less than 7.5 is equal to
A $\Pr(Z > 2)$ **B** $\Pr(Z < -1.5)$ **C** $\Pr(Z < 1)$ **D** $\Pr(Z \geq 1.5)$ **E** $\Pr(Z < -4)$

- 15 The graph of $y = 2kx - 2$ intersects the graph of $y = x^2 + 12x$ at two points for
A $k = 12$ **B** $k > 6 + \sqrt{2}$ or $k < 6 - \sqrt{2}$ **C** $4 < k < 7$
D $5 < k < 7$ **E** $6 - \sqrt{2} < k < 6 + \sqrt{2}$
- 16 The set of solutions to the equation $e^{4x} - 7e^{2x} + 12 = 0$ is
A $\{3, 4\}$ **B** $\{-4, -3\}$ **C** $\{-2, -\sqrt{3}, \sqrt{3}, 2\}$
D $\{\log_e \sqrt{3}, \log_e 2\}$ **E** $\{-\log_e \sqrt{3}, \log_e \sqrt{3}, \log_e 2\}$
- 17 The graph of the function $f: [0, \infty) \rightarrow \mathbb{R}$ where $f(x) = 7x^{\frac{3}{2}}$ is reflected in the x -axis and then translated 3 units to the right and 4 units down. The equation of the new graph is
A $y = 7(x-3)^{\frac{3}{2}} + 4$ **B** $y = -7(x-3)^{\frac{3}{2}} - 4$ **C** $y = -7(x+3)^{\frac{3}{2}} - 1$
D $y = -7(x-4)^{\frac{3}{2}} + 3$ **E** $y = 7(x-4)^{\frac{3}{2}} + 3$
- 18 If a random variable X has probability density function $f(x) = \begin{cases} \frac{1}{8}x & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$
then $E(X)$ is equal to
A $\frac{1}{2}$ **B** 1 **C** $\frac{8}{3}$ **D** $\frac{16}{3}$ **E** 2
- 19 Which of the following is **not** true about the function $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x^2 - 4| - 2$?
A The graph of f is continuous everywhere
B $f(-2) = -2$
C $f(x) \geq -2$ for all values of x
D $f'(x) = 2x$ for all $x > 0$
E $f'(x) = 2x$ for all $x < -2$
- 20 For the system of simultaneous equations

$$\begin{aligned} z &= 1 \\ x + y &= 3 \\ x - y &= 5 \end{aligned}$$
an equivalent matrix equation is
A $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$ **B** $\begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix}$
C $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 2 \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$ **D** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$
E $\begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$

- 21 The function $f: [a, \infty) \rightarrow \mathbb{R}$, with rule $f(x) = \log_e((x-2)^4)$ will have an inverse function if
A $a \geq 3$ **B** $a \leq -2$ **C** $a < 2$ **D** $a \geq 0$ **E** $a \geq -1$
- 22 The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which maps the curve with equation $y = e^x$ to the curve with equation $y = e^{(2x+4)} - 3$ could have rule $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) =$
A $\begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ **B** $\begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
C $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ **D** $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -2 \\ 3 \end{bmatrix}$
E $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 4 \\ 3 \end{bmatrix}$
- 23 Let $f'(x) = g'(x) + 4$ and $f(1) = 2$ and $g(x) = -xf(x)$. Then $f(x) =$
A $g(x) + 4x + 4$ **B** $g'(x) + 4$ **C** $g(x) + 4x$
D $\frac{4-4x}{x+1}$ **E** $g(x) + 4$
- 24 The function f satisfies the equation $f\left(\frac{xy}{2}\right) = 2f(x)f(y)$ for x and y any non-zero real numbers. A possible rule for the function is
A $f(x) = \frac{1}{x}$ **B** $f(x) = \log_e |2x|$ **C** $f(x) = 2x$
D $f(x) = 2^x$ **E** $f(x) = \cos(2x)$
- 25 The random variable X has the following probability distribution

X	0	1	2
$\Pr(X = x)$	a	b	0.6

If the mean of X is 1.6 then

- A** $a = 0.3$ and $b = 0.7$ **B** $a = 0.2$ and $b = 0.2$ **C** $a = 0.4$ and $b = 0.4$
D $a = 0.1$ and $b = 0.3$ **E** $a = 0$ and $b = 0.4$
- 26 The simultaneous equations $(m-4)x + 6y = 6$ and $2x + (m-3)y = 2m - 10$ have no solution for
A $m \in \mathbb{R} \setminus \{0, 7\}$ **B** $m \in \mathbb{R} \setminus \{0\}$ **C** $m \in \mathbb{R} \setminus \{7\}$
D $m = 7$ **E** $m = 0$

20.2 Short-answer questions (technology free)

1 Let $f(x) = x^2 + 6$ and $g(x) = 3x + 1$. Write down the rule of $f(g(x))$.

2 For the simultaneous linear equations

$$\begin{aligned} kx + 3y &= 0 \\ 4x + (k + 2)y &= 0 \end{aligned}$$

where k is a real constant, find the value(s) of k for which there are infinitely many solutions.

3 Find the equation of the image of $y = \frac{1}{x}$ under the transformation defined by the matrix $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$ and describe a sequence of transformations that maps the graph of $y = \frac{1}{x}$ onto the graph of its image.

4 a Let $f(x) = (5x^3 - 3x)^7$. Find $f'(x)$.

b Let $f(x) = 2xe^{4x}$. Evaluate $f'(0)$.

5 a Differentiate $x^2 \log_e(2x)$ with respect to x .

b For $f(x) = \frac{\sin(x)}{2x + 1}$ find $f'\left(\frac{\pi}{2}\right)$.

6 a Let $f(x) = e^{\sin(2x)}$. Find $f'(x)$.

b Let $f(x) = 3x \tan(2x)$. Evaluate $f'\left(\frac{\pi}{3}\right)$.

7 Find the general solution to the equation $\sin(2x) - \cos(2x) = 0$.

8 For the function $f: [-\pi, \pi] \rightarrow R, f(x) = 4 \sin\left(2\left(x + \frac{\pi}{6}\right)\right)$:

a Write down the amplitude and period of the function.

b Sketch the graph of the function f . Label axes intercepts and endpoints with their coordinates.

9 Sketch the graph of $f: [-1, \infty) \rightarrow R, f(x) = 1 - \frac{4}{x - 2}$. Label all axis intercepts. Label each asymptote with its equation.

10 For the function $f: R \rightarrow R, f(x) = 5e^{x-1} - 3$

a find the rule for the inverse function f^{-1}

b find the domain of the inverse function f^{-1} .

11 Solve the equation $\cos\left(\frac{5x}{2}\right) = \frac{1}{2}$ for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

12 Let $g: R \rightarrow R, g(x) = 5x^2$. Show that $g(u + v) + g(u - v) = 2(g(u) + g(v))$.

13 Find the average value of $y = e^x$ over the interval $[0, 4]$.

- 14** The graph of $y = ax^3 + bx + c$ has intercepts $(0, 6)$ and $(-2, 0)$ and has a turning point where $x = -1$.

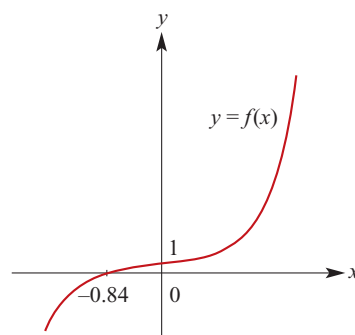
- a** Find the value of c .
- b** Write down two simultaneous equations in a and b from the given information.
- c** Hence find the values of a and b .

- 15** If $g: R \rightarrow R$, $g(x) = 3 - e^{2x}$

- a** Find the rule and domain of the function g^{-1} .
- b** Sketch the graph of $y = g(g^{-1}(x))$ for its maximal domain.

- 16** The graph of the hybrid function $f(x) = \begin{cases} -2x^4 + 1 & x \leq 0 \\ 2x^4 + 1 & \text{otherwise} \end{cases}$ is shown below.

- a** Draw the graph of the derivative function f' .
- b** Write down a rule for the derivative function.



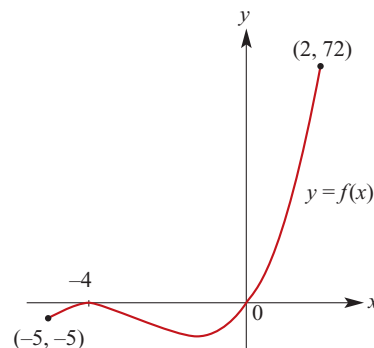
- 17** Find an antiderivative of $\frac{1}{1-3x}$ with respect to x .
- 18** Let $f: R \setminus \left\{\frac{1}{2}\right\} \rightarrow R$ where $f(x) = \frac{3}{2x-1} + 3$. Find f^{-1} , the inverse function of f .
- 19** Solve the equation $\tan(2x) = -\sqrt{3}$ for $x \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$.
- 20** Let X be a normally distributed random variable with a mean of 84 and a standard deviation of 6. Let Z be the standard normal random variable.
- a** Find the probability that X is greater than 84.
 - b** Use the result that $\Pr(Z < 1) = 0.84$ to find the probability that $78 < X < 90$.
 - c** Find the probability that $X < 78$ given that $X < 84$.
- 21** The probability density function of a random variable X is given by

$$f(x) = \begin{cases} \frac{x}{24} & 1 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

- a** Find $\Pr(X < 3)$
- b** If $b \in [1, 7]$ and $\Pr(X \geq b) = \frac{3}{8}$, find b .

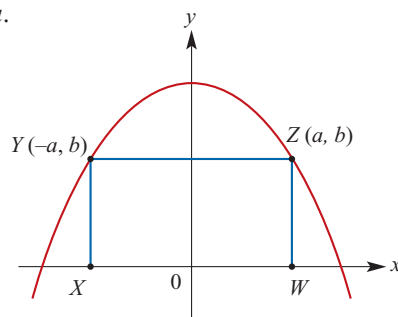
- 22 The graph of $f: [-5, 2] \rightarrow \mathbb{R}, f(x) = x^3 + 8x^2 + 16x$ is as shown.

- a Sketch the graph of $y = |f(x)|$.
 b State the range of the function with rule $|f(x)|$.



- 23 A tangent to the graph of $y = x^{\frac{1}{3}}$ has equation $y = \frac{1}{3}x + a$. Find the value(s) of a .
 24 A rectangle $XYZW$ has two vertices on the x -axis and the other two vertices on the graph of $y = 16 - 4x^2$, as shown in the diagram.

- a Find the area, A , of rectangle $XYZW$ in terms of a .
 b Find the maximum value of A and the value of a for which this occurs.



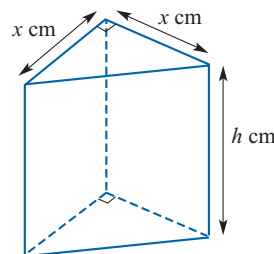
- 25 Let $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -3x^2 + 2bx + 9$ with $\int_{-1}^3 f(x) dx = 32$. Find the value of b .
 26 The area between the x -axis, the curve $y = \log_e(x)$ and the line $x = a$ where $a > 1$ is known to be 2. Find a , correct to two decimal places.
 27 Simone has either a sandwich or pasta for lunch. If she has a sandwich one morning, the probability she has a sandwich for lunch the next day is 0.6. If she has pasta, the probability she has pasta for lunch the next day is 0.3. Suppose she has a sandwich for lunch on a Monday. What is the probability that she has pasta for lunch on the following Wednesday?
 28 A player of a game of chance can win \$0, \$1, \$2 or \$3. The amount won is a random variable with probability distribution given by

\$x	0	1	2	3
$\Pr(X = x)$	0.6	0.2	0.15	0.05

- a What is the mode of X ?
 b Find the mean of X .
 c What is the probability that a player wins the same amount from two games?

- 29** Every Thursday night Chris goes to the gym or goes for a run. If he goes to the gym one Thursday night, the probability that he goes to the gym the next Thursday night is 0.5. If he goes for a run one Thursday night, then the probability that he goes for a run the next Thursday night is 0.6. If he goes to the gym one Thursday night what is the probability that he goes for a run on exactly two of the next three Thursdays?
- 30** A brick is made in the shape of a right triangular prism. The triangular end is a right-angled triangle with the equal sides of length x cm. The height of the brick is h cm. The volume of the brick is 2000 cm^3 .

- a** Find an expression for h in terms of x .
- b** Show that the total surface area, $A \text{ cm}^2$, of the brick is given by $A = \frac{4000\sqrt{2} + 8000}{x} + x^2$
- c** Find the cube of the length of the side of the triangle if the brick has minimum surface area.



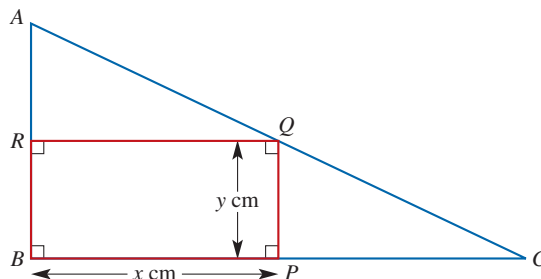
20.3 Extended-response questions

- 1 a i** Find the coordinates of the stationary point for the curve with equation:

$$y = \frac{16x^3 + 4x^2 + 1}{2x^2}$$

- ii** Determine the nature of this stationary point.

- b** In the triangle ABC , the magnitude of $\angle ABC = 90^\circ$, $AB = 5$ cm and $AC = 13$ cm. The rectangle $BPQR$ is such that its vertices P , Q and R lie on BC , CA and AB respectively.



- i** Given that $BP = x$ cm and $PQ = y$ cm, show that $y = \frac{60 - 5x}{12}$
- ii** Find the area of the rectangle, $A \text{ cm}^2$, in terms of x .
- iii** Find the maximum value of this area as x varies.
- 2** A theoretical model of the relationship between two variables x and y predicts the values in the following table:

x	0	1	3
y	6	0	0

- a** An equation of the form

$$y = k(x - p)(x - q)$$

is suggested, where k , p and q are constants and $p < q$.

Use the information in the table to find p , q and k .

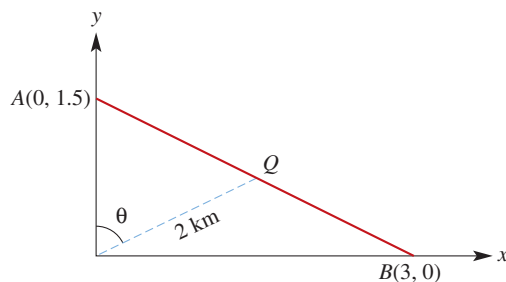
- b** A series of experiments is carried out to test this model. The values of y when $x = 0, 1$ and 3 are found to be as predicted, but when $x = 2$ the value of y is found to be 2 . After further discussion a new model is proposed with an equation of the form

$$y = m(x - p)^2(x - q)$$

where p and q have the values already calculated and m is a constant.

- i** Find the value of m .
 - ii** Obtain the equation of this new model in the form, $y = ax^3 + bx^2 + cx + d$
 - iii** Sketch the graph of y against x . State the coordinates of the stationary points and the nature of each of these points.
- 3** A curve C has equation $y = ax - x^2$, where a is a positive constant.
- a** Sketch C , showing clearly the coordinate of the points of intersection with the axes.
 - b** Calculate the area of the finite region bounded by C and the x -axis, giving your answer in terms of a .
 - c** The lines with equations $x = \frac{1}{3}a$ and $x = \frac{2}{3}a$ intersect C at the points A and B respectively.
 - i** Find, in terms of a , the y -coordinates of A and B .
 - ii** Calculate the area of the finite region bounded by C and the straight line AB , giving your answer in terms of a .
- 4** **a** Find the equation of the straight line joining the points $A(0, 1.5)$ and $B(3, 0)$.
- b** Let $y = \sin \theta + 2 \cos \theta$
- i** Find $\frac{dy}{d\theta}$.
 - ii** Solve the equation $\frac{dy}{d\theta} = 0$ for $\theta, 0^\circ \leq \theta \leq 90^\circ$.
 - iii** State the coordinates of the stationary point of $y = \sin \theta + 2 \cos \theta$ for $0^\circ \leq \theta \leq 90^\circ$.
 - iv** It can be shown that $\sin \theta + 2 \cos \theta$ can be written in the form $r \sin(\theta + \alpha)$. Use the result of **iii** and the fact that when $\theta = 0$, $y = 2$ to find the values of r and α .
 - v** Use addition of ordinates and the result of **iv** to sketch the graph of y against θ for $0^\circ \leq \theta \leq 90^\circ$.
- c** The figure shows a map of a region of wetland. The units of the coordinates are kilometres, and the y -axis points due north. A walker leaves her car somewhere on the straight road between A and B . She walks in a straight line for distance of 2 km to a monument at the origin O . While she is looking at it fog comes down, so that she cannot see her way back to her car. She needs to work out the bearing on which she should walk.
- i** Write down the coordinates of a point Q which is 2 km from O on a bearing of θ .
 - ii** Show that for Q to be on the road between A and B , θ must satisfy the equation:

$$2 \sin \theta + 4 \cos \theta = 3$$
 - iii** Use the result of **b iv** to solve the equation for θ , where θ is between 0° and 90° .

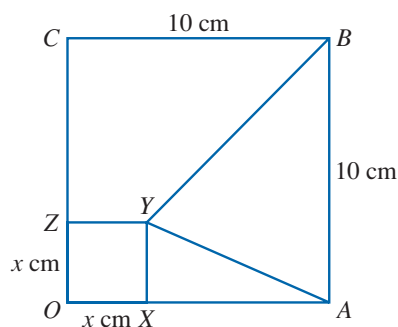


- 5 The number of people unemployed in a particular population can be modelled by the function

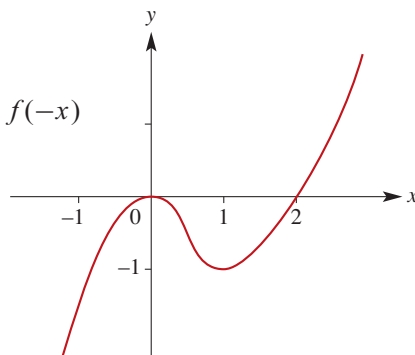
$$f(t) = 1000(t^2 - 10t + 44)e^{\frac{-t}{10}}$$

where t is the number of months after January 1997 and $0 \leq t \leq 35$.

- a Use this function to find expression for:
- the rate of increase of the number unemployed
 - the rate of increase of this rate of increase
- b Find the values of t for which:
- the number unemployed was increasing
 - the rate of increase of the number unemployed was going down
 - the number unemployed was increasing and the rate of increase of the number unemployed was going down
- 6 A square piece of card $OABC$, of side 10 cm, is cut into four pieces by removing a square $OXYZ$ of side x cm, as shown, and then cutting out the triangle ABY .

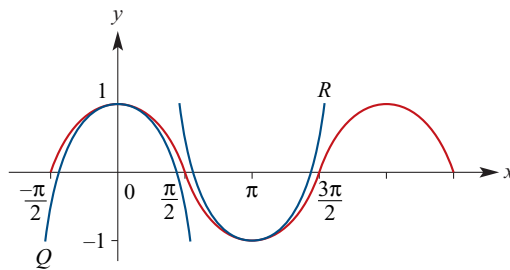


- a
- Find A cm², the sum of the areas $OXYZ$ and ABY , in terms of x .
 - Find the domain of the function which determines this area.
 - Sketch the graph of the function, with domain determined in ii.
 - State the minimum value of this area.
- b
- Find the rule for the function of x which represents the area of triangle AXY .
 - Sketch the graph of this function for a suitable domain.
- c Find the ratio of the areas of the four pieces when the area of triangle AXY is a maximum.
- 7 The graph of $y = f(x)$ is shown.

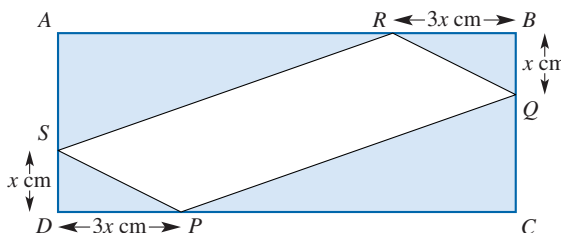


- a Sketch the graph of:
- $y = 2f(x)$
 - $y = f(2x)$
 - $y = f(-x)$
 - $y = -f(x)$
 - $y = f(x + 2)$
- b Explain why f does not have an inverse function.
- c
- Sketch the graph of $g: (2, \infty) \rightarrow \mathbb{R}$, $g(x) = f(x)$
 - Sketch the graph of g^{-1} .
- d
- Given that $g(x) = x^2(x - 2)$ and $g: (2, \infty) \rightarrow \mathbb{R}$, calculate the gradient of the graph of $y = g(x)$ at the point $(3, 9)$.
 - Hence find the gradient of $y = g^{-1}(x)$ at the point $(9, 3)$.

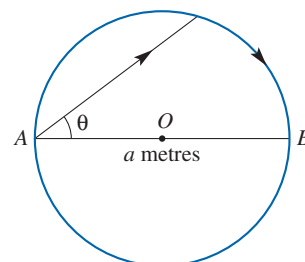
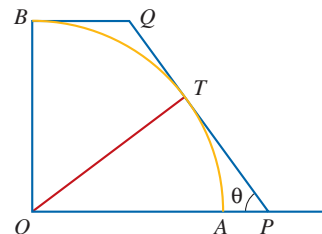
- 8 The diagram shows part of the graph of $y = \cos x$ and the graphs of two quadratic functions, denoted by Q and R , which approximate to the cosine function around $x = 0$ and $x = \pi$ respectively. It is known that the equation of Q is $y = 1 - \frac{1}{2}x^2$



- a i** Find an estimate of $\cos 0.1$ by using the approximation $y = 1 - \frac{1}{2}x^2$
- ii** Find an approximation for the solution of the equation $\cos x = 0.98$, by solving the quadratic equation $1 - \frac{1}{2}x^2 = 0.98$ $\left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right)$
- b i** The graph Q can be transformed into R by a reflection in the x -axis, followed by a translation. Use this fact to find an equation for the graph R .
- ii** Estimate the value of $\cos 3$ using this approximation.
- 9 In the figure, $ABCD$ is a rectangle with $AB = 30$ cm, $AD = 10$ cm. The portions shaded are cut away, leaving the parallelogram $PQRS$. $BQ = SD = x$ cm and $RB = PD = 3x$ cm.



- a** Find the area, S cm², of the parallelogram in terms of x .
- b** Find the allowable values of x .
- c** Find the value of x for which S is a maximum.
- d** Sketch the graph of S against x for a suitable domain.
- 10 In the figure, OAB is a quadrant of a circle of radius 1 unit. OA is produced to a varying point P . From P a tangent to the quadrant is drawn touching it at T and meeting another tangent BQ at Q . Let $\angle OPQ = \theta$.
- a i** Find the length OP as a function of θ .
- ii** Find the length BQ as a function of θ .
- b** Show that the area S of trapezium $OPBQ$ is $\frac{2 - \cos \theta}{2 \sin \theta}$.
- c** Show that $\frac{dS}{d\theta} = \frac{2 - 4 \cos \theta}{4 \sin^2 \theta}$
- d** Find the minimum value of S and the distance AP when S is a minimum.
- 11 A dog is at A on the edge of a certain circular lake of diameter a metres and she wishes to reach her owner who is at the diametrically opposite point B . The dog can swim at $\frac{1}{2}$ m/s and run at 1 m/s.



- a** If she swims in a direction making an angle of θ with AB and then runs round the edge of the lake to B , show that the time taken, T s, is given by:

$$T = a(\theta + 2 \cos \theta)$$

- b** On the one set of axes sketch the graphs:

$$y = 200\theta \text{ and } y = 400 \cos \theta \text{ for } 0 \leq \theta \leq \frac{\pi}{2}$$

Using addition of ordinates sketch the graph of:

$$y = 200(\theta + 2 \cos \theta)$$

(Find the maximum value of y for $0 \leq \theta \leq \frac{\pi}{2}$ by finding $\frac{dy}{d\theta}$ and solving the equation

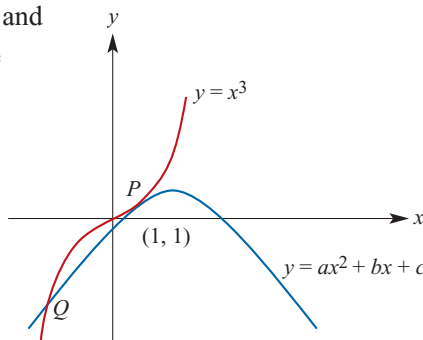
$$\frac{dy}{d\theta} = 0.)$$

- c** Sketch the graph of $T = a(\theta + 2 \cos \theta)$ for $0 \leq \theta \leq \frac{\pi}{2}$ and state the minimum value of T .
- 12 a i** Show that if $f(x) = (x - 1)g(x)$ and $f'(x) = (x - 1)h(x)$ where $g(x)$ and $h(x)$ are polynomials, then $(x - 1)$ must be a factor of $g(x)$.
- ii** Let $F(x) = x^3 - kx^2 - (3 - 2k)x - (k - 2)$
Show that $F(1) = F'(1) = 0$

iii Using the results of **i** and **ii** solve the equation $F(x) = 0$

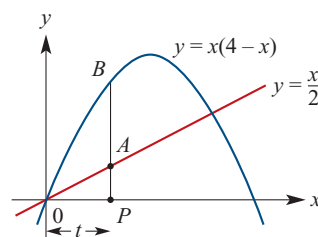
- b** The parabola with equation $y = ax^2 + bx + c$ and the cubic $y = x^3$ touch at $P(1, 1)$ (and have the same gradient at this point). The curves also meet at Q .

- i** Find b and c in terms of a .
- ii** If the coordinates of Q are (h, k) , find h in terms of a (use the result of **a**).
- iii** If Q has coordinates $(-2, -8)$ find the values of a, b and c .
- iv** If Q has coordinates $(-3, -27)$ find the values of a, b and c .



- 13** P is the point with coordinates $(t, 0)$, where $0 < t < \frac{7}{2}$.
The line PAB is parallel to the y -axis.

- a** Let Z be the length of AB . Find Z in terms of t .
- b** Sketch the graph of Z against t .
- c** State the maximum value of Z and the value of t for which it occurs.

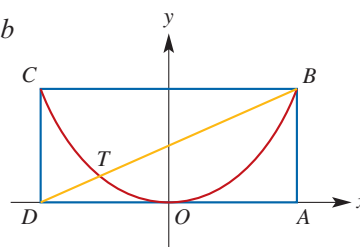


- 14** A study of the numbers of male and female children in families in a certain population is being carried out.
- a** A simple model is that each child in any family is equally likely to be male or female, and the sex of each child is independent of the sex of any previous children in the

family. Using this model calculate the probability that, in a randomly chosen family of four children:

- i there will be two males and two females
 - ii there will be exactly one female given that there is at least one female
- b** An alternative model is that the first child in any family is equally likely to be male or female, but that, for any subsequent children, the probability that they will be of the same sex as the previous child is $\frac{3}{5}$. Using this model calculate the probability that in a randomly chosen family of four children:
- i all four will be of the same sex
 - ii no two consecutive children will be of the same sex
 - iii there will be two males and two females

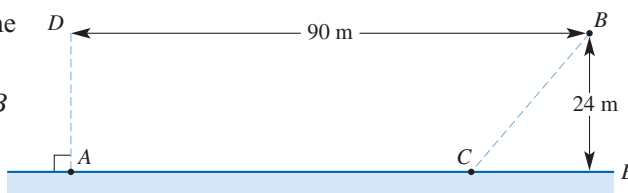
- 15** In the figure $ABCD$ is a rectangle. $OA = OD = a$, $AB = b$
The equation of the parabola BOC is $y = kx^2$



- a** Express k in terms of a and b .
- b** If BD cuts the parabola at T , find:
 - i the equation of the straight line BD
 - ii the coordinates of T
- c** Show that the area bounded by the parabola and the line BC is $\frac{4}{3}ab$ square units.
- d** Let S_1 be the area of the region bounded by the line segment BT and the curve BOT .
Let S_2 be the area of the region bounded by curve CT and the line segments BC and BT .
Find the ratio $S_1 : S_2$.

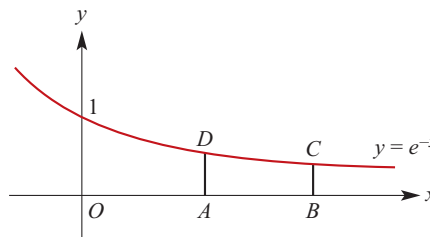
- 16** A certain type of brass washer is manufactured as follows. A length of brass rod is cut, cross-sectionally, into pieces of mean thickness 0.25 cm and with standard deviation 0.002. These brass slices are then put through a machine that punches out a circular hole of mean diameter 0.5 cm through the middle of the slice, with a standard deviation of 0.05 cm. The thickness of the washers and the diameters of the holes are known to be normally distributed and do not depend on each other.
- a** Find the probability that a washer, selected at random, will:
 - i have a thickness of less than 0.253 cm
 - ii have a thickness of less than 0.247 cm
 - iii have a hole punched with a diameter greater than 0.56
 - iv have a hole punched with a diameter less than 0.44 cm
 - b** If the brass washers are acceptable only if they are between 0.247 cm and 0.253 cm in thickness with a hole of diameter between 0.44 cm and 0.56 cm, find:
 - i the percentage of washers that are rejected
 - ii the expected number of washers of acceptable thickness in a batch of 1000 washers
 - iii the expected number of washers of acceptable thickness that will be rejected in a batch of 1000 washers

- 17** A ditch is to be dug to connect the points A and B in the figure. The earth on the same side of AE as B is hard and on the opposite side is soft.



The cost of digging hard earth is \$200 per metre and soft earth is \$100 per metre.
Where should C be chosen? (C is the point where a turn is made.)

- 18** The diagram shows a sketch of the graph of $y = e^{-x}$. The points A and B have coordinates $(n, 0)$ and $(n + 1, 0)$ respectively, and the points C and D on the curve are such that AD and BC are parallel to the y -axis.



- a i** Find the equation of the tangent to $y = e^{-x}$ at the point D .
 - ii** Find the intercept of the tangent with the x -axis.
 - b i** Find the area of the region $ABCD$.
 - ii** The line BD divides the region into two parts. Find the ratio of the two areas of these parts.
- 19** A closed capsule is to be constructed as shown in the diagram. It consists of a circular cylinder of height h cm which has flat base of radius r cm. It is surmounted by a hemispherical cap.
-
- a i** Show that the volume, $V \text{ cm}^3$, of the capsule is given by $V = \frac{\pi r^2}{3}(3h + 2r)$
 - ii** Show that the surface area of the capsule, $S \text{ cm}^2$, is given by $S = \pi r(2h + 3r)$
 - b i** If $V = \pi a^3$, where a is a positive constant, find h in terms of a and r .
 - ii** Hence find S in terms of a and r .
 - c i** By using addition of ordinates sketch the graph of S against r for a suitable domain.
 - ii** Find the coordinates of the turning point by finding $\frac{dS}{dr}$ and solving the equation $\frac{dS}{dr} = 0$ for r and determining the corresponding value of S .
- 20** A manufacturer sells cylinders whose diameters are normally distributed with mean 3 cm and standard deviation 0.002 cm. The selling price is \$ s per cylinder and the cost of manufacture of each cylinder is \$1. A cylinder is returned and the purchase money is refunded if the diameter of the cylinder is found to differ by more than d cm. A returned cylinder is regarded as a total loss to the manufacturer. The probability that a cylinder is returned is 0.25.

- a** Find d .
- b** The profit, Q , per cylinder is a random variable. Give the possible values of Q in terms of s , and the probabilities of these values.
- c** Express the mean and standard deviation of Q in terms of s .
- 21** The length of a certain species of worm has a normal distribution with mean 20 cm and standard deviation 1.5 cm.
- a** Find the probability that a randomly selected worm has a length greater than 22 cm.
- b** If the lengths of the worms are measured to the nearest centimetre, find the probability that a randomly selected worm has a length of 20 cm.
- c** If five worms are randomly selected, find the probability that exactly two will have their lengths measured as 20 cm (measurement to the nearest centimetre).

- 22** The number of tonnes of coal, P , produced by x miners per shift is given by the rule:

$$P = \frac{x^2}{90}(56 - x) \text{ where } 1 \leq x \leq 40$$

- a** Find $\frac{dP}{dx}$.
- b** **i** Sketch the graph of P against x for $1 \leq x \leq 40$.
ii State the maximum value of P .
- c** Write down an expression in terms of x for the average production per man in the shift. Denote the average production per man by A (tonnes).
i Sketch the graph of A against x for $1 \leq x \leq 40$.
ii State the maximum value of A and the value of x for which it occurs.

- 23** Consider the family of quadratic functions with rule

$$f(x) = (k + 2)x^2 + (6k - 4)x + 2$$

where k is an arbitrary constant.

- a** Sketch the graph of f when:
i $k = 0$ **ii** $k = -2$ **iii** $k = -4$
- b** Find the coordinates of the turning point of the graph of $y = f(x)$ in terms of k . If the coordinates of the turning point are (a, b) find:
i $\{k: a > 0\}$ **ii** $\{k: a = 0\}$ **iii** $\{k: b > 0\}$ **iv** $\{k: b < 0\}$
- c** For what values of k is the turning point a local maximum?
- d** By using the discriminant state the values of k for which:
i $f(x)$ is a perfect square
ii there are no solutions for the equation $f(x) = 0$
- 24** **a** Find the solution to the equation $e^{2-2x} = 2e^{-x}$
b For $y = e^{2-2x} - 2e^{-x}$:
i Find $\frac{dy}{dx}$. **ii** Solve the equation $\frac{dy}{dx} = 0$

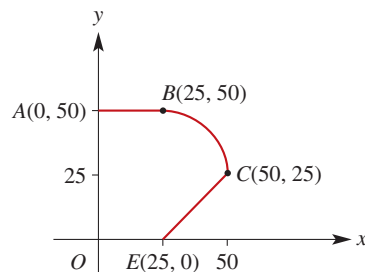
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- iii State the coordinates of the turning points of $y = e^{2-2x} - 2e^{-x}$
- iv Sketch the graph of $y = e^{2-2x} - 2e^{-x}$ for $x \geq 0$.
- c State the set of values of k for which the equation $e^{2-2x} - 2e^{-x} = k$ has two distinct positive solutions.
- 25 a Sketch, on a single clear diagram, the graph of:
- i $y = x^2$ ii $y = (x + a)^2$ iii $y = b(x + a)^2$ iv $y = b(x + a)^2 + c$
 where a , b , and c are positive constants with $b > 1$.
- b Show that
- $$\frac{2x^2 + 4x + 5}{x^2 + 2x + 1} = \frac{3}{(x + 1)^2} + 2$$
- for all values except $x = -1$.
- c Hence state precisely a sequence of transformations by which the graph of $y = \frac{2x^2 + 4x + 5}{x^2 + 2x + 1}$ may be obtained from the graph of $y = \frac{1}{x^2}$
- d Evaluate $\int_0^1 \frac{2x^2 + 4x + 5}{x^2 + 2x + 1} dx$.
- e Sketch the graphs of $y = \frac{1}{x^2}$ and $y = \frac{3}{(x + 1)^2} + 2$ on the one set of axes and indicate the region for which the area has been determined in d.

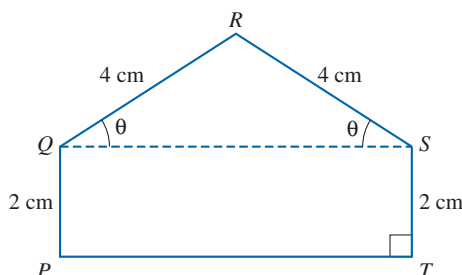
- 26 A real estate agent has a block of land to sell. An x - y coordinate grid is placed with the origin at O , as shown in the diagram.

The block of land is $OABCE$. OA , AB , CE and EO are straight line segments.

Points B and C lie on a parabola with equation of the form $y = ax^2 + 4x + c$



- a Find the equation of line segments:
- i AB ii EC
- b Find the values of a and c and hence the equation of the parabola through points B and C .
- c Find the area of:
- i rectangle $OEBA$ ii region EBC (with boundaries as defined above)
- iii the block of land
- 27 In the diagram $PQRST$ is a thin metal plate. $PQST$ is a rectangle with $PQ = 2$ cm and QRS is an isosceles triangle with $QR = RS = 4$ cm.
- a Show that the area of the metal plate A cm² is given by:



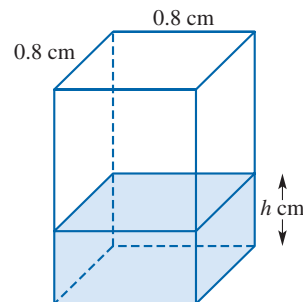
$$A = 16(\cos \theta + \cos \theta \sin \theta), 0 < \theta < \frac{\pi}{2}$$

- (cont'd.)

- e i** Find $A'(p)$, $0 \leq p \leq 1$. **ii** Sketch the graph of $A'(p)$ against p .
iii For what value of p is $A'(p)$ a minimum?
iv Describe what the result of **iii** means.

- 31** A liquid is contained in a tank which is cuboid with square cross-section as shown in the diagram. The depth of liquid, h cm, in the tank at time t minutes is given by the function with the rule:

$$h(t) = (4.5 - 0.3t)^3$$



- a** State the depth of the liquid at time $t = 0$.
b State the practical domain for the function h .
c State the rule for the volume, $V \text{ cm}^3$, of water in the tank at time t .
d Explain briefly why an inverse function h^{-1} exists and find its rule and domain.
e Draw graphs of both h and h^{-1} on the one set of axes.
- 32** A machine produces ball-bearings with a mean diameter of 3 mm. It is found that 6.3% of the production is being rejected as below the lower tolerance limit of 2.9 mm and a further 6.3% is being rejected as being above the upper tolerance level of 3.1 mm. Assume that the diameters are normally distributed.
- a** Calculate the standard deviation of the distribution.
b A sample of eight ball bearings is taken. Find the probability that:
i at least one is rejected **ii** two are rejected
c The setting of the machine now ‘wanders’ so that the standard deviation remains the same, but the mean changes to 3.05.
i Calculate the total percentage of the production that will now fall outside the given tolerance levels.
ii Find c such that the probability that the diameter lies in the interval $(3.05 - c, 3.05 + c)$ is 0.9.
- 33** There is a probability of 0.8 that a boarding student will miss breakfast if he oversleeps. There is a probability of 0.3 that the student will miss breakfast even when he does not oversleep. The student has a probability of 0.4 of oversleeping.
- a** On a random day what is the probability of:
i the student oversleeping and missing breakfast?
ii the student not oversleeping and missing breakfast?
iii the student not missing breakfast?
b Given that the student misses breakfast find the probability that he overslept.
c It is found that 10 students in the boarding house have identical probabilities for sleeping in and missing breakfast to the student mentioned above.
Find the probability that:
i exactly two of the 10 students miss breakfast
ii at least one of the 10 students misses breakfast
iii at least eight of the students don’t miss breakfast

- 34 a** On the one set of axes sketch the graphs of $y = \frac{1}{x}$ and $y = e^x$ for $x > 0$.
- b** Using addition of ordinates sketch the graph of $y = \frac{1}{x} + e^x$ for $x > 0$.
(Do not attempt at this stage to find the coordinates of the turning points.)
- c** Find $\frac{dy}{dx}$ for $y = \frac{1}{x} + e^x$
- d i** Show that $\frac{dy}{dx} = 0 \Leftrightarrow 2 \log_e x = -x, x > 0$
- ii** Explain why this means that the local minimum of $y = \frac{1}{x} + e^x$ lies in the interval $(0, 1)$.
- iii** Using a careful sketch graph or a calculator show that the point of intersection of the graphs of $y = 2 \log_e x$ and $y = -x$ is at $(0.70, -0.70)$ (correct to two decimal places).
- iv** Hence find the coordinates of the local minimum of $y = \frac{1}{x} + e^x$ (correct to one decimal place).

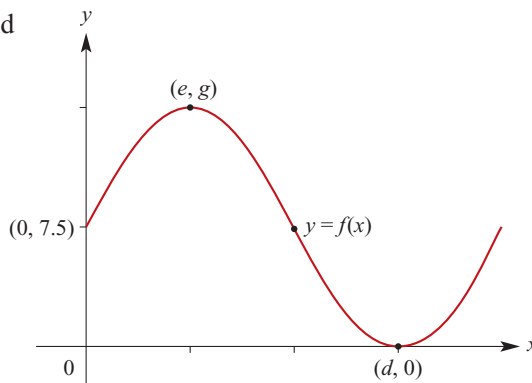
- 35** A section of a creek bank can be modelled by the function $f: [0, 50] \rightarrow R$,

$$f(x) = a + b \sin\left(\frac{2\pi x}{50}\right)$$
 where x is measured in metres.

a i Find the values of a, b, d, e , and g .

ii The other bank of the river can be modelled by the function $y = f(x) + 4$

Sketch the graph of this new function.



- b** Find the coordinates of the points on the first bank which have y -coordinate 10.
- c** A particular river has a less severe bend than this creek. It is found that a section of the bank of the river can be modelled by the function:

$$g: [0, 250] \rightarrow R, g(x) = 2f\left(\frac{x}{5}\right)$$

Sketch the graph of this function. Clearly indicate the coordinates of the turning points.

- d** Over the years the river bank moves. The shape of the bends are maintained but there is translation of 10 metres in the positive direction of the x -axis.
- i** Give the rule that describes this section of the river bank, after the translation (relative to the original axes).
- ii** Sketch the graph of this new function.

- 36** The continuous random variable X has probability density function f with rule given by:

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq 2 \\ k(5 - 2x) & \text{if } 2 < x \leq \frac{5}{2} \\ 0 & \text{if } x > \frac{5}{2} \end{cases}$$

- a Find the value of k .
- b Find:
 - i $E(X)$
 - ii the median of X
 - iii σ , the standard deviation of X , correct to two decimal places
- c Find $\Pr(X < (\mu - \sigma))$ where $\mu = E(X)$

37 The lifetime, X in days, of a large batch of computer components has a probability density function given by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ k(a - x) & \text{if } 0 < x \leq a \\ 0 & \text{if } x > a \end{cases}$$

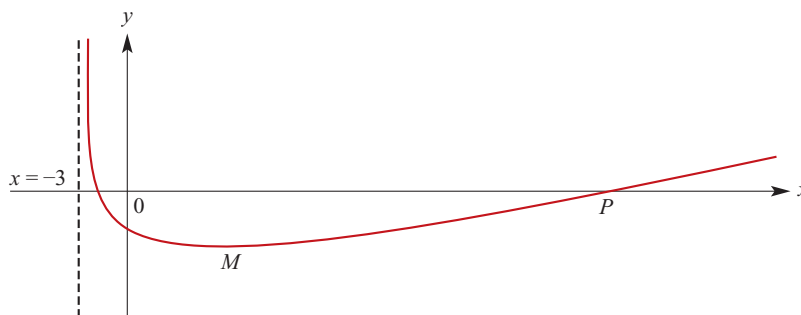
where k and a are positive constants.

Find:

- k in terms of a
- the mean, μ , and the variance, σ^2 , of X in terms of a
- $\Pr(X > \mu + 2\sigma)$, giving your answer to two three decimal places
- the value of a if the median lifetime is 1000 days

38 The diagram shows a sketch graph of:

$$y = \frac{x}{10} - \log_e(x + 3), x > -3$$



- a** Find the x -coordinate of the local minimum at M .
- b** Show that the gradient of the curve is always less than $\frac{1}{10}$.
- c** Find the equation of the line through M with a gradient of $\frac{1}{10}$.
- d** **i** Hence show that the value of the x -axis intercept at P is greater than $10 \log_e 10$.
ii Find, correct to three decimal places, the value of the intercept at P .

39 The random variable X has the probability density function given by

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ kx^n & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x > 1 \end{cases}$$

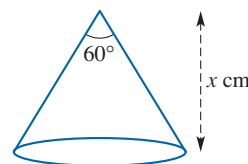
where n and k are constants and $n > 0$

Find in terms of n :

- a** k **b** $E(X)$ **c** $\text{Var}(X)$
d the median of X **e** the mode of X

- 40** A particle is moving along a path with equation $y = \sqrt{x^2 + 24}$
- Find $\frac{dy}{dx}$.
 - Find the coordinates of the local minimum of the curve.
 - Is the function with this rule even?
 - As $x \rightarrow \infty$, it is evident that $y \rightarrow x$ and that as $x \rightarrow -\infty$, $y \rightarrow -x$. Sketch the graph of $y = \sqrt{x^2 + 24}$ showing the asymptotes.
 - Find the equation of the normal to the curve at the point with coordinates (1, 5) and sketch the graph of this normal with the graph of **d**.
 - When the particle is at the point with coordinates (5, 7), y is increasing at a rate of 10 units per second. At what rate is x increasing?
 - Show that $\frac{d}{dx} \left(12 \log_e |\sqrt{x^2 + 24} + x| + \frac{x\sqrt{x^2 + 24}}{2} \right) = \sqrt{x^2 + 24}$
 - Use this result to find the area of the region contained between the curve and the lines $x = 2$, $x = 5$ and the x -axis.

- 41** The volume, $V \text{ cm}^3$, of right circular cone of height h cm and radius r cm is given by $V = \frac{1}{3} \pi r^2 h$ and the curved surface area, $S \text{ cm}^2$, of the cone is given by $S = \pi r \sqrt{r^2 + h^2}$. Sand falls onto a horizontal floor at a rate of $20 \text{ cm}^3/\text{s}$. The sand falls in a pile so that it is in the shape of a right circular cone with vertical angle 60° . The height of the sand t seconds after the sand starts to fall is x cm.

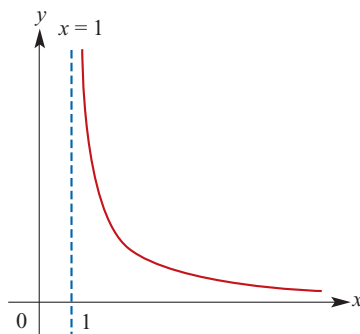


- Find the radius of the base of the pile of sand at time t seconds in terms of x .
 - Find V in terms of x .
 - Find $\frac{dV}{dx}$ in terms of x .
 - Find $\frac{dx}{dt}$ in terms of x .
 - Find $\frac{dt}{dx}$ in terms of x and, given that $x = 0$ when $t = 0$, find an expression for x^3 in terms of t .
 - Find $\frac{dS}{dt}$ when $t = 60$.
- 42** The boxplot is a display used to describe the distribution of a data set. Located on the boxplot are the minimum, the lower quartile, the median, the upper quartile and the maximum. Boxplots also show outliers. These are values which are more than 1.5 interquartile ranges below the lower quartile or above the upper quartile.
- Suppose that a random variable Z is normally distributed with a mean of 0 and a standard deviation of 1.
 - Find the value of the median, i.e. m such that $\Pr(Z \leq m) = 0.5$
 - Find the value of the lower quartile, i.e. q_1 such that $\Pr(Z \leq q_1) = 0.25$
 - Find the value of the upper quartile, i.e. q_3 such that $\Pr(Z \leq q_3) = 0.75$
 - Hence find the interquartile range (IQR) for this distribution.

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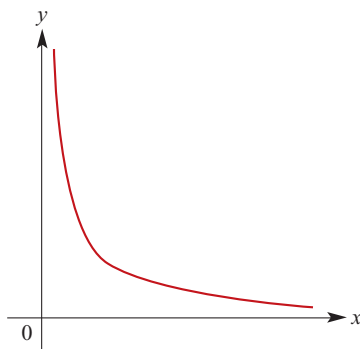
- v Find $\Pr(q_1 - 1.5 \times \text{IQR} < Z < q_3 + 1.5 \times \text{IQR})$.
 - vi What percentage of data values would you expect to be designated as outliers for this distribution?
- b Suppose that a random variable X is normally distributed with a mean of μ and a standard deviation of σ .
- i Find the value of the median, i.e. m such the $\Pr(X \leq m) = 0.5$
 - ii Find the value of the lower quartile, i.e. q_1 such that $\Pr(X \leq q_1) = 0.25$
 - iii Find the value of the upper quartile, i.e. q_3 such that $\Pr(X \leq q_3) = 0.75$
 - iv Hence find the interquartile range (IQR) for this distribution.
 - v Find $\Pr(q_1 - 1.5 \times \text{IQR} < X < q_3 + 1.5 \times \text{IQR})$.
 - vi What percentage of data values would you expect to be designated as outliers for this distribution?

- 43 The diagram shows the graph of the function $g : (1, \infty) \rightarrow \mathbb{R}, g(x) = \frac{1}{x-1}$. The line segment AB is drawn from the point $A(2, 1)$ to the point $B(b, g(b))$ where $b > 2$.



- a i What is the gradient of AB ?
 - ii At what value of x between 1 and b does the tangent to the graph of g have the same gradient as AB ?
- b i Calculate $\int_2^{e+1} g(x) dx$.
- ii Let c be a real number with $1 < c < 2$. Find the exact value of c such that $\int_c^{1+e} g(x) dx$ is equal to 8.
- c i What is the area of the trapezium bounded by line segment AB , the x -axis, the line $x = 2$ and the line $x = b$?
- ii For what exact value of b does this area equal 8?
- d $\int_2^{mn+1} g(x) dx + \int_2^{\frac{m}{n}+1} g(x) dx = 2$ where $n > 0$. Find the value of m .

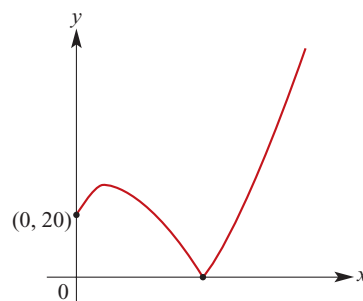
- 44 The diagram below shows the graph of the function $f: \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \frac{1}{x^2}$. The line segment AB is drawn from the point $A(1, 1)$ to the point $B(b, f(b))$ where $b > 1$.



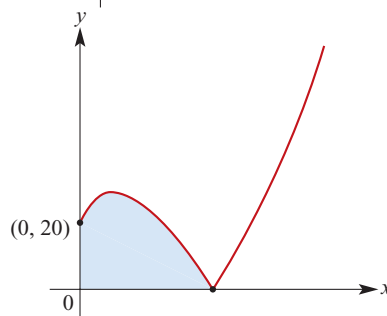
- a i What is the gradient of AB ?
 - ii At what value of x between 1 and b does the tangent to the graph of f have the same gradient as AB ?
- b i What is the area, $S(b)$, of the trapezium bounded by line segment AB , the x -axis, the line $x = 1$ and the line $x = b$?
- ii For what exact value of b does this area equal $\frac{10}{9}$?
 - iii Show that for $b > 1$, $\int_1^b f(x) dx < 1$.
- c Show that the function $D(b) = S(b) - \int_1^b f(x) dx$ is strictly increasing for $b > 1$.

- 45 Let $f: R^+ \cup \{0\} \rightarrow R, f(x) = |x - 8\sqrt{x} - 20|$. The graph of $y = f(x)$ is shown below.

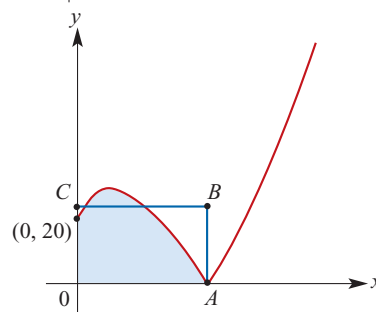
- a** Point $A(a, 0)$ is the x -axis intercept of $y = f(x)$.
Find the value of a .
- b** State the interval for which the graph of f has negative gradient.



- c** Find the area of the shaded region of the graph.

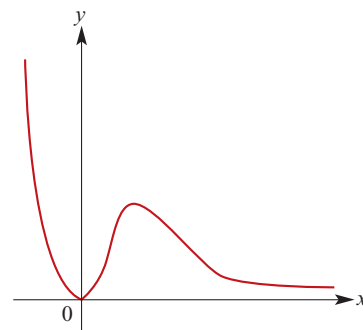


- d** O is the origin, C is a point on the positive y -axis and B is a point such that $OABC$ is a rectangle.



Find the length OC such that the area of the rectangle $OABC$ is the same as the area of the shaded region and explain why this length is the average value of the function for the interval $[0, a]$.

- e** Let $g: R \rightarrow R, g(x) = x^2$.
- i** Find the rule of $f(g(x))$.
Let $h(x) = f(g(x))$.
- ii** Find the derivative of $h(x)$ with respect to x .
- 46 For the function $f: R \rightarrow R, f(x) = x^m e^{-nx+n}$ where m and n are positive integers. The graph of $y = f(x)$ is as shown.
- a** Find the coordinates of the stationary point not at the origin in terms of n and state its nature.
- b** Find the coordinates of the point on the graph at which the tangent of f passes through the origin.



- c** Consider the continuous probability density function with rule

$$f(x) = \begin{cases} kx^2e^{-2x+2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } k \text{ is a positive real number.}$$

- i** Find the value of k .
- ii** Find $\Pr(X < 1)$ where X is the associated random variable.
- iii** Find the mode of the distribution.

- 47** Let X be a continuous random variable with probability density function given by

$$f(x) = \begin{cases} ke^{-qx} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } q \text{ is a positive real number.}$$

- a**
 - i** Find the value of k in terms of q .
 - ii** Find $E(X)$ in terms of q .
 - iii** Find $\text{var}(X)$ in terms of q .
 - iv** Show that the median of the distribution is $m = \frac{1}{q} \log_e(2)$.
- b** Find $\Pr\left(X > \frac{1}{q} \log_e(3) \mid X > \frac{1}{q} \log_e(2)\right)$.
- c** The distance X metres between flaws in a certain type of yarn is the random variable of a continuous probability distribution with probability density function $f(x) = 0.01e^{-0.01x}$ for $x \geq 0$.
 - i** Sketch the graph of $y = f(x)$.
 - ii** Find the probability, correct to two decimal places, that the distance between consecutive flaws is more than 100 m.
 - iii** Find the median value of this distribution, correct to two decimal places.

- 48** Sue has two favourite types of potato crisps, Crunchy Tasty and Spicy Lovely. She has exactly one packet a day. If she has Crunchy Tasty one day she will have Crunchy Tasty the next day with a probability of p . If she has Spicy Lovely one day she will have Crunchy Tasty the next day with a probability of $4p^2$.

- a**
 - i** If Sue has Crunchy Tasty on a Monday, what is the probability in terms of p that she will have them on the following Wednesday?
 - ii** If the probability of **i** is 0.544 find the value of p .
- b** Use the value of p found in part **a** to find the following given that on Monday she had Crunchy Tasty potato crisps.
 - i** The probability that she has Crunchy Tasty chips on Monday, Tuesday, Wednesday and Thursday.
 - ii** The probability that she has Crunchy Tasty on exactly 3 of the days Monday, Tuesday, Wednesday and Thursday.
- c** Write down the transition matrix for this information in terms of p .
- d** Write down the 'steady state' probabilities in terms of p .
- e** Sue decides to stop having a packet of chips every day. The probability of her buying a packet on any day is 0.8. Find the probability of her buying
 - i** at least one packet in the next 3 days
 - ii** exactly 3 packets in a 7 day week.

Glossary*

A

absolute maximum: [p. 370] M is the **absolute maximum** value of a continuous function f for an interval $[a, b]$ if $f(x) \leq M$ for all $x \in [a, b]$.

absolute minimum: [p. 370] N is the **absolute minimum** value of a continuous function f for an interval $[a, b]$ if $f(x) \geq N$ for all $x \in [a, b]$.

absolute value function (or modulus function), $|x|$:

$$[p. 20] |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

acceleration: [p. 380] the rate of change of a particle's velocity with respect to time.

acceleration, average: [MM1&2] for the time interval $[t_1, t_2]$, is defined by $\frac{v_2 - v_1}{t_2 - t_1}$ where v_2 is the velocity at time t_2 and v_1 is the velocity at time t_1

acceleration, instantaneous: [MM1&2] $a = \frac{dv}{dt}$

addition rule (for probability): [p. 511] the probability of A or B or both occurring, given by the rule $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

amplitude of circular functions: [p. 223] the distance between the mean position and the maximum position, e.g. the graph of $y = \sin x$ has an amplitude of 1

antiderivative: [p. 447] the general antiderivative F , where $F'(x) = f(x)$, found using $\int f(x)dx = F(x) + c$, where c is an arbitrary real number

arrangements: [p. 696] the number of arrangements of n objects in groups of size r , given by

$$\frac{n!}{(n-r)!} = n \times (n-1) \times (n-2) \dots (n-r+1)$$

average value: [p. 475] The **average value of a function** f with rule $y = f(x)$ for an interval $[a, b]$ is defined as $\frac{1}{b-a} \int_a^b f(x)dx$.

B

Bernoulli sequence: [p. 542] describes the repetitions of an experiment where:

- each trial results in one of two outcomes
- the probability for success in a trial is a constant
- the trials are independent

binomial distribution: [p. 543] the probability of observing X successes in n binomial trials each with probability of success p , given by

$$\Pr(X = x) = \binom{n}{x} (p)^x (1-p)^{n-x} \quad x = 0, 1, \dots, n$$

$$\text{where } \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

binomial expansion: [p. 699]

$$\begin{aligned} (x+a)^n &= x^n + \binom{n}{1} x^{n-1} a + \binom{n}{2} x^{n-2} a^2 + \dots + a^n \\ &= \sum_{k=0}^n \binom{n}{k} x^{n-k} a^k \end{aligned}$$

The $(r+1)$ th term is $\binom{n}{r} x^{n-r} a^r$

binomial experiment: [p. 543] an experiment consisting of a number, n , of identical trials. Each trial results in one of two outcomes, usually designated either a **success**, S , or a **failure**, F . The probability of success on a single trial, p , is constant for all trials. The trials are independent (so that the outcome on any trial is not affected by the outcome of any previous trial).

* The glossary contains some terms which were introduced in Mathematical Methods CAS Units 1 and 2, but which are not explicitly mentioned in the Mathematical Methods CAS Units 3 and 4 study design. The reference for these is given as [MM1&2].



chain rule: [p. 311] often used to differentiate some more complicated functions by transforming the original function into two simpler functions, e.g., $f(x)$ is transformed to $h(x)$ and $g(u)$ which are ‘chained’ together as $x \xrightarrow{h} u \xrightarrow{g} y$. Using Leibniz notation the chain rule is stated as $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

change of base: [p. 195] $\log_a b = \frac{\log_c b}{\log_c a}$

circle, general equation of: [p. 5] the general equation for a circle is $(x-h)^2 + (y-k)^2 = r^2$. The **centre** of the circle is the point (h, k) and the **radius** is r .

circular functions: [p. 213] the sine, cosine and tangent functions

complement: [p. 511] denoted, for example, A' , the set of points that are in ϵ but not in A . Similarly, B' is the set of points that are in ϵ but not in B .

composite functions: [p. 25] for functions f and g with domains X and Y respectively, and such that the range of $f \subseteq Y$, the composite function of g with f is defined as $g \circ f: X \rightarrow R$, where $g \circ f(x) = g(f(x))$

compound event: [MM1&2] several outcomes from a random experiment, such as observing an even number when a die is rolled

conditional probability: [p. 512] the probability of an event A occurring when it is known that some event B has occurred, written as $\Pr(A|B)$

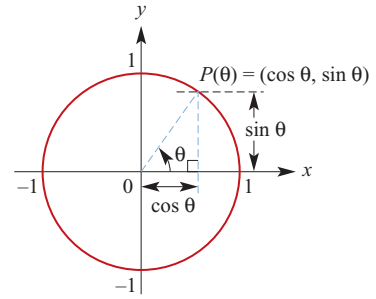
constant function: [MM1&2] a function f given by $f: R \rightarrow R, f(x) = a, a \in R$

continuous function: [p. 331] a function f is defined as continuous at the point $x = a$ if the following three conditions are met: $f(x)$ is defined at $x = a$; $\lim_{x \rightarrow a} f(x)$ exists; and $\lim_{x \rightarrow a} f(x) = f(a)$

continuous random variable: [p. 594] a random variable that can take any value in an interval of the real line

coordinates: [MM1&2] a unique ordered pair of numbers x, y that identifies a point on the coordinate plane. The first number in the ordered pair identifies the position with regard to the x -axis whereas the second number identifies the position with regard to the y -axis.

cosine function: [p. 213] cosine θ , or $\cos \theta$, defined as the x -coordinate of the point P on the unit circle where OP forms an angle of θ radians with the positive ray of the x -axis.



cubic function: [p. 128] a third degree polynomial function, f , with rule

$$f(x) = ax^3 + bx^2 + cx + d, a \neq 0$$

cumulative distribution function: [p. 604] usually denoted $F(x)$ and gives the probability that the random variable X takes a value less than or equal to x ; that is, $F(x) = \Pr(X \leq x) = \int_{-\infty}^x f(t)dt$



definite integral: [p. 444] $\int_a^b f(x)dx$ is called the definite integral from a to b

degree of polynomial: [p. 128] given by the value of n , the highest power of x with a non-zero coefficient

dependent trials: [MM1&2] sampling without replacement

derivative of cosine function: [p. 417] for $f: R \rightarrow R, f(\theta) = \cos \theta$, the derivative f' is given by $f': R \rightarrow R, f'(\theta) = -\sin \theta$

derivative of exponential function: [p. 402] for $f(x) = e^{kx}, k \in R$, the derivative f' is given by $f'(x) = ke^{kx}$

derivative of logarithm function: [p. 406] for $f: R^+ \rightarrow R, f(x) = \log_e(kx)$, the derivative f' is given by $f': R^+ \rightarrow R, f'(x) = \frac{1}{x}$

derivative of sine function: [p. 416] for $f: R \rightarrow R, f(\theta) = \sin \theta$, the derivative f' is given by $f': R \rightarrow R, f'(\theta) = \cos \theta$

derivative of tangent function: [p. 418] for $f(\theta) = \tan k\theta$, the derivative f' is given by $f'(\theta) = k \sec^2 k\theta$

determinant of a 2×2 matrix: [p. 57] If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \det(A) = ad - cb$$

difference of two cubes: [MM1&2] $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

difference of two squares: [MM1&2] $x^2 - y^2 = (x - y)(x + y)$

differentiability: [p. 333] A function f is said to be differentiable at a point $(a, f(a))$ if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists.}$$

differentiation rules: [p. 302] the general result for any non-zero real power gives: (1) for $f(x) = x^a$, the derivative f' is given by $f'(x) = ax^{a-1}$, for $x > 0$ and $a \in \mathbb{R}$ and (2) for $f(x) = 1$, $f'(x) = 0$

dilation from the x-axis: [p. 87] in general a dilation of a units from the x -axis is described by the rule $(x, y) \rightarrow (x, ay)$. In general the curve with equation $y = f(x)$ is mapped to the curve with equation $y = af(x)$ by the transformation with rule $(x, y) \rightarrow (x, ay)$.

dilation from the y-axis: [p. 87] in general a dilation of a units from the y -axis is described by the rule $(x, y) \rightarrow (ax, y)$. In general the curve with equation $y = f(x)$ is mapped to the curve with equation $y = f\left(\frac{x}{a}\right)$ by the transformation with rule $(x, y) \rightarrow (ax, y)$.

dimension of a matrix: [p. 56] The size, or **dimension**, of the matrix is described by specifying the number of rows (horizontal lines) and columns (vertical lines) that occur in the matrix. A matrix with m rows and n columns is said to be an $m \times n$ matrix.

discontinuity at a point: [p. 331] a function is said to be discontinuous at a point if it is not continuous at that point

discrete random variable: [p. 519] a random variable X which can take only a countable number of values, usually whole numbers

discriminant of a quadratic function: [p. 137] the expression $b^2 - 4ac$, which is part of the quadratic formula. For the quadratic equation

$$0 = ax^2 + bx + c,$$

- if $b^2 - 4ac > 0$, there are two solutions
- if $b^2 - 4ac = 0$, there is one solution
- if $b^2 - 4ac < 0$, there are no real solutions.

disjoint: [p. 2] two sets A and B are said to be **disjoint** if they have no elements in common, i.e. $A \cap B = \emptyset$

distance between two given points: [p. 51] the distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

domain: [p. 5] the set of all the first elements of ordered pairs

E

element (of a set): [p. 1] a member of, or an object in, a set. If x is an element of a set A , this is written $x \in A$.

empirical probability: [MM1&2] the probability assigned to an event on the basis of repeated

experimentation, i.e.

$$\Pr(A) \approx \frac{\text{number of times the event occurs}}{\text{number of trials}}$$

for large n

empty set, $\emptyset$: [p. 2] the set that has no members

equating coefficients: [p. 129] if $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ and $Q(x) = b_0 + b_1x + b_2x^2 + \dots + b_mx^m$ are equal, i.e. if $P(x) = Q(x)$ for all x , then the degree of $P(x)$ = degree of $Q(x)$, and $a_0 = b_0, a_1 = b_1, a_2 = b_2, \dots, a_n = b_m$

Euler's number, e : [p. 180] $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

even function: [p. 16] A function f is even if $f(x) = f(-x)$ for all x in the domain of f .

expected value of a random variable, $E(X)$ or μ : [pp. 524, 608] the (true) mean value of the random variable X , where $E(X) = \sum_x x \cdot \Pr(X = x) =$

$$\sum_x x \cdot p(x) \text{ for a discrete random variable, and } E(X) = \int_{-\infty}^{\infty} xf(x) dx \text{ for a continuous random variable}$$

exponential function (or index function):

[p. 174] the function, $f(x) = ka^x$, where k is a non-zero constant and the base a is a positive real number other than 1

F

factor: [MM1&2] a number or expression that divides another number or expression without remainder

factor theorem: [p. 132] if $ax + b$ is a factor of $P(x)$ then $P\left(-\frac{b}{a}\right) = 0$. Conversely, if $P\left(-\frac{b}{a}\right) = 0$ then $ax + b$ is a factor of P

factorise: [MM1&2] express as a product of factors

formula: [MM1&2] an equation containing symbols that states a relationship between two or more quantities, e.g. $A = lw$ (area = length $\times$ width). The value of A , called the 'subject' of the formula, can be found by substituting in given values of l and w .

function: [p. 7] a relation for which each x -value of an ordered pair has a unique y -value, i.e. if (a, b) and (a, c) are ordered pairs of a function, then $b = c$

function, many-to-one: [p. 14] a function for which more than one x -value maps onto the same y -value, e.g. $y = x^2$

function, one-to-one: [p. 14] a function for which each y -value has a unique x -value, e.g. $y = 2x$

fundamental theorem of calculus: [p. 456] the

theorem that states that $\int_a^b f(x)dx = G(b) - G(a)$, where G is any antiderivative of f

G

gradient function: [p. 301] for a function f , the gradient or derived function is denoted by f' , where $f': R \rightarrow R$ and $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

gradient of a line, m : [p. 49] $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are coordinates of points on the line. The gradient of a vertical line (parallel to the y -axis) is undefined.

H

horizontal line test: [p. 14] if a horizontal line can be drawn anywhere on the graph of a function and only ever intersects the graph a maximum of once, the function is **one-to-one**

hybrid function: [p. 15] a function that has different rules for different subsets of the domain

I

independence: [p. 514] A and B are independent events if $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$

independent trials: [p. 512] sampling with replacement

index function (or exponential function):

[p. 174] the function, $f(x) = ka^x$, where k is a non-zero constant and the base a is a positive real number other than 1

index laws: [p. 183]

- to multiply two numbers in exponent form with the same base, **add** the exponents:
 $a^m \times a^n = a^{m+n}$
- to divide two numbers in exponent form with the same base, **subtract** the exponents:
 $a^m \div a^n = a^{m-n}$
- to raise the power of a to another power, **multiply** the exponents: $(a^m)^n = a^{m \times n}$
- If $a^x = a^y$ then $x = y$
- for rational exponents: $a^{\frac{1}{n}} = \sqrt[n]{a}$

inequation: [MM1&2] a mathematical statement that contains an 'inequality' symbol rather than an 'equals' sign, e.g. $2x + 1 < 4$

integers: [p. 2] the elements of $\{\dots, -2, -1, 0, 1, 2, \dots\}$

integration rules: [pp. 448–464]

$$\begin{aligned} \int x^r dx &= \frac{x^{r+1}}{r+1} + c, r \in \mathbb{Q} \setminus \{-1\} \\ \int f(x) + g(x) dx &= \int f(x) dx + \int g(x) dx \\ \int kf(x) dx &= k \int f(x) dx \text{ where } k \text{ is a real number} \\ \int e^{kx} dx &= \frac{1}{k} e^{kx} + c \\ \int \frac{1}{ax+b} dx &= \frac{1}{a} \log_e(ax+b) + c \\ \int \sin(kx+d) dx &= \frac{-1}{k} \cos(kx+d) + c \\ \int \cos(kx+d) dx &= \frac{1}{k} \sin(kx+d) + c \end{aligned}$$

intersection of sets, $\cap$: [pp. 2, 510] the intersection of sets A and B , written $A \cap B$, represents the elements common to both A and B . Thus $x \in A \cap B$ if and only if $x \in A$ and $x \in B$.

inverse function: [p. 31] if f is a one-to-one function, then the inverse function f^{-1} is defined by $f^{-1}(x) = y$ if $f(y) = x$, for $x \in \text{ran } f$, $y \in \text{dom } f$

irrational numbers: [p. 2] the real numbers that are not rationals, e.g. π and $\sqrt{2}$

K

Karnaugh map: [p. 512] a probability table

L

law of total probability: [MM1&2] in the case of two events, A and B : $\Pr(A) = \Pr(A|B)\Pr(B) + \Pr(A|B')\Pr(B')$

limit of a function: [p. 328] the value that a function with rule $f(x)$ approaches as x approaches a given value. $\lim_{x \rightarrow a} f(x) = p$ means that as x approaches a , $f(x)$ approaches p .

limits, properties of: [p. 329]

- $\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$, i.e. the limit of the sum is the sum of the limits
- $\lim_{x \rightarrow c} (kf(x)) = k \lim_{x \rightarrow c} f(x)$, k being a given number
- $\lim_{x \rightarrow c} (f(x)g(x)) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x)$, i.e. the limit of the product is the product of the limits
- $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided $\lim_{x \rightarrow c} g(x) \neq 0$
i.e. the limit of the quotient is the quotient of the limits

linear equation: [pp. 45, 128] a polynomial equation of degree 1, e.g. $2x + 1 = 0$

linear function: [p. 128] a function with rule $f(x) = mx + c$, e.g. $f(x) = 3x + 1$

literal equation: [MM1&2] an equation for the variable x in which the coefficients of x , including the constants, are pronumerals, e.g. $ax + b = c$

logarithm: [p. 184] if $a \in \mathbb{R}^+ \setminus \{1\}$ and $x \in \mathbb{R}$ then the statements $a^x = n$ and $\log_a n = x$ are equivalent

logarithm laws: [p. 185]

- $\log_a(mn) = \log_a m + \log_a n$
- $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$
- $\log_a\left(\frac{1}{n}\right) = -\log_a n$
- $\log_a(m^p) = p \log_a m$



Markov sequence (chain): [p. 569] describes the repetitions of an experiment where:

- each trial results in one of two outcomes
- the probability of each possible outcome for a trial is conditional only on the trial immediately preceding it
- the conditional probabilities are the same for each occasion.

matrices addition and subtraction: [p. 56]

Addition will be defined for two matrices **only** when they have the same number of rows and the same number of columns. In this case the sum of two matrices is found by adding corresponding elements, for example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -3 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 7 & 3 \end{bmatrix}$$

matrices, equal: [p. 56] Two matrices **A**, **B** are **equal**, and can be written as **A** = **B** when:

- each has the same number of rows and the same number of columns
- they have the same number or element at corresponding positions.

matrix, inverse of a square matrix: [p. 57] **B** is said to be the **inverse** of **A** when **AB** = **BA** = **I**. The inverse of a square matrix **A** is denoted by **A**⁻¹. The inverse is unique. It does not exist for every square matrix.

matrix multiplication: [p. 57] If **A** is an $m \times n$ matrix and **B** is an $n \times r$ matrix, then the product **AB** is the $m \times r$ matrix whose entries are determined as follows.

To find the entry in row i and column j of **AB** single out row i in matrix **A** and column j in matrix **B**. Multiply the corresponding entries from the row and column and then add up the resulting products.

Note: The product **AB** is defined only if the number of columns of **A** is the same as the number of rows of **B**.

matrix multiplication by a scalar: [p. 57] It is useful to define **multiplication of a matrix by a real number**. If **A** is an $m \times n$ matrix, and k is a real number, then $k\mathbf{A}$ is an $m \times n$ matrix whose elements are k times the corresponding elements of **A**, for example:

$$3 \begin{bmatrix} 2 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & -6 \\ 0 & 3 \end{bmatrix}$$

matrix, multiplicative identity: [p. 57] For square matrices of a given dimension, e.g. 2×2 , a multiplicative identity **I** exists.

For example, for 2×2 matrices

I = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, **AI** = **IA** = **A**, and this result holds for any square matrix multiplied by the appropriate multiplicative identity.

matrix, regular: [p. 57] A square matrix is said to be **regular** if its inverse exists.

matrix, singular: [p. 57] a square matrix that does not have an inverse

matrix, square: [p. 57] a matrix with the same number of rows and columns

matrix, zero: [MM1&2] the $m \times n$ matrix with all elements equal to zero

maximal domain (implied domain): [p. 15] When the rule for a relation is written and no domain is stipulated, then it is understood that the domain taken is the largest for which the rule has meaning. This domain is called the maximal or implied domain.

mean of a random variable, E(X) or μ : [pp. 524, 608] expected value

median of a random variable, m : [pp. 527, 610] the middle value of the distribution such that:

$$\int_{-\infty}^m f(x) dx = 0.5 \text{ for a continuous random variable}$$

midpoint of a line segment: [p. 51] Let $M(x, y)$ be the midpoint of the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$. The coordinates of M are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

mode of a random variable, M : [pp. 528, 611] the value of the random variable which has the highest probability of occurring such that $\Pr(X = M) \geq \Pr(X = x)$ for all other values of X for a discrete random variable and $f(M) \geq f(x)$ for all other x for a continuous random variable

modulus function (or absolute value function), $|x|$:

$$[p. 20] |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

modulus function, properties of the: [p. 20]

- $|ab| = |a| |b|$
- $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

- $|a + b| \leq |a| + |b|$. If a and b are both non-negative or both non-positive then equality holds.
- If $a \geq 0$, $|x| \leq a$ is equivalent to $-a \leq x \leq a$.
- If $a \geq 0$, $|x - k| \leq a$ is equivalent to $k - a \leq x \leq k + a$.

multiplication rule (for choices): [p. 695] when sequential choices are involved, the total number of possibilities is found by multiplying the number of options at each successive stage

multiplication rule (for probability): [p. 513] the rule to determine the probability of events A and B occurring; $\Pr(A \cap B) = \Pr(A|B) \cdot \Pr(B)$

multi-stage experiment: [p. 513] an experiment that could be considered to take place in more than one stage, such as tossing two coins

mutually exclusive: [p. 511] said of two sets with no elements in common



n factorial, $n!$: [p. 696] an abbreviation for the product of all the integers from n down to 1; $n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 2 \times 1$

natural numbers: [p. 2] the elements of $\{1, 2, 3, 4, \dots\}$

nC_r (or $\binom{n}{r}$): [p. 696] ${}^nC_r = \frac{n!}{r!(n-r)!}$; the number of combinations of n objects in groups of size r

normal distribution: [p. 633] the distribution of a continuous random variable with probability density function $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ $\mu \in R, \sigma > 0$, where μ is the mean and σ the standard deviation of the random variable

normal, equation of the: [p. 348] Let P be the point on the curve $y = f(x)$ with coordinates (x_1, y_1) . Then, if f is differentiable for $x = x_1$, the equation of the normal at (x_1, y_1) is $y - y_1 = -\frac{1}{f'(x_1)}(x - x_1)$

not an element (of a set): [p. 1] the notation $x \notin A$ means x is **not** an element of A



odd function: [p. 16] A function f is odd if $f(-x) = -f(x)$ for all x in the domain of f .

ordered pair, (x, y) : [p. 5] a pair of elements x and y in which x is considered to be the first element and y the second



period of a function: [p. 223] the period of a function f with domain R is a positive number a such

that $f(x+a) = f(x)$ for a in R , e.g. the period of the sine function is 2π as $\sin(x+2\pi) = \sin x$. For functions of the form $y = a \cos(nx + \xi) + c$ or $y = a \sin(nx + \xi) + c$ the period is given by $\frac{2\pi}{n}$. The period of a tangent is π .

polynomial: [p. 128] a rule of the type $y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ ($n \in N$) where $a_0, a_1, \dots, a_n$ are numbers called coefficients

power function: [p. 81] a rule of the type $f(x) = x^n$, where n is a rational number

probability: [p. 510] a numerical value assigned to the likelihood of an event occurring. If the event A is impossible then $\Pr(A) = 0$, and if event A is certain then $\Pr(A) = 1$. Otherwise $0 < \Pr(A) < 1$

probability density function: [p. 596] usually denoted $f(x)$; describes the probability distribution of a continuous random variable, X , such that

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx$$

probability distribution: [p. 521] denoted $p(x)$ or $\Pr(X=x)$, a function that assigns probabilities to each value of X . It can be represented by a rule, a table or a graph, and must give a probability $p(x)$ for every value x that X can take.

probability table: [p. 511] a table used for illustrating a probability problem diagrammatically

product of functions: [p. 24] $f g(x) = f(x) g(x)$ and domain $(fg) = \text{dom } f \cap \text{dom } g$

product rule: [p. 317] Let $F(x) = f(x) \cdot g(x)$. If $f'(x)$ and $g'(x)$ exist then $F'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$. The product rule may also be stated in Leibniz notation: if $y = uv$ where u and v are functions of x , then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$



q th percentile: [p. 610] the value of X , p , such that $\Pr(X \leq p) = q\%$, given by the solution of the equation $\int_{-\infty}^p f(x) dx = \frac{q}{100}$

quadratic formula: [p. 136] $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

is the solution of the quadratic equation $ax^2 + bx + c = 0$

quadratic function, general equation of: [pp. 128, 136] a second degree polynomial function, f , with rule $f(x) = ax^2 + bx + c$, where a, b and c are constants and $a \neq 0$. The y -axis intercept is given by c .

quadratic function, turning point form of: [p. 136] a second degree polynomial function, f , with rule $f(x) = a(x-h)^2 + k$, $a \neq 0$, where (h, k) is the turning point

quartic function: [p. 128] a fourth degree polynomial function, f , with rule $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, $a \neq 0$

quotient rule: [p. 320] Let $F(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$. If $f'(x)$ and $g'(x)$ exist, then $F'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$. The quotient rule may be also stated in Leibniz notation: if $y = \frac{u}{v}$, $v \neq 0$, where u and v are functions of x , $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

R

R^+ : [p. 3] $\{x: x > 0\}$

R^- : [p. 3] $\{x: x < 0\}$

$R \setminus \{0\}$: [p. 3] the set of real numbers excluding 0

R^2 : [p. 3] $\{(x, y): x, y \in R\}$; the set of all ordered pairs of real numbers

radian: [p. 211] 1 radian (written 1°) is the angle subtended at the centre of the unit circle by an arc of length 1 unit

random experiment: [p. 510] an experiment, such as the tossing of a die, where the outcome of a single trial is uncertain but observable

random number tables: [MM1&2] a table that contains the digits 0, 1, ..., 9 in random order, and that can be used to generate random sequences of numbers

random variable: [p. 519] a variable that takes its value from the outcome of a random experiment, such as the number of heads observed when a coin is tossed three times

range: [p. 5] the set of all the second elements of ordered pairs of a relation

rational numbers: [p. 2] numbers of the form $\frac{p}{q}$ with p and q integers, $q \neq 0$

rectangular hyperbola: [p. 81] the basic rectangular hyperbola has equation $y = \frac{1}{x}$

reflection in the x -axis: [p. 90] In general a reflection in the x -axis is described by the rule $(x, y) \rightarrow (x, -y)$. In general the curve with equation $y = f(x)$ is mapped to the curve with equation $y = -f(x)$ by the transformation with rule $(x, y) \rightarrow (x, -y)$.

reflection in the y -axis: [p. 90] In general a reflection in the y -axis is described by the rule $(x, y) \rightarrow (-x, y)$. In general the curve with equation $y = f(x)$ is mapped to the curve with equation

$y = f(-x)$ by the transformation with rule $(x, y) \rightarrow (-x, y)$.

relation: [p. 5] a set of ordered pairs

relation, many-to-many: [p. 5] a relation for which more than one x -value maps onto more than one y -value, e.g. $x^2 + y^2 = 4$

relation, one-to-many: [p. 14] a relation for which an x -value maps onto more than one y -value, e.g. $y = \pm \sqrt{x}$

remainder theorem: [p. 132] when the polynomial $P(x)$ is divided by $ax + b$ the remainder is $P\left(-\frac{b}{a}\right)$

run length: [p. 585] the number of times that the same outcome is observed in a Bernoulli sequence

S

sample space, ϵ : [p. 509] the set of possible outcomes for an experiment

sampling with replacement: [p. 542] the process of selecting individual objects sequentially from a group of objects, and replacing the selected object, so that the probability of obtaining a particular object does not change with each successive selection.

sampling without replacement: [MM1&2] the process of selecting individual objects sequentially from a group of objects, and not replacing the selected object, so that the probability of obtaining a particular object changes with each successive selection.

selections: [p. 696] the number of combinations of n objects in groups of size r is

$$\begin{aligned} {}^nC_r &= \frac{{}^nP_r}{r!} = \frac{n \times (n-1) \times (n-2) \dots (n-r+1)}{r!} \\ &= \frac{n!}{r!(n-r)!} \end{aligned}$$

set difference: [p. 2] the set difference of two sets A and B is given by $A \setminus B = \{x: x \in A, x \notin B\}$

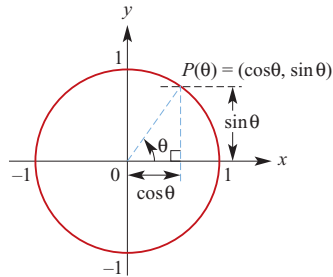
simple event: [p. 510] a single outcome from a random experiment, such as observing a two when a die is rolled

simulation: [MM1&2] the process of finding an approximate solution to a probability problem by repeated trials using a simulation model

simulation model: [MM1&2] a simple model that is analogous to a real world situation. For example, the outcomes from a toss of a coin (head, tail) could be used as a simulation model for the sex of a child (male, female) under the assumption that in both situations the probabilities are 0.5 for each outcome.

simultaneous equations: [p. 45] equations of two or more lines or curves in a cartesian plane, the solution of which is the point of intersection of the lines or curves

sine function: [p. 213] $\sin \theta$, or $\sin \theta$, defined as the y -coordinate of the point P on the unit circle where OP forms an angle of θ radians with the positive ray of the x -axis



standard deviation of a random variable, σ or $\text{sd}(X)$:

[p. 529] a measure of the spread or variability such that $\text{sd}(X) = \sqrt{\text{Var}(X)}$

standard normal distribution: [p. 633] a special case of the normal distribution where $\mu = 0$ and $\sigma = 1$

stationary point: [p. 358] a point with coordinates $(a, g(a))$ on a curve $y = g(x)$ where $g'(a) = 0$

steady state: [p. 548] Many two-state Markov chains will converge to a steady state, independent of the initial state.

straight line, equation of (given two points):

[pp. 50] $y - y_1 = m(x - x_1)$, where $m = \frac{y_2 - y_1}{x_2 - x_1}$

straight line, equation of (gradient–intercept form):

[p. 50] $y = mx + c$, where m is the gradient of the line

straight lines, perpendicular: [p. 50] If two straight lines are perpendicular, the product of their gradients is -1 . Conversely, if the product of the gradients of two lines is -1 , then the two lines are perpendicular.

Strictly decreasing: [p. 341] A function f is said to be strictly decreasing on a given set if for all a and b in the set $b > a$ implies $f(b) < f(a)$.

Strictly increasing: [p. 341] A function f is said to be strictly increasing on a given set if for all a and b in the set $b > a$ implies $f(b) > f(a)$.

subjective probability: [MM1&2] the probability assigned to an event on the basis of prior experience

subset: [p. 1] A set B is called a subset of set A , if and only if $x \in B$ implies $x \in A$. To indicate that B is a subset of A , we write $B \subseteq A$.

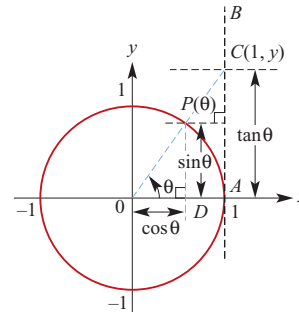
sum of functions: [p. 24] $(f + g)(x) = f(x) + g(x)$, domain of $(f + g) = \text{dom } f \cap \text{dom } g$

sum of two cubes: [MM1&2]
 $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

T

tangent, equation of: [p. 347] Let P be the point on the curve $y = f(x)$ with coordinates (x_1, y_1) . Then, if f is differentiable for $x = x_1$, the equation of the tangent at (x_1, y_1) is given by $(y - y_1) = f'(x_1)(x - x_1)$

tangent function: [p. 214] If a tangent to the unit circle, at A , is drawn then the y -coordinate of C (the point of intersection of the extension of OP and the tangent) is called tangent θ , or $\tan \theta$.



transition matrix: [p. 570] a square matrix that gives the probability for each of the possible outcomes at each step of a Markov chain conditional on each of the possible outcomes at the previous step

tree diagram: [p. 515] a diagram representing the outcomes of a multi-stage experiment

U

union of sets, $\cup$: [pp. 2, 511] the union of sets A and B , written $A \cup B$, is the set of elements that are either in A or in B . This does not exclude objects that are elements of both A and B .

V

variance of a random variable, σ^2 or $\text{Var}(X)$:

[p. 529] a measure of the spread or variability such that $\text{Var}(X) = E[(X - \mu)^2]$

Venn diagram: [MM1&2] a diagram showing sets and the relationships between sets

vertical line test: [p. 7] one way to identify if a relation is a function; draw a graph of the relation and if a vertical line can be drawn anywhere on the graph and only ever intersects the graph a maximum of once, the relation is a function

Counting methods and the binomial theorem

A.1 Counting methods

The addition rule

In general, to choose among alternatives simply add up the available number for each alternative.

Example 1

At the library Alan is having trouble deciding which book to borrow. He has a choice between three mystery novels, three biographies or two science fiction novels. How many choices of book does he have?

Solution

Since he is choosing between alternatives (mystery novels **or** biographies **or** science fiction), he has a total of $3 + 3 + 2 = 8$ choices.

The multiplication rule

When sequential choices are involved, the total number of possibilities is found by multiplying the number of options at each successive stage.

Example 2

Sandi has six choices of windcheaters or jackets, and seven choices of jeans or skirts. How many choices does she have for a complete outfit?

Solution

Since Sandi will wear both a windcheater or jacket **and** jeans or a skirt, we cannot consider these to be alternative choices. We could draw a tree diagram to list the possibilities, but this would be arduous. Using the multiplication rule, however, we can quickly determine the number of choices as $6 \times 7 = 42$.

Permutation or arrangements

The number of permutations of n objects in groups of size r is denoted nP_r and:

$$\begin{aligned} {}^nP_r &= \frac{n!}{(n-r)!} \\ &= n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) \end{aligned}$$

Example 3

How many different four-figure numbers can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 if each digit may be used only once?

Solution

Since we are arranging nine objects in groups of four, the answer is:

$$\begin{aligned} {}^9P_4 &= \frac{9!}{5!} \\ &= 9 \times 8 \times 7 \times 6 \\ &= 3024 \end{aligned}$$

Combinations or selections

In general, the number of combinations of n objects in groups of size r is:

$$\begin{aligned} {}^nC_r &= \frac{{}^nP_r}{r!} \\ &= \frac{n \times (n-1) \times (n-2) \times \cdots \times (n-r+1)}{r!} \\ &= \frac{n!}{r!(n-r)!} \end{aligned}$$

A commonly used alternative notation for nC_r is $\binom{n}{r}$.

Example 4

Four flavours of ice cream are available at the school canteen: vanilla, chocolate, strawberry and caramel. How many different double cone selections are possible if two different flavours must be used?

Solution

The number of combinations of four ice creams in groups of size two is:

$$\begin{aligned} {}^4C_2 &= \frac{4!}{2!2!} \\ &= \frac{4 \times 3}{2 \times 1} \\ &= 6 \end{aligned}$$

Example 5

A team of three boys and three girls is to be chosen from a group of eight boys and five girls. How many different teams are possible?

Solution

We can choose three boys from eight in 8C_3 ways, **and** three girls from five in 5C_3 ways. Thus the total number of possible teams is:

$$\begin{aligned} {}^8C_3 \times {}^5C_3 &= 56 \times 10 \\ &= 560 \end{aligned}$$

Exercise A1

- 1 A student needs to select a two unit study for her course, one unit in each semester. In semester 1 she must choose one of two mathematics units, three language units and four science units. In semester 2 she has a choice of two history units, three geography units and two art units. How many choices does she have for the complete course?
- 2 In order to travel from Melbourne to Brisbane, Dominic is given the following choices. He can fly directly from Melbourne to Brisbane on one of three airlines, or he can fly from Melbourne to Sydney on one of four airlines and then travel from Sydney to Brisbane with one of five bus lines, or he can go on one of three bus lines directly from Melbourne to Brisbane. In how many ways could he travel from Melbourne to Brisbane?
- 3 If there are eight swimmers in the final of the 1500 m freestyle event, in how many ways can the first three places be filled?
- 4 In how many ways can the letters of the word TROUBLE be arranged:
a if they are all used **b** in groups of three?
- 5 In how many ways can the letters of the word PANIC be arranged:
a if they are all used **b** in groups of four?
- 6 A student has the choice of three mathematics subjects and four science subjects. In how many ways can she choose to study one mathematics and two science subjects?
- 7 A survey is to be conducted, and eight people are to be chosen from a group of 30.
a How many different groups of eight people could be chosen?
b If the group contains 10 men and 20 women, how many groups of eight people containing exactly two men are possible?
- 8 From a standard 52-card deck, how many seven-card hands have exactly five spades and two hearts?

- 9 In how many ways can a committee of five be selected from eight women and four men:
a without restriction? **b** if there must be three women on the committee?
- 10 Six females and five males are interviewed for five positions. If all are found to be acceptable for any position, in how many ways could the following combinations be selected?
a three females and two males **b** four females and one male
c five females **d** five people regardless of sex
e at least four females

A.2 Summation notation

Suppose m and n are integers with $m < n$. Then:

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + \cdots + a_n$$

This notation, which is called **summation notation** or **sigma notation**, is very convenient for concisely representing sums. These sums will arise throughout the course. The notation uses the symbol Σ , the upper case form of the Greek letter sigma.

The symbol $\sum_{i=m}^n a_i$ is read ‘the sum of the numbers a_i from i equals m to i equals n ’.

The expression $a_m + a_{m+1} + \cdots + a_n$ is called the *expanded form* of $\sum_{i=m}^n a_i$.

Example 1

Write $\sum_{i=1}^5 2^i$ in expanded form and evaluate.

Solution

$$\begin{aligned}\sum_{i=1}^5 2^i &= 2^1 + 2^2 + 2^3 + 2^4 + 2^5 \\ &= 2 + 4 + 8 + 16 + 32 \\ &= 62\end{aligned}$$

Example 2

Write $1^2 + 2^2 + 3^2 + \cdots + 30^2$ using summation notation.

Solution

$$1^2 + 2^2 + 3^2 + \cdots + 30^2 = \sum_{k=1}^{30} k^2$$

Example 3

Write $x_1 + x_2 + x_3 + \cdots + x_{11}$ using summation notation.

Solution

$$x_1 + x_2 + x_3 + \cdots + x_{11} = \sum_{i=1}^{11} x_i$$

Exercise A2

1 Write each of the following in expanded form and evaluate:

$$\begin{array}{llll} \text{a} \sum_{i=1}^4 i^3 & \text{b} \sum_{k=1}^5 k^3 & \text{c} \sum_{i=1}^5 (-1)^i i & \text{d} \frac{1}{5} \sum_{i=1}^5 i \\ \text{e} \sum_{i=1}^6 i & \text{f} \sum_{k=1}^4 (k-1)^2 & \text{g} \frac{1}{3} \sum_{i=1}^4 (i-2)^2 & \text{h} \sum_{i=1}^6 i^2 \end{array}$$

2 Write each of the following using summation notation:

$$\begin{array}{ll} \text{a} 1 + 2 + 3 + \cdots + n & \text{b} x_1 + x_2 + x_3 + \cdots + x_{11} \\ \text{c} \frac{x_1 + x_2 + x_3 + \cdots + x_{10}}{10} & \text{d} 1^4 + 2^4 + 3^4 + \cdots + n^4 + (n+1)^4 \\ \text{e} 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} & \end{array}$$

3 Write each of the following in expanded form:

$$\begin{array}{llll} \text{a} \sum_{i=1}^n x^i & \text{b} \sum_{i=0}^5 x^i \cdot 2^{5-i} & \text{c} \sum_{i=0}^6 x^i \cdot 2^i \cdot 3^{6-i} & \text{d} \sum_{i=0}^4 (x - x_i)^i \end{array}$$

4 Write each of the following using summation notation:

$$\begin{array}{ll} \text{a} x^5 + 3x^4 + 9x^3 + 27x^2 + 81x + 243 & \text{b} x^5 - 3x^4 + 9x^3 - 27x^2 + 81x - 243 \\ \text{c} 4x^2 + 2x + 1 & \text{d} 8x^3 + 12x^2 + 18x + 27 \end{array}$$

A.3 The binomial theorem

Consider the expansion of binomial powers shown below:

$$\begin{aligned} (x+b)^0 &= 1 \\ (x+b)^1 &= 1x + 1b \\ (x+b)^2 &= 1x^2 + 2xb + 1b^2 \\ (x+b)^3 &= 1x^3 + 3x^2b + 3xb^2 + 1b^3 \\ (x+b)^4 &= 1x^4 + 4x^3b + 6x^2b^2 + 4xb^3 + 1b^4 \\ (x+b)^5 &= 1x^5 + 5x^4b + 10x^3b^2 + 10x^2b^3 + 5xb^4 + 1b^5 \end{aligned}$$

The coefficients can be arranged as shown:

$$\begin{array}{ccccccccccc} & & & & & & 1 & & & & & \\ & & & & & & & 1 & & 1 & & \\ & & & & & 1 & & 2 & & 1 & & \\ & & & 1 & & 3 & & 3 & & 1 & & \\ & & 1 & & 4 & & 6 & & 4 & & 1 & \\ & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

This array is known as Pascal's triangle. It was observed in *Essential Mathematical Methods 1 & 2 CAS* that the array can be constructed from combinations:

								Row
				$\binom{0}{0}$				0
			$\binom{1}{0}$		$\binom{1}{1}$			1
		$\binom{2}{0}$		$\binom{2}{1}$		$\binom{2}{2}$		2
	$\binom{3}{0}$		$\binom{3}{1}$		$\binom{3}{2}$		$\binom{3}{3}$	3
$\binom{4}{0}$		$\binom{4}{1}$		$\binom{4}{2}$		$\binom{4}{3}$	$\binom{4}{4}$	4

Remember: $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

The expansion of $(x + b)^8$ can be written by utilising this observation:

$$\begin{aligned}(x + b)^8 = & \binom{8}{0} x^8 + \binom{8}{1} x^7 b + \binom{8}{2} x^6 b^2 + \binom{8}{3} x^5 b^3 + \binom{8}{4} x^4 b^4 \\ & + \binom{8}{5} x^3 b^5 + \binom{8}{6} x^2 b^6 + \binom{8}{7} x b^7 + \binom{8}{8} b^8\end{aligned}$$

In summation notation:

$$(x + b)^8 = \sum_{k=0}^8 \binom{8}{k} x^{8-k} b^k$$

In general:

$$(x + b)^n = \sum_{k=0}^n \binom{n}{k} (x)^{n-k} b^k$$

and $(ax + b)^n = \sum_{k=0}^n \binom{n}{k} (ax)^{n-k} b^k$

The first term of the expansion of $(ax + b)^n$ is $\binom{n}{0} (ax)^n$

The second term is $\binom{n}{1} (ax)^{n-1} b$

The $(r + 1)$ th term is $\binom{n}{r} (ax)^{n-r} b^r$

It is convention that the binomial expansion of $(ax + b)^n$ is written in decreasing exponents of x .

Example 4

Find the coefficient of x^{20} in the expansion of $(x + 2)^{30}$.

Solution

The $(r + 1)$ th term is $\binom{30}{r} x^{30-r} 2^r$

When $30 - r = 20$, $r = 10$

$\therefore$ the term with x^{20} is $\binom{30}{20} 2^{10} x^{20}$

The coefficient of x^{20} is $\binom{30}{20} 2^{10}$

Example 5

Expand $(2x + 3)^5$.

Solution

$$\begin{aligned} (2x + 3)^5 &= \sum_{k=0}^5 \binom{5}{k} (2x)^{5-k} 3^k \\ &= \binom{5}{0} (2x)^5 + \binom{5}{1} (2x)^4 \cdot 3 + \binom{5}{2} (2x)^3 \cdot 3^2 + \binom{5}{3} (2x)^2 \cdot 3^3 \\ &\quad + \binom{5}{4} (2x) \cdot 3^4 + 3^5 \\ &= 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243 \end{aligned}$$

Example 6

Find the eighth term in the expansion of $(2x - 4)^{10}$.

Solution

The $(r + 1)$ th term is $\binom{10}{r} (2x)^{10-r} \cdot (-4)^r$

$$\begin{aligned} \therefore \text{the 8th term is } &\binom{10}{7} (2x)^3 \cdot (-4)^7 \\ &= -15\,728\,640x^3 \end{aligned}$$

Exercise A3

1 Expand each of the following using the binomial theorem:

a $(x + 6)^6$ **b** $(2x + 1)^5$ **c** $(2x - 1)^5$ **d** $(2x + 3)^6$
e $(2x - 6)^6$ **f** $(2x - 3)^4$ **g** $(x - 2)^6$ **h** $(x + 1)^{10}$

2 Find the eighth term in the expansion of:

a $(2x - 1)^{10}$ **b** $(2x + 1)^{10}$ **c** $(1 - 2x)^{10}$
d $(3x + 1)^{12}$ **e** $(x + 3)^{12}$ **f** $(2x - b)^{12}$

(Descending powers of x are assumed.)

3 Find the third term in the expansion of $\left(2 - \frac{1}{3}x\right)^9$ assuming descending powers of x .

4 Find the sixth term in the expansion of $(3x - 1)^{11}$ assuming descending powers of x .

5 Expand $(1 - x)^{11}$.

6 Find the coefficient of x^3 in the expansion of each of the following:

a $(x + 2)^5$ **b** $(2x - 1)^6$ **c** $(1 - 2x)^5$
d $(4x - 3)^7$ **e** $(3x + 4)^4$ **f** $(3x - 2)^5$

7 Find the coefficient of x^{10} in the expansion of $(2x - 3)^{14}$.

8 Find the coefficient of x^5 in the expansion of $(4 - 2x)^6$.

Computer Algebra System (TI-Nspire OS3.0)

B.1 Introduction to the TI-Nspire

This appendix is written for the TI-Nspire Clickpad, Touchpad and CX CAS calculators. The CX calculator and the function of some of the keys is shown in the diagram below. Similar diagrams for the Clickpad and Touchpad calculators are available at <http://www.cambridge.edu.au/go/>.



The Home screen

When pressing the home key () you always arrive at the Home screen which displays the **Scratchpad** and **Documents** platforms.

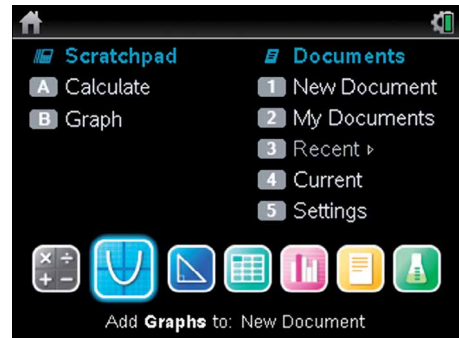
The **Scratchpad** platform enables you to:

- A.** Open the Calculate screen in the Scratchpad.
- B.** Open the Graph screen in the Scratchpad.

The **Documents** platform enables you to:

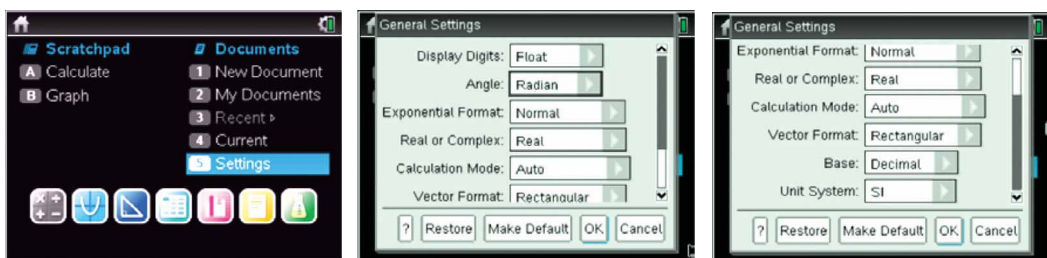
- 1.** Open a New Document
- 2.** Go to the My Documents folder
- 3.** Access Recent Documents
- 4.** Return to the current document.
- 5.** View the System Information and change the Settings

Below the **Scratchpad** and **Documents** sections of the Home screen are icons that add Calculator, Graphs, Geometry, Lists & Spreadsheet, Data & Statistics and Notes pages to an existing document.



System Settings

To change the system settings go to the Home screen () and select **Settings**. Select **Settings**>**General**. Use the **tab** key () to move between the different categories and use the **selection tool** () to open a category. Use the **up arrow** () or **down arrow** () on the touchpad to move between the different options within a category. When finished, move to **Make Default** and press **enter** () to confirm. The settings shown below are the recommended settings for Mathematical Methods.

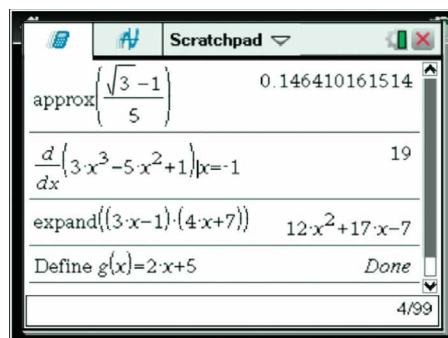


Scratchpad

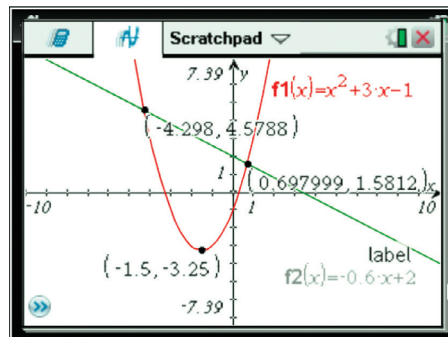
All instructions given in the text, and in the Appendices, are based on the **Documents** platform. However, all instructions relating to a **Calculator** application (page) or a **Graphs** application (page) can be transferred to the **Scratchpad:Calculate** or **Scratchpad:Graph** platform respectively.

The **Scratchpad** consists of two applications;

A: Calculate – this is a fully functional CAS calculation platform that allows for quick and easy access to the home screen and menus.



B: Graph – this is a fully functional graphing platform that allows for quick and easy access to the graphing screen and menus.



Note: when using **Scratchpad:Calculate** you need to be careful of previously defined variables. If in doubt, press **(var)**, to see what variables are defined, and delete them if necessary using **(menu) > Actions > Clear a-z** or **(menu) > Actions > Delete Variable** e. g. DelVar f

Note that **(menu) > Actions > Clear History** only clears the screen and does not clear stored variables.

It is possible to switch between the **Scratchpad** and the **Documents** platforms by pressing **(on)** (also **(=)** on the Touchpad & CX). This might be useful to do a quick trial, or check, calculation or graph in the **Scratchpad** whilst working in the **Documents** platform. Select **Current Document** on the **Home** page to return to the current document.

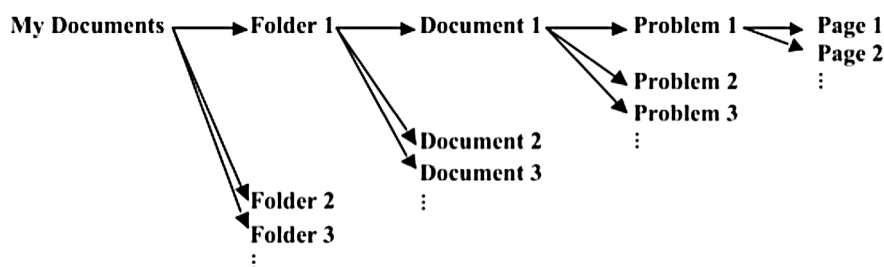
It is possible to complete the majority of the VCE Mathematical Methods course using only the **Scratchpad** platform if preferred.

It should be noted that any defined variables created in the **Scratchpad** are not available in the **Documents** platform.

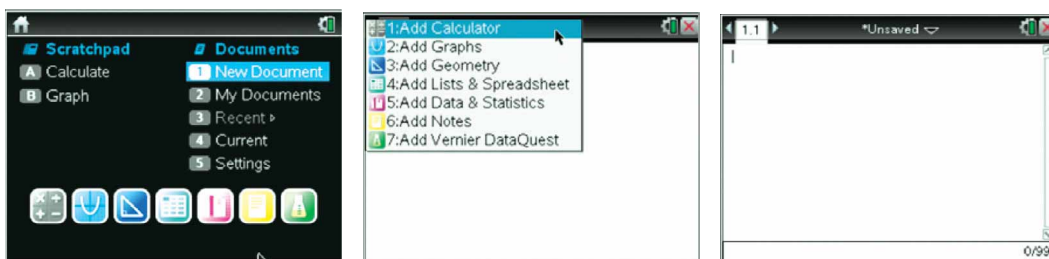
To move between the **Calculate** and **Graph** screens whilst in the **Scratchpad** either press **(on)** and select from this screen or press **(ctrl) [T]** or **(ctrl) [G]** (also **(=)** on the Touchpad & CX).

Documents – organisation and navigation

The calculator allows you to save documents very much like on a computer. Documents are saved in the **My Documents** folder ($\text{[on]} > \text{My Documents}$). In this folder you can create your own folders. Each document can contain a number of problems, and each problem can contain a number of pages.



Select **New Document** ($\text{[on]} > \text{New Document}$) from the home screen to create a new document. You will be prompted to select a Calculator application, a Graphs application, a Geometry application, a Lists & Spreadsheet application, a Data & Statistics application or a Notes application for the first page of this document.



The numbering 1.1 in the top left hand corner indicates that you are in problem 1, on page 1. In general, the numbering $m \cdot n$ indicates problem m , on page n .

There are several ways of adding another page (application) to the document. Either go to the home screen ([on]) and select the appropriate application, [ctrl] and [I] and select the application, or [ctrl] and [doc] (Touchpad and CX only).

To insert an additional five pages, each one with one of the remaining five applications, press [on] [V] [on] [G] [on] [L] [on] [D] [on] [N] . The five pages will look as shown below.

Note: the DataQuest application has not been included here. It is used for datalogging.

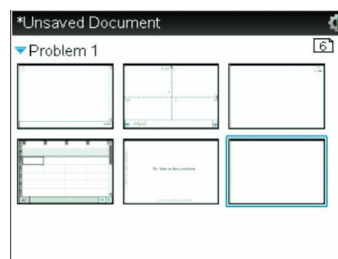


Notice how the new pages have numbers 1.2, 1.3, 1.4, 1.5 and 1.6 as explained above.

In Mathematical Methods we will mainly be using the Calculator application, the Graphs application and the Lists & Spreadsheet application.

Navigation within a document can be done in two ways.

One way is to press ctrl $\left[\text{left arrow}\right]$ to move to the page on the left; and press ctrl $\left[\text{right arrow}\right]$ to move to the page on the right. Alternatively use the cursor (Touchpad & CX only) to point to the page tabs and click to open. Another way is to get an overview of the document by pressing ctrl $\left[\text{document icon}\right]$ and navigate using the **arrows** ($\blacktriangleleft$ $\blacktriangleright$ $\blacktriangleup$ $\blacktriangledown$) on the Touchpad. Press **enter** (enter) to select a highlighted page. For the six pages above, the overview will look as shown at right.



From this view it is easy to **insert** (ctrl $\left[\text{I}\right]$) a new page (after a highlighted page) and **delete** (del or clear on the Clickpad) a highlighted page.

To get back to the individual page view either press **enter** (enter) on a highlighted page or press ctrl $\left[\text{document icon}\right]$.

There is also the opportunity to add another problem (doc) $\text{> Insert > Problem}$ (ctrl $\left[\text{doc icon}\right]$ $\text{> Insert > Problem}$ on the Clickpad) to a document. Within this problem new pages can be inserted as explained above.

However, despite all the opportunities for sophisticated organisation of documents, most students of Mathematical Methods can, to begin with, create one document, with one **Calculator** application and one **Graphs** application, and use that for all their calculations and graphing (or use the **Scratchpad** as mentioned earlier)

B.2 Using the Calculator Application

The Calculator Application works like a basic calculator and it is here that you will complete most of your arithmetical calculations. The following instructions are also suitable for the Scratchpad:Calculate platform.

Entering and editing expressions

The following example shows how to enter, evaluate and edit the expression $\frac{100(1.1^5 - 1)}{0.1}$.

To enter and evaluate the expression $\frac{100(1.1^5 - 1)}{0.1}$, press the following keys:

(ctrl) ÷ (1) (0) (0) × (1) (1) (1) ^ (5) tab (−) (1) ▼ (0) (1) enter

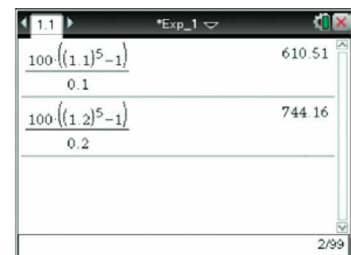
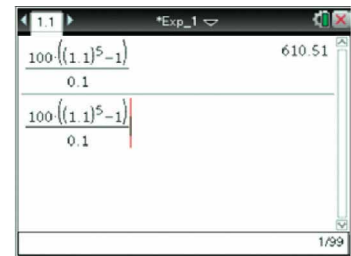
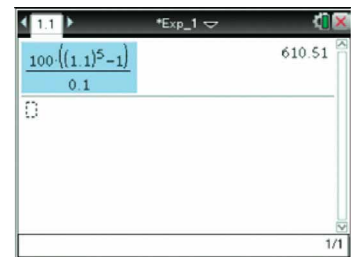
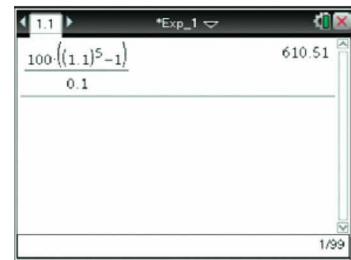
Notice the use of the division template and that each opening bracket is automatically followed by a closing bracket.

To enter and evaluate the expression $\frac{100(1.2^5 - 1)}{0.2}$, notice the similarity with the one we entered above and move upwards by pressing the **up arrow** (▲) twice until the previous expression is highlighted as shown.

Press **enter** (enter) to paste this expression in the new line as shown.

Use the **arrows** (◀ ▶ ▲ ▼) on the touchpad to move into the expression and use the **delete** key (clear) or (del) on Clickpad to delete the two 1s before replacing them by 2s. Press **enter** (enter) to evaluate this new expression.

Rather than copy and paste an entire expression as explained above, it is also possible to copy and paste a part of an expression. To do this, move to the expression as explained above, move into it using the **arrows** (◀ ▶ ▲ ▼) on the touchpad and select the part you want to copy by pressing and holding down the (shift) (or (ctrl) on Clickpad) key while using the **arrows** (◀ ▶ ▲ ▼) on the touchpad to select the part you want to copy. Press **enter** (enter) to paste this selection in the new line.



The calculator can also perform more complicated mathematical calculations, all of which can be found in the **menu** (menu). The most frequently used in Mathematical Methods are explained below, in order of appearance.

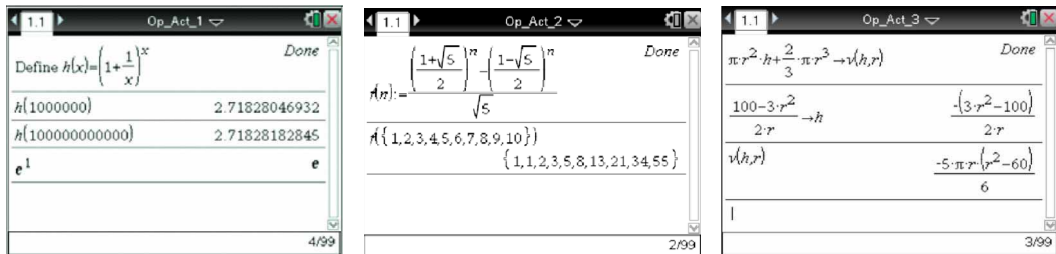
Operations in the Actions (Tools) menu

Define

This command is used for defining functions of one or more variables.

There are two other ways of defining functions; either use the notation $:=$ using $\text{(ctrl)} \text{(=)}$ on the Touchpad or (=) on the clickpad or use **store** ($\text{(ctrl)} \text{(var)}$).

The following screens illustrate their use.



Clear a-z

This command is used to clear the one-character variables a-z.

It is highly recommended to use mainly one-character variables and use this command frequently to clear these variables.

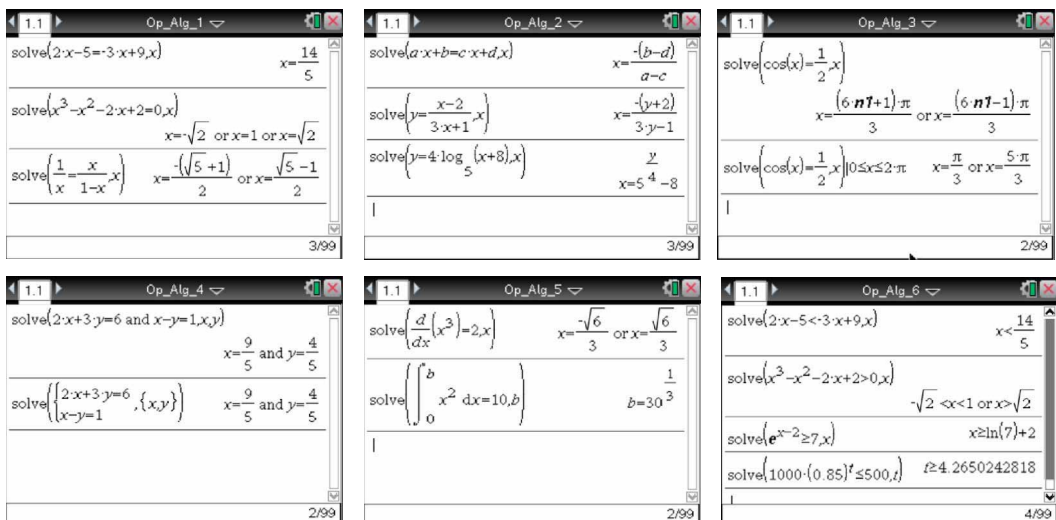
Operations in the Algebra menu

Solve

This command is used to solve equations, simultaneous equations and some inequalities.

An approximate (decimal) answer can be obtained by pressing $\text{(ctrl)} + \text{(enter)}$ or by including a decimal number in the expression.

The following screens illustrate its use.



Factor

This command is used for factorisation.

Factorisation over the rational numbers is obtained by not specifying the variable, whereas factorisation over the real numbers is obtained by specifying the variable.

The following screens illustrate its use.

$\text{factor}(2x^4 - x^2) \quad x^2 \cdot (2x^2 - 1)$ $\text{factor}(2x^4 - x^2, x) \quad x^2 \cdot (\sqrt{2} \cdot x - 1) \cdot (\sqrt{2} \cdot x + 1)$ $\text{factor}(x^3 - 9x^2 + 13x - 5) \quad (x-1) \cdot (x^2 - 8x + 5)$ $\text{factor}(x^3 - 9x^2 + 13x - 5, x) \quad (x-1) \cdot (x + \sqrt{11} - 4) \cdot (x - \sqrt{11} - 4)$	$\text{factor}(a^2 - b^2) \quad (a+b) \cdot (a-b)$ $\text{factor}(a^3 - b^3) \quad (a-b) \cdot (a^2 + a \cdot b + b^2)$ $\text{factor}\left(\frac{2}{x-1} + \frac{1}{(x-1)^2} + 1\right) \quad \frac{x^2}{(x-1)^2}$	$\text{factor}(24) \quad 2^3 \cdot 3$ $\text{factor}(-24) \quad -1 \cdot 2^3 \cdot 3$ $\text{factor}(1024) \quad 2^{10}$ $\text{factor}(1001) \quad 7 \cdot 11 \cdot 13$ $\text{factor}(201) \quad 2^{18} \cdot 3^8 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19$
---	---	--

Expand

This command is used for expansion of expressions.

By specifying the variable, the expanded expression will be ordered in decreasing powers of that variable. Symbolic expressions can only be expanded for an appropriate domain.

The following screens illustrate its use.

$\text{expand}((a+b)^3) \quad a^3 + 3a^2b + 3ab^2 + b^3$ $\text{expand}((a+b)^4) \quad a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$ $\text{expand}((a+b)^4, b) \quad b^4 + 4b^3a + 6b^2a^2 + 4ba^3 + a^4$	$\text{expand}\left(\frac{1}{x^2-1}\right) \quad \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$ $\text{expand}\left(\frac{x^3+2x+1}{x^2-1}\right) \quad \frac{1}{x+1} + \frac{2}{x-1} + x$	$\text{expand}((a^m)^n) \quad (a^m)^n$ $\text{expand}((a^m)^n) a > 0 \quad a^{m \cdot n}$ $\text{expand}(\ln(a \cdot b)) \quad \ln(a \cdot b)$ $\text{expand}(\ln(a \cdot b)) a > 0 \text{ and } b > 0 \quad \ln(a) + \ln(b)$
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Operations in the Calculus menu

Derivative & Derivative at a Point

These operations are used to differentiate expressions and/or give numerical derivatives. The following screens illustrate their use.

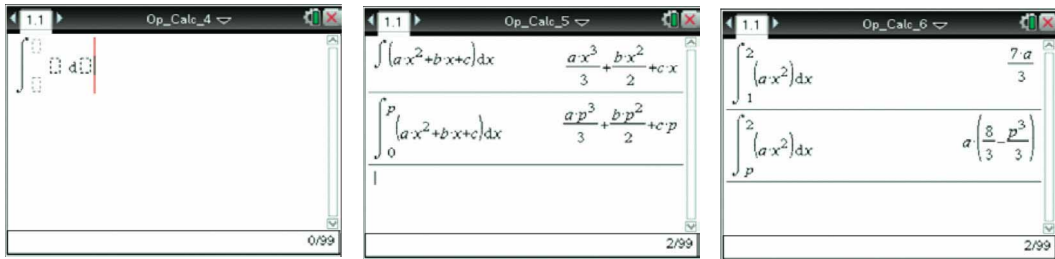
$\frac{d}{dx}(\square)$	$\frac{d}{dx}(\square) x=1$	$\frac{d}{dx}(4x^3 + 2x^2 - 7x + 8) \quad 12x^2 + 4x - 7$ $\frac{d}{dx}(4x^3 + 2x^2 - 7x + 8) x=1 \quad 9$ $\frac{d}{dx}(ax^2 + bx + c) \quad 2ax + b$	$\frac{d}{dx}(4x^3 + 2x^2 - 7x + 8) \quad 12x^2 + 4x - 7$ $\frac{d}{dx}(4x^3 + 2x^2 - 7x + 8) x=1 \quad 9$ $\frac{d}{dx}(ax^2 + bx + c) \quad 2ax + b$
-------------------------	-------------------------------	--	--

The differentiation template can also be accessed from the templates menu (Ⓜ or $\text{ctr} + \text{Ⓜ}$ on Clickpad).

In the templates menu there is also a template for the n th derivative.

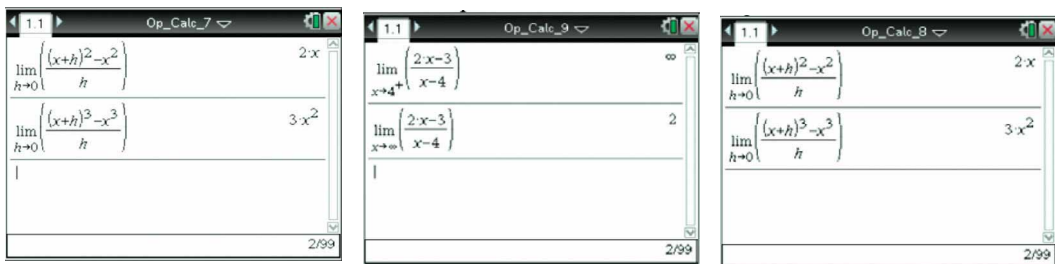
Integral

This operation is used to integrate expressions. The following screens illustrate its use.



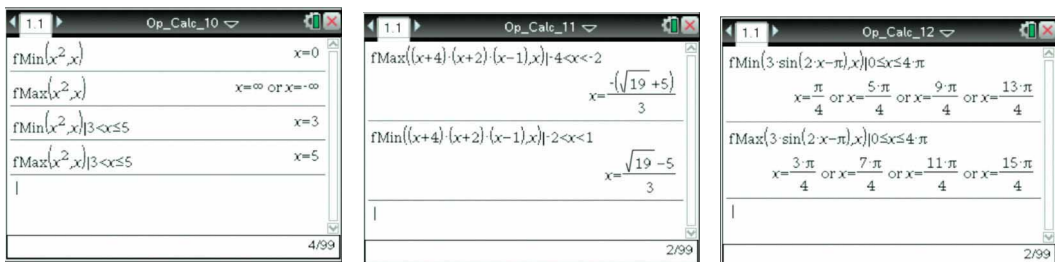
Limit

This operation is used to find limits of expressions. The following screens illustrate its use.



Function Minimum and Function Maximum

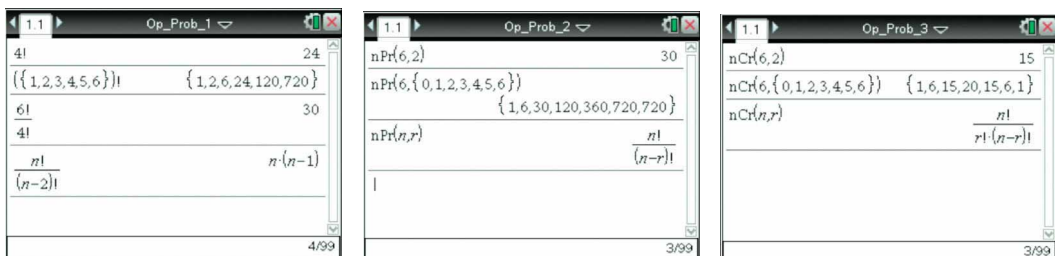
These operations return the value for which the minimum or maximum value (or the greatest lower bound or least upper bound) of a function occurs. The following screens illustrate their use.



Operations in the Probability menu

Factorial (!), Permutations and Combinations

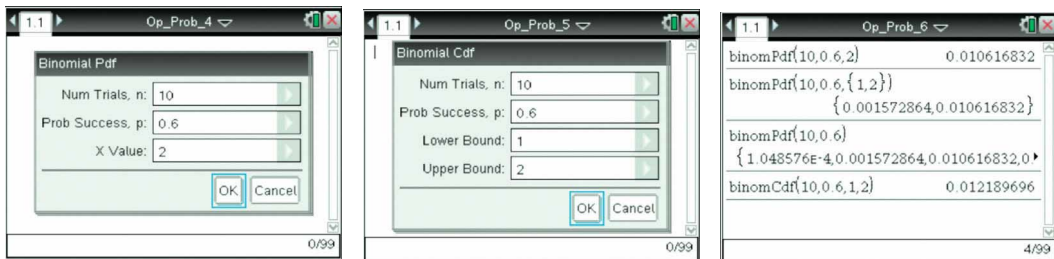
These functions are used to solve counting method problems. The following screens illustrate their use.



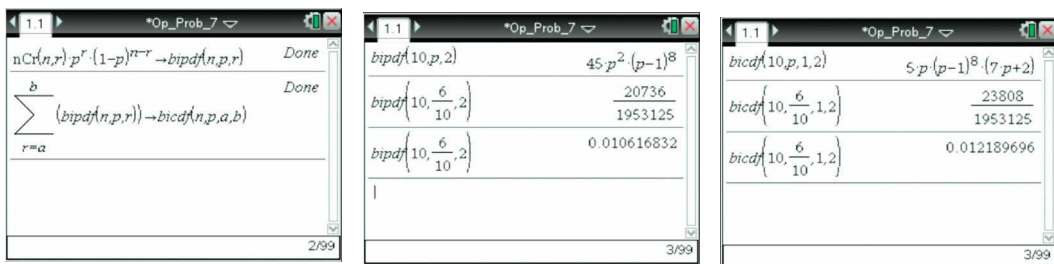
Distributions

This menu contains a list of distributions. For us, especially the Binomial Distribution and the Normal Distribution are of interest.

For calculations using the Binomial Distribution we use either **Binomial Pdf** (binomial probability distribution function) or **Binomial Cdf** (binomial cumulative distribution function). Use the **tab** key (␣) to move between the number fields. The following screens illustrate their use.



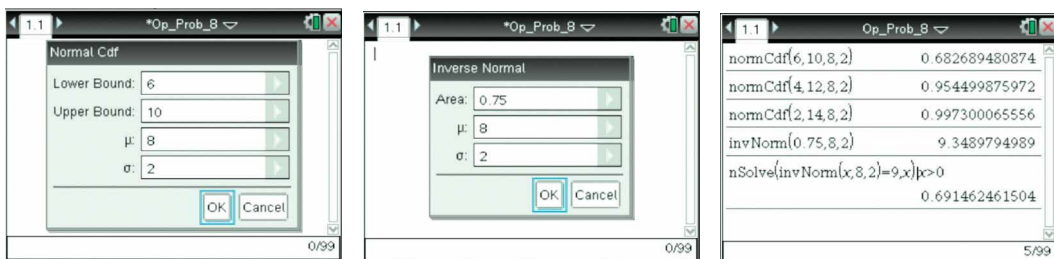
Unfortunately, these functions can only return an approximate (decimal) answer, so to obtain an exact answer consider defining the functions *bipdf* and *bicdf* as shown below.



For calculations using the Normal Distribution we use either **Normal Cdf** (normal cumulative distribution function) or **Inverse Normal**.

When using these functions you can solve for any of the variables using **nSolve()** (Numerical Solve) and an appropriate domain.

The following screens illustrate their use.

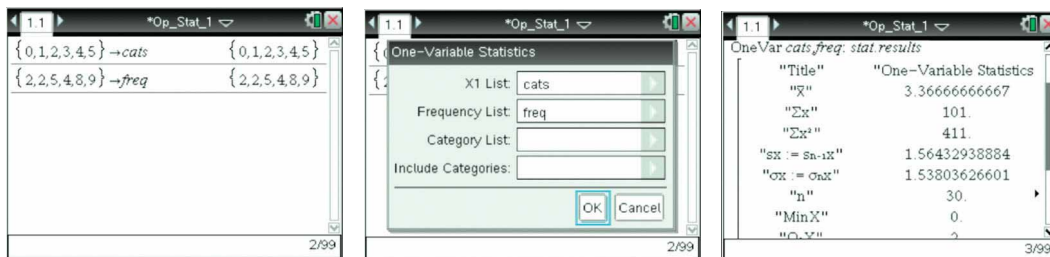


Operations in the Statistics menu

Stat Calculations

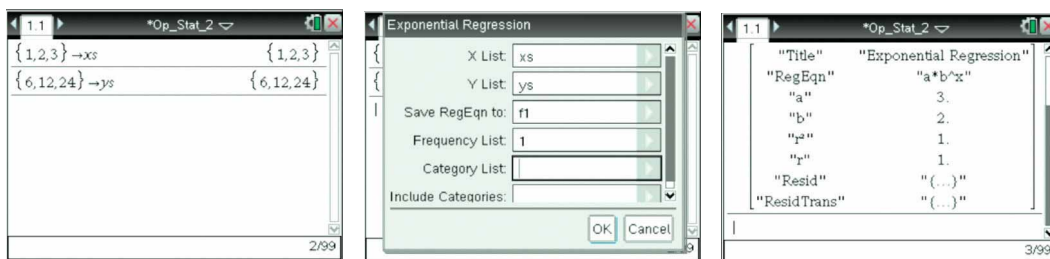
This menu contains a list of operations on data.

The following screens illustrate the use of the **One-Variable Statistics** command.



Note that the commands to find the *mean*, *median*, *variance* and *standard deviation* of a list of numbers are available from **menu** > **Statistics** > **List Maths**

The following screens illustrate the use of the **Exponential Regression** command. The last screen shows that the exponential function going through the points (1,6), (2,12) and (3,24) has equation $f(x) = 3 \times 2^x$.



B.3 Using the Graphs Application

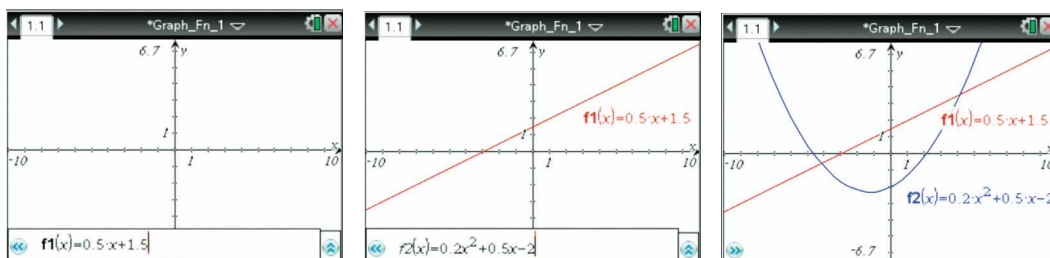
The following instructions are also suitable for the **Scratchpad:Graph** platform.

As the Points & Lines menu is not available in the Scratchpad, use Analyze Graph to find the intersection point/s.

Plotting functions

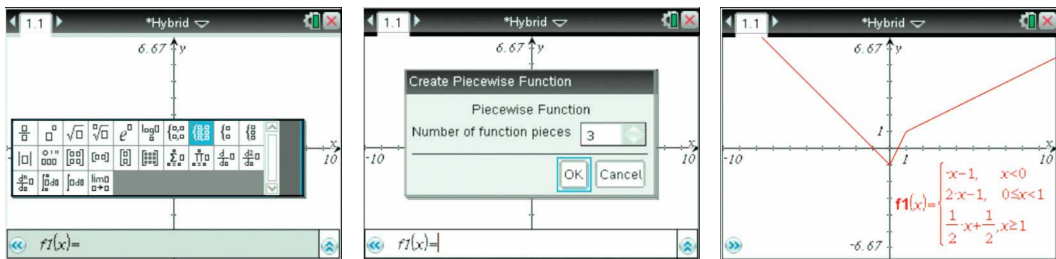
To graph a function simply type the expression (in terms of x) in the Entry Line for one of the predefined functions $f1(x) - f99(x)$ followed by **enter** as shown in the screens below.

If necessary, you can **hide/show the entry line** by pressing **ctrl** + **G**.



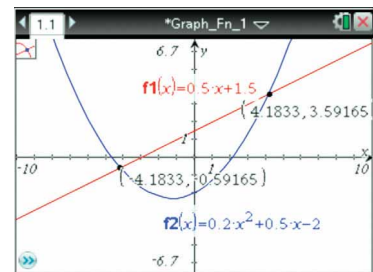
Plotting hybrid functions

In a Graphs application with the cursor in the entry line, select the hybrid (piecewise) function template, $\left(\frac{\text{w}}{\text{a}}\right)$ (ctrl + $\left(\frac{\text{w}}{\text{a}}\right)$ on the Clickpad) and enter as shown.



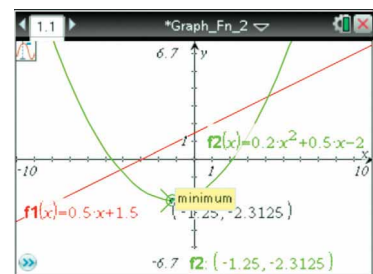
Finding intersection points

Use $\left(\frac{\text{w}}{\text{a}}\right)$ > **Points & Lines** > **Intersection Point/s** to display the approximate (decimal) coordinates of the intersection point(s). Select each of the two graphs (or a graph and an axes) using the **arrow keys** ($\leftarrow$ $\rightarrow$ $\uparrow$ $\downarrow$) (or the touchpad) followed by **enter**. Press **escape** (esc) to exit the command. On the screen to the right, Intersection Point(s) has been used to find the intersection points of f_1 and f_2 . Alternatively use $\left(\frac{\text{w}}{\text{a}}\right)$ > **Analyze Graph** > **Intersection**.



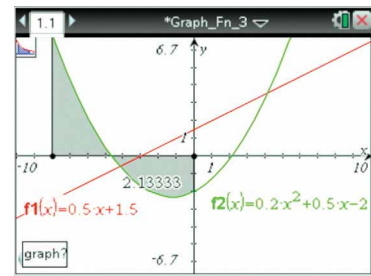
Finding zero(s) and local minimum(s)/maximum(s)

Either $\left(\frac{\text{w}}{\text{a}}\right)$ > **Trace** > **Graph Trace** or $\left(\frac{\text{w}}{\text{a}}\right)$ > **Analyze Graph** > can be used to display the approximate (decimal) coordinates of points on the graph. In Graph Trace the tracing point ($\times$) can either be moved using the **arrow keys** ($\leftarrow$ $\rightarrow$) or by typing a specific x -value followed by **enter**. Use $\uparrow$ $\downarrow$ to navigate between graphs. When trace reaches a zero it displays **zero**, when it reaches a local minimum it displays **minimum** etc. Press **escape** (esc) to exit the command. On the screen to the right, Graph Trace has been used to find the turning point of f_2 . When using Analyze Graph you select the lower bound (to the left of the key point) by clicking ($\leftarrow$) and move to the right ($\rightarrow$) of the key point (to select the upper bound) and click.



Finding integrals

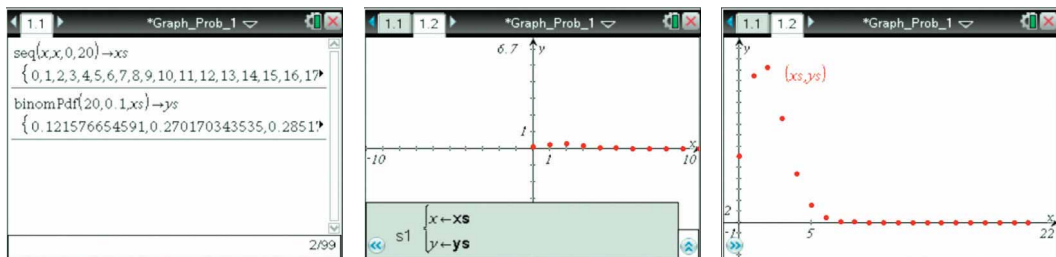
Use **(menu) > Analyze Graph > Integral** to display the approximate (decimal) integral. Select the graph, if prompted, then type an open parenthesis and type the lower limit, followed by **enter**, move cursor to the right, type an open parenthesis again and type the upper limit followed by **enter**. On the screen to the right, Integral has been used to find the integral of f_2 between -8 and 0 .



Plotting probability distributions

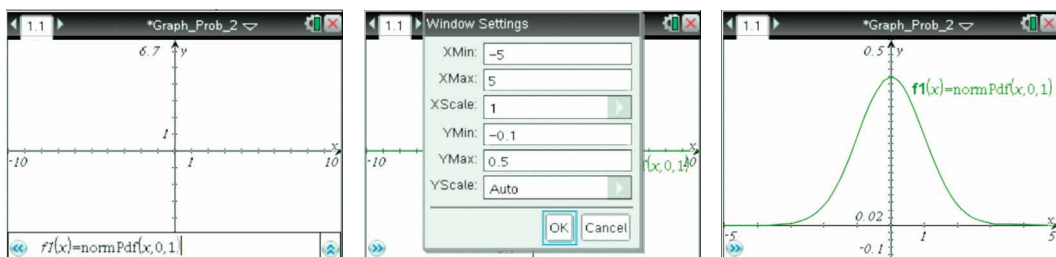
To graph a (discrete) Binomial Distribution first create a sequence of the outcomes in a Calculator application and use the binomial probability distribution function command **binomPdf(n, p, x)** on that list. In a Graphs application select **Scatter Plot** using **(menu) > Graph Type > Scatter Plot** and define the x -values to be the sequence of the outcomes and the y -values to be the corresponding binomial probability distribution function values. Finally, use **(menu) > Window/Zoom > Zoom – Data** to get a window that best fits the data.

The following screens illustrate how to graph the binomial distribution with parameters $n = 20$ and $p = 0.1$.



To graph a (continuous) Normal Distribution curve we use the normal probability distribution function command **normPdf(x, μ, σ)** from the catalog (**(2ND) (1) (N)**).

The following screens illustrate how to graph the standard normal distribution in the window $-5 \leq x \leq 5$ and $-0.1 \leq y \leq 0.5$.



B.4 Using the Lists & Spreadsheet Application

The Spreadsheet is in many ways similar to a Microsoft Excel® spreadsheet. Cells can be filled down, sorted etc., and it will handle both absolute and relative references.

Absolute and relative references

Simple interest and compound interest is used here to illustrate the use of absolute and relative cell references.

Enter 0 in cell A1, 100 in cell B1 and 100 in cell C1. Use the **arrows** (◀ ▶ ▲ ▼) on the NavPad to move between cells. Enter $=a1 + 1$ in cell A2, $=b1 + 0.1 \times b\$1$ in cell B2 and $=c1 + 0.1 \times c1$ in cell C2. The \$ symbol is found using (ctrl) (\$) (also (?) on the Touchpad & CX)

Highlight the cells A2, B2 and C2 using (shift) (↵) on the Clickpad) and the **arrows** (◀ ▶ ▲ ▼) and use (menu) > **Data** > **Fill** and the **down arrow key** (▼), followed by **enter**, to generate the numbers in the next three rows.

The following screens illustrate the process.

	A	B	C
1	0	100	100
2	1	110.	110.
3	2	121.	123.1
4	3	133.1	136.61
5	4	146.41	150.071

The resulting numbers are the value of 100 dollars invested at 10% p.a. simple interest (column B) and 10% p.a. compounded interest (column C) over the next four years.

Note the important difference in the formulas in cells B5 ($=B4 + 0.1 \times B\$1$) and C5 ($=C4 + 0.1 \times C4$). In the B-column, the use of the \$ symbol in front of 1 in cell B2 fixed this value such that when filled down it did not change, still adding 10% of the value in cell B1. This is an example of an *absolute* reference. In the C-column, the cell references are slightly different, always adding 10% of the value in the previous cell. This is an example of a *relative* reference.

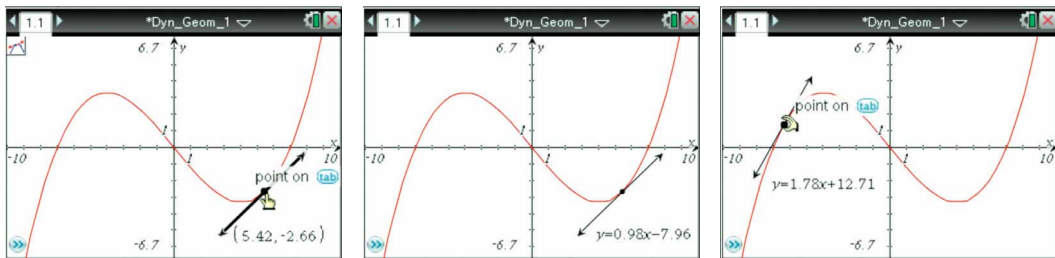
B.5 Dynamic Geometry

The calculator can dynamically change objects and corresponding calculations and thus provide ample opportunities for teachers and students to dynamically illustrate mathematical concepts and ideas. The Geometry tools are not available in the **Scratchpad:Graph** platform.

Example 1 – a moving tangent

This example shows how the calculator can be used to dynamically illustrate the concept of *instantaneous rate of change* as the gradient of the tangent.

1. Open a Graphs application ((**ctrl** on ).
 2. Type the expression of the function $f(x) = x(x - 7)(x + 7)/40$ in the Entry Line and press (**enter**).
 3. Hide the function label using (**menu**)>**Actions**>**Hide/Show**
 4. Place a **tangent** ((**menu**)>**Points & Lines**>**Tangent**) at any point on the graph using the NavPad. Press (**enter**) twice.
 5. Show the equation of the tangent ((**menu**)>**Actions**>**Coordinates & Equations**). Double click on the tangent line. Press (**esc**) to exit the **show equation** tool.
 6. Move the open hand () to the point. Press and hold down () when the point is flashing in the hand. The hand is now closed ()
 7. Move the point using the touchpad to dynamically illustrate the gradient of the curve.
- The following screens show the tangent placed at a point on the curve and the point moved along the curve.

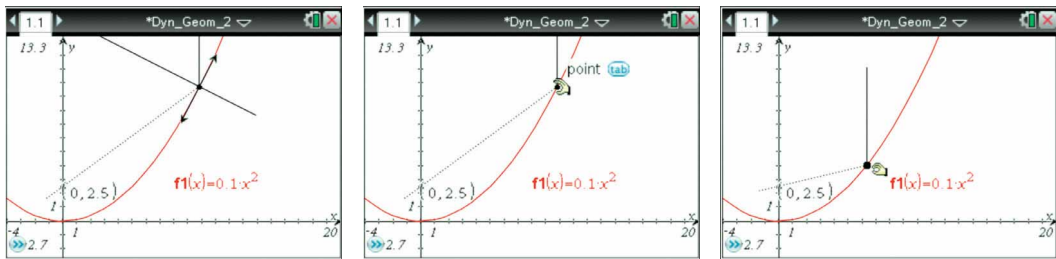


Example 2 – a parabola with a point

This example shows how the calculator can be used to dynamically illustrate why parabolas play an important part in the design of torches and satellite dishes.

1. Open a Graphs application ((**ctrl** on ).
2. Type in the expression of the quadratic $f(x) = 0.1x^2$ in the Entry Line. Press (**enter**)
3. Select the **first quadrant** window ((**menu**)>**Window/Zoom**>**Zoom-Quadrant 1**).
4. Place a **ray** ((**menu**)>**Points & Lines**>**Ray**) on a point on the graph, pointing upwards, parallel to the y -axis. Hold down (**shift**) () on the Clickpad) to move in increments of 15° .
5. Place a **tangent** ((**menu**)>**Points & Lines**>**Tangent**) at the point on the graph using the NavPad.
6. Construct the **perpendicular** line ((**menu**)>**Construction**>**Perpendicular**) to the tangent at this point.
7. **Reflect** ((**menu**)>**Transformations**>**Reflection**) the ray in this line. Select the line of reflection first.
8. Place a **point of intersection** ((**menu**)>**Points & Lines**>**Intersection Point/s**) between the y -axis and this new ray.
9. **Show the coordinates** ((**menu**)>**Actions**>**Coordinates & Equations**) of this point. (double click)
10. **Hide** ((**menu**)>**Actions**>**Hide/Show**) the tangent and the perpendicular line.

11. Change the **attribute** (**menu** > **Actions** > **Attributes**) of the ray and the reflected ray to **dashed line style**
12. Move the open hand () to the point. If necessary, press (**tab**) until **point on** is displayed. Press and hold down  when the point is flashing in the hand. The hand is now closed ()
13. Move the point using the touchpad to dynamically illustrate the unchanging *focal point*. The following screens show the ray, tangent and reflected ray at a point on the curve and the point moved along the curve.



B.6 Learning More

There is much to be learned from reading the manual that comes with the calculator, but there are also some highly recommended free online tutorials on the internet, including:

■ **Atomic Learning TI-Nspire tutorials**

http://www.atomiclearning.com/k12/en/ti_nspire/

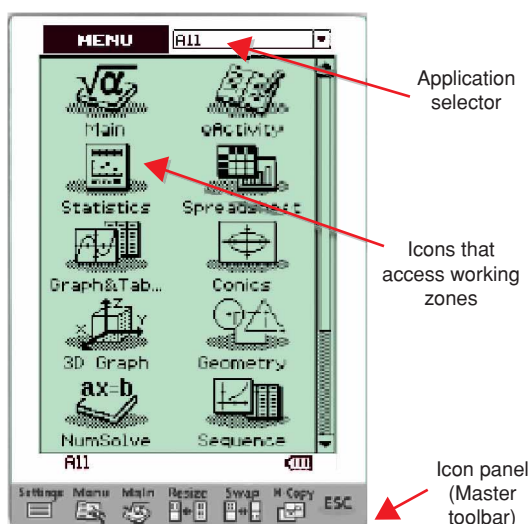
Computer Algebra System (ClassPad 330)

C.1 Introduction

For reference material on basic operations of the calculator, refer to the free downloadable documentation provided by Casio on the website:

http://www.casioed.net.au/downloads/classpad_download.php.

The Classpad calculator operates in many ways like your computer, with various drop down menus which are accessed using the stylus and tapping items on the Menu bar. Shortcuts in particular applications are found in the pictorial toolbar. You will find that Cut, Copy and Paste in the **Edit** menu very useful.



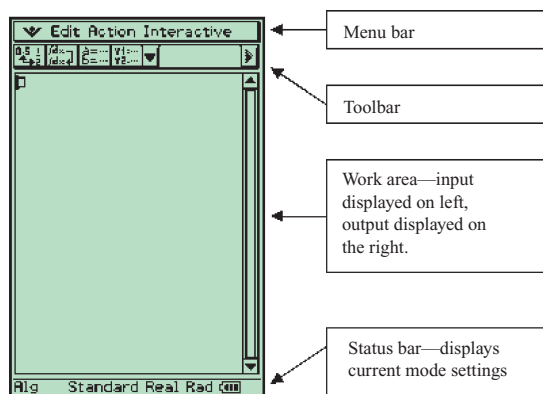
The  screen provides access to various in-built programs of the machine, selected by tapping the icons on the touch-sensitive screen using the stylus.

To stop a process that is in action, tap **ESC**.

The screen

The various areas of the screen referred to in these notes are shown in this diagram.

In some programs, such as  the screen is split into two halves. If you wish one of these to fill the whole screen, tap to ensure it is selected (bold border) and tap  on the **Master toolbar**. Tapping it again will re-split the screen.



Settings

The basic settings for the calculator are managed by tapping  from any of the program screens. Quick changes can be made by tapping the appropriate item on the **Status Bar**.

In  they appear as shown at right.



Tapping any of them toggles between the modes.

- **Alg**—algebra mode automatically simplifies expressions. (Alternative **Assist**—assistant mode does not automatically simplify expressions.)
- **Standard**—results given in exact form (fraction, square root, etc.) (Alternative **Decimal**—converts all results to decimal (approximate value).)
- **Real**—real mode for real number calculations. (Alternate **Cplx**—for complex number calculations.)
- **Rad**—angles displayed in radians. (Alternative **Deg**—angles displayed in degrees.)

Executing in commands in

Operator entries are shown on the left of the screen. Once entered, press  and the answer will be displayed right-justified on the screen. Scroll arrows will appear if your entry, or the answer, is too wide for a single screen. Additional input options, such as square roots, alternative variables and trigonometric or calculus operations, are accessed by pressing the  button on the calculator's keypad. If an answer is given in exact mode and you wish a decimal answer, highlight the answer with the stylus and tap . To convert from decimal to exact, follow the same steps.

Errors, deleting, inserting and clearing

The  button deletes the last character entered. You can use the central multi-directional arrow key, or the stylus, to return the cursor to the place where you wish to insert extra

characters. If you have already executed a command, you can still use the stylus to go back and make a correction and then execute again. To clear the screen (in any program section of the calculator, select **Edit—Clear all**. To clear values or functions which have been saved in memory locations, select **Edit—Clear all variables**. To clear individual variables, without clearing them all, use —**Variable manager**. You can double-click on various subfolders to expand them and then select items to delete using the **Edit** submenu items.

Note also that, like using a Windows computer program, you can highlight, copy, paste and drag-and-drop expressions, or parts of expressions, to create new expressions in the entry line.

C.2 Using the Main menu

In this section the Main menu is explored. Work through it to become acquainted with the menu.

There are two ways to operate in this menu, but the Interactive menu is the simplest when used in conjunction with the stylus. In this section we shall demonstrate some examples of how this is used.

Solve

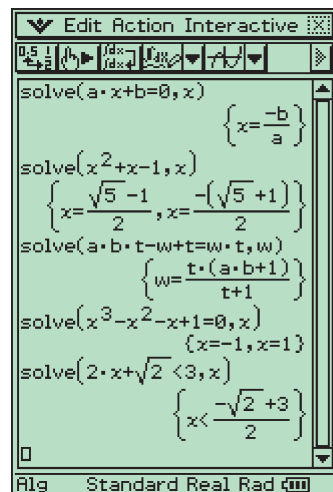
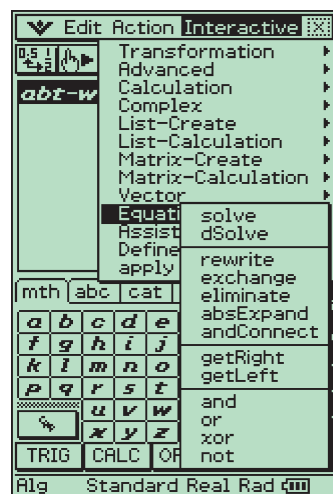
This is used to solve equation and inequalities. The variables x , y and z are found on the hard keyboard.

Other variables may be entered using the  and selecting **VAR**. Variables are shown in bold italics. The  keyboard allows you to type sentences, etc.; however, the letters are not always recognised as variables. If you choose to use the  keyboard, you must type $a \times x$, for example, as ax will be treated as text.

For example:

- Enter $ax + b = 0$, highlight it with the stylus, tap **Interactive—Equation/inequality—solve** and ensure the variable selected is x . Solution returned is $x = -\frac{b}{a}$.
- Enter $x^2 + x - 1 = 0$, and follow the same instructions as above. The answer is as shown.

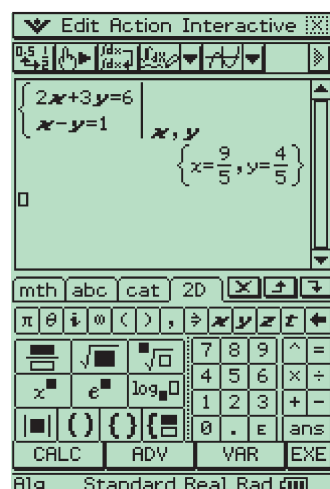
Note: $x^2 + x - 1 = 0$ is entered, but the calculator converts this to standard algebraic notation when **EXE** is pressed. Also note in this example that ‘ $= 0$ ’ has been omitted. If the right-hand side of an equation is zero, it is unnecessary to enter it.



- Enter $abt - w + t = wt$ for w ; follow the instructions above and select w as the variable.
- Solve $x^3 - x^2 - x + 1 = 0$ for x .
- Solve $2x + \sqrt{2} < 3$ for x . See the screen above for solutions returned.

Note: $\sqrt{\square}$ is found in $\boxed{2D}$ when $\boxed{\text{Keyboard}}$ is activated and $<$ is in $\boxed{\text{mth}}$ OPTN. If the answer is not in the form required, it is often possible to cut and paste it in the next entry line and use **Interactive—Transformation—simplify** as shown here.

- Solve simultaneous equations uses a dedicated entry style. Note carefully the screen at right. To enter, tap $\left\{ \begin{array}{l} \square \\ \square \end{array} \right\}$ in the $\boxed{2D}$ window and enter the equations and variables as shown. For more than two equations, tap $\left\{ \begin{array}{l} \square \\ \square \end{array} \right\}$ until the number of equations required is given.

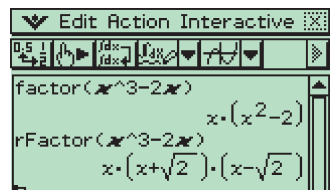


Factor

To factorise is to transform the expression to a different form. This command is found in **Interactive—Transformation—factor**.

Examples:

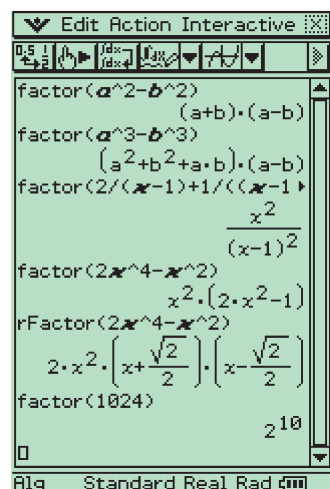
- To factorise $x^3 - 2x$ over the rational numbers, use **factor**.
- To factorise over the real numbers, select **rFactor**.



Some further examples are provided here. See the screens for the results.

- factor $a^2 - b^2$.
- factor $a^3 - b^3$.
- factor $\left(\frac{2}{x-1} + \frac{1}{(x-1)^2} + 1 \right)$.
- factor $(2x^4 - x^2)$ over the rationals.
- factor $(2x^4 - x^2)$ over the reals.

This command can also be used to give the prime decomposition (factors) of integers.



Expand

To expand expressions, use **Interactive**—

Transformation—expand.

For example:

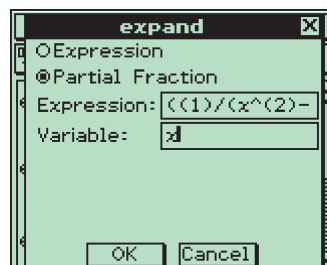
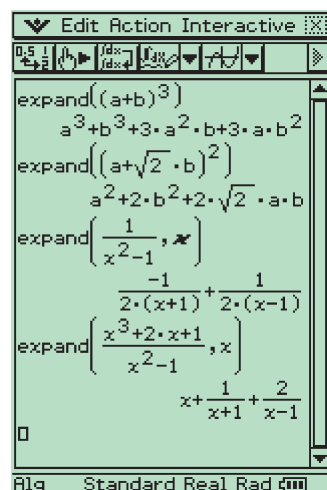
- $\text{expand}(a + b)^3$
- $\text{expand}(a + \sqrt{2}b)^2$.

The **expand** command can also be used to form partial fractions. In this case, enter and highlight the expression, select **Interactive**—**Transformation—expand**, select the **Partial Fraction** option and set the variable as x .

For example:

- $\text{expand} \frac{1}{x^2 - 1}$
- $\text{expand} \frac{x^3 + 2x + 1}{x^2 - 1}$.

Note: The top screen shows all the examples; the bottom screen shows how to enter for partial fractions.

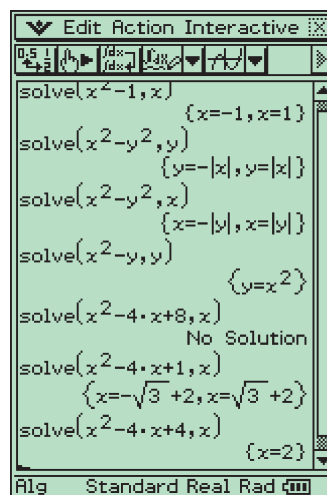


Zeros

To find the zeros of an expression in the **Main** menu, use **Interactive**—**Equation/inequality—solve** and ensure you set the variable. The calculator assumes you are solving an equation for which one side is zero.

For example:

- Zeros of $x^2 - 1$ for x .
- Zeros of $x^2 - y^2$ for y .
- Zeros of $x^2 - y^2$ for x .
- Zeros of $x^2 - y$ for y .
- Zeros of $x^2 - 4x + 8$ for x . No solutions.
- Zeros of $x^2 - 4x + 1$ for x . Two solutions.
- Zeros of $x^2 - 4x + 4$ for x . One solution.



Approximate

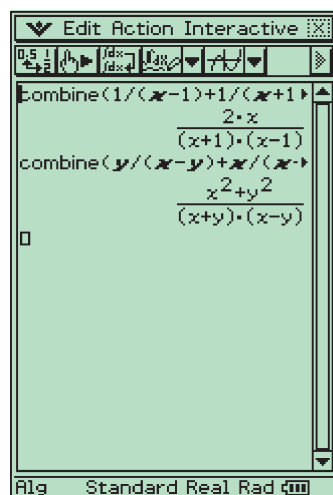
Switch mode in the status bar to Decimal. If an answer is given in Standard (Exact) mode, it can be converted by highlighting the answer and tapping $\frac{0.5}{1}$ in the toolbar.

Combining fractions

This command gives an expression with a common denominator. The denominator is returned in factored form.

For example:

- Enter and highlight $1/(x-1) + 1/(x+1)$. Then select **Interactive—Transformation—combine**.
- Enter and highlight $y/(x-y) + y/(x+y)$. Then select **Interactive—Transformation—combine**.



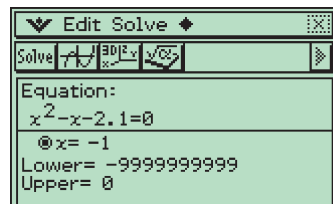
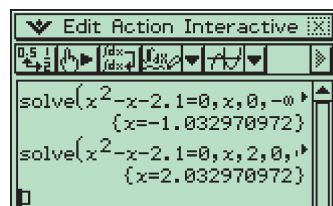
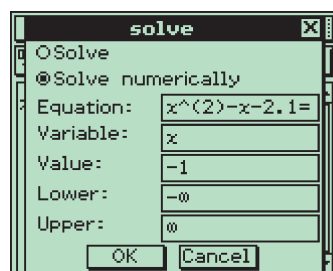
Solve numerically

There are several ways to find numerical solutions to equations. In each of these ways, only one solution is given. You can vary the guess (**Value**) or the bounds of the search (**Lower** and **Upper**) to find particular solutions. If an expression such as $x^2 - x - 2.1$ is entered without an equals sign, the calculator will assume the expression is equal to zero and solve the equation.

For example:

- Enter and highlight $x^2 - x - 2.1 = 0$, select **Interactive—Equation/inequality—solve** and tap the **solve numerically** option. Note that the lower and upper bounds are set to ∞ and a guess of -1 has been entered to return the first solution: 1.03297.
- For the second solution shown, the first line has been copied and pasted (or dragged) to the next entry line and the guess $x = 2$ has been entered to return the solution $x = 2.03297$.
- Alternatively, use  from the Main menu and enter Lower and Upper bounds. A guess may also be entered, but is not necessary.

Note: The bounds selected arise from a quick sketch which indicated that the quadratic equation has one positive and one negative solution.
- A third method involves using the Graph program and finding the solution using the GSolve application.



C.3 Graphing

Graphing using the Casio Classpad is usually done in the  menu. The screen is split with the equation entry screen and the graph screen both shown.

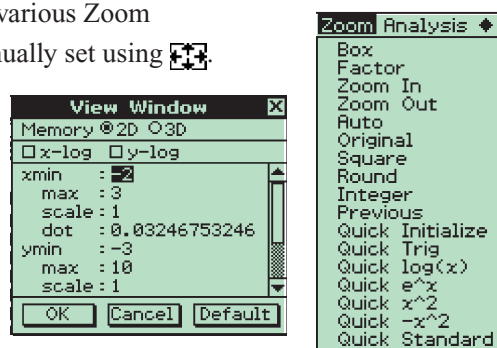
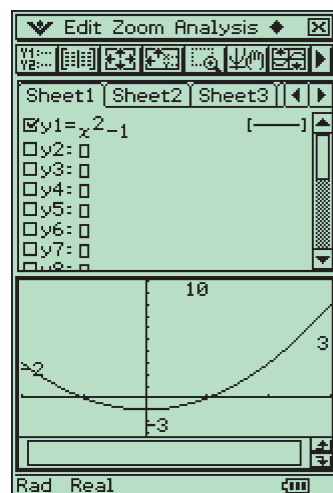
The normal configuration is the $y =$ form; however, by tapping the arrow $y=$  in the menu bar, alternative forms of equation and inequality may be selected.

After entering the equation, tap the select box to the left of the equation and then tap  to produce the graph in the lower screen. More than one equation can be entered and you will note tabs allowing more than one set of equations to be entered and stored for other uses.

Note: In the example, the upper box is selected—shown by a bold boundary. Tapping the lower half of the screen will select the graph section and the menu bar will change to the operations available on the graph.

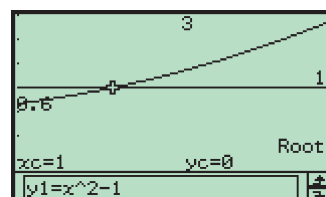
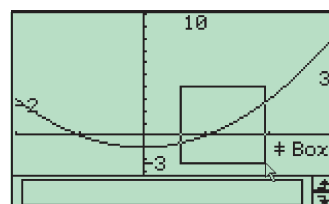
To alter the viewing window for the graph, various Zoom menus are available or the window can be manually set using .

Tap  on the **icon panel** to allow the selected window (graph or equation) to be re-sized to fill the screen. Tapping again returns both windows to the screen.



To find features of the graph, ensure the graph window is selected (bold frame), then tap **Analysis—GSolve** and select the menu item you wish. If seeking the root, the calculator will return the left-hand root shown in the graph window by default. To find the other root, use **Zoom-Box** to zoom into an area containing only the desired root. **Analysis—GSolve** will not return this root.

Use **Zoom—Previous** to return to the previous view. **Analysis—Trace** allows use of the round arrow cursor on the calculator to move along the curve and display grid points in the lower part of the screen.

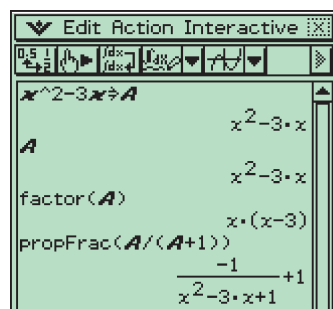


C.4 Defining functions

Expressions can be stored in memory location.

For this the $\Rightarrow$ is used from **Keyboard** **[mth]** and the variable A is entered using **[VAR]**.

For example $x^2 - 3x \Rightarrow A$. The operations discussed previously can now be applied to A .



C.5 Probability and counting

Basic probability functions are found using

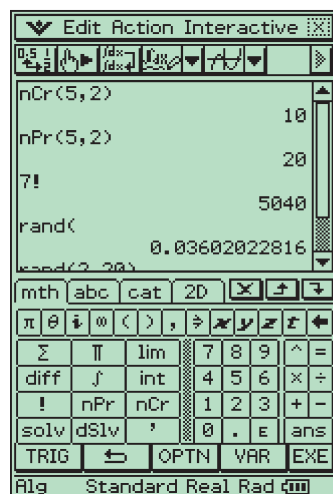
the **Keyboard** **[mth]** **CALC** menu.

To find nC_r and nP_r tap the key and then enter the values of n and r as shown.

To find $n!$, type the value of n and tap the **!** key.

Generate a random number between 0 and 1 by typing `rand()`.

Continue to tap **EXE** to produce more numbers.



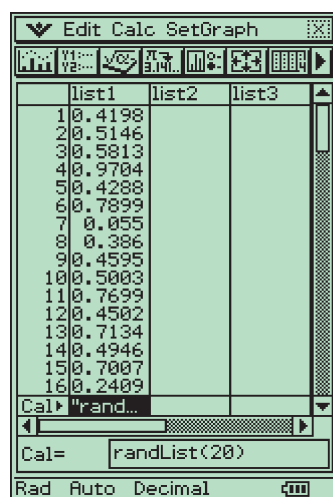
The function `randList(20)` will create a set of 20 random numbers between 0 and 1.

To generate a random integer between 25 and 50, type `rand(2,20)` as shown.

To create a random list of numbers between 0 and 1, the Stat editor can also be used. In the **Cal** line for the required list type `randList(20)` to create a list of 20 random numbers.

Note: The mode should be set to **Decimal** in the **status bar** at the bottom of the screen.

The functions referred to here can be found in the catalogue. Go to **Keyboard** **cat** and scroll to and select the function required.

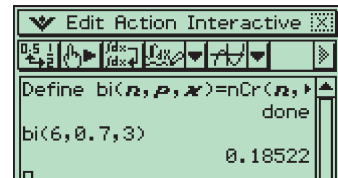
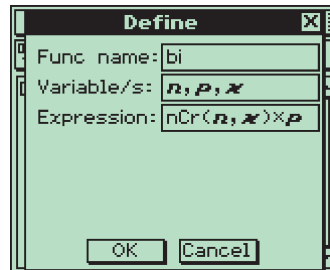


To find the value of a term in a binomial distribution, go to the Stat screen and enter **Calc—Distribution** and scroll to **Binomial PD**. Enter the values for n , p and x and tap **Next >>**. To find more values, use **<< Back** and enter the required values.

Alternatively, define a function

$$bi(n, p, x) = nCr(n, x) \times p^x \times (1 - p)^{(n - x)}$$

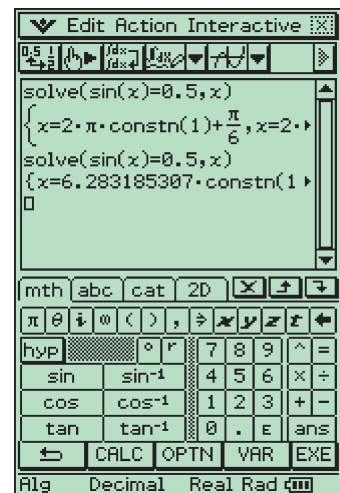
This can be used for any binomial distribution problem.



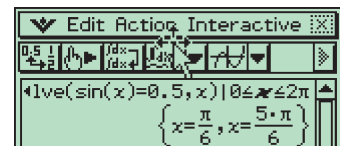
C.6 Trigonometric functions

The choice of radian or degree measure is made by tapping the appropriate section of the Status bar at the bottom of the screen.

Trigonometric equations are solved using the Interactive functions in . Initially, we shall consider the solution of the equation $\sin x = 0.5$. The exact solution is given as $x = 2k\pi + \frac{5\pi}{6}$ or $x = 2k\pi + \frac{\pi}{6}$. The parameter k is shown as $\text{constn}(1)$ and $\text{constn}(2)$ on the screen. This indicates that these are the first and second parameters to be used. The screen shows the solution when the status bar is set to Standard or to Decimal. Tap on  to scroll across for the rest of the solution.



In cases where the domain is limited, enter the equation with the domain $\sin(x) = 0.5 \mid 0 \leq x \leq 2\pi$, highlight *the equation part only* and then tap **Interactive—Equation/inequality—solve** and ensure the variable is set to x . After tapping **EXE** the calculator should read $\text{solve}(\sin(x) = 0.5, x) \mid 0 \leq x \leq 2\pi$.



C.7 Using the Calculus menu

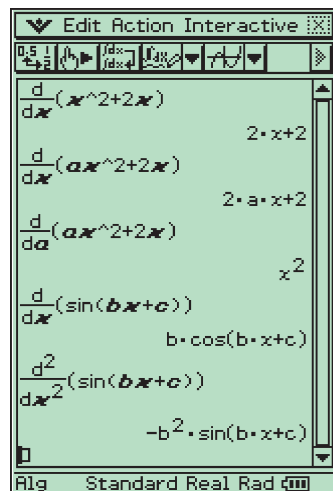
Four of the operations of the calculus menu are considered here. These functions appear on the keyboard **2D** **CALC** screen and variables and expressions can be entered directly. (It is also possible to use the options in the **Interactive—Calculation** drop-down menu.)

Differentiate

This operation is used to differentiate expressions. Several examples are shown in the screen here.

For example:

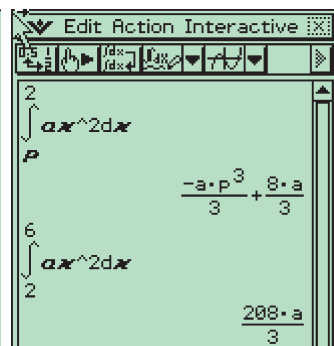
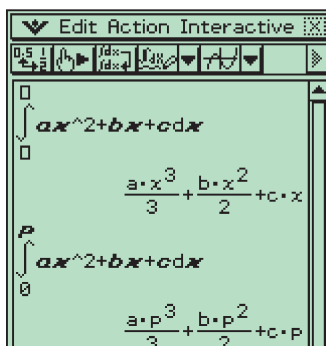
- In **2D** CALC tap $\frac{d}{dx}$, enter x in the darkened entry box and the expression $x^2 + 2x$ in the empty box.
- Change the variable in the darkened box when differentiating with respect to a variable other than x .
- If a second derivative is required, tap $\frac{d^2}{dx^2}$ and set the order to 2 in the small raised entry box beside d .



Integrate

For indefinite integrals, in **2D** CALC tap $\int$ and enter $ax^2 + bx + c$ in the space provided for the expression, then enter x as the variable and tap **EXE**.

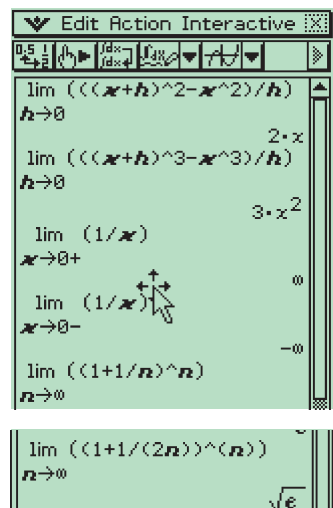
For definite integrals, the same process is followed except that the upper and lower bounds are entered.



Limit

The limits considered in this section will be those that are considered in the senior years of school mathematics. Right and left limits are shown in the sample screen.

To enter limits use **2D** CALC, tap $\lim_{x \rightarrow 0}$ and enter the required information in the entry boxes provided. Remember to use the **VAR** screen to enter variables. $x^3 - 3x^2$.



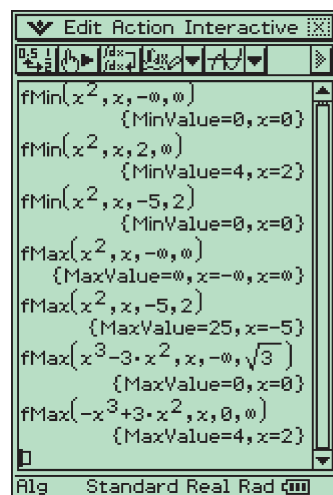
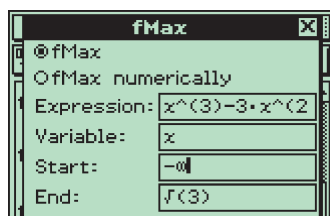
Minimum (fMin) and Maximum (fMax)

These operations return the value for which the maximum or minimum value (or the least upper bound or greatest lower bound of a function occurs). **fMax** returns the value for which a local maximum occurs only if this is the actual maximum for the interval being considered.

Similarly for **fMin**.

- Use **Interactive–Calculate–fMax** or **fMin**

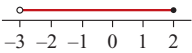
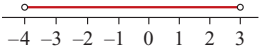
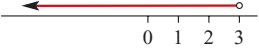
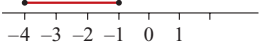
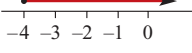
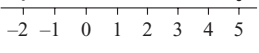
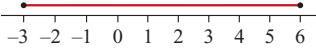
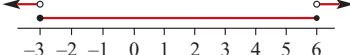
A number of examples are shown in the screen on the right. The entries for the example seeking the maximum value of $x^3 - 3x^2$ for $x < \sqrt{3}$ are shown below.

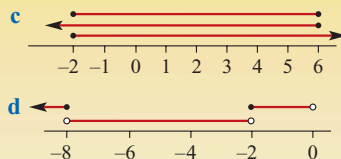


Answers

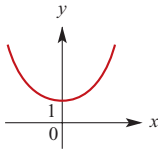
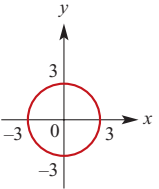
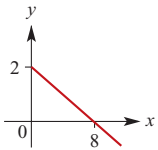
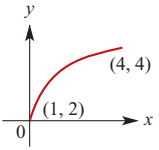
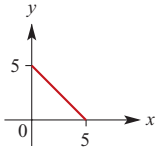
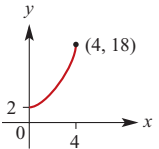
Chapter 1

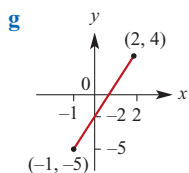
Exercise 1A

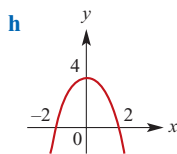
- 1 a $\{7, 9\}$ b $\{7, 9\}$
 c $\{2, 3, 5, 7, 9, 11, 15, 19, 23\}$
 d $\{2, 3, 5, 11\}$ e $\{2\}$ f $\{2, 7, 9\}$
 g $\{2, 3, 5, 7\}$ h $\{7\}$
 i $\{7, 9, 15, 19, 23\}$ j $\{3, \infty\}$
- 2 a $\{a, e\}$ b $\{a, b, c, d, e, i, o, u\}$
 c $\{b, c, d\}$ d $\{i, o, u\}$
- 3 a $\{6\}$ b $\{2, 4, 8, 10\}$ c $\{1, 3, 5, 7, 9\}$
 d $\{1, 2, 3, 4, 5, 7, 8, 9, 10\}$
 e $\{1, 2, 3, 4, 5, 7, 8, 9, 10\}$
 f $\{5, 7\}$ g $\{5, 7\}$ h $\{6\}$
- 4 a $[-3, 1)$ b $(-4, 5]$ c $(-\sqrt{2}, 0)$
 d $(-\frac{1}{\sqrt{2}}, \sqrt{3})$ e $(-\infty, -3)$
 f $(0, \infty)$ g $(-\infty, 0)$ h $[-2, \infty)$
- 5 a $(-2, 3)$ b $[-4, 1)$
 c $[-1, 5]$ d $(-3, 2]$
- 6 a 
 b 
 c 
 d 
 e 
 f 
- 7 a 
 b 



Exercise 1B

- 1 a Domain = R Range = $[-2, \infty)$
 b Domain = $(-\infty, 2]$ Range = R
 c Domain = $(-2, 3)$ Range = $[0, 9)$
 d Domain = $(-3, 1)$ Range = $(-6, 2)$
 e Domain = $[-4, 0]$ Range = $[0, 4]$
 f Domain = R Range = $(-\infty, 2)$
- 2 a 
 Domain = R
 Range = $[1, \infty)$
- b 
 Domain = $[-3, 3]$
 Range = $[-3, 3]$
- c 
 Domain = $R^+ \cup \{0\}$
 Range = $(-\infty, 2]$
- d 
 Domain = $[0, \infty)$
 Range = $[0, \infty)$
- e 
 Domain = $[0, 5]$
 Range = $[0, 5]$
- f 
 Domain = $[0, 4]$
 Range = $[2, 18]$


 Domain = $[-1, 2]$

 Range = $[-5, 4]$

 Domain = R

 Range = $(-\infty, 4]$

3 a Not a function

 Domain = $\{-1, 1, 2, 3\}$

 Range = $\{1, 2, 3, 4\}$

b A function

 Domain = $\{-2, -1, 0, 1, 2\}$

 Range = $\{-4, -1, 0, 3, 5\}$

c Not a function

 Domain = $\{-2, -1, 2, 4\}$

 Range = $\{-2, 1, 2, 4, 6\}$

d A function

 Domain = $\{-1, 0, 1, 2, 3\}$

 Range = $\{4\}$

e A function

 Domain = R

 Range = $\{4\}$

f Not a function

 Domain = $\{2\}$

 Range = Z

g A function

 Domain = R

 Range = R

h Not a function

 Domain = R

 Range = R

i Not a function

 Domain = $[-4, 4]$

 Range = $[-4, 4]$

4 a $g(-2) = 10$, $g(4) = 46$ **b** $[-2, \infty)$

5 a $f(-1) = -2$, $f(2) = 16$, $f(-3) = 6$

b $g(-1) = -10$, $g(2) = 14$, $g(3) = 54$

c i $f(-2x) = 8x^2 - 8x$

ii $f(x - 2) = 2x^2 - 4x$

iii $g(-2x) = -16x^3 - 4x - 6$

iv $g(x + 2) = 2x^3 + 12x^2 + 26x + 14$

v $g(x^2) = 2x^6 + 2x^2 - 6$

6 a 3 **b** 7 **c** $-\frac{3}{2}$

7 a $x = -3$ **b** $x > -3$ **c** $x = \frac{2}{3}$

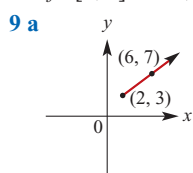
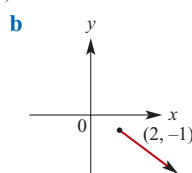
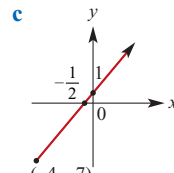
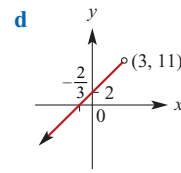
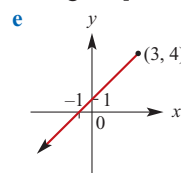
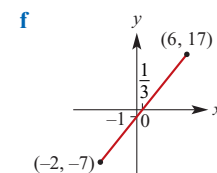
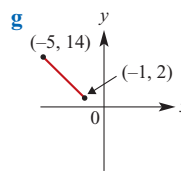
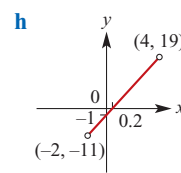
8 a $f: R \rightarrow R$, $f(x) = 2x + 3$

b $f: R \rightarrow R$, $f(x) = -\frac{4}{3}x + 4$

c $f: [0, \infty) \rightarrow R$, $f(x) = 2x - 3$

d $f: R \rightarrow R$, $f(x) = x^2 - 9$

e $f: [0, 2] \rightarrow R$, $f(x) = 5x - 3$


 Range = $[3, \infty)$

 Range = $(-\infty, -1]$

 Range = $[-7, \infty)$

 Range = $(-\infty, 11)$

 Range = $(-\infty, 4]$

 Range = $[-7, 17]$

 Range = $[2, 14]$

 Range = $(-11, 19)$

10 a $f(2) = -3$, $f(-3) = 37$, $f(-2) = 21$

b $g(-2) = 7$, $g(1) = 1$, $g(-3) = 9$

c i $f(a) = 2a^2 - 6a + 1$

ii $f(a + 2) = 2a^2 + 2a - 3$

iii $g(-a) = 3 + 2a$

iv $g(2a) = 3 - 2(2a) = 3 - 4a$

v $f(5 - a) = 21 - 14a + 2a^2$

vi $f(2a) = 8a^2 - 12a + 1$

vii $g(a) + f(a) = 2a^2 - 8a + 4$

viii $g(a) - f(a) = 2 + 4a - 2a^2$

11 a $\left\{\frac{2}{3}, -1\right\}$ **b** $\left\{-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right\}$

c $\left\{0, -\frac{1}{3}\right\}$ **d** $(-\infty, -1) \cup \left(\frac{2}{3}, \infty\right)$

e $(-\infty, -\sqrt{\frac{2}{3}}) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

f $\left[-\frac{1}{3}, 0\right]$

12 a $f(-2) = 2$

b $f(2) = 6$

c $f(-a) = a^2 - a$

d $f(a) + f(-a) = 2a^2$

e $f(a) - f(-a) = 2a$

f $a^4 + a^2$

13 a $\{2\}$

b $\{x : x > 2\}$

c $\left\{\frac{a+2}{3}\right\}$

d $\left\{-\frac{8}{3}\right\}$

e $\{1\}$

f $\left\{\frac{13}{18}\right\}$

14 a $k = \frac{4}{3}$

b $k = 6$

c $k = -\frac{7}{3}$

d $k = 9$

e $k = \frac{1}{3}$

f $k = -\frac{2}{9}$

15 a $\frac{6}{5}$

b $\frac{1}{5}$

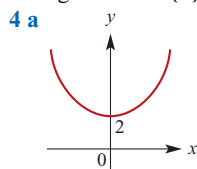
c $\pm \frac{1}{3}$

d 1

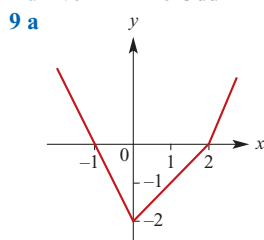
e -1, 2

Exercise 1C

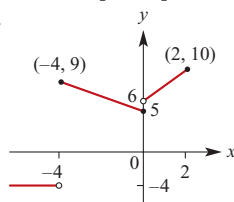
- 1 One-to-one functions are **b, d** and **f**.
 2 **a** Functions are **i, iii, iv, vi, vii** and **viii**.
b One-to-one functions are **iii** and **vii**.
 3 $y = \sqrt{x+2}, x \geq -2$; range = $R^+ \cup \{0\}$
 $y = -\sqrt{x+2}, x \geq -2$;
 range = $R^- \cup \{0\}$



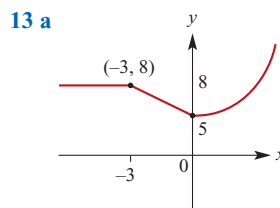
- b** $g_1(x) = x^2 + 2, x \geq 0$
 $g_2(x) = x^2 + 2, x < 0$
 5 **a** Domain = R Range = R
b Domain = $[0, \infty)$ Range = $[0, \infty)$
c Domain = R Range = $[-2, \infty)$
d Domain = $[-4, 4]$ Range = $[0, 4]$
e Domain = $R \setminus \{0\}$ Range = $R \setminus \{0\}$
f Domain = R Range = $(-\infty, 4]$
g Domain = $[3, \infty)$ Range = $[0, \infty)$
 6 **a** Domain = R Range = R
b Domain = R Range = $[-2, \infty)$
c Domain = $[-3, 3]$ Range = $[0, 3]$
d Domain = $R \setminus \{1\}$ Range = $R \setminus \{0\}$
 7 **a** $R \setminus \{3\}$ **b** $(-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$
c R **d** $[4, 11]$ **e** $R \setminus \{-1\}$
f $(-\infty, -1] \cup [2, \infty)$ **g** $R \setminus \{-1, 2\}$
h $(-\infty, -2) \cup [1, \infty)$ **i** $\left[0, \frac{1}{3}\right]$
j $[-5, 5]$ **k** $[3, 12]$
 8 **a** Even **b** Odd **c** Neither
d Even **e** Odd **f** Neither



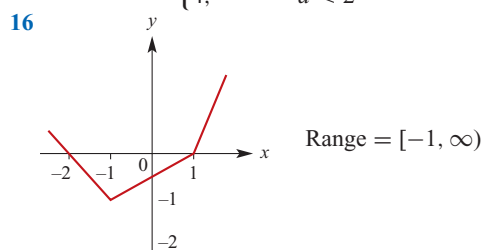
- b** Range = $[-2, \infty)$
 10 Domain = $(-3, 0] \cup [1, 3)$ Range = $[-2, 3]$
 11 Domain = $[-5, 4]$ Range = $[-4, 0) \cup [2, 5]$
 12 **a**



- b** Domain = $(-\infty, 2]$
 Range = $[5, 10] \cup \{-4\}$



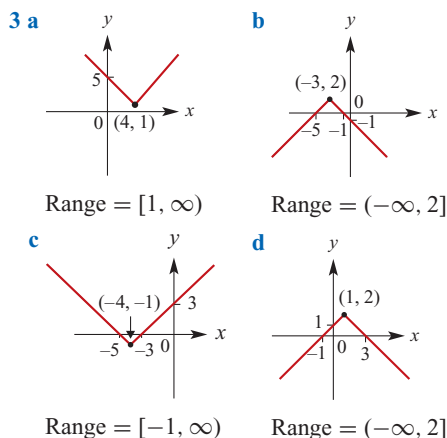
- b** Range = $[5, \infty)$
 14 **a** $f(-4) = -8$ **b** $f(0) = 0$
c $f(4) = \frac{1}{4}$
d $f(a+3) = \begin{cases} \frac{1}{a+3}, & a > 0 \\ 2(a+3), & a \leq 0 \end{cases}$
e $f(2a) = \begin{cases} \frac{1}{2a}, & a > \frac{3}{2} \\ 4a, & a \leq \frac{3}{2} \end{cases}$
f $f(a-3) = \begin{cases} \frac{1}{a-3}, & a > 6 \\ 2(a-3), & a \leq 6 \end{cases}$
 15 **a** $f(0) = 4$ **b** $f(3) = \sqrt{2}$
c $f(8) = \sqrt{7}$
d $f(a+1) = \begin{cases} \sqrt{a}, & a \geq 0 \\ 4, & a < 0 \end{cases}$
e $f(a-1) = \begin{cases} \sqrt{a-2}, & a \geq 2 \\ 4, & a < 2 \end{cases}$



- 17 $y = \begin{cases} -x-4, & x < -2 \\ \frac{1}{2}x-1, & -2 \leq x \leq 3 \\ -\frac{1}{2}x+2, & x > 3 \end{cases}$

Exercise 1D

- 1 **a** 8 **b** 8 **c** 2
d -2 **e** -2 **f** 4
 2 **a** $(-3, 3)$
b $(-\infty, -5] \cup [5, \infty)$
c $[1, 3]$
d $(-1, 5)$
e $(-\infty, -8] \cup [2, \infty)$
f $[-3, -1]$



Exercise 1E

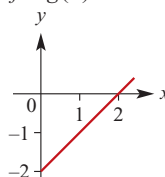
- 1 a** $(f+g)(x) = 4x + 2$
 $(fg)(x) = 3x^2 + 6x$
 $\text{dom} = R$
- b** $(f+g)(x) = 1$
 $(fg)(x) = x^2 - x^4$
 $\text{dom} = (0, 2]$
- c** $(f+g)(x) = \frac{x+1}{\sqrt{x}}$
 $(fg)(x) = 1$
 $\text{dom} = [1, \infty)$
- d** $(f+g)(x) = x^2 + \sqrt{4-x}$
 $(fg)(x) = x^2\sqrt{4-x}$
 $\text{dom} = [0, 4]$
- 2 a** i Even ii Odd
 iii Even iv Odd
- b** $(f+h)(x) = x^2 + 1 + \frac{1}{x^2}$ even;
 $(gk)(x) = 1$ even;
 $(fh)(x) = 1 + \frac{1}{x^2}$ even;
 $(f+g)(x) = x^2 + x + 1$ neither;
 $(g+k)(x) = x + \frac{1}{x}$ odd;
 $(fg)(x) = x^3 + x$ odd

Exercise 1F

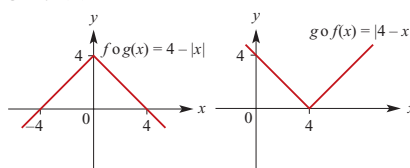
- 1 a** $f(g(x)) = 4x - 1, g(f(x)) = 4x - 2$
b $f(g(x)) = 8x + 5, g(f(x)) = 8x + 3$
c $f(g(x)) = 4x - 7, g(f(x)) = 4x - 5$
d $f(g(x)) = 2x^2 - 1, g(f(x)) = (2x - 1)^2$
e $f(g(x)) = 2(x - 5)^2 + 1,$
 $g(f(x)) = 2x^2 - 4$
f $f(g(x)) = 2|x| + 1, g(f(x)) = |2x + 1|$
- 2 a** $f \circ h(x) = 6x + 3$
b $h(f(x)) = 6x - 1$ **c** $f \circ h(2) = 15$
d 11 **e** 21 **f** -7 **g** 3
3 a $9x^2 + 12x + 3$ **b** $3x^2 + 6x + 1$
c 120 **d** 46 **e** 3 **f** 1

- 4 a** $h(g(x)) = \frac{1}{(3x+2)^2}$
 $\text{Domain} = R^+$
b $g(h(x)) = \frac{3}{x^2} + 2$
 $\text{Domain} = R \setminus \{0\}$
c $\frac{1}{25}$ **d** 5
- 5 a** $\text{ran } f = [-4, \infty), \text{ran } g = [0, \infty)$
b $f \circ g(x) = x - 4, \text{ran} = [-4, \infty)$
c $\text{ran } f \not\subset \text{dom } g$
- 6 a** $f \circ g(x) = x, \text{domain} = R \setminus \left\{\frac{1}{2}\right\},$
 $\text{range} = R \setminus \left\{\frac{1}{2}\right\}$
b $g \circ f(x) = x, \text{domain} = R \setminus \{0\},$
 $\text{range} = R \setminus \{0\}$

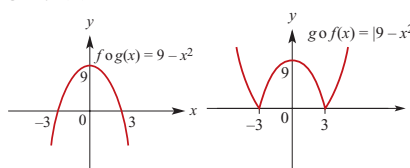
- 7 a** $\text{ran } f = [-2, \infty) \not\subset \text{dom } g = R^+ \cup \{0\}$
b $f \circ g(x) = x - 2, x \geq 0$



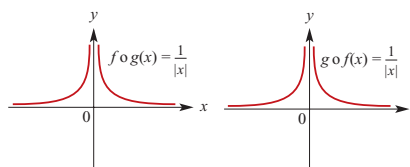
- 8 a** $f \circ g(x) = 4 - |x|$ and
 $g \circ f(x) = |4 - x|$



- b** $f \circ g(x) = 9 - |x|^2 = 9 - x^2$ and
 $g \circ f(x) = |9 - x^2|$



- c** $f \circ g(x) = \frac{1}{|x|}$ and $g \circ f(x) = \left|\frac{1}{x}\right| = \frac{1}{|x|}$



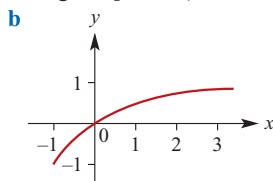
- 9 a** $f \circ g$ is not defined since
 $\text{ran } g = [-1, \infty) \not\subset \text{dom } f = (-\infty, 3]$
b $g^*: [-2, 2] \rightarrow R, g^*(x) = x^2 - 1$
 $f \circ g^*: \{x: -2 \leq x \leq 2\} \rightarrow R,$
 $f \circ g^*(x) = 4 - x^2$
- 10 a** $f \circ g$ is not defined since $\text{ran } g \not\subset \text{dom } f$
b $g_1: \{x: x < 3\} \rightarrow R, \text{ where } g_1(x) = 3 - x$

	Domain	Range
f	R	$[0, \infty)$
g	$(-\infty, 3]$	$[0, \infty)$

- 11**
a $\text{ran } g \subset \text{domain } f$, $\therefore f \circ g$ exists
b $\text{ran } f \not\subset \text{domain } g$, $\therefore g \circ f$ does not exist
12 a $S = [-2, 2]$
b $\text{ran } f = [0, 2]$, $\text{ran } g = [1, \infty)$
c $\text{ran } f = [0, 2] \subset \text{dom } g = R$,
 $\therefore g \circ f$ is defined
 $\text{ran } g = [1, \infty) \not\subset \text{dom } f = [-2, 2]$,
 $\therefore f \circ g$ is not defined
13 a $a \in [2, 3]$

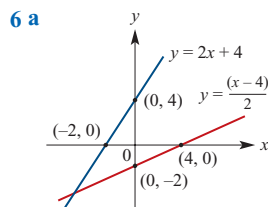
Exercise 1G

- 1 a** $f^{-1}(x) = x + 4$ **b** $f^{-1}(x) = \frac{x}{2}$
c $f^{-1}(x) = \frac{4x}{3}$ **d** $f^{-1}(x) = \frac{4x+2}{3}$
2 a $f^{-1}(x) = \frac{1}{2}(x+4)$, domain = $[-8, 8]$,
range = $[-2, 6]$
b $g^{-1}(x) = 9 - \frac{1}{x}$, domain = $(-\infty, 0)$,
range = $(9, \infty)$
c $h^{-1}(x) = \sqrt{x-2}$, domain = $[2, \infty)$,
range = $R^+ \cup \{0\}$
d $f^{-1}(x) = \frac{1}{5}(x+2)$, domain = $[-17, 28]$,
range = $[-3, 6]$
e $g^{-1}(x) = \sqrt{x+1}$, domain = $(0, \infty)$,
range = $(1, \infty)$
f $h^{-1}(x) = x^2$, domain = $(0, \infty)$,
range = $(0, \infty)$
3 a $g^{-1}(x) = \sqrt{x+1} - 1$, domain = $[-1, \infty)$,
range = $[-1, \infty)$

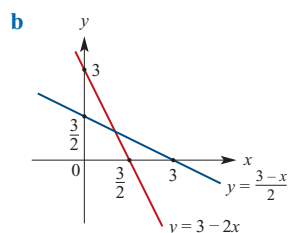


- 4** $f^{-1}(2) = \frac{1}{2}$, $\text{dom } f^{-1} = [-3, 3]$
5 a $f^{-1}(x) = \frac{x}{2}$, $\text{dom } f^{-1} = [-2, 6]$,
 $\text{ran } f^{-1} = [-1, 3]$
b $f^{-1}(x) = \sqrt{\frac{x+4}{2}}$, $\text{dom } f^{-1} = [-4, \infty)$,
 $\text{ran } f^{-1} = [0, \infty)$
c $\{(6, 1), (4, 2), (8, 3), (11, 5)\}$,
 $\text{dom} = \{6, 4, 8, 11\}$, $\text{ran} = \{1, 2, 3, 5\}$
d $h^{-1}(x) = -x^2$, $\text{dom } h^{-1} = R^+$,
 $\text{ran } h^{-1} = R^-$
e $f^{-1}(x) = \sqrt[3]{x-1}$, $\text{dom } f^{-1} = R$,
 $\text{ran } f^{-1} = R$

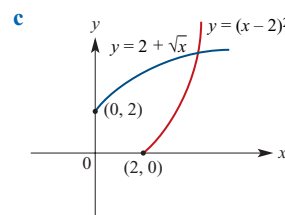
- f** $g^{-1}(x) = -1 + \sqrt{x}$, $\text{dom } g^{-1} = (0, 16)$,
 $\text{ran } g^{-1} = (-1, 3)$
g $g^{-1}(x) = x^2 + 1$, $\text{dom } g^{-1} = R^+ \cup \{0\}$,
 $\text{ran } g^{-1} = [1, \infty)$
h $h^{-1}(x) = \sqrt{4-x^2}$, $\text{dom } h^{-1} = [0, 2]$,
 $\text{ran } h^{-1} = [0, 2]$



$$y = \frac{x-4}{2}, \text{dom} = R, \text{ran} = R$$

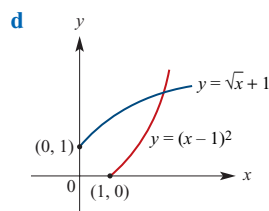


$$f^{-1}(x) = 3 - 2x, \text{dom} = R, \text{ran} = R$$



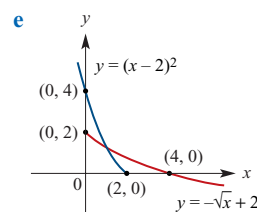
$$f^{-1}(x) = \sqrt{x} + 2, \text{dom} = R^+ \cup \{0\},$$

$$\text{ran} = [2, \infty)$$



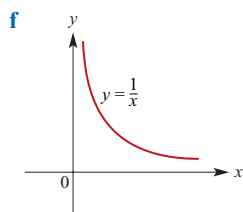
$$f^{-1}(x) = \sqrt{x} + 1, \text{dom} = R^+ \cup \{0\},$$

$$\text{ran} = [1, \infty)$$

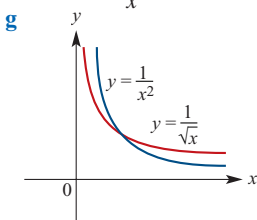


$$f^{-1}(x) = 2 - \sqrt{x}$$

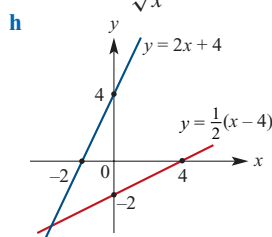
$$\text{dom} = R^+ \cup \{0\}, \text{ran} = (-\infty, 2]$$



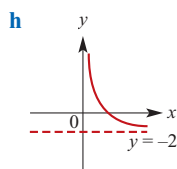
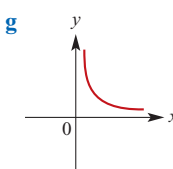
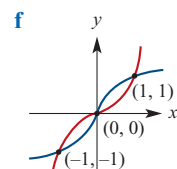
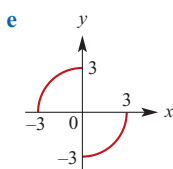
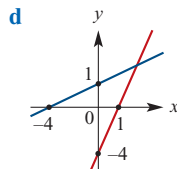
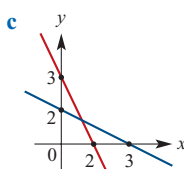
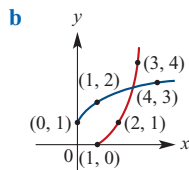
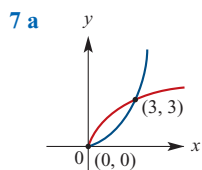
$$f^{-1}(x) = \frac{1}{x}, \text{ dom} = R^+, \text{ ran} = R^+$$



$$f^{-1}(x) = \frac{1}{\sqrt{x}}, \text{ dom} = R^+, \text{ ran} = R^+$$



$$h^{-1}(x) = 2x + 4, \text{ dom} = R, \text{ ran} = R$$



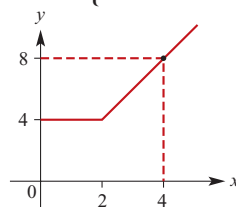
8 a C **b** B **c** D **d** A

9 a $A = (-\infty, 3]$

b $b = 0, g^{-1}(x) = \sqrt{1-x}, x \in [-3, 1]$

Exercise 1H

$$\begin{aligned} 1 \quad f(x) &= \begin{cases} 4 & \text{if } 0 \leq x \leq 2 \\ 4 + 2(x-2) & \text{if } x > 2 \end{cases} \\ &= \begin{cases} 4 & \text{if } 0 \leq x \leq 2 \\ 2x & \text{if } x > 2 \end{cases} \end{aligned}$$

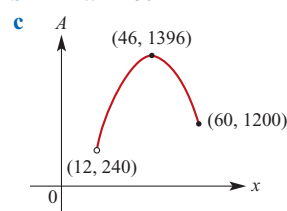


$$2 \quad V(x) = 4x(10-x)(18-x)$$

Domain = (0, 10)

$$3 \quad a \quad A(x) = -x^2 + 92x - 720$$

$$b \quad 12 < x < 60$$



d Maximum area = 1396 m²
occurs when $x = 46$ and $y = 34$

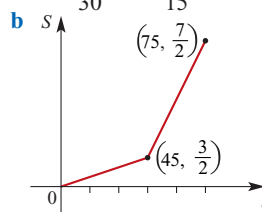
$$4 \quad a \quad i \quad S = 2x^2 + 6xh$$

$$ii \quad S = 2x^2 + \frac{3V}{x}$$

b Maximal domain = (0, ∞)

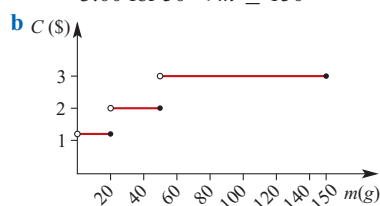
c Maximum value of $S = 1508 \text{ m}^2$

$$5 \quad a \quad \frac{1}{30}, b \quad \frac{1}{15}, c \quad 45, d \quad -\frac{3}{2}, e \quad 75$$



$$c \quad \text{Range} = \left[0, \frac{7}{2}\right]$$

$$6 \quad a \quad C = 1.20 \text{ for } 0 < m \leq 20 \\ = 2.00 \text{ for } 20 < m \leq 50 \\ = 3.00 \text{ for } 50 < m \leq 150$$



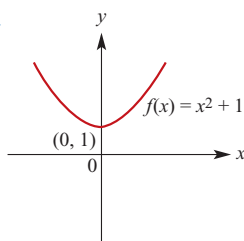
Domain = (0, 150]
Range = {1.20, 2.00, 3.00}

Multiple-choice questions

- 1 E 2 B 3 E 4 C 5 E
6 C 7 D 8 B 9 B 10 B

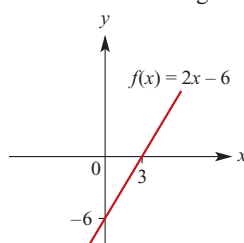
Short-answer questions (technology-free)

1 a



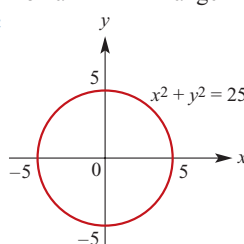
Domain = $\mathbb{R}$ Range = $[1, \infty)$

b



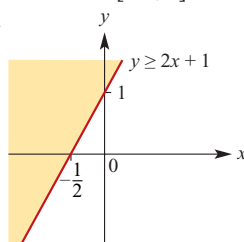
Domain = $\mathbb{R}$ Range = $\mathbb{R}$

c



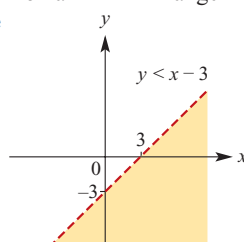
Domain = $[-5, 5]$ Range = $[-5, 5]$

d



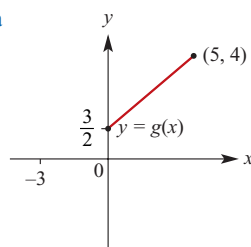
Domain = $\mathbb{R}$ Range = $\mathbb{R}$

e



Domain = $\mathbb{R}$ Range = $\mathbb{R}$

2 a



b Range of $g = \left[\frac{3}{2}, 4\right]$

c $g^{-1}: \left[\frac{3}{2}, 4\right] \rightarrow \mathbb{R}$

$$g^{-1}(x) = 2x - 3$$

Range = $[0, 5]$

Domain = $\left[\frac{3}{2}, 4\right]$

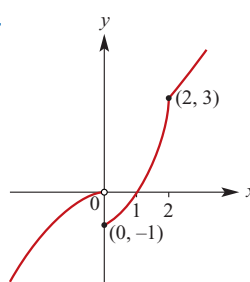
d $\{5\}$ e $\left\{\frac{7}{2}\right\}$

3 a $\left\{\frac{1}{5}\right\}$

b $\{11\}$

c $\left\{-\frac{1}{10}\right\}$

4



5 a $\mathbb{R} \setminus \{3\}$ b $\mathbb{R} \setminus (-\sqrt{5}, \sqrt{5})$ c $\mathbb{R} \setminus \{1, -2\}$

d $[-5, 5]$ e $[5, 15]$ f $\mathbb{R} \setminus \{2\}$

6 $(f + g)(x) = x^2 + 5x + 1$

$$(fg)(x) = (x - 3)(x + 2)^2$$

7 $(f + g): [1, 5] \rightarrow \mathbb{R}$, $(f + g)(x) = x^2 + 1$

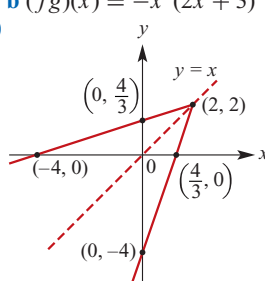
$$(fg): [1, 5] \rightarrow \mathbb{R}, (fg)(x) = 2x(x - 1)^2$$

8 $f^{-1}: [8, \infty) \rightarrow \mathbb{R}$, $f^{-1}(x) = \sqrt{x + 1}$

9 a $(f + g)(x) = -x^2 + 2x + 3$

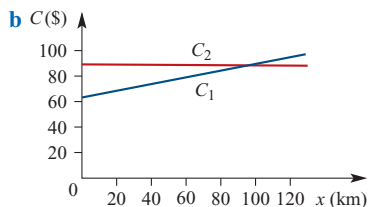
b $(fg)(x) = -x^2(2x + 3)$ c $\{-1, 3\}$

10



Extended-response questions

1 a $C_1 = 64 + 0.25x$
 $C_2 = 89$



c $x > 100$ km

2 a $S = 6x^2$

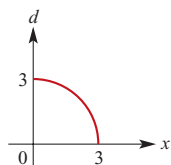
b $S = 6V^{\frac{2}{3}}$

3 a $A = \frac{\sqrt{3}s^2}{4}$

b $A = \frac{\sqrt{3}h^2}{3}$

4 a $d(x) = \sqrt{9 - x^2}$,

b $\text{dom} = [0, 3]$
 $\text{ran} = [0, 3]$



5 $S(x) = \frac{160x}{x + 80}$

6 a $V_1: (0, 12) \rightarrow R$,

$V_1(h) = \pi \left(36 - \frac{h^2}{4} \right) h$

b $V_2: (0, 6) \rightarrow R$, $V_2(r) = 2\pi r^2 \sqrt{36 - r^2}$

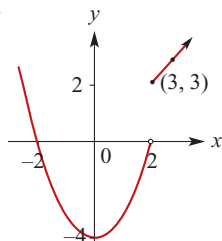
	domain	range
f	R	R
g	R	R

$\text{ran } f = \text{dom } g$, $\therefore g \circ f$ exists

$g \circ f(x) = 2 + (1 + x)^3$

b $g \circ f$ is one-to-one and therefore a function.
 $\therefore (g \circ f)^{-1}$ is defined, $(g \circ f)^{-1}(10) = 1$

8 a



b i -3

ii 3

c $S = (-\infty, 0]$

d $f \circ h(x) = \begin{cases} 4x^2 - 4 & \text{if } x < 1 \\ 2x & \text{if } x \geq 1 \end{cases}$

$h \circ f(x) = \begin{cases} 2x^2 - 8 & \text{if } x < 2 \\ 2x & \text{if } x \geq 2 \end{cases}$

9 $A(t) = \begin{cases} \frac{3t^2}{2}, & 0 \leq t \leq 1 \\ 3t - \frac{3}{2}, & t > 1 \end{cases}$

Domain = $[0, \infty)$

Range = $[0, \infty)$

10 a $f: R \setminus \left\{ \frac{a}{c} \right\} \rightarrow R$, $f^{-1}(x) = \frac{b - xd}{xc - a}$

b i $f^{-1}: R \setminus \{1\} \rightarrow R$, $f^{-1}(x) = \frac{2 - x}{3x - 3}$

ii $f^{-1}: R \setminus \left\{ \frac{3}{2} \right\} \rightarrow R$,
 $f^{-1}(x) = \frac{3x + 2}{2x - 3}$

iii $f^{-1}: R \setminus \{-1\} \rightarrow R$, $f^{-1}(x) = \frac{1 - x}{x + 1}$

iv $f^{-1}: R \setminus \{-1\} \rightarrow R$, $f^{-1}(x) = \frac{1 - x}{x + 1}$

c If $a, b, c, d \in R \setminus \{0\}$, $f = f^{-1}$ when
 $a = -d$.

11 a i $YB = r$

ii $ZB = r$

iii $AZ = x - r$

iv $CY = 3 - r$

b $r = \frac{x + 3 - \sqrt{x^2 + 9}}{2}$

c i $r = 1$

ii 1.25

12 b $f(x) = \frac{q}{x}$

c i $f^{-1}(x) = \frac{3x + 8}{x - 3} = f(x)$

ii $x = 3 \pm \sqrt{17}$

13 a i $f(2) = 3$, $f(f(2)) = 2$

$f(f(f(2))) = 3$

ii $f(f(x)) = x$

b $f(f(x)) = \frac{-x - 3}{x - 1}$, $f(f(f(x))) = x$,
i.e. $f(f(x)) = f^{-1}(x)$

Chapter 2

Exercise 2A

1 a 10 **b** 1 **c** 4 **d** 28 **e** $8\frac{1}{2}$

f $\frac{17}{9}$ **g** $\frac{7}{5}$ **h** 21 **i** 2 **j** $\frac{7}{2}$

2 a $x = 12$, $y = 8$ **b** $x = 5$, $y = -8$

c $x = 3$, $y = 1$ **d** $x = 2$, $y = 1$

e $x = 17$, $y = -19$ **f** $x = 10$, $y = 6$

3 80 km **4** 96 km

5 Width = 6 cm, length = 10 cm

6 John scored 4, David 8

7 a $w = 20n + 800$ **b** \$1400 **c** 41 units

8 a $V = 15t + 250$ **b** 1150 litres

c 5 hours, 16 minutes and 40 seconds

9 a $V = 10\,000 - 10t$ **b** 9400 litres

c 16 hours and 40 minutes

10 a $C = 25t + 100$

b i \$150 **ii** \$162.50

c i 11 hours **ii** 12 hours

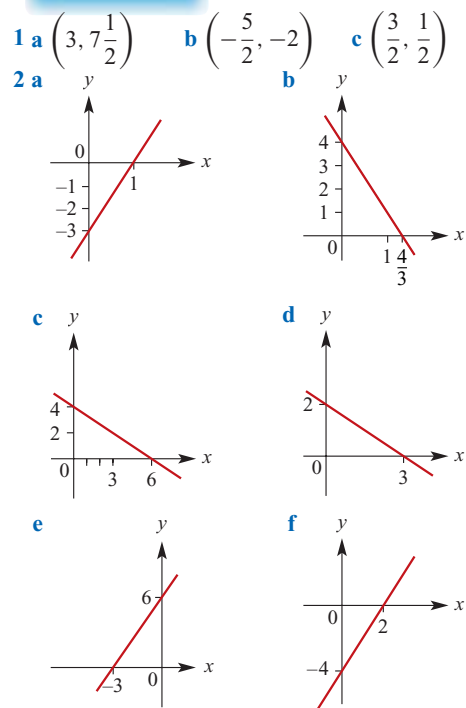
Exercise 2B

1 a $x = \frac{m-n}{a}$ **b** $x = \frac{b}{b-a}$ **c** $x = -\frac{bc}{a}$
d $x = \frac{5}{p-q}$ **e** $x = \frac{m+n}{n-m}$ **f** $x = \frac{ab}{1-b}$
g $x = 3a$ **h** $x = -mn$ **i** $x = \frac{1-b}{2ab}$
j $x = \frac{p-q}{p+q}$ **k** $x = \frac{3ab}{b-a}$ **l** $x = \frac{1}{3a-b}$
m $x = \frac{p^2 + p^2t + t^2}{q(p+t)}$ **n** $x = -\frac{5a}{3}$

4 a $x = \frac{d-bc}{1-ab}, y = \frac{c-ad}{1-ab}$
b $x = \frac{a^2+ab+b^2}{a+b}, y = \frac{ab}{a+b}$
c $x = \frac{t+s}{2a}, y = \frac{t-s}{2b}$
d $x = a+b, y = a-b$
e $x = c, y = -a$
f $x = a+1, y = a-1$

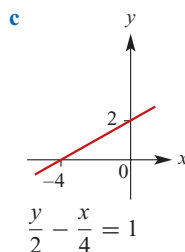
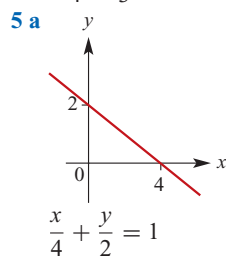
5 a $s = a(2a+1)$ **b** $s = \frac{2a^2}{1-a}$
c $s = \frac{a^2+a+1}{a(a+1)}$ **d** $s = \frac{a}{(a-1)^2}$
e $s = 3a^3(3a+1)$ **f** $s = \frac{3a}{a+2}$
g $s = 2a^2 - 1 + \frac{1}{a^2}$ **h** $s = \frac{5a^2}{a^2+6}$

Exercise 2C



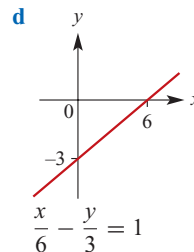
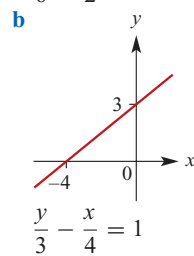
3 a $2x - 6 = y$
c $5 = 3y - 4x$

4 a $\frac{y}{2} - \frac{x}{3} = 1$
c $-\frac{x}{4} - \frac{y}{3} = 1$



b $-3x - 5 = y$
d $y = 2x + 1$

b $\frac{x}{4} + \frac{y}{6} = 1$
d $\frac{x}{6} - \frac{y}{2} = 1$



6 $C = \frac{11}{200}n + 2, \57

7 a $C = 5n + 175$

b Yes **c** \$175

8 a $\sqrt{5} \approx 2.236$

b $\sqrt{2} \approx 1.414$

c $\sqrt{29} \approx 5.385$

d $2\sqrt{82} \approx 18.111$

e $\sqrt{20} \approx 4.472$

f 5

9 a i $y = 2x + 4$

ii $2y + x = 13$

b i $y = -2x + 7$

ii $2y - x = 4$

10 $y = 2x - 3$

11 $y = -5$ or 3

12 a i $5y + 2x = 13$

ii $5y + 4x = 11$

b i $2y - 5x = 11$

ii $4y - 5x = 17$

13 a 32.01°

b 153.43°

c 56.31°

d 120.96°

14 45°

15 $a = -12$ or $a = 8$

16 a $y = 3x - 6$

b $(2, 0)$

17 $k = 5$ and $h = 4$ or $k = -2$ and $h = -3$

18 a $a + 2$

b $\frac{4}{5}$

19 a $m = \frac{1}{2}$

b $(5, 7)$

c $AB = \sqrt{13}, AC = 2\sqrt{13}$

20 a $3y - x = 22$

b $(14, 12)$

c $(16, 6)$

d 80 square units

21 a $(2, 3)$

b $y + 5x = 13$

c i $2y = 3x - 13$

ii $(3, -2)$

iii $(1, 8)$

Exercise 2D

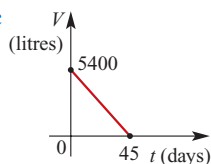
1 a $4 - T$

b i $90T$

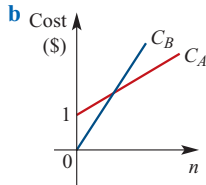
c i $T = 1$

ii $70(4 - T)$

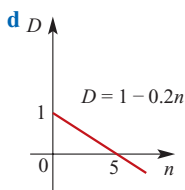
- ii 90 km on freeway, 210 km on country roads
 2 a $L = -120t + 5400$ b 5400 litres
 c d $[0, 45]$



- e 45 days f 120 litres per day
 3 a $C_A = 0.4n + 1$ b Cost (\$)
 $C_B = 0.6n$



c $n = 5$



The difference in charges against kilometres travelled.

- 4 a $y = \frac{9}{4}x$ b 24 622 m
 c $y = -\frac{27}{26}x + \frac{855}{26}$ d $\frac{5393}{108}$
 5 a i -4 ii $\frac{4}{9}$
 b i $y = \frac{4}{9}x + \frac{10}{3}$ ii $y = -4x + 30$
 c Equation of AC: $y = x$, equation of BD: $y = 4$
 d (4, 4)

Exercise 2E

- 1 a $\begin{bmatrix} 0 & -1 \\ 6 & 4 \end{bmatrix}$ b $\begin{bmatrix} -1 & -2 \\ 0 & -2 \end{bmatrix}$ c $\begin{bmatrix} -10 & -6 \\ 12 & 7 \end{bmatrix}$
 d $\begin{bmatrix} 4 & 3 \\ 0 & 0 \end{bmatrix}$ e $\begin{bmatrix} 2k & k \\ 3k & 2k \end{bmatrix}$ f $\begin{bmatrix} -2 & -4 \\ 15 & 10 \end{bmatrix}$
 g $\begin{bmatrix} 6 & 5 \\ -3 & -2 \end{bmatrix}$ h 1 i $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$
 j 2 k $\begin{bmatrix} 1 & 1 \\ -3 & -1 \end{bmatrix}$ l 2
 2 a $\begin{bmatrix} -1 & 1 \\ -4 & 3 \end{bmatrix}$ b $\begin{bmatrix} 2 & -1 \\ 7 & 14 \end{bmatrix}$ c $\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

- 3 $A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & -1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$
 $AB = \begin{bmatrix} 5 & 1 \\ -3 & -1 \end{bmatrix}$, $(AB)^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -3 & -5 \end{bmatrix}$
 $A^{-1}B^{-1} = \begin{bmatrix} -1 & \frac{1}{2} \\ 3 & -1 \end{bmatrix}$, $B^{-1}A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -3 & -5 \end{bmatrix}$
 $(AB)^{-1} = B^{-1}A^{-1}$
 4 a $\begin{bmatrix} -\frac{1}{2} & \frac{3}{2} \\ 1 & -2 \end{bmatrix}$ b $\begin{bmatrix} 0 & 7 \\ 1 & -8 \end{bmatrix}$ c $\begin{bmatrix} \frac{5}{2} & -7 \\ \frac{11}{2} & -21 \end{bmatrix}$
 5 a $\begin{bmatrix} -3 & \frac{11}{8} \\ \frac{1}{16} & \frac{7}{16} \end{bmatrix}$ b $\begin{bmatrix} -\frac{11}{16} & \frac{17}{16} \\ -1 & \frac{3}{4} \end{bmatrix}$
 7 a $\begin{bmatrix} 5 & 7 & 7 \\ -6 & 11 & 12 \\ 14 & 2 & 0 \end{bmatrix}$ b $\begin{bmatrix} 16 & 14 & 42 \\ 0 & 10 & 48 \\ 24 & 13 & 30 \end{bmatrix}$
 c $\begin{bmatrix} 24 & 24 & 16 \\ 66 & 16 & 6 \\ 0 & 22 & 16 \end{bmatrix}$ d $\begin{bmatrix} 9 & 18 & 15 \\ -18 & 30 & 18 \\ 36 & 3 & 0 \end{bmatrix}$
 e $\begin{bmatrix} \frac{3}{2} & -\frac{1}{2} & -1 \\ -3 & 1 & 3 \\ \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix}$ f $\begin{bmatrix} 1 & 5 & 3 \\ -6 & 9 & 0 \\ 10 & 0 & 0 \end{bmatrix}$
 g $\begin{bmatrix} -1 & 4 & 1 \\ -6 & 8 & -6 \\ 8 & -1 & 0 \end{bmatrix}$ h $\begin{bmatrix} 9 & 9 & 11 \\ -6 & 13 & 24 \\ 18 & 4 & 0 \end{bmatrix}$
 i $\begin{bmatrix} -3 & \frac{7}{8} & \frac{7}{4} \\ \frac{8}{4} & \frac{48}{-11} & \frac{24}{-8} \\ \frac{3}{-5} & \frac{18}{4} & \frac{9}{5} \end{bmatrix}$ j $\begin{bmatrix} -3 & \frac{7}{8} & \frac{7}{4} \\ \frac{8}{4} & \frac{48}{-11} & \frac{24}{-8} \\ \frac{3}{-5} & \frac{18}{4} & \frac{9}{5} \end{bmatrix}$
 8 a $\begin{bmatrix} 17 & -3 & -23 & 8 \\ 24 & 95 & 54 & 22 \\ 9 & 52 & 58 & 1 \\ 1 & 13 & 10 & 1 \end{bmatrix}$
 b $\begin{bmatrix} 10 & 10 & 14 & 8 \\ 16 & 26 & 24 & 1 \\ 63 & 50 & 80 & 13 \\ 70 & 74 & 92 & 55 \end{bmatrix}$
 c $\begin{bmatrix} 9 & 18 & 15 & -18 \\ 30 & 18 & 36 & 3 \\ 0 & 12 & 3 & 18 \\ 0 & 0 & 3 & 3 \end{bmatrix}$

$$\mathbf{d} \begin{bmatrix} \frac{-1}{12} & \frac{11}{24} & \frac{1}{12} & \frac{-1}{8} \\ \frac{-2}{9} & \frac{-1}{9} & \frac{2}{9} & 0 \\ \frac{7}{7} & \frac{7}{7} & \frac{-7}{7} & \frac{1}{1} \\ \frac{36}{37} & \frac{72}{1} & \frac{36}{-1} & \frac{8}{-1} \\ \frac{36}{36} & \frac{72}{72} & \frac{36}{36} & \frac{8}{8} \end{bmatrix}$$

$$\mathbf{e} \begin{bmatrix} 3 & 5 & 4 & -7 \\ 8 & 5 & 10 & 1 \\ -1 & -2 & -1 & 5 \\ 0 & -7 & -7 & 1 \end{bmatrix}$$

$$\mathbf{f} \begin{bmatrix} 3 & 4 & 3 & -8 \\ 6 & 4 & 8 & 1 \\ -2 & -8 & -3 & 4 \\ 0 & -14 & -15 & 1 \end{bmatrix}$$

$$9 \mathbf{a} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{b} \begin{bmatrix} \frac{11}{2} \\ -12 \\ 2 \end{bmatrix}$$

$$10 \mathbf{a} \begin{bmatrix} \frac{-7}{24} \\ \frac{2}{9} \\ \frac{13}{72} \\ \frac{43}{72} \end{bmatrix}$$

$$\mathbf{b} \begin{bmatrix} \frac{-1}{2} \\ -2 \\ \frac{3}{5} \\ \frac{6}{11} \\ \frac{6}{6} \end{bmatrix}$$

Exercise 2F

$$1 \mathbf{a} x = 4, y = -3 \quad \mathbf{b} x = \frac{-3}{2}, y = \frac{1}{2}$$

$$\mathbf{c} x = \frac{51}{38}, y = \frac{-31}{38} \quad \mathbf{d} x = \frac{37}{10}, y = \frac{7}{5}$$

2 Their graphs are parallel straight lines that do not coincide.

$$3 x = t + 6, y = t \quad \text{where } t \in \mathbb{R}$$

$$4 m = 9$$

$$5 \mathbf{a} m = -5 \quad \mathbf{b} m = 3$$

$$6 \mathbf{a} \mathbf{i} m = -2 \quad \mathbf{ii} m = 4$$

$$\mathbf{b} x = \frac{4}{m+2}, y = \frac{2(m+4)}{m+2}$$

$$7 \mathbf{a} x = 2, y = 0 \quad k \neq \frac{-3}{2}$$

$$\mathbf{b} k = \frac{-3}{2}$$

$$8 \mathbf{a} b \in \mathbb{R} \setminus \{10\}$$

$$\mathbf{b} b = 10, c = 8$$

$$\mathbf{c} b = 10, c \neq 8$$

Exercise 2G

$$1 \mathbf{a} x = 2, y = 3, z = 1 \quad \mathbf{b} x = -3, y = 5, z = 2$$

$$\mathbf{c} x = 5, y = 0, z = 7 \quad \mathbf{d} x = 6, y = 5, z = 1$$

$$2 \mathbf{a} y = 4z - 2$$

$$\mathbf{b} x = 8 - 5\lambda, y = 4\lambda - 2, z = \lambda$$

$$3 \mathbf{a} -y + 5z = 15, -y + 5z = 15$$

$\mathbf{b}$ The two equations are the same.

$$\mathbf{c} y = 5\lambda - 15$$

$$\mathbf{d} x = 43 - 13\lambda$$

$$4 \mathbf{a} x = \lambda - 1, y = \lambda, z = 5$$

$$\mathbf{b} x = \lambda + 3, y = 3\lambda, z = \lambda$$

$$\mathbf{c} x = \frac{14 - 3\lambda}{6}, y = \frac{10 - 3\lambda}{6}, z = \lambda$$

$$5 z = t, y = \frac{-3(t+2)}{4}, x = \frac{26 - 3t}{4},$$

$$w = \frac{t-2}{2}; t \in \mathbb{R}$$

$$x = -4, y = -12, z = 14$$

$$6 \mathbf{a} x = 1, y = 2, z = 3$$

$$\mathbf{b} x = \frac{-5}{3}, y = \frac{-(\lambda+5)}{3}, z = \lambda$$

$$\mathbf{c} z = t, y = -2(t-1), x = \frac{2-3t}{2}$$

$$7 \mathbf{a} \begin{bmatrix} \frac{8-12a}{13a(a-2)} & \frac{1}{13} - \frac{2a-8}{13a(a-2)} & \frac{3a-4}{13a(a-2)} + \frac{1}{13} \\ \frac{13a-2}{13a(a-2)} & \frac{-2(a+1)}{13a(a-2)} & \frac{1-2a}{13a(a-2)} \\ \frac{16}{13a(a-2)} & \frac{16-10a}{13a(a-2)} & \frac{3a-8}{13a(a-2)} \end{bmatrix}$$

$$\mathbf{b} a = 0, 2 \quad \mathbf{c} a = 0$$

$$\mathbf{d} x = \frac{t}{2}, y = \frac{26-t}{8}, z = t; t \in \mathbb{R}$$

$$8 p = 3; z = t, y = 5 - 4t, x = 3(t-1)$$

Multiple-choice questions

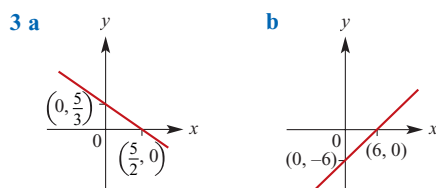
$$1 \mathbf{E} \quad 2 \mathbf{E} \quad 3 \mathbf{D} \quad 4 \mathbf{C} \quad 5 \mathbf{B}$$

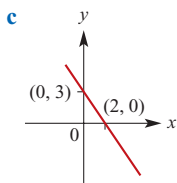
$$6 \mathbf{A} \quad 7 \mathbf{A} \quad 8 \mathbf{C} \quad 9 \mathbf{B} \quad 10 \mathbf{C}$$

Short-answer questions (technology-free)

$$1 \mathbf{a} -8 \quad \mathbf{b} \frac{7}{5} \quad \mathbf{c} 30 \quad \mathbf{d} 7$$

$$2 \mathbf{a} x = -2, y = 2 \quad \mathbf{b} x = -44, y = -39$$





4 a $y = -2x + 5$

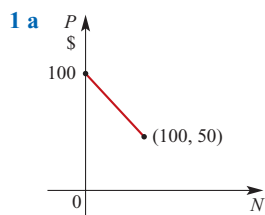
b $y = 2x + 2$

c $y = \frac{1}{2}x + \frac{1}{2}$

d $y = -2x + 3$

5 $\sqrt{13}$

Extended-response questions

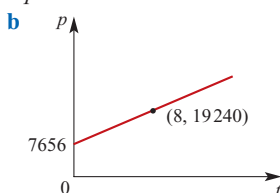


b $P = -\frac{1}{2}N + 100$

c i \$56

ii $N = 80$

2 a $p = 1448t + 7656$



c 22 136

d 1448 people per year

3 a $y = \frac{5}{3}x - 4$

b $\left(\frac{66}{7}, \frac{82}{7}\right)$

c $\frac{5}{3}$

d 15

e $\frac{629}{14}$ square units

4 a $y = \frac{4}{7}x + \frac{31}{14}$

b $\frac{59}{14}$

c $\sqrt{65}$

d $\frac{65}{28}$ square units

5 a $\left(1, -\frac{1}{2}\right)$

b $\sqrt{269}$

c $\sqrt{269}$

d $y = -\frac{13}{10}x + \frac{4}{5}$

e $y = \frac{10}{13}x - \frac{33}{26}$

f $\left(\frac{7}{2}, -\frac{15}{4}\right)$

g (26, -33)

6 a 125 litres

b $x = 291\frac{2}{3}, y = 208\frac{1}{3}$

7 Brad 20, Flynn 10, Elizabeth 15

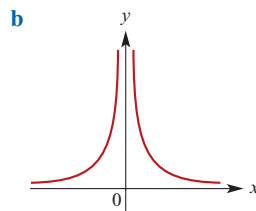
Chapter 3

Exercise 3A

1 a $f(x) = \frac{1}{x^4}$

Maximal domain = $R \setminus \{0\}$, range = R^+

Asymptotes at $x = 0, y = 0$

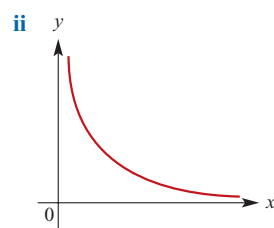


2 a i $f(x) = \frac{1}{\sqrt{x}}$

Maximal domain = R^+ ,

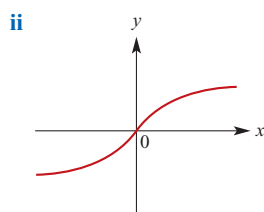
range = R^+

Asymptotes at $x = 0, y = 0$



b i $f(x) = x^{\frac{3}{5}}$

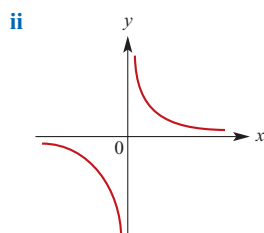
Maximal domain = R , range = R



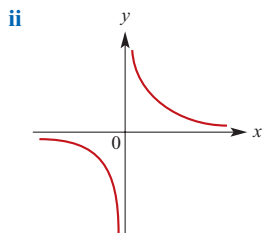
c i $f(x) = x^{-\frac{3}{5}}$

Maximal domain = $R \setminus \{0\}$, range = $R \setminus \{0\}$

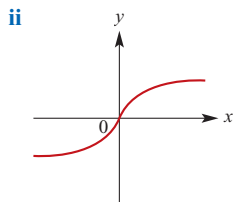
Asymptotes at $x = 0, y = 0$



- d i** $f(x) = \frac{1}{\sqrt[3]{x}}$
Maximal domain = $R \setminus \{0\}$, range = $R \setminus \{0\}$
Asymptotes at $x = 0$, $y = 0$



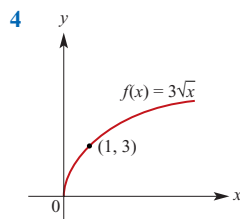
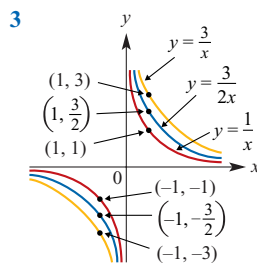
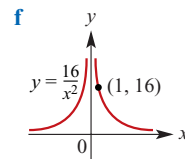
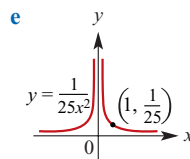
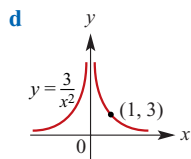
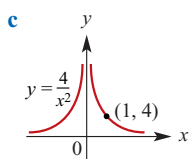
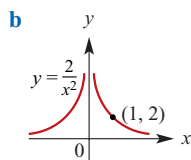
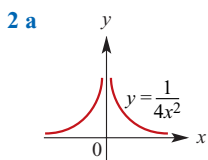
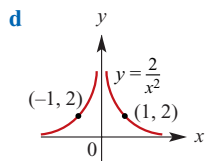
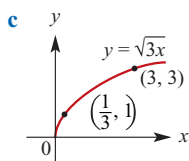
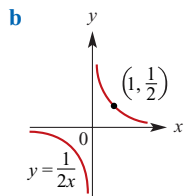
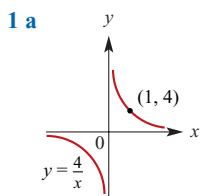
- e i** $f(x) = \sqrt[5]{x}$
Maximal domain = R , range = R



- 3 a** $\{x: 0 < x < 1\}$ **b** $\{x: 0 < x < 1\}$

- 4 a** Odd **b** Even **c** Odd
d Odd **e** Even **f** Odd

Exercise 3B



Range = R^+

- 5 a** Dilation of factor 5 from the x -axis

- b** Dilation of factor 4 from the x -axis

- c** Dilation of factor $\frac{1}{5}$ from the y -axis

- 6 a i** $y = 4|x|$ **ii** $y = \frac{2}{3}|x|$

- iii** $y = 2|x|$ **iv** $y = \frac{1}{5}|x|$

- b i** $y = 4\sqrt[3]{x}$ **ii** $y = \frac{2}{3} \times \sqrt[3]{x}$

- iii** $y = \sqrt[3]{2x}$ **iv** $y = \sqrt[3]{\frac{x}{5}}$

- c i** $y = \frac{4}{x^3}$ **ii** $y = \frac{2}{3x^3}$

- iii** $y = \frac{1}{8x^3}$ **iv** $y = \frac{125}{x^3}$

- d i** $y = \frac{4}{x^4}$ **ii** $y = \frac{2}{3x^4}$

- iii** $y = \frac{1}{16x^4}$ **iv** $y = \frac{625}{x^4}$

- e i** $y = \frac{4}{\sqrt[3]{x}}$ **ii** $y = \frac{2}{3\sqrt[3]{x}}$

- iii** $y = \frac{1}{\sqrt[3]{2x}}$ **iv** $y = \frac{1}{\sqrt[3]{5x}}$

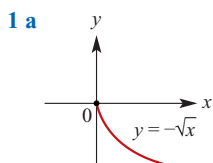
- f i** $y = 4(x)^{\frac{2}{3}}$ **ii** $y = \frac{2}{3}(x)^{\frac{2}{3}}$

- iii** $y = (2x)^{\frac{2}{3}}$ **iv** $y = \left(\frac{x}{5}\right)^{\frac{2}{3}}$

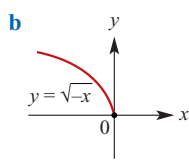
- g i** $y = \frac{4}{x^{\frac{3}{4}}}$ **ii** $y = \frac{2}{3x^{\frac{3}{4}}}$

- iii** $y = \frac{1}{(2x)^{\frac{3}{4}}}$ **iv** $y = \left(\frac{5}{x}\right)^{\frac{3}{4}}$

Exercise 3C



dom = $[0, \infty)$



dom = $(-\infty, 0]$

2 Reflection in the y -axis

3 a i $y = -|x|$

ii $y = |x|$

b i $y = -\sqrt[3]{x}$

ii $y = \sqrt[3]{x}$

c i $y = \frac{-1}{x^3}$

ii $y = \frac{-1}{x^3}$

d i $y = \frac{-1}{x^4}$

ii $y = \frac{1}{x^4}$

e i $y = \frac{-1}{\sqrt[3]{x}}$

ii $y = \frac{-1}{\sqrt[3]{x}}$

f i $y = -(x)^{\frac{2}{3}}$

ii $y = x^{\frac{2}{3}}$

g i $y = \frac{-1}{\frac{3}{x^4}}$

ii $y = \frac{1}{(-x)^4}$

4 a Even

b Odd

c Odd

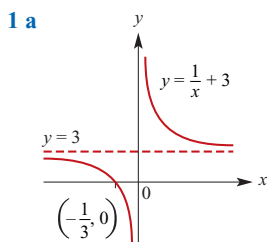
d Even

e Odd

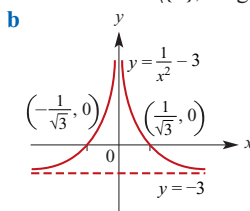
f Even

g Neither odd nor even

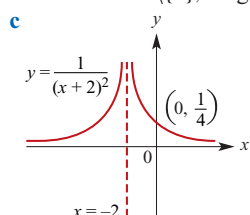
Exercise 3D



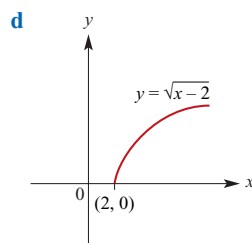
Domain = $R \setminus \{0\}$, range = $R \setminus \{3\}$



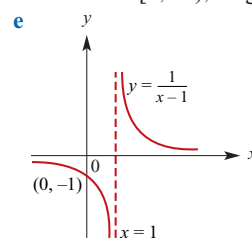
Domain = $R \setminus \{0\}$, range = $(-3, \infty)$



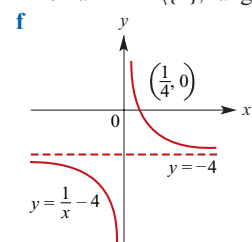
Domain = $R \setminus \{-2\}$, range = R^+



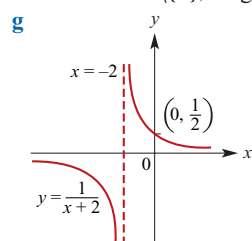
Domain = $[2, \infty)$, range = $R^+ \cup \{0\}$



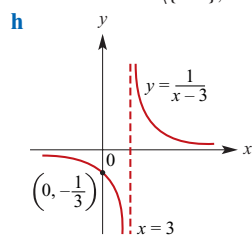
Domain = $R \setminus \{1\}$, range = $R \setminus \{0\}$



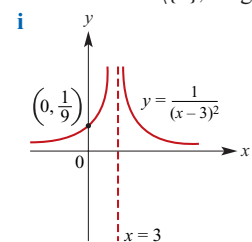
Domain = $R \setminus \{0\}$, range = $R \setminus \{-4\}$



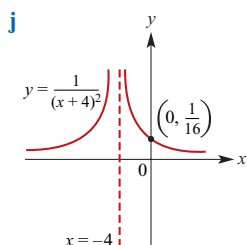
Domain = $R \setminus \{-2\}$, range = $R \setminus \{0\}$



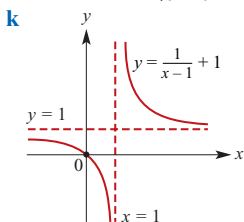
Domain = $R \setminus \{3\}$, range = $R \setminus \{0\}$



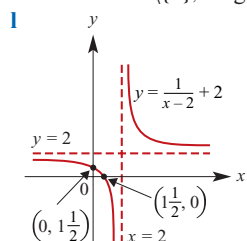
Domain = $R \setminus \{3\}$, range = R^+



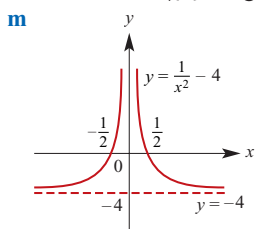
Domain = $\mathbb{R} \setminus \{-4\}$, range = $\mathbb{R}^+$



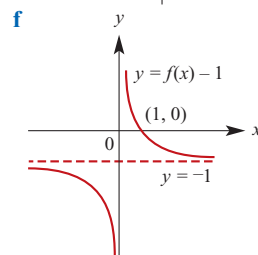
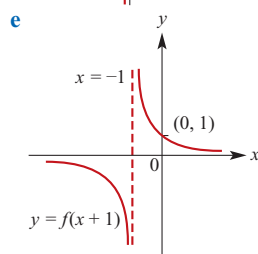
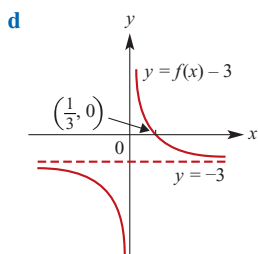
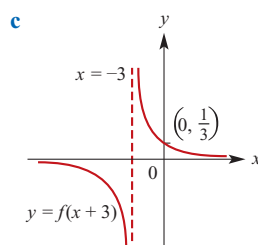
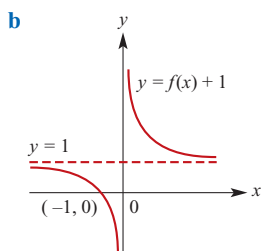
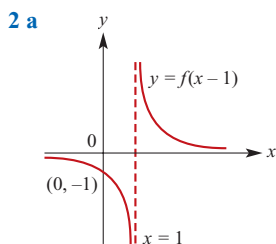
Domain = $\mathbb{R} \setminus \{1\}$, range = $\mathbb{R} \setminus \{1\}$



Domain = $\mathbb{R} \setminus \{2\}$, range = $\mathbb{R} \setminus \{2\}$



Domain = $\mathbb{R} \setminus \{0\}$, range = $(-4, \infty)$



3 a Translation with mapping

$$(x, y) \rightarrow (x - 5, y)$$

b Translation with mapping

$$(x, y) \rightarrow (x, y + 2)$$

c Translation with mapping

$$(x, y) \rightarrow (x, y + 4)$$

4 a i $y = |x - 7| + 1$

ii $y = |x + 2| - 6$

iii $y = |x - 2| - 3$

iv $y = |x + 1| + 4$

b i $y = \sqrt[3]{x - 7} + 1$

ii $y = \sqrt[3]{x + 2} - 6$

iii $y = \sqrt[3]{x - 2} - 3$

iv $y = \sqrt[3]{x + 1} + 4$

c i $y = \frac{1}{(x - 7)^3} + 1$

ii $y = \frac{1}{(x + 2)^3} - 6$

iii $y = \frac{1}{(x - 2)^3} - 3$

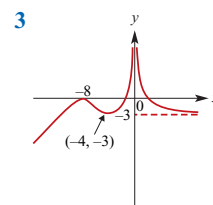
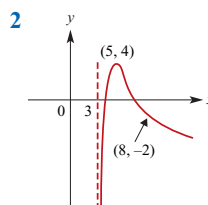
iv $y = \frac{1}{(x + 1)^3} + 4$

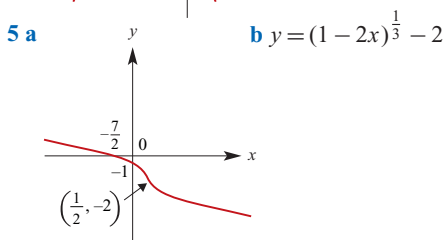
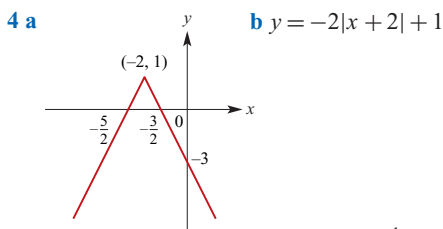
d i $y = \frac{1}{(x-7)^4} + 1$
ii $y = \frac{1}{(x+2)^4} - 6$
iii $y = \frac{1}{(x-2)^4} - 3$
iv $y = \frac{1}{(x+1)^4} + 4$
e i $y = \frac{1}{\sqrt[3]{x-7}} + 1$
ii $y = \frac{1}{\sqrt[3]{x+2}} - 6$
iii $y = \frac{1}{\sqrt[3]{x-2}} - 3$
iv $y = \frac{1}{\sqrt[3]{x+1}} + 4$
f i $y = (x-7)^{\frac{2}{3}} + 1$
ii $y = (x+2)^{\frac{2}{3}} - 6$
iii $y = (x-2)^{\frac{2}{3}} - 3$
iv $y = (x+1)^{\frac{2}{3}} + 4$
g i $y = \frac{1}{(x-7)^{\frac{3}{4}}} + 1$
ii $y = \frac{1}{(x+2)^{\frac{3}{4}}} - 6$
iii $y = \frac{1}{(x-2)^{\frac{3}{4}}} - 3$
iv $y = \frac{1}{(x+1)^{\frac{3}{4}}} + 4$

Exercise 3E

1 a i $y = -2|x-3| - 4$
ii $y = -2|x-3| + 4$
iii $y = -2|x-3| - 4$
iv $y = -2|x-3| - 8$
v $y = -2|x-3| + 8$
vi $y = -2|x-3| + 8$
b i $y = -2\sqrt[3]{x-3} - 4$
ii $y = -2\sqrt[3]{x-3} + 4$
iii $y = -2\sqrt[3]{x-3} - 4$
iv $y = -2\sqrt[3]{x-3} - 8$
v $y = -2\sqrt[3]{x-3} + 8$
vi $y = -2\sqrt[3]{x-3} + 8$
c i $y = \frac{-2}{(x-3)^3} - 4$
ii $y = \frac{-2}{(x-3)^3} + 4$
iii $y = \frac{-2}{(x-3)^3} - 4$
iv $y = \frac{-2}{(x-3)^3} - 8$
v $y = \frac{-2}{(x-3)^3} + 8$

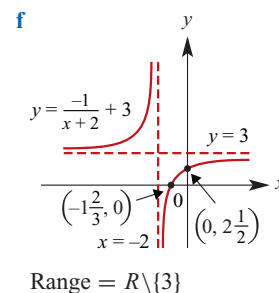
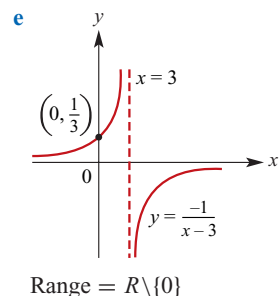
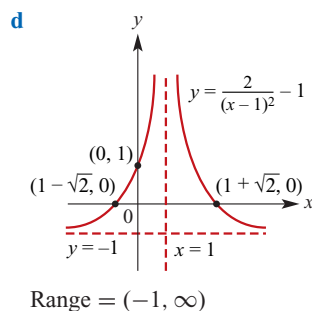
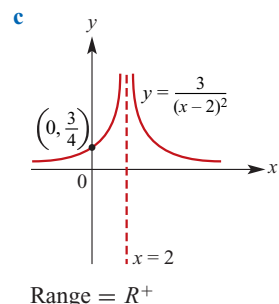
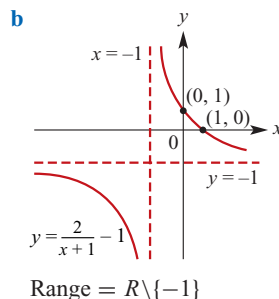
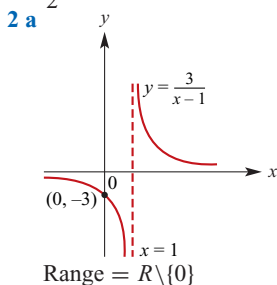
vi $y = \frac{-2}{(x-3)^3} + 8$
d i $y = \frac{-2}{(x-3)^4} - 4$
ii $y = \frac{-2}{(x-3)^4} + 4$
iii $y = \frac{-2}{(x-3)^4} - 4$
iv $y = \frac{-2}{(x-3)^4} - 8$
v $y = \frac{-2}{(x-3)^4} + 8$
vi $y = \frac{-2}{(x-3)^4} + 8$
e i $y = \frac{-2}{\sqrt[3]{x-3}} - 4$
ii $y = \frac{-2}{\sqrt[3]{x-3}} + 4$
iii $y = \frac{-2}{\sqrt[3]{x-3}} - 4$
iv $y = \frac{-2}{\sqrt[3]{x-3}} - 8$
v $y = \frac{-2}{\sqrt[3]{x-3}} + 8$
vi $y = \frac{-2}{\sqrt[3]{x-3}} + 8$
f i $y = -2(x-3)^{\frac{2}{3}} - 4$
ii $y = -2(x-3)^{\frac{2}{3}} + 4$
iii $y = -2(x-3)^{\frac{2}{3}} - 4$
iv $y = -2(x-3)^{\frac{2}{3}} - 8$
v $y = -2(x-3)^{\frac{2}{3}} + 8$
vi $y = -2(x-3)^{\frac{2}{3}} + 8$
g i $y = \frac{-2}{(x-3)^{\frac{3}{4}}} - 4$
ii $y = \frac{-2}{(x-3)^{\frac{3}{4}}} + 4$
iii $y = \frac{-2}{(x-3)^{\frac{3}{4}}} - 4$
iv $y = \frac{-2}{(x-3)^{\frac{3}{4}}} - 8$
v $y = \frac{-2}{(x-3)^{\frac{3}{4}}} + 8$
vi $y = \frac{-2}{(x-3)^{\frac{3}{4}}} + 8$

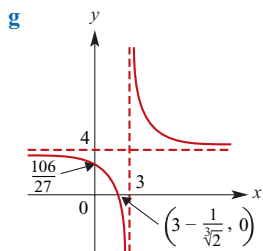




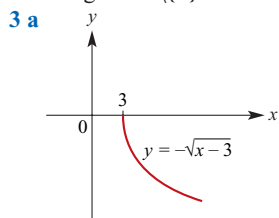
Exercise 3F

- 1 a** A dilation of factor 2 from the x -axis, followed by a translation of 1 unit in the positive direction of the x -axis and 3 units in the positive direction of the y -axis.
- b** A reflection in the x -axis and a dilation of factor 3 from the x -axis (in either order), followed by a translation of 4 units in the negative direction of the x -axis and 7 units in the negative direction of the y -axis.
- c** A reflection in the y -axis and a dilation of factor 4 from the x -axis (in either order), followed by a translation of 1 unit in the positive direction of the x -axis and 5 units in the negative direction of the y -axis.
- d** A reflection in the x -axis, followed by a translation of 1 unit in the negative direction of the x -axis and 2 units in the positive direction of the y -axis.
- e** A reflection in the y -axis and a dilation of factor 2 from the x -axis (in either order), followed by a translation of 3 units in the positive direction of the y -axis.
- f** A translation of 3 units in the negative direction of the x -axis and 4 units in the negative direction of the y -axis, followed by a reflection in either axis and a dilation of factor $\frac{1}{2}$ from the x -axis (in either order).

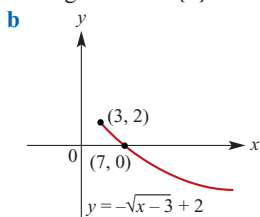




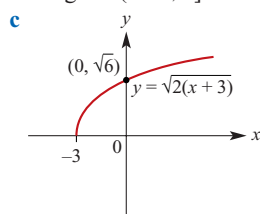
$$\text{Range} = \mathbb{R} \setminus \{4\}$$



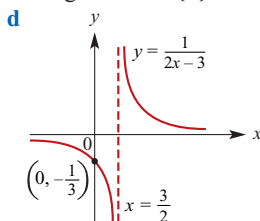
$$\text{Range} = \mathbb{R}^- \cup \{0\}$$



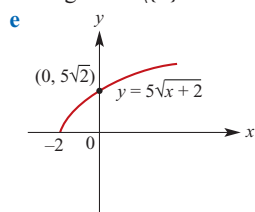
$$\text{Range} = (-\infty, 2]$$



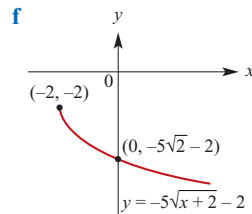
$$\text{Range} = \mathbb{R}^+ \cup \{0\}$$



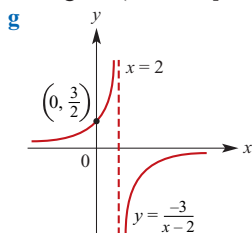
$$\text{Range} = \mathbb{R} \setminus \{0\}$$



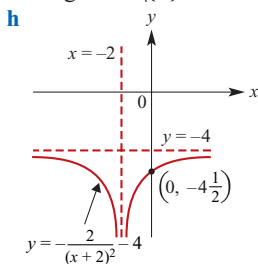
$$\text{Range} = \mathbb{R}^+ \cup \{0\}$$



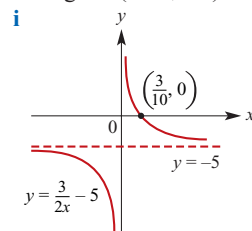
$$\text{Range} = (-\infty, -2]$$



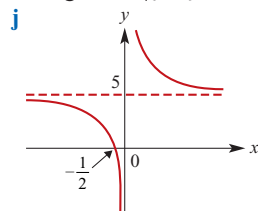
$$\text{Range} = \mathbb{R} \setminus \{0\}$$



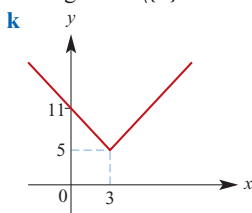
$$\text{Range} = (-\infty, -4)$$



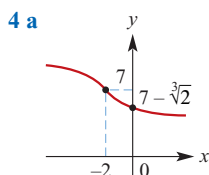
$$\text{Range} = \mathbb{R} \setminus \{-5\}$$



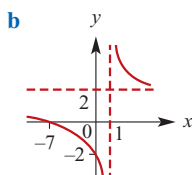
$$\text{Range} = \mathbb{R} \setminus \{5\}$$



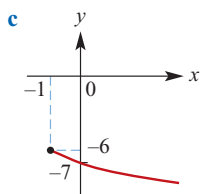
$$\text{Range} = [5, \infty)$$



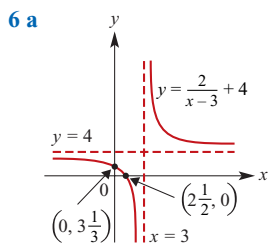
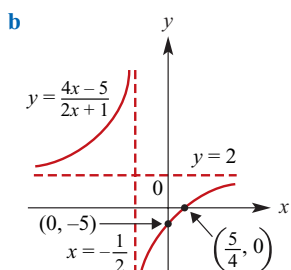
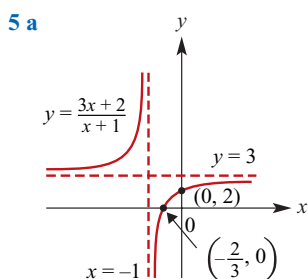
Range = R



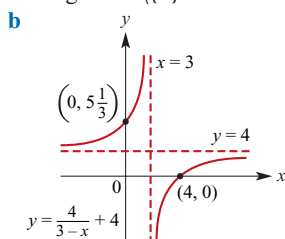
Range = $R \setminus \{2\}$



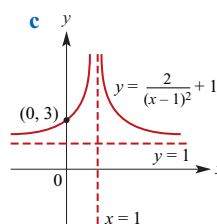
Range = $(-\infty, -6]$



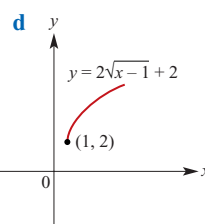
Range = $R \setminus \{4\}$



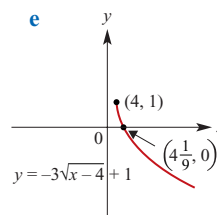
Range = $R \setminus \{4\}$



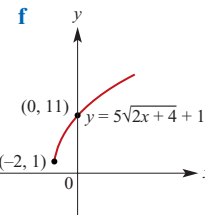
Range = $(1, \infty)$



Range = $[2, \infty)$



Range = $(-\infty, 1]$



Range = $[1, \infty)$

Exercise 3G

1 a $(1, -6)$ **b** $(2, -2)$ **c** $(1, 2)$

d $(-1, -2)$ **e** $(-2, 1)$

2 $\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

3 a $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ **b** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **c** $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

d $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ **e** $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

4 $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$
 $x = x' - 2; y = \frac{y' - 1}{3}$

5 $y = 2 \left(\frac{x^2}{16} - \frac{x}{4} + 2 \right) = \frac{x^2}{8} - \frac{x}{2} + 4$

6 $y = -2(x^3 + 2x)$

7 $y = \frac{-x}{3} + 3$ **8** $y = \frac{3}{2} \left(\frac{x}{2} - 4 \right)$

9 $y = \frac{3}{2}x - 9$

10 $y = \frac{1}{6}x + \frac{11}{2}$

11 $\frac{y-4}{2} = -2 \left(\frac{x+1}{4} \right)^3 + 6 \left(\frac{x+1}{4} \right)$

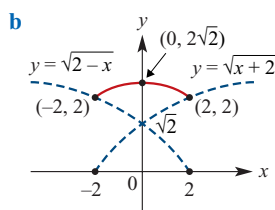
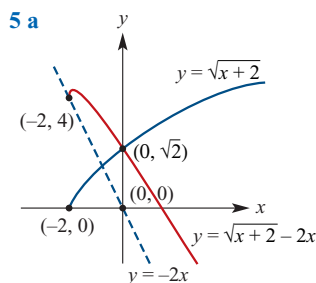
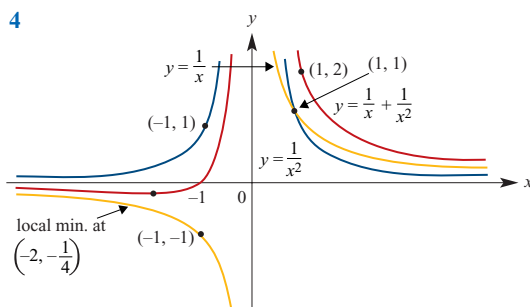
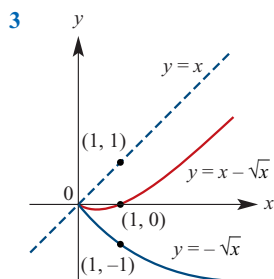
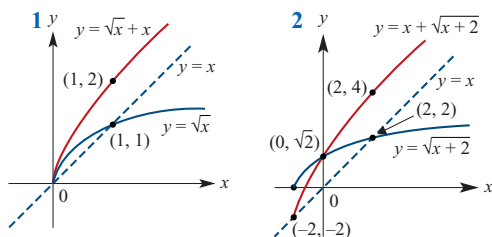
12 $\frac{2-y}{2} = -2 \left(\frac{x+1}{3} \right)^3 + 6 \left(\frac{x+1}{3} \right)^2 + 2$

Exercise 3H

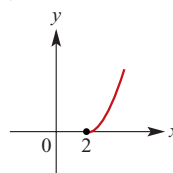
1 $b = -1, B = 2, A = 1$ **2** $A = 2, B = 3$

3 $A = 2, B = -1$ **4** $b = 2, B = -3, A = 8$

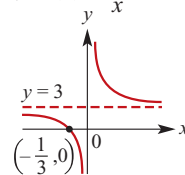
5 $a = -2, b = 1$ **6** $a = 3, b = -5$

Exercise 3I

Exercise 3J

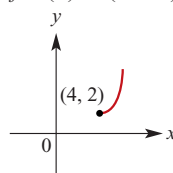
1 a $f^{-1}: [2, \infty) \rightarrow R$,
 $f^{-1}(x) = (x-2)^2$



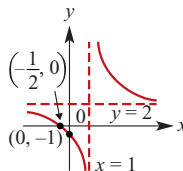
b $f^{-1}: R \setminus \{0\} \rightarrow R$,
 $f^{-1}(x) = \frac{1}{x} + 3$



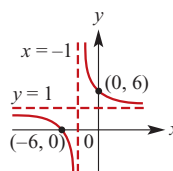
c $f^{-1}: [4, \infty) \rightarrow R$,
 $f^{-1}(x) = (x-4)^2 + 2$



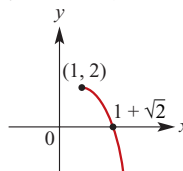
d $f^{-1}: R \setminus \{1\} \rightarrow R$, $f^{-1}(x) = \frac{3}{x-1} + 2$



e $f^{-1}: R \setminus \{-1\} \rightarrow R$, $f^{-1}(x) = \frac{5}{x+1} + 1$



f $f^{-1}: [1, \infty) \rightarrow R$, $f^{-1}(x) = 2 - (x-1)^2$



2 a $y = \frac{3}{x}$, domain = $R \setminus \{0\}$

b $y = (x+4)^3 - 2$, domain = R

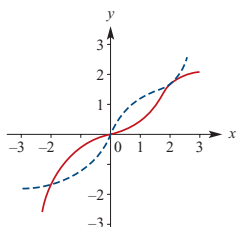
c $y = (2-x)^2$, domain = $(-\infty, 2]$

d $y = \frac{3}{x-1}$, domain = $R \setminus \{1\}$

e $y = \sqrt[3]{\frac{2}{5-x}} + 6$, domain = $R \setminus \{5\}$

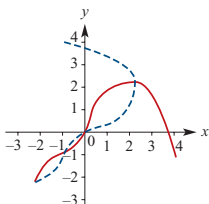
f $y = \frac{1}{(x-2)^{4/3}} + 1$, domain = $(2, \infty)$

3 a



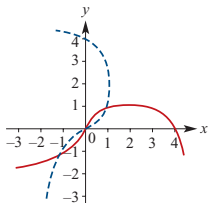
Inverse is a function.

b



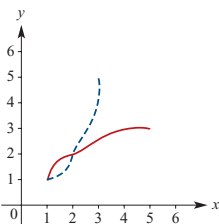
Inverse is not a function.

c



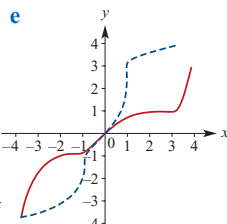
Inverse is not a function.

d



Inverse is a function.

e



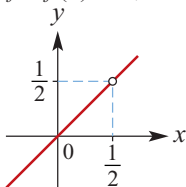
Inverse is not a function.

- 4 a $f^{-1}: R \setminus \{1\} \rightarrow R$ $f^{-1}(x) = \frac{x+1}{x-1}$
 b $f^{-1}: R^+ \cup \{0\} \rightarrow R$ $f^{-1}(x) = x^2 + 2$
 c $f^{-1}: R \setminus \left\{\frac{2}{3}\right\} \rightarrow R$ $f^{-1}(x) = \frac{2x+3}{3x-2}$

5

	domain	range	
a	$R \setminus \left\{\frac{1}{2}\right\}$	$R \setminus \left\{\frac{1}{2}\right\}$	Since
	$\text{ran } f = \text{dom } f, f \circ f \text{ is defined.}$		

- b $f \circ f(x) = x, x \in R \setminus \left\{\frac{1}{2}\right\}$



- c $f^{-1}: R \setminus \left\{\frac{1}{2}\right\} \rightarrow R, f^{-1}(x) = \frac{x+3}{2x-1}$

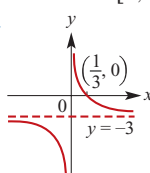
Multiple-choice questions

- 1 C 2 E 3 D 4 A 5 D
6 B 7 B 8 C 9 A 10 A

Short-answer questions (technology-free)

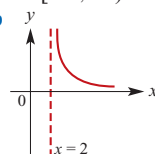
- 1 a Domain = $R \setminus \{0\}$, range = $R \setminus \{2\}$
 b Domain = $\left[\frac{2}{3}, \infty\right)$, range = $(-\infty, 3]$
 c Domain = $R \setminus \{2\}$, range = $(3, \infty)$
 d Domain = $R \setminus \{2\}$, range = $R \setminus \{4\}$
 e Domain = $[2, \infty)$, range = $[-5, \infty)$

2 a



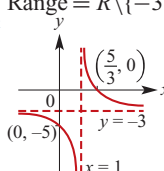
Range = $R \setminus \{-3\}$

b



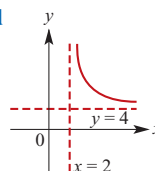
Range = $(0, \infty)$

c



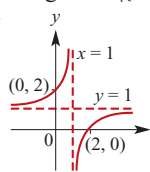
Range = $R \setminus \{-3\}$

d



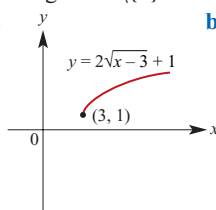
Range = $(4, \infty)$

e

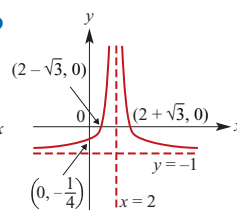


Range = $R \setminus \{1\}$

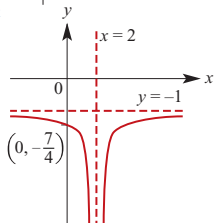
3 a



b

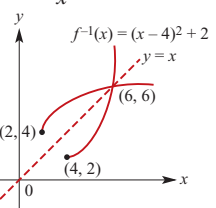


c



- 4 $y = -\frac{6}{x} + 9, a = -6, b = 9$

5



- 6 $f^{-1}(x) = \frac{x+2}{1-x}$

Extended-response questions

1 a $R \setminus \{-2\}$

- b ■ Dilation of factor 24 from the x -axis
- Translation of 2 units in the negative direction of the x -axis
- Translation of 6 units in the negative direction of the y -axis

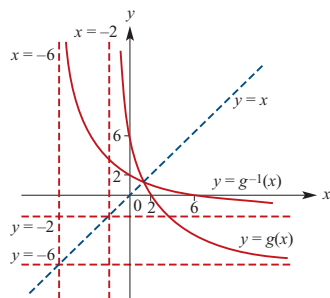
c $(0, 6), (2, 0)$

d i $g^{-1}(x) = \frac{24}{x+6} - 2$

ii Domain of g^{-1} = range of $g = (-6, \infty)$

iii $x = -4 + 2\sqrt{7}$

iv



2 a $[-3, \infty)$

- b ■ Dilation of factor $\frac{1}{2}$ from the y -axis
- Dilation of factor 2 from the x -axis
- Reflection in the x -axis
- Translation of 3 units in the negative direction of the x -axis
- Translation of 4 units in the positive direction of the y -axis

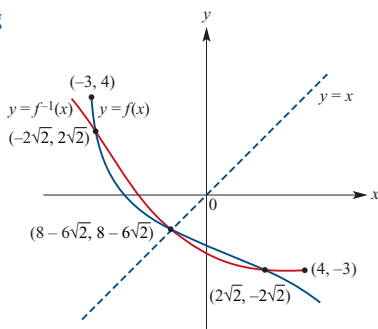
c $(0, 4 - 2\sqrt{6}), (-1, 0)$

d $f^{-1}(x) = \frac{(x-4)^2}{8} - 3$

e Domain of f^{-1} = range of $f = (-\infty, 4]$

f $x = 8 - 6\sqrt{2}$ or $x = 2\sqrt{2}$ or $x = -2\sqrt{2}$

g



3 a i $\frac{3}{125}$

ii $(x, y) \rightarrow (x, -y)$

iii $(x, y) \rightarrow (x + 25, y + 15)$

iv $(x, y) \rightarrow (x + 25, -\frac{3}{125}y + 15)$

b i $y = -\frac{3}{125}(x - 25)^2 + 15$

ii $(x, y) \rightarrow (x + 50, y)$

iii $y = -\frac{3}{125}(x - 75)^2 + 15$

c i $(x, y) \rightarrow \left(x + \frac{m}{2}, -\frac{4n}{m^2}y + n\right)$

ii $y = -\frac{4n}{m^2}\left(x - \frac{m}{2}\right)^2 + n$

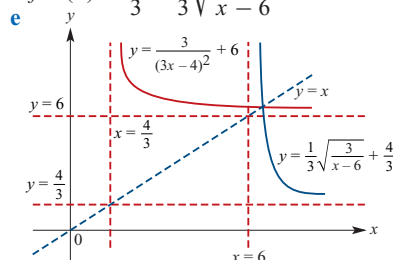
iii $y = -\frac{4n}{m^2}\left(x - \frac{3m}{2}\right)^2 + n$

4 a $R \setminus \left\{\frac{4}{3}\right\}$

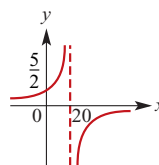
b $\frac{4}{3}$

c $f^{-1}(x) = \frac{4}{3} + \frac{1}{3}\sqrt{\frac{3}{x-6}}$

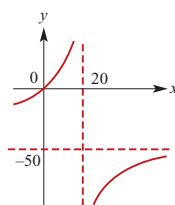
d $x = 6.015$



5 a $f(x) = \frac{50}{20-x}$



b ii



c $g^{-1}(x) = \frac{20x}{50+x}$

6 a i $y = f^{-1}(x - 5) + 3$

ii $y = f^{-1}(x - 3) + 5$

iii $y = 5f^{-1}\left(\frac{x}{3}\right)$

iv $y = 3f^{-1}\left(\frac{x}{5}\right)$

b $y = cf^{-1}\left(\frac{x-b}{a}\right) + d$. The graph of

$y = f(x)$ has undergone the following sequence of transformations: a reflection in the line $y = x$, then a dilation of factor c from the x -axis and factor a from the y -axis, and a translation of b units in the positive direction of the x -axis and d units in the positive direction of the y -axis.

Chapter 4

Exercise 4A

1 $A = 1, B = 3$

2 a $A = 1, B = -2, C = 6$

b $A = 4, B = -\frac{3}{2}, C = 5$

c $A = 1, B = -3, C = 5$

3 a $2x^2 - x + 12 + \frac{33}{x-3}$

b $5x^4 + 8x^3 - 8x^2 + 6x - 6$

c $x^2 - 9x + 27 - \frac{26(x-2)}{x^2-2}$

4 a -16 b $a = 4$

5 a 28 b 0

c $f(x) = (x+2)(3x+1)(2x-3)$

6 b $k = \frac{11}{2}$

7 a $a = 3, b = 8$ b $(2x-1), (x-1)$

8 a $a = \frac{-92}{9}, b = 9$

9 81 10 b $6x - 4$

11 b $x^2 - 3, x^2 + x + 2$

12 a $2, -4, 2, 3$ b $0, 2$ c $\frac{1}{2}, 2$

d $-2, 2$ e $0, -2, 2$ f $0, -3, 3$

g $1, -2, -\frac{1}{4}, \frac{1}{3}$ h $1, -2$ i $1, -2, \frac{1}{3}, \frac{3}{2}$

13 a $(-1, 0), (0, 0), (2, 0)$

b $(-2, 0), (0, 6), (1, 0), (3, 0)$

c $(-1, 0), (0, 6), (2, 0), (3, 0)$

d $(-\frac{1}{2}, 0), (0, 2), (1, 0), (2, 0)$

e $(-2, 0), (-1, 0), (0, -2), (1, 0)$

f $(-1, 0), (-\frac{2}{3}, 0), (0, -6), (3, 0)$

g $(-4, 0), (0, -16), (-\frac{2}{5}, 0), (2, 0)$

h $(-\frac{1}{2}, 0), (0, 1), (\frac{1}{3}, 0), (1, 0)$

i $(-2, 0), (-\frac{3}{2}, 0), (0, -30), (5, 0)$

14 $p = 1, q = -6$ 15 -33

16 a $(x-9)(x-13)(x+11)$

b $(x+11)(x-9)(x-11)$

c $(x+11)(2x-9)(x-11)$

d $(x+11)(2x-13)(2x-9)$

17 a $(x-1)(x+1)(x-7)(x+6)$

b $(x-3)(x+4)(x^2+3x+9)$

18 a $(x-9)(x-5)(2x^2+3x+9)$

b $(x+5)(x+9)(x^2-x+9)$

c $(x-3)(x+5)(x^2+x+9)$

d $(x-4)(x-3)(x+5)(x+6)$

Exercise 4B

1 a Crosses the x -axis

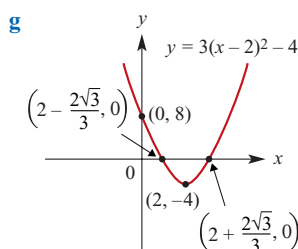
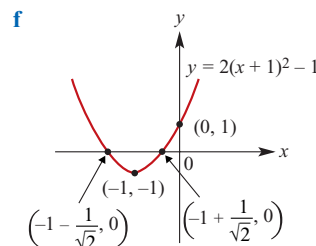
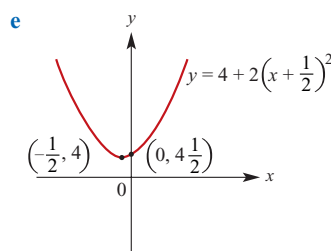
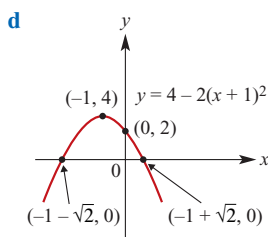
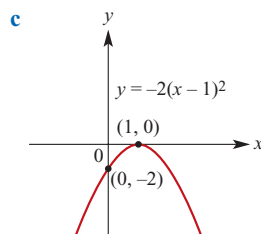
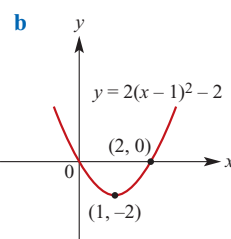
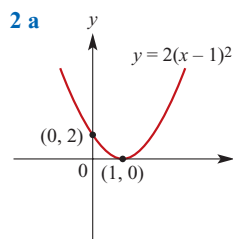
b Does not cross the x -axis

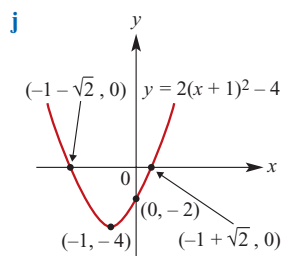
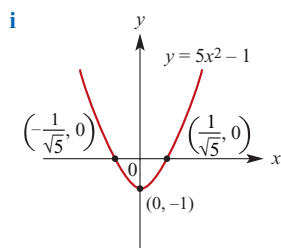
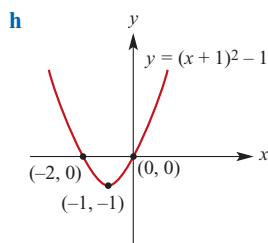
c Just touches the x -axis

d Crosses the x -axis

e Does not cross the x -axis

f Does not cross the x -axis





3 a $f(x) = \left(x + \frac{3}{2}\right)^2 - 4\frac{1}{4}$
 Min. $= -4\frac{1}{4}$, range $= \left[-4\frac{1}{4}, \infty\right)$

b $f(x) = (x - 3)^2 - 1$
 Min. $= -1$, range $= [-1, \infty)$

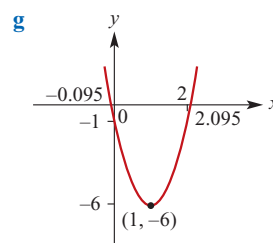
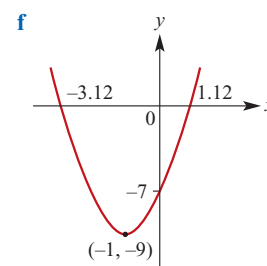
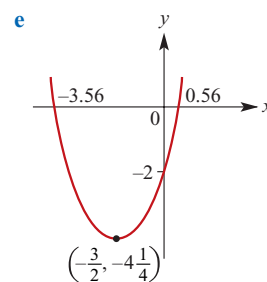
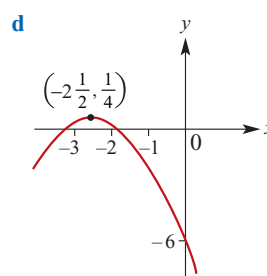
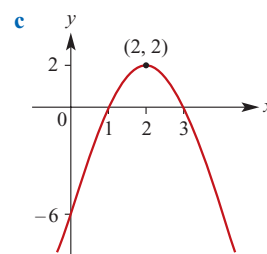
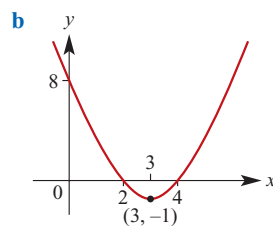
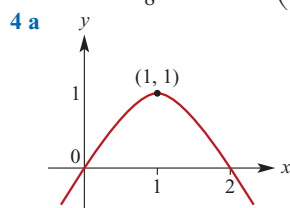
c $f(x) = 2(x + 2)^2 - 14$
 Min. $= -14$, range $= [-14, \infty)$

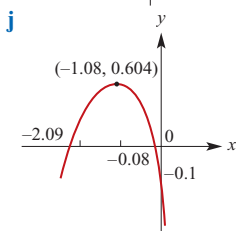
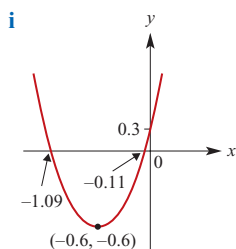
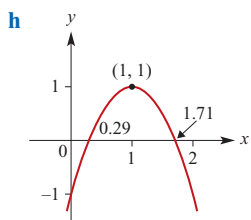
d $f(x) = 4(x + 1)^2 - 11$
 Min. $= -11$, range $= [-11, \infty)$

e $f(x) = 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8}$
 Min. $= -\frac{25}{8}$, range $= \left[-\frac{25}{8}, \infty\right)$

f $f(x) = -3\left(x + \frac{1}{3}\right)^2 + \frac{22}{3}$
 Max. $= \frac{22}{3}$, range $= \left(-\infty, \frac{22}{3}\right]$

g $f(x) = -2\left(x - \frac{9}{4}\right)^2 + \frac{169}{8}$
 Max. $= \frac{169}{8}$, range $= \left(-\infty, \frac{169}{8}\right]$





5 a B **b** D

6 a C **b** B **c** D

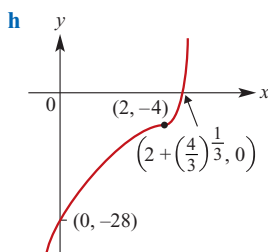
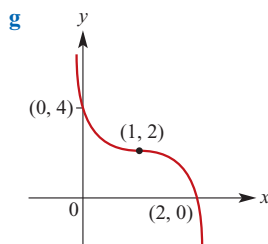
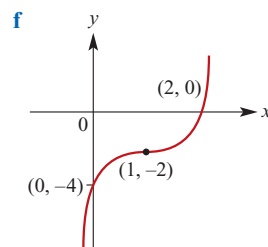
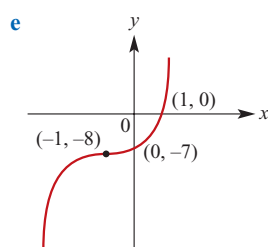
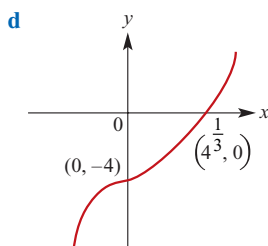
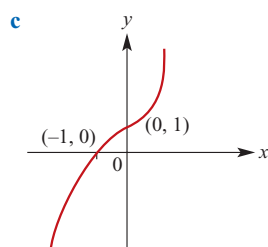
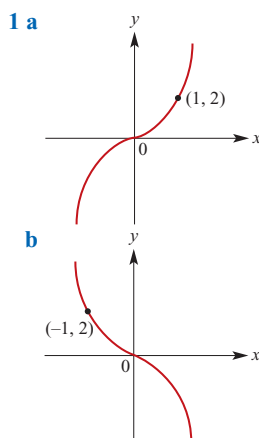
7 a $m > 3$ or $m < 0$ **b** $m = 3$

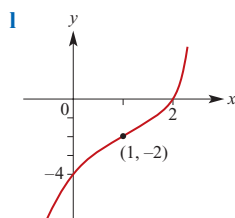
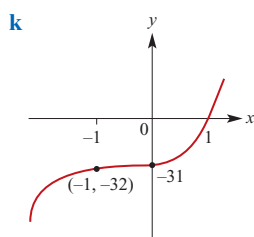
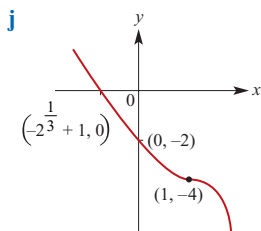
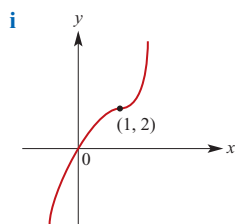
9 a $k < -5$ or $k > 0$ **b** $k = -5$

Exercise 4C

- 1 a** $y = 4 - \frac{4}{25}x^2$ **b** $y = -x^2$
c $y = x^2 + 2x$ **d** $y = 2x - x^2$
e $y = x^2 - 5x + 4$ **f** $y = x^2 - 4x - 5$
g $y = x^2 - 2x - 1$ **h** $y = x^2 - 4x + 6$
- 2** $y = -\frac{1}{8}x^2 + x + 1$, $y = \frac{1}{8}x^2 + x - 5$
- 3** $b = 2$, $B = 4$, $A = 1$

Exercise 4D





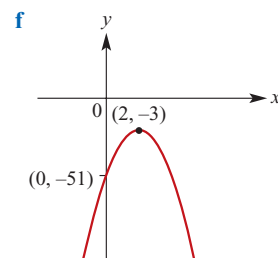
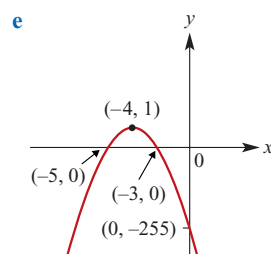
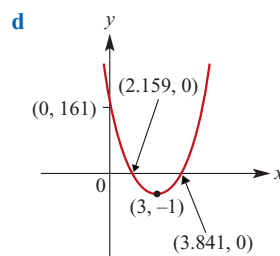
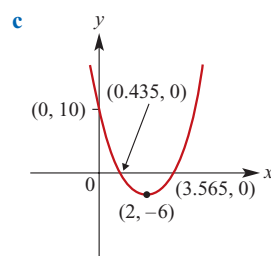
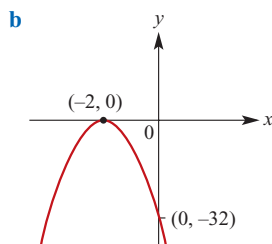
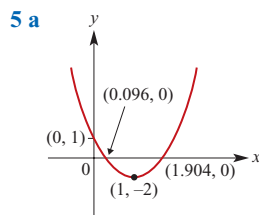
2 $a = -3, h = 0, k = 4$

3 $a = 16, h = -1, k = 7$

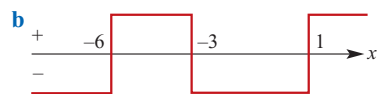
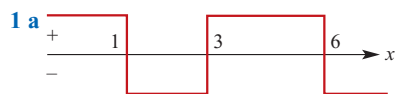
4 a $y = 3x^3$ **b** $y = (x + 1)^3 + 1$

c $y = -(x - 2)^3 - 3$

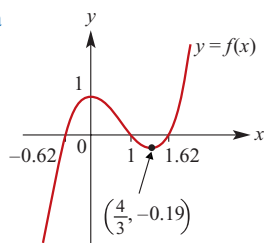
d $y = 2(x + 1)^3 - 2$ **e** $y = \frac{x^3}{27}$



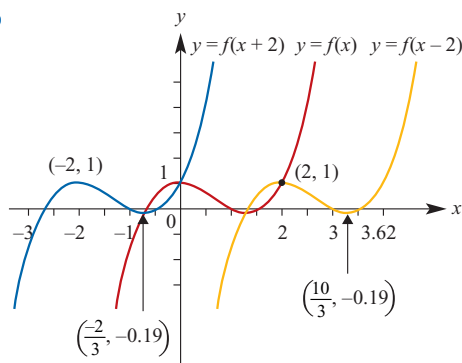
Exercise 4E



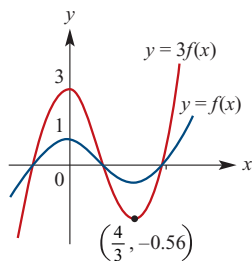
2 a



b

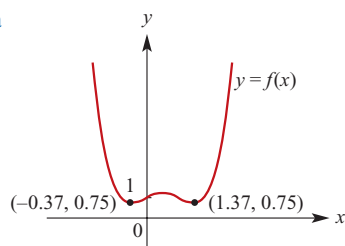


For clarity the graph of $y = 3f(x)$ is shown on separate axes.

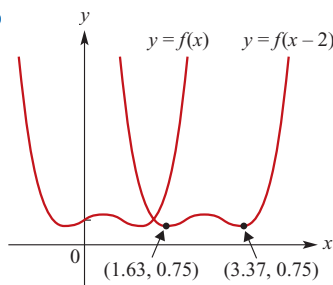


Exercise 4F

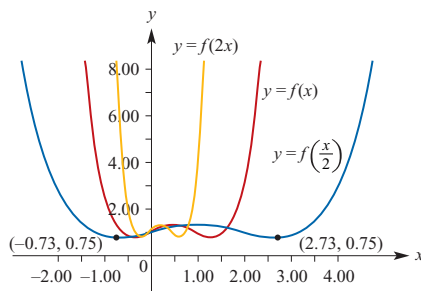
1 a



b

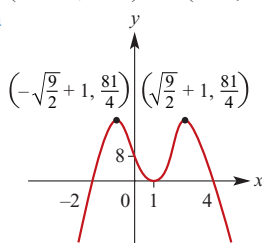


Graphs of dilations shown on separate axes for clarity.

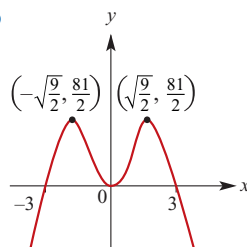


Turning points for $y = f(2x)$ are at $(-0.18, 0.75)$ and $(0.68, 0.75)$.

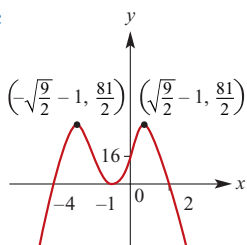
2 a



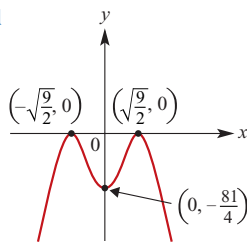
b



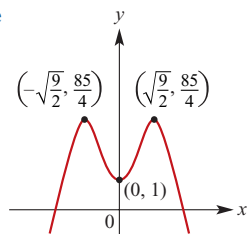
c



d



e



Exercise 4G

- 1 $y = \frac{11}{60}(x+5)(x+2)(x-6)$
- 2 $y = \frac{5}{9}(x+1)(x-3)^2$
- 3 a $y = x^3 + x + 1$
 b $y = x^3 - x + 1$
 c $y = 2x^3 - x^2 + x - 2$
- 4 a $y = (2x+1)(x-1)(x-2)$
 b $y = \frac{1}{4}x(x^2+2)$
 c $y = x^2(x+1)$
 d $y = x^3 + 2x^2 - x - 2$
 e $y = (x+2)(x-3)^2$
- 5 a $y = -2x^3 - 25x^2 + 48x + 135$
 b $y = 2x^3 - 30x^2 + 40x + 13$
- 6 a $y = -2x^4 + 22x^3 - 10x^2 - 37x + 40$
 b $y = x^4 - x^3 + x^2 + 2x + 8$
 c $y = \frac{31}{36}x^4 + \frac{5}{4}x^3 - \frac{157}{36}x^2 - \frac{5}{4}x + \frac{11}{2}$

Exercise 4H

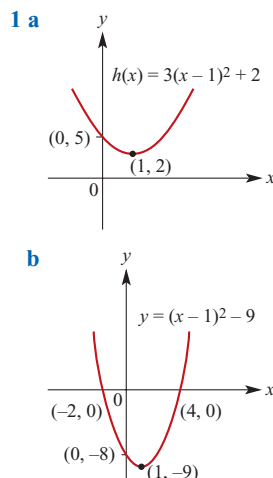
- 1 a $x = \frac{-1 \pm \sqrt{1-4k^2}}{2k}, \quad k \in \left[-\frac{1}{2}, \frac{1}{2}\right] \setminus \{0\}$
 b $x = 4a$ or $3a$ or 0 c $x = 0$ or $x = a^{\frac{1}{3}}$
 d $x = \frac{k \pm \sqrt{k^2-4k}}{2}, \quad k > 4$ or $k < 0$
 e $x = 0$ or $x = \pm\sqrt{a}$ f $x = \pm a$
 g $x = a$ or $x = b$
 h $x = a$ or $x = a^{\frac{1}{3}}$ or $x = \pm\sqrt{a}$ if $a > 0$
- 2 a $x = \left(\frac{b+c}{a}\right)^{\frac{1}{3}}$ b $x = (a)^{\frac{7}{3}}$
 c $x = (a-c)^n$ d $x = \left(\frac{c}{a}\right)^{\frac{1}{3}} - b$
 e $x = \left(\frac{b}{a}\right)^3$ f $x = (c+d)^{\frac{1}{3}}$
- 3 a $(0, 0), (1, 1)$ b $\left(\frac{1}{2}, \frac{1}{2}\right), (0, 0)$
 c $\left(\frac{3+\sqrt{13}}{2}, \sqrt{13}+4\right); \left(\frac{3-\sqrt{13}}{2}, 4-\sqrt{13}\right)$
- 4 a $(13, 3), (3, 13)$ b $(10, 5), (5, 10)$
 c $(11, 8), (-8, -11)$ d $(9, 4), (4, 9)$
 e $(9, 5), (-5, -9)$
- 5 a $(17, 11), (11, 17)$ b $(37, 14), (14, 37)$
 c $(14, 9), (-9, -14)$
- 6 $(2, 4), (0, 0)$
- 7 $\left(\frac{\sqrt{5}+5}{2}, \frac{\sqrt{5}+5}{2}\right), \left(\frac{5-\sqrt{5}}{2}, \frac{5-\sqrt{5}}{2}\right)$

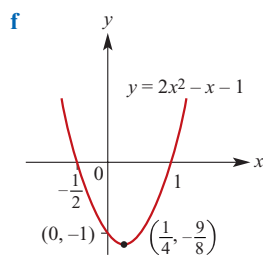
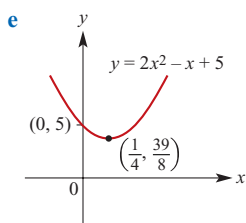
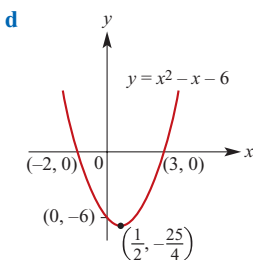
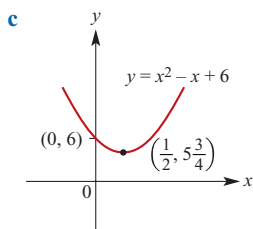
- 8 $\left(\frac{-130-80\sqrt{2}}{41}, \frac{60-64\sqrt{2}}{41}\right), \left(\frac{80\sqrt{2}-130}{41}, \frac{64\sqrt{2}+60}{41}\right)$
- 9 $\left(\frac{1+\sqrt{21}}{2}, \frac{-1+\sqrt{21}}{2}\right), \left(\frac{1-\sqrt{21}}{2}, \frac{-1+\sqrt{21}}{2}\right)$
- 10 $\left(\frac{4}{9}, 2\right)$ 11 $\left(\frac{-6\sqrt{5}}{5}, \frac{3\sqrt{5}}{5}\right)$
- 12 $\left(-2, \frac{1}{2}\right)$ 13 $(3, 2), (0, -1)$
- 14 a $(3, 2), \left(\frac{-8}{5}, \frac{-15}{4}\right)$ b $\left(\frac{27}{2}, \frac{10}{3}\right), (5, 9)$
 c $(6, 4), \left(\frac{-12}{5}, -10\right)$
- 15 $c^2 - ac + b = 0$
- 16 $(-1 - \sqrt{161}, 1 - \sqrt{161}), (\sqrt{161} - 1, \sqrt{161} + 1)$
- 17 a $x = \frac{\sqrt{4m+25}+5}{2m}, y = \frac{\sqrt{4m+25}+5}{2}$
 or $x = \frac{5-\sqrt{4m+25}}{2m}, y = \frac{5-\sqrt{4m+25}}{2}$
 b $m = \frac{-25}{4}, \left(-\frac{2}{5}, \frac{5}{2}\right)$
 c $m < \frac{-25}{4}$ and $m \neq 0$

Multiple-choice questions

- 1 E 2 D 3 E 4 C 5 E
 6 A 7 C 8 E 9 C 10 C

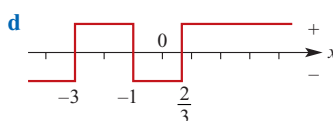
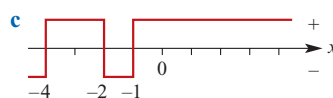
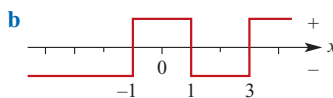
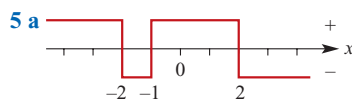
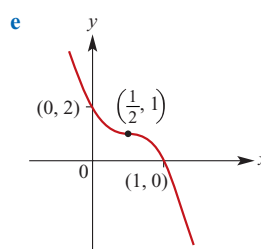
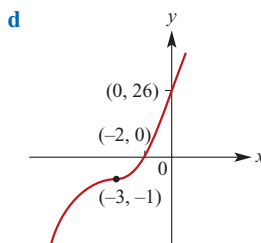
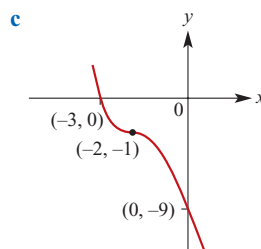
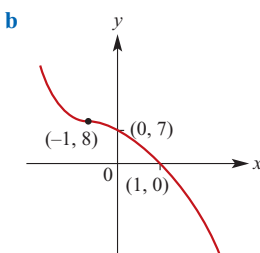
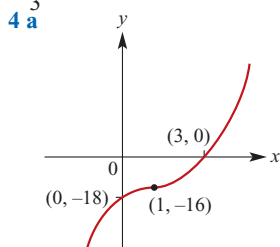
Short-answer questions (technology-free)





2 $y = \frac{4}{3}x^2 - \frac{1}{3}$; $a = \frac{4}{3}$, $b = -\frac{1}{3}$

3 $\frac{1}{3}(1 \pm \sqrt{31})$

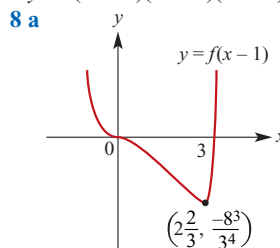


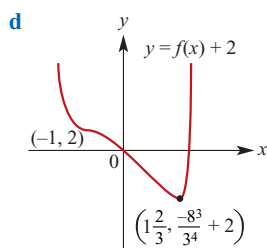
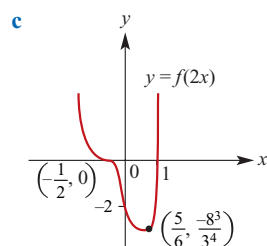
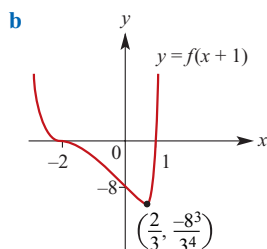
6 a 8

b 0

c 0

7 $y = (x - 7)(x + 3)(x + 2)$

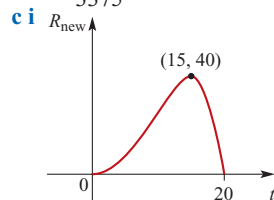




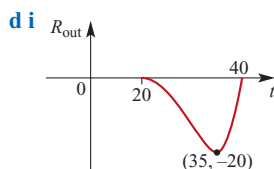
9 $y = -x^3 + 7x^2 - 11x + 6$

Extended-response questions

1 a $k = \frac{4}{3375}$ **b** 11.852 mL/min

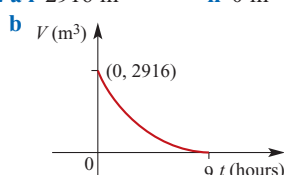


ii 23.704 mL/min



ii 11.852 mL/minute out

2 a i 2916 m³ **ii** 0 m³



c 3.96 hours

3 a $V = x(96 - 4x)(48 - 2x) = 8x(24 - x)^2$

b i $0 < x < 24$

ii $x \approx 8$,
maximum volume $\approx 16\,500$ cm³

c When $x = 10$, $V = 15\,680$ cm³.

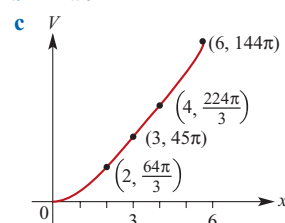
d Max. volume = 14 440 cm³

e Min. volume = 9720 cm³

4 $V = \frac{1}{3}\pi x^2(18 - x)$

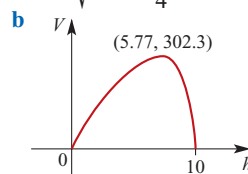
a i $\frac{64\pi}{3}$ cm³ **ii** 45π cm³ **iii** $\frac{224\pi}{3}$ cm³

b 144π cm³



d $x = 5$. The depth is 5 cm.

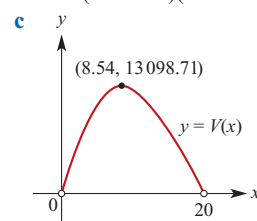
5 a $r = \sqrt{25 - \frac{h^2}{4}}$



c $V = 96\pi$ cm³

d $h = 2$, $r = 2\sqrt{6}$, i.e. height = 2 cm and
radius = $2\sqrt{6}$ cm, or $h = 8.85$ and $r = 2.33$

6 a $V = (84 - 2x)(40 - 2x)x$ **b** (0, 20)



d i $x = 2$, $V = 5760$ **ii** $x = 6$, $V = 12\,096$

iii $x = 8$, $V = 13\,056$ **iv** $x = 10$, $V = 12\,800$

e $x = 13.50$ or $x = 4.18$ **f** 13 098.71 cm³

7 a i $A = 2x(16 - x^2)$ **ii** (0, 4)

b i 42 **ii** $x = 0.82$ or $x = 3.53$

c i $V = 2x^2(16 - x^2)$

ii $x = 2.06$ or $x = 3.43$

8 a $A = \frac{\pi}{2}x^2 + yx$

b i $y = 100 - \pi x$

ii $A = 100x - \frac{\pi}{2}x^2$ **iii** $\left(0, \frac{100}{\pi}\right)$

c $x = 12.43$

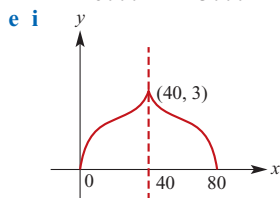
d i $V = \frac{x^2}{50} \left(100 - \frac{\pi}{2}x\right)$, $x \in \left(0, \frac{100}{\pi}\right)$

ii 248.5 m³ **iii** $x = 18.84$

9 a $y = \frac{1}{12\,000}x^3 - \frac{1}{200}x^2 + \frac{17}{120}x$

b $x = 20$

d $y = -\frac{1}{6000}x^3 + \frac{29}{3000}x^2 - \frac{1}{20}x$

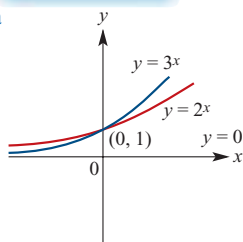


- ii Second section of graph is formed by a reflection of the graph of $y = f(x)$, $x \in [0, 40]$, in the line $x = 40$.

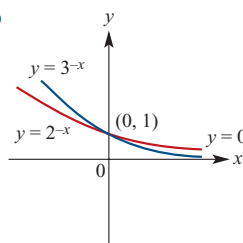
Chapter 5

Exercise 5A

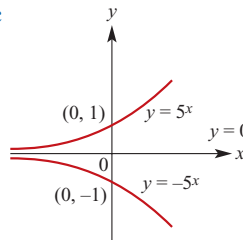
1 a



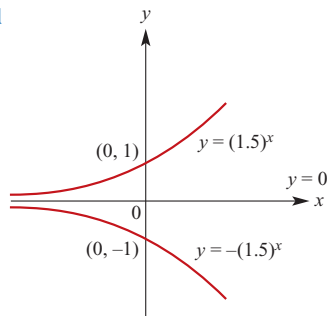
b



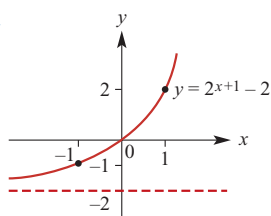
c



d

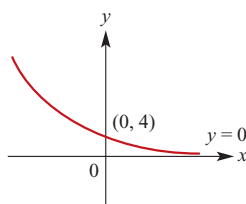


2 a



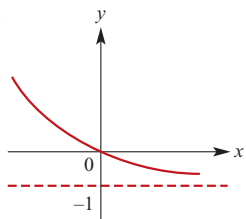
Range = $(-2, \infty)$

b



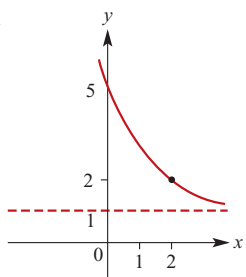
Range = R^+

c



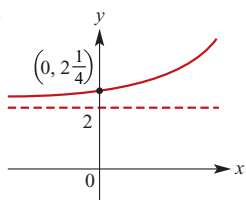
Range = $(-1, \infty)$

d



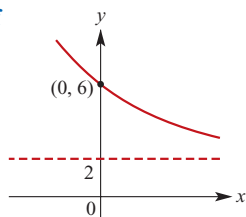
Range = $(1, \infty)$

e

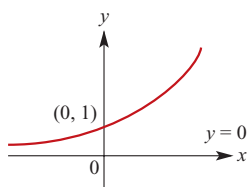
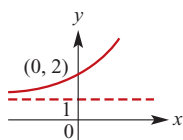
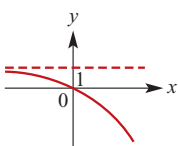
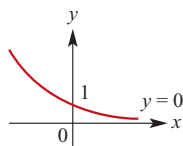
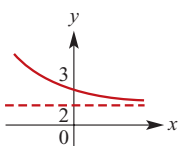
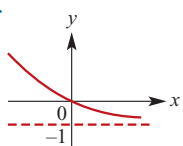
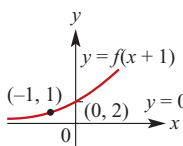
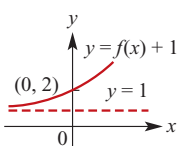
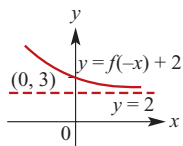
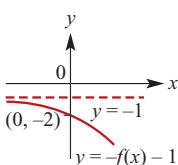
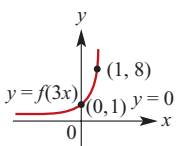
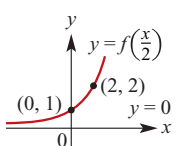
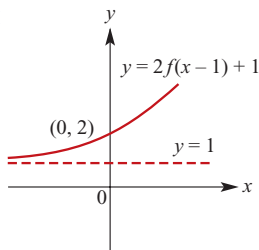
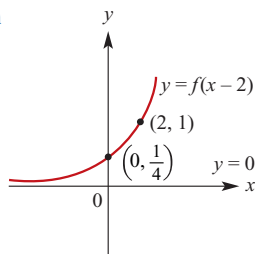
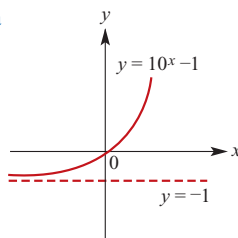
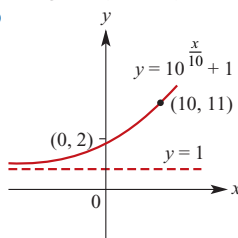
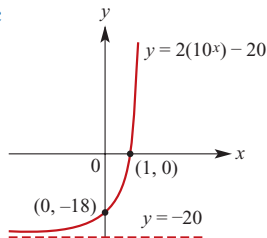
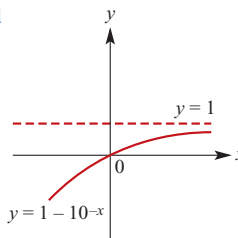
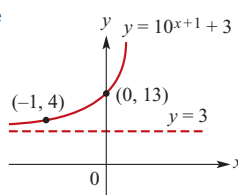


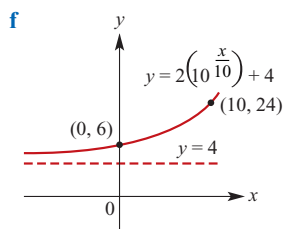
Range = $(2, \infty)$

f

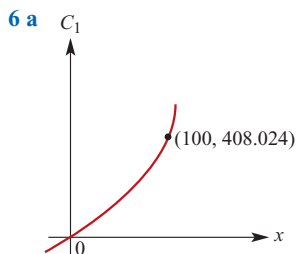


Range = $(2, \infty)$

3 a

 Range = R^+
b

 Range = $(1, \infty)$
c

 Range = $(-\infty, 1)$
d

 Range = R^+
e

 Range = $(2, \infty)$
f

 Range = $(-1, \infty)$
4 a

b

c

d

e

f

g

h

5 a

 Range = $(-1, \infty)$
b

 Range = $(1, \infty)$
c

 Range = $(-20, \infty)$
d

 Range = $(-\infty, 1)$
e

 Range = $(3, \infty)$



Range = $(4, \infty)$



b i \$408.02

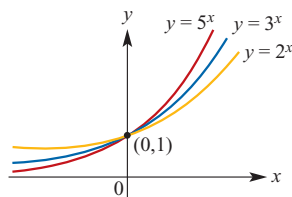
ii \$1274.70

c 239 days

d ii 302 days

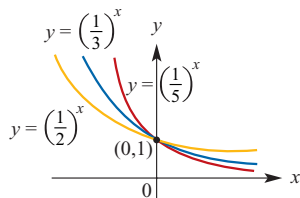
7 36 days

8 a i



ii $x < 0$ **iii** $x > 0$ **iv** $x = 0$

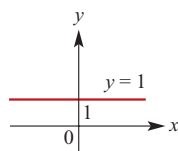
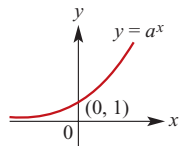
b i



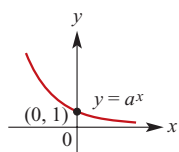
ii $x > 0$ **iii** $x < 0$ **iv** $x = 0$

c i $a > 1$

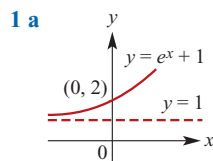
ii $a = 1$



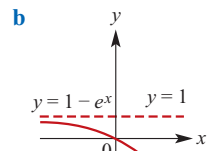
iii $0 < a < 1$



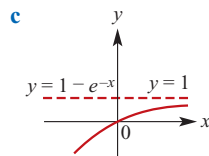
Exercise 5B



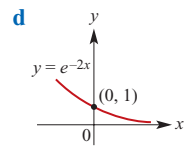
Range = $(1, \infty)$



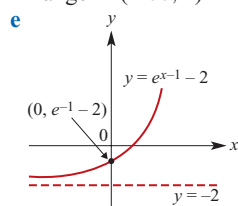
Range = $(-\infty, 1)$



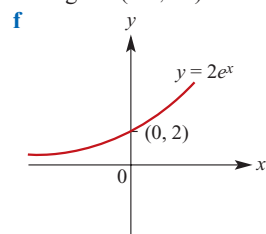
Range = $(-\infty, 1)$



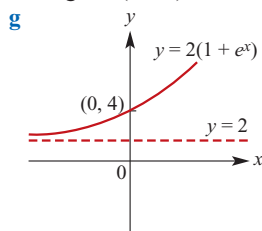
Range = $(0, \infty)$



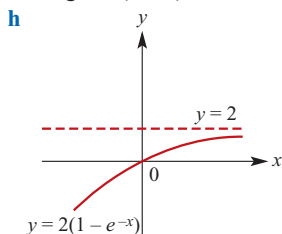
Range = $(-2, \infty)$



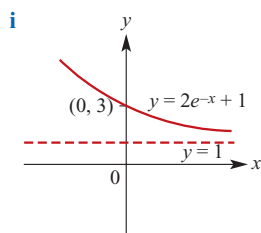
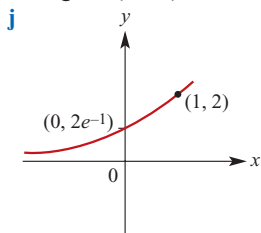
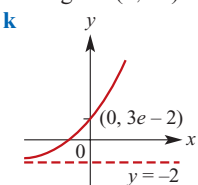
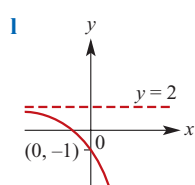
Range = $(0, \infty)$



Range = $(2, \infty)$



Range = $(-\infty, 2)$


 Range = $(1, \infty)$

 Range = $(0, \infty)$

 Range = $(-2, \infty)$

 Range = $(-\infty, 2)$

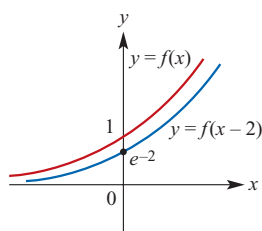
2 a $x = 1.146$ or $x = -1.841$

b $x = -0.443$

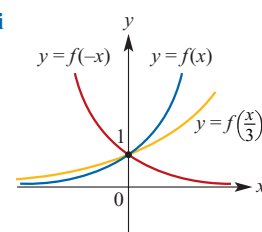
c $x = -0.703$

d $x = 1.857$ or $x = 4.536$

3 a, b i



ii, iii



Exercise 5C

1 a $6x^6y^9$

b $3x^6$

c $\frac{6y^2}{x^2}$

d $18x^8y^4$

e 16

f $\frac{5x^{28}}{y^6}$

g $24x^5y^{10}$

h $2xy^2$

i x^2y^2

2 a 4

b $\frac{1}{2}$

c 4

d $\frac{1}{4}$

e $\frac{3}{5}$

f 3

g $\frac{5}{2}$

h 6

i 4

3 a 1

b 1

c $-\frac{3}{2}$

d 3

e -2

f 4

g $-\frac{10}{3}$

h $-\frac{3}{2}$

i 6

j $\frac{3}{5}$

k $\pm \frac{1}{2}$

4 a 1, 2

b 1

c 2

d 1

e 1, 2

f 0, 1

g 2, 4

h 0, 1

i -1, 2

j -1, 0

Exercise 5D

1 a 3

b -4

c -3

d 6

e 6

f -7

2 a $\log_e 6$

b $\log_e 4$

c $\log_e 10^6 (= 6 \log_e 10)$

d $\log_e 7$

e $\log_e \frac{1}{60} (= -\log_e 60)$

f $\log_e u^3 v^6 (= 3 \log_e uv^2)$

g $7 \log_e x = \log_e x^7$

h $\log_e 1 = 0$

3 a $x = 100$

b $x = 16$

c $x = 6$

d $x = 64$

e $x = e^3 - 5 \approx 15.086$

f $x = \frac{1}{2}$

g $x = -1$

h $x = 10^{-3} = \frac{1}{1000}$

i $x = 36$

4 a $x = 15$

b $x = 5$

c $x = 4$

d $x = 1 (x = -\frac{1}{2} \text{ is not an allowable solution})$

e $x = \frac{3}{2}$

5 a $\log_{10} 27$

b $\log_2 4 (= 2)$

c $\frac{1}{2} \log_{10} \frac{a}{b} = \log_{10} \left(\frac{a}{b} \right)^{\frac{1}{2}}$

d $\log_{10} \left(\frac{10a}{b^{\frac{1}{3}}} \right)$

e $\log_{10} \left(\frac{1}{8} \right) (= -3 \log_{10} 2)$

6 a 1

b 1

c $2\frac{1}{2}$

d 3

e 0

7 a $-x$

b $2 \log_2 x$

c 0

8 a $x = 4$

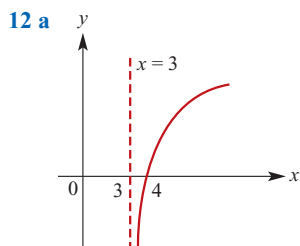
b $x = \frac{3e}{5+2e} \approx 0.7814$

9 a $x = \frac{-1 + \sqrt{1+12e}}{6}$, i.e. $x \approx 0.7997$

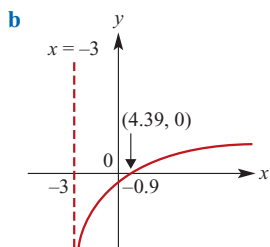
b $x = \log_e 2 \approx 0.6931$

10 $\frac{1}{4}, 2$

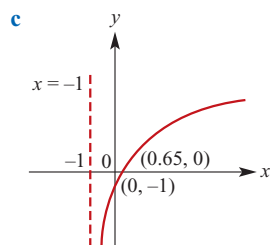
11 $N = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$



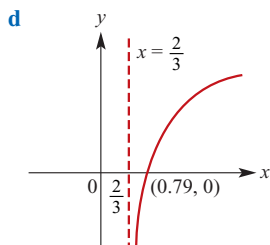
Domain = $(3, \infty)$, range = R



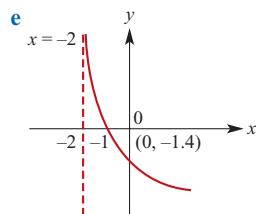
Domain = $(-3, \infty)$, range = R



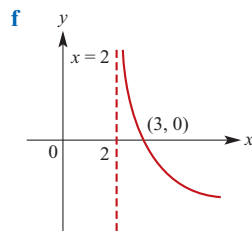
Domain = $(-1, \infty)$, range = R



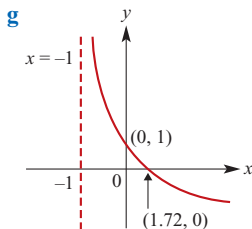
Domain = $\left(\frac{2}{3}, \infty\right)$, range = R



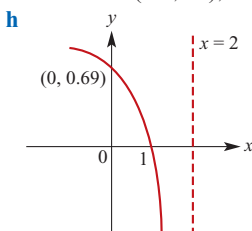
Domain = $(-2, \infty)$, range = R



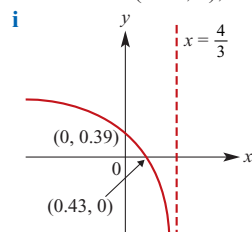
Domain = $(2, \infty)$, range = R



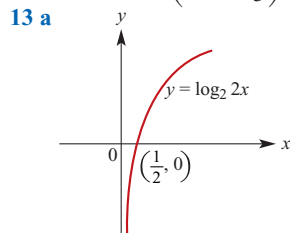
Domain = $(-1, \infty)$, range = R



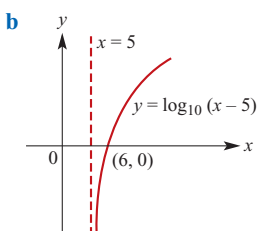
Domain = $(-\infty, 2)$, range = R



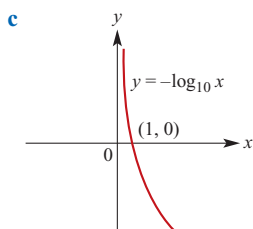
Domain = $\left(-\infty, \frac{4}{3}\right)$, range = R



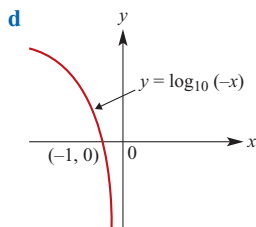
Domain = R^+



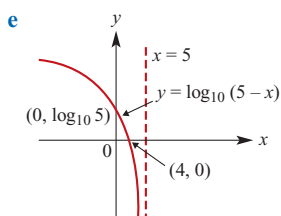
Domain = $(5, \infty)$



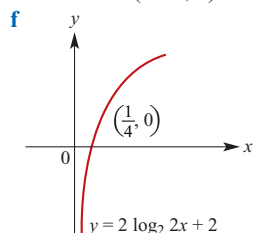
Domain = R^+



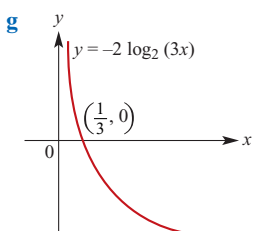
Domain = R^-



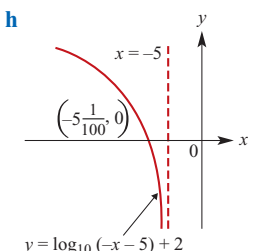
Domain = $(-\infty, 5)$



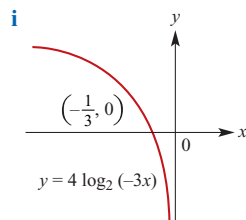
Domain = R^+



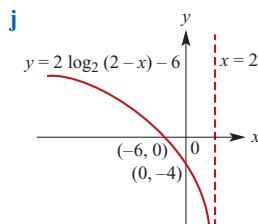
Domain = R^+



Domain = $(-\infty, -5)$



Domain = R^-

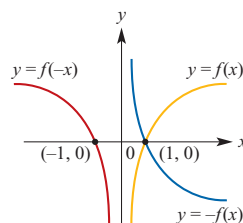


Domain = $(-\infty, 2)$

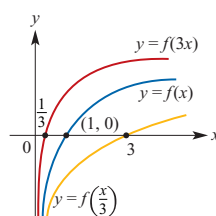
14 a $x = 1.557$

b $x = 1.189$

15 b i, ii



iii, iv



Exercise 5E

1 $a = 2, b = 4$

2 $a = \frac{14}{e-1}, b = \frac{14}{1-e}$ ($a \approx 8.148$,
 $b \approx -8.148$)

3 $a = 200, b = 500$ **4** $a = 250, b = \frac{1}{3} \log_e 5$

5 $a = 3, b = 5$ **6** $a = 2, b = \frac{1}{3} \log_e 5$

7 $a = 2, b = 4$ **8** $a = 2, b = 3$

9 $b = 1, a = \frac{2}{\log_e 2}, c = 8$ ($a \approx 2.885$)

10 $a = \frac{2}{\log_e 2}, b = 4$

Exercise 5F

1 a 2.58

b -0.32

c 2.18

d 1.16

e -2.32

f -0.68

g -2.15

h -1.38

i 2.89

j -1.70

k -4.42

l 5.76

m -6.21

n 2.38

o 2.80

2 a $x < 2.81$ b $x > 1.63$ c $x < -0.68$

d $x \leq 3.89$ e $x \geq 0.57$

3 a $x = \frac{\log_e 5}{\log_e 2} \approx 2.3219$

b $x = \frac{\log_e 7}{\log_e 3} \approx 1.7712$

c $x = 0$ or $x = \frac{\log_e 8}{\log_e 3} \approx 1.8928$

4 a 10

b 3

c 8

5 a 2.9656

b 5.8329

c 2.0850

d 2.8551

e -4.6674

f -0.8273

6 0.544

b 549.3

9 a 9^u

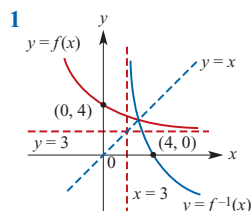
b $u + \frac{1}{2}$

c $\frac{2}{u}$

10 625, $\frac{1}{625}$

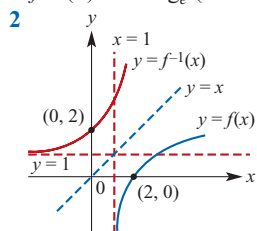
11 $\frac{2}{p}$

Exercise 5G



$f(x) = e^{-x} + 3$

$f^{-1}(x) = -\log_e(x - 3)$



$f(x) = \log_e(x - 1)$

$f^{-1}(x) = e^x + 1$

3 a $f^{-1}(x) = \frac{1}{2}e^x$; domain = R , range = R^+

b $f^{-1}(x) = \frac{1}{2}e^{\frac{x-1}{3}}$; domain = R , range = R^+

c $f^{-1}(x) = \log_e(x - 2)$; domain = $(2, \infty)$, range = R

d $f^{-1}(x) = \log_e(x) - 2$; domain = R^+ , range = R

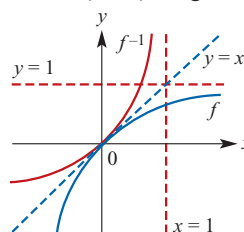
e $f^{-1}(x) = \frac{1}{2}(e^x - 1)$; domain = R , range = $(-\frac{1}{2}, \infty)$

f $f^{-1}(x) = \frac{1}{3}(e^{\frac{x}{4}} - 2)$; domain = R , range = $(-\frac{2}{3}, \infty)$

g $f^{-1}(x) = 10^x - 1$; domain = R , range = $(-1, \infty)$

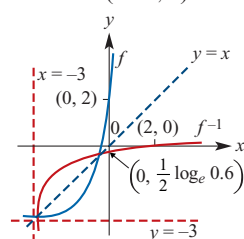
h $f^{-1}(x) = \log_e \frac{1}{2}x + 1$; domain = $(0, \infty)$, range = R

4 a, c



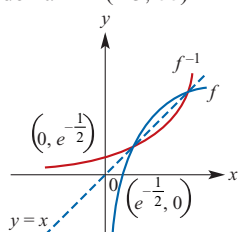
b $f^{-1}(x) = -\log_e(1 - x)$, domain = $(-\infty, 1)$

5 a, c



b $f^{-1}(x) = \frac{1}{2} \log_e \left(\frac{x+3}{5} \right)$, domain = $(-3, \infty)$

6 a, c



b $f^{-1}(x) = e^{\frac{x-1}{2}}$, range = R^+

7 a $x = e^{\frac{y-5}{2}}$ b $x = -\frac{1}{6} \log_e \left(\frac{P}{A} \right)$

c $n = \frac{\log_e \left(\frac{y}{a} \right)}{\log_e x}$ d $x = \log_{10} \left(\frac{y}{5} \right)$

e $x = \frac{1}{2}e^{\frac{5-y}{3}}$

f $n = \frac{1}{2} \left(\frac{\log_e \left(\frac{y}{6} \right)}{\log_e x} \right)$

g $x = \frac{1}{2}(e^y + 1)$

h $x = \log_e \left(\frac{5}{5-y} \right)$

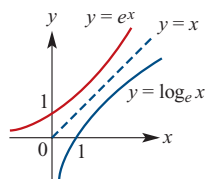
8 a $f^{-1}(x) = \log_e \left(\frac{x+4}{2} \right)$

b (0.895, 0.895), (-3.962, -3.962)

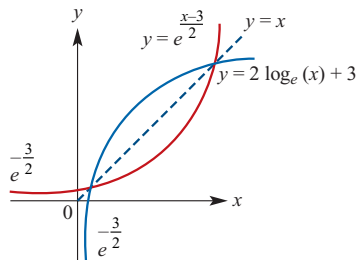
9 a $f^{-1}(x) = e^{\frac{x-4}{2}} - 3$

b (8.964, 8.964), (-2.969, -2.969)

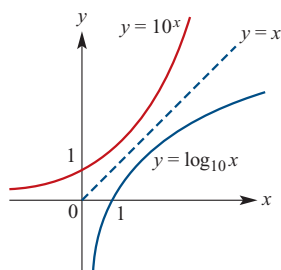
10 a i



ii



iii


 b $f(x)$ and $g(x)$ are inverse functions

Exercise 5H

 1 $m = 0.094$, $d_0 = 41.9237$ 2 $k = 0.349$, $N_0 = 50.25$

 3 a i $N_0 = 20\,000$ ii -0.223

b 6.2 years

 4 a $M_0 = 10$, $k = 4.95 \times 10^{-3}$

b 7.07 grams

c 325 days

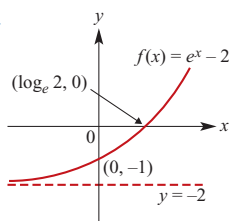
Multiple-choice questions

1 C 2 D 3 B 4 E 5 A

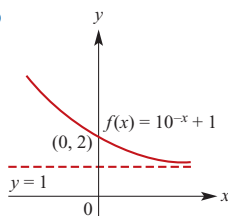
6 C 7 B 8 A 9 C 10 D

Short-answer questions (technology-free)

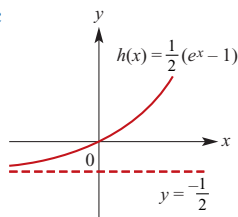
1 a



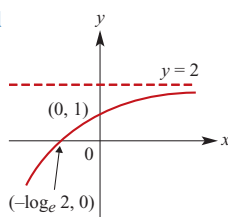
b



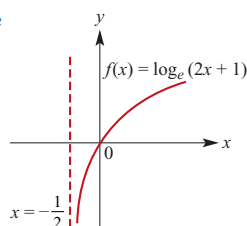
c



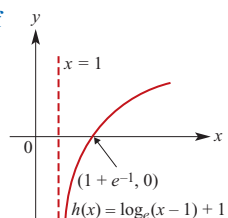
d



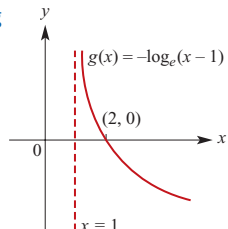
e



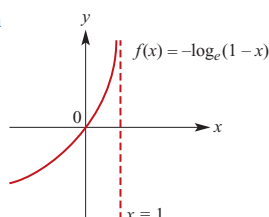
f



g



h



2 a $f^{-1}: (-1, \infty) \rightarrow \mathbb{R}, f^{-1}(x) = \frac{1}{2} \log_e(x+1)$

b $f^{-1}: \mathbb{R} \rightarrow \mathbb{R}, f^{-1}(x) = e^{\frac{x}{3}} + 2$

c $f^{-1}: \mathbb{R} \rightarrow \mathbb{R}, f^{-1}(x) = 10^x - 1$

d $f^{-1}: (2, \infty) \rightarrow \mathbb{R}, f^{-1}(x) = \log_2(x-1)$

3 a $y = e^{2x}$ b $y = 10x$ c $y = 16x^3$

d $y = \frac{x^5}{10}$ e $y = \frac{e^3}{x}$ f $y = e^{2x-3}$

4 a $x = \frac{\log_e 11}{\log_e 3}$ b $x = \frac{\log_e(0.8)}{\log_e 2}$

c $x = \frac{\log_e 3}{\log_e(\frac{2}{3})}$

5 a $x = 1$ b $x = \frac{2}{3}$ c $x = \frac{1}{20}$

d $x = \log_{10} 3$ or $x = \log_{10} 4$

6 a $a = 2, b = 2^{-\frac{2}{3}} - 1$ 7 $10^{\frac{6}{5}} - 1$

8 $\frac{1}{3} \log_e\left(\frac{287}{4}\right)$ 9 2a 10 R

11 a $a = \log_e 5, b = 5, k = 2$

Extended-response questions

1 a 73.5366°C

b 59.5946

2 a 770

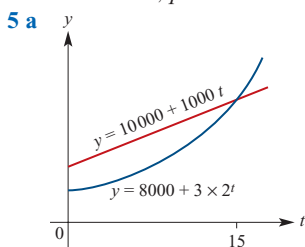
b 1840

3 a $k = 22\,497, \lambda = 0.22$

b \$11\,627

4 a $A = 65\,000, p = 0.064$

b \$47\,200



b i (12.210, 22\,209.62)

ii $t = 12.21$

iii 22\,210

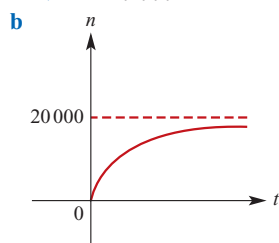
c ii (12.21, 12.21)

d $c = 0.52$

6 a iii a = $\frac{1}{2}$ or a = 1

iv If $a = 1, e^{-2B} = 1$ and $B = 0, A \in \mathbb{R}^+$. If $a = \frac{1}{2}, B = \frac{1}{2} \log_e 2.$

v $A = 20\,000$

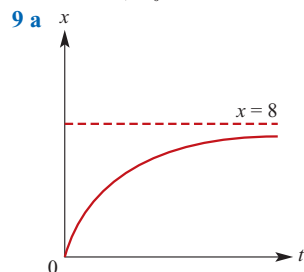


c $\frac{\log_e(0.1)}{\frac{1}{2} \log_e(\frac{1}{2})} = \frac{2 \log_e 10}{\log_e 2} \approx 6.644$

After 6.65 hours the population is 18 000.

7 a 75 b 2.37 c 0.646

8 $k = -0.5, A_0 = 100$



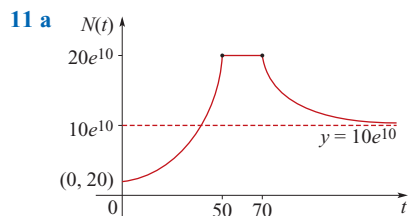
b i 0 grams

ii 2.64 grams

iii 6.92 grams

c 10.4 minutes

10 a $k = 0.235$ b 22.7° c 7.17 minutes



b i $N(10) = 147.78$

ii $N(40) = 59\,619.16$

iii $N(60) = 20e^{10} = 440\,529.32$

iv $N(80) = 220\,274.66$

c i 25 days

ii 35 days

Chapter 6

Exercise 6A

1 a $\frac{5\pi}{18}$

b $\frac{34\pi}{45}$

c $\frac{25\pi}{18}$

d $\frac{17\pi}{9}$

e $\frac{7\pi}{3}$

f $\frac{49\pi}{18}$

2 a 60°

b 150°

c 240°

d 140°

e 630°

f 252°

3 a 45.84°

b 93.97°

c 143.24°

d 226.89°

e 239.50°

f 340.91°

4 a 0.65

b 1.29

c 2.01

d 2.13

e 5.93

f 2.31

Exercise 6B

1 a 0, 1

b -1, 0

c -1, 0

d 1, 0

e 0, -1

f 0, 1

2 a 0

b Not defined

c Not defined

d 0

e Not defined

3 a -1

b 0

c -1

d 0

e 1

f 1

4 a 0.99

b 0.52

c -0.87

d 0.92

e -0.67

f -0.23

g -0.99

h 0.44

i -34.23

j -2.57

k 0.95

l 0.75

Exercise 6C

- 1 a 0.52 b -0.68 c 0.52 d 0.4
 e -0.52 f 0.68 g -0.4 h -0.68
 2 a 0.64, 2.5 b 0.64, 5.64
 c 3.61, 5.816 d 1.77, 4.51
 3 a 17.46, 162.54 b 66.42, 293.58
 c 233.13, 306.87 d 120, 240

Exercise 6D

- 1 0.6 2 0.6 3 -0.7 4 0.3 5 -0.3
 6 $1\frac{3}{7}$ 7 -0.3 8 0.6 9 -0.6 10 -0.3

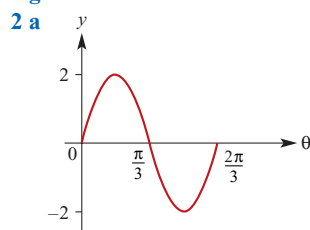
Exercise 6E

- 1 a $\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\frac{1}{\sqrt{3}}$ b $-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 1$
 c $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1$ d $-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3}$
 e $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1$ f $-\frac{1}{2}, \frac{\sqrt{3}}{2}, -\frac{1}{\sqrt{3}}$
 2 a $\frac{1}{\sqrt{2}}$ b $-\frac{1}{2}$ c $-\frac{\sqrt{3}}{2}$ d $\frac{1}{2}$
 e $-\frac{1}{2}$ f $-\frac{1}{\sqrt{2}}$ g $-\frac{1}{\sqrt{2}}$ h $\frac{1}{2}$
 i $\frac{1}{2}$ j $\frac{\sqrt{3}}{2}$ k $\frac{1}{2}$ l $\frac{\sqrt{3}}{2}$
 m $\frac{1}{\sqrt{2}}$ n $-\frac{1}{2}$ o $\frac{1}{\sqrt{2}}$ p $-\frac{1}{2}$
 3 a $60^\circ, 300^\circ$ b $60^\circ, 120^\circ$ c $210^\circ, 330^\circ$
 d $120^\circ, 240^\circ$ e $60^\circ, 120^\circ$ f $150^\circ, 210^\circ$
 4 a $-\frac{\pi}{6}$ or $-\frac{5\pi}{6}$ b $-\frac{\pi}{6}$ or $\frac{\pi}{6}$ c $-\frac{5\pi}{6}$ or $\frac{5\pi}{6}$
 5 a $\frac{5\pi}{6}$ b $\frac{5\pi}{6}$ c $\frac{2\pi}{3}$
 d $\frac{\pi}{2} + b$ e $\frac{\pi}{2} + b$
 6 a $\frac{7\pi}{6}$ b $\frac{4\pi}{3}$ c $\frac{7\pi}{6}$
 d $\frac{3\pi}{2} - b$ e $b + \pi$ f $b + \pi$
 7 a $\frac{7\pi}{4}, \frac{3\pi}{4}$ b $\frac{3\pi}{4}, \frac{5\pi}{4}$ c $\frac{5\pi}{6}, \frac{7\pi}{6}$
 d $\frac{7\pi}{6}, \frac{11\pi}{6}$ e $\frac{\pi}{4}, \frac{7\pi}{4}$ f $\frac{2\pi}{3}, \frac{4\pi}{3}$
 8 a $\frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$
 b $\frac{5\pi}{18}, \frac{7\pi}{18}, \frac{17\pi}{18}, \frac{19\pi}{18}, \frac{29\pi}{18}, \frac{31\pi}{18}$
 c $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

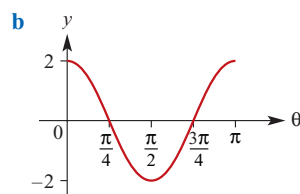
- d $\frac{\pi}{12}, \frac{7\pi}{12}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{17\pi}{12}, \frac{23\pi}{12}$
 e $\frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}$
 f $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$
 g $\frac{5\pi}{18}, \frac{7\pi}{18}, \frac{17\pi}{18}, \frac{19\pi}{18}, \frac{29\pi}{18}, \frac{31\pi}{18}$
 h $\frac{5\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{23\pi}{12}$
 i $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

Exercise 6F

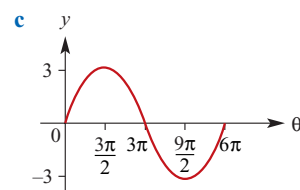
- 1 a i $\frac{2\pi}{3}$ ii 3
 b i $\frac{2\pi}{3}$ ii 2
 c i π ii $\frac{1}{2}$
 d i $\frac{6\pi}{2}$ ii 2
 e i $\frac{\pi}{2}$ ii 3
 f i 2π ii $\frac{1}{2}$
 g i 4π ii $\frac{2}{3}$



Amplitude = 2

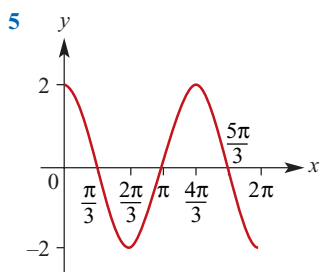
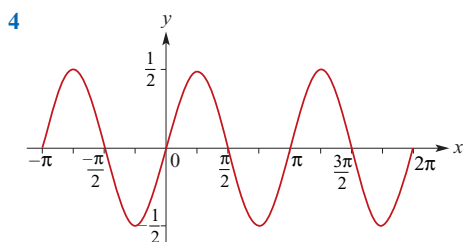
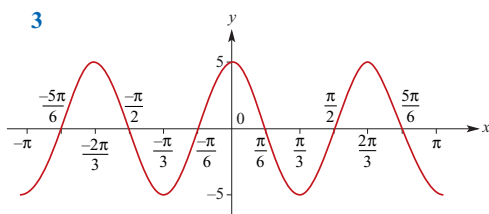
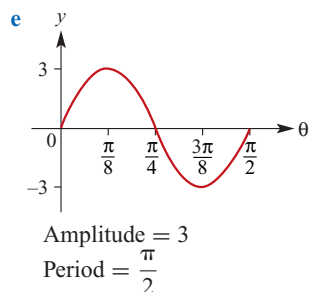
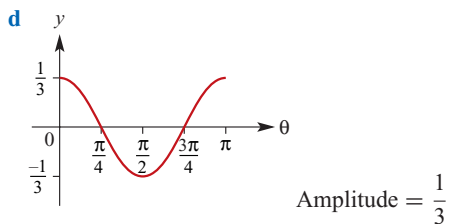
Period = $\frac{2\pi}{3}$ 

Amplitude = 2

Period = π 

Amplitude = 3

Period = 6π

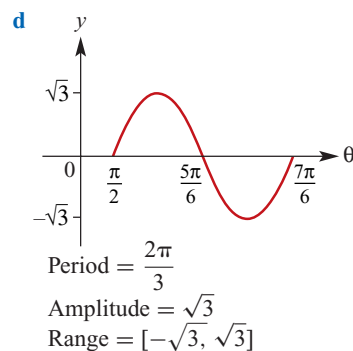
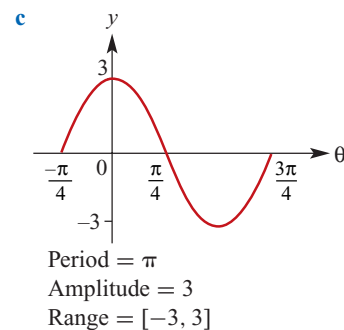
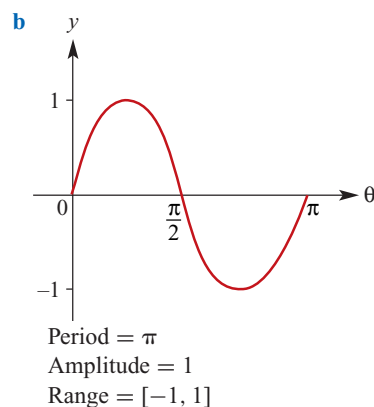
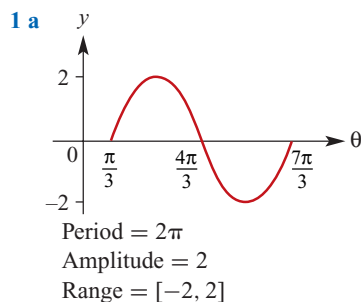


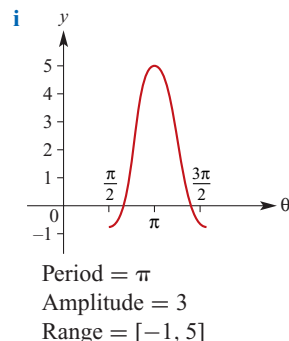
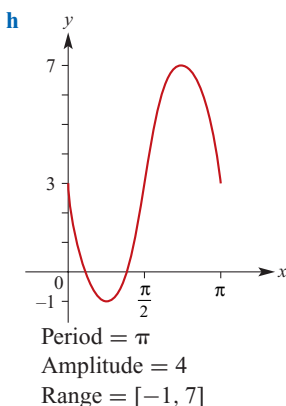
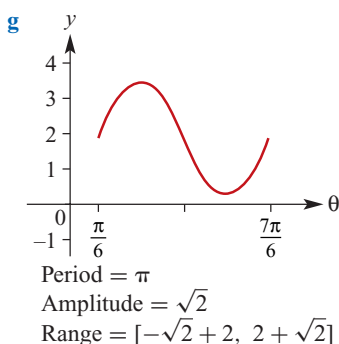
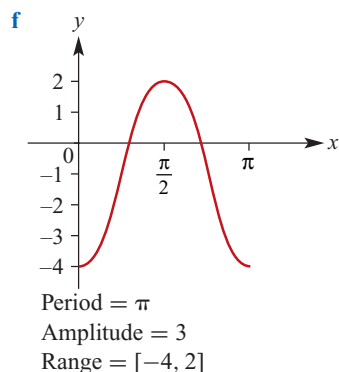
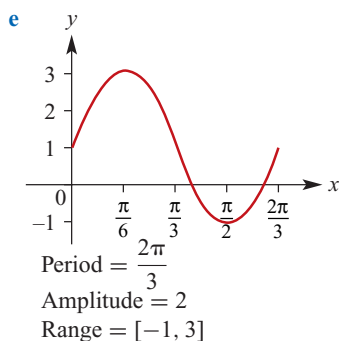
6 $y = 2 \sin \frac{x}{3}$

7 $y = \frac{1}{2} \cos \frac{x}{3}$

8 $y = \frac{1}{2} \sin \frac{x}{2}$

Exercise 6G





2 a $y = \frac{1}{2} \cos\left(\frac{1}{3}\left(x - \frac{\pi}{4}\right)\right)$

b $y = 2 \cos\left(x - \frac{\pi}{4}\right)$

c $y = -\frac{1}{3} \cos\left(x - \frac{\pi}{3}\right)$

3 a ■ Dilation of factor 3 from the x -axis

■ Dilation of factor $\frac{1}{2}$ from the y -axis

■ Reflection in the x -axis

b ■ Dilation of factor 3 from the x -axis

■ Dilation of factor $\frac{1}{2}$ from the y -axis

■ Reflection in the x -axis

■ Translation of $\frac{\pi}{3}$ units in the positive direction of the x -axis

c ■ Dilation of factor 3 from the x -axis

■ Dilation of factor $\frac{1}{2}$ from the y -axis

■ Translation of $\frac{\pi}{3}$ units in the positive direction of the x -axis

■ Translation of 2 units in the positive direction of the y -axis

d ■ Dilation of factor 2 from the x -axis

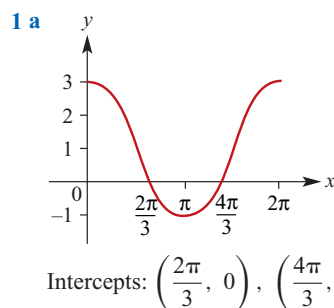
■ Dilation of factor $\frac{1}{2}$ from the y -axis

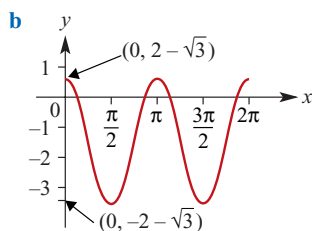
■ Reflection in the x -axis

■ Translation of $\frac{\pi}{3}$ units in the positive direction of the x -axis

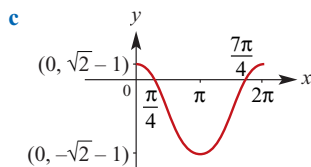
■ Translation of 5 units in the positive direction of the y -axis

Exercise 6H

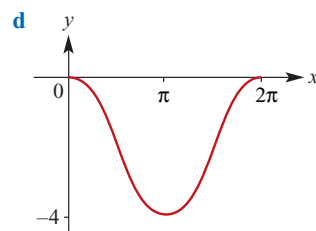




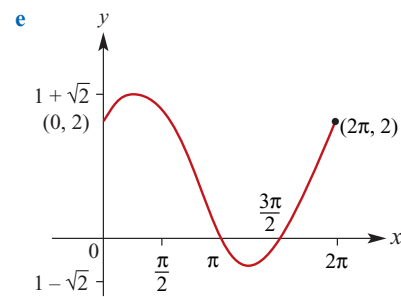
Intercepts: $\left(\frac{\pi}{12}, 0\right), \left(\frac{11\pi}{12}, 0\right),$
 $\left(\frac{13\pi}{12}, 0\right), \left(\frac{23\pi}{12}, 0\right)$



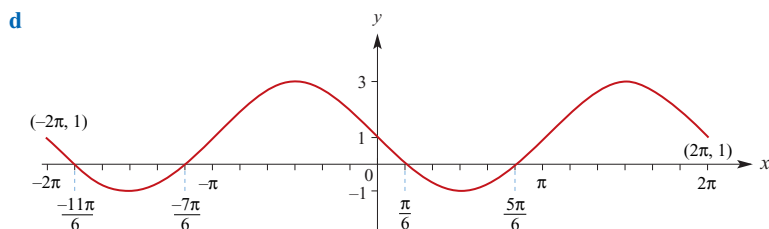
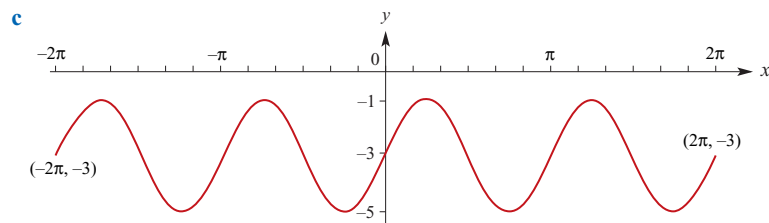
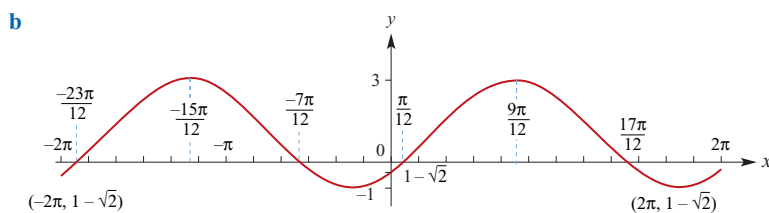
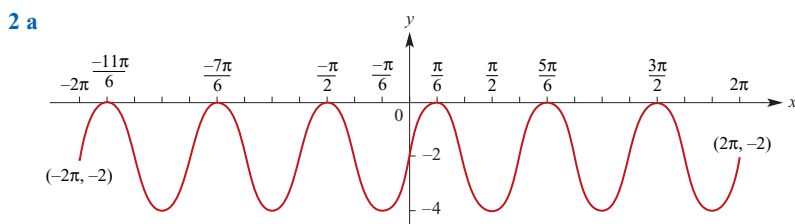
Intercepts: $\left(\frac{\pi}{4}, 0\right), \left(\frac{7\pi}{4}, 0\right)$



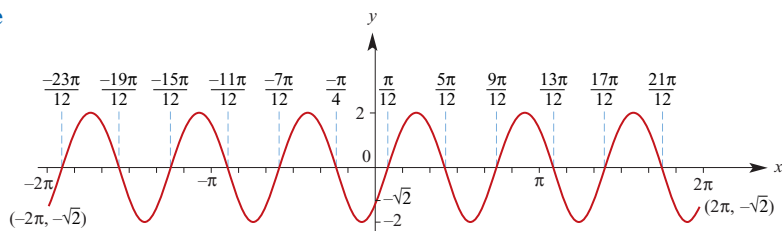
Intercepts: $(0, 0), (2\pi, 0)$



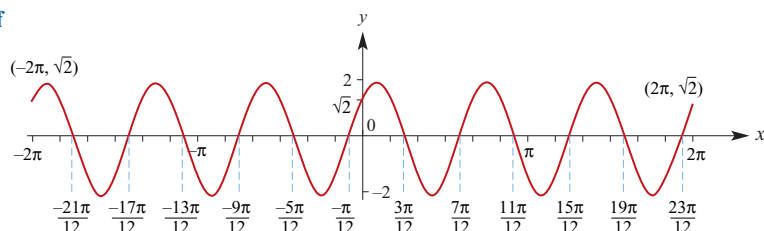
Intercepts: $(\pi, 0), \left(\frac{3\pi}{2}, 0\right)$



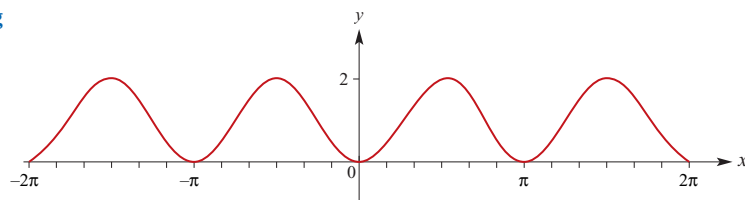
e



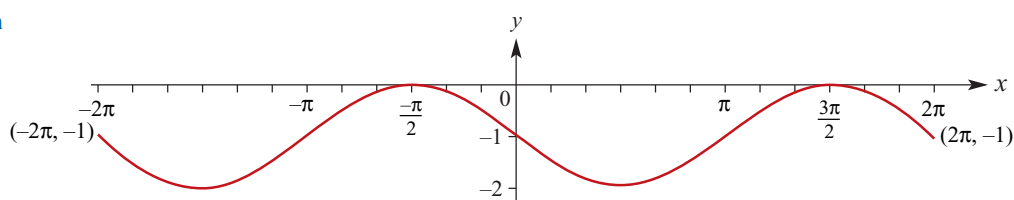
f



g

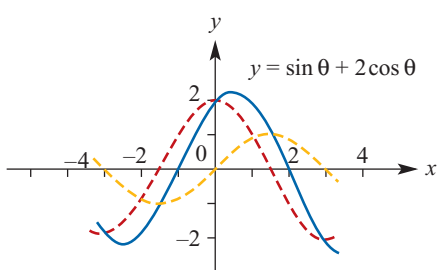


h

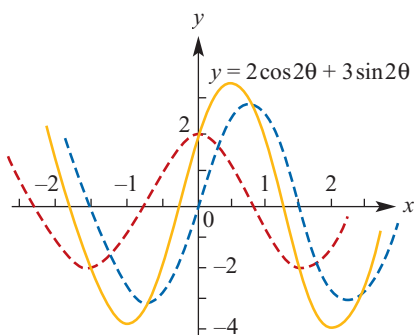


Exercise 6I

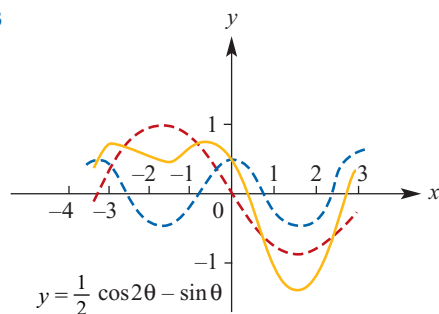
1



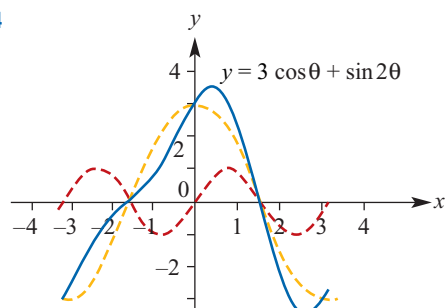
2



3

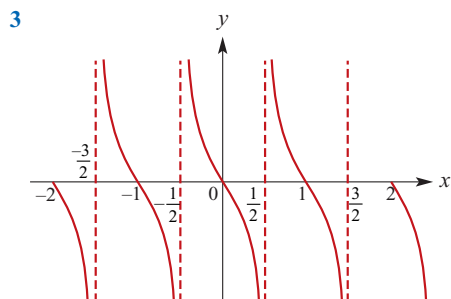
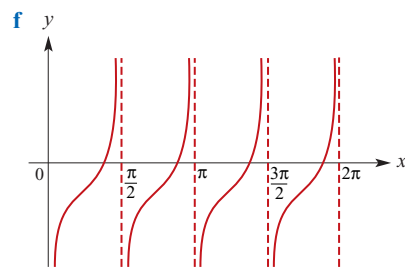
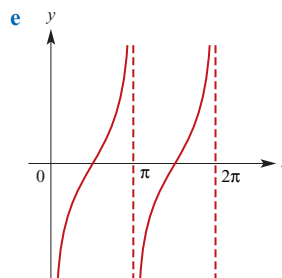
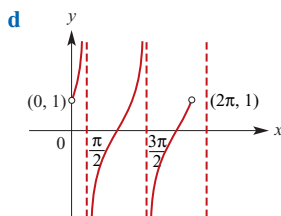
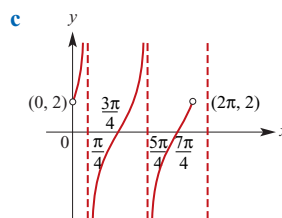
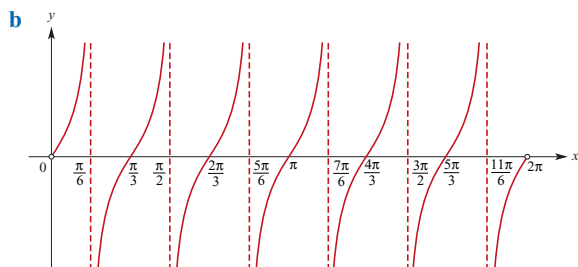
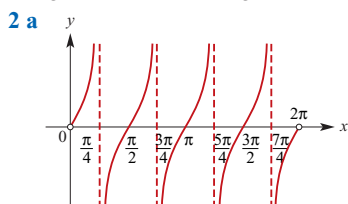


4

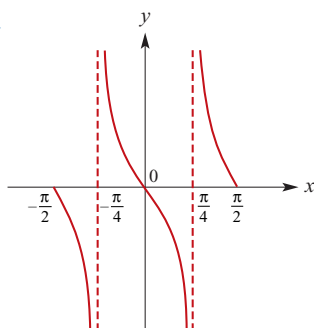


- 1 $A = 3, n = \frac{\pi}{4}$
- 2 $A = 0.5, \varepsilon = \frac{-\pi}{3}$
- 3 $A = 3, b = 5, n = 3$
- 4 $A = -4, n = \frac{\pi}{6}$
- 5 $A = 4, n = \frac{\pi}{4}, \varepsilon = \frac{-\pi}{2}$
(Note: ε can take infinitely many values.)
- 6 $A = 2, n = \frac{\pi}{3}, \varepsilon = \frac{-\pi}{6}$
(Note: ε can take infinitely many values.)
- 7 $A = 4, n = \frac{\pi}{4}, \varepsilon = \frac{-\pi}{2}$ and $d = 2$
(Note: ε can take infinitely many values.)
- 8 $A = 2, n = \frac{\pi}{3}, \varepsilon = \frac{-\pi}{6}$ and $d = 2$
(Note: ε can take infinitely many values.)

1 a $\frac{\pi}{3}$ **b** 2π **c** $\frac{2\pi}{3}$ **d** 1 **e** 2



4 a



5 a $\frac{\pi}{6}, \frac{7\pi}{6}, \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}, \frac{17\pi}{16}, \frac{21\pi}{16}, \frac{25\pi}{16}, \frac{29\pi}{16}$

b $\frac{\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}$

c $\frac{5\pi}{12}, \frac{11\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}$

d $\frac{\pi}{4}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{5\pi}{4}, \frac{19\pi}{12}, \frac{23\pi}{12}$

e 0.4636, 3.6052

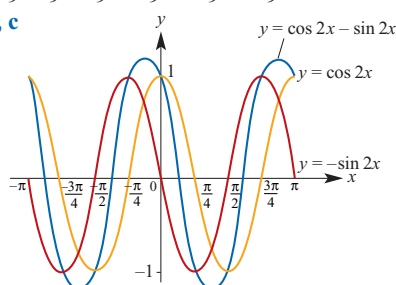
f 1.1071, 4.2487

g $\frac{3\pi}{8}, \frac{7\pi}{8}, \frac{11\pi}{8}, \frac{15\pi}{8}$

h $\frac{\pi}{18}, \frac{7\pi}{18}, \frac{13\pi}{18}, \frac{19\pi}{18}, \frac{25\pi}{18}, \frac{31\pi}{18}$

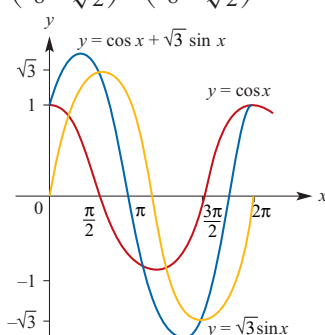
i $\frac{\pi}{9}, \frac{4\pi}{9}, \frac{7\pi}{9}, \frac{10\pi}{9}, \frac{13\pi}{9}, \frac{16\pi}{9}$

6 a, c



b $\left(\frac{-5\pi}{8}, \frac{-1}{\sqrt{2}}\right), \left(\frac{-\pi}{8}, \frac{1}{\sqrt{2}}\right), \left(\frac{3\pi}{8}, \frac{-1}{\sqrt{2}}\right), \left(\frac{7\pi}{8}, \frac{1}{\sqrt{2}}\right)$

7 a, c



b $\left(\frac{\pi}{6}, \frac{\sqrt{3}}{2}\right), \left(\frac{7\pi}{6}, -\frac{\sqrt{3}}{2}\right)$

8 a $\frac{7\pi}{24}, \frac{19\pi}{24}, \frac{31\pi}{24}, \frac{43\pi}{24}$

b $\frac{5\pi}{12}, \frac{11\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}$

c $\frac{11\pi}{36}, \frac{23\pi}{36}, \frac{35\pi}{36}, \frac{47\pi}{36}, \frac{59\pi}{36}, \frac{71\pi}{36}$

9 $n = 3, A = 5$ 10 $n = \frac{\pi}{2}, A = 6$

Exercise 6L

1 a i 2π ii 4π iii -4π

b i $\frac{4\pi}{3}, \frac{8\pi}{3}$ ii $\frac{14\pi}{3}, \frac{10\pi}{3}$ iii $\frac{-14\pi}{3}, \frac{-10\pi}{3}$

2 a $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$

b $\frac{2\pi n}{3} + \frac{\pi}{9}$ or $\frac{2\pi n}{3} + \frac{2\pi}{9}, n \in \mathbb{Z}$

c $x = n\pi + \pi/3, n \in \mathbb{Z}$

3 a $\frac{\pi}{6}, \frac{5\pi}{6}$ b $\frac{\pi}{12}, \frac{11\pi}{12}$ c $\frac{\pi}{3}, \frac{5\pi}{6}$

4 a $\frac{-11\pi}{6}, \frac{-7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

5 $\frac{-\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}$

6 a $x = n\pi - \frac{\pi}{6}$ or $x = n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$

b $x = \frac{n\pi}{2} - \frac{\pi}{12}, n \in \mathbb{Z}$

c $x = 2n\pi + \frac{5\pi}{6}$ or $x = 2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$

7 $x = \frac{(4n-1)\pi}{4}$ or $x = n\pi, n \in \mathbb{Z}$,

$\left\{\frac{-5\pi}{4}, -\pi, \frac{\pi}{4}, 0, \frac{3\pi}{4}, \pi, \frac{7\pi}{4}\right\}$

8 $x = \frac{n\pi}{3}; x = \frac{-\pi}{3}$ or $\frac{-2\pi}{3}$ or $-\pi$ or 0

9 $x = \frac{6n-1}{12}$ or $x = \frac{3n+2}{6};$

$x = \frac{-2}{3}, \frac{-7}{12}, \frac{-1}{6}, \frac{-1}{12}, \frac{1}{3}, \frac{5}{12}, \frac{5}{6}, \frac{11}{12},$

Exercise 6M

1 $-\frac{4}{5}, -\frac{4}{3}$

2 $-\frac{12}{13}, -\frac{5}{12}$

3 $-\frac{2\sqrt{6}}{5}, -2\sqrt{6}$

Exercise 6N

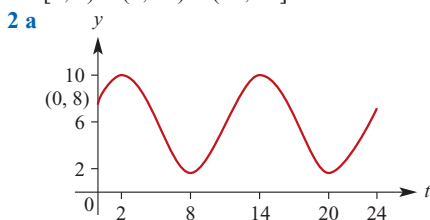
1 a i Amplitude = $1\frac{1}{2}$

ii Period = 12

iii $d(t) = 3.5 - 1.5 \cos\left(\frac{\pi}{6}t\right)$

iv 1.5 m

b $[0, 3) \cup (9, 15) \cup (21, 24]$

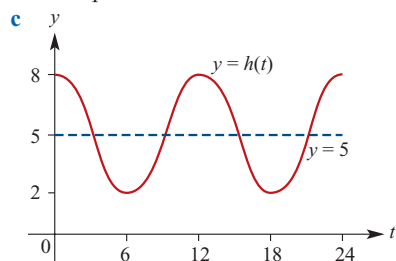


b 2:00

c 8:00, 20:00

3 a $A = 3, n = \frac{\pi}{6}, \varepsilon = \frac{\pi}{2}, b = 5$

b i 2:21 a.m., 9:39 a.m., 2:21 p.m., 9:39 p.m.



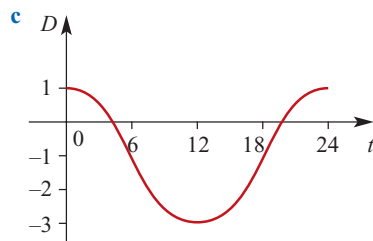
4 a 5 b 1

c $t = 0.524$ s, 2.618 s, 4.712 s

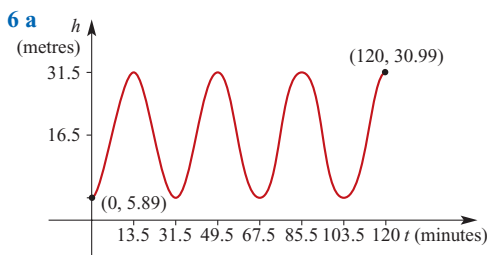
d $t = 0$ s, 1.047 s, 2.094 s

e Particle oscillates about the point $x = 3$ from $x = 1$ to $x = 5$.

5 a 19.5°C b $D = -1 + 2 \cos\left(\frac{\pi t}{12}\right)$



d $\{t : 4 < t < 20\}$.



b 5.89 metres

d 6 times

f 4.21 metres

c 27.51 seconds

e 20 times

g 13.9 metres

Multiple-choice questions

1 C 2 E 3 D 4 A 5 A
6 C 7 B 8 C 9 E 10 B

Short-answer questions (technology-free)

1 a $\frac{\pi}{6}, \frac{5\pi}{6}$ b $-\frac{2\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}$

c $-\frac{\pi}{6}, \frac{\pi}{6}, \frac{11\pi}{6}$

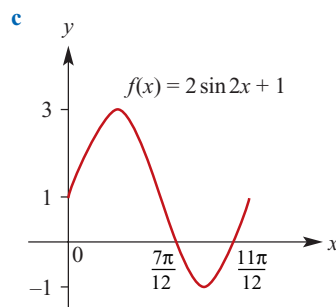
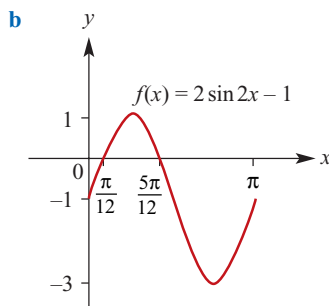
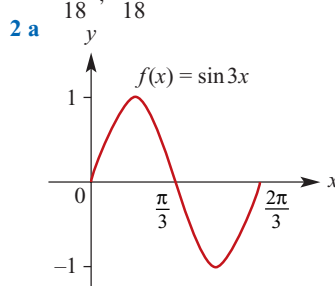
d $-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

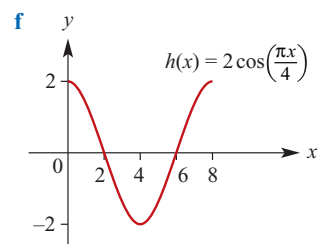
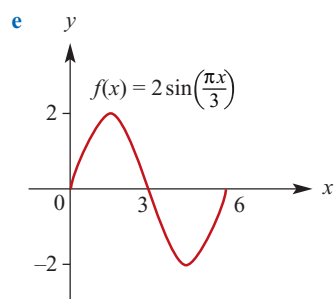
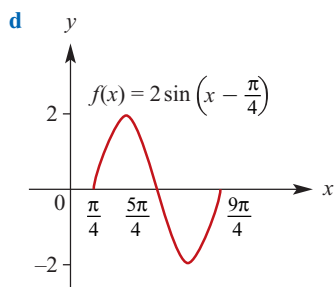
e $-\frac{\pi}{6}, -\frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ f $-\frac{\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}$

g $-\frac{3\pi}{8}, -\frac{5\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}$

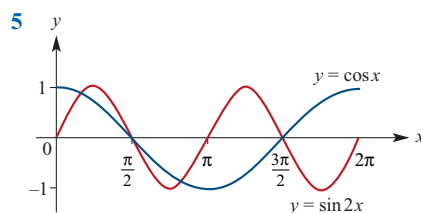
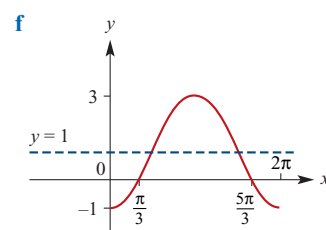
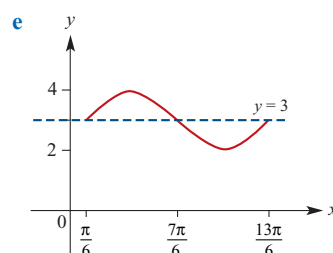
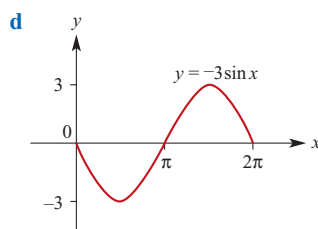
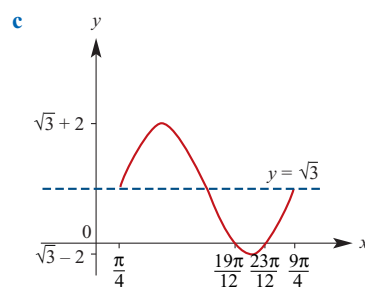
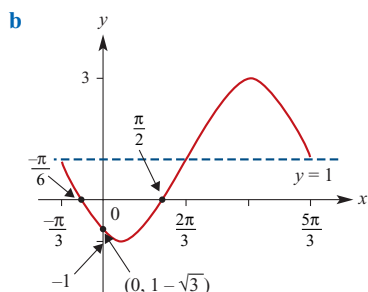
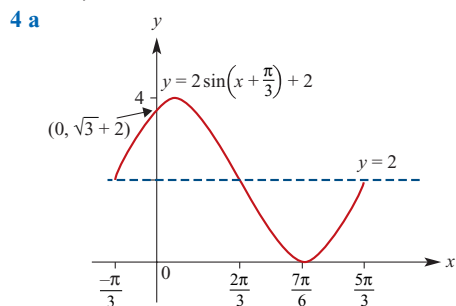
h $-\frac{7\pi}{18}, -\frac{11\pi}{18}, \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}$

$\frac{25\pi}{18}, \frac{29\pi}{18}$

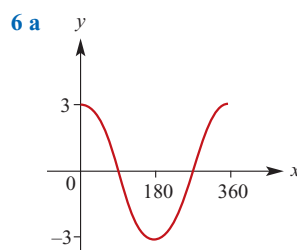


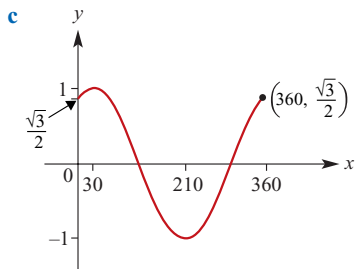
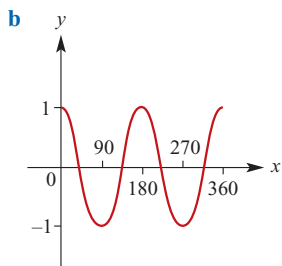


- 3 a** 30, 150
b 45, 135, 225, 315
c 240, 300
d 90, 120, 270, 300
e 120, 240



- a** 4 solutions
c 2 solutions
b 4 solutions



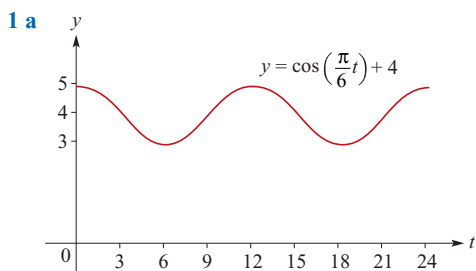


7 a $-\frac{2\pi}{3}$ or $\frac{\pi}{3}$ **b** $-\frac{\pi}{4}$ or $\frac{3\pi}{4}$

c $-\frac{5\pi}{8}$ or $-\frac{\pi}{8}$ or $\frac{3\pi}{8}$ or $\frac{7\pi}{8}$

d $-\frac{2\pi}{3}$ or $-\frac{\pi}{6}$ or $\frac{\pi}{3}$ or $\frac{5\pi}{3}$

Extended-response questions



b 9:00, 15:00 **c** 8:00, 16:00

2 a Maximum = 210 cm, minimum = 150 cm, mean = 180 cm

b $y = 30 \sin\left(\frac{\pi t}{6} - \frac{\pi}{2}\right) + 180$;

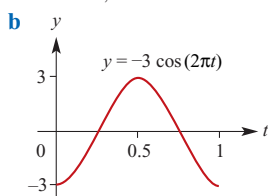
$A = 30, n = \frac{\pi}{6}, b = 180, \epsilon = -\frac{\pi}{2}$

c i 165 cm

ii $180 - 15\sqrt{3} \approx 154$ cm

d $\approx 4:24, \approx 7:36$

3 a $a = -3, n = 2\pi$



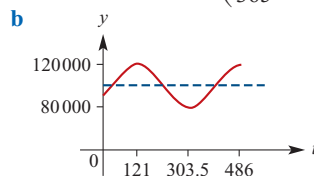
c i $t = \frac{1}{3}$ second **ii** $t = \frac{1}{6}$ second

d $t = 0.196$ second

4 a $a = 20\,000, b = 100\,000, n = \frac{2\pi}{365}$,

$\epsilon \approx 5.77$,

i.e. $y = 20\,000 \sin\left(\frac{2\pi t}{365} + 5.77\right) + 100\,000$



c i $t = 242.7, t = 364.3$

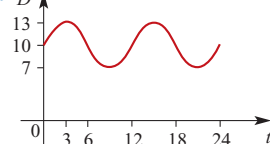
ii $t = 60.2, t = 181.8$

d $\approx 117\,219 \text{ m}^3/\text{day}$

5 a i 1.83×10^{-3} hours **ii** 11.79 hours

b April 25 ($t = 3.856$), August 14 ($t = 7.477$)

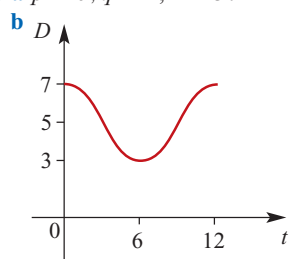
6 a D



b $\{t: D(t) \geq 8.5\} = \{t: 0 \leq t \leq 7\} \cup \{t: 11 \leq t \leq 19\} \cup \{t: 23 \leq t \leq 24\}$

c 12.898 m

7 a $p = 5, q = 2, r = 30$



c A ship can enter 2 hours after low tide.

8 a i $25\sqrt{3}$

ii 30

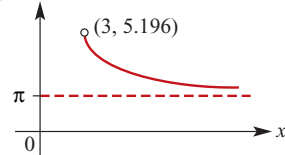
b 2.27, 0.53

d $b = 8$

e $\theta = 0.927$ or 1.837

f $a = 4\sqrt{3}$

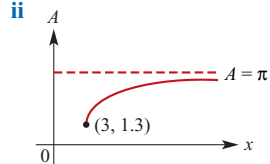
9 c



d i 2.055 **ii** 0.858 **iii** 0.0738 **iv** 0.0041

e $nr \tan\left(\frac{\pi}{n}\right)$

f i $n \sin\left(\frac{\pi}{n}\right) \cos\left(\frac{\pi}{n}\right)$



Chapter 7

Exercise 7A

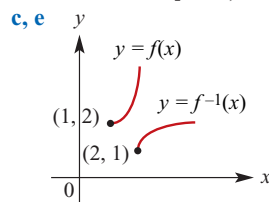
- 1 a i $f \circ g(x) = 3 \sin(2|x|)$
 $g \circ f(x) = 3 |\sin 2x|$
 ii Range of $f \circ g = [-3, 3]$,
 domain of $f \circ g = R$,
 range of $g \circ f = [0, 3]$,
 domain of $g \circ f = R$
- b i $f \circ g(x) = -2 \cos(2|x|)$
 $g \circ f(x) = 2 |\cos 2x|$
 ii Range of $f \circ g = [-2, 2]$,
 domain of $f \circ g = R$,
 range of $g \circ f = [0, 2]$,
 domain of $g \circ f = R$
- c i $f \circ g(x) = e^{|x|}$, $g \circ f(x) = e^x$
 ii Range of $f \circ g = [1, \infty)$,
 domain of $f \circ g = R$,
 range of $g \circ f = (0, \infty)$,
 domain of $g \circ f = R$
- d i $f \circ g(x) = e^{2|x|} - 1$, $g \circ f(x) = |e^{2x} - 1|$
 ii Range of $f \circ g = [0, \infty)$,
 domain of $f \circ g = R$,
 range of $g \circ f = [0, \infty)$,
 domain of $g \circ f = R$
- e i $f \circ g(x) = -2e^{|x|} - 1$, $g \circ f(x) = 2e^x + 1$
 ii Range of $f \circ g = (-\infty, -3]$,
 domain of $f \circ g = R$,
 range of $g \circ f = (1, \infty)$,
 domain of $g \circ f = R$
- f i $f \circ g(x) = \log_e(2|x|)$,
 $g \circ f(x) = |\log_e 2x|$
 ii Range of $f \circ g = R$,
 domain of $f \circ g = R \setminus \{0\}$,
 range of $g \circ f = [0, \infty)$,
 domain of $g \circ f = R^+$
- g i $f \circ g(x) = \log_e(|x| - 1)$,
 $g \circ f(x) = |\log_e(x - 1)|$
 ii Range of $f \circ g = R$,
 domain of $f \circ g = R \setminus [-1, 1]$,
 range of $g \circ f = [0, \infty)$,
 domain of $g \circ f = (1, \infty)$
- h i $f \circ g(x) = -\log_e(|x|)$,
 $g \circ f(x) = |\log_e x|$
 ii Range of $f \circ g = R$,
 domain of $f \circ g = R \setminus \{0\}$,
 range of $g \circ f = [0, \infty)$,
 domain of $g \circ f = R^+$
- 2 a $h(x) = f \circ g(x)$ where $f(x) = e^x$ and
 $g(x) = x^3$
 b $h(x) = f \circ g(x)$ where $f(x) = \cos x$ and
 $g(x) = |2x|$
 c $h(x) = f \circ g(x)$ where $f(x) = x^n$ and
 $g(x) = x^2 - 2x$

- d $h(x) = f \circ g(x)$ where $f(x) = \cos x$ and
 $g(x) = x^2$
 e $h(x) = f \circ g(x)$ where $f(x) = x^2$ and
 $g(x) = \cos x$
 f $h(x) = f \circ g(x)$ where $f(x) = x^4$ and
 $g(x) = x^2 - 1$
 g $h(x) = f \circ g(x)$ where $f(x) = \log_e x$ and
 $g(x) = x^2$
 h $h(x) = f \circ g(x)$ where $f(x) = |x|$ and
 $g(x) = \cos 2x$
 i $h(x) = f \circ g(x)$ where $f(x) = x^3 - 2x$ and
 $g(x) = x^2 - 2x$

- 3 a $2e^{2x}$ b $\frac{1}{2} \log_e \left(\frac{x}{2}\right)$ c e^{x^2}
- 4 a $f^{-1}(x) = -\frac{1}{2} \log_e x$, $g^{-1}(x) = (x - 1)^{\frac{1}{3}}$
 b $f \circ g(x) = e^{-2(x^3+1)}$, $\text{ran } f \circ g = R^+$,
 $g \circ f(x) = e^{-6x} + 1$, $\text{ran } g \circ f = (1, \infty)$
- 5 a $f^{-1}(x) = \frac{1}{x} - 1$ b $x = \frac{\sqrt{5} - 1}{2}$
- 6 a $f^{-1}(x) = e^x - 1$, $\text{dom } f^{-1} = R$,
 $g^{-1}(x) = \sqrt{x + 1} - 1$, $\text{dom } g^{-1} = (-1, \infty)$
 b $\log_e(x^2 + 2x + 1)$
- 7 $f \circ g(x) = \log_e \left(\frac{1}{x}\right) f(x) + f \circ g(x) = 0$
- 8 x
- 9 a $f(g(x)) = (x^2 - 8)(x^2 - 10)$,
 $g(f(x)) = (x - 4)^2(x - 6)^2 - 4$
 b $x = 1$
- 10 $x = \pm\sqrt{6}$ or $x = \pm\sqrt{2}$
- 12 a = 6 b = 0 $g(x) = e^{6x}$

Exercise 7B

- 1 a a = 1 b $[2, \infty)$



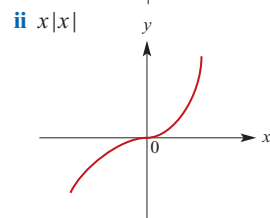
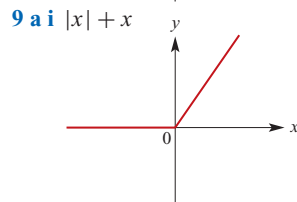
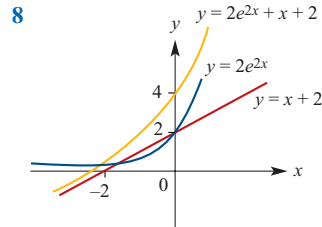
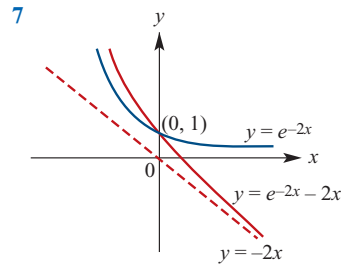
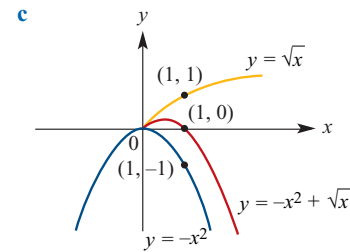
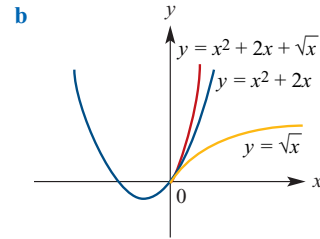
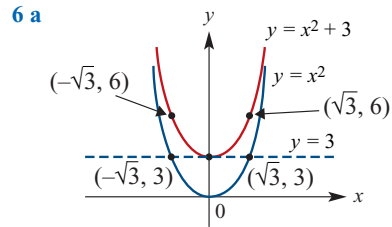
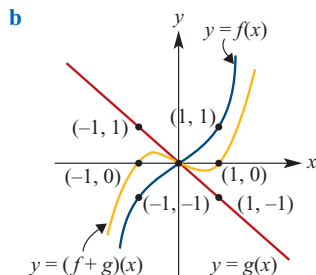
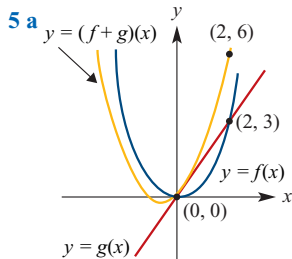
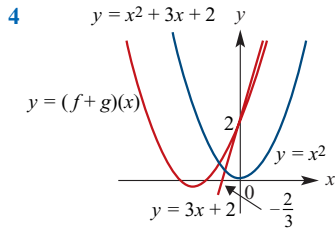
- d $f^{-1}: [2, \infty) \rightarrow R$, $f^{-1}(x) = \sqrt{x - 2} + 1$,
 domain of $f^{-1} = [2, \infty)$, range = $[1, \infty)$
- 2 a $\pm\sqrt{\frac{x}{2}} = y$, domain = $R^+ \cup \{0\}$
- b $\pm\sqrt{\frac{x+1}{2}} = y$, domain = $[-1, \infty)$
- c $y = \pm\sqrt{x} - 1$, domain = $R^+ \cup \{0\}$
- d $y = \pm\sqrt{x+1} - 1$, domain = $[-1, \infty)$
- e $y = \pm\sqrt{\frac{x-1}{2}} + 1$, domain = $[1, \infty)$
- f $y = \pm\sqrt{1-x} + 1$, domain = $(-\infty, 1]$

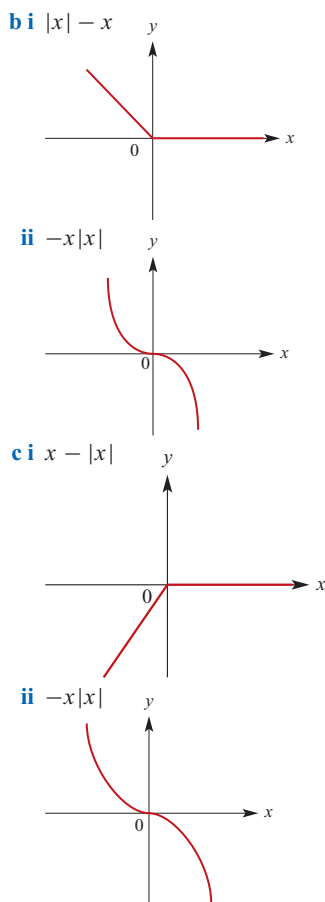
- 3 a** $y = 2 \pm \frac{1}{\sqrt{x}}$, domain = $\mathbb{R}^+$
b $y = 1 \pm \frac{1}{\sqrt{x-1}}$, domain = $(1, \infty)$
c $y = 1 \pm \frac{2}{\sqrt{x}}$, domain = $\mathbb{R}^+$
d $y = \begin{cases} 1+x, & y \geq 1 \\ 1-x, & y < 1 \end{cases}$, domain = $[0, \infty)$
e $y = \begin{cases} x-1, & y \geq 2 \\ 5-x, & y < 2 \end{cases}$, domain = $[3, \infty)$
f $y = \begin{cases} 1+x, & y \geq -1 \\ -3-x, & y < -1 \end{cases}$,
 domain = $[-2, \infty)$

- 4 a** $f^{-1}(x) = \left(\frac{x-6}{2}\right)^{\frac{5}{3}}$, domain = $\mathbb{R}$
b $y = \pm \left(\frac{x-6}{2}\right)^{\frac{5}{2}}$, domain = $[6, \infty)$

Exercise 7C

- 1 a i** $e^{-2x} - 2x$ **ii** $-2xe^{-2x}$
b i $e + 1$ **ii** e
2 a i $\sin\left(\frac{\pi x}{2}\right) - 2x$
ii $-2x \sin\left(\frac{\pi x}{2}\right)$
b i -1 **ii** -2
3 a i $\cos\left(\frac{\pi x}{2}\right) + e^x$ **ii** $e^x \cos\left(\frac{\pi x}{2}\right)$
b i 2 **ii** 1





Exercise 7D

- 1 $f(x - y) = f(x) - f(y)$
- 2 $f(x + y) = f(x) + f(y) - 3$; $a = -3$
- 3 $g(x) = 0$ or $g(x) = 1$
- 4 $f(x + y) = |x + y|$. Let $x = 2, y = -4$.
 $f(x + y) = 2 \neq |2| + |-4| = f(x) + f(y)$
- 5 $f(x + y) = \sin(x + y)$. Let $x = \frac{5\pi}{2}, y = \frac{\pi}{2}$.
 $f(x + y) = \sin(3\pi) = 0 \neq \sin\left(\frac{5\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right)$
 $= 2$
- 7 **a** $h(x + y) = (x + y)^2$. Let $x = 1, y = -1$.
 $h(x + y) = 0 \neq h(1) + h(-1) = 2$
- 10 $f(xy) = axy$. Let $x = 2$ and $y = 3$.
 $f(xy) = 6a$ and $f(x) = 2a$ and $f(y) = 3a$
 $f(x)f(y) = 6a^2$
 $6a^2 = 6a$ implies $a = 0$ or 1

Exercise 7E

- 1 **a** $\frac{4}{m}$ **b** $m \geq 4$ or $m < 0$
- c** $f^{-1}(x) = \frac{x+4}{m}$, domain = R
- d** $\left(\frac{4}{m-1}, \frac{4}{m-1}\right), m \in R \setminus \{1\}$

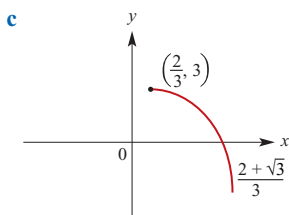
- e** $y = -\frac{1}{m}x - 4$
- 2 **a** $x = \frac{c}{2}$ **b** $c \leq 2$
- d** $f^{-1}(x) = \frac{c-x}{2}$, domain = R
- e** $\left(\frac{c}{3}, \frac{c}{3}\right)$ **f** $y = \frac{1}{2}x + c$
- 3 **a** 0 and b **b** $\left(\frac{b}{2}, \frac{-b}{4}\right)$
- c i** $(0, 0)$ and $(b-1, 1-b)$ **ii** $b = 1$
iii $b \in R \setminus \{1\}$
- 4 **a** $-1 \pm 2\sqrt{2}$ **b** $\pm 2\sqrt{2}$
- c** $a = -8, b = 16$
- 5 **a** $(-\infty, 2a]$
- b** $\left(\frac{-1 + \sqrt{1+8a}}{2}, \frac{-1 + \sqrt{1+8a}}{2}\right)$
- c** $a = 1$ **d** $a = 3$ **e** $a = \frac{c^2 + c}{2}$
- 6 **a** $(0, 0)$ and $(a, 0)$ **b** $(0, 0)$ **c** $\frac{a^2}{4}$
- d** $a = 3$ or $a = -5$
- 7 **a** $\frac{1}{b} \log_e \left(\frac{c}{a}\right)$ **b** $e^{\frac{b}{c}} - a$
- c** $\frac{a+1}{c}$ **d** $\frac{\log_e(c) - b}{a}$
- 8 **a** $x = a$ **b** $(a+1, 0)$
- c** $(a + e^{\frac{1}{c}}, 1)$ **d** $c = \frac{1}{\log_e(2-a)}$
- 9 **a** $y = -b$ **b** $(\log_e(b) + 1, 0)$
- c i** $b = \frac{1}{e}$ **ii** $0 < b < \frac{1}{e}$
- 10 $a = 5 - c, b = -1$ where the rule is $y = ax^2 + bx + c$
- 11 $a = \frac{3d+4}{6}, b = 2-d$ and
 $c = -\frac{(3d+28)}{6}$ where
 $y = ax^3 + bx^2 + cx + d$
- 12 **a** $c = 28 - 8\sqrt{6}$ or $c = 28 + 8\sqrt{6}$
- b** $(-\infty, 8) \cup (28 + 8\sqrt{6}, \infty) \cup (8, 28 - 8\sqrt{6})$
- 13 $a = \frac{5d-9}{30}, b = \frac{41-10d}{30}$,
 $c = \frac{-25d-2}{30}$ where $y = ax^3 + bx^2 + cx + d$
- 14 **a** $\left[\frac{x'-3}{y'-2}, \frac{k}{a-x'}\right]$ **b** $y = \frac{4k}{3-x} + 2$ **c** $\frac{-3}{2}$
- 15 **a** $\left[\frac{4}{y'+2}, \frac{4}{y'+2}\right]$
- b** $y = 2 \times 2^{\frac{x-a}{4}} - 2$ **c** $a = 0$

Multiple-choice questions

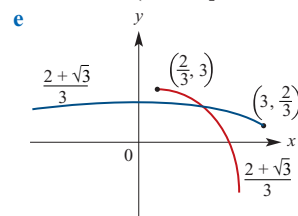
- 1 B 2 E 3 E 4 D 5 B
- 6 E 7 D 8 E 9 E 10 B

Short-answer questions (technology-free)

- 1 a i** $f \circ g(x) = 3 \cos(2|x|)$
 $g \circ f(x) = |3 \cos(2x)|$
ii maximal domain = R ;
range = $[-3, 3]$
maximal domain = R ;
range = $[0, 3]$
- b i** $f \circ g(x) = \log_e(3|x|)$
 $g \circ f(x) = |\log_e(3x)|$
ii maximal domain = $R \setminus \{0\}$;
range = R
maximal domain = R^+ ;
range = $[0, \infty)$
- c i** $f \circ g(x) = \log_e(2 - |x|)$
 $g \circ f(x) = |\log_e(2 - x)|$
ii maximal domain = $(-2, 2)$;
range = $(-\infty, \log_e 2)$
maximal domain = $(-\infty, 2)$;
range = $[0, \infty)$
- d i** $f \circ g(x) = -\log_e(2|x|)$
 $g \circ f(x) = |\log_e(2x)|$
ii maximal domain = $R \setminus \{0\}$; range = R
maximal domain = $(0, \infty)$; range = $[0, \infty)$
- 2 a** $h(x) = f \circ g(x)$ where $g(x) = |2x|$ and $f(x) = \cos x$
(Note: answer is not unique.)
- b** $h(x) = f \circ g(x)$ where $g(x) = x^2 - x$ and $f(x) = x^n$
(Note: answer is not unique.)
- c** $h(x) = f \circ g(x)$ where $g(x) = \sin x$ and $f(x) = \log_e x$
(Note: answer is not unique.)
- d** $h(x) = f \circ g(x)$ where $g(x) = \sin 2x$ and $f(x) = -2|x|$
(Note: answer is not unique.)
- e** $h(x) = f \circ g(x)$ where $g(x) = x^2 - 3x$ and $f(x) = x^4 - 2x^2$
(Note: answer is not unique.)
- 3 a i** $(f + g)(x) = 2 \cos\left(\frac{\pi x}{2}\right) + e^{-x}$
ii $(fg)(x) = 2e^{-x} \cos\left(\frac{\pi x}{2}\right)$
- b i** $(f + g)(0) = 3$ **ii** $(fg)(0) = 2$
- 4 a** $\frac{2}{3}$ **b** $(-\infty, 3]$



d $f^{-1}(x) = \frac{2 + \sqrt{3-x}}{3}$, range = $\left[\frac{2}{3}, \infty\right)$,
domain = $(-\infty, 3]$



5 a $y = \frac{2 \pm \sqrt{x-4}}{3}$ **b** $y = \pm \sqrt{\frac{3-x}{2}}$

Extended-response questions

1 a $D = 0.05t^2 - 0.25t + 1.8$ **b** \$3 000 000

2 a $a = \frac{-7}{48}$, $b = \frac{23}{24}$, $c = 7.5$

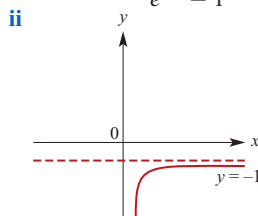
At 12:00 noon, rate of rainfall = $\frac{35}{6}$ mm/h

Rainfall was greatest $\frac{23}{7}$ hours after
4:00 a.m., i.e. at $\approx 7:17$ a.m.

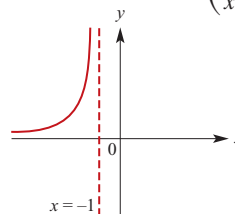
3 a $(0, 1)$, $(-\infty, 0)$

b $f^{-1}(x) = -\log_e x$, $g^{-1}(x) = \frac{1}{x} + 1$

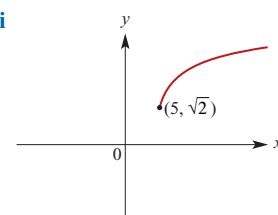
c i $g \circ f(x) = \frac{1}{e^{-x} - 1} = \frac{e^x}{1 - e^x}$



d i $(g \circ f)^{-1}(x) = \log_e\left(\frac{x}{x+1}\right)$

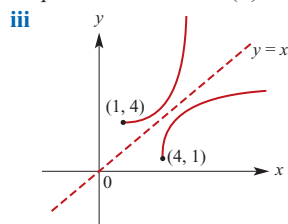


4 a i



ii $[\sqrt{2}, \infty)$ **iii** $f^{-1}(x) = x^2 + 3$,
domain of $f^{-1} = [\sqrt{2}, \infty)$

b i $p = 3$ **ii** $h^{-1}(x) = x^2 + 3$



5 a $(0, 1]$ **b** $(0, 1]$

c range of $g \subseteq$ domain of f , $f \circ g(x) = \sin\left(\frac{1}{x}\right)$

d Not defined as range of f is not contained in the domain of g

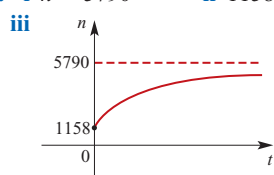
e $g^{-1}(x) = \frac{1}{x}$, domain of $g^{-1} = (0, 1]$, range of $g^{-1} = [1, \infty)$

f range of $f =$ domain of g^{-1}

$g^{-1} \circ f(x) = \frac{1}{\sin x}$

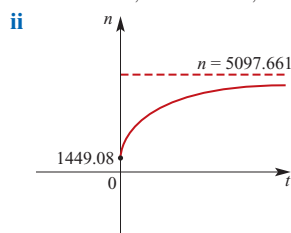
domain $= (0, \pi)$, range $= [1, \infty)$

6 a i $n = 5790$ **ii** 1158



iv $t = \frac{-100}{3} \log_e\left(\frac{179}{1600}\right)$

b i $a = 2.518$, $b = 0.049$, $c = 5097.661$



Chapter 8

Multiple-choice questions

- | | | | | | |
|-------------|-------------|-------------|-------------|-------------|-------------|
| 1 D | 2 A | 3 B | 4 E | 5 A | 6 A |
| 7 C | 8 A | 9 B | 10 B | 11 C | 12 C |
| 13 A | 14 E | 15 D | 16 C | 17 E | 18 C |
| 19 B | 20 D | 21 C | 22 A | 23 C | 24 D |
| 25 D | 26 E | 27 A | 28 D | 29 D | 30 A |
| 31 E | 32 D | 33 E | 34 A | 35 A | 36 E |
| 37 C | 38 D | 39 C | 40 B | 41 E | 42 B |
| 43 C | 44 D | 45 B | 46 B | 47 E | 48 A |
| 49 C | 50 D | 51 B | 52 C | 53 C | 54 A |
| 55 B | 56 A | 57 B | 58 D | 59 D | 60 D |
| 61 A | 62 A | 63 D | 64 B | 65 D | 66 B |
| 67 E | 68 C | 69 C | 70 A | | |

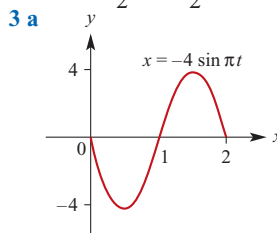
Extended-response questions

1 a $a = -0.09$, $b = 9$ **b** $DE = 2.79$ m

c Length of bar $= 2 \times \sqrt{30} \approx 10.95$ m

2 a $a = -3$ **b** $x = -1$, $x = -\frac{1}{2}$, $x = 2$

c ii $b = \frac{7}{2}$, $c = \frac{3}{2}$



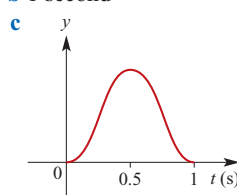
b i $x = 0$ **ii** $x = -4$ **iii** $x = 0$

c $t = \frac{7}{6}$

d Period $= \frac{2\pi}{\pi} = 2$ seconds

4 a i 0 **ii** 2.5 **iii** 0

b 1 second



d $t = 0.35$ seconds

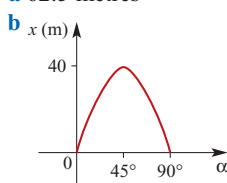
5 a $k = 0.0292$

b 150×10^6

c 6.4494×10^8

d 23.417 years

6 a 62.5 metres



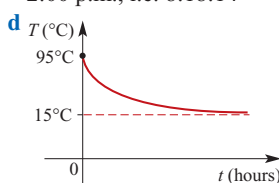
c $24^\circ 18'$ or 65.7°

7 a $A = 80$, $k = 0.3466$

b 17.5°C

c 6 hours 18 minutes and 14 seconds after

2:00 p.m., i.e. 8:18:14



8 a Carriage A $(0.83)^n I$, carriage B $0.66(0.89)^n I$

b 6 stations

9 a Area $= 0.02(0.92)^{\frac{x}{10}}$

b 0.0199 mm^2

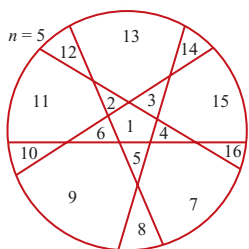
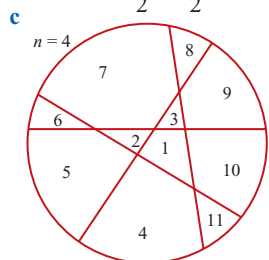
c Load $= 0.02(0.92)^{10-2.9x}$

d $x < 2.59$ m

10 a $f(n) = \frac{1}{2}n^2 + \frac{1}{2}n + 1$

b i $f(4) = 8 + 2 + 1 = 11$

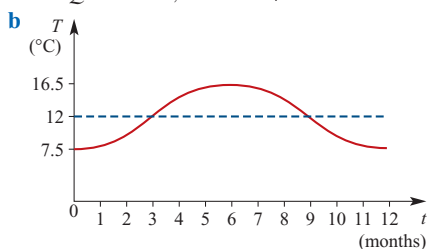
ii $f(5) = \frac{25}{2} + \frac{5}{2} + 1 = 16$



11 a $y = h - k \cos\left(\frac{\pi t}{6}\right)$

i 12 units

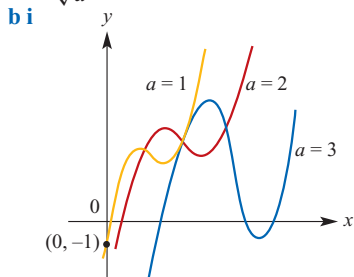
ii $OQ = h - k$, $OR = h + k$



c $h = 12$, $k = 4.5$

12 a i $\left(3 + \frac{1}{\sqrt{a}}, 0\right) \left(3 - \frac{1}{\sqrt{a}}, 0\right)$

ii $\frac{1}{\sqrt{a}}$

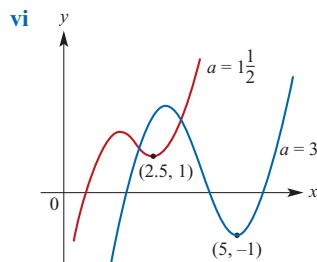


ii $a = \frac{3\sqrt{3}}{2}$

iii $a > \frac{3\sqrt{3}}{2}$

iv $a = 3$

v $a = 1\frac{1}{2}$



c i $\left(a, \frac{-4}{27}a^3 + a\right)$

ii $\frac{4}{27}a^3, \frac{2a}{3}$

iii $A = \frac{4a^4}{81}$

iv $a = 3$

v $3\sqrt[4]{375}$

Chapter 9

Exercise 9A

1 a $h + 9$

b 9.1

c 9

2 a $x + 1$

b $2x^3 + 1$

c 40

d 0

e 5

f 1

g $2x + 1$

h $3x$

i $3x^3 + x$

j $6x$

3 a $2 + 3h + h^2$

b 2

4 $2x + h, 2x$

5 $h + 6, 6$

Exercise 9B

1 a $10x$

b 3

c 0

d $6x + 4$

e $15x^2$

f $10x - 6$

2 a $5x^4$

b $28x^6$

c 6

d $10x - 4$

e $12x^2 + 12x + 2$

f $20x^3 + 9x^2$

g $-4x + 4$

h $18x^2 - 4x + 4$

3 a -2

b 0

c $15x^2 - 6x + 2$

d $\frac{6x^2 - 8}{5}$

e $4x - 5$

f $12x - 12$

g $50x^4$

h $27x^2 + 3$

4 a $8x - 4$

b $2x + 2$

c $6x^2 - 12x + 18$

d $x^2 - 2x + 1$

5 a $(3, 16)$, gradient = 8

b $(0, -1)$, gradient = -1

c $(-1, 6)$, gradient = -8

d $(4, 594)$, gradient = 393

e $(1, -28)$, gradient = -92

f $\left(2\frac{1}{2}, 0\right)$, gradient = 0

6 a 1

b 1

c $(1, \infty)$

d $(-\infty, 1)$

e $2\frac{2}{3}$

f 4 or -2

7 a $\{x: x < -1\} \cup \{x: x > 1\}$

b $\{x: -1 < x < 1\}$

c $\{1, -1\}$

8 a $(-1, 0.5) \cup (2, \infty)$

b $(-\infty, -1) \cup (0.5, 2)$

c $\{-1, 0.5, 2\}$

9 a $\left(-\frac{1}{4}, 2\right) \cup (2, \infty)$ b $\left(-\infty, -\frac{1}{4}\right)$

c $\left\{-\frac{1}{4}, 2\right\}$

10 a $(1, -9)$ b $(2, -8)$ c $\left(\frac{1}{4}, -8\frac{7}{16}\right)$

11 a $\left(-\frac{1}{2}, 3\frac{1}{2}\right)$ b $(2, 32)$ or $(-2, -32)$

c $(2, 6)$ d $(0, 0)$ or $(2, -4)$

Exercise 9C

1 a $-6x^{-3} - 5x^{-2}$ b $12x - \frac{15}{x^4}$

c $\frac{15}{x^4} - \frac{8}{x^3}$

d $-18x^{-4} - 6x^{-3}$

e $-\frac{2}{x^2}$

2 a $\frac{4}{z^2}$

b $\frac{-18 - 2z}{z^4}$

c $3z^{-4}$

d $\frac{-2z^3 + z^2 - 4}{z^2}$

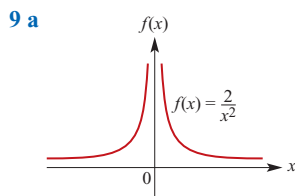
e $\frac{6 - 12z}{z^4}$

f $-6x - \frac{6}{x^2}$

3 $\pm \frac{1}{2}$ 4 a = -1, b = 4 5 $\frac{1}{2}$

6 a = -9, b = 1 7 k = 0 or $\frac{3}{2}$

8 a $11\frac{3}{4}$ b $\frac{1}{32}$ c -1 d 5



b $PQ = \frac{-2(2+h)}{(1+h)^2}$ c -4

Exercise 9D

1 a $8x(x^2 + 1)^3$ b $20x(2x^2 - 3)^4$

c $24(6x + 1)^3$

d $an(ax + b)^{n-1}$

e $2anx(ax^2 + b)^{n-1}$

f $\frac{6x}{(1 - x^2)^4}$

g $-3\left(x^2 - \frac{1}{x^2}\right)^{-4}\left(2x + \frac{2}{x^3}\right)$

h $(1 - x)^{-2}$

2 a $6(x + 1)^5$ b $4x^3(3x + 1)(x + 1)^7$

c $4\left(6x^3 + \frac{2}{x}\right)^3\left(18x^2 - \frac{2}{x^2}\right)$ d $-4(x + 1)^{-5}$

3 a $n[f(x)]^{n-1}f'(x)$ b $\frac{-f'(x)}{[f(x)]^2}$

Exercise 9E

1 a $\frac{x^{-4}}{5}$

b $\frac{5}{2}x^{\frac{3}{2}}$

c $\frac{5}{2}x^{\frac{3}{2}} - \frac{3}{2}x^{\frac{1}{2}}$

d $\frac{3}{2}x^{-\frac{1}{2}} - \frac{20}{3}x^{\frac{2}{3}}$

e $-\frac{6}{7}x^{-\frac{13}{7}}$

f $-\frac{1}{4}x^{-\frac{5}{4}} + 2x^{-\frac{1}{2}}$

2 a $\frac{1}{27}$ b $\frac{1}{12}$

c $\frac{2}{9}$ d $\frac{5}{2}$

3 a $\frac{1}{\sqrt{2x+1}}$

b $\frac{-3}{2\sqrt{4-3x}}$

c $\frac{x}{\sqrt{x^2+2}}$

d $\frac{-1}{\sqrt[3]{(4-3x)^2}}$

e $\frac{3}{2}\sqrt{x} - \frac{1}{\sqrt{x^3}}$

f $3\sqrt{x}\left(\frac{5x+6}{2}\right)$

6 a $\frac{x}{\sqrt{x^2+2}}$

b $\frac{3x^2-5}{3\sqrt[3]{(x^3-5x)^2}}$

c $\frac{2x+2}{5\sqrt[5]{(x^2+2x)^4}}$

Exercise 9F

1 $20x^4 + 36x^2 + 4x$ 2 $9x^{\frac{1}{2}} + \frac{3}{2}x^{-\frac{1}{2}}$

3 $3(2x - 1)^2(8x - 1)$

4 $8x(2x^2 + 1)(6x^2 + 1)$

5 $5(3x + 1)^{\frac{1}{2}}(3x + 4)$ 6 $\frac{5x^2 - 8x + 1}{\sqrt{2x - 4}}$

7 $x^2(3x^2 + 4x + 3)(3x^2 + 2x + 1)^{-2}$

8 $2x^3(5x^2 - 2)(2x^2 - 1)^{-\frac{1}{2}}$

9 $2x\sqrt[3]{x^2+2x} + \frac{2x^2(x+1)}{3\sqrt[3]{(x^2+2x)^2}}$

Exercise 9G

1 a $\frac{4}{(x+4)^2}$

b $\frac{4x}{(x^2+1)^2}$

c $\frac{1}{2}\left(x^{-\frac{1}{2}} - x^{\frac{1}{2}}\right)$

d $\frac{(x+2)^2(x-3)(x-1)}{(x^2+1)^2}$

e $\frac{2+2x-x^2}{(x^2+2)^2}$

f $\frac{-4x}{(x^2-1)^2}$

g $\frac{x^2+4x+1}{(x^2+x+1)^2}$

h $\frac{-2(4x^3+3x^2+1)}{(2x^3+2x)^2}$

2 a 81, gradient = 378 b 0, gradient = 0

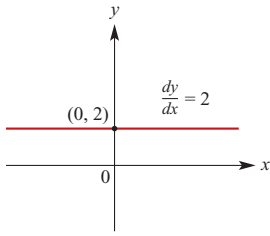
c 0, gradient = 0 d $\frac{1}{2}$, gradient = 0

e $\frac{3}{2}$, gradient = $-\frac{1}{2}$

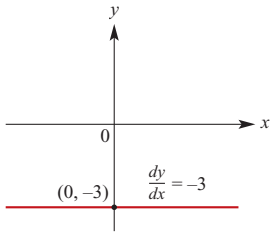
3 a $\frac{2x^2+x+1}{\sqrt{x^2+1}}$ b $x\frac{(7x^3+3x+4)}{2\sqrt{x^3+1}}$ c $\frac{5}{(x+3)^2}$

Exercise 9H

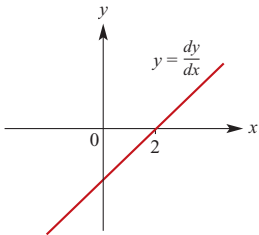
1 a



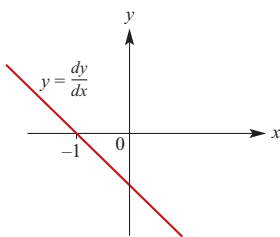
b



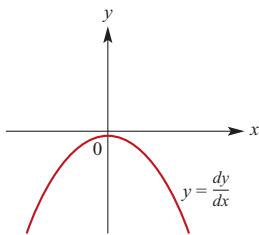
c



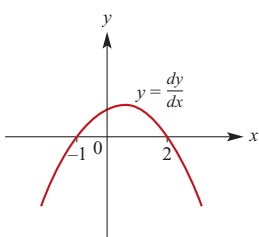
d



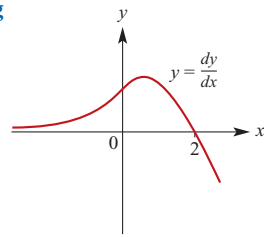
e



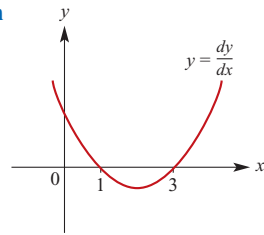
f



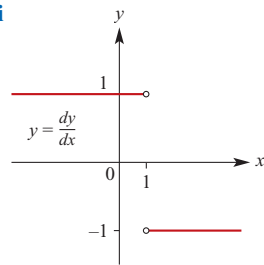
g



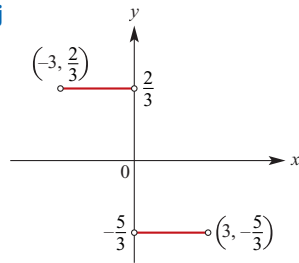
h



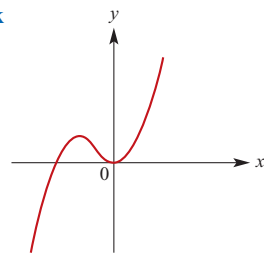
i



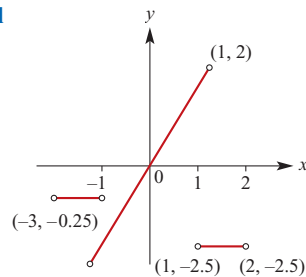
j

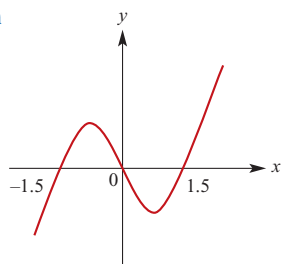
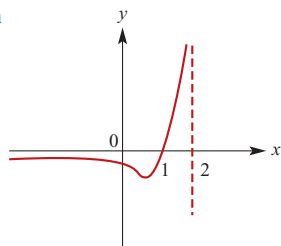
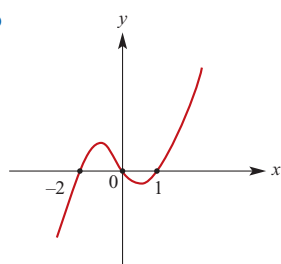
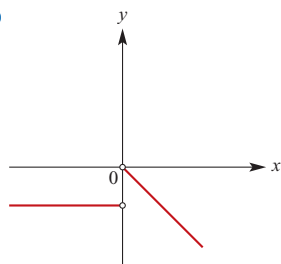
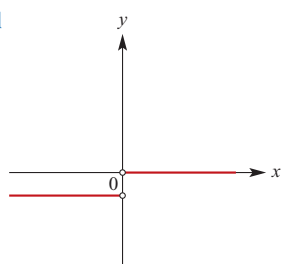
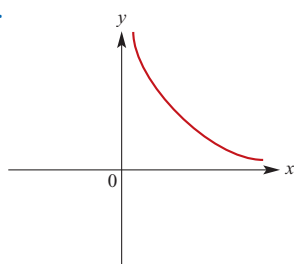
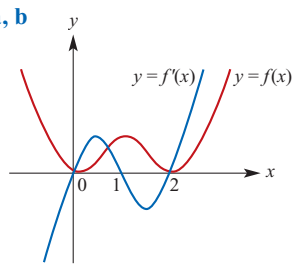


k

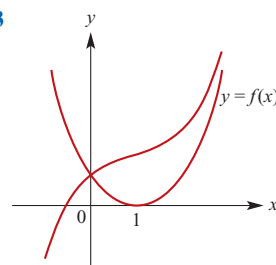


l

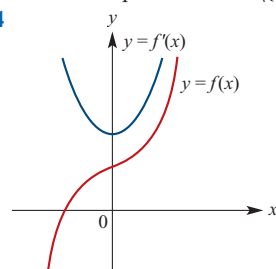


m

n

o

p

q

r

2 a, b


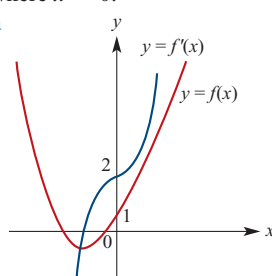
c i 0 **ii** 0 **iii** 0 **iv** 96
d i 1 **ii** 0.423

3


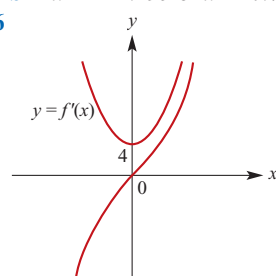
Gradient is 0 at $\left(1, \frac{4}{3}\right)$.
 Gradient is positive for $R \setminus \{1\}$.

4


Gradient is positive always; minimum gradient where $x = 0$.

5 a


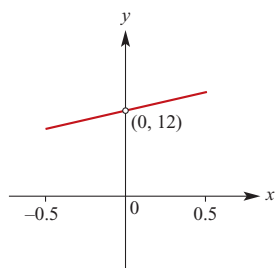
b i $x = -1.495$ or $x = 0.798$ **ii** $x = 0.630$

6


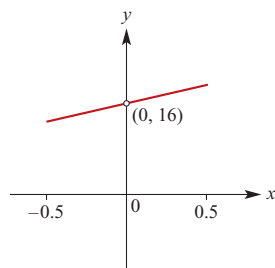
a Minimum value of $f'(x)$ is 4.

b $x = -0.471$ or $x = 0.471$

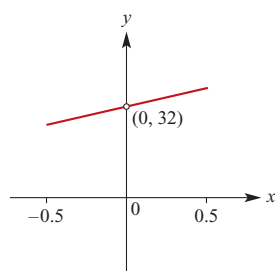
7 a i



ii



iii



b Graphs and tables illustrate that as $x \rightarrow 0$ from the left and from the right the value of $g(x)$ approaches the gradient of $y = f(x)$ when $x = 2$.

Exercise 9I

- 1 a** 17 **b** 3 **c** -4 **d** $\frac{1}{8}$
e 3 **f** 4 **g** 2 **h** $2\sqrt{3}$
i -2 **j** 12 **k** $\frac{11}{9}$ **l** $\frac{1}{4}$

2 a 3, 4 **b** 7

3 a 0 as $f(0) = 0$, $\lim_{x \rightarrow 0^+} f(x) = 0$, but

$$\lim_{x \rightarrow 0^-} f(x) = 2$$

b 1 as $f(1) = 3$, $\lim_{x \rightarrow 1^+} f(x) = 3$, but

$$\lim_{x \rightarrow 1^-} f(x) = -1$$

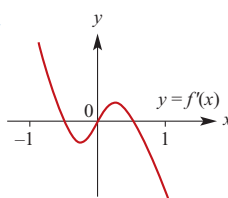
c 0 as $f(0) = 1$, $\lim_{x \rightarrow 0^+} f(x) = 1$, but

$$\lim_{x \rightarrow 0^-} f(x) = 0$$

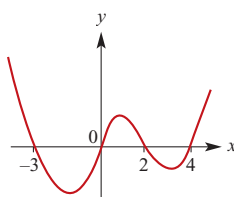
4 $R \setminus \{1\}$

Exercise 9J

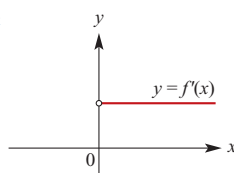
1 a



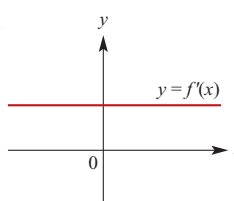
b



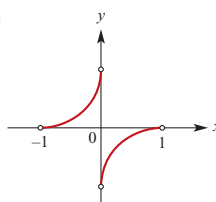
c



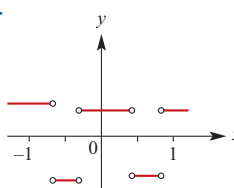
d



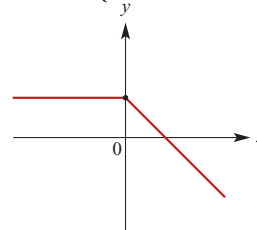
e



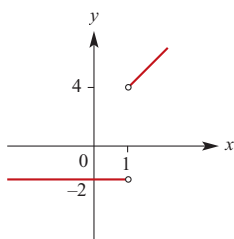
f



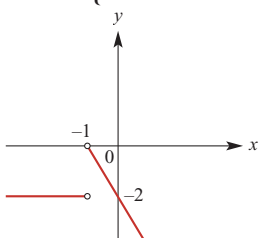
$$2 \quad f'(x) = \begin{cases} -2x + 3 & \text{if } x \geq 0 \\ 3 & \text{if } x < 0 \end{cases}$$



- 3 Defined for $R \setminus \{1\}$ $f'(x) = \begin{cases} 2x + 2 & \text{if } x > 1 \\ -2 & \text{if } x < 1 \end{cases}$



- 4 Defined for $R \setminus \{-1\}$ $f'(x) = \begin{cases} -2x - 2 & \text{if } x > -1 \\ -2 & \text{if } x < -1 \end{cases}$



- 5 a $R \setminus \{1\}; f'(x) = \frac{1}{3}(x-1)^{-\frac{2}{3}}$
 b $R \setminus \{0\}; f'(x) = \frac{1}{5}(x)^{-\frac{4}{5}}$
 c $R \setminus \{0\}; f'(x) = \frac{2}{3}x^{-\frac{1}{3}}$
 d $R \setminus \{-2\}; f'(x) = \frac{2}{5}(x+2)^{-\frac{3}{5}}$
 6 a $f'(x) = \begin{cases} 2x - 4 & \text{if } x > 4 \text{ or } x < 0 \\ 4 - 2x & \text{if } 0 < x < 4 \end{cases}$
 b $f'(x) = \begin{cases} 2x - 4 & \text{if } x > 0 \\ 2x + 4 & \text{if } x < 0 \end{cases}$

Exercise 9K

- 1 a 8 b -8
 2 a D b F c B d C e A f E
 3 a $\frac{-2x^5 + 4x^3 - 2x}{(x^4 - 1)^2}$ b $\{0\}$
 4 a -7 b -14
 5 a $5\frac{1}{2}$ b $\frac{1}{8}$
 6 $2x\sqrt{3x^2 + 1}$
 7 a $4x - 3$ b -3 c $\{1\}$
 8 $-\frac{3}{25}$ 10 24π 11 $(0.28, 0.14)$ 12 $\frac{2}{3}$
 14 $-\frac{8x}{(x^2 - 2)^2}$ 15 $24x - 12$
 16 $-63(5 - 7x)^8$ 17 $\frac{1}{9}$ 18 $\frac{2}{3}$
 19 -70 20 0 21 -1
 22 a $-\frac{2}{(2x + 1)^2}$ b -2
 23 a $x = 0$ or $x = -2$ b $x > 0$ or $x < -2$
 c $-2 < x < 0$

- 24 a $\frac{1}{(1-x)^2}$ b $(y+1)^2$

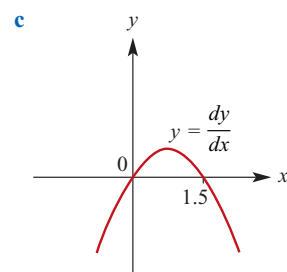
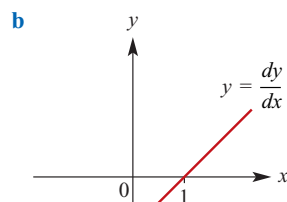
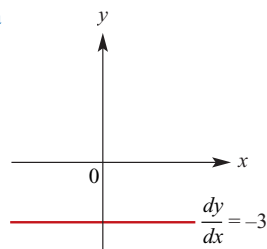
- 25 $-3x(x^2 + 1)^{-\frac{5}{2}}$ 26 3

Multiple-choice questions

- 1 B 2 C 3 A 4 A 5 B
 6 C 7 D 8 E 9 A 10 B

Short-answer questions (technology-free)

- 1 a $1 - \frac{x}{\sqrt{1-x^2}}$ b $\frac{-4x^2 - 2x + 12}{(x^2 + 3)^2}$
 c $\frac{3}{2\sqrt{1+3x}}$ d $\frac{-2}{x^2} - \frac{1}{2x^{\frac{3}{2}}}$
 e $\frac{3x-15}{2\sqrt{x-3}}$ f $\frac{1+2x^2}{\sqrt{1+x^2}}$
 g $\frac{4x}{(x^2+1)^2}$ h $\frac{-x^2+1}{(x^2+1)^2}$
 i $\frac{10x}{3}(2+5x^2)^{-\frac{2}{3}}$ j $\frac{-2x^2-2x+4}{(x^2+2)^2}$
 k $4x(3x^2+2)^{-\frac{1}{3}}$
 2 a -6 b 1 c 5 d $\frac{1}{6}$
 3 a



- 4 $2\left(4 - \frac{9}{x^2}\right)\left(4x + \frac{9}{x}\right), x = \pm \frac{3}{2}$

5 b $\left(\frac{3}{2}, \infty\right) \cap (-1, 4) = \left(\frac{3}{2}, 4\right)$

6 a $xf'(x) + f(x)$

b $f'(x)$ if $x \in [0, 4]$ and $-f'(x)$ if $x \in (-\infty, 0) \cup (4, \infty)$

c $\frac{2xf(x) - 2x^2 f'(x)}{[f(x)]^3}$

Extended-response questions

1 a i -4 ii -6 iii -18

iv -18 v 6 vi $-\frac{1}{6}$

b $a = \frac{5}{2}, b = 1, c = -\frac{7}{2}, d = 6$

2 a i -1 and 3 ii $x > 3$ and $x < -1$

b i 3 and 7 ii (3, 6) and (7, 1)

c i $\frac{1}{2}$ and $\frac{5}{2}$ ii $\left(\frac{1}{2}, 6\right), \left(\frac{5}{2}, 1\right)$

d i 2 and 10 ii (2, 6) and (10, 1)

e i 2 and 10 ii (2, 18) and (10, 3)

3 a $x = \alpha$ or $x = \beta$

b $(x - \beta)^{m-1}(x - \alpha)^{n-1}((m+n)x - \alpha m - \beta n)$

c $x = \alpha$ or $x = \beta$ or $x = \frac{\alpha m + \beta n}{m+n}$

d i $x > \frac{\alpha m + \beta n}{m+n}, x \neq \beta$

ii $x < \alpha$ or $x > \frac{\alpha m + \beta n}{m+n}$

4 b $\frac{nx^{n-1}}{(x^n + 1)^2}$ d 0 e $x > 0$

Chapter 10

Exercise 10A

1 $y = 4x - 5, y = -\frac{1}{4}x + 3\frac{1}{2}$

2 $y = -\frac{1}{3}x - 1$

3 $y = x - 2$ and $y = -x + 3$

4 $y = 18x + 1, y = -\frac{1}{18}x + 1$

5 $\left(\frac{3}{2}, -\frac{11}{4}\right), c = -\frac{29}{4}$

6 a i $y = 2x - 3$ ii $y = -\frac{1}{2}x - \frac{1}{2}$

b i $y = -3x - 1$ ii $y = \frac{1}{3}x - 1$

c i $y = -x - 2$ ii $y = x$

d i $y = 8x + 2$ ii $y = -\frac{1}{8}x - \frac{49}{8}$

e i $y = \frac{3}{2}x + 1$ ii $y = -\frac{2}{3}x + 1$

f i $y = \frac{1}{2}x + \frac{1}{2}$ ii $y = -2x + 3$

g i $y = \frac{2}{3}x + \frac{4}{3}$ ii $y = -\frac{3}{2}x + \frac{7}{2}$

h i $y = 4x - 16$

ii $y = -\frac{1}{4}x - \frac{15}{2}$

i i $y = -2$

ii $x = 2$

j i $y = 4x - 4$

ii $y = -\frac{1}{4}x + \frac{1}{4}$

7 a $y = -1$

b $y = \frac{3}{2}x + \frac{1}{2}$

c $y = -2x - 1$

d $y = -4x + 5$

8 a $x = 4$

b $x = -5$

c $x = -\frac{1}{2}$

d $x = -5$

Exercise 10B

1 11.31° or $11^\circ 19'$

2 30.96° or $30^\circ 58'$

3 $161^\circ 5'$

4 a $\tan^{-1}(0.5) = 26^\circ 34'$

b $\tan^{-1}(0.33) = 18^\circ 26'$

c $\tan^{-1}(2.83) = 70^\circ 39'$

5 a (1, 1), (0, 0)

b $8^\circ 8'$ and 0°

6 $71^\circ 34'$

7 a (0, 0) $\alpha = 45^\circ$, (1, 1) and (-1, -1) $\alpha = 26^\circ 34'$

b (0, 0) $\alpha = 18^\circ 26'$, (1, 2) and (-1, -2) $\alpha = 12^\circ 32'$

c (1, 1) and (-1, 1) $\alpha = 63^\circ 26'$ d $14^\circ 2'$

Exercise 10C

1 a $\frac{dy}{dx} = 12x^2 - 16x$

b 16

c 0.32

2 a $-\frac{1}{2}x^{-\frac{3}{2}}$

b $-\frac{1}{2000}$

c $-\frac{3}{2000}$

d 0.0985

e i $\delta y \approx -\frac{h}{2}a^{-\frac{3}{2}}$

ii $\frac{1}{\sqrt{a+h}} \approx \frac{1}{\sqrt{a}} - \frac{h}{2a^{\frac{3}{2}}}$

3 -28q

4 a $\frac{23p}{10}$

b $\frac{230p}{11}\%$

5 $\frac{dy}{dx} = 5 - \frac{8}{x^2}; \delta y = 3p$

6 a 3p

b $\frac{p}{2\sqrt{a}}$

c $\frac{-6p}{(3a+1)^2}$

d $8(2a+1)^3 p$

e $36a(6a^2 - 1)^2 p$

f $\frac{(-2a^2 - 10a + 2)p}{(a^2 + 1)^2}$

g $\frac{2a}{3(a^2 + 10)^{\frac{2}{3}}} p$

h $\frac{p}{(a+1)^2}$

7 $\frac{1}{20}$ cm

8 $\frac{31}{30}$

9 -0.48

10 $10p\pi$ cm²

11 a $\frac{dT}{dl} = \frac{\pi}{\sqrt{lg}}$ b $\frac{\pi\sqrt{10}}{40\sqrt{g}}$ seconds

12 4% 13 4%

14 a i $\frac{1}{(1-x)^2}$ ii 1

15 a i $f'(x) = \frac{1}{2(1+x)^{\frac{1}{2}}}$ ii $f'(0) = \frac{1}{2}$

17 a $\frac{8\pi}{5}$; 2.5% b $\frac{16\pi}{5}$; 3.75%

Exercise 10D

1 a (2, -16), (-2, 16) b (1, -2)

c (0, 0), (1, 1) d (4, 48)

e (0, 0), $\left(\frac{2}{\sqrt{3}}, \frac{16}{3}\right)$, $\left(\frac{-2}{\sqrt{3}}, \frac{16}{3}\right)$

f $\left(\frac{1}{3}, 4\frac{2}{3}\right)$ g (3, 2)

h (0, -10), (2, 6)

2 a a = 6 b b = 3

3 a = 2, b = -4, c = -1

4 a = $\frac{2}{3}$, b = $-2\frac{1}{2}$, c = -3, d = $7\frac{1}{2}$

5 a a = 2 and b = 9 b (-1, -5)

6 $x = \frac{1}{2}$ or $x = \frac{1-4n}{2n+2}$

7 $x = \pm 1$ or $x = 0$ 8 $\left(1, \frac{1}{2}\right)$ or $\left(-1, -\frac{1}{2}\right)$

9 a (2, 4) b $\left(\frac{2\sqrt{3}}{3}, \frac{16\sqrt{3}}{9}\right)$, $\left(-\frac{2\sqrt{3}}{3}, \frac{16\sqrt{3}}{9}\right)$

c (0, 0), $\left(\frac{8}{3}, \frac{256}{27}\right)$ d (0, 0), $(\sqrt{2}, 4)$, $(-\sqrt{2}, 4)$

Exercise 10E

1 a x = 0

	0	
+	0	+
inflexion		

b x = 2, x = -5

	-5		2	
+	0	-	0	+
max.		min.		

c x = -1, x = $\frac{1}{2}$

	-1		$\frac{1}{2}$	
+	0	-	0	+
max.		min.		

d x = -3, x = 4

	-3		4	
-	0	+	0	-
min.		max.		

e x = -3, x = 4

	-3		4	
+	0	-	0	+
max.		min.		

f x = 0, x = $\frac{27}{5}$

	0		$\frac{27}{5}$	
+	0	-	0	+
max.		min.		

g x = 1, x = 3

	1		3	
+	0	-	0	+
max.		min.		

h x = 1, x = 3

	1		3	
-	0	+	0	-
min.		max.		

2 a x = -2(max.), x = 2(min.)

b x = 0(min.), x = 2(max.)

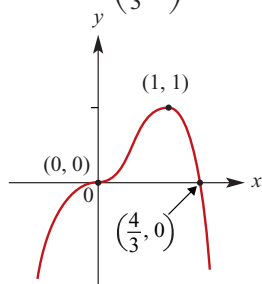
c x = $\frac{1}{3}$ (max.), x = 3(min.)

d x = 0(inflexion)

e x = -2(inflexion), x = 0(min.)

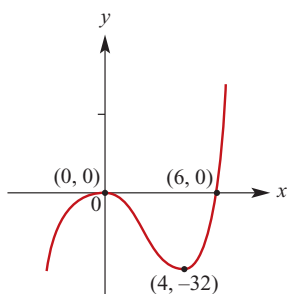
f x = $-\frac{1}{\sqrt{3}}$ (max.), x = $\frac{1}{\sqrt{3}}$ (min.)

3 a i $(0, 0)$ and $(\frac{4}{3}, 0)$



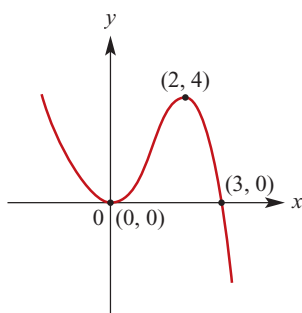
ii $x = 0$ (inflexion), $x = 1$ (max.)

b i $(0, 0)$ and $(6, 0)$



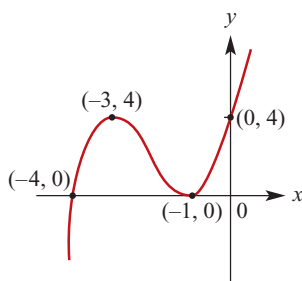
ii $x = 0$ (max.), $x = 4$ (min.)

c i $(0, 0)$ and $(3, 0)$



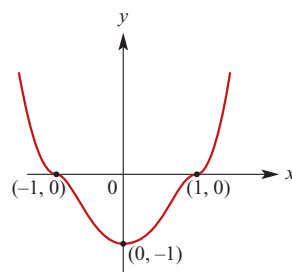
ii $x = 0$ (min.), $x = 2$ (max.)

d i $(-4, 0)$, $(-1, 0)$, $(0, 4)$



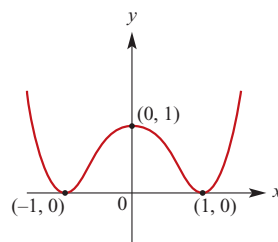
ii $x = -3$ (max.), $x = -1$ (min.)

e i $(-1, 0)$, $(0, -1)$, $(1, 0)$



ii $(-1, 0)$ (inflexion), $(0, -1)$ (min.), $(1, 0)$ (inflexion)

f i $(-1, 0)$, $(0, 1)$, $(1, 0)$

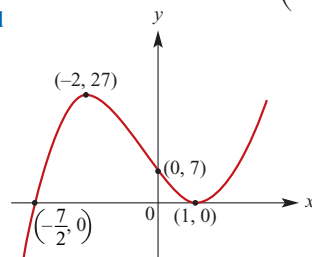


ii $(-1, 0)$ (min.), $(0, 1)$ (max.), $(1, 0)$ (min.)

4 a $(-2, 27)$ (max.), $(1, 0)$ (min.)

b $(1, 0)$ is a turning point. c $(-\frac{7}{2}, 0)$, $(0, 7)$

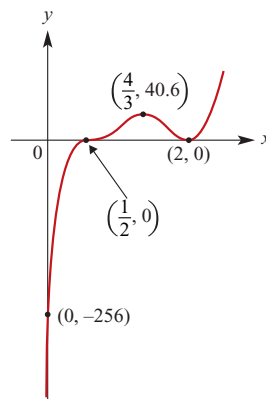
d

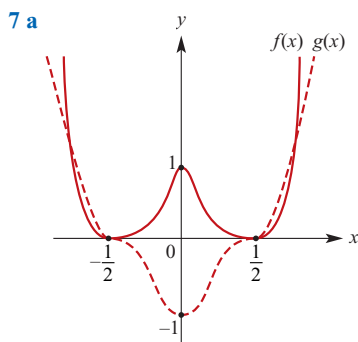


5 b $a = 3$, $b = 2$, $(0, 2)$ (min.), $(-2, 6)$ (max.)

6 a $(0, -256)$, $(\frac{1}{2}, 0)$, $(2, 0)$

b $(\frac{1}{2}, 0)$ (inflexion), $(\frac{4}{3}, 40.6)$ (max.), $(2, 0)$ (min.)





b i $\left\{x: x > \frac{1}{\sqrt{2}}\right\} \cup \left\{x: -\frac{1}{2} < x < \frac{1}{2}\right\} \cup \left\{x: x < -\frac{1}{\sqrt{2}}\right\}$

ii $\left\{x: x > \frac{\sqrt{66}}{12}\right\} \cup \left\{x: \frac{-\sqrt{66}}{12} < x < -\frac{1}{2}\right\} \cup \left\{x: -\frac{1}{2} < x < 0\right\}$

8 a $(-2, 0)$ (max.), $\left(\frac{4}{3}, -18\frac{14}{27}\right)$ (min.)

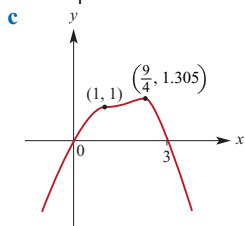
b No stationary points

9 a $(0, -1)$ stationary point of inflexion, $(-1, -1)$ minimum

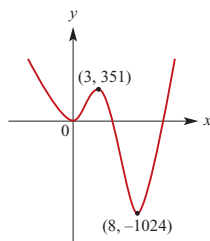
b $(0, 0)$ stationary point of inflexion, $(-1.5, -2.6875)$ minimum

c No stationary points, gradient is always positive

10 b $x \leq \frac{9}{4}$



11



12 a $x = -1$, stationary point of inflexion

$x = 1$, minimum

$x = 5$, maximum

b $x = 0$, maximum
 $x = 2$, minimum

c $x = -4$, minimum

$x = 0$, maximum

d $x = -3$, minimum

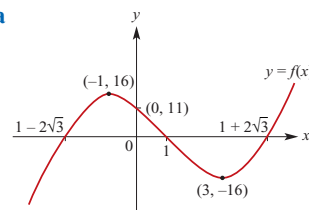
$x = 2$, stationary point of inflexion

13 a $(0, 0)$ local maximum; $(2\sqrt{2}, -64)$ and $(-2\sqrt{2}, -64)$ local minima

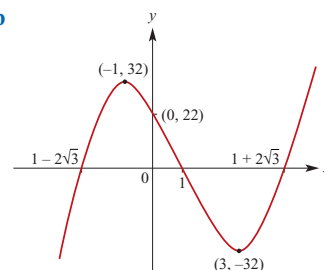
b $(2\sqrt{2}, 64)$ and $(-2\sqrt{2}, 64)$ local maxima, $(0, 0)$ local minimum

c $\left(\pm 4\sqrt{\frac{m-1}{m}}, \frac{16^m(m-1)}{m^m}\right)^{m-1}$ local maxima, $(0, 0)$ local minimum

14 a

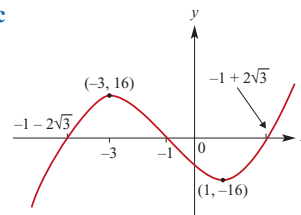


b



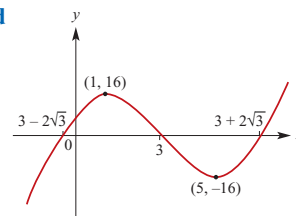
Graph dilated by factor 2 from x -axis

c



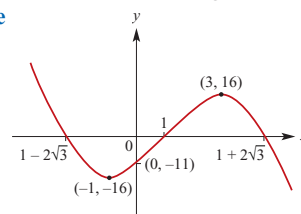
Translated 2 to the left

d



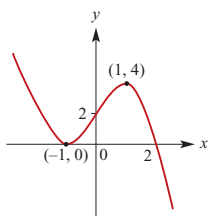
Translated 2 to the right

e

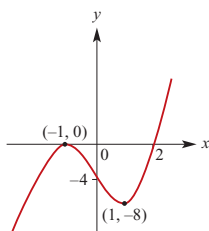


Reflected in the x -axis

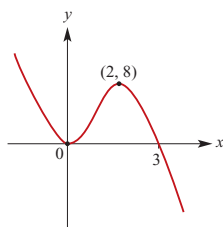
15 a



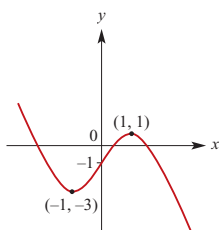
b



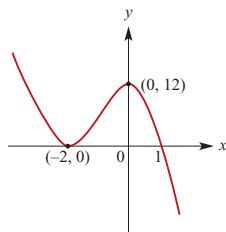
c



d



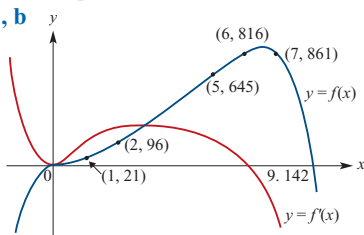
e



16 a $(a + l, 0)$ and $(b + l, 0)$

b $(h + l, kp)$

17 a, b



$$y = -x^4 + 8x^3 + 10x^2 + 4x$$

local max. at $(6.761, 867.07)$, no stationary

point of inflexion, $\frac{dy}{dx} = 4$ when $x = 0$

c -960 d $x = 4.317$ or $x = 8.404$

Exercise 10F

1 Absolute minimum $= -30$

Absolute maximum $= 15$

2 Absolute minimum $= -20.25$

Absolute maximum $= 0$

3 Absolute minimum $= -8$

Absolute maximum $= 2304$

4 Absolute minimum $= -70$

Absolute maximum $= 2$

5 b $\frac{dV}{dx} = 30x - 36x^2$

c Local maximum $= \left(\frac{5}{6}, \frac{125}{36}\right)$

d Absolute maximum value is 3.456 when $x = 0.8$

e Absolute maximum is $\frac{125}{36}$ when $x = \frac{5}{6}$

6 a $25 \leq y \leq 28$

b Absolute maximum value is 125.

Absolute minimum value is 56.

7 b $\frac{dA}{dx} = \frac{1}{4}(x - 5)$ c $x = 5$ d $\frac{61}{8}$ metres

8 Absolute maximum $= 12.1$

Absolute minimum $= 4$

9 a $\frac{1}{(x - 4)^2} - \frac{1}{(x - 1)^2}$ b $\left(\frac{5}{2}, \frac{4}{3}\right)$

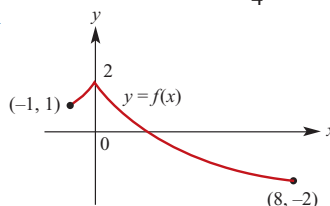
c Absolute minimum $= \frac{4}{3}$
Absolute maximum $= \frac{1}{2}$

10 a $\frac{1}{(x - 4)^2} - \frac{1}{(x + 1)^2}$

b $\left(\frac{3}{2}, \frac{4}{5}\right)$

c Absolute minimum $= \frac{4}{5}$
Absolute maximum $= \frac{1}{4}$

11



Absolute maximum $= 2$
Absolute minimum $= -2$

Exercise 10G

1 First $= \frac{4}{3}$, second $= \frac{8}{3}$

2 $(1, 1)$

3 12°C

4 8 mm for maximum, $\frac{4}{3}$ mm for minimum

5 625 m²

6 Maximum $P = 2500$

7 $2 \text{ km} \times 1 \text{ km} = \text{maximum of } 2 \text{ km}^2$

$$8 \ p = \frac{3}{2}, q = \frac{8}{3}$$

$$9 \ b \ V = \frac{75x - x^3}{2}$$

$$c \ 125 \text{ cm}^2$$

Exercise 10H

$$1 \ a \ \frac{dV}{dt} \quad b \ \frac{dS}{dr} \quad c \ \frac{dV}{dx}$$

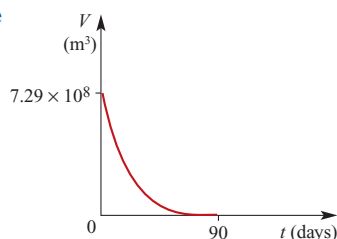
$$d \ \frac{dA}{dt} \quad e \ \frac{dV}{dh}$$

2 Wanes by 0.006 units per day

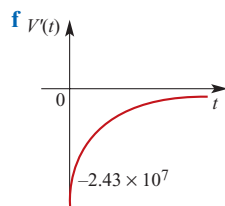
$$3 \ a \ -3 \times 10^3(90 - t)^2 \quad b \ 90 \text{ days}$$

$$c \ 7.29 \times 10^8 \text{ m}^3 \quad d \ 80 \text{ days}$$

e

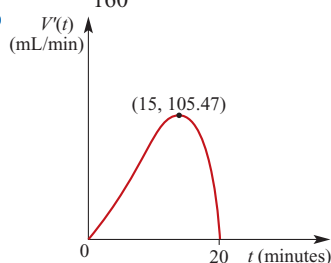


f



$$4 \ a \ V'(t) = \frac{t^3}{160}(20 - t)$$

b



$$c \ t = 15$$

$$5 \ a \ t \approx 100, t \approx 250, t \approx 500$$

$$b \approx 430\,000 \text{ m}^3/\text{day}$$

$$c \approx 270\,000 \text{ m}^3/\text{day}$$

$$d \ (100, 250) \cup (500, 600)$$

$$6 \ a \ V = 6t^2 - 18t + 12$$

$$b \ t = 1, x = 5; t = 2, x = 4$$

$$c \ -6 \text{ m/s}^2, 6 \text{ m/s}^2 \quad d \ -1.5 \text{ m/s}$$

$$7 \ a \ 8 \text{ cm from } O \quad b \ 2 \text{ cm/s}$$

$$c \ 9 \text{ cm from } O \text{ after } 1 \text{ s} \quad d \ -2 \text{ cm/s}^2$$

$$8 \ a \ \frac{\sqrt{2}t}{\sqrt{t^2 + 1}} \quad b \ \frac{\sqrt{2}}{(t^2 + 1)^{\frac{3}{2}}}$$

$$c \ 1 \text{ cm/s}, \frac{1}{2} \text{ cm/s}^2$$

Exercise 10I

$$1 \ \frac{3}{10\pi} \text{ cm/s} \quad 2 \ 8 \text{ cm}^2/\text{s} \quad 3 \ \frac{1}{8\pi} \text{ cm/s}$$

$$4 \ \frac{1}{133} \text{ cm/s} \quad 5 \ 2\sqrt{10} \text{ cm}^3/\text{s}$$

$$6 \ \frac{15}{62} \text{ cm/s} \quad 7 \ \frac{1}{8\pi} \text{ cm/minute}$$

$$8 \ a \ 20x(x^2 - 5)^9 \quad b \ 80 \times 11^9$$

$$9 \ \frac{\pi}{50} \text{ m}^3/\text{year} \quad 10 \ a \ V = \frac{\pi x^3}{48} \quad b \ \frac{64}{5\pi} \text{ cm/s}$$

Exercise 10J

$$1 \ a \ f'(x) = (x - 1)(3x - 2b - 1)$$

$$b \ \left(\frac{2b + 1}{3}, \frac{-4(b - 1)^3}{27} \right) \text{ and } (1, 0)$$

$$c \ \frac{2b + 1}{3} > 1 \text{ as } b > 1 \quad d \ b = \frac{11}{2}$$

$$2 \ a \ \text{Increasing } \left(0, \frac{2}{3a} \right) \text{ Decreasing } \left(\frac{2}{3a}, \infty \right)$$

$$b \ y = \frac{-1}{a}x + \frac{1}{a^2} \quad c \ y = ax - 1$$

$$d \ \left(-\infty, \frac{4}{27a^2} \right]$$

$$3 \ a \ i \ 2(a - 3) \quad ii \ m = 2(a - 3)$$

$$b \ (a, (a - 3)^2)$$

$$c \ y = 2(a - 3)x - a^2 + 9$$

$$d \ \frac{3 + a}{2}$$

$$4 \ a \ h = 1 \text{ or } -3 \quad b \ a = 2$$

$$c \ a = -48, b = -64$$

$$5 \ a \ (0, 0), (\sqrt{2}, -4), (-\sqrt{2}, -4)$$

$$b \ (a, b), (\sqrt{2} + a, -4 + b),$$

$$(-\sqrt{2} + a, -4 + b)$$

$$6 \ a \ (a, 0) \text{ and}$$

$$\left(\frac{a + 2}{3}, \frac{4(a - 1)^3}{27} \right)$$

$$b \ (a, 0) \text{ local minimum,}$$

$$\left(\frac{a + 2}{3}, \frac{4(a - 1)^3}{27} \right) \text{ local maximum}$$

$$c \ i \ y = (a - 1)^2x - (a - 1)^2$$

$$ii \ y = 0$$

$$iii \ y = -\frac{(a - 1)^2}{4}(x - a)$$

$$7 \ a \ 2(x - 1)(x - b)(2x - b - 1)$$

$$b \ (1, 0), (b, 0), \left(\frac{b + 1}{2}, \frac{(b - 1)^4}{16} \right) \quad c \ b = 3$$

$$8 \ a \ a = \frac{1}{486}, b = 0, c = \frac{-1}{162} \text{ and } d = \frac{1459}{243}$$

$$b \ \frac{1 - x}{2} = \frac{1}{486}[(y - 3)^3 - 3(y - 3) + 2918]$$

$$\text{or } x = \frac{-1}{243}[(y - 3)^3 - 3(y - 3) + 2918] + 1$$

$$9 \ a \ a = c - 20 \text{ and } b = 30 - 2c \quad b \ \frac{45}{2}$$

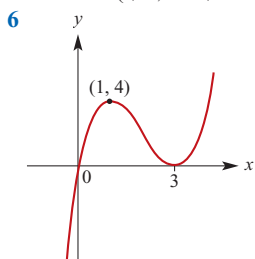
10 a $a = \frac{4d-1}{4}$, $b = \frac{-(2d+3)}{2}$, $c = \frac{11-4d}{4}$
 b $d = \frac{38}{67}$

Multiple-choice questions

- 1 A 2 E 3 E 4 B 5 A
 6 E 7 B 8 C 9 B 10 B

Short-answer questions (technology-free)

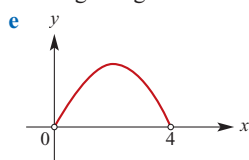
- 1 a $y = -x$ b $(0, 0)$
 2 $-3a^2$ and $y = 6ax - 3a^2$
 3 $y = 3x - 3$, $x = \frac{11}{3}$
 4 a 5π sq. units/unit b 6π sq. units/unit
 5 a Max. at $(1, 1)$; stationary point of inflexion $(0, 0)$
 b Max. at $(-1, 0)$; min. at $(1, -4)$
 c Max. at $(-\sqrt{3}, 6\sqrt{3} + 1)$; min. at $(\sqrt{3}, -6\sqrt{3} + 1)$



- 7 $x = 1$ stationary point of inflexion; $x = 2$ minimum
 8 $y = -9x + 7$ 9 $(0, 2)$ 10 $x = \frac{-1}{2}$
 11 a $R \setminus \{1\}$ $f'(x) = \frac{4}{5}(x-1)^{-\frac{1}{5}}$
 b $y = \frac{4}{5}x - \frac{3}{5}$ and $y = -\frac{4}{5}x + 1$
 c $(1, \frac{1}{5})$
 12 a 64π cm³/cm b 64π cm³/s

Extended-response questions

- 1 a $y = 4x - x^2$ b $0 < x < 4$ c $y = 4$, $x = 2$
 d Gradient to the left of $x = 2$ is positive and to the right negative.

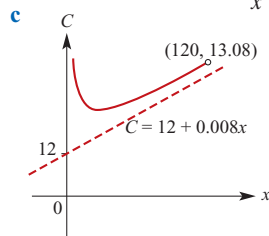


- f $0 < y < 4$
 2 a $A = 4xy$ b $y = -\frac{2}{3}x + 8$
 c $A = 4x \left(8 - \frac{2}{3}x\right) = 32x - \frac{8}{3}x^2$

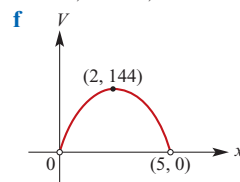
- d $x = 6$, $y = 4$
 e $A = 96$. The maximum area is 96 m².

- 3 a i \$12.68 ii \$12.74

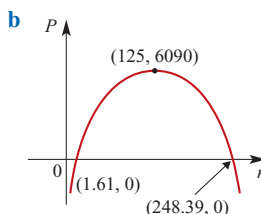
b $C = 12 + 0.008x + \frac{14.40}{x}$



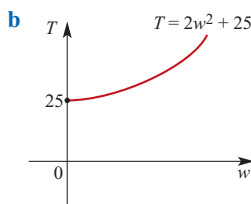
- d 42.43 km/h
 4 a $V = 4(x^3 - 13x^2 + 40x)$
 b $0 < x < 5$ c $x = 2$
 d 2 cm, 12 cm, 6 cm e 144 cm³



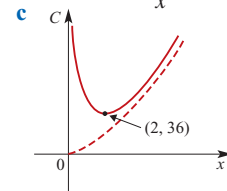
- 5 a i $n = 125$ ii Maximum daily profit = \$6090



- c $2 \leq n \leq 248$ d $n = 20$
 6 32
 7 a $T = 2w^2 + 25$



- c $A = \frac{25}{w} + 2w$
 d i $\frac{5\sqrt{2}}{2}$ kg ≈ 3.54 kg ii $A = 10\sqrt{2} \approx 14.14$
 8 b $C = 3x^2 + \frac{48}{x}$



- d i $x = 2$, $h = 3$ i.e. 2 m, 2 m, 3 m
 ii 36 m²

9 10 m, 10 m, 5 m

10 a $A = \frac{1}{2}a^2\theta$ b $A = \frac{1}{2}\left(\frac{100}{\theta+2}\right)^2\theta$

c $\theta = 2$ d $\frac{1}{16} \text{ m}^2 = 625 \text{ cm}^2$

11 b i $r = \frac{L}{4}$ ii $\theta = 2$ iii maximum

12 b $\frac{dT}{dx} = \frac{x}{\sqrt{x^2+900}} - \frac{3}{5}$

c i $x = 22.5$ ii 71 seconds

d 63 seconds

13 b $T = \frac{20+16\sqrt{2}}{15} \approx 2.84, 2.84$ hours

14 1.16 hours after noon, 1.2 km apart

15 b $0 < x < 1$ c $x = \frac{1}{\sqrt{2}}, y = \pm 1$

d $A = 2\sqrt{2}$, i.e. the maximum cross-sectional area is $2\sqrt{2}$

16 c ii $\frac{dA}{dx} = -3x^2 - 2ax + a^2$

17 a $b = 5, c = 6$

b i 6 weeks ii 3.852 weeks

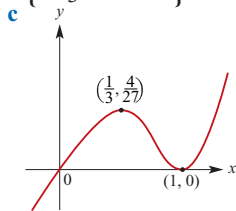
c $190\frac{2}{3} \text{ cm}^2$

18 a $(1, -6)$ b $3(x-1)^2 + 3$

c $3(x-1)^2 + 3 > 3$ for all $x \in \mathbb{R} \setminus \{1\}$

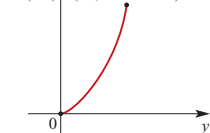
19 a $a = 1, c = 1, b = -2, d = 0$

b $\left\{x: \frac{1}{3} < x < 1\right\}$



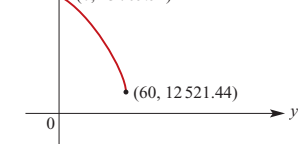
20 a $53\,109\,671 \text{ m}^3$ b $\frac{dV}{dy} = \pi(y+630)^2$

c $V(\text{m}^3)$ (60, 82 165 214)



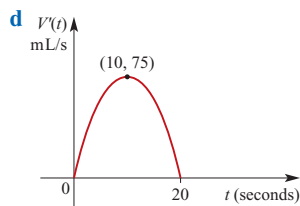
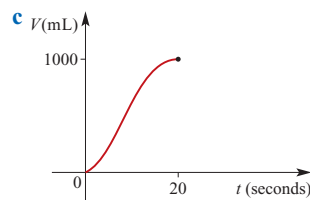
d Capacity = 82 165 214 m^3

e $\frac{dV}{dt}$ (0, 13 765.51)



21 a i $V(0) = 0$ ii $V(20) = 1\,000 \text{ mL}$

b $\frac{dV}{dt} = \frac{3}{4}(20t - t^2)$



e $t = 10$ seconds, 75 mL/s

22 a i $r = \frac{2\pi - \theta}{2\pi}$

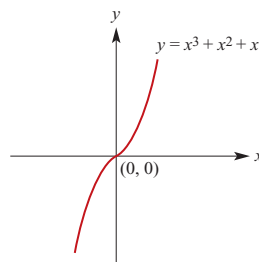
ii $h = \sqrt{1 - \left(\frac{2\pi - \theta}{2\pi}\right)^2}$

b $\frac{49\sqrt{15}\pi}{1536}$ c 0.3281, 2.5271

d i $\theta = 1.153$ ii $V_{\max} = 0.403 \text{ m}^3$

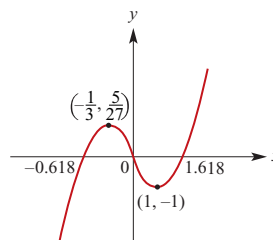
e 0.403 m^3

23 a i

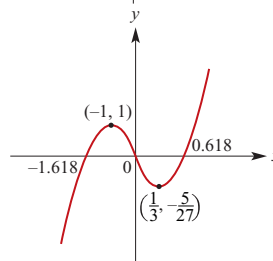


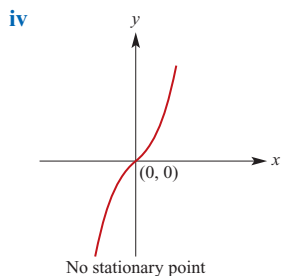
No stationary points

ii



iii



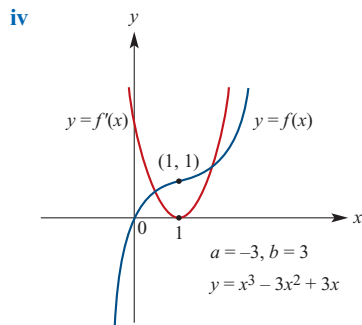
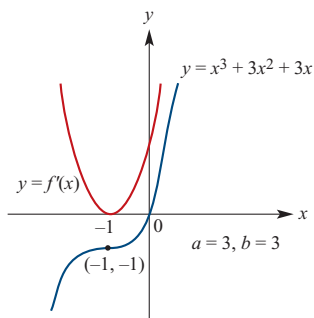


b i $f'(x) = 3x^2 + 2ax + b$

ii $x = \frac{-a \pm \sqrt{a^2 - 3b}}{3}$

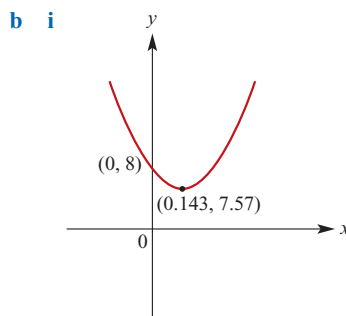
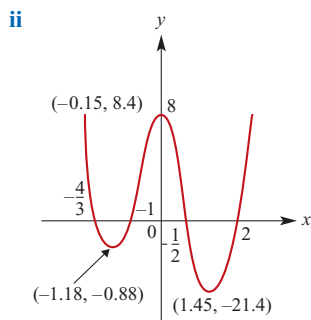
c ii $(-1, -1), (1, 1)$ stationary points of inflexion
 $a = -3$ or $a = 3$

iii, iv



d $a^2 < 3b$

24 a i $a = -21$



ii Min. at $(0.143, 7.57)$

iii $g'(x) = 24x^3 - 3x^2 + 42x - 6$

iv 0.1427

v $g'(0) = -6; g'(10) = 24114$

vi $g''(x) = 72x^2 - 6x + 42$

vii $g''(x) > 0$ for all x . Therefore $g'(x)$ has no turning points. $\therefore$ only one solution

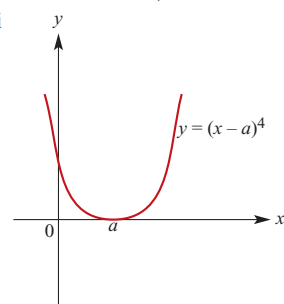
25 b i $x = a$ or $x = b$ or $x = \frac{b+a}{2}$

ii $x = a$ or $x = b$

c Stationary points $(a, 0), (b, 0);$

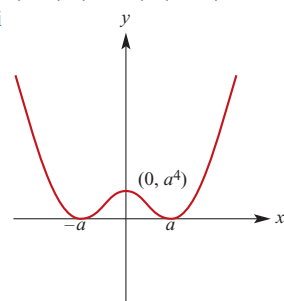
$\left(\frac{b+a}{2}, \frac{(a-b)^4}{16}\right)$

e i



ii $(a, 0), (-a, 0), (0, a^4)$

iii



26 b i $x = a$ or $x = \frac{3b+a}{4}$

ii $x = a$ or $x = b$

c $(a, 0), \left(\frac{3b+a}{4}, \frac{-27}{256}(b-a)^4\right)$, local min.

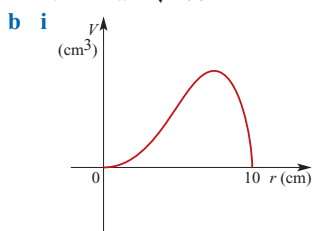
$(a, 0)$ is a stationary point of inflexion

e $\left(-\frac{a}{2}, \frac{-27a^4}{16}\right)$ and $(a, 0)$

f i $b = -\frac{a}{3}$

27 a i $y = \sqrt{100 - r^2}$ and $h = 2\sqrt{100 - r^2}$

ii $V = 2\pi r^2 \sqrt{100 - r^2}$



ii $V = 2418.4$, $r = 8.165$, $h = 11.55$

iii $r = 6.456$ or $r = 9.297$

c i $\frac{dV}{dr} = \frac{400\pi r - 6\pi r^3}{(100 - r^2)^{\frac{1}{2}}}$

ii $V_{\max} = \frac{4000\pi\sqrt{3}}{9}$ occurs when
 $r = \frac{20\sqrt{6}}{6} = \frac{10\sqrt{6}}{3}$

d ii $\frac{dV}{dr} > 0$ for $r \in \left(0, \frac{20\sqrt{6}}{6}\right)$

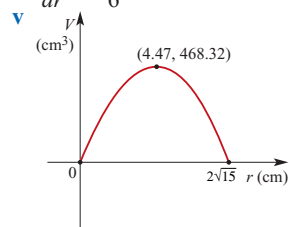
iii $\frac{dV}{dr}$ is increasing for $r \in (0, 5.21)$

28 a i $h = \frac{100 - 3r^2}{2r}$

ii $V = \frac{\pi r}{6}(300 - 5r^2)$

iii defined for $0 < r < 2\sqrt{15}$

iv $\frac{dV}{dr} = \frac{\pi}{6}(300 - 15r^2)$



29 a $\frac{dy}{dt} = \frac{2}{27}$

b $y = \frac{4}{225}x^2$

c i $\frac{dx}{dt} = \frac{5}{54}$

ii $\frac{ds}{dt} = \frac{50}{9}$

30 a i $y = \frac{100}{x^2}$

ii $s = \frac{3000}{x} + 60x^2$

b i $\frac{ds}{dx} = -\frac{3000}{x^2} + 120x$

ii 1538.99 cm^2

c $585 \text{ cm}^2/\text{s}$

Chapter 11

Exercise 11A

1 a $5e^{5x}$

c $-12e^{-4x} + e^x - 2x$

e $(2x + 3)e^{x^2 + 3x + 1}$

g $2e^{2x} - 2e^{-2x}$

b $-21e^{-3x}$

d $\frac{e^{3x} - e^x}{e^{2x}}$

f $(6x - 1)e^{3x^2 - x}$

2 a $e^x(x^2 + 2x + 1)$

b $e^{2x}(2x^3 + 3x^2 + 6x + 5)$

c $2e^{4x+1}(x+1)(2x+3)$

d $\frac{-8x-7}{2e^{4x}\sqrt{x+1}}$

3 a $\frac{3e^x - 2e^{4x}}{(e^{3x} + 3)^2}$

b $\frac{-2e^x}{(e^x - 1)^2}$

c $\frac{2e^{2x} - 2}{(e^{2x} - 2)^2}$

4 a $\frac{2x^3(2-x)}{e^{2x}}$

b $2e^{2x+3}$

c $\frac{3}{2}(2e^{2x} + 1)(e^{2x} + x)^{\frac{1}{2}}$

d $\frac{e^x(x-1)}{x^2}$

e $xe^{\frac{1}{2}x^2}$

f $-x^2e^{-x}$

5 a $e^x(f'(x) + f(x))$

b $\frac{e^x(f(x) - f'(x))}{[f(x)]^2}$

c $f'(x)e^{f(x)}$

d $2e^x f'(x)f(x) + [f(x)]^2 e^x$

Exercise 11B

1 a $\frac{2}{x}$

b $\frac{2}{x}$

c $2x + \frac{3}{x}$

d $\frac{3x-1}{x^2}$

e $\frac{3+x}{x}$

f $\frac{1}{x+1}$

g $\frac{2}{2x+3}$

h $\frac{-2}{-2x+3}$

i $\frac{6}{2x-3}$

j $\frac{3}{15-x}$

k $4 - \frac{3}{x-6}$

2 a $\frac{2x}{x^2+1}$

b $\log_e x + 1$

c $\frac{1 - \log_e x}{x^2}$

d $2x(2 \log_e x + 1)$

e $e^x \left(\log_e x + \frac{1}{x} \right)$

f $\log_e(-x) + 1$

g 1

h $\frac{x^2 + 1 - 2x^2 \log_e x}{x(x^2 + 1)^2}$

3 a $(e, 1)$, $m = \frac{1}{e}$

b (e, e) , $m = 2$

c $(e, \log(e^2 + 1))$, $m = \frac{2e}{e^2 + 1}$

d $(-e, 1)$, $m = -\frac{1}{e}$

e $(1, 1)$, $m = 2$

f $(1, 0)$, $m = -1$

g $(0, 0)$, $m = -2$

h $(0, 0)$, $m = 0$

4 $\frac{1}{2}$

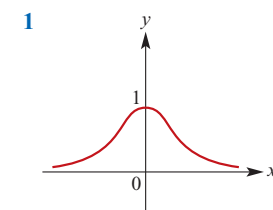
5 $\frac{1+2x}{1+x+x^2}$

6 $\frac{3}{5}$

7 2

8 2

Exercise 11C



2 $\{x: -2 < x < 0\}$

3 $x < 1$, max. value $= \frac{100}{e^4} \approx 1.83$

4 a Min. $f(x) = f(0) = 0$

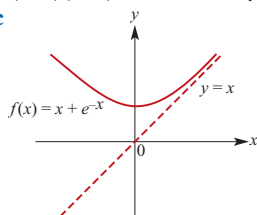
5 a $y = 2$

c $y = 4e^2x - 3e^2$

e $y = 3xe - 2e$

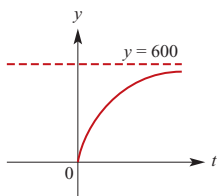
6 a $(0, 1)(\min)$

c



7 $t = 0$, $v = 0.4$ m/s; $t = 1$,
 $v = 0.4e \approx 1.087$ m/s; $t = 2$,
 $v = 0.4e^2 \approx 2.956$ m/s

8 a



b 2.02

9 Maximum population is 44 (44.107) at $t = 5$

10 a $-2y$

b ky

11 a 0.18 kg

b 3.47 hours

c i 6.93 hours

ii 10.4 hours

d 0.2 m

12 $2q$

13 a $f'(x) = ae^{ax}$

b $f(h) \approx ha + 1$

c $f(b+h) \approx e^{ba}(ha+1)$

14 a $\delta y \approx pe^{\frac{a}{2}}$

b $\delta y \approx -2e^ap$

c $\delta y \approx e^a(1+a)p$

d $\delta y \approx e^{-a}(1-a)p$

15 a $\frac{dy}{dt} = 2e^{2t} \frac{dx}{dt} = e^t$

b 2

16 $p = 1$, $q = -6$, $r = 9$

17 a $\frac{5}{\pi e^4}$

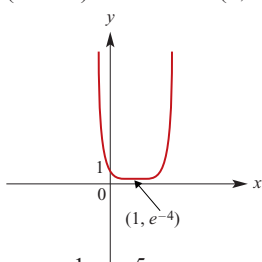
b $\frac{5}{\pi e^5}$

18 $\frac{2e^2}{e^4 - 1}$

19 a $(8x - 8)e^{4x^2 - 8x}$

b $(1, e^{-4})$

c



d $y = -\frac{1}{8}x + \frac{5}{4}$

20 a Tangent $y = x - 1$, normal $y = 1 - x$

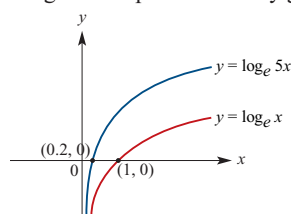
b Tangent $y = -x - 1$, normal $y = x + 1$

21 a $y = 2x - 1$

b $y = kx - 1$

c $y = kx - 1$

22 Tangents are parallel for any given value of x .



23 a p

b $\frac{p}{a}$

24 0

25 $\frac{p}{2}$

26 0.01

27 $\frac{dy}{dt} = \frac{1}{t}$, $\frac{dy}{dx} = \frac{t^2 + 1}{2t^2}$, $\frac{dy}{dx} = 1$, $\frac{dx}{dt} = \frac{2t}{t^2 + 1}$

28 a $\frac{dV}{dh} = \pi([\log_e(h+1)]^2 + h + 2\log_e(h+1) + h + 1)$.

When $h = 2$,

$\frac{dV}{dh} = \pi([\log_e(3)]^2 + 2\log_e(3) + 5)$ and

$\frac{dh}{dt} = 5 \times \frac{dh}{dV}$

b When $h = 10$,

$\frac{dV}{dh} = \pi([\log_e(11)]^2 + 2\log_e(11) + 21)$ and

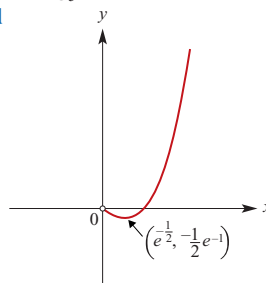
$\frac{dh}{dt} = 5 \times \frac{dh}{dV}$

29 a $2x \log_e x + x$

b $x = 1$

c $x = e^{-\frac{1}{2}}$

d



Exercise 11D

1 a $5 \cos 5x$

b $-5 \sin 5x$

c $5 \sec^2 5x$

d $2 \sin x \cos x = \sin 2x$

e $3 \sec^2(3x+1)$

f $-2x \sin(x^2+1)$

g $2 \sin\left(x - \frac{\pi}{4}\right) \cos\left(x - \frac{\pi}{4}\right)$
 $= \sin\left(2x - \frac{\pi}{2}\right) = \cos 2x$

h $-\frac{\pi}{90} \sin x^\circ$

i $\frac{\pi}{60} \cos x^\circ$

j $\frac{\pi}{60} \sec^2(3x)^\circ$

2 a $\frac{1}{\sqrt{2}}; \sqrt{2}$

b 1; 0

c 2; 0

d 0; 0

e 1; 0

f 1; 4

- 3 a** $-5 \sin x - 6 \cos 3x$ **b** $-\sin x + \cos x$
c $\cos x + \sec^2 x$ **d** $2 \tan x \sec^2 x$
4 a $3x^2 \cos x - x^3 \sin x$
b $\frac{-(1+x) \sin x - \cos x}{(1+x)^2}$
c $e^{-x}(\cos x - \sin x)$ **d** $3 - 2 \sin x$
e $3 \cos 3x \cos 4x - 4 \sin 4x \sin 3x$
f $2 \cos 2x \tan 2x + 2 \sec^2 2x \sin 2x$
g $12 \sin x + 12x \cos x$ **h** $e^{\sin x}(x^2 \cos x + 2x)$
i $2x \cos^2 x - 2x^2 \cos x \sin x$
j $e^x(\sec^2 x + \tan x)$
5 a -2 **b** -6π **c** $-e^\pi$
d $-e^\pi$ **e** $-\frac{1}{\pi}$ **f** 0
6 a $\frac{\tan x}{2}$ **b** $\tan x$
7 a $\frac{\sin 2x - 1}{\sin 2x - 1} = \frac{2 \sin x \cos x - 1}{2 \sin x \cos x - 1}$
b $\sec^2 x e^{\tan x}$ **c** $\frac{7}{2} x^{\frac{5}{2}} \sin 3x + 3x^{\frac{7}{2}} \cos 3x$

Exercise 11E

- 1 a** $y = 2x$ **b** $y = -x + \frac{\pi}{2}$
c $y = 2x - \frac{\pi}{2} + 1$ **d** $y = 2x$
e $y = 3x$ **f** $y = -x - 1 + \frac{\pi}{2}$
2 $3 - 2 \sin x$, gradient always positive
3 Distance $= \frac{\pi}{2} - 1$ **4** 45° **5** $18^\circ 26'$
6 a 4.197 **b** -0.4
7 a i $\frac{p}{\cos^2 \theta}$ **ii** $2p$
b $\frac{\pi}{90} + 1$
8 a $\frac{1}{\sqrt{2}}; -\frac{1}{\sqrt{2}}$
b i $\frac{1}{\sqrt{2}}(1 - h)$ **ii** 0.6967
9 a $-2 \sin(2a)p$ **b** $\frac{1}{2} \cos\left(\frac{a}{2}\right)p$
c $\frac{2p}{\cos^2(2a)}$ **d** $-\frac{1}{2 \cos^2\left(\frac{a}{2}\right)}p$
e $\sin\left(\frac{\pi}{4} - a\right)p$ **f** $-\frac{1}{2} \cos\left(\frac{a}{2}\right)p$
10 a Max. at $x = \frac{\pi}{3}$ and $\frac{5\pi}{3}$,
 min. at $x = 0, \pi$ and 2π
b Max. at $x = \frac{\pi}{6}$, min. at $x = \frac{5\pi}{6}$,
 point of inflexion at $\frac{3\pi}{2}$
c Max. at $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$,
 min. at $x = \frac{7\pi}{6}$ and $\frac{11\pi}{6}$

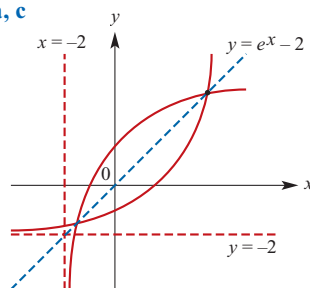
- d** Max. at $x = \frac{\pi}{3}$, point of inflexion at $x = \pi$,
 min. at $\frac{5\pi}{3}$

11 a $\frac{12}{5}$ m/s **b** $-\frac{1}{5}$ rad/s

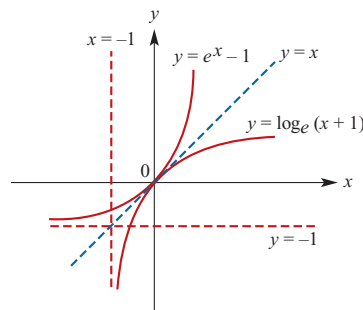
Exercise 11F

- 1 a** 0.69 **b** 1 **c** -1.82 **d** 1.54
2 a Local max. (2.07, 1.61);
 local min. (4.49, 0.53)
b Local max. (1.57, 0)
c Local max. (0.56, 0.55), (3.22, 2.85);
 local min. (1.39, 0.087) (4.66, 1.24)
d Local min. (0, 0)
e Local max. (4.27, -3.56);
 local min. (5.15, 3.56)
f Local max. (0.32, 0.20), $\left(\frac{\pi}{2}, 1\right)$, (2.82, 0.20),
 (3.93, 0), (5.5, 0);
 local min. (0.79, 0), (2.36, 0), (3.46, -0.202),
 (4.71, 1), (5.96, -0.202)

5 a, c



- b** $(-1.841, -1.841), (1.146, 1.146)$
c $y = \log_e(x + 2)$ **d** $R \setminus [-1.841, 1.146]$
e i $a = -1$ **ii** $y = x$
iii (0, 0) **iv** $y = \log_e(x + 1)$
v



- 6 b** $\frac{dy}{dx} = 2$ for $y = 2e^x - 2$
 $\frac{dy}{dx} = \frac{1}{2}$ for $y = \log_e\left(\frac{x+2}{2}\right)$
c $y = 2x$ and $y = \frac{1}{2}x$

- d i** $f^{-1}: (-a, \infty) \rightarrow \mathbb{R}$,
 $f^{-1}(x) = \log_e \left(\frac{x+a}{a} \right)$
ii $y = ax$ and $y = \frac{1}{a}x$
iv Pairs of graphs of this form intersect at the origin.

Exercise 11G

- 1 a** $\lambda = 0.1373$, $P_0 = 30$
b 9.625 hours
c i 4.120 units/hour **ii** 1.373 units/hour
2 a $\frac{1}{30}$ rad/s = 1.9099°/s **b** $\frac{200\sqrt{3}}{3}$ m/s
3 a 8 cos θ
b Area = 16(1 + cos θ) sin θ ;
 maximum area = $12\sqrt{3}$ square units
4 a $\frac{75}{\cos \theta}$ seconds
b 220 - 60 tan θ seconds
d $\frac{dT}{d\theta} = \frac{75 \sin \theta - 60}{\cos^2 \theta}$
e $\theta = \sin^{-1} \frac{4}{5} \approx 53^\circ 8'$
f $T = 265$ seconds; P is 400 metres from B for minimum time

Multiple-choice questions

- 1 C** **2 B** **3 D** **4 E** **5 D**
6 C **7 A** **8 E** **9 B** **10 C**

Short-answer questions (technology-free)

- 1 a** $\frac{2x}{x^2 + 2}$ **b** 3 cos(3x + 2)
c $-\frac{1}{2} \sin \left(\frac{x}{2} \right)$ **d** $(2x - 2)e^{x^2 - 2x}$
e $\frac{1}{x - 3}$ **f** $2\pi \cos(2\pi x)$
g 6 sin(3x + 1) cos(3x + 1)
 $= 3 \sin(6x + 2)$
h $\frac{1}{2x\sqrt{\log_e x}}$ **i** $\frac{2 - 2 \log_e 2x}{x^2}$
j $2x \sin(2\pi x) + 2\pi x^2 \sin(2\pi x)$
2 a $e^x \sin 2x + 2e^x \cos 2x$
b $4x \log_e x + 2x$ **c** $\frac{1 - 3 \log_e x}{x^4}$
d $2 \cos 2x \cos 3x - 3 \sin 2x \sin 3x$
e $\frac{2}{\cos^2 2x} = 2 \sec^2 2x$
f $-9 \cos^2(3x + 2) \sin(3x + 2)$
g $2x \sin^2 3x + 6x^2 \cos 3x \sin 3x$
3 a $2e^2 \approx 14.78$ **b** 0
c $15e^3 + 2 \approx 303.28$ **d** 1

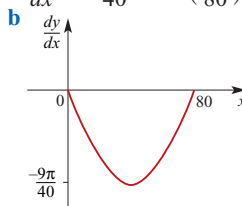
- 4 a** ae^{ax} **b** ae^{ax+b} **c** $-be^{a-bx}$
d $abe^{ax} - bae^{bx}$ **e** $(a-b)e^{(a-b)x}$
5 0.25 m/s; 0.25e m/s; 0.25e² m/s; 0.25e⁴ m/s
6 a $25e^{100t}$ °C/second **b** $25e^5$ °C/second
7 $y = ex$
8 b 20 cm/year

- 10** 2 **11** $a = 2$ or $a = 1$

- 12 a** $y = \frac{1}{e}x$ **b** $y = \frac{x}{\sqrt{2}} - \frac{\pi}{2\sqrt{2}} + \sqrt{2}$
c $y = x - \frac{3\pi}{2}$ **d** $y = \left(\frac{-1}{e} \right)x$

Extended-response questions

- 1 a** $\frac{dy}{dx} = \frac{-9\pi}{40} \sin \left(\frac{\pi x}{80} \right)$



- c** Magnitude of gradient is a maximum at the point (40, 12)

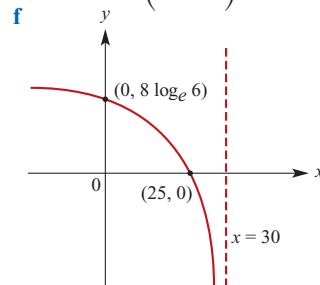
- 2 a** $f'(x) = \frac{3\,000\,000e^{-0.3x}}{(1 + 100e^{-0.3x})^2}$

- b i** 294 kangaroos/year
ii 933 kangaroos/year

- 3 a** $a = 30$ **b** $(0, 8 \log_e 6)(25, 0)$

- c** $f'(20) = -0.8$

- d** $f^{-1}(x) = 5 \left(6 - e^{\frac{x}{8}} \right)$ **e** R

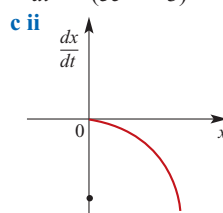


- 4 b** $\left(\frac{\pi}{2}, e \right), \left(\frac{3\pi}{2}, \frac{1}{e} \right)$ **c** $\left[\frac{1}{e}, e \right]$

- d** 2π as $g(x + 2\pi) = g(x)$

- 6 a i** 30 g **ii** 12.28 g

- b** $\frac{dx}{dt} = \frac{-300\lambda e^{\lambda t}}{(5e^{\lambda t} - 3)^2}$



7 b $MP = \frac{2}{\tan \theta}$ c $NQ = 8 \tan \theta$

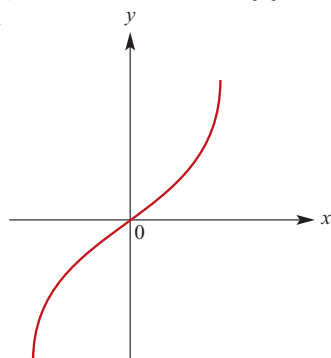
d $OP + OQ = \frac{2}{\tan \theta} + 8 \tan \theta + 10$

e $\frac{dx}{d\theta} = -2 \operatorname{cosec}^2 \theta + 8 \sec^2 \theta$

f $x = 18$, $\theta = 26^\circ 34'$

8 a $f'(x) = e^x + e^{-x}$ b $\{0\}$

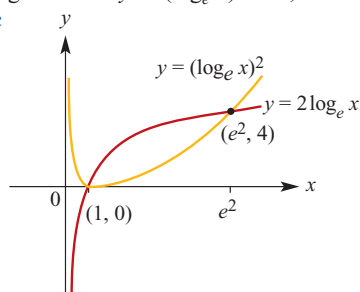
d



9 a $x = 1$ or $x = e^2$

b Gradient of $y = 2 \log_e x$ is 2, when $x = 1$;
gradient of $y = (\log_e x)^2$ is 0, when $x = 1$

c



d $\{x: 2 \log_e x > (\log_e x)^2\} = [x: 1 < x < e^2]$

10 $x = e$

11 a $h = a(1 + \cos \theta)$ b $r = a \sin \theta$

d $\frac{dV}{d\theta} = \frac{1}{3} \pi a^3 [-\sin^3 \theta$

$+ 2 \sin \theta \cos \theta (1 + \cos \theta)]$

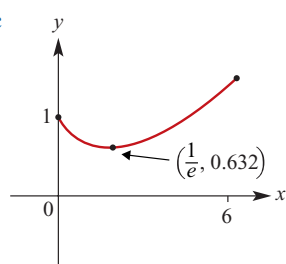
$\theta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70^\circ 32'$

e $V = \frac{32}{81} \pi a^3 \text{ cm}^3$

12 a $f'(x) = \log_e x + 1$

b $x \approx 0.37$, i.e. during the fourth month of its life

c



d When $x = 6$

13 b $\frac{dy}{dt} = \frac{bAe^{bt}}{(1 + Ae^{bt})^2}$

e After 7 hours (to the nearest hour)

14 $\frac{144}{25\pi}$ degrees per second $= 1.833$ degrees per second

15 0.1 km/s

16 a $y = ex$ b $y = 2ex$ c $y = kex$

e i A unique real root $k = \frac{1}{e}$ or $k \leq 0$

ii $k > \frac{1}{e}$

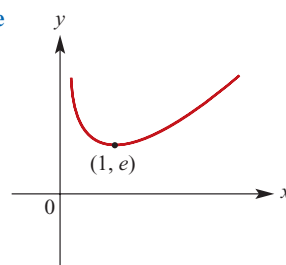
17 a $f'(x) = \frac{xe^x - e^x}{x^2}$

b $x = 1$ c $(1, e)$; minimum

d i $\frac{f'(x)}{f(x)} = \frac{x-1}{x}$

ii $\lim_{x \rightarrow \infty} \frac{f'(x)}{f(x)} = 1$, i.e. $f(x) \rightarrow f'(x)$ as $x \rightarrow \infty$

e



f $t = \frac{1}{k} \approx 45.27$ years, minimum occurred in 1945

18 a $A = 1000$, $k = \frac{1}{5} \log_e 10 \approx 0.46$

b $\frac{dN}{dt} = kAe^{kt}$, where $A = 1000$ and

$k = \frac{1}{5} \log_e 10$

c $\frac{dN}{dt} = kN$

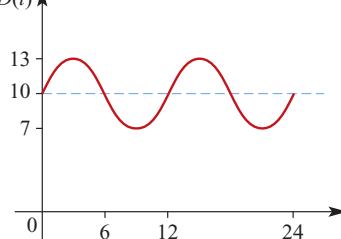
d i $\frac{dN}{dt} \approx 2905.7$

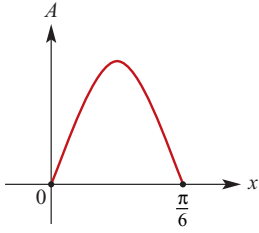
ii $\frac{dN}{dt} \approx 4.61 \times 10^{12}$

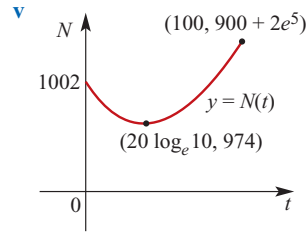
19 a $t \approx 34.66$ years

b $t \approx 9.12$ years

20 a $D(t)$



- b** $\{t: D(t) \geq 8.5\} = \{t: 0 \leq t \leq 7\} \cup \{t: 11 \leq t \leq 19\} \cup \{t: 23 \leq t \leq 24\}$
- c i** 0 metres/hour
ii $-\frac{\pi}{2}$ metres/hour
iii $\frac{\pi}{2}$ metres/hour
- d i** $t = 0, 12, 24$ **ii** $t = 6, 18$
- 21 a** The height is reached $\frac{1}{6}$ th of a second after $t = 0$.
b $\frac{2}{3}$
c When $t = \frac{1}{3}$, speed = 0.6π m/s (moving downwards).
 When $t = \frac{2}{3}$, speed = 0.6π m/s (moving upwards).
 When $t = \frac{1}{6}$, speed = 0 m/s.
- 22 a** $p = 12, q = 8, r = \frac{1}{6}$
b $T'(3) = -\frac{4}{3}\pi$; hours of night decreasing by $\frac{4\pi}{3}$ hours/month $T'(9) = \frac{4\pi}{3}$, hours of night increasing by $\frac{4\pi}{3}$ hours/month
c $-\frac{8}{3}$ hours/month
d $t = 9$ (after 9 months)
- 23 a** $A = 2x \cos(3x)$
b i $\frac{dA}{dx} = 2 \cos(3x) - 6x \sin(3x)$
ii When $x = 0$; $\frac{dA}{dx} = 2$.
 When $x = \frac{\pi}{6}$, $\frac{dA}{dx} = -\pi$.
- c i**
- 
- ii** $x = 0.105$ or $x = 0.449$
iii Maximum area = 0.374 when $x = 0.287$
- 24 a i** $N'(t) = -1 + \frac{1}{10}e^{\frac{t}{20}}$
ii Minimum population is 974 and occurs when $t = 20 \log_e 10$.
iii $N(0) = 1002$
iv $N(100) = 900 + 2e^5$



- b i** $N_2(0) = 1002$
ii $N_2(100) = 990 + 2e^{\frac{1}{2}}$
iv Minimum population is 974 and occurs when $t = (20 \log_e 10)^2$.
c ii Minimum population is 297 and occurs when $t = 100.24$.
d i $N'_3(t) = -\frac{3}{2}t^{\frac{1}{2}} + \frac{1}{10}e^{\frac{t}{20}}$
- 25 c** $\frac{\pi}{180} \sec^2 \alpha$
- 26 a i** $BX = h \tan \alpha$
ii $R = \frac{h}{2}(1 + \tan^2 \alpha)$
b i $\delta R = \frac{\pi}{180} \times \frac{h \sin \alpha}{\cos^3 \alpha} \delta \alpha$
ii -0.1209
- 27 a** $\frac{1}{3} \log_e \left(\frac{10}{3}\right)$
b i $x = 0$ and $x = \frac{5}{2}$
ii $x = \frac{-4 + 5a \pm \sqrt{25a^2 + 16}}{4a}$

Chapter 12

Exercise 12A

- 1 a** 3.81 square units **b** 1.34 square units
c $\frac{35}{2}$ square units
- 2 a** 13.2 square units **b** 10.2 square units
- 3 a** 10 square units **b** 10.64 square units
- 4 a** 0.72 square units
b 2.88 square units, decrease strip width
- 5 a** 36.8 square units **b** 36.7 square units
- 6 a** ≈ 48 square units **b** Distance travelled
- 7** 11.9 square units
- 8 a** $\frac{9}{2}$ square units **b** 9 square units
c 4 square units

Exercise 12B

- 1 a** $\frac{x^4}{8} + c$ **b** $\frac{5}{4}x^4 - x^2 + c$
c $\frac{x^4}{5} - x^3 + c$ **d** $2z + \frac{5}{2}z^2 - z^3 + c$

$$2 \text{ a } -3x^{-1} + c = \frac{-3}{x} + c$$

$$\text{b } -\frac{2}{3x^3} + 3x^2 + c$$

$$\text{c } \frac{4x^{\frac{3}{2}}}{3} + \frac{2x^{\frac{5}{2}}}{5} + c = \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + c$$

$$\text{d } \frac{9}{4}x^{\frac{4}{3}} - \frac{20}{9}x^{\frac{9}{4}} + c \quad \text{e } \frac{3z^2}{2} - \frac{2}{z} + c$$

$$\text{f } \frac{12}{7}x^{\frac{7}{4}} - \frac{14}{3}x^{\frac{3}{2}} + c$$

$$3 \text{ a } y = x^2 - 3x + 3 \quad \text{b } y = \frac{x^4}{4} + 6$$

$$\text{c } y = \frac{2}{3}x^{\frac{3}{2}} + \frac{1}{2}x^2 - \frac{22}{3}$$

$$4 \text{ f}(x) = x^3 + \frac{1}{x} - 8 \quad \text{b } 201 \quad \text{c } s = \frac{3}{2}t^2 + \frac{8}{t} - 8$$

$$6 \text{ a } k = -32$$

$$\text{b } 201$$

Exercise 12C

$$1 \text{ a } \frac{1}{6}(2x-1)^3 + c \quad \text{b } -\frac{1}{4}(t-2)^4 + c$$

$$\text{c } \frac{1}{20}(5x-2)^4 + c \quad \text{d } \frac{2}{9}(3x+6)^{\frac{3}{2}} + c$$

$$\text{e } \frac{2}{3}(3x+6)^{\frac{1}{2}} + c \quad \text{f } \frac{1}{9}(2x-4)^{\frac{9}{2}} + c$$

$$\text{g } \frac{1}{7}(3x+11)^{\frac{7}{3}} + c \quad \text{h } -\frac{2}{9}(2-3x)^{\frac{3}{2}} + c$$

$$\text{i } -\frac{1}{10}(5-2x)^5 + c \quad \text{j } \frac{1}{16}(3-4x)^{-4} + c$$

$$2 \text{ a } \frac{1}{2} \log_e |x| + c \quad \text{b } \frac{1}{3} \log_e |3x+2| + c$$

$$\text{c } -\frac{3}{4} \log_e |1-4x| + c \quad \text{d } x - \log_e |x+1| + c$$

$$\text{e } x + \log_e |x| + c \quad \text{f } \frac{-1}{x+1} + c$$

$$\text{g } \frac{x^2}{2} + 2x + \log_e |x+1| + c$$

$$\text{h } 2x - \log_e |x+1| + c$$

$$\text{i } -\frac{3}{2(x-1)^2} + c$$

$$3 \text{ a } 5 \log_e |x| + c \quad \text{b } 3 \log_e |x-4| + c$$

$$\text{c } 5 \log_e |2x+1| + c$$

$$\text{d } -3 \log_e |2x-5| + c \quad \text{e } \frac{-3}{2x-1}$$

$$\text{f } -\frac{1}{3} \log_e |3x-4| + c$$

$$4 \text{ a } y = \frac{1}{2} \log_e |x| + 1 \quad \text{b } y = 10 - \log_e |5-2x|$$

$$5 \text{ y} = 10 \log_e |x-5|$$

$$6 \text{ y} = 3 \log_e \left(\frac{|x-2|}{2} \right) + 10$$

$$7 \text{ y} = \frac{5}{4} \log_e \left(\frac{5}{|2x-1|} \right) + 10$$

Exercise 12D

$$1 \text{ a } \frac{1}{2}e^{2x} - 2e^{\frac{x}{2}} + c \quad \text{b } e^x - e^{-x} + c$$

$$\text{c } \frac{2}{3}e^{3x} + e^{-x} + c \quad \text{d } 15e^{\frac{x}{3}} - 15e^{\frac{x}{5}} + c$$

$$2 \text{ a } y = \frac{1}{2}(e^{2x} - x^2 + 9)$$

$$\text{b } y = -\frac{3}{e^x} - e^x + 8$$

$$3 \text{ y} = 9 - 2e^{-2}$$

$$4 \text{ a } k = 2 \quad \text{b } y = \frac{1}{2}e^{2x} + \frac{1}{2}e^2$$

Exercise 12E

$$1 \text{ a } \frac{7}{3} \quad \text{b } 20 \quad \text{c } -\frac{1}{4} \quad \text{d } 9$$

$$\text{e } \frac{1}{2} \quad \text{f } \frac{140}{3} \quad \text{g } 15\frac{1}{3} \quad \text{h } 343\frac{11}{20}$$

$$2 \text{ a } 10 \quad \text{b } 1 \quad \text{c } \frac{13}{3} \quad \text{d } \frac{1}{3} \quad \text{e } \frac{10}{441}$$

$$\text{f } 34 \quad \text{g } \frac{2}{3}(2^{\frac{3}{2}} - 1) \quad \text{h } 2 - 2^{\frac{1}{2}} \quad \text{i } \frac{1}{15}$$

$$3 \text{ a } y = \frac{1}{2}(e^2 - 1) \quad \text{b } y = \frac{1}{2}(3 - e^{-2})$$

$$\text{c } y = 6e^{\frac{1}{3}} - 4 \quad \text{d } y = e^2 - e^{-2}$$

$$4 \text{ a } 10 \quad \text{b } 17 \quad \text{c } -5 \quad \text{d } 9 \quad \text{e } -3$$

$$5 \text{ a } \log_e \left(\frac{1}{3} \right) \quad \text{b } \frac{1}{2} \log_e (3) \quad \text{c } \frac{3}{2} \log_e \left(\frac{19}{17} \right)$$

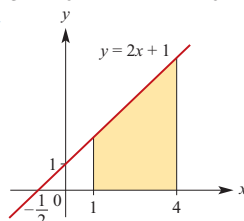
Exercise 12F

$$1 \text{ a } 3 \quad \text{b } 44$$

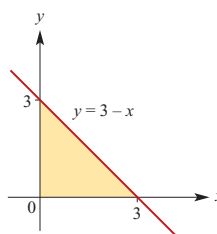
$$\text{c } 8 \quad \text{d } 10$$

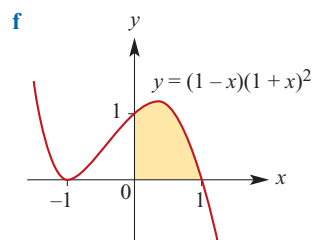
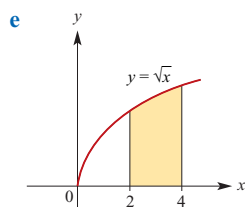
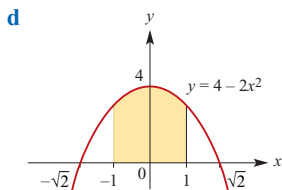
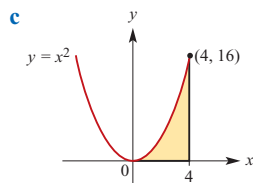
$$2 \text{ a } \frac{4}{3} \quad \text{b } \frac{1}{6} \quad \text{c } 121\frac{1}{2} \quad \text{d } \frac{1}{6} \quad \text{e } 4\sqrt{3} \quad \text{f } 108$$

$$3 \text{ a }$$

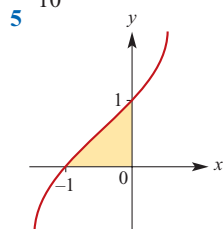


$$\text{b }$$





4 $\frac{321}{10}$ square units

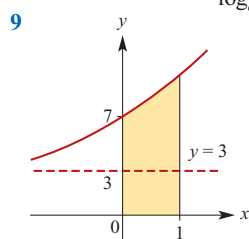


$$\int_{-1}^0 1 + x^3 dx = \frac{3}{4} \text{ sq. units}$$

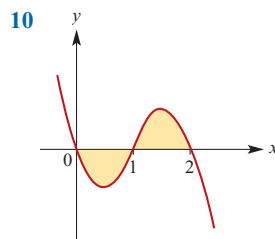
6 a $\frac{5}{6}$ sq. units **b** $8\frac{1}{6}$ sq. units

7 a $A(0, 3), B(1, 0)$ **b** 2 sq. units

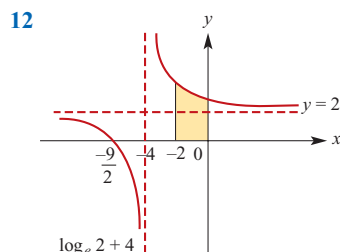
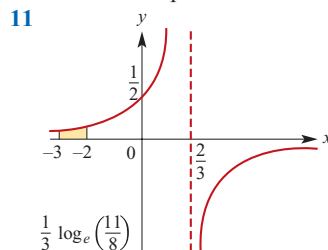
8 b Derivative = $(\log_e a)e^{x \log_e a}$,
antiderivative = $\frac{e^{x \log_e a}}{\log_e a}$



Area = $2e^2 + 1 \approx 15.78$ square units



Area = 0.5 square units



Exercise 12G

1 a $\frac{1}{3} \sin 3x$

b $-2 \cos \frac{1}{2}x$

c $\sin 3x$

d $-4 \cos \frac{1}{2}x$

e $-\frac{1}{2} \cos \left(2x - \frac{\pi}{3}\right)$

f $\frac{1}{3} \sin 3x - \frac{1}{2} \cos 2x$

g $\frac{1}{4} \sin 4x + \frac{1}{4} \cos 4x$

h $\frac{1}{4} \cos 2x + \frac{1}{3} \sin 3x$

i $-\frac{1}{4} \sin \left(2x + \frac{\pi}{3}\right)$

j $-\frac{1}{\pi} \cos \pi x$

2 a $1 - \frac{1}{\sqrt{2}}$

b $\frac{2}{3}$

c $1 + \frac{1}{\sqrt{2}}$ **d** 2

e 1

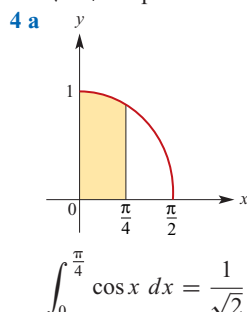
f $\frac{2}{3}$

g $-\frac{1}{2}$ **h** 4

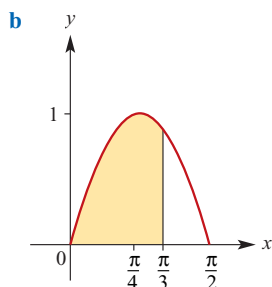
i $\frac{1 - \sqrt{3}}{4}$

j -2

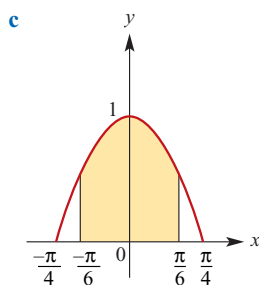
3 $-\sqrt{2} + 2$ square units



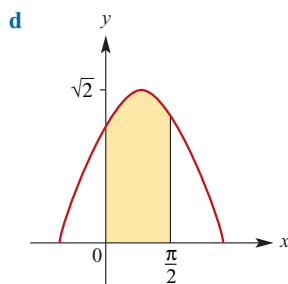
$$\int_0^{\pi/4} \cos x dx = \frac{1}{\sqrt{2}}$$



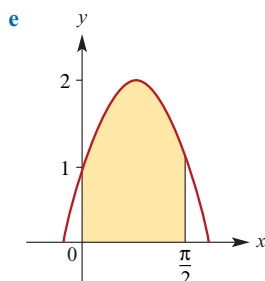
$$\int_0^{\pi/3} \sin 2x \, dx = \frac{3}{4}$$



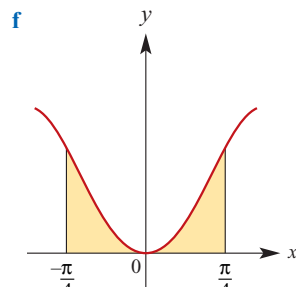
$$\int_{-\pi/6}^{\pi/6} \cos 2x \, dx = \frac{\sqrt{3}}{2}$$



$$\int_0^{\pi/2} \cos \theta + \sin \theta \, d\theta = 2$$

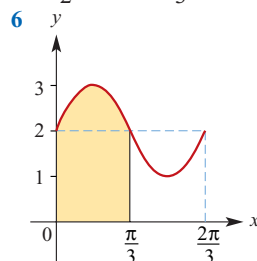


$$\int_0^{\pi/2} \sin 2\theta + 1 \, d\theta = 1 + \frac{\pi}{2}$$



$$\int_{-\pi/4}^{\pi/4} 1 - \cos 2\theta \, d\theta = \frac{\pi}{2} - 1$$

5 a $\frac{\sqrt{2}}{2}$ **b** $-\frac{1}{3}$ **c** $-\frac{\sqrt{3}}{3}$ **d** $-\frac{1}{\sqrt{2}}$



$$\int_0^{\pi/3} 2 + \sin 3x \, dx = \frac{2\pi}{3} + \frac{2}{3}$$

Exercise 12H

1 a $4\frac{2}{3}$ **b** $2\frac{2}{3}$ **c** 12 **d** $\frac{5\sqrt{3}}{4} - 2$

e $\frac{e^4}{2} + 4 \log_e 2 - \frac{e^2}{2}$ **f** $\frac{2}{3}$ **g** 4

h $\frac{5\pi^2}{8} + 1$ **i** $8 \log_e 2 + \frac{51}{4}$ **j** $\frac{1}{12}$

2 0.5 sq. units

3 a 139.68 **b** 18.50 **c** -0.66

d -23.76 **e** 2.06 **f** 0.43

4 b $5 \log_e 3 + 4$ **5 b** $5 + 6 \log_e 2$

6 a $\frac{dy}{dx} = -4 \left(1 - \frac{1}{2}x\right)^7$ Hence

$$\int \left(1 - \frac{1}{2}x\right)^7 dx = -\frac{1}{4} \left(1 - \frac{1}{2}x\right)^8 + c$$

b $\frac{dy}{dx} = -\tan x$.

Hence $\int_0^{\pi/3} \tan x \, dx = \log_e 2$

7 $f(x) = 1 - 2 \cos \frac{1}{2}x$

8 a $f(x) = \frac{1}{2} \sin 2x + 1$

b $f(x) = 3 \log_e x + 6$

c $f(x) = 2e^{\frac{x}{2}} - 1$

9 $\sin 3x + 3x \cos 3x$. Hence

$$\int_0^{\frac{\pi}{6}} x \cos 3x \, dx = \frac{\pi}{18} - \frac{1}{9}$$

10 $a = 1$, $b = -2$; area = 3 square units

11 a 1.450 square units

b 1.716 square units

12 0.1345

13 $f(x) = \frac{1}{2}(x^2 - \cos 2x + 3)$

14 a $\int f(x) \, dx = (x^2 + 1)^3 + c$

b $\int h(x) \, dx = \sin(x^2) + c$

c $\int (f(x) + h(x)) \, dx = (x^2 + 1)^3 + \sin x^2 + c$

d $\int -f(x) \, dx = -(x^2 + 1)^3 + c$

e $\int f(x) - 4 \, dx = (x^2 + 1)^3 - 4x + c$

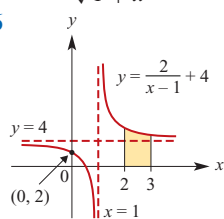
f $\int 3h(x) \, dx = 3 \sin x^2 + c$

15 a $\frac{1}{\cos^2 x}$, $\tan x$

b $\frac{6x}{3x^2 + 7}$, $\frac{1}{6} \log_e \frac{19}{7}$

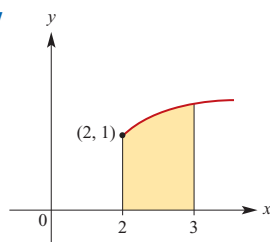
c $1 + \frac{x}{\sqrt{1+x^2}}$, $\log_e(1 + \sqrt{2})$

16



$$\int_0^3 \left(\frac{2}{x-1} + 4 \right) dx = (2 \log_e 2) + 4$$

17



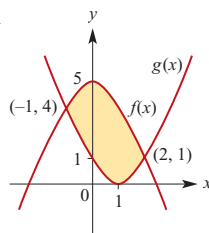
$$\int_2^3 \sqrt{2x-4} + 1 \, dx = \frac{1}{3} \times 2^{\frac{3}{2}} + 1$$

18 a $\frac{4\sqrt{2}}{3} - \frac{2}{3}$ b $2\frac{5}{3}$ c $\frac{1}{3} \log_e 4$

d $\frac{1}{2} \log_e 3 + 3$ e $-2\frac{2}{3}$ f $2\sqrt{2} - 2$

Exercise 12I

1



Area = 9 square units

2 36 square units

3 a 36 square units b $20\frac{5}{6}$ square units

c 4 square units d $4\frac{1}{2}$ square units

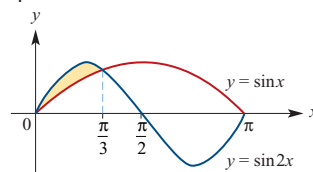
e $4\frac{1}{2}$ square units

4 a 2 square units

b $e + e^{-1} - 2 \approx 1.086$ square units

5 3.699 square units

6 $\frac{1}{4}$ square units



7 $\frac{1}{2}$ square units

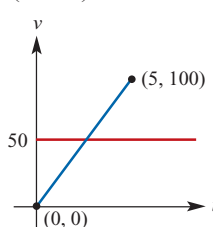
8 $P(\log_e 3, 3)$; area ≈ 2.197 square units

Exercise 12J

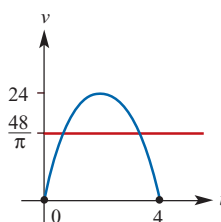
1 a $\frac{2}{3}$ b $\frac{2}{\pi}$ c $\frac{2}{\pi}$ d 0 e $\frac{1}{2}(e^2 - e^{-2})$

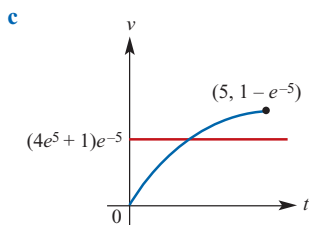
2 $10(e^5 - 1)e^{-5} \approx 9.93^\circ\text{C}$

3 a

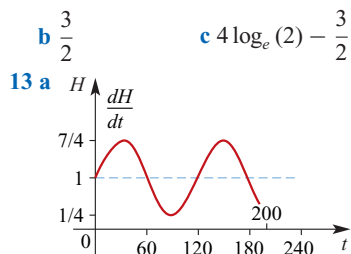
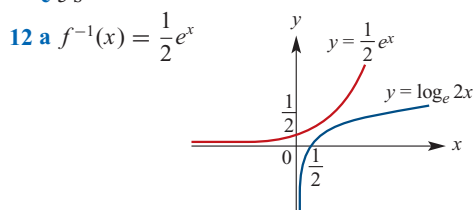


b





- 4** $\frac{147}{10}$ m/s **5** $\frac{a^2}{6}$
6 a $3000(2 - 2^{0.9})$ N/m²
b $1000(4^{0.1} - 1)$ N/m²
7 a $x = t^2 - 3t$ **b** $x = 0$ **c** 0
d $\frac{9}{2}$ metres **e** $\frac{3}{2}$ m/s
8 a Displacement = $\frac{2t^3}{3} - 4t^2 + 6t + 4$
 Acceleration = $4t - 8$
b When $t = 1$, displacement = $\frac{20}{3}$ m
 When $t = 3$, displacement = 4 m
c When $t = 1$, acceleration = -4 m/s²
 When $t = 3$, acceleration = 4 m/s²
9 Initial displacement -3 m
10 Velocity = 73 m/s
 Position = $\frac{646}{3}$ m
11 a Velocity = $-10t + 25$
b Height = $-5t^2 + 25t$
c $\frac{5}{2}$ s
d $\frac{125}{4}$ m
e 5 s



- b** $t \in (10, 50) \cup (130, 170)$
c $t = 30$ or $t = 150$
d i 120 kilojoules
ii 221.48 kilojoules
14 $71\,466\frac{2}{3}$ m³
15 a 46 5 m² **b** 46 500 m³
16 1.26 m

- 17 a** 6 metres **b** 18π m²
c i $y - 3 + 3 \cos\left(\frac{a}{3}\right)$
 $= \frac{-1}{\sin\left(\frac{a}{3}\right)}(x - a)$
ii 5.409
18 a i 9 **ii** $\frac{3(\sqrt{2} + 2)}{2}$ **iii** 12
b Maximum value is 12, minimum value is 0.834
c $\frac{48(\pi + 1)}{\pi}$ litres

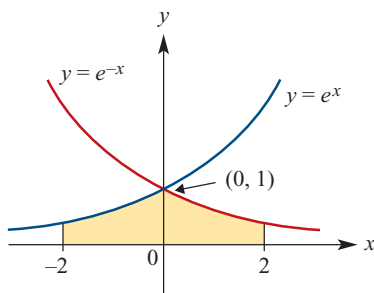
Multiple-choice questions

- 1** C **2** C **3** B **4** B **5** A
6 D **7** E **8** C **9** C **10** D

Short-answer questions (technology-free)

- 1 a** $\frac{65}{4}$ **b** 0 **c** $-\frac{5a^{\frac{3}{2}}}{3}$ **d** $-\frac{55}{3}$
e $\frac{1}{2}$ **f** 1 **g** 0 **h** 0
2 $\frac{23}{2}$ **3** 3 **4** 4 **5** 820
6 $\frac{85}{4}$ **7** $\frac{5}{3}$ **8** $\frac{5}{3}$
9 $\int_c^d f(x) - g(x)dx + \int_b^c g(x) - f(x)dx + \int_a^b f(x) - g(x)dx$
10 a $P(3, 9), Q(7.5, 0)$ **b** 29.25 square units
11 a 5 **b** $p = \frac{20}{7}$
12 3.45 square units
13 a $A(0, 6), B(5, 5)$ **b** $15\frac{1}{6}$ square units
c $\frac{125}{6}$ square units
14 a 2 square units
b $e + e^{-1} - 2 \approx 1.086$ square units
15 a
-
- b** $e^2 + 1 \approx 8.39$

16 a



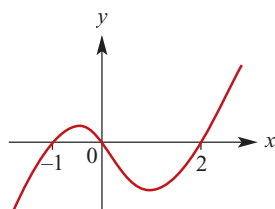
b $2 - 2e^{-2}$

17 a $e - 1 \approx 1.72$

b $2(e - 1) \approx 3.44$ square units

18 $2 + e^2 \approx 9.39$ square units

19 $3\frac{1}{12}$ square units



20 a $e^2 + 1$ b $\log_e\left(\frac{2}{3}\right) - \frac{3}{2}$

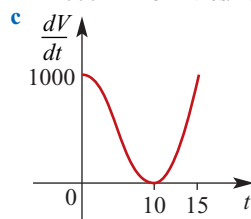
c $\frac{\pi^2}{8} + 1$ d $\frac{1}{2}\left(\log_e\left(\frac{5}{6}\right) - \frac{1}{e^8} + \frac{1}{e^{10}}\right)$

Extended-response questions

1 a When $t = 0$, 1000 million litres/hour.
When $t = 2$, 896 million litres/hour.

b i $t = 0$ and $t = 15$

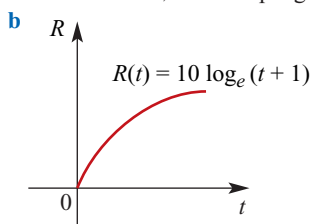
ii 1000 million litres/hour



d i 5000

ii 5000 million litres flowed out in the first 10 hours.

2 a When $t = 5$, ≈ 17.9 penguins per year;
when $t = 10$, ≈ 23.97 penguins per year;
when $t = 100$, ≈ 46.15 penguins per year.



c $t = e^{\frac{R}{10}} - 1; R \geq 0$
 $(R^{-1}(t) = e^{\frac{t}{10}} - 1)$

d i 3661

ii The growth in the size of the penguin population over 100 years (assuming zero death rate).

3 a $4y - 5x = -3$

b $\left(\frac{3}{5}, 0\right)$ c $(1, 0)$ d $\frac{9}{40}$ e $9 : 49$

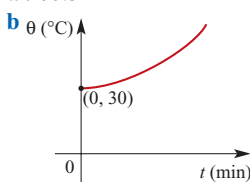
5 a $\frac{1}{3}$ square units

d $1 - \frac{n-1}{n+1} = \frac{2}{n+1}$ square units

e $\frac{9}{11}, \frac{99}{101}, \frac{999}{1001}$

f area between curves approaches 1

6 a 968.3°



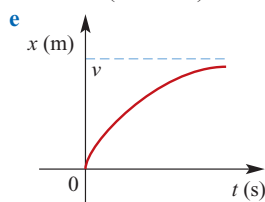
c 2.7 min

d 64.5°C/min

7 a $5 \times 10^4 \text{ m/s}^2$

b Magnitude of velocity becomes very small

c $5 \times 10^4 (1 - e^{-20}) \text{ m}$ d $v(1 - e^{-t})$



8 a $\frac{d}{dx}(e^{-3x} \sin 2x) = -3e^{-3x} \sin 2x + 2e^{-3x} \cos 2x$

c $\int e^{-3x} \sin 2x dx = \frac{-1}{13}(3e^{-3x} \sin 2x + 2e^{-3x} \cos 2x)$

9 a i $\tan a = \frac{4}{3}$ ii $\sin a = \frac{4}{5}$ $\cos a = \frac{3}{5}$

b 2 square units

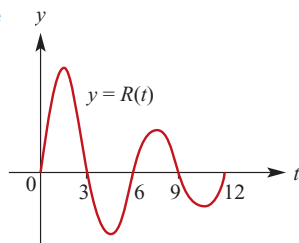
10 a $\frac{dy}{dx} = \log_e x + 1, \int_1^e (\log_e x) dx = 1$

b $\frac{dy}{dx} = (\log_e x)^n + n(\log_e x)^{n-1}$

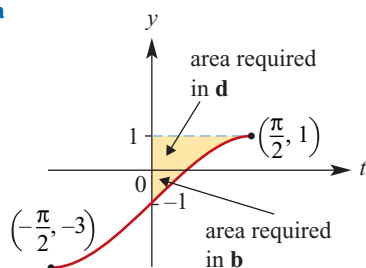
d $\int_1^e (\log_e x)^3 dx = 6 - 2e$

11 $s = \sqrt[3]{ba^2}$
 $r = \sqrt[3]{b^2a}$

- 12 a** $\frac{dy}{dx} = -\frac{x}{10}e^{\frac{x}{10}}$ and
 $\frac{dy}{dx} = -x(100 - x^2)^{-\frac{1}{2}}$
- b** When $x = 0$, $\frac{dy}{dx} = 0$ for both functions
- c** $-e$ **d** 6.71 square units
e 8.55% **f** $(25\pi - 50)$ square units
- g i** $10(10e - 20)$
ii $(25\pi - 100e + 200)$ square units
- 13 a i** $R(0) = 0$ **ii** $R(3) = 0$
- b** $R'(t) = e^{-\frac{t}{10}} \left(\frac{10\pi}{3} \cos\left(\frac{\pi t}{3}\right) - \sin\left(\frac{\pi t}{3}\right) \right)$
- c i** 1.41, 4.41, 7.41, 10.41
ii (1.41, 8.65) (7.41, 4.75) local maxima
 (4.41, -6.41) (10.41, -3.52) local minima
- d** $t = 0$ or 3 or 6 or 9 or 12
- e**



- f i** 16.47 litres **ii** 12.20 litres
iii 8.27 litres
g 12.99 litres
- 14 a**



- b** $\int_0^{\frac{\pi}{6}} f(x) dx = 2 - \sqrt{3} - \frac{\pi}{6}$
- c** $f^{-1}: [-3, 1] \rightarrow R$,
 $f^{-1}(x) = \sin^{-1} \frac{(x+1)}{2}$
- d** $\int_0^1 f^{-1}(x) dx = \frac{\pi}{2} - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(x) dx$
 $= \frac{5\pi}{6} - \sqrt{3}$

- 15 b** $1 - \frac{\pi}{4}$

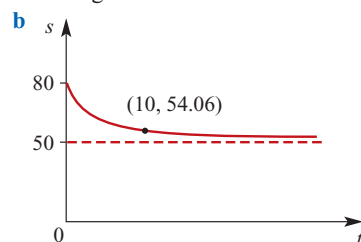
Chapter 13

Multiple-choice questions

- | | | | | | |
|------|------|------|------|------|------|
| 1 E | 2 D | 3 E | 4 D | 5 B | 6 C |
| 7 D | 8 A | 9 D | 10 D | 11 A | 12 B |
| 13 B | 14 E | 15 D | 16 E | 17 B | 18 C |
| 19 E | 20 D | 21 B | 22 E | 23 B | 24 D |
| 25 B | 26 A | 27 D | 28 D | 29 A | 30 C |
| 31 A | 32 C | 33 D | 34 B | 35 A | 36 E |
| 37 D | 38 A | 39 A | 40 A | 41 B | 42 D |
| 43 B | 44 B | 45 C | 46 D | 47 B | 48 D |
| 49 C | 50 A | 51 C | 52 B | 53 E | 54 C |
| 55 C | 56 D | 57 C | 58 B | 59 D | 60 C |
| 61 B | 62 B | 63 D | 64 E | 65 B | 66 D |

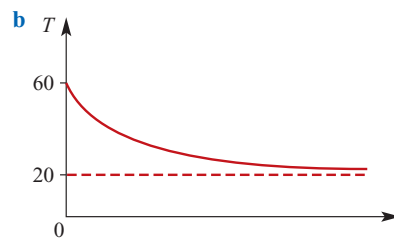
Extended-response questions

- 1 a** 54.06 g



- b**
- c** $\frac{ds}{dt} = -6e^{-\frac{1}{5}t}$ **d** $\frac{ds}{dt} = -\frac{1}{5}(s - 50)$
- e** 0.8 g/litre **f** 17 seconds

- 2 a** 60°C

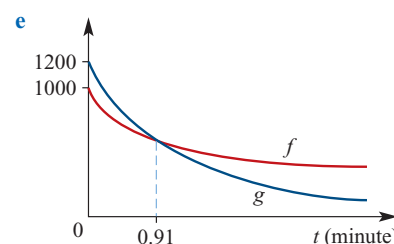


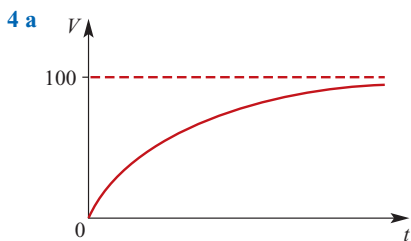
- c** $\frac{dT}{dt} = -14.4e^{-0.36t}$
- d** $\frac{dT}{dt} = -0.36(T - 20)$

- 3 a** 1.386 minutes **b** 2200, 5.38%

- c** 66.4 spores/minute

- d** 0.9116 minutes





b i $20e^{-0.2t} \text{ m/s}^2$ **ii** $20 - \frac{V}{5} \text{ m/s}^2$

c 8.05 s

5 b 0.028

c $0.846^\circ\text{C/minute}$

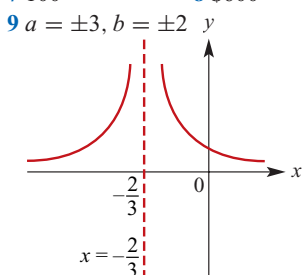
6 a i 0.1155

ii 0.2

b 13.86 days

7 100

8 \$600

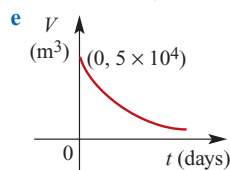


10 a $5 \times 10^4 \text{ m}^3$

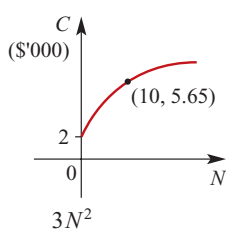
b $-12\,500 \text{ m}^3/\text{day}$

c $-3500 \text{ m}^3/\text{day}$

d After 222.61 days



11 a

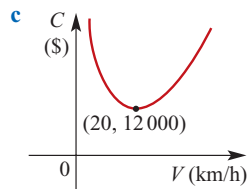


b $\frac{3N^2}{4(N^3 + 16)^{\frac{3}{4}}}$

c Rate of change of cost in \$1000s with respect to the increase in the number of bottle tops produced

12 a \$17 000

b $C = \frac{160\,000}{V} + 10V^2$



d $V = 20, C = 12\,000$

e 12 560

13 a $\frac{60}{\sqrt{39}} \approx 9.61 \text{ km}$ **b** $\frac{60}{\sqrt{39}} \approx 9.61 \text{ km}$

14 $\left(4 - \sqrt{\frac{12}{7}}\right) \text{ m} \approx 2.7 \text{ m}$

15 a $\{x: x > 1\}$

b $\{x: 0 < x < 2\}$

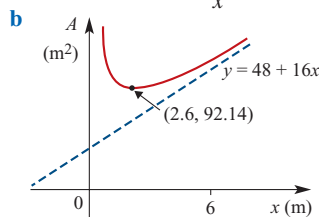
c $\{x: x > 1\}, \left\{x: 0 < x < \frac{3}{2}\right\}$

d $\{x: x > 1\}, \left\{x: 0 < x < \frac{n+1}{n}\right\}$

16 a $(1, 1), (-1, -1)$ **b** $(\pm\sqrt[6]{2}, 2^{-\frac{1}{3}})$

c $(\pm\sqrt[2n+2]{n}, n^{-\frac{n}{2n+2}})$

17 a $A = 48 + 16x + \frac{108}{x}$

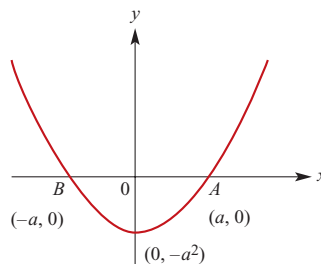


c Height = $\frac{3\sqrt{3}}{2} \text{ m}$, width = $\frac{4\sqrt{3}}{3} \text{ m}$

d 172 m^2

18 $p = 4$, number of items = 50

19 a $(a, 0), (-a, 0)$



b $\frac{4}{3}a^3$

c $\frac{2a^2}{3}$

d 2 : 3

20 a $-5\frac{1}{3}$

c $a = 1$ or $a = -2$

21 a i $50e^{-1}$ litres/minute

ii $t = 5$

iii 2 minutes 18 seconds

iv 3 minutes 48 seconds

b 14.74 litres

c 53 seconds

22 a $S = \frac{\pi h^2 \sin \alpha}{\cos^2 \alpha}$

b i $\frac{100\pi(1 + \sin^2 \alpha)}{\cos^3 \alpha}$

ii $\delta S = \frac{100\pi(2 - \cos^2 \alpha)}{\cos^3 \alpha} \delta \alpha$

iii $\frac{2 - \cos^2 \alpha}{\cos \alpha \sin \alpha} \times \frac{100}{1}$ **v** 29.1°

Chapter 14

Exercise 14A

- 1 a $\frac{1}{2}$ b $\frac{2}{3}$
 2 a $\frac{17}{500}$ b $\frac{9}{250}$ c $\frac{41}{125}$ d $\frac{41}{500}$
 3 a 0.65 b 0.35
 4 a 0.067 b 0.047
 5 5% 6 $\frac{6}{7}$
 7 a 0.5 b 0.385 c $\frac{200}{385} = 0.519$ d 0.4
 8 0.39 9 $\frac{1}{7}$
 10 0.48 11 0.0479
 12 a 0.486 b 0.012 c 0.138
 13 a $\frac{2}{5}$ b $\frac{1}{15}$ c $\frac{8}{15}$
 14 a $\frac{2}{5}$ b $\frac{7}{40}$ c $\frac{7}{16}$ d $\frac{7}{15}$
 15 a $\frac{5}{14}$ b $\frac{3}{5}$
 16 a 0.735 b $\frac{24}{53}$
 17 $\frac{3}{44}$

Exercise 14B

- 1 a Discrete b Not discrete
 c Discrete d Discrete
 2 a Not discrete b Discrete
 c Not discrete d Discrete
 3 a {HHH, THH, HTH, HHT, HTT, THT, TTH, TTT}

Experimental outcome	Value of x
HHH	3
THH	2
HTH	2
HHT	2
HTT	1
THT	1
TTH	1
TTT	0

- c $\frac{1}{2}$
 4 a {(1, 1), (1, 2), (1, 3), ..., (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)}

- b $Y = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$

Second die	First die					
	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

- c $\Pr(Y < 5) = \frac{6}{36} = \frac{1}{6}$
 d $\Pr(Y = 3 | Y < 5) = \frac{1}{3}$
 5 a

	1	2	3	4	5	6
1	1	1	1	1	1	1
2	1	2	2	2	2	2
3	1	2	3	3	3	3
4	1	2	3	4	4	4
5	1	2	3	4	5	5
6	1	2	3	4	5	6

 b 1, 2, 3, 4, 5, 6 c 0.19
 6 a 0.288 b 0.064 c 0.352 d 0.182
 7 a See 4a.
 b $\Pr(A) = \frac{1}{6}$, $\Pr(B) = \frac{1}{6}$, $\Pr(C) = \frac{5}{12}$,
 $\Pr(D) = \frac{1}{6}$
 c $\Pr(A|B) = \frac{1}{6}$, $\Pr(A|C) = \frac{1}{5}$,
 $\Pr(A|D) = \frac{1}{6}$
 d A and B are independent. A and D are independent.

Exercise 14C

- 1 a c
 2 a

x	1	2	3	4	5
$p(x)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

 b $\frac{3}{5}$ c $\frac{1}{3}$
 3

x	0	1	2
$p(x)$	0.36	0.48	0.16

 4 a

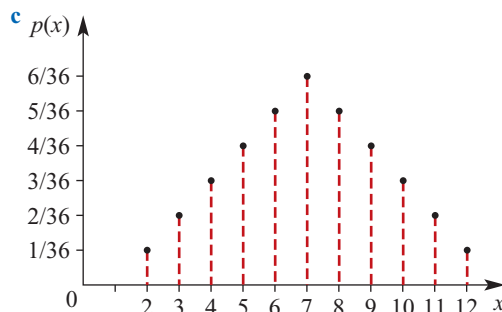
x	0	1	2	3
$p(x)$	$\frac{27}{125}$	$\frac{54}{125}$	$\frac{36}{125}$	$\frac{8}{125}$

 b

x	0	1	2	3
$p(x)$	$\frac{5}{30}$	$\frac{15}{30}$	$\frac{9}{30}$	$\frac{1}{30}$

5 a See 4a of Exercise 14B.

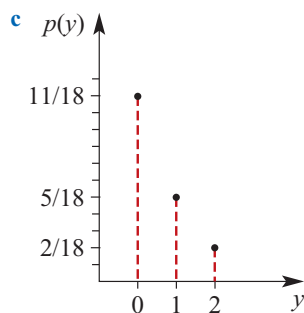
x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$



d $\frac{5}{18}$ e $\frac{7}{10}$

6 a $\{(1, 1), (1, 2), (1, 3), \dots, (6, 4), (6, 5), (6, 6)\}$

y	0	1	2
$p(y)$	$\frac{11}{18}$	$\frac{5}{18}$	$\frac{2}{18}$



7 a

x	0	1	2
$\Pr(X=x)$	$\frac{1}{3}$	$\frac{8}{15}$	$\frac{2}{15}$

b $\frac{7}{15}$

8 a

x	10	20	100
$\Pr(X=x)$	$\frac{3}{4}$	$\frac{6}{25}$	$\frac{1}{100}$

b

y	20	30	40	110	120	200
$\Pr(Y=y)$	$\frac{9}{16}$	$\frac{9}{25}$	$\frac{36}{625}$	$\frac{3}{200}$	$\frac{3}{625}$	$\frac{1}{10000}$

9 a $\frac{1}{4}$

b $\{EENE, ENEE, ENNN, NEEE, NENN, NNEN\}, \frac{3}{8}$

c $\frac{3}{8}$

Exercise 14D

1 a No unique mode, median = 5, $E(X) = 4.6$

b No unique mode, median = $\frac{1}{2}$,
 $E(X) = 0.5$

c Mode = 2, median = 2, $E(X) = 2.38$

d Mode = 0.6, median = 0.6, $E(X) = 0.569$

e Mode = 7, median = 7, $E(X) = 7$

f No unique mode, median = 0, $E(X) = 0$

2 a $c = 0.35$ b $X = 1$ c 2

d $E(X) = 2.3$

e $\text{Var}(X) = 1.61$; $\text{sd}(X) = 1.27$

f 0.95

3 a $k = \frac{1}{15}$ b 5 c 4

d $E(X) = 3.667$ e $\text{Var}(X) = 1.556$

f 0.9333

4 Expected profit = \$3000

5 A loss of 17c 6 1.54

7 a $E(X) = 7$ b $\text{Var}(X) = 5.83$ c 0.944

8 a $E(X) = 4.11$ b $E(X^3) = 78.57$

c $E(5X - 4) = 16.55$ d $E\left(\frac{1}{X}\right) = 0.255$

9 a $\text{Var}(2X) = 64$ b $\text{Var}(X + 2) = 16$

c $\text{Var}(1 - X) = 16$ d $\text{sd}(3X) = 12$

10 a 3 b 1.5 c 0.9688

11 a $p = \frac{1}{16}$ b $E(X) = 2$ c $\text{Var}(X) = 3.5$

12 a $k = \frac{1}{21}$ b $\frac{91}{21}$ c $\frac{20}{9}$

13 a

x	1	2	3	4	6	8	9	12	16
$\Pr(X=x)$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

b i $\frac{1}{4}$ ii $\frac{25}{4}$ iii $\frac{275}{16}$

14 a $\frac{21}{4}$ b $\frac{7}{12}$ c $\frac{497}{48}$

15

x	1	2	3	4	5	6	7	8	9	10	11	12
$\Pr(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

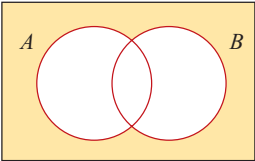
$E(X) = \frac{49}{12}$

Multiple-choice questions

1 A 2 D 3 B 4 A 5 D
6 E 7 C 8 C 9 B 10 D

Short-answer questions (technology-free)

1 Yes, as $\Pr(A \cap B) = 0$

2  $\Pr(A' \cap B') = \Pr(A \cup B)'$

3 a $\frac{40}{81}$

b $\frac{5}{9}$

4 0.4

5 a 0.1

b 1

c 1

d 1.3

e 2.01

6 a 21.5

b $\frac{53}{256}$

c 630.75

d $\frac{29\sqrt{3}}{2}$

7 a

p	$x - 2$	-2
$\Pr(P = p)$	$\frac{4}{5}$	$\frac{1}{5}$

b $\frac{4}{5}x - 2$

c $x > \$2.50$

8 a 0.47

b $\frac{47}{70}$

9 21.5%

10 a $\frac{1}{24}$

b $\frac{17}{24}$

c $\frac{5}{6}$

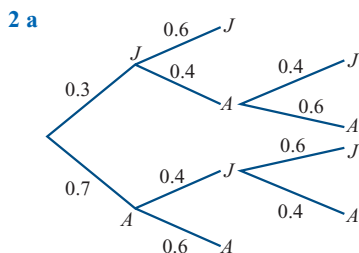
d $\frac{11}{18}$

Extended-response questions

1 a 0.1

b 0.2

c 4



b i 0.396

ii 0.604

c i

x	2	3
$\Pr(X = x)$	0.6	0.4

ii 2.4

d 0.46

3 a 0.5

b 0.05

c 0.033

d $\frac{25}{33}$

4 a i 1.21

ii $\text{Var}(P) = 1.6659$, $\text{sd}(P) = 1.2907$

iii 0.94

b i

t	1	0.40	0
$p(t)$	0.39	0.27	0.34

ii $E(T) = 0.498 \approx 0.50$

iii 1

5 \$14

6 a $E(Y) = 2.002$

b $\text{Var}(X) = 2.014$; $\text{sd}(X) = 1.419$

c i

b	0	100	200
$p(b)$	0.677	0.270	0.053

ii $E(B) = \$37.60$

7 a mean = 13.5%, sd = 16.2%

b 0.95

c $E(G) = 6.9\%$, $\text{sd}(G) = 9.726\%$

8 Yes

9 \$1.00

10 a i 0.65

ii 0.2275

iii 0.079625

iv 0.042875

b Expected cost = 8.439 375 million dollars

c Expected profit is 10.703 125 million dollars.

11 a $\frac{1}{3}(304 - 2x)$

b $x = 2$

c $27\frac{2}{9}$

12 b $x = \frac{1}{2}, \frac{49}{288}$

13 a i $\frac{1}{81}$

ii $\frac{8}{81}$

iii $\frac{4}{81}$

iv $\frac{4}{27}$

v $\frac{56}{81}$

b 4.4197 cents. (As the lowest value coin is 5¢, he can settle for that.)

Chapter 15

Exercise 15A

1 a and b

2 a 0.0595

b 0.0512

c 0.0081

3 a 0.05358

b 0.0087

c 0.0623

4 a $\frac{5^4}{6^5} \approx 0.0804$

b $\frac{5^5}{6^5} \approx 0.4019$

5 a 0.1156

b 0.7986

c 0.3170

6 0.6791

7 a 0.1123

b 0.5561

c 0.000 01

d 0.000 01

8 0.6836

9 0.544

10 0.624

11 0.1356

12 a 0.0138

b 0.2765

c 0.8208

d 0.3368

13 a $\left(\frac{1}{4}\right)^6 = 0.000 24$

b 0.1694

14 0.9744

15 0.5432

16 a $(0.8)^8 \approx 0.168$

b 0.001 23

c 0.002 1

17 a $(0.15)^{10} \approx 0.000 000 006$

b $1 - (0.85)^{10} \approx 0.803 1$

c 0.567 4

18 0.962

19 a 0.002 455

b 0.003 37

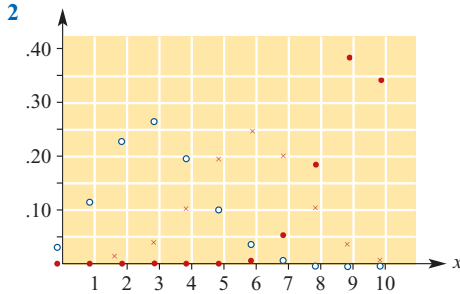
20 a 0.011 529

b 0.002 59

c 0.0392

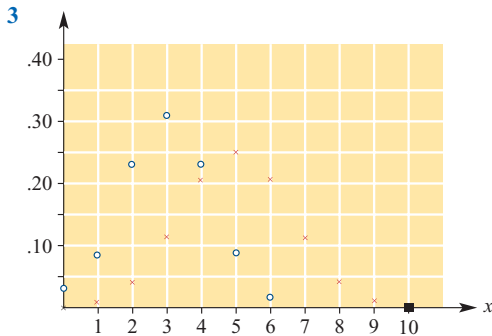
Exercise 15B

1 a 4 b 10 c 16



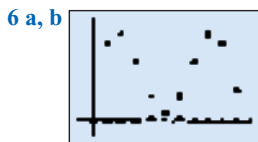
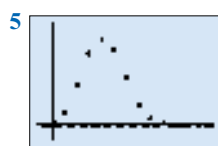
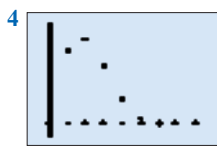
Legend:

- $p = 0.3$, skewed positively
- × $p = 0.6$, 'slightly' skewed positively
- $p = 0.9$, skewed negatively

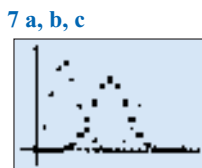


Legend:

- $p = 0.5; n = 6$
 - × $p = 0.5; n = 10$
- Both plots are symmetrical:
- ($p = 0.5; n = 6$) is symmetrical about $x = 3$
 - ($p = 0.5; n = 10$) is symmetrical about $x = 5$



c The distribution in part b is a reflection of the distribution in part a in the line $X = 5$.



d When $p = 0.5$ the distribution is symmetric, when $p = 0.2$ it is positively skewed and when $p = 0.8$ negatively skewed.

Exercise 15C

- 1 a Mean = 5, variance = 4
b Mean = 6, variance = 2.4
c Mean = $\frac{500}{3}$, variance = $\frac{1000}{9}$
d Mean = 8, variance = 6.4
- 2 a 1 b 0.2632 c 37.5
- 4 $n = 48, p = \frac{1}{4}, \Pr(X = 7) = 0.0339$
- 5 $n = 100, p = \frac{3}{10}, \Pr(X = 20) = 0.0076$
- 6 Mean = 10, sd = $\sqrt{5}$
The probability of obtaining between 6 and 14 heads is 0.95.
- 7 Mean = 120, sd = $4\sqrt{3}$
The probability that between 107 and 133 students attend a state school is 0.95.

Exercise 15D

- 1 a i $(0.8)^5 \approx 0.3277$ ii 0.6723 b 14
- 2 a i 0.1937 ii $1 - (0.9)^{10} \approx 0.6513$
- b 12
- 3 7 4 7
- 5 10 6 42
- 7 86

Multiple-choice questions

- 1 D 2 A 3 E 4 B 5 A
6 B 7 C 8 C 9 E 10 B

Short-answer questions (technology-free)

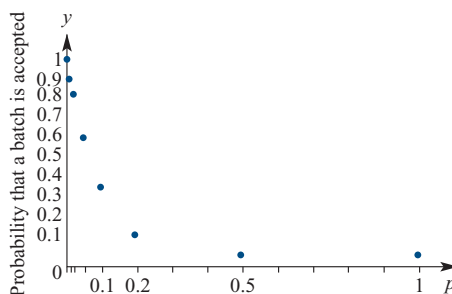
- 1 a $\frac{16}{81}$ b $\frac{32}{81}$ c $\frac{16}{27}$ d $\frac{65}{81}$
- 2 $\frac{54}{125}$ 3 0.40951
- 4 a 2 b $\frac{3\sqrt{5}}{5}$
- 5 a $(1 - p)^4$ b $4p(1 - p)^3$
c $1 - (1 - p)^4$ d p^4
e $1 - (1 - p)^4 - 4p(1 - p)^3$
- 6 120
- 7 $\frac{5p(1 - p)^4}{1 - (1 - p)^5}$
- 8 $\frac{5}{16}$
- 9 $\frac{32}{625}$

Extended-response questions

- 1 a 0.0173 b 0.2131

2

p	Probability that a batch is accepted
0	1
0.01	0.9044
0.02	0.8171
0.05	0.5987
0.1	0.3487
0.2	0.1074
0.5	0.00098
1	0



- 3 a 0.0582
-
- b Mean = 0.4, sd = 0.6197,
-
- $\mu \pm 2\sigma = 0.4 \pm 1.2394$

c Yes

- 4 0.0327

- 5 a i 0.0819 ii 0.9011

- b i
- $15p^2(1-p)^4$

ii $\frac{dP}{dp} = 30p(1-p)^3(1-3p)$

- 6 a 2 b
- $n = 6, p = \frac{1}{3}$

c

No. in group wearing glasses	0	1	2	3	4	5	6
Theoretical frequencies	17.56	52.68	65.84	43.90	16.46	3.29	0.27

- 7 a 0.9139 b 0.04145 c 10.702

- 8 a 0.0735 b 0.5015 c 27

- 9
- $\frac{1}{3} \leq q \leq 1$

Exercise 16A

- 1 a 0.36 b 0.26

- 2 a
- $\begin{bmatrix} 0.65 & 0.44 \\ 0.35 & 0.56 \end{bmatrix}$
- b 0.5492

- 3 a
- $\begin{bmatrix} \Pr(W_{i+1}) \\ \Pr(L_{i+1}) \end{bmatrix} = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \times \begin{bmatrix} \Pr(W_i) \\ \Pr(L_i) \end{bmatrix}$
-
- b 0.517

- 4 a
- $\begin{bmatrix} \Pr(L_{i+1}) \\ \Pr(T_{i+1}) \end{bmatrix} = \begin{bmatrix} 0.35 & 0.1 \\ 0.65 & 0.9 \end{bmatrix} \times \begin{bmatrix} \Pr(L_i) \\ \Pr(T_i) \end{bmatrix}$
- b 0.8

- 5 a 0.214 b 0.096

- c
- $\begin{bmatrix} \Pr(A_{i+1}) \\ \Pr(E_{i+1}) \end{bmatrix} = \begin{bmatrix} 0.57 & 0.47 \\ 0.43 & 0.53 \end{bmatrix}^i \times \begin{bmatrix} \Pr(A_1) \\ \Pr(E_1) \end{bmatrix}$
-
- i 0.536 ii 0.5236

- 6 a 0.007 b 0.398

- c
- $\begin{bmatrix} \Pr(R_{i+1}) \\ \Pr(F_{i+1}) \end{bmatrix} = \begin{bmatrix} 0.43 & 0.16 \\ 0.57 & 0.84 \end{bmatrix}^i \times \begin{bmatrix} \Pr(R_1) \\ \Pr(F_1) \end{bmatrix}$
-
- i 0.781 ii 0.219

Exercise 16B

- 1 a i
- $\begin{bmatrix} 59 \\ 41 \end{bmatrix}$
- ii
- $\begin{bmatrix} 67.7 \\ 32.3 \end{bmatrix}$
- iii
- $\begin{bmatrix} 70.31 \\ 29.69 \end{bmatrix}$

- b
- $\begin{bmatrix} 0.7150 & 0.7126 \\ 0.2850 & 0.2875 \end{bmatrix}$

- c i
- $\begin{bmatrix} 67.7 \\ 32.3 \end{bmatrix}$
- ii
- $\begin{bmatrix} 70.31 \\ 29.69 \end{bmatrix}$
- iii
- $\begin{bmatrix} 71.4195 \\ 28.5805 \end{bmatrix}$

- 2 a i
- $\begin{bmatrix} 194.8 \\ 185.2 \end{bmatrix}$
- ii
- $\begin{bmatrix} 204.568 \\ 175.432 \end{bmatrix}$
- iii
- $\begin{bmatrix} 211.015 \\ 168.985 \end{bmatrix}$

- b
- $\begin{bmatrix} 0.6223 & 0.5396 \\ 0.3777 & 0.4604 \end{bmatrix}$

- c i
- $\begin{bmatrix} 204.568 \\ 175.432 \end{bmatrix}$
- ii
- $\begin{bmatrix} 211.015 \\ 168.985 \end{bmatrix}$
- iii
- $\begin{bmatrix} 218.078 \\ 161.922 \end{bmatrix}$

- 3 a i
- $\begin{bmatrix} 0.23 \\ 0.77 \end{bmatrix}$
- ii
- $\begin{bmatrix} 0.2714 \\ 0.7286 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.280435 \\ 0.715965 \end{bmatrix}$

- b i
- $\begin{bmatrix} 0.41 \\ 0.59 \end{bmatrix}$
- ii
- $\begin{bmatrix} 0.3038 \\ 0.6962 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.280624 \\ 0.719376 \end{bmatrix}$

- 4 a i
- $\begin{bmatrix} 2 \\ 3 \\ 1 \\ 3 \\ 3 \end{bmatrix}$
- ii
- $\begin{bmatrix} 17 \\ 36 \\ 19 \\ 36 \\ 107 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.5162 \\ 0.4838 \end{bmatrix}$

- b i
- $\begin{bmatrix} 3 \\ 8 \\ 5 \\ 8 \end{bmatrix}$
- ii
- $\begin{bmatrix} 107 \\ 192 \\ 85 \\ 192 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.5160 \\ 0.4840 \end{bmatrix}$

- 5 a i
- $\begin{bmatrix} 0.28 \\ 0.04 \\ 0.68 \end{bmatrix}$
- ii
- $\begin{bmatrix} 0.636 \\ 0.1596 \\ 0.2044 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.4718 \\ 0.1664 \\ 0.3618 \end{bmatrix}$

- b i
- $\begin{bmatrix} 0.34 \\ 0.48 \\ 0.18 \end{bmatrix}$
- ii
- $\begin{bmatrix} 0.4024 \\ 0.2782 \\ 0.3194 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.4761 \\ 0.1673 \\ 0.3566 \end{bmatrix}$

- c i
- $\begin{bmatrix} 0.80 \\ 0.19 \\ 0.01 \end{bmatrix}$
- ii
- $\begin{bmatrix} 0.2966 \\ 0.1251 \\ 0.5783 \end{bmatrix}$
- iii
- $\begin{bmatrix} 0.4808 \\ 0.1679 \\ 0.3513 \end{bmatrix}$

- 6 a
- $\begin{bmatrix} 0.85 & 0.2 \\ 0.15 & 0.8 \end{bmatrix}$

- b 186 at Bayside, 179 at Surfside

- 7 a
- $\begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{bmatrix}$
- b 57 200

- 8 a
- $\begin{bmatrix} 0.92 & 0.88 \\ 0.08 & 0.12 \end{bmatrix}$
- b 91.7%

9 a $\begin{bmatrix} 0.91 & 0.13 \\ 0.09 & 0.87 \end{bmatrix}$

b 53.6% at school A, 46.4% at school B

10 a $\begin{bmatrix} 0.5 & 0.28 \\ 0.5 & 0.72 \end{bmatrix}$

b i 0.359 ii 0.359

11 a $\begin{bmatrix} 0.2 & 0.7 \\ 0.8 & 0.3 \end{bmatrix}$

b i 0.6 ii 0.475

12 a 2000 young, 18 000 middle-aged

b 6500 young, 3420 middle-aged, 10 080 old

13 a $\begin{bmatrix} 0.80 & 0.40 & 0.35 \\ 0.15 & 0.30 & 0.30 \\ 0.05 & 0.30 & 0.35 \end{bmatrix}$ b $\begin{bmatrix} 60 \\ 25 \\ 15 \end{bmatrix}$

c $\begin{bmatrix} 63.3 \\ 21.0 \\ 15.8 \end{bmatrix}$ d $\begin{bmatrix} 65.4 \\ 20.2 \\ 14.4 \end{bmatrix}$

14 a 0.279

b 0.285

Exercise 16C

1 a $\begin{bmatrix} 0.6 & 0.5 \\ 0.4 & 0.5 \end{bmatrix}$

b

Game (n)	$\Pr(X_n = 1 X_0 = 0)$	$\Pr(X_n = 1 X_0 = 1)$
1	0.4	0.5
2	0.44	0.45
3	0.444	0.445
4	0.4444	0.4445
5	0.4444	0.4445
6	0.4444	0.4444

c 0.4444, 0.5556 d $\frac{4}{9}, \frac{5}{9}$

2 a $\begin{bmatrix} 0.85 & 0.95 \\ 0.15 & 0.05 \end{bmatrix}$

b

Day (n)	$\Pr(X_n = 0 X_0 = 0)$	$\Pr(X_n = 0 X_0 = 1)$
1	0.85	0.95
2	0.865	0.855
3	0.8635	0.8645
4	0.8637	0.8636
5	0.8636	0.8636
6	0.8636	0.8636

c 0.8636, 0.1364 d $\frac{19}{22}, \frac{3}{22}$

3 64.3%, 35.7%

4 a $\begin{bmatrix} 0.74 & 0.14 \\ 0.26 & 0.86 \end{bmatrix}$

b 338 at store A, 451 at store B

c 276 at store A, 513 at store B

5 a $\begin{bmatrix} 18.1818 \\ 31.8182 \end{bmatrix}$ b $\begin{bmatrix} 200 \\ 11 \\ 350 \\ 11 \end{bmatrix}$

6 a $\begin{bmatrix} 0.4167 \\ 0.5833 \end{bmatrix}$

b $\begin{bmatrix} 5 \\ 12 \end{bmatrix}$

7 a $\begin{bmatrix} 0.2743 \\ 0.3521 \\ 0.3735 \end{bmatrix}$

b $\begin{bmatrix} 0.2743 \\ 0.3521 \\ 0.3735 \end{bmatrix}$

c $\begin{bmatrix} 0.2743 \\ 0.3521 \\ 0.3735 \end{bmatrix}$

d All the same

8 a $\begin{bmatrix} 0.70 & 0.25 \\ 0.30 & 0.75 \end{bmatrix}$

b 105 in Melbourne, 125 in Ballarat

9 80% sad, 20% happy

10 5405 young, 4865 middle-aged, 9730 old

11 28.4% cool, 22.3% mild, 21.6% warm, 27.6% hot

Exercise 16D

1 a 0.0034 2 0.0154

3 a 0.0864 b 0.0384

4 a 0.1029 b 0.0353

c 0.0189 d 0.0005

5 a 0.0625 b 0.0366

6 a 0.0655 b 0.0439

7 a 0.0696 b 0.0700

8 a 0.0226 b 0.0656

Multiple-choice questions

1 C 2 A 3 C 4 B 5 B

6 C 7 E 8 B 9 D 10 D

Short-answer questions (technology-free)

1 a $\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$ b $\begin{bmatrix} 0.34 \\ 0.66 \end{bmatrix}$

2 a $\begin{bmatrix} 0.1 \\ 0.3 \end{bmatrix}$ b $\begin{bmatrix} 0.64 \\ 0.36 \end{bmatrix}$

3 $\begin{bmatrix} 4 & 2 \\ 7 & 5 \\ 3 & 3 \\ 7 & 5 \end{bmatrix}$ 4 0.34 5 0.36

6 59.3% club A, 40.7% club B

7 0.1029

Extended-response questions

1 a $\frac{20}{81}$ b $\frac{1}{9}$ c i $\frac{5}{12}$ ii $\frac{7}{18}$ d $\frac{3}{5}$

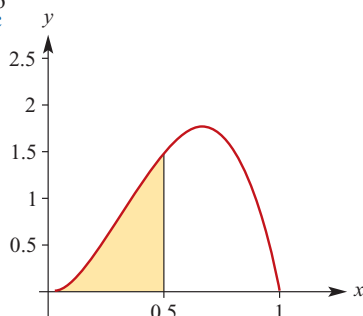
- 2 a $\begin{bmatrix} 0.84 & 0.23 \\ 0.16 & 0.77 \end{bmatrix}$ b 356 bus, 317 train
 c 397 bus, 276 train d 0.0218
 3 a $\begin{bmatrix} 0.56 & 0.26 \\ 0.44 & 0.74 \end{bmatrix}$
 b i 36.95% in Camberwell, 63.05% in Hawthorn
 ii 37.14% in Camberwell, 62.86% in Hawthorn

Chapter 17

Exercise 17A

2 $-\frac{11}{6}$

3 a, c

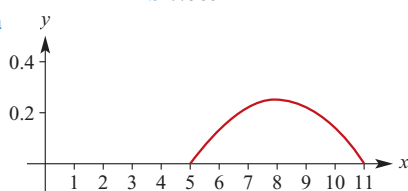


b $\Pr(X < 0.5) = \frac{5}{16}$

4 a 1

b 0.865

5 a



b 0.259

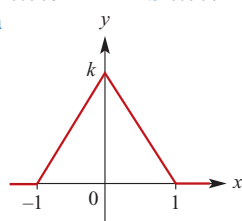
6 b i 0.024

ii 0.155

7 a 0.005

b 0.007

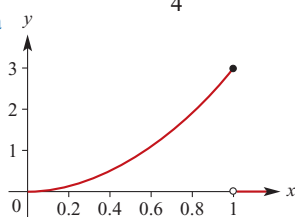
8 a



b $k = 1$

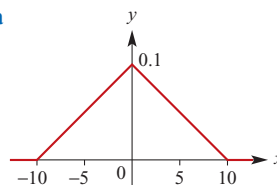
c $\frac{3}{4}$

9 a



b 0.406

10 a



b 0.190

11 a $k = 1000$

b 0.5

12 a $\frac{2}{3}$

b $\frac{17}{30}$

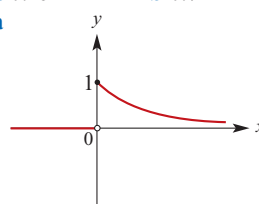
13 a 0.202

b 0.449

14 a 0.45

b 0.711

15 a

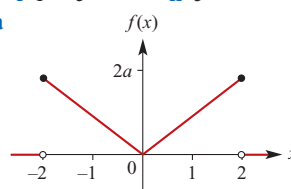


b i $1 - e^{-\frac{1}{2}}$

ii e^{-1}

iii $e^{-\frac{1}{2}}$

16 a



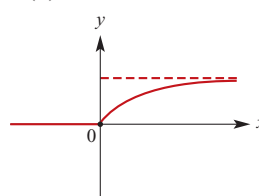
b $a = \frac{1}{4}$

Exercise 17B

1 a $F(x) = \frac{1}{5}x$ if $0 < x \leq 5$ and 0 if $x \leq 0$ and $x > 5$

b $\frac{3}{5}$

2 a $F(x) = 1 - e^{-x^2}$ if $x \geq 0$ and 0 if $x < 0$



b e^{-4}

c 0.0182

3 a $k = \frac{1}{36}$

b $\frac{1}{48}$

Exercise 17C

1 a $\frac{2}{3}$

b $\frac{1}{3}$

c $\frac{1}{2}$

d Does not exist

2 a 1

b 2.097

c 1.132

d 0.4444

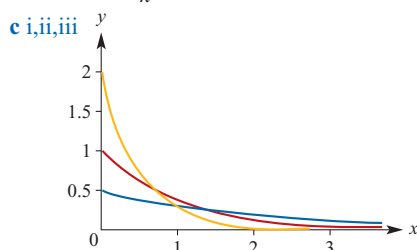
3 a 0.567

b 0.458

- 4 0 5 $A = \frac{2}{9}, B = 3$
 6 a 2 b 1.858
 7 a 1 b 0.5
 8 a 0.632 b 0.233 c 0.693
 9 0.1294 10 2.773 minutes
 11 a $f(x) = 12x^2(1-x), 0 \leq x \leq 1$ b $\frac{2}{3}$

- 12 $\frac{\pi}{2}$
 13 a 1 b 1 c 1
 14 a 0.714 b 0.736
 15 a 12 b 12
 16 a 0.4 b $\frac{\sqrt{19}-1}{6}$ c 1

- 17 a $ke^{-kx}, \frac{-(kx+1)}{k}e^{-kx}$



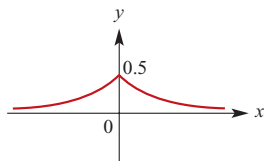
- d The graph of e^{-x} is dilated by a factor of $\frac{1}{\lambda}$ from the x -axis and λ from the y -axis.

Exercise 17D

- 1 a 0.630 b 0.909 c 0.279

- 2 a 0.366 b $\text{Var}(X) = \frac{1}{18}, \text{sd}(X) = \frac{\sqrt{2}}{6}$

- 3 a b 1.386



- 4 a $\frac{1}{\log_e 9}$ b $E(X) = 3.641, \text{Var}(X) = 4.948$

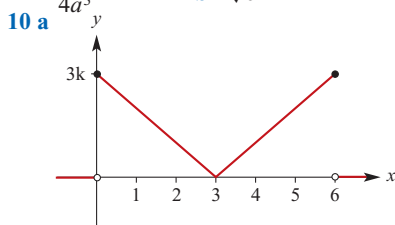
- 5 a 0.366 b $E(X) = 0.333, \text{Var}(X) = 0.056$

- 6 0.641

- 7 a 0.732 b $E(X) = \frac{4}{3}, \text{Var}(X) = \frac{2}{9}$

- 8 a 0.0004 b $\frac{16}{3}$ c 2.21

- 9 a $\frac{3}{4a^3}$ b $2\sqrt{5}$



- b $k = \frac{1}{9}$ d 4.5

Exercise 17E

- 1 1300
 2 a 0.708, 0.048 b \$98.94, \$0.33
 3 a 0, 5.4 b 3, 0.6
 c $g(x) = \frac{x^2}{18}$ if $-3 \leq x \leq 3$ and 0 otherwise
 4 a 7.5, 8.5 b (1.669, 13.331)
 5 35, 10

Multiple-choice questions

- 1 B 2 D 3 D 4 A 5 E
 6 B 7 C 8 E 9 A 10 A

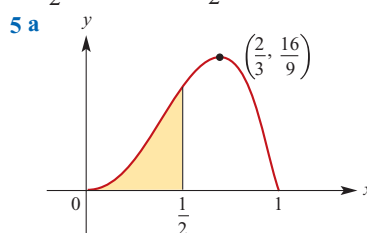
Short-answer questions (technology-free)

- 1 a 2 b 0.21 c 0.44

- 2 a $\frac{1}{3}, b = 2$

- 3 mean = median = mode = $\frac{\pi}{2}$

- 4 a $\frac{1}{2}$ b $\frac{1}{2}$ c $\frac{1}{2}$



- b $\Pr(X < 0.5) = \frac{5}{16}$

- 6 a $k = 12$ b $x = \frac{2}{3}$ c $\Pr\left(X < \frac{2}{3}\right) = \frac{16}{27}$

- d $\Pr\left(X < \frac{1}{3} | X < \frac{2}{3}\right) = \frac{3}{16}$

- 7 a $1 - e^{-2}$ b $\frac{e^2 - 1}{e^2 - 3}$

- 8 a $e^{\frac{1}{2}}$ b $e^{\frac{3}{4}} - e^{\frac{1}{4}}$

- c $f(x) = \frac{1}{x}, 1 < x < e$

- 9 (320, 340) 10 (246, 254)

Extended-response questions

- 1 a $\frac{-2}{81}$ b 700 hours c $\frac{x^2 - 200x + 10\,000}{810\,000}$

- 2 a $\frac{1}{4}$ b $\frac{5 - \sqrt{5}}{5}$ c $\frac{1}{3}, \frac{8}{15}$

3 a Median = 6, $\text{IQR} = \frac{10}{3}$
 b $E(X) = 6$, $\text{Var}(X) = 4.736$

4 a $\frac{7}{25}$ b \$22.13

5 c $\frac{8}{3}, 4$

6 b $E(X) = 2$, $\text{Var}(X) = 0.2$ c $\frac{5}{32}$

7 a 25 b $\frac{2}{3}$ d $\frac{2}{3}$

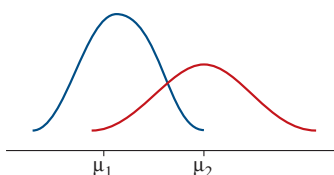
8 a $k = \frac{1}{4}$ b $E(X) = 2$, $\text{Var}(X) = \frac{2}{3}$
 c $\frac{3}{4}$ d $\frac{4\sqrt{5}}{5}$

Chapter 18

Exercise 18A

1 c

2



3 a 1

b i $E(X) = \int_{-\infty}^{\infty} x \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2} dx$
 ii 2

c i $E(X^2) = \int_{-\infty}^{\infty} x^2 \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2} dx$
 ii 13 iii 3

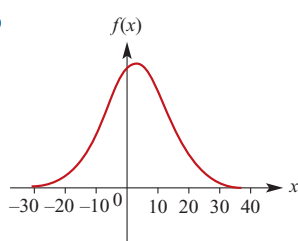
4 a 1

b i $\int_{-\infty}^{\infty} x \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x+4}{5}\right)^2} dx$
 ii -4

c i $\int_{-\infty}^{\infty} x^2 \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x+4}{5}\right)^2} dx$
 ii 41 iii 5

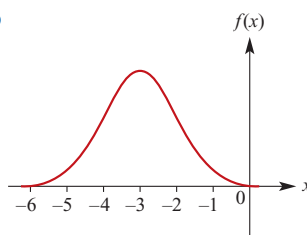
5 a $\mu = 3$, $\sigma = 10$

b



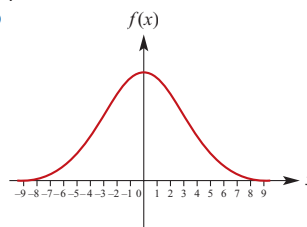
6 a $\mu = -3$, $\sigma = 1$

b



7 a $\mu = 0$, $\sigma = 3$

b



8 a Translation of 3 units in the positive direction of the x -axis; dilation of factor 2 from the y -axis; dilation of factor $\frac{1}{2}$ from the x -axis

b Translation of 3 units in the positive direction of the x -axis; dilation of factor $\frac{1}{2}$ from the y -axis; dilation of factor 2 from the x -axis

c Translation of 3 units in the negative direction of the x -axis; dilation of factor 2 from the y -axis; dilation of factor $\frac{1}{2}$ from the x -axis

9 a Translation of 3 units in the negative direction of the x -axis; dilation of factor $\frac{1}{2}$ from the y -axis; dilation of factor 2 from the x -axis

b Translation of 3 units in the negative direction of the x -axis; dilation of factor 2 from the y -axis; dilation of factor $\frac{1}{2}$ from the x -axis

c Translation of 3 units in the positive direction of the x -axis; dilation of factor $\frac{1}{2}$ from the y -axis; dilation of factor 2 from the x -axis

Exercise 18B

1 Mean = 135, sd = 5

2 Mean = 10, sd = $\frac{4}{3}$

3 16%

4 a 68% b 16% c 0.15%

5 21.1 and 33.5

6 68% of the values lie within one standard deviation of the mean;
 95% of the values lie within two standard deviations of the mean;
 99.7% of the values lie within three standard deviations of the mean.

7 2.5%

8 a 16% b 16%

9 a 68% b 16% c 2.5%

10 a 95% b 16% c 50% d 99.7%

- 11 a 0 b $-\frac{5}{4}$ c 1.5
 12 a -1.4 b 1.1 c 3.5
 13 Michael 1.4, Cheryl 1.5; Cheryl
 14 Biology 1.73, History 0.90; Biology
 15 a Mary: French 1, English 0.875,
 Mathematics 0
 Steve: French -0.5 , English -1 ,
 Mathematics 1.25
 Sue: French 0, English 0.7,
 Mathematics -0.2
 b i Mary ii Mary iii Steve
 c Mary

Exercise 18C

- 1 a 0.9772 b 0.9938 c 0.9938
 d 0.9943 e 0.0228 f 0.0668
 g 0.3669 h 0.1562
 2 a 0.9772 b 0.6915 c 0.9938
 d 0.9003 e 0.0228 f 0.0099
 g 0.0359 h 0.1711
 3 a 0.6827 b 0.9545 c 0.9973
 4 a 0.0214 b 0.9270 c 0.0441
 d 0.1311
 5 c = 1.2816 6 c = 0.6745 7 c = 1.96
 8 -1.6449 9 -0.8416 10 -1.2816
 11 -1.9600
 12 a 0.9522 b 0.7977 c 0.0478 d 0.1547
 13 a 0.9452 b 0.2119 c 0.9452 d 0.1571
 14 a 9.2897 b 8.5631
 15 a c = 10 b k = 15.88
 16 a 0.7161 b 0.0966 c 0.5204
 d c = 33.5143 e k = 13.02913
 f $c_1 = 8.28; c_2 = 35.72$
 17 a 0.9772 b 0.9772 c 10.822
 d 9.5792 e $c_2 = 10.98; c_1 = 9.02$

Exercise 18D

- 1 a i 0.2525 ii 0.0478 iii 0.0901
 b 124.7
 2 a i 0.7340 ii 0.8944 iii 0.5530
 b 170.25 cm c 153.267
 3 a i 0.0766 ii 0.9998 iii 0.153
 b 57.3
 4 a 10.56% b 78.51%
 5 mean = 1.55 kg; sd = 0.194 kg
 6 a 36.9% b 69
 7 a 0.0228 b 0.0005 c 0.0206
 8 1004 ml
 9 a small 0.1587
 medium 0.7745
 large 0.0668
 b \$348.92

- 10 a i 0.1169 ii 17.7
 b 0.0284
 11 a 0.0228, 0.1587 b $\frac{34}{3}$

Multiple-choice questions

- 1 A 2 C 3 B 4 B 5 E
 6 E 7 D 8 C 9 A 10 D

Short-answer questions (technology-free)

- 1 a $1 - p$ b $1 - p$ c $2p - 1$
 2 a $a = -1$ b $b = -1$
 c 0.5
 3 $(x, y) \rightarrow \left(\frac{x-8}{3}, 3y\right)$
 4 a $\frac{q}{p}$ b $1 - q$ c $\frac{1-p}{1-q}$
 5 a $\Pr(Z < \frac{1}{2})$ b $\Pr(Z < -\frac{1}{2})$ c $\Pr(Z > \frac{1}{2})$
 d $\Pr(-\frac{1}{2} < Z < \frac{1}{2})$ e $\Pr(-\frac{1}{2} < Z < 1)$
 6 a 0.84 b 0.5 c 0.16 d 0.68
 7 a 0.16 b 0.34 c 0.32 d 0.02
 8 a 0.69 b 0.19 c 0.15 d 0.68
 9 Best C, worst B

Extended-response questions

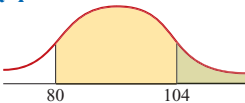
- 1 > 63 $\rightarrow$ high
 [56, 62] $\rightarrow$ moderate
 [45, 55] $\rightarrow$ average
 [37, 44] $\rightarrow$ little
 < 37 $\rightarrow$ low
 2 3.92
 3 a i 0.1587 ii 0.9747 iii 0.0164
 b 53 592 c 3.7×10^{-11}
 4 a 3.17×10^{-5} b False
 c $c_1 = 13.53, c_2 = 16.47$
 5 0.0802 6 0.92%
 7 a 0.9044 b 5.88 c 9.044
 d 0.2651 e \$17.61
 8 a $\mu = 0, \sigma = 2.658$ b 0.882
 9 a $\mu = 60.058; \sigma = 0.2$ b 10%
 10 a 0.1056 b 0.0803 c 0.5944

Chapter 19

Multiple-choice questions

- 1 E 2 D 3 A 4 B 5 E
 6 D 7 D 8 C 9 E 10 C
 11 B 12 D 13 C 14 B 15 E
 16 E 17 A 18 A 19 D 20 B
 21 A 22 D 23 C 24 A 25 D
 26 C 27 E 28 B 29 B 30 C
 31 D 32 C 33 E 34 A 35 B
 36 E 37 C 38 A 39 B 40 B
 41 B

Extended-response questions

- 1 b $\frac{9}{16}$ 2 \$0.76
- 3 a 0.27 b 0.3025
c i 0.78 ii 0.27
d 0.4995
- 4 a $P = \begin{cases} 0.75x - 0.5s, & x \leq s \\ 0.5s - 0.25x, & x > s \end{cases}$
b \$5.95
c $E(P) = \sum_{x=24}^s (0.75x - 0.5s)p(x) + \sum_{x=s+1}^{30} (0.5s - 0.25x)p(x)$
- d 27
- 6 a i $\frac{1}{6}$ ii $\frac{1}{36}$ iii $\frac{1}{6}$
b i $\frac{4}{25}$ ii $\frac{41}{100}$
- c $\frac{121}{600}$
- 7 a $\begin{bmatrix} 3 & 1 \\ 4 & 2 \\ 1 & 1 \\ 4 & 2 \end{bmatrix}$ i $\frac{11}{16}$ ii $\frac{85}{128}$
b $\frac{1}{3}$
- 8 a $\begin{bmatrix} 0.85 & 0.06 \\ 0.15 & 0.94 \end{bmatrix}$
b i 51% Dr Laslett, 49% Dr Kildare
c Yes, Dr Laslett during the 9th year
- 9 a 0.6915 b 0.1365
10 a 0.0436 b 0.2667 c 183
d 59 271
- 11 a i 0.1587 ii 511.63
b 0.1809
- 12 a -1 b 28.847 c 0.5276
- 13 a i $\frac{1}{8}$ ii $X: 0.6915, Y: 0.5625$
iii $E(X) = 10, E(Y) = 10.67,$
machine 1
b $\frac{3}{4}$
- 14 a i $\frac{1}{2500}$ ii $\frac{16}{3}$ iii 0.8281 iv 0.7677
b 0.9971
- 15 a i 
ii $\mu = 92.956, \sigma = 6.3084$
b i 16.73% of sensors ii 81° C
- 16 a 0.1056
b Machine should be set at 1027.92.

- 17 a $C = \frac{20}{49}$ b $E(X) = \frac{120}{49}$
c $\text{Var}(X) = \frac{6180}{2401}$
- 18 a i 0.2 ii 0.7 iii $\frac{0.125}{128}$ iv $\frac{3}{160}$
b i 0.360 15 ii $\frac{625}{625}$
c 0.163 08
- 19 a i 0.0105 ii 0.0455
b 0.4396 c $\frac{1149}{1909}$
- 20 a i $\mu = 4.25$ ii $\sigma = 0.9421$
iii 0.94 iv 0.9
b i Binomial ii 18
iii 1.342 iv 0.3917

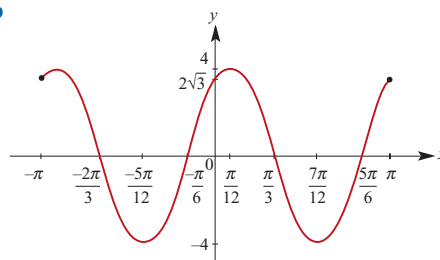
Chapter 20

Multiple choice questions

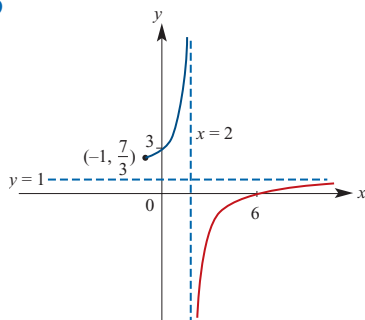
- 1 B 2 A 3 D 4 B 5 C
6 D 7 A 8 D 9 C 10 D
11 C 12 A 13 A 14 A 15 B
16 D 17 B 18 C 19 D 20 E
21 A 22 A 23 E 24 A 25 E
26 E

Short-answer questions (technology – free)

- 1 $f(g(x)) = (3x + 1)^2 + 6 = 9x^2 + 6x + 7$
2 $k = -1 - \sqrt{13}$ or $k = -1 + \sqrt{13}$
3 $y = -\frac{6}{x}$
Reflection in x -axis, dilation of factor 2 from y -axis, and dilation of factor 3 from the x -axis or a reflection in the x -axis followed by a dilation of factor 6 from the x -axis.
- 4 a $f'(x) = 21x^6 (5x^2 - 3)^6 (5x^2 - 1)$
b $f'(0) = 2$
- 5 a $x(1 + 2 \log_e(2x))$ b $f'\left(\frac{\pi}{2}\right) = \frac{-2}{(\pi + 1)^2}$
- 6 a $f'(x) = 2 \cos(2x) e^{\sin(2x)}$
b $f'\left(\frac{\pi}{3}\right) = 8\pi - 3\sqrt{3}$
- 7 $x = \frac{\pi(4n + 1)}{8}$
- 8 a Amplitude = 4, period = π
b



9



10 a $f^{-1}(x) = \log_e \left(\frac{x+3}{5} \right) + 1$

b $\text{dom } f^{-1} = (-3, \infty)$

11 $x = -\frac{2\pi}{15}, \frac{2\pi}{15}$

12 Required to show

13 $\frac{1}{4}(e^4 - 1)$

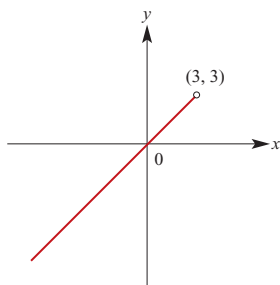
14 a $c = 6$

b $4a + b - 3 = 0, 3a + b = 0$

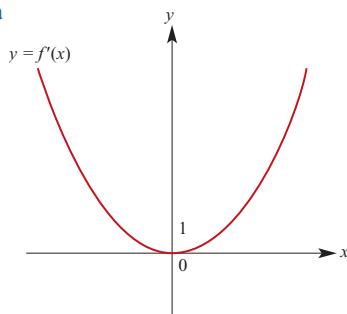
c $a = 3, b = -9$

15 a $g^{-1}(x) = \frac{1}{2} \log_e (3 - x), \text{dom } g^{-1} = (-\infty, 3)$

b



16 a



b $f'(x) = \begin{cases} -8x^3 & x \leq 0 \\ 8x^3 & \text{otherwise} \end{cases}$

17 $\frac{-1}{3} \log_e |1 - 3x| + c$

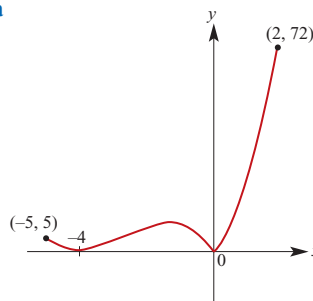
18 $f^{-1}(x) = \frac{x}{2(x-3)}$

19 $-\frac{\pi}{6}, \frac{\pi}{3}$

20 a 0.5 b 0.68 c 0.32

21 a $\frac{1}{6}$ b $\sqrt{31}$

22 a



b $[0, 72]$

23 $\frac{-2}{3}, \frac{2}{3}$

24 a $32a - 8a^3$

b Maximum value of $A = \frac{128\sqrt{3}}{9}$ when

$a = \frac{2\sqrt{3}}{3}$

25 $b = 3$

26 3.59

27 0.36

28 a \$0 b \$0.65 c 0.425

29 0.37

30 a $\frac{4000}{x^2}$ b Required to show

c $2000(2 + \sqrt{2})$

Extended-response questions

1 a i $\left(\frac{1}{2}, 8\right)$ ii Minimum

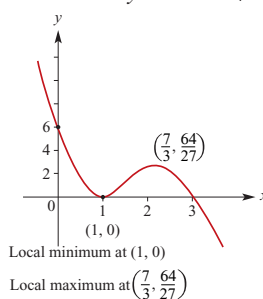
b ii $A = \frac{x}{12}(60 - 5x)$

iii Maximum area is 15 cm^2 .

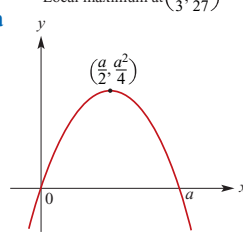
2 a $p = 1, q = 3, k = 2$

b i $m = -2$ ii $y = -2x^3 + 10x^2 - 14x + 6$

iii



3 a



b $\frac{a^3}{6}$ square units

c i $y = \frac{2a^2}{9}, y = \frac{2a^2}{9}$ iii $\frac{a^3}{162}$ square units

4 a $y = -\frac{1}{2}x + \frac{3}{2}$

b i $\frac{dy}{d\theta} = \cos \theta - 2 \sin \theta$

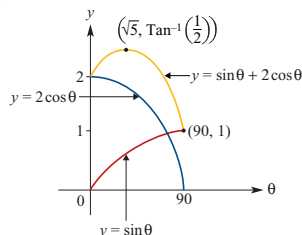
ii $\theta = \tan^{-1}\left(\frac{1}{2}\right) = 26.57^\circ$

iii (26.57, 2.2361), exact values

$\left(\tan^{-1}\left(\frac{1}{2}\right), \sqrt{5}\right)$

iv $r = \sqrt{5}$, $\alpha = 63.435^\circ$

v



c i $Q(2 \sin \theta, 2 \cos \theta)$ ii $\theta = 74.4346^\circ$

5 a i $f'(t) = -100e^{-\frac{t}{10}}(t^2 - 30t + 144)$

ii $f''(t) = 10e^{-\frac{t}{10}}(t^2 - 50t + 444)$

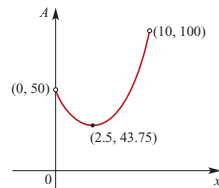
b i $t \in (6, 24)$

ii $t \in (25 - \sqrt{181}, 25 + \sqrt{181})$
 $\approx (11.546, 35)$

iii $t \in (11.546, 24)$

6 a i $A = x^2 - 5x + 50$ ii (0, 10)

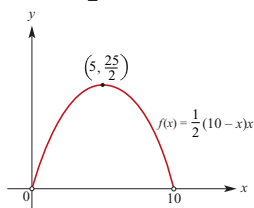
iii



iv Minimum area = 43.75 cm²

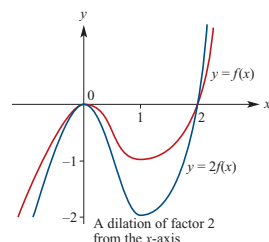
b i $f(x) = \frac{1}{2}(10 - x)x$

ii

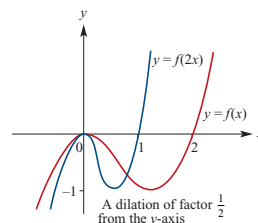


c $AYX : OXYZ : ABY : CBYZ = 1 : 2 : 2 : 3$

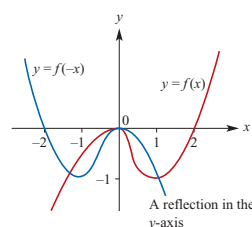
7 a i



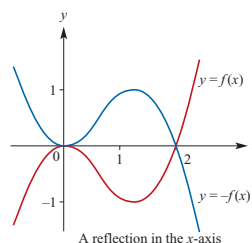
ii



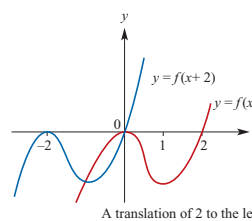
iii



iv

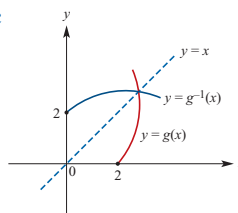


v



b f does not have an inverse function as it is not one-to-one.

c



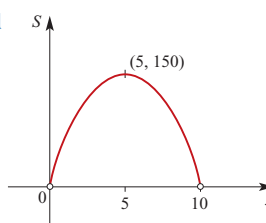
d i Gradient = 15 ii Gradient = $\frac{1}{15}$

8 a i 0.995 ii $x = \pm 0.2$

b i $h(x) = \frac{1}{2}(x - \pi)^2 - 1$
 ≈ -0.98999

9 a $S = (60x - 6x^2)$ b $0 < x < 10$ c $x = 5$

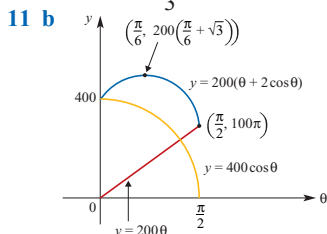
d



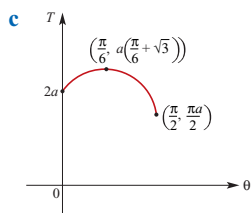
10 a i $OP = \frac{1}{\sin \theta}$ ii $BQ = \frac{1 - \cos \theta}{\sin \theta}$

d Minimum value of $S = \frac{\sqrt{3}}{2}$

$AP = \frac{2\sqrt{3} - 3}{3}$ units



Maximum at $(\frac{\pi}{6}, 200(\frac{\pi}{6} + \sqrt{3}))$



Minimum value for T is $\frac{\pi a}{2}$.

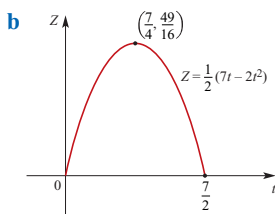
12 a iii $x = 1$ or $x = k - 2$

b i $b = 3 - 2a$; $c = a - 2$ ii $h = a - 2$

iii $a = 0, b = 3, c = -2$

iv $a = -1, b = 5, c = -3$

13 a $Z = \frac{1}{2}(7t - 2t^2)$



c Maximum value of $Z = \frac{49}{16}$, when $t = \frac{7}{4}$

14 a i $\frac{3}{8}$ ii $\frac{4}{15}$

b i $\frac{27}{125}$ ii $\frac{8}{125}$ iii $\frac{38}{125}$

15 a $k = \frac{b}{a^2}$

b i $y = \frac{b}{2a}x + \frac{b}{2}$ ii $(-\frac{a}{2}, \frac{b}{4})$

d $S_1 : S_2 = 27 : 37$

16 a i 0.9332 ii 0.0668 iii 0.1151 iv 0.1151

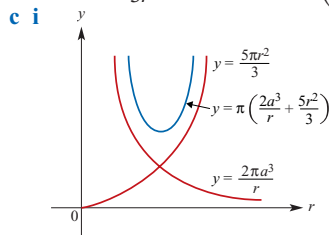
b i 33.3% ii 866.4 iii 199.4

17 $90 - 8\sqrt{3}$ metres from A towards E

18 a i $y = -e^{-n}x + e^{-n}n + e^{-n}$ ii $x = n + 1$

b i $\frac{1}{e^n} \left(1 - \frac{1}{e}\right)$ ii $e : e - 2$

19 b i $h = \frac{3a^3 - 2r^3}{3r^2}$ ii $S = \pi \left(\frac{2a^3}{r} + \frac{5r^2}{3}\right)$



ii Coordinates of local minimum

$\left(\left(\sqrt[3]{0.6}\right)a, \pi a^2 \left(\frac{2}{\sqrt[3]{0.6}} + \frac{5}{3} \left(\sqrt[3]{0.6}\right)^2\right)\right)$

20 a 0.0023

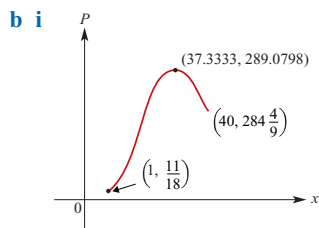
b

Q	$s - 1$	-1
$\Pr(Q = q)$	$\frac{3}{4}$	$\frac{1}{4}$

c $E(Q) = \frac{3}{4}s - 1$ $\text{sd}(Q) = \frac{\sqrt{3}}{4}s$

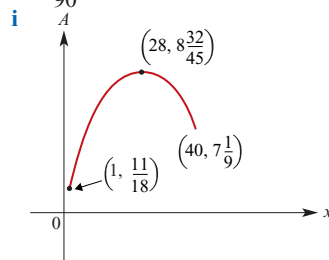
21 a 0.09121 b 0.2611 c 0.275

22 a $\frac{dP}{dx} = \frac{1}{90}(112x - 3x^2)$

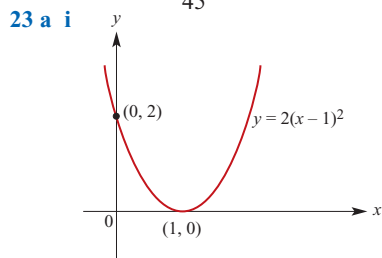


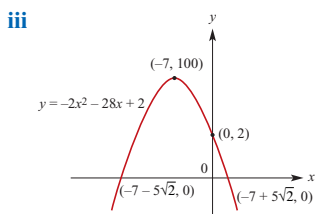
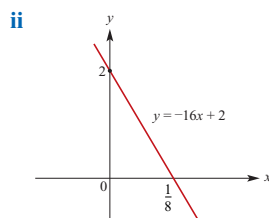
ii The maximum value of P is 289.0798 tonnes.

c $A = \frac{x}{90}(56 - x)$



ii $A_{\max} = 8 \frac{32}{45}$ tonnes/man, $x = 28$





b $\left(\frac{2-3k}{k+2}, \frac{-(2-3k)^2 + 2(k+2)}{k+2} \right)$

i $\left\{ k: -2 < k < \frac{2}{3} \right\}$ **ii** $\left\{ \frac{2}{3} \right\}$

iii $\left\{ k: 0 < k < \frac{14}{9} \right\} \cup \left\{ k: k < -2 \right\}$

iv $\left\{ k: -2 < k < 0 \right\} \cup \left\{ k: k > \frac{14}{9} \right\}$

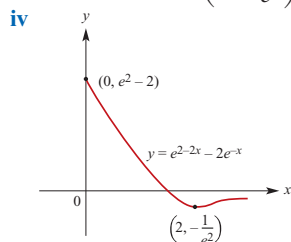
c $k < -2$

d i $k = 0$ or $k = \frac{14}{9}$ **ii** $0 < k < \frac{14}{9}$

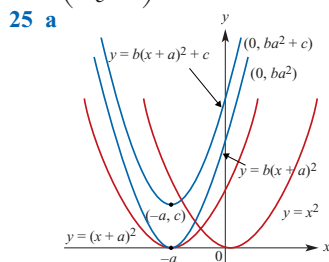
24 a $x = 2 - \log_e 2$

b i $\frac{dy}{dx} = -2e^{2-2x} + 2e^{-x}$

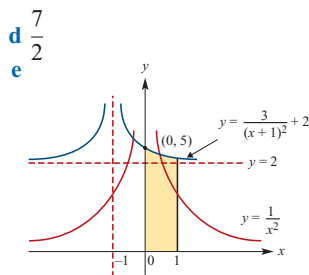
ii $x = 2$ **iii** $\left(2, -\frac{1}{e^2} \right)$



c $\left(-\frac{1}{e^2}, 0 \right)$



- c**
- A dilation of factor 3 from the x -axis
 - A translation of 1 unit in the negative direction of the x -axis
 - A translation of 2 units in the positive direction of the y -axis



26 a i $y = 50$ **ii** $y = x - 25$

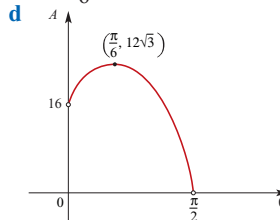
b $a = -\frac{1}{15}$ $c = -\frac{25}{3}$

c i 1250 square units

ii $\frac{14\,375}{18}$ square units

iii $\frac{36\,875}{18}$ square units

27 c $\theta = \frac{\pi}{6}$

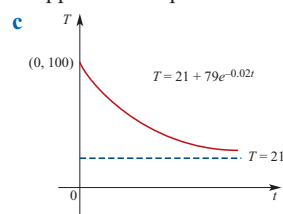


The maximum value of A is $12\sqrt{3}$.

28 a $\mu = 5.0290$, $\sigma = 0.0909$ **b** \$409.28

29 a $k = \frac{1}{10} \log_e \frac{79}{63} \approx 0.02$, $A = 79$

b Approx 2 : 44 p.m.

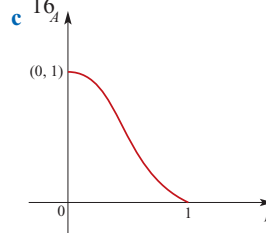


d Average rate of change = $-1.6^\circ \text{C/minute}$.

e i $2.0479^\circ \text{C/minute}$

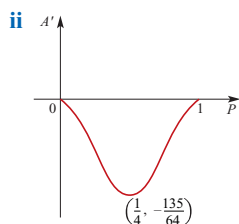
ii $-0.8826^\circ \text{C/minute}$

30 a $\frac{3}{16}$ **b** $b = 4$



d i 0.076 **ii** 0.657

e i $A'(p) = -20p(1-p)^3$



iii $p = \frac{1}{4}$

iv Most rapid rate of change of probabilities is occurring when $p = \frac{1}{4}$.

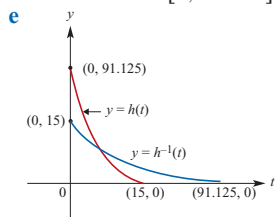
31 a 91.125 cm

c $V = 0.64(4.5 - 0.3t)^3$

d h is a one-to-one function.

$h^{-1}(t) = 15 - \frac{10t^{\frac{1}{3}}}{3}$

dom of $h^{-1} = [0, 91.125]$



32 a 0.065 36

b i 0.6595

ii 0.198 14

c i 23.3%

ii $c = 0.1075$

33 a i 0.32

ii 0.18

iii 0.5

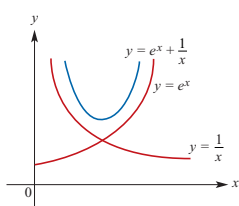
b 0.64

c i 0.043 95

ii 0.999

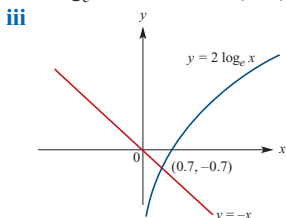
iii $\frac{7}{128}$

34 a, b



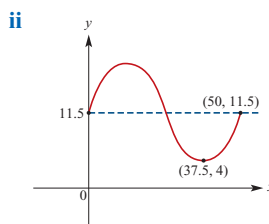
c $y = \frac{1}{x} + e^x$
 $\frac{dy}{dx} = -\frac{1}{x^2} + e^x$

d ii $2 \log_e x < 0, \therefore x \in (0, 1)$

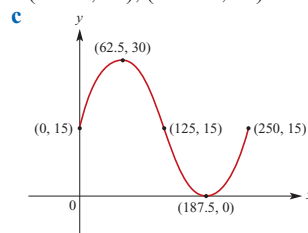


iv (0.7, 3.4)

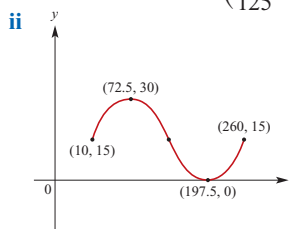
35 a i $e = 12.5, g = 15, d = 37.5,$
 $a = 7.5, b = 7.5$



b (2.704, 10), (22.296, 10)



d i $h(x) = 15 + 15 \sin\left(\frac{\pi}{125}(x - 10)\right)$



36 a $k = 4$

b i $E(X) = \frac{13}{6}$

ii $\frac{10 - \sqrt{2}}{4}$

iii $\frac{\sqrt{2}}{12}$

c 0.1857

37 a $k = \frac{2}{a^2}$

b $E(X) = \frac{a}{3}, \text{Var}(X) = \frac{a^2}{18}$

c $\frac{6 - 4\sqrt{2}}{9}$

d $a = 1000(\sqrt{2} + 2)$

38 a 7

c $y = \frac{1}{10}x - \log_e 10$

d ii 36.852

39 a $k = n + 1$

b $E(X) = \frac{n+1}{n+2}$

c $\frac{n+1}{(n+2)^2(n+3)}$

d Median = $\frac{n+1}{2}\sqrt{\frac{1}{2}}$

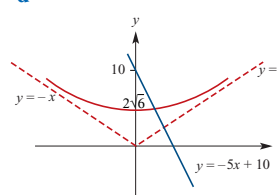
e Mode = 1

40 a $\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 24}}$

b (0, 2\sqrt{6})

c Even

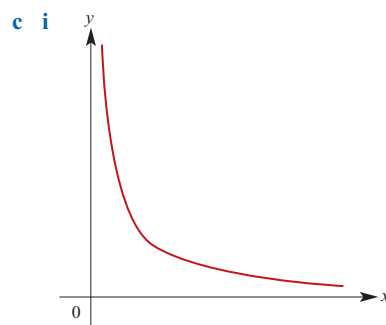
d



e $y = -5x + 10$

f 14 units/second

- h** $12 \log_e (\sqrt{7} - 1) - 2\sqrt{7} + \frac{35}{2}$
- 41 a** $r = \frac{\sqrt{3}x}{3}$ **b** $V = \frac{\pi x^3}{9}$
- c** $\frac{dV}{dx} = \frac{1}{3}\pi x^2$ **d** $\frac{dx}{dt} = \frac{60}{\pi x^2}$
- e** $x^3 = \frac{180t}{\pi}$ **f** $\frac{5^{\frac{1}{3}} 2^{\frac{8}{3}} \pi^{\frac{1}{3}}}{3}$
- 42 a i** 0 **ii** -0.6745 **iii** 0.6745
- iv** 1.3490 **v** 99.3% **vi** 0.7%
- b i** μ **ii** $\mu - 0.6745\sigma$
- iii** $\mu + 0.6745\sigma$ **iv** 1.3490σ
- v** 0.9930 **vi** 0.7%
- 43 a i** $\frac{1}{1-b}$ **ii** $1 + \sqrt{b-1}$
- b i** 1 **ii** $1 + e^{-7}$
- c i** $\frac{b(b-2)}{2(b-1)}$ **ii** $9 + \sqrt{65}$
- d** e
- 44 a i** $\frac{-(b+1)}{b^2}$ **ii** $\left(\frac{2b^2}{b+1}\right)^{\frac{1}{3}}$
- b i** $\frac{(b^2+1)(b-1)}{2b^2}$
- ii** 3
- iii** Required to show
- c** Required to show
- 45 a** 100
- b** (16, 100)
- c** $\frac{7000}{3}$ square units
- d** $OC = \frac{70}{3}$. The average value of a function over the interval $[0, a]$ is defined as
- $$\frac{\int_0^a f(x) dx}{a} = \frac{\text{area of rectangle } OABC}{\text{width of rectangle } OABC} = \frac{\text{height of rectangle } OABC}{\text{width of rectangle } OABC}$$
- e i** $f(g(x)) = |x^2 - 8|x| - 20|$
- ii** $h'(x) = \begin{cases} 2x + 8 & x < -10 \\ -2x - 8 & -10 < x < 0 \\ -2x + 8 & 0 < x < 10 \\ 2x - 8 & x > 10 \end{cases}$
- 46 a** $\left(\frac{m}{n}, \left(\frac{m}{n}\right)^m e^{n-m}\right)$, local maximum turning point
- b** $\left(\frac{m-1}{n}, \left(\frac{m-1}{n}\right)^m e^{n-m+1}\right)$
- c i** $\frac{4}{e^2}$ **ii** $1 - \frac{5}{e^2}$ **iii** 1
- 47 a i** q **ii** $\frac{1}{q}$ **iii** $\frac{1}{q^2}$
- iv** Required to show
- b** $\frac{1}{3}$



- ii** 0.37
- iii** 69.31
- 48 a i** $5p^2 - 4p^3$
- ii** 0.4
- b i** 0.064
- ii** 0.4032
- c** $\begin{bmatrix} p & 4p^2 \\ 1-p & 1-4p^2 \end{bmatrix}$
- d** $\frac{4p^2}{1-p+4p^2}, \frac{1-p}{1-p+4p^2}$
- e i** 0.992
- ii** 0.028672

Appendix A

Exercise A1

- 1** 63 **2** 26 **3** 336
- 4 a** 5040 **b** 210
- 5 a** 120 **b** 120
- 6** 18
- 7 a** 5 852 925 **b** 1 744 200
- 8** 100 386
- 9 a** 792 **b** 336
- 10 a** 200 **b** 75 **c** 6
- d** 462 **e** 81

Exercise A2

- 1 a** $\sum_{i=1}^4 i^3 = 1 + 8 + 27 + 64 = 100$
- b** $\sum_{k=1}^5 k^3 = 1 + 8 + 27 + 64 + 125 = 225$
- c** $\sum_{i=1}^5 (-1)^i i = -1 + 2 - 3 + 4 - 5 = -3$
- d** $\frac{1}{5} \sum_{i=1}^5 i = \frac{1}{5}(1 + 2 + 3 + 4 + 5) = 3$
- e** $\sum_{i=1}^6 i = 1 + 2 + 3 + 4 + 5 + 6 = 21$
- f** $\sum_{k=1}^4 (k-1)^2 = 0 + 1 + 4 + 9 = 14$

$$\text{g } \frac{1}{3} \sum_{i=1}^4 (i-2)^2 = \frac{1}{3}(1+0+1+4) = 2$$

$$\text{h } \sum_{i=1}^6 i^2 = 1+4+9+16+25+36 = 91$$

$$2 \text{ a } \sum_{i=1}^n i \quad \text{b } \sum_{i=1}^{11} x_i \quad \text{c } \frac{1}{10} \sum_{i=1}^{10} x_i$$

$$\text{d } \sum_{i=1}^{n+1} i^4 \quad \text{e } \sum_{i=1}^5 \frac{1}{i}$$

$$3 \text{ a } \sum_{i=1}^n x^i = x + x^2 + x^3 + \cdots + x^n$$

$$\text{b } \sum_{i=0}^5 x^i \cdot 2^{5-i} = 32 + 16x + 8x^2 + 4x^3 + 2x^4 + x^5$$

$$\text{c } \sum_{i=0}^6 x^i \times 2^i \times 3^{6-i} = 3^6 + 2 \times 3^5 \times x + 2^2 \times 3^4 \times x^2 + 2^3 \times 3^3 \times x^3 + 2^4 \times 3^2 \times x^4 + 2^5 \times 3 \times x^5 + 2^6 \times x^6$$

$$\text{d } \sum_{i=0}^4 (x-x_i)^i = (x-x_1) + (x-x_2)^2 + (x-x_3)^3 + (x-x_4)^4$$

$$4 \text{ a } \sum_{i=0}^5 x^{5-i} \cdot 3^i \quad \text{b } \sum_{i=0}^5 x^{5-i} \cdot (-3)^i$$

$$\text{c } \sum_{i=0}^2 x^{2-i} \cdot 2^{2-i} \quad \text{d } \sum_{i=0}^3 (2x)^{3-i} \cdot 3^i$$

Exercise A3

$$1 \text{ a } x^6 + 36x^5 + 540x^4 + 4320x^3 + 19440x^2 + 46656x + 46656$$

$$\text{b } 32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$$

$$\text{c } 32x^5 - 80x^4 + 80x^3 - 40x^2 + 10x - 1$$

$$\text{d } 64x^6 + 576x^5 + 2160x^4 + 4320x^3 + 4860x^2 + 2916x + 729$$

$$\text{e } 64x^6 - 1152x^5 + 8640x^4 - 34560x^3 + 77760x^2 - 93312x + 46656$$

$$\text{f } 16x^4 - 96x^3 + 216x^2 - 216x + 81$$

$$\text{g } x^6 - 12x^5 + 60x^4 - 160x^3 + 240x^2 - 192x + 64$$

$$\text{h } x^{10} + 10x^9 + 45x^8 + 120x^7 + 210x^6 + 252x^5 + 210x^4 + 120x^3 + 45x^2 + 10x + 1$$

$$2 \text{ a } -960x^3 \quad \text{b } 960x^3 \quad \text{c } -960x^3$$

$$\text{d } 192456x^5 \quad \text{e } 1732104x^5 \quad \text{f } -25344b^7x^5$$

$$3 - \frac{16}{243}x^7 \quad 4 - 336798x^6$$

$$5 (-x+1)^{11} = -x^{11} + 11x^{10} - 55x^9 + 165x^8 - 330x^7 + 462x^6 - 462x^5 + 330x^4 - 165x^3 + 55x^2 - 11x + 1$$

$$6 \text{ a } 40 \quad \text{b } -160 \quad \text{c } -80$$

$$\text{d } 181440 \quad \text{e } 432 \quad \text{f } 1080$$

$$7 \text{ 83026944} \quad 8 - 768$$